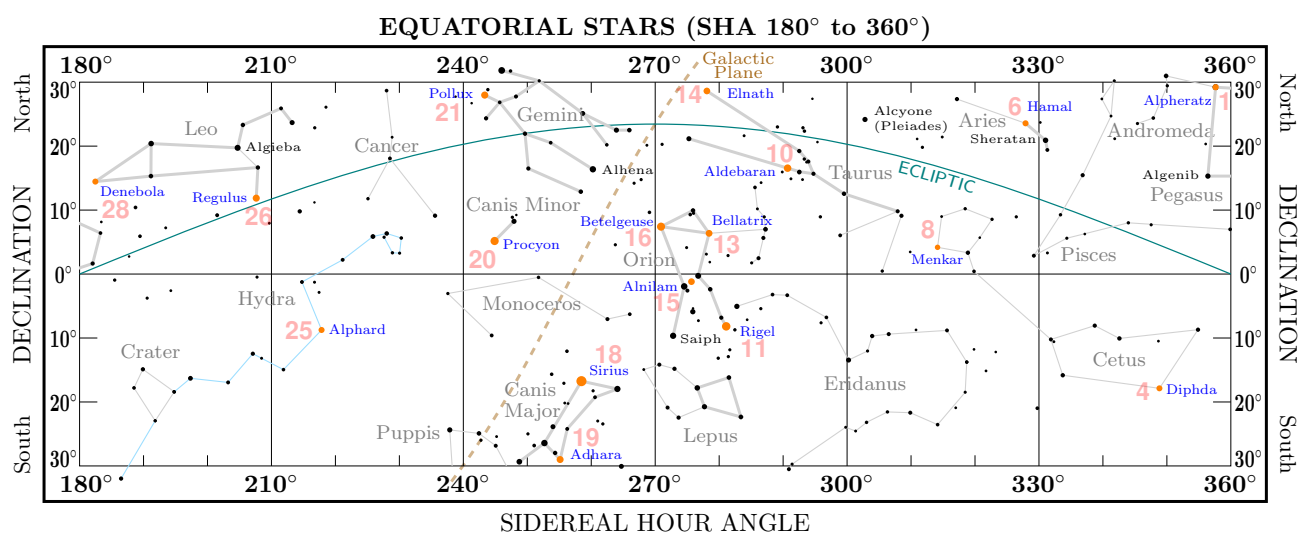
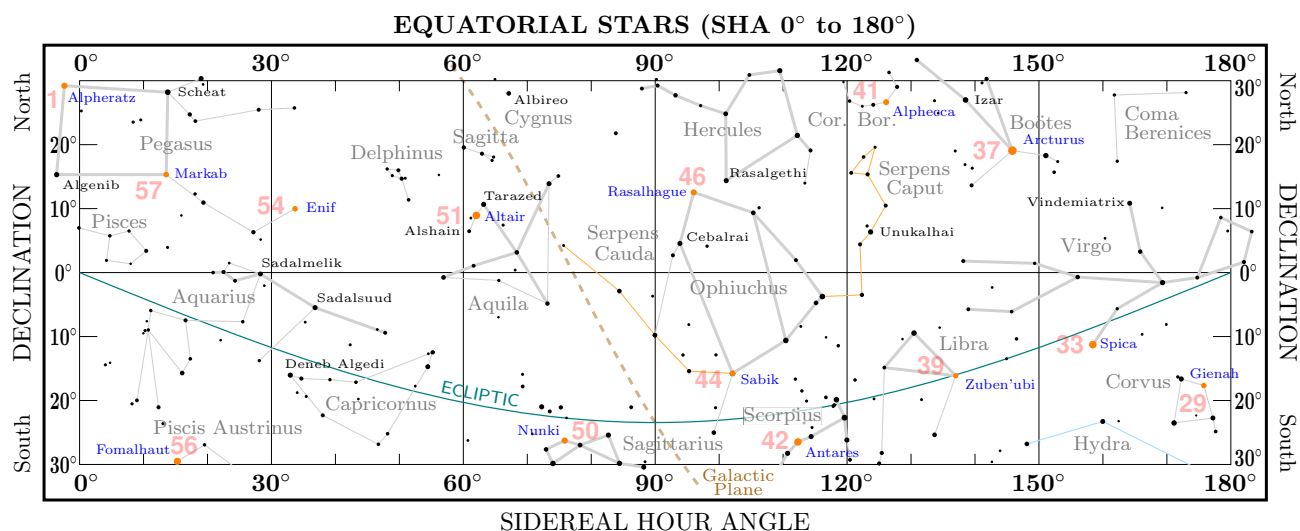




THE NAUTICAL ALMANAC

2029

Author: Andrew BAUER
Original concept from: Enno RODEGERDTS

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Disclaimer: These are computer generated tables - use them at your own risk. The accuracy has been randomly checked with JPL HORIZONS System, but cannot be guaranteed. The author claims no liability for any consequences arising from use of these tables. Besides, this publication only contains the 'daily pages' of the Nautical Almanac: an official version of the Nautical Almanac is indispensable.

Information in the data page footers

Information pertaining to the IERS EOP data has been added to the odd data page footers if using MiKTeX or TeX Live (2020 or later). The International Earth Rotation Service (IERS) provides accurate data (updated weekly) on the Earth Orientation Parameters (EOP).

Earth's speed of rotation is not constant, i.e. the day length fluctuates.¹ This is due to *internal torques* caused by relative movements and mass redistribution of Earth's core, mantle, oceans, atmosphere, and cryosphere. This has an immediate impact on the GHA values of all celestial objects.

The IERS monitors and measures several parameters taking the actual speed of Earth's rotation into account. Their measured data begins on 2nd January 1973. Predictive data begins following the last day of (obtained) data and extends about 360 days into the future. (The IERS results are published with a delay of about 18-hours between the date of publication and the last available date with measured EOP.²) These Nautical Almanac daily pages take the (measured or predicted) UT1-UTC values into account providing highly accurate navigational data especially if the predictions are fairly recent.

As long as either measured or predicted data is available the footer will show:

[IERS Earth Orientation data as of dd-mmm-yyyy](#)

This indicates that IERS EOP data is in use - older dates are measured; newer dates are predictions.

If the final date of IERS prediction data is on the current data page, the footer shows:

[IERS Earth Orientation predictions end dd-mmm-yyyy](#)

Pages with dates beyond the final date of IERS prediction data have the following footer:

[No IERS EOP prediction data available](#)

Skyfield then defaults to using the ΔT and leap second files that ship with Skyfield internally.

The footers mentioned are only displayed as long as `'useIERS = True'` is set in `config.py` to enable use of IERS EOP data.

Brief historical overview

The story begins with the XEphem astronomical library, which is declared 'end of life' by its author, Elwood Charles Downey, as no further updates are planned. He generously gave permission for use of XEphem code in Ephem (also known as Pyephem), an astronomical library authored by Brandon Rhodes. Enno Rodegerdts (<https://sv-inua.net/>) created the original Nautical Almanac 'daily pages' in Pyalmanac using Python 2 and LaTeX. After contacting him I obtained permission for its future enhancement and maintenance. Pyalmanac uses Ephem.

Meanwhile Brandon Rhodes was working on a far more sophisticated astronomical library, Skyfield. This was 'state of the art' and clearly surpassed the 'Jean Meeus'-based Pyephem/Ephem. Skyfield uses NASA's NAIF (Navigation and Ancillary Information Facility) SPICE algorithms. The results agree with those from the HORIZONS System (*operated by NASA JPL (Jet Propulsion Laboratory) SSD (Solar System Dynamics) group, not by NAIF*). This in turn implies that celestial positions calculated by Skyfield agree with those generated by the United States Naval Observatory and their *Astronomical Almanac* to within 0.0005 arcseconds (half a milliarcsecond).

Pyephem was then in 'maintenance mode'. Clearly Pyalmanac needed adaptation to use Skyfield, and thus SFalmanac was born. However its performance was poor regarding the calculation of 'events' such as: sunrise, sunset, moonrise, moonset, civil twilight start/end and nautical twilight start/end. An interim (faster) solution was required.

A hybrid application, originally named Skyalmanac, was developed using Ephem to calculate 'events' and Skyfield for the rest. This was indeed much faster at the cost of poorer 'event time' data. It took a while to find a better solution: multiprocessing, which was built into SFalmanac. This now could compare to the execution times in Pyalmanac but with improved data.

New functionality was added to SFalmanac, e.g. lunar phase as a graphic; Lunar Distance tables and charts. The original Skyalmanac is deprecated and has now been replaced with the latest SFalmanac code, so Skyalmanac and SFalmanac are now identical apart from the name. Since April 2019 <http://thenauticalalmanac.com> has been publishing Celestial Navigation related material with software provided here.

¹https://en.wikipedia.org/wiki/Day_length_fluctuations

²<https://hpiers.obspm.fr/eoppc/bul/bulb/explanatory.html>

January 01, 02, 03 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	100°56.1	200°35.7	S22°31.2	276°00.5	N00°33.2	257°20.6	S08°29.1	67°54.2	N10°38.1	Alpheratz	357°31.4 29°15.3
1	115°58.5	215°34.8		291°02.1	32.8	272°22.8	29.2	82°56.7	38.1	Ankaa	353°04.1 -42°09.0
2	131°01.0	230°33.9	31.8	306°03.7	32.5	287°24.9	29.3	97°59.2	38.1	Schedar	349°27.2 56°42.1
3	146°03.5	245°32.9	· · 32.1	321°05.3	· · 32.2	302°27.1	· · 29.4	113°01.7	· · 38.1	Diphda	348°44.1 -17°49.7
4	161°05.9	260°32.0	32.4	336°06.9	31.8	317°29.3	29.5	128°04.2	38.1	Achernar	335°17.8 -57°05.6
5	176°08.4	275°31.1	32.7	351°08.4	31.5	332°31.5	29.6	143°06.6	38.1	Hamal	327°47.3 23°36.1
6	191°10.9	290°30.2	S22°33.0	6°10.0	N00°31.1	347°33.7	S08°29.7	158°09.1	N10°38.1	Polaris	311°54.8 89°23.3
7	206°13.3	305°29.3	33.3	21°11.6	30.8	2°35.9	29.8	173°11.6	38.1	Acamar	315°09.1 -40°11.5
8	221°15.8	320°28.4	33.6	36°13.2	30.5	17°38.1	29.9	188°14.1	38.1	Menkar	314°02.5 4°12.2
9	236°18.3	335°27.5	· · 33.9	51°14.8	· · 30.1	32°40.3	· · 30.0	203°16.6	· · 38.1	Mirfak	308°23.2 49°58.0
10	251°20.7	350°26.6	34.2	66°16.3	29.8	47°42.5	30.1	218°19.1	38.1	Aldebaran	290°35.6 16°34.0
11	266°23.2	5°25.6	34.5	81°17.9	29.5	62°44.7	30.2	233°21.6	38.1	Rigel	281°00.4 -8°10.2
12	281°25.7	20°24.7	S22°34.8	96°19.5	N00°29.1	77°46.8	S08°30.3	248°24.0	N10°38.1	Capella	280°16.6 46°01.6
13	296°28.1	35°23.8	35.1	111°21.1	28.8	92°49.0	30.4	263°26.5	38.1	Bellatrix	278°19.1 6°22.4
14	311°30.6	50°22.9	35.4	126°22.7	28.5	107°51.2	30.5	278°29.0	38.1	Elnath	277°57.4 28°37.8
15	326°33.0	65°22.0	· · 35.7	141°24.3	· · 28.1	122°53.4	· · 30.6	293°31.5	· · 38.1	Alnilam	275°34.1 -1°11.1
16	341°35.5	80°21.1	36.0	156°25.8	27.8	137°55.6	30.7	308°34.0	38.1	Betelgeuse	270°48.2 7°24.6
17	356°38.0	95°20.2	36.3	171°27.4	27.5	152°57.8	30.8	323°36.5	38.1	Canopus	263°50.4 -52°42.8
18	11°40.4	110°19.2	S22°36.6	186°29.0	N00°27.1	168°00.0	S08°30.9	338°39.0	N10°38.1	Sirius	258°23.0 -16°45.5
19	26°42.9	125°18.3	36.9	201°30.6	26.8	183°02.2	31.0	353°41.4	38.2	Adhara	255°02.9 -29°00.8
20	41°45.4	140°17.4	37.2	216°32.2	26.5	198°04.4	31.1	8°43.9	38.2	Procyon	244°47.1 5°08.8
21	56°47.8	155°16.5	· · 37.5	231°33.8	· · 26.1	213°06.6	· · 31.2	23°46.4	· · 38.2	Pollux	243°13.0 27°57.1
22	71°50.3	170°15.6	37.7	246°35.4	25.8	228°08.8	31.3	38°48.9	38.2	Avior	234°12.7 -59°36.1
23	86°52.8	185°14.7	38.0	261°37.0	25.5	243°11.0	31.4	53°51.4	38.2	Suhail	222°43.5 -43°33.0
Mer.pass. 17:13		ν -0.9' d 0.3' m -3.91		ν 1.6' d -0.3' m 0.61		ν 2.2' d 0.1' m -1.91		ν 2.5' d 0.0' m 0.21		Miaplacidus	221°36.6 -69°50.1
										Alphard	217°44.3 -8°47.2
										Regulus	207°30.8 11°49.3
										Dubhe	193°37.0 61°35.2
										Denebola	182°21.6 14°24.4
										Gienah	175°40.2 -17°42.2
										Acrux	172°56.2 -63°15.4
										Gacrux	171°47.8 -57°16.4
										Alioth	166°10.3 55°47.8
										Spica	158°18.9 -11°18.8
										Alkaid	152°49.7 49°09.8
										Hadar	148°31.5 -60°30.5
										Menkent	147°53.9 -36°30.6
										Arcturus	145°45.1 19°01.7
										Rigel Kent.	139°36.0 -60°57.1
										Kochab	137°20.8 74°01.8
										Zuben'ubi	136°52.6 -16°09.7
										Alphecca	126°01.3 26°36.9
										Antares	112°12.1 -26°29.6
										Atria	107°03.9 -69°04.5
										Sabik	101°59.4 -15°45.5
										Shaula	96°06.4 -37°07.3
										Rasalhague	95°55.9 12°32.4
										Eltanin	90°41.2 51°29.1
										Kaus Aust.	83°28.6 -34°22.1
										Vega	80°31.5 38°48.7
										Nunki	75°44.1 -26°15.5
										Altair	61°57.1 8°56.8
										Peacock	53°01.3 -56°38.5
										Deneb	49°23.9 45°23.3
										Enif	33°35.8 10°00.6
										Al Na'ir	27°29.3 -46°49.3
										Fomalhaut	15°11.2 -29°28.1
										Scheat	13°42.2 28°14.7
										Markab	13°26.8 15°21.9
										Jan 01 Mon	SHA Mer.pass
										Venus	99°39.6 10:38
										Mars	175°04.5 05:35
										Jupiter	156°24.5 06:50
										Saturn	326°58.1 19:25
										Jan 02 Tue	SHA Mer.pass
										Venus	98°18.5 10:40
										Mars	174°43.3 05:33
										Jupiter	156°17.9 06:46
										Saturn	326°58.6 19:21
										Jan 03 Wed	SHA Mer.pass
										Venus	96°57.3 10:41
										Mars	174°22.5 05:30
										Jupiter	156°11.5 06:43
										Saturn	326°59.1 19:17
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1
Mer.pass. 17:06		ν -0.9' d 0.3' m -3.91		ν 1.6' d -0.3' m 0.57		ν 2.2' d 0.1' m -1.92		ν 2.5' d 0.0' m 0.22			

Sun							Moon						
h													
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°08.0	S22°59.6	355°02.5	6.9'	N22°34.2	-7.0'	58.2'						
1	194°07.7	59.3	9°28.4	6.9'	22°27.2	-7.1'	58.2'						
2	209°07.4	59.1	23°54.4	7.0'	22°20.1	-7.2'	58.2'						
3	224°07.1	· · 58.9	38°20.3	7.0'	22°12.8	-7.4'	58.2'						
4	239°06.8	58.7	52°46.4	7.1'	22°05.5	-7.5'	58.3'						
5	254°06.5	58.5	67°12.4	7.1'	21°57.9	-7.7'	58.3'						
6	269°06.2	S22°58.3	81°38.5	7.1'	N21°50.3	-7.8'	58.3'						
7	284°05.9	58.1	96°04.6	7.2'	21°42.5	-7.9'	58.3'						
8	299°05.6	57.9	110°30.8	7.2'	21°34.5	-8.1'	58.4'						
9	314°05.3	· · 57.7	124°57.0	7.3'	21°26.4	-8.2'	58.4'						
10	329°05.1	57.4	139°23.3	7.3'	21°18.2	-8.4'	58.4'						
11	344°04.8	57.2	153°49.6	7.3'	21°09.9	-8.5'	58.4'						
12	359°04.5	S22°57.0	168°15.9	7.4'	N21°01.4	-8.6'	58.4'						
13	14°04.2	56.8	182°42.3	7.4'	20°52.8	-8.8'	58.5'						
14	29°03.9	56.6	197°08.7	7.5'	20°44.0	-8.9'	58.5'						
15	44°03.6	· · 56.4	211°35.2	7.5'	20°35.1	-9.0'	58.5'						
16	59°03.3	56.1	226°01.7	7.6'	20°26.1	-9.1'	58.5'						
17	74°03.0	55.9	240°28.2	7.6'	20°17.0	-9.3'	58.5'						
18	89°02.7	S22°55.7	254°54.9	7.7'	N20°07.7	-9.4'	58.6'						
19	104°02.4	55.5	269°21.5	7.7'	19°58.3	-9.5'	58.6'						
20	119°02.1	55.3	283°48.2	7.8'	19°48.8	-9.6'	58.6'						
21	134°01.8	· · 55.0	298°15.0	7.8'	19°39.1	-9.8'	58.6'						
22	149°01.6	54.8	312°41.8	7.9'	19°29.4	-9.9'	58.6'						
23	164°01.3	54.6	327°08.6	7.9'	19°19.5	-10.0'	58.7'						
SD = 16.3' $d = -0.2'$							SD = 15.9'						

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179°01.0	S22°54.4	341°35.6	8.0'	N19°09.5	-10.1'	58.7'
1	194°00.7	54.1	356°02.5	8.0'	18°59.3	-10.2'	58.7'
2	209°00.4	53.9	10°29.5	8.1'	18°49.1	-10.4'	58.7'
3	224°00.1	· · 53.7	24°56.6	8.1'	18°38.7	-10.5'	58.7'
4	238°59.8	53.4	39°23.7	8.2'	18°28.2	-10.6'	58.8'
5	253°59.5	53.2	53°50.9	8.2'	18°17.6	-10.7'	58.8'
6	268°59.2	S22°53.0	68°18.1	8.3'	N18°06.9	-10.8'	58.8'
7	283°58.9	52.8	82°45.4	8.3'	17°56.1	-10.9'	58.8'
8	298°58.7	52.5	97°12.7	8.4'	17°45.2	-11.0'	58.8'
9	313°58.4	· · 52.3	111°40.1	8.4'	17°34.1	-11.2'	58.8'
10	328°58.1	52.1	126°07.5	8.5'	17°23.0	-11.3'	58.9'
11	343°57.8	51.8	140°35.0	8.5'	17°11.7	-11.4'	58.9'
12	358°57.5	S22°51.6	155°02.6	8.6'	N17°00.3	-11.5'	58.9'
13	13°57.2	51.3	169°30.2	8.7'	16°48.9	-11.6'	58.9'
14	28°56.9	51.1	183°57.8	8.7'	16°37.3	-11.7'	58.9'
15	43°56.6	· · 50.9	198°25.6	8.8'	16°25.6	-11.8'	58.9'
16	58°56.4	50.6	212°53.3	8.8'	16°13.8	-11.9'	58.9'
17	73°56.1	50.4	227°21.1	8.9'	16°02.0	-12.0'	59.0'
18	88°55.8	S22°50.2	241°49.0	8.9'	N15°50.0	-12.1'	59.0'
19	103°55.5	49.9	256°16.9	9.0'	15°37.9	-12.2'	59.0'
20	118°55.2	49.7	270°44.9	9.0'	15°25.8	-12.3'	59.0'
21	133°54.9	· · 49.4	285°13.0	9.1'	15°13.5	-12.3'	59.0'
22	148°54.6	49.2	299°41.1	9.1'	15°01.2	-12.4'	59.0'
23	163°54.3	48.9	314°09.2	9.2'	14°48.7	-12.5'	59.0'
	SD = 16.3'	$d = -0.2'$			SD = 16.0'		

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°54.1	S22°48.7	328°37.4	9.3'	N14°36.2	-12.6'	59.1'
1	193°53.8	48.5	343°05.7	9.3'	14°23.6	-12.7'	59.1'
2	208°53.5	48.2	357°34.0	9.4'	14°10.9	-12.8'	59.1'
3	223°53.2	· · 48.0	12°02.3	9.4'	13°58.1	-12.9'	59.1'
4	238°52.9	47.7	26°30.7	9.5'	13°45.2	-13.0'	59.1'
5	253°52.6	47.5	40°59.2	9.5'	13°32.3	-13.0'	59.1'
6	268°52.3	S22°47.2	55°27.7	9.6'	N13°19.2	-13.1'	59.1'
7	283°52.1	47.0	69°56.3	9.6'	13°06.1	-13.2'	59.1'
8	298°51.8	46.7	84°24.9	9.7'	12°52.9	-13.3'	59.1'
9	313°51.5	· · 46.5	98°53.5	9.7'	12°39.7	-13.3'	59.2'
10	328°51.2	46.2	113°22.3	9.8'	12°26.3	-13.4'	59.2'
11	343°50.9	46.0	127°51.0	9.8'	12°12.9	-13.5'	59.2'
12	358°50.6	S22°45.7	142°19.8	9.9'	N11°59.4	-13.6'	59.2'
13	13°50.3	45.4	156°48.7	9.9'	11°45.9	-13.6'	59.2'
14	28°50.1	45.2	171°17.6	10.0'	11°32.3	-13.7'	59.2'
15	43°49.8	· · 44.9	185°46.6	10.0'	11°18.6	-13.8'	59.2'
16	58°49.5	44.7	200°15.6	10.0'	11°04.8	-13.8'	59.2'
17	73°49.2	44.4	214°44.6	10.1'	10°51.0	-13.9'	59.2'
18	88°48.9	S22°44.2	229°13.7	10.1'	N10°37.1	-13.9'	59.2'
19	103°48.6	43.9	243°42.8	10.2'	10°23.2	-14.0'	59.2'
20	118°48.4	43.6	258°12.0	10.2'	10°09.2	-14.1'	59.3'
21	133°48.1	· · 43.4	272°41.2	10.3'	09°55.1	-14.1'	59.3'
22	148°47.8	43.1	287°10.5	10.3'	09°41.0	-14.2'	59.3'
23	163°47.5	42.9	301°39.8	10.3'	09°26.8	-14.2'	59.3'
	SD = 16.3'	$d = -0.2'$			SD = 16.1'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:23	10:38	■	■	13:30	15:46
N 70°	08:04	09:47	■	■	14:21	16:05
68°	07:49	09:15	■	■	14:53	16:20
66°	07:37	08:52	10:25	13:43	15:17	16:32
64°	07:26	08:33	09:48	14:21	15:35	16:43
62°	07:17	08:18	09:22	14:47	15:51	16:52
60°	07:09	08:05	09:02	15:07	16:03	17:00
N 58°	07:02	07:54	08:45	15:24	16:15	17:07
56°	06:55	07:44	08:31	15:38	16:24	17:13
54°	06:50	07:35	08:19	15:50	16:33	17:19
52°	06:44	07:27	08:08	16:01	16:41	17:24
50°	06:39	07:20	07:58	16:10	16:48	17:30
45°	06:28	07:05	07:38	16:30	17:04	17:41
N 40°	06:18	06:52	07:22	16:46	17:17	17:51
35°	06:09	06:40	07:08	17:00	17:28	18:00
30°	06:00	06:30	06:56	17:12	17:38	18:08
20°	05:44	06:12	06:36	17:33	17:57	18:24
N 10°	05:28	05:55	06:17	17:51	18:14	18:40
0°	05:12	05:38	06:00	18:08	18:30	18:56
S 10°	04:53	05:20	05:43	18:25	18:48	19:15
20°	04:31	05:00	05:25	18:43	19:08	19:37
30°	04:03	04:36	05:03	19:05	19:32	20:05
35°	03:45	04:21	04:51	19:18	19:47	20:23
40°	03:22	04:03	04:36	19:32	20:05	20:46
45°	02:53	03:41	04:18	19:50	20:26	21:15
S 50°	02:09	03:13	03:57	20:11	20:55	21:58
52°	01:43	02:58	03:46	20:22	21:09	22:24
54°	01:04	02:41	03:34	20:34	21:27	23:02
56°	—:—	02:20	03:21	20:47	21:47	—:—
58°	—:—	01:52	03:05	21:03	22:14	—:—
S 60°	—:—	01:10	02:45	21:22	22:56	—:—

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°		15:35	18:27		12:14	11:13
N 70°		16:21	18:45		11:26	10:53
68°	14:20	16:51	18:59	11:31	10:55	10:37
66°	15:07	17:13	19:10	10:44	10:32	10:23
64°	15:37	17:30	19:20	10:13	10:13	10:12
62°	16:00	17:44	19:28	09:49	09:58	10:03
60°	16:18	17:56	19:35	09:30	09:45	09:55
N 58°	16:33	18:06	19:41	09:15	09:34	09:48
56°	16:46	18:15	19:46	09:01	09:24	09:42
54°	16:57	18:23	19:51	08:50	09:16	09:36
52°	17:07	18:30	19:55	08:39	09:08	09:31
50°	17:15	18:37	19:59	08:30	09:01	09:26
45°	17:34	18:50	20:07	08:10	08:46	09:16
N 40°	17:49	19:01	20:14	07:54	08:33	09:07
35°	18:02	19:11	20:20	07:41	08:23	09:00
30°	18:13	19:19	20:25	07:29	08:13	08:53
20°	18:32	19:34	20:34	07:09	07:57	08:42
N 10°	18:48	19:46	20:42	06:51	07:43	08:32
0°	19:03	19:58	20:50	06:34	07:29	08:23
S 10°	19:19	20:09	20:57	06:17	07:16	08:13
20°	19:35	20:21	21:05	05:59	07:01	08:03
30°	19:53	20:35	21:13	05:38	06:44	07:51
35°	20:04	20:43	21:19	05:26	06:35	07:44
40°	20:16	20:53	21:24	05:12	06:23	07:36
45°	20:31	21:03	21:31	04:55	06:10	07:27
S 50°	20:49	21:16	21:39	04:34	05:54	07:16
52°	20:57	21:22	21:43	04:24	05:46	07:10
54°	21:06	21:29	21:47	04:13	05:37	07:05
56°	21:16	21:36	21:51	04:00	05:28	06:58
58°	21:28	21:44	21:56	03:45	05:16	06:51
S 60°	21:42	21:53	22:01	03:27	05:04	06:43

January 04, 05, 06 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	103°53.5	199°29.4	S22°50.6	277°55.5	N00°09.5	259°58.7	S08°35.9	70°52.9	N10°38.6	Alpheratz	357°31.4	29°15.3											
1	118°56.0	214°28.5	50.8	292°57.1	09.2	275°00.9	36.0	85°55.4	38.6	Ankaa	353°04.1	-42°09.0											
2	133°58.4	229°27.6	51.0	307°58.7	08.9	290°03.1	36.1	100°57.8	38.6	Schedar	349°27.2	56°42.1											
3	149°00.9	244°26.7	· · 51.3	323°00.3	· · 08.6	305°05.3	· · 36.2	116°00.3	· · 38.6	Diphda	348°44.1	-17°49.7											
4	164°03.4	259°25.7	51.5	338°01.9	08.2	320°07.5	36.3	131°02.8	38.6	Achernar	335°17.8	-57°05.6											
5	179°05.8	274°24.8	51.7	353°03.5	07.9	335°09.8	36.4	146°05.3	38.6	Hamal	327°47.4	23°36.1											
6	194°08.3	289°23.9	S22°51.9	8°05.1	N00°07.6	350°12.0	S08°36.5	161°07.7	N10°38.6	Polaris	311°56.0	89°23.4											
7	209°10.8	304°22.9	52.1	23°06.8	07.3	5°14.2	36.6	176°10.2	38.6	Acamar	315°09.1	-40°11.5											
8	224°13.2	319°22.0	52.4	38°08.4	07.0	20°16.4	36.7	191°12.7	38.6	Menkar	314°02.6	4°12.2											
9	239°15.7	334°21.1	· · 52.6	53°10.0	· · 06.7	35°18.6	· · 36.7	206°15.2	· · 38.6	Mirfak	308°23.2	49°58.0											
10	254°18.1	349°20.2	52.8	68°11.6	06.3	50°20.8	36.8	221°17.6	38.6	Aldebaran	290°35.6	16°34.0											
11	269°20.6	4°19.2	53.0	83°13.2	06.0	65°23.0	36.9	236°20.1	38.7	Rigel	281°00.4	-8°10.2											
12	284°23.1	19°18.3	S22°53.2	98°14.9	N00°05.7	80°25.2	S08°37.0	251°22.6	N10°38.7	Capella	280°16.6	46°01.6											
13	299°25.5	34°17.4	53.4	113°16.5	05.4	95°27.4	37.1	266°25.0	38.7	Bellatrix	278°19.1	6°22.4											
14	314°28.0	49°16.4	53.6	128°18.1	05.1	110°29.6	37.2	281°27.5	38.7	Elnath	277°57.4	28°37.8											
15	329°30.5	64°15.5	· · 53.9	143°19.7	· · 04.8	125°31.8	· · 37.3	296°30.0	· · 38.7	Alnilam	275°34.1	-1°11.1											
16	344°32.9	79°14.6	54.1	158°21.3	04.4	140°34.0	37.4	311°32.5	38.7	Betelgeuse	270°48.2	7°24.6											
17	359°35.4	94°13.6	54.3	173°23.0	04.1	155°36.2	37.5	326°34.9	38.7	Canopus	263°50.4	-52°42.8											
18	14°37.9	109°12.7	S22°54.5	188°24.6	N00°03.8	170°38.5	S08°37.5	341°37.4	N10°38.7	Sirius	258°23.0	-16°45.5											
19	29°40.3	124°11.8	54.7	203°26.2	03.5	185°40.7	37.6	356°39.9	38.7	Adhara	255°02.9	-29°00.8											
20	44°42.8	139°10.8	54.9	218°27.8	03.2	200°42.9	37.7	11°42.3	38.7	Procyon	244°47.1	5°08.8											
21	59°45.3	154°09.9	· · 55.1	233°29.5	· · 02.9	215°45.1	· · 37.8	26°44.8	· · 38.8	Pollux	243°13.0	27°57.1											
22	74°47.7	169°09.0	55.3	248°31.1	02.6	230°47.3	37.9	41°47.3	38.8	Avior	234°12.7	-59°36.2											
23	89°50.2	184°08.1	55.5	263°32.7	02.2	245°49.5	38.0	56°49.8	38.8	Suhail	222°43.5	-43°33.0											
Mer.pass. 17:02				ν -0.9' d 0.2' m -3.91				ν 1.6' d -0.3' m 0.55				ν 2.2' d 0.1' m -1.93				ν 2.5' d 0.0' m 0.22							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	104°52.6	199°07.1	S22°55.7	278°34.3	N00°01.9	260°51.7	S08°38.1	71°52.2	N10°38.8										
1	119°55.1	214°06.2	55.9	293°36.0	01.6	275°53.9	38.2	86°54.7	38.8										
2	134°57.6	229°05.3	56.1	308°37.6	01.3	290°56.1	38.3	101°57.2	38.8										
3	150°00.0	244°04.3	· · 56.3	323°39.2	· · 01.0	305°58.3	· · 38.3	116°59.6	· · 38.8										
4	165°02.5	259°03.4	56.5	338°40.9	00.7	321°00.6	38.4	132°02.1	38.8										
5	180°05.0	274°02.5	56.7	353°42.5	00.4	336°02.8	38.5	147°04.6	38.8										
6	195°07.4	289°01.5	S22°56.8	8°44.1	N00°00.0	351°05.0	S08°38.6	162°07.1	N10°38.9										
7	210°09.9	304°00.6	57.0	23°45.7	S00°00.3	6°07.2	38.7	177°09.5	38.9										
8	225°12.4	318°59.7	57.2	38°47.4	00.6	21°09.4	38.8	192°12.0	38.9										
9	240°14.8	333°58.7	· · 57.4	53°49.0	· · 00.9	36°11.6	· · 38.9	207°14.5	· · 38.9										
10	255°17.3	348°57.8	57.6	68°50.6	01.2	51°13.8	39.0	222°16.9	38.9										
11	270°19.7	3°56.8	57.8	83°52.3	01.5	66°16.0	39.0	237°19.4	38.9										
12	285°22.2	18°55.9	S22°58.0	98°53.9	S00°01.8	81°18.3	S08°39.1	252°21.9	N10°38.9										
13	300°24.7	33°55.0	58.2	113°55.5	02.1	96°20.5	39.2	267°24.3	38.9										
14	315°27.1	48°54.0	58.3	128°57.2	02.4	111°22.7	39.3	282°26.8	38.9										
15	330°29.6	63°53.1	· · 58.5	143°58.8	· · 02.7	126°24.9	· · 39.4	297°29.3	· · 39.0										
16	345°32.1	78°52.2	58.7	159°00.4	03.1	141°27.1	39.5	312°31.7	39.0										
17	0°34.5	93°51.2	58.9	174°02.1	03.4	156°29.3	39.6	327°34.2	39.0										
18	15°37.0	108°50.3	S22°59.0	189°03.7	S00°03.7	171°31.5	S08°39.6	342°36.7	N10°39.0										
19	30°39.5	123°49.4	59.2	204°05.4	04.0	186°33.8	39.7	357°39.1	39.0										
20	45°41.9	138°48.4	59.4	219°07.0	04.3	201°36.0	39.8	12°41.6	39.0										
21	60°44.4	153°47.5	· · 59.6	234°08.6	· · 04.6	216°38.2	· · 39.9	27°44.1	· · 39.0										
22	75°46.9	168°46.6	59.7	249°10.3	04.9	231°40.4	40.0	42°46.5	39.0										
23	90°49.3	183°45.6	59.9	264°11.9	05.2	246°42.6	40.1	57°49.0	39.1										
Mer.pass. 16:58				ν -0.9' d 0.2' m -3.91				ν 1.6' d -0.3' m 0.54				ν 2.2' d 0.1' m -1.93				ν 2.5' d 0.0' m 0.22			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	105°51.8	198°44.7	S23°00.1	279°13.5	S00°05.5	261°44.8	S08°40.2	72°51.5	N10°39.1										
1	120°54.2	213°43.7	00.2	294°15.2	05.8	276°47.1	40.3	87°53.9	39.1										
2	135°56.7	228°42.8	00.4	309°16.8	06.1	291°49.3	40.3	102°56.4	39.1										
3	150°59.2	243°41.9	· · 00.6	324°18.5	· · 06.4	306°51.5	· · 40.4	117°58.9	· · 39.1										
4	166°01.6	258°40.9	00.7	339°20.1	06.8	321°53.7	40.5	133°01.3	39.1										
5	181°04.1	273°40.0	00.9	354°21.8	07.1	336°55.9	40.6	148°03.8	39.1										
6	196°06.6	288°39.1	S23°01.1	9°23.4	S00°07.4	351°58.1	S08°40.7	163°06.3	N10°39.1										
7	211°09.0	303°38.1	01.2	24°25.0	07.7	7°00.4	40.8	178°08.7	39.2										
8	226°11.5	318°37.2	01.4	39°26.7	08.0	22°02.6	40.8	193°11.2	39.2										
9	241°14.0	333°36.2	· · 01.6	54°28.3	· · 08.3	37°04.8	· · 40.9	208°13.7	· · 39.2										
10	256°16.4	348°35.3	01.7	69°30.0	08.6	52°07.0	41.0	223°16.1	39.2										
11	271°18.9	3°34.4	01.9	84°31.6	08.9	67°09.2	41.1	238°18.6	39.2										
12	286°21.3	18°33.4	S23°02.0	99°33.3	S00°09.2	82°11.5	S08°41.2	253°21.1	N10°39.2										
13	301°23.8	33°32.5	02.2	114°34.9	09.5	97°13.7	41.3	268°23.5	39.2										
14	316°26.3	48°31.6	02.3	129°36.6	09.8	112°15.9	41.4	283°26.0	39.2										
15	331°28.7	63°30.6	· · 02.5	144°38.2	· · 10.1	127°18.1	· · 41.4	298°28.4	· · 39.3										
16	346°31.2	78°29.7	02.6	159°39.9	10.4	142°20.3	41.5	313°30.9	39.3										
17	1°33.7	93°28.7	02.8	174°41.5	10.7	157°22.6	41.6	328°33.4	39.3										
18	16°36.1	108°27.8	S23°02.9	189°43.2	S00°11.0	172°24.8	S08°41.7	343°35.8	N10°39.3										
19	31°38.6	123°26.9	03.1	204°44.8	11.3	187°27.0	41.8	358°38.3	39.3										
20	46°41.1	138°25.9	03.2	219°46.5	11.6	202°29.2	41.9	13°40.8	39.3										
21	61°43.5	153°25.0	· · 03.4	234°48.1	· · 11.9	217°31.4	· · 41.9	28°43.2	· · 39.3										
22	76°46.0	168°24.0	03.5	249°49.8	12.2	232°33.7	42.0	43°45.7	39.4										
23	91°48.5	183°23.1	03.6	264°51.4	12.5	247°35.9	42.1	58°48.1	39.4										
Mer.pass. 16:54				ν -0.9' d 0.2' m -3.91				ν 1.6' d 0.3' m 0.52				ν 2.2' d 0.1' m -1.94				ν 2.5' d 0.0' m 0.23			

Jan 04 Thu	SHA	Mer.pass
Venus	95°35.9	10:43
Mars	174°01.9	05:28
Jupiter	156°05.2	06:39
Saturn	326°59.4	19:13

Jan 05 Fri	SHA	Mer.pass
Venus	94°14.5	10:44
Mars	173°41.7	05:25
Jupiter	155°59.1	06:36
Saturn	326°59.6	19:09

Jan 06 Sat	SHA	
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DUT1 = UT1-UTC = +0.2053 sec ΔT = TT-UT1 = +68.9787 sec

2029 January 04 to Jan. 06 UT

h Sun Moon						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	178°47.2	S22°42.6	316°09.1	10.4'	N09°12.6	-14.3' 59.3'
1	193°46.9	42.3	330°38.5	10.4'	08°58.3	-14.3' 59.3'
2	208°46.7	42.1	345°08.0	10.5'	08°44.0	-14.4' 59.3'
3	223°46.4	· · 41.8	359°37.4	10.5'	08°29.6	-14.4' 59.3'
4	238°46.1	41.5	14°06.9	10.5'	08°15.2	-14.5' 59.3'
5	253°45.8	41.3	28°36.5	10.6'	08°00.7	-14.5' 59.3'
6	268°45.5	S22°41.0	43°06.0	10.6'	N07°46.2	-14.6' 59.3'
7	283°45.3	40.7	57°35.6	10.6'	07°31.6	-14.6' 59.3'
8	298°45.0	40.5	72°05.3	10.7'	07°17.0	-14.6' 59.3'
9	313°44.7	· · 40.2	86°35.0	10.7'	07°02.4	-14.7' 59.3'
10	328°44.4	39.9	101°04.7	10.7'	06°47.7	-14.7' 59.3'
11	343°44.1	39.6	115°34.4	10.8'	06°33.0	-14.8' 59.3'
12	358°43.9	S22°39.4	130°04.2	10.8'	N06°18.2	-14.8' 59.3'
13	13°43.6	39.1	144°34.0	10.8'	06°03.4	-14.8' 59.3'
14	28°43.3	38.8	159°03.8	10.9'	05°48.6	-14.9' 59.3'
15	43°43.0	· · 38.5	173°33.7	10.9'	05°33.7	-14.9' 59.3'
16	58°42.7	38.3	188°03.6	10.9'	05°18.9	-14.9' 59.4'
17	73°42.5	38.0	202°33.5	10.9'	05°03.9	-14.9' 59.4'
18	88°42.2	S22°37.7	217°03.4	11.0'	N04°49.0	-15.0' 59.4'
19	103°41.9	37.4	231°33.4	11.0'	04°34.0	-15.0' 59.4'
20	118°41.6	37.2	246°03.3	11.0'	04°19.0	-15.0' 59.4'
21	133°41.3	· · 36.9	260°33.3	11.0'	04°04.0	-15.0' 59.4'
22	148°41.1	36.6	275°03.4	11.0'	03°49.0	-15.1' 59.4'
23	163°40.8	36.3	289°33.4	11.1'	03°33.9	-15.1' 59.4'
SD = 16.3' d = -0.3' SD = 16.2'						
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	178°40.5	S22°36.0	304°03.5	11.1'	N03°18.9	-15.1' 59.4'
1	193°40.2	35.7	318°33.5	11.1'	03°03.8	-15.1' 59.4'
2	208°39.9	35.5	333°03.6	11.1'	02°48.7	-15.1' 59.4'
3	223°39.7	· · 35.2	347°33.8	11.1'	02°33.5	-15.1' 59.4'
4	238°39.4	34.9	2°03.9	11.1'	02°18.4	-15.1' 59.4'
5	253°39.1	34.6	16°34.0	11.2'	02°03.3	-15.2' 59.4'
6	268°38.8	S22°34.3	31°04.2	11.2'	N01°48.1	-15.2' 59.4'
7	283°38.6	34.0	45°34.3	11.2'	01°32.9	-15.2' 59.4'
8	298°38.3	33.7	60°04.5	11.2'	01°17.8	-15.2' 59.4'
9	313°38.0	· · 33.5	74°34.7	11.2'	01°02.6	-15.2' 59.4'
10	328°37.7	33.2	89°04.9	11.2'	00°47.4	-15.2' 59.4'
11	343°37.5	32.9	103°35.1	11.2'	00°32.2	-15.2' 59.4'
12	358°37.2	S22°32.6	118°05.3	11.2'	N00°17.1	-15.2' 59.4'
13	13°36.9	32.3	132°35.5	11.2'	N00°01.9	-15.2' 59.4'
14	28°36.6	32.0	147°05.7	11.2'	S00°13.3	15.2' 59.4'
15	43°36.3	· · 31.7	161°36.0	11.2'	00°28.5	15.2' 59.4'
16	58°36.1	31.4	176°06.2	11.2'	00°43.7	15.2' 59.4'
17	73°35.8	31.1	190°36.4	11.2'	00°58.8	15.2' 59.4'
18	88°35.5	S22°30.8	205°06.6	11.2'	S01°14.0	15.2' 59.3'
19	103°35.2	30.5	219°36.8	11.2'	01°29.1	15.1' 59.3'
20	118°35.0	30.2	234°07.1	11.2'	01°44.3	15.1' 59.3'
21	133°34.7	· · 29.9	248°37.3	11.2'	01°59.4	15.1' 59.3'
22	148°34.4	29.6	263°07.5	11.2'	02°14.5	15.1' 59.3'
23	163°34.2	29.3	277°37.7	11.2'	02°29.6	15.1' 59.3'
SD = 16.3' d = -0.3' SD = 16.2'						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	178°33.9	S22°29.0	292°07.9	11.2'	S02°44.7	15.1' 59.3'
1	193°33.6	28.7	306°38.1	11.2'	02°59.8	15.1' 59.3'
2	208°33.3	28.4	321°08.3	11.2'	03°14.8	15.0' 59.3'
3	223°33.1	· · 28.1	335°38.4	11.2'	03°29.9	15.0' 59.3'
4	238°32.8	27.8	350°08.6	11.1'	03°44.9	15.0' 59.3'
5	253°32.5	27.5	4°38.7	11.1'	03°59.9	15.0' 59.3'
6	268°32.2	S22°27.2	19°08.9	11.1'	S04°14.8	14.9' 59.3'
7	283°32.0	26.9	33°39.0	11.1'	04°29.8	14.9' 59.3'
8	298°31.7	26.6	48°09.1	11.1'	04°44.7	14.9' 59.3'
9	313°31.4	· · 26.3	62°39.2	11.1'	04°59.6	14.9' 59.3'
10	328°31.1	26.0	77°09.2	11.1'	05°14.4	14.8' 59.3'
11	343°30.9	25.7	91°39.3	11.0'	05°29.2	14.8' 59.3'
12	358°30.6	S22°25.4	106°09.3	11.0'	S05°44.0	14.8' 59.3'
13	13°30.3	25.0	120°39.3	11.0'	05°58.8	14.7' 59.3'
14	28°30.1	24.7	135°09.3	11.0'	06°13.5	14.7' 59.3'
15	43°29.8	· · 24.4	149°39.3	10.9'	06°28.2	14.6' 59.3'
16	58°29.5	24.1	164°09.2	10.9'	06°42.8	14.6' 59.3'
17	73°29.3	23.8	178°39.2	10.9'	06°57.4	14.6' 59.2'
18	88°29.0	S22°23.5	193°09.0	10.9'	S07°12.0	14.5' 59.2'
19	103°28.7	23.2	207°38.9	10.8'	07°26.5	14.5' 59.2'
20	118°28.4	22.8	222°08.7	10.8'	07°41.0	14.4' 59.2'
21	133°28.2	· · 22.5	236°38.6	10.8'	07°55.4	14.4' 59.2'
22	148°27.9	22.2	251°08.3	10.7'	08°09.8	14.3' 59.2'
23	163°27.6	21.9	265°38.1	10.7'	08°24.2	14.3' 59.2'
SD = 16.3' d = -0.3' SD = 16.2'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:19	10:29	■	■	13:43	15:53
N 70°	08:01	09:42	■	■	14:30	16:10
68°	07:47	09:12	11:27	12:45	15:00	16:25
66°	07:35	08:49	10:19	13:52	15:22	16:37
64°	07:25	08:31	09:44	14:27	15:40	16:47
62°	07:16	08:16	09:19	14:52	15:55	16:56
60°	07:08	08:04	08:59	15:12	16:08	17:03
N 58°	07:01	07:53	08:43	15:28	16:19	17:10
56°	06:55	07:43	08:29	15:42	16:28	17:17
54°	06:49	07:35	08:18	15:54	16:37	17:22
52°	06:44	07:27	08:07	16:04	16:44	17:28
50°	06:39	07:20	07:58	16:14	16:51	17:33
45°	06:28	07:05	07:38	16:33	17:07	17:44
N 40°	06:18	06:52	07:22	16:49	17:19	17:53
35°	06:09	06:41	07:09	17:03	17:31	18:02
30°	06:01	06:31	06:57	17:14	17:41	18:10
20°	05:45	06:12	06:36	17:35	17:59	18:26
N 10°	05:30	05:56	06:19	17:53	18:15	18:41
0°	05:13	05:39	06:02	18:09	18:32	18:58
S 10°	04:55	05:22	05:45	18:26	18:49	19:16
20°	04:33	05:02	05:27	18:44	19:09	19:38
30°	04:05	04:38	05:06	19:05	19:32	20:05
35°	03:47	04:24	04:53	19:18	19:47	20:23
40°	03:25	04:06	04:39	19:32	20:04	20:45
45°	02:56	03:45	04:21	19:49	20:26	21:14
S 50°	02:14	03:17	04:00	20:10	20:54	21:56
52°	01:50	03:02	03:50	20:21	21:08	22:20
54°	01:14	02:46	03:38	20:32	21:24	22:55
56°	—:—	02:25	03:25	20:45	21:45	—:—
58°	—:—	01:59	03:09	21:01	22:10	—:—
S 60°	—:—	01:20	02:50	21:20	22:48	—:—

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	20:48	23:02	· · ·	10:39	10:11	09:44
N 70°	20:54	22:59	· · ·	10:30	10:10	09:51
68°	20:59	22:56	· · ·	10:22	10:10	09:57
66°	21:03	22:53	· · ·	10:16	10:09	10:02
64°	21:06	22:51	· · ·	10:11	10:09	10:07
62°	21:09	22:49	· · ·	10:06	10:08	10:10
60°	21:11	22:47	· · ·	10:02	10:08	10:14
N 58°	21:14	22:46	· · ·	09:58	10:08	10:17
56°	21:16	22:44	· · ·	09:55	10:07	10:19
54°	21:17	22:43	· · ·	09:52	10:07	10:22
52°	21:19	22:42	· · ·	09:50	10:07	10:24
50°	21:20	22:41	· · ·	09:47	10:07	10:26
45°	21:24	22:39	23:55	09:42	10:06	10:30
N 40°	21:26	22:38	23:49	09:37	10:06	10:34
35°	21:28	22:36	23:44	09:33	10:05	10:37
30°	21:30	22:35	23:39	09:30	10:05	10:39
20°	21:34	22:33	23:32	09:24	10:04	10:44
N 10°	21:37	22:31	23:25	09:19	10:04	10:49
0°	21:40	22:29	23:19	09:14	10:03	10:53
S 10°	21:43	22:27	23:12	09:08	10:03	10:57
20°	21:46	22:26	23:06	09:03	10:02	11:01
30°	21:49	22:24	22:59	08:56	10:01	11:06
35°	21:51	22:22	22:54	08:53	10:01	11:08
40°	21:53	22:21	22:50	08:49	10:00	11:12
45°	21:56	22:20	22:44	08:44	10:00	11:15
S 50°	21:59	22:18	22:38	08:38	09:59	11:20
52°	22:00	22:17	22:35	08:35	09:59	11:22
54°	22:02	22:16	22:31	08:32	09:58	11:24
56°	22:04	22:15	22:28	08:29	09:58	11:27
58°	22:05	22:14	22:24	08:25	09:57	11:29
S 60°	22:08	22:13	22:19	08:21	09:57	11:32

Day	Sun		Mer. Pass	Moon		Age 19-21 87-68%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
04	04:51	05:05	12:05	03:02	15:27	
05	05:18	05:31	12:06	03:51	16:16	
06	05:44	05:58	12:06	04:41	17:06	

January 07, 08, 09 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	106°50.9	198°22.2	S23°03.8	279°53.1	S00°12.8	262°38.1	S08°42.2	73°50.6	N10°39.4					Alpheratz	357°31.4	29°15.3	
1	121°53.4	213°21.2	03.9	294°54.7	13.1	277°40.3	42.3	88°53.1	39.4					Ankaa	353°04.1	-42°09.0	
2	136°55.8	228°20.3	04.1	309°56.4	13.4	292°42.5	42.4	103°55.5	39.4					Schedar	349°27.2	56°42.1	
3	151°58.3	243°19.3	· · 04.2	324°58.0	· · 13.7	307°44.8	· · 42.4	118°58.0	· · 39.4					Diphda	348°44.1	-17°49.7	
4	167°00.8	258°18.4	04.3	339°59.7	14.0	322°47.0	42.5	134°00.5	39.4					Achernar	335°17.8	-57°05.6	
5	182°03.2	273°17.5	04.5	355°01.3	14.3	337°49.2	42.6	149°02.9	39.5					Hamal	327°47.4	23°36.1	
6	197°05.7	288°16.5	S23°04.6	10°03.0	S00°14.6	352°51.4	S08°42.7	164°05.4	N10°39.5					Polaris	311°57.2	89°23.4	
7	212°08.2	303°15.6	04.7	25°04.6	14.9	7°53.7	42.8	179°07.8	39.5					Acamar	315°09.1	-40°11.5	
8	227°10.6	318°14.6	04.9	40°06.3	15.2	22°55.9	42.9	194°10.3	39.5					Menkar	314°02.6	4°12.2	
9	242°13.1	333°13.7	· · 05.0	55°08.0	· · 15.5	37°58.1	· · 42.9	209°12.8	· · 39.5					Mirfak	308°23.2	49°58.0	
10	257°15.6	348°12.7	05.1	70°09.6	15.8	53°00.3	43.0	224°15.2	39.5					Aldebaran	290°35.6	16°34.0	
11	272°18.0	3°11.8	05.2	85°11.3	16.1	68°02.6	43.1	239°17.7	39.5					Rigel	281°00.4	-8°10.2	
12	287°20.5	18°10.9	S23°05.4	100°12.9	S00°16.4	83°04.8	S08°43.2	254°20.1	N10°39.6					Capella	280°16.6	46°01.6	
13	302°23.0	33°09.9	05.5	115°14.6	16.7	98°07.0	43.3	269°22.6	39.6					Bellatrix	278°19.1	6°22.4	
14	317°25.4	48°09.0	05.6	130°16.3	17.0	113°09.2	43.4	284°25.1	39.6					Elnath	277°57.4	28°37.8	
15	332°27.9	63°08.0	· · 05.7	145°17.9	· · 17.3	128°11.5	· · 43.4	299°27.5	· · 39.6					Alnilam	275°34.1	-1°11.1	
16	347°30.3	78°07.1	05.9	160°19.6	17.6	143°13.7	43.5	314°30.0	39.6					Betelgeuse	270°48.2	7°24.6	
17	2°32.8	93°06.2	06.0	175°21.2	17.9	158°15.9	43.6	329°32.4	39.6					Canopus	263°50.4	-52°42.8	
18	17°35.3	108°05.2	S23°06.1	190°22.9	S00°18.2	173°18.1	S08°43.7	344°34.9	N10°39.6					Sirius	258°23.0	-16°45.5	
19	32°37.7	123°04.3	06.2	205°24.6	18.5	188°20.4	43.8	359°37.3	39.7					Adhara	255°02.9	-29°00.8	
20	47°40.2	138°03.3	06.3	220°26.2	18.8	203°22.6	43.8	14°39.8	39.7					Procyon	244°47.1	5°08.8	
21	62°42.7	153°02.4	· · 06.4	235°27.9	· · 19.1	218°24.8	· · 43.9	29°42.3	· · 39.7					Pollux	243°13.0	27°57.1	
22	77°45.1	168°01.4	06.6	250°29.6	19.4	233°27.1	44.0	44°44.7	39.7					Avior	234°12.7	-59°36.2	
23	92°47.6	183°00.5	06.7	265°31.2	19.7	248°29.3	44.1	59°47.2	39.7					Suhail	222°43.4	-43°33.0	
Mer.pass. 16:50			ν -0.9' d 0.1' m -3.91	ν 1.7' d 0.3' m 0.50	ν 2.2' d 0.1' m -1.94	ν 2.5' d 0.0' m 0.23											

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	107°50.1	197°59.6	S23°06.8	280°32.9	S00°20.0	263°31.5	S08°44.2	74°49.6	N10°39.7								
1	122°52.5	212°58.6	06.9	295°34.6	20.3	278°33.7	44.2	89°52.1	39.8								
2	137°55.0	227°57.7	07.0	310°36.2	20.6	293°36.0	44.3	104°54.5	39.8								
3	152°57.4	242°56.7	· · 07.1	325°37.9	· · 20.9	308°38.2	· · 44.4	119°57.0	· · 39.8								
4	167°59.9	257°55.8	07.2	340°39.6	21.2	323°40.4	44.5	134°59.5	39.8								
5	183°02.4	272°54.8	07.3	355°41.2	21.5	338°42.7	44.6	150°01.9	39.8								
6	198°04.8	287°53.9	S23°07.4	10°42.9	S00°21.7	353°44.9	S08°44.7	165°04.4	N10°39.8								
7	213°07.3	302°52.9	07.5	25°44.6	22.0	8°47.1	44.7	180°06.8	39.9								
8	228°09.8	317°52.0	07.6	40°46.2	22.3	23°49.3	44.8	195°09.3	39.9								
9	243°12.2	332°51.1	· · 07.7	55°47.9	· · 22.6	38°51.6	· · 44.9	210°11.7	· · 39.9								
10	258°14.7	347°50.1	07.8	70°49.6	22.9	53°53.8	45.0	225°14.2	39.9								
11	273°17.2	2°49.2	07.9	85°51.3	23.2	68°56.0	45.1	240°16.7	39.9								
12	288°19.6	17°48.2	S23°08.0	100°52.9	S00°23.5	83°58.3	S08°45.1	255°19.1	N10°39.9								
13	303°22.1	32°47.3	08.1	115°54.6	23.8	99°00.5	45.2	270°21.6	39.9								
14	318°24.6	47°46.3	08.2	130°56.3	24.1	114°02.7	45.3	285°24.0	40.0								
15	333°27.0	62°45.4	· · 08.3	145°58.0	· · 24.4	129°05.0	· · 45.4	300°26.5	· · 40.0								
16	348°29.5	77°44.4	08.4	160°59.6	24.7	144°07.2	45.5	315°28.9	40.0								
17	3°31.9	92°43.5	08.5	176°01.3	25.0	159°09.4	45.5	330°31.4	40.0								
18	18°34.4	107°42.6	S23°08.6	191°03.0	S00°25.2	174°11.7	S08°45.6	345°33.8	N10°40.0								
19	33°36.9	122°41.6	08.6	206°04.7	25.5	189°13.9	45.7	0°36.3	40.0								
20	48°39.3	137°40.7	08.7	221°06.3	25.8	204°16.1	45.8	15°38.7	40.1								
21	63°41.8	152°39.7	· · 08.8	236°08.0	· · 26.1	219°18.4	· · 45.8	30°41.2	· · 40.1								
22	78°44.3	167°38.8	08.9	251°09.7	26.4	234°20.6	45.9	45°43.6	40.1								
23	93°46.7	182°37.8	09.0	266°11.4	26.7	249°22.8	46.0	60°46.1	40.1								
Mer.pass. 16:46			ν -0.9' d 0.1' m -3.91	ν 1.7' d 0.3' m 0.48	ν 2.2' d 0.1' m -1.95	ν 2.5' d 0.0' m 0.24											

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	108°49.2	197°36.9	S23°09.1	281°13.1	S00°27.0	264°25.1	S08°46.1	75°48.6	N10°40.1		
1	123°51.7	212°35.9	09.1	296°14.7	27.3	279°27.3	46.2	90°51.0	40.1		
2	138°54.1	227°35.0	09.2	311°16.4	27.6	294°29.5	46.2	105°53.5	40.2		
3	153°56.6	242°34.0	· · 09.3	326°18.1	· · 27.8	309°31.8	· · 46.3	120°55.9	· · 40.2		
4	168°59.1	257°33.1	09.4	341°19.8	28.1	324°34.0	46.4	135°58.4	40.2		
5	184°01.5	272°32.2	09.4	356°21.5	28.4	339°36.2	46.5	151°00.8	40.2		
6	199°04.0	287°31.2	S23°09.5	11°23.2	S00°28.7	354°38.5	S08°46.6	166°03.3	N10°40.2		
7	214°06.4	302°30.3	09.6	26°24.8	29.0	9°40.7	46.6	181°05.7	40.3		
8	229°08.9	317°29.3	09.7	41°26.5	29.3	24°42.9	46.7	196°08.2	40.3		
9	244°11.4	332°28.4	· · 09.7	56°28.2	· · 29.6	39°45.2	· · 46.8	211°10.6	· · 40.3		
10	259°13.8	347°27.4	09.8	71°29.9	29.8	54°47.4	46.9	226°13.1	40.3		
11	274°16.3	2°26.5	09.9	86°31.6	30.1	69°49.6	46.9	241°15.5	40.3		
12	289°18.8	17°25.5	S23°09.9	101°33.3	S00°30.4	84°51.9	S08°47.0	256°18.0	N10°40.3		
13	304°21.2	32°24.6	10.0	116°35.0	30.7	99°54.1	47.1	271°20.4	40.4		
14	319°23.7	47°23.6	10.1	131°36.6	31.0	114°56.4	47.2	286°22.9	40.4		
15	334°26.2	62°22.7	· · 10.1	146°38.3	· · 31.3	129°58.6	· · 47.2	301°25.3	· · 40.4		
16	349°28.6	77°21.7	10.2	161°40.0	31.6	145°00.8	47.3	316°27.8	40.4		
17	4°31.1	92°20.8	10.2	176°41.7	31.8	160°03.1	47.4	331°30.2	40.4		
18	19°33.6	107°19.9	S23°10.3	191°43.4	S00°32.1	175°05.3	S08°47.5	346°32.7	N10°40.5		
19	34°36.0	122°18.9	10.4	206°45.1	32.4	190°07.6	47.6	1°35.1	40.5		
20	49°38.5	137°18.0	10.4	221°46.8	32.7	205°09.8	47.6	16°37.6	40.5		
21	64°40.9	152°17.0	· · 10.5	236°48.5	· · 33.0	220°12.0	· · 47.7	31°40.0	· · 40.5		
22	79°43.4	167°16.1	10.5	251°50.2	33.3	235°14.3	47.8	46°42.5	40.5		
23	94°45.9	182°15.1									

DUT1 = UT1-UTC = +0.2054 sec ΔT = TT-UT1 = +68.9786 sec

2029 January 07 to Jan. 09 UT

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°27.4	S22°21.6	280°07.8	10.7'	S08°38.5	14.2'	59.2'
1	193°27.1	21.3	294°37.5	10.7'	08°52.7	14.2'	59.2'
2	208°26.8	20.9	309°07.1	10.6'	09°06.9	14.1'	59.2'
3	223°26.6	· · 20.6	323°36.8	10.6'	09°21.0	14.1'	59.2'
4	238°26.3	20.3	338°06.3	10.5'	09°35.1	14.0'	59.2'
5	253°26.0	20.0	352°35.9	10.5'	09°49.2	14.0'	59.2'
6	268°25.8	S22°19.6	7°05.4	10.5'	S10°03.1	13.9'	59.2'
7	283°25.5	19.3	21°34.9	10.4'	10°17.0	13.9'	59.1'
8	298°25.2	19.0	36°04.3	10.4'	10°30.9	13.8'	59.1'
9	313°25.0	· · 18.7	50°33.7	10.3'	10°44.7	13.7'	59.1'
10	328°24.7	18.3	65°03.0	10.3'	10°58.4	13.7'	59.1'
11	343°24.4	18.0	79°32.3	10.3'	11°12.1	13.6'	59.1'
12	358°24.2	S22°17.7	94°01.6	10.2'	S11°25.7	13.5'	59.1'
13	13°23.9	17.4	108°30.8	10.2'	11°39.2	13.5'	59.1'
14	28°23.6	17.0	123°00.0	10.1'	11°52.7	13.4'	59.1'
15	43°23.4	· · 16.7	137°29.1	10.1'	12°06.1	13.3'	59.1'
16	58°23.1	16.4	151°58.2	10.0'	12°19.4	13.3'	59.1'
17	73°22.8	16.0	166°27.3	10.0'	12°32.7	13.2'	59.1'
18	88°22.6	S22°15.7	180°56.3	9.9'	S12°45.9	13.1'	59.0'
19	103°22.3	15.4	195°25.2	9.9'	12°59.0	13.0'	59.0'
20	118°22.0	15.0	209°54.1	9.9'	13°12.0	13.0'	59.0'
21	133°21.8	· · 14.7	224°23.0	9.8'	13°25.0	12.9'	59.0'
22	148°21.5	14.4	238°51.8	9.8'	13°37.8	12.8'	59.0'
23	163°21.2	14.0	253°20.5	9.7'	13°50.6	12.7'	59.0'
SD = 16.3' d = -0.3'			SD = 16.1'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.0	S22°13.7	267°49.2	9.6'	S14°03.4	12.6'	59.0'
1	193°20.7	13.3	282°17.9	9.6'	14°16.0	12.6'	59.0'
2	208°20.4	13.0	296°46.4	9.5'	14°28.5	12.5'	59.0'
3	223°20.2	· · 12.7	311°15.0	9.5'	14°41.0	12.4'	59.0'
4	238°19.9	12.3	325°43.5	9.4'	14°53.4	12.3'	58.9'
5	253°19.7	12.0	340°11.9	9.4'	15°05.7	12.2'	58.9'
6	268°19.4	S22°11.6	354°40.3	9.3'	S15°17.9	12.1'	58.9'
7	283°19.1	11.3	9°08.6	9.3'	15°30.0	12.0'	58.9'
8	298°18.9	11.0	23°36.9	9.2'	15°42.0	11.9'	58.9'
9	313°18.6	· · 10.6	38°05.1	9.2'	15°53.9	11.8'	58.9'
10	328°18.3	10.3	52°33.3	9.1'	16°05.8	11.7'	58.9'
11	343°18.1	09.9	67°01.4	9.1'	16°17.5	11.6'	58.9'
12	358°17.8	S22°09.6	81°29.5	9.0'	S16°29.2	11.5'	58.9'
13	13°17.6	09.2	95°57.4	8.9'	16°40.7	11.4'	58.8'
14	28°17.3	08.9	110°25.4	8.9'	16°52.2	11.3'	58.8'
15	43°17.0	· · 08.5	124°53.3	8.8'	17°03.5	11.2'	58.8'
16	58°16.8	08.2	139°21.1	8.8'	17°14.7	11.1'	58.8'
17	73°16.5	07.8	153°48.8	8.7'	17°25.9	11.0'	58.8'
18	88°16.3	S22°07.5	168°16.6	8.6'	S17°36.9	10.9'	58.8'
19	103°16.0	07.1	182°44.2	8.6'	17°47.8	10.8'	58.8'
20	118°15.7	06.8	197°11.8	8.5'	17°58.7	10.7'	58.8'
21	133°15.5	· · 06.4	211°39.3	8.5'	18°09.4	10.6'	58.8'
22	148°15.2	06.1	226°06.8	8.4'	18°20.0	10.5'	58.7'
23	163°15.0	05.7	240°34.2	8.4'	18°30.5	10.4'	58.7'
SD = 16.3' d = -0.3'			SD = 16.1'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°14.7	S22°05.3	255°01.6	8.3'	S18°40.9	10.3'	58.7'
1	193°14.4	05.0	269°28.9	8.2'	18°51.1	10.2'	58.7'
2	208°14.2	04.6	283°56.1	8.2'	19°01.3	10.0'	58.7'
3	223°13.9	· · 04.3	298°23.3	8.1'	19°11.4	9.9'	58.7'
4	238°13.7	03.9	312°50.4	8.1'	19°21.3	9.8'	58.7'
5	253°13.4	03.6	327°17.5	8.0'	19°31.1	9.7'	58.7'
6	268°13.1	S22°03.2	341°44.5	8.0'	S19°40.8	9.6'	58.6'
7	283°12.9	02.8	356°11.5	7.9'	19°50.3	9.4'	58.6'
8	298°12.6	02.5	10°38.3	7.8'	19°59.8	9.3'	58.6'
9	313°12.4	· · 02.1	25°05.2	7.8'	20°09.1	9.2'	58.6'
10	328°12.1	01.7	39°32.0	7.7'	20°18.3	9.1'	58.6'
11	343°11.9	01.4	53°58.7	7.7'	20°27.4	9.0'	58.6'
12	358°11.6	S22°01.0	68°25.4	7.6'	S20°36.4	8.8'	58.6'
13	13°11.4	00.7	82°52.0	7.6'	20°45.2	8.7'	58.5'
14	28°11.1	22°00.3	97°18.5	7.5'	20°53.9	8.6'	58.5'
15	43°10.8	21°59.9	111°45.0	7.5'	21°02.5	8.4'	58.5'
16	58°10.6	59.6	126°11.5	7.4'	21°10.9	8.3'	58.5'
17	73°10.3	59.2	140°37.9	7.3'	21°19.2	8.2'	58.5'
18	88°10.1	S21°58.8	155°04.2	7.3'	S21°27.4	8.0'	58.5'
19	103°09.8	58.4	169°30.5	7.2'	21°35.4	7.9'	58.5'
20	118°09.6	58.1	183°56.8	7.2'	21°43.4	7.8'	58.5'
21	133°09.3	· · 57.7	198°22.9	7.1'	21°51.1	7.6'	58.4'
22	148°09.1	57.3	212°49.1	7.1'	21°58.8	7.5'	58.4'
23	163°08.8	57.0	227°15.2	7.0'	22°06.3	7.4'	58.4'
SD = 16.3' d = -0.4'			SD = 16.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:14	10:18	■	■	13:57	16:00
N 70°	07:57	09:35	■	■	14:39	16:17
68°	07:44	09:07	11:07	13:07	15:07	16:31
66°	07:32	08:45	10:12	14:02	15:29	16:42
64°	07:22	08:28	09:39	14:35	15:46	16:52
62°	07:14	08:14	09:15	14:59	16:00	17:00
60°	07:06	08:02	08:56	15:18	16:12	17:08
N 58°	07:00	07:51	08:41	15:33	16:23	17:14
56°	06:54	07:42	08:28	15:46	16:32	17:20
54°	06:48	07:33	08:16	15:58	16:41	17:26
52°	06:43	07:26	08:06	16:08	16:48	17:31
50°	06:38	07:19	07:57	16:17	16:55	17:36
45°	06:27	07:04	07:37	16:36	17:10	17:46
N 40°	06:18	06:52	07:22	16:52	17:22	17:56
35°	06:09	06:41	07:09	17:05	17:33	18:04
30°	06:01	06:31	06:57	17:17	17:43	18:13
20°	05:46	06:13	06:37	17:37	18:01	18:28
N 10°	05:31	05:57	06:20	17:54	18:17	18:43
0°	05:15	05:41	06:03	18:11	18:33	18:59
S 10°	04:57	05:24	05:47	18:27	18:50	19:17
20°	04:35	05:04	05:29	18:45	19:09	19:38
30°	04:08	04:41	05:08	19:05	19:33	20:05
35°	03:50	04:26	04:56	19:18	19:47	20:23
40°	03:29	04:09	04:42	19:32	20:04	20:44
45°	03:00	03:48	04:25	19:48	20:25	21:12
S 50°	02:20	03:21	04:04	20:09	20:52	21:53
52°	01:56	03:07	03:54	20:19	21:06	22:16
54°	01:23	02:51	03:42	20:30	21:22	22:48
56°	—:—	02:31	03:29	20:43	21:41	—:—
58°	—:—	02:06	03:14	20:58	22:05	—:—
S 60°	—:—	01:31	02:56	21:16	22:40	—:—

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	01:21	04:00	■	09:12	08:21	■
N 70°	01:06	03:24	■	09:29	08:59	■
68°	00:54	02:59	05:22	09:43	09:26	08:55
66°	00:44	02:40	04:43	09:55	09:46	09:35
64°	00:36	02:24	04:16	10:05	10:03	10:02
62°	00:29	02:11	03:55	10:13	10:17	10:24
60°	00:23	02:01	03:39	10:20	10:29	10:41
N 58°	00:18	01:51	03:25	10:26	10:39	10:56
56°	00:13	01:43	03:13	10:32	10:48	11:09
54°	00:09	01:36	03:02	10:37	10:56	11:20
52°	00:06	01:29	02:53	10:42	11:03	11:29
50°	00:02	01:24	02:45	10:46	11:10	11:38
45°	· · ·	01:11	02:27	10:55	11:24	11:57
N 40°	· · ·	01:01	02:13	11:03	11:35	12:12
35°	· · ·	00:52	02:01	11:09	11:45	12:25
30°	· · ·	00:45	01:50	11:15	11:54	12:37
20°	· · ·	00:32	01:33	11:25	12:09	12:56
N 10°	· · ·	00:20	01:17	11:34	12:22	13:13
0°	· · ·	00:10	01:03	11:43	12:35	13:29
S 10°	23:59	· · ·	00:49	11:51	12:47	13:45
20°	23:48	· · ·	00:33	12:00	13:01	14:02
30°	23:36	· · ·	00:16	12:10	13:16	14:22
35°	23:28	· · ·	00:06	12:16	13:25	14:34
40°	23:20	23:55	· · ·	12:23	13:35	14:47
45°	23:11	23:41	· · ·	12:31	13:47	15:03
S 50°	22:59	23:25	23:58	12:41	14:02	15:22
52°	22:54	23:18	23:48	12:45	14:09	15:32
54°	22:48	23:09	23:37	12:50	14:17	15:42
56°	22:42	23:00	23:25	12:56	14:25	15:54
58°	22:35	22:49	23:10	13:02	14:35	16:08
S 60°	22:27	22:37	22:54	13:09	14:47	16:24

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	22-24
				hh:mm	hh:mm	57-34%
07	06:11	06:23	12:06	05:31	17:56	
08	06:36	06:49	12:07	06:22	18:49	
09	07:01	07:14	12:07	07:16	19:44	

January 10, 11, 12 UT (Wed., Thu., Fri.)

	Aries	Venus	Mars	Jupiter	Saturn				
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec		
0	109° 48.3	197° 14.2	S23° 10.6	281° 53.6	S00° 33.8	265° 18.8	S08° 47.9	76° 47.4	N10° 40.6
1	124° 50.8	212° 13.2	10.7	296° 55.3	34.1	280° 21.0	48.0	91° 49.8	40.6
2	139° 53.3	227° 12.3	10.7	311° 57.0	34.4	295° 23.2	48.1	106° 52.3	40.6
3	154° 55.7	242° 11.3	•• 10.8	326° 58.7	•• 34.7	310° 25.5	•• 48.2	121° 54.7	•• 40.6
4	169° 58.2	257° 10.4	10.8	342° 00.3	34.9	325° 27.7	48.2	136° 57.2	40.6
5	185° 00.7	272° 09.4	10.9	357° 02.0	35.2	340° 30.0	48.3	151° 59.6	40.7
6	200° 03.1	287° 08.5	S23° 10.9	12° 03.7	S00° 35.5	355° 32.2	S08° 48.4	167° 02.1	N10° 40.7
7	215° 05.6	302° 07.5	11.0	27° 05.4	35.8	10° 34.4	48.5	182° 04.5	40.7
8	230° 08.0	317° 06.6	11.0	42° 07.1	36.1	25° 36.7	48.5	197° 06.9	40.7
9	245° 10.5	332° 05.6	•• 11.0	57° 08.8	•• 36.3	40° 38.9	•• 48.6	212° 09.4	•• 40.7
10	260° 13.0	347° 04.7	11.1	72° 10.5	36.6	55° 41.2	48.7	227° 11.8	40.8
11	275° 15.4	2° 03.7	11.1	87° 12.2	36.9	70° 43.4	48.8	242° 14.3	40.8
12	290° 17.9	17° 02.8	S23° 11.2	102° 13.9	S00° 37.2	85° 45.7	S08° 48.8	257° 16.7	N10° 40.8
13	305° 20.4	32° 01.9	11.2	117° 15.6	37.5	100° 47.9	48.9	272° 19.2	40.8
14	320° 22.8	47° 00.9	11.2	132° 17.4	37.7	115° 50.1	49.0	287° 21.6	40.8
15	335° 25.3	62° 00.0	•• 11.3	147° 19.1	•• 38.0	130° 52.4	•• 49.1	302° 24.1	•• 40.9
16	350° 27.8	76° 59.0	11.3	162° 20.8	38.3	145° 54.6	49.1	317° 26.5	40.9
17	5° 30.2	91° 58.1	11.3	177° 22.5	38.6	160° 56.9	49.2	332° 29.0	40.9
18	20° 32.7	106° 57.1	S23° 11.3	192° 24.2	S00° 38.8	175° 59.1	S08° 49.3	347° 31.4	N10° 40.9
19	35° 35.2	121° 56.2	11.4	207° 25.9	39.1	191° 01.4	49.4	2° 33.8	40.9
20	50° 37.6	136° 55.2	11.4	222° 27.6	39.4	206° 03.6	49.4	17° 36.3	40.9
21	65° 40.1	151° 54.3	•• 11.4	237° 29.3	•• 39.7	221° 05.9	•• 49.5	32° 38.7	•• 41.0
22	80° 42.5	166° 53.3	11.5	252° 31.0	39.9	236° 08.1	49.6	47° 41.2	41.0
23	95° 45.0	181° 52.4	11.5	267° 32.7	40.2	251° 10.3	49.7	62° 43.6	41.0
Mer.pass. 16:38		ν -0.9' d 0.1' m -3.91		ν 1.7' d 0.3' m 0.44		ν 2.2' d 0.1' m -1.96		ν 2.4' d 0.0' m 0.25	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	110°47.5	196°51.4	S23°11.5	282°34.4	S00°40.5	266°12.6	S08°49.7	77°46.1	N10°41.0
1	125°49.9	211°50.5	11.5	297°36.1	40.8	281°14.8	49.8	92°48.5	41.1
2	140°52.4	226°49.5	11.5	312°37.8	41.0	296°17.1	49.9	107°51.0	41.1
3	155°54.9	241°48.6	· · 11.6	327°39.5	· · 41.3	311°19.3	· · 50.0	122°53.4	· · 41.1
4	170°57.3	256°47.6	11.6	342°41.3	41.6	326°21.6	50.0	137°55.8	41.1
5	185°59.8	271°46.7	11.6	357°43.0	41.9	341°23.8	50.1	152°58.3	41.1
6	201°02.3	286°45.7	S23°11.6	12°44.7	S00°42.1	356°26.1	S08°50.2	168°00.7	N10°41.2
7	216°04.7	301°44.8	11.6	27°46.4	42.4	11°28.3	50.2	183°03.2	41.2
8	231°07.2	316°43.8	11.6	42°48.1	42.7	26°30.6	50.3	198°05.6	41.2
9	246°09.7	331°42.9	· · 11.6	57°49.8	· · 43.0	41°32.8	· · 50.4	213°08.1	· · 41.2
10	261°12.1	346°41.9	11.6	72°51.5	43.2	56°35.1	50.5	228°10.5	41.2
11	276°14.6	1°41.0	11.7	87°53.3	43.5	71°37.3	50.5	243°12.9	41.3
12	291°17.0	16°40.0	S23°11.7	102°55.0	S00°43.8	86°39.6	S08°50.6	258°15.4	N10°41.3
13	306°19.5	31°39.1	11.7	117°56.7	44.0	101°41.8	50.7	273°17.8	41.3
14	321°22.0	46°38.1	11.7	132°58.4	44.3	116°44.1	50.8	288°20.3	41.3
15	336°24.4	61°37.2	· · 11.7	148°00.1	· · 44.6	131°46.3	· · 50.8	303°22.7	· · 41.3
16	351°26.9	76°36.3	11.7	163°01.8	44.9	146°48.6	50.9	318°25.1	41.4
17	6°29.4	91°35.3	11.7	178°03.6	45.1	161°50.8	51.0	333°27.6	41.4
18	21°31.8	106°34.4	S23°11.7	193°05.3	S00°45.4	176°53.1	S08°51.0	348°30.0	N10°41.4
19	36°34.3	121°33.4	11.7	208°07.0	45.7	191°55.3	51.1	3°32.5	41.4
20	51°36.8	136°32.5	11.7	223°08.7	45.9	206°57.6	51.2	18°34.9	41.5
21	66°39.2	151°31.5	· · 11.7	238°10.4	· · 46.2	221°59.8	· · 51.3	33°37.3	· · 41.5
22	81°41.7	166°30.6	11.7	253°12.2	46.5	237°02.1	51.3	48°39.8	41.5
23	96°44.2	181°29.6	11.7	268°13.9	46.7	252°04.3	51.4	63°42.2	41.5
Mer.pass. 16:34		ν -0.9' d 0.0' m -3.91		ν 1.7' d 0.3' m 0.42		ν 2.2' d 0.1' m -1.97		ν 2.4' d 0.0' m 0.25	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	111°46.6	196°28.7	S23°11.6	283°15.6	S00°47.0	267°06.6	S08°51.5	78°44.7	N10°41.5
1	126°49.1	211°27.7	11.6	298°17.3	47.3	282°08.8	51.5	93°47.1	41.6
2	141°51.5	226°26.8	11.6	313°19.1	47.5	297°11.1	51.6	108°49.5	41.6
3	156°54.0	241°25.8	•• 11.6	328°20.8	•• 47.8	312°13.3	•• 51.7	123°52.0	•• 41.6
4	171°56.5	256°24.9	11.6	343°22.5	48.1	327°15.6	51.8	138°54.4	41.6
5	186°58.9	271°23.9	11.6	358°24.3	48.3	342°17.8	51.8	153°56.9	41.6
6	202°01.4	286°23.0	S23°11.6	13°26.0	S00°48.6	357°20.1	S08°51.9	168°59.3	N10°41.7
7	217°03.9	301°22.0	11.6	28°27.7	48.9	12°22.4	52.0	184°01.7	41.7
8	232°06.3	316°21.1	11.5	43°29.4	49.1	27°24.6	52.0	199°04.2	41.7
9	247°08.8	331°20.1	•• 11.5	58°31.2	•• 49.4	42°26.9	•• 52.1	214°06.6	•• 41.7
10	262°11.3	346°19.2	11.5	73°32.9	49.7	57°29.1	52.2	229°09.0	41.8
11	277°13.7	1°18.2	11.5	88°34.6	49.9	72°31.4	52.2	244°11.5	41.8
12	292°16.2	16°17.3	S23°11.5	103°36.4	S00°50.2	87°33.6	S08°52.3	259°13.9	N10°41.8
13	307°18.7	31°16.3	11.4	118°38.1	50.5	102°35.9	52.4	274°16.4	41.8
14	322°21.1	46°15.4	11.4	133°39.8	50.7	117°38.1	52.5	289°18.8	41.9
15	337°23.6	61°14.4	•• 11.4	148°41.6	•• 51.0	132°40.4	•• 52.5	304°21.2	•• 41.9
16	352°26.0	76°13.5	11.4	163°43.3	51.2	147°42.7	52.6	319°23.7	41.9
17	7°28.5	91°12.5	11.3	178°45.0	51.5	162°44.9	52.7	334°26.1	41.9
18	22°31.0	106°11.6	S23°11.3	193°46.8	S00°51.8	177°47.2	S08°52.7	349°28.5	N10°41.9
19	37°33.4	121°10.6	11.3	208°48.5	52.0	192°49.4	52.8	4°31.0	42.0
20	52°35.9	136°09.7	11.2	223°50.2	52.3	207°51.7	52.9	19°33.4	42.0
21	67°38.4	151°08.8	•• 11.2	238°52.0	•• 52.6	222°53.9	•• 52.9	34°35.8	•• 42.0
22	82°40.8	166°07.8	11.2	253°53.7	52.8	237°56.2	53.0	49°38.3	42.0
23	97°43.3	181°06.9	11.1	268°55.5	53.1	252°58.5	53.1	64°40.7	42.1
Mer.pass. 16:30		ν -0.9' d -0.0' m -3.91		ν 1.7' d 0.3' m 0.40		ν 2.3' d 0.1' m -1.97		ν 2.4' d 0.0' m 0.25	

	SHA	Dec
Alpheratz	357° 31.4	29° 15.3
Ankaa	353° 04.1	-42° 09.0
Schedar	340° 27.2	56° 42.1
Diphda	348° 44.1	-17° 49.7
Achernar	335° 17.8	-57° 05.6
Hamal	327° 47.4	23° 36.1
Polaris	311° 58.2	89° 23.4
Acamar	315° 09.2	-40° 11.5
Menkar	314° 02.6	4° 12.2
Mirfak	308° 23.2	49° 58.0
Aldebaran	290° 35.6	16° 34.0
Rigel	281° 00.4	-8° 10.2
Capella	280° 16.6	46° 01.6
Bellatrix	278° 19.1	6° 22.4
Elnath	277° 57.4	28° 37.8
Anilam	275° 34.1	-1° 11.2
Betelgeuse	270° 48.2	7° 24.6
Canopus	263° 50.4	-52° 42.8
Sirius	258° 23.0	-16° 45.5
Adhara	255° 02.9	-29° 00.9
Procyon	244° 47.1	5° 08.8
Pollux	243° 13.0	27° 57.1
Avior	232° 12.7	-59° 36.2
Suhail	222° 43.4	-43° 33.0
Miaplacidus	221° 36.5	-69° 50.2
Alphard	217° 44.3	-8° 47.2
Regulus	207° 30.7	11° 49.3
Dubhe	193° 36.8	61° 35.2
Denebola	182° 21.5	14° 24.3
Gienah	175° 40.1	-17° 42.2
Acrux	172° 56.1	-63° 15.4
Gacrux	171° 47.7	-57° 16.4
Alioth	166° 10.2	55° 47.7
Spica	158° 18.8	-11° 18.8
Alkaid	152° 49.6	49° 09.7
Hadar	148° 31.3	-60° 30.5
Menkent	147° 53.8	-36° 30.6
Arcturus	145° 45.0	19° 01.7
Rigel Kent.	139° 35.9	-60° 57.1
Kochab	137° 20.7	74° 01.8
Zuben'ubi	136° 52.5	-16° 09.7
Alphecca	126° 01.2	26° 36.9
Antares	112° 12.1	-26° 29.6
Atria	107° 03.8	-69° 04.5
Sabik	101° 59.3	-15° 45.5
Shaula	96° 06.3	-37° 07.3
Rasalhague	95° 55.8	12° 32.3
Eltanin	90° 41.2	51° 29.0
Kaus Aust.	83° 28.6	-34° 22.1
Vega	80° 31.4	38° 48.7
Nunki	75° 44.1	-26° 15.5
Altair	61° 57.1	8° 56.8
Peacock	53° 01.3	-56° 38.4
Deneb	49° 24.0	45° 23.2
Enif	33° 35.9	10° 00.6
Al Na'ir	27° 29.3	-46° 49.3
Fomalhaut	15° 11.2	-29° 28.1
Scheat	13° 42.2	28° 14.6
Markab	13° 26.8	15° 21.9

Jan 10 Wed	SHA	Mer.pass
Venus	87°25.8	10:52
Mars	172°05.2	05:12
Jupiter	155°30.4	06:18
Saturn	326°59.0	18:50

Jan 11 Thu	SHA	Mer.pass
Venus	86°03.9	10:53
Mars	171°46.9	05:09
Jupiter	155°25.1	06:14
Saturn	326°58.6	18:46

Jan 12 Fri	SHA	Mer.pass
Venus	84°42.0	10:55
Mars	171°29.0	05:06
Jupiter	155°20.0	06:11
Saturn	326°58.0	18:42

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°08.6	S21°56.6	241°41.2	7.0'	S22°13.7	7.2'	58.4'
1	193°08.3	56.2	256°07.2	6.9'	22°20.9	7.1'	58.4'
2	208°08.0	55.8	270°33.2	6.9'	22°28.0	7.0'	58.4'
3	223°07.8	· · 55.5	284°59.1	6.9'	22°34.9	6.8'	58.4'
4	238°07.5	55.1	299°24.9	6.8'	22°41.7	6.7'	58.3'
5	253°07.3	54.7	313°50.7	6.8'	22°48.4	6.5'	58.3'
6	268°07.0	S21°54.3	328°16.5	6.7'	S22°54.9	6.4'	58.3'
7	283°06.8	53.9	342°42.2	6.7'	23°01.3	6.2'	58.3'
8	298°06.5	53.6	357°07.9	6.6'	23°07.6	6.1'	58.3'
9	313°06.3	· · 53.2	11°33.5	6.6'	23°13.6	5.9'	58.3'
10	328°06.0	52.8	25°59.1	6.6'	23°19.6	5.8'	58.2'
11	343°05.8	52.4	40°24.7	6.5'	23°25.4	5.7'	58.2'
12	358°05.5	S21°52.0	54°50.2	6.5'	S23°31.1	5.5'	58.2'
13	13°05.3	51.7	69°15.7	6.5'	23°36.6	5.4'	58.2'
14	28°05.0	51.3	83°41.2	6.4'	23°41.9	5.2'	58.2'
15	43°04.8	· · 50.9	98°06.6	6.4'	23°47.1	5.1'	58.2'
16	58°04.5	50.5	112°32.0	6.4'	23°52.2	4.9'	58.2'
17	73°04.3	50.1	126°57.4	6.3'	23°57.1	4.8'	58.1'
18	88°04.0	S21°49.7	141°22.7	6.3'	S24°01.9	4.6'	58.1'
19	103°03.8	49.3	155°48.0	6.3'	24°06.5	4.5'	58.1'
20	118°03.5	49.0	170°13.3	6.3'	24°10.9	4.3'	58.1'
21	133°03.3	· · 48.6	184°38.6	6.2'	24°15.2	4.1'	58.1'
22	148°03.0	48.2	199°03.8	6.2'	24°19.4	4.0'	58.1'
23	163°02.8	47.8	213°29.0	6.2'	24°23.4	3.8'	58.0'
	SD = 16.3'	d = -0.4'	SD = 15.9'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°02.6	S21°47.4	227°54.2	6.2'	S24°27.2	3.7'	58.0'
1	193°02.3	47.0	242°19.4	6.2'	24°30.9	3.5'	58.0'
2	208°02.1	46.6	256°44.6	6.1'	24°34.4	3.4'	58.0'
3	223°01.8	· · 46.2	271°09.7	6.1'	24°37.8	3.2'	58.0'
4	238°01.6	45.8	285°34.8	6.1'	24°41.0	3.1'	58.0'
5	253°01.3	45.4	299°59.9	6.1'	24°44.1	2.9'	57.9'
6	268°01.1	S21°45.0	314°25.0	6.1'	S24°47.0	2.8'	57.9'
7	283°00.8	44.6	328°50.1	6.1'	24°49.8	2.6'	57.9'
8	298°00.6	44.2	343°15.2	6.1'	24°52.4	2.4'	57.9'
9	313°00.3	· · 43.8	357°40.3	6.1'	24°54.8	2.3'	57.9'
10	328°00.1	43.4	12°05.4	6.1'	24°57.1	2.1'	57.9'
11	342°59.8	43.0	26°30.4	6.1'	24°59.2	2.0'	57.8'
12	357°59.6	S21°42.6	40°55.5	6.1'	S25°01.2	1.8'	57.8'
13	12°59.4	42.2	55°20.6	6.1'	25°03.0	1.7'	57.8'
14	27°59.1	41.8	69°45.7	6.1'	25°04.7	1.5'	57.8'
15	42°58.9	· · 41.4	84°10.7	6.1'	25°06.2	1.3'	57.8'
16	57°58.6	41.0	98°35.8	6.1'	25°07.5	1.2'	57.7'
17	72°58.4	40.6	113°00.9	6.1'	25°08.7	1.0'	57.7'
18	87°58.1	S21°40.2	127°26.0	6.1'	S25°09.7	0.9'	57.7'
19	102°57.9	39.8	141°51.1	6.1'	25°10.6	0.7'	57.7'
20	117°57.7	39.4	156°16.2	6.1'	25°11.3	0.6'	57.7'
21	132°57.4	· · 39.0	170°41.4	6.2'	25°11.9	0.4'	57.7'
22	147°57.2	38.6	185°06.5	6.2'	25°12.3	0.3'	57.6'
23	162°56.9	38.2	199°31.7	6.2'	25°12.6	0.1'	57.6'
	SD = 16.3'	d = -0.4'	SD = 15.8'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	177°56.7	S21°37.8	213°56.9	6.2'	S25°12.7	-0.1'	57.6'
1	192°56.4	37.4	228°22.1	6.2'	25°12.6	-0.2'	57.6'
2	207°56.2	36.9	242°47.3	6.3'	25°12.4	-0.4'	57.6'
3	222°56.0	· · 36.5	257°12.6	6.3'	25°12.0	-0.5'	57.5'
4	237°55.7	36.1	271°37.9	6.3'	25°11.5	-0.7'	57.5'
5	252°55.5	35.7	286°03.2	6.3'	25°10.8	-0.8'	57.5'
6	267°55.2	S21°35.3	300°28.5	6.4'	S25°10.0	-1.0'	57.5'
7	282°55.0	34.9	314°53.9	6.4'	25°09.0	-1.1'	57.5'
8	297°54.8	34.5	329°19.3	6.4'	25°07.8	-1.3'	57.4'
9	312°54.5	· · 34.1	343°44.8	6.5'	25°06.5	-1.4'	57.4'
10	327°54.3	33.6	358°10.2	6.5'	25°05.1	-1.6'	57.4'
11	342°54.1	33.2	12°35.8	6.6'	25°03.5	-1.7'	57.4'
12	357°53.8	S21°32.8	27°01.3	6.6'	S25°01.8	-1.9'	57.4'
13	12°53.6	32.4	41°26.9	6.6'	24°59.9	-2.0'	57.3'
14	27°53.3	32.0	55°52.5	6.7'	24°57.8	-2.2'	57.3'
15	42°53.1	· · 31.5	70°18.2	6.7'	24°55.6	-2.3'	57.3'
16	57°52.9	31.1	84°44.0	6.8'	24°53.3	-2.5'	57.3'
17	72°52.6	30.7	99°09.7	6.8'	24°50.8	-2.6'	57.3'
18	87°52.4	S21°30.3	113°35.6	6.9'	S24°48.1	-2.8'	57.3'
19	102°52.2	29.9	128°01.4	6.9'	24°45.3	-2.9'	57.2'
20	117°51.9	29.4	142°27.3	7.0'	24°42.4	-3.1'	57.2'
21	132°51.7	· · 29.0	156°53.3	7.0'	24°39.3	-3.2'	57.2'
22	147°51.4	28.6	171°19.4	7.1'	24°36.1	-3.4'	57.2'
23	162°51.2	28.2	185°45.4	7.1'	24°32.7	-3.5'	57.1'
	SD = 16.3'	d = -0.4'	SD = 15.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:09	10:06	■	■	14:11	16:08
N 70°	07:53	09:28	■	■	14:49	16:24
68°	07:40	09:01	10:51	13:26	15:15	16:37
66°	07:29	08:41	10:04	14:13	15:36	16:48
64°	07:19	08:24	09:34	14:43	15:52	16:57
62°	07:11	08:11	09:11	15:06	16:06	17:05
60°	07:04	07:59	08:53	15:24	16:18	17:12
N 58°	06:58	07:49	08:38	15:39	16:28	17:19
56°	06:52	07:40	08:25	15:51	16:37	17:24
54°	06:47	07:32	08:14	16:03	16:45	17:30
52°	06:42	07:24	08:04	16:12	16:52	17:35
50°	06:37	07:18	07:55	16:21	16:59	17:39
45°	06:27	07:03	07:36	16:40	17:13	17:50
N 40°	06:18	06:51	07:21	16:55	17:25	17:59
35°	06:09	06:41	07:08	17:08	17:36	18:07
30°	06:01	06:31	06:57	17:19	17:45	18:15
20°	05:47	06:14	06:37	17:39	18:02	18:30
N 10°	05:32	05:58	06:20	17:56	18:18	18:44
0°	05:16	05:42	06:04	18:12	18:34	19:00
S 10°	04:59	05:25	05:48	18:28	18:51	19:17
20°	04:38	05:06	05:31	18:45	19:09	19:38
30°	04:11	04:43	05:10	19:05	19:32	20:05
35°	03:54	04:29	04:58	19:17	19:46	20:22

January 13, 14, 15 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	112°45.8	196°05.9	S23°11.1	283°57.2	S00°53.3	268°00.7	S08°53.1	79°43.2	N10°42.1	Alpheratz	357°31.4	29°15.3
1	127°48.2	211°05.0	11.0	298°58.9	53.6	283°03.0	53.2	94°45.6	42.1	Ankaa	353°04.1	-42°09.0
2	142°50.7	226°04.0	11.0	314°00.7	53.9	298°05.2	53.3	109°48.0	42.1	Schedar	349°27.2	56°42.1
3	157°53.2	241°03.1	•• 11.0	329°02.4	•• 54.1	313°07.5	•• 53.3	124°50.5	•• 42.2	Diphda	348°44.1	-17°49.7
4	172°55.6	256°02.1	10.9	344°04.2	54.4	328°09.8	53.4	139°52.9	42.2	Achernar	335°17.9	-57°05.6
5	187°58.1	271°01.2	10.9	359°05.9	54.6	343°12.0	53.5	154°55.3	42.2	Hamal	327°47.4	23°36.1
6	203°00.5	286°00.2	S23°10.8	14°07.6	S00°54.9	358°14.3	S08°53.6	169°57.8	N10°42.2	Polaris	311°59.3	89°23.4
7	218°03.0	300°59.3	10.8	29°09.4	55.2	13°16.5	53.6	185°00.2	42.2	Acamar	315°09.2	-40°11.5
8	233°05.5	315°58.3	10.7	44°11.1	55.4	28°18.8	53.7	200°02.6	42.3	Menkar	314°02.6	4°12.2
9	248°07.9	330°57.4	•• 10.7	59°12.9	•• 55.7	43°21.1	•• 53.8	215°05.1	•• 42.3	Mirfak	308°23.2	49°58.0
10	263°10.4	345°56.4	10.6	74°14.6	55.9	58°23.3	53.8	230°07.5	42.3	Aldebaran	290°35.6	16°34.0
11	278°12.9	0°55.5	10.6	89°16.4	56.2	73°25.6	53.9	245°09.9	42.3	Rigel	281°00.4	-8°10.2
12	293°15.3	15°54.5	S23°10.5	104°18.1	S00°56.4	88°27.9	S08°54.0	260°12.4	N10°42.4	Capella	280°16.6	46°01.6
13	308°17.8	30°53.6	10.5	119°19.9	56.7	103°30.1	54.0	275°14.8	42.4	Bellatrix	278°19.1	6°22.4
14	323°20.3	45°52.6	10.4	134°21.6	57.0	118°32.4	54.1	290°17.2	42.4	Elnath	277°57.4	28°37.8
15	338°22.7	60°51.7	•• 10.4	149°23.4	•• 57.2	133°34.6	•• 54.2	305°19.7	•• 42.4	Alnilam	275°34.1	-1°11.2
16	353°25.2	75°50.7	10.3	164°25.1	57.5	148°36.9	54.2	320°22.1	42.5	Betelgeuse	270°48.2	7°24.6
17	8°27.7	90°49.8	10.3	179°26.9	57.7	163°39.2	54.3	335°24.5	42.5	Canopus	263°50.4	-52°42.8
18	23°30.1	105°48.8	S23°10.2	194°28.6	S00°58.0	178°41.4	S08°54.4	350°26.9	N10°42.5	Sirius	258°23.0	-16°45.6
19	38°32.6	120°47.9	10.1	209°30.4	58.2	193°43.7	54.4	5°29.4	42.5	Adhara	255°02.9	-29°00.9
20	53°35.0	135°47.0	10.1	224°32.1	58.5	208°46.0	54.5	20°31.8	42.6	Procyon	244°47.1	5°08.8
21	68°37.5	150°46.0	•• 10.0	239°33.9	•• 58.7	223°48.2	•• 54.6	35°34.2	•• 42.6	Pollux	243°12.9	27°57.1
22	83°40.0	165°45.1	09.9	254°35.6	59.0	238°50.5	54.6	50°36.7	42.6	Avior	234°12.7	-59°36.2
23	98°42.4	180°44.1	09.9	269°37.4	59.3	253°52.8	54.7	65°39.1	42.6	Suhail	222°43.4	-43°33.1
Mer.pass. 16:26		ν -0.9' d -0.0' m -3.91		ν 1.7' d 0.3' m 0.38		ν 2.3' d 0.1' m -1.98		ν 2.4' d 0.0' m 0.26		Miaplacidus	221°36.5	-69°50.2
										Alphard	217°44.2	-8°47.3
										Regulus	207°30.7	11°49.3
										Dubhe	193°36.8	61°35.2
										Denebola	182°21.5	14°24.3
										Gienah	175°40.1	-17°42.3
										Acrux	172°56.0	-63°15.4
										Gacrux	171°47.7	-57°16.4
										Alioth	166°10.2	55°47.7
										Spica	158°18.8	-11°18.8
										Alkaid	152°49.6	49°09.7
										Hadar	148°31.3	-60°30.5
										Menkent	147°53.7	-36°30.6
										Arcturus	145°45.0	19°01.7
										Rigel Kent.	139°35.8	-60°57.1
										Kochab	137°20.6	74°01.8
										Zuben'ubi	136°52.5	-16°09.7
										Alphecca	126°01.2	26°36.8
										Antares	112°12.1	-26°29.6
										Atria	107°03.7	-69°04.5
										Sabik	101°59.3	-15°45.6
										Shaula	96°06.3	-37°07.3
										Rasalhague	95°55.8	12°32.3
										Eltanin	90°41.2	51°29.0
										Kaus Aust.	83°28.6	-34°22.1
										Vega	80°31.4	38°48.6
										Nunki	75°44.1	-26°15.5
										Altair	61°57.1	8°56.8
										Peacock	53°01.3	-56°38.4
										Deneb	49°24.0	45°23.2
										Enif	33°35.8	10°00.6
										Al Na'ir	27°29.3	-46°49.3
										Fomalhaut	15°11.2	-29°28.1
										Scheat	13°42.2	28°14.6
										Markab	13°26.8	15°21.8
Jan 13 Sat												
		SHA		Mer.pass								
		Venus		83°20.1		10:56						
		Mars		171°11.4		05:04						
		Jupiter		155°15.0		06:07						
		Saturn		326°57.4		18:38						
Jan 14 Sun												
		SHA		Mer.pass								
		Venus		81°58.3		10:58						
		Mars		170°54.2		05:01						
		Jupiter		155°10.1		06:03						
		Saturn		326°56.6		18:34						
Jan 15 Mon												
		SHA		Mer.pass								
		Venus		80°36.4		10:59						
		Mars		170°37.4		04:58						
		Jupiter		155°05.4		06:00						
		Saturn		326°55.8		18:30						
Horizontal parallax												
		Venus:		0.1								
		Mars:		0.1								

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	177°51.0	S21°27.7	200°11.6	7.2'	S24°29.2	-3.7'	57.1'
1	192°50.7	27.3	214°37.8	7.3'	24°25.6	-3.8'	57.1'
2	207°50.5	26.9	229°04.1	7.3'	24°21.8	-3.9'	57.1'
3	222°50.3	· · 26.4	243°30.4	7.4'	24°17.8	-4.1'	57.1'
4	237°50.0	26.0	257°56.8	7.4'	24°13.7	-4.2'	57.0'
5	252°49.8	25.6	272°23.2	7.5'	24°09.5	-4.4'	57.0'
6	267°49.6	S21°25.2	286°49.7	7.6'	S24°05.2	-4.5'	57.0'
7	282°49.3	24.7	301°16.3	7.6'	24°00.7	-4.6'	57.0'
8	297°49.1	24.3	315°43.0	7.7'	23°56.1	-4.8'	57.0'
9	312°48.9	· · 23.9	330°09.7	7.8'	23°51.3	-4.9'	56.9'
10	327°48.6	23.4	344°36.5	7.9'	23°46.4	-5.0'	56.9'
11	342°48.4	23.0	359°03.3	7.9'	23°41.4	-5.2'	56.9'
12	357°48.2	S21°22.6	13°30.2	8.0'	S23°36.2	-5.3'	56.9'
13	12°47.9	22.1	27°57.2	8.1'	23°31.0	-5.4'	56.9'
14	27°47.7	21.7	42°24.3	8.1'	23°25.5	-5.5'	56.8'
15	42°47.5	· · 21.2	56°51.4	8.2'	23°20.0	-5.7'	56.8'
16	57°47.3	20.8	71°18.7	8.3'	23°14.3	-5.8'	56.8'
17	72°47.0	20.4	85°46.0	8.4'	23°08.5	-5.9'	56.8'
18	87°46.8	S21°19.9	100°13.3	8.4'	S23°02.6	-6.0'	56.7'
19	102°46.6	19.5	114°40.8	8.5'	22°56.5	-6.2'	56.7'
20	117°46.3	19.0	129°08.3	8.6'	22°50.4	-6.3'	56.7'
21	132°46.1	· · 18.6	143°35.9	8.7'	22°44.1	-6.4'	56.7'
22	147°45.9	18.2	158°03.6	8.8'	22°37.7	-6.5'	56.7'
23	162°45.7	17.7	172°31.4	8.8'	22°31.1	-6.7'	56.6'
SD = 16.3'			SD = 15.6'				
$d = -0.4'$							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	177°45.4	S21°17.3	186°59.2	8.9'	S22°24.5	-6.8'	56.6'
1	192°45.2	16.8	201°27.1	9.0'	22°17.7	-6.9'	56.6'
2	207°45.0	16.4	215°55.1	9.1'	22°10.8	-7.0'	56.6'
3	222°44.7	· · 15.9	230°23.2	9.2'	22°03.8	-7.1'	56.6'
4	237°44.5	15.5	244°51.4	9.3'	21°56.7	-7.2'	56.5'
5	252°44.3	15.0	259°19.7	9.3'	21°49.5	-7.3'	56.5'
6	267°44.1	S21°14.6	273°48.0	9.4'	S21°42.1	-7.5'	56.5'
7	282°43.8	14.2	288°16.4	9.5'	21°34.7	-7.6'	56.5'
8	297°43.6	13.7	302°44.9	9.6'	21°27.1	-7.7'	56.4'
9	312°43.4	· · 13.3	317°13.5	9.7'	21°19.4	-7.8'	56.4'
10	327°43.2	12.8	331°42.2	9.8'	21°11.7	-7.9'	56.4'
11	342°42.9	12.4	346°11.0	9.9'	21°03.8	-8.0'	56.4'
12	357°42.7	S21°11.9	0°39.8	9.9'	S20°55.8	-8.1'	56.4'
13	12°42.5	11.4	15°08.8	10.0'	20°47.7	-8.2'	56.3'
14	27°42.3	11.0	29°37.8	10.1'	20°39.5	-8.3'	56.3'
15	42°42.0	· · 10.5	44°06.9	10.2'	20°31.2	-8.4'	56.3'
16	57°41.8	10.1	58°36.1	10.3'	20°22.8	-8.5'	56.3'
17	72°41.6	09.6	73°05.4	10.4'	20°14.3	-8.6'	56.2'
18	87°41.4	S21°09.2	87°34.8	10.5'	S20°05.8	-8.7'	56.2'
19	102°41.1	08.7	102°04.2	10.5'	19°57.1	-8.8'	56.2'
20	117°40.9	08.3	116°33.8	10.6'	19°48.3	-8.9'	56.2'
21	132°40.7	· · 07.8	131°03.4	10.7'	19°39.4	-9.0'	56.2'
22	147°40.5	07.3	145°33.1	10.8'	19°30.5	-9.1'	56.1'
23	162°40.3	06.9	160°02.9	10.9'	19°21.4	-9.1'	56.1'
SD = 16.3'			SD = 15.4'				
$d = -0.4'$							

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	177°40.0	S21°06.4	174°32.8	11.0'	S19°12.3	-9.2'	56.1'
1	192°39.8	06.0	189°02.8	11.1'	19°03.0	-9.3'	56.1'
2	207°39.6	05.5	203°32.9	11.2'	18°53.7	-9.4'	56.0'
3	222°39.4	· · 05.0	218°03.0	11.2'	18°44.3	-9.5'	56.0'
4	237°39.1	04.6	232°33.2	11.3'	18°34.8	-9.6'	56.0'
5	252°38.9	04.1	247°03.6	11.4'	18°25.3	-9.7'	56.0'
6	267°38.7	S21°03.6	261°34.0	11.5'	S18°15.6	-9.7'	56.0'
7	282°38.5	03.2	276°04.5	11.6'	18°05.9	-9.8'	55.9'
8	297°38.3	02.7	290°35.0	11.7'	17°56.1	-9.9'	55.9'
9	312°38.1	· · 02.2	305°05.7	11.7'	17°46.2	-10.0'	55.9'
10	327°37.8	01.8	319°36.5	11.8'	17°36.2	-10.0'	55.9'
11	342°37.6	01.3	334°07.3	11.9'	17°26.2	-10.1'	55.8'
12	357°37.4	S21°00.8	348°38.2	12.0'	S17°16.1	-10.2'	55.8'
13	12°37.2	21°00.4	3°09.2	12.1'	17°05.9	-10.3'	55.8'
14	27°37.0	20°59.9	17°40.3	12.2'	16°55.6	-10.3'	55.8'
15	42°36.7	· · 59.4	32°11.4	12.2'	16°45.3	-10.4'	55.8'
16	57°36.5	58.9	46°42.7	12.3'	16°34.9	-10.5'	55.7'
17	72°36.3	58.5	61°14.0	12.4'	16°24.4	-10.5'	55.7'
18	87°36.1	S20°58.0	75°45.4	12.5'	S16°13.9	-10.6'	55.7'
19	102°35.9	57.5	90°16.9	12.6'	16°03.3	-10.7'	55.7'
20	117°35.7	57.1	104°48.5	12.7'	15°52.6	-10.7'	55.6'
21	132°35.5	· · 56.6	119°20.1	12.7'	15°41.8	-10.8'	55.6'
22	147°35.2	56.1	133°51.9	12.8'	15°31.0	-10.9'	55.6'
23	162°35.0	55.6	148°23.7	12.9'	15°20.2	-10.9'	55.6'
SD = 16.3'			SD = 15.3'				
$d = -0.5'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:02	09:54	■	■	14:25	16:17
N 70°	07:47	09:20	■	■	14:59	16:32
68°	07:35	08:55	10:36	13:43	15:24	16:44
66°	07:25	08:36	09:55	14:24	15:43	16:54
64°	07:16	08:20	09:27	14:52	15:59	17:03
62°	07:08	08:07	09:06	15:13	16:12	17:11
60°	07:02	07:56	08:49	15:30	16:23	17:17
N 58°	06:55	07:46	08:34	15:44	16:33	17:23
56°	06:50	07:37	08:22	15:57	16:41	17:29
54°	06:45	07:30	08:11	16:07	16:49	17:34
52°	06:40	07:23	08:02	16:17	16:56	17:39
50°	06:36	07:16	07:53	16:26	17:03	17:43
45°	06:26	07:02	07:35	16:44	17:16	17:53
N 40°	06:17	06:50	07:20	16:58	17:28	18:02
35°	06:09	06:40	07:08	17:11	17:38	18:10
30°	06:01	06:31	06:57	17:22	17:48	18:17
20°	05:47	06:14	06:38	17:41	18:04	18:31
N 10°	05:33	05:59	06:21	17:57	18:20	18:46
0°	05:17	05:43	06:05	18:13	18:35	19:01
S 10°	05:00	05:27	05:50	18:29	18:51	19:18
20°	04:40	05:09	05:33	18:45	19:10	19:38
30°	04:14	04:46	05:13	19:05	19:32	20:04
35°	03:57	04:32	05:01	19:17	19:45	20:21
40°	03:36	04:16	04:48	19:30	20:02	20:41
45°	03:10	03:56	04:32	19:46	20:21	21:08
S 50°	02:32	03:30	04:12	20:05	20:47	21:45
52°	02:11	03:18	04:03	20:15	21:00	22:05
54°	01:44	03:03	03:52	20:25	21:14	22:32
56°	00:59	02:45	03:40	20:37	21:32	23:14
58°	—:—	02:23	03:26	20:51	21:54	—:—
S 60°	—:—	01:53	03:10	21:07	22:22	—:—

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	■	11:45	■	■	14:08
N 70°	■	■	10:45	■	■	15:06
68°	■	10:50	10:10	■	13:17	15:40
66°	10:35	09:58	09:45	11:40	14:08	16:04
64°	09:23	09:26	09:25	12:52	14:40	16:23
62°	08:46	09:02	09:09	13:28	15:03	16:38
60°	08:20	08:42	08:56	13:54	15:22	16:51
N 58°	08:00	08:26	08:44	14:14	15:37	17:02
56°	07:43	08:13	08:34	14:31	15:50	17:11
54°	07:28	08:01	08:25	14:45	16:02	17:20
52°	07:15	07:51	08:17	14:57	16:12	17:27
50°	07:04	07:41	08:09	15:08	16:21	17:34
45°	06:41	07:21	07:54	15:31	16:39	17:48
N 40°	06:22	07:05	07:41	15:49	16:55	18:00
35°	06:07	06:52	07:30	16:04	17:07	18:10
30°	05:53	06:40	07:20	16:17	17:19	18:18
20°	05:30	06:19	07:04	16:40	17:38	18:33
N 10°	05:10	06:01	06:49	16:59	17:54	18:46
0°	04:51	05:45	06:35	17:17	18:10	18:58
S 10°	04:32	05:28	06:22	17:35	18:25	19:10
20°	04:12	05:10	06:07	17:54	18:41	19:23
30°	03:48	04:49	05:50	18:16	19:00	19:38
35°	03:34	04:37	05:40	18:29	19:10	19:46
40°	03:18	04:23	05:29	18:43	19:23	19:55
45°	02:59	04:07	05:15	19:01	19:37	20:06
S 50°	02:36	03:46	04:59	19:22	19:54	20:20
52°	02:24	03:36	04:51	19:32	20:03	20:26
54°	02:11	03:25	04:43	19:44	20:12	20:33
56°	01:56	03:13	04:33	19:57	20:22	20:40
58°	01:38	02:58	04:22	20:12	20:34	20:49
S 60°	01:17	02:40	04:09	20:30	20:47	20:58

Day	Sun			Moon		Age 28-1 3-0%
	Eqn. of Time		Mer.	Mer. Pass.		
	00 ^h	12 ^h	Pass	Upper	Lower	
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	
13	08:36	08:47	12:09	11:04	23:31	
14	08:58	09:09	12:09	11:57	—:—	
15	09:20	09:30	12:10	12:47	00:23	

January 16, 17, 18 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	115°43.2	194°57.8	S23°05.1	286°04.2	S01°11.3	270°44.1	S08°57.8	82°38.0	N10°43.9	Alpheratz	357°31.5	29°15.3
1	130°45.6	209°56.8	05.0	301°06.0	11.6	285°46.4	57.9	97°40.4	44.0	Ankaa	353°04.1	-42°09.0
2	145°48.1	224°55.9	04.9	316°07.7	11.8	300°48.6	57.9	112°42.8	44.0	Schedar	349°27.3	56°42.1
3	160°50.6	239°55.0	•• 04.7	331°09.5	•• 12.0	315°50.9	•• 58.0	127°45.2	•• 44.0	Diphda	348°44.1	-17°49.7
4	175°53.0	254°54.0	04.6	346°11.3	12.3	330°53.2	58.0	142°47.7	44.1	Achernar	335°17.9	-57°05.6
5	190°55.5	269°53.1	04.5	1°13.1	12.5	345°55.5	58.1	157°50.1	44.1	Hamal	327°47.4	23°36.1
6	205°58.0	284°52.1	S23°04.3	16°14.9	S01°12.7	0°57.7	S08°58.2	172°52.5	N10°44.1	Polaris	312°00.7	89°23.4
7	221°00.4	299°51.2	04.2	31°16.7	13.0	16°00.0	58.2	187°54.9	44.1	Acamar	315°09.2	-40°11.5
8	236°02.9	314°50.3	04.1	46°18.5	13.2	31°02.3	58.3	202°57.3	44.2	Menkar	314°02.6	4°12.2
9	251°05.4	329°49.3	•• 03.9	61°20.3	•• 13.5	46°04.6	•• 58.3	217°59.8	•• 44.2	Mirfak	308°23.2	49°58.0
10	266°07.8	344°48.4	03.8	76°22.1	13.7	61°06.9	58.4	233°02.2	44.2	Aldebaran	290°35.6	16°34.0
11	281°10.3	359°47.4	03.6	91°23.9	13.9	76°09.2	58.5	248°04.6	44.3	Rigel	281°00.4	-8°10.3
12	296°12.7	14°46.5	S23°03.5	106°25.7	S01°14.2	91°11.4	S08°58.5	263°07.0	N10°44.3	Capella	280°16.6	46°01.6
13	311°15.2	29°45.5	03.4	121°27.5	14.4	106°13.7	58.6	278°09.4	44.3	Bellatrix	278°19.1	6°22.4
14	326°17.7	44°44.6	03.2	136°29.3	14.6	121°16.0	58.6	293°11.8	44.3	Elnath	277°57.4	28°37.8
15	341°20.1	59°43.7	•• 03.1	151°31.1	•• 14.9	136°18.3	•• 58.7	308°14.3	•• 44.4	Alnilam	275°34.1	-1°11.2
16	356°22.6	74°42.7	02.9	166°32.9	15.1	151°20.6	58.7	323°16.7	44.4	Betelgeuse	270°48.2	7°24.6
17	11°25.1	89°41.8	02.8	181°34.7	15.3	166°22.8	58.8	338°19.1	44.4	Canopus	263°50.4	-52°42.9
18	26°27.5	104°40.8	S23°02.6	196°36.5	S01°15.6	181°25.1	S08°58.9	353°21.5	N10°44.5	Sirius	258°23.0	-16°45.6
19	41°30.0	119°39.9	02.5	211°38.3	15.8	196°27.4	58.9	8°23.9	44.5	Adhara	255°02.9	-29°00.9
20	56°32.5	134°39.0	02.3	226°40.1	16.0	211°29.7	59.0	23°26.4	44.5	Procyon	244°47.1	5°08.8
21	71°34.9	149°38.0	•• 02.2	241°41.9	•• 16.3	226°32.0	•• 59.0	38°28.8	•• 44.5	Pollux	243°12.9	27°57.1
22	86°37.4	164°37.1	02.0	256°43.7	16.5	241°34.3	59.1	53°31.2	44.6	Avior	234°12.6	-59°36.2
23	101°39.9	179°36.1	01.9	271°45.5	16.7	256°36.5	59.2	68°33.6	44.6	Suhail	222°43.4	-43°33.1
Mer.pass. 16:14		ν -0.9' d -0.1' m -3.90		ν 1.8' d 0.2' m 0.32		ν 2.3' d 0.1' m -2.00		ν 2.4' d 0.0' m 0.27				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	116°42.3	194°35.2	S23°01.7	286°47.3	S01°17.0	271°38.8	S08°59.2	83°36.0	N10°44.6	Alphard	217°44.2	-8°47.3
1	131°44.8	209°34.3	01.6	301°49.1	17.2	286°41.1	59.3	98°38.4	44.7	Regulus	207°30.7	11°49.3
2	146°47.2	224°33.3	01.4	316°50.9	17.4	301°43.4	59.3	113°40.9	44.7	Dubhe	193°36.8	61°35.3
3	161°49.7	239°32.4	•• 01.2	331°52.7	•• 17.6	316°45.7	•• 59.4	128°43.3	•• 44.7	Denebola	182°21.5	14°24.3
4	176°52.2	254°31.4	01.1	346°54.5	17.9	331°48.0	59.4	143°45.7	44.7	Gienah	175°40.0	-17°42.3
5	191°54.6	269°30.5	00.9	1°56.3	18.1	346°50.3	59.5	158°48.1	44.8	Acrux	172°56.0	-63°15.4
6	206°57.1	284°29.6	S23°00.7	16°58.1	S01°18.3	1°52.5	S08°59.6	173°50.5	N10°44.8	Gacrux	171°47.6	-57°16.4
7	221°59.6	299°28.6	00.6	31°59.9	18.6	16°54.8	59.6	188°52.9	44.8	Alioth	166°10.1	55°47.7
8	237°02.0	314°27.7	00.4	47°01.7	18.8	31°57.1	59.7	203°55.4	44.9	Spica	158°18.8	-11°18.8
9	252°04.5	329°26.7	•• 00.2	62°03.5	•• 19.0	46°59.4	•• 59.7	218°57.8	•• 44.9	Alkaid	152°49.5	-49°09.7
10	267°07.0	344°25.8	23°00.1	77°05.4	19.2	62°01.7	59.8	234°00.2	44.9	Hadar	148°31.2	-60°30.5
11	282°09.4	359°24.9	22°59.9	92°07.2	19.5	77°04.0	59.8	249°02.6	45.0	Menkent	147°53.7	-36°30.7
12	297°11.9	14°23.9	S22°59.7	107°09.0	S01°19.7	92°06.3	S08°59.9	264°05.0	N10°45.0	Arcturus	145°45.0	19°01.7
13	312°14.4	29°23.0	59.6	122°10.8	19.9	107°08.6	09°00.0	279°07.4	45.0	Rigel Kent.	139°35.8	-60°57.1
14	327°16.8	44°22.0	59.4	137°12.6	20.2	122°10.9	00.0	294°09.8	45.1	Kochab	137°20.5	74°01.8
15	342°19.3	59°21.1	•• 59.2	152°14.4	•• 20.4	137°13.1	•• 00.1	309°12.3	•• 45.1	Zuben'ubi	136°52.5	-16°09.7
16	357°21.7	74°20.2	59.0	167°16.2	20.6	152°15.4	00.1	324°14.7	45.1	Alphecca	126°01.2	26°36.8
17	12°24.2	89°19.2	58.9	182°18.1	20.8	167°17.7	00.2	339°17.1	45.1	Antares	112°12.0	-26°29.7
18	27°26.7	104°18.3	S22°58.7	197°19.9	S01°21.1	182°20.0	S09°00.2	354°19.5	N10°45.2	Atria	107°03.7	-69°04.5
19	42°29.1	119°17.4	58.5	212°21.7	21.3	197°22.3	00.3	9°21.9	45.2	Sabik	101°59.3	-15°45.6
20	57°31.6	134°16.4	58.3	227°23.5	21.5	212°24.6	00.4	24°24.3	45.2	Shaula	96°06.3	-37°07.3
21	72°34.1	149°15.5	•• 58.1	242°25.3	•• 21.7	227°26.9	•• 00.4	39°26.7	•• 45.3	Rasalhague	95°55.8	12°32.3
22	87°36.5	164°14.5	58.0	257°27.1	22.0	242°29.2	00.5	54°29.1	45.3	Eltanin	90°41.1	51°29.0
23	102°39.0	179°13.6	57.8	272°29.0	22.2	257°31.5	00.5	69°31.6	45.3	Kaus Aust.	83°28.6	-34°22.1
Mer.pass. 16:11		ν -0.9' d -0.2' m -3.90		ν 1.8' d 0.2' m 0.30		ν 2.3' d 0.1' m -2.01		ν 2.4' d 0.0' m 0.28		Vega	80°31.4	38°48.6

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	117°41.5	194°12.7	S22°57.6	287°30.8	S01°22.4	272°33.8	S09°00.6	84°34.0	N10°45.4	Nunki	75°44.1	-26°15.5
1	132°43.9	209°11.7	57.4	302°32.6	22.6	287°36.0	00.6	99°36.4	45.4	Altair	61°57.1	8°56.8
2	147°46.4	224°10.8	57.2	317°34.4	22.9	302°38.3	00.7	114°38.8	45.4	Peacock	53°01.3	-56°38.4
3	162°48.8	239°09.9	•• 57.0	332°36.3	•• 23.1	317°40.6	•• 00.7	129°41.2	•• 45.5	Deneb	49°24.0	45°23.2
4	177°51.3	254°08.9	56.8	347°38.1	23.3	332°42.9	00.8	144°43.6	45.5	Enif	33°35.8	10°00.6
5	192°53.8	269°08.0	56.6	2°39.9	23.5	347°45.2	00.9	159°46.0	45.5	Al Na'ir	27°29.3	-46°49.2
6	207°56.2	284°07.1	S22°56.4	17°41.7	S01°23.7	2°47.5	S09°00.9	174°48.4	N10°45.5	Fomalhaut	15°11.2	-29°28.1
7	222°58.7	299°06.1	56.3	32°43.6	24.0	17°49.8	01.0	189°50.9	45.6	Scheat	13°42.2	28°14.6
8	238°01.2	314°05.2	56.1	47°45.4	24.2	32°52.1	01.0	204°53.3	45.6	Markab	13°26.8	15°21.8
9	253°03.6	329°04.3	•• 55.9	62°47.2	•• 24.4	47°54.4	•• 01.1	219°55.7	•• 45.6			
10	268°06.1	344°03.3	55.7	77°49.0	24.6	62°56.7	01.1	234°58.1	45.7	Jan 16 Tue	SHA	Mer.pass
11	283°08.6	359°02.4	55.5	92°50.9	24.8	77°59.0	01.2	250°00.5	45.7	Venus	79°14.6	11:01
12	298°11.0	14°01.4	S22°55.3	107°52.7	S01°25.1	93°01.3	S09°01.2	265°02.9	N10°45.7	Mars	170°21.0	04:55
13	313°13.5	29°00.5	55.1	122°54.5	25.3	108°03.6	01.3	280°05.3	45.8	Jupiter	155°00.9	05:56
14	328°16.0	43°59.6	54.9	137°56.4	25.5	123°05.9	01.3	295°07.7	45.8	Saturn	326°54.8	18:26
15	343°18.4	58°58.6	•• 54.7	152°58.2	•• 25.7	138°08.2	•• 01.4	310°10.1	•• 45.8			
16	358°20.9	73°57.7	54.4	168°00.0	25.9	153°10.5	01.5	325°12.5	45.9	Jan 17 Wed	SHA	Mer.pass
17	13°23.3	88°56.8	54.2	183°01.9	26.2	168°12.8	01.5	340°15.0	45.9	Venus	76°31.2	11:04
18	28°25.8	103°55.8	S22°54.0	198°03.7	S01°26.4	183°15.0	S09°01.6	355°17.4	N10°45.9	Mars	169°49.3	04:49
19	43°28.3	118°54.9	53.8	213°05.5	26.6	198°17.3	01.6	10°19.8	46.0	Jupiter	154°52.3	05:49
20	58°30.7	133°54.0	53.6	228°07.4	26.8	213°19.6	01.7	25°22.2	46.0	Saturn	326°52.5	18:19
21	73°33.2	148°53.0	•• 53.4	243°09.2								

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	177°34.8	S20°55.1	162°55.6	13.0'	S15°09.3	-11.0'	55.6'
1	192°34.6	54.7	177°27.5	13.0'	14°58.3	-11.0'	55.5'
2	207°34.4	54.2	191°59.6	13.1'	14°47.2	-11.1'	55.5'
3	222°34.2	· · 53.7	206°31.7	13.2'	14°36.1	-11.2'	55.5'
4	237°34.0	53.2	221°03.9	13.3'	14°25.0	-11.2'	55.5'
5	252°33.7	52.8	235°36.2	13.3'	14°13.8	-11.3'	55.5'
6	267°33.5	S20°52.3	250°08.5	13.4'	S14°02.5	-11.3'	55.4'
7	282°33.3	51.8	264°40.9	13.5'	13°51.2	-11.4'	55.4'
8	297°33.1	51.3	279°13.4	13.6'	13°39.8	-11.4'	55.4'
9	312°32.9	· · 50.8	293°46.0	13.6'	13°28.4	-11.5'	55.4'
10	327°32.7	50.3	308°18.6	13.7'	13°17.0	-11.5'	55.4'
11	342°32.5	49.9	322°51.3	13.8'	13°05.4	-11.6'	55.3'
12	357°32.3	S20°49.4	337°24.1	13.8'	S12°53.9	-11.6'	55.3'
13	12°32.0	48.9	351°56.9	13.9'	12°42.3	-11.7'	55.3'
14	27°31.8	48.4	6°29.8	14.0'	12°30.6	-11.7'	55.3'
15	42°31.6	· · 47.9	21°02.8	14.0'	12°18.9	-11.7'	55.3'
16	57°31.4	47.4	35°35.9	14.1'	12°07.2	-11.8'	55.2'
17	72°31.2	46.9	50°09.0	14.2'	11°55.4	-11.8'	55.2'
18	87°31.0	S20°46.4	64°42.2	14.2'	S11°43.6	-11.9'	55.2'
19	102°30.8	45.9	79°15.4	14.3'	11°31.7	-11.9'	55.2'
20	117°30.6	45.5	93°48.7	14.4'	11°19.8	-11.9'	55.2'
21	132°30.4	· · 45.0	108°22.1	14.4'	11°07.9	-12.0'	55.1'
22	147°30.2	44.5	122°55.5	14.5'	10°55.9	-12.0'	55.1'
23	162°30.0	44.0	137°29.0	14.6'	10°43.9	-12.0'	55.1'
SD = 16.3'			SD = 15.2'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	177°29.8	S20°43.5	152°02.6	14.6'	S10°31.8	-12.1'	55.1'
1	192°29.6	43.0	166°36.2	14.7'	10°19.8	-12.1'	55.1'
2	207°29.3	42.5	181°09.9	14.7'	10°07.6	-12.1'	55.0'
3	222°29.1	· · 42.0	195°43.6	14.8'	09°55.5	-12.2'	55.0'
4	237°28.9	41.5	210°17.4	14.8'	09°43.3	-12.2'	55.0'
5	252°28.7	41.0	224°51.3	14.9'	09°31.1	-12.2'	55.0'
6	267°28.5	S20°40.5	239°25.2	15.0'	S09°18.9	-12.3'	55.0'
7	282°28.3	40.0	253°59.1	15.0'	09°06.6	-12.3'	54.9'
8	297°28.1	39.5	268°33.1	15.1'	08°54.3	-12.3'	54.9'
9	312°27.9	· · 39.0	283°07.2	15.1'	08°42.0	-12.3'	54.9'
10	327°27.7	38.5	297°41.3	15.2'	08°29.6	-12.4'	54.9'
11	342°27.5	38.0	312°15.5	15.2'	08°17.3	-12.4'	54.9'
12	357°27.3	S20°37.5	326°49.7	15.3'	S08°04.9	-12.4'	54.8'
13	12°27.1	37.0	341°24.0	15.3'	07°52.4	-12.4'	54.8'
14	27°26.9	36.5	355°58.3	15.4'	07°40.0	-12.5'	54.8'
15	42°26.7	· · 36.0	10°32.6	15.4'	07°27.5	-12.5'	54.8'
16	57°26.5	35.5	25°07.1	15.5'	07°15.0	-12.5'	54.8'
17	72°26.3	35.0	39°41.5	15.5'	07°02.5	-12.5'	54.8'
18	87°26.1	S20°34.5	54°16.0	15.5'	S06°50.0	-12.5'	54.7'
19	102°25.9	34.0	68°50.6	15.6'	06°37.5	-12.6'	54.7'
20	117°25.7	33.5	83°25.2	15.6'	06°24.9	-12.6'	54.7'
21	132°25.5	· · 33.0	97°59.8	15.7'	06°12.3	-12.6'	54.7'
22	147°25.3	32.4	112°34.5	15.7'	05°59.7	-12.6'	54.7'
23	162°25.1	31.9	127°09.2	15.8'	05°47.1	-12.6'	54.7'
SD = 16.2'			SD = 15.0'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	177°24.9	S20°31.4	141°43.9	15.8'	S05°34.5	-12.6'	54.6'
1	192°24.7	30.9	156°18.7	15.8'	05°21.9	-12.6'	54.6'
2	207°24.5	30.4	170°53.5	15.9'	05°09.2	-12.7'	54.6'
3	222°24.3	· · 29.9	185°28.4	15.9'	04°56.6	-12.7'	54.6'
4	237°24.1	29.4	200°03.3	15.9'	04°43.9	-12.7'	54.6'
5	252°23.9	28.9	214°38.2	16.0'	04°31.3	-12.7'	54.6'
6	267°23.7	S20°28.3	229°13.2	16.0'	S04°18.6	-12.7'	54.6'
7	282°23.5	27.8	243°48.2	16.0'	04°05.9	-12.7'	54.5'
8	297°23.3	27.3	258°23.2	16.1'	03°53.2	-12.7'	54.5'
9	312°23.1	· · 26.8	272°58.3	16.1'	03°40.5	-12.7'	54.5'
10	327°22.9	26.3	287°33.4	16.1'	03°27.7	-12.7'	54.5'
11	342°22.7	25.8	302°08.5	16.1'	03°15.0	-12.7'	54.5'
12	357°22.5	S20°25.3	316°43.6	16.2'	S03°02.3	-12.7'	54.5'
13	12°22.3	24.7	331°18.8	16.2'	02°49.6	-12.7'	54.5'
14	27°22.1	24.2	345°54.0	16.2'	02°36.8	-12.7'	54.4'
15	42°21.9	· · 23.7	0°29.2	16.2'	02°24.1	-12.7'	54.4'
16	57°21.7	23.2	15°04.5	16.3'	02°11.4	-12.7'	54.4'
17	72°21.5	22.6	29°39.8	16.3'	01°58.6	-12.7'	54.4'
18	87°21.4	S20°22.1	44°15.0	16.3'	S01°45.9	-12.7'	54.4'
19	102°21.2	21.6	58°50.4	16.3'	01°33.2	-12.7'	54.4'
20	117°21.0	21.1	73°25.7	16.4'	01°20.4	-12.7'	54.4'
21	132°20.8	· · 20.6	88°01.0	16.4'	01°07.7	-12.7'	54.4'
22	147°20.6	20.0	102°36.4	16.4'	00°55.0	-12.7'	54.3'
23	162°20.4	19.5	117°11.8	16.4'	00°42.2	-12.7'	54.3'
SD = 16.2'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:55	09:42	■	■	14:39	16:26
N 70°	07:41	09:11	11:36	12:45	15:10	16:40
68°	07:30	08:48	10:22	13:59	15:33	16:52
66°	07:20	08:30	09:46	14:35	15:51	17:01
64°	07:12	08:15	09:20	15:01	16:06	17:09
62°	07:05	08:03	09:00	15:21	16:18	17:16
60°	06:58	07:52	08:44	15:37	16:29	17:23
N 58°	06:53	07:43	08:30	15:51	16:38	17:28
56°	06:47	07:34	08:18	16:02	16:47	17:34
54°	06:43	07:27	08:08	16:13	16:54	17:38
52°	06:38	07:20	07:59	16:22	17:01	17:43
50°	06:34	07:14	07:51	16:30	17:07	17:47
45°	06:25	07:01	07:33	16:47	17:20	17:56
N 40°	06:16	06:49	07:19	17:02	17:31	18:05
35°	06:08	06:39	07:07	17:14	17:41	18:12
30°	06:01	06:30	06:56	17:24	17:50	18:19
20°	05:47	06:14	06:38	17:43	18:06	18:33
N 10°	05:33	05:59	06:22	17:59	18:21	18:47
0°	05:19	05:44	06:07	18:14	18:36	19:02
S 10°	05:02	05:29	05:51	18:29	18:52	19:18
20°	04:42	05:11	05:35	18:45	19:09	19:38
30°	04:17	04:49	05:16	19:05	19:31	20:03
35°	04:00	04:36	05:04	19:16	19:44	20:19
40°	03:41	04:20	04:51	19:29	20:00	20:39
45°	03:15	04:01	04:36	19:44	20:19	21:04
S 50°	02:39	03:36	04:17	20:03	20:44	21:40
52°	02:19	03:23	04:08	20:12	20:56	21:59
54°	01:54	03:09	03:58	20:22	21:10	22:23
56°	01:17	02:52	03:46	20:33	21:26	22:59
58°	—:—	02:31	03:33	20:46	21:47	—:—
S 60°	—:—	02:04	03:17	21:02	22:13	—:—

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	10:34	10:00	09:34	16:55	18:59	20:52
N 70°	10:10	09:48	09:30	17:17	19:10	20:54
68°	09:51	09:37	09:25	17:34	19:18	20:56
66°	09:36	09:29	09:22	17:48	19:25	20:57
64°	09:24	09:22	09:19	17:59	19:30	20:58
62°	09:13	09:15	09:17	18:09	19:35	20:59
60°	09:04	09:10	09:14	18:17	19:40	20:59
N 58°	08:56	09:05	09:12	18:24	19:43	21:00
56°	08:49	09:01	09:11	18:30	19:47	21:01
54°	08:43	08:57	09:09	18:36	19:50	21:01
52°	08:37	08:53	09:08	18:41	19:52	21:02
50°	08:32	08:50	09:06	18:45	19:55	21:02
45°	08:20	08:43	09:03	18:55	20:00	21:03
N 40°	08:11	08:37	09:01	19:03	20:04	21:04
35°	08:03	08:32	08:59	19:10	20:08	21:04
30°	07:56	08:28	08:57	19:16	20:11	21:05
20°	07:43	08:20	08:54	19:26	20:17	21:06
N 10°	07:33	08:13	08:51	19:35	20:22	21:07
0°	07:22	08:06	08:48	19:44	20:27	21:08
S 10°	07:12	08:00	08:45	19:52	20:31	21:09
20°	07:01	07:53	08:43	20:01	20:36	21:09
30°	06:49	07:45	08:39	20:11	20:42	21:10
35°	06:41	07:40	08:37	20:17	20:45	21:11
40°	06:33	07:35	08:35	20:23	20:48	21:11
45°	06:23	07:29	08:33	20:31	20:52	21:12
S 50°	06:11	07:22	08:30	20:40	20:57	21:13
52°	06:06	07:18	08:28	20:44	20:59	21:13
54°	06:00	07:14	08:27	20:49	21:02	21:14
56°	05:53	07:10	08:25	20:54	21:04	21:14
58°	05:45	07:06	08:23	20:59	21:07	21:15
S 60°	05:36	07:00	08:21	21:05	21:11	21:15

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12 ^h	Pass	Mer.Pass.	Lower	2-4
	00 ^h	mm:ss	hh:mm	Upper	hh:mm	2-10%
16	09:41	09:51	12:10	13:33	01:10	
17	10:01	10:11	12:10	14:17	01:55	
18	10:20	10:30	12:10	14:58	02:37	

January 19, 20, 21 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	118°40.6	193°50.2	S22°52.8	288°14.7	S01°27.7	273°28.8	S09°01.9	85°31.8	N10°46.1	Alpheratz	357°31.5	29°15.3											
1	133°43.1	208°49.3	52.5	303°16.6	27.9	288°31.1	01.9	100°34.2	46.2	Ankaa	353°04.2	-42°09.0											
2	148°45.5	223°48.4	52.3	318°18.4	28.1	303°33.4	02.0	115°36.6	46.2	Schedar	349°27.3	56°42.1											
3	163°48.0	238°47.4	•• 52.1	333°20.2	•• 28.3	318°35.7	•• 02.0	130°39.0	•• 46.2	Diphda	348°44.1	-17°49.7											
4	178°50.5	253°46.5	51.9	348°22.1	28.5	333°38.0	02.1	145°41.4	46.3	Achernar	335°17.9	-57°05.6											
5	193°52.9	268°45.6	51.7	3°23.9	28.7	348°40.3	02.1	160°43.9	46.3	Hamal	327°47.4	23°36.1											
6	208°55.4	283°44.7	S22°51.4	18°25.8	S01°29.0	3°42.6	S09°02.2	175°46.3	N10°46.3	Polaris	312°02.2	89°23.4											
7	223°57.8	298°43.7	51.2	33°27.6	29.2	18°44.9	02.2	190°48.7	46.4	Acamar	315°09.2	-40°11.5											
8	239°00.3	313°42.8	51.0	48°29.5	29.4	33°47.2	02.3	205°51.1	46.4	Menkar	314°02.6	4°12.2											
9	254°02.8	328°41.9	•• 50.8	63°31.3	•• 29.6	48°49.5	•• 02.3	220°53.5	•• 46.4	Mirfak	308°23.2	49°58.0											
10	269°05.2	343°40.9	50.5	78°33.1	29.8	63°51.8	02.4	235°55.9	46.5	Aldebaran	290°35.6	16°34.0											
11	284°07.7	358°40.0	50.3	93°35.0	30.0	78°54.1	02.5	250°58.3	46.5	Rigel	281°00.4	-8°10.3											
12	299°10.2	13°39.1	S22°50.1	108°36.8	S01°30.2	93°56.4	S09°02.5	266°00.7	N10°46.5	Capella	280°16.6	46°01.6											
13	314°12.6	28°38.1	49.9	123°38.7	30.4	108°58.7	02.6	281°03.1	46.6	Bellatrix	278°19.1	6°22.4											
14	329°15.1	43°37.2	49.6	138°40.5	30.7	124°01.0	02.6	296°05.5	46.6	Elnath	277°57.4	28°37.8											
15	344°17.6	58°36.3	•• 49.4	153°42.4	•• 30.9	139°03.3	•• 02.7	311°07.9	•• 46.6	Alnilam	275°34.1	-1°11.2											
16	359°20.0	73°35.4	49.2	168°44.2	31.1	154°05.6	02.7	326°10.3	46.7	Betelgeuse	270°48.2	7°24.6											
17	14°22.5	88°34.4	48.9	183°46.1	31.3	169°07.9	02.8	341°12.7	46.7	Canopus	263°50.4	-52°42.9											
18	29°24.9	103°33.5	S22°48.7	198°47.9	S01°31.5	184°10.3	S09°02.8	356°15.1	N10°46.7	Sirius	258°23.0	-16°45.6											
19	44°27.4	118°32.6	48.4	213°49.8	31.7	199°12.6	02.9	11°17.5	46.8	Adhara	255°02.9	-29°00.9											
20	59°29.9	133°31.6	48.2	228°51.6	31.9	214°14.9	02.9	26°19.9	46.8	Procyon	244°47.1	5°08.8											
21	74°32.3	148°30.7	•• 48.0	243°53.5	•• 32.1	229°17.2	•• 03.0	41°22.3	•• 46.8	Pollux	243°12.9	27°57.1											
22	89°34.8	163°29.8	47.7	258°55.4	32.3	244°19.5	03.0	56°24.7	46.9	Avior	234°12.6	-59°36.3											
23	104°37.3	178°28.9	47.5	273°57.2	32.5	259°21.8	03.1	71°27.2	46.9	Suhail	222°43.4	-43°33.1											
Mer.pass. 16:03				ν -0.9' d -0.2' m -3.90				ν 1.8' d 0.2' m 0.25				ν 2.3' d 0.1' m -2.02				ν 2.4' d 0.0' m 0.28							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	119°39.7	193°27.9	S22°47.2	288°59.1	S01°32.7	274°24.1	S09°03.1	86°29.6	N10°46.9										
1	134°42.2	208°27.0	47.0	304°00.9	33.0	289°26.4	03.2	101°32.0	47.0										
2	149°44.7	223°26.1	46.7	319°02.8	33.2	304°28.7	03.2	116°34.4	47.0										
3	164°47.1	238°25.1	•• 46.5	334°04.6	•• 33.4	319°31.0	•• 03.3	131°36.8	•• 47.0										
4	179°49.6	253°24.2	46.2	349°06.5	33.6	334°33.3	03.3	146°39.2	47.1										
5	194°52.1	268°23.3	46.0	4°08.4	33.8	349°35.6	03.4	161°41.6	47.1										
6	209°54.5	283°22.4	S22°45.7	19°10.2	S01°34.0	4°37.9	S09°03.4	176°44.0	N10°47.1										
7	224°57.0	298°21.4	45.5	34°12.1	34.2	19°40.2	03.5	191°46.4	47.2										
8	239°59.4	313°20.5	45.2	49°13.9	34.4	34°42.5	03.5	206°48.8	47.2										
9	255°01.9	328°19.6	•• 45.0	64°15.8	•• 34.6	49°44.8	•• 03.6	221°51.2	•• 47.2										
10	270°04.4	343°18.7	44.7	79°17.7	34.8	64°47.1	03.6	236°53.6	47.3										
11	285°06.8	358°17.7	44.5	94°19.5	35.0	79°49.4	03.7	251°56.0	47.3										
12	300°09.3	13°16.8	S22°44.2	109°21.4	S01°35.2	94°51.8	S09°03.7	266°58.4	N10°47.3										
13	315°11.8	28°15.9	43.9	124°23.3	35.4	109°54.1	03.8	282°00.8	47.4										
14	330°14.2	43°15.0	43.7	139°25.1	35.6	124°56.4	03.8	297°03.2	47.4										
15	345°16.7	58°14.0	•• 43.4	154°27.0	•• 35.8	139°58.7	•• 03.9	312°05.6	•• 47.4										
16	0°19.2	73°13.1	43.2	169°28.9	36.0	155°01.0	03.9	327°08.0	47.5										
17	15°21.6	88°12.2	42.9	184°30.7	36.2	170°03.3	03.9	342°10.4	47.5										
18	30°24.1	103°11.3	S22°42.6	199°32.6	S01°36.4	185°05.6	S09°04.0	357°12.8	N10°47.5										
19	45°26.5	118°10.3	42.4	214°34.5	36.6	200°07.9	04.0	12°15.2	47.6										
20	60°29.0	133°09.4	42.1	229°36.4	36.8	215°10.2	04.1	27°17.6	47.6										
21	75°31.5	148°08.5	•• 41.8	244°38.2	•• 37.0	230°12.5	•• 04.1	42°20.0	•• 47.7										
22	90°33.9	163°07.6	41.5	259°40.1	37.2	245°14.9	04.2	57°22.4	47.7										
23	105°36.4	178°06.7	41.3	274°42.0	37.4	260°17.2	04.2	72°24.8	47.7										
Mer.pass. 15:59				ν -0.9' d -0.2' m -3.90				ν 1.9' d 0.2' m 0.23				ν 2.3' d 0.1' m -2.03				ν 2.4' d 0.0' m 0.29			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	120°38.9	193°05.7	S22°41.0	289°43.8	S01°37.6	275°19.5	S09°04.3	87°27.2	N10°47.8										
1	135°41.3	208°04.8	40.7	304°45.7	37.8	290°21.8	04.3	102°29.6	47.8										
2	150°43.8	223°03.9	40.5	319°47.6	38.0	305°24.1	04.4	117°32.0	47.8										
3	165°46.3	238°03.0	•• 40.2	334°49.5	•• 38.2	320°26.4	•• 04.4	132°34.4	•• 47.9										
4	180°48.7	253°02.0	39.9	349°51.4	38.4	335°28.7	04.5	147°36.8	47.9										
5	195°51.2	268°01.1	39.6	4°53.2	38.6	350°31.0	04.5	162°39.2	47.9										
6	210°53.7	283°00.2	S22°39.3	19°55.1	S01°38.8	5°33.4	S09°04.6	177°41.6	N10°48.0										
7	225°56.1	297°59.3	39.1	34°57.0	39.0	20°35.7	04.6	192°44.0	48.0										
8	240°58.6	312°58.4	38.8	49°58.9	39.2	35°38.0	04.7	207°46.4	48.0										
9	256°01.0	327°57.4	•• 38.5	65°00.8	•• 39.4	50°40.3	•• 04.7	222°48.8	•• 48.1										
10	271°03.5	342°56.5	38.2	80°02.6	39.6	65°42.6	04.8	237°51.2	48.1										
11	286°06.0	357°55.6	37.9	95°04.5	39.8	80°44.9	04.8	252°53.6	48.2										
12	301°08.4	12°54.7	S22°37.6	110°06.4	S01°40.0	95°47.2	S09°04.8	267°56.0	N10°48.2										
13	316°10.9	27°53.8	37.3	125°08.3	40.2	110°49.6	04.9	282°58.4	48.2										
14	331°13.4	42°52.8	37.0	140°10.2	40.4	125°51.9	04.9	298°00.8	48.3										
15	346°15.8	57°51.9	•• 36.8	155°12.1	•• 40.6	140°54.2	•• 05.0	313°03.2	•• 48.3										
16	1°18.3	72°51.0	36.5	170°13.9	40.8	155°56.5	05.0	328°05.5	48.3										
17	16°20.8	87°50.1	36.2	185°15.8	41.0	170°58.8	05.1	343°07.9	48.4										
18	31°23.2	102°49.2	S22°35.9	200°17.7	S01°41.2	186°01.1	S09°05.1	358°10.3	N10°48.4										
19	46°25.7	117°48.3	35.6	215°19.6	41.4	201°03.5	05.2	13°12.7	48.4										
20	61°28.2	132°47.3	35.3	230°21.5	41.5	216°05.8	05.2	28°15.1	48.5										
21	76°30.6	147°46.4	•• 35.0	245°23.4	•• 41.7	231°08.1	•• 05.3	43°17.5	•• 48.5										
22	91°33.1	162°45.5	34.7	260°25.3	41.9	246°10.4	05.3	58°19.9	48.6										
23	106°35.5	177°44.6	34.4	275°27.2	42.1	261°12.7	05.3	73°22.3	48.6										
Mer.pass. 15:55				ν -0.9' d -0.3' m -3.90				ν 1.9' d 0.2' m 0.21				ν 2.3' d 0.0' m -2.03				ν 2.4' d 0.0' m 0.29			

Jan 19 Fri	SHA	Mer.pass
Venus	75°09.7	11:05
Mars	169°34.1	04:46
Jupiter	154°48.2	05:45
Saturn	326°51.2	18:15

Jan 20 Sat	SHA	Mer.pass
Venus	73°48.2	11:07
Mars	169°19.3	04:43
Jupiter	154°44.3	05:42
Saturn	326°49.8	18:11

Jan 21 Sun	SHA	Mer.pass
Venus	72°26.9	11:08
Mars	169°05.0	04:40
Jupiter	154°40.6	05:38
Saturn	326°48.3	18:07

Horizontal parallax		
Venus:		0.1
Mars:		0.1

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	177°20.2	S20°19.0	131°47.2	16.4'	S00°29.5	-12.7'	54.3'
1	192°20.0	18.5	146°22.6	16.4'	00°16.8	-12.7'	54.3'
2	207°19.8	17.9	160°58.1	16.4'	S00°04.1	-12.7'	54.3'
3	222°19.6	· · 17.4	175°33.5	16.5'	N00°08.6	12.7'	54.3'
4	237°19.4	16.9	190°09.0	16.5'	00°21.3	12.7'	54.3'
5	252°19.2	16.3	204°44.4	16.5'	00°34.0	12.7'	54.3'
6	267°19.1	S20°15.8	219°19.9	16.5'	N00°46.7	12.7'	54.3'
7	282°18.9	15.3	233°55.4	16.5'	00°59.4	12.7'	54.3'
8	297°18.7	14.7	248°30.9	16.5'	01°12.1	12.7'	54.2'
9	312°18.5	· · 14.2	263°06.4	16.5'	01°24.7	12.6'	54.2'
10	327°18.3	13.7	277°41.9	16.5'	01°37.4	12.6'	54.2'
11	342°18.1	13.1	292°17.4	16.5'	01°50.0	12.6'	54.2'
12	357°17.9	S20°12.6	306°52.9	16.5'	N02°02.6	12.6'	54.2'
13	12°17.7	12.1	321°28.5	16.5'	02°15.2	12.6'	54.2'
14	27°17.5	11.5	336°04.0	16.5'	02°27.8	12.6'	54.2'
15	42°17.4	· · 11.0	350°39.5	16.5'	02°40.4	12.6'	54.2'
16	57°17.2	10.5	5°15.0	16.5'	02°53.0	12.6'	54.2'
17	72°17.0	09.9	19°50.6	16.5'	03°05.6	12.5'	54.2'
18	87°16.8	S20°09.4	34°26.1	16.5'	N03°18.1	12.5'	54.2'
19	102°16.6	08.9	49°01.6	16.5'	03°30.6	12.5'	54.2'
20	117°16.4	08.3	63°37.1	16.5'	03°43.1	12.5'	54.2'
21	132°16.2	· · 07.8	78°12.7	16.5'	03°55.6	12.5'	54.1'
22	147°16.1	07.2	92°48.2	16.5'	04°08.1	12.5'	54.1'
23	162°15.9	06.7	107°23.7	16.5'	04°20.6	12.4'	54.1'
SD = 16.2' $d = -0.5'$ SD = 14.8'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	177°15.7	S20°06.2	121°59.2	16.5'	N04°33.0	12.4'	54.1'
1	192°15.5	05.6	136°34.7	16.5'	04°45.4	12.4'	54.1'
2	207°15.3	05.1	151°10.1	16.5'	04°57.8	12.4'	54.1'
3	222°15.1	· · 04.5	165°45.6	16.5'	05°10.2	12.3'	54.1'
4	237°15.0	04.0	180°21.1	16.4'	05°22.5	12.3'	54.1'
5	252°14.8	03.4	194°56.5	16.4'	05°34.8	12.3'	54.1'
6	267°14.6	S20°02.9	209°31.9	16.4'	N05°47.1	12.3'	54.1'
7	282°14.4	02.3	224°07.4	16.4'	05°59.4	12.3'	54.1'
8	297°14.2	01.8	238°42.8	16.4'	06°11.7	12.2'	54.1'
9	312°14.1	· · 01.2	253°18.1	16.4'	06°23.9	12.2'	54.1'
10	327°13.9	00.7	267°53.5	16.4'	06°36.1	12.2'	54.1'
11	342°13.7	20°00.1	282°28.9	16.3'	06°48.3	12.2'	54.1'
12	357°13.5	S19°59.6	297°04.2	16.3'	N07°00.4	12.1'	54.1'
13	12°13.3	59.0	311°39.5	16.3'	07°12.6	12.1'	54.1'
14	27°13.2	58.5	326°14.8	16.3'	07°24.7	12.1'	54.1'
15	42°13.0	· · 57.9	340°50.1	16.2'	07°36.7	12.0'	54.1'
16	57°12.8	57.4	355°25.3	16.2'	07°48.8	12.0'	54.1'
17	72°12.6	56.8	10°00.5	16.2'	08°00.8	12.0'	54.1'
18	87°12.4	S19°56.3	24°35.7	16.2'	N08°12.7	11.9'	54.1'
19	102°12.3	55.7	39°10.9	16.1'	08°24.7	11.9'	54.1'
20	117°12.1	55.2	53°46.1	16.1'	08°36.6	11.9'	54.1'
21	132°11.9	· · 54.6	68°21.2	16.1'	08°48.5	11.8'	54.1'
22	147°11.7	54.1	82°56.3	16.1'	09°00.3	11.8'	54.1'
23	162°11.6	53.5	97°31.3	16.0'	09°12.1	11.8'	54.1'
SD = 16.2' $d = -0.5'$ SD = 14.8'							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	177°11.4	S19°52.9	112°06.4	16.0'	N09°23.9	11.7'	54.1'
1	192°11.2	52.4	126°41.4	16.0'	09°35.6	11.7'	54.1'
2	207°11.0	51.8	141°16.3	15.9'	09°47.4	11.7'	54.1'
3	222°10.9	· · 51.3	155°51.3	15.9'	09°59.0	11.6'	54.1'
4	237°10.7	50.7	170°26.2	15.9'	10°10.7	11.6'	54.1'
5	252°10.5	50.1	185°01.0	15.8'	10°22.2	11.6'	54.1'
6	267°10.3	S19°49.6	199°35.9	15.8'	N10°33.8	11.5'	54.1'
7	282°10.2	49.0	214°10.7	15.8'	10°45.3	11.5'	54.1'
8	297°10.0	48.5	228°45.4	15.7'	10°56.8	11.4'	54.1'
9	312°09.8	· · 47.9	243°20.1	15.7'	11°08.2	11.4'	54.1'
10	327°09.6	47.3	257°54.8	15.6'	11°19.6	11.4'	54.1'
11	342°09.5	46.8	272°29.4	15.6'	11°31.0	11.3'	54.1'
12	357°09.3	S19°46.2	287°04.0	15.6'	N11°42.3	11.3'	54.1'
13	12°09.1	45.6	301°38.6	15.5'	11°53.6	11.2'	54.1'
14	27°09.0	45.1	316°13.1	15.5'	12°04.8	11.2'	54.1'
15	42°08.8	· · 44.5	330°47.6	15.4'	12°16.0	11.1'	54.1'
16	57°08.6	43.9	345°22.0	15.4'	12°27.1	11.1'	54.2'
17	72°08.4	43.4	359°56.4	15.3'	12°38.2	11.0'	54.2'
18	87°08.3	S19°42.8	14°30.7	15.3'	N12°49.2	11.0'	54.2'
19	102°08.1	42.2	29°05.0	15.2'	13°00.2	10.9'	54.2'
20	117°07.9	41.7	43°39.2	15.2'	13°11.1	10.9'	54.2'
21	132°07.8	· · 41.1	58°13.4	15.1'	13°22.0	10.8'	54.2'
22	147°07.6	40.5	72°47.5	15.1'	13°32.9	10.8'	54.2'
23	162°07.4	39.9	87°21.6	15.0'	13°43.7	10.7'	54.2'
SD = 16.2' $d = -0.6'$ SD = 14.8'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:47	09:29	■	■	14:54	16:36
N 70°	07:34	09:01	11:02	13:21	15:22	16:49
68°	07:24	08:40	10:08	14:15	15:43	16:59
66°	07:15	08:23	09:36	14:47	16:00	17:08
64°	07:07	08:09	09:13	15:10	16:14	17:16
62°	07:01	07:58	08:54	15:29	16:25	17:22
60°	06:55	07:48	08:39	15:44	16:35	17:28
N 58°	06:49	07:39	08:26	15:57	16:44	17:34
56°	06:44	07:31	08:14	16:08	16:52	17:38
54°	06:40	07:24	08:05	16:18	16:59	17:43
52°	06:36	07:18	07:56	16:27	17:05	17:47
50°	06:32	07:12	07:48	16:35	17:11	17:51
45°	06:23	06:59	07:31	16:51	17:24	18:00
N 40°	06:15	06:48	07:17	17:05	17:34	18:08
35°	06:08	06:39	07:06	17:17	17:44	18:15
30°	06:01	06:30	06:55	17:27	17:53	18:22
20°	05:47	06:14	06:38	17:45	18:08	18:35
N 10°	05:34	06:00	06:22	18:00	18:22	18:48
0°	05:20	05:45	06:07	18:15	18:37	19:02
S 10°	05:04	05:30	05:53	18:29	18:52	19:18
20°	04:45	05:13	05:37	18:45	19:09	19:37
30°	04:20	04:52	05:18	19:04	19:30	20:02
35°	04:04	04:39	05:07	19:14	19:43	20:17
40°	03:45	04:24	04:55	19:27	19:58	20:37
45°	03:20	04:05	04:40	19:42	20:16	21:01
S 50°	02:46	03:41	04:22	20:00	20:40	21:35
52°	02:28	03:29	04:13	20:08	20:52	21:53
54°	02:05	03:16	04:03	20:18	21:05	22:15
56°	01:32	03:00	03:52	20:29	21:21	22:46
58°	—:—	02:40	03:40	20:41	21:40	23:51
S 60°	—:—	02:16	03:25	20:56	22:04	—:—

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	09:11	08:47	08:20	22:42	· · ·	00:35
N 70°	09:13	08:56	08:37	22:36	· · ·	00:20
68°	09:14	09:03	08:51	22:31	· · ·	00:08
66°	09:16	09:09	09:03	22:27	23:58	· · ·
64°	09:17	09:14	09:12	22:23	23:49	· · ·
62°	09:18	09:19	09:21	22:20	23:42	· · ·
60°	09:19	09:23	09:28	22:18	23:36	· · ·
N 58°	09:19	09:26	09:34	22:15	23:31	· · ·
56°	09:20	09:29	09:40	22:13	23:26	· · ·
54°	09:21	09:32	09:45	22:12	23:22	· · ·
52°	09:21	09:35	09:49	22:10	23:18	· · ·
50°	09:22	09:37	09:54	22:08	23:15	· · ·
45°	09:23	09:42	10:03	22:05	23:07	· · ·
N 40°	09:24	09:46	10:10	22:02	23:01	· · ·
35°	09:24	09:50	10:17	22:00	22:55	23:51
30°	09:25	09:53	10:23	21:58	22:51	23:44
20°	09:26	09:59	10:33	21:54	22:42	23:32
N 10°	09:28	10:04	10:42	21:51	22:35	23:21
0°	09:29	10:09	10:50	21:48	22:29	23:10
S 10°	09:30	10:14	10:59	21:45	22:22	23:00
20°	09:31	10:19	11:08	21:42	22:15	22:49
30°	09:32	10:25	11:18	21:38	22:07	22:37
35°	09:33	10:29	11:24	21:36	22:02	22:30
40°	09:34	10:32	11:31	21:34	21:57	22:22
45°	09:35	10:37	11:39	21:31	21:51	22:13
S 50°	09:36	10:43	11:49	21:28	21:44	22:01
52°	09:37	10:45	11:54	21:27	21:41	21:56
54°	09:38	10:48	11:59	21:25	21:37	21:51
56°	09:38	10:51	12:04	21:23	21:33	21:44
58°	09:39	10:55	12:10	21:21	21:29	21:37
S 60°	09:40	10:58	12:18	21:19	21:24	21:29

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	5-7 17-33%
19	10:39	10:48	12:11	15:38	03:18	
20	10:57	11:06	12:11	16:19	03:59	
21	11:14	11:23	12:11	17:00	04:39	

Aries		Venus		Mars		Jupiter		Saturn	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	121°38.0	192°43.7	S22°34.1	290°29.1	S01°42.3	276°15.0	S09°05.4	88°24.7	N10°48.6
1	136°40.5	207°42.8	33.8	305°31.0	42.5	291°17.4	05.4	103°27.1	48.7
2	151°42.9	222°41.8	33.5	320°32.9	42.7	306°19.7	05.5	118°29.5	48.7
3	166°45.4	237°40.9	•• 33.2	335°34.8	•• 42.9	321°22.0	•• 05.5	133°31.9	•• 48.7
4	181°47.9	252°40.0	32.9	350°36.7	43.1	336°24.3	05.6	148°34.3	48.8
5	196°50.3	267°39.1	32.5	5°38.5	43.3	351°26.6	05.6	163°36.7	48.8
6	211°52.8	282°38.2	S22°32.2	20°40.4	S01°43.5	6°29.0	S09°05.7	178°39.1	N10°48.9
7	226°55.3	297°37.3	31.9	35°42.3	43.6	21°31.3	05.7	193°41.5	48.9
8	241°57.7	312°36.3	31.6	50°44.2	43.8	36°33.6	05.7	208°43.9	48.9
9	257°00.2	327°35.4	•• 31.3	65°46.1	•• 44.0	51°35.9	•• 05.8	223°46.3	•• 49.0
10	272°02.6	342°34.5	31.0	80°48.0	44.2	66°38.3	05.8	238°48.6	49.0
11	287°05.1	357°33.6	30.7	95°49.9	44.4	81°40.6	05.9	253°51.0	49.0
12	302°07.6	12°32.7	S22°30.4	110°51.8	S01°44.6	96°42.9	S09°05.9	268°53.4	N10°49.1
13	317°10.0	27°31.8	30.0	125°53.8	44.8	111°45.2	06.0	283°55.8	49.1
14	332°12.5	42°30.9	29.7	140°55.7	45.0	126°47.5	06.0	298°58.2	49.2
15	347°15.0	57°30.0	•• 29.4	155°57.6	•• 45.1	141°49.9	•• 06.0	314°00.6	•• 49.2
16	2°17.4	72°29.0	29.1	170°59.5	45.3	156°52.2	06.1	329°03.0	49.2
17	17°19.9	87°28.1	28.8	186°01.4	45.5	171°54.5	06.1	344°05.4	49.3
18	32°22.4	102°27.2	S22°28.4	201°03.3	S01°45.7	186°56.8	S09°06.2	359°07.8	N10°49.3
19	47°24.8	117°26.3	28.1	216°05.2	45.9	201°59.2	06.2	14°10.2	49.3
20	62°27.3	132°25.4	27.8	231°07.1	46.1	217°01.5	06.3	29°12.6	49.4
21	77°29.8	147°24.5	•• 27.5	246°09.0	•• 46.3	232°03.8	•• 06.3	44°15.0	•• 49.4
22	92°32.2	162°23.6	27.1	261°10.9	46.4	247°06.1	06.3	59°17.4	49.5
23	107°34.7	177°22.7	26.8	276°12.8	46.6	262°08.5	06.4	74°19.7	49.5
Mer.pass. 15:51		ν -0.9' d -0.3' m -3.90		ν 1.9' d 0.2' m 0.19		ν 2.3' d 0.0' m -2.04		ν 2.4' d 0.0' m 0.29	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	122°37.1	192°21.8	S22°26.5	291°14.7	S01°46.8	277°10.8	S09°06.4	89°22.1	N10°49.5
1	137°39.6	207°20.8	26.1	306°16.7	47.0	292°13.1	06.5	104°24.5	49.6
2	152°42.1	222°19.9	25.8	321°18.6	47.2	307°15.4	06.5	119°26.9	49.6
3	167°44.5	237°19.0	· · 25.5	336°20.5	· · 47.4	322°17.8	· · 06.6	134°29.3	· · 49.7
4	182°47.0	252°18.1	25.1	351°22.4	47.5	337°20.1	06.6	149°31.7	49.7
5	197°49.5	267°17.2	24.8	6°24.3	47.7	352°22.4	06.6	164°34.1	49.7
6	212°51.9	282°16.3	S22°24.5	21°26.2	S01°47.9	7°24.7	S09°06.7	179°36.5	N10°49.8
7	227°54.4	297°15.4	24.1	36°28.1	48.1	22°27.1	06.7	194°38.9	49.8
8	242°56.9	312°14.5	23.8	51°30.1	48.3	37°29.4	06.8	209°41.3	49.8
9	257°59.3	327°13.6	· · 23.4	66°32.0	· · 48.4	52°31.7	· · 06.8	224°43.6	· · 49.9
10	273°01.8	342°12.7	23.1	81°33.9	48.6	67°34.1	06.8	239°46.0	49.9
11	288°04.3	357°11.8	22.7	96°35.8	48.8	82°36.4	06.9	254°48.4	50.0
12	303°06.7	12°10.9	S22°22.4	111°37.7	S01°49.0	97°38.7	S09°06.9	269°50.8	N10°50.0
13	318°09.2	27°09.9	22.1	126°39.7	49.2	112°41.0	07.0	284°53.2	50.0
14	333°11.6	42°09.0	21.7	141°41.6	49.3	127°43.4	07.0	299°55.6	50.1
15	348°14.1	57°08.1	· · 21.4	156°43.5	· · 49.5	142°45.7	· · 07.0	314°58.0	· · 50.1
16	3°16.6	72°07.2	21.0	171°45.4	49.7	157°48.0	07.1	330°00.4	50.2
17	18°19.0	87°06.3	20.7	186°47.4	49.9	172°50.4	07.1	345°02.7	50.2
18	33°21.5	102°05.4	S22°20.3	201°49.3	S01°50.0	187°52.7	S09°07.2	0°05.1	N10°50.2
19	48°24.0	117°04.5	20.0	216°51.2	50.2	202°55.0	07.2	15°07.5	50.3
20	63°26.4	132°03.6	19.6	231°53.1	50.4	217°57.4	07.2	30°09.9	50.3
21	78°28.9	147°02.7	· · 19.2	246°55.1	· · 50.6	232°59.7	· · 07.3	45°12.3	· · 50.4
22	93°31.4	162°01.8	18.9	261°57.0	50.7	248°02.0	07.3	60°14.7	50.4
23	108°33.8	177°00.9	18.5	276°58.9	50.9	263°04.3	07.4	75°17.6	50.4
Mer.pass. 15:47		$\nu -0.9^{\circ} d -0.3^{\circ}$ m -3.90		$\nu 1.9^{\circ} d 0.2'$ m 0.17		$\nu 2.3' d 0.0'$ m -2.05		$\nu 2.4' d 0.0'$ m 0.30	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	123°36.3	192°00.0	S22°18.2	292°00.9	S01°51.1	278°06.7	S09°07.4	90°19.5	N10°50.5
1	138°38.7	206°59.1	17.8	307°02.8	51.3	293°09.0	07.4	105°21.8	50.5
2	153°41.2	221°58.2	17.4	322°04.7	51.4	308°11.3	07.5	120°22.2	50.6
3	168°43.7	236°57.3	· · 17.1	337°06.7	· · 51.6	323°13.7	· · 07.5	135°26.6	· · 50.6
4	183°46.1	251°56.4	16.7	352°08.6	51.8	338°16.0	07.6	150°29.0	50.6
5	198°48.6	266°55.5	16.4	7°10.5	52.0	353°18.3	07.6	165°31.4	50.7
6	213°51.1	281°54.6	S22°16.0	22°12.5	S01°52.1	8°20.7	S09°07.6	180°33.8	N10°50.7
7	228°53.5	296°53.7	15.6	37°14.4	52.3	23°23.0	07.7	195°36.1	50.8
8	243°56.0	311°52.8	15.2	52°16.3	52.5	38°25.3	07.7	210°38.5	50.8
9	258°58.5	326°51.9	· · 14.9	67°18.3	· · 52.6	53°27.7	· · 07.8	225°40.9	· · 50.8
10	274°00.9	341°51.0	14.5	82°20.2	52.8	68°30.0	07.8	240°43.3	50.9
11	289°03.4	356°50.1	14.1	97°22.2	53.0	83°32.4	07.8	255°45.7	50.9
12	304°05.9	11°49.2	S22°13.8	112°24.1	S01°53.2	98°34.7	S09°07.9	270°48.1	N10°51.0
13	319°08.3	26°48.3	13.4	127°26.0	53.3	113°37.0	07.9	285°50.5	51.0
14	334°10.8	41°47.4	13.0	142°28.0	53.5	128°39.4	07.9	300°52.8	51.0
15	349°13.2	56°46.5	· · 12.6	157°29.9	· · 53.7	143°41.7	· · 08.0	315°55.2	· · 51.1
16	4°15.7	71°45.6	12.3	172°31.9	53.8	158°44.0	08.0	330°57.6	51.1
17	19°18.2	86°44.7	11.9	187°33.8	54.0	173°46.4	08.1	346°00.0	51.2
18	34°20.6	101°43.8	S22°11.5	202°35.8	S01°54.2	188°48.7	S09°08.1	1°02.4	N10°51.2
19	49°23.1	116°42.9	11.1	217°37.7	54.3	203°51.1	08.1	16°04.8	51.2
20	64°25.6	131°42.0	10.7	232°39.7	54.5	218°53.4	08.2	31°07.1	51.3
21	79°28.0	146°41.1	· · 10.3	247°41.6	· · 54.7	233°55.7	· · 08.2	46°09.5	· · 51.3
22	94°30.5	161°40.2	10.0	262°43.6	54.8	248°58.1	08.2	61°11.9	51.4
23	109°33.0	176°39.3	09.6	277°45.5	55.0	264°00.4	08.3	76°14.7	51.4
Mer.pass. 15:43		ν -0.9' d -0.4' m -3.90		ν 1.9' d 0.2' m 0.14		ν 2.3' d 0.0' m -2.05		ν 2.4' d 0.0' m 0.30	

	Stars	
	SHA	Dec
Alpheratz	357° 31.5	29° 15.2
Ankaa	353° 04.2	-42° 09.0
Schedar	349° 27.3	56° 42.1
Diphda	348° 44.1	-17° 49.7
Achernar	335° 17.9	-57° 05.6
Hamal	237° 47.4	23° 36.1
Polaris	312° 03.6	89° 23.4
Acamar	315° 09.2	-40° 11.5
Menkar	314° 02.6	4° 12.2
Mirfak	308° 23.2	49° 58.0
Aldebaran	209° 35.6	16° 34.0
Rigel	281° 00.4	-8° 10.3
Capella	280° 16.6	46° 01.6
Bellatrix	278° 19.1	6° 22.4
Elnath	277° 57.4	28° 37.8
Anilam	275° 34.1	-1° 11.2
Betelgeuse	270° 48.2	7° 24.6
Canopus	263° 50.4	-52° 42.9
Sirius	258° 23.0	-16° 45.6
Adhara	255° 02.9	-29° 00.9
Procyon	244° 47.1	5° 08.8
Pollux	243° 12.9	27° 57.1
Avior	234° 12.6	-59° 36.3
Suhail	222° 43.4	-43° 33.1
Miaplacidus	221° 36.5	-69° 50.2
Alphard	217° 44.2	-8° 47.3
Regulus	207° 30.7	11° 49.2
Dubhe	193° 36.7	61° 35.3
Denebola	182° 21.4	14° 24.3
Gienah	175° 40.0	-17° 42.3
Acrux	172° 55.9	-63° 15.5
Gacrux	171° 47.6	-57° 16.4
Alioth	166° 10.1	55° 47.7
Spica	158° 18.8	-11° 18.9
Alkaid	152° 49.5	49° 09.7
Hadar	148° 31.1	-60° 30.6
Menkent	147° 53.7	-36° 30.7
Arcturus	145° 44.9	19° 01.6
Rigel Kent.	139° 35.7	-60° 57.1
Kochab	137° 20.4	74° 01.8
Zuben'ubi	136° 52.4	-16° 09.7
Alphecca	126° 01.1	26° 36.8
Antares	112° 12.0	-26° 29.7
Atria	107° 03.6	-69° 04.4
Sabik	101° 59.2	-15° 45.6
Shaula	96° 06.2	-37° 07.3
Rasalhague	95° 55.8	12° 32.3
Eltanin	90° 41.1	51° 29.0
Kaus Aust.	83° 28.5	-34° 22.1
Vega	80° 31.4	38° 48.6
Nunki	75° 44.1	-26° 15.5
Altair	61° 57.1	8° 56.8
Peacock	53° 01.3	-56° 38.4
Deneb	49° 24.0	45° 23.2
Enif	33° 35.9	10° 00.6
Al Na'ir	27° 29.3	-46° 49.2
Fomalhaut	15° 11.2	-29° 28.1
Scheat	13° 42.2	28° 14.6
Markab	13° 26.8	15° 21.8

Jan 22 Mon	SHA	Mer.pass
Venus	71°05.7	11:10
Mars	168°51.1	04:37
Jupiter	154°37.0	05:34
Saturn	326°46.7	18:03

Jan 23 Tue	SHA	Mer.pass
Venus	69°44.6	11:11
Mars	168°37.6	04:34
Jupiter	154°33.6	05:30
Saturn	326°45.0	18:00

Jan 24 Wed	SHA	Mer.pass
Venus	68°23.7	11:13
Mars	168°24.6	04:31
Jupiter	154°30.4	05:27
Saturn	326°43.2	17:56

Horizontal parallax	
Venus:	0.1
Mars:	0.1

DUT1 = UT1-UTC = +0.2060 sec $\Delta T = TT-UT1 = +68.9780$ sec

2029 January 22 to Jan. 24 UT

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	177°07.3	S19°39.4	101°55.7	15.0'	N13°54.4	10.7' 54.2'
1	192°07.1	38.8	116°29.6	14.9'	14°05.1	10.6' 54.2'
2	207°06.9	38.2	131°03.6	14.9'	14°15.8	10.6' 54.2'
3	222°06.8	· · 37.7	145°37.4	14.8'	14°26.3	10.5' 54.2'
4	237°06.6	37.1	160°11.3	14.8'	14°36.9	10.5' 54.3'
5	252°06.4	36.5	174°45.0	14.7'	14°47.4	10.4' 54.3'
6	267°06.3	S19°35.9	189°18.7	14.6'	N14°57.8	10.4' 54.3'
7	282°06.1	35.3	203°52.4	14.6'	15°08.1	10.3' 54.3'
8	297°05.9	34.8	218°26.0	14.5'	15°18.5	10.3' 54.3'
9	312°05.8	· · 34.2	232°59.5	14.5'	15°28.7	10.2' 54.3'
10	327°05.6	33.6	247°33.0	14.4'	15°38.9	10.1' 54.3'
11	342°05.4	33.0	262°06.4	14.3'	15°49.0	10.1' 54.3'
12	357°05.3	S19°32.4	276°39.7	14.3'	N15°59.1	10.0' 54.3'
13	12°05.1	31.9	291°13.0	14.2'	16°09.1	10.0' 54.4'
14	27°05.0	31.3	305°46.2	14.2'	16°19.1	9.9' 54.4'
15	42°04.8	· · 30.7	320°19.4	14.1'	16°29.0	9.8' 54.4'
16	57°04.6	30.1	334°52.5	14.0'	16°38.8	9.8' 54.4'
17	72°04.5	29.5	349°25.5	14.0'	16°48.5	9.7' 54.4'
18	87°04.3	S19°29.0	3°58.5	13.9'	N16°58.2	9.6' 54.4'
19	102°04.1	28.4	18°31.4	13.8'	17°07.9	9.6' 54.4'
20	117°04.0	27.8	33°04.2	13.8'	17°17.4	9.5' 54.5'
21	132°03.8	· · 27.2	47°37.0	13.7'	17°26.9	9.4' 54.5'
22	147°03.7	26.6	62°09.7	13.6'	17°36.4	9.4' 54.5'
23	162°03.5	26.0	76°42.3	13.6'	17°45.7	9.3' 54.5'
SD = 16.2' $d = -0.6'$ SD = 14.8'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	177°03.3	S19°25.4	91°14.9	13.5'	N17°55.0	9.2' 54.5'
1	192°03.2	24.8	105°47.4	13.4'	18°04.2	9.1' 54.5'
2	207°03.0	24.3	120°19.8	13.3'	18°13.4	9.1' 54.6'
3	222°02.9	· · 23.7	134°52.1	13.3'	18°22.5	9.0' 54.6'
4	237°02.7	23.1	149°24.4	13.2'	18°31.5	8.9' 54.6'
5	252°02.6	22.5	163°56.6	13.1'	18°40.4	8.9' 54.6'
6	267°02.4	S19°21.9	178°28.7	13.1'	N18°49.2	8.8' 54.6'
7	282°02.2	21.3	193°00.8	13.0'	18°58.0	8.7' 54.6'
8	297°02.1	20.7	207°32.8	12.9'	19°06.7	8.6' 54.7'
9	312°01.9	· · 20.1	222°04.7	12.8'	19°15.3	8.5' 54.7'
10	327°01.8	19.5	236°36.5	12.8'	19°23.9	8.5' 54.7'
11	342°01.6	18.9	251°08.3	12.7'	19°32.3	8.4' 54.7'
12	357°01.5	S19°18.3	265°39.9	12.6'	N19°40.7	8.3' 54.7'
13	12°01.3	17.7	280°11.6	12.5'	19°49.0	8.2' 54.8'
14	27°01.2	17.1	294°43.1	12.4'	19°57.2	8.1' 54.8'
15	42°01.0	· · 16.5	309°14.5	12.4'	20°05.4	8.1' 54.8'
16	57°00.8	15.9	323°45.9	12.3'	20°13.4	8.0' 54.8'
17	72°00.7	15.3	338°17.2	12.2'	20°21.4	7.9' 54.8'
18	87°00.5	S19°14.7	352°48.4	12.1'	N20°29.3	7.8' 54.9'
19	102°00.4	14.1	7°19.5	12.1'	20°37.1	7.7' 54.9'
20	117°00.2	13.5	21°50.6	12.0'	20°44.8	7.6' 54.9'
21	132°00.1	· · 12.9	36°21.6	11.9'	20°52.4	7.5' 54.9'
22	146°59.9	12.3	50°52.5	11.8'	20°59.9	7.4' 54.9'
23	161°59.8	11.7	65°23.3	11.7'	21°07.4	7.3' 55.0'
SD = 16.2' $d = -0.6'$ SD = 14.9'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	176°59.6	S19°11.1	79°54.0	11.7'	N21°14.7	7.3' 55.0'
1	191°59.5	10.5	94°24.7	11.6'	21°22.0	7.2' 55.0'
2	206°59.3	09.9	108°55.3	11.5'	21°29.1	7.1' 55.0'
3	221°59.2	· · 09.3	123°25.8	11.4'	21°36.2	7.0' 55.1'
4	236°59.0	08.7	137°56.2	11.3'	21°43.1	6.9' 55.1'
5	251°58.9	08.1	152°26.5	11.3'	21°50.0	6.8' 55.1'
6	266°58.7	S19°07.5	166°56.8	11.2'	N21°56.8	6.7' 55.1'
7	281°58.6	06.9	181°26.9	11.1'	22°03.5	6.6' 55.2'
8	296°58.4	06.3	195°57.0	11.0'	22°10.0	6.5' 55.2'
9	311°58.3	· · 05.7	210°27.0	10.9'	22°16.5	6.4' 55.2'
10	326°58.1	05.1	224°57.0	10.8'	22°22.9	6.3' 55.2'
11	341°58.0	04.5	239°26.8	10.8'	22°29.2	6.2' 55.3'
12	356°57.8	S19°03.9	253°56.6	10.7'	N22°35.3	6.1' 55.3'
13	11°57.7	03.3	268°26.3	10.6'	22°41.4	6.0' 55.3'
14	26°57.5	02.6	282°55.9	10.5'	22°47.4	5.9' 55.3'
15	41°57.4	· · 02.0	297°25.4	10.4'	22°53.2	5.7' 55.4'
16	56°57.3	01.4	311°54.8	10.4'	22°59.0	5.6' 55.4'
17	71°57.1	00.8	326°24.2	10.3'	23°04.6	5.5' 55.4'
18	86°57.0	S19°00.2	340°53.5	10.2'	N23°10.1	5.4' 55.5'
19	101°56.8	18°59.6	355°22.7	10.1'	23°15.5	5.3' 55.5'
20	116°56.7	59.0	9°51.8	10.0'	23°20.8	5.2' 55.5'
21	131°56.5	· · 58.3	24°20.8	10.0'	23°26.0	5.1' 55.5'
22	146°56.4	57.7	38°49.8	9.9'	23°31.1	5.0' 55.6'
23	161°56.2	57.1	53°18.6	9.8'	23°36.1	4.9' 55.6'
SD = 16.2' $d = -0.6'$ SD = 15.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:38	09:17	■	■	15:08	16:47
N 70°	07:27	08:51	10:39	13:46	15:34	16:58
68°	07:17	08:32	09:55	14:30	15:53	17:08
66°	07:09	08:16	09:26	14:58	16:09	17:16
64°	07:02	08:03	09:05	15:20	16:21	17:23
62°	06:56	07:52	08:47	15:37	16:32	17:29
60°	06:50	07:43	08:33	15:52	16:42	17:34
N 58°	06:46	07:35	08:21	16:04	16:50	17:39
56°	06:41	07:27	08:10	16:14	16:57	17:43
54°	06:37	07:20	08:01	16:24	17:04	17:48
52°	06:33	07:14	07:52	16:32	17:10	17:51
50°	06:29	07:09	07:45	16:40	17:15	17:55
45°	06:21	06:57	07:29	16:55	17:28	18:03
N 40°	06:14	06:46	07:16	17:09	17:38	18:11
35°	06:07	06:37	07:04	17:20	17:47	18:18
30°	06:00	06:29	06:55	17:30	17:55	18:24
20°	05:47	06:14	06:37	17:47	18:10	18:37
N 10°	05:35	06:00	06:22	18:01	18:24	18:49
0°	05:21	05:46	06:08	18:15	18:37	19:03
S 10°	05:05	05:32	05:54	18:30	18:52	19:18
20°	04:47	05:15	05:39	18:45	19:09	19:37
30°	04:23	04:55	05:21	19:03	19:29	20:00
35°	04:08	04:42	05:10	19:13	19:41	20:15
40°	03:49	04:27	04:58	19:25	19:56	20:34
45°	03:26	04:10	04:44	19:39	20:13	20:57
S 50°	02:53	03:47	04:27	19:56	20:36	21:29
52°	02:36	03:36	04:18	20:04	20:47	21:46
54°	02:15	03:23	04:09	20:14	21:00	22:07
56°	01:46	03:08	03:59	20:24	21:14	22:34
58°	00:58	02:49	03:47	20:36	21:32	23:18
S 60°	—:—	02:27	03:33	20:50	21:54	—:—

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	07:40	□	□	02:44	□	□
N 70°	08:13	07:27	□	02:12	04:31	□
68°	08:37	08:15	07:07	01:50	03:43	06:31
66°	08:55	08:47	08:34	01:32	03:13	05:05
64°	09:11	09:10	09:12	01:18	02:50	04:27
62°	09:23	09:29	09:38	01:06	02:33	04:01
60°	09:34	09:44	09:59	00:56	02:18	03:41
N 58°	09:44	09:57	10:17	00:47	02:05	03:24
56°	09:52	10:09	10:31	00:40	01:54	03:10
54°	10:00	10:19	10:44	00:33	01:45	02:57
52°	10:06	10:28	10:55	00:27	01:37	02:47
50°	10:13	10:36	11:05	00:21	01:29	02:37
45°	10:26	10:53	11:26	00:09	01:13	02:17
N 40°	10:37	11:07	11:43	00:00	01:00	02:01
35°	10:46	11:19	11:57	· · ·	00:49	01:47
30°	10:54	11:29	12:09	· · ·	00:39	01:35
20°	11:09	11:48	12:31	· · ·	00:22	01:15
N 10°	11:21	12:03	12:49	· · ·	00:08	00:58
0°	11:33	12:18	13:07	23:54	· · ·	00:41
S 10°	11:45	12:33	13:24	23:41	· · ·	00:25
20°	11:58	12:49	13:43	23:27	· · ·	00:08
30°	12:12	13:08	14:05	23:10	23:48	· · ·
35°	12:21	13:19	14:18	23:01	23:36	· · ·
40°	12:31	13:31	14:33	22:50	23:23	· · ·
45°	12:42	13:46	14:50	22:37	23:07	23:44
S 50°	12:56	14:05	15:12	22:22	22:48	23:21
52°	13:03	14:13	15:23	22:15	22:39	23:10
54°	13:10	14:23	15:35	22:07	22:29	22:58
56°	13:19	14:34	15:49	21:58	22:17	22:44
58°	13:28	14:47	16:05	21:48	22:04	22:27
S 60°	13:39	15:01	16:24	21:37	21:49	22:08

Day	Sun		Mer. Pass	Moon		Age 8-10 42-62%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
22	11:31	11:39	12:12	17:44	05:22	
23	11:47	11:54	12:12	18:30	06:06	
24	12:02	12:09	12:12	19:19	06:54	

January 25, 26, 27 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	124°35.4	191°38.4	S22°09.2	292°47.5	S01°55.2	279°02.7	S09°08.3	91°16.7	N10°51.5					Alpheratz	357°31.5	29°15.2	
1	139°37.9	206°37.5	08.8	307°49.4	55.3	294°05.1	08.4	106°19.0	51.5					Ankaa	353°04.2	-42°09.0	
2	154°40.4	221°36.6	08.4	322°51.4	55.5	309°07.4	08.4	121°21.4	51.5					Schedar	349°27.3	56°42.1	
3	169°42.8	236°35.7	•• 08.0	337°53.3	•• 55.7	324°09.8	•• 08.4	136°23.8	•• 51.6					Diphda	348°44.2	-17°49.7	
4	184°45.3	251°34.8	07.6	352°55.3	55.8	339°12.1	08.5	151°26.2	51.6					Achernar	335°18.0	-57°05.6	
5	199°47.7	266°33.9	07.2	7°57.2	56.0	354°14.4	08.5	166°28.6	51.7					Hamal	327°47.4	23°36.1	
6	214°50.2	281°33.1	S22°06.8	22°59.2	S01°56.2	9°16.8	S09°08.5	181°31.0	N10°51.7					Polaris	312°04.8	89°23.4	
7	229°52.7	296°32.2	06.4	38°01.1	56.3	24°19.1	08.6	196°33.3	51.7					Acamar	315°09.2	-40°11.5	
8	244°55.1	311°31.3	06.0	53°03.1	56.5	39°21.5	08.6	211°35.7	51.8					Menkar	314°02.6	4°12.2	
9	259°57.6	326°30.4	•• 05.6	68°05.1	•• 56.7	54°23.8	•• 08.6	226°38.1	•• 51.8					Mirfak	308°23.3	49°58.0	
10	275°00.1	341°29.5	05.3	83°07.0	56.8	69°26.2	08.7	241°40.5	51.9					Aldebaran	290°35.6	16°34.0	
11	290°02.5	356°28.6	04.8	98°09.0	57.0	84°28.5	08.7	256°42.9	51.9					Rigel	281°00.5	-8°10.3	
12	305°05.0	11°27.7	S22°04.4	113°10.9	S01°57.1	99°30.8	S09°08.8	271°45.2	N10°52.0					Capella	280°16.6	46°01.6	
13	320°07.5	26°26.8	04.0	128°12.9	57.3	114°33.2	08.8	286°47.6	52.0					Bellatrix	278°19.1	6°22.4	
14	335°09.9	41°25.9	03.6	143°14.9	57.5	129°35.5	08.8	301°50.0	52.0					Elnath	277°57.4	28°37.8	
15	350°12.4	56°25.0	•• 03.2	158°16.8	•• 57.6	144°37.9	•• 08.9	316°52.4	•• 52.1					Alnilam	275°34.1	-1°11.2	
16	5°14.9	71°24.1	02.8	173°18.8	57.8	159°40.2	08.9	331°54.7	52.1					Betelgeuse	270°48.2	7°24.6	
17	20°17.3	86°23.2	02.4	188°20.8	57.9	174°42.6	08.9	346°57.1	52.2					Canopus	263°50.4	-52°42.9	
18	35°19.8	101°22.4	S22°02.0	203°22.7	S01°58.1	189°44.9	S09°09.0	1°59.5	N10°52.2					Sirius	258°23.0	-16°45.6	
19	50°22.2	116°21.5	01.6	218°24.7	58.3	204°47.3	09.0	17°01.9	52.3					Adhara	255°02.9	-29°00.9	
20	65°24.7	131°20.6	01.2	233°26.7	58.4	219°49.6	09.0	32°04.3	52.3					Procyon	244°47.1	5°08.8	
21	80°27.2	146°19.7	•• 00.8	248°28.6	•• 58.6	234°51.9	•• 09.1	47°06.6	•• 52.3					Pollux	243°12.9	27°57.1	
22	95°29.6	161°18.8	00.4	263°30.6	58.7	249°54.3	09.1	62°09.0	52.4					Avior	234°12.6	-59°36.3	
23	110°32.1	176°17.9	00.0	278°32.6	58.9	264°56.6	09.1	77°11.4	52.4					Suhail	222°43.4	-43°33.1	
Mer.pass. 15:39			ν -0.9' d -0.4' m -3.90	ν 2.0' d 0.2' m 0.12	ν 2.3' d 0.0' m -2.06	ν 2.4' d 0.0' m 0.30											

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	125°34.6	191°17.0	S21°59.5	293°34.5	S01°59.1	279°59.0	S09°09.2	92°13.8	N10°52.5
1	140°37.0	206°16.1	59.1	308°36.5	59.2	295°01.3	09.2	107°16.2	52.5
2	155°39.5	221°15.3	58.7	323°38.5	59.4	310°03.7	09.2	122°18.5	52.6
3	170°42.0	236°14.4	•• 58.3	338°40.5	•• 59.5	325°06.0	•• 09.3	137°20.9	•• 52.6
4	185°44.4	251°13.5	57.9	353°42.4	59.7	340°08.4	09.3	152°23.3	52.6
5	200°46.9	266°12.6	57.4	8°44.4	01°59.8	355°10.7	09.3	167°25.7	52.7
6	215°49.4	281°11.7	S21°57.0	23°46.4	S02°00.0	10°13.1	S09°09.4	182°28.0	N10°52.7
7	230°51.8	296°10.8	56.6	38°48.4	00.1	25°15.4	09.4	197°30.4	52.8
8	245°54.3	311°09.9	56.2	53°50.3	00.3	40°17.8	09.4	212°32.8	52.8
9	260°56.7	326°09.1	•• 55.7	68°52.3	•• 00.5	55°20.1	•• 09.5	227°35.2	•• 52.9
10	275°59.2	341°08.2	55.3	83°54.3	00.6	70°22.5	09.5	242°37.5	52.9
11	291°01.7	356°07.3	54.9	98°56.3	00.8	85°24.8	09.5	257°39.9	52.9
12	306°04.1	11°06.4	S21°54.5	113°58.3	S02°00.9	100°27.2	S09°09.6	272°42.3	N10°53.0
13	321°06.6	26°05.5	54.0	129°00.2	01.1	115°29.5	09.6	287°44.7	53.0
14	336°09.1	41°04.6	53.6	144°02.2	01.2	130°31.9	09.6	302°47.0	53.1
15	351°11.5	56°03.8	•• 53.2	159°04.2	•• 01.4	145°34.2	•• 09.7	317°49.4	•• 53.1
16	6°14.0	71°02.9	52.7	174°06.2	01.5	160°36.6	09.7	332°51.8	53.2
17	21°16.5	86°02.0	52.3	189°08.2	01.7	175°38.9	09.7	347°54.2	53.2
18	36°18.9	101°01.1	S21°51.9	204°10.2	S02°01.8	190°41.3	S09°09.8	2°56.5	N10°53.2
19	51°21.4	116°00.2	51.4	219°12.1	02.0	205°43.6	09.8	17°58.9	53.3
20	66°23.8	130°59.4	51.0	234°14.1	02.1	220°46.0	09.8	33°01.3	53.3
21	81°26.3	145°58.5	•• 50.5	249°16.1	•• 02.3	235°48.3	•• 09.9	48°03.7	•• 53.4
22	96°28.8	160°57.6	50.1	264°18.1	02.4	250°50.7	09.9	63°06.0	53.4
23	111°31.2	175°56.7	49.7	279°20.1	02.6	265°53.0	09.9	78°08.4	53.5
Mer.pass. 15:35			ν -0.9' d -0.4' m -3.90	ν 2.0' d 0.2' m 0.10	ν 2.3' d 0.0' m -2.07	ν 2.4' d 0.0' m 0.31			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	126°33.7	190°55.8	S21°49.2	294°22.1	S02°02.7	280°55.4	S09°10.0	93°10.8	N10°53.5
1	141°36.2	205°55.0	48.8	309°24.1	02.9	295°57.7	10.0	108°13.2	53.6
2	156°38.6	220°54.1	48.3	324°26.1	03.0	311°00.1	10.0	123°15.5	53.6
3	171°41.1	235°53.2	•• 47.9	339°28.1	•• 03.2	326°02.4	•• 10.0	138°17.9	•• 53.6
4	186°43.6	250°52.3	47.4	354°30.1	03.3	341°04.8	10.1	153°20.3	53.7
5	201°46.0	265°51.4	47.0	9°32.1	03.5	356°07.2	10.1	168°22.6	53.7
6	216°48.5	280°50.6	S21°46.5	24°34.1	S02°03.6	11°09.5	S09°10.1	183°25.0	N10°53.8
7	231°51.0	295°49.7	46.1	39°36.1	03.8	26°11.9	10.2	198°27.4	53.8
8	246°53.4	310°48.8	45.6	54°38.1	03.9	41°14.2	10.2	213°29.8	53.9
9	261°55.9	325°47.9	•• 45.2	69°40.0	•• 04.0	56°16.6	•• 10.2	228°32.1	•• 53.9
10	276°58.3	340°47.1	44.7	84°42.0	04.2	71°18.9	10.3	243°34.5	54.0
11	292°00.8	355°46.2	44.3	99°44.1	04.3	86°21.3	10.3	258°36.9	54.0
12	307°03.3	10°45.3	S21°43.8	114°46.1	S02°04.5	101°23.6	S09°10.3	273°39.3	N10°54.0
13	322°05.7	25°44.4	43.4	129°48.1	04.6	116°26.0	10.4	288°41.6	54.1
14	337°08.2	40°43.6	42.9	144°50.1	04.8	131°28.4	10.4	303°44.0	54.1
15	352°10.7	55°42.7	•• 42.4	159°52.1	•• 04.9	146°30.7	•• 10.4	318°46.4	•• 54.2
16	7°13.1	70°41.8	42.0	174°54.1	05.1	161°33.1	10.4	333°48.7	54.2
17	22°15.6	85°40.9	41.5	189°56.1	05.2	176°35.4	10.5	348°51.1	54.3
18	37°18.1	100°40.1	S21°41.1	204°58.1	S02°05.3	191°37.8	S09°10.5	3°53.5	N10°54.3
19	52°20.5	115°39.2	40.6	220°00.1	05.5	206°40.2	10.5	18°55.8	54.4
20	67°23.0	130°38.3	40.1	235°02.1	05.6	221°42.5	10.6	33°58.2	54.4
21	82°25.5	145°37.5	•• 39.7	250°04.1	•• 05.8	236°44.9	•• 10.6	49°00.6	•• 54.5
22	97°27.9	160°36.6	39.2	265°06.1	05.9	251°47.2	10.6	64°03.0	54.5
23	112°30.4	175°35.7	38.7	280°08.1	06.0	266°49.6	10.6	79°05.3	54.5
Mer.pass. 15:31			ν -0.9' d -0.4' m -3.90	ν 2.0' d 0.1' m 0.07	ν 2.4' d 0.0' m -2.07	ν 2.4' d 0.0' m 0.31			

Jan 25 Thu	SHA	Mer.pass
Venus	67°03.0	11:14
Mars	168°12.0	04:28
Jupiter	154°27.3	05:23
Saturn	326°41.2	17:52

Jan 26 Fri	SHA	Mer.pass
Venus	65°42.5	11:16
Mars	168°00.0	04:25
Jupiter	154°24.4	05:19
Saturn	326°39.2	17:48

Jan 27 Sat	SHA	Mer.pass
Venus	64°22.1	11:17
Mars	167°48.4	04:22
Jupiter	154°21.7	05:15
Saturn	326°37.1	17:44

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

DUT1 = UT1-UTC = +0.2061 sec $\Delta T = TT-UT1 = +68.9779$ sec

2029 January 25 to Jan. 27 UT

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°56.1	S18°56.5	67°47.4	9.7'	N23°40.9	4.7'	55.6'
1	191°56.0	55.9	82°16.2	9.6'	23°45.7	4.6'	55.7'
2	206°55.8	55.3	96°44.8	9.6'	23°50.3	4.5'	55.7'
3	221°55.7	· · 54.6	111°13.4	9.5'	23°54.8	4.4'	55.7'
4	236°55.5	54.0	125°41.9	9.4'	23°59.2	4.3'	55.7'
5	251°55.4	53.4	140°10.3	9.3'	24°03.4	4.1'	55.8'
6	266°55.3	S18°52.8	154°38.6	9.3'	N24°07.6	4.0'	55.8'
7	281°55.1	52.2	169°06.9	9.2'	24°11.6	3.9'	55.8'
8	296°55.0	51.5	183°35.0	9.1'	24°15.5	3.8'	55.9'
9	311°54.8	· · 50.9	198°03.1	9.0'	24°19.3	3.6'	55.9'
10	326°54.7	50.3	212°31.2	9.0'	24°22.9	3.5'	55.9'
11	341°54.6	49.7	226°59.1	8.9'	24°26.4	3.4'	56.0'
12	356°54.4	S18°49.0	241°27.0	8.8'	N24°29.8	3.3'	56.0'
13	11°54.3	48.4	255°54.8	8.7'	24°33.1	3.1'	56.0'
14	26°54.1	47.8	270°22.6	8.7'	24°36.2	3.0'	56.0'
15	41°54.0	· · 47.2	284°50.3	8.6'	24°39.2	2.9'	56.1'
16	56°53.9	46.5	299°17.9	8.5'	24°42.1	2.7'	56.1'
17	71°53.7	45.9	313°45.4	8.5'	24°44.9	2.6'	56.1'
18	86°53.6	S18°45.3	328°12.9	8.4'	N24°47.5	2.5'	56.2'
19	101°53.5	44.7	342°40.3	8.3'	24°50.0	2.4'	56.2'
20	116°53.3	44.0	357°07.6	8.3'	24°52.3	2.2'	56.2'
21	131°53.2	· · 43.4	11°34.9	8.2'	24°54.5	2.1'	56.3'
22	146°53.1	42.8	26°02.1	8.1'	24°56.6	1.9'	56.3'
23	161°52.9	42.1	40°29.2	8.1'	24°58.6	1.8'	56.3'
	SD = 16.2'	d = -0.6'	SD = 15.2'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°52.8	S18°41.5	54°56.3	8.0'	N25°00.4	1.7'	56.4'
1	191°52.7	40.9	69°23.3	8.0'	25°02.0	1.5'	56.4'
2	206°52.5	40.2	83°50.2	7.9'	25°03.6	1.4'	56.4'
3	221°52.4	· · 39.6	98°17.1	7.8'	25°05.0	1.3'	56.5'
4	236°52.3	39.0	112°44.0	7.8'	25°06.2	1.1'	56.5'
5	251°52.1	38.3	127°10.7	7.7'	25°07.3	1.0'	56.5'
6	266°52.0	S18°37.7	141°37.5	7.7'	N25°08.3	0.8'	56.6'
7	281°51.9	37.1	156°04.1	7.6'	25°09.1	0.7'	56.6'
8	296°51.7	36.4	170°30.7	7.6'	25°09.8	0.5'	56.6'
9	311°51.6	· · 35.8	184°57.3	7.5'	25°10.4	0.4'	56.7'
10	326°51.5	35.2	199°23.8	7.5'	25°10.8	0.3'	56.7'
11	341°51.3	34.5	213°50.3	7.4'	25°11.0	0.1'	56.8'
12	356°51.2	S18°33.9	228°16.7	7.4'	N25°11.1	-0.0'	56.8'
13	11°51.1	33.2	242°43.0	7.3'	25°11.1	-0.2'	56.8'
14	26°50.9	32.6	257°09.3	7.3'	25°10.9	-0.3'	56.9'
15	41°50.8	· · 32.0	271°35.6	7.2'	25°10.6	-0.5'	56.9'
16	56°50.7	31.3	286°01.8	7.2'	25°10.1	-0.6'	56.9'
17	71°50.6	30.7	300°28.0	7.1'	25°09.5	-0.8'	57.0'
18	86°50.4	S18°30.0	314°54.2	7.1'	N25°08.7	-0.9'	57.0'
19	101°50.3	29.4	329°20.3	7.1'	25°07.8	-1.1'	57.0'
20	116°50.2	28.8	343°46.3	7.0'	25°06.8	-1.2'	57.1'
21	131°50.1	· · 28.1	358°12.4	7.0'	25°05.5	-1.4'	57.1'
22	146°49.9	27.5	12°38.4	7.0'	25°04.2	-1.5'	57.1'
23	161°49.8	26.8	27°04.3	6.9'	25°02.7	-1.7'	57.2'
	SD = 16.2'	d = -0.6'	SD = 15.4'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°49.7	S18°26.2	41°30.2	6.9'	N25°01.0	-1.8'	57.2'
1	191°49.6	25.5	55°56.1	6.9'	24°59.2	-2.0'	57.2'
2	206°49.4	24.9	70°22.0	6.8'	24°57.2	-2.1'	57.3'
3	221°49.3	· · 24.2	84°47.8	6.8'	24°55.1	-2.3'	57.3'
4	236°49.2	23.6	99°13.7	6.8'	24°52.8	-2.4'	57.4'
5	251°49.1	22.9	113°39.4	6.8'	24°50.4	-2.6'	57.4'
6	266°48.9	S18°22.3	128°05.2	6.7'	N24°47.8	-2.7'	57.4'
7	281°48.8	21.6	142°30.9	6.7'	24°45.1	-2.9'	57.5'
8	296°48.7	21.0	156°56.7	6.7'	24°42.2	-3.0'	57.5'
9	311°48.6	· · 20.3	171°22.4	6.7'	24°39.1	-3.2'	57.5'
10	326°48.4	19.7	185°48.0	6.7'	24°35.9	-3.3'	57.6'
11	341°48.3	19.0	200°13.7	6.6'	24°32.6	-3.5'	57.6'
12	356°48.2	S18°18.4	214°39.4	6.6'	N24°29.1	-3.7'	57.6'
13	11°48.1	17.7	229°05.0	6.6'	24°25.5	-3.8'	57.7'
14	26°48.0	17.1	243°30.6	6.6'	24°21.7	-4.0'	57.7'
15	41°47.8	· · 16.4	257°56.2	6.6'	24°17.7	-4.1'	57.7'
16	56°47.7	15.8	272°21.8	6.6'	24°13.6	-4.3'	57.8'
17	71°47.6	15.1	286°47.4	6.6'	24°09.3	-4.4'	57.8'
18	86°47.5	S18°14.5	301°13.0	6.6'	N24°04.9	-4.6'	57.8'
19	101°47.4	13.8	315°38.6	6.6'	24°00.3	-4.7'	57.9'
20	116°47.2	13.1	330°04.2	6.6'	23°55.6	-4.9'	57.9'
21	131°47.1	· · 12.5	344°29.8	6.6'	23°50.7	-5.0'	58.0'
22	146°47.0	11.8	358°55.4	6.6'	23°45.7	-5.2'	58.0'
23	161°46.9	11.2	13°20.9	6.6'	23°40.5	-5.3'	58.0'
	SD = 16.2'	d = -0.6'	SD = 15.6'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:29	09:04	11:32	12:54	15:23	16:58
N 70°	07:19	08:41	10:19	14:08	15:46	17:08
68°	07:10	08:23	09:42	14:44	16:03	17:16
66°	07:03	08:09	09:16	15:10	16:18	17:23
64°	06:56	07:57	08:56	15:30	16:29	17:30
62°	06:51	07:46	08:40	15:46	16:40	17:35
60°	06:46	07:38	08:27	15:59	16:48	17:40
N 58°	06:41	07:30	08:15	16:11	16:56	17:45
56°	06:37	07:23	08:05	16:21	17:03	17:49
54°	06:33	07:17	07:56	16:29	17:09	17:53
52°	06:30	07:11	07:48	16:37	17:15	17:56
50°	06:26	07:06	07:41	16:44	17:20	17:59
45°	06:19	06:54	07:26	17:00	17:31	18:07
N 40°	06:12	06:44	07:13	17:12	17:41	18:14
35°	06:05	06:36	07:03	17:23	17:50	18:20
30°	05:59	06:28	06:53	17:32	17:58	18:26
20°	05:47	06:14	06:37	17:48	18:12	18:38
N 10°	05:35	06:00	06:23	18:03	18:25	18:50
0°	05:22	05:47	06:09	18:16	18:38	19:03
S 10°	05:07	05:33	05:55	18:30	18:52	19:18
20°	04:49	05:17	05:41	18:44	19:08	19:36
30°	04:26	04:57	05:23	19:01	19:27	19:59
35°	04:11	04:45	05:13	19:11	19:39	20:13
40°	03:54	04:31	05:02	19:23	19:53	20:31
45°	03:31	04:14	04:48	19:36	20:10	20:53
S 50°	03:00	03:52	04:32	19:53	20:32	21:23
52°	02:44	03:42	04:24	20:00	20:42	21:39
54°	02:25	03:30	04:15	20:09	20:54	21:58
56°	01:59	03:16	04:05	20:19	21:08	22:22
58°	01:21	02:59	03:54	20:30	21:24	22:58
S 60°	—:—	02:38	03:41	20:43	21:45	—:—

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	□	□	□	□	□	□
64°	09:19	09:48	11:05	06:06	07:30	08:11
62°	09:58	10:38	11:47	05:27	06:41	07:29
60°	10:25	11:09	12:16	05:00	06:10	07:00
N 58°	10:46	11:32	12:38	04:39	05:46	06:38
56°	11:04	11:51	12:56	04:22	05:27	06:20
54°	11:19	12:07	13:11	04:08	05:11	06:04
52°	11:32	12:21	13:24	03:55	04:58	05:51
50°	11:43	12:33	13:35	03:44	04:46	05:40
45°	12:07	12:58	13:59	03:21	04:21	05:15
N 40°	12:26	13:18	14:18	03:02	04:01	04:56
35°	12:42	13:34	14:34	02:46	03:44	04:40
30°	12:56	13:49	14:48	02:33	03:30	04:25
20°	13:19	14:13	15:12	02:10	03:06	04:01
N 10°	13:40	14:34	15:32	01:50	02:45	03:41
0°	13:59	14:54	15:51	01:32	02:25	03:21
S 10°	14:18	15:14	16:10	01:13	02:05	03:01
20°	14:39	15:35	16:30	00:53	01:44	02:41
30°	15:03	15:59	16:53	00:31	01:20	02:16
35°	15:17	16:14	17:07	00:17	01:06	02:02
40°	15:33	16:31	17:23	00:02	00:49	01:45
45°	15:53	16:51	17:42	· · ·	00:29	01:25
S 50°	16:17	17:16	18:05	· · ·	00:04	01:00
52°	16:29	17:28	18:16	23:52	· · ·	00:48
54°	16:43	17:42	18:29	23:39	· · ·	00:34
56°	16:59	17:58	18:44	23:23	· · ·	00:18
58°	17:17	18:17	19:01	23:04	23:59	· · ·
S 60°	17:40	18:41	19:22	22:41	23:35	· · ·

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
25	12:16	12:22	12:12	20:12	07:45	
26	12:29	12:35	12:13	21:07	08:39	
27	12:41	12:47	12:13	22:04	09:36	

January 28, 29, 30 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	127°32.8	190°34.8	S21°38.2	295°10.1	S02°06.2	281°52.0	S09°10.7	94°07.7	N10°54.6	Alpheratz	357°31.5	29°15.2
1	142°35.3	205°34.0	37.8	310°12.2	06.3	296°54.3	10.7	109°10.1	54.6	Ankaa	353°04.2	-42°09.0
2	157°37.8	220°33.1	37.3	325°14.2	06.5	311°56.7	10.7	124°12.4	54.7	Schedar	349°27.4	56°42.1
3	172°40.2	235°32.2	•• 36.8	340°16.2	•• 06.6	326°59.0	•• 10.8	139°14.8	•• 54.7	Diphda	348°44.2	-17°49.7
4	187°42.7	250°31.4	36.4	355°18.2	06.7	342°01.4	10.8	154°17.2	54.8	Achernar	335°18.0	-57°05.6
5	202°45.2	265°30.5	35.9	10°20.2	06.9	357°03.8	10.8	169°19.5	54.8	Hamal	327°47.4	23°36.1
6	217°47.6	280°29.6	S21°35.4	25°22.2	S02°07.0	12°06.1	S09°10.8	184°21.9	N10°54.9	Polaris	312°06.1	89°23.4
7	232°50.1	295°28.8	34.9	40°24.2	07.2	27°08.5	10.9	199°24.3	54.9	Acamar	315°09.3	-40°11.5
8	247°52.6	310°27.9	34.4	55°26.3	07.3	42°10.9	10.9	214°26.6	55.0	Menkar	314°02.6	4°12.2
9	262°55.0	325°27.0	•• 34.0	70°28.3	•• 07.4	57°13.2	•• 10.9	229°29.0	•• 55.0	Mirfak	308°23.3	49°58.0
10	277°57.5	340°26.2	33.5	85°30.3	07.6	72°15.6	11.0	244°31.4	55.0	Aldebaran	290°35.6	16°34.0
11	293°00.0	355°25.3	33.0	100°32.3	07.7	87°17.9	11.0	259°33.7	55.1	Rigel	281°00.5	-8°10.3
12	308°02.4	10°24.4	S21°32.5	115°34.4	S02°07.8	102°20.3	S09°11.0	274°36.1	N10°55.1	Capella	280°16.6	46°01.6
13	323°04.9	25°23.6	32.0	130°36.4	08.0	117°22.7	11.0	289°38.5	55.2	Bellatrix	278°19.1	6°22.4
14	338°07.3	40°22.7	31.5	145°38.4	08.1	132°25.0	11.1	304°40.8	55.2	Elnath	277°57.4	28°37.8
15	353°09.8	55°21.8	•• 31.1	160°40.4	•• 08.2	147°27.4	•• 11.1	319°43.2	•• 55.3	Alnilam	275°34.1	-1°11.2
16	8°12.3	70°21.0	30.6	175°42.5	08.4	162°29.8	11.1	334°45.6	55.3	Betelgeuse	270°48.2	7°24.6
17	23°14.7	85°20.1	30.1	190°44.5	08.5	177°32.1	11.1	349°47.9	55.4	Canopus	263°50.4	-52°42.9
18	38°17.2	100°19.2	S21°29.6	205°46.5	S02°08.6	192°34.5	S09°11.2	4°50.3	N10°55.4	Sirius	258°23.0	-16°45.6
19	53°19.7	115°18.4	29.1	220°48.5	08.8	207°36.9	11.2	19°52.7	55.5	Adhara	255°02.9	-29°00.9
20	68°22.1	130°17.5	28.6	235°50.6	08.9	222°39.2	11.2	34°55.0	55.5	Procyon	244°47.0	5°08.8
21	83°24.6	145°16.7	•• 28.1	250°52.6	•• 09.0	237°41.6	•• 11.2	49°57.4	•• 55.6	Pollux	243°12.9	27°57.1
22	98°27.1	160°15.8	27.6	265°54.6	09.2	252°44.0	11.3	64°59.8	55.6	Avior	234°12.6	-59°36.3
23	113°29.5	175°14.9	27.1	280°56.7	09.3	267°46.3	11.3	80°02.1	55.7	Suhail	222°43.4	-43°33.2
Mer.pass. 15:27		ν -0.9' d -0.5' m -3.90		ν 2.0' d 0.1' m 0.05		ν 2.4' d 0.0' m -2.08		ν 2.4' d 0.0' m 0.31				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	128°32.0	190°14.1	S21°26.6	295°58.7	S02°09.4	282°48.7	S09°11.3	95°04.5	N10°55.7
1	143°34.5	205°13.2	26.1	311°00.7	09.6	297°51.1	11.4	110°06.9	55.7
2	158°36.9	220°12.3	25.6	326°02.8	09.7	312°53.4	11.4	125°09.2	55.8
3	173°39.4	235°11.5	•• 25.1	341°04.8	•• 09.8	327°55.8	•• 11.4	140°11.6	•• 55.8
4	188°41.8	250°10.6	24.6	356°06.8	09.9	342°58.2	11.4	155°14.0	55.9
5	203°44.3	265°09.8	24.1	11°08.9	10.1	358°00.5	11.5	170°16.3	55.9
6	218°46.8	280°08.9	S21°23.6	26°10.9	S02°10.2	13°02.9	S09°11.5	185°18.7	N10°56.0
7	233°49.2	295°08.1	23.1	41°13.0	10.3	28°05.3	11.5	200°21.0	56.0
8	248°51.7	310°07.2	22.6	56°15.0	10.5	43°07.7	11.5	215°23.4	56.1
9	263°54.2	325°06.3	•• 22.1	71°17.0	•• 10.6	58°10.0	•• 11.6	230°25.8	•• 56.1
10	278°56.6	340°05.5	21.6	86°19.1	10.7	73°12.4	11.6	245°28.1	56.2
11	293°59.1	355°04.6	21.1	101°21.1	10.8	88°14.8	11.6	260°30.5	56.2
12	309°01.6	10°03.8	S21°20.6	116°23.2	S02°11.0	103°17.1	S09°11.6	275°32.9	N10°56.3
13	324°04.0	25°02.9	20.1	131°25.2	11.1	118°19.5	11.7	290°35.2	56.3
14	339°06.5	40°02.1	19.5	146°27.3	11.2	133°21.9	11.7	305°37.6	56.4
15	354°09.0	55°01.2	•• 19.0	161°29.3	•• 11.3	148°24.3	•• 11.7	320°40.0	•• 56.4
16	9°11.4	70°00.3	18.5	176°31.4	11.5	163°26.6	11.7	335°42.3	56.5
17	24°13.9	84°59.5	18.0	191°33.4	11.6	178°29.0	11.7	350°44.7	56.5
18	39°16.3	99°58.6	S21°17.5	206°35.4	S02°11.7	193°31.4	S09°11.8	5°47.0	N10°56.6
19	54°18.8	114°57.8	17.0	221°37.5	11.8	208°33.7	11.8	20°49.4	56.6
20	69°21.3	129°56.9	16.4	236°39.6	12.0	223°36.1	11.8	35°51.8	56.7
21	84°23.7	144°56.1	•• 15.9	251°41.6	•• 12.1	238°38.5	•• 11.8	50°54.1	•• 56.7
22	99°26.2	159°55.2	15.4	266°43.7	12.2	253°40.9	11.9	65°56.5	56.7
23	114°28.7	174°54.4	14.9	281°45.7	12.3	268°43.2	11.9	80°58.8	56.8
Mer.pass. 15:23		ν -0.9' d -0.5' m -3.90		ν 2.0' d 0.1' m 0.03		ν 2.4' d 0.0' m -2.09		ν 2.4' d 0.0' m 0.32	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	129°31.1	189°53.5	S21°14.3	296°47.8	S02°12.4	283°45.6	S09°11.9	96°01.2	N10°56.8
1	144°33.6	204°52.7	13.8	311°49.8	12.6	298°48.0	11.9	111°03.6	56.9
2	159°36.1	219°51.8	13.3	326°51.9	12.7	313°50.4	12.0	126°05.9	56.9
3	174°38.5	234°51.0	•• 12.8	341°53.9	•• 12.8	328°52.7	•• 12.0	141°08.3	•• 57.0
4	189°41.0	249°50.1	12.2	356°56.0	12.9	343°55.1	12.0	156°10.6	57.0
5	204°43.4	264°49.3	11.7	11°58.1	13.0	358°57.5	12.0	171°13.0	57.1
6	219°45.9	279°48.4	S21°11.2	27°00.1	S02°13.2	13°59.9	S09°12.0	186°15.4	N10°57.1
7	234°48.4	294°47.6	10.6	42°02.2	13.3	29°02.2	12.1	201°17.7	57.2
8	249°50.8	309°46.7	10.1	57°04.2	13.4	44°04.6	12.1	216°20.1	57.2
9	264°53.3	324°45.9	•• 09.6	72°06.3	•• 13.5	59°07.0	•• 12.1	231°22.4	•• 57.3
10	279°55.8	339°45.0	09.0	87°08.4	13.6	74°09.4	12.1	246°24.8	57.3
11	294°58.2	354°44.2	08.5	102°10.4	13.8	89°11.8	12.2	261°27.2	57.4
12	310°00.7	9°43.3	S21°08.0	117°12.5	S02°13.9	104°14.1	S09°12.2	276°29.5	N10°57.4
13	325°03.2	24°42.5	07.4	132°14.6	14.0	119°16.5	12.2	291°31.9	57.5
14	340°05.6	39°41.6	06.9	147°16.6	14.1	134°18.9	12.2	306°34.2	57.5
15	355°08.1	54°40.8	•• 06.4	162°18.7	•• 14.2	149°21.3	•• 12.2	321°36.6	•• 57.6
16	10°10.6	69°39.9	05.8	177°20.8	14.3	164°23.7	12.3	336°39.0	57.6
17	25°13.0	84°39.1	05.3	192°22.8	14.4	179°26.0	12.3	351°41.3	57.7
18	40°15.5	99°38.3	S21°04.7	207°24.9	S02°14.6	194°28.4	S09°12.3	6°43.7	N10°57.7
19	55°17.9	114°37.4	04.2	222°27.0	14.7	209°30.8	12.3	21°46.0	57.8
20	70°20.4	129°36.6	03.6	237°29.1	14.8	224°33.2	12.3	36°48.4	57.8
21	85°22.9	144°35.7	•• 03.1	252°31.1	•• 14.9	239°35.6	•• 12.4	51°50.7	•• 57.9
22	100°25.3	159°34.9	02.5	267°33.2	15.0	254°37.9	12.4	66°53.1	57.9
23	115°27.8	174°34.0	02.0	282°35.3	15.1	269°40.3	12.4	81°55.5	58.0
Mer.pass. 15:19		ν -0.9' d -0.5' m -3.90		ν 2.1' d 0.1' m 0.00		ν 2.4' d 0.0' m -2.10		ν 2.4' d 0.0' m 0.32	

Jan 28 Sun	SHA	Mer.pass
Venus	63°02.0	11:18
Mars	167°37.3	04:19
Jupiter	154°19.1	05:12
Saturn	326°34.8	17:41

Jan 29 Mon	SHA	Mer.pass
Venus	61°42.1	11:20
Mars	167°26.7	04:16
Jupiter	154°16.7	05:08
Saturn	326°32.5	17:37

Jan 30 Tue	SHA	Mer.pass
Venus	60°22.4	11:21
Mars	167°16.6	04:12
Jupiter	154°14.5	05:04
Saturn	326°30.1	17:33

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.2062 sec ΔT = TT-UT1 = +68.9778 sec

2029 January 28 to Jan. 30 UT

h	Sun				Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	176°46.8	S18°10.5	27°46.5	6.6'	N23°35.2	-5.5'	58.1'	
1	191°46.7	09.9	42°12.1	6.6'	23°29.7	-5.6'	58.1'	
2	206°46.5	09.2	56°37.7	6.6'	23°24.1	-5.8'	58.1'	
3	221°46.4	· · 08.5	71°03.3	6.6'	23°18.3	-5.9'	58.2'	
4	236°46.3	07.9	85°28.9	6.6'	23°12.4	-6.1'	58.2'	
5	251°46.2	07.2	99°54.5	6.6'	23°06.3	-6.2'	58.2'	
6	266°46.1	S18°06.5	114°20.2	6.6'	N23°00.0	-6.4'	58.3'	
7	281°46.0	05.9	128°45.8	6.7'	22°53.6	-6.5'	58.3'	
8	296°45.8	05.2	143°11.5	6.7'	22°47.1	-6.7'	58.3'	
9	311°45.7	· · 04.6	157°37.1	6.7'	22°40.4	-6.8'	58.4'	
10	326°45.6	03.9	172°02.8	6.7'	22°33.6	-7.0'	58.4'	
11	341°45.5	03.2	186°28.5	6.7'	22°26.6	-7.1'	58.4'	
12	356°45.4	S18°02.6	200°54.2	6.7'	N22°19.5	-7.3'	58.5'	
13	11°45.3	01.9	215°20.0	6.8'	22°12.2	-7.4'	58.5'	
14	26°45.2	01.2	229°45.7	6.8'	22°04.8	-7.6'	58.5'	
15	41°45.1	18°00.6	244°11.5	6.8'	21°57.2	-7.7'	58.6'	
16	56°44.9	17°59.9	258°37.3	6.8'	21°49.5	-7.9'	58.6'	
17	71°44.8	59.2	273°03.1	6.9'	21°41.7	-8.0'	58.6'	
18	86°44.7	S17°58.6	287°29.0	6.9'	N21°33.7	-8.1'	58.7'	
19	101°44.6	57.9	301°54.9	6.9'	21°25.5	-8.3'	58.7'	
20	116°44.5	57.2	316°20.8	6.9'	21°17.3	-8.4'	58.7'	
21	131°44.4	· · 56.5	330°46.7	7.0'	21°08.8	-8.6'	58.8'	
22	146°44.3	55.9	345°12.7	7.0'	21°00.3	-8.7'	58.8'	
23	161°44.2	55.2	359°38.6	7.0'	20°51.6	-8.8'	58.8'	
	SD = 16.2'	$d = -0.7'$	SD = 15.8'					
Mon	GHA	Dec	GHA	ν	Dec	d	HP	
0	176°44.1	S17°54.5	14°04.7	7.1'	N20°42.7	-9.0'	58.9'	
1	191°44.0	53.9	28°30.7	7.1'	20°33.8	-9.1'	58.9'	
2	206°43.8	53.2	42°56.8	7.1'	20°24.7	-9.2'	58.9'	
3	221°43.7	· · 52.5	57°22.9	7.2'	20°15.4	-9.4'	58.9'	
4	236°43.6	51.8	71°49.1	7.2'	20°06.0	-9.5'	59.0'	
5	251°43.5	51.2	86°15.3	7.2'	19°56.5	-9.6'	59.0'	
6	266°43.4	S17°50.5	100°41.5	7.3'	N19°46.9	-9.8'	59.0'	
7	281°43.3	49.8	115°07.7	7.3'	19°37.1	-9.9'	59.1'	
8	296°43.2	49.1	129°34.0	7.3'	19°27.2	-10.0'	59.1'	
9	311°43.1	· · 48.4	144°00.4	7.4'	19°17.2	-10.2'	59.1'	
10	326°43.0	47.8	158°26.7	7.4'	19°07.0	-10.3'	59.2'	
11	341°42.9	47.1	172°53.1	7.4'	18°56.7	-10.4'	59.2'	
12	356°42.8	S17°46.4	187°19.6	7.5'	N18°46.3	-10.5'	59.2'	
13	11°42.7	45.7	201°46.1	7.5'	18°35.7	-10.7'	59.2'	
14	26°42.6	45.1	216°12.6	7.6'	18°25.1	-10.8'	59.3'	
15	41°42.5	· · 44.4	230°39.2	7.6'	18°14.3	-10.9'	59.3'	
16	56°42.4	43.7	245°05.8	7.7'	18°03.4	-11.0'	59.3'	
17	71°42.3	43.0	259°32.4	7.7'	17°52.3	-11.1'	59.3'	
18	86°42.2	S17°42.3	273°59.1	7.7'	N17°41.2	-11.3'	59.4'	
19	101°42.1	41.6	288°25.8	7.8'	17°29.9	-11.4'	59.4'	
20	116°42.0	41.0	302°52.6	7.8'	17°18.5	-11.5'	59.4'	
21	131°41.9	· · 40.3	317°19.4	7.9'	17°07.0	-11.6'	59.4'	
22	146°41.8	39.6	331°46.3	7.9'	16°55.4	-11.7'	59.5'	
23	161°41.7	38.9	346°13.2	7.9'	16°43.7	-11.8'	59.5'	
	SD = 16.2'	$d = -0.7'$	SD = 16.1'					
Tue	GHA	Dec	GHA	ν	Dec	d	HP	
0	176°41.6	S17°38.2	0°40.2	8.0'	N16°31.9	-11.9'	59.5'	
1	191°41.5	37.5	15°07.2	8.0'	16°19.9	-12.1'	59.5'	
2	206°41.4	36.8	29°34.2	8.1'	16°07.9	-12.2'	59.6'	
3	221°41.3	· · 36.2	44°01.3	8.1'	15°55.7	-12.3'	59.6'	
4	236°41.2	35.5	58°28.4	8.2'	15°43.5	-12.4'	59.6'	
5	251°41.1	34.8	72°55.6	8.2'	15°31.1	-12.5'	59.6'	
6	266°41.0	S17°34.1	87°22.8	8.3'	N15°18.6	-12.6'	59.7'	
7	281°40.9	33.4	101°50.0	8.3'	15°06.1	-12.7'	59.7'	
8	296°40.8	32.7	116°17.3	8.3'	14°53.4	-12.8'	59.7'	
9	311°40.7	· · 32.0	130°44.7	8.4'	14°40.6	-12.9'	59.7'	
10	326°40.6	31.3	145°12.1	8.4'	14°27.7	-13.0'	59.7'	
11	341°40.5	30.6	159°39.5	8.5'	14°14.8	-13.1'	59.8'	
12	356°40.4	S17°29.9	174°07.0	8.5'	N14°01.7	-13.1'	59.8'	
13	11°40.3	29.3	188°34.5	8.6'	13°48.6	-13.2'	59.8'	
14	26°40.2	28.6	203°02.1	8.6'	13°35.4	-13.3'	59.8'	
15	41°40.1	· · 27.9	217°29.7	8.7'	13°22.0	-13.4'	59.8'	
16	56°40.0	27.2	231°57.3	8.7'	13°08.6	-13.5'	59.9'	
17	71°39.9	26.5	246°25.0	8.7'	12°55.1	-13.6'	59.9'	
18	86°39.8	S17°25.8	260°52.8	8.8'	N12°41.5	-13.7'	59.9'	
19	101°39.7	25.1	275°20.5	8.8'	12°27.9	-13.7'	59.9'	
20	116°39.6	24.4	289°48.3	8.9'	12°14.1	-13.8'	59.9'	
21	131°39.5	· · 23.7	304°16.2	8.9'	12°00.3	-13.9'	60.0'	
22	146°39.5	23.0	318°44.1	8.9'	11°46.4	-14.0'	60.0'	
23	161°39.4	22.3	333°12.1	9.0'	11°32.4	-14.1'	60.0'	
	SD = 16.2'	$d = -0.7'$	SD = 16.2'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:19	08:50	10:52	13:35	15:37	17:09
N 70°	07:10	08:30	10:01	14:27	15:58	17:18
68°	07:02	08:14	09:29	14:58	16:14	17:25
66°	06:56	08:01	09:06	15:22	16:27	17:32
64°	06:50	07:50	08:48	15:40	16:38	17:37
62°	06:45	07:40	08:33	15:54	16:47	17:42
60°	06:41	07:32	08:20	16:07	16:55	17:46
N 58°	06:37	07:25	08:09	16:18	17:02	17:50
56°	06:33	07:18	08:00	16:27	17:09	17:54
54°	06:29	07:12	07:52	16:35	17:15	17:58
52°	06:26	07:07	07:44	16:43	17:20	18:01
50°	06:23	07:02	07:37	16:49	17:25	18:04
45°	06:16	06:51	07:23	17:04	17:35	18:11
N 40°	06:10	06:42	07:11	17:16	17:45	18:17
35°	06:04	06:34	07:01	17:26	17:53	18:23
30°	05:58	06:27	06:52	17:35	18:00	18:29
20°	05:47	06:13	06:36	17:50	18:13	18:40
N 10°	05:35	06:01	06:23	18:04	18:26	18:51
0°	05:23	05:48	06:10	18:17	18:38	19:04
S 10°	05:08	05:34	05:57	18:30	18:52	19:18
20°	04:51	05:19	05:42	18:44	19:07	19:35
30°	04:29	05:00	05:26	19:00	19:26	19:57
35°	04:15	04:49	05:17	19:09	19:37	20:10
40°	03:58	04:35	05:06	19:20	19:50	20:27
45°	03:37	04:19	04:53	19:33	20:06	20:49
S 50°	03:08	03:58	04:37	19:48	20:27	21:17
52°	02:53	03:48	04:29	19:56	20:37	21:32
54°	02:34	03:37	04:21	20:04	20:48	21:49
56°	02:12	03:23	04:12	20:13	21:01	22:11
58°	01:40	03:08	04:01	20:24	21:16	22:41
S 60°	00:32	02:49	03:49	20:36	21:35	23:36

January 31, 01, 02 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	130°30.3	189°33.2	S21°01.4	297°37.4	S02°15.2	284°42.7	S09°12.4	96°57.8	N10°58.0	Alpheratz	357°31.5	29°15.2
1	145°32.7	204°32.4	00.9	312°39.4	15.4	299°45.1	12.5	112°00.2	58.1	Ankaa	353°04.2	-42°09.0
2	160°35.2	219°31.5	21°00.3	327°41.5	15.5	314°47.5	12.5	127°02.5	58.1	Schedar	349°27.4	56°42.1
3	175°37.7	234°30.7	20°59.8	342°43.6	•• 15.6	329°49.9	•• 12.5	142°04.9	•• 58.2	Diphda	348°44.2	-17°49.7
4	190°40.1	249°29.8	59.2	357°45.7	15.7	344°52.2	12.5	157°07.2	58.2	Achernar	335°18.0	-57°05.6
5	205°42.6	264°29.0	58.7	12°47.8	15.8	359°54.6	12.5	172°09.6	58.3	Hamal	327°47.4	23°36.1
6	220°45.1	279°28.2	S20°58.1	27°49.8	S02°15.9	14°57.0	S09°12.6	187°11.9	N10°58.3	Polaris	312°07.7	89°23.4
7	235°47.5	294°27.3	57.6	42°51.9	16.0	29°59.4	12.6	202°14.3	58.4	Acamar	315°09.3	-40°11.5
8	250°50.0	309°26.5	57.0	57°54.0	16.1	45°01.8	12.6	217°16.7	58.4	Menkar	314°02.6	4°12.2
9	265°52.4	324°25.6	•• 56.4	72°56.1	•• 16.2	60°04.2	•• 12.6	232°19.0	•• 58.5	Mirfak	308°23.3	49°58.0
10	280°54.9	339°24.8	55.9	87°58.2	16.3	75°06.5	12.6	247°21.4	58.5	Aldebaran	290°35.6	16°34.0
11	295°57.4	354°24.0	55.3	103°00.3	16.4	90°08.9	12.6	262°23.7	58.6	Rigel	281°00.5	-8°10.3
12	310°59.8	9°23.1	S20°54.8	118°02.4	S02°16.6	105°11.3	S09°12.7	277°26.1	N10°58.6	Capella	280°16.6	46°01.6
13	326°02.3	24°22.3	54.2	133°04.5	16.7	120°13.7	12.7	292°28.4	58.7	Bellatrix	278°19.1	6°22.4
14	341°04.8	39°21.5	53.6	148°06.5	16.8	135°16.1	12.7	307°30.8	58.7	Elnath	277°57.4	28°37.8
15	356°07.2	54°20.6	•• 53.1	163°08.6	•• 16.9	150°18.5	•• 12.7	322°33.1	•• 58.8	Alnilam	275°34.1	-1°11.2
16	11°09.7	69°19.8	52.5	178°10.7	17.0	165°20.9	12.7	337°35.5	58.8	Betelgeuse	270°48.2	7°24.6
17	26°12.2	84°19.0	51.9	193°12.8	17.1	180°23.2	12.8	352°37.9	58.9	Canopus	263°50.5	-52°42.9
18	41°14.6	99°18.1	S20°51.3	208°14.9	S02°17.2	195°25.6	S09°12.8	7°40.2	N10°58.9	Sirius	258°23.0	-16°45.6
19	56°17.1	114°17.3	50.8	223°17.0	17.3	210°28.0	12.8	22°42.6	59.0	Adhara	255°02.9	-29°01.0
20	71°19.5	129°16.5	50.2	238°19.1	17.4	225°30.4	12.8	37°44.9	59.0	Procyon	244°47.0	5°08.8
21	86°22.0	144°15.6	•• 49.6	253°21.2	•• 17.5	240°32.8	•• 12.8	52°47.3	•• 59.1	Pollux	243°12.9	27°57.1
22	101°24.5	159°14.8	49.1	268°23.3	17.6	255°35.2	12.8	67°49.6	59.1	Avior	234°12.6	-59°36.3
23	116°26.9	174°14.0	48.5	283°25.4	17.7	270°37.6	12.9	82°52.0	59.2	Suhail	222°43.4	-43°33.2
Mer.pass. 15:15		ν -0.8' d -0.6' m -3.90			ν 2.1' d 0.1' m -0.02		ν 2.4' d 0.0' m -2.10		ν 2.4' d 0.0' m 0.32			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	131°29.4	189°13.1	S20°47.9	298°27.5	S02°17.8	285°40.0	S09°12.9	97°54.3	N10°59.2			
1	146°31.9	204°12.3	47.3	313°29.6	17.9	300°42.4	12.9	112°56.7	59.3			
2	161°34.3	219°11.5	46.7	328°31.7	18.0	315°44.7	12.9	127°59.0	59.3			
3	176°36.8	234°10.6	•• 46.2	343°33.8	•• 18.1	330°47.1	•• 12.9	143°01.4	•• 59.4			
4	191°39.3	249°09.8	45.6	358°35.9	18.2	345°49.5	13.0	158°03.7	59.4			
5	206°41.7	264°09.0	45.0	13°38.0	18.3	0°51.9	13.0	173°06.1	59.5			
6	221°44.2	279°08.2	S20°44.4	28°40.1	S02°18.4	15°54.3	S09°13.0	188°08.4	N10°59.5			
7	236°46.7	294°07.3	43.8	43°42.2	18.5	30°56.7	13.0	203°10.8	59.6			
8	251°49.1	309°06.5	43.3	58°44.3	18.6	45°59.1	13.0	218°13.1	59.6			
9	266°51.6	324°05.7	•• 42.7	73°46.4	•• 18.7	61°01.5	•• 13.0	233°15.5	•• 59.7			
10	281°54.0	339°04.8	42.1	88°48.5	18.8	76°03.9	13.1	248°17.8	59.7			
11	296°56.5	354°04.0	41.5	103°50.7	18.9	91°06.3	13.1	263°20.2	59.8			
12	311°59.0	9°03.2	S20°40.9	118°52.8	S02°19.0	106°08.7	S09°13.1	278°22.5	N10°59.8			
13	327°01.4	24°02.4	40.3	133°54.9	19.1	121°11.1	13.1	293°24.9	59.9			
14	342°03.9	39°01.5	39.7	148°57.0	19.2	136°13.4	13.1	308°27.2	10°59.9			
15	357°06.4	54°00.7	•• 39.1	163°59.1	•• 19.3	151°15.8	•• 13.1	323°29.6	11°00.0			
16	12°08.8	68°59.9	38.5	179°01.2	19.4	166°18.2	13.1	338°31.9	00.1			
17	27°11.3	83°59.1	37.9	194°03.3	19.5	181°20.6	13.2	353°34.3	00.1			
18	42°13.8	98°58.2	S20°37.3	209°05.5	S02°19.6	196°23.0	S09°13.2	8°36.6	N11°00.2			
19	57°16.2	113°57.4	36.7	224°07.6	19.7	211°25.4	13.2	23°39.0	00.2			
20	72°18.7	128°56.6	36.2	239°09.7	19.8	226°27.8	13.2	38°41.3	00.3			
21	87°21.2	143°55.8	•• 35.6	254°11.8	•• 19.9	241°30.2	•• 13.2	53°43.7	•• 00.3			
22	102°23.6	158°55.0	35.0	269°13.9	20.0	256°32.6	13.2	68°46.0	00.4			
23	117°26.1	173°54.1	34.4	284°16.1	20.1	271°35.0	13.3	83°48.4	00.4			
Mer.pass. 15:12		ν -0.8' d -0.6' m -3.90			ν 2.1' d 0.1' m -0.05		ν 2.4' d 0.0' m -2.11		ν 2.4' d 0.1' m 0.33			

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	132°28.5	188°53.3	S20°33.7	299°18.2	S02°20.2	286°37.4	S09°13.3	98°50.7	N11°00.5			
1	147°31.0	203°52.5	33.1	314°20.3	20.2	301°39.8	13.3	113°53.1	00.5			
2	162°33.5	218°51.7	32.5	329°22.4	20.3	316°42.2	13.3	128°55.4	00.6			
3	177°35.9	233°50.9	•• 31.9	344°24.6	•• 20.4	331°44.6	•• 13.3	143°57.8	•• 00.6			
4	192°38.4	248°50.0	31.3	359°26.7	20.5	346°47.0	13.3	159°00.1	00.7			
5	207°40.9	263°49.2	30.7	14°28.8	20.6	1°49.4	13.3	174°02.5	00.7			
6	222°43.3	278°48.4	S20°30.1	29°30.9	S02°20.7	16°51.8	S09°13.4	189°04.8	N11°00.8			
7	237°45.8	293°47.6	29.5	44°33.1	20.8	31°54.2	13.4	204°07.2	00.8			
8	252°48.3	308°46.8	28.9	59°35.2	20.9	46°56.6	13.4	219°09.5	00.9			
9	267°50.7	323°45.9	•• 28.3	74°37.3	•• 21.0	61°59.0	•• 13.4	234°11.9	•• 00.9			
10	282°53.2	338°45.1	27.7	89°39.5	21.1	77°01.4	13.4	249°14.2	01.0			
11	297°55.6	353°44.3	27.1	104°41.6	21.1	92°03.8	13.4	264°16.6	01.0			
12	312°58.1	8°43.5	S20°26.4	119°43.7	S02°21.2	107°06.2	S09°13.4	279°18.9	N11°01.1			
13	328°00.6	23°42.8	25.8	134°45.9	21.3	122°08.6	13.5	294°21.2	01.2			
14	343°03.0	38°41.9	25.2	149°48.0	21.4	137°11.0	13.5	309°23.6	01.2			
15	358°05.5	53°41.1	•• 24.6	164°50.1	•• 21.5	152°13.4	•• 13.5	324°25.9	•• 01.3			
16	13°08.0	68°40.2	24.0	179°52.3	21.6	167°15.8	13.5	339°28.3	01.3			
17	28°10.4	83°39.4	23.4	194°54.4	21.7	182°18.2	13.5	354°30.6	01.4			
18	43°12.9	98°38.6	S20°22.7	209°56.6	S02°21.8	197°20.6	S09°13.5	9°33.0	N11°01.4			
19	58°15.4	113°37.8	22.1	224°58.7	21.8	212°23.0	13.5	24°35.3	01.5			
20	73°17.8	128°37.0	21.5	240°00.8	21.9	227°25.4	13.5	39°37.7	01.5			
21	88°20.3	143°36.2	•• 20.9	255°03.0	•• 22.0	242°27.8	•• 13.6	54°40.0	•• 01.6			
22	103°22.8	158°35.4	20.2	270°05.1	22.1	257°30.2	13.6	69°42.4	01.6			
23	118°25.2	173°34.6	19.6	285°07.3	22.2	272°32.6	13.6	84°44.7	01.7			
Mer.pass. 15:08		ν -0.8' d -0.6' m -3.90			ν 2.1' d 0.1' m -0.07		ν 2.4' d 0.0' m -2.12		ν 2.3' d 0.1' m 0.33			

Jan 31 Wed	SHA	Mer.pass		
Venus	59°02.9	11:22		
Mars	167°07.1	04:09		
Jupiter	154°12.4	05:00		
Saturn	326°27.5	17:29		
Feb 01 Thu	SHA	Mer.pass		
Venus	57°43.7	11:24		
Mars	166°58.1	04:06		
Jupiter	154°10.6	04:57		
Saturn	326°24.9	17:26		
Feb 02 Fri	SHA	Mer.pass		
Venus	56°24.8	11:25		
Mars	166°49.6	04:02		
Jupiter	154°08.9	04:53		
Saturn	326°22.2	17:22		
Horizontal parallax				
Venus:			0.1	
Mars:			0.2	

DUT1 = UT1-UTC = +0.2063 sec ΔT = TT-UT1 = +68.9777 sec

2029 January 31 to Feb. 02 UT

h Sun Moon						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	176°39.3	S17°21.6	347°40.0	9.0'	N11°18.4	-14.1' 60.0'
1	191°39.2	20.9	2°08.1	9.1'	11°04.2	-14.2' 60.0'
2	206°39.1	20.2	16°36.1	9.1'	10°50.0	-14.3' 60.0'
3	221°39.0	· · 19.5	31°04.2	9.1'	10°35.8	-14.3' 60.0'
4	236°38.9	18.8	45°32.4	9.2'	10°21.4	-14.4' 60.1'
5	251°38.8	18.1	60°00.5	9.2'	10°07.0	-14.5' 60.1'
6	266°38.7	S17°17.4	74°28.8	9.3'	N09°52.6	-14.5' 60.1'
7	281°38.6	16.7	88°57.0	9.3'	09°38.0	-14.6' 60.1'
8	296°38.6	16.0	103°25.3	9.3'	09°23.4	-14.7' 60.1'
9	311°38.5	· · 15.3	117°53.6	9.4'	09°08.8	-14.7' 60.1'
10	326°38.4	14.6	132°22.0	9.4'	08°54.1	-14.8' 60.1'
11	341°38.3	13.9	146°50.4	9.4'	08°39.3	-14.8' 60.1'
12	356°38.2	S17°13.2	161°18.8	9.5'	N08°24.5	-14.9' 60.2'
13	11°38.1	12.5	175°47.3	9.5'	08°09.6	-14.9' 60.2'
14	26°38.0	11.8	190°15.8	9.5'	07°54.7	-15.0' 60.2'
15	41°37.9	· · 11.1	204°44.3	9.6'	07°39.7	-15.0' 60.2'
16	56°37.9	10.3	219°12.8	9.6'	07°24.7	-15.1' 60.2'
17	71°37.8	09.6	233°41.4	9.6'	07°09.6	-15.1' 60.2'
18	86°37.7	S17°08.9	248°10.0	9.6'	N06°54.5	-15.2' 60.2'
19	101°37.6	08.2	262°38.7	9.7'	06°39.4	-15.2' 60.2'
20	116°37.5	07.5	277°07.4	9.7'	06°24.2	-15.2' 60.2'
21	131°37.4	· · 06.8	291°36.1	9.7'	06°08.9	-15.3' 60.2'
22	146°37.3	06.1	306°04.8	9.8'	05°53.7	-15.3' 60.2'
23	161°37.3	05.4	320°33.6	9.8'	05°38.4	-15.3' 60.2'
SD = 16.2' d = -0.7' SD = 16.4'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	176°37.2	S17°04.7	335°02.3	9.8'	N05°23.0	-15.4' 60.2'
1	191°37.1	04.0	349°31.1	9.8'	05°07.6	-15.4' 60.3'
2	206°37.0	03.2	4°00.0	9.9'	04°52.2	-15.4' 60.3'
3	221°36.9	· · 02.5	18°28.8	9.9'	04°36.8	-15.5' 60.3'
4	236°36.9	01.8	32°57.7	9.9'	04°21.3	-15.5' 60.3'
5	251°36.8	01.1	47°26.6	9.9'	04°05.9	-15.5' 60.3'
6	266°36.7	S17°00.4	61°55.5	9.9'	N03°50.4	-15.5' 60.3'
7	281°36.6	16°59.7	76°24.4	10.0'	03°34.8	-15.6' 60.3'
8	296°36.5	59.0	90°53.4	10.0'	03°19.3	-15.6' 60.3'
9	311°36.5	· · 58.2	105°22.4	10.0'	03°03.7	-15.6' 60.3'
10	326°36.4	57.5	119°51.3	10.0'	02°48.1	-15.6' 60.3'
11	341°36.3	56.8	134°20.3	10.0'	02°32.5	-15.6' 60.3'
12	356°36.2	S16°56.1	148°49.4	10.0'	N02°16.9	-15.6' 60.3'
13	11°36.1	55.4	163°18.4	10.0'	02°01.3	-15.6' 60.3'
14	26°36.1	54.7	177°47.4	10.1'	01°45.6	-15.6' 60.3'
15	41°36.0	· · 53.9	192°16.5	10.1'	01°30.0	-15.7' 60.3'
16	56°35.9	53.2	206°45.6	10.1'	01°14.3	-15.7' 60.3'
17	71°35.8	52.5	221°14.6	10.1'	00°58.7	-15.7' 60.3'
18	86°35.7	S16°51.8	235°43.7	10.1'	N00°43.0	-15.7' 60.3'
19	101°35.7	51.1	250°12.8	10.1'	00°27.3	-15.7' 60.3'
20	116°35.6	50.3	264°41.9	10.1'	N00°11.7	-15.7' 60.3'
21	131°35.5	· · 49.6	279°11.0	10.1'	S00°04.0	15.7' 60.3'
22	146°35.4	48.9	293°40.2	10.1'	00°19.6	15.7' 60.3'
23	161°35.4	48.2	308°09.3	10.1'	00°35.3	15.6' 60.3'
SD = 16.2' d = -0.7' SD = 16.4'						
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	176°35.3	S16°47.4	322°38.4	10.1'	S00°50.9	15.6' 60.3'
1	191°35.2	46.7	337°07.5	10.1'	01°06.6	15.6' 60.2'
2	206°35.1	46.0	351°36.7	10.1'	01°22.2	15.6' 60.2'
3	221°35.1	· · 45.3	6°05.8	10.1'	01°37.8	15.6' 60.2'
4	236°35.0	44.5	20°34.9	10.1'	01°53.4	15.6' 60.2'
5	251°34.9	43.8	35°04.1	10.1'	02°09.0	15.6' 60.2'
6	266°34.9	S16°43.1	49°33.2	10.1'	S02°24.6	15.6' 60.2'
7	281°34.8	42.4	64°02.3	10.1'	02°40.1	15.5' 60.2'
8	296°34.7	41.6	78°31.4	10.1'	02°55.7	15.5' 60.2'
9	311°34.6	· · 40.9	93°00.6	10.1'	03°11.2	15.5' 60.2'
10	326°34.6	40.2	107°29.7	10.1'	03°26.7	15.5' 60.2'
11	341°34.5	39.4	121°58.8	10.1'	03°42.1	15.4' 60.2'
12	356°34.4	S16°38.7	136°27.9	10.1'	S03°57.6	15.4' 60.2'
13	11°34.4	38.0	150°57.0	10.1'	04°13.0	15.4' 60.2'
14	26°34.3	37.3	165°26.0	10.1'	04°28.3	15.3' 60.2'
15	41°34.2	· · 36.5	179°55.1	10.1'	04°43.7	15.3' 60.1'
16	56°34.1	35.8	194°24.2	10.0'	04°59.0	15.3' 60.1'
17	71°34.1	35.1	208°53.2	10.0'	05°14.3	15.2' 60.1'
18	86°34.0	S16°34.3	223°22.3	10.0'	S05°29.5	15.2' 60.1'
19	101°33.9	33.6	237°51.3	10.0'	05°44.7	15.2' 60.1'
20	116°33.9	32.9	252°20.3	10.0'	05°59.9	15.1' 60.1'
21	131°33.8	· · 32.1	266°49.3	10.0'	06°15.0	15.1' 60.1'
22	146°33.7	31.4	281°18.2	10.0'	06°30.1	15.0' 60.1'
23	161°33.7	30.6	295°47.2	9.9'	06°45.1	15.0' 60.1'
SD = 16.2' d = -0.7' SD = 16.4'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:08	08:37	10:25	14:04	15:52	17:20
N 70°	07:01	08:19	09:44	14:45	16:10	17:28
68°	06:54	08:04	09:16	15:12	16:24	17:34
66°	06:49	07:52	08:55	15:33	16:36	17:40
64°	06:44	07:42	08:39	15:50	16:46	17:45
62°	06:39	07:34	08:25	16:03	16:55	17:49
60°	06:35	07:26	08:13	16:15	17:02	17:53
N 58°	06:32	07:19	08:03	16:25	17:09	17:56
56°	06:28	07:13	07:54	16:34	17:15	18:00
54°	06:25	07:08	07:47	16:41	17:20	18:03
52°	06:22	07:03	07:40	16:48	17:25	18:06
50°	06:20	06:58	07:33	16:55	17:29	18:08
45°	06:13	06:48	07:20	17:08	17:39	18:15
N 40°	06:07	06:40	07:08	17:19	17:48	18:20
35°	06:02	06:32	06:59	17:29	17:55	18:26
30°	05:56	06:25	06:50	17:37	18:02	18:31
20°	05:46	06:12	06:35	17:52	18:15	18:41
N 10°	05:35	06:00	06:22	18:05	18:27	18:52
0°	05:23	05:48	06:10	18:17	18:39	19:04
S 10°	05:10	05:36	05:58	18:29	18:51	19:17
20°	04:53	05:21	05:44	18:43	19:06	19:33
30°	04:32	05:03	05:29	18:58	19:24	19:54
35°	04:19	04:52	05:20	19:07	19:34	20:08
40°	04:03	04:39	05:09	19:17	19:47	20:24
45°	03:42	04:24	04:57	19:29	20:02	20:44
S 50°	03:15	04:04	04:42	19:44	20:22	21:11
52°	03:01	03:55	04:35	19:51	20:31	21:24
54°	02:44	03:44	04:27	19:59	20:42	21:41
56°	02:23	03:31	04:19	20:07	20:54	22:01
58°	01:56	03:17	04:09	20:17	21:08	22:27
S 60°	01:13	02:59	03:57	20:28	21:25	23:06

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	18:02	20:23	22:43	09:05	08:35	08:07
N 70°	18:13	20:23	22:32	08:51	08:31	08:12
68°	18:21	20:23	22:24	08:41	08:27	08:15
66°	18:28	20:23	22:17	08:32	08:25	08:18
64°	18:34	20:23	22:11	08:24	08:22	08:20
62°	18:39	20:23	22:06	08:17	08:20	08:22
60°	18:43	20:23	22:02	08:12	08:18	08:24
N 58°	18:47	20:23	21:58	08:06	08:17	08:26
56°	18:50	20:23	21:55	08:02	08:15	08:28
54°	18:53	20:23	21:51	07:58	08:14	08:29
52°	18:56	20:23	21:49	07:54	08:13	08:30
50°	18:58	20:23	21:46	07:51	08:11	08:31
45°	19:04	20:23	21:41	07:43	08:09	08:34
N 40°	19:08	20:23	21:36	07:37	08:07	08:36
35°	19:12	20:23	21:33	07:31	08:05	08:38
30°	19:15	20:23	21:29	07:27	08:03	08:39
20°	19:21	20:23	21:23	07:18	08:01	08:42
N 10°	19:26	20:23	21:18	07:11	07:58	08:45
0°	19:31	20:23	21:14	07:04	07:56	08:47
S 10°	19:36	20:23	21:09	06:57	07:53	08:49
20°	19:41	20:23	21:04	06:49	07:51	08:52
30°	19:47	20:23	20:59	06:40	07:47	08:54
35°	19:50	20:23	20:56	06:35	07:46	08:56
40°	19:54	20:23	20:52	06:29	07:44	08:58
45°	19:58	20:23	20:48	06:22	07:41	09:00
S 50°	20:03	20:23	20:43	06:14	07:38	09:02
52°	20:06	20:24	20:41	06:10	07:37	09:03
54°	20:08	20:24	20:39	06:06	07:36	09:05
56°	20:11	20:24	20:36	06:01	07:34	09:06
58°	20:14	20:24	20:33	05:56	07:32	09:08
S 60°	20:18	20:24	20:30	05:50	07:30	09:09

Day	Sun		Mer. Pass	Moon		Age 17-19 99-90%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
31	13:23	13:27	12:13	00:51	13:17	
01	13:31	13:35	12:14	01:43	14:09	
02	13:39	13:42	12:14	02:35	15:00	

February 03, 04, 05 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	133°27.7	188°33.7	S20°19.0	300°09.4	S02°22.3	287°35.0	S09°13.6	99°47.0	N11°01.7	Alpheratz	357°31.5	29°15.2
1	148°30.1	203°32.9	18.4	315°11.6	22.3	302°37.4	13.6	114°49.4	01.8	Ankaa	353°04.2	-42°09.0
2	163°32.6	218°32.1	17.7	330°13.7	22.4	317°39.8	13.6	129°51.7	01.8	Schedar	349°27.4	56°42.1
3	178°35.1	233°31.3	•• 17.1	345°15.9	•• 22.5	332°42.2	•• 13.6	144°54.1	•• 01.9	Diphda	348°44.2	-17°49.7
4	193°37.5	248°30.5	16.5	0°18.0	22.6	347°44.6	13.6	159°56.4	02.0	Achernar	335°18.0	-57°05.6
5	208°40.0	263°29.7	15.8	15°20.2	22.7	2°47.0	13.6	174°58.8	02.0	Hamal	327°47.5	23°36.1
6	223°42.5	278°28.9	S20°15.2	30°22.3	S02°22.7	17°49.4	S09°13.7	190°01.1	N11°02.1	Polaris	312°09.3	89°23.4
7	238°44.9	293°28.1	14.6	45°24.5	22.8	32°51.9	13.7	205°03.5	02.1	Acamar	315°09.3	-40°11.5
8	253°47.4	308°27.3	13.9	60°26.6	22.9	47°54.3	13.7	220°05.8	02.2	Menkar	314°02.6	4°12.2
9	268°49.9	323°26.5	•• 13.3	75°28.8	•• 23.0	62°56.7	•• 13.7	235°08.1	•• 02.2	Mirfak	308°23.3	49°58.0
10	283°52.3	338°25.7	12.7	90°30.9	23.1	77°59.1	13.7	250°10.5	02.3	Aldebaran	290°35.6	16°34.0
11	298°54.8	353°24.9	12.0	105°33.1	23.1	93°01.5	13.7	265°12.8	02.3	Rigel	281°00.5	-8°10.3
12	313°57.2	8°24.1	S20°11.4	120°35.3	S02°23.2	108°03.9	S09°13.7	280°15.2	N11°02.4	Capella	280°16.7	46°01.6
13	328°59.7	23°23.3	10.7	135°37.4	23.3	123°06.3	13.7	295°17.5	02.4	Bellatrix	278°19.1	6°22.4
14	344°02.2	38°22.5	10.1	150°39.6	23.4	138°08.7	13.7	310°19.9	02.5	Elnath	277°57.4	28°37.8
15	359°04.6	53°21.6	•• 09.4	165°41.7	•• 23.4	153°11.1	•• 13.8	325°22.2	•• 02.5	Alnilam	275°34.1	-1°11.2
16	14°07.1	68°20.8	08.8	180°43.9	23.5	168°13.5	13.8	340°24.5	02.6	Betelgeuse	270°48.2	7°24.6
17	29°09.6	83°20.0	08.2	195°46.1	23.6	183°15.9	13.8	355°26.9	02.7	Canopus	263°50.5	-52°42.9
18	44°12.0	98°19.2	S20°07.5	210°48.2	S02°23.7	198°18.3	S09°13.8	10°29.2	N11°02.7	Sirius	258°23.0	-16°45.6
19	59°14.5	113°18.4	06.9	225°50.4	23.8	213°20.7	13.8	25°31.6	02.8	Adhara	255°02.9	-29°01.0
20	74°17.0	128°17.6	06.2	240°52.6	23.8	228°23.2	13.8	40°33.9	02.8	Procyon	244°47.1	5°08.8
21	89°19.4	143°16.8	•• 05.6	255°54.7	•• 23.9	243°25.6	•• 13.8	55°36.2	•• 02.9	Pollux	243°12.9	27°57.1
22	104°21.9	158°16.0	04.9	270°56.9	24.0	258°28.0	13.8	70°38.6	02.9	Avior	234°12.6	-59°36.4
23	119°24.4	173°15.2	04.3	285°59.1	24.0	273°30.4	13.8	85°40.9	03.0	Suhail	222°43.4	-43°33.2
Mer.pass. 15:04		ν -0.8' d -0.6' m -3.90		ν 2.1' d 0.1' m -0.10		ν 2.4' d 0.0' m -2.12		ν 2.3' d 0.1' m 0.33				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	134°26.8	188°14.4	S20°03.6	301°01.2	S02°24.1	288°32.8	S09°13.8	100°43.3	N11°03.0
1	149°29.3	203°13.6	03.0	316°03.4	24.2	303°35.2	13.9	115°45.6	03.1
2	164°31.7	218°12.8	02.3	331°05.6	24.3	318°37.6	13.9	130°47.9	03.2
3	179°34.2	233°12.0	•• 01.7	346°07.8	•• 24.3	333°40.0	•• 13.9	145°50.3	•• 03.2
4	194°36.7	248°11.2	01.0	1°09.9	24.4	348°42.5	13.9	160°52.6	03.3
5	209°39.1	263°10.5	20°00.3	16°12.1	24.5	3°44.9	13.9	175°55.0	03.3
6	224°41.6	278°09.7	S19°59.7	31°14.3	S02°24.6	18°47.3	S09°13.9	190°57.3	N11°03.4
7	239°44.1	293°08.9	59.0	46°16.5	24.6	33°49.7	13.9	205°59.6	03.4
8	254°46.5	308°08.1	58.4	61°18.6	24.7	48°52.1	13.9	221°02.0	03.5
9	269°49.0	323°07.3	•• 57.7	76°20.8	•• 24.8	63°54.5	•• 13.9	236°04.3	•• 03.5
10	284°51.5	338°06.5	57.0	91°23.0	24.8	78°56.9	13.9	251°06.7	03.6
11	299°53.9	353°05.7	56.4	106°25.2	24.9	93°59.3	13.9	266°09.0	03.6
12	314°56.4	8°04.9	S19°55.7	121°27.4	S02°25.0	109°01.8	S09°13.9	281°11.3	N11°03.7
13	329°58.9	23°04.1	55.0	136°29.5	25.0	124°04.2	14.0	296°13.7	03.8
14	345°01.3	38°03.3	54.4	151°31.7	25.1	139°06.6	14.0	311°16.0	03.8
15	0°03.8	53°02.5	•• 53.7	166°33.9	•• 25.2	154°09.0	•• 14.0	326°18.3	•• 03.9
16	15°06.2	68°01.7	53.0	181°36.1	25.2	169°11.4	14.0	341°20.7	03.9
17	30°08.7	83°00.9	52.4	196°38.3	25.3	184°13.8	14.0	356°23.0	04.0
18	45°11.2	98°00.1	S19°51.7	211°40.5	S02°25.4	199°16.3	S09°14.0	11°25.4	N11°04.0
19	60°13.6	112°59.3	51.0	226°42.7	25.4	214°18.7	14.0	26°27.7	04.1
20	75°16.1	127°58.6	50.4	241°44.9	25.5	229°21.1	14.0	41°30.0	04.1
21	90°18.6	142°57.8	•• 49.7	256°47.1	•• 25.6	244°23.5	•• 14.0	56°32.4	•• 04.2
22	105°21.0	157°57.0	49.0	271°49.2	25.6	259°25.9	14.0	71°34.7	04.3
23	120°23.5	172°56.2	48.3	286°51.4	25.7	274°28.3	14.0	86°37.0	04.3
Mer.pass. 15:00		ν -0.8' d -0.7' m -3.90		ν 2.2' d 0.1' m -0.12		ν 2.4' d 0.0' m -2.13		ν 2.3' d 0.1' m 0.34	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	135°26.0	187°55.4	S19°47.7	301°53.6	S02°25.7	289°30.8	S09°14.0	101°39.4	N11°04.4
1	150°28.4	202°54.6	47.0	316°55.8	25.8	304°33.2	14.0	116°41.7	04.4
2	165°30.9	217°53.8	46.3	331°58.0	25.9	319°35.6	14.0	131°44.1	04.5
3	180°33.3	232°53.0	•• 45.6	347°00.2	•• 25.9	334°38.0	•• 14.0	146°46.4	•• 04.5
4	195°35.8	247°52.3	44.9	2°02.4	26.0	349°40.4	14.1	161°48.7	04.6
5	210°38.3	262°51.5	44.3	17°04.6	26.1	4°42.9	14.1	176°51.1	04.7
6	225°40.7	277°50.7	S19°43.6	32°06.8	S02°26.1	19°45.3	S09°14.1	191°53.4	N11°04.7
7	240°43.2	292°49.9	42.9	47°09.0	26.2	34°47.7	14.1	206°55.7	04.8
8	255°45.7	307°49.1	42.2	62°11.2	26.2	49°50.1	14.1	221°58.1	04.8
9	270°48.1	322°48.3	•• 41.5	77°13.4	•• 26.3	64°52.6	•• 14.1	237°00.4	•• 04.9
10	285°50.6	337°47.6	40.8	92°15.6	26.4	79°55.0	14.1	252°02.7	04.9
11	300°53.1	352°46.8	40.1	107°17.9	26.4	94°57.4	14.1	267°05.1	05.0
12	315°55.5	7°46.0	S19°39.5	122°20.1	S02°26.5	109°59.8	S09°14.1	282°07.4	N11°05.0
13	330°58.0	22°45.2	38.8	137°22.3	26.5	125°02.2	14.1	297°09.7	05.1
14	346°00.5	37°44.4	38.1	152°24.5	26.6	140°04.7	14.1	312°12.1	05.2
15	1°02.9	52°43.6	•• 37.4	167°26.7	•• 26.6	155°07.1	•• 14.1	327°14.4	•• 05.2
16	16°05.4	67°42.9	36.7	182°28.9	26.7	170°09.5	14.1	342°16.7	05.3
17	31°07.8	82°42.1	36.0	197°31.1	26.7	185°11.9	14.1	357°19.1	05.3
18	46°10.3	97°41.3	S19°35.3	212°33.3	S02°26.8	200°14.4	S09°14.1	12°21.4	N11°05.4
19	61°12.8	112°40.5	34.6	227°35.5	26.9	215°16.8	14.1	27°23.7	05.4
20	76°15.2	127°39.8	33.9	242°37.8	26.9	230°19.2	14.1	42°26.1	05.5
21	91°17.7	142°39.0	•• 33.2	257°40.0	•• 27.0	245°21.6	•• 14.1	57°28.4	•• 05.6
22	106°20.2	157°38.2	32.5	272°42.2	27.0	260°24.1	14.1	72°30.7	05.6
23	121°22.6	172°37.4	31.8	287°44.4	27.1	275°26.5	14.1	87°33.1	05.7
Mer.pass. 14:56		ν -0.8' d -0.7' m -3.90		ν 2.2' d 0.1' m -0.15		ν 2.4' d 0.0' m -2.14		ν 2.3' d 0.1' m 0.34	

Feb 03 Sat			SHA	Mer.pass
Venus	55°06.1		11:26	
Mars	166°41.7		03:59	
Jupiter	154°07.3		04:49	
Saturn	326°19.4		17:18	

Feb 04 Sun			SHA	Mer.pass
Venus	53°47.6		11:28	
Mars	166°34.4		03:55	
Jupiter	154°06.0		04:45	
Saturn	326°16.4		17:14	

Feb 05 Mon			SHA	Mer.pass
Venus	52°29.4		11:29	
Mars	166°27.7		03:52	
Jupiter	154°04.8		04:41	
Saturn	326°13.4		17:11	

Horizontal parallax			
Venus:		0.1	
Mars:		0.2	

DUT1 = UT1-UTC = +0.2064 sec $\Delta T = TT-UT1 = +68.9776$ sec

2029 February 03 to Feb. 05 UT

Sun							Moon						
h	Sat	GHA	Dec	GHA	ν	Dec	d	HP					
0		176°33.6	S16°29.9	310°16.1	9.9'	S07°00.1	14.9'	60.0'					
1		191°33.5	29.2	324°45.1	9.9'	07°15.1	14.9'	60.0'					
2		206°33.5	28.4	339°13.9	9.9'	07°30.0	14.8'	60.0'					
3		221°33.4	· · 27.7	353°42.8	9.9'	07°44.8	14.8'	60.0'					
4		236°33.3	27.0	8°11.7	9.8'	07°59.6	14.7'	60.0'					
5		251°33.3	26.2	22°40.5	9.8'	08°14.3	14.7'	60.0'					
6		266°33.2	S16°25.5	37°09.3	9.8'	S08°29.0	14.6'	60.0'					
7		281°33.2	24.7	51°38.1	9.8'	08°43.7	14.6'	59.9'					
8		296°33.1	24.0	66°06.9	9.7'	08°58.2	14.5'	59.9'					
9		311°33.0	· · 23.3	80°35.6	9.7'	09°12.7	14.5'	59.9'					
10		326°33.0	22.5	95°04.3	9.7'	09°27.2	14.4'	59.9'					
11		341°32.9	21.8	109°33.0	9.7'	09°41.6	14.3'	59.9'					
12		356°32.8	S16°21.0	124°01.6	9.6'	S09°55.9	14.3'	59.9'					
13		11°32.8	20.3	138°30.3	9.6'	10°10.2	14.2'	59.8'					
14		26°32.7	19.6	152°58.9	9.6'	10°24.4	14.1'	59.8'					
15		41°32.7	· · 18.8	167°27.4	9.5'	10°38.5	14.1'	59.8'					
16		56°32.6	18.1	181°56.0	9.5'	10°52.5	14.0'	59.8'					
17		71°32.5	17.3	196°24.5	9.5'	11°06.5	13.9'	59.8'					
18		86°32.5	S16°16.6	210°52.9	9.4'	S11°20.4	13.8'	59.8'					
19		101°32.4	15.8	225°21.4	9.4'	11°34.3	13.8'	59.7'					
20		116°32.4	15.1	239°49.8	9.4'	11°48.0	13.7'	59.7'					
21		131°32.3	· · 14.3	254°18.1	9.3'	12°01.7	13.6'	59.7'					
22		146°32.2	13.6	268°46.5	9.3'	12°15.3	13.5'	59.7'					
23		161°32.2	12.8	283°14.8	9.3'	12°28.8	13.4'	59.7'					
SD = 16.2'							SD = 16.4'						
d = -0.7'													
Sun	GHA	Dec	GHA	ν	Dec	d	HP						
0	176°32.1	S16°12.1	297°43.0	9.2'	S12°42.3	13.4'	59.7'						
1	191°32.1	11.4	312°11.3	9.2'	12°55.7	13.3'	59.6'						
2	206°32.0	10.6	326°39.4	9.2'	13°08.9	13.2'	59.6'						
3	221°31.9	· · 09.9	341°07.6	9.1'	13°22.1	13.1'	59.6'						
4	236°31.9	09.1	355°35.7	9.1'	13°35.2	13.0'	59.6'						
5	251°31.8	08.4	10°03.8	9.0'	13°48.3	12.9'	59.6'						
6	266°31.8	S16°07.6	24°31.8	9.0'	S14°01.2	12.8'	59.5'						
7	281°31.7	06.9	38°59.8	9.0'	14°14.0	12.7'	59.5'						
8	296°31.7	06.1	53°27.8	8.9'	14°26.8	12.7'	59.5'						
9	311°31.6	· · 05.3	67°55.7	8.9'	14°39.4	12.6'	59.5'						
10	326°31.6	04.6	82°23.5	8.8'	14°52.0	12.5'	59.5'						
11	341°31.5	03.8	96°51.4	8.8'	15°04.5	12.4'	59.4'						
12	356°31.4	S16°03.1	111°19.2	8.7'	S15°16.8	12.3'	59.4'						
13	11°31.4	02.3	125°46.9	8.7'	15°29.1	12.2'	59.4'						
14	26°31.3	01.6	140°14.6	8.7'	15°41.3	12.1'	59.4'						
15	41°31.3	· · 00.8	154°42.3	8.6'	15°53.3	12.0'	59.4'						
16	56°31.2	16°00.1	169°09.9	8.6'	16°05.3	11.9'	59.3'						
17	71°31.2	15°59.3	183°37.5	8.5'	16°17.2	11.8'	59.3'						
18	86°31.1	S15°58.6	198°05.0	8.5'	S16°28.9	11.7'	59.3'						
19	101°31.1	57.8	212°32.5	8.4'	16°40.6	11.5'	59.3'						
20	116°31.0	57.0	227°00.0	8.4'	16°52.1	11.4'	59.2'						
21	131°31.0	· · 56.3	241°27.4	8.4'	17°03.6	11.3'	59.2'						
22	146°30.9	55.5	255°54.7	8.3'	17°14.9	11.2'	59.2'						
23	161°30.9	54.8	270°22.0	8.3'	17°26.1	11.1'	59.2'						
SD = 16.2'							SD = 16.3'						
d = -0.7'													
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	176°30.8	S15°54.0	284°49.3	8.2'	S17°37.2	11.0'	59.2'						
1	191°30.8	53.2	299°16.5	8.2'	17°48.2	10.9'	59.1'						
2	206°30.7	52.5	313°43.7	8.1'	17°59.1	10.8'	59.1'						
3	221°30.7	· · 51.7	328°10.9	8.1'	18°09.8	10.6'	59.1'						
4	236°30.6	51.0	342°38.0	8.1'	18°20.5	10.5'	59.1'						
5	251°30.6	50.2	357°05.0	8.0'	18°31.0	10.4'	59.0'						
6	266°30.5	S15°49.4	11°32.0	8.0'	S18°41.4	10.3'	59.0'						
7	281°30.5	48.7	25°59.0	7.9'	18°51.7	10.2'	59.0'						
8	296°30.4	47.9	40°25.9	7.9'	19°01.9	10.1'	59.0'						
9	311°30.4	· · 47.1	54°52.8	7.8'	19°11.9	9.9'	59.0'						
10	326°30.3	46.4	69°19.6	7.8'	19°21.9	9.8'	58.9'						
11	341°30.3	45.6	83°46.4	7.7'	19°31.7	9.7'	58.9'						
12	356°30.3	S15°44.9	98°13.2	7.7'	S19°41.4	9.6'	58.9'						
13	11°30.2	44.1	112°39.9	7.7'	19°50.9	9.4'	58.9'						
14	26°30.2	43.3	127°06.5	7.6'	20°00.3	9.3'	58.8'						
15	41°30.1	· · 42.6	141°33.2	7.6'	20°09.6	9.2'	58.8'						
16	56°30.1	41.8	155°59.7	7.5'	20°18.8	9.0'	58.8'						
17	71°30.0	41.0	170°26.3	7.5'	20°27.8	8.9'	58.8'						
18	86°30.0	S15°40.2	184°52.8	7.5'	S20°36.8	8.8'	58.7'						
19	101°29.9	39.5	199°19.2	7.4'	20°45.5	8.6'	58.7'						
20	116°29.9	38.7	213°45.7	7.4'	20°54.2	8.5'	58.7'						
21	131°29.9	· · 37.9	228°12.0	7.3'	21°02.7	8.4'	58.7'						
22	146°29.8	37.2	242°38.4	7.3'	21°11.1	8.2'	58.6'						
23	161°29.8	36.4	257°04.7	7.3'	21°19.3	8.1'	58.6'						
SD = 16.2'							SD = 16.1'						
d = -0.8'													

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:57	08:24	10:02	14:27	16:06	17:32
N 70°	06:51	08:07	09:28	15:01	16:22	17:38
68°	06:46	07:54	09:03	15:26	16:35	17:44
66°	06:41	07:43	08:44	15:44	16:45	17:48
64°	06:37	07:34	08:29	15:59	16:55	17:52
62°	06:33	07:26	08:17	16:12	17:02	17:56
60°	06:29	07:20	08:06	16:23	17:09	17:59
N 58°	06:26	07:13	07:57	16:32	17:15	18:02
56°	06:23	07:08	07:48	16:40	17:21	18:05
54°	06:21	07:03	07:41	16:47	17:26	18:08
52°	06:18	06:58	07:35	16:54	17:30	18:10
50°	06:16	06:54	07:29	17:00	17:34	18:13
45°	06:10	06:45	07:16	17:12	17:43	18:18
N 40°	06:05	06:37	07:05	17:23	17:51	18:24
35°	06:00	06:30	06:56	17:32	17:58	18:29
30°	05:55	06:23	06:48	17:40	18:05	18:33
20°	05:45	06:11	06:34	17:54	18:17	18:43
N 10°	05:35	06:00	06:22	18:06	18:28	18:53
0°	05:24	05:49	06:10	18:17	18:39	19:04
S 10°	05:11	05:37	05:59	18:29	18:51	19:17
20°	04:55	05:23	05:46	18:42	19:05	19:32
30°	04:35	05:06	05:31	18:56	19:21	19:52
35°	04:23	04:55	05:23	19:04	19:32	20:05
40°	04:07	04:43	05:13	19:14	19:44	20:20
45°	03:48	04:29	05:01	19:26	19:58	20:39
S 50°	03:22	04:10	04:47	19:39	20:16	21:04
52°	03:09	04:01	04:41	19:46	20:25	21:17
54°	02:53	03:51	04:33	19:53	20:35	21:32
56°	02:35	03:39	04:25	20:01	20:47	21:50
58°	02:11	03:26	04:16	20:10	21:00	22:13
S 60°	01:37	03:10	04:05	20:21	21:15	22:45

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	· ·	01:16	■	07:37	06:56	■
N 70°	· ·	00:49	03:33	07:51	07:24	06:33
68°	· ·	00:29	02:46	08:02	07:46	07:21
66°	· ·	00:14	02:16	08:11	08:03	07:53
64°	· ·	00:01	01:53	08:18	08:17	08:16
62°	23:50	· ·	01:36	08:25	08:29	08:35
60°	23:41	· ·	01:21	08:31	08:39	08:50
N 58°	23:33	· ·	01:09	08:36	08:48	09:04
56°	23:26	· ·	00:58	08:41	08:56	09:15
54°	23:20	· ·	00:48	08:45	09:03	09:25
52°	23:15	· ·	00:40	08:48	09:09	09:34
50°	23:10	· ·	00:32	08:52	09:15	09:42
45°	22:59	· ·	00:17	08:59	09:27	09:59
N 40°	22:50	· ·	00:04	09:06	09:38	10:14
35°	22:43	23:53	· ·	09:11	09:46	10:26
30°	22:36	23:43	· ·	09:16	09:54	10:36
20°	22:25	23:27	· ·	09:24	10:08	10:54
N 10°	22:15	23:12	· ·	09:31	10:20	11:10
0°	22:06	22:59	23:54	09:38	10:31	11:25
S 10°	21:57	22:46	23:38	09:45	10:42	11:40
20°	21:47	22:32	23:20	09:53	10:54	11:56
30°	21:36	22:16	23:00	10:01	11:08	12:15
35°	21:30	22:07	22:49	10:06	11:16	12:26
40°	21:23	21:57	22:35	10:12	11:25	12:38
45°	21:15	21:45	22:20	10:18	11:36	12:53
S 50°	21:05	21:30	22:00	10:26	11:49	13:11
52°	21:01	21:23	21:51	10:30	11:55	13:19
54°	20:56	21:16	21:41	10:34	12:02	13:29
56°	20:50	21:07	21:30	10:38	12:10	13:40
58°	20:44	20:58	21:17	10:43	12:18	13:52
S 60°	20:37	20:47	21:02	10:49	12:28	14:07

February 06, 07, 08 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	136°25.1	187°36.6	S19°31.1	302°46.6	S02°27.1	290°28.9	S09°14.1	102°35.4	N11°05.7	Alpheratz	357°31.5	29°15.2
1	151°27.6	202°35.9		317°48.9	27.2	305°31.3	14.1	117°37.7	05.8	Ankaa	353°04.2	-42°09.0
2	166°30.0	217°35.1		332°51.1	27.2	320°33.8	14.2	132°40.1	05.9	Schedar	349°27.4	56°42.1
3	181°32.5	232°34.3	•• 29.0	347°53.3	•• 27.3	335°36.2	•• 14.2	147°42.4	•• 05.9	Diphda	348°44.2	-17°49.7
4	196°35.0	247°33.5		2°55.5	27.3	350°38.6	14.2	162°44.7	06.0	Achernar	335°18.1	-57°05.6
5	211°37.4	262°32.8		17°57.8	27.4	5°41.0	14.2	177°47.1	06.0	Hamal	327°47.5	23°36.1
6	226°39.9	277°32.0	S19°26.9	33°00.0	S02°27.4	20°43.5	S09°14.2	192°49.4	N11°06.1	Polaris	312°10.6	89°23.4
7	241°42.3	292°31.2		48°02.2	27.5	35°45.9	14.2	207°51.7	06.1	Acamar	315°09.3	-40°11.5
8	256°44.8	307°30.5		63°04.4	27.5	50°48.3	14.2	222°54.1	06.2	Menkar	314°02.7	4°12.2
9	271°47.3	322°29.7	•• 24.8	78°06.7	•• 27.6	65°50.8	•• 14.2	237°56.4	•• 06.3	Mirfak	308°23.3	49°58.0
10	286°49.7	337°28.9		93°08.9	27.6	80°53.2	14.2	252°58.7	06.3	Aldebaran	290°35.6	16°34.0
11	301°52.2	352°28.1		108°11.1	27.7	95°55.6	14.2	268°01.1	06.4	Rigel	281°00.5	-8°10.3
12	316°54.7	7°27.4	S19°22.6	123°13.4	S02°27.7	110°58.1	S09°14.2	283°03.4	N11°06.4	Capella	280°16.7	46°01.6
13	331°57.1	22°26.6		138°15.6	27.8	126°00.5	14.2	298°05.7	06.5	Bellatrix	278°19.1	6°22.4
14	346°59.6	37°25.8		153°17.8	27.8	141°02.9	14.2	313°08.0	06.5	Elnath	277°57.4	28°37.8
15	2°02.1	52°25.1	•• 20.5	168°20.1	•• 27.9	156°05.3	•• 14.2	328°10.4	•• 06.6	Alnilam	275°34.1	-1°11.2
16	17°04.5	67°24.3		183°22.3	27.9	171°07.8	14.2	343°12.7	06.7	Betelgeuse	270°48.3	7°24.6
17	32°07.0	82°23.5		198°24.5	28.0	186°10.2	14.2	358°15.0	06.7	Canopus	263°50.5	-52°43.0
18	47°09.4	97°22.8	S19°18.3	213°26.8	S02°28.0	201°12.6	S09°14.2	13°17.4	N11°06.8	Sirius	258°23.0	-16°45.6
19	62°11.9	112°22.0		228°29.0	28.0	216°15.1	14.2	28°19.7	06.8	Adhara	255°02.9	-29°01.0
20	77°14.4	127°21.2		243°31.3	28.1	231°17.5	14.2	43°22.0	06.9	Procyon	244°47.1	5°08.8
21	92°16.8	142°20.5	•• 16.2	258°33.5	•• 28.1	246°19.9	•• 14.2	58°24.3	•• 07.0	Pollux	243°12.9	27°57.1
22	107°19.3	157°19.7		273°35.8	28.2	261°22.4	14.2	73°26.7	07.0	Avior	234°12.7	-59°36.4
23	122°21.8	172°18.9		288°38.0	28.2	276°24.8	14.2	88°29.0	07.1	Suhail	222°43.3	-43°33.2
Mer.pass. 14:52		ν -0.8' d -0.7' m -3.90		ν 2.2' d 0.1' m -0.17		ν 2.4' d 0.0' m -2.15		ν 2.3' d 0.1' m 0.34				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	137°24.2	187°18.2	S19°14.0	303°40.2	S02°28.3	291°27.2	S09°14.2	103°31.3	N11°07.1
1	152°26.7	202°17.4	13.3	318°42.5	28.3	306°29.7	14.2	118°33.7	07.2
2	167°29.2	217°16.6	12.5	333°44.7	28.3	321°32.1	14.2	133°36.0	07.2
3	182°31.6	232°15.9	•• 11.8	348°47.0	•• 28.4	336°34.5	•• 14.2	148°38.3	•• 07.3
4	197°34.1	247°15.1	11.1	3°49.2	28.4	351°37.0	14.2	163°40.6	07.4
5	212°36.6	262°14.3	10.4	18°51.5	28.5	6°39.4	14.2	178°43.0	07.4
6	227°39.0	277°13.6	S19°09.6	33°53.7	S02°28.5	21°41.8	S09°14.2	193°45.3	N11°07.5
7	242°41.5	292°12.8	08.9	48°56.0	28.5	36°44.3	14.2	208°47.6	07.5
8	257°43.9	307°12.1	08.2	63°58.2	28.6	51°46.7	14.2	223°50.0	07.6
9	272°46.4	322°11.3	•• 07.4	79°00.5	•• 28.6	66°49.2	•• 14.2	238°52.3	•• 07.7
10	287°48.9	337°10.5	06.7	94°02.8	28.7	81°51.6	14.2	253°54.6	07.7
11	302°51.3	352°09.8	06.0	109°05.0	28.7	96°54.0	14.2	268°56.9	07.8
12	317°53.8	7°09.0	S19°05.2	124°07.3	S02°28.7	111°56.5	S09°14.2	283°59.3	N11°07.8
13	332°56.3	22°08.3	04.5	139°09.5	28.8	126°58.9	14.2	299°01.6	07.9
14	347°58.7	37°07.5	03.8	154°11.8	28.8	142°01.3	14.2	314°03.9	08.0
15	3°01.2	52°06.8	•• 03.0	169°14.1	•• 28.8	157°03.8	•• 14.2	329°06.2	•• 08.0
16	18°03.7	67°06.0	02.3	184°16.3	28.9	172°06.2	14.2	344°08.6	08.1
17	33°06.1	82°05.2	01.5	199°18.6	28.9	187°08.7	14.2	359°10.9	08.1
18	48°08.6	97°04.5	S19°00.8	214°20.9	S02°28.9	202°11.1	S09°14.2	14°13.2	N11°08.2
19	63°11.1	112°03.7	19°00.1	229°23.1	29.0	217°13.5	14.2	29°15.5	08.3
20	78°13.5	127°03.0	18°59.3	244°25.4	29.0	232°16.0	14.2	44°17.9	08.3
21	93°16.0	142°02.2	•• 58.6	259°27.7	•• 29.0	247°18.4	•• 14.2	59°20.2	•• 08.4
22	108°18.4	157°01.5	57.8	274°29.9	29.1	262°20.9	14.2	74°22.5	08.4
23	123°20.9	172°00.7	57.1	289°32.2	29.1	277°23.3	14.2	89°24.8	08.5
Mer.pass. 14:48		ν -0.8' d -0.7' m -3.90		ν 2.2' d 0.0' m -0.20		ν 2.4' d 0.0' m -2.15		ν 2.3' d 0.1' m 0.34	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	138°23.4	187°00.0	S18°56.3	304°34.5	S02°29.1	292°25.7	S09°14.2	104°27.2	N11°08.6
1	153°25.8	201°59.2	55.6	319°36.7	29.2	307°28.2	14.2	119°29.5	08.6
2	168°28.3	216°58.5	54.8	334°39.0	29.2	322°30.6	14.2	134°31.8	08.7
3	183°30.8	231°57.7	•• 54.1	349°41.3	•• 29.2	337°33.1	•• 14.2	149°34.1	•• 08.7
4	198°33.2	246°57.0	53.3	4°43.6	29.3	352°35.5	14.2	164°36.5	08.8
5	213°35.7	261°56.2	52.6	19°45.8	29.3	7°37.9	14.2	179°38.8	08.9
6	228°38.2	276°55.5	S18°51.8	34°48.1	S02°29.3	22°40.4	S09°14.2	194°41.1	N11°08.9
7	243°40.6	291°54.7	51.1	49°50.4	29.3	37°42.8	14.2	209°43.4	09.0
8	258°43.1	306°54.0	50.3	64°52.7	29.4	52°45.3	14.2	224°45.8	09.0
9	273°45.6	321°53.2	•• 49.6	79°55.0	•• 29.4	67°47.7	•• 14.2	239°48.1	•• 09.1
10	288°48.0	336°52.5	48.8	94°57.3	29.4	82°50.2	14.2	254°50.4	09.2
11	303°50.5	351°51.7	48.0	109°59.5	29.5	97°52.6	14.1	269°52.7	09.2
12	318°52.9	6°51.0	S18°47.3	125°01.8	S02°29.5	112°55.1	S09°14.1	284°55.1	N11°09.3
13	333°55.4	21°50.2	46.5	140°04.1	29.5	127°57.5	14.1	299°57.4	09.3
14	348°57.9	36°49.5	45.8	155°06.4	29.5	142°59.9	14.1	314°59.7	09.4
15	4°00.3	51°48.7	•• 45.0	170°08.7	•• 29.6	158°02.4	•• 14.1	330°02.0	•• 09.5
16	19°02.8	66°48.0	44.2	185°11.0	29.6	173°04.8	14.1	345°04.3	09.5
17	34°05.3	81°47.2	43.5	200°13.3	29.6	188°07.3	14.1	0°06.7	09.6
18	49°07.7	96°46.5	S18°42.7	215°15.6	S02°29.6	203°09.7	S09°14.1	15°09.0	N11°09.6
19	64°10.2	111°45.8	41.9	230°17.8	29.6	218°12.2	14.1	30°11.3	09.7
20	79°12.7	126°45.0	41.2	245°20.1	29.7	233°14.6	14.1	45°13.6	09.8
21	94°15.1	141°44.3	•• 40.4	260°22.4	•• 29.7	248°17.1	•• 14.1	60°15.9	•• 09.8
22	109°17.6	156°43.5	39.6	275°24.7	29.7	263°19.5	14.1	75°18.3	09.9
23	124°20.1	171°42.8	38.9	290°27.0	29.7	278°22.0	14.1	90°20.6	09.9
Mer.pass. 14:44		ν -0.8' d -0.7' m -3.90		ν 2.3' d 0.0' m -0.22		ν 2.4' d -0.0' m -2.16		ν 2.3' d 0.1' m 0.35	

Feb 06 Tue	SHA	Mer.pass
Venus	51°11.5	11:30
Mars	166°21.5	03:48
Jupiter	154°03.8	04:37
Saturn	326°10.3	17:07

Feb 07 Wed	SHA	Mer.pass
Venus	49°53.9	11:31
Mars	166°16.0	03:45
Jupiter	154°03.0	04:33
Saturn	326°07.1	17:03

Feb 08 Thu	SHA	Mer.pass
Venus	48°36.6	11:33
Mars	166°11.1	03:41
Jupiter	154°02.4	04:30
Saturn	326°03.8	17:00

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.2065 sec ΔT = TT-UT1 = +68.9775 sec

2029 February 06 to Feb. 08 UT

h		Sun			Moon		
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°29.7	S15°35.6	271°30.9	7.2'	S21°27.4	8.0'	58.6'
1	191°29.7	34.9	285°57.2	7.2'	21°35.4	7.8'	58.6'
2	206°29.6	34.1	300°23.4	7.2'	21°43.2	7.7'	58.5'
3	221°29.6	· · 33.3	314°49.5	7.1'	21°50.9	7.6'	58.5'
4	236°29.6	32.5	329°15.6	7.1'	21°58.5	7.4'	58.5'
5	251°29.5	31.8	343°41.7	7.1'	22°05.9	7.3'	58.5'
6	266°29.5	S15°31.0	358°07.8	7.0'	S22°13.2	7.1'	58.4'
7	281°29.4	30.2	12°33.8	7.0'	22°20.3	7.0'	58.4'
8	296°29.4	29.4	26°59.8	7.0'	22°27.3	6.9'	58.4'
9	311°29.4	· · 28.7	41°25.8	6.9'	22°34.2	6.7'	58.4'
10	326°29.3	27.9	55°51.7	6.9'	22°40.9	6.6'	58.4'
11	341°29.3	27.1	70°17.6	6.9'	22°47.5	6.4'	58.3'
12	356°29.3	S15°26.3	84°43.5	6.8'	S22°53.9	6.3'	58.3'
13	11°29.2	25.6	99°09.3	6.8'	23°00.2	6.1'	58.3'
14	26°29.2	24.8	113°35.1	6.8'	23°06.3	6.0'	58.3'
15	41°29.1	· · 24.0	128°00.9	6.8'	23°12.3	5.8'	58.2'
16	56°29.1	23.2	142°26.7	6.7'	23°18.1	5.7'	58.2'
17	71°29.1	22.5	156°52.5	6.7'	23°23.8	5.5'	58.2'
18	86°29.0	S15°21.7	171°18.2	6.7'	S23°29.4	5.4'	58.2'
19	101°29.0	20.9	185°43.9	6.7'	23°34.8	5.2'	58.1'
20	116°29.0	20.1	200°09.6	6.7'	23°40.0	5.1'	58.1'
21	131°28.9	· · 19.3	214°35.2	6.6'	23°45.1	5.0'	58.1'
22	146°28.9	18.6	229°00.9	6.6'	23°50.1	4.8'	58.1'
23	161°28.9	17.8	243°26.5	6.6'	23°54.9	4.7'	58.0'
SD = 16.2'		$d = -0.8'$	SD = 16.0'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	176°28.8	S15°17.0	257°52.1	6.6'	S23°59.5	4.5'	58.0'
1	191°28.8	16.2	272°17.7	6.6'	24°04.0	4.3'	58.0'
2	206°28.8	15.4	286°43.3	6.6'	24°08.4	4.2'	58.0'
3	221°28.7	· · 14.6	301°08.9	6.6'	24°12.6	4.0'	57.9'
4	236°28.7	13.9	315°34.5	6.6'	24°16.6	3.9'	57.9'
5	251°28.7	13.1	330°00.0	6.6'	24°20.5	3.7'	57.9'
6	266°28.6	S15°12.3	344°25.6	6.5'	S24°24.2	3.6'	57.9'
7	281°28.6	11.5	358°51.1	6.5'	24°27.8	3.4'	57.8'
8	296°28.6	10.7	13°16.7	6.5'	24°31.3	3.3'	57.8'
9	311°28.5	· · 09.9	27°42.2	6.5'	24°34.6	3.1'	57.8'
10	326°28.5	09.1	42°07.7	6.5'	24°37.7	3.0'	57.8'
11	341°28.5	08.4	56°33.3	6.5'	24°40.7	2.8'	57.7'
12	356°28.5	S15°07.6	70°58.8	6.5'	S24°43.5	2.7'	57.7'
13	11°28.4	06.8	85°24.3	6.5'	24°46.2	2.5'	57.7'
14	26°28.4	06.0	99°49.9	6.5'	24°48.7	2.4'	57.7'
15	41°28.4	· · 05.2	114°15.4	6.5'	24°51.1	2.2'	57.6'
16	56°28.3	04.4	128°40.9	6.6'	24°53.3	2.1'	57.6'
17	71°28.3	03.6	143°06.5	6.6'	24°55.4	1.9'	57.6'
18	86°28.3	S15°02.8	157°32.1	6.6'	S24°57.3	1.8'	57.6'
19	101°28.3	02.0	171°57.6	6.6'	24°59.0	1.6'	57.6'
20	116°28.2	01.3	186°23.2	6.6'	25°00.6	1.5'	57.5'
21	131°28.2	15°00.5	200°48.8	6.6'	25°02.1	1.3'	57.5'
22	146°28.2	14°59.7	215°14.4	6.6'	25°03.4	1.2'	57.5'
23	161°28.2	58.9	229°40.1	6.6'	25°04.6	1.0'	57.5'
SD = 16.2'		$d = -0.8'$	SD = 15.8'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°28.1	S14°58.1	244°05.7	6.7'	S25°05.6	0.8'	57.4'
1	191°28.1	57.3	258°31.4	6.7'	25°06.4	0.7'	57.4'
2	206°28.1	56.5	272°57.0	6.7'	25°07.1	0.5'	57.4'
3	221°28.1	· · 55.7	287°22.8	6.7'	25°07.6	0.4'	57.4'
4	236°28.0	54.9	301°48.5	6.8'	25°08.0	0.2'	57.3'
5	251°28.0	54.1	316°14.2	6.8'	25°08.3	0.1'	57.3'
6	266°28.0	S14°53.3	330°40.0	6.8'	S25°08.4	-0.1'	57.3'
7	281°28.0	52.5	345°05.8	6.8'	25°08.3	-0.2'	57.3'
8	296°27.9	51.7	359°31.6	6.9'	25°08.1	-0.4'	57.2'
9	311°27.9	· · 50.9	13°57.5	6.9'	25°07.7	-0.5'	57.2'
10	326°27.9	50.1	28°23.4	6.9'	25°07.2	-0.7'	57.2'
11	341°27.9	49.3	42°49.3	7.0'	25°06.6	-0.8'	57.2'
12	356°27.8	S14°48.5	57°15.3	7.0'	S25°05.8	-1.0'	57.2'
13	11°27.8	47.7	71°41.3	7.0'	25°04.8	-1.1'	57.1'
14	26°27.8	46.9	86°07.3	7.1'	25°03.7	-1.2'	57.1'
15	41°27.8	· · 46.1	100°33.4	7.1'	25°02.4	-1.4'	57.1'
16	56°27.8	45.3	114°59.5	7.2'	25°01.1	-1.5'	57.1'
17	71°27.7	44.5	129°25.7	7.2'	24°59.5	-1.7'	57.0'
18	86°27.7	S14°43.7	143°51.9	7.2'	S24°57.8	-1.8'	57.0'
19	101°27.7	42.9	158°18.1	7.3'	24°56.0	-2.0'	57.0'
20	116°27.7	42.1	172°44.4	7.3'	24°54.0	-2.1'	57.0'
21	131°27.7	· · 41.3	187°10.7	7.4'	24°51.9	-2.3'	56.9'
22	146°27.6	40.5	201°37.1	7.4'	24°49.6	-2.4'	56.9'
23	161°27.6	39.7	216°03.5	7.5'	24°47.2	-2.5'	56.9'
SD = 16.2'		$d = -0.8'$	SD = 15.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:46	08:10	09:41	14:49	16:20	17:44
N 70°	06:41	07:56	09:12	15:18	16:34	17:49
68°	06:36	07:44	08:51	15:39	16:45	17:53
66°	06:33	07:34	08:34	15:56	16:55	17:57
64°	06:29	07:26	08:20	16:09	17:03	18:00
62°	06:26	07:19	08:08	16:21	17:10	18:03
60°	06:23	07:13	07:58	16:31	17:16	18:06
N 58°	06:21	07:07	07:50	16:39	17:22	18:09
56°	06:18	07:02	07:42	16:47	17:27	18:11
54°	06:16	06:58	07:36	16:53	17:31	18:13
52°	06:14	06:54	07:29	16:59	17:35	18:15
50°	06:12	06:50	07:24	17:05	17:39	18:17
45°	06:07	06:41	07:12	17:17	17:47	18:22
N 40°	06:02	06:34	07:02	17:27	17:55	18:27
35°	05:57	06:27	06:54	17:35	18:01	18:31
30°	05:53	06:21	06:46	17:42	18:07	18:36
20°	05:44	06:10	06:33	17:55	18:18	18:44
N 10°	05:35	06:00	06:22	18:07	18:28	18:54
0°	05:24	05:49	06:11	18:18	18:39	19:04
S 10°	05:12	05:38	06:00	18:28	18:50	19:16
20°	04:57	05:25	05:48	18:40	19:03	19:30
30°	04:38	05:09	05:34	18:54	19:19	19:49
35°	04:26	04:59	05:26	19:02	19:29	20:01
40°	04:12	04:47	05:17	19:11	19:40	20:16
45°	03:53	04:33	05:06	19:21	19:54	20:34
S 50°	03:29	04:16	04:53	19:34	20:11	20:58
52°	03:17	04:08	04:47	19:40	20:19	21:09
54°	03:03	03:58	04:40	19:47	20:28	21:23
56°	02:46	03:47	04:32	19:55	20:39	21:40
58°	02:24	03:35	04:24	20:03	20:51	22:00
S 60°	01:56	03:20	04:14	20:13	21:06	22:28

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	04:29	■	■	07:36	■	■
64°	03:48	05:38	07:01	08:18	08:26	09:03
62°	03:20	04:55	06:08	08:46	09:10	09:56
60°	02:58	04:26	05:36	09:08	09:39	10:28
N 58°	02:41	04:04	05:12	09:26	10:01	10:52
56°	02:26	03:46	04:52	09:42	10:19	11:12
54°	02:13	03:31	04:36	09:55	10:35	11:28
52°	02:02	03:18	04:22	10:06	10:48	11:42
50°	01:53	03:06	04:10	10:16	11:00	11:54
45°	01:32	02:42	03:45	10:38	11:24	12:19
N 40°	01:15	02:23	03:25	10:55	11:44	12:39
35°	01:01	02:07	03:08	11:10	12:00	12:56
30°	00:49	01:54	02:54	11:23	12:14	13:10
20°	00:29	01:30	02:29	11:45	12:38	13:35
N 10°	00:11	01:10	02:08	12:04	12:59	13:56
0°	· ·	00:51	01:48	12:21	13:18	14:16
S 10°	· ·	00:32	01:29	12:39	13:38	14:35
20°	· ·	00:12	01:08	12:58	13:59	14:56
30°	23:49	· ·	00:43	13:20	14:23	15:20
35°	23:36	· ·	00:29	13:33	14:37	15:35
40°	23:20	· ·	00:12	13:48	14:53	15:51
45°	23:02	23:53	· ·	14:06	15:13	16:11
S 50°	22:39	23:28	· ·	14:28	15:38	16:35
52°	22:28	23:16	· ·	14:39	15:49	16:47
54°	22:16	23:02	· ·	14:51	16:03	17:01
56°	22:02	22:47	23:46	15:05	16:19	17:16
58°	21:45	22:28	23:28	15:21	16:37	17:35
S 60°	21:25	22:05	23:05	15:40	17:00	17:58

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer. Pass. hh:mm	Lower Mer. Pass. hh:mm	
06	14:01	14:03	12:14	06:08	18:36	
07	14:05	14:06	12:14	07:05	19:33	
08	14:07	14:09	12:14	08:02	20:30	

Aries Venus M

<https://pypi.org/project/sfalmanac/>

DUT1 = UT1-UTC = +0.2066 sec $\Delta T = TT-UT1 = +68.9774$ sec

2029 February 09 to Feb. 11 UT

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.6	S14°38.9	230°30.0	7.5'	S24°44.7	-2.7'	56.9'
1	191°27.6	38.1	244°56.5	7.6'	24°42.0	-2.8'	56.9'
2	206°27.6	37.3	259°23.1	7.6'	24°39.2	-3.0'	56.8'
3	221°27.6	· · 36.5	273°49.7	7.7'	24°36.2	-3.1'	56.8'
4	236°27.5	35.7	288°16.4	7.7'	24°33.1	-3.2'	56.8'
5	251°27.5	34.9	302°43.1	7.8'	24°29.9	-3.4'	56.8'
6	266°27.5	S14°34.1	317°09.9	7.9'	S24°26.5	-3.5'	56.7'
7	281°27.5	33.3	331°36.8	7.9'	24°23.0	-3.7'	56.7'
8	296°27.5	32.5	346°03.7	8.0'	24°19.3	-3.8'	56.7'
9	311°27.5	· · 31.7	0°30.6	8.0'	24°15.5	-3.9'	56.7'
10	326°27.5	30.9	14°57.7	8.1'	24°11.6	-4.1'	56.7'
11	341°27.4	30.1	29°24.8	8.2'	24°07.5	-4.2'	56.6'
12	356°27.4	S14°29.3	43°51.9	8.2'	S24°03.3	-4.3'	56.6'
13	11°27.4	28.5	58°19.1	8.3'	23°59.0	-4.5'	56.6'
14	26°27.4	27.6	72°46.4	8.3'	23°54.6	-4.6'	56.6'
15	41°27.4	· · 26.8	87°13.7	8.4'	23°50.0	-4.7'	56.5'
16	56°27.4	26.0	101°41.2	8.5'	23°45.3	-4.8'	56.5'
17	71°27.4	25.2	116°08.6	8.5'	23°40.4	-5.0'	56.5'
18	86°27.4	S14°24.4	130°36.2	8.6'	S23°35.5	-5.1'	56.5'
19	101°27.3	23.6	145°03.8	8.7'	23°30.4	-5.2'	56.5'
20	116°27.3	22.8	159°31.5	8.7'	23°25.2	-5.3'	56.4'
21	131°27.3	· · 22.0	173°59.2	8.8'	23°19.8	-5.5'	56.4'
22	146°27.3	21.2	188°27.0	8.9'	23°14.4	-5.6'	56.4'
23	161°27.3	20.3	202°54.9	9.0'	23°08.8	-5.7'	56.4'
SD = 16.2'			d = -0.8'				
			SD = 15.5'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.3	S14°19.5	217°22.9	9.0'	S23°03.1	-5.8'	56.4'
1	191°27.3	18.7	231°50.9	9.1'	22°57.2	-5.9'	56.3'
2	206°27.3	17.9	246°19.0	9.2'	22°51.3	-6.1'	56.3'
3	221°27.3	· · 17.1	260°47.2	9.3'	22°45.2	-6.2'	56.3'
4	236°27.3	16.3	275°15.5	9.3'	22°39.0	-6.3'	56.3'
5	251°27.3	15.5	289°43.8	9.4'	22°32.7	-6.4'	56.2'
6	266°27.2	S14°14.6	304°12.2	9.5'	S22°26.3	-6.5'	56.2'
7	281°27.2	13.8	318°40.7	9.6'	22°19.8	-6.6'	56.2'
8	296°27.2	13.0	333°09.2	9.6'	22°13.2	-6.8'	56.2'
9	311°27.2	· · 12.2	347°37.9	9.7'	22°06.4	-6.9'	56.2'
10	326°27.2	11.4	2°06.6	9.8'	21°59.5	-7.0'	56.1'
11	341°27.2	10.6	16°35.3	9.9'	21°52.6	-7.1'	56.1'
12	356°27.2	S14°09.7	31°04.2	9.9'	S21°45.5	-7.2'	56.1'
13	11°27.2	08.9	45°33.1	10.0'	21°38.3	-7.3'	56.1'
14	26°27.2	08.1	60°02.2	10.1'	21°31.0	-7.4'	56.1'
15	41°27.2	· · 07.3	74°31.2	10.2'	21°23.6	-7.5'	56.0'
16	56°27.2	06.5	89°00.4	10.2'	21°16.1	-7.6'	56.0'
17	71°27.2	05.6	103°29.7	10.3'	21°08.4	-7.7'	56.0'
18	86°27.2	S14°04.8	117°59.0	10.4'	S21°00.7	-7.8'	56.0'
19	101°27.2	04.0	132°28.4	10.5'	20°52.9	-7.9'	56.0'
20	116°27.2	03.2	146°57.9	10.6'	20°45.0	-8.0'	55.9'
21	131°27.2	· · 02.4	161°27.4	10.6'	20°37.0	-8.1'	55.9'
22	146°27.2	01.5	175°57.1	10.7'	20°28.8	-8.2'	55.9'
23	161°27.2	00.7	190°26.8	10.8'	20°20.6	-8.3'	55.9'
SD = 16.2'			d = -0.8'				
			SD = 15.4'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.2	S13°59.9	204°56.6	10.9'	S20°12.3	-8.4'	55.9'
1	191°27.2	59.1	219°26.5	11.0'	20°03.9	-8.5'	55.8'
2	206°27.2	58.2	233°56.4	11.0'	19°55.4	-8.6'	55.8'
3	221°27.2	· · 57.4	248°26.5	11.1'	19°46.8	-8.7'	55.8'
4	236°27.2	56.6	262°56.6	11.2'	19°38.2	-8.8'	55.8'
5	251°27.2	55.8	277°26.8	11.3'	19°29.4	-8.9'	55.8'
6	266°27.2	S13°54.9	291°57.1	11.4'	S19°20.5	-8.9'	55.7'
7	281°27.2	54.1	306°27.4	11.4'	19°11.6	-9.0'	55.7'
8	296°27.2	53.3	320°57.8	11.5'	19°02.5	-9.1'	55.7'
9	311°27.2	· · 52.5	335°28.3	11.6'	18°53.4	-9.2'	55.7'
10	326°27.2	51.6	349°58.9	11.7'	18°44.2	-9.3'	55.7'
11	341°27.2	50.8	4°29.6	11.7'	18°34.9	-9.4'	55.7'
12	356°27.2	S13°50.0	19°00.3	11.8'	S18°25.5	-9.5'	55.6'
13	11°27.2	49.2	33°31.2	11.9'	18°16.1	-9.5'	55.6'
14	26°27.2	48.3	48°02.1	12.0'	18°06.5	-9.6'	55.6'
15	41°27.2	· · 47.5	62°33.0	12.0'	17°56.9	-9.7'	55.6'
16	56°27.2	46.7	77°04.1	12.1'	17°47.2	-9.8'	55.6'
17	71°27.2	45.8	91°35.2	12.2'	17°37.5	-9.8'	55.5'
18	86°27.2	S13°45.0	106°06.4	12.3'	S17°27.6	-9.9'	55.5'
19	101°27.2	44.2	120°37.7	12.4'	17°17.7	-10.0'	55.5'
20	116°27.2	43.3	135°09.0	12.4'	17°07.7	-10.1'	55.5'
21	131°27.2	· · 42.5	149°40.5	12.5'	16°57.6	-10.1'	55.5'
22	146°27.2	41.7	164°12.0	12.6'	16°47.5	-10.2'	55.4'
23	161°27.2	40.8	178°43.5	12.6'	16°37.3	-10.3'	55.4'
SD = 16.2'			d = -0.8'				
			SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:34	07:56	09:22	15:08	16:34	17:56
N 70°	06:30	07:44	08:57	15:33	16:46	18:00
68°	06:27	07:34	08:38	15:52	16:56	18:03
66°	06:24	07:25	08:23	16:07	17:05	18:06
64°	06:21	07:18	08:10	16:19	17:12	18:08
62°	06:19	07:11	08:00	16:30	17:18	18:11
60°	06:17	07:06	07:51	16:39	17:24	18:13
N 58°	06:15	07:01	07:43	16:46	17:28	18:15
56°	06:13	06:56	07:36	16:53	17:33	18:17
54°	06:11	06:52	07:30	16:59	17:37	18:19
52°	06:09	06:49	07:24	17:05	17:41	18:20
50°	06:07	06:45	07:19	17:10	17:44	18:22
45°	06:03	06:37	07:08	17:21	17:52	18:26
N 40°	05:59	06:31	06:59	17:30	17:58	18:30
35°	05:55	06:25	06:51	17:38	18:04	18:34
30°	05:51	06:19	06:44	17:45	18:10	18:38
20°	05:43	06:09	06:32	17:57	18:20	18:46
N 10°	05:34	05:59	06:21	18:08	18:29	18:54
0°	05:25	05:49	06:11	18:18	18:39	19:04
S 10°	05:13	05:39	06:00	18:28	18:50	19:15
20°	04:59	05:26	05:49	18:39	19:02	19:29
30°	04:41	05:11	05:36	18:51	19:17	19:46
35°	04:30	05:02	05:29	18:59	19:26	19:58
40°	04:16	04:51	05:20	19:07	19:36	20:11
45°	03:59	04:38	05:10	19:17	19:49	20:29
S 50°	03:36	04:22	04:58	19:29	20:05	20:51
52°	03:25	04:14	04:52	19:35	20:13	21:02
54°	03:12	04:05	04:46	19:41	20:22	21:15
56°	02:56	03:55	04:39	19:48	20:31	21:30
58°	02:37	03:44	04:31	19:56	20:43	21:48
S 60°	02:12	03:30	04:22	20:05	20:56	22:12

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	■	■	■	■
N 70°	■	■	09:21	■	■	12:18
68°	■	■	08:33	■	■	13:05
66°	■	08:16	08:01	■	11:36	13:35
64°	07:30	07:37	07:38	10:31	12:15	13:58
62°	06:49	07:09	07:19	11:11	12:42	14:16
60°	06:21	06:48	07:04	11:39	13:03	14:31
N 58°	05:59	06:30	06:51	12:01	13:20	14:43
56°	05:42	06:16	06:39	12:18	13:34	14:54
54°	05:26	06:03	06:29	12:33	13:47	15:03
52°	05:13	05:52	06:20	12:46	13:58	15:12
50°	05:02	05:42	06:12	12:58	14:07	15:19
45°	04:38	05:21	05:55	13:21	14:27	15:35
N 40°	04:18	05:03	05:41	13:40	14:44	15:48
35°	04:02	04:49	05:29	13:56	14:58	15:59
30°	03:48	04:36	05:18	14:09	15:09	16:09
20°	03:25	04:15	05:00	14:32	15:30	16:25
N 10°	03:04	03:56	04:45	14:52	15:47	16:40
0°	02:45	03:39	04:30	15:11	16:04	16:53
S 10°	02:25	03:21	04:15	15:29	16:20	17:06
20°	02:05	03:02	03:59	15:49	16:37	17:20
30°	01:41	02:41	03:40	16:12	16:57	17:36
35°	01:27	02:28	03:30	16:25	17:08	17:46
40°	01:11	02:13	03:17	16:40	17:22	17:56
45°	00:51	01:56	03:03	16:59	17:37	18:08
S 50°	00:27	01:34	02:45	17:21	17:56	18:23
52°	00:15	01:23	02:36	17:32	18:05	18:30
54°	00:02	01:12	02:27	17:44	18:15	18:38
56°	· ·	00:58	02:16	17:58	18:26	18:46
58°	· ·	00:42	02:04	18:14	18:39	18:56
S 60°	· ·	00:23	01:49	18:33	18:54	19:07

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
09	14:10	14:10	12:14	08:58	21:25	
10	14:11	14:11	12:14	09:51	22:17	
11	14:11	14:11	12:14	10:41	23:05	

February 12, 13, 14 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	142°19.9	185°50.1	S17°40.3	308°17.8	S02°30.1	296°21.6	S09°13.4	108°09.6	N11°14.5	Alpheratz	357°31.5	29°15.2	
1	157°22.4	200°49.4	39.4	323°20.2	30.1	311°24.0	13.4	123°11.9	14.6	Ankaa	353°04.2	-42°09.0	
2	172°24.9	215°48.7	38.6	338°22.6	30.0	326°26.5	13.4	138°14.2	14.7	Schedar	349°27.4	56°42.1	
3	187°27.3	230°48.0	· · 37.8	353°25.0	· · 30.0	341°29.0	· · 13.4	153°16.5	· · 14.7	Diphda	348°44.2	-17°49.7	
4	202°29.8	245°47.3	36.9	8°27.3	30.0	356°31.4	13.4	168°18.8	14.8	Achernar	335°18.1	-57°05.6	
5	217°32.3	260°46.5	36.1	23°29.7	30.0	11°33.9	13.4	183°21.1	14.9	Hamal	327°47.5	23°36.1	
6	232°34.7	275°45.8	S17°35.2	38°32.1	S02°30.0	26°36.4	S09°13.4	198°23.4	N11°14.9	Polaris	312°13.5	89°23.4	
7	247°37.2	290°45.1	34.4	53°34.5	30.0	41°38.9	13.3	213°25.7	15.0	Acamar	315°09.3	-40°11.5	
8	262°39.6	305°44.4	33.6	68°36.9	30.0	56°41.3	13.3	228°28.1	15.1	Menkar	314°02.7	4°12.2	
9	277°42.1	320°43.7	· · 32.7	83°39.3	· · 29.9	71°43.8	· · 13.3	243°30.4	· · 15.1	Mirfak	308°23.4	49°58.0	
10	292°44.6	335°43.1	31.9	98°41.7	29.9	86°46.3	13.3	258°32.7	15.2	Aldebaran	290°35.7	16°34.0	
11	307°47.0	350°42.4	31.0	113°44.1	29.9	101°48.8	13.3	273°35.0	15.2	Rigel	281°00.5	-8°10.3	
12	322°49.5	5°41.7	S17°30.2	128°46.5	S02°29.9	116°51.2	S09°13.3	288°37.3	N11°15.3	Capella	280°16.7	46°01.6	
13	337°52.0	20°41.0	29.3	143°48.9	29.9	131°53.7	13.3	303°39.6	15.4	Bellatrix	278°19.1	6°22.4	
14	352°54.4	35°40.3	28.5	158°51.3	29.9	146°56.2	13.2	318°41.9	15.4	Elnath	277°57.4	28°37.8	
15	7°56.9	50°39.6	· · 27.6	173°53.7	· · 29.8	161°58.7	· · 13.2	333°44.2	· · 15.5	Alnilam	275°34.2	-1°11.2	
16	22°59.4	65°38.9	26.8	188°56.1	29.8	177°01.1	13.2	348°46.5	15.6	Betelgeuse	270°48.3	7°24.6	
17	38°01.8	80°38.2	25.9	203°58.5	29.8	192°03.6	13.2	3°48.8	15.6	Canopus	263°50.5	-52°43.0	
18	53°04.3	95°37.5	S17°25.1	219°00.9	S02°29.8	207°06.1	S09°13.2	18°51.1	N11°15.7	Sirius	258°23.0	-16°45.6	
19	68°06.8	110°36.8	24.2	234°03.3	29.8	222°08.6	13.2	33°53.4	15.8	Adhara	255°02.9	-29°01.0	
20	83°09.2	125°36.1	23.4	249°05.7	29.7	237°11.1	13.1	48°55.7	15.8	Procyon	244°47.0	5°08.8	
21	98°11.7	140°35.4	· · 22.5	264°08.1	· · 29.7	252°13.5	· · 13.1	63°58.0	· · 15.9	Pollux	243°12.9	27°57.1	
22	113°14.1	155°34.7	21.7	279°10.5	29.7	267°16.0	13.1	79°00.3	16.0	Avior	234°12.7	-59°36.4	
23	128°16.6	170°34.0	20.8	294°12.9	29.7	282°18.5	13.1	94°02.7	16.0	Suhail	222°43.3	-43°33.2	
Mer.pass. 14:28		ν -0.7' d -0.8' m -3.89		ν 2.4' d -0.0' m -0.33		ν 2.5' d -0.0' m -2.19		ν 2.3' d 0.1' m 0.36		Miaplacidus	221°36.5	-69°50.4	
										Alphard	217°44.1	-8°47.4	
										Regulus	207°30.6	11°49.2	
										Dubhe	193°36.5	61°35.3	
										Denebola	182°21.3	14°24.3	
										Gienah	175°39.9	-17°42.4	
										Acrux	172°55.6	-63°15.6	
										Gacrux	171°47.3	-57°16.5	
										Alioth	166°09.8	55°47.7	
										Spica	158°18.6	-11°18.9	
										Alkaid	152°49.3	49°09.7	
										Hadar	148°30.8	-60°30.6	
										Menkent	147°53.5	-36°30.7	
										Arcturus	145°44.8	19°01.6	
										Rigil Kent.	139°35.4	-60°57.1	
										Kochab	137°20.0	74°01.7	
										Zuben'ubi	136°52.2	-16°09.8	
										Alphecca	126°00.9	26°36.8	
										Antares	112°11.8	-26°29.7	
										Atria	107°03.2	-69°04.4	
										Sabik	101°59.1	-15°45.6	
										Shaula	96°06.0	-37°07.3	
										Rasalhague	95°55.6	12°32.2	
										Eltanin	90°40.9	51°28.9	
										Kaus Aust.	83°28.4	-34°22.1	
										Vega	80°31.3	38°48.5	
										Nunki	75°43.9	-26°15.5	
										Altair	61°57.0	8°56.7	
										Peacock	53°01.1	-56°38.3	
										Deneb	49°23.9	45°23.1	
										Enif	33°35.8	10°00.6	
										Al Na'ir	27°29.3	-46°49.2	
										Fomalhaut	15°11.2	-29°28.1	
										Scheat	13°42.2	28°14.6	
										Markab	13°26.8	15°21.8	
										Feb 12 Mon		SHA	Mer.pass
										Venus	43°30.1	11:37	
										Mars	165°57.9	03:26	
										Jupiter	154°01.6	04:14	
										Saturn	325°49.6	16:45	
										Feb 13 Tue		SHA	Mer.pass
										Venus	42°14.3	11:38	
										Mars	165°56.2	03:22	
										Jupiter	154°01.9	04:10	
										Saturn	325°45.9	16:41	
										Feb 14 Wed		SHA	Mer.pass
										Venus	40°58.7	11:39	
										Mars	165°55.3	03:19	
										Jupiter	154°02.3	04:06	
										Saturn	325°42.0	16:37	
										Horizontal parallax			
										Venus:	0.1		
										Mars:	0.2		
Mer.pass. 14:20		ν -0.7' d -0.9' m -3.89		ν 2.4' d -0.0' m -0.38		ν 2.5' d -0.0' m -2.21		ν 2.3' d 0.1' m 0.36					

DUT1 = UT1-UTC = +0.2068 sec $\Delta T = TT-UT1 = +68.9772$ sec

2029 February 12 to Feb. 14 UT

h	Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.2	S13°40.0	193°15.2	12.7'	S16°27.0	-10.3'	55.4'
1	191°27.2	39.2	207°46.9	12.8'	16°16.7	-10.4'	55.4'
2	206°27.2	38.3	222°18.7	12.9'	16°06.2	-10.5'	55.4'
3	221°27.2	· · 37.5	236°50.6	12.9'	15°55.8	-10.5'	55.4'
4	236°27.2	36.7	251°22.5	13.0'	15°45.2	-10.6'	55.3'
5	251°27.3	35.8	265°54.5	13.1'	15°34.6	-10.7'	55.3'
6	266°27.3	S13°35.0	280°26.6	13.1'	S15°23.9	-10.7'	55.3'
7	281°27.3	34.2	294°58.7	13.2'	15°13.2	-10.8'	55.3'
8	296°27.3	33.3	309°30.9	13.3'	15°02.4	-10.9'	55.3'
9	311°27.3	· · 32.5	324°03.2	13.4'	14°51.6	-10.9'	55.2'
10	326°27.3	31.7	338°35.6	13.4'	14°40.6	-11.0'	55.2'
11	341°27.3	30.8	353°08.0	13.5'	14°29.7	-11.0'	55.2'
12	356°27.3	S13°30.0	7°40.5	13.6'	S14°18.6	-11.1'	55.2'
13	11°27.3	29.2	22°13.1	13.6'	14°07.6	-11.1'	55.2'
14	26°27.3	28.3	36°45.7	13.7'	13°56.4	-11.2'	55.2'
15	41°27.4	· · 27.5	51°18.4	13.8'	13°45.2	-11.2'	55.1'
16	56°27.4	26.6	65°51.1	13.8'	13°34.0	-11.3'	55.1'
17	71°27.4	25.8	80°24.0	13.9'	13°22.7	-11.3'	55.1'
18	86°27.4	S13°25.0	94°56.8	13.9'	S13°11.3	-11.4'	55.1'
19	101°27.4	24.1	109°29.8	14.0'	13°00.0	-11.4'	55.1'
20	116°27.4	23.3	124°02.8	14.1'	12°48.5	-11.5'	55.1'
21	131°27.4	· · 22.4	138°35.9	14.1'	12°37.0	-11.5'	55.0'
22	146°27.4	21.6	153°09.0	14.2'	12°25.5	-11.6'	55.0'
23	161°27.5	20.8	167°42.2	14.3'	12°13.9	-11.6'	55.0'
SD = 16.2' $d = -0.8'$			SD = 15.1'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.5	S13°19.9	182°15.5	14.3'	S12°02.3	-11.7'	55.0'
1	191°27.5	19.1	196°48.8	14.4'	11°50.6	-11.7'	55.0'
2	206°27.5	18.2	211°22.1	14.4'	11°38.9	-11.8'	55.0'
3	221°27.5	· · 17.4	225°55.6	14.5'	11°27.1	-11.8'	54.9'
4	236°27.5	16.5	240°29.0	14.5'	11°15.3	-11.8'	54.9'
5	251°27.5	15.7	255°02.6	14.6'	11°03.5	-11.9'	54.9'
6	266°27.6	S13°14.9	269°36.2	14.6'	S10°51.6	-11.9'	54.9'
7	281°27.6	14.0	284°09.8	14.7'	10°39.7	-11.9'	54.9'
8	296°27.6	13.2	298°43.5	14.8'	10°27.8	-12.0'	54.9'
9	311°27.6	· · 12.3	313°17.3	14.8'	10°15.8	-12.0'	54.9'
10	326°27.6	11.5	327°51.1	14.9'	10°03.8	-12.1'	54.8'
11	341°27.6	10.6	342°25.0	14.9'	09°51.7	-12.1'	54.8'
12	356°27.7	S13°09.8	356°58.9	15.0'	S09°39.6	-12.1'	54.8'
13	11°27.7	08.9	11°32.8	15.0'	09°27.5	-12.1'	54.8'
14	26°27.7	08.1	26°06.8	15.1'	09°15.4	-12.2'	54.8'
15	41°27.7	· · 07.2	40°40.9	15.1'	09°03.2	-12.2'	54.8'
16	56°27.7	06.4	55°15.0	15.2'	08°51.0	-12.2'	54.7'
17	71°27.8	05.5	69°49.2	15.2'	08°38.7	-12.3'	54.7'
18	86°27.8	S13°04.7	84°23.4	15.2'	S08°26.5	-12.3'	54.7'
19	101°27.8	03.9	98°57.6	15.3'	08°14.2	-12.3'	54.7'
20	116°27.8	03.0	113°31.9	15.3'	08°01.9	-12.3'	54.7'
21	131°27.8	· · 02.2	128°06.2	15.4'	07°49.5	-12.4'	54.7'
22	146°27.9	01.3	142°40.6	15.4'	07°37.2	-12.4'	54.7'
23	161°27.9	00.5	157°15.0	15.5'	07°24.8	-12.4'	54.6'
SD = 16.2' $d = -0.8'$			SD = 15.0'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.9	S12°59.6	171°49.5	15.5'	S07°12.3	-12.4'	54.6'
1	191°27.9	58.7	186°24.0	15.5'	06°59.9	-12.5'	54.6'
2	206°27.9	57.9	200°58.5	15.6'	06°47.5	-12.5'	54.6'
3	221°28.0	· · 57.0	215°33.1	15.6'	06°35.0	-12.5'	54.6'
4	236°28.0	56.2	230°07.7	15.7'	06°22.5	-12.5'	54.6'
5	251°28.0	55.3	244°42.4	15.7'	06°10.0	-12.5'	54.6'
6	266°28.0	S12°54.5	259°17.0	15.7'	S05°57.5	-12.5'	54.5'
7	281°28.1	53.6	273°51.8	15.8'	05°44.9	-12.6'	54.5'
8	296°28.1	52.8	288°26.5	15.8'	05°32.4	-12.6'	54.5'
9	311°28.1	· · 51.9	303°01.3	15.8'	05°19.8	-12.6'	54.5'
10	326°28.1	51.1	317°36.1	15.9'	05°07.2	-12.6'	54.5'
11	341°28.2	50.2	332°11.0	15.9'	04°54.6	-12.6'	54.5'
12	356°28.2	S12°49.4	346°45.9	15.9'	S04°42.0	-12.6'	54.5'
13	11°28.2	48.5	1°20.8	15.9'	04°29.4	-12.6'	54.5'
14	26°28.2	47.7	15°55.8	16.0'	04°16.7	-12.6'	54.4'
15	41°28.3	· · 46.8	30°30.7	16.0'	04°04.1	-12.7'	54.4'
16	56°28.3	45.9	45°05.7	16.0'	03°51.4	-12.7'	54.4'
17	71°28.3	45.1	59°40.8	16.1'	03°38.8	-12.7'	54.4'
18	86°28.3	S12°44.2	74°15.8	16.1'	S03°26.1	-12.7'	54.4'
19	101°28.4	43.4	88°50.9	16.1'	03°13.4	-12.7'	54.4'
20	116°28.4	42.5	103°26.0	16.1'	03°00.7	-12.7'	54.4'
21	131°28.4	· · 41.7	118°01.1	16.2'	02°48.1	-12.7'	54.4'
22	146°28.5	40.8	132°36.3	16.2'	02°35.4	-12.7'	54.3'
23	161°28.5	39.9	147°11.5	16.2'	02°22.7	-12.7'	54.3'
SD = 16.2' $d = -0.9'$			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:22	07:42	09:04	15:26	16:48	18:09
N 70°	06:19	07:32	08:42	15:48	16:58	18:11
68°	06:17	07:23	08:25	16:04	17:07	18:13
66°	06:15	07:15	08:12	16:18	17:14	18:15
64°	06:13	07:09	08:01	16:29	17:20	18:17
62°	06:11	07:03	07:51	16:38	17:26	18:18
60°	06:10	06:59	07:43	16:47	17:31	18:20
N 58°	06:08	06:54	07:36	16:54	17:35	18:21
56°	06:07	06:50	07:29	17:00	17:39	18:23
54°	06:05	06:47	07:24	17:06	17:43	18:24
52°	06:04	06:43	07:18	17:11	17:46	18:25
50°	06:02	06:40	07:14	17:15	17:49	18:27
45°	05:59	06:33	07:04	17:25	17:56	18:30
N 40°	05:56	06:27	06:55	17:34	18:02	18:33
35°	05:52	06:22	06:48	17:41	18:07	18:37
30°	05:49	06:17	06:41	17:47	18:12	18:40
20°	05:42	06:08	06:30	17:58	18:21	18:47
N 10°	05:34	05:59	06:20	18:08	18:30	18:55
0°	05:25	05:50	06:11	18:18	18:39	19:03
S 10°	05:14	05:40	06:01	18:27	18:49	19:14
20°	05:01	05:28	05:51	18:37	19:00	19:27
30°	04:44	05:14	05:39	18:49	19:14	19:44
35°	04:33	05:05	05:32	18:56	19:22	19:54
40°	04:20	04:55	05:24	19:04	19:32	20:07
45°	04:04	04:43	05:15	19:13	19:44	20:23
S 50°	03:43	04:28	05:03	19:24	19:59	20:44
52°	03:32	04:20	04:58	19:29	20:07	20:54
54°	03:20	04:12	04:52	19:35	20:15	21:06
56°	03:06	04:03	04:46	19:41	20:24	21:20
58°	02:49	03:52	04:38	19:48	20:34	21:37
S 60°	02:27	03:40	04:30	19:56	20:46	21:57

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	09:05	08:25	07:57	14:13	16:25	18:21
N 70°	08:33	08:08	07:49	14:43	16:39	18:26
68°	08:10	07:55	07:43	15:05	16:51	18:30
66°	07:52	07:44	07:38	15:22	17:00	18:33
64°	07:37	07:35	07:33	15:35	17:08	18:36
62°	07:24	07:27	07:29	15:47	17:14	18:38
60°	07:14	07:20	07:26	15:57	17:20	18:40
N 58°	07:04	07:14	07:22	16:05	17:25	18:42
56°	06:56	07:09	07:20	16:13	17:29	18:44
54°	06:49	07:04	07:17	16:19	17:33	18:46
52°	06:42	07:00	07:15	16:25	17:37	18:47
50°	06:36	06:56	07:13	16:30	17:40	18:48
45°	06:23	06:47	07:08	16:42	17:47	18:51
N 40°	06:13	06:40	07:04	16:51	17:53	18:53
35°	06:03	06:34	07:01	17:00	17:58	18:55
30°	05:55	06:28	06:58	17:07	18:02	18:57
20°	05:41	06:18	06:53	17:19	18:10	18:59
N 10°	05:29	06:10	06:49	17:29	18:17	19:02
0°	05:17	06:02	06:44	17:39	18:23	19:04
S 10°	05:06	05:54	06:40	17:49	18:29	19:07
20°	04:53	05:46	06:36	17:59	18:35	19:09
30°	04:39	05:36	06:31	18:11	18:43	19:12
35°	04:31	05:30	06:28	18:18	18:47	19:13
40°	04:21	05:24	06:24	18:25	18:51	19:15
45°	04:10	05:16	06:20	18:34	18:57	19:17
S 50°	03:56	05:07	06:16	18:45	19:03	19:20
52°	03:50	05:03	06:13	18:50	19:06	19:21
54°	03:43	04:58	06:11	18:55	19:10	19:22
56°	03:35	04:53	06:08	19:01	19:13	19:23
58°	03:26	04:47	06:05	19:08	19:17	19:25
S 60°	03:16	04:40	06:02	19:15	19:22	19:26

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Upper hh:mm	Lower hh:mm	29-1 2-0%
12	14:11	14:11	12:14	11:28	23:51	
13	14:10	14:09	12:14	12:12	—:—	
14	14:08	14:07	12:14	12:54	00:34	

February 15, 16, 17 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	145°17.3	185°00.7	S16°37.9	311°12.3	S02°28.0	299°20.3	S09°12.2	110°55.4	N11°19.3	Alpheratz	357°31.5	29°15.2
1	160°19.8	200°00.1	37.0	326°14.8	28.0	314°22.8	12.1	125°57.7	19.4	Ankaa	353°04.3	-42°09.0
2	175°22.3	214°59.4	36.1	341°17.3	27.9	329°25.3	12.1	141°00.0	19.4	Schedar	349°27.5	56°42.1
3	190°24.7	229°58.7	· · 35.2	356°19.7	· · 27.9	344°27.8	· · 12.1	156°02.3	· · 19.5	Diphda	348°44.2	-17°49.7
4	205°27.2	244°58.1	34.3	11°22.2	27.8	359°30.3	12.1	171°04.6	19.6	Achernar	335°18.1	-57°05.6
5	220°29.7	259°57.4	33.4	26°24.7	27.8	14°32.8	12.1	186°06.9	19.6	Hamal	327°47.5	23°36.1
6	235°32.1	274°56.8	S16°32.5	41°27.2	S02°27.7	29°35.3	S09°12.0	201°09.2	N11°19.7	Polaris	312°15.2	89°23.4
7	250°34.6	289°56.1	31.6	56°29.6	27.7	44°37.8	12.0	216°11.5	19.8	Acamar	315°09.4	-40°11.5
8	265°37.1	304°55.4	30.7	71°32.1	27.6	59°40.3	12.0	231°13.8	19.9	Menkar	314°02.7	4°12.2
9	280°39.5	319°54.8	· · 29.8	86°34.6	· · 27.6	74°42.8	· · 12.0	246°16.1	· · 19.9	Mirfak	308°23.4	49°58.0
10	295°42.0	334°54.1	28.9	101°37.1	27.5	89°45.3	11.9	261°18.4	20.0	Aldebaran	290°35.7	16°34.0
11	310°44.5	349°53.4	28.0	116°39.6	27.5	104°47.8	11.9	276°20.7	20.1	Rigel	281°00.5	-8°10.3
12	325°46.9	4°52.8	S16°27.1	131°42.0	S02°27.4	119°50.3	S09°11.9	291°23.0	N11°20.1	Capella	280°16.7	46°01.6
13	340°49.4	19°52.1	26.2	146°44.5	27.4	134°52.8	11.9	306°25.3	20.2	Bellatrix	278°19.1	6°22.4
14	355°51.8	34°51.5	25.3	161°47.0	27.3	149°55.3	11.9	321°27.6	20.3	Elnath	277°57.5	28°37.8
15	10°54.3	49°50.8	· · 24.4	176°49.5	· · 27.3	164°57.8	· · 11.8	336°29.9	· · 20.3	Alnilam	275°34.2	-1°11.2
16	25°56.8	64°50.1	23.5	191°52.0	27.2	180°00.3	11.8	351°32.2	20.4	Betelgeuse	270°48.3	7°24.6
17	40°59.2	79°49.5	22.6	206°54.5	27.2	195°02.8	11.8	6°34.5	20.5	Canopus	263°50.5	-52°43.0
18	56°01.7	94°48.8	S16°21.6	221°56.9	S02°27.1	210°05.3	S09°11.8	21°36.8	N11°20.5	Sirius	258°23.0	-16°45.7
19	71°04.2	109°48.2	20.7	236°59.4	27.1	225°07.8	11.7	36°39.1	20.6	Adhara	255°02.9	-29°01.0
20	86°06.6	124°47.5	19.8	252°01.9	27.0	240°10.3	11.7	51°41.4	20.7	Procyon	244°47.1	5°08.8
21	101°09.1	139°46.9	· · 18.9	267°04.4	· · 27.0	255°12.8	· · 11.7	66°43.7	· · 20.7	Pollux	243°12.9	27°57.2
22	116°11.6	154°46.2	18.0	282°06.9	26.9	270°15.3	11.7	81°46.0	20.8	Avior	234°12.7	-59°36.4
23	131°14.0	169°45.6	17.1	297°09.4	26.8	285°17.8	11.6	96°48.3	20.9	Suhail	222°43.3	-43°33.3
Mer.pass. 14:16		ν -0.7' d -0.9' m -3.89		ν 2.5' d -0.0' m -0.41		ν 2.5' d -0.0' m -2.21		ν 2.3' d 0.1' m 0.36				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	146°16.5	184°44.9	S16°16.2	312°11.9	S02°26.8	300°20.3	S09°11.6	111°50.5	N11°21.0	Denebola	182°21.3	14°24.3
1	161°18.9	199°44.2	15.2	327°14.4	26.7	315°22.8	11.6	126°52.8	21.0	Gienah	175°39.8	-17°42.4
2	176°21.4	214°43.6	14.3	342°16.9	26.7	330°25.3	11.6	141°55.1	21.1	Acrux	172°55.6	-63°15.6
3	191°23.9	229°42.9	· · 13.4	357°19.4	· · 26.6	345°27.8	· · 11.5	156°57.4	· · 21.2	Gacrux	171°47.3	-57°16.6
4	206°26.3	244°42.3	12.5	12°21.9	26.6	0°30.3	11.5	171°59.7	21.2	Alioth	166°09.8	55°47.7
5	221°28.8	259°41.6	11.6	27°24.4	26.5	15°32.8	11.5	187°02.0	21.3	Spica	158°18.6	-11°18.9
6	236°31.3	274°41.0	S16°10.7	42°26.9	S02°26.4	30°35.3	S09°11.5	202°04.3	N11°21.4	Alkaid	152°49.2	49°09.7
7	251°33.7	289°40.3	09.7	57°29.4	26.4	45°37.8	11.4	217°06.6	21.4	Hadar	148°30.8	-60°30.6
8	266°36.2	304°39.7	08.8	72°31.9	26.3	60°40.3	11.4	232°08.9	21.5	Menkent	147°53.4	-36°30.7
9	281°38.7	319°39.0	· · 07.9	87°34.4	· · 26.3	75°42.8	· · 11.4	247°11.2	· · 21.6	Arcturus	145°44.8	19°01.6
10	296°41.1	334°38.4	07.0	102°36.9	26.2	90°45.3	11.4	262°13.5	21.6	Rigel Kent.	139°35.3	-60°57.1
11	311°43.6	349°37.7	06.0	117°39.4	26.1	105°47.8	11.3	277°15.8	21.7	Kochab	137°19.9	74°01.7
12	326°46.1	4°37.1	S16°05.1	132°41.9	S02°26.1	120°50.3	S09°11.3	292°18.1	N11°21.8	Zuben'ubi	136°52.2	-16°09.8
13	341°48.5	19°36.4	04.2	147°44.5	26.0	135°52.8	11.3	307°20.4	21.9	Alphecca	126°00.9	26°36.7
14	356°51.0	34°35.8	03.3	162°47.0	25.9	150°55.3	11.3	322°22.7	21.9	Antares	112°11.8	-26°29.7
15	11°53.4	49°35.1	· · 02.3	177°49.5	· · 25.9	165°57.8	· · 11.2	337°24.9	· · 22.0	Atria	107°03.1	-69°04.4
16	26°55.9	64°34.5	01.4	192°52.0	25.8	181°00.3	11.2	352°27.2	22.1	Sabik	101°59.1	-15°45.6
17	41°58.4	79°33.9	16°00.5	207°54.5	25.8	196°02.8	11.2	7°29.5	22.1	Shaula	96°06.0	-37°07.3
18	57°00.8	94°33.2	S15°59.6	222°57.0	S02°25.7	211°05.4	S09°11.1	22°31.8	N11°22.2	Rasalhague	95°55.6	12°32.2
19	72°03.3	109°32.6	58.6	237°59.5	25.6	226°07.9	11.1	37°34.1	22.3	Eltanin	90°40.9	51°28.9
20	87°05.8	124°31.9	57.7	253°02.1	25.6	241°10.4	11.1	52°36.4	22.3	Kaus Aust.	83°28.3	-34°22.1
21	102°08.2	139°31.3	· · 56.8	268°04.6	· · 25.5	256°12.9	· · 11.1	67°38.7	· · 22.4	Vega	80°31.2	38°48.5
22	117°10.7	154°30.6	55.8	283°07.1	25.4	271°15.4	11.0	82°41.0	22.5	Nunki	75°43.9	-26°15.5
23	132°13.2	169°30.0	54.9	298°09.6	25.4	286°17.9	11.0	97°43.3	22.5	Altair	61°57.0	8°56.7
Mer.pass. 14:13		ν -0.7' d -0.9' m -3.89		ν 2.5' d -0.1' m -0.44		ν 2.5' d -0.0' m -2.22		ν 2.3' d 0.1' m 0.36		Peacock	53°01.1	-56°38.3

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	147°15.6	184°29.4	S15°54.0	313°12.2	S02°25.3	301°20.4	S09°11.0	112°45.6	N11°22.6	Deneb	49°23.9	45°23.0
1	162°18.1	199°28.7	53.0	328°14.7	25.2	316°22.9	11.0	127°47.9	22.7	Enif	33°35.8	10°00.6
2	177°20.6	214°28.1	52.1	343°17.2	25.2	331°25.4	10.9	142°50.2	22.8	Al Na'ir	27°29.3	-46°49.1
3	192°23.0	229°27.4	· · 51.2	358°19.7	· · 25.1	346°27.9	· · 10.9	157°52.4	· · 22.8	Fomalhaut	15°11.2	-29°28.1
4	207°25.5	244°26.8	50.2	13°22.3	25.0	1°30.4	10.9	172°54.7	22.9	Scheat	13°42.2	28°14.5
5	222°27.9	259°26.2	49.3	28°24.8	25.0	16°32.9	10.8	187°57.0	23.0	Markab	13°26.8	15°21.8
6	237°30.4	274°25.5	S15°48.4	43°27.3	S02°24.9	31°35.5	S09°10.8	202°59.3	N11°23.0	Feb 15 Thu SHA Mer.pass		
7	252°32.9	289°24.9	47.4	58°29.9	24.8	46°38.0	10.8	218°01.6	23.1	Venus	39°43.4	11:40
8	267°35.3	304°24.2	46.5	73°32.4	24.7	61°40.5	10.8	233°03.9	23.2	Mars	165°55.0	03:15
9	282°37.8	319°23.6	· · 45.5	88°34.9	· · 24.7	76°43.0	· · 10.7	248°06.2	· · 23.3	Jupiter	154°03.0	04:02
10	297°40.3	334°23.0	44.6	103°37.5	24.6	91°45.5	10.7	263°08.5	23.3	Saturn	325°38.1	16:34
11	312°42.7	349°22.3	43.7	118°40.0	24.5	106°48.0	10.7	278°10.8	23.4	Feb 16 Fri SHA Mer.pass		
12	327°45.2	4°21.7	S15°42.7	133°42.6	S02°24.4	121°50.5	S09°10.7	293°13.1	N11°23.5	Venus	38°28.4	11:42
13	342°47.7	19°21.1	41.8	148°45.1	24.4	136°53.0	10.6	308°15.3	23.5	Mars	165°55.4	03:11
14	357°50.1	34°20.4	40.8	163°47.6	24.3	151°55.6	10.6	323°17.6	23.6	Jupiter	154°03.8	03:58
15	12°52.6	49°19.8	· · 39.9	178°50.2	· · 24.2	166°58.1	· · 10.6	338°19.9	· · 23.7	Saturn	325°34.1	16:30
16	27°55.0	64°19.2	38.9	193°52.7	24.2	182°00.6	10.5	353°22.2	23.7	Feb 17 Sat SHA Mer.pass		
17	42°57.5	79°18.5	38.0	208°55.3	24.1	197°03.1	10.5	8°24.5	23.8	Venus	37°13.7	11:43
18	58°00.0	94°17.9	S15°37.0	223°57.8	S02°24.0	212°05.6	S09°10.5	23°26.8	N11°23.9	Mars	165°56.5	03:07
19	73°02.4	109°17.3	36.1	239°00.4	23.9	227°08.1	10.4	38°29.1	24.0	Jupiter	154°04.8	03:54
20	88°04.9	124°16.6	35.1	254°02.9	233							

DUT1 = UT1-UTC = +0.2069 sec $\Delta T = TT-UT1 = +68.9771$ sec

2029 February 15 to Feb. 17 UT

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°28.5	S12°39.1	161°46.7	16.2'	S02°10.0	-12.7'	54.3'
1	191°28.5	38.2	176°21.9	16.2'	01°57.3	-12.7'	54.3'
2	206°28.6	37.4	190°57.1	16.3'	01°44.6	-12.7'	54.3'
3	221°28.6	· · 36.5	205°32.4	16.3'	01°31.9	-12.7'	54.3'
4	236°28.6	35.6	220°07.7	16.3'	01°19.2	-12.7'	54.3'
5	251°28.7	34.8	234°42.9	16.3'	01°06.5	-12.7'	54.3'
6	266°28.7	S12°33.9	249°18.2	16.3'	S00°53.8	-12.7'	54.3'
7	281°28.7	33.0	263°53.6	16.3'	00°41.1	-12.7'	54.2'
8	296°28.8	32.2	278°28.9	16.3'	00°28.4	-12.7'	54.2'
9	311°28.8	· · 31.3	293°04.2	16.4'	00°15.7	-12.7'	54.2'
10	326°28.8	30.5	307°39.6	16.4'	S00°03.0	-12.7'	54.2'
11	341°28.9	29.6	322°15.0	16.4'	N00°09.7	12.7'	54.2'
12	356°28.9	S12°28.7	336°50.3	16.4'	N00°22.3	12.7'	54.2'
13	11°28.9	27.9	351°25.7	16.4'	00°35.0	12.7'	54.2'
14	26°29.0	27.0	6°01.1	16.4'	00°47.7	12.6'	54.2'
15	41°29.0	· · 26.1	20°36.5	16.4'	01°00.3	12.6'	54.2'
16	56°29.0	25.3	35°11.9	16.4'	01°13.0	12.6'	54.2'
17	71°29.1	24.4	49°47.4	16.4'	01°25.6	12.6'	54.2'
18	86°29.1	S12°23.5	64°22.8	16.4'	N01°38.2	12.6'	54.1'
19	101°29.1	22.7	78°58.2	16.4'	01°50.8	12.6'	54.1'
20	116°29.2	21.8	93°33.7	16.4'	02°03.4	12.6'	54.1'
21	131°29.2	· · 20.9	108°09.1	16.4'	02°16.0	12.6'	54.1'
22	146°29.2	20.1	122°44.5	16.4'	02°28.6	12.6'	54.1'
23	161°29.3	19.2	137°20.0	16.4'	02°41.1	12.5'	54.1'
SD = 16.2' d = -0.9' SD = 14.8'							

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°29.3	S12°18.3	151°55.4	16.4'	N02°53.7	12.5'	54.1'
1	191°29.4	17.5	166°30.8	16.4'	03°06.2	12.5'	54.1'
2	206°29.4	16.6	181°06.3	16.4'	03°18.7	12.5'	54.1'
3	221°29.4	· · 15.7	195°41.7	16.4'	03°31.2	12.5'	54.1'
4	236°29.5	14.9	210°17.1	16.4'	03°43.7	12.5'	54.1'
5	251°29.5	14.0	224°52.6	16.4'	03°56.1	12.4'	54.1'
6	266°29.5	S12°13.1	239°28.0	16.4'	N04°08.6	12.4'	54.1'
7	281°29.6	12.3	254°03.4	16.4'	04°21.0	12.4'	54.1'
8	296°29.6	11.4	268°38.8	16.4'	04°33.4	12.4'	54.0'
9	311°29.7	· · 10.5	283°14.2	16.4'	04°45.8	12.4'	54.0'
10	326°29.7	09.6	297°49.6	16.4'	04°58.1	12.3'	54.0'
11	341°29.7	08.8	312°25.0	16.4'	05°10.5	12.3'	54.0'
12	356°29.8	S12°07.9	327°00.4	16.4'	N05°22.8	12.3'	54.0'
13	11°29.8	07.0	341°35.7	16.4'	05°35.1	12.3'	54.0'
14	26°29.9	06.2	356°11.1	16.3'	05°47.4	12.2'	54.0'
15	41°29.9	· · 05.3	10°46.4	16.3'	05°59.6	12.2'	54.0'
16	56°29.9	04.4	25°21.7	16.3'	06°11.8	12.2'	54.0'
17	71°30.0	03.5	39°57.1	16.3'	06°24.0	12.2'	54.0'
18	86°30.0	S12°02.7	54°32.3	16.3'	N06°36.2	12.1'	54.0'
19	101°30.1	01.8	69°07.6	16.3'	06°48.3	12.1'	54.0'
20	116°30.1	00.9	83°42.9	16.2'	07°00.4	12.1'	54.0'
21	131°30.2	12°00.0	98°18.1	16.2'	07°12.5	12.0'	54.0'
22	146°30.2	11°59.2	112°53.4	16.2'	07°24.5	12.0'	54.0'
23	161°30.2	58.3	127°28.6	16.2'	07°36.5	12.0'	54.0'
SD = 16.2' d = -0.9' SD = 14.8'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°30.3	S11°57.4	142°03.8	16.2'	N07°48.5	12.0'	54.0'
1	191°30.3	56.5	156°38.9	16.1'	08°00.5	11.9'	54.0'
2	206°30.4	55.7	171°14.1	16.1'	08°12.4	11.9'	54.0'
3	221°30.4	· · 54.8	185°49.2	16.1'	08°24.3	11.9'	54.0'
4	236°30.5	53.9	200°24.3	16.1'	08°36.1	11.8'	54.0'
5	251°30.5	53.0	214°59.4	16.0'	08°48.0	11.8'	54.0'
6	266°30.6	S11°52.2	229°34.4	16.0'	N08°59.8	11.8'	54.0'
7	281°30.6	51.3	244°09.4	16.0'	09°11.5	11.7'	54.0'
8	296°30.7	50.4	258°44.4	16.0'	09°23.2	11.7'	54.0'
9	311°30.7	· · 49.5	273°19.4	15.9'	09°34.9	11.6'	54.0'
10	326°30.8	48.6	287°54.3	15.9'	09°46.5	11.6'	54.0'
11	341°30.8	47.8	302°29.2	15.9'	09°58.1	11.6'	54.0'
12	356°30.8	S11°46.9	317°04.1	15.8'	N10°09.7	11.5'	54.0'
13	11°30.9	46.0	331°39.0	15.8'	10°21.2	11.5'	54.0'
14	26°30.9	45.1	346°13.8	15.8'	10°32.7	11.4'	54.0'
15	41°31.0	· · 44.2	0°48.6	15.8'	10°44.1	11.4'	54.0'
16	56°31.0	43.4	15°23.3	15.7'	10°55.5	11.4'	54.0'
17	71°31.1	42.5	29°58.0	15.7'	11°06.9	11.3'	54.0'
18	86°31.1	S11°41.6	44°32.7	15.6'	N11°18.2	11.3'	54.0'
19	101°31.2	40.7	59°07.4	15.6'	11°29.5	11.2'	54.0'
20	116°31.2	39.8	73°42.0	15.6'	11°40.7	11.2'	54.0'
21	131°31.3	· · 39.0	88°16.6	15.5'	11°51.9	11.1'	54.0'
22	146°31.3	38.1	102°51.1	15.5'	12°03.0	11.1'	54.0'
23	161°31.4	37.2	117°25.6	15.5'	12°14.1	11.0'	54.0'
SD = 16.2' d = -0.9' SD = 14.7'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:09	07:29	08:46	15:43	17:01	18:21
N 70°	06:08	07:19	08:27	16:02	17:10	18:22
68°	06:07	07:12	08:13	16:17	17:18	18:23
66°	06:06	07:06	08:01	16:29	17:24	18:24
64°	06:05	07:00	07:51	16:39	17:29	18:25
62°	06:04	06:55	07:42	16:47	17:34	18:26
60°	06:02	06:51	07:35	16:54	17:38	18:27
N 58°	06:01	06:47	07:28	17:01	17:42	18:28
56°	06:00	06:44	07:22	17:07	17:45	18:29
54°	05:59	06:41	07:17	17:12	17:48	18:30
52°	05:58	06:38	07:13	17:16	17:51	18:31
50°	05:57	06:35	07:08	17:20	17:54	18:32
45°	05:55	06:29	06:59	17:30	18:00	18:34
N 40°	05:52	06:24	06:51	17:37	18:05	18:37
35°	05:49	06:19	06:45	17:44	18:10	18:39
30°	05:46	06:14	06:39	17:50	18:14	18:42
20°	05:40	06:06	06:28	18:00	18:22	18:48
N 10°	05:33	05:58	06:19	18:09	18:30	18:55
0°	05:25	05:49	06:11	18:17	18:39	19:03
S 10°	05:15	05:40	06:02	18:26	18:48	19:13
20°	05:03	05:30	05:52	18:35	18:58	19:25
30°	04:47	05:16	05:41	18:46	19:11	19:40
35°	04:37	05:08	05:35	18:53	19:19	19:50
40°	04:25	04:59	05:28	19:00	19:28	20:03
45°	04:09	04:48	05:19	19:08	19:39	20:18
S 50°	03:49	04:33	05:09	19:18	19:53	20:37
52°	03:40	04:27	05:04	19:23	20:00	20:46
54°	03:29	04:19	04:58	19:28	20:07	20:57
56°	03:16	04:11	04:53	19:34	20:16	21:10
58°	03:00	04:01	04:46	19:41	20:25	21:25
S 60°	02:41	03:49	04:38	19:48	20:36	21:44

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	07:33	07:11	06:45	20:11	22:02	· · ·
N 70°	07:33	07:16	06:59	20:08	21:50	23:38
68°	07:32	07:21	07:10	20:06	21:41	23:20
66°	07:31	07:25	07:19	20:04	21:34	23:06
64°	07:31	07:29	07:27	20:02	21:28	22:55
62°	07:30	07:32	07:33	20:01	21:22	22:45
60°	07:30	07:34	07:39	19:59	21:18	22:37
N 58°	07:30	07:37	07:44	19:58	21:13	22:29
56°	07:29	07:39	07:49	19:57	21:10	22:23
54°	07:29	07:41	07:53	19:56	21:07	22:17
52°	07:29	07:42	07:57	19:56	21:04	22:12
50°	07:29	07:44	08:00	19:55	21:01	22:07
45°	07:28	07:47	08:08	19:53	20:55	21:57
N 40°	07:28	07:50	08:14	19:52	20:50	21:49
35°	07:27	07:53	08:19	19:51	20:46	21:42
30°	07:27	07:55	08:24	19:50	20:43	21:36
20°	07:26	07:59	08:32	19:48	20:36	21:25
N 10°	07:26	08:03	08:40	19:46	20:31	21:15
0°	07:25	08:06	08:47	19:45	20:25	21:07
S 10°	07:25	08:09	08:54	19:44	20:20	20:58
20°	07:25	08:13	09:01	19:42	20:15	20:49
30°	07:24	08:17	09:10	19:40	20:09	20:38
35°	07:24	08:19	09:15	19:39	20:05	20:32
40°	07:24	08:22	09:21	19:38	20:01	20:25
45°	07:23	08:25	09:27	19:37	19:56	20:17
S 50°	07:23	08:29	09:35	19:35	19:51	20:08
52°	07:23	08:31	09:39	19:34	19:48	20:03
54°	07:22	08:33	09:43	19:34	19:45	19:58
56°	07:22	08:35	09:48	19:33	19:42	19:53
58°	07:22	08:37	09:53	19:32	19:39	19:47
S 60°	07:22	08:40	09:58	19:31	19:35	19:40

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss	hh:mm	Upper hh:mm	Lower hh:mm	2-4 2-11%
15	14:06	14:04	12:14	13:35	01:15	
16	14:03	14:01	12:14	14:16	01:55	
17	13:59	13:57	12:14	14:57	02:36	

February 18, 19, 20 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	148°14.8	184°14.1	S15°31.3	314°13.1	S02°23.5	302°20.7	S09°10.3	113°40.5	N11°24.3	Alpheratz	357°31.5	29°15.2
1	163°17.2	199°13.5	30.4	329°15.7	23.5	317°23.2	10.3	128°42.8	24.4	Ankaa	353°04.3	-42°09.0
2	178°19.7	214°12.8	29.4	344°18.2	23.4	332°25.7	10.2	143°45.1	24.5	Schedar	349°27.5	56°42.0
3	193°22.2	229°12.2	•• 28.5	359°20.8	•• 23.3	347°28.3	•• 10.2	158°47.4	•• 24.5	Diphda	348°44.2	-17°49.7
4	208°24.6	244°11.6	27.5	14°23.4	23.2	2°30.8	10.2	173°49.7	24.6	Achernar	335°18.2	-57°05.5
5	223°27.1	259°11.0	26.6	29°25.9	23.1	17°33.3	10.1	188°51.9	24.7	Hamal	327°47.5	23°36.1
6	238°29.5	274°10.3	S15°25.6	44°28.5	S02°23.1	32°35.8	S09°10.1	203°54.2	N11°24.7	Polaris	312°16.7	89°23.4
7	253°32.0	289°09.7	24.7	59°31.0	23.0	47°38.3	10.1	218°56.5	24.8	Acamar	315°09.4	-40°11.5
8	268°34.5	304°09.1	23.7	74°33.6	22.9	62°40.9	10.1	233°58.8	24.9	Menkar	314°02.7	4°12.1
9	283°36.9	319°08.5	•• 22.7	89°36.2	•• 22.8	77°43.4	•• 10.0	249°01.1	•• 25.0	Mirfak	308°23.4	49°58.0
10	298°39.4	334°07.8	21.8	104°38.7	22.7	92°45.9	10.0	264°03.4	25.0	Aldebaran	290°35.7	16°34.0
11	313°41.9	349°07.2	20.8	119°41.3	22.6	107°48.4	10.0	279°05.7	25.1	Rigel	281°00.5	-8°10.3
12	328°44.3	4°06.6	S15°19.9	134°43.9	S02°22.6	122°50.9	S09°09.9	294°07.9	N11°25.2	Capella	280°16.7	46°01.6
13	343°46.8	19°06.0	18.9	149°46.5	22.5	137°53.5	09.9	309°10.2	25.2	Bellatrix	278°19.2	6°22.4
14	358°49.3	34°05.3	17.9	164°49.0	22.4	152°56.0	09.9	324°12.5	25.3	Elnath	277°57.5	28°37.8
15	13°51.7	49°04.7	•• 17.0	179°51.6	•• 22.3	167°58.5	•• 09.8	339°14.8	•• 25.4	Alnilam	275°34.2	-1°11.2
16	28°54.2	64°04.1	16.0	194°54.2	22.2	183°01.0	09.8	354°17.1	25.5	Betelgeuse	270°48.3	7°24.6
17	43°56.6	79°03.5	15.0	209°56.7	22.1	198°03.5	09.8	9°19.4	25.5	Canopus	263°50.6	-52°43.0
18	58°59.1	94°02.9	S15°14.1	224°59.3	S02°22.0	213°06.1	S09°09.7	24°21.7	N11°25.6	Sirius	258°23.1	-16°45.7
19	74°01.6	109°02.2	13.1	240°01.9	21.9	228°08.6	09.7	39°23.9	25.7	Adhara	255°03.0	-29°01.0
20	89°04.0	124°01.6	12.1	255°04.5	21.9	243°11.1	09.7	54°26.2	25.7	Procyon	244°47.1	5°08.8
21	104°06.5	139°01.0	•• 11.2	270°07.1	•• 21.8	258°13.6	•• 09.6	69°28.5	•• 25.8	Pollux	243°12.9	27°57.2
22	119°09.0	154°00.4	10.2	285°09.6	21.7	273°16.1	09.6	84°30.8	25.9	Avior	234°12.7	-59°36.4
23	134°11.4	168°59.8	09.2	300°12.2	21.6	288°18.7	09.6	99°33.1	26.0	Suhail	222°43.3	-43°33.3
Mer.pass. 14:05		ν -0.6' d -1.0' m -3.89		ν 2.6' d -0.1' m -0.49		ν 2.5' d -0.0' m -2.23		ν 2.3' d 0.1' m 0.37		Miaplacidus	221°36.5	-69°50.4
										Alphard	217°44.1	-8°47.4
										Regulus	207°30.6	11°49.2
										Dubhe	193°36.5	61°35.3
										Denebola	182°21.3	14°24.3
										Gienah	175°39.8	-17°42.4
										Acrux	172°55.6	-63°15.6
										Gacrux	171°47.3	-57°16.6
										Alioth	166°09.8	55°47.7
										Spica	158°18.6	-11°18.9
										Alkaid	152°49.2	49°09.7
										Hadar	148°30.8	-60°30.6
										Menkent	147°53.4	-36°30.8
										Arcturus	145°44.7	19°01.6
										Rigil Kent.	139°35.3	-60°57.2
										Kochab	137°19.8	74°01.7
										Zuben'ubi	136°52.2	-16°09.8
										Alphecca	126°00.9	26°36.7
										Antares	112°11.8	-26°29.7
										Atria	107°03.1	-69°04.4
										Sabik	101°59.0	-15°45.6
										Shaula	96°06.0	-37°07.3
										Rasalhague	95°55.6	12°32.2
										Eltanin	90°40.9	51°28.9
										Kaus Aust.	83°28.3	-34°22.1
										Vega	80°31.2	38°48.5
										Nunki	75°43.9	-26°15.5
										Altair	61°57.0	8°56.7
										Peacock	53°01.1	-56°38.3
										Deneb	49°23.9	45°23.0
										Enif	33°35.8	10°00.5
										Al Na'ir	27°29.3	-46°49.1
										Fomalhaut	15°11.2	-29°28.1
										Scheat	13°42.2	28°14.5
										Markab	13°26.8	15°21.8
										Feb 18 Sun SHA Mer.pass		
										Venus	35°59.3	11:44
										Mars	165°58.4	03:03
										Jupiter	154°05.9	03:50
										Saturn	325°25.8	16:23
										Feb 19 Mon SHA Mer.pass		
										Venus	34°45.2	11:45
										Mars	166°00.9	02:58
										Jupiter	154°07.3	03:46
										Saturn	325°21.5	16:19
										Feb 20 Tue SHA Mer.pass		
										Venus	33°31.4	11:46
										Mars	166°04.2	02:54
										Jupiter	154°08.8	03:42
										Saturn	325°17.1	16:16
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.2	
Mer.pass. 13:57		ν -0.6' d -1.0' m -3.89		ν 2.6' d -0.1' m -0.55		ν 2.5' d -0.0' m -2.25		ν 2.3' d 0.1' m 0.37				

DUT1 = UT1-UTC = +0.2070 sec ΔT = TT-UT1 = +68.9770 sec

2029 February 18 to Feb. 20 UT

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
0	176°31.4	S11°36.3	132°00.1	15.4'	N12°25.1	11.0'	54.0'						
1	191°31.5	35.4	146°34.5	15.4'	12°36.1	10.9'	54.0'						
2	206°31.6	34.5	161°08.9	15.3'	12°47.1	10.9'	54.0'						
3	221°31.6	· · 33.6	175°43.2	15.3'	12°58.0	10.8'	54.0'						
4	236°31.7	32.8	190°17.5	15.3'	13°08.8	10.8'	54.0'						
5	251°31.7	31.9	204°51.7	15.2'	13°19.6	10.7'	54.0'						
6	266°31.8	S11°31.0	219°26.0	15.2'	N13°30.3	10.7'	54.0'						
7	281°31.8	30.1	234°00.1	15.1'	13°41.0	10.6'	54.0'						
8	296°31.9	29.2	248°34.2	15.1'	13°51.7	10.6'	54.0'						
9	311°31.9	· · 28.3	263°08.3	15.0'	14°02.2	10.5'	54.0'						
10	326°32.0	27.4	277°42.3	15.0'	14°12.8	10.5'	54.0'						
11	341°32.0	26.6	292°16.3	14.9'	14°23.2	10.4'	54.0'						
12	356°32.1	S11°25.7	306°50.2	14.9'	N14°33.7	10.4'	54.0'						
13	11°32.1	24.8	321°24.1	14.8'	14°44.0	10.3'	54.0'						
14	26°32.2	23.9	335°58.0	14.8'	14°54.3	10.2'	54.0'						
15	41°32.3	· · 23.0	350°31.7	14.7'	15°04.6	10.2'	54.1'						
16	56°32.3	22.1	5°05.5	14.7'	15°14.7	10.1'	54.1'						
17	71°32.4	21.2	19°39.2	14.6'	15°24.9	10.1'	54.1'						
18	86°32.4	S11°20.3	34°12.8	14.6'	N15°34.9	10.0'	54.1'						
19	101°32.5	19.5	48°46.4	14.5'	15°44.9	9.9'	54.1'						
20	116°32.5	18.6	63°19.9	14.5'	15°54.9	9.9'	54.1'						
21	131°32.6	· · 17.7	77°53.4	14.4'	16°04.8	9.8'	54.1'						
22	146°32.7	16.8	92°26.8	14.4'	16°14.6	9.8'	54.1'						
23	161°32.7	15.9	107°00.1	14.3'	16°24.3	9.7'	54.1'						
SD = 16.2' $d = -0.9'$							SD = 14.7'						

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	176°32.8	S11°15.0	121°33.4	14.2'	N16°34.0	9.6'	54.1'
1	191°32.8	14.1	136°06.7	14.2'	16°43.7	9.6'	54.1'
2	206°32.9	13.2	150°39.9	14.1'	16°53.2	9.5'	54.1'
3	221°33.0	· · 12.3	165°13.0	14.1'	17°02.7	9.4'	54.1'
4	236°33.0	11.4	179°46.1	14.0'	17°12.1	9.4'	54.2'
5	251°33.1	10.5	194°19.1	14.0'	17°21.5	9.3'	54.2'
6	266°33.1	S11°09.7	208°52.0	13.9'	N17°30.8	9.2'	54.2'
7	281°33.2	08.8	223°24.9	13.8'	17°40.0	9.1'	54.2'
8	296°33.3	07.9	237°57.8	13.8'	17°49.1	9.1'	54.2'
9	311°33.3	· · 07.0	252°30.5	13.7'	17°58.2	9.0'	54.2'
10	326°33.4	06.1	267°03.3	13.7'	18°07.2	8.9'	54.2'
11	341°33.4	05.2	281°35.9	13.6'	18°16.1	8.9'	54.2'
12	356°33.5	S11°04.3	296°08.5	13.5'	N18°25.0	8.8'	54.3'
13	11°33.6	03.4	310°41.0	13.5'	18°33.8	8.7'	54.3'
14	26°33.6	02.5	325°13.5	13.4'	18°42.5	8.6'	54.3'
15	41°33.7	· · 01.6	339°45.9	13.3'	18°51.1	8.6'	54.3'
16	56°33.8	11°00.7	354°18.2	13.3'	18°59.6	8.5'	54.3'
17	71°33.8	10°59.8	8°50.5	13.2'	19°08.1	8.4'	54.3'
18	86°33.9	S10°58.9	23°22.7	13.1'	N19°16.5	8.3'	54.3'
19	101°33.9	58.0	37°54.8	13.1'	19°24.8	8.2'	54.3'
20	116°34.0	57.1	52°26.9	13.0'	19°33.1	8.2'	54.4'
21	131°34.1	· · 56.2	66°58.9	12.9'	19°41.2	8.1'	54.4'
22	146°34.1	55.3	81°30.9	12.9'	19°49.3	8.0'	54.4'
23	161°34.2	54.4	96°02.7	12.8'	19°57.3	7.9'	54.4'
	SD = 16.2'	$d = -0.9'$			SD = 14.8'		

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°34.3	S10°53.5	110°34.6	12.7'	N20°05.2	7.8'	54.4'
1	191°34.3	52.6	125°06.3	12.7'	20°13.0	7.7'	54.4'
2	206°34.4	51.7	139°38.0	12.6'	20°20.7	7.6'	54.5'
3	221°34.5	· · 50.8	154°09.6	12.5'	20°28.4	7.6'	54.5'
4	236°34.5	49.9	168°41.1	12.5'	20°35.9	7.5'	54.5'
5	251°34.6	49.0	183°12.6	12.4'	20°43.4	7.4'	54.5'
6	266°34.7	S10°48.1	197°44.0	12.3'	N20°50.8	7.3'	54.5'
7	281°34.7	47.2	212°15.3	12.3'	20°58.1	7.2'	54.5'
8	296°34.8	46.3	226°46.6	12.2'	21°05.3	7.1'	54.6'
9	311°34.9	· · 45.4	241°17.7	12.1'	21°12.4	7.0'	54.6'
10	326°35.0	44.5	255°48.9	12.0'	21°19.5	6.9'	54.6'
11	341°35.0	43.6	270°19.9	12.0'	21°26.4	6.8'	54.6'
12	356°35.1	S10°42.7	284°50.9	11.9'	N21°33.2	6.7'	54.6'
13	11°35.2	41.8	299°21.8	11.8'	21°40.0	6.6'	54.7'
14	26°35.2	40.9	313°52.6	11.8'	21°46.6	6.6'	54.7'
15	41°35.3	· · 40.0	328°23.4	11.7'	21°53.2	6.5'	54.7'
16	56°35.4	39.1	342°54.1	11.6'	21°59.6	6.4'	54.7'
17	71°35.4	38.2	357°24.7	11.6'	22°06.0	6.3'	54.7'
18	86°35.5	S10°37.3	11°55.3	11.5'	N22°12.2	6.2'	54.8'
19	101°35.6	36.4	26°25.8	11.4'	22°18.4	6.1'	54.8'
20	116°35.7	35.5	40°56.2	11.3'	22°24.5	6.0'	54.8'
21	131°35.7	· · 34.6	55°26.5	11.3'	22°30.4	5.9'	54.8'
22	146°35.8	33.7	69°56.8	11.2'	22°36.3	5.8'	54.8'
23	161°35.9	32.8	84°27.0	11.1'	22°42.0	5.7'	54.9'
	SD = 16.2'	$d = -0.9'$			SD = 14.8'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:56	07:14	08:29	16:00	17:15	18:34
N 70°	05:56	07:07	08:13	16:16	17:22	18:34
68°	05:56	07:01	08:00	16:29	17:28	18:33
66°	05:56	06:55	07:50	16:39	17:34	18:33
64°	05:56	06:51	07:41	16:48	17:38	18:33
62°	05:55	06:47	07:33	16:56	17:42	18:34
60°	05:55	06:43	07:26	17:02	17:45	18:34
N 58°	05:55	06:40	07:21	17:08	17:49	18:34
56°	05:54	06:37	07:15	17:13	17:51	18:35
54°	05:53	06:34	07:11	17:18	17:54	18:35
52°	05:53	06:32	07:07	17:22	17:56	18:36
50°	05:52	06:30	07:03	17:26	17:59	18:36
45°	05:50	06:24	06:54	17:34	18:04	18:38
N 40°	05:48	06:20	06:47	17:41	18:08	18:40
35°	05:46	06:16	06:41	17:47	18:12	18:42
30°	05:44	06:12	06:36	17:52	18:16	18:44
20°	05:38	06:04	06:27	18:01	18:24	18:49
N 10°	05:32	05:57	06:18	18:09	18:31	18:55
0°	05:25	05:49	06:10	18:17	18:38	19:03
S 10°	05:16	05:41	06:02	18:25	18:46	19:11
20°	05:05	05:31	05:54	18:34	18:56	19:23
30°	04:50	05:19	05:44	18:43	19:08	19:37
35°	04:40	05:12	05:38	18:49	19:15	19:47
40°	04:29	05:03	05:31	18:56	19:24	19:58
45°	04:14	04:52	05:23	19:03	19:34	20:12
S 50°	03:56	04:39	05:14	19:13	19:47	20:30
52°	03:47	04:33	05:10	19:17	19:53	20:39
54°	03:37	04:26	05:05	19:22	20:00	20:49
56°	03:25	04:18	04:59	19:27	20:08	21:00
58°	03:11	04:09	04:53	19:33	20:16	21:14
S 60°	02:53	03:59	04:46	19:39	20:27	21:31

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	06:13	05:09	□	00:01	02:35	□
N 70°	06:38	06:06	□	· · ·	01:40	□
68°	06:57	06:40	06:10	· · ·	01:07	03:12
66°	07:12	07:05	06:55	· · ·	00:43	02:27
64°	07:25	07:24	07:25	· · ·	00:25	01:58
62°	07:36	07:40	07:48	· · ·	00:09	01:36
60°	07:45	07:54	08:06	23:57	· · ·	01:18
N 58°	07:53	08:05	08:22	23:46	· · ·	01:03
56°	08:01	08:15	08:35	23:37	· · ·	00:51
54°	08:07	08:24	08:46	23:28	· · ·	00:40
52°	08:13	08:32	08:56	23:21	· · ·	00:30
50°	08:18	08:39	09:05	23:14	· · ·	00:21
45°	08:30	08:55	09:25	23:00	· · ·	00:03
N 40°	08:39	09:07	09:40	22:48	23:48	· · ·
35°	08:47	09:18	09:54	22:38	23:36	· · ·
30°	08:55	09:28	10:05	22:30	23:25	· · ·
20°	09:07	09:44	10:25	22:15	23:06	23:59
N 10°	09:18	09:59	10:43	22:02	22:50	23:40
0°	09:29	10:13	10:59	21:50	22:35	23:23
S 10°	09:39	10:26	11:15	21:37	22:20	23:05
20°	09:50	10:41	11:33	21:25	22:03	22:46
30°	10:03	10:58	11:54	21:10	21:45	22:25
35°	10:11	11:08	12:06	21:01	21:34	22:12
40°	10:20	11:19	12:19	20:52	21:22	21:58
45°	10:30	11:33	12:36	20:41	21:08	21:41
S 50°	10:42	11:49	12:56	20:27	20:50	21

February 21, 22, 23 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	151°12.2	183°30.1	S14°20.9	317°20.3	S02°16.6	305°22.7	S09°07.8	116°24.8	N11°29.5	Alpheratz	357°31.5	29°15.2	
1	166°14.6	198°29.5	19.9	332°23.0	16.5	320°25.2	07.8	131°27.1	29.6	Ankaa	353°04.3	-42°09.0	
2	181°17.1	213°28.9	18.9	347°25.6	16.4	335°27.8	07.8	146°29.4	29.7	Schedar	349°27.5	56°42.0	
3	196°19.6	228°28.3	•• 17.9	2°28.3	•• 16.3	350°30.3	•• 07.7	161°31.6	•• 29.8	Diphda	348°44.2	-17°49.7	
4	211°22.0	243°27.7	16.9	17°30.9	16.2	5°32.9	07.7	176°33.9	29.8	Achernar	335°18.2	-57°05.5	
5	226°24.5	258°27.1	15.9	32°33.6	16.1	20°35.4	07.6	191°36.2	29.9	Hamal	327°47.5	23°36.0	
6	241°27.0	273°26.5	S14°14.9	47°36.2	S02°16.0	35°37.9	S09°07.6	206°38.5	N11°30.0	Polaris	312°18.0	89°23.4	
7	256°29.4	288°26.0	13.9	62°38.9	15.8	50°40.5	07.6	221°40.8	30.1	Acamar	315°09.4	-40°11.5	
8	271°31.9	303°25.4	12.9	77°41.5	15.7	65°43.0	07.5	236°43.0	30.1	Menkar	314°02.7	4°12.1	
9	286°34.4	318°24.8	•• 11.9	92°44.2	•• 15.6	80°45.6	•• 07.5	251°45.3	•• 30.2	Mirfak	308°23.4	49°58.0	
10	301°36.8	333°24.2	10.8	107°46.9	15.5	95°48.1	07.4	266°47.6	30.3	Aldebaran	290°35.7	16°34.0	
11	316°39.3	348°23.6	09.8	122°49.5	15.4	110°50.6	07.4	281°49.9	30.4	Rigel	281°00.5	-8°10.3	
12	331°41.7	3°23.0	S14°08.8	137°52.2	S02°15.3	125°53.2	S09°07.4	296°52.1	N11°30.4	Capella	280°16.8	46°01.6	
13	346°44.2	18°22.4	07.8	152°54.8	15.1	140°55.7	07.3	311°54.4	30.5	Bellatrix	278°19.2	6°22.4	
14	1°46.7	33°21.8	06.8	167°57.5	15.0	155°58.3	07.3	326°56.7	30.6	Elnath	277°57.5	28°37.8	
15	16°49.1	48°21.3	•• 05.8	183°00.2	•• 14.9	171°00.8	•• 07.2	341°59.0	•• 30.7	Alnilam	275°34.2	-1°11.2	
16	31°51.6	63°20.7	04.8	198°02.8	14.8	186°03.4	07.2	357°01.2	30.7	Betelgeuse	270°48.3	7°24.6	
17	46°54.1	78°20.1	03.8	213°05.5	14.7	201°05.9	07.2	12°03.5	30.8	Canopus	263°50.6	-52°43.0	
18	61°56.5	93°19.5	S14°02.7	228°08.2	S02°14.5	216°08.4	S09°07.1	27°05.8	N11°30.9	Sirius	258°23.1	-16°45.7	
19	76°59.0	108°18.9	01.7	243°10.8	14.4	231°11.0	07.1	42°08.1	31.0	Adhara	255°03.0	-29°01.0	
20	92°01.5	123°18.3	14°00.7	258°13.5	14.3	246°13.5	07.0	57°10.3	31.0	Procyon	244°47.1	5°08.8	
21	107°03.9	138°17.7	13°59.7	273°16.2	•• 14.2	261°16.1	•• 07.0	72°12.6	•• 31.1	Pollux	243°12.9	27°57.2	
22	122°06.4	153°17.2	58.7	288°18.8	14.0	276°18.6	07.0	87°14.9	31.2	Avior	234°12.7	-59°36.5	
23	137°08.8	168°16.6	57.7	303°21.5	13.9	291°21.2	06.9	102°17.2	31.3	Suhail	222°43.4	-43°33.3	
Mer.pass. 13:53		ν -0.6' d -1.0' m -3.89		ν 2.6' d -0.1' m -0.58		ν 2.5' d -0.0' m -2.26		ν 2.3' d 0.1' m 0.37		Miaplacidus	221°36.5	-69°50.4	
										Alphard	217°44.1	-8°47.4	
										Regulus	207°30.6	11°49.2	
										Dubhe	193°36.5	61°35.4	
										Denebola	182°21.3	14°24.3	
										Gienah	175°39.8	-17°42.4	
										Acrux	172°55.6	-63°15.6	
										Gacrux	171°47.3	-57°16.6	
										Alioth	166°09.8	55°47.8	
										Spica	158°18.5	-11°18.9	
										Alkaid	152°49.2	49°09.7	
										Hadar	148°30.7	-60°30.7	
										Menkent	147°53.4	-36°30.8	
										Arcturus	145°44.7	19°01.6	
										Rigel Kent.	139°35.3	-60°57.2	
										Kochab	137°19.8	74°01.7	
										Zuben'ubi	136°52.2	-16°09.8	
										Alphecca	126°00.9	26°36.7	
										Antares	112°11.7	-26°29.7	
										Atria	107°03.0	-69°04.4	
										Sabik	101°59.0	-15°45.6	
										Shaula	96°06.0	-37°07.3	
										Rasalhague	95°55.6	12°32.2	
										Eltanin	90°40.9	51°28.9	
										Kaus Aust.	83°28.3	-34°22.0	
										Vega	80°31.2	38°48.5	
										Nunki	75°43.9	-26°15.5	
										Altair	61°57.0	8°56.7	
										Peacock	53°01.1	-56°38.2	
										Deneb	49°23.9	45°23.0	
										Enif	33°35.8	10°00.5	
										Al Na'ir	27°29.3	-46°49.1	
										Fomalhaut	15°11.2	-29°28.1	
										Scheat	13°42.2	28°14.5	
										Markab	13°26.8	15°21.8	
										Feb 21 Wed		SHA	Mer.pass
										Venus	32°17.9	11:46	
										Mars	166°08.2	02:50	
										Jupiter	154°10.5	03:38	
										Saturn	325°12.6	16:12	
										Feb 22 Thu		SHA	Mer.pass
										Venus	31°04.7	11:47	
										Mars	166°12.9	02:46	
										Jupiter	154°12.4	03:34	
										Saturn	325°08.1	16:08	
										Feb 23 Fri		SHA	Mer.pass
										Venus	29°51.7	11:48	
										Mars	166°18.3	02:42	
										Jupiter	154°14.5	03:30	
										Saturn	325°03.5	16:05	
										Horizontal parallax			
										Venus:	0.1		
										Mars:	0.2		
Mer.pass. 13:45		ν -0.6' d -1.0' m -3.89		ν 2.7' d -0.1' m -0.64		ν 2.6' d -0.0' m -2.27		ν 2.3' d 0.1' m 0.38					

DUT1 = UT1-UTC = +0.2071 sec ΔT = TT-UT1 = +68.9769 sec

2029 February 21 to Feb. 23 UT

h Sun Moon						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	176°35.9	S10°31.9	98°57.1	11.1'	N22°47.7	5.5' 54.9'
1	191°36.0	31.0	113°27.2	11.0'	22°53.2	5.4' 54.9'
2	206°36.1	30.1	127°57.2	10.9'	22°58.7	5.3' 54.9'
3	221°36.2	· · 29.2	142°27.1	10.9'	23°04.0	5.2' 55.0'
4	236°36.2	28.3	156°57.0	10.8'	23°09.3	5.1' 55.0'
5	251°36.3	27.4	171°26.7	10.7'	23°14.4	5.0' 55.0'
6	266°36.4	S10°26.5	185°56.4	10.6'	N23°19.4	4.9' 55.0'
7	281°36.5	25.6	200°26.1	10.6'	23°24.3	4.8' 55.0'
8	296°36.5	24.6	214°55.7	10.5'	23°29.1	4.7' 55.1'
9	311°36.6	· · 23.7	229°25.2	10.4'	23°33.8	4.6' 55.1'
10	326°36.7	22.8	243°54.6	10.4'	23°38.3	4.5' 55.1'
11	341°36.8	21.9	258°24.0	10.3'	23°42.8	4.3' 55.2'
12	356°36.8	S10°21.0	272°53.3	10.2'	N23°47.1	4.2' 55.2'
13	11°36.9	20.1	287°22.5	10.2'	23°51.4	4.1' 55.2'
14	26°37.0	19.2	301°51.7	10.1'	23°55.5	4.0' 55.2'
15	41°37.1	· · 18.3	316°20.7	10.0'	23°59.5	3.9' 55.3'
16	56°37.1	17.4	330°49.8	10.0'	24°03.4	3.8' 55.3'
17	71°37.2	16.5	345°18.7	9.9'	24°07.1	3.6' 55.3'
18	86°37.3	S10°15.6	359°47.6	9.8'	N24°10.8	3.5' 55.3'
19	101°37.4	14.6	14°16.5	9.8'	24°14.3	3.4' 55.4'
20	116°37.5	13.7	28°45.2	9.7'	24°17.7	3.3' 55.4'
21	131°37.5	· · 12.8	43°13.9	9.6'	24°21.0	3.2' 55.4'
22	146°37.6	11.9	57°42.6	9.6'	24°24.2	3.0' 55.5'
23	161°37.7	11.0	72°11.1	9.5'	24°27.2	2.9' 55.5'
SD = 16.2' d = -0.9' SD = 15.0'						

Thu	GHA	Dec	GHA	ν	Dec	d HP
0	176°37.8	S10°10.1	86°39.7	9.4'	N24°30.1	2.8' 55.5'
1	191°37.9	09.2	101°08.1	9.4'	24°32.9	2.7' 55.5'
2	206°37.9	08.3	115°36.5	9.3'	24°35.6	2.5' 55.6'
3	221°38.0	· · 07.4	130°04.8	9.3'	24°38.1	2.4' 55.6'
4	236°38.1	06.4	144°33.1	9.2'	24°40.6	2.3' 55.6'
5	251°38.2	05.5	159°01.3	9.1'	24°42.9	2.2' 55.7'
6	266°38.3	S10°04.6	173°29.4	9.1'	N24°45.0	2.0' 55.7'
7	281°38.3	03.7	187°57.5	9.0'	24°47.1	1.9' 55.7'
8	296°38.4	02.8	202°25.6	9.0'	24°49.0	1.8' 55.8'
9	311°38.5	· · 01.9	216°53.6	8.9'	24°50.8	1.7' 55.8'
10	326°38.6	01.0	231°21.5	8.9'	24°52.4	1.5' 55.8'
11	341°38.7	10°00.0	245°49.3	8.8'	24°53.9	1.4' 55.8'
12	356°38.8	S09°59.1	260°17.1	8.8'	N24°55.3	1.3' 55.9'
13	11°38.8	58.2	274°44.9	8.7'	24°56.6	1.1' 55.9'
14	26°38.9	57.3	289°12.6	8.7'	24°57.7	1.0' 55.9'
15	41°39.0	· · 56.4	303°40.3	8.6'	24°58.7	0.9' 56.0'
16	56°39.1	55.5	318°07.9	8.6'	24°59.5	0.7' 56.0'
17	71°39.2	54.6	332°35.4	8.5'	25°00.3	0.6' 56.0'
18	86°39.3	S09°53.6	347°02.9	8.5'	N25°00.9	0.5' 56.1'
19	101°39.3	52.7	1°30.4	8.4'	25°01.3	0.3' 56.1'
20	116°39.4	51.8	15°57.8	8.4'	25°01.6	0.2' 56.1'
21	131°39.5	· · 50.9	30°25.1	8.3'	25°01.8	0.0' 56.2'
22	146°39.6	50.0	44°52.5	8.3'	25°01.8	-0.1' 56.2'
23	161°39.7	49.1	59°19.7	8.2'	25°01.7	-0.2' 56.2'
SD = 16.2' d = -0.9' SD = 15.1'						

Fri	GHA	Dec	GHA	ν	Dec	d HP
0	176°39.8	S09°48.1	73°47.0	8.2'	N25°01.5	-0.4' 56.3'
1	191°39.9	47.2	88°14.1	8.1'	25°01.1	-0.5' 56.3'
2	206°39.9	46.3	102°41.3	8.1'	25°00.6	-0.7' 56.4'
3	221°40.0	· · 45.4	117°08.4	8.1'	25°00.0	-0.8' 56.4'
4	236°40.1	44.5	131°35.5	8.0'	24°59.2	-0.9' 56.4'
5	251°40.2	43.5	146°02.5	8.0'	24°58.2	-1.1' 56.5'
6	266°40.3	S09°42.6	160°29.5	8.0'	N24°57.2	-1.2' 56.5'
7	281°40.4	41.7	174°56.4	7.9'	24°56.0	-1.4' 56.5'
8	296°40.5	40.8	189°23.3	7.9'	24°54.6	-1.5' 56.6'
9	311°40.6	· · 39.9	203°50.2	7.8'	24°53.1	-1.6' 56.6'
10	326°40.6	38.9	218°17.1	7.8'	24°51.4	-1.8' 56.6'
11	341°40.7	38.0	232°43.9	7.8'	24°49.7	-1.9' 56.7'
12	356°40.8	S09°37.1	247°10.7	7.8'	N24°47.7	-2.1' 56.7'
13	11°40.9	36.2	261°37.4	7.7'	24°45.7	-2.2' 56.8'
14	26°41.0	35.3	276°04.2	7.7'	24°43.4	-2.4' 56.8'
15	41°41.1	· · 34.3	290°30.9	7.7'	24°41.1	-2.5' 56.8'
16	56°41.2	33.4	304°57.5	7.7'	24°38.6	-2.7' 56.9'
17	71°41.3	32.5	319°24.2	7.6'	24°35.9	-2.8' 56.9'
18	86°41.4	S09°31.6	333°50.8	7.6'	N24°33.1	-2.9' 56.9'
19	101°41.5	30.7	348°17.4	7.6'	24°30.2	-3.1' 57.0'
20	116°41.5	29.7	2°44.0	7.6'	24°27.1	-3.2' 57.0'
21	131°41.6	· · 28.8	17°10.6	7.5'	24°23.9	-3.4' 57.1'
22	146°41.7	27.9	31°37.1	7.5'	24°20.5	-3.5' 57.1'
23	161°41.8	27.0	46°03.6	7.5'	24°17.0	-3.7' 57.1'
SD = 16.2' d = -0.9' SD = 15.4'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:42	07:00	08:13	16:16	17:29	18:47
N 70°	05:44	06:54	07:59	16:30	17:34	18:45
68°	05:45	06:49	07:48	16:41	17:39	18:44
66°	05:46	06:45	07:38	16:50	17:43	18:43
64°	05:47	06:41	07:30	16:58	17:47	18:42
62°	05:47	06:38	07:24	17:04	17:50	18:41
60°	05:47	06:35	07:18	17:10	17:53	18:41
N 58°	05:47	06:33	07:13	17:15	17:55	18:41
56°	05:47	06:30	07:08	17:20	17:58	18:41
54°	05:47	06:28	07:04	17:24	18:00	18:41
52°	05:47	06:26	07:00	17:27	18:02	18:41
50°	05:47	06:24	06:57	17:31	18:04	18:41
45°	05:46	06:20	06:49	17:38	18:08	18:42
N 40°	05:44	06:16	06:43	17:44	18:12	18:43
35°	05:43	06:12	06:38	17:49	18:15	18:45
30°	05:41	06:09	06:33	17:54	18:18	18:46
20°	05:37	06:02	06:25	18:02	18:25	18:50
N 10°	05:31	05:56	06:17	18:10	18:31	18:56
0°	05:25	05:49	06:10	18:17	18:38	19:02
S 10°	05:17	05:41	06:03	18:24	18:45	19:10
20°	05:06	05:33	05:55	18:32	18:54	19:20
30°	04:52	05:21	05:46	18:40	19:05	19:34
35°	04:44	05:15	05:41	18:46	19:12	19:43
40°	04:33	05:07	05:35	18:51	19:19	19:53
45°	04:20	04:57	05:28	18:58	19:29	20:06
S 50°	04:02	04:45	05:19	19:07	19:41	20:23
52°	03:54	04:39	05:15	19:11	19:46	20:31
54°	03:45	04:33	05:11	19:15	19:53	20:40
56°	03:34	04:25	05:06	19:19	20:00	20:51
58°	03:21	04:17	05:01	19:25	20:08	21:03
S 60°	03:05	04:08	04:54	19:31	20:17	21:18

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	06:37	□	□	04:26	□	□
64°	07:30	07:48	08:39	03:34	05:04	06:05
62°	08:03	08:31	09:25	03:02	04:20	05:19
60°	08:27	09:01	09:55	02:38	03:51	04:49
N 58°	08:46	09:23	10:18	02:19	03:28	04:26
56°	09:02	09:41	10:36	02:03	03:10	04:07
54°	09:16	09:57	10:52	01:50	02:55	03:52
52°	09:28	10:10	11:05	01:38	02:42	03:38
50°	09:39	10:22	11:17	01:27	02:30	03:26
45°	10:01	10:46	11:41	01:06	02:06	03:02
N 40°	10:19	11:06	12:01	00:48	01:47	02:42
35°	10:35	11:22	12:17	00:33	01:31	02:26
30°	10:48	11:36	12:31	00:21	01:17	02:12
20°	11:10	12:00	12:55	· · ·	00:53	01:47
N 10°	11:30	12:21	13:16	· · ·	00:32	01:26
0°	11:49	12:41	13:35	· · ·	00:13	01:07
S 10°	12:07	13:00	13:55	23:54	· · ·	00:47
20°	12:27	13:21	14:15	23:34	· · ·	00:26
30°	12:50	13:46	14:39	23:10	· · ·	00:02
35°	13:03	14:00	14:54	22:56	23:47	· · ·
40°	13:19	14:16	15:10	22:40	23:31	· · ·
45°	13:38	14:36	15:29	22:21	23:11	· · ·
S 50°	14:01	15:01	15:54	21:57	22:46	23:47
52°	14:12	15:13	16:05	21:46	22:34	23:35
54°	14:25	15:27	16:19	21:33	22:20	23:22
56°	14:40	15:43	16:34	21:18	22:04	23:07
58°	14:57	16:02	16:52	21:00	21:45	22:49
S 60°	15:19	16:26	17:14	20:39	21:21	22:27

Day	Sun		Mer. Pass	Moon		Age 8-10 44-64%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
21	13:36	13:33	12:14	18:01	05:35	
22	13:29	13:25	12:13	18:54	06:27	
23	13:21	13:17	12:13	19:49	07:21	

February 24, 25, 26 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	154°09.6	182°48.6	S13°07.0	320°34.1	S02°07.3	308°26.3	S09°04.8	119°08.4	N11°35.0	Alpheratz	357°31.5	29°15.2	
1	169°12.1	197°48.1	05.9	335°36.8	07.2	323°28.8	04.7	134°10.7	35.1	Ankaa	353°04.3	-42°08.9	
2	184°14.5	212°47.5	04.9	350°39.5	07.0	338°31.4	04.7	149°12.9	35.1	Schedar	349°27.5	56°42.0	
3	199°17.0	227°47.0	•• 03.8	5°42.3	•• 06.9	353°34.0	•• 04.7	164°15.2	•• 35.2	Diphda	348°44.2	-17°49.7	
4	214°19.4	242°46.4	02.8	20°45.0	06.7	8°36.5	04.6	179°17.5	35.3	Achernar	335°18.2	-57°05.5	
5	229°21.9	257°45.8	01.7	35°47.8	06.6	23°39.1	04.6	194°19.7	35.4	Hamal	327°47.5	23°36.0	
6	244°24.4	272°45.3	S13°00.6	50°50.5	S02°06.4	38°41.6	S09°04.5	209°22.0	N11°35.4	Polaris	312°19.3	89°23.4	
7	259°26.8	287°44.7	12°59.6	65°53.2	06.3	53°44.2	04.5	224°24.3	35.5	Acamar	315°09.4	-40°11.5	
8	274°29.3	302°44.2	58.5	80°56.0	06.1	68°46.8	04.4	239°26.5	35.6	Menkar	314°02.7	4°12.1	
9	289°31.8	317°43.6	•• 57.5	95°58.7	•• 06.0	83°49.3	•• 04.4	254°28.8	•• 35.7	Mirfak	308°23.4	49°58.0	
10	304°34.2	332°43.1	56.4	111°01.5	05.8	98°51.9	04.3	269°31.1	35.8	Aldebaran	290°35.7	16°34.0	
11	319°36.7	347°42.5	55.4	126°04.2	05.7	113°54.5	04.3	284°33.3	35.8	Rigel	281°00.6	-8°10.3	
12	334°39.2	2°42.0	S12°54.3	141°07.0	S02°05.5	128°57.0	S09°04.2	299°35.6	N11°35.9	Capella	280°16.8	46°01.6	
13	349°41.6	17°41.4	53.3	156°09.7	05.4	143°59.6	04.2	314°37.9	36.0	Bellatrix	278°19.2	6°22.4	
14	4°44.1	32°40.9	52.2	171°12.5	05.2	159°02.2	04.1	329°40.1	36.1	Elnath	277°57.5	28°37.8	
15	19°46.6	47°40.3	•• 51.1	186°15.2	•• 05.1	174°04.7	•• 04.1	344°42.4	•• 36.1	Alnilam	275°34.2	-1°11.2	
16	34°49.0	62°39.8	50.1	201°18.0	04.9	189°07.3	04.0	359°44.7	36.2	Betelgeuse	270°48.3	7°24.6	
17	49°51.5	77°39.2	49.0	216°20.8	04.8	204°09.9	04.0	14°46.9	36.3	Canopus	263°50.6	-52°43.0	
18	64°53.9	92°38.7	S12°48.0	231°23.5	S02°04.6	219°12.4	S09°03.9	29°49.2	N11°36.4	Sirius	258°23.1	-16°45.7	
19	79°56.4	107°38.1	46.9	246°26.3	04.5	234°15.0	03.9	44°51.4	36.5	Adhara	255°03.0	-29°01.0	
20	94°58.9	122°37.6	45.8	261°29.0	04.3	249°17.5	03.8	59°53.7	36.5	Procyon	244°47.1	5°08.8	
21	110°01.3	137°37.0	•• 44.8	276°31.8	•• 04.2	264°20.1	•• 03.8	74°56.0	•• 36.6	Pollux	243°12.9	27°57.2	
22	125°03.8	152°36.5	43.7	291°34.6	04.0	279°22.7	03.7	89°58.2	36.7	Avior	234°12.7	-59°36.5	
23	140°06.3	167°35.9	42.6	306°37.3	03.9	294°25.2	03.7	105°00.5	36.8	Suhail	222°43.4	-43°33.3	
Mer.pass. 13:41		ν -0.6' d -1.1' m -3.89		ν 2.7' d -0.1' m -0.66		ν 2.6' d -0.0' m -2.28		ν 2.3' d 0.1' m 0.38		Miaplacidus	221°36.5	-69°50.4	
										Alphard	217°44.1	-8°47.4	
										Regulus	207°30.6	11°49.2	
										Dubhe	193°36.4	61°35.4	
										Denebola	182°21.2	14°24.3	
										Gienah	175°39.8	-17°42.4	
										Acrux	172°55.5	-63°15.6	
										Gacrux	171°47.2	-57°16.6	
										Alioth	166°09.7	55°47.8	
										Spica	158°18.5	-11°19.0	
										Alkaid	152°49.2	49°09.7	
										Hadar	148°30.7	-60°30.7	
										Menkent	147°53.4	-36°30.8	
										Arcturus	145°44.7	19°01.6	
										Rigil Kent.	139°35.2	-60°57.2	
										Kochab	137°19.7	74°01.7	
										Zuben'ubi	136°52.1	-16°09.8	
										Alphecca	126°00.9	26°36.7	
										Antares	112°11.7	-26°29.7	
										Atria	107°02.9	-69°04.4	
										Sabik	101°59.0	-15°45.6	
										Shaula	96°05.9	-37°07.3	
										Rasalhague	95°55.5	12°32.2	
										Eltanin	90°40.8	51°28.8	
										Kaus Aust.	83°28.3	-34°22.0	
										Vega	80°31.2	38°48.5	
										Nunki	75°43.8	-26°15.5	
										Altair	61°56.9	8°56.7	
										Peacock	53°01.1	-56°38.2	
										Deneb	49°23.9	45°23.0	
										Enif	33°35.8	10°00.5	
										Al Na'ir	27°29.3	-46°49.1	
										Fomalhaut	15°11.2	-29°28.1	
										Scheat	13°42.2	28°14.5	
										Markab	13°26.8	15°21.8	
										Feb 24 Sat		SHA	Mer.pass
										Venus	28°39.0	11:49	
										Mars	166°24.5	02:37	
										Jupiter	154°16.7	03:26	
										Saturn	324°58.8	16:01	
										Feb 25 Sun		SHA	Mer.pass
										Venus	27°26.6	11:50	
										Mars	166°31.4	02:33	
										Jupiter	154°19.1	03:22	
										Saturn	324°54.0	15:57	
										Feb 26 Mon		SHA	Mer.pass
										Venus	26°14.5	11:51	
										Mars	166°39.0	02:28	
										Jupiter	154°21.7	03:17	
										Saturn	324°49.2	15:54	
										Horizontal parallax			
										Venus:	0.1		
										Mars:	0.2		
Mer.pass. 13:33		ν -0.5' d -1.1' m -3.89		ν 2.8' d -0.2' m -0.72		ν 2.6' d -0.1' m -2.29		ν 2.3' d 0.1' m 0.38					

DUT1 = UT1-UTC = +0.2072 sec ΔT = TT-UT1 = +68.9768 sec

2029 February 24 to Feb. 26 UT

Sun				Moon				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	176°41.9	S09°26.0		60°30.1	7.5'	N24°13.3	-3.8'	57.2'
1	191°42.0	25.1		74°56.6	7.5'	24°09.5	-4.0'	57.2'
2	206°42.1	24.2		89°23.1	7.5'	24°05.5	-4.1'	57.3'
3	221°42.2	· · 23.3		103°49.6	7.4'	24°01.4	-4.3'	57.3'
4	236°42.3	22.3		118°16.0	7.4'	23°57.1	-4.4'	57.3'
5	251°42.4	21.4		132°42.4	7.4'	23°52.7	-4.5'	57.4'
6	266°42.5	S09°20.5		147°08.9	7.4'	N23°48.2	-4.7'	57.4'
7	281°42.6	19.6		161°35.3	7.4'	23°43.5	-4.8'	57.4'
8	296°42.7	18.6		176°01.7	7.4'	23°38.7	-5.0'	57.5'
9	311°42.8	· · 17.7		190°28.1	7.4'	23°33.7	-5.1'	57.5'
10	326°42.9	16.8		204°54.5	7.4'	23°28.6	-5.3'	57.6'
11	341°43.0	15.9		219°20.9	7.4'	23°23.3	-5.4'	57.6'
12	356°43.0	S09°14.9		233°47.3	7.4'	N23°17.9	-5.6'	57.6'
13	11°43.1	14.0		248°13.6	7.4'	23°12.3	-5.7'	57.7'
14	26°43.2	13.1		262°40.0	7.4'	23°06.6	-5.9'	57.7'
15	41°43.3	· · 12.2		277°06.4	7.4'	23°00.7	-6.0'	57.8'
16	56°43.4	11.2		291°32.8	7.4'	22°54.7	-6.1'	57.8'
17	71°43.5	10.3		305°59.2	7.4'	22°48.6	-6.3'	57.8'
18	86°43.6	S09°09.4		320°25.6	7.4'	N22°42.3	-6.4'	57.9'
19	101°43.7	08.4		334°51.9	7.4'	22°35.9	-6.6'	57.9'
20	116°43.8	07.5		349°18.3	7.4'	22°29.3	-6.7'	58.0'
21	131°43.9	· · 06.6		3°44.7	7.4'	22°22.6	-6.9'	58.0'
22	146°44.0	05.7		18°11.1	7.4'	22°15.7	-7.0'	58.0'
23	161°44.1	04.7		32°37.5	7.4'	22°08.7	-7.1'	58.1'
SD = 16.2'		$d = -0.9'$		SD = 15.6'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	176° 44.2	S09° 03.8	47° 04.0	7.4'	N22° 01.5	-7.3'	58.1'
1	191° 44.3	02.9	61° 30.4	7.4'	21° 54.3	-7.4'	58.2'
2	206° 44.4	01.9	75° 56.8	7.4'	21° 46.8	-7.6'	58.2'
3	221° 44.5	· · 01.0	90° 23.3	7.5'	21° 39.3	-7.7'	58.2'
4	236° 44.6	09° 00.1	104° 49.7	7.5'	21° 31.5	-7.9'	58.3'
5	251° 44.7	08° 59.1	119° 16.2	7.5'	21° 23.7	-8.0'	58.3'
6	266° 44.8	S08° 58.2	133° 42.7	7.5'	N21° 15.7	-8.1'	58.4'
7	281° 44.9	57.3	148° 09.2	7.5'	21° 07.6	-8.3'	58.4'
8	296° 45.0	56.4	162° 35.7	7.5'	20° 59.3	-8.4'	58.4'
9	311° 45.1	· · 55.4	177° 02.2	7.5'	20° 50.9	-8.5'	58.5'
10	326° 45.2	54.5	191° 28.7	7.6'	20° 42.4	-8.7'	58.5'
11	341° 45.3	53.6	205° 55.3	7.6'	20° 33.7	-8.8'	58.6'
12	356° 45.4	S08° 52.6	220° 21.9	7.6'	N20° 24.9	-8.9'	58.6'
13	11° 45.5	51.7	234° 48.5	7.6'	20° 15.9	-9.1'	58.6'
14	26° 45.6	50.8	249° 15.1	7.6'	20° 06.8	-9.2'	58.7'
15	41° 45.7	· · 49.8	263° 41.7	7.6'	19° 57.6	-9.3'	58.7'
16	56° 45.8	48.9	278° 08.4	7.7'	19° 48.3	-9.5'	58.8'
17	71° 45.9	48.0	292° 35.0	7.7'	19° 38.8	-9.6'	58.8'
18	86° 46.0	S08° 47.0	307° 01.7	7.7'	N19° 29.2	-9.7'	58.8'
19	101° 46.1	46.1	321° 28.4	7.7'	19° 19.5	-9.9'	58.9'
20	116° 46.2	45.2	335° 55.2	7.8'	19° 09.6	-10.0'	58.9'
21	131° 46.3	· · 44.2	350° 21.9	7.8'	18° 59.6	-10.1'	59.0'
22	146° 46.5	43.3	4° 48.7	7.8'	18° 49.5	-10.3'	59.0'
23	161° 46.6	42.4	19° 15.5	7.8'	18° 39.2	-10.4'	59.0'
	SD = 16.2'	$d = -0.9'$			SD = 15.9'		

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	176°46.7	S08°41.4	33°42.3	7.8'	N18°28.8	-10.5'	59.1'
1	191°46.8	40.5	48°09.2	7.9'	18°18.3	-10.6'	59.1'
2	206°46.9	39.5	62°36.0	7.9'	18°07.7	-10.8'	59.1'
3	221°47.0	· · 38.6	77°02.9	7.9'	17°56.9	-10.9'	59.2'
4	236°47.1	37.7	91°29.9	7.9'	17°46.1	-11.0'	59.2'
5	251°47.2	36.7	105°56.8	8.0'	17°35.1	-11.1'	59.3'
6	266°47.3	S08°35.8	120°23.8	8.0'	N17°24.0	-11.2'	59.3'
7	281°47.4	34.9	134°50.8	8.0'	17°12.7	-11.3'	59.3'
8	296°47.5	33.9	149°17.8	8.1'	17°01.4	-11.5'	59.4'
9	311°47.6	· · 33.0	163°44.9	8.1'	16°49.9	-11.6'	59.4'
10	326°47.7	32.1	178°11.9	8.1'	16°38.3	-11.7'	59.4'
11	341°47.8	31.1	192°39.0	8.1'	16°26.6	-11.8'	59.5'
12	356°47.9	S08°30.2	207°06.2	8.2'	N16°14.8	-11.9'	59.5'
13	11°48.0	29.2	221°33.3	8.2'	16°02.9	-12.0'	59.5'
14	26°48.2	28.3	236°00.5	8.2'	15°50.9	-12.1'	59.6'
15	41°48.3	· · 27.4	250°27.7	8.2'	15°38.7	-12.3'	59.6'
16	56°48.4	26.4	264°54.9	8.3'	15°26.5	-12.4'	59.7'
17	71°48.5	25.5	279°22.2	8.3'	15°14.1	-12.5'	59.7'
18	86°48.6	S08°24.6	293°49.5	8.3'	N15°01.7	-12.6'	59.7'
19	101°48.7	23.6	308°16.8	8.3'	14°49.1	-12.7'	59.8'
20	116°48.8	22.7	322°44.2	8.4'	14°36.4	-12.8'	59.8'
21	131°48.9	· · 21.7	337°11.5	8.4'	14°23.7	-12.9'	59.8'
22	146°49.0	20.8	351°38.9	8.4'	14°10.8	-13.0'	59.9'
23	161°49.1	19.9	6°06.3	8.4'	13°57.8	-13.1'	59.9'
	SD = 16.1'	$d = -0.9'$			SD = 16.1'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:28	06:46	07:57	16:31	17:42	19:00
N 70°	05:31	06:41	07:45	16:43	17:46	18:57
68°	05:34	06:38	07:35	16:52	17:50	18:54
66°	05:36	06:35	07:27	17:00	17:53	18:52
64°	05:37	06:32	07:20	17:07	17:56	18:51
62°	05:38	06:29	07:14	17:13	17:58	18:49
60°	05:39	06:27	07:09	17:18	18:00	18:48
N 58°	05:40	06:25	07:05	17:22	18:02	18:47
56°	05:40	06:23	07:01	17:26	18:04	18:47
54°	05:41	06:22	06:57	17:30	18:05	18:46
52°	05:41	06:20	06:54	17:33	18:07	18:46
50°	05:41	06:18	06:51	17:36	18:08	18:46
45°	05:41	06:15	06:44	17:42	18:12	18:46
N 40°	05:40	06:12	06:39	17:48	18:15	18:46
35°	05:39	06:09	06:34	17:52	18:18	18:47
30°	05:38	06:06	06:30	17:56	18:20	18:48
20°	05:35	06:00	06:23	18:04	18:26	18:52
N 10°	05:30	05:55	06:16	18:10	18:31	18:56
0°	05:24	05:49	06:10	18:16	18:37	19:01
S 10°	05:17	05:42	06:03	18:23	18:44	19:09
20°	05:08	05:34	05:56	18:29	18:52	19:18
30°	04:55	05:24	05:48	18:37	19:02	19:30
35°	04:47	05:18	05:43	18:42	19:08	19:38
40°	04:37	05:10	05:38	18:47	19:15	19:48
45°	04:25	05:01	05:32	18:53	19:24	20:00
S 50°	04:09	04:50	05:24	19:01	19:34	20:16
52°	04:01	04:45	05:21	19:04	19:40	20:23
54°	03:52	04:39	05:17	19:08	19:45	20:32
56°	03:42	04:33	05:13	19:12	19:52	20:42
58°	03:31	04:25	05:08	19:17	19:59	20:53
S 60°	03:17	04:17	05:02	19:22	20:07	21:06

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	☐	☐	12:01	☐	☐	08:36
N 70°	☐	☐	12:47	☐	☐	07:49
68°	☐	10:46	13:16	☐	07:52	07:18
66°	09:12	11:32	13:38	07:28	07:05	06:55
64°	10:12	12:02	13:55	06:28	06:35	06:36
62°	10:46	12:24	14:09	05:53	06:12	06:21
60°	11:11	12:42	14:21	05:28	05:53	06:08
N 58°	11:31	12:57	14:31	05:08	05:37	05:57
56°	11:47	13:10	14:40	04:52	05:24	05:48
54°	12:01	13:21	14:48	04:38	05:13	05:39
52°	12:13	13:31	14:55	04:25	05:02	05:31
50°	12:23	13:39	15:01	04:14	04:53	05:24
45°	12:46	13:58	15:15	03:51	04:34	05:09
N 40°	13:04	14:13	15:26	03:33	04:18	04:57
35°	13:19	14:25	15:35	03:17	04:04	04:46
30°	13:32	14:36	15:44	03:04	03:52	04:37
20°	13:54	14:55	15:58	02:41	03:32	04:21
N 10°	14:13	15:12	16:10	02:21	03:14	04:07
0°	14:31	15:27	16:22	02:02	02:58	03:53
S 10°	14:49	15:42	16:33	01:43	02:41	03:40
20°	15:08	15:58	16:46	01:23	02:23	03:25
30°	15:30	16:17	16:59	01:00	02:02	03:08
35°	15:43	16:27	17:07	00:46	01:50	02:59
40°	15:58	16:40	17:17	00:30	01:36	02:47
45°	16:15	16:54	17:27	00:11	01:19	02:34
S 50°	16:37	17:12	17:40		00:58	02:18
52°	16:47	17:20	17:46		00:48	02:10
54°	16:59	17:29	17:53		00:37	02:01
56°	17:12	17:39	18:00		00:24	01:52
58°	17:27	17:51	18:08		00:09	01:41
S 60°	17:46	18:05	18:17	23:52		01:28

February 27, 28, 01 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	157°07.0	182°09.6	S11°49.8	323°54.3	S01°55.6	311°31.4	S09°01.2	121°51.3	N11°40.6	Alpheratz	357°31.6	29°15.2
1	172°09.5	197°09.1	48.7	338°57.2	55.5	326°34.0	01.1	136°53.5	40.7	Ankaa	353°04.3	-42°08.9
2	187°11.9	212°08.5	47.6	354°00.0	55.3	341°36.6	01.1	151°55.8	40.8	Schedar	349°27.5	56°42.0
3	202°14.4	227°08.0	· · 46.5	9°02.8	· · 55.1	356°39.1	· · 01.0	166°58.0	· · 40.9	Diphda	348°44.2	-17°49.7
4	217°16.9	242°07.5	45.4	24°05.6	54.9	11°41.7	01.0	182°00.3	40.9	Achernar	335°18.2	-57°05.5
5	232°19.3	257°07.0	44.3	39°08.5	54.7	26°44.3	00.9	197°02.6	41.0	Hamal	327°47.5	23°36.0
6	247°21.8	272°06.4	S11°43.3	54°11.3	S01°54.6	41°46.9	S09°00.9	212°04.8	N11°41.1	Polaris	312°20.8	89°23.4
7	262°24.3	287°05.9	42.2	69°14.1	54.4	56°49.5	00.8	227°07.1	41.2	Acamar	315°09.4	-40°11.5
8	277°26.7	302°05.4	41.1	84°17.0	54.2	71°52.1	00.7	242°09.3	41.3	Menkar	314°02.7	4°12.1
9	292°29.2	317°04.9	· · 40.0	99°19.8	· · 54.0	86°54.7	· · 00.7	257°11.6	· · 41.3	Mirfak	308°23.5	49°58.0
10	307°31.7	332°04.4	38.9	114°22.7	53.8	101°57.2	00.6	272°13.8	41.4	Aldebaran	290°35.7	16°34.0
11	322°34.1	347°03.8	37.8	129°25.5	53.7	116°59.8	00.6	287°16.1	41.5	Rigel	281°00.6	-8°10.3
12	337°36.6	2°03.3	S11°36.7	144°28.3	S01°53.5	132°02.4	S09°00.5	302°18.4	N11°41.6	Capella	280°16.8	46°01.7
13	352°39.0	17°02.8	35.6	159°31.2	53.3	147°05.0	00.5	317°20.6	41.7	Bellatrix	278°19.2	6°22.4
14	7°41.5	32°02.3	34.5	174°34.0	53.1	162°07.6	00.4	332°22.9	41.7	Elnath	277°57.5	28°37.8
15	22°44.0	47°01.8	· · 33.4	189°36.9	· · 52.9	177°10.2	· · 00.4	347°25.1	· · 41.8	Alnilam	275°34.2	-1°11.2
16	37°46.4	62°01.2	32.3	204°39.7	52.7	192°12.7	00.3	2°27.4	41.9	Betelgeuse	270°48.3	7°24.6
17	52°48.9	77°00.7	31.2	219°42.6	52.5	207°15.3	00.2	17°29.6	42.0	Canopus	263°50.6	-52°43.0
18	67°51.4	92°00.2	S11°30.1	234°45.4	S01°52.4	222°17.9	S09°00.2	32°31.9	N11°42.1	Sirius	258°23.1	-16°45.7
19	82°53.8	106°59.7	29.0	249°48.3	52.2	237°20.5	00.1	47°34.1	42.1	Adhara	255°03.0	-29°01.0
20	97°56.3	121°59.2	27.9	264°51.1	52.0	252°23.1	00.1	62°36.4	42.2	Procyon	244°47.1	5°08.8
21	112°58.8	136°58.6	· · 26.8	279°54.0	· · 51.8	267°25.7	· · 00.0	77°38.7	· · 42.3	Pollux	243°12.9	27°57.2
22	128°01.2	151°58.1	25.7	294°56.8	51.6	282°28.3	09°00.0	92°40.9	42.4	Avior	234°12.7	-59°36.5
23	143°03.7	166°57.6	24.6	309°59.7	51.4	297°30.9	08°59.9	107°43.2	42.5	Suhail	222°43.4	-43°33.3
Mer.pass. 13:29		ν -0.5' d -1.1' m -3.89		ν 2.8' d -0.2' m -0.75		ν 2.6' d -0.1' m -2.30		ν 2.3' d 0.1' m 0.38		Miaplacidus	221°36.5	-69°50.5
										Alphard	217°44.1	-8°47.4
										Regulus	207°30.6	11°49.2
										Dubhe	193°36.4	61°35.4
										Denebola	182°21.2	14°24.3
										Gienah	175°39.8	-17°42.4
										Acrux	172°55.5	-63°15.7
										Gacrux	171°47.2	-57°16.6
										Alioth	166°09.7	55°47.8
										Spica	158°18.5	-11°19.0
										Alkaid	152°49.1	49°09.7
										Hadar	148°30.7	-60°30.7
										Menkent	147°53.3	-36°30.8
										Arcturus	145°44.7	19°01.6
										Rigil Kent.	139°35.2	-60°57.2
										Kochab	137°19.6	74°01.7
										Zuben'ubi	136°52.1	-16°09.8
										Alphecca	126°00.8	26°36.7
										Antares	112°11.7	-26°29.7
										Atria	107°02.9	-69°04.4
										Sabik	101°59.0	-15°45.6
										Shaula	96°05.9	-37°07.3
										Rasalhague	95°55.5	12°32.2
										Eltanin	90°40.8	51°28.8
										Kaus Aust.	83°28.2	-34°22.0
										Vega	80°31.1	38°48.5
										Nunki	75°43.8	-26°15.5
										Altair	61°56.9	8°56.7
										Peacock	53°01.0	-56°38.2
										Deneb	49°23.8	45°23.0
										Enif	33°35.8	10°00.5
										Al Na'ir	27°29.3	-46°49.1
										Fomalhaut	15°11.2	-29°28.1
										Scheat	13°42.2	28°14.5
										Markab	13°26.8	15°21.8
Feb 27 Tue		SHA		Mer.pass								
Venus		25°02.6		11:52								
Mars		166°47.3		02:24								
Jupiter		154°24.4		03:13								
Saturn		324°44.3		15:50								
Feb 28 Wed		SHA		Mer.pass								
Venus		23°50.9		11:53								
Mars		166°56.4		02:19								
Jupiter		154°27.3		03:09								
Saturn		324°39.3		15:47								
Mar 01 Thu		SHA		Mer.pass								
Venus		22°39.5		11:53								
Mars		167°06.2		02:15								
Jupiter		154°30.4		03:05								
Saturn		324°34.2		15:43								
Horizontal parallax												
Venus:												0.1
Mars:												0.2

DUT1 = UT1-UTC = +0.2073 sec $\Delta T = TT-UT1 = +68.9767$ sec

2029 February 27 to Mar. 01 UT

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	176°49.3	S08°18.9	20°33.8	8.5'	N13°44.7	-13.2' 59.9'
1	191°49.4	18.0	35°01.3	8.5'	13°31.6	-13.3' 60.0'
2	206°49.5	17.0	49°28.8	8.5'	13°18.3	-13.4' 60.0'
3	221°49.6	· · 16.1	63°56.3	8.6'	13°04.9	-13.4' 60.0'
4	236°49.7	15.2	78°23.8	8.6'	12°51.5	-13.5' 60.0'
5	251°49.8	14.2	92°51.4	8.6'	12°38.0	-13.6' 60.1'
6	266°49.9	S08°13.3	107°19.0	8.6'	N12°24.3	-13.7' 60.1'
7	281°50.0	12.3	121°46.6	8.6'	12°10.6	-13.8' 60.1'
8	296°50.1	11.4	136°14.3	8.7'	11°56.8	-13.9' 60.2'
9	311°50.3	· · 10.4	150°42.0	8.7'	11°42.9	-14.0' 60.2'
10	326°50.4	09.5	165°09.7	8.7'	11°28.9	-14.1' 60.2'
11	341°50.5	08.6	179°37.4	8.7'	11°14.9	-14.1' 60.2'
12	356°50.6	S08°07.6	194°05.1	8.8'	N11°00.8	-14.2' 60.3'
13	11°50.7	06.7	208°32.9	8.8'	10°46.5	-14.3' 60.3'
14	26°50.8	05.7	223°00.7	8.8'	10°32.2	-14.4' 60.3'
15	41°50.9	· · 04.8	237°28.5	8.8'	10°17.9	-14.4' 60.4'
16	56°51.1	03.8	251°56.3	8.9'	10°03.4	-14.5' 60.4'
17	71°51.2	02.9	266°24.2	8.9'	09°48.9	-14.6' 60.4'
18	86°51.3	S08°02.0	280°52.1	8.9'	N09°34.3	-14.7' 60.4'
19	101°51.4	01.0	295°19.9	8.9'	09°19.7	-14.7' 60.5'
20	116°51.5	00°00.1	309°47.9	8.9'	09°05.0	-14.8' 60.5'
21	131°51.6	07°59.1	324°15.8	8.9'	08°50.2	-14.9' 60.5'
22	146°51.7	58.2	338°43.7	9.0'	08°35.3	-14.9' 60.5'
23	161°51.9	57.2	353°11.7	9.0'	08°20.4	-15.0' 60.6'
SD = 16.1' $d = -0.9'$			SD = 16.3'			
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	176°52.0	S07°56.3	7°39.7	9.0'	N08°05.4	-15.0' 60.6'
1	191°52.1	55.3	22°07.7	9.0'	07°50.4	-15.1' 60.6'
2	206°52.2	54.4	36°35.7	9.0'	07°35.3	-15.1' 60.6'
3	221°52.3	· · 53.5	51°03.7	9.0'	07°20.2	-15.2' 60.6'
4	236°52.4	52.5	65°31.8	9.1'	07°05.0	-15.3' 60.7'
5	251°52.6	51.6	79°59.8	9.1'	06°49.7	-15.3' 60.7'
6	266°52.7	S07°50.6	94°27.9	9.1'	N06°34.4	-15.4' 60.7'
7	281°52.8	49.7	108°56.0	9.1'	06°19.0	-15.4' 60.7'
8	296°52.9	48.7	123°24.1	9.1'	06°03.6	-15.4' 60.7'
9	311°53.0	· · 47.8	137°52.2	9.1'	05°48.2	-15.5' 60.8'
10	326°53.2	46.8	152°20.3	9.1'	05°32.7	-15.5' 60.8'
11	341°53.3	45.9	166°48.4	9.1'	05°17.2	-15.6' 60.8'
12	356°53.4	S07°44.9	181°16.5	9.1'	N05°01.6	-15.6' 60.8'
13	11°53.5	44.0	195°44.7	9.2'	04°46.0	-15.6' 60.8'
14	26°53.6	43.0	210°12.8	9.2'	04°30.3	-15.7' 60.8'
15	41°53.8	· · 42.1	224°41.0	9.2'	04°14.7	-15.7' 60.9'
16	56°53.9	41.1	239°09.2	9.2'	03°59.0	-15.7' 60.9'
17	71°54.0	40.2	253°37.3	9.2'	03°43.2	-15.8' 60.9'
18	86°54.1	S07°39.2	268°05.5	9.2'	N03°27.4	-15.8' 60.9'
19	101°54.2	38.3	282°33.7	9.2'	03°11.6	-15.8' 60.9'
20	116°54.4	37.4	297°01.9	9.2'	02°55.8	-15.8' 60.9'
21	131°54.5	· · 36.4	311°30.1	9.2'	02°40.0	-15.9' 60.9'
22	146°54.6	35.5	325°58.2	9.2'	02°24.1	-15.9' 61.0'
23	161°54.7	34.5	340°26.4	9.2'	02°08.2	-15.9' 61.0'
SD = 16.1' $d = -0.9'$			SD = 16.5'			
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	176°54.8	S07°33.6	354°54.6	9.2'	N01°52.3	-15.9' 61.0'
1	191°55.0	32.6	9°22.8	9.2'	01°36.4	-15.9' 61.0'
2	206°55.1	31.7	23°51.0	9.2'	01°20.4	-15.9' 61.0'
3	221°55.2	· · 30.7	38°19.2	9.2'	01°04.5	-16.0' 61.0'
4	236°55.3	29.8	52°47.3	9.2'	00°48.5	-16.0' 61.0'
5	251°55.5	28.8	67°15.5	9.2'	00°32.6	-16.0' 61.0'
6	266°55.6	S07°27.9	81°43.7	9.2'	N00°16.6	-16.0' 61.0'
7	281°55.7	26.9	96°11.9	9.2'	N00°00.6	-16.0' 61.0'
8	296°55.8	25.9	110°40.0	9.1'	S00°15.4	16.0' 61.0'
9	311°55.9	· · 25.0	125°08.2	9.1'	00°31.4	16.0' 61.0'
10	326°56.1	24.0	139°36.3	9.1'	00°47.3	16.0' 61.0'
11	341°56.2	23.1	154°04.4	9.1'	01°03.3	16.0' 61.1'
12	356°56.3	S07°22.1	168°32.6	9.1'	S01°19.3	16.0' 61.1'
13	11°56.4	21.2	183°00.7	9.1'	01°35.3	16.0' 61.1'
14	26°56.6	20.2	197°28.8	9.1'	01°51.2	15.9' 61.1'
15	41°56.7	· · 19.3	211°56.9	9.1'	02°07.2	15.9' 61.1'
16	56°56.8	18.3	226°24.9	9.1'	02°23.1	15.9' 61.1'
17	71°56.9	17.4	240°53.0	9.0'	02°39.0	15.9' 61.1'
18	86°57.1	S07°16.4	255°21.0	9.0'	S02°54.9	15.9' 61.1'
19	101°57.2	15.5	269°49.1	9.0'	03°10.8	15.9' 61.1'
20	116°57.3	14.5	284°17.1	9.0'	03°26.7	15.8' 61.1'
21	131°57.4	· · 13.6	298°45.1	9.0'	03°42.5	15.8' 61.1'
22	146°57.6	12.6	313°13.1	9.0'	03°58.3	15.8' 61.1'
23	161°57.7	11.7	327°41.0	8.9'	04°14.1	15.8' 61.1'
SD = 16.1' $d = -0.9'$			SD = 16.6'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:14	06:31	07:41	16:46	17:56	19:14
N 70°	05:18	06:28	07:31	16:56	17:58	19:09
68°	05:22	06:26	07:23	17:04	18:01	19:05
66°	05:25	06:24	07:16	17:11	18:03	19:02
64°	05:27	06:22	07:10	17:16	18:04	18:59
62°	05:29	06:20	07:05	17:21	18:06	18:57
60°	05:31	06:19	07:01	17:25	18:07	18:56
N 58°	05:32	06:17	06:57	17:29	18:09	18:54
56°	05:33	06:16	06:53	17:32	18:10	18:53
54°	05:34	06:15	06:50	17:36	18:11	18:52
52°	05:35	06:14	06:48	17:38	18:12	18:51
50°	05:35	06:12	06:45	17:41	18:13	18:51
45°	05:36	06:10	06:39	17:46	18:16	18:50
N 40°	05:36	06:07	06:35	17:51	18:18	18:49
35°	05:36	06:05	06:30	17:55	18:20	18:50
30°	05:35	06:03	06:27	17:58	18:23	18:50
20°	05:33	05:58	06:20	18:05	18:27	18:53
N 10°	05:29	05:53	06:15	18:10	18:32	18:56
0°	05:24	05:48	06:09	18:16	18:37	19:01
S 10°	05:18	05:42	06:04	18:21	18:42	19:07
20°	05:09	05:35	05:57	18:27	18:49	19:16
30°	04:57	05:26	05:50	18:34	18:58	19:27
35°	04:50	05:20	05:46	18:38	19:04	19:34
40°	04:41	05:14	05:42	18:43	19:10	19:43
45°	04:29	05:06	05:36	18:48	19:18	19:54
S 50°	04:15	04:56	05:29	18:54	19:28	20:09
52°	04:08	04:51	05:26	18:57	19:33	20:16
54°	04:00	04:46	05:23	19:01	19:38	20:23
56°	03:51	04:40	05:19	19:04	19:43	20:32
58°	03:40	04:33	05:15	19:08	19:50	20:43
S 60°	03:27	04:25	05:10	19:13	19:57	20:55

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	14:56	17:24	19:47	07:35	07:02	06:34
N 70°	15:14	17:29	19:41	07:16	06:53	06:34
68°	15:27	17:33	19:37	07:00	06:46	06:34
66°	15:38	17:36	19:34	06:47	06:40	06:34
64°	15:48	17:39	19:31	06:36	06:35	06:34
62°	15:55	17:42	19:28	06:27	06:31	06:34
60°	16:02	17:44	19:26	06:19	06:27	06:34
N 58°	16:08	17:46	19:24	06:12	06:24	06:34
56°	16:13	17:48	19:22	06:06	06:20	06:34
54°	16:18	17:49	19:21	06:00	06:18	06:34
52°	16:22	17:50	19:19	05:55	06:15	06:34
50°	16:26	17:52	19:18	05:50	06:13	06:34
45°	16:34	17:54	19:15	05:40	06:08	06:34
N 40°	16:41	17:57	19:13	05:32	06:04	06:34
35°	16:47	17:59	19:11	05:24	06:00	06:34
30°	16:52	18:00	19:09	05:18	05:57	06:34
20°	17:01	18:03	19:06	05:07	05:51	06:34
N 10°	17:08	18:06	19:04	04:57	05:46	06:34
0°	17:16	18:09	19:01	04:47	05:41	06:34
S 10°	17:23	18:11	18:59	04:38	05:36	06:33
20°	17:30	18:14	18:57	04:28	05:31	06:33
30°	17:39	18:17	18:54	04:16	05:24	06:33
35°	17:44	18:19	18:52	04:09	05:21	06:33
40°	17:50	18:21	18:51	04:01	05:17	06:33
45°	17:56	18:23	18:49	03:52	05:12	06:33
S 50°	18:04	18:26	18:46	03:41	05:06	06:33
52°	18:08	18:27	18:45	03:36	05:04	06:32
54°	18:12	18:28	18:44	03:30	05:01	06:32
56°	18:16	18:30	18:43	03:24	04:58	06:32
58°	18:21	18:31	18:41	03:17	04:54	06:32
S 60°	18:26	18:33	18:40	03:08	04:50	06:32

Day	Sun		Mer. Pass	Moon		Age 14-16 96-100%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer. Pass. hh:mm	Lower Mer. Pass. hh:mm	
27	12:43	12:38	12:13	23:28	11:02	
28	12:32	12:26	12:12	—:—	11:55	
01	12:21	12:15	12:12	00:21	12:48	

March 02, 03, 04 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	SHA	Dec				
0	160°04.4	181°32.8	S10°29.8	327°21.1	S01°41.6	314°38.0	S08°57.0	124°33.5	N11°46.4	Alpheratz	357°31.6	29°15.1											
1	175°06.9	196°32.3	28.7	342°24.0	41.4	329°40.6	57.0	139°35.7	46.5	Ankaa	353°04.3	-42°08.9											
2	190°09.4	211°31.8	27.6	357°26.9	41.2	344°43.2	56.9	154°38.0	46.6	Schedar	349°27.5	56°42.0											
3	205°11.8	226°31.3	•• 26.4	12°29.9	•• 41.0	359°45.8	•• 56.8	169°40.2	•• 46.7	Diphda	348°44.2	-17°49.7											
4	220°14.3	241°30.8	25.3	27°32.8	40.8	14°48.4	56.8	184°42.5	46.8	Achernar	335°18.2	-57°05.5											
5	235°16.7	256°30.3	24.2	42°35.7	40.6	29°51.1	56.7	199°44.7	46.9	Hamal	327°47.6	23°36.0											
6	250°19.2	271°29.8	S10°23.0	57°38.6	S01°40.4	44°53.7	S08°56.6	214°47.0	N11°46.9	Polaris	312°22.3	89°23.4											
7	265°21.7	286°29.3	21.9	72°41.5	40.2	59°56.3	56.6	229°49.2	47.0	Acamar	315°09.4	-40°11.5											
8	280°24.1	301°28.8	20.8	87°44.5	40.0	74°58.9	56.5	244°51.5	47.1	Menkar	314°02.7	4°12.1											
9	295°26.6	316°28.4	•• 19.6	102°47.4	•• 39.7	90°01.5	•• 56.5	259°53.7	•• 47.2	Mirfak	308°23.5	49°58.0											
10	310°29.1	331°27.9	18.5	117°50.3	39.5	105°04.1	56.4	274°56.0	47.3	Aldebaran	290°35.7	16°34.0											
11	325°31.5	346°27.4	17.4	132°53.3	39.3	120°06.7	56.3	289°58.2	47.3	Rigel	281°00.6	-8°10.3											
12	340°34.0	1°26.9	S10°16.2	147°56.2	S01°39.1	135°09.3	S08°56.3	305°00.5	N11°47.4	Capella	280°16.8	46°01.7											
13	355°36.5	16°26.4	15.1	162°59.1	38.9	150°11.9	56.2	320°02.7	47.5	Bellatrix	278°19.2	6°22.4											
14	10°38.9	31°25.9	14.0	178°02.1	38.7	165°14.5	56.1	335°04.9	47.6	Elnath	277°57.5	28°37.8											
15	25°41.4	46°25.4	•• 12.8	193°05.0	•• 38.4	180°17.1	•• 56.1	350°07.2	•• 47.7	Alnilam	275°34.2	-1°11.2											
16	40°43.8	61°24.9	11.7	208°07.9	38.2	195°19.7	56.0	5°09.4	47.8	Betelgeuse	270°48.3	7°24.6											
17	55°46.3	76°24.4	10.5	223°10.9	38.0	210°22.3	56.0	20°11.7	47.8	Canopus	263°50.7	-52°43.0											
18	70°48.8	91°23.9	S10°09.4	238°13.8	S01°37.8	225°24.9	S08°55.9	35°13.9	N11°47.9	Sirius	258°23.1	-16°45.7											
19	85°51.2	106°23.4	08.3	253°16.7	37.6	240°27.5	55.8	50°16.2	48.0	Adhara	255°03.0	-29°01.0											
20	100°53.7	121°23.0	07.1	268°19.7	37.4	255°30.1	55.8	65°18.4	48.1	Procyon	244°47.1	5°08.8											
21	115°56.2	136°22.5	•• 06.0	283°22.6	•• 37.1	270°32.8	•• 55.7	80°20.7	•• 48.2	Pollux	243°13.0	27°57.2											
22	130°58.6	151°22.0	04.9	298°25.6	36.9	285°35.4	55.6	95°22.9	48.3	Avior	234°12.8	-59°36.5											
23	146°01.1	166°21.5	03.7	313°28.5	36.7	300°38.0	55.6	110°25.2	48.3	Suhail	222°43.4	-43°33.3											
Mer.pass. 13:18				ν -0.5' d -1.1' m -3.89				ν 2.9' d -0.2' m -0.83				ν 2.6' d -0.1' m -2.32				ν 2.2' d 0.1' m 0.38							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	161°03.6	181°21.0	S10°02.6	328°31.5	S01°36.5	315°40.6	S08°55.5	125°27.4	N11°48.4										
1	176°06.0	196°20.5	01.4	343°34.4	36.3	330°43.2	55.4	140°29.7	48.5										
2	191°08.5	211°20.0	10°00.3	358°37.4	36.0	345°45.8	55.4	155°31.9	48.6										
3	206°11.0	226°19.5	09°59.1	13°40.3	•• 35.8	0°48.4	•• 55.3	170°34.1	•• 48.7										
4	221°13.4	241°19.1	58.0	28°43.2	35.6	15°51.0	55.3	185°36.4	48.7										
5	236°15.9	256°18.6	56.9	43°46.2	35.4	30°53.6	55.2	200°38.6	48.8										
6	251°18.3	271°18.1	S09°55.7	58°49.2	S01°35.2	45°56.2	S08°55.1	215°40.9	N11°48.9										
7	266°20.8	286°17.6	54.6	73°52.1	34.9	60°58.8	55.1	230°43.1	49.0										
8	281°23.3	301°17.1	53.4	88°55.1	34.7	76°01.5	55.0	245°45.4	49.1										
9	296°25.7	316°16.6	•• 52.3	103°58.0	•• 34.5	91°04.1	•• 54.9	260°47.6	•• 49.2										
10	311°28.2	331°16.2	51.1	119°01.0	34.3	106°06.7	54.9	275°49.9	49.2										
11	326°30.7	346°15.7	50.0	134°03.9	34.0	121°09.3	54.8	290°52.1	49.3										
12	341°33.1	1°15.2	S09°48.9	149°06.9	S01°33.8	136°11.9	S08°54.7	305°54.3	N11°49.4										
13	356°35.6	16°14.7	47.7	164°09.9	33.6	151°14.5	54.7	320°56.6	49.5										
14	11°38.1	31°14.2	46.6	179°12.8	33.4	166°17.1	54.6	335°58.8	49.6										
15	26°40.5	46°13.7	•• 45.4	194°15.8	•• 33.1	181°19.7	•• 54.5	351°01.1	•• 49.7										
16	41°43.0	61°13.3	44.3	209°18.8	32.9	196°22.4	54.5	6°03.3	49.7										
17	56°45.4	76°12.8	43.1	224°21.7	32.7	211°25.0	54.4	21°05.6	49.8										
18	71°47.9	91°12.3	S09°42.0	239°24.7	S01°32.4	226°27.6	S08°54.3	36°07.8	N11°49.9										
19	86°50.4	106°11.8	40.8	254°27.7	32.2	241°30.2	54.3	51°10.0	50.0										
20	101°52.8	121°11.3	39.7	269°30.6	32.0	256°32.8	54.2	66°12.3	50.1										
21	116°55.3	136°10.9	•• 38.5	284°33.6	•• 31.8	271°35.4	•• 54.1	81°14.5	•• 50.2										
22	131°57.8	151°10.4	37.4	299°36.6	31.5	286°38.0	54.1	96°16.8	50.2										
23	147°00.2	166°09.9	36.2	314°39.5	31.3	301°40.7	54.0	111°19.0	50.3										
Mer.pass. 13:14				ν -0.5' d -1.1' m -3.89				ν 2.9' d -0.2' m -0.86				ν 2.6' d -0.1' m -2.32				ν 2.2' d 0.1' m 0.38			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	162°02.7	181°09.4	S09°35.1	329°42.5	S01°31.1	316°43.3	S08°53.9	126°21.3	N11°50.4										
1	177°05.2	196°08.9	33.9	344°45.5	30.8	331°45.9	53.9	141°23.5	50.5										
2	192°07.6	211°08.5	32.8	359°48.5	30.6	346°48.5	53.8	156°25.7	50.6										
3	207°10.1	226°08.0	•• 31.6	14°51.4	•• 30.4	1°51.1	•• 53.7	171°28.0	•• 50.7										
4	222°12.6	241°07.5	30.5	29°54.4	30.2	16°53.7	53.7	186°30.2	50.7										
5	237°15.0	256°07.0	29.3	44°57.4	29.9	31°56.4	53.6	201°32.5	50.8										
6	252°17.5	271°06.6	S09°28.1	60°00.4	S01°29.7	46°59.0	S08°53.5	216°34.7	N11°50.9										
7	267°19.9	286°06.1	27.0	75°03.4	29.5	62°01.6	53.5	231°37.0	51.0										
8	282°22.4	301°05.6	25.8	90°06.4	29.2	77°04.2	53.4	246°39.2	51.1										
9	297°24.9	316°05.1	•• 24.7	105°09.3	•• 29.0	92°06.8	•• 53.3	261°41.4	•• 51.2										
10	312°27.3	331°04.7	23.5	120°12.3	28.8	107°09.5	53.3	276°43.7	51.3										
11	327°29.8	346°04.2	22.4	135°15.3	28.5	122°12.1	53.2	291°45.9	51.3										
12	342°32.3	1°03.7	S09°21.2	150°18.3	S01°28.3	137°14.7	S08°53.1	306°48.2	N11°51.4										
13	357°34.7	16°03.2	20.1	165°21.3	28.0	152°17.3	53.1	321°50.4	51.5										
14	12°37.2	31°02.8	18.9	180°24.3	27.8	167°19.9	53.0	336°52.6	51.6										
15	27°39.7	46°02.3	•• 17.7	195°27.3	•• 27.6	182°22.5	•• 52.9	351°54.9	•• 51.7										
16	42°42.1	61°01.8	16.6	210°30.3	27.3	197°25.2	52.9	6°57.1	51.8										
17	57°44.6	76°01.4	15.4	225°33.3	27.1	212°27.8	52.8	21°59.4	51.8										
18	72°47.1	91°00.9	S09°14.3	240°36.3	S01°26.9	227°30.4	S08°52.7	37°01.6	N11°51.9										
19	87°49.5	106°00.4	13.1	255°39.3	26.6	242°33.0	52.7	52°03.8	52.0										
20	102°52.0	120°59.9	11.9	270°42.3	26.4	257°35.7	52.6	67°06.1	52.1										
21	117°54.4	135°59.5	•• 10.8	285°45.3	•• 26.1	272°38.3	•• 52.5	82°08.3	•• 52.2										
22	132°56.9	150°59.0	09.6	300°48.3	25.9	287°40.9	52.5	97°10.6	52.3										
23	147°59.4	165°58.5	08.5	315°51.3	25.7	302°43.5	52.4	112°12.8	52.3										
Mer.pass. 13:10				ν -0.5' d -1.2' m -3.89				ν 3.0' d -0.2' m -0.89				ν 2.6' d -0.1' m -2.33				ν 2.2' d 0.1' m 0.39			

Mar 02 Fri	SHA	Mer.pass
Venus	21°28.4	11:54
Mars	167°16.7	02:10
Jupiter	154°33.6	03:01
Saturn	324°29.1	15:39

Mar 03 Sat	SHA	Mer.pass
Venus	20°17.4	11:55
Mars	167°27.9	02:05
Jupiter	154°37.0	02:57
Saturn	324°23.8	15:36

Mar 04 Sun	SHA	Mer.pass
Venus	19°06.7	11:56
Mars	167°39.8	02:01
Jupiter	154°40.6	02:53
Saturn	324°18.6	15:32

Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

DUT1 = UT1-UTC = +0.2074 sec ΔT = TT-UT1 = +68.9766 sec

2029 March 02 to Mar. 04 UT

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°57.8	S07°10.7	342°09.0	8.9'	S04°29.9	15.7'	61.1'
1	191°58.0	09.8	356°36.9	8.9'	04°45.6	15.7'	61.1'
2	206°58.1	08.8	11°04.8	8.9'	05°01.3	15.7'	61.1'
3	221°58.2	· · 07.8	25°32.7	8.9'	05°17.0	15.6'	61.0'
4	236°58.3	06.9	40°00.5	8.8'	05°32.6	15.6'	61.0'
5	251°58.5	05.9	54°28.4	8.8'	05°48.2	15.6'	61.0'
6	266°58.6	S07°05.0	68°56.2	8.8'	S06°03.8	15.5'	61.0'
7	281°58.7	04.0	83°24.0	8.8'	06°19.3	15.5'	61.0'
8	296°58.9	03.1	97°51.7	8.7'	06°34.8	15.4'	61.0'
9	311°59.0	· · 02.1	112°19.5	8.7'	06°50.2	15.4'	61.0'
10	326°59.1	01.2	126°47.2	8.7'	07°05.6	15.3'	61.0'
11	341°59.2	07°00.2	141°14.8	8.7'	07°20.9	15.3'	61.0'
12	356°59.4	S06°59.2	155°42.5	8.6'	S07°36.2	15.2'	61.0'
13	11°59.5	58.3	170°10.1	8.6'	07°51.5	15.2'	61.0'
14	26°59.6	57.3	184°37.7	8.6'	08°06.6	15.1'	61.0'
15	41°59.8	· · 56.4	199°05.3	8.5'	08°21.8	15.1'	61.0'
16	56°59.9	55.4	213°32.8	8.5'	08°36.8	15.0'	60.9'
17	72°00.0	54.5	228°00.3	8.5'	08°51.8	14.9'	60.9'
18	87°00.1	S06°53.5	242°27.8	8.4'	S09°06.8	14.9'	60.9'
19	102°00.3	52.5	256°55.2	8.4'	09°21.7	14.8'	60.9'
20	117°00.4	51.6	271°22.6	8.4'	09°36.5	14.8'	60.9'
21	132°00.5	· · 50.6	285°50.0	8.3'	09°51.2	14.7'	60.9'
22	147°00.7	49.7	300°17.3	8.3'	10°05.9	14.6'	60.9'
23	162°00.8	48.7	314°44.6	8.3'	10°20.5	14.5'	60.9'
SD = 16.1' d = -1.0' SD = 16.7'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	177°00.9	S06°47.8	329°11.9	8.2'	S10°35.1	14.5'	60.8'
1	192°01.1	46.8	343°39.1	8.2'	10°49.5	14.4'	60.8'
2	207°01.2	45.8	358°06.3	8.2'	11°03.9	14.3'	60.8'
3	222°01.3	· · 44.9	12°33.5	8.1'	11°18.2	14.2'	60.8'
4	237°01.5	43.9	27°00.6	8.1'	11°32.5	14.2'	60.8'
5	252°01.6	43.0	41°27.6	8.0'	11°46.6	14.1'	60.8'
6	267°01.7	S06°42.0	55°54.7	8.0'	S12°00.7	14.0'	60.7'
7	282°01.9	41.0	70°21.7	8.0'	12°14.7	13.9'	60.7'
8	297°02.0	40.1	84°48.7	7.9'	12°28.6	13.8'	60.7'
9	312°02.1	· · 39.1	99°15.6	7.9'	12°42.4	13.7'	60.7'
10	327°02.3	38.2	113°42.5	7.8'	12°56.1	13.6'	60.7'
11	342°02.4	37.2	128°09.3	7.8'	13°09.8	13.5'	60.6'
12	357°02.5	S06°36.2	142°36.1	7.8'	S13°23.3	13.5'	60.6'
13	12°02.7	35.3	157°02.9	7.7'	13°36.8	13.4'	60.6'
14	27°02.8	34.3	171°29.6	7.7'	13°50.1	13.3'	60.6'
15	42°02.9	· · 33.4	185°56.3	7.6'	14°03.4	13.2'	60.6'
16	57°03.1	32.4	200°22.9	7.6'	14°16.5	13.1'	60.5'
17	72°03.2	31.4	214°49.5	7.6'	14°29.6	13.0'	60.5'
18	87°03.3	S06°30.5	229°16.1	7.5'	S14°42.6	12.9'	60.5'
19	102°03.5	29.5	243°42.6	7.5'	14°55.4	12.8'	60.5'
20	117°03.6	28.6	258°09.1	7.4'	15°08.2	12.6'	60.5'
21	132°03.7	· · 27.6	272°35.5	7.4'	15°20.8	12.5'	60.4'
22	147°03.9	26.6	287°01.9	7.4'	15°33.4	12.4'	60.4'
23	162°04.0	25.7	301°28.3	7.3'	15°45.8	12.3'	60.4'
SD = 16.1' d = -1.0' SD = 16.6'							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	177°04.2	S06°24.7	315°54.6	7.3'	S15°58.1	12.2'	60.4'
1	192°04.3	23.7	330°20.9	7.2'	16°10.3	12.1'	60.3'
2	207°04.4	22.8	344°47.1	7.2'	16°22.4	12.0'	60.3'
3	222°04.6	· · 21.8	359°13.3	7.1'	16°34.4	11.9'	60.3'
4	237°04.7	20.9	13°39.4	7.1'	16°46.2	11.7'	60.3'
5	252°04.8	19.9	28°05.5	7.1'	16°58.0	11.6'	60.2'
6	267°05.0	S06°18.9	42°31.6	7.0'	S17°09.6	11.5'	60.2'
7	282°05.1	18.0	56°57.6	7.0'	17°21.1	11.4'	60.2'
8	297°05.3	17.0	71°23.6	6.9'	17°32.5	11.3'	60.1'
9	312°05.4	· · 16.0	85°49.5	6.9'	17°43.8	11.1'	60.1'
10	327°05.5	15.1	100°15.4	6.9'	17°54.9	11.0'	60.1'
11	342°05.7	14.1	114°41.3	6.8'	18°05.9	10.9'	60.1'
12	357°05.8	S06°13.1	129°07.1	6.8'	S18°16.8	10.8'	60.0'
13	12°05.9	12.2	143°32.9	6.7'	18°27.6	10.6'	60.0'
14	27°06.1	11.2	157°58.6	6.7'	18°38.2	10.5'	60.0'
15	42°06.2	· · 10.3	172°24.3	6.7'	18°48.7	10.4'	60.0'
16	57°06.4	09.3	186°50.0	6.6'	18°59.1	10.2'	59.9'
17	72°06.5	08.3	201°15.6	6.6'	19°09.3	10.1'	59.9'
18	87°06.6	S06°07.4	215°41.2	6.6'	S19°19.4	10.0'	59.9'
19	102°06.8	06.4	230°06.8	6.5'	19°29.4	9.8'	59.8'
20	117°06.9	05.4	244°32.3	6.5'	19°39.2	9.7'	59.8'
21	132°07.1	· · 04.5	258°57.8	6.4'	19°48.9	9.6'	59.8'
22	147°07.2	03.5	273°23.2	6.4'	19°58.4	9.4'	59.7'
23	162°07.3	02.5	287°48.6	6.4'	20°07.9	9.3'	59.7'
SD = 16.1' d = -1.0' SD = 16.5'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:58	06:17	07:25	17:01	18:09	19:28
N 70°	05:05	06:15	07:17	17:09	18:10	19:21
68°	05:10	06:14	07:10	17:15	18:11	19:16
66°	05:14	06:13	07:04	17:21	18:12	19:12
64°	05:17	06:12	07:00	17:25	18:13	19:08
62°	05:20	06:11	06:56	17:29	18:14	19:05
60°	05:22	06:10	06:52	17:33	18:15	19:03
N 58°	05:24	06:09	06:49	17:36	18:15	19:01
56°	05:26	06:09	06:46	17:39	18:16	18:59
54°	05:27	06:08	06:43	17:41	18:17	18:58
52°	05:28	06:07	06:41	17:44	18:17	18:57
50°	05:29	06:06	06:39	17:46	18:18	18:56
45°	05:31	06:05	06:34	17:50	18:20	18:54
N 40°	05:32	06:03	06:30	17:54	18:21	18:53
35°	05:32	06:01	06:27	17:58	18:23	18:52
30°	05:32	06:00	06:24	18:01	18:25	18:52
20°	05:30	05:56	06:18	18:06	18:28	18:54
N 10°	05:28	05:52	06:13	18:11	18:32	18:56
0°	05:24	05:48	06:09	18:15	18:36	19:00
S 10°	05:18	05:43	06:04	18:20	18:41	19:06
20°	05:10	05:36	05:59	18:25	18:47	19:13
30°	05:00	05:28	05:52	18:31	18:55	19:23
35°	04:53	05:23	05:49	18:34	19:00	19:30
40°	04:45	05:17	05:45	18:38	19:06	19:38
45°	04:34	05:10	05:40	18:43	19:13	19:48
S 50°	04:21	05:01	05:34	18:48	19:21	20:01
52°	04:14	04:57	05:32	18:51	19:26	20:08
54°	04:07	04:52	05:29	18:54	19:30	20:15
56°	03:59	04:47	05:26	18:57	19:35	20:23
58°	03:49	04:41	05:22	19:00	19:41	20:32
S 60°	03:38	04:34	05:18	19:04	19:48	20:43

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	22:16	· · ·	01:19	06:05	05:31	04:25
N 70°	21:58	· · ·	00:30	06:14	05:51	05:16
68°	21:44	23:59	· · ·	06:21	06:07	05:48
66°	21:33	23:37	· · ·	06:27	06:20	06:12
64°	21:24	23:19	· · ·	06:32	06:31	06:31
62°	21:16	23:04	· · ·	06:37	06:41	06:46
60°	21:09	22:52	· · ·	06:41	06:49	06:59
N 58°	21:03	22:42	· · ·	06:44	06:56	07:11
56°	20:57	22:33	· · ·	06:47	07:02	07:21
54°	20:53	22:25	23:54	06:50	07:08	07:29
52°	20:49	22:18	23:44	06:53	07:13	07:37
50°	20:45	22:11	23:35	06:55	07:18	07:44
45°	20:36	21:57	23:17	07:00	07:28	08:00
N 40°	20:30	21:46	23:01	07:04	07:36	08:12
35°	20:24	21:37	22:49	07:08	07:44	08:23
30°	20:19	21:28	22:37	07:11	07:50	08:33
20°	20:10	21:14	22:18	07:17	08:02	08:49
N 10°	20:02	21:01	22:02	07:22	08:11	09:03
0°	19:55	20:50	21:47	07:27	08:21	09:17
S 10°	19:48	20:38	21:31	07:31	08:30	09:30
20°	19:41	20:26	21:15	07:36	08:40	09:45
30°	19:32	20:13	20:57	07:42	08:52	10:01
35°	19:27	20:05	20:46	07:46	08:58	10:11
40°	19:22	19:56	20:34	07:49	09:06	10:22
45°	19:16	19:45	20:20	07:54	09:15	10:35
S 50°	19:08	19:33	20:02	07:59	09:26	10:51
52°	19:05	19:27	19:54	08:01	09:31	10:59
54°	19:01	19:21	19:45	08:04	09:36	11:07
56°	18:57	19:14	19:35	08:07	09:42	11:17
58°	18:52	19:06	19:23	08:10	09:49	11:28
S 60°	18:47	18:57	19:10	08:14	09:57	11:40

Day	Sun		Mer. Pass	Moon		Age 17-19 97-84%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
02	12:09	12:03	12:12	01:14	13:41	
03	11:56	11:50	12:12	02:08	14:35	
04	11:43	11:37	12:12	03:03	15:32	

March 05, 06, 07 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	163°01.8	180°58.1	S09°07.3	330°54.3	S01°25.4	317°46.1	S08°52.3	127°15.0	N11°52.4	Alpheratz	357°31.6	29°15.1
1	178°04.3	195°57.6	06.1	345°57.3	25.2	332°48.8	52.2	142°17.3	52.5	Ankaa	353°04.3	-42°08.9
2	193°06.8	210°57.1	05.0	1°00.3	25.0	347°51.4	52.2	157°19.5	52.6	Schedar	349°27.5	56°42.0
3	208°09.2	225°56.7	•• 03.8	16°03.3	•• 24.7	2°54.0	•• 52.1	172°21.8	•• 52.7	Diphda	348°44.2	-17°49.7
4	223°11.7	240°56.2	02.6	31°06.3	24.5	17°56.6	52.0	187°24.0	52.8	Achernar	335°18.3	-57°05.5
5	238°14.2	255°55.7	01.5	46°09.3	24.2	32°59.3	52.0	202°26.2	52.8	Hamal	327°47.6	23°36.0
6	253°16.6	270°55.2	S09°00.3	61°12.3	S01°24.0	48°01.9	S08°51.9	217°28.5	N11°52.9	Polaris	312°23.5	89°23.4
7	268°19.1	285°54.8	08°59.1	76°15.3	23.7	63°04.5	51.8	232°30.7	53.0	Acamar	315°09.5	-40°11.5
8	283°21.5	300°54.3	58.0	91°18.3	23.5	78°07.1	51.8	247°33.0	53.1	Menkar	314°02.8	4°12.1
9	298°24.0	315°53.8	•• 56.8	106°21.4	•• 23.3	93°09.8	•• 51.7	262°35.2	•• 53.2	Mirfak	308°23.5	49°58.0
10	313°26.5	330°53.4	55.7	121°24.4	23.0	108°12.4	51.6	277°37.4	53.3	Aldebaran	290°35.8	16°34.0
11	328°28.9	345°52.9	54.5	136°27.4	22.8	123°15.0	51.5	292°39.7	53.4	Rigel	281°00.6	-8°10.3
12	343°31.4	0°52.4	S08°53.3	151°30.4	S01°22.5	138°17.6	S08°51.5	307°41.9	N11°53.4	Capella	280°16.8	46°01.6
13	358°33.9	15°52.0	52.2	166°33.4	22.3	153°20.3	51.4	322°44.1	53.5	Bellatrix	278°19.2	6°22.4
14	13°36.3	30°51.5	51.0	181°36.5	22.0	168°22.9	51.3	337°46.4	53.6	Elnath	277°57.5	28°37.8
15	28°38.8	45°51.1	•• 49.8	196°39.5	•• 21.8	183°25.5	•• 51.3	352°48.6	•• 53.7	Alnilam	275°34.2	-1°11.2
16	43°41.3	60°50.6	48.6	211°42.5	21.5	198°28.1	51.2	7°50.9	53.8	Betelgeuse	270°48.3	7°24.5
17	58°43.7	75°50.1	47.5	226°45.5	21.3	213°30.8	51.1	22°53.1	53.9	Canopus	263°50.7	-52°43.0
18	73°46.2	90°49.7	S08°46.3	241°48.5	S01°21.0	228°33.4	S08°51.1	37°55.3	N11°54.0	Sirius	258°23.1	-16°45.7
19	88°48.7	105°49.2	45.1	256°51.6	20.8	243°36.0	51.0	52°57.6	54.0	Adhara	255°03.0	-29°01.1
20	103°51.1	120°48.7	44.0	271°54.6	20.6	258°38.6	50.9	67°59.8	54.1	Procyon	244°47.1	5°08.7
21	118°53.6	135°48.3	•• 42.8	286°57.6	•• 20.3	273°41.3	•• 50.8	83°02.0	•• 54.2	Pollux	243°13.0	27°57.2
22	133°56.0	150°47.8	41.6	302°00.7	20.1	288°43.9	50.8	98°04.3	54.3	Avior	234°12.8	-59°36.5
23	148°58.5	165°47.4	40.5	317°03.7	19.8	303°46.5	50.7	113°06.5	54.4	Suhail	222°43.4	-43°33.3
Mer.pass. 13:06		ν -0.5' d -1.2' m -3.89		ν 3.0' d -0.2' m -0.92		ν 2.6' d -0.1' m -2.34		ν 2.2' d 0.1' m 0.39		Miaplacidus	221°36.6	-69°50.5
										Alphard	217°44.1	-87°47.4
										Regulus	207°30.6	11°49.2
										Dubhe	193°36.4	61°35.4
										Denebola	182°21.2	14°24.3
										Gienah	175°39.8	-17°42.4
										Acrux	172°55.5	-63°15.7
										Gacrux	171°47.2	-57°16.7
										Alioth	166°09.7	55°47.8
										Spica	158°18.5	-11°19.0
										Alkaid	152°49.1	49°09.7
										Hadar	148°30.6	-60°30.7
										Menkent	147°53.3	-36°30.8
										Arcturus	145°44.6	19°01.6
										Rigel Kent.	139°35.1	-60°57.2
										Kochab	137°19.5	74°01.8
										Zuben'ubi	136°52.1	-16°09.8
										Alphecca	126°00.8	26°36.7
										Antares	112°11.6	-26°29.7
										Atria	107°02.7	-69°04.4
										Sabik	101°58.9	-15°45.6
										Shaula	96°05.8	-37°07.3
										Rasalhague	95°55.5	12°32.2
										Eltanin	90°40.7	51°28.8
										Kaus Aust.	83°28.2	-34°22.0
										Vega	80°31.1	38°48.4
										Nunki	75°43.8	-26°15.5
										Altair	61°56.9	8°56.7
										Peacock	53°01.0	-56°38.2
										Deneb	49°23.8	45°23.0
										Enif	33°35.8	10°00.5
										Al Na'ir	27°29.2	-46°49.1
										Fomalhaut	15°11.2	-29°28.0
										Scheat	13°42.2	28°14.5
										Markab	13°26.8	15°21.7
Mar 05 Mon												
		SHA		Mer.pass								
		Venus		17°56.2		11:56						
		Mars		167°52.4		01:56						
		Jupiter		154°44.3		02:48						
		Saturn		324°13.2		15:29						
Mar 06 Tue												
		SHA		Mer.pass								
		Venus		16°45.9		11:57						
		Mars		168°05.7		01:51						
		Jupiter		154°48.2		02:44						
		Saturn		324°07.8		15:25						
Mar 07 Wed												
		SHA		Mer.pass								
		Venus		15°35.8		11:58						
		Mars		168°19.7		01:46						
		Jupiter		154°52.2		02:40						
		Saturn		324°02.3		15:22						
Horizontal parallax												
		Venus:		0.1								
		Mars:		0.2								

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	177°07.5	S06°01.6	302°14.0	6.3'	S20°17.1	9.1' 59.7'
1	192°07.6	06°00.6	316°39.4	6.3'	20°26.3	9.0' 59.7'
2	207°07.8	05°59.6	331°04.7	6.3'	20°35.3	8.9' 59.6'
3	222°07.9	· · 58.7	345°30.0	6.3'	20°44.1	8.7' 59.6'
4	237°08.1	57.7	359°55.2	6.2'	20°52.9	8.6' 59.6'
5	252°08.2	56.7	14°20.5	6.2'	21°01.4	8.4' 59.5'
6	267°08.3	S05°55.8	28°45.7	6.2'	S21°09.8	8.3' 59.5'
7	282°08.5	54.8	43°10.8	6.1'	21°18.1	8.1' 59.5'
8	297°08.6	53.8	57°36.0	6.1'	21°26.3	8.0' 59.4'
9	312°08.8	· · 52.9	72°01.1	6.1'	21°34.2	7.8' 59.4'
10	327°08.9	51.9	86°26.2	6.1'	21°42.1	7.7' 59.4'
11	342°09.0	50.9	100°51.2	6.0'	21°49.8	7.5' 59.3'
12	357°09.2	S05°50.0	115°16.3	6.0'	S21°57.3	7.4' 59.3'
13	12°09.3	49.0	129°41.3	6.0'	22°04.7	7.2' 59.3'
14	27°09.5	48.0	144°06.3	6.0'	22°11.9	7.1' 59.2'
15	42°09.6	· · 47.1	158°31.3	6.0'	22°19.0	6.9' 59.2'
16	57°09.8	46.1	172°56.3	5.9'	22°25.9	6.8' 59.2'
17	72°09.9	45.1	187°21.2	5.9'	22°32.7	6.6' 59.1'
18	87°10.1	S05°44.2	201°46.1	5.9'	S22°39.3	6.5' 59.1'
19	102°10.2	43.2	216°11.0	5.9'	22°45.8	6.3' 59.1'
20	117°10.3	42.2	230°35.9	5.9'	22°52.1	6.2' 59.0'
21	132°10.5	· · 41.2	245°00.8	5.9'	22°58.3	6.0' 59.0'
22	147°10.6	40.3	259°25.7	5.9'	23°04.3	5.9' 59.0'
23	162°10.8	39.3	273°50.6	5.9'	23°10.2	5.7' 58.9'
SD = 16.1' d = -1.0' SD = 16.3'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	177°10.9	S05°38.3	288°15.4	5.8'	S23°15.9	5.5' 58.9'
1	192°11.1	37.4	302°40.3	5.8'	23°21.4	5.4' 58.9'
2	207°11.2	36.4	317°05.1	5.8'	23°26.8	5.2' 58.8'
3	222°11.4	· · 35.4	331°29.9	5.8'	23°32.0	5.1' 58.8'
4	237°11.5	34.5	345°54.8	5.8'	23°37.1	4.9' 58.8'
5	252°11.6	33.5	0°19.6	5.8'	23°42.0	4.8' 58.7'
6	267°11.8	S05°32.5	14°44.4	5.8'	S23°46.8	4.6' 58.7'
7	282°11.9	31.5	29°09.2	5.8'	23°51.4	4.4' 58.7'
8	297°12.1	30.6	43°34.1	5.8'	23°55.8	4.3' 58.6'
9	312°12.2	· · 29.6	57°58.9	5.8'	24°00.1	4.1' 58.6'
10	327°12.4	28.6	72°23.7	5.8'	24°04.2	4.0' 58.6'
11	342°12.5	27.7	86°48.6	5.8'	24°08.2	3.8' 58.5'
12	357°12.7	S05°26.7	101°13.4	5.9'	S24°12.0	3.7' 58.5'
13	12°12.8	25.7	115°38.3	5.9'	24°15.6	3.5' 58.5'
14	27°13.0	24.8	130°03.1	5.9'	24°19.1	3.3' 58.4'
15	42°13.1	· · 23.8	144°28.0	5.9'	24°22.5	3.2' 58.4'
16	57°13.3	22.8	158°52.9	5.9'	24°25.6	3.0' 58.4'
17	72°13.4	21.8	173°17.8	5.9'	24°28.7	2.9' 58.3'
18	87°13.6	S05°20.9	187°42.7	5.9'	S24°31.5	2.7' 58.3'
19	102°13.7	19.9	202°07.6	6.0'	24°34.2	2.5' 58.3'
20	117°13.9	18.9	216°32.6	6.0'	24°36.8	2.4' 58.2'
21	132°14.0	· · 17.9	230°57.6	6.0'	24°39.1	2.2' 58.2'
22	147°14.2	17.0	245°22.6	6.0'	24°41.4	2.1' 58.1'
23	162°14.3	16.0	259°47.6	6.0'	24°43.4	1.9' 58.1'
SD = 16.1' d = -1.0' SD = 16.1'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	177°14.5	S05°15.0	274°12.6	6.1'	S24°45.4	1.8' 58.1'
1	192°14.6	14.1	288°37.7	6.1'	24°47.1	1.6' 58.0'
2	207°14.8	13.1	303°02.8	6.1'	24°48.7	1.4' 58.0'
3	222°14.9	· · 12.1	317°27.9	6.2'	24°50.1	1.3' 58.0'
4	237°15.1	11.1	331°53.1	6.2'	24°51.4	1.1' 57.9'
5	252°15.2	10.2	346°18.2	6.2'	24°52.6	1.0' 57.9'
6	267°15.4	S05°09.2	0°43.5	6.3'	S24°53.5	0.8' 57.9'
7	282°15.5	08.2	15°08.7	6.3'	24°54.4	0.7' 57.8'
8	297°15.7	07.2	29°34.0	6.3'	24°55.0	0.5' 57.8'
9	312°15.8	· · 06.3	43°59.3	6.4'	24°55.5	0.4' 57.8'
10	327°16.0	05.3	58°24.7	6.4'	24°55.9	0.2' 57.7'
11	342°16.1	04.3	72°50.1	6.4'	24°56.1	0.1' 57.7'
12	357°16.3	S05°03.4	87°15.5	6.5'	S24°56.1	-0.1' 57.7'
13	12°16.4	02.4	101°41.0	6.5'	24°56.0	-0.3' 57.6'
14	27°16.6	01.4	116°06.5	6.6'	24°55.8	-0.4' 57.6'
15	42°16.7	05°00.4	130°32.1	6.6'	24°55.4	-0.6' 57.6'
16	57°16.9	04°59.5	144°57.7	6.7'	24°54.8	-0.7' 57.5'
17	72°17.0	58.5	159°23.4	6.7'	24°54.1	-0.9' 57.5'
18	87°17.2	S04°57.5	173°49.1	6.8'	S24°53.3	-1.0' 57.5'
19	102°17.3	56.5	188°14.9	6.8'	24°52.3	-1.2' 57.4'
20	117°17.5	55.6	202°40.7	6.9'	24°51.1	-1.3' 57.4'
21	132°17.6	· · 54.6	217°06.6	6.9'	24°49.8	-1.4' 57.4'
22	147°17.8	53.6	231°32.5	7.0'	24°48.4	-1.6' 57.4'
23	162°17.9	52.6	245°58.5	7.0'	24°46.8	-1.7' 57.3'
SD = 16.1' d = -1.0' SD = 15.8'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:43	06:02	07:09	17:15	18:23	19:43
N 70°	04:51	06:02	07:03	17:21	18:23	19:34
68°	04:57	06:02	06:57	17:27	18:22	19:27
66°	05:03	06:02	06:53	17:31	18:22	19:22
64°	05:07	06:02	06:49	17:35	18:22	19:17
62°	05:10	06:02	06:46	17:38	18:22	19:14
60°	05:13	06:02	06:43	17:40	18:22	19:10
N 58°	05:16	06:01	06:40	17:43	18:22	19:08
56°	05:18	06:01	06:38	17:45	18:22	19:05
54°	05:20	06:01	06:36	17:47	18:23	19:04
52°	05:22	06:01	06:34	17:49	18:23	19:02
50°	05:23	06:00	06:33	17:51	18:23	19:00
45°	05:25	05:59	06:29	17:54	18:24	18:58
N 40°	05:27	05:58	06:25	17:57	18:25	18:56
35°	05:28	05:57	06:23	18:00	18:25	18:55
30°	05:29	05:56	06:20	18:03	18:27	18:54
20°	05:28	05:54	06:16	18:07	18:29	18:54
N 10°	05:26	05:51	06:12	18:11	18:32	18:56
0°	05:23	05:47	06:08	18:14	18:35	18:59
S 10°	05:18	05:43	06:04	18:18	18:39	19:04
20°	05:11	05:37	06:00	18:22	18:45	19:10
30°	05:02	05:30	05:54	18:27	18:51	19:20
35°	04:56	05:26	05:52	18:30	18:56	19:26
40°	04:48	05:21	05:48	18:33	19:01	19:33
45°	04:39	05:14	05:44	18:37	19:07	19:42
S 50°	04:27	05:06	05:39	18:42	19:15	19:54
52°	04:21	05:03	05:37	18:44	19:18	20:00
54°	04:14	04:58	05:35	18:46	19:23	20:07
56°	04:06	04:54	05:32	18:49	19:27	20:14
58°	03:58	04:48	05:29	18:52	19:32	20:23
S 60°	03:47	04:42	05:26	18:55	19:38	20:32

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	02:39	■	■	05:08	■	■
66°	01:48	■	■	06:00	■	■
64°	01:16	03:11	04:46	06:32	06:40	07:06
62°	00:53	02:35	03:57	06:57	07:16	07:55
60°	00:34	02:09	03:26	07:16	07:42	08:26
N 58°	00:19	01:49	03:03	07:32	08:03	08:50
56°	00:06	01:32	02:44	07:45	08:20	09:09
54°	· ·	01:17	02:28	07:57	08:35	09:25
52°	· ·	01:05	02:15	08:08	08:47	09:38
50°	· ·	00:54	02:03	08:17	08:59	09:50
45°	· ·	00:31	01:38	08:37	09:22	10:15
N 40°	· ·	00:13	01:18	08:53	09:41	10:35
35°	23:58	· ·	01:02	09:07	09:57	10:52
30°	23:45	· ·	00:48	09:19	10:10	11:06
20°	23:22	· ·	00:23	09:40	10:34	11:30
N 10°	23:03	· ·	00:03	09:57	10:54	11:51
0°	22:45	23:43	· ·	10:14	11:13	12:11
S 10°	22:27	23:24	· ·	10:31	11:32	12:30
20°	22:08	23:03	· ·	10:49	11:52	12:51
30°	21:46	22:39	23:36	11:10	12:15	13:15
35°	21:33	22:25	23:22	11:22	12:29	13:30
40°	21:18	22:09	23:06	11:36	12:45	13:46
45°	21:01	21:50	22:46	11:53	13:04	14:06
S 50°	20:39	21:26	22:22	12:13	13:28	14:30
52°	20:29	21:14	22:10	12:23	13:39	14:42
54°	20:17	21:01	21:57	12:34	13:52	14:55
56°	20:04	20:46	21:41	12:47	14:07	15:11
58°	19:49	20:28	21:23	13:02	14:25	15:29
S 60°	19:31	20:06	21:00	13:19	14:46	15:52

Day	Sun		Mer. Pass	Moon		Age 20-22 75-54%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer. Pass. hh:mm	Lower Mer. Pass. hh:mm	
05	11:30	11:23	12:11	04:00	16:29	
06	11:16	11:09	12:11	04:59	17:28	
07	11:02	10:55	12:11	05:57	18:26	

March 08, 09, 10 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	165°59.3	180°25.1	S07°42.6	334°33.6	S01°07.1	320°55.7	S08°47.1	129°56.0	N11°58.6					Alpheratz	357°31.6	29°15.1	
1	181°01.7	195°24.7	41.4	349°36.7	06.9	335°58.3	47.0	144°58.2	58.7					Ankaa	353°04.3	-42°08.9	
2	196°04.2	210°24.2	40.2	4°39.8	06.6	351°00.9	46.9	160°00.4	58.8					Schedar	349°27.5	56°42.0	
3	211°06.6	225°23.8	· · 39.0	19°42.9	· · 06.3	6°03.6	· · 46.9	175°02.7	· · 58.8					Diphda	348°44.2	-17°49.7	
4	226°09.1	240°23.4	37.8	34°46.0	06.1	21°06.2	46.8	190°04.9	58.9					Achernar	335°18.3	-57°05.5	
5	241°11.6	255°22.9	36.6	49°49.1	05.8	36°08.9	46.7	205°07.1	59.0					Hamal	327°47.6	23°36.0	
6	256°14.0	270°22.5	S07°35.4	64°52.2	S01°05.5	51°11.5	S08°46.6	220°09.4	N11°59.1					Polaris	312°24.6	89°23.4	
7	271°16.5	285°22.0	34.2	79°55.3	05.3	66°14.2	46.6	235°11.6	59.2					Acamar	315°09.5	-40°11.5	
8	286°19.0	300°21.6	33.0	94°58.4	05.0	81°16.8	46.5	250°13.8	59.3					Menkar	314°02.8	4°12.1	
9	301°21.4	315°21.1	· · 31.9	110°01.5	· · 04.7	96°19.5	· · 46.4	265°16.1	· · 59.4					Mirfak	308°23.5	49°58.0	
10	316°23.9	330°20.7	30.7	125°04.6	04.5	111°22.1	46.3	280°18.3	59.4					Aldebaran	290°35.8	16°34.0	
11	331°26.4	345°20.2	29.5	140°07.7	04.2	126°24.7	46.3	295°20.5	59.5					Rigel	281°00.6	-8°10.3	
12	346°28.8	0°19.8	S07°28.3	155°10.8	S01°03.9	141°27.4	S08°46.2	310°22.7	N11°59.6					Capella	280°16.8	46°01.6	
13	1°31.3	15°19.4	27.1	170°13.9	03.6	156°30.0	46.1	325°25.0	59.7					Bellatrix	278°19.2	6°22.4	
14	16°33.8	30°18.9	25.9	185°17.0	03.4	171°32.7	46.0	340°27.2	59.8					Elnath	277°57.5	28°37.8	
15	31°36.2	45°18.5	· · 24.7	200°20.1	· · 03.1	186°35.3	· · 45.9	355°29.4	11°59.9					Anilam	275°34.3	-1°11.2	
16	46°38.7	60°18.0	23.5	215°23.2	02.8	201°38.0	45.9	10°31.7	12°00.0					Betelgeuse	270°48.4	7°24.5	
17	61°41.1	75°17.6	22.3	230°26.3	02.5	216°40.6	45.8	25°33.9	00.1					Canopus	263°50.7	-52°43.1	
18	76°43.6	90°17.2	S07°21.1	245°29.4	S01°02.3	231°43.3	S08°45.7	40°36.1	N12°00.1					Sirius	258°23.1	-16°45.7	
19	91°46.1	105°16.7	19.9	260°32.5	02.0	246°45.9	45.6	55°38.3	00.2					Adhara	255°03.0	-29°01.1	
20	106°48.5	120°16.3	18.7	275°35.6	01.7	261°48.6	45.6	70°40.6	00.3					Procyon	244°47.1	5°08.7	
21	121°51.0	135°15.8	· · 17.5	290°38.7	· · 01.4	276°51.2	· · 45.5	85°42.8	· · 00.4					Pollux	243°13.0	27°57.2	
22	136°53.5	150°15.4	16.3	305°41.8	01.2	291°53.8	45.4	100°45.0	00.5					Avior	234°12.8	-59°36.5	
23	151°55.9	165°15.0	15.1	320°45.0	00.9	306°56.5	45.3	115°47.3	00.6					Suhail	222°43.4	-43°33.4	
Mer.pass. 12:54			ν -0.4' d -1.2' m -3.89	ν 3.1' d -0.3' m -1.00	ν 2.6' d -0.1' m -2.35	ν 2.2' d 0.1' m 0.39											

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	166°58.4	180°14.5	S07°13.9	335°48.1	S01°00.6	321°59.1	S08°45.2	130°49.5	N12°00.7
1	182°00.9	195°14.1	12.7	350°51.2	00.3	337°01.8	45.2	145°51.7	00.7
2	197°03.3	210°13.6	11.5	5°54.3	01°00.1	352°04.4	45.1	160°53.9	00.8
3	212°05.8	225°13.2	· · 10.3	20°57.4	00°59.8	7°07.1	· · 45.0	175°56.2	· · 00.9
4	227°08.3	240°12.8	09.1	36°00.5	59.5	22°09.7	44.9	190°58.4	01.0
5	242°10.7	255°12.3	07.9	51°03.7	59.2	37°12.4	44.9	206°00.6	01.1
6	257°13.2	270°11.9	S07°06.7	66°06.8	S00°59.0	52°15.0	S08°44.8	221°02.9	N12°01.2
7	272°15.6	285°11.5	05.5	81°09.9	58.7	67°17.7	44.7	236°05.1	01.3
8	287°18.1	300°11.0	04.3	96°13.0	58.4	82°20.3	44.6	251°07.3	01.4
9	302°20.6	315°10.6	· · 03.1	111°16.1	· · 58.1	97°23.0	· · 44.5	266°09.5	· · 01.4
10	317°23.0	330°10.1	01.9	126°19.3	57.8	112°25.6	44.5	281°11.8	01.5
11	332°25.5	345°09.7	07°00.7	141°22.4	57.6	127°28.3	44.4	296°14.0	01.6
12	347°28.0	0°09.3	S06°59.5	156°25.5	S00°57.3	142°30.9	S08°44.3	311°16.2	N12°01.7
13	2°30.4	15°08.8	58.3	171°28.7	57.0	157°33.6	44.2	326°18.4	01.8
14	17°32.9	30°08.4	57.1	186°31.8	56.7	172°36.2	44.1	341°20.7	01.9
15	32°35.4	45°08.0	· · 55.9	201°34.9	· · 56.4	187°38.9	· · 44.1	356°22.9	· · 02.0
16	47°37.8	60°07.5	54.7	216°38.0	56.2	202°41.5	44.0	11°25.1	02.1
17	62°40.3	75°07.1	53.5	231°41.2	55.9	217°44.2	43.9	26°27.4	02.1
18	77°42.7	90°06.7	S06°52.3	246°44.3	S00°55.6	232°46.9	S08°43.8	41°29.6	N12°02.2
19	92°45.2	105°06.2	51.1	261°47.4	55.3	247°49.5	43.7	56°31.8	02.3
20	107°47.7	120°05.8	49.9	276°50.6	55.0	262°52.2	43.7	71°34.0	02.4
21	122°50.1	135°05.4	· · 48.7	291°53.7	· · 54.8	277°54.8	· · 43.6	86°36.3	· · 02.5
22	137°52.6	150°04.9	47.5	306°56.9	54.5	292°57.5	43.5	101°38.5	02.6
23	152°55.1	165°04.5	46.3	322°00.0	54.2	308°00.1	43.4	116°40.7	02.7
Mer.pass. 12:50			ν -0.4' d -1.2' m -3.89	ν 3.1' d -0.3' m -1.03	ν 2.6' d -0.1' m -2.36	ν 2.2' d 0.1' m 0.39			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	167°57.5	180°04.1	S06°45.1	337°03.1	S00°53.9	323°02.8	S08°43.3	131°42.9	N12°02.8
1	183°00.0	195°03.6	43.8	352°06.3	53.6	338°05.4	43.3	146°45.2	02.8
2	198°02.5	210°03.2	42.6	7°09.4	53.3	353°08.1	43.2	161°47.4	02.9
3	213°04.9	225°02.8	· · 41.4	22°12.6	· · 53.1	8°10.7	· · 43.1	176°49.6	· · 03.0
4	228°07.4	240°02.4	40.2	37°15.7	52.8	23°13.4	43.0	191°51.8	03.1
5	243°09.9	255°01.9	39.0	52°18.9	52.5	38°16.0	42.9	206°54.1	03.2
6	258°12.3	270°01.5	S06°37.8	67°22.0	S00°52.2	53°18.7	S08°42.9	221°56.3	N12°03.3
7	273°14.8	285°01.1	36.6	82°25.1	51.9	68°21.4	42.8	236°58.5	03.4
8	288°17.2	300°00.6	35.4	97°28.3	51.6	83°24.0	42.7	252°00.7	03.5
9	303°19.7	315°00.2	· · 34.2	112°31.4	· · 51.3	98°26.7	· · 42.6	267°03.0	· · 03.6
10	318°22.2	329°59.8	33.0	127°34.6	51.0	113°29.3	42.5	282°05.2	03.6
11	333°24.6	344°59.3	31.8	142°37.7	50.8	128°32.0	42.5	297°07.4	03.7
12	348°27.1	359°58.9	S06°30.6	157°40.9	S00°50.5	143°34.6	S08°42.4	312°09.6	N12°03.8
13	3°29.6	14°58.5	29.3	172°44.1	50.2	158°37.3	42.3	327°11.9	03.9
14	18°32.0	29°58.1	28.1	187°47.2	49.9	173°40.0	42.2	342°14.1	04.0
15	33°34.5	44°57.6	· · 26.9	202°50.4	· · 49.6	188°42.6	· · 42.1	357°16.3	· · 04.1
16	48°37.0	59°57.2	25.7	217°53.5	49.3	203°45.3	42.0	12°18.5	04.2
17	63°39.4	74°56.8	24.5	232°56.7	49.0	218°47.9	42.0	27°20.8	04.3
18	78°41.9	89°56.3	S06°23.3	247°59.8	S00°48.7	233°50.6	S08°41.9	42°23.0	N12°04.3
19	93°44.4	104°55.9	22.1	263°03.0	48.5	248°53.2	41.8	57°25.2	04.4
20	108°46.8	119°55.5	20.9	278°06.2	48.2	263°55.9	41.7	72°27.4	04.5
21	123°49.3	134°55.1	· · 19.7	293°09.3	· · 47.9	278°58.6	· · 41.6	87°29.6	· · 04.6
22	138°51.7	149°54.6	18.4	308°12.5	47.6	294°01.2	41.6	102°31.9	04.7
23	153°54.2	164°54.2	17.2	323°15.6	47.3	309°03.9	41.5	117°34.1	04.8
Mer.pass. 12:46			ν -0.4' d -1.2' m -3.89	ν 3.1' d -0.3' m -1.06	ν 2.7' d -0.1' m -2.37	ν 2.2' d 0.1' m 0.39			

Mar 08 Thu	SHA	Mer.pass
Venus	14°25.9	11:59
Mars	168°34.4	01:41
Jupiter	154°56.4	02:36
Saturn	323°56.7	15:18

Mar 09 Fri	SHA	Mer.pass
Venus	13°16.1	11:59
Mars	168°49.7	01:36
Jupiter	155°00.7	02:32
Saturn	323°51.1	15:14

Mar 10 Sat	SHA	Mer.pass
Venus	12°06.5	12:00
Mars	169°05.6	01:31
Jupiter	155°05.2	02:27
Saturn	323°45.4	15:11

Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

DUT1 = UT1-UTC = +0.2076 sec ΔT = TT-UT1 = +68.9764 sec

2029 March 08 to Mar. 10 UT

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	177°18.1	S04°51.7	260°24.5	7.1'	S24°45.1	-1.9'	57.3'
1	192°18.2	50.7	274°50.6	7.2'	24°43.2	-2.0'	57.3'
2	207°18.4	49.7	289°16.8	7.2'	24°41.1	-2.2'	57.2'
3	222°18.5	· · 48.7	303°43.0	7.3'	24°39.0	-2.3'	57.2'
4	237°18.7	47.8	318°09.3	7.3'	24°36.7	-2.5'	57.2'
5	252°18.8	46.8	332°35.6	7.4'	24°34.2	-2.6'	57.1'
6	267°19.0	S04°45.8	347°02.0	7.5'	S24°31.6	-2.7'	57.1'
7	282°19.2	44.8	1°28.5	7.5'	24°28.9	-2.9'	57.1'
8	297°19.3	43.8	15°55.0	7.6'	24°26.0	-3.0'	57.0'
9	312°19.5	· · 42.9	30°21.6	7.7'	24°23.0	-3.1'	57.0'
10	327°19.6	41.9	44°48.3	7.7'	24°19.9	-3.3'	57.0'
11	342°19.8	40.9	59°15.0	7.8'	24°16.6	-3.4'	56.9'
12	357°19.9	S04°39.9	73°41.8	7.9'	S24°13.2	-3.6'	56.9'
13	12°20.1	39.0	88°08.6	7.9'	24°09.6	-3.7'	56.9'
14	27°20.2	38.0	102°35.6	8.0'	24°05.9	-3.8'	56.9'
15	42°20.4	· · 37.0	117°02.6	8.1'	24°02.1	-3.9'	56.8'
16	57°20.5	36.0	131°29.7	8.1'	23°58.2	-4.1'	56.8'
17	72°20.7	35.1	145°56.8	8.2'	23°54.1	-4.2'	56.8'
18	87°20.9	S04°34.1	160°24.0	8.3'	S23°49.9	-4.3'	56.7'
19	102°21.0	33.1	174°51.3	8.4'	23°45.6	-4.5'	56.7'
20	117°21.2	32.1	189°18.7	8.4'	23°41.1	-4.6'	56.7'
21	132°21.3	· · 31.1	203°46.1	8.5'	23°36.5	-4.7'	56.6'
22	147°21.5	30.2	218°13.7	8.6'	23°31.8	-4.8'	56.6'
23	162°21.6	29.2	232°41.3	8.7'	23°26.9	-5.0'	56.6'
SD = 16.1'			SD = 15.6'				
$d = -1.0'$							
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	177°21.8	S04°28.2	247°08.9	8.8'	S23°22.0	-5.1'	56.6'
1	192°22.0	27.2	261°36.7	8.8'	23°16.9	-5.2'	56.5'
2	207°22.1	26.3	276°04.5	8.9'	23°11.7	-5.3'	56.5'
3	222°22.3	· · 25.3	290°32.4	9.0'	23°06.3	-5.5'	56.5'
4	237°22.4	24.3	305°00.4	9.1'	23°00.9	-5.6'	56.4'
5	252°22.6	23.3	319°28.5	9.1'	22°55.3	-5.7'	56.4'
6	267°22.7	S04°22.3	333°56.6	9.2'	S22°49.6	-5.8'	56.4'
7	282°22.9	21.4	348°24.8	9.3'	22°43.8	-5.9'	56.4'
8	297°23.1	20.4	2°53.2	9.4'	22°37.9	-6.0'	56.3'
9	312°23.2	· · 19.4	17°21.5	9.5'	22°31.8	-6.2'	56.3'
10	327°23.4	18.4	31°50.0	9.5'	22°25.7	-6.3'	56.3'
11	342°23.5	17.4	46°18.6	9.6'	22°19.4	-6.4'	56.3'
12	357°23.7	S04°16.5	60°47.2	9.7'	S22°13.1	-6.5'	56.2'
13	12°23.8	15.5	75°15.9	9.8'	22°06.6	-6.6'	56.2'
14	27°24.0	14.5	89°44.7	9.9'	22°00.0	-6.7'	56.2'
15	42°24.2	· · 13.5	104°13.6	10.0'	21°53.3	-6.8'	56.1'
16	57°24.3	12.5	118°42.5	10.0'	21°46.5	-6.9'	56.1'
17	72°24.5	11.6	133°11.6	10.1'	21°39.6	-7.0'	56.1'
18	87°24.6	S04°10.6	147°40.7	10.2'	S21°32.5	-7.1'	56.1'
19	102°24.8	9.6	162°09.9	10.3'	21°25.4	-7.2'	56.0'
20	117°25.0	8.6	176°39.2	10.4'	21°18.2	-7.3'	56.0'
21	132°25.1	· · 07.7	191°08.5	10.5'	21°10.9	-7.4'	56.0'
22	147°25.3	06.7	205°38.0	10.5'	21°03.4	-7.5'	56.0'
23	162°25.4	05.7	220°07.5	10.6'	20°55.9	-7.6'	55.9'
SD = 16.1'			SD = 15.4'				
$d = -1.0'$							
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	177°25.6	S04°04.7	234°37.1	10.7'	S20°48.3	-7.7'	55.9'
1	192°25.8	03.7	249°06.8	10.8'	20°40.5	-7.8'	55.9'
2	207°25.9	02.7	263°36.6	10.9'	20°32.7	-7.9'	55.9'
3	222°26.1	· · 01.8	278°06.5	10.9'	20°24.8	-8.0'	55.8'
4	237°26.2	04°00.8	292°36.4	11.0'	20°16.8	-8.1'	55.8'
5	252°26.4	03°59.8	307°06.5	11.1'	20°08.7	-8.2'	55.8'
6	267°26.6	S03°58.8	321°36.6	11.2'	S20°00.5	-8.3'	55.8'
7	282°26.7	57.8	336°06.8	11.3'	19°52.2	-8.4'	55.7'
8	297°26.9	56.9	350°37.0	11.4'	19°43.8	-8.5'	55.7'
9	312°27.0	· · 55.9	5°07.4	11.4'	19°35.4	-8.6'	55.7'
10	327°27.2	54.9	19°37.8	11.5'	19°26.8	-8.6'	55.7'
11	342°27.4	53.9	34°08.3	11.6'	19°18.2	-8.7'	55.6'
12	357°27.5	S03°52.9	48°38.9	11.7'	S19°09.4	-8.8'	55.6'
13	12°27.7	52.0	63°09.6	11.8'	19°00.6	-8.9'	55.6'
14	27°27.8	51.0	77°40.4	11.8'	18°51.7	-9.0'	55.6'
15	42°28.0	· · 50.0	92°11.2	11.9'	18°42.8	-9.1'	55.6'
16	57°28.2	49.0	106°42.1	12.0'	18°33.7	-9.1'	55.5'
17	72°28.3	48.0	121°13.1	12.1'	18°24.6	-9.2'	55.5'
18	87°28.5	S03°47.0	135°44.2	12.1'	S18°15.3	-9.3'	55.5'
19	102°28.7	46.1	150°15.3	12.2'	18°06.0	-9.4'	55.5'
20	117°28.8	45.1	164°46.5	12.3'	17°56.7	-9.4'	55.4'
21	132°29.0	· · 44.1	179°17.8	12.4'	17°47.2	-9.5'	55.4'
22	147°29.1	43.1	193°49.2	12.5'	17°37.7	-9.6'	55.4'
23	162°29.3	42.1	208°20.7	12.5'	17°28.1	-9.7'	55.4'
SD = 16.1'			SD = 15.2'				
$d = -1.0'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:27	05:47	06:54	17:29	18:37	19:58
N 70°	04:37	05:48	06:49	17:34	18:35	19:47
68°	04:45	05:50	06:45	17:38	18:33	19:39
66°	04:51	05:51	06:42	17:41	18:32	19:32
64°	04:56	05:52	06:39	17:44	18:31	19:27
62°	05:01	05:52	06:36	17:46	18:30	19:22
60°	05:04	05:53	06:34	17:48	18:29	19:18
N 58°	05:08	05:53	06:32	17:50	18:29	19:15
56°	05:10	05:54	06:30	17:51	18:29	19:12
54°	05:13	05:54	06:29	17:53	18:28	19:09
52°	05:15	05:54	06:27	17:54	18:28	19:07
50°	05:17	05:54	06:26	17:56	18:28	19:05
45°	05:20	05:54	06:23	17:58	18:28	19:02
N 40°	05:23	05:54	06:21	18:01	18:28	18:59
35°	05:24	05:53	06:19	18:03	18:28	18:57
30°	05:25	05:53	06:17	18:05	18:28	18:56
20°	05:26	05:51	06:13	18:08	18:30	18:55
N 10°	05:25	05:49	06:10	18:11	18:32	18:56
0°	05:22	05:46	06:07	18:14	18:34	18:58
S 10°	05:18	05:43	06:04	18:17	18:38	19:02
20°	05:13	05:38	06:01	18:20	18:42	19:08
30°	05:04	05:32	05:57	18:24	18:48	19:16
35°	04:59	05:29	05:54	18:26	18:52	19:21
40°	04:52	05:24	05:51	18:29	18:56	19:28
45°	04:43	05:19	05:48	18:32	19:01	19:37
S 50°	04:32	05:12	05:44	18:35	19:08	19:47
52°	04:27	05:08	05:43	18:37	19:11	19:53
54°	04:21	05:05	05:41	18:39	19:15	19:58
56°	04:14	05:00	05:38	18:41	19:19	20:05
58°	04:06	04:56	05:36	18:43	19:24	20:13
S 60°	03:57	04:50	05:33	18:46	19:29	20:22

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	08:06	■	■	09:26
68°	■	■	06:52	■	■	10:39
66°	■	06:31	06:15	■	09:13	11:15
64°	05:32	05:45	05:48	08:19	09:59	11:41
62°	04:49	05:15	05:28	09:03	10:29	12:01
60°	04:20	04:52	05:11	09:31	10:52	12:17
N 58°	03:57	04:33	04:56	09:54	11:10	12:31
56°	03:39	04:18	04:44	10:12	11:25	12:43
54°	03:24	04:04	04:33	10:27	11:38	12:53
52°	03:10	03:53	04:24	10:40	11:49	13:02
50°	02:59	03:42	04:15	10:52	11:59	13:10
45°	02:34	03:20	03:57	11:16	12:21	13:27
N 40°	02:15	03:03	03:42	11:35	12:38	13:41
35°	01:59	02:48	03:29	11:51	12:52	13:53
30°	01:45	02:35	03:18	12:05	13:04	14:03
20°	01:21	02:13	02:59	12:28	13:25	14:21
N 10°	01:00	01:53	02:43	12:48	13:44	14:36
0°	00:40	01:35	02:27	13:07	14:01	14:50
S 10°	00:21	01:17	02:11	13:26	14:17	15:05
20°	00:00	00:58	01:55	13:46	14:35	15:20
30°	· · ·	00:36	01:35	14:09	14:56	15:37
35°	· · ·	00:23	01:24	14:23	15:08	15:47
40°	· · ·	00:07	01:11	14:38	15:22	15:58
45°	23:49	· · ·	00:55	14:57	15:38	16:11
S 50°	23:27	· · ·	00:36	15:20	15:58	16:27
52°	23:16	· · ·	00:27	15:31	16:07	16:35
54°	23:04	· · ·	00:17	15:43	16:18	16:43
56°	22:50	· · ·	00:06	15:58	16:30	16:52
58°	22:33	23:52	· · ·	16:14	16:44	17:03
S 60°	22:14	23:37	· · ·	16:35	17:00	17:15

Day	Sun		Mer. Pass	Moon		Age 23-25 43-24%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
08	10:48	10:40	12:11	06:54	19:21	
09	10:33	10:25	12:10	07:48	20:14	
10	10:18	10:10	12:10	08:39	21:03	

March 11, 12, 13 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	168°56.7	179°53.8	S06°16.0	338°18.8	S00°47.0	324°06.5	S08°41.4	132°36.3	N12°04.9	Alpheratz	357°31.6	29°15.1	
1	183°59.1	194°53.4	14.8	353°22.0	46.7	339°09.2	41.3	147°38.5	05.0	Ankaa	353°04.3	-42°08.9	
2	199°01.6	209°52.9	13.6	8°25.1	46.4	354°11.9	41.2	162°40.8	05.1	Schedar	349°27.5	56°42.0	
3	214°04.1	224°52.5	•• 12.4	23°28.3	•• 46.1	9°14.5	•• 41.1	177°43.0	•• 05.1	Diphda	348°44.2	-17°49.7	
4	229°06.5	239°52.1	11.2	38°31.5	45.8	24°17.2	41.1	192°45.2	05.2	Achernar	335°18.3	-57°05.5	
5	244°09.0	254°51.7	09.9	53°34.7	45.5	39°19.9	41.0	207°47.4	05.3	Hamal	327°47.6	23°36.0	
6	259°11.5	269°51.2	S06°08.7	68°37.8	S00°45.2	54°22.5	S08°40.9	222°49.7	N12°05.4	Polaris	312°26.0	89°23.4	
7	274°13.9	284°50.8	07.5	83°41.0	45.0	69°25.2	40.8	237°51.9	05.5	Acamar	315°09.5	-40°11.5	
8	289°16.4	299°50.4	06.3	98°44.2	44.7	84°27.8	40.7	252°54.1	05.6	Menkar	314°02.8	4°12.1	
9	304°18.8	314°50.0	•• 05.1	113°47.3	•• 44.4	99°30.5	•• 40.6	267°56.3	•• 05.7	Mirfak	308°23.5	49°58.0	
10	319°21.3	329°49.6	03.9	128°50.5	44.1	114°33.2	40.6	282°58.5	05.8	Aldebaran	290°35.8	16°34.0	
11	334°23.8	344°49.1	02.6	143°53.7	43.8	129°35.8	40.5	298°00.8	05.9	Rigel	281°00.6	-8°10.3	
12	349°26.2	359°48.7	S06°01.4	158°56.9	S00°43.5	144°38.5	S08°40.4	313°03.0	N12°05.9	Capella	280°16.9	46°01.7	
13	4°28.7	14°48.3	06°00.2	174°00.0	43.2	159°41.2	40.3	328°05.2	06.0	Bellatrix	278°19.2	6°22.4	
14	19°31.2	29°47.9	05°59.0	189°03.2	42.9	174°43.8	40.2	343°07.4	06.1	Elnath	277°57.6	28°37.8	
15	34°33.6	44°47.4	•• 57.8	204°06.4	•• 42.6	189°46.5	•• 40.1	358°09.6	•• 06.2	Alnilam	275°34.3	-1°11.2	
16	49°36.1	59°47.0	56.6	219°09.6	42.3	204°49.1	40.1	13°11.9	06.3	Betelgeuse	270°48.4	7°24.5	
17	64°38.6	74°46.6	55.3	234°12.8	42.0	219°51.8	40.0	28°14.1	06.4	Canopus	263°50.7	-52°43.1	
18	79°41.0	89°46.2	S05°54.1	249°16.0	S00°41.7	234°54.5	S08°39.9	43°16.3	N12°06.5	Sirius	258°23.1	-16°45.7	
19	94°43.5	104°45.8	52.9	264°19.1	41.4	249°57.1	39.8	58°18.5	06.6	Adhara	255°03.0	-29°01.1	
20	109°46.0	119°45.3	51.7	279°22.3	41.1	264°59.8	39.7	73°20.8	06.7	Procyon	244°47.1	5°08.7	
21	124°48.4	134°44.9	•• 50.5	294°25.5	•• 40.8	280°02.5	•• 39.6	88°23.0	•• 06.7	Pollux	243°13.0	27°57.2	
22	139°50.9	149°44.5	49.2	309°28.7	40.5	295°05.1	39.5	103°25.2	06.8	Avior	234°12.8	-59°36.5	
23	154°53.3	164°44.1	48.0	324°31.9	40.2	310°07.8	39.5	118°27.4	06.9	Suhail	222°43.4	-43°33.4	
Mer.pass. 12:42		ν -0.4' d -1.2' m -3.89		ν 3.2' d -0.3' m -1.09		ν 2.7' d -0.1' m -2.37		ν 2.2' d 0.1' m 0.39		Miaplacidus	221°36.6	-69°50.5	
										Alphard	217°44.1	-8°47.4	
										Regulus	207°30.6	11°49.2	
										Dubhe	193°36.4	61°35.4	
										Denebola	182°21.2	14°24.3	
										Gienah	175°39.7	-17°42.5	
										Acrux	172°55.4	-63°15.7	
										Gacrux	171°47.1	-57°16.7	
										Alioth	166°09.6	55°47.8	
										Spica	158°18.4	-11°19.0	
										Alkaid	152°49.0	49°09.7	
										Hadar	148°30.5	-60°30.7	
										Menkent	147°53.2	-36°30.8	
										Arcturus	145°44.6	19°01.6	
										Rigel Kent.	139°35.0	-60°57.2	
										Kochab	137°19.4	74°01.8	
										Zuben'ubi	136°52.0	-16°09.9	
										Alphecca	126°00.7	26°36.7	
										Antares	112°11.6	-26°29.7	
										Atria	107°02.6	-69°04.4	
										Sabik	101°58.9	-15°45.6	
										Shaula	96°05.8	-37°07.3	
										Rasalhague	95°55.4	12°32.2	
										Eltanin	90°40.7	51°28.8	
										Kaus Aust.	83°28.1	-34°22.0	
										Vega	80°31.0	38°48.4	
										Nunki	75°43.7	-26°15.5	
										Altair	61°56.8	8°56.7	
										Peacock	53°00.9	-56°38.2	
										Deneb	49°23.8	45°23.0	
										Enif	33°35.7	10°00.5	
										Al Na'ir	27°29.2	-46°49.0	
										Fomalhaut	15°11.2	-29°28.0	
										Scheat	13°42.2	28°14.5	
										Markab	13°26.8	15°21.7	
										Mar 11 Sun		SHA	Mer.pass
										Venus	10°57.1	12:01	
										Mars	169°22.1	01:26	
										Jupiter	155°09.9	02:23	
										Saturn	323°39.6	15:07	
										Mar 12 Mon		SHA	Mer.pass
										Venus	9°47.8	12:01	
										Mars	169°39.3	01:21	
										Jupiter	155°14.7	02:19	
										Saturn	323°33.8	15:04	
										Mar 13 Tue		SHA	Mer.pass
										Venus	8°38.7	12:02	
										Mars	169°56.9	01:16	
										Jupiter	155°19.6	02:15	
										Saturn	323°27.9	15:00	
										Horizontal parallax			
										Venus:	0.1		
										Mars:	0.2		

DUT1 = UT1-UTC = +0.2077 sec ΔT = TT-UT1 = +68.9763 sec

2029 March 11 to Mar. 13 UT

h Sun Moon						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	177°29.5	S03°41.2	222°52.2	12.6'	S17°18.4	-9.7' 55.4'
1	192°29.6	40.2	237°23.8	12.7'	17°08.7	-9.8' 55.3'
2	207°29.8	39.2	251°55.5	12.8'	16°58.9	-9.9' 55.3'
3	222°30.0	· · 38.2	266°27.2	12.8'	16°49.0	-10.0' 55.3'
4	237°30.1	37.2	280°59.1	12.9'	16°39.0	-10.0' 55.3'
5	252°30.3	36.2	295°31.0	13.0'	16°29.0	-10.1' 55.3'
6	267°30.5	S03°35.3	310°02.9	13.0'	S16°18.9	-10.2' 55.2'
7	282°30.6	34.3	324°35.0	13.1'	16°08.8	-10.2' 55.2'
8	297°30.8	33.3	339°07.1	13.2'	15°58.5	-10.3' 55.2'
9	312°30.9	· · 32.3	353°39.3	13.3'	15°48.3	-10.3' 55.2'
10	327°31.1	31.3	8°11.5	13.3'	15°37.9	-10.4' 55.2'
11	342°31.3	30.3	22°43.9	13.4'	15°27.5	-10.5' 55.1'
12	357°31.4	S03°29.4	37°16.3	13.5'	S15°17.0	-10.5' 55.1'
13	12°31.6	28.4	51°48.7	13.5'	15°06.5	-10.6' 55.1'
14	27°31.8	27.4	66°21.2	13.6'	14°55.9	-10.6' 55.1'
15	42°31.9	· · 26.4	80°53.8	13.7'	14°45.3	-10.7' 55.1'
16	57°32.1	25.4	95°26.5	13.7'	14°34.6	-10.8' 55.0'
17	72°32.3	24.4	109°59.2	13.8'	14°23.8	-10.8' 55.0'
18	87°32.4	S03°23.5	124°32.0	13.9'	S14°13.0	-10.9' 55.0'
19	102°32.6	22.5	139°04.9	13.9'	14°02.1	-10.9' 55.0'
20	117°32.8	21.5	153°37.8	14.0'	13°51.2	-11.0' 55.0'
21	132°32.9	· · 20.5	168°10.8	14.0'	13°40.2	-11.0' 55.0'
22	147°33.1	19.5	182°43.8	14.1'	13°29.2	-11.1' 54.9'
23	162°33.3	18.5	197°16.9	14.2'	13°18.1	-11.1' 54.9'
SD = 16.1' d = -1.0' SD = 15.1'						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	177°33.4	S03°17.6	211°50.1	14.2'	S13°07.0	-11.2' 54.9'
1	192°33.6	16.6	226°23.3	14.3'	12°55.8	-11.2' 54.9'
2	207°33.8	15.6	240°56.6	14.3'	12°44.6	-11.3' 54.9'
3	222°33.9	· · 14.6	255°30.0	14.4'	12°33.3	-11.3' 54.8'
4	237°34.1	13.6	270°03.4	14.5'	12°22.0	-11.4' 54.8'
5	252°34.3	12.6	284°36.9	14.5'	12°10.6	-11.4' 54.8'
6	267°34.4	S03°11.7	299°10.4	14.6'	S11°59.2	-11.4' 54.8'
7	282°34.6	10.7	313°43.9	14.6'	11°47.8	-11.5' 54.8'
8	297°34.8	09.7	328°17.6	14.7'	11°36.3	-11.5' 54.8'
9	312°34.9	· · 08.7	342°51.3	14.7'	11°24.8	-11.6' 54.8'
10	327°35.1	07.7	357°25.0	14.8'	11°13.2	-11.6' 54.7'
11	342°35.3	06.7	11°58.8	14.8'	11°01.6	-11.7' 54.7'
12	357°35.4	S03°05.7	26°32.6	14.9'	S10°49.9	-11.7' 54.7'
13	12°35.6	04.8	41°06.5	14.9'	10°38.2	-11.7' 54.7'
14	27°35.8	03.8	55°40.5	15.0'	10°26.5	-11.8' 54.7'
15	42°35.9	· · 02.8	70°14.4	15.0'	10°14.8	-11.8' 54.7'
16	57°36.1	01.8	84°48.5	15.1'	10°03.0	-11.8' 54.6'
17	72°36.3	03°00.8	99°22.6	15.1'	09°51.1	-11.9' 54.6'
18	87°36.4	S02°59.8	113°56.7	15.2'	S09°39.3	-11.9' 54.6'
19	102°36.6	58.8	128°30.9	15.2'	09°27.4	-11.9' 54.6'
20	117°36.8	57.9	143°05.1	15.3'	09°15.4	-12.0' 54.6'
21	132°36.9	· · 56.9	157°39.4	15.3'	09°03.5	-12.0' 54.6'
22	147°37.1	55.9	172°13.7	15.4'	08°51.5	-12.0' 54.6'
23	162°37.3	54.9	186°48.1	15.4'	08°39.5	-12.0' 54.5'
SD = 16.1' d = -1.0' SD = 15.0'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	177°37.4	S02°53.9	201°22.5	15.4'	S08°27.4	-12.1' 54.5'
1	192°37.6	52.9	215°56.9	15.5'	08°15.3	-12.1' 54.5'
2	207°37.8	52.0	230°31.4	15.5'	08°03.2	-12.1' 54.5'
3	222°38.0	· · 51.0	245°05.9	15.6'	07°51.1	-12.2' 54.5'
4	237°38.1	50.0	259°40.5	15.6'	07°39.0	-12.2' 54.5'
5	252°38.3	49.0	274°15.1	15.6'	07°26.8	-12.2' 54.5'
6	267°38.5	S02°48.0	288°49.7	15.7'	S07°14.6	-12.2' 54.4'
7	282°38.6	47.0	303°24.4	15.7'	07°02.3	-12.2' 54.4'
8	297°38.8	46.0	317°59.1	15.7'	06°50.1	-12.3' 54.4'
9	312°39.0	· · 45.1	332°33.8	15.8'	06°37.8	-12.3' 54.4'
10	327°39.1	44.1	347°08.6	15.8'	06°25.5	-12.3' 54.4'
11	342°39.3	43.1	1°43.4	15.8'	06°13.2	-12.3' 54.4'
12	357°39.5	S02°42.1	16°18.2	15.9'	S06°00.9	-12.3' 54.4'
13	12°39.7	41.1	30°53.1	15.9'	05°48.6	-12.4' 54.4'
14	27°39.8	40.1	45°28.0	15.9'	05°36.2	-12.4' 54.4'
15	42°40.0	· · 39.1	60°02.9	16.0'	05°23.8	-12.4' 54.3'
16	57°40.2	38.1	74°37.9	16.0'	05°11.4	-12.4' 54.3'
17	72°40.3	37.2	89°12.9	16.0'	04°59.0	-12.4' 54.3'
18	87°40.5	S02°36.2	103°47.9	16.0'	S04°46.6	-12.4' 54.3'
19	102°40.7	35.2	118°23.0	16.1'	04°34.2	-12.4' 54.3'
20	117°40.9	34.2	132°58.0	16.1'	04°21.7	-12.5' 54.3'
21	132°41.0	· · 33.2	147°33.1	16.1'	04°09.3	-12.5' 54.3'
22	147°41.2	32.2	162°08.2	16.1'	03°56.8	-12.5' 54.3'
23	162°41.4	31.2	176°43.4	16.2'	03°44.3	-12.5' 54.3'
SD = 16.1' d = -1.0' SD = 14.9'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:10	05:31	06:38	17:43	18:51	20:13
N 70°	04:22	05:35	06:35	17:46	18:47	20:01
68°	04:31	05:37	06:32	17:49	18:44	19:51
66°	04:39	05:39	06:30	17:51	18:42	19:43
64°	04:45	05:41	06:28	17:53	18:40	19:36
62°	04:51	05:43	06:27	17:54	18:38	19:30
60°	04:55	05:44	06:25	17:55	18:37	19:26
N 58°	04:59	05:45	06:24	17:57	18:36	19:22
56°	05:02	05:46	06:23	17:58	18:35	19:18
54°	05:05	05:46	06:22	17:59	18:34	19:15
52°	05:08	05:47	06:21	18:00	18:33	19:13
50°	05:10	05:48	06:20	18:00	18:33	19:10
45°	05:15	05:49	06:18	18:02	18:32	19:06
N 40°	05:18	05:49	06:16	18:04	18:31	19:02
35°	05:20	05:49	06:15	18:05	18:30	19:00
30°	05:22	05:49	06:13	18:06	18:30	18:58
20°	05:23	05:49	06:11	18:09	18:31	18:56
N 10°	05:23	05:48	06:09	18:11	18:32	18:56
0°	05:22	05:46	06:06	18:13	18:34	18:58
S 10°	05:19	05:43	06:04	18:15	18:36	19:01
20°	05:14	05:39	06:02	18:17	18:40	19:05
30°	05:06	05:34	05:58	18:20	18:44	19:12
35°	05:01	05:31	05:57	18:22	18:47	19:17
40°	04:55	05:27	05:55	18:24	18:51	19:23
45°	04:48	05:23	05:52	18:26	18:56	19:31
S 50°	04:38	05:17	05:49	18:29	19:02	19:40
52°	04:33	05:14	05:48	18:30	19:04	19:45
54°	04:27	05:11	05:46	18:32	19:07	19:50
56°	04:21	05:07	05:45	18:33	19:11	19:56
58°	04:14	05:03	05:43	18:35	19:15	20:03
S 60°	04:06	04:58	05:41	18:37	19:19	20:11

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	07:35	06:47	06:18	11:37	13:57	15:55
N 70°	06:55	06:27	06:07	12:16	14:15	16:03
68°	06:27	06:11	05:59	12:42	14:30	16:09
66°	06:06	05:58	05:52	13:02	14:41	16:14
64°	05:49	05:48	05:46	13:18	14:51	16:19
62°	05:34	05:38	05:41	13:32	14:59	16:23
60°	05:22	05:30	05:36	13:43	15:06	16:26
N 58°	05:12	05:23	05:32	13:52	15:12	16:29
56°	05:03	05:17	05:29	14:01	15:17	16:31
54°	04:55	05:11	05:25	14:08	15:22	16:34
52°	04:48	05:06	05:22	14:15	15:26	16:36
50°	04:41	05:02	05:20	14:21	15:30	16:38
45°	04:27	04:52	05:14	14:34	15:38	16:42
N 40°	04:15	04:44	05:09	14:44	15:45	16:45
35°	04:05	04:36	05:05	14:53	15:51	16:48
30°	03:56	04:30	05:01	15:01	15:57	16:51
20°	03:41	04:19	04:54	15:14	16:06	16:55
N 10°	03:28	04:09	04:48	15:26	16:14	16:59
0°	03:15	04:00	04:43	15:37	16:21	17:03
S 10°	03:02	03:51	04:37	15:48	16:28	17:06
20°	02:49	03:41	04:32	15:59	16:36	17:10
30°	02:33	03:30	04:25	16:13	16:45	17:15
35°	02:24	03:24	04:21	16:20	16:50	17:17
40°	02:14	03:16	04:17	16:29	16:55	17:20
45°	02:02	03:07	04:11	16:39	17:02	17:23
S 50°	01:47	02:57	04:05	16:50	17:10	17:27
52°	01:40	02:52	04:02	16:56	17:13	17:28
54°	01:32	02:47	03:59	17:02	17:17	17:30
56°	01:24	02:41	03:56	17:09	17:22	17:32
58°	01:14	02:34	03:52	17:16	17:26	17:35
S 60°	01:02	02:26	03:48	17:25	17:32	17:37

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
11	10:02	09:54	12:10	09:26	21:49	
12	09:46	09:38	12:10	10:11	22:32	
13	09:30	09:22	12:09	10:53	23:13	

March 14, 15, 16 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	171°54.1	179°23.8	S04°47.9	342°09.2	S00°25.3	327°18.7	S08°35.2	135°16.1	N12°11.3	Alpheratz	357°31.6	29°15.1
1	186°56.5	194°23.4	46.7	357°12.5	25.0	342°21.4	35.1	150°18.3	11.4	Ankaa	353°04.3	-42°08.9
2	201°59.0	209°23.0	45.5	12°15.7	24.7	357°24.1	35.0	165°20.5	11.5	Schedar	349°27.6	56°41.9
3	217°01.5	224°22.6	•• 44.2	27°19.0	•• 24.3	12°26.8	•• 34.9	180°22.7	•• 11.6	Diphda	348°44.2	-17°49.6
4	232°03.9	239°22.2	43.0	42°22.2	24.0	27°29.4	34.9	195°25.0	11.7	Achernar	335°18.3	-57°05.4
5	247°06.4	254°21.8	41.8	57°25.4	23.7	42°32.1	34.8	210°27.2	11.8	Hamal	327°47.6	23°36.0
6	262°08.9	269°21.4	S04°40.5	72°28.7	S00°23.4	57°34.8	S08°34.7	225°29.4	N12°11.9	Polaris	312°27.4	89°23.4
7	277°11.3	284°21.0	39.3	87°31.9	23.1	72°37.5	34.6	240°31.6	11.9	Acamar	315°09.5	-40°11.5
8	292°13.8	299°20.5	38.1	102°35.1	22.8	87°40.1	34.5	255°33.8	12.0	Menkar	314°02.8	4°12.1
9	307°16.3	314°20.1	•• 36.8	117°38.4	•• 22.5	102°42.8	•• 34.4	270°36.0	•• 12.1	Mirfak	308°23.6	49°58.0
10	322°18.7	329°19.7	35.6	132°41.6	22.2	117°45.5	34.3	285°38.2	12.2	Aldebaran	290°35.8	16°34.0
11	337°21.2	344°19.3	34.4	147°44.9	21.8	132°48.2	34.2	300°40.5	12.3	Rigel	281°00.6	-8°10.3
12	352°23.7	359°18.9	S04°33.1	162°48.1	S00°21.5	147°50.9	S08°34.1	315°42.7	N12°12.4	Capella	280°16.9	46°01.7
13	7°26.1	14°18.5	31.9	177°51.4	21.2	162°53.5	34.0	330°44.9	12.5	Bellatrix	278°19.2	6°22.4
14	22°28.6	29°18.1	30.7	192°54.6	20.9	177°56.2	34.0	345°47.1	12.6	Elnath	277°57.6	28°37.8
15	37°31.0	44°17.7	•• 29.4	207°57.9	•• 20.6	192°58.9	•• 33.9	0°49.3	•• 12.7	Alnilam	275°34.3	-1°11.2
16	52°33.5	59°17.3	28.2	223°01.1	20.3	208°01.6	33.8	15°51.5	12.8	Betelgeuse	270°48.4	7°24.5
17	67°36.0	74°16.9	27.0	238°04.3	20.0	223°04.3	33.7	30°53.7	12.8	Canopus	263°50.8	-52°43.1
18	82°38.4	89°16.5	S04°25.7	253°07.6	S00°19.6	238°07.0	S08°33.6	45°55.9	N12°12.9	Sirius	258°23.1	-16°45.7
19	97°40.9	104°16.1	24.5	268°10.8	19.3	253°09.6	33.5	60°58.2	13.0	Adhara	255°03.1	-29°01.1
20	112°43.4	119°15.7	23.2	283°14.1	19.0	268°12.3	33.4	76°00.4	13.1	Procyon	244°47.1	5°08.7
21	127°45.8	134°15.3	•• 22.0	298°17.4	•• 18.7	283°15.0	•• 33.3	91°02.6	•• 13.2	Pollux	243°13.0	27°57.2
22	142°48.3	149°14.9	20.8	313°20.6	18.4	298°17.7	33.2	106°04.8	13.3	Avior	234°12.9	-59°36.6
23	157°50.8	164°14.5	19.5	328°23.9	18.1	313°20.4	33.1	121°07.0	13.4	Suhail	222°43.4	-43°33.4
Mer.pass. 12:30		ν -0.4' d -1.2' m -3.89		ν 3.2' d -0.3' m -1.17		ν 2.7' d -0.1' m -2.39		ν 2.2' d 0.1' m 0.39				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	172°53.2	179°14.1	S04°18.3	343°27.1	S00°17.8	328°23.0	S08°33.1	136°09.2	N12°13.5
1	187°55.7	194°13.7	17.1	358°30.4	17.4	343°25.7	33.0	151°11.4	13.6
2	202°58.1	209°13.3	15.8	13°33.6	17.1	358°28.4	32.9	166°13.7	13.7
3	218°00.6	224°12.9	•• 14.6	28°36.9	•• 16.8	13°31.1	•• 32.8	181°15.9	•• 13.8
4	233°03.1	239°12.5	13.3	43°40.1	16.5	28°33.8	32.7	196°18.1	13.8
5	248°05.5	254°12.0	12.1	58°43.4	16.2	43°36.5	32.6	211°20.3	13.9
6	263°08.0	269°11.6	S04°10.9	73°46.7	S00°15.9	58°39.2	S08°32.5	226°22.5	N12°14.0
7	278°10.5	284°11.2	09.6	88°49.9	15.5	73°41.8	32.4	241°24.7	14.1
8	293°12.9	299°10.8	08.4	103°53.2	15.2	88°44.5	32.3	256°26.9	14.2
9	308°15.4	314°10.4	•• 07.2	118°56.4	•• 14.9	103°47.2	•• 32.2	271°29.1	•• 14.3
10	323°17.9	329°10.0	05.9	133°59.7	14.6	118°49.9	32.1	286°31.3	14.4
11	338°20.3	344°09.6	04.7	149°03.0	14.3	133°52.6	32.0	301°33.6	14.5
12	353°22.8	359°09.2	S04°03.4	164°06.2	S00°13.9	148°55.3	S08°32.0	316°35.8	N12°14.6
13	8°25.3	14°08.8	02.2	179°09.5	13.6	163°57.9	31.9	331°38.0	14.7
14	23°27.7	29°08.4	04°01.0	194°12.8	13.3	179°00.6	31.8	346°40.2	14.8
15	38°30.2	44°08.0	03°59.7	209°16.0	•• 13.0	194°03.3	•• 31.7	1°42.4	•• 14.8
16	53°32.6	59°07.6	58.5	224°19.3	12.7	209°06.0	31.6	16°44.6	14.9
17	68°35.1	74°07.2	57.2	239°22.6	12.3	224°08.7	31.5	31°46.8	15.0
18	83°37.6	89°06.8	S03°56.0	254°25.8	S00°12.0	239°11.4	S08°31.4	46°49.0	N12°15.1
19	98°40.0	104°06.4	54.7	269°29.1	11.7	254°14.1	31.3	61°51.3	15.2
20	113°42.5	119°06.0	53.5	284°32.4	11.4	269°16.8	31.2	76°53.5	15.3
21	128°45.0	134°05.6	•• 52.3	299°35.6	•• 11.1	284°19.4	•• 31.1	91°55.7	•• 15.4
22	143°47.4	149°05.2	51.0	314°38.9	10.7	299°22.1	31.0	106°57.9	15.5
23	158°49.9	164°04.8	49.8	329°42.2	10.4	314°24.8	30.9	122°00.1	15.6
Mer.pass. 12:26		ν -0.4' d -1.2' m -3.89		ν 3.3' d -0.3' m -1.20		ν 2.7' d -0.1' m -2.39		ν 2.2' d 0.1' m 0.39	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	173°52.4	179°04.4	S03°48.5	344°45.4	S00°10.1	329°27.5	S08°30.8	137°02.3	N12°15.7
1	188°54.8	194°04.0	47.3	359°48.7	09.8	344°30.2	30.8	152°04.5	15.8
2	203°57.3	209°03.6	46.1	14°52.0	09.5	359°32.9	30.7	167°06.7	15.8
3	218°59.8	224°03.2	•• 44.8	29°55.3	•• 09.1	14°35.6	•• 30.6	182°08.9	•• 15.9
4	234°02.2	239°02.8	43.6	44°58.5	08.8	29°38.3	30.5	197°11.1	16.0
5	249°04.7	254°02.4	42.3	60°01.8	08.5	44°41.0	30.4	212°13.4	16.1
6	264°07.1	269°02.0	S03°41.1	75°05.1	S00°08.2	59°43.6	S08°30.3	227°15.6	N12°16.2
7	279°09.6	284°01.6	39.8	90°08.4	07.8	74°46.3	30.2	242°17.8	16.3
8	294°12.1	299°01.3	38.6	105°11.7	07.5	89°49.0	30.1	257°20.0	16.4
9	309°14.5	314°00.9	•• 37.4	120°14.9	•• 07.2	104°51.7	•• 30.0	272°22.2	•• 16.5
10	324°17.0	329°00.5	36.1	135°18.2	06.9	119°54.4	29.9	287°24.4	16.6
11	339°19.5	344°00.1	34.9	150°21.5	06.6	134°57.1	29.8	302°26.6	16.7
12	354°21.9	358°59.7	S03°33.6	165°24.8	S00°06.2	149°59.8	S08°29.7	317°28.8	N12°16.8
13	9°24.4	13°59.3	32.4	180°28.1	05.9	165°02.5	29.6	332°31.0	16.9
14	24°26.9	28°58.9	31.1	195°31.4	05.6	180°05.2	29.5	347°33.2	16.9
15	39°29.3	43°58.5	•• 29.9	210°34.6	•• 05.3	195°07.9	•• 29.4	2°35.4	•• 17.0
16	54°31.8	58°58.1	28.6	225°37.9	04.9	210°10.6	29.4	17°37.7	17.1
17	69°34.2	73°57.7	27.4	240°41.2	04.6	225°13.2	29.3	32°39.9	17.2
18	84°36.7	88°57.3	S03°26.2	255°44.5	S00°04.3	240°15.9	S08°29.2	47°42.1	N12°17.3
19	99°39.2	103°56.9	24.9	270°47.8	04.0	255°18.6	29.1	62°44.3	17.4
20	114°41.6	118°56.5	23.7	285°51.1	03.6	270°21.3	29.0	77°46.5	17.5
21	129°44.1	133°56.1	•• 22.4	300°54.4	•• 03.3	285°24.0	•• 28.9	92°48.7	•• 17.6
22	144°46.6	148°55.7	21.2	315°57.6	03.0	300°26.7	28.8	107°50.9	17.7
23	159°49.0	163°55.3	19.9	331°00.9	02.7	315°29.4	28.7	122°53.1	17.8
Mer.pass. 12:22		ν -0.4' d -1.2' m -3.89		ν 3.3' d -0.3' m -1.22		ν 2.7' d -0.1' m -2.40		ν 2.2' d 0.1' m 0.39	

Mar 14 Wed	SHA	Mer.pass
Venus	7°29.7	12:03
Mars	170°15.2	01:11
Jupiter	155°24.6	02:10
Saturn	323°22.0	14:57

Mar 15 Thu	SHA	Mer.pass
Venus	6°20.8	12:03
Mars	170°33.9	01:06
Jupiter	155°29.8	02:06
Saturn	323°16.0	14:53

Mar 16 Fri	SHA	Mer.pass
Venus	5°12.1	12:04
Mars	170°53.1	01:01
Jupiter	155°35.1	02:02
Saturn	323°09.9	14:50

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.2078 sec ΔT = TT-UT1 = +68.9762 sec

2029 March 14 to Mar. 16 UT

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	177°41.5	S02°30.3	191°18.5	16.2'	S03°31.8	-12.5'	54.2'
1	192°41.7	29.3	205°53.7	16.2'	03°19.3	-12.5'	54.2'
2	207°41.9	28.3	220°28.9	16.2'	03°06.8	-12.5'	54.2'
3	222°42.1	· · 27.3	235°04.2	16.2'	02°54.3	-12.5'	54.2'
4	237°42.2	26.3	249°39.4	16.3'	02°41.8	-12.5'	54.2'
5	252°42.4	25.3	264°14.7	16.3'	02°29.3	-12.5'	54.2'
6	267°42.6	S02°24.3	278°49.9	16.3'	S02°16.7	-12.5'	54.2'
7	282°42.7	23.3	293°25.2	16.3'	02°04.2	-12.5'	54.2'
8	297°42.9	22.4	308°00.5	16.3'	01°51.6	-12.5'	54.2'
9	312°43.1	· · 21.4	322°35.9	16.3'	01°39.1	-12.5'	54.2'
10	327°43.3	20.4	337°11.2	16.4'	01°26.6	-12.5'	54.1'
11	342°43.4	19.4	351°46.5	16.4'	01°14.0	-12.5'	54.1'
12	357°43.6	S02°18.4	6°21.9	16.4'	S01°01.5	-12.5'	54.1'
13	12°43.8	17.4	20°57.3	16.4'	00°48.9	-12.5'	54.1'
14	27°44.0	16.4	35°32.7	16.4'	00°36.4	-12.5'	54.1'
15	42°44.1	· · 15.4	50°08.1	16.4'	00°23.8	-12.5'	54.1'
16	57°44.3	14.5	64°43.5	16.4'	S00°11.3	-12.5'	54.1'
17	72°44.5	13.5	79°18.9	16.4'	N00°01.2	12.5'	54.1'
18	87°44.6	S02°12.5	93°54.3	16.4'	N00°13.8	12.5'	54.1'
19	102°44.8	11.5	108°29.7	16.4'	00°26.3	12.5'	54.1'
20	117°45.0	10.5	123°05.2	16.4'	00°38.8	12.5'	54.1'
21	132°45.2	· · 09.5	137°40.6	16.4'	00°51.3	12.5'	54.1'
22	147°45.3	08.5	152°16.0	16.4'	01°03.8	12.5'	54.0'
23	162°45.5	07.5	166°51.5	16.4'	01°16.3	12.5'	54.0'
	SD = 16.1'	$d = -1.0'$	SD = 14.8'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	177°45.7	S02°06.6	181°26.9	16.4'	N01°28.8	12.5'	54.0'
1	192°45.9	05.6	196°02.4	16.4'	01°41.3	12.5'	54.0'
2	207°46.0	04.6	210°37.8	16.4'	01°53.8	12.5'	54.0'
3	222°46.2	· · 03.6	225°13.3	16.4'	02°06.2	12.5'	54.0'
4	237°46.4	02.6	239°48.7	16.4'	02°18.7	12.4'	54.0'
5	252°46.6	01.6	254°24.2	16.4'	02°31.1	12.4'	54.0'
6	267°46.7	S02°00.6	268°59.6	16.4'	N02°43.6	12.4'	54.0'
7	282°46.9	01°59.6	283°35.0	16.4'	02°56.0	12.4'	54.0'
8	297°47.1	58.7	298°10.5	16.4'	03°08.4	12.4'	54.0'
9	312°47.3	· · 57.7	312°45.9	16.4'	03°20.8	12.4'	54.0'
10	327°47.4	56.7	327°21.3	16.4'	03°33.1	12.4'	54.0'
11	342°47.6	55.7	341°56.7	16.4'	03°45.5	12.3'	54.0'
12	357°47.8	S01°54.7	356°32.2	16.4'	N03°57.8	12.3'	54.0'
13	12°48.0	53.7	11°07.6	16.4'	04°10.2	12.3'	54.0'
14	27°48.1	52.7	25°43.0	16.4'	04°22.5	12.3'	54.0'
15	42°48.3	· · 51.7	40°18.3	16.4'	04°34.7	12.3'	53.9'
16	57°48.5	50.8	54°53.7	16.4'	04°47.0	12.2'	53.9'
17	72°48.7	49.8	69°29.1	16.4'	04°59.2	12.2'	53.9'
18	87°48.8	S01°48.8	84°04.4	16.3'	N05°11.5	12.2'	53.9'
19	102°49.0	47.8	98°39.8	16.3'	05°23.7	12.2'	53.9'
20	117°49.2	46.8	113°15.1	16.3'	05°35.8	12.2'	53.9'
21	132°49.4	· · 45.8	127°50.4	16.3'	05°48.0	12.1'	53.9'
22	147°49.5	44.8	142°25.7	16.3'	06°00.1	12.1'	53.9'
23	162°49.7	43.8	157°01.0	16.3'	06°12.2	12.1'	53.9'
	SD = 16.1'	$d = -1.0'$	SD = 14.7'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	177°49.9	S01°42.8	171°36.2	16.2'	N06°24.3	12.1'	53.9'
1	192°50.1	41.9	186°11.5	16.2'	06°36.4	12.0'	53.9'
2	207°50.3	40.9	200°46.7	16.2'	06°48.4	12.0'	53.9'
3	222°50.4	· · 39.9	215°21.9	16.2'	07°00.4	12.0'	53.9'
4	237°50.6	38.9	229°57.1	16.2'	07°12.3	11.9'	53.9'
5	252°50.8	37.9	244°32.3	16.1'	07°24.3	11.9'	53.9'
6	267°51.0	S01°36.9	259°07.4	16.1'	N07°36.2	11.9'	53.9'
7	282°51.1	35.9	273°42.5	16.1'	07°48.1	11.8'	53.9'
8	297°51.3	34.9	288°17.6	16.1'	07°59.9	11.8'	53.9'
9	312°51.5	· · 34.0	302°52.7	16.1'	08°11.7	11.8'	53.9'
10	327°51.7	33.0	317°27.8	16.0'	08°23.5	11.8'	53.9'
11	342°51.8	32.0	332°02.8	16.0'	08°35.3	11.7'	53.9'
12	357°52.0	S01°31.0	346°37.8	16.0'	N08°47.0	11.7'	53.9'
13	12°52.2	30.0	1°12.8	16.0'	08°58.7	11.6'	53.9'
14	27°52.4	29.0	15°47.7	15.9'	09°10.3	11.6'	53.9'
15	42°52.6	· · 28.0	30°22.6	15.9'	09°21.9	11.6'	53.9'
16	57°52.7	27.0	44°57.5	15.9'	09°33.5	11.5'	53.9'
17	72°52.9	26.0	59°32.4	15.8'	09°45.1	11.5'	53.9'
18	87°53.1	S01°25.1	74°07.2	15.8'	N09°56.6	11.5'	53.9'
19	102°53.3	24.1	88°42.0	15.8'	10°08.0	11.4'	53.9'
20	117°53.4	23.1	103°16.8	15.7'	10°19.4	11.4'	53.9'
21	132°53.6	· · 22.1	117°51.6	15.7'	10°30.8	11.3'	53.9'
22	147°53.8	21.1	132°26.3	15.7'	10°42.1	11.3'	53.9'
23	162°54.0	20.1	147°00.9	15.6'	10°53.4	11.3'	53.9'
	SD = 16.1'	$d = -1.0'$	SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:52	05:16	06:23	17:57	19:05	20:30
N 70°	04:06	05:21	06:21	17:59	18:59	20:15
68°	04:18	05:25	06:20	18:00	18:55	20:03
66°	04:27	05:28	06:19	18:01	18:52	19:54
64°	04:34	05:31	06:18	18:01	18:49	19:46
62°	04:40	05:33	06:17	18:02	18:46	19:39
60°	04:46	05:35	06:16	18:03	18:44	19:34
N 58°	04:50	05:36	06:15	18:03	18:43	19:29
56°	04:54	05:38	06:15	18:04	18:41	19:25
54°	04:58	05:39	06:14	18:04	18:40	19:21
52°	05:01	05:40	06:14	18:05	18:39	19:18
50°	05:04	05:41	06:13	18:05	18:37	19:15
45°	05:09	05:43	06:12	18:06	18:35	19:10
N 40°	05:13	05:44	06:11	18:07	18:34	19:06
35°	05:16	05:45	06:10	18:08	18:33	19:02
30°	05:18	05:46	06:10	18:08	18:32	19:00
20°	05:21	05:46	06:08	18:10	18:32	18:57
N 10°	05:22	05:46	06:07	18:11	18:32	18:56
0°	05:21	05:45	06:06	18:12	18:33	18:57
S 10°	05:19	05:43	06:04	18:13	18:34	18:59
20°	05:15	05:40	06:02	18:15	18:37	19:03
30°	05:08	05:36	06:00	18:17	18:41	19:09
35°	05:04	05:34	05:59	18:18	18:43	19:13
40°	04:59	05:31	05:58	18:19	18:46	19:18
45°	04:52	05:27	05:56	18:21	18:50	19:25
S 50°	04:43	05:22	05:54	18:23	18:55	19:33
52°	04:39	05:19	05:53	18:23	18:57	19:37
54°	04:34	05:17	05:52	18:24	19:00	19:42
56°	04:28	05:13	05:51	18:25	19:03	19:48
58°	04:22	05:10	05:50	18:27	19:06	19:54
S 60°	04:15	05:06	05:48	18:28	19:10	20:01

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	05:54	05:32	05:09	17:45	19:34	21:29
N 70°	05:51	05:35	05:19	17:45	19:26	21:11
68°	05:48	05:38	05:28	17:45	19:20	20:57
66°	05:46	05:40	05:35	17:45	19:14	20:45
64°	05:44	05:42	05:41	17:45	19:10	20:36
62°	05:43	05:44	05:46	17:45	19:06	20:28
60°	05:41	05:46	05:51	17:44	19:02	20:21
N 58°	05:40	05:47	05:55	17:44	18:59	20:15
56°	05:39	05:48	05:58	17:44	18:57	20:09
54°	05:38	05:50	06:02	17:44	18:54	20:05
52°	05:37	05:51	06:05	17:44	18:52	20:00
50°	05:36	05:51	06:08	17:44	18:50	19:56
45°	05:34	05:54	06:14	17:44	18:46	19:48
N 40°	05:32	05:55	06:19	17:44	18:42	19:41
35°	05:31	05:57	06:23	17:44	18:39	19:35
30°	05:30	05:58	06:27	17:44	18:37	19:30
20°	05:28	06:00	06:34	17:44	18:32	19:20
N 10°	05:26	06:03	06:39	17:44	18:28	19:12
0°	05:24	06:05	06:45	17:44	18:24	19:05
S 10°	05:22	06:07	06:51	17:43	18:20	18:58
20°	05:20	06:09	06:57	17:43	18:16	18:50
30°	05:18	06:11	07:04	17:43	18:11	18:41
35°	05:17	06:13	07:08	17:43	18:09	18:36
40°	05:16	06:14	07:13	17:43	18:06	18:30
45°	05:14	06:16	07:18	17:43	18:02	18:23
S 50°	05:12	06:18	07:24	17:43	17:58	18:15
52°	05:11	06:19	07:27	17:42	17:56	18:11
54°	05:10	06:21	07:31	17:42	17:54	18:07
56°	05:09	06:22	07:34	17:42	17:52	18:03
58°	05:08	06:23	07:38	17:42	17:50	17:58
S 60°	05:07	06:25	07:43	17:42	17:47	17:52

Day	Sun		Mer.	Moon		Age
	Eqn. of Time	12h	Pass	Mer.Pass.	Upper	Lower
	00h		hh:mm	hh:mm	hh:mm	hh:mm
	mm:ss	mm:ss				
14	09:14	09:06	12:09	11:34	23:54	
15	08:57	08:49	12:09	12:14	—:—	
16	08:40	08:32	12:09	12:55	00:35	



March 17, 18, 19 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	174°51.5	178°54.9	S03°18.7	346°04.2	S00°02.3	330°32.1	S08°28.6	137°55.3	N12°17.9	Alpheratz	357°31.6	29°15.1
1	189°54.0	193°54.5	17.4	1°07.5	02.0	345°34.8	28.5	152°57.5	18.0	Ankaa	353°04.3	-42°08.9
2	204°56.4	208°54.1	16.2	16°10.8	01.7	0°37.5	28.4	167°59.7	18.0	Schedar	349°27.6	56°41.9
3	219°58.9	223°53.7	•• 14.9	31°14.1	•• 01.4	15°40.2	•• 28.3	183°01.9	•• 18.1	Diphda	348°44.2	-17°49.6
4	235°01.4	238°53.3	13.7	46°17.4	01.0	30°42.9	28.2	198°04.2	18.2	Achernar	335°18.3	-57°05.4
5	250°03.8	253°52.9	12.4	61°20.7	00.7	45°45.6	28.1	213°06.4	18.3	Hamal	327°47.6	23°36.0
6	265°06.3	268°52.5	S03°11.2	76°24.0	S00°00.4	60°48.3	S08°28.0	228°08.6	N12°18.4	Polaris	312°28.6	89°23.4
7	280°08.7	283°52.2	10.0	91°27.3	S00°00.1	75°51.0	27.9	243°10.8	18.5	Acamar	315°09.5	-40°11.5
8	295°11.2	298°51.8	08.7	106°30.6	N00°00.3	90°53.6	27.8	258°13.0	18.6	Menkar	314°02.8	4°12.1
9	310°13.7	313°51.4	•• 07.5	121°33.9	•• 00.6	105°56.3	•• 27.7	273°15.2	•• 18.7	Mirfak	308°23.6	49°58.0
10	325°16.1	328°51.0	06.2	136°37.2	00.9	120°59.0	27.6	288°17.4	18.8	Aldebaran	290°35.8	16°34.0
11	340°18.6	343°50.6	05.0	151°40.5	01.2	136°01.7	27.6	303°19.6	18.9	Rigel	281°00.7	-8°10.3
12	355°21.1	358°50.2	S03°03.7	166°43.8	N00°01.6	151°04.4	S08°27.5	318°21.8	N12°19.0	Capella	280°16.9	46°01.6
13	10°23.5	13°49.8	02.5	181°47.1	01.9	166°07.1	27.4	333°24.0	19.1	Bellatrix	278°19.3	6°22.4
14	25°26.0	28°49.4	01.2	196°50.4	02.2	181°09.8	27.3	348°26.2	19.1	Elnath	277°57.6	28°37.8
15	40°28.5	43°49.0	03°00.0	211°53.7	•• 02.6	196°12.5	•• 27.2	3°28.4	•• 19.2	Alnilam	275°34.3	-1°11.2
16	55°30.9	58°48.6	02°58.7	226°57.0	02.9	211°15.2	27.1	18°30.6	19.3	Betelgeuse	270°48.4	7°24.5
17	70°33.4	73°48.2	57.5	242°00.3	03.2	226°17.9	27.0	33°32.8	19.4	Canopus	263°50.8	-52°43.1
18	85°35.8	88°47.8	S02°56.2	257°03.6	N00°03.5	241°20.6	S08°26.9	48°35.1	N12°19.5	Sirius	258°23.2	-16°45.7
19	100°38.3	103°47.4	55.0	272°06.9	03.9	256°23.3	26.8	63°37.3	19.6	Adhara	255°03.1	-29°01.1
20	115°40.8	118°47.0	53.7	287°10.2	04.2	271°26.0	26.7	78°39.5	19.7	Procyon	244°47.1	5°08.7
21	130°43.2	133°46.7	•• 52.5	302°13.5	•• 04.5	286°28.7	•• 26.6	93°41.7	•• 19.8	Pollux	243°13.0	27°57.2
22	145°45.7	148°46.3	51.2	317°16.8	04.8	301°31.4	26.5	108°43.9	19.9	Avior	234°12.9	-59°36.6
23	160°48.2	163°45.9	50.0	332°20.1	05.2	316°34.1	26.4	123°46.1	20.0	Suhail	222°43.4	-43°33.4
Mer.pass. 12:19		ν -0.4' d -1.2' m -3.89		ν 3.3' d -0.3' m -1.25		ν 2.7' d -0.1' m -2.40		ν 2.2' d 0.1' m 0.39				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	175°50.6	178°45.5	S02°48.7	347°23.4	N00°05.5	331°36.8	S08°26.3	138°48.3	N12°20.1
1	190°53.1	193°45.1	47.5	2°26.7	05.8	346°39.5	26.2	153°50.5	20.2
2	205°55.6	208°44.7	46.2	17°30.0	06.2	1°42.2	26.1	168°52.7	20.3
3	220°58.0	223°44.3	•• 45.0	32°33.3	•• 06.5	16°44.9	•• 26.0	183°54.9	•• 20.3
4	236°00.5	238°43.9	43.7	47°36.7	06.8	31°47.6	25.9	198°57.1	20.4
5	251°03.0	253°43.5	42.5	62°40.0	07.1	46°50.3	25.8	213°59.3	20.5
6	266°05.4	268°43.1	S02°41.2	77°43.3	N00°07.5	61°53.0	S08°25.7	229°01.5	N12°20.6
7	281°07.9	283°42.7	40.0	92°46.6	07.8	76°55.7	25.6	244°03.7	20.7
8	296°10.3	298°42.4	38.7	107°49.9	08.1	91°58.4	25.5	259°05.9	20.8
9	311°12.8	313°42.0	•• 37.5	122°53.2	•• 08.5	107°01.1	•• 25.4	274°08.1	•• 20.9
10	326°15.3	328°41.6	36.2	137°56.5	08.8	122°03.8	25.3	289°10.3	21.0
11	341°17.7	343°41.2	35.0	152°59.8	09.1	137°06.5	25.2	304°12.5	21.1
12	356°20.2	358°40.8	S02°33.7	168°03.2	N00°09.5	152°09.2	S08°25.1	319°14.7	N12°21.2
13	11°22.7	13°40.4	32.5	183°06.5	09.8	167°11.9	25.0	334°17.0	21.3
14	26°25.1	28°40.0	31.2	198°09.8	10.1	182°14.6	24.9	349°19.2	21.4
15	41°27.6	43°39.6	•• 30.0	213°13.1	•• 10.4	197°17.3	•• 24.9	4°21.4	•• 21.5
16	56°30.1	58°39.2	28.7	228°16.4	10.8	212°20.0	24.8	19°23.6	21.5
17	71°32.5	73°38.8	27.5	243°19.7	11.1	227°22.7	24.7	34°25.8	21.6
18	86°35.0	88°38.5	S02°26.2	258°23.1	N00°11.4	242°25.4	S08°24.6	49°28.0	N12°21.7
19	101°37.4	103°38.1	25.0	273°26.4	11.8	257°28.1	24.5	64°30.2	21.8
20	116°39.9	118°37.7	23.7	288°29.7	12.1	272°30.8	24.4	79°32.4	21.9
21	131°42.4	133°37.3	•• 22.4	303°33.0	•• 12.4	287°33.5	•• 24.3	94°34.6	•• 22.0
22	146°44.8	148°36.9	21.2	318°36.3	12.8	302°36.2	24.2	109°36.8	22.1
23	161°47.3	163°36.5	19.9	333°39.7	13.1	317°38.9	24.1	124°39.0	22.2
Mer.pass. 12:15		ν -0.4' d -1.2' m -3.89		ν 3.3' d 0.3' m -1.27		ν 2.7' d -0.1' m -2.41		ν 2.2' d 0.1' m 0.39	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	176°49.8	178°36.1	S02°18.7	348°43.0	N00°13.4	332°41.6	S08°24.0	139°41.2	N12°22.3
1	191°52.2	193°35.7	17.4	3°46.3	13.8	347°44.3	23.9	154°43.4	22.4
2	206°54.7	208°35.4	16.2	18°49.6	14.1	2°47.0	23.8	169°45.6	22.5
3	221°57.2	223°35.0	•• 14.9	33°52.9	•• 14.4	17°49.7	•• 23.7	184°47.8	•• 22.6
4	236°59.6	238°34.6	13.7	48°56.3	14.8	32°52.4	23.6	199°50.0	22.7
5	252°02.1	253°34.2	12.4	63°59.6	15.1	47°55.1	23.5	214°52.2	22.8
6	267°04.6	268°33.8	S02°11.2	79°02.9	N00°15.4	62°57.9	S08°23.4	229°54.4	N12°22.8
7	282°07.0	283°33.4	09.9	94°06.2	15.7	78°00.6	23.3	244°56.6	22.9
8	297°09.5	298°33.0	08.7	109°09.6	16.1	93°03.3	23.2	259°58.8	23.0
9	312°11.9	313°32.6	•• 07.4	124°12.9	•• 16.4	108°06.0	•• 23.1	275°01.0	•• 23.1
10	327°14.4	328°32.3	06.2	139°16.2	16.7	123°08.7	23.0	290°03.2	23.2
11	342°16.9	343°31.9	04.9	154°19.6	17.1	138°11.4	22.9	305°05.4	23.3
12	357°19.3	358°31.5	S02°03.6	169°22.9	N00°17.4	153°14.1	S08°22.8	320°07.6	N12°23.4
13	12°21.8	13°31.1	02.4	184°26.2	17.7	168°16.8	22.7	335°09.8	23.5
14	27°24.3	28°30.7	02°01.1	199°29.5	18.1	183°19.5	22.6	350°12.0	23.6
15	42°26.7	43°30.3	01°59.9	214°32.9	•• 18.4	198°22.2	•• 22.5	5°14.2	•• 23.7
16	57°29.2	58°29.9	58.6	229°36.2	18.7	213°24.9	22.4	20°16.4	23.8
17	72°31.7	73°29.6	57.4	244°39.5	19.1	228°27.6	22.3	35°18.6	23.9
18	87°34.1	88°29.2	S01°56.1	259°42.9	N00°19.4	243°30.3	S08°22.2	50°20.8	N12°24.0
19	102°36.6	103°28.8	54.9	274°46.2	19.7	258°33.0	22.1	65°23.0	24.1
20	117°39.1	118°28.4	53.6	289°49.5	20.1	273°35.7	22.0	80°25.2	24.1
21	132°41.5	133°28.0	•• 52.4	304°52.9	•• 20.4	288°38.4	•• 21.9	95°27.4	•• 24.2
22	147°44.0	148°27.6	51.1	319°56.2	20.7	303°41.2	21.8	110°29.6	24.3
23	162°46.4	163°27.2	49.8	334°59.5	21.1	318°43.9	21.7	125°31.9	24.4
Mer.pass. 12:11		ν -0.4' d -1.3' m -3.89		ν 3.3' d 0.3' m -1.30		ν 2.7' d -0.1' m -2.41		ν 2.2' d 0.1' m 0.39	

Mar 17 Sat		SHA	Mer.pass
Venus		4°03.4	12:05
Mars		171°12.7	00:56
Jupiter		155°40.6	01:58
Saturn		323°03.8	14:46

Mar 18 Sun		SHA	Mer.pass
Venus		2°54.8	12:05
Mars		171°32.8	00:50
Jupiter		155°46.2	01:53
Saturn		322°57.7	14:43

Mar 19 Mon		SHA	Mer.pass
Venus		1°46.4	12:06
Mars		171°53.2	00:45
Jupiter		155°51.9	01:49
Saturn		322°51.4	14:39

Horizontal parallax		
Venus:		0.1
Mars:		0.2

DUT1 = UT1-UTC = +0.2079 sec $\Delta T = TT-UT1 = +68.9761$ sec

2029 March 17 to Mar. 19 UT

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	177°54.2	S01°19.1	161°35.6	15.6'	N11°04.7	11.2'	53.9'
1	192°54.3	18.1	176°10.2	15.6'	11°15.9	11.2'	53.9'
2	207°54.5	17.1	190°44.8	15.5'	11°27.1	11.1'	53.9'
3	222°54.7	· · 16.2	205°19.3	15.5'	11°38.2	11.1'	53.9'
4	237°54.9	15.2	219°53.8	15.5'	11°49.3	11.0'	53.9'
5	252°55.1	14.2	234°28.3	15.4'	12°00.3	11.0'	53.9'
6	267°55.2	S01°13.2	249°02.7	15.4'	N12°11.3	10.9'	53.9'
7	282°55.4	12.2	263°37.1	15.3'	12°22.2	10.9'	53.9'
8	297°55.6	11.2	278°11.4	15.3'	12°33.1	10.8'	53.9'
9	312°55.8	· · 10.2	292°45.7	15.3'	12°43.9	10.8'	53.9'
10	327°56.0	09.2	307°20.0	15.2'	12°54.7	10.7'	53.9'
11	342°56.1	08.2	321°54.2	15.2'	13°05.5	10.7'	53.9'
12	357°56.3	S01°07.3	336°28.4	15.1'	N13°16.1	10.6'	53.9'
13	12°56.5	06.3	351°02.5	15.1'	13°26.8	10.6'	53.9'
14	27°56.7	05.3	5°36.6	15.1'	13°37.4	10.5'	53.9'
15	42°56.9	· · 04.3	20°10.7	15.0'	13°47.9	10.5'	53.9'
16	57°57.0	03.3	34°44.7	15.0'	13°58.4	10.4'	53.9'
17	72°57.2	02.3	49°18.7	14.9'	14°08.8	10.4'	53.9'
18	87°57.4	S01°01.3	63°52.6	14.9'	N14°19.1	10.3'	53.9'
19	102°57.6	01°00.3	78°26.5	14.8'	14°29.4	10.2'	53.9'
20	117°57.8	00°59.3	93°00.3	14.8'	14°39.7	10.2'	53.9'
21	132°57.9	· · 58.4	107°34.1	14.7'	14°49.9	10.1'	53.9'
22	147°58.1	57.4	122°07.8	14.7'	15°00.0	10.1'	53.9'
23	162°58.3	56.4	136°41.5	14.6'	15°10.1	10.0'	53.9'
SD = 16.1'			SD = 14.7'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	177°58.5	S00°55.4	151°15.2	14.6'	N15°20.1	9.9'	53.9'
1	192°58.7	54.4	165°48.8	14.5'	15°30.0	9.9'	53.9'
2	207°58.8	53.4	180°22.3	14.5'	15°39.9	9.8'	53.9'
3	222°59.0	· · 52.4	194°55.8	14.4'	15°49.7	9.8'	53.9'
4	237°59.2	51.4	209°29.2	14.4'	15°59.5	9.7'	54.0'
5	252°59.4	50.4	224°02.6	14.3'	16°09.2	9.6'	54.0'
6	267°59.6	S00°49.5	238°36.0	14.3'	N16°18.8	9.6'	54.0'
7	282°59.7	48.5	253°09.2	14.2'	16°28.4	9.5'	54.0'
8	297°59.9	47.5	267°42.5	14.2'	16°37.9	9.4'	54.0'
9	313°00.1	· · 46.5	282°15.7	14.1'	16°47.3	9.4'	54.0'
10	328°00.3	45.5	296°48.8	14.1'	16°56.7	9.3'	54.0'
11	343°00.5	44.5	311°21.9	14.0'	17°06.0	9.2'	54.0'
12	358°00.6	S00°43.5	325°54.9	14.0'	N17°15.2	9.2'	54.0'
13	13°00.8	42.5	340°27.9	13.9'	17°24.4	9.1'	54.0'
14	28°01.0	41.5	355°00.8	13.9'	17°33.4	9.0'	54.0'
15	43°01.2	· · 40.6	9°33.6	13.8'	17°42.5	8.9'	54.0'
16	58°01.4	39.6	24°06.4	13.7'	17°51.4	8.9'	54.0'
17	73°01.6	38.6	38°39.2	13.7'	18°00.3	8.8'	54.0'
18	88°01.7	S00°37.6	53°11.9	13.6'	N18°09.1	8.7'	54.0'
19	103°01.9	36.6	67°44.5	13.6'	18°17.8	8.6'	54.1'
20	118°02.1	35.6	82°17.1	13.5'	18°26.4	8.6'	54.1'
21	133°02.3	· · 34.6	96°49.6	13.5'	18°35.0	8.5'	54.1'
22	148°02.5	33.6	111°22.1	13.4'	18°43.5	8.4'	54.1'
23	163°02.6	32.6	125°54.5	13.3'	18°51.9	8.3'	54.1'
SD = 16.1'			SD = 14.7'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°02.8	S00°31.7	140°26.8	13.3'	N19°00.2	8.3'	54.1'
1	193°03.0	30.7	154°59.1	13.2'	19°08.5	8.2'	54.1'
2	208°03.2	29.7	169°31.3	13.2'	19°16.6	8.1'	54.1'
3	223°03.4	· · 28.7	184°03.5	13.1'	19°24.7	8.0'	54.1'
4	238°03.6	27.7	198°35.6	13.1'	19°32.7	7.9'	54.1'
5	253°03.7	26.7	213°07.7	13.0'	19°40.7	7.8'	54.2'
6	268°03.9	S00°25.7	227°39.7	12.9'	N19°48.5	7.8'	54.2'
7	283°04.1	24.7	242°11.6	12.9'	19°56.3	7.7'	54.2'
8	298°04.3	23.7	256°43.5	12.8'	20°03.9	7.6'	54.2'
9	313°04.5	· · 22.8	271°15.3	12.8'	20°11.5	7.5'	54.2'
10	328°04.7	21.8	285°47.1	12.7'	20°19.0	7.4'	54.2'
11	343°04.8	20.8	300°18.8	12.6'	20°26.4	7.3'	54.2'
12	358°05.0	S00°19.8	314°50.4	12.6'	N20°33.8	7.2'	54.2'
13	13°05.2	18.8	329°22.0	12.5'	20°41.0	7.1'	54.2'
14	28°05.4	17.8	343°53.5	12.5'	20°48.2	7.1'	54.3'
15	43°05.6	· · 16.8	358°24.9	12.4'	20°55.2	7.0'	54.3'
16	58°05.8	15.8	12°56.3	12.3'	21°02.2	6.9'	54.3'
17	73°05.9	14.9	27°27.7	12.3'	21°09.1	6.8'	54.3'
18	88°06.1	S00°13.9	41°58.9	12.2'	N21°15.8	6.7'	54.3'
19	103°06.3	12.9	56°30.1	12.1'	21°22.5	6.6'	54.3'
20	118°06.5	11.9	71°01.3	12.1'	21°29.1	6.5'	54.3'
21	133°06.7	· · 10.9	85°32.4	12.0'	21°35.6	6.4'	54.4'
22	148°06.9	09.9	100°03.4	12.0'	21°42.0	6.3'	54.4'
23	163°07.0	08.9	114°34.4	11.9'	21°48.3	6.2'	54.4'
SD = 16.1'			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:33	05:00	06:07	18:11	19:19	20:47
N 70°	03:50	05:06	06:07	18:11	19:12	20:30
68°	04:03	05:12	06:07	18:11	19:06	20:16
66°	04:14	05:16	06:07	18:10	19:02	20:05
64°	04:23	05:20	06:07	18:10	18:58	19:56
62°	04:30	05:23	06:07	18:10	18:55	19:48
60°	04:36	05:26	06:07	18:10	18:52	19:42
N 58°	04:42	05:28	06:07	18:10	18:49	19:36
56°	04:46	05:30	06:07	18:10	18:47	19:31
54°	04:50	05:32	06:07	18:10	18:45	19:27
52°	04:54	05:33	06:07	18:10	18:44	19:24
50°	04:57	05:35	06:07	18:10	18:42	19:20
45°	05:03	05:37	06:07	18:10	18:39	19:14
N 40°	05:08	05:39	06:06	18:10	18:37	19:09
35°	05:12	05:41	06:06	18:10	18:35	19:05
30°	05:14	05:42	06:06	18:10	18:34	19:02
20°	05:18	05:44	06:06	18:11	18:33	18:58
N 10°	05:20	05:44	06:05	18:11	18:32	18:56
0°	05:20	05:44	06:05	18:11	18:32	18:56
S 10°	05:19	05:43	06:04	18:12	18:33	18:57
20°	05:16	05:41	06:03	18:12	18:34	19:00
30°	05:10	05:38	06:02	18:13	18:37	19:05
35°	05:07	05:36	06:02	18:14	18:39	19:08
40°	05:02	05:34	06:01	18:14	18:41	19:13
45°	04:56	05:31	06:00	18:15	18:44	19:19
S 50°	04:48	05:27	05:59	18:16	18:48	19:26
52°	04:44	05:25	05:58	18:16	18:50	19:30
54°	04:40	05:22	05:58	18:17	18:52	19:34
56°	04:35	05:20	05:57	18:18	18:55	19:39
58°	04:30	05:17	05:56	18:18	18:57	19:44
S 60°	04:23	05:14	05:56	18:19	19:00	19:51

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	04:41	03:57	☐	23:41	☐	☐
N 70°	05:00	04:35	03:43	23:05	· ·	01:30
68°	05:16	05:02	04:41	22:39	· ·	00:33
66°	05:29	05:23	05:15	22:20	23:59	· ·
64°	05:40	05:39	05:40	22:04	23:35	· ·
62°	05:49	05:53	06:00	21:51	23:16	· ·
60°	05:57	06:05	06:16	21:40	23:01	· ·
N 58°	06:04	06:15	06:30	21:31	22:47	· ·
56°	06:10	06:24	06:42	21:23	22:36	23:49
54°	06:15	06:32	06:52	21:15	22:26	23:36
52°	06:20	06:39	07:01	21:09	22:17	23:25
50°	06:25	06:45	07:10	21:03	22:10	23:16
45°	06:35	06:59	07:27	20:50	21:53	22:55
N 40°	06:43	07:11	07:42	20:40	21:39	22:39
35°	06:50	07:20	07:54	20:31	21:28	22:25
30°	06:57	07:29	08:05	20:23	21:18	22:13
20°	07:08	07:44	08:24	20:10	21:00	21:52
N 10°	07:17	07:57	08:40	19:58	20:45	21:35
0°	07:27	08:10	08:55	19:47	20:31	21:18
S 10°	07:36	08:22	09:11	19:36	20:17	21:01
20°	07:46	08:36	09:27	19:25	20:03	20:44
30°	07:57	08:51	09:46	19:12	19:46	20:24
35°	08:04	09:00	09:57	19:04	19:36	20:12
40°	08:11	09:11	10:10	18:56	19:25	19:58
45°	08:20	09:23	10:25	18:46	19:11	19:42
S 50°	08:31	09:38	10:44	18:34	18:55	19:23
52°	08:36	09:44	10:53	18:28	18:48	19:13
54°	08:41	09:52	11:03	18:22	18:40	19:03
56°	08:47	10:01	11:14	18:15	18:30	18:51
58°	08:54	10:11	11:27	18:07	18:20	18:38
S 60°	09:02	10:22	11:43	17:59	18:08	18:22

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	2-4 3-13%
17	08:23	08:15	12:08	13:37	01:16	
18	08:06	07:57	12:08	14:21	01:58	
19	07:49	07:40	12:08	15:07	02:43	

March 20, 21, 22 UT (Tue., Wed., Thu.)

Aries				Venus				Mars				Jupiter				Saturn				Stars																																																																																																																																																																																																																																																																																																	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec																																																																																																																																																																																																																																																																																																				
0	177°48.9	178°26.9	S01°48.6	350°02.9	N00°21.4	333°46.6	S08°21.6	140°34.1	N12°24.5	Alpheratz	357°31.6	29°15.1	1	192°51.4	193°26.5	47.3	5°06.2	21.7	155°36.3	24.6	Ankaa	353°04.3	-42°08.9	2	207°53.8	208°26.1	46.1	20°09.5	22.1	33°52.0	21.4	170°38.5	24.7	Schedar	349°27.6	56°41.9	3	222°56.3	223°25.7	•• 44.8	35°12.9	•• 22.4	18°54.7	•• 21.3	185°40.7	•• 24.8	Diphda	348°44.2	-17°49.6	4	237°58.8	238°25.3	43.6	50°16.2	22.7	33°57.4	21.2	200°42.9	24.9	Achernar	335°18.3	-57°05.4	5	253°01.2	253°24.9	42.3	65°19.6	23.1	49°00.1	21.1	215°45.1	25.0	Hamal	327°47.6	23°36.0	6	268°03.7	268°24.5	S01°41.1	80°22.9	N00°23.4	64°02.8	S08°21.0	230°47.3	N12°25.1	Polaris	312°29.5	89°23.4	7	283°06.2	283°24.2	39.8	95°26.2	23.7	79°05.5	20.9	245°49.5	25.2	Acamar	315°09.5	-40°11.5	8	298°08.6	298°23.8	38.5	110°29.6	24.1	94°08.2	20.8	260°51.7	25.3	Menkar	314°02.8	4°12.1	9	313°11.1	313°23.4	•• 37.3	125°32.9	•• 24.4	109°11.0	•• 20.7	275°53.9	•• 25.4	Mirfak	308°23.6	49°58.0	10	328°13.5	328°23.0	36.0	140°36.3	24.7	124°13.7	20.6	290°56.1	25.5	Aldebaran	290°35.8	16°34.0	11	343°16.0	343°22.6	34.8	155°39.6	25.1	139°16.4	20.5	305°58.3	25.5	Rigel	281°00.7	-8°10.3	12	358°18.5	358°22.2	S01°33.5	170°42.9	N00°25.4	154°19.1	S08°20.4	321°00.5	N12°25.6	Capella	280°16.9	46°01.6	13	13°20.9	13°21.9	32.3	185°46.3	25.7	169°21.8	20.3	336°02.7	25.7	Bellatrix	278°19.3	6°22.4	14	28°23.4	28°21.5	31.0	200°49.6	26.1	184°24.5	20.2	351°04.9	25.8	Elnath	277°57.6	28°37.8	15	43°25.9	43°21.1	•• 29.7	215°53.0	•• 26.4	199°27.2	•• 20.1	6°07.1	•• 25.9	Alnilam	275°34.3	-1°11.2	16	58°28.3	58°20.7	28.5	230°56.3	26.7	214°29.9	20.0	21°09.3	26.0	Betelgeuse	270°48.4	7°24.5	17	73°30.8	73°20.3	27.2	245°59.7	27.1	229°32.6	19.9	36°11.5	26.1	Canopus	263°50.8	-52°43.1	18	88°33.3	88°19.9	S01°26.0	261°03.0	N00°27.4	244°35.3	S08°19.8	51°13.7	N12°26.2	Sirius	258°23.2	-16°45.7	19	103°35.7	103°19.6	24.7	276°06.3	27.8	259°38.1	19.7	66°15.9	26.3	Adhara	255°03.1	-29°01.1	20	118°38.2	118°19.2	23.5	291°09.7	28.1	274°40.8	19.6	81°18.1	26.4	Procyon	244°47.2	5°08.7	21	133°40.7	133°18.8	•• 22.2	306°13.0	•• 28.4	289°43.5	•• 19.5	96°20.3	•• 26.5	Pollux	243°13.0	27°57.2	22	148°43.1	148°18.4	20.9	321°16.4	28.8	304°46.2	19.4	111°22.5	26.6	Avior	234°12.9	-59°36.6	23	163°45.6	163°18.0	19.7	336°19.7	29.1	319°48.9	19.3	126°24.7	26.7	Suhail	222°43.4	-43°33.4
Mer.pass. 12:07		ν -0.4' d -1.3' m -3.89			ν 3.3' d 0.3' m -1.32			ν 2.7' d -0.1' m -2.42			ν 2.2' d 0.1' m 0.39																																																																																																																																																																																																																																																																																																										

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec																																																																																																																																																																																																																																																																																																				
0	178°48.0	178°17.6	S01°18.4	351°23.1	N00°29.4	334°51.6	S08°19.2	141°26.9	N12°26.8	Denebola	182°21.2	14°24.3	1	193°50.5	193°17.3	17.2	6°26.4	29.8	156°29.1	26.9	Gienah	175°39.7	-17°42.5	2	208°53.0	208°16.9	15.9	21°29.8	30.1	4°57.0	19.0	171°31.3	27.0	Acrux	172°55.4	-63°15.8	3	223°55.4	223°16.5	•• 14.7	36°33.1	•• 30.4	19°59.8	•• 18.9	186°33.4	•• 27.0	Gacrux	171°47.1	-57°16.7	4	238°57.9	238°16.1	13.4	51°36.5	30.8	35°02.5	18.8	201°35.6	27.1	Alioth	166°09.6	55°47.8	5	254°00.4	253°15.7	12.1	66°39.8	31.1	50°05.2	18.7	216°37.8	27.2	Spica	158°18.4	-11°19.0	6	269°02.8	268°15.3	S01°10.9	81°43.2	N00°31.4	65°07.9	S08°18.6	231°40.0	N12°27.3	Alkaid	152°49.0	49°09.8	7	284°05.3	283°15.0	09.6	96°46.5	31.8	80°10.6	18.5	246°42.2	27.4	Hadar	148°30.5	-60°30.8	8	299°07.8	298°14.6	08.4	111°49.9	32.1	95°13.3	18.4	261°44.4	27.5	Menkent	147°53.2	-36°30.9	9	314°10.2	313°14.2	•• 07.1	126°53.2	•• 32.4	110°16.0	•• 18.3	276°46.6	•• 27.6	Arcturus	145°44.6	19°01.6	10	329°12.7	328°13.8	05.8	141°56.6	32.8	125°18.8	18.2	291°48.8	27.7	Rigel Kent.	139°35.0	-60°57.3	11	344°15.2	343°13.4	04.6	156°59.9	33.1	140°21.5	18.1	306°51.0	27.8	Kochab	137°19.3	74°01.8	12	359°17.6	358°13.1	S01°03.3	172°03.3	N00°33.4	155°24.2	S08°18.0	321°53.2	N12°27.9	Zuben'ubi	136°52.0	-16°09.9	13	14°20.1	13°12.7	02.1	187°06.6	33.8	170°26.9	17.9	336°55.4	28.0	Alphecca	126°00.7	26°36.7	14	29°22.5	28°12.3	01°00.8	202°10.0	34.1	185°29.6	17.8	351°57.6	28.1	Antares	112°11.5	-26°29.7	15	44°25.0	43°11.9	00°59.5	217°13.3	•• 34.5	200°32.3	•• 17.7	6°59.8	•• 28.2	Atria	107°02.5	-69°04.4	16	59°27.5	58°11.5	58.3	232°16.7	34.8	215°35.0	17.6	22°02.0	28.3	Sabik	101°58.8	-15°45.6	17	74°29.9	73°11.1	57.0	247°20.0	35.1	230°37.8	17.5	37°04.2	28.4	Shaula	96°05.7	-37°07.3	18	89°32.4	88°10.8	S00°55.8	262°23.4	N00°35.5	245°40.5	S08°17.3	52°06.4	N12°28.5	Rasalhague	95°55.4	12°32.2	19	104°34.9	103°10.4	54.5	277°26.7	35.8	260°43.2	17.2	67°08.6	28.5	Eltanin	90°40.6	51°28.8	20	119°37.3	118°10.0	53.3	292°30.1	36.1	275°45.9	17.1	82°10.8	28.6	Kaus Aust.	83°28.1	-34°22.0	21	134°39.8	133°09.6	•• 52.0	307°33.5	•• 36.5	290°48.6	•• 17.0	97°13.0	•• 28.7	Vega	80°31.0	38°48.4	22	149°42.3	148°09.2	50.7	322°36.8	36.8	305°51.3	16.9	112°15.2	28.8	Nunki	75°43.6	-26°15.5	23	164°44.7	163°08.9	49.5	337°40.2	37.1	320°54.1	16.8	127°17.4	28.9	Altair	61°56.8	8°56.7
Mer.pass. 12:03		ν -0.4' d -1.3' m -3.89			ν 3.3' d 0.3' m -1.35			ν 2.7' d -0.1' m -2.42			ν 2.2' d 0.1' m 0.39																																																																																																																																																																																																																																																																																																										

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec																																																																																																																																																																																																																																																																																																															
0	179°47.2	178°08.5	S00°48.2	352°43.5	N00°37.5	335°56.8	S08°16.7	142°19.6	N12°29.0	Enif	33°35.7	10°00.5	1	194°49.7	193°08.1	47.0	7°46.9	37.8	157°21.8	29.1	Al Na'ir	27°29.2	-46°49.0	2	209°52.1	208°07.7	45.7	22°50.2	38.1	6°02.2	16.5	172°24.0	29.2	Fomalhaut	15°11.2	-29°28.0	3	224°54.6	223°07.3	•• 44.4	37°53.6	•• 38.5	21°04.9	•• 16.4	187°26.2	•• 29.3	Scheat	13°42.2	28°14.5	4	239°57.0	238°07.0	43.2	52°57.0	38.8	36°07.6	16.3	202°28.4	29.4	Markab	13°26.8	15°21.7	5	254°59.5	253°06.6	41.9	68°00.3	39.2	51°10.4	16.2	217°30.6	29.5	Mar 20 Tue				SHA	Mer.pass	6	270°02.0	268°06.2	S00°40.7	83°03.7	N00°39.5	66°13.1	S08°16.1	232°32.8	N12°29.6	Venus	0°37.9	12:07	7	285°04.4	283°05.8	39.4	98°07.0	39.8	81°15.8	16.0	247°35.0	29.7	Mars	172°14.0	00:40	8	300°06.9	298°05.4	38.1	113°10.4	40.2	96°18.5	15.9	262°37.2	29.8	Jupiter	155°57.7	01:45	9	315°09.4	313°05.1	•• 36.9	128°13.8	•• 40.5	111°21.2	•• 15.8	277°39.4	•• 29.9	Saturn	322°45.1	14:36	10	330°11.8	328°04.7	35.6	143°17.1	40.8	126°24.0	15.7	292°41.6	30.0	Mar 21 Wed				SHA	Mer.pass	11	345°14.3	343°04.3	34.4	158°20.5	41.2	141°26.7	15.6	307°43.8	30.1	Venus	359°29.6	12:07	12	0°16.8	358°03.9	S00°33.1	173°23.8	N00°41.5	156°29.4	S08°15.5	322°46.0	N12°30.1	Mars	172°35.0	00:34	13	15°19.2	13°03.5	31.8	188°27.2	41.8	171°32.1	15.4	337°48.2	30.2	Jupiter	156°03.6	01:40	14	30°21.7	28°03.2	30.6	203°30.6	42.2	186°34.8	15.3	352°50.4	30.3	Saturn	322°38.8	14:32	15	45°24.1	43°02.8	•• 29.3	218°33.9	•• 42.5	201°37.6	•• 15.2	7°52.5	•• 30.4	Mar 22 Thu				SHA	Mer.pass	16	60°26.6	58°02.4	28.1	233°37.3	42.8	216°40.3	15.1	22°54.7	30.5	Venus	358°21.3	12:08	17	75°29.1	73°02.0	26.8	248°40.6	43.2	231°43.0	15.0	37°56.9	30.6	Mars	172°56.3	00:29	18	90°31.5	88°01.6	S00°25.5	263°44.0	N00°43.5	246°45.7	S08°14.9	52°59.1	N12°30.7	Jupiter	156°09.6	01:36	19	105°34.0	103°01.3	24.3	278°47.4	43.8	261°48.4	14.8	68°01.3	30.8	Saturn	322°32.4	14:29	20	120°36.5	118°00.9	23.0	293°50.7	44.2	276°51.2	14.7	83°03.5	30.9	Horizontal parallax				Venus:	0.1	21	135°38.9	133°00.5	•• 21.8	308°54.1	•• 44.5	291°53.9	•• 14.6	98°05.7	•• 31.0	Mars:	0.2	22	150°41.4	148°00.1	20.5	323°57.5	44.9	306°56.6	14.5	113°07.9	31.1	Mer.pass. 11:59				ν -0.4' d -1.3' m -3.89			ν 3.4' d 0.3' m -1.37			ν 2.7' d -0.1' m -2.43			ν 2.2' d 0.1' m 0.39		

DUT1 = UT1-UTC = +0.2080 sec $\Delta T = TT-UT1 = +68.9760$ sec

2029 March 20 to Mar. 22 UT

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	HP
Tue							
0	178°07.2	S00°07.9		129°05.3	11.8'	N21°54.5	6.1'
1	193°07.4	06.9		143°36.1	11.8'	22°00.7	6.0'
2	208°07.6	06.0		158°06.9	11.7'	22°06.7	5.9'
3	223°07.8	· · 05.0		172°37.6	11.7'	22°12.6	5.8'
4	238°08.0	04.0		187°08.3	11.6'	22°18.4	5.7'
5	253°08.1	03.0		201°38.9	11.5'	22°24.1	5.6'
6	268°08.3	S00°02.0		216°09.5	11.5'	N22°29.7	5.5'
7	283°08.5	01.0		230°39.9	11.4'	22°35.2	5.4'
8	298°08.7	S00°00.0		245°10.4	11.4'	22°40.7	5.3'
9	313°08.9	N00°01.0		259°40.7	11.3'	22°46.0	5.2'
10	328°09.1	02.0		274°11.0	11.2'	22°51.2	5.1'
11	343°09.3	02.9		288°41.3	11.2'	22°56.3	5.0'
12	358°09.4	N00°03.9		303°11.5	11.1'	N23°01.2	4.9'
13	13°09.6	04.9		317°41.6	11.1'	23°06.1	4.8'
14	28°09.8	05.9		332°11.7	11.0'	23°10.9	4.7'
15	43°10.0	· · 06.9		346°41.7	11.0'	23°15.6	4.6'
16	58°10.2	07.9		1°11.7	10.9'	23°20.1	4.4'
17	73°10.4	08.9		15°41.6	10.8'	23°24.6	4.3'
18	88°10.5	N00°09.9		30°11.4	10.8'	N23°28.9	4.2'
19	103°10.7	10.8		44°41.2	10.7'	23°33.1	4.1'
20	118°10.9	11.8		59°10.9	10.7'	23°37.3	4.0'
21	133°11.1	· · 12.8		73°40.6	10.6'	23°41.3	3.9'
22	148°11.3	13.8		88°10.2	10.6'	23°45.2	3.8'
23	163°11.5	14.8		102°39.8	10.5'	23°48.9	3.7'
	SD = 16.1'	d = -1.0'		SD = 14.8'			

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°11.7	N00°15.8	117°09.3	10.5'	N23°52.6	3.5'	54.8'
1	193°11.8	16.8	131°38.8	10.4'	23°56.1	3.4'	54.9'
2	208°12.0	17.8	146°08.2	10.4'	23°59.6	3.3'	54.9'
3	223°12.2	· · 18.7	160°37.6	10.3'	24°02.9	3.2'	54.9'
4	238°12.4	19.7	175°06.9	10.3'	24°06.1	3.1'	54.9'
5	253°12.6	20.7	189°36.1	10.2'	24°09.2	3.0'	54.9'
6	268°12.8	N00°21.7	204°05.3	10.2'	N24°12.1	2.8'	55.0'
7	283°13.0	22.7	218°34.5	10.1'	24°15.0	2.7'	55.0'
8	298°13.1	23.7	233°03.6	10.1'	24°17.7	2.6'	55.0'
9	313°13.3	· · 24.7	247°32.6	10.0'	24°20.3	2.5'	55.0'
10	328°13.5	25.7	262°01.6	10.0'	24°22.8	2.4'	55.1'
11	343°13.7	26.6	276°30.6	9.9'	24°25.2	2.2'	55.1'
12	358°13.9	N00°27.6	290°59.5	9.9'	N24°27.4	2.1'	55.1'
13	13°14.1	28.6	305°28.4	9.8'	24°29.5	2.0'	55.1'
14	28°14.3	29.6	319°57.2	9.8'	24°31.6	1.9'	55.1'
15	43°14.4	· · 30.6	334°25.9	9.7'	24°33.4	1.8'	55.2'
16	58°14.6	31.6	348°54.7	9.7'	24°35.2	1.6'	55.2'
17	73°14.8	32.6	3°23.3	9.6'	24°36.8	1.5'	55.2'
18	88°15.0	N00°33.6	17°52.0	9.6'	N24°38.3	1.4'	55.2'
19	103°15.2	34.5	32°20.6	9.6'	24°39.7	1.3'	55.3'
20	118°15.4	35.5	46°49.1	9.5'	24°41.0	1.1'	55.3'
21	133°15.6	· · 36.5	61°17.6	9.5'	24°42.1	1.0'	55.3'
22	148°15.7	37.5	75°46.1	9.4'	24°43.1	0.9'	55.4'
23	163°15.9	38.5	90°14.5	9.4'	24°44.0	0.7'	55.4'
	SD = 16.0'	d = 1.0'	SD = 15.0'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°16.1	N00°39.5	104°42.9	9.4'	N24°44.7	0.6'	55.4'
1	193°16.3	40.5	119°11.3	9.3'	24°45.3	0.5'	55.4'
2	208°16.5	41.4	133°39.6	9.3'	24°45.8	0.4'	55.5'
3	223°16.7	· · 42.4	148°07.9	9.2'	24°46.2	0.2'	55.5'
4	238°16.9	43.4	162°36.1	9.2'	24°46.4	0.1'	55.5'
5	253°17.1	44.4	177°04.4	9.2'	24°46.5	-0.0'	55.5'
6	268°17.2	N00°45.4	191°32.5	9.1'	N24°46.5	-0.2'	55.6'
7	283°17.4	46.4	206°00.7	9.1'	24°46.3	-0.3'	55.6'
8	298°17.6	47.4	220°28.8	9.1'	24°46.0	-0.4'	55.6'
9	313°17.8	· · 48.4	234°56.9	9.1'	24°45.6	-0.6'	55.7'
10	328°18.0	49.3	249°24.9	9.0'	24°45.1	-0.7'	55.7'
11	343°18.2	50.3	263°52.9	9.0'	24°44.4	-0.8'	55.7'
12	358°18.4	N00°51.3	278°20.9	9.0'	N24°43.6	-0.9'	55.7'
13	13°18.5	52.3	292°48.9	8.9'	24°42.6	-1.1'	55.8'
14	28°18.7	53.3	307°16.8	8.9'	24°41.5	-1.2'	55.8'
15	43°18.9	· · 54.3	321°44.7	8.9'	24°40.3	-1.3'	55.8'
16	58°19.1	55.3	336°12.6	8.9'	24°39.0	-1.5'	55.9'
17	73°19.3	56.2	350°40.5	8.8'	24°37.5	-1.6'	55.9'
18	88°19.5	N00°57.2	5°08.3	8.8'	N24°35.9	-1.7'	55.9'
19	103°19.7	58.2	19°36.1	8.8'	24°34.1	-1.9'	56.0'
20	118°19.9	00°59.2	34°03.9	8.8'	24°32.3	-2.0'	56.0'
21	133°20.0	01°00.2	48°31.7	8.8'	24°30.3	-2.2'	56.0'
22	148°20.2	01.2	62°59.5	8.7'	24°28.1	-2.3'	56.1'
23	163°20.4	02.2	77°27.2	8.7'	24°25.8	-2.4'	56.1'
	SD = 16.0'	d = 1.0'	SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:13	04:44	05:52	18:25	19:34	21:06
N 70°	03:33	04:52	05:53	18:23	19:25	20:45
68°	03:48	04:59	05:55	18:21	19:18	20:29
66°	04:01	05:04	05:56	18:20	19:12	20:16
64°	04:11	05:09	05:57	18:19	19:07	20:06
62°	04:19	05:13	05:57	18:18	19:03	19:57
60°	04:26	05:16	05:58	18:18	18:59	19:50
N 58°	04:32	05:19	05:59	18:17	18:56	19:43
56°	04:38	05:22	05:59	18:16	18:54	19:38
54°	04:42	05:24	05:59	18:16	18:51	19:33
52°	04:46	05:26	06:00	18:15	18:49	19:29
50°	04:50	05:28	06:00	18:15	18:47	19:25
45°	04:57	05:32	06:01	18:14	18:43	19:18
N 40°	05:03	05:35	06:02	18:13	18:40	19:12
35°	05:07	05:37	06:02	18:13	18:38	19:07
30°	05:11	05:39	06:02	18:12	18:36	19:04
20°	05:16	05:41	06:03	18:11	18:33	18:59
N 10°	05:18	05:43	06:04	18:11	18:32	18:56
0°	05:19	05:43	06:04	18:10	18:31	18:55
S 10°	05:19	05:43	06:04	18:10	18:31	18:55
20°	05:17	05:42	06:04	18:10	18:32	18:57
30°	05:12	05:40	06:04	18:10	18:33	19:01
35°	05:09	05:39	06:04	18:09	18:35	19:04
40°	05:05	05:37	06:04	18:09	18:36	19:08
45°	05:00	05:35	06:04	18:09	18:39	19:13
S 50°	04:53	05:31	06:04	18:09	18:42	19:20
52°	04:50	05:30	06:04	18:10	18:43	19:23
54°	04:46	05:28	06:03	18:10	18:45	19:26
56°	04:42	05:26	06:03	18:10	18:47	19:31
58°	04:37	05:24	06:03	18:10	18:49	19:35
S 60°	04:31	05:21	06:03	18:10	18:51	19:41

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	05:05	□	□	01:48	□	□
64°	05:45	05:57	06:34	01:08	02:39	03:51
62°	06:12	06:35	07:18	00:41	02:01	03:07
60°	06:34	07:02	07:47	00:20	01:35	02:37
N 58°	06:51	07:23	08:09	00:03	01:14	02:15
56°	07:06	07:40	08:28	· · ·	00:57	01:57
54°	07:19	07:55	08:43	· · ·	00:42	01:41
52°	07:30	08:08	08:56	· · ·	00:30	01:28
50°	07:40	08:19	09:08	· · ·	00:19	01:16
45°	08:01	08:42	09:32	23:56	· · ·	00:52
N 40°	08:18	09:01	09:52	23:37	· · ·	00:33
35°	08:33	09:17	10:08	23:22	· · ·	00:16
30°	08:45	09:31	10:22	23:08	· · ·	00:02
20°	09:07	09:54	10:46	22:45	23:38	· · ·
N 10°	09:26	10:14	11:07	22:26	23:18	· · ·
0°	09:43	10:33	11:26	22:07	22:59	23:52
S 10°	10:01	10:53	11:45	21:49	22:39	23:32
20°	10:20	11:13	12:06	21:29	22:18	23:12
30°	10:42	11:37	12:30	21:06	21:55	22:48
35°	10:55	11:51	12:44	20:53	21:40	22:34
40°	11:09	12:07	13:00	20:38	21:24	22:18
45°	11:27	12:26	13:20	20:19	21:05	22:19
S 50°	11:49	12:50	13:44	19:57	20:40	21:35
52°	12:00	13:02	13:56	19:46	20:29	21:23
54°	12:12	13:15	14:09	19:34	20:15	21:10
56°	12:26	13:30	14:25	19:20	20:00	20:54
58°	12:42	13:49	14:43	19:03	19:41	20:36
S 60°	13:01	14:11	15:06	18:44	19:19	20:14

Day	Sun		Mer. Pass	Moon		Age 5-7 20-37%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
20	07:31	07:22	12:07	15:55	03:30	
21	07:13	07:04	12:07	16:46	04:20	
22	06:56	06:47	12:07	17:39	05:12	

March 23, 24, 25 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	SHA	Dec		
0	180°46.3	177°59.4	S00°18.0	354°04.2	N00°45.5	337°02.0	S08°14.2	143°12.3	N12°31.3	Alpheratz	357°31.5	29°15.1					
1	195°48.8	192°59.0	16.7	9°07.6	45.9	352°04.8	14.1	158°14.5	31.4	Ankaa	353°04.3	-42°08.8					
2	210°51.3	207°58.6	15.5	24°10.9	46.2	7°07.5	14.0	173°16.7	31.5	Schedar	349°27.6	56°41.9					
3	225°53.7	222°58.2	•• 14.2	39°14.3	•• 46.5	22°10.2	•• 13.9	188°18.9	•• 31.6	Diphda	348°44.2	-17°49.6					
4	240°56.2	237°57.8	12.9	54°17.7	46.9	37°12.9	13.8	203°21.1	31.7	Achernar	335°18.3	-57°05.4					
5	255°58.6	252°57.5	11.7	69°21.0	47.2	52°15.6	13.7	218°23.3	31.8	Hamal	327°47.6	23°36.0					
6	271°01.1	267°57.1	S00°10.4	84°24.4	N00°47.5	67°18.4	S08°13.6	233°25.5	N12°31.8	Polaris	312°30.4	89°23.3					
7	286°03.6	282°56.7	09.2	99°27.8	47.9	82°21.1	13.5	248°27.7	31.9	Acamar	315°09.6	-40°11.5					
8	301°06.0	297°56.3	07.9	114°31.1	48.2	97°23.8	13.4	263°29.9	32.0	Menkar	314°02.8	4°12.1					
9	316°08.5	312°56.0	•• 06.6	129°34.5	•• 48.5	112°26.5	•• 13.3	278°32.1	•• 32.1	Mirfak	308°23.6	49°57.9					
10	331°11.0	327°55.6	05.4	144°37.9	48.9	127°29.3	13.2	293°34.2	32.2	Aldebaran	290°35.8	16°34.0					
11	346°13.4	342°55.2	04.1	159°41.2	49.2	142°32.0	13.1	308°36.4	32.3	Rigel	281°00.7	-8°10.3					
12	1°15.9	357°54.8	S00°02.9	174°44.6	N00°49.6	157°34.7	S08°13.0	323°38.6	N12°32.4	Capella	280°16.9	46°01.6					
13	16°18.4	12°54.4	01.6	189°48.0	49.9	172°37.4	12.9	338°40.8	32.5	Bellatrix	278°19.3	6°22.4					
14	31°20.8	27°54.1	S00°00.3	204°51.3	50.2	187°40.2	12.8	353°43.0	32.6	Elnath	277°57.6	28°37.8					
15	46°23.3	42°53.7	N00°00.9	219°54.7	•• 50.6	202°42.9	•• 12.7	8°45.2	•• 32.7	Alnilam	275°34.3	-1°11.2					
16	61°25.8	57°53.3	02.2	234°58.1	50.9	217°45.6	12.6	23°47.4	32.8	Betelgeuse	270°48.4	7°24.5					
17	76°28.2	72°52.9	03.5	250°01.5	51.2	232°48.3	12.5	38°49.6	32.9	Canopus	263°50.8	-52°43.1					
18	91°30.7	87°52.5	N00°04.7	265°04.8	N00°51.6	247°51.0	S08°12.4	53°51.8	N12°33.0	Sirius	258°23.2	-16°45.7					
19	106°33.1	102°52.2	06.0	280°08.2	51.9	262°53.8	12.3	68°54.0	33.1	Adhara	255°03.1	-29°01.1					
20	121°35.6	117°51.8	07.2	295°11.6	52.2	277°56.5	12.1	83°56.2	33.2	Procyon	244°47.2	5°08.7					
21	136°38.1	132°51.4	•• 08.5	310°14.9	•• 52.6	292°59.2	•• 12.0	98°58.4	•• 33.3	Pollux	243°13.0	27°57.2					
22	151°40.5	147°51.0	09.8	325°18.3	52.9	308°01.9	11.9	114°00.6	33.4	Avior	234°12.9	-59°36.6					
23	166°43.0	162°50.7	11.0	340°21.7	53.2	323°04.7	11.8	129°02.8	33.5	Suhail	222°43.4	-43°33.4					
Mer.pass. 11:55			ν -0.4' d -1.3' m -3.89			ν 3.4' d 0.3' m -1.39			ν 2.7' d -0.1' m -2.43			ν 2.2' d 0.1' m 0.39					

Sat	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec					
0	181°45.5	177°50.3	N00°12.3	355°25.1	N00°53.6	338°07.4	S08°11.7	144°05.0	N12°33.6					
1	196°47.9	192°49.9	13.5	10°28.4	53.9	353°10.1	11.6	159°07.1	33.6					
2	211°50.4	207°49.5	14.8	25°31.8	54.2	8°12.8	11.5	174°09.3	33.7					
3	226°52.9	222°49.1	•• 16.1	40°35.2	•• 54.6	23°15.6	•• 11.4	189°11.5	•• 33.8					
4	241°55.3	237°48.8	17.3	55°38.5	54.9	38°18.3	11.3	204°13.7	33.9					
5	256°57.8	252°48.4	18.6	70°41.9	55.2	53°21.0	11.2	219°15.9	34.0					
6	272°00.3	267°48.0	N00°19.8	85°45.3	N00°55.6	68°23.8	S08°11.1	234°18.1	N12°34.1					
7	287°02.7	282°47.6	21.1	100°48.7	55.9	83°26.5	11.0	249°20.3	34.2					
8	302°05.2	297°47.3	22.4	115°52.0	56.2	98°29.2	10.9	264°22.5	34.3					
9	317°07.6	312°46.9	•• 23.6	130°55.4	•• 56.6	113°31.9	•• 10.8	279°24.7	•• 34.4					
10	332°10.1	327°46.5	24.9	145°58.8	56.9	128°34.7	10.7	294°26.9	34.5					
11	347°12.6	342°46.1	26.2	161°02.2	57.3	143°37.4	10.6	309°29.1	34.6					
12	2°15.0	357°45.7	N00°27.4	176°05.5	N00°57.6	158°40.1	S08°10.5	324°31.3	N12°34.7					
13	17°17.5	12°45.4	28.7	191°08.9	57.9	173°42.8	10.3	339°33.5	34.8					
14	32°20.0	27°45.0	29.9	206°12.3	58.3	188°45.6	10.2	354°35.6	34.9					
15	47°22.4	42°44.6	•• 31.2	221°15.7	•• 58.6	203°48.3	•• 10.1	9°37.8	•• 35.0					
16	62°24.9	57°44.2	32.5	236°19.0	58.9	218°51.0	10.0	24°40.0	35.1					
17	77°27.4	72°43.9	33.7	251°22.4	59.3	233°53.8	09.9	39°42.2	35.2					
18	92°29.8	87°43.5	N00°35.0	266°25.8	N00°59.6	248°56.5	S08°09.8	54°44.4	N12°35.3					
19	107°32.3	102°43.1	36.2	281°29.2	00°59.9	263°59.2	09.7	69°46.6	35.4					
20	122°34.7	117°42.7	37.5	296°32.5	01°00.3	279°01.9	09.6	84°48.8	35.5					
21	137°37.2	132°42.3	•• 38.8	311°35.9	•• 00.6	294°04.7	•• 09.5	99°51.0	•• 35.5					
22	152°39.7	147°42.0	40.0	326°39.3	00.9	309°07.4	09.4	114°53.2	35.6					
23	167°42.1	162°41.6	41.3	341°42.7	01.3	324°10.1	09.3	129°55.4	35.7					
Mer.pass. 11:51			ν -0.4' d 1.3' m -3.89			ν 3.4' d 0.3' m -1.40			ν 2.7' d -0.1' m -2.43			ν 2.2' d 0.1' m 0.39		

Sun	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec					
0	182°44.6	177°41.2	N00°42.5	356°46.0	N01°01.6	339°12.9	S08°09.2	144°57.6	N12°35.8					
1	197°47.1	192°40.8	43.8	11°49.4	01.9	354°15.6	09.1	159°59.7	35.9					
2	212°49.5	207°40.5	45.1	26°52.8	02.3	9°18.3	09.0	175°01.9	36.0					
3	227°52.0	222°40.1	•• 46.3	41°56.2	•• 02.6	24°21.0	•• 08.9	190°04.1	•• 36.1					
4	242°54.5	237°39.7	47.6	56°59.5	02.9	39°23.8	08.7	205°06.3	36.2					
5	257°56.9	252°39.3	48.9	72°02.9	03.3	54°26.5	08.6	220°08.5	36.3					
6	272°59.4	267°39.0	N00°50.1	87°06.3	N01°03.6	69°29.2	S08°08.5	235°10.7	N12°36.4					
7	288°01.9	282°38.6	51.4	102°09.7	03.9	84°32.0	08.4	250°12.9	36.5					
8	303°04.3	297°38.2	52.6	117°13.1	04.3	99°34.7	08.3	265°15.1	36.6					
9	318°06.8	312°37.8	•• 53.9	132°16.4	•• 04.6	114°37.4	•• 08.2	280°17.3	•• 36.7					
10	333°09.2	327°37.4	55.2	147°19.8	04.9	129°40.1	08.1	295°19.5	36.8					
11	348°11.7	342°37.1	56.4	162°23.2	05.3	144°42.9	08.0	310°21.6	36.9					
12	3°14.2	357°36.7	N00°57.7	177°26.6	N01°05.6	159°45.6	S08°07.9	325°23.8	N12°37.0					
13	18°16.6	12°36.3	00°58.9	192°30.0	05.9	174°48.3	07.8	340°26.0	37.1					
14	33°19.1	27°35.9	01°00.2	207°33.3	06.3	189°51.1	07.7	355°28.2	37.2					
15	48°21.6	42°35.6	•• 01.5	222°36.7	•• 06.6	204°53.8	•• 07.6	10°30.4	•• 37.3					
16	63°24.0	57°35.2	02.7	237°40.1	06.9	219°56.5	07.5	25°32.6	37.4					
17	78°26.5	72°34.8	04.0	252°43.5	07.2	234°59.3	07.3	40°34.8	37.5					
18	93°29.0	87°34.4	N01°05.2	267°46.8	N01°07.6	250°02.0	S08°07.2	55°37.0	N12°37.5					
19	108°31.4	102°34.1	06.5	282°50.2	07.9	265°04.7	07.1	70°39.2	37.6					
20	123°33.9	117°33.7	07.8	297°53.6	08.2	280°07.5	07.0	85°41.4	37.7					
21	138°36.4	132°33.3	•• 09.0	312°57.0	•• 08.6	295°10.2	•• 06.9	100°43.5	•• 37.8					
22	153°38.8	147°32.9	10.3	328°00.4	08.9	310°12.9	06.8	115°45.7	37.9					
23	168°41.3	162°32.5	11.6	343°03.7	09.2	325°15.7	06.7	130°47.9	38.0					
Mer.pass. 11:47			ν -0.4' d 1.3' m -3.89			ν 3.4' d 0.3' m -1.41			ν 2.7' d -0.1' m -2.44			ν 2.2' d 0.1' m 0.38		

Mar 23 Fri			SHA	Mer.pass
Venus			357°13.0	12:08
Mars			173°17.9	00:24
Jupiter			156°15.7	01:32
Saturn			322°26.0	14:25

Mar 24 Sat			SHA	Mer.pass
Venus			356°04.8	12:09
Mars			173°39.6	00:18
Jupiter			156°21.9	01:27
Saturn			322°19.5	14:22

Mar 25 Sun			SHA	Mer.pass
Venus			354°56.6	12:10

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°20.6	N01°03.1	91°54.9	8.7'	N24°23.4	-2.6'	56.1'
1	193°20.8	04.1	106°22.6	8.7'	24°20.8	-2.7'	56.2'
2	208°21.0	05.1	120°50.3	8.7'	24°18.1	-2.8'	56.2'
3	223°21.2	· · 06.1	135°17.9	8.7'	24°15.3	-3.0'	56.2'
4	238°21.4	07.1	149°45.6	8.6'	24°12.4	-3.1'	56.3'
5	253°21.5	08.1	164°13.2	8.6'	24°09.3	-3.2'	56.3'
6	268°21.7	N01°09.1	178°40.9	8.6'	N24°06.0	-3.4'	56.3'
7	283°21.9	10.0	193°08.5	8.6'	24°02.7	-3.5'	56.4'
8	298°22.1	11.0	207°36.1	8.6'	23°59.2	-3.6'	56.4'
9	313°22.3	· · 12.0	222°03.7	8.6'	23°55.5	-3.8'	56.4'
10	328°22.5	13.0	236°31.3	8.6'	23°51.7	-3.9'	56.5'
11	343°22.7	14.0	250°58.8	8.6'	23°47.8	-4.0'	56.5'
12	358°22.9	N01°15.0	265°26.4	8.6'	N23°43.8	-4.2'	56.5'
13	13°23.0	16.0	279°53.9	8.6'	23°39.6	-4.3'	56.6'
14	28°23.2	16.9	294°21.5	8.5'	23°35.3	-4.5'	56.6'
15	43°23.4	· · 17.9	308°49.0	8.5'	23°30.8	-4.6'	56.6'
16	58°23.6	18.9	323°16.6	8.5'	23°26.2	-4.7'	56.7'
17	73°23.8	19.9	337°44.1	8.5'	23°21.5	-4.9'	56.7'
18	88°24.0	N01°20.9	352°11.7	8.5'	N23°16.7	-5.0'	56.7'
19	103°24.2	21.9	6°39.2	8.5'	23°11.7	-5.1'	56.8'
20	118°24.4	22.8	21°06.7	8.5'	23°06.5	-5.3'	56.8'
21	133°24.6	· · 23.8	35°34.3	8.5'	23°01.3	-5.4'	56.9'
22	148°24.7	24.8	50°01.8	8.5'	22°55.9	-5.5'	56.9'
23	163°24.9	25.8	64°29.3	8.5'	22°50.3	-5.7'	56.9'
SD = 16.0'			SD = 15.3'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°25.1	N01°26.8	78°56.9	8.5'	N22°44.7	-5.8'	57.0'
1	193°25.3	27.8	93°24.4	8.5'	22°38.9	-5.9'	57.0'
2	208°25.5	28.7	107°51.9	8.5'	22°32.9	-6.1'	57.0'
3	223°25.7	· · 29.7	122°19.5	8.5'	22°26.9	-6.2'	57.1'
4	238°25.9	30.7	136°47.0	8.6'	22°20.7	-6.3'	57.1'
5	253°26.1	31.7	151°14.6	8.6'	22°14.3	-6.5'	57.2'
6	268°26.2	N01°32.7	165°42.1	8.6'	N22°07.9	-6.6'	57.2'
7	283°26.4	33.7	180°09.7	8.6'	22°01.3	-6.7'	57.2'
8	298°26.6	34.7	194°37.3	8.6'	21°54.5	-6.9'	57.3'
9	313°26.8	· · 35.6	209°04.9	8.6'	21°47.7	-7.0'	57.3'
10	328°27.0	36.6	223°32.5	8.6'	21°40.7	-7.1'	57.3'
11	343°27.2	37.6	238°00.1	8.6'	21°33.5	-7.3'	57.4'
12	358°27.4	N01°38.6	252°27.7	8.6'	N21°26.3	-7.4'	57.4'
13	13°27.6	39.6	266°55.3	8.6'	21°18.9	-7.5'	57.5'
14	28°27.8	40.6	281°22.9	8.6'	21°11.4	-7.6'	57.5'
15	43°27.9	· · 41.5	295°50.5	8.6'	21°03.7	-7.8'	57.5'
16	58°28.1	42.5	310°18.2	8.7'	20°55.9	-7.9'	57.6'
17	73°28.3	43.5	324°45.8	8.7'	20°48.0	-8.0'	57.6'
18	88°28.5	N01°44.5	339°13.5	8.7'	N20°40.0	-8.2'	57.7'
19	103°28.7	45.5	353°41.2	8.7'	20°31.8	-8.3'	57.7'
20	118°28.9	46.4	8°08.9	8.7'	20°23.5	-8.4'	57.7'
21	133°29.1	· · 47.4	22°36.6	8.7'	20°15.1	-8.5'	57.8'
22	148°29.3	48.4	37°04.3	8.7'	20°06.6	-8.7'	57.8'
23	163°29.5	49.4	51°32.1	8.8'	19°57.9	-8.8'	57.9'
SD = 16.0'			SD = 15.5'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°29.6	N01°50.4	65°59.8	8.8'	N19°49.1	-8.9'	57.9'
1	193°29.8	51.4	80°27.6	8.8'	19°40.2	-9.0'	57.9'
2	208°30.0	52.3	94°55.4	8.8'	19°31.2	-9.2'	58.0'
3	223°30.2	· · 53.3	109°23.1	8.8'	19°22.0	-9.3'	58.0'
4	238°30.4	54.3	123°51.0	8.8'	19°12.7	-9.4'	58.1'
5	253°30.6	55.3	138°18.8	8.8'	19°03.3	-9.5'	58.1'
6	268°30.8	N01°56.3	152°46.6	8.9'	N18°53.8	-9.6'	58.1'
7	283°31.0	57.3	167°14.5	8.9'	18°44.1	-9.8'	58.2'
8	298°31.2	58.2	181°42.4	8.9'	18°34.4	-9.9'	58.2'
9	313°31.3	01°59.2	196°10.2	8.9'	18°24.5	-10.0'	58.3'
10	328°31.5	02°00.2	210°38.2	8.9'	18°14.5	-10.1'	58.3'
11	343°31.7	01.2	225°06.1	8.9'	18°04.4	-10.2'	58.3'
12	358°31.9	N02°02.2	239°34.0	9.0'	N17°54.1	-10.4'	58.4'
13	13°32.1	03.1	254°02.0	9.0'	17°43.8	-10.5'	58.4'
14	28°32.3	04.1	268°30.0	9.0'	17°33.3	-10.6'	58.5'
15	43°32.5	· · 05.1	282°57.9	9.0'	17°22.7	-10.7'	58.5'
16	58°32.7	06.1	297°25.9	9.0'	17°12.0	-10.8'	58.5'
17	73°32.9	07.1	311°54.0	9.0'	17°01.2	-10.9'	58.6'
18	88°33.0	N02°08.1	326°22.0	9.1'	N16°50.3	-11.0'	58.6'
19	103°33.2	09.0	340°50.1	9.1'	16°39.2	-11.1'	58.7'
20	118°33.4	10.0	355°18.1	9.1'	16°28.1	-11.3'	58.7'
21	133°33.6	· · 11.0	9°46.2	9.1'	16°16.8	-11.4'	58.7'
22	148°33.8	12.0	24°14.3	9.1'	16°05.5	-11.5'	58.8'
23	163°34.0	13.0	38°42.5	9.1'	15°54.0	-11.6'	58.8'
SD = 16.0'			SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:51	04:27	05:36	18:39	19:49	21:27
N 70°	03:15	04:37	05:40	18:35	19:38	21:02
68°	03:33	04:46	05:42	18:32	19:29	20:43
66°	03:47	04:52	05:44	18:30	19:22	20:28
64°	03:59	04:58	05:46	18:28	19:16	20:16
62°	04:08	05:03	05:48	18:26	19:11	20:06
60°	04:16	05:07	05:49	18:25	19:07	19:58
N 58°	04:23	05:11	05:50	18:24	19:03	19:51
56°	04:29	05:14	05:51	18:22	19:00	19:45
54°	04:34	05:17	05:52	18:21	18:57	19:39
52°	04:39	05:19	05:53	18:20	18:54	19:35
50°	04:43	05:21	05:54	18:20	18:52	19:31
45°	04:51	05:26	05:55	18:18	18:47	19:22
N 40°	04:58	05:30	05:57	18:16	18:43	19:15
35°	05:03	05:33	05:58	18:15	18:40	19:10
30°	05:07	05:35	05:59	18:14	18:38	19:06
20°	05:13	05:38	06:00	18:12	18:34	19:00
N 10°	05:16	05:41	06:02	18:11	18:32	18:56
0°	05:18	05:42	06:03	18:09	18:30	18:54
S 10°	05:19	05:43	06:04	18:08	18:29	18:54
20°	05:17	05:43	06:05	18:07	18:29	18:55
30°	05:14	05:42	06:06	18:06	18:30	18:57
35°	05:12	05:41	06:07	18:05	18:30	19:00
40°	05:09	05:40	06:07	18:05	18:32	19:03
45°	05:04	05:38	06:08	18:04	18:33	19:07
S 50°	04:58	05:36	06:08	18:03	18:35	19:13
52°	04:56	05:35	06:09	18:03	18:36	19:16
54°	04:52	05:34	06:09	18:02	18:37	19:19
56°	04:49	05:32	06:09	18:02	18:39	19:22
58°	04:44	05:31	06:10	18:01	18:40	19:26
S 60°	04:39	05:29	06:10	18:01	18:42	19:31

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	□	□	□	□	□	□
N 70°	□	□	09:39	□	□	06:23
68°	□	□	10:24	□	□	05:38
66°	06:18	08:50	10:53	05:58	05:18	05:08
64°	07:48	09:27	11:14	04:27	04:41	04:45
62°	08:26	09:54	11:32	03:50	04:14	04:27
60°	08:52	10:14	11:46	03:23	03:53	04:12
N 58°	09:13	10:31	11:58	03:02	03:36	03:59
56°	09:30	10:45	12:09	02:45	03:21	03:48
54°	09:44	10:57	12:18	02:30	03:08	03:38
52°	09:57	11:08	12:27	02:17	02:57	03:29
50°	10:08	11:17	12:34	02:06	02:47	03:21
45°	10:31	11:37	12:50	01:43	02:27	03:04
N 40°	10:49	11:54	13:03	01:24	02:10	02:50
35°	11:05	12:07	13:13	01:08	01:55	02:38
30°	11:18	12:19	13:23	00:54	01:43	02:28
20°	11:41	12:39	13:39	00:31	01:21	02:10
N 10°	12:01	12:57	13:54	00:11	01:03	01:54
0°	12:19	13:13	14:07	· ·	00:45	01:39
S 10°	12:38	13:30	14:20	· ·	00:28	01:24
20°	12:58	13:47	14:34	· ·	00:09	01:08
30°	13:20	14:07	14:50	23:47	· ·	00:49
35°	13:34	14:19	15:00	23:34	· ·	00:39
40°	13:49	14:32	15:10	23:19	· ·	00:26
45°	14:07	14:48	15:22	23:02	· ·	00:11
S 50°	14:30	15:07	15:37	22:40	23:53	· ·
52°	14:41	15:16	15:44	22:29	23:44	· ·
54°	14:53	15:26	15:52	22:17	23:35	· ·
56°	15:07	15:38	16:01	22:03	23:24	· ·
58°	15:23	15:51	16:10	21:47	23:11	· ·
S 60°	15:43	16:06	16:22	21:28	22:56	· ·

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12h	Pass	Mer.Pass.	Lower	8-10
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	47-68%
23	06:38	06:29	12:06	18:32	06:05	
24	06:20	06:10	12:06	19:26	06:59	
25	06:01	05:52	12:06	20:19	07:53	

March 26, 27, 28 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	SHA	Dec		
0	183°43.7	177°32.2	N01°12.8	358°07.1	N01°09.6	340°18.4	S08°06.6	145°50.1	N12°38.1	Alpheratz	357°31.5	29°15.1					
1	198°46.2	192°31.8	14.1	13°10.5	09.9	355°21.1	06.5	160°52.3	38.2	Ankaa	353°04.3	-42°08.8					
2	213°48.7	207°31.4	15.3	28°13.9	10.2	10°23.9	06.4	175°54.5	38.3	Schedar	349°27.5	56°41.9					
3	228°51.1	222°31.0	•• 16.6	43°17.3	•• 10.6	25°26.6	•• 06.3	190°56.7	•• 38.4	Diphda	348°44.2	-17°49.6					
4	243°53.6	237°30.7	17.9	58°20.6	10.9	40°29.3	06.2	205°58.9	38.5	Achernar	335°18.3	-57°05.4					
5	258°56.1	252°30.3	19.1	73°24.0	11.2	55°32.1	06.0	221°01.1	38.6	Hamal	327°47.6	23°36.0					
6	273°58.5	267°29.9	N01°20.4	88°27.4	N01°11.6	70°34.8	S08°05.9	236°03.2	N12°38.7	Polaris	312°31.5	89°23.3					
7	289°01.0	282°29.5	21.6	103°30.8	11.9	85°37.5	05.8	251°05.4	38.8	Acamar	315°09.6	-40°11.5					
8	304°03.5	297°29.1	22.9	118°34.2	12.2	100°40.3	05.7	266°07.6	38.9	Menkar	314°02.8	4°12.1					
9	319°05.9	312°28.8	•• 24.2	133°37.6	•• 12.5	115°43.0	•• 05.6	281°09.8	•• 39.0	Mirfak	308°23.6	49°57.9					
10	334°08.4	327°28.4	25.4	148°40.9	12.9	130°45.7	05.5	296°12.0	39.1	Aldebaran	290°35.8	16°34.0					
11	349°10.8	342°28.0	26.7	163°44.3	13.2	145°48.5	05.4	311°14.2	39.2	Rigel	281°00.7	-8°10.3					
12	4°13.3	357°27.6	N01°27.9	178°47.7	N01°13.5	160°51.2	S08°05.3	326°16.4	N12°39.3	Capella	280°17.0	46°01.6					
13	19°15.8	12°27.3	29.2	193°51.1	13.9	175°53.9	05.2	341°18.6	39.4	Bellatrix	278°19.3	6°22.4					
14	34°18.2	27°26.9	30.5	208°54.5	14.2	190°56.7	05.1	356°20.7	39.5	Elnath	277°57.6	28°37.8					
15	49°20.7	42°26.5	•• 31.7	223°57.8	•• 14.5	205°59.4	•• 05.0	11°22.9	•• 39.6	Alnilam	275°34.3	-1°11.2					
16	64°23.2	57°26.1	33.0	239°01.2	14.9	221°02.1	04.8	26°25.1	39.7	Betelgeuse	270°48.4	7°24.5					
17	79°25.6	72°25.8	34.2	254°04.6	15.2	236°04.9	04.7	41°27.3	39.7	Canopus	263°50.9	-52°43.1					
18	94°28.1	87°25.4	N01°35.5	269°08.0	N01°15.5	251°07.6	S08°04.6	56°29.5	N12°39.8	Sirius	258°23.2	-16°45.7					
19	109°30.6	102°25.0	36.8	284°11.4	15.8	266°10.3	04.5	71°31.7	39.9	Adhara	255°03.1	-29°01.1					
20	124°33.0	117°24.6	38.0	299°14.7	16.2	281°13.1	04.4	86°33.9	40.0	Procyon	244°47.2	5°08.7					
21	139°35.5	132°24.2	•• 39.3	314°18.1	•• 16.5	296°15.8	•• 04.3	101°36.1	•• 40.1	Pollux	243°13.0	27°57.2					
22	154°38.0	147°23.9	40.5	329°21.5	16.8	311°18.5	04.2	116°38.2	40.2	Avior	234°13.0	-59°36.6					
23	169°40.4	162°23.5	41.8	344°24.9	17.2	326°21.3	04.1	131°40.4	40.3	Suhail	222°43.5	-43°33.4					
Mer.pass. 11:43			ν -0.4' d 1.3' m -3.90			ν 3.4' d 0.3' m -1.41			ν 2.7' d -0.1' m -2.44			ν 2.2' d 0.1' m 0.38					

Tue	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec									
0	184°42.9	177°23.1	N01°43.1	359°28.3	N01°17.5	341°24.0	S08°04.0	146°42.6	N12°40.4	1	199°45.3	192°22.7	44.3	14°31.7	17.8									
1	199°45.3	192°22.7	44.3	14°31.7	17.8	356°26.8	03.9	161°44.8	40.5	2	214°47.8	207°22.4	45.6	29°35.0	18.1									
2	214°47.8	207°22.4	45.6	29°35.0	18.1	11°29.5	03.8	176°47.0	40.6	3	229°50.3	222°22.0	•• 46.8	44°38.4	•• 18.5									
3	229°50.3	222°22.0	•• 46.8	44°38.4	•• 18.5	26°32.2	•• 03.6	191°49.2	•• 40.7	4	244°52.7	237°21.6	48.1	59°41.8	18.8									
4	244°52.7	237°21.6	48.1	59°41.8	18.8	41°35.0	03.5	206°51.4	40.8	5	259°55.2	252°21.2	49.4	74°45.2	19.1									
5	259°55.2	252°21.2	49.4	74°45.2	19.1	56°37.7	03.4	221°53.6	40.9	6	274°57.7	267°20.9	N01°50.6	89°48.6	N01°19.5									
6	274°57.7	267°20.9	N01°50.6	89°48.6	N01°19.5	71°40.4	S08°03.3	236°55.7	N12°41.0	7	290°00.1	282°20.5	51.9	104°51.9	19.8									
7	290°00.1	282°20.5	51.9	104°51.9	19.8	86°43.2	03.2	251°57.9	41.1	8	305°02.6	297°20.1	53.1	119°55.3	20.1									
8	305°02.6	297°20.1	53.1	119°55.3	20.1	101°45.9	03.1	267°00.1	41.2	9	320°05.1	312°19.7	•• 54.4	134°58.7	•• 20.4									
9	320°05.1	312°19.7	•• 54.4	134°58.7	•• 20.4	116°48.6	•• 03.0	282°02.3	•• 41.3	10	335°07.5	327°19.3	55.7	150°02.1	20.8									
10	335°07.5	327°19.3	55.7	150°02.1	20.8	131°51.4	02.9	297°04.5	41.4	11	350°10.0	342°19.0	56.9	165°05.5	21.1									
11	350°10.0	342°19.0	56.9	165°05.5	21.1	146°54.1	02.8	312°06.7	41.5	12	5°12.5	357°18.6	N01°58.2	180°08.8	N01°21.4									
12	5°12.5	357°18.6	N01°58.2	180°08.8	N01°21.4	161°56.9	S08°02.7	327°08.9	N12°41.6	13	20°14.9	12°18.2	01°59.4	195°12.2	21.7									
13	20°14.9	12°18.2	01°59.4	195°12.2	21.7	176°59.6	02.5	342°11.0	41.7	14	35°17.4	27°17.8	02°00.7	210°15.6	22.1									
14	35°17.4	27°17.8	02°00.7	210°15.6	22.1	192°02.3	02.4	357°13.2	41.8	15	50°19.8	42°17.5	•• 02.0	225°19.0	•• 22.4									
15	50°19.8	42°17.5	•• 02.0	225°19.0	•• 22.4	207°05.1	•• 02.3	12°15.4	•• 41.9	16	65°22.3	57°17.1	03.2	240°22.4	22.7									
16	65°22.3	57°17.1	03.2	240°22.4	22.7	222°07.8	02.2	27°17.6	42.0	17	80°24.8	72°16.7	04.5	255°25.8	23.1									
17	80°24.8	72°16.7	04.5	255°25.8	23.1	237°10.6	02.1	42°19.8	42.0	18	95°27.2	87°16.3	N02°05.7	270°29.1	N01°23.4									
18	95°27.2	87°16.3	N02°05.7	270°29.1	N01°23.4	252°13.3	S08°02.0	57°22.0	N12°42.1	19	110°29.7	102°15.9	07.0	285°32.5	23.7									
19	110°29.7	102°15.9	07.0	285°32.5	23.7	267°16.0	01.9	72°24.2	42.2	20	125°32.2	117°15.6	08.2	300°35.9	24.0									
20	125°32.2	117°15.6	08.2	300°35.9	24.0	282°18.8	01.8	87°26.3	42.3	21	140°34.6	132°15.2	•• 09.5	315°39.3	•• 24.4									
21	140°34.6	132°15.2	•• 09.5	315°39.3	•• 24.4	297°21.5	•• 01.7	102°28.5	•• 42.4	22	155°37.1	147°14.8	10.8	330°42.7	24.7									
22	155°37.1	147°14.8	10.8	330°42.7	24.7	312°24.2	01.6	117°30.7	42.5	23	170°39.6	162°14.4	12.0	345°46.0	25.0									
23	170°39.6	162°14.4	12.0	345°46.0	25.0	327°27.0	01.4	132°32.9	42.6	Mer.pass. 11:39			ν -0.4' d 1.3' m -3.90			ν 3.4' d 0.3' m -1.41			ν 2.7' d -0.1' m -2.44			ν 2.2' d 0.1' m 0.38		

Wed	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec
0	185°42.0	177°14.1	N02°13.3	0°49.4	N01°25.3	342°29.7	S08°01.3	147°35.1	N12°42.7	1	200°44.5	192°13.7	14.5	15°52.8	25.7
1	200°44.5	192°13.7	14.5	15°52.8	25.7	357°32.5	01.2	162°37.3	42.8	2	215°46.9	207°13.3	15.8	30°56.2	26.0
2	215°46.9	207°13.3	15.8	30°56.2	26.0	12°35.2	01.1	177°39.5	42.9	3	230°49.4	222°12.9	•• 17.1	45°59.6	•• 26.3
3	230°49.4	222°12.9	•• 17.1	45°59.6	•• 26.3	27°37.9	•• 01.0	192°41.6	•• 43.0	4	245°51.9	237°12.5	18.3	61°03.0	26.6
4	245°51.9	237°12.5	18.3	61°03.0	26.6	42°40.7	00.9	207°43.8	43.1	5	260°54.3	252°12.2	19.6	76°06.3	27.0
5	260°54.3	252°12.2	19.6	76°06.3	27.0	57°43.4	00.8	222°46.0	43.2	6	275°56.8	267°11.8	N02°20.8	91°09.7	N01°27.3
6	275°56.8	267°11.8	N02°20.8	91°09.7	N01°27.3	72°46.2	S08°00.7	237°48.2	N12°43.3	7	290°59.3	282°11.4	22.1	106°13.1	27.6
7	290°59.3	282°11.4	22.1	106°13.1	27.6	87°48.9	00.6	252°50.4	43.4	8	306°01.7	297°11.0	23.3	121°16.5	27.9
8	306°01.7	297°11.0	23.3	121°16.5	27.9	102°51.6	00.4	267°52.6	43.5	9	321°04.2	312°10.7	•• 24.6	136°19.9	•• 28.2
9	321°04.2	312°10.7	•• 24.6	136°19.9	•• 28.2	117°54.4	•• 00.3	282°54.8	•• 43.6	10	336°06.7	327°10.3	25.9	151°23.2	28.6
10	336°06.7	327°10.3	25.9	151°23.2	28.6	132°57.1	00.2	297°56.9	43.7	11	351°0				

DUT1 = UT1-UTC = +0.2082 sec ΔT = TT-UT1 = +68.9758 sec

2029 March 26 to Mar. 28 UT

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	d
Mon							HP
0	178°34.2	N02°13.9		53°10.6	9.2'	N15°42.4	-11.7'
1	193°34.4	14.9		67°38.8	9.2'	15°30.7	-11.8'
2	208°34.6	15.9		82°06.9	9.2'	15°18.9	-11.9'
3	223°34.8	16.9	· ·	96°35.1	9.2'	15°07.1	-12.0'
4	238°34.9	17.9		111°03.3	9.2'	14°55.1	-12.1'
5	253°35.1	18.8		125°31.6	9.2'	14°43.0	-12.2'
6	268°35.3	N02°19.8		139°59.8	9.2'	N14°30.8	-12.3'
7	283°35.5	20.8		154°28.0	9.3'	14°18.5	-12.4'
8	298°35.7	21.8		168°56.3	9.3'	14°06.1	-12.5'
9	313°35.9	22.8	· ·	183°24.6	9.3'	13°53.6	-12.6'
10	328°36.1	23.7		197°52.9	9.3'	13°41.0	-12.7'
11	343°36.3	24.7		212°21.2	9.3'	13°28.3	-12.8'
12	358°36.5	N02°25.7		226°49.5	9.3'	N13°15.5	-12.9'
13	13°36.6	26.7		241°17.8	9.3'	13°02.7	-13.0'
14	28°36.8	27.7		255°46.2	9.4'	12°49.7	-13.1'
15	43°37.0	28.6	· ·	270°14.5	9.4'	12°36.6	-13.1'
16	58°37.2	29.6		284°42.9	9.4'	12°23.5	-13.2'
17	73°37.4	30.6		299°11.3	9.4'	12°10.3	-13.3'
18	88°37.6	N02°31.6		313°39.7	9.4'	N11°56.9	-13.4'
19	103°37.8	32.6		328°08.1	9.4'	11°43.5	-13.5'
20	118°38.0	33.5		342°36.5	9.4'	11°30.0	-13.6'
21	133°38.2	34.5	· ·	357°04.9	9.4'	11°16.5	-13.7'
22	148°38.3	35.5		11°33.3	9.4'	11°02.8	-13.7'
23	163°38.5	36.5		26°01.8	9.4'	10°49.1	-13.8'
SD = 16.0'				SD = 16.1'			
d = 1.0'							

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°38.7	N02°37.4	40°30.2	9.5'	N10°35.3	-13.9'	59.8'
1	193°38.9	38.4	54°58.7	9.5'	10°21.4	-14.0'	59.8'
2	208°39.1	39.4	69°27.1	9.5'	10°07.4	-14.0'	59.9'
3	223°39.3	40.4	83°55.6	9.5'	09°53.3	-14.1'	59.9'
4	238°39.5	41.4	98°24.1	9.5'	09°39.2	-14.2'	59.9'
5	253°39.7	42.3	112°52.5	9.5'	09°25.0	-14.3'	60.0'
6	268°39.9	N02°43.3	127°21.0	9.5'	N09°10.8	-14.3'	60.0'
7	283°40.1	44.3	141°49.5	9.5'	08°56.4	-14.4'	60.0'
8	298°40.2	45.3	156°18.0	9.5'	08°42.0	-14.5'	60.1'
9	313°40.4	46.2	170°46.5	9.5'	08°27.5	-14.5'	60.1'
10	328°40.6	47.2	185°15.0	9.5'	08°13.0	-14.6'	60.1'
11	343°40.8	48.2	199°43.5	9.5'	07°58.4	-14.7'	60.2'
12	358°41.0	N02°49.2	214°12.0	9.5'	N07°43.7	-14.7'	60.2'
13	13°41.2	50.2	228°40.5	9.5'	07°29.0	-14.8'	60.2'
14	28°41.4	51.1	243°08.9	9.5'	07°14.2	-14.8'	60.3'
15	43°41.6	52.1	257°37.4	9.5'	06°59.4	-14.9'	60.3'
16	58°41.8	53.1	272°05.9	9.5'	06°44.5	-15.0'	60.3'
17	73°41.9	54.1	286°34.4	9.5'	06°29.5	-15.0'	60.4'
18	88°42.1	N02°55.0	301°02.9	9.5'	N06°14.5	-15.1'	60.4'
19	103°42.3	56.0	315°31.4	9.5'	05°59.4	-15.1'	60.4'
20	118°42.5	57.0	329°59.8	9.5'	05°44.3	-15.2'	60.5'
21	133°42.7	58.0	344°28.3	9.5'	05°29.1	-15.2'	60.5'
22	148°42.9	58.9	358°56.7	9.4'	05°13.9	-15.3'	60.5'
23	163°43.1	59.9	13°25.2	9.4'	04°58.7	-15.3'	60.6'
SD = 16.0'				SD = 16.3'			
d = 1.0'							

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°43.3	N03°00.9	27°53.6	9.4'	N04°43.4	-15.3'	60.6'
1	193°43.5	01.9	42°22.1	9.4'	04°28.0	-15.4'	60.6'
2	208°43.7	02.8	56°50.5	9.4'	04°12.6	-15.4'	60.6'
3	223°43.8	03.8	71°18.9	9.4'	03°57.2	-15.5'	60.7'
4	238°44.0	04.8	85°47.3	9.4'	03°41.8	-15.5'	60.7'
5	253°44.2	05.8	100°15.7	9.4'	03°26.3	-15.5'	60.7'
6	268°44.4	N03°06.8	114°44.0	9.4'	N03°10.7	-15.6'	60.7'
7	283°44.6	07.7	129°12.4	9.3'	02°55.1	-15.6'	60.8'
8	298°44.8	08.7	143°40.7	9.3'	02°39.6	-15.6'	60.8'
9	313°45.0	09.7	158°09.1	9.3'	02°23.9	-15.7'	60.8'
10	328°45.2	10.7	172°37.4	9.3'	02°08.3	-15.7'	60.8'
11	343°45.4	11.6	187°05.7	9.3'	01°52.6	-15.7'	60.9'
12	358°45.5	N03°12.6	201°33.9	9.3'	N01°36.9	-15.7'	60.9'
13	13°45.7	13.6	216°02.2	9.2'	01°21.2	-15.7'	60.9'
14	28°45.9	14.6	230°30.4	9.2'	01°05.4	-15.8'	60.9'
15	43°46.1	15.5	244°58.6	9.2'	00°49.6	-15.8'	61.0'
16	58°46.3	16.5	259°26.8	9.2'	00°33.9	-15.8'	61.0'
17	73°46.5	17.5	273°55.0	9.1'	00°18.1	-15.8'	61.0'
18	88°46.7	N03°18.4	288°23.1	9.1'	N00°02.3	-15.8'	61.0'
19	103°46.9	19.4	302°51.3	9.1'	S00°13.6	15.8'	61.0'
20	118°47.1	20.4	317°19.4	9.1'	00°29.4	15.8'	61.1'
21	133°47.2	21.4	331°47.4	9.0'	00°45.2	15.8'	61.1'
22	148°47.4	22.3	346°15.5	9.0'	01°01.1	15.8'	61.1'
23	163°47.6	23.3	0°43.5	9.0'	01°16.9	15.8'	61.1'
SD = 16.0'				SD = 16.5'			
d = 1.0'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:26	04:10	05:21	18:52	20:04	21:51
N 70°	02:56	04:22	05:26	18:47	19:51	21:20
68°	03:17	04:32	05:29	18:43	19:41	20:58
66°	03:33	04:40	05:33	18:40	19:33	20:41
64°	03:46	04:47	05:35	18:37	19:26	20:27
62°	03:57	04:53	05:38	18:34	19:20	20:16
60°	04:06	04:58	05:40	18:32	19:15	20:07
N 58°	04:14	05:02	05:42	18:30	19:10	19:59
56°	04:20	05:06	05:43	18:29	19:06	19:52
54°	04:26	05:09	05:45	18:27	19:03	19:46
52°	04:31	05:12	05:46	18:26	19:00	19:40
50°	04:36	05:15	05:47	18:24	18:57	19:36
45°	04:45	05:20	05:50	18:22	18:51	19:26
N 40°	04:53	05:25	05:52	18:19	18:47	19:19
35°	04:59	05:28	05:54	18:17	18:43	19:13
30°	05:03	05:31	05:55	18:16	18:40	19:08
20°	05:10	05:36	05:58	18:13	18:35	19:01
N 10°	05:15	05:39	06:00	18:11	18:32	18:56
0°	05:17	05:41	06:02	18:09	18:29	18:53
S 10°	05:19	05:43	06:04	18:07	18:27	18:52
20°	05:18	05:44	06:06	18:04	18:26	18:52
30°	05:16	05:44	06:08	18:02	18:26	18:54
35°	05:14	05:44	06:09	18:01	18:26	18:56
40°	05:12	05:43	06:10	18:00	18:27	18:58
45°	05:08	05:42	06:12	17:58	18:27	19:01
S 50°	05:03	05:41	06:13	17:56	18:29	19:06
52°	05:01	05:40	06:14	17:56	18:29	19:08
54°	04:58	05:39	06:15	17:55	18:30	19:11
56°	04:55	05:39	06:16	17:54	18:31	19:14
58°	04:51	05:37	06:17	17:53	18:32	19:18
S 60°	04:47	05:36	06:18	17:52	18:33	19:22

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	11:48	14:19	16:41	06:08	05:28	04:59
N 70°	12:15	14:30	16:41	05:39	05:15	04:55
68°	12:35	14:39	16:42	05:18	05:04	04:51
66°	12:50	14:46	16:42	05:01	04:54	04:48
64°	13:03	14:52	16:42	04:47	04:46	04:46
62°	13:14	14:57	16:42	04:35	04:40	04:43
60°	13:23	15:02	16:42	04:25	04:34	04:41
N 58°	13:31	15:06	16:43	04:16	04:29	04:40
56°	13:38	15:09	16:43	04:08	04:24	04:38
54°	13:44	15:13	16:43	04:01	04:20	04:37
52°	13:50	15:15	16:43	03:54	04:16	04:35
50°	13:55	15:18	16:43	03:49	04:12	04:34
45°	14:05	15:23	16:43	03:36	04:05	04:31
N 40°	14:14	15:28	16:43	03:26	03:58	04:29
35°	14:22	15:32	16:44	03:17	03:53	04:27
30°	14:29	15:36	16:44	03:09	03:48	04:25
20°	14:40	15:42	16:44	02:55	03:39	04:22
N 10°	14:50	15:47	16:44	02:43	03:32	04:19
0°	15:00	15:52	16:44	02:32	03:25	04:17
S 10°	15:09	15:57	16:45	02:21	03:17	04:14
20°	15:19	16:02	16:45	02:08	03:09	04:11
30°	15:30	16:08	16:45	01:54	03:00	04:08
35°	15:37	16:12	16:46	01:46	02:55	04:06
40°	15:44	16:16	16:46	01:36	02:49	04:04
45°	15:53	16:20	16:46	01:25	02:42	04:01
S 50°	16:03	16:25	16:47	01:12	02:34	03:58
52°	16:08	16:28	16:47	01:05	02:30	03:57
54°	16:13	16:31	16:47	00:58	02:26	03:55
56°	16:19	16:33	16:47	00:50	02:21	03:53
58°	16:25	16:37	16:47	00:42	02:15	03:51
S 60°	16:32	16:40	16:48	00:31	02:09	03:49

Day	Sun			Moon		Age 11-13 77-93%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	Mer. Pass hh:mm	Mer.Pass. Upper hh:mm	Lower hh:mm	
26	05:43	05:34	12:06	21:12	08:46	
27	05:25	05:16	12:05	22:04	09:38	
28	05:07	04:58	12:05	22:57	10:31	

March 29, 30, 31 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	186°41.2	177°05.0	N02°43.5	2°10.6	N01°33.1	343°35.5	S07°58.7	148°27.5	N12°45.0					Alpheratz	357°31.5	29°15.1	
1	201°43.6	192°04.6	44.7	17°13.9	33.4	358°38.2	58.6	163°29.7	45.1					Ankaa	353°04.3	-42°08.8	
2	216°46.1	207°04.2	46.0	32°17.3	33.7	13°41.0	58.4	178°31.9	45.2					Schedar	349°27.6	56°41.9	
3	231°48.5	222°03.8	· · 47.2	47°20.7	· · 34.0	28°43.7	· · 58.3	193°34.1	· · 45.3					Diphda	348°44.2	-17°49.6	
4	246°51.0	237°03.5	48.5	62°24.1	34.4	43°46.5	58.2	208°36.2	45.4					Achernar	335°18.3	-57°05.4	
5	261°53.5	252°03.1	49.7	77°27.5	34.7	58°49.2	58.1	223°38.4	45.5					Hamal	327°47.6	23°36.0	
6	276°55.9	267°02.7	N02°51.0	92°30.8	N01°35.0	73°52.0	S07°58.0	238°40.6	N12°45.6					Polaris	312°32.5	89°23.3	
7	291°58.4	282°02.3	52.3	107°34.2	35.3	88°54.7	57.9	253°42.8	45.7					Acamar	315°09.6	-40°11.4	
8	307°00.9	297°01.9	53.5	122°37.6	35.6	103°57.4	57.8	268°45.0	45.8					Menkar	314°02.8	4°12.1	
9	322°03.3	312°01.6	· · 54.8	137°41.0	· · 36.0	119°00.2	· · 57.7	283°47.2	· · 45.9					Mirfak	308°23.6	49°57.9	
10	337°05.8	327°01.2	56.0	152°44.4	36.3	134°02.9	57.5	298°49.3	46.0					Aldebaran	290°35.9	16°34.0	
11	352°08.3	342°00.8	57.3	167°47.7	36.6	149°05.7	57.4	313°51.5	46.1					Rigel	281°00.7	-8°10.3	
12	7°10.7	357°00.4	N02°58.5	182°51.1	N01°36.9	164°08.4	S07°57.3	328°53.7	N12°46.2					Capella	280°17.0	46°01.6	
13	22°13.2	12°00.0	02°59.8	197°54.5	37.2	179°11.2	57.2	343°55.9	46.3					Bellatrix	278°19.3	6°22.4	
14	37°15.7	26°59.7	03°01.0	212°57.9	37.6	194°13.9	57.1	358°58.1	46.4					Elnath	277°57.7	28°37.8	
15	52°18.1	41°59.3	· · 02.3	228°01.2	· · 37.9	209°16.7	· · 57.0	14°00.3	· · 46.5					Alnilam	275°34.3	-1°11.2	
16	67°20.6	56°58.9	03.6	243°04.6	38.2	224°19.4	56.9	29°02.4	46.6					Betelgeuse	270°48.4	7°24.5	
17	82°23.0	71°58.5	04.8	258°08.0	38.5	239°22.1	56.8	44°04.6	46.7					Canopus	263°50.9	-52°43.1	
18	97°25.5	86°58.1	N03°06.1	273°11.4	N01°38.8	254°24.9	S07°56.6	59°06.8	N12°46.8					Sirius	258°23.2	-16°45.7	
19	112°28.0	101°57.8	07.3	288°14.7	39.1	269°27.6	56.5	74°09.0	46.9					Adhara	255°03.1	-29°01.1	
20	127°30.4	116°57.4	08.6	303°18.1	39.5	284°30.4	56.4	89°11.2	47.0					Procyon	244°47.2	5°08.7	
21	142°32.9	131°57.0	· · 09.8	318°21.5	· · 39.8	299°33.1	· · 56.3	104°13.3	· · 47.1					Pollux	243°13.1	27°57.2	
22	157°35.4	146°56.6	11.1	333°24.9	40.1	314°35.9	56.2	119°15.5	47.2					Avior	234°13.0	-59°36.6	
23	172°37.8	161°56.2	12.3	348°28.3	40.4	329°38.6	56.1	134°17.7	47.2					Suhail	222°43.5	-43°33.4	
Mer.pass. 11:31			ν -0.4' d 1.3' m -3.90			ν 3.4' d 0.3' m -1.39			ν 2.7' d -0.1' m -2.45			ν 2.2' d 0.1' m 0.38					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	187°40.3	176°55.9	N03°13.6	3°31.6	N01°40.7	344°41.4	S07°56.0	149°19.9	N12°47.3					
1	202°42.8	191°55.5	14.8	18°35.0	41.0	359°44.1	55.9	164°22.1	47.4					
2	217°45.2	206°55.1	16.1	33°38.4	41.4	14°46.8	55.7	179°24.3	47.5					
3	232°47.7	221°54.7	· · 17.4	48°41.8	· · 41.7	29°49.6	· · 55.6	194°26.4	· · 47.6					
4	247°50.1	236°54.3	18.6	63°45.1	42.0	44°52.3	55.5	209°28.6	47.7					
5	262°52.6	251°54.0	19.9	78°48.5	42.3	59°55.1	55.4	224°30.8	47.8					
6	277°55.1	266°53.6	N03°21.1	93°51.9	N01°42.6	74°57.8	S07°55.3	239°33.0	N12°47.9					
7	292°57.5	281°53.2	22.4	108°55.3	42.9	90°00.6	55.2	254°35.2	48.0					
8	308°00.0	296°52.8	23.6	123°58.6	43.3	105°03.3	55.1	269°37.3	48.1					
9	323°02.5	311°52.4	· · 24.9	139°02.0	· · 43.6	120°06.1	· · 55.0	284°39.5	· · 48.2					
10	338°04.9	326°52.1	26.1	154°05.4	43.9	135°08.8	54.8	299°41.7	48.3					
11	353°07.4	341°51.7	27.4	169°08.8	44.2	150°11.6	54.7	314°43.9	48.4					
12	8°09.9	356°51.3	N03°28.6	184°12.1	N01°44.5	165°14.3	S07°54.6	329°46.1	N12°48.5					
13	23°12.3	11°50.9	29.9	199°15.5	44.8	180°17.1	54.5	344°48.2	48.6					
14	38°14.8	26°50.5	31.1	214°18.9	45.1	195°19.8	54.4	359°50.4	48.7					
15	53°17.3	41°50.1	· · 32.4	229°22.3	· · 45.4	210°22.5	· · 54.3	14°52.6	· · 48.8					
16	68°19.7	56°49.8	33.6	244°25.6	45.8	225°25.3	54.2	29°54.8	48.9					
17	83°22.2	71°49.4	34.9	259°29.0	46.1	240°28.0	54.1	44°57.0	49.0					
18	98°24.6	86°49.0	N03°36.1	274°32.4	N01°46.4	255°30.8	S07°53.9	59°59.1	N12°49.1					
19	113°27.1	101°48.6	37.4	289°35.7	46.7	270°33.5	53.8	75°01.3	49.2					
20	128°29.6	116°48.2	38.7	304°39.1	47.0	285°36.3	53.7	90°03.5	49.3					
21	143°32.0	131°47.9	· · 39.9	319°42.5	· · 47.3	300°39.0	· · 53.6	105°05.7	· · 49.4					
22	158°34.5	146°47.5	41.2	334°45.9	47.6	315°41.8	53.5	120°07.9	49.5					
23	173°37.0	161°47.1	42.4	349°49.2	47.9	330°44.5	53.4	135°10.0	49.6					
Mer.pass. 11:27			ν -0.4' d 1.3' m -3.90			ν 3.4' d 0.3' m -1.37			ν 2.7' d -0.1' m -2.45			ν 2.2' d 0.1' m 0.38		

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	188°39.4	176°46.7	N03°43.7	4°52.6	N01°48.3	345°47.3	S07°53.3	150°12.2	N12°49.7
1	203°41.9	191°46.3	44.9	19°56.0	48.6	0°50.0	53.1	165°14.4	49.8
2	218°44.4	206°45.9	46.2	34°59.3	48.9	15°52.8	53.0	180°16.6	49.9
3	233°46.8	221°45.6	· · 47.4	50°02.7	· · 49.2	30°55.5	· · 52.9	195°18.8	· · 50.0
4	248°49.3	236°45.2	48.7	65°06.1	49.5	45°58.3	52.8	210°20.9	50.1
5	263°51.7	251°44.8	49.9	80°09.5	49.8	61°01.0	52.7	225°23.1	50.2
6	278°54.2	266°44.4	N03°51.2	95°12.8	N01°50.1	76°03.8	S07°52.6	240°25.3	N12°50.3
7	293°56.7	281°44.0	52.4	110°16.2	50.4	91°06.5	52.5	255°27.5	50.3
8	308°59.1	296°43.6	53.7	125°19.6	50.7	106°09.3	52.4	270°29.7	50.4
9	324°01.6	311°43.3	· · 54.9	140°22.9	· · 51.0	121°12.0	· · 52.2	285°31.8	· · 50.5
10	339°04.1	326°42.9	56.2	155°26.3	51.4	136°14.8	52.1	300°34.0	50.6
11	354°06.5	341°42.5	57.4	170°29.7	51.7	151°17.5	52.0	315°36.2	50.7
12	9°09.0	356°42.1	N03°58.7	185°33.0	N01°52.0	166°20.3	S07°51.9	330°38.4	N12°50.8
13	24°11.5	11°41.7	03°59.9	200°36.4	52.3	181°23.0	51.8	345°40.6	50.9
14	39°13.9	26°41.3	04°01.2	215°39.8	52.6	196°25.8	51.7	0°42.7	51.0
15	54°16.4	41°41.0	· · 02.4	230°43.1	· · 52.9	211°28.5	· · 51.6	15°44.9	· · 51.1
16	69°18.9	56°40.6	03.7	245°46.5	53.2	226°31.3	51.4	30°47.1	51.2
17	84°21.3	71°40.2	04.9	260°49.9	53.5	241°34.0	51.3	45°49.3	51.3
18	99°23.8	86°39.8	N04°06.2	275°53.2	N01°53.8	256°36.8	S07°51.2	60°51.5	N12°51.4
19	114°26.2	101°39.4	07.4	290°56.6	54.1	271°39.5	51.1	75°53.6	51.5
20	129°28.7	116°39.0	08.7	306°00.0	54.4	286°42.3	51.0	90°55.8	51.6
21	144°31.2	131°38.6	· · 09.9	321°03.3	· · 54.7	301°45.0	· · 50.9	105°58.0	· · 51.7
22	159°33.6	146°38.3	11.2	336°06.7	55.0	316°47.8	50.8	121°0	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°47.8	N03°24.3	15°11.5	9.0'	S01°32.8	15.8'	61.1'
1	193°48.0	25.3	29°39.4	8.9'	01°48.6	15.8'	61.2'
2	208°48.2	26.2	44°07.4	8.9'	02°04.4	15.8'	61.2'
3	223°48.4	· · 27.2	58°35.2	8.9'	02°20.3	15.8'	61.2'
4	238°48.6	28.2	73°03.1	8.8'	02°36.1	15.8'	61.2'
5	253°48.8	29.2	87°30.9	8.8'	02°52.0	15.8'	61.2'
6	268°48.9	N03°30.1	101°58.7	8.8'	S03°07.8	15.8'	61.2'
7	283°49.1	31.1	116°26.5	8.7'	03°23.6	15.8'	61.2'
8	298°49.3	32.1	130°54.2	8.7'	03°39.4	15.8'	61.3'
9	313°49.5	· · 33.0	145°21.9	8.7'	03°55.1	15.8'	61.3'
10	328°49.7	34.0	159°49.6	8.6'	04°10.9	15.7'	61.3'
11	343°49.9	35.0	174°17.2	8.6'	04°26.6	15.7'	61.3'
12	358°50.1	N03°36.0	188°44.8	8.5'	S04°42.4	15.7'	61.3'
13	13°50.3	36.9	203°12.3	8.5'	04°58.0	15.7'	61.3'
14	28°50.5	37.9	217°39.8	8.5'	05°13.7	15.6'	61.3'
15	43°50.6	· · 38.9	232°07.3	8.4'	05°29.4	15.6'	61.3'
16	58°50.8	39.9	246°34.7	8.4'	05°45.0	15.6'	61.3'
17	73°51.0	40.8	261°02.0	8.3'	06°00.5	15.5'	61.4'
18	88°51.2	N03°41.8	275°29.4	8.3'	S06°16.1	15.5'	61.4'
19	103°51.4	42.8	289°56.7	8.2'	06°31.6	15.5'	61.4'
20	118°51.6	43.7	304°23.9	8.2'	06°47.1	15.4'	61.4'
21	133°51.8	· · 44.7	318°51.1	8.2'	07°02.5	15.4'	61.4'
22	148°52.0	45.7	333°18.3	8.1'	07°17.9	15.3'	61.4'
23	163°52.2	46.6	347°45.4	8.1'	07°33.2	15.3'	61.4'
SD = 16.0'			SD = 16.7'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°52.3	N03°47.6	2°12.4	8.0'	S07°48.5	15.3'	61.4'
1	193°52.5	48.6	16°39.4	8.0'	08°03.8	15.2'	61.4'
2	208°52.7	49.6	31°06.4	7.9'	08°19.0	15.2'	61.4'
3	223°52.9	· · 50.5	45°33.3	7.9'	08°34.1	15.1'	61.4'
4	238°53.1	51.5	60°00.2	7.8'	08°49.2	15.0'	61.4'
5	253°53.3	52.5	74°27.0	7.8'	09°04.3	15.0'	61.4'
6	268°53.5	N03°53.4	88°53.8	7.7'	S09°19.3	14.9'	61.4'
7	283°53.7	54.4	103°20.5	7.7'	09°34.2	14.9'	61.4'
8	298°53.8	55.4	117°47.2	7.6'	09°49.0	14.8'	61.4'
9	313°54.0	· · 56.3	132°13.8	7.6'	10°03.8	14.7'	61.4'
10	328°54.2	57.3	146°40.3	7.5'	10°18.6	14.7'	61.4'
11	343°54.4	58.3	161°06.8	7.5'	10°33.2	14.6'	61.4'
12	358°54.6	N03°59.3	175°33.3	7.4'	S10°47.8	14.5'	61.4'
13	13°54.8	04°00.2	189°59.7	7.3'	11°02.3	14.4'	61.4'
14	28°55.0	01.2	204°26.0	7.3'	11°16.8	14.4'	61.4'
15	43°55.2	· · 02.2	218°52.3	7.2'	11°31.2	14.3'	61.4'
16	58°55.4	03.1	233°18.6	7.2'	11°45.5	14.2'	61.4'
17	73°55.5	04.1	247°44.8	7.1'	11°59.7	14.1'	61.4'
18	88°55.7	N04°05.1	262°10.9	7.1'	S12°13.8	14.0'	61.4'
19	103°55.9	06.0	276°37.0	7.0'	12°27.9	14.0'	61.3'
20	118°56.1	07.0	291°03.0	7.0'	12°41.8	13.9'	61.3'
21	133°56.3	· · 08.0	305°29.0	6.9'	12°55.7	13.8'	61.3'
22	148°56.5	08.9	319°54.9	6.9'	13°09.5	13.7'	61.3'
23	163°56.7	09.9	334°20.7	6.8'	13°23.2	13.6'	61.3'
SD = 16.0'			SD = 16.7'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°56.9	N04°10.9	348°46.5	6.7'	S13°36.8	13.5'	61.3'
1	193°57.0	11.8	3°12.3	6.7'	13°50.3	13.4'	61.3'
2	208°57.2	12.8	17°37.9	6.6'	14°03.7	13.3'	61.3'
3	223°57.4	· · 13.8	32°03.6	6.6'	14°17.0	13.2'	61.3'
4	238°57.6	14.7	46°29.1	6.5'	14°30.2	13.1'	61.3'
5	253°57.8	15.7	60°54.7	6.5'	14°43.3	13.0'	61.2'
6	268°58.0	N04°16.7	75°20.1	6.4'	S14°56.3	12.9'	61.2'
7	283°58.2	17.6	89°45.5	6.3'	15°09.2	12.8'	61.2'
8	298°58.4	18.6	104°10.9	6.3'	15°22.0	12.7'	61.2'
9	313°58.5	· · 19.6	118°36.2	6.2'	15°34.7	12.6'	61.2'
10	328°58.7	20.5	133°01.4	6.2'	15°47.2	12.5'	61.2'
11	343°58.9	21.5	147°26.6	6.1'	15°59.7	12.3'	61.1'
12	358°59.1	N04°22.5	161°51.7	6.1'	S16°12.0	12.2'	61.1'
13	13°59.3	23.4	176°16.8	6.0'	16°24.2	12.1'	61.1'
14	28°59.5	24.4	190°41.8	6.0'	16°36.3	12.0'	61.1'
15	43°59.7	· · 25.4	205°06.7	5.9'	16°48.3	11.9'	61.1'
16	58°59.8	26.3	219°31.7	5.9'	17°00.2	11.7'	61.1'
17	74°00.0	27.3	233°56.5	5.8'	17°11.9	11.6'	61.0'
18	89°00.2	N04°28.3	248°21.3	5.8'	S17°23.5	11.5'	61.0'
19	104°00.4	29.2	262°46.1	5.7'	17°35.0	11.4'	61.0'
20	119°00.6	30.2	277°10.8	5.6'	17°46.4	11.2'	61.0'
21	134°00.8	· · 31.2	291°35.4	5.6'	17°57.6	11.1'	61.0'
22	149°01.0	32.1	306°00.0	5.5'	18°08.7	11.0'	60.9'
23	164°01.2	33.1	320°24.6	5.5'	18°19.7	10.8'	60.9'
SD = 16.0'			SD = 16.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:58	03:52	05:05	19:06	20:21	22:20
N 70°	02:34	04:07	05:12	19:00	20:05	21:40
68°	02:59	04:18	05:17	18:54	19:53	21:14
66°	03:18	04:28	05:21	18:50	19:43	20:54
64°	03:33	04:36	05:25	18:46	19:35	20:38
62°	03:45	04:42	05:28	18:42	19:28	20:26
60°	03:55	04:48	05:31	18:39	19:22	20:15
N 58°	04:04	04:53	05:33	18:37	19:17	20:07
56°	04:12	04:58	05:35	18:35	19:13	19:59
54°	04:18	05:01	05:37	18:33	19:09	19:52
52°	04:24	05:05	05:39	18:31	19:05	19:46
50°	04:29	05:08	05:41	18:29	19:02	19:41
45°	04:39	05:15	05:44	18:25	18:55	19:30
N 40°	04:48	05:20	05:47	18:22	18:50	19:22
35°	04:54	05:24	05:49	18:20	18:45	19:15
30°	04:59	05:28	05:52	18:18	18:42	19:10
20°	05:07	05:33	05:55	18:14	18:36	19:02
N 10°	05:13	05:37	05:58	18:11	18:32	18:56
0°	05:16	05:40	06:01	18:08	18:28	18:52
S 10°	05:18	05:43	06:04	18:05	18:26	18:50
20°	05:19	05:45	06:07	18:02	18:24	18:49
30°	05:18	05:46	06:10	17:59	18:23	18:50
35°	05:17	05:46	06:11	17:57	18:22	18:51
40°	05:15	05:46	06:13	17:55	18:22	18:53
45°	05:12	05:46	06:15	17:53	18:22	18:56
S 50°	05:08	05:46	06:18	17:50	18:22	19:00
52°	05:06	05:45	06:19	17:49	18:22	19:01
54°	05:04	05:45	06:20	17:47	18:23	19:04
56°	05:01	05:45	06:22	17:46	18:23	19:06
58°	04:58	05:44	06:23	17:44	18:23	19:09
S 60°	04:55	05:43	06:25	17:43	18:24	19:12

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	19:06	21:46	■	04:33	04:03	03:21
N 70°	18:55	21:19	· · ·	04:37	04:16	03:50
68°	18:47	20:59	23:24	04:40	04:27	04:12
66°	18:40	20:43	22:53	04:42	04:36	04:30
64°	18:34	20:30	22:29	04:45	04:44	04:44
62°	18:29	20:19	22:11	04:47	04:51	04:56
60°	18:25	20:10	21:56	04:49	04:56	05:06
N 58°	18:21	20:02	21:43	04:50	05:02	05:16
56°	18:18	19:55	21:32	04:52	05:06	05:24
54°	18:15	19:49	21:22	04:53	05:10	05:31
52°	18:12	19:43	21:14	04:54	05:14	05:37
50°	18:10	19:38	21:06	04:55	05:17	05:43
45°	18:05	19:27	20:50	04:57	05:25	05:56
N 40°	18:00	19:18	20:37	04:59	05:31	06:06
35°	17:56	19:11	20:25	05:01	05:37	06:15
30°	17:53	19:04	20:16	05:03	05:41	06:23
20°	17:47	18:52	19:59	05:05	05:50	06:37
N 10°	17:43	18:42	19:44	05:07	05:57	06:49
0°	17:38	18:33	19:31	05:10	06:04	07:00
S 10°	17:33	18:24	19:17	05:12	06:11	07:12
20°	17:29	18:14	19:03	05:14	06:18	07:24
30°	17:23	18:03	18:47	05:17	06:27	07:38
35°	17:20	17:57	18:38	05:18	06:31	07:46
40°	17:17	17:50	18:27	05:20	06:37	07:55
45°	17:13	17:42	18:15	05:22	06:44	08:06
S 50°	17:08	17:32	18:00	05:24	06:51	08:20
52°	17:06	17:28	17:53	05:25	06:55	08:26
54°	17:04	17:23	17:46	05:26	06:59	08:33
56°	17:01	17:17	17:37	05:27	07:03	08:41
58°	16:58	17:11	17:27	05:29	07:08	08:49
S 60°	16:55	17:04	17:16	05:30	07:14	08:59

Day	Sun		Mer. Pass hh:mm	Moon		Age 14-16 98-99%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
29	04:49	04:40	12:05	23:51	11:24	
30	04:31	04:22	12:04	—:—	12:18	
31	04:13	04:04	12:04	00:47	13:15	

April 01, 02, 03 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	189°38.6	176°37.5	N04°13.6	6°13.4	N01°55.6	346°53.3	S07°50.5	151°04.5	N12°52.0	Alpheratz	357°31.5	29°15.1
1	204°41.0	191°37.1	14.9	21°16.8	55.9	1°56.0	50.4	166°06.7	52.1	Ankaa	353°04.3	-42°08.8
2	219°43.5	206°36.7	16.1	36°20.2	56.3	16°58.8	50.3	181°08.9	52.2	Schedar	349°27.5	56°41.9
3	234°46.0	221°36.3	•• 17.4	51°23.5	•• 56.6	32°01.5	•• 50.2	196°11.1	•• 52.3	Diphda	348°44.2	-17°49.6
4	249°48.4	236°36.0	18.6	66°26.9	56.9	47°04.3	50.1	211°13.2	52.4	Achernar	335°18.4	-57°05.4
5	264°50.9	251°35.6	19.9	81°30.2	57.2	62°07.0	50.0	226°15.4	52.5	Hamal	327°47.6	23°36.0
6	279°53.4	266°35.2	N04°21.1	96°33.6	N01°57.5	77°09.8	S07°49.8	241°17.6	N12°52.6	Polaris	312°33.2	89°23.3
7	294°55.8	281°34.8	22.4	111°37.0	57.8	92°12.5	49.7	256°19.8	52.7	Acamar	315°09.6	-40°11.4
8	309°58.3	296°34.4	23.6	126°40.3	58.1	107°15.3	49.6	271°21.9	52.8	Menkar	314°02.8	4°12.1
9	325°00.7	311°34.0	•• 24.9	141°43.7	•• 58.4	122°18.0	•• 49.5	286°24.1	•• 52.9	Mirfak	308°23.6	49°57.9
10	340°03.2	326°33.6	26.1	156°47.1	58.7	137°20.8	49.4	301°26.3	53.0	Aldebaran	290°35.9	16°34.0
11	355°05.7	341°33.2	27.4	171°50.4	59.0	152°23.5	49.3	316°28.5	53.1	Rigel	281°00.7	-8°10.3
12	10°08.1	356°32.9	N04°28.6	186°53.8	N01°59.3	167°26.3	S07°49.2	331°30.7	N12°53.2	Capella	280°17.0	46°01.6
13	25°10.6	11°32.5	29.9	201°57.1	59.6	182°29.0	49.0	346°32.8	53.3	Bellatrix	278°19.3	6°22.4
14	40°13.1	26°32.1	31.1	217°00.5	01°59.9	197°31.8	48.9	1°35.0	53.4	Elnath	277°57.7	28°37.8
15	55°15.5	41°31.7	•• 32.4	232°03.9	02°00.2	212°34.5	•• 48.8	16°37.2	•• 53.5	Alnilam	275°34.4	-1°11.2
16	70°18.0	56°31.3	33.6	247°07.2	00.5	227°37.3	48.7	31°39.4	53.6	Betelgeuse	270°48.5	7°24.5
17	85°20.5	71°30.9	34.8	262°10.6	00.8	242°40.0	48.6	46°41.5	53.6	Canopus	263°50.9	-52°43.1
18	100°22.9	86°30.5	N04°36.1	277°13.9	N02°01.1	257°42.8	S07°48.5	61°43.7	N12°53.7	Sirius	258°23.2	-16°45.7
19	115°25.4	101°30.2	37.3	292°17.3	01.4	272°45.5	48.3	76°45.9	53.8	Adhara	255°03.2	-29°01.1
20	130°27.8	116°29.8	38.6	307°20.6	01.7	287°48.3	48.2	91°48.1	53.9	Procyon	244°47.2	5°08.7
21	145°30.3	131°29.4	•• 39.8	322°24.0	•• 02.0	302°51.0	•• 48.1	106°50.2	•• 54.0	Pollux	243°13.1	27°57.2
22	160°32.8	146°29.0	41.1	337°27.4	02.3	317°53.8	48.0	121°52.4	54.1	Avior	234°13.0	-59°36.6
23	175°35.2	161°28.6	42.3	352°30.7	02.6	332°56.5	47.9	136°54.6	54.2	Suhail	222°43.5	-43°33.4
Mer.pass. 11:20		ν -0.4' d 1.2' m -3.90		ν 3.4' d 0.3' m -1.34		ν 2.8' d -0.1' m -2.46		ν 2.2' d 0.1' m 0.38		Miaplacidus	221°36.9	-69°50.6
										Alphard	217°44.2	-8°47.4
										Regulus	207°30.6	11°49.2
										Dubhe	193°36.4	61°35.5
										Denebola	182°21.2	14°24.3
										Gienah	175°39.7	-17°42.5
										Acrux	172°55.4	-63°15.9
										Gacrux	171°47.1	-57°16.8
										Alioth	166°09.6	55°47.9
										Spica	158°18.4	-11°19.0
										Alkaid	152°49.0	49°09.8
										Hadar	148°30.4	-60°30.8
										Menkent	147°53.1	-36°30.9
										Arcturus	145°44.5	19°01.6
										Rigil Kent.	139°34.8	-60°57.3
										Kochab	137°19.2	74°01.9
										Zuben'ubi	136°51.9	-16°09.9
										Alphecca	126°00.6	26°36.8
										Antares	112°11.4	-26°29.7
										Atria	107°02.2	-69°04.5
										Sabik	101°58.7	-15°45.6
										Shaula	96°05.6	-37°07.3
										Rasalhague	95°55.3	12°32.2
										Eltanin	90°40.5	51°28.8
										Kaus Aust.	83°27.9	-34°22.0
										Vega	80°30.9	38°48.4
										Nunki	75°43.5	-26°15.4
										Altair	61°56.7	8°56.7
										Peacock	53°00.7	-56°38.1
										Deneb	49°23.6	45°22.9
										Enif	33°35.6	10°00.5
										Al Na'ir	27°29.1	-46°48.9
										Fomalhaut	15°11.1	-29°27.9
										Scheat	13°42.2	28°14.4
										Markab	13°26.8	15°21.7
Apr 01 Sun		SHA		Mer.pass								
Venus		346°58.9		12:14								
Mars		176°34.9		23:30								
Jupiter		157°14.7		00:52								
Saturn		321°26.0		13:54								
Apr 02 Mon		SHA		Mer.pass								
Venus		345°50.5		12:14								
Mars		176°56.4		23:24								
Jupiter		157°21.6		00:48								
Saturn		321°19.1		13:50								
Apr 03 Tue		SHA		Mer.pass								
Venus		344°42.0		12:15								
Mars		177°17.7		23:19								
Jupiter		157°28.5		00:44								
Saturn		321°12.1		13:47								
Horizontal parallax												
Venus:				0.1								
Mars:				0.2								
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	191°36.8	176°18.9	N05°13.4	8°54.5	N02°10.0	349°05.4	S07°45.0	152°49.0	N12°56.7			
1	206°39.3	191°18.5	14.6	23°57.8	10.2	4°08.2	44.9	167°51.2	56.8			
2	221°41.8	206°18.1	15.8	39°01.2	10.5	19°10.9	44.8	182°53.3	56.9			
3	236°44.2	221°17.7	•• 17.1	54°04.5	•• 10.8	34°13.7	•• 44.7	197°55.5	•• 57.0			
4	251°46.7	236°17.3	18.3	69°07.9	11.1	49°16.4	44.5	212°57.7	57.1			
5	266°49.2	251°16.9	19.6	84°11.2	11.4	64°19.2	44.4	227°59.9	57.2			
6	281°51.6	266°16.5	N05°20.8	99°14.6	N02°11.7	79°21.9	S07°44.3	243°02.0	N12°57.3			
7	296°54.1	281°16.1	22.0	114°17.9	12.0	94°24.7	44.2	258°04.2	57.4			
8	311°56.6	296°15.7	23.3	129°21.3	12.3	109°27.4	44.1	273°06.4	57.5			
9	326°59.0	311°15.3	•• 24.5	144°24.6	•• 12.6	124°30.2	•• 44.0	288°08.6	•• 57.5			
10	342°01.5	326°14.9	25.8	159°27.9	12.9	139°32.9	43.8	303°10.7	57.6			
11	357°04.0	341°14.5	27.0	174°31.3	13.1	154°35.7	43.7	318°12.9	57.7			
12	12°06.4	356°14.2	N05°28.2	189°34.6	N02°13.4	169°38.5	S07°43.6	333°15.1	N12°57.8			
13	27°08.9	11°13.8	29.5	204°38.0	13.7	184°41.2	43.5	348°17.3	57.9			
14	42°11.3	26°13.4	30.7	219°41.3	14.0	199°44.0	43.4	3°19.4	58.0			
15	57°13.8	41°13.0	•• 31.9	234°44.6	•• 14.3	214°46.7	•• 43.3	18°21.6	•• 58.1			
16	72°16.3	56°12.6	33.2	249°48.0	14.6	229°49.5	43.1	33°23.8	58.2			
17	87°18.7	71°12.2	34.4	264°51.3	14.9	244°52.2	43.0	48°26.0	58.3			
18	102°21.2	86°11.8	N05°35.6	279°54.7	N02°15.1	259°55.0	S07°42.9	63°28.1	N12°58.4			
19	117°23.7	101°11.4	36.9	294°58.0	15.4	274°57.8	42.8	78°30.3	58.5			
20	132°26.1	116°11.0	38.1	310°01.3	15.7	290°00.5	42.7	93°32.5	58.6			
21	147°28.6	131°10.6	•• 39.4	325°04.7	•• 16.0	305°03.3	•• 42.6	108°34.7	•• 58.7			
22	162°31.1	146°10.2	40.6	340°08.0	16.3	320°06.0	42.4	123°36.8	58.8			
23	177°33.5	161°09.8	41.8	355°11.3	16.6	335°08.8	42.3	138°39.0	58.9			
Mer.pass. 11:12		ν -0.4' d 1.2' m -3.90		ν 3.3' d 0.3' m -1.30		ν 2.8' d -0.1' m -2.46		ν 2.2' d 0.1' m 0.37				

DUT1 = UT1-UTC = +0.2084 sec $\Delta T = TT-UT1 = +68.9756$ sec

2029 April 01 to Apr. 03 UT

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°01.3	N04°34.1	334°49.1	5.5'	S18°30.5	10.7'	60.9'						
1	194°01.5	35.0	349°13.5	5.4'	18°41.2	10.6'	60.9'						
2	209°01.7	36.0	3°37.9	5.4'	18°51.7	10.4'	60.8'						
3	224°01.9	· · 37.0	18°02.3	5.3'	19°02.2	10.3'	60.8'						
4	239°02.1	37.9	32°26.6	5.3'	19°12.4	10.1'	60.8'						
5	254°02.3	38.9	46°50.9	5.2'	19°22.6	10.0'	60.8'						
6	269°02.5	N04°39.8	61°15.1	5.2'	S19°32.6	9.9'	60.7'						
7	284°02.6	40.8	75°39.3	5.1'	19°42.4	9.7'	60.7'						
8	299°02.8	41.8	90°03.4	5.1'	19°52.1	9.6'	60.7'						
9	314°03.0	· · 42.7	104°27.5	5.1'	20°01.7	9.4'	60.7'						
10	329°03.2	43.7	118°51.5	5.0'	20°11.1	9.3'	60.6'						
11	344°03.4	44.7	133°15.6	5.0'	20°20.4	9.1'	60.6'						
12	359°03.6	N04°45.6	147°39.5	4.9'	S20°29.5	9.0'	60.6'						
13	14°03.8	46.6	162°03.5	4.9'	20°38.4	8.8'	60.5'						
14	29°03.9	47.5	176°27.4	4.9'	20°47.2	8.7'	60.5'						
15	44°04.1	· · 48.5	190°51.3	4.8'	20°55.9	8.5'	60.5'						
16	59°04.3	49.5	205°15.1	4.8'	21°04.4	8.3'	60.5'						
17	74°04.5	50.4	219°38.9	4.8'	21°12.7	8.2'	60.4'						
18	89°04.7	N04°51.4	234°02.7	4.7'	S21°20.9	8.0'	60.4'						
19	104°04.9	52.4	248°26.4	4.7'	21°29.0	7.9'	60.4'						
20	119°05.1	53.3	262°50.1	4.7'	21°36.8	7.7'	60.3'						
21	134°05.2	· · 54.3	277°13.8	4.7'	21°44.6	7.6'	60.3'						
22	149°05.4	55.2	291°37.5	4.6'	21°52.1	7.4'	60.3'						
23	164°05.6	56.2	306°01.1	4.6'	21°59.5	7.2'	60.2'						
SD = 16.0' d = 1.0'							SD = 16.6'						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°05.8	N04°57.2	320°24.8	4.6'	S22°06.8	7.1'	60.2'						
1	194°06.0	58.1	334°48.4	4.6'	22°13.8	6.9'	60.2'						
2	209°06.2	04°59.1	349°11.9	4.5'	22°20.8	6.8'	60.1'						
3	224°06.4	05°00.0	3°35.5	4.5'	22°27.5	6.6'	60.1'						
4	239°06.5	01.0	17°59.1	4.5'	22°34.1	6.4'	60.1'						
5	254°06.7	02.0	32°22.6	4.5'	22°40.5	6.3'	60.0'						
6	269°06.9	N05°02.9	46°46.1	4.5'	S22°46.8	6.1'	60.0'						
7	284°07.1	03.9	61°09.6	4.5'	22°52.9	5.9'	60.0'						
8	299°07.3	04.8	75°33.1	4.5'	22°58.8	5.8'	59.9'						
9	314°07.5	· · 05.8	89°56.6	4.5'	23°04.5	5.6'	59.9'						
10	329°07.6	06.8	104°20.1	4.5'	23°10.1	5.4'	59.9'						
11	344°07.8	07.7	118°43.6	4.5'	23°15.6	5.3'	59.8'						
12	359°08.0	N05°08.7	133°07.1	4.5'	S23°20.8	5.1'	59.8'						
13	14°08.2	09.6	147°30.5	4.5'	23°25.9	4.9'	59.8'						
14	29°08.4	10.6	161°54.0	4.5'	23°30.8	4.8'	59.7'						
15	44°08.6	· · 11.6	176°17.5	4.5'	23°35.6	4.6'	59.7'						
16	59°08.8	12.5	190°41.0	4.5'	23°40.2	4.4'	59.6'						
17	74°08.9	13.5	205°04.5	4.5'	23°44.6	4.2'	59.6'						
18	89°09.1	N05°14.4	219°28.0	4.5'	S23°48.8	4.1'	59.6'						
19	104°09.3	15.4	233°51.5	4.5'	23°52.9	3.9'	59.5'						
20	119°09.5	16.3	248°15.0	4.5'	23°56.8	3.7'	59.5'						
21	134°09.7	· · 17.3	262°38.5	4.5'	24°00.6	3.6'	59.5'						
22	149°09.9	18.3	277°02.1	4.6'	24°04.1	3.4'	59.4'						
23	164°10.0	19.2	291°25.6	4.6'	24°07.5	3.2'	59.4'						
SD = 16.0' d = 1.0'							SD = 16.4'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°10.2	N05°20.2	305°49.2	4.6'	S24°10.8	3.1'	59.3'						
1	194°10.4	21.1	320°12.8	4.6'	24°13.8	2.9'	59.3'						
2	209°10.6	22.1	334°36.5	4.7'	24°16.7	2.7'	59.3'						
3	224°10.8	· · 23.0	349°00.1	4.7'	24°19.5	2.6'	59.2'						
4	239°11.0	24.0	3°23.8	4.7'	24°22.0	2.4'	59.2'						
5	254°11.1	25.0	17°47.5	4.7'	24°24.4	2.2'	59.2'						
6	269°11.3	N05°25.9	32°11.2	4.8'	S24°26.6	2.1'	59.1'						
7	284°11.5	26.9	46°35.0	4.8'	24°28.7	1.9'	59.1'						
8	299°11.7	27.8	60°58.8	4.8'	24°30.6	1.7'	59.0'						
9	314°11.9	· · 28.8	75°22.6	4.9'	24°32.3	1.6'	59.0'						
10	329°12.1	29.7	89°46.5	4.9'	24°33.9	1.4'	59.0'						
11	344°12.2	30.7	104°10.4	5.0'	24°35.3	1.2'	58.9'						
12	359°12.4	N05°31.7	118°34.4	5.0'	S24°36.5	1.1'	58.9'						
13	14°12.6	32.6	132°58.4	5.0'	24°37.5	0.9'	58.8'						
14	29°12.8	33.6	147°22.4	5.1'	24°38.4	0.7'	58.8'						
15	44°13.0	· · 34.5	161°46.5	5.1'	24°39.2	0.6'	58.8'						
16	59°13.1	35.5	176°10.6	5.2'	24°39.7	0.4'	58.7'						
17	74°13.3	36.4	190°34.8	5.2'	24°40.2	0.2'	58.7'						
18	89°13.5	N05°37.4	204°59.0	5.3'	S24°40.4	0.1'	58.6'						
19	104°13.7	38.3	219°23.3	5.3'	24°40.5	-0.1'	58.6'						
20	119°13.9	39.3	233°47.6	5.4'	24°40.4	-0.2'	58.6'						
21	134°14.1	· · 40.2	248°12.0	5.4'	24°40.2	-0.4'	58.5'						
22	149°14.2	41.2	262°36.5	5.5'	24°39.8	-0.6'	58.5'						
23	164°14.4	42.2	277°01.0	5.6'	24°39.2	-0.7'	58.4'						
SD = 16.0' d = 1.0'							SD = 16.2'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:21	03:33	04:49	19:21	20:38	23:00
N 70°	02:11	03:51	04:57	19:12	20:20	22:03
68°	02:41	04:04	05:04	19:05	20:06	21:31
66°	03:03	04:15	05:09	18:59	19:54	21:08
64°	03:20	04:24	05:14	18:55	19:45	20:50
62°	03:33	04:32	05:18	18:50	19:37	20:36
60°	03:45	04:39	05:22	18:47	19:30	20:24
N 58°	03:54	04:44	05:25	18:44	19:24	20:15
56°	04:03	04:49	05:27	18:41	19:19	20:06
54°	04:10	04:54	05:30	18:38	19:14	19:59
52°	04:16	04:58	05:32	18:36	19:10	19:52
50°	04:22	05:01	05:34	18:34	19:07	19:46
45°	04:33	05:09	05:39	18:29	18:59	19:35
N 40°	04:43	05:15	05:42	18:25	18:53	19:25
35°	04:50	05:20	05:45	18:22	18:48	19:18
30°	04:56	05:24	05:48	18:19	18:44	19:12
20°	05:05	05:30	05:53	18:15	18:37	19:03
N 10°	05:11	05:35	05:57	18:10	18:32	18:56
0°	05:15	05:39	06:00	18:07	18:27	18:52
S 10°	05:18	05:43	06:04	18:03	18:24	18:49
20°	05:20	05:45	06:07	17:59	18:21	18:47
30°	05:20	05:47	06:11	17:55	18:19	18:47
35°	05:19	05:48	06:14	17:53	18:18	18:47
40°	05:18	05:49	06:16	17:50	18:17	18:48
45°	05:16	05:50	06:19	17:47	18:16	18:50
S 50°	05:13	05:50	06:23	17:44	18:16	18:53
52°	05:11	05:50	06:24	17:42	18:16	18:55
54°	05:09	05:50	06:26	17:40	18:15	18:56
56°	05:07	05:51	06:28	17:38	18:15	18:58
58°	05:05	05:51	06:30	17:36	18:15	19:01
S 60°	05:02	05:51	06:32	17:34	18:15	19:03

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	■	■	■	■
N 70°	00:16	■	■	02:56	■	■
68°	■	■	■	03:48	■	■
66°		01:13	■	04:21	04:07	■
64°		00:29	02:17	04:46	04:52	05:11
62°		00:00	01:35	05:05	05:21	05:53
60°	23:38		01:06	05:21	05:44	06:22
N 58°	23:20		00:45	05:34	06:02	06:44
56°	23:05		00:27	05:46	06:18	07:02
54°	22:52		00:11	05:57	06:31	07:18
52°	22:41	23:58		06:06	06:43	07:31
50°	22:31	23:47		06:14	06:53	07:43
45°	22:10	23:23		06:32	07:15	08:07
N 40°	21:53	23:04		06:46	07:33	08:26
35°	21:39	22:48	23:50	06:58	07:47	08:42
30°	21:26	22:34	23:36	07:09	08:00	08:56
20°	21:05	22:10	23:12	07:28	08:22	09:20
N 10°	20:47	21:50	22:51	07:44	08:42	09:41
0°	20:31	21:31	22:31	07:59	09:00	10:00
S 10°	20:14	21:13	22:12	08:14	09:17	10:20
20°	19:56	20:53	21:51	08:31	09:37	10:40
30°	19:36	20:29	21:28	08:49	09:59	11:04
35°	19:24	20:16	21:13	09:00	10:12	11:18
40°	19:10	20:00	20:57	09:13	10:27	11:34
45°	18:54	19:42	20:38	09:28	10:45	11:53
S 50°	18:35	19:19	20:14	09:47	11:07	12:18
52°	18:26	19:08	20:02	09:55	11:18	12:29
54°	18:15	18:56	19:49	10:05	11:30	12:42
56°	18:04	18:41	19:34	10:16	11:44	12:58
58°	17:50	18:25	19:15	10:29	12:01	13:16
S 60°	17:35	18:05	18:53	10:44	12:20	13:38

April 04, 05, 06 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	192°36.0	176°09.4	N05°43.1	10°14.7	N02°16.8	350°11.5	S07°42.2	153°41.2	N12°59.0	Alpheratz	357°31.5	29°15.1
1	207°38.5	191°09.0	44.3	25°18.0	17.1	5°14.3	42.1	168°43.3	59.1	Ankaa	353°04.3	-42°08.8
2	222°40.9	206°08.6	45.5	40°21.3	17.4	20°17.1	42.0	183°45.5	59.2	Schedar	349°27.5	56°41.9
3	237°43.4	221°08.2	· · 46.8	55°24.7	· · 17.7	35°19.8	· · 41.9	198°47.7	· · 59.3	Diphda	348°44.2	-17°49.6
4	252°45.8	236°07.8	48.0	70°28.0	18.0	50°22.6	41.8	213°49.9	59.4	Achernar	335°18.4	-57°05.3
5	267°48.3	251°07.4	49.2	85°31.3	18.3	65°25.3	41.6	228°52.0	59.5	Hamal	327°47.6	23°36.0
6	282°50.8	266°07.0	N05°50.5	100°34.7	N02°18.5	80°28.1	S07°41.5	243°54.2	N12°59.6	Polaris	312°33.7	89°23.3
7	297°53.2	281°06.7	51.7	115°38.0	18.8	95°30.8	41.4	258°56.4	59.7	Acamar	315°09.6	-40°11.4
8	312°55.7	296°06.3	52.9	130°41.3	19.1	110°33.6	41.3	273°58.6	59.8	Menkar	314°02.8	4°12.1
9	327°58.2	311°05.9	· · 54.2	145°44.7	· · 19.4	125°36.4	· · 41.2	289°00.7	12°59.9	Mirfak	308°23.6	49°57.9
10	343°00.6	326°05.5	55.4	160°48.0	19.7	140°39.1	41.1	304°02.9	13°00.0	Aldebaran	290°35.9	16°34.0
11	358°03.1	341°05.1	56.6	175°51.3	19.9	155°41.9	40.9	319°05.1	00.1	Rigel	281°00.7	-8°10.3
12	13°05.6	356°04.7	N05°57.9	190°54.7	N02°20.2	170°44.6	S07°40.8	334°07.3	N13°00.2	Capella	280°17.0	46°01.6
13	28°08.0	11°04.3	05°59.1	205°58.0	20.5	185°47.4	40.7	349°09.4	00.3	Bellatrix	278°19.3	6°22.4
14	43°10.5	26°03.9	06°00.3	221°01.3	20.8	200°50.1	40.6	4°11.6	00.4	Elnath	277°57.7	28°37.8
15	58°12.9	41°03.5	· · 01.6	236°04.6	· · 21.1	215°52.9	· · 40.5	19°13.8	· · 00.5	Alnilam	275°34.4	-1°11.2
16	73°15.4	56°03.1	02.8	251°08.0	21.3	230°55.7	40.4	34°15.9	00.6	Betelgeuse	270°48.5	7°24.5
17	88°17.9	71°02.7	04.0	266°11.3	21.6	245°58.4	40.2	49°18.1	00.7	Canopus	263°51.0	-52°43.1
18	103°20.3	86°02.3	N06°05.3	281°14.6	N02°21.9	261°01.2	S07°40.1	64°20.3	N13°00.8	Sirius	258°23.2	-16°45.7
19	118°22.8	101°01.9	06.5	296°17.9	22.2	276°03.9	40.0	79°22.5	00.9	Adhara	255°03.2	-29°01.1
20	133°25.3	116°01.5	07.7	311°21.3	22.4	291°06.7	39.9	94°24.6	01.0	Procyon	244°47.2	5°08.7
21	148°27.7	131°01.1	· · 08.9	326°24.6	· · 22.7	306°09.5	· · 39.8	109°26.8	· · 01.1	Pollux	243°13.1	27°57.2
22	163°30.2	146°00.7	10.2	341°27.9	23.0	321°12.2	39.7	124°29.0	01.2	Avior	234°13.0	-59°36.6
23	178°32.7	161°00.3	11.4	356°31.2	23.3	336°15.0	39.5	139°31.1	01.3	Suhail	222°43.5	-43°33.5
Mer.pass. 11:08		ν -0.4' d 1.2' m -3.90		ν 3.3' d 0.3' m -1.28		ν 2.8' d -0.1' m -2.46		ν 2.2' d 0.1' m 0.37		Miaplacidus	221°36.9	-69°50.6
										Alphard	217°44.2	-8°47.4
										Regulus	207°30.6	11°49.2
										Dubhe	193°36.5	61°35.5
										Denebola	182°21.2	14°24.3
										Gienah	175°39.7	-17°42.5
										Acrux	172°55.4	-63°15.9
										Gacrux	171°47.1	-57°16.8
										Alioth	166°09.6	55°47.9
										Spica	158°18.3	-11°19.0
										Alkaid	152°48.9	49°09.8
										Hadar	148°30.3	-60°30.9
										Menkent	147°53.1	-36°30.9
										Arcturus	145°44.5	19°01.6
										Rigel Kent.	139°34.8	-60°57.3
										Kochab	137°19.1	74°01.9
										Zuben'ubi	136°51.9	-16°09.9
										Alphecca	126°00.6	26°36.8
										Antares	112°11.4	-26°29.7
										Atria	107°02.2	-69°04.5
										Sabik	101°58.7	-15°45.6
										Shaula	96°05.6	-37°07.3
										Rasalhague	95°55.2	12°32.2
										Eltanin	90°40.4	51°28.8
										Kaus Aust.	83°27.9	-34°22.0
										Vega	80°30.8	38°48.4
										Nunki	75°43.5	-26°15.4
										Altair	61°56.7	8°56.7
										Peacock	53°00.7	-56°38.1
										Deneb	49°23.6	45°22.9
										Enif	33°35.6	10°00.5
										Al Na'ir	27°29.1	-46°48.9
										Fomalhaut	15°11.1	-29°27.9
										Scheat	13°42.1	28°14.4
										Markab	13°26.7	15°21.7
										Apr 04 Wed	SHA	Mer.pass
										Venus	343°33.4	12:16
										Mars	177°38.7	23:14
										Jupiter	157°35.6	00:39
										Saturn	321°05.2	13:43
										Apr 05 Thu	SHA	Mer.pass
										Venus	342°24.8	12:16
										Mars	177°59.4	23:09
										Jupiter	157°42.6	00:35
										Saturn	320°58.2	13:40
										Apr 06 Fri	SHA	Mer.pass
										Venus	341°16.0	12:17
										Mars	178°19.9	23:03
										Jupiter	157°49.7	00:30
										Saturn	320°51.2	13:36
										Horizontal parallax		
										Venus:		0.1
										Mars:		0.2
Mer.pass. 11:00		ν -0.4' d 1.2' m -3.90		ν 3.3' d 0.3' m -1.25		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.37				

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 14.6	N05° 43.1	291° 25.6	5.6'	S24° 38.5	-0.9'	58.4'
1	194° 14.8	44.1	305° 50.2	5.7'	24° 37.6	-1.0'	58.4'
2	209° 15.0	45.0	320° 14.9	5.8'	24° 36.6	-1.2'	58.3'
3	224° 15.1	46.0	334° 39.6	5.8'	24° 35.4	-1.3'	58.3'
4	239° 15.3	46.9	349° 04.5	5.9'	24° 34.1	-1.5'	58.2'
5	254° 15.5	47.9	3° 29.4	6.0'	24° 32.6	-1.6'	58.2'
6	269° 15.7	N05° 48.8	17° 54.3	6.0'	S24° 30.9	-1.8'	58.2'
7	284° 15.9	49.8	32° 19.3	6.1'	24° 29.1	-1.9'	58.1'
8	299° 16.1	50.7	46° 44.4	6.2'	24° 27.2	-2.1'	58.1'
9	314° 16.2	51.7	61° 09.6	6.2'	24° 25.1	-2.2'	58.0'
10	329° 16.4	52.6	75° 34.9	6.3'	24° 22.8	-2.4'	58.0'
11	344° 16.6	53.6	90° 00.2	6.4'	24° 20.5	-2.5'	58.0'
12	359° 16.8	N05° 54.5	104° 25.6	6.5'	S24° 17.9	-2.7'	57.9'
13	14° 17.0	55.5	118° 51.1	6.6'	24° 15.2	-2.8'	57.9'
14	29° 17.1	56.4	133° 16.6	6.6'	24° 12.4	-3.0'	57.9'
15	44° 17.3	57.4	147° 42.2	6.7'	24° 09.4	-3.1'	57.8'
16	59° 17.5	58.3	162° 08.0	6.8'	24° 06.3	-3.3'	57.8'
17	74° 17.7	05° 59.3	176° 33.7	6.9'	24° 03.0	-3.4'	57.7'
18	89° 17.9	N06° 00.2	190° 59.6	7.0'	S23° 59.6	-3.5'	57.7'
19	104° 18.0	01.2	205° 25.6	7.0'	23° 56.1	-3.7'	57.7'
20	119° 18.2	02.1	219° 51.6	7.1'	23° 52.4	-3.8'	57.6'
21	134° 18.4	03.1	234° 17.8	7.2'	23° 48.6	-4.0'	57.6'
22	149° 18.6	04.0	248° 44.0	7.3'	23° 44.6	-4.1'	57.5'
23	164° 18.8	05.0	263° 10.3	7.4'	23° 40.5	-4.2'	57.5'
	SD = 16.0'	$d = 1.0'$	SD = 15.9'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 18.9	N06° 05.9	277° 36.7	7.5'	S23° 36.3	-4.4'	57.5'
1	194° 19.1	06.9	292° 03.1	7.6'	23° 32.0	-4.5'	57.4'
2	209° 19.3	07.8	306° 29.7	7.7'	23° 27.5	-4.6'	57.4'
3	224° 19.5	08.8	320° 56.4	7.7'	23° 22.9	-4.7'	57.4'
4	239° 19.7	09.7	335° 23.1	7.8'	23° 18.1	-4.9'	57.3'
5	254° 19.8	10.7	349° 49.9	7.9'	23° 13.2	-5.0'	57.3'
6	269° 20.0	N06° 11.6	4° 16.9	8.0'	S23° 08.2	-5.1'	57.2'
7	284° 20.2	12.6	18° 43.9	8.1'	23° 03.1	-5.3'	57.2'
8	299° 20.4	13.5	33° 11.0	8.2'	22° 57.9	-5.4'	57.2'
9	314° 20.6	14.5	47° 38.2	8.3'	22° 52.5	-5.5'	57.1'
10	329° 20.7	15.4	62° 05.5	8.4'	22° 47.0	-5.6'	57.1'
11	344° 20.9	16.4	76° 32.9	8.5'	22° 41.4	-5.7'	57.1'
12	359° 21.1	N06° 17.3	91° 00.4	8.6'	S22° 35.6	-5.9'	57.0'
13	14° 21.3	18.3	105° 28.0	8.7'	22° 29.8	-6.0'	57.0'
14	29° 21.4	19.2	119° 55.6	8.8'	22° 23.8	-6.1'	56.9'
15	44° 21.6	20.1	134° 23.4	8.9'	22° 17.7	-6.2'	56.9'
16	59° 21.8	21.1	148° 51.3	9.0'	22° 11.5	-6.3'	56.9'
17	74° 22.0	22.0	163° 19.2	9.1'	22° 05.2	-6.4'	56.8'
18	89° 22.2	N06° 23.0	177° 47.3	9.1'	S21° 58.7	-6.5'	56.8'
19	104° 22.3	23.9	192° 15.4	9.2'	21° 52.2	-6.7'	56.8'
20	119° 22.5	24.9	206° 43.7	9.3'	21° 45.5	-6.8'	56.7'
21	134° 22.7	25.8	221° 12.0	9.4'	21° 38.8	-6.9'	56.7'
22	149° 22.9	26.8	235° 40.4	9.5'	21° 31.9	-7.0'	56.7'
23	164° 23.0	27.7	250° 09.0	9.6'	21° 24.9	-7.1'	56.6'
	SD = 16.0'	$d = 0.9'$	SD = 15.7'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 23.2	N06° 28.7	264° 37.6	9.7'	S21° 17.9	-7.2'	56.6'
1	194° 23.4	29.6	279° 06.3	9.8'	21° 10.7	-7.3'	56.6'
2	209° 23.6	30.5	293° 35.1	9.9'	21° 03.4	-7.4'	56.5'
3	224° 23.8	31.5	308° 04.1	10.0'	20° 56.0	-7.5'	56.5'
4	239° 23.9	32.4	322° 33.1	10.1'	20° 48.5	-7.6'	56.5'
5	254° 24.1	33.4	337° 02.2	10.2'	20° 40.9	-7.7'	56.4'
6	269° 24.3	N06° 34.3	351° 31.4	10.3'	S20° 33.3	-7.8'	56.4'
7	284° 24.5	35.3	6° 00.7	10.4'	20° 25.5	-7.9'	56.4'
8	299° 24.6	36.2	20° 30.1	10.5'	20° 17.6	-8.0'	56.3'
9	314° 24.8	37.1	34° 59.5	10.6'	20° 09.6	-8.1'	56.3'
10	329° 25.0	38.1	49° 29.1	10.7'	20° 01.6	-8.2'	56.3'
11	344° 25.2	39.0	63° 58.8	10.8'	19° 53.4	-8.2'	56.2'
12	359° 25.3	N06° 40.0	78° 28.6	10.9'	S19° 45.2	-8.3'	56.2'
13	14° 25.5	40.9	92° 58.4	11.0'	19° 36.8	-8.4'	56.2'
14	29° 25.7	41.9	107° 28.4	11.0'	19° 28.4	-8.5'	56.1'
15	44° 25.9	42.8	121° 58.4	11.1'	19° 19.9	-8.6'	56.1'
16	59° 26.1	43.7	136° 28.5	11.2'	19° 11.3	-8.7'	56.1'
17	74° 26.2	44.7	150° 58.8	11.3'	19° 02.6	-8.8'	56.0'
18	89° 26.4	N06° 45.6	165° 29.1	11.4'	S18° 53.9	-8.8'	56.0'
19	104° 26.6	46.6	179° 59.5	11.5'	18° 45.0	-8.9'	56.0'
20	119° 26.8	47.5	194° 30.0	11.6'	18° 36.1	-9.0'	55.9'
21	134° 26.9	48.4	209° 00.6	11.7'	18° 27.1	-9.1'	55.9'
22	149° 27.1	49.4	223° 31.3	11.8'	18° 18.0	-9.2'	55.9'
23	164° 27.3	50.3	238° 02.0	11.9'	18° 08.8	-9.2'	55.9'
	SD = 16.0'	$d = 0.9'$	SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	03:14	04:33	19:35	20:56	—:—
N 70°	01:43	03:34	04:43	19:25	20:35	22:32
68°	02:21	03:50	04:51	19:16	20:18	21:50
66°	02:46	04:02	04:58	19:09	20:05	21:23
64°	03:06	04:13	05:04	19:04	19:55	21:03
62°	03:21	04:22	05:08	18:59	19:46	20:47
60°	03:34	04:29	05:13	18:54	19:38	20:34
N 58°	03:44	04:35	05:16	18:50	19:31	20:23
56°	03:54	04:41	05:20	18:47	19:26	20:13
54°	04:02	04:46	05:23	18:44	19:20	20:05
52°	04:09	04:51	05:25	18:41	19:16	19:58
50°	04:15	04:55	05:28	18:38	19:12	19:52
45°	04:27	05:03	05:33	18:33	19:03	19:39
N 40°	04:37	05:10	05:37	18:28	18:56	19:29
35°	04:45	05:16	05:41	18:25	18:50	19:21
30°	04:52	05:20	05:44	18:21	18:45	19:14
20°	05:02	05:28	05:50	18:15	18:38	19:04
N 10°	05:09	05:34	05:55	18:10	18:32	18:56
0°	05:14	05:39	05:59	18:06	18:27	18:51
S 10°	05:18	05:43	06:04	18:01	18:22	18:47
20°	05:20	05:46	06:08	17:57	18:19	18:44
30°	05:21	05:49	06:13	17:52	18:16	18:43
35°	05:21	05:51	06:16	17:49	18:14	18:43
40°	05:21	05:52	06:19	17:45	18:12	18:44
45°	05:19	05:53	06:23	17:42	18:11	18:45
S 50°	05:17	05:55	06:27	17:37	18:09	18:47
52°	05:16	05:55	06:29	17:35	18:09	18:48
54°	05:15	05:56	06:31	17:33	18:08	18:49
56°	05:13	05:56	06:34	17:30	18:08	18:50
58°	05:12	05:57	06:36	17:28	18:07	18:52
S 60°	05:09	05:58	06:39	17:25	18:06	18:54

Aries				Venus		Mars		Jupiter		Saturn	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	195°33.4	175°40.5	N07°11.4	14°13.3	N02°36.3	353°30.2	S07°33.8	156°17.5	N13°06.1		
1	210°35.9	190°40.1	12.6	29°16.6	36.6	8°33.0	33.7	171°19.7	06.2		
2	225°38.3	205°39.7	13.8	44°19.9	36.8	23°35.7	33.6	186°21.8	06.3		
3	240°40.8	220°39.3	· · 15.0	59°23.2	· · 37.1	38°38.5	· · 33.4	201°24.0	· · 06.4		
4	255°43.3	235°38.8	16.2	74°26.5	37.4	53°41.3	33.3	216°26.2	06.5		
5	270°45.7	250°38.4	17.5	89°29.8	37.6	68°44.0	33.2	231°28.3	06.6		
6	285°48.2	265°38.0	N07°18.7	104°33.0	N02°37.9	83°46.8	S07°33.1	246°30.5	N13°06.7		
7	300°50.7	280°37.6	19.9	119°36.3	38.1	98°49.5	33.0	261°32.7	06.8		
8	315°53.1	295°37.2	21.1	134°39.6	38.4	113°52.3	32.8	276°34.8	06.9		
9	330°55.6	310°36.8	· · 22.3	149°42.9	· · 38.6	128°55.1	· · 32.7	291°37.0	· · 07.0		
10	345°58.0	325°36.4	23.5	164°46.2	38.9	143°57.8	32.6	306°39.2	07.1		
11	1°00.5	340°36.0	24.7	179°49.5	39.1	159°00.6	32.5	321°41.4	07.2		
12	16°03.0	355°35.6	N07°26.0	194°52.8	N02°39.4	174°03.4	S07°32.4	336°43.5	N13°07.3		
13	31°05.4	10°35.1	27.2	209°56.0	39.6	189°06.1	32.3	351°45.7	07.3		
14	46°07.9	25°34.7	28.4	224°59.3	39.9	204°08.9	32.1	6°47.9	07.4		
15	61°10.4	40°34.3	· · 29.6	240°02.6	· · 40.1	219°11.6	· · 32.0	21°50.0	· · 07.5		
16	76°12.8	55°33.9	30.8	255°05.9	40.4	234°14.4	31.9	36°52.2	07.6		
17	91°15.3	70°33.5	32.0	270°09.2	40.6	249°17.2	31.8	51°54.4	07.7		
18	106°17.8	85°33.1	N07°33.2	285°12.4	N02°40.9	264°19.9	S07°31.7	66°56.5	N13°07.8		
19	121°20.2	100°32.7	34.4	300°15.7	41.1	279°22.7	31.6	81°58.7	07.9		
20	136°22.7	115°32.3	35.7	315°19.0	41.4	294°25.5	31.4	97°00.9	08.0		
21	151°25.2	130°31.8	· · 36.9	330°22.3	· · 41.6	309°28.2	· · 31.3	112°03.0	· · 08.1		
22	166°27.6	145°31.4	38.1	345°25.6	41.9	324°31.0	31.2	127°05.2	08.2		
23	181°30.1	160°31.0	39.3	0°28.8	42.1	339°33.7	31.1	142°07.4	08.3		
Mer.pass. 10:56				ν -0.4' d 1.2' m -3.90		ν 3.3' d 0.3' m -1.22		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.37	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	196°32.5	175°30.6	N07°40.5	15°32.1	N02°42.4	354°36.5	S07°31.0	157°09.5	N13°08.4
1	211°35.0	190°30.2	41.7	30°35.4	42.6	9°39.3	30.8	172°11.7	08.5
2	226°37.5	205°29.8	42.9	45°38.7	42.9	24°42.0	30.7	187°13.9	08.6
3	241°39.9	220°29.4	· · 44.1	60°41.9	· · 43.1	39°44.8	· · 30.6	202°16.0	· · 08.7
4	256°42.4	235°28.9	45.3	75°45.2	43.4	54°47.6	30.5	217°18.2	08.8
5	271°44.9	250°28.5	46.6	90°48.5	43.6	69°50.3	30.4	232°20.4	08.9
6	286°47.3	265°28.1	N07°47.8	105°51.7	N02°43.8	84°53.1	S07°30.3	247°22.5	N13°09.0
7	301°49.8	280°27.7	49.0	120°55.0	44.1	99°55.8	30.1	262°24.7	09.1
8	316°52.3	295°27.3	50.2	135°58.3	44.3	114°58.5	30.0	277°26.9	09.2
9	331°54.7	310°26.9	· · 51.4	151°01.5	· · 44.6	130°01.4	· · 29.9	292°29.0	· · 09.3
10	346°57.2	325°26.4	52.6	166°04.8	44.8	145°04.1	29.8	307°31.2	09.4
11	1°59.6	340°26.0	53.8	181°08.1	45.1	160°06.9	29.7	322°33.4	09.5
12	17°02.1	355°25.6	N07°55.0	196°11.3	N02°45.3	175°09.7	S07°29.5	337°35.5	N13°09.6
13	32°04.6	10°25.2	56.2	211°14.6	45.6	190°12.4	29.4	352°37.7	09.7
14	47°07.0	25°24.8	57.4	226°17.9	45.8	205°15.2	29.3	7°39.9	09.8
15	62°09.5	40°24.4	· · 58.6	241°21.1	· · 46.0	220°18.0	· · 29.2	22°42.0	· · 09.9
16	77°12.0	55°23.9	07°59.8	256°24.4	46.3	235°20.7	29.1	37°44.2	10.0
17	92°14.4	70°23.5	08°01.0	271°27.7	46.5	250°23.5	29.0	52°46.4	10.1
18	107°16.9	85°23.1	N08°02.2	286°30.9	N02°46.8	265°26.2	S07°28.8	67°48.5	N13°10.2
19	122°19.4	100°22.7	03.4	301°34.2	47.0	280°29.0	28.7	82°50.7	10.3
20	137°21.8	115°22.3	04.7	316°37.4	47.2	295°31.8	28.6	97°52.9	10.4
21	152°24.3	130°21.8	· · 05.9	331°40.7	· · 47.5	310°34.5	· · 28.5	112°55.0	· · 10.5
22	167°26.8	145°21.4	07.1	346°44.0	47.7	325°37.3	28.4	127°57.2	10.6
23	182°29.2	160°21.0	08.3	1°47.2	47.9	340°40.1	28.2	142°59.4	10.7
Mer.pass. 10:52		ν -0.4' d 1.2' m -3.90		ν 3.3' d 0.2' m -1.20		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.37	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	197°31.7	175°20.6	N08°09.5	16°50.5	N02°48.2	355°42.8	S07°28.1	158°01.5	N13°10.8
1	212°34.1	190°20.2	10.7	31°53.7	48.4	10°45.6	28.0	173°03.7	10.9
2	227°36.6	205°19.7	11.9	46°57.0	48.7	25°48.4	27.9	188°05.9	11.0
3	242°39.1	220°19.3	· · 13.1	62°00.2	· · 48.9	40°51.1	· · 27.8	203°08.0	· · 11.1
4	257°41.5	235°18.9	14.3	77°03.5	49.1	55°53.9	27.7	218°10.2	11.2
5	272°44.0	250°18.5	15.5	92°06.7	49.4	70°56.6	27.5	233°12.4	11.3
6	287°46.5	265°18.0	N08°16.7	107°10.0	N02°49.6	85°59.4	S07°27.4	248°14.5	N13°11.4
7	302°48.9	280°17.6	17.9	122°13.2	49.8	101°02.2	27.3	263°16.7	11.5
8	317°51.4	295°17.2	19.1	137°16.5	50.1	116°04.9	27.2	278°18.9	11.6
9	332°53.9	310°16.8	· · 20.3	152°19.7	· · 50.3	131°07.7	· · 27.1	293°21.0	· · 11.7
10	347°56.3	325°16.4	21.5	167°23.0	50.5	146°10.5	26.9	308°23.2	11.8
11	2°58.8	340°15.9	22.7	182°26.2	50.8	161°13.2	26.8	323°25.4	11.9
12	18°01.2	355°15.5	N08°23.9	197°29.5	N02°51.0	176°16.0	S07°26.7	338°27.5	N13°12.0
13	33°03.7	10°15.1	25.1	212°32.7	51.2	191°18.8	26.6	353°29.7	12.1
14	48°06.2	25°14.7	26.3	227°36.0	51.5	206°21.5	26.5	8°31.9	12.2
15	63°08.6	40°14.2	· · 27.5	242°39.2	· · 51.7	221°24.3	· · 26.4	23°34.0	· · 12.3
16	78°11.1	55°13.8	28.7	257°42.4	51.9	236°27.0	26.2	38°36.2	12.4
17	93°13.6	70°13.4	29.9	272°45.7	52.1	251°29.8	26.1	53°38.4	12.5
18	108°16.0	85°13.0	N08°31.1	287°48.9	N02°52.4	266°32.6	S07°26.0	68°40.5	N13°12.6
19	123°18.5	100°12.5	32.3	302°52.2	52.6	281°35.3	25.9	83°42.7	12.7
20	138°21.0	115°12.1	33.5	317°55.4	52.8	296°38.1	25.8	98°44.9	12.8
21	153°23.4	130°11.7	· · 34.7	332°58.6	· · 53.1	311°40.9	· · 25.6	113°47.0	· · 12.9
22	168°25.9	145°11.3	35.9	348°01.9	53.3	326°43.6	25.5	128°49.2	13.0
23	183°28.4	160°10.8	37.0	3°05.1	53.5	341°46.4	25.4	143°51.4	13.1
Mer.pass. 10:48		ν -0.4' d 1.2' m -3.90		ν 3.3' d 0.2' m -1.18		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.37	

	Stars	
	SHA	Dec
Alpheratz	357° 31.5	29° 15.1
Ankaa	353° 04.3	-42° 08.8
Schedar	349° 27.5	56° 41.8
Diphda	348° 44.2	-17° 49.6
Achernar	335° 18.4	-57° 05.3
Hamal	327° 47.6	23° 36.0
Polaris	312° 34.4	89° 23.3
Acamar	315° 09.6	-40° 11.4
Menkar	314° 02.8	4° 12.1
Mirfak	308° 23.7	49° 57.9
Aldebaran	290° 35.9	16° 34.0
Rigel	281° 00.7	-8° 10.3
Capella	280° 17.0	46° 01.6
Bellatrix	278° 19.3	6° 22.4
Elnath	277° 57.7	28° 37.8
Anilam	275° 34.4	-1° 11.2
Betelgeuse	270° 48.5	7° 24.5
Canopus	263° 51.0	-52° 43.1
Sirius	258° 23.3	-16° 45.7
Adhara	255° 03.2	-29° 01.1
Procyon	244° 47.2	5° 08.7
Pollux	243° 13.1	27° 57.2
Avior	234° 13.1	-59° 36.6
Suhail	222° 43.5	-43° 33.5
Miaplacidus	221° 36.9	-69° 50.6
Alphard	217° 44.2	-8° 47.4
Regulus	207° 30.6	11° 49.2
Dubhe	193° 36.5	61° 35.5
Denebola	182° 21.2	14° 24.3
Gienah	175° 39.7	-17° 42.5
Acrux	172° 55.4	-63° 15.9
Gacrux	171° 47.1	-57° 16.8
Alioth	166° 09.6	55° 47.9
Spica	158° 18.3	-11° 19.0
Alkaid	152° 48.9	49° 09.8
Hadar	148° 30.3	-60° 30.9
Menkent	147° 53.1	-36° 30.9
Arcturus	145° 44.5	19° 01.6
Rigel Kent.	139° 34.8	-60° 57.4
Kochab	137° 19.1	74° 01.9
Zuben 'ubi	136° 51.9	-16° 09.9
Alphecca	126° 00.6	26° 36.8
Antares	112° 11.4	-26° 29.7
Atria	107° 02.1	-69° 04.5
Sabik	101° 58.7	-15° 45.6
Shaula	96° 05.5	-37° 07.3
Rasalhague	95° 55.2	12° 32.2
Eltanin	90° 40.4	51° 28.8
Kaus Aust.	83° 27.9	-34° 22.0
Vega	80° 30.8	38° 48.4
Nunki	75° 43.5	-26° 15.4
Altair	61° 56.7	8° 56.7
Peacock	53° 00.6	-56° 38.1
Deneb	49° 23.6	45° 22.9
Enif	33° 35.6	10° 00.5
Al Na'ir	27° 29.0	-46° 48.9
Fomalhaut	15° 11.1	-29° 27.9
Scheat	13° 42.1	28° 14.4
Makrab	13° 26.7	15° 21.7

Apr 07 Sat	SHA	Mer.pass
Venus	340° 07.1	12:18
Mars	178° 39.9	22:58
Jupiter	157° 56.8	00:26
Saturn	320° 44.1	13:33

Apr 08 Sun	SHA	Mer.pass
Venus	338° 58.1	12:18
Mars	178° 59.6	22:53
Jupiter	158° 04.0	00:21
Saturn	320° 37.0	13:29

Apr 09 Mon	SHA	Mer.pass
Venus	337° 48.9	12:19
Mars	179° 18.8	22:48
Jupiter	158° 11.1	00:17
Saturn	320° 29.9	13:26

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.2086 sec ΔT = TT-UT1 = +68.9754 sec

2029 April 07 to Apr. 09 UT

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	179°27.5	N06°51.3	252°32.9	11.9'	S17°59.6	-9.3' 55.8'
1	194°27.6	52.2	267°03.8	12.0'	17°50.3	-9.4' 55.8'
2	209°27.8	53.1	281°34.8	12.1'	17°40.9	-9.5' 55.8'
3	224°28.0	· · 54.1	296°06.0	12.2'	17°31.4	-9.5' 55.7'
4	239°28.2	55.0	310°37.2	12.3'	17°21.9	-9.6' 55.7'
5	254°28.3	56.0	325°08.4	12.4'	17°12.3	-9.7' 55.7'
6	269°28.5	N06°56.9	339°39.8	12.5'	S17°02.6	-9.7' 55.6'
7	284°28.7	57.8	354°11.3	12.5'	16°52.9	-9.8' 55.6'
8	299°28.9	58.8	8°42.8	12.6'	16°43.1	-9.9' 55.6'
9	314°29.0	06°59.7	23°14.4	12.7'	16°33.2	-9.9' 55.6'
10	329°29.2	07°00.7	37°46.1	12.8'	16°23.2	-10.0' 55.5'
11	344°29.4	01.6	52°17.9	12.9'	16°13.2	-10.1' 55.5'
12	359°29.5	N07°02.5	66°49.8	12.9'	S16°03.1	-10.1' 55.5'
13	14°29.7	03.5	81°21.7	13.0'	15°53.0	-10.2' 55.5'
14	29°29.9	04.4	95°53.7	13.1'	15°42.8	-10.3' 55.4'
15	44°30.1	· · 05.3	110°25.8	13.2'	15°32.5	-10.3' 55.4'
16	59°30.2	06.3	124°58.0	13.3'	15°22.2	-10.4' 55.4'
17	74°30.4	07.2	139°30.2	13.3'	15°11.8	-10.4' 55.4'
18	89°30.6	N07°08.2	154°02.6	13.4'	S15°01.4	-10.5' 55.3'
19	104°30.8	09.1	168°35.0	13.5'	14°50.9	-10.6' 55.3'
20	119°30.9	10.0	183°07.4	13.5'	14°40.4	-10.6' 55.3'
21	134°31.1	· · 11.0	197°40.0	13.6'	14°29.8	-10.7' 55.3'
22	149°31.3	11.9	212°12.6	13.7'	14°19.1	-10.7' 55.2'
23	164°31.5	12.8	226°45.3	13.8'	14°08.4	-10.8' 55.2'
SD = 16.0' $d = 0.9'$			SD = 15.2'			
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	179°31.6	N07°13.8	241°18.1	13.8'	S13°57.6	-10.8' 55.2'
1	194°31.8	14.7	255°50.9	13.9'	13°46.8	-10.9' 55.2'
2	209°32.0	15.6	270°23.8	14.0'	13°35.9	-10.9' 55.1'
3	224°32.1	· · 16.6	284°56.8	14.0'	13°25.0	-11.0' 55.1'
4	239°32.3	17.5	299°29.8	14.1'	13°14.1	-11.0' 55.1'
5	254°32.5	18.4	314°02.9	14.2'	13°03.1	-11.1' 55.1'
6	269°32.7	N07°19.4	328°36.1	14.2'	S12°52.0	-11.1' 55.0'
7	284°32.8	20.3	343°09.3	14.3'	12°40.9	-11.1' 55.0'
8	299°33.0	21.2	357°42.6	14.4'	12°29.7	-11.2' 55.0'
9	314°33.2	· · 22.2	12°16.0	14.4'	12°18.6	-11.2' 55.0'
10	329°33.3	23.1	26°49.4	14.5'	12°07.3	-11.3' 55.0'
11	344°33.5	24.0	41°22.9	14.5'	11°56.0	-11.3' 54.9'
12	359°33.7	N07°25.0	55°56.4	14.6'	S11°44.7	-11.4' 54.9'
13	14°33.9	25.9	70°30.1	14.7'	11°33.4	-11.4' 54.9'
14	29°34.0	26.8	85°03.7	14.7'	11°22.0	-11.4' 54.9'
15	44°34.2	· · 27.8	99°37.4	14.8'	11°10.5	-11.5' 54.8'
16	59°34.4	28.7	114°11.2	14.8'	10°59.1	-11.5' 54.8'
17	74°34.5	29.6	128°45.1	14.9'	10°47.6	-11.5' 54.8'
18	89°34.7	N07°30.6	143°18.9	14.9'	S10°36.0	-11.6' 54.8'
19	104°34.9	31.5	157°52.9	15.0'	10°24.4	-11.6' 54.8'
20	119°35.1	32.4	172°26.9	15.1'	10°12.8	-11.6' 54.7'
21	134°35.2	· · 33.4	187°00.9	15.1'	10°01.2	-11.7' 54.7'
22	149°35.4	34.3	201°35.0	15.2'	09°49.5	-11.7' 54.7'
23	164°35.6	35.2	216°09.2	15.2'	09°37.8	-11.7' 54.7'
SD = 16.0' $d = 0.9'$			SD = 15.0'			
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	179°35.7	N07°36.1	230°43.4	15.3'	S09°26.0	-11.8' 54.7'
1	194°35.9	37.1	245°17.6	15.3'	09°14.2	-11.8' 54.7'
2	209°36.1	38.0	259°51.9	15.3'	09°02.4	-11.8' 54.6'
3	224°36.2	· · 38.9	274°26.3	15.4'	08°50.6	-11.9' 54.6'
4	239°36.4	39.9	289°00.7	15.4'	08°38.7	-11.9' 54.6'
5	254°36.6	40.8	303°35.1	15.5'	08°26.9	-11.9' 54.6'
6	269°36.7	N07°41.7	318°09.6	15.5'	S08°14.9	-11.9' 54.6'
7	284°36.9	42.6	332°44.1	15.6'	08°03.0	-12.0' 54.5'
8	299°37.1	43.6	347°18.7	15.6'	07°51.0	-12.0' 54.5'
9	314°37.3	· · 44.5	1°53.3	15.6'	07°39.0	-12.0' 54.5'
10	329°37.4	45.4	16°27.9	15.7'	07°27.0	-12.0' 54.5'
11	344°37.6	46.4	31°02.6	15.7'	07°15.0	-12.1' 54.5'
12	359°37.8	N07°47.3	45°37.3	15.8'	S07°02.9	-12.1' 54.5'
13	14°37.9	48.2	60°12.1	15.8'	06°50.9	-12.1' 54.5'
14	29°38.1	49.1	74°46.9	15.8'	06°38.8	-12.1' 54.4'
15	44°38.3	· · 50.1	89°21.7	15.9'	06°26.6	-12.1' 54.4'
16	59°38.4	51.0	103°56.6	15.9'	06°14.5	-12.2' 54.4'
17	74°38.6	51.9	118°31.5	15.9'	06°02.3	-12.2' 54.4'
18	89°38.8	N07°52.8	133°06.4	16.0'	S05°50.2	-12.2' 54.4'
19	104°38.9	53.8	147°41.4	16.0'	05°38.0	-12.2' 54.4'
20	119°39.1	54.7	162°16.4	16.0'	05°25.8	-12.2' 54.3'
21	134°39.3	· · 55.6	176°51.4	16.1'	05°13.6	-12.2' 54.3'
22	149°39.4	56.5	191°26.5	16.1'	05°01.3	-12.2' 54.3'
23	164°39.6	57.5	206°01.6	16.1'	04°49.1	-12.3' 54.3'
SD = 16.0' $d = 0.9'$			SD = 14.9'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:53	04:17	19:50	21:16	—:—
N 70°	01:05	03:17	04:29	19:38	20:51	23:15
68°	01:58	03:35	04:38	19:28	20:32	22:13
66°	02:29	03:49	04:46	19:19	20:17	21:40
64°	02:51	04:01	04:53	19:13	20:05	21:16
62°	03:08	04:11	04:59	19:07	19:55	20:58
60°	03:23	04:19	05:04	19:01	19:46	20:44
N 58°	03:34	04:27	05:08	18:57	19:39	20:31
56°	03:44	04:33	05:12	18:53	19:32	20:21
54°	03:53	04:39	05:15	18:49	19:26	20:12
52°	04:01	04:44	05:18	18:46	19:21	20:04
50°	04:07	04:48	05:21	18:43	19:17	19:57
45°	04:21	04:57	05:28	18:37	19:07	19:43
N 40°	04:32	05:05	05:33	18:32	18:59	19:32
35°	04:41	05:11	05:37	18:27	18:53	19:23
30°	04:48	05:17	05:41	18:23	18:47	19:16
20°	04:59	05:25	05:48	18:16	18:39	19:05
N 10°	05:07	05:32	05:53	18:10	18:32	18:56
0°	05:13	05:38	05:58	18:05	18:26	18:50
S 10°	05:18	05:42	06:04	18:00	18:21	18:45
20°	05:21	05:47	06:09	17:54	18:16	18:42
30°	05:23	05:51	06:15	17:48	18:12	18:40
35°	05:24	05:53	06:18	17:45	18:10	18:39
40°	05:24	05:55	06:22	17:41	18:08	18:39
45°	05:23	05:57	06:27	17:36	18:06	18:40
S 50°	05:22	05:59	06:32	17:31	18:03	18:41
52°	05:21	06:00	06:34	17:28	18:02	18:41
54°	05:20	06:01	06:37	17:26	18:01	18:42
56°	05:19	06:02	06:40	17:23	18:00	18:43
58°	05:18	06:03	06:43	17:19	17:59	18:44
S 60°	05:16	06:05	06:46	17:16	17:58	18:46

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	06:08	05:08	04:36	08:59	11:33	13:34
N 70°	05:15	04:44	04:24	09:50	11:55	13:44
68°	04:42	04:26	04:13	10:22	12:12	13:52
66°	04:17	04:11	04:05	10:45	12:25	13:59
64°	03:58	03:58	03:58	11:03	12:36	14:05
62°	03:43	03:48	03:51	11:18	12:46	14:10
60°	03:30	03:39	03:46	11:30	12:54	14:14
N 58°	03:18	03:31	03:41	11:41	13:01	14:18
56°	03:08	03:24	03:37	11:50	13:07	14:21
54°	02:59	03:18	03:33	11:58	13:13	14:24
52°	02:52	03:12	03:29	12:06	13:17	14:27
50°	02:45	03:07	03:26	12:12	13:22	14:30
45°	02:29	02:56	03:19	12:26	13:32	14:35
N 40°	02:17	02:47	03:13	12:38	13:40	14:39
35°	02:06	02:39	03:08	12:48	13:46	14:43
30°	01:57	02:32	03:03	12:56	13:52	14:47
20°	01:40	02:20	02:56	13:11	14:03	14:52
N 10°	01:26	02:09	02:49	13:23	14:12	14:58
0°	01:13	01:59	02:42	13:35	14:20	15:02
S 10°	00:59	01:49	02:36	13:47	14:28	15:07
20°	00:45	01:38	02:29	13:59	14:37	15:12
30°	00:28	01:25	02:21	14:14	14:47	15:17
35°	00:19	01:18	02:16	14:22	14:53	15:21
40°	00:07	01:10	02:11	14:31	14:59	15:24
45°	· · ·	01:00	02:05	14:42	15:07	15:28
S 50°	· · ·	00:49	01:57	14:55	15:16	15:33
52°	· · ·	00:43	01:54	15:01	15:20	15:36
54°	· · ·	00:37	01:50	15:08	15:24	15:38
56°	· · ·	00:31	01:46	15:15	15:29	15:41
58°	· · ·	00:23	01:41	15:23	15:35	15:44
S 60°	· · ·	00:14	01:36	15:33	15:41	15:47

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
07	02:10	02:02	12:02	07:24	19:47	
08	01:54	01:45	12:02	08:09	20:31	
09	01:37	01:29	12:01	08:52	21:13	

April 10, 11, 12 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	198°30.8	175°10.4	N08°38.2	18°08.4	N02°53.7	356°49.2	S07°25.3	158°53.5	N13°13.2	Alpheratz	357°31.5	29°15.1
1	213°33.3	190°10.0	39.4	33°11.6	54.0	11°51.9	25.2	173°55.7	13.3	Ankaa	353°04.3	-42°08.7
2	228°35.7	205°09.6	40.6	48°14.8	54.2	26°54.7	25.1	188°57.8	13.4	Schedar	349°27.5	56°41.8
3	243°38.2	220°09.1	· · 41.8	63°18.1	· · 54.4	41°57.5	· · 24.9	204°00.0	· · 13.4	Diphda	348°44.2	-17°49.6
4	258°40.7	235°08.7	43.0	78°21.3	54.6	57°00.2	24.8	219°02.2	13.5	Achernar	335°18.4	-57°05.3
5	273°43.1	250°08.3	44.2	93°24.5	54.9	72°03.0	24.7	234°04.3	13.6	Hamal	327°47.6	23°36.0
6	288°45.6	265°07.8	N08°45.4	108°27.8	N02°55.1	87°05.7	S07°24.6	249°06.5	N13°13.7	Polaris	312°35.2	89°23.3
7	303°48.1	280°07.4	46.6	123°31.0	55.3	102°08.5	24.5	264°08.7	13.8	Acamar	315°09.6	-40°11.4
8	318°50.5	295°07.0	47.8	138°34.2	55.5	117°11.3	24.3	279°10.8	13.9	Menkar	314°02.8	4°12.1
9	333°53.0	310°06.6	· · 49.0	153°37.4	· · 55.8	132°14.0	· · 24.2	294°13.0	· · 14.0	Mirfak	308°23.7	49°57.9
10	348°55.5	325°06.1	50.2	168°40.7	56.0	147°16.8	24.1	309°15.2	14.1	Aldebaran	290°35.9	16°34.0
11	3°57.9	340°05.7	51.4	183°43.9	56.2	162°19.6	24.0	324°17.3	14.2	Rigel	281°00.7	-8°10.3
12	19°00.4	355°05.3	N08°52.6	198°47.1	N02°56.4	177°22.3	S07°23.9	339°19.5	N13°14.3	Capella	280°17.0	46°01.6
13	34°02.9	10°04.8	53.8	213°50.3	56.6	192°25.1	23.8	354°21.7	14.4	Bellatrix	278°19.4	6°22.4
14	49°05.3	25°04.4	54.9	228°53.6	56.9	207°27.9	23.6	9°23.8	14.5	Elnath	277°57.7	28°37.8
15	64°07.8	40°04.0	· · 56.1	243°56.8	· · 57.1	222°30.6	· · 23.5	24°26.0	· · 14.6	Alnilam	275°34.4	-1°11.2
16	79°10.2	55°03.5	57.3	259°00.0	57.3	237°33.4	23.4	39°28.1	14.7	Betelgeuse	270°48.5	7°24.5
17	94°12.7	70°03.1	58.5	274°03.2	57.5	252°36.2	23.3	54°30.3	14.8	Canopus	263°51.0	-52°43.1
18	109°15.2	85°02.7	N08°59.7	289°06.5	N02°57.7	267°38.9	S07°23.2	69°32.5	N13°14.9	Sirius	258°23.3	-16°45.7
19	124°17.6	100°02.3	09°00.9	304°09.7	57.9	282°41.7	23.0	84°34.6	15.0	Adhara	255°03.2	-29°01.1
20	139°20.1	115°01.8	02.1	319°12.9	58.2	297°44.4	22.9	99°36.8	15.1	Procyon	244°47.2	5°08.7
21	154°22.6	130°01.4	· · 03.3	334°16.1	· · 58.4	312°47.2	· · 22.8	114°39.0	· · 15.2	Pollux	243°13.1	27°57.2
22	169°25.0	145°01.0	04.5	349°19.3	58.6	327°50.0	22.7	129°41.1	15.3	Avior	234°13.1	-59°36.6
23	184°27.5	160°00.5	05.6	4°22.5	58.8	342°52.7	22.6	144°43.3	15.4	Suhail	222°43.5	-43°33.5
Mer.pass. 10:44		ν -0.4' d 1.2' m -3.90		ν 3.2' d 0.2' m -1.16		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.36		Miaplacidus	221°37.0	-69°50.7
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	199°30.0	175°00.1	N09°06.8	19°25.8	N02°59.0	357°55.5	S07°22.5	159°45.5	N13°15.5	Alphard	217°44.2	-8°47.4
1	214°32.4	189°59.7	08.0	34°29.0	59.2	12°58.3	22.3	174°47.6	15.6	Regulus	207°30.6	11°49.2
2	229°34.9	204°59.2	09.2	49°32.2	59.5	28°01.0	22.2	189°49.8	15.7	Dubhe	193°36.5	61°35.6
3	244°37.3	219°58.8	· · 10.4	64°35.4	· · 59.7	43°03.8	· · 22.1	204°52.0	· · 15.8	Denebola	182°21.2	14°24.3
4	259°39.8	234°58.4	11.6	79°38.6	02°59.9	58°06.6	22.0	219°54.1	15.9	Gienah	175°39.7	-17°42.5
5	274°42.3	249°57.9	12.8	94°41.8	03°00.1	73°09.3	21.9	234°56.3	16.0	Acrux	172°55.4	-63°15.9
6	289°44.7	264°57.5	N09°13.9	109°45.0	N03°00.3	88°12.1	S07°21.7	249°58.4	N13°16.1	Gacrux	171°47.1	-57°16.8
7	304°47.2	279°57.0	15.1	124°48.2	00.5	103°14.9	21.6	265°00.6	16.2	Alioth	166°09.6	55°47.9
8	319°49.7	294°56.6	16.3	139°51.4	00.7	118°17.6	21.5	280°02.8	16.3	Spica	158°18.3	-11°19.0
9	334°52.1	309°56.2	· · 17.5	154°54.6	· · 00.9	133°20.4	· · 21.4	295°04.9	· · 16.4	Alkaid	152°48.9	49°09.9
10	349°54.6	324°55.7	18.7	169°57.9	01.2	148°23.2	21.3	310°07.1	16.5	Hadar	148°30.3	-60°30.9
11	4°57.1	339°55.3	19.9	185°01.1	01.4	163°25.9	21.2	325°09.3	16.6	Menkent	147°53.1	-36°30.9
12	19°59.5	354°54.9	N09°21.0	200°04.3	N03°01.6	178°28.7	S07°21.0	340°11.4	N13°16.7	Arcturus	145°44.5	19°01.6
13	35°02.0	9°54.4	22.2	215°07.5	01.8	193°31.4	20.9	355°13.6	16.8	Rigel Kent.	139°34.8	-60°57.4
14	50°04.5	24°54.0	23.4	230°10.7	02.0	208°34.2	20.8	10°15.7	16.9	Kochab	137°19.1	74°01.9
15	65°06.9	39°53.6	· · 24.6	245°13.9	· · 02.2	223°37.0	· · 20.7	25°17.9	· · 17.0	Zuben'ubi	136°51.9	-16°09.9
16	80°09.4	54°53.1	25.8	260°17.1	02.4	238°39.7	20.6	40°20.1	17.1	Alphecca	126°00.6	26°36.8
17	95°11.8	69°52.7	26.9	275°20.3	02.6	253°42.5	20.4	55°22.2	17.2	Antares	112°11.3	-26°29.7
18	110°14.3	84°52.2	N09°28.1	290°23.5	N03°02.8	268°45.3	S07°20.3	70°24.4	N13°17.3	Atria	107°02.1	-69°04.5
19	125°16.8	99°51.8	29.3	305°26.7	03.0	283°48.0	20.2	85°26.6	17.4	Sabik	101°58.6	-15°45.6
20	140°19.2	114°51.4	30.5	320°29.9	03.2	298°50.8	20.1	100°28.7	17.5	Shaula	96°05.5	-37°07.3
21	155°21.7	129°50.9	· · 31.7	335°33.1	· · 03.4	313°53.6	· · 20.0	115°30.9	· · 17.6	Rasalhague	95°55.2	12°32.2
22	170°24.2	144°50.5	32.8	350°36.2	03.6	328°56.3	19.9	130°33.0	17.7	Eltanin	90°40.4	51°28.9
23	185°26.6	159°50.0	34.0	5°39.4	03.8	343°59.1	19.7	145°35.2	17.8	Kaus Aust.	83°27.9	-34°22.0
Mer.pass. 10:40		ν -0.4' d 1.2' m -3.90		ν 3.2' d 0.2' m -1.14		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.36		Vega	80°30.8	38°48.4
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	200°29.1	174°49.6	N09°35.2	20°42.6	N03°04.0	359°01.9	S07°19.6	160°37.4	N13°17.9	Nunki	75°43.5	-26°15.4
1	215°31.6	189°49.2	36.4	35°45.8	04.2	14°04.6	19.5	175°39.5	18.0	Altair	61°56.6	8°56.7
2	230°34.0	204°48.7	37.6	50°49.0	04.5	29°07.4	19.4	190°41.7	18.1	Peacock	53°00.6	-56°38.1
3	245°36.5	219°48.3	· · 38.7	65°52.2	· · 04.7	44°10.2	· · 19.3	205°43.9	· · 18.2	Deneb	49°23.5	45°22.9
4	260°38.9	234°47.8	39.9	80°55.4	04.9	59°12.9	19.1	220°46.0	18.3	Enif	33°35.6	10°00.5
5	275°41.4	249°47.4	41.1	95°58.6	05.1	74°15.7	19.0	235°48.2	18.4	Al Na'ir	27°29.0	-46°48.9
6	290°43.9	264°47.0	N09°42.3	111°01.8	N03°05.3	89°18.5	S07°18.9	250°50.3	N13°18.5	Fomalhaut	15°11.1	-29°27.9
7	305°46.3	279°46.5	43.4	126°05.0	05.5	104°21.2	18.8	265°52.5	18.6	Scheat	13°42.1	28°14.4
8	320°48.8	294°46.1	44.6	141°08.1	05.7	119°24.0	18.7	280°54.7	18.7	Markab	13°26.7	15°21.7
9	335°51.3	309°45.6	· · 45.8	156°11.3	· · 05.9	134°26.7	· · 18.6	295°56.8	· · 18.8	Apr 10 Tue SHA Mer.pass		
10	350°53.7	324°45.2	47.0	171°14.5	06.1	149°29.5	18.4	310°59.0	18.9	Venus	336°39.6	12:20
11	5°56.2	339°44.7	48.1	186°17.7	06.3	164°32.3	18.3	326°01.2	19.0	Mars	179°37.5	22:43
12	20°58.7	354°44.3	N09°49.3	201°20.9	N03°06.5	179°35.0	S07°18.2	341°03.3	N13°19.1	Jupiter	158°18.3	00:13
13	36°01.1	9°43.9	50.5	216°24.0	06.6	194°37.8	18.1	356°05.5	19.2	Saturn	320°22.7	13:22
14	51°03.6	24°43.4	51.6	231°27.2	06.8	209°40.6	18.0	11°07.6	19.3	Apr 11 Wed SHA Mer.pass		
15	66°06.1	39°43.0	· · 52.8	246°30.4	· · 07.0	224°43.3	· · 17.8	26°09.8	· · 19.4	Venus	335°30.1	12:20
16	81°08.5	54°42.5	54.0	261°33.6	07.2	239°46.1	17.7	41°12.0	19.5	Mars	179°55.8	22:37
17	96°11.0	69°42.1	55.2	276°36.8	07.4	254°48.9	17.6	56°14.1	19.6	Jupiter	158°25.6	00:08
18	111°13.4	84°41.6	N09°56.3	291°39.9	N03°07.6	269°51.6	S07°17.5	71°16.3	N13°19.7	Saturn	320°15.5	13:19
19	126°15.9	99°41.2	57.5	306°43.1	07.8	284°54.4	17.4	86°18.5	19.8	Apr 12 Thu SHA Mer.pass		
20	141°18.4	114°40.7	58.7	321°46.3	08.0	299°57.2						

DUT1 = UT1-UTC = +0.2087 sec ΔT = TT-UT1 = +68.9753 sec

2029 Apr 10 to Apr. 12 UT

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 39.8	N07° 58.4	220° 36.7	16.1'	S04° 36.8	-12.3'	54.3'
1	194° 39.9	07° 59.3	235° 11.8	16.2'	04° 24.5	-12.3'	54.3'
2	209° 40.1	08° 00.2	249° 47.0	16.2'	04° 12.2	-12.3'	54.3'
3	224° 40.3	· · 01.2	264° 22.2	16.2'	04° 00.0	-12.3'	54.3'
4	239° 40.4	02.1	278° 57.4	16.2'	03° 47.6	-12.3'	54.2'
5	254° 40.6	03.0	293° 32.6	16.3'	03° 35.3	-12.3'	54.2'
6	269° 40.8	N08° 03.9	308° 07.9	16.3'	S03° 23.0	-12.3'	54.2'
7	284° 40.9	04.9	322° 43.2	16.3'	03° 10.7	-12.3'	54.2'
8	299° 41.1	05.8	337° 18.5	16.3'	02° 58.3	-12.3'	54.2'
9	314° 41.3	· · 06.7	351° 53.8	16.3'	02° 46.0	-12.4'	54.2'
10	329° 41.4	07.6	6° 29.1	16.4'	02° 33.6	-12.4'	54.2'
11	344° 41.6	08.5	21° 04.5	16.4'	02° 21.3	-12.4'	54.2'
12	359° 41.8	N08° 09.5	35° 39.8	16.4'	S02° 08.9	-12.4'	54.2'
13	14° 41.9	10.4	50° 15.2	16.4'	01° 56.5	-12.4'	54.1'
14	29° 42.1	11.3	64° 50.6	16.4'	01° 44.2	-12.4'	54.1'
15	44° 42.3	· · 12.2	79° 26.0	16.4'	01° 31.8	-12.4'	54.1'
16	59° 42.4	13.2	94° 01.5	16.4'	01° 19.4	-12.4'	54.1'
17	74° 42.6	14.1	108° 36.9	16.4'	01° 07.0	-12.4'	54.1'
18	89° 42.8	N08° 15.0	123° 12.3	16.5'	S00° 54.7	-12.4'	54.1'
19	104° 42.9	15.9	137° 47.8	16.5'	00° 42.3	-12.4'	54.1'
20	119° 43.1	16.8	152° 23.3	16.5'	00° 29.9	-12.4'	54.1'
21	134° 43.2	· · 17.8	166° 58.7	16.5'	00° 17.5	-12.4'	54.1'
22	149° 43.4	18.7	181° 34.2	16.5'	S00° 05.2	-12.4'	54.1'
23	164° 43.6	19.6	196° 09.7	16.5'	N00° 07.2	-12.4'	54.0'
SD = 16.0'			SD = 14.8'				
$d = 0.9'$							

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 43.7	N08° 20.5	210° 45.2	16.5'	N00° 19.6	12.4'	54.0'
1	194° 43.9	21.4	225° 20.7	16.5'	00° 31.9	12.4'	54.0'
2	209° 44.1	22.3	239° 56.2	16.5'	00° 44.3	12.3'	54.0'
3	224° 44.2	· · 23.3	254° 31.7	16.5'	00° 56.6	12.3'	54.0'
4	239° 44.4	24.2	269° 07.2	16.5'	01° 09.0	12.3'	54.0'
5	254° 44.6	25.1	283° 42.7	16.5'	01° 21.3	12.3'	54.0'
6	269° 44.7	N08° 26.0	298° 18.2	16.5'	N01° 33.6	12.3'	54.0'
7	284° 44.9	26.9	312° 53.7	16.5'	01° 45.9	12.3'	54.0'
8	299° 45.0	27.8	327° 29.3	16.5'	01° 58.3	12.3'	54.0'
9	314° 45.2	· · 28.8	342° 04.8	16.5'	02° 10.6	12.3'	54.0'
10	329° 45.4	29.7	356° 40.3	16.5'	02° 22.9	12.3'	54.0'
11	344° 45.5	30.6	11° 15.8	16.5'	02° 35.1	12.3'	54.0'
12	359° 45.7	N08° 31.5	25° 51.3	16.5'	N02° 47.4	12.3'	54.0'
13	14° 45.9	32.4	40° 26.7	16.5'	02° 59.7	12.2'	54.0'
14	29° 46.0	33.3	55° 02.2	16.5'	03° 11.9	12.2'	53.9'
15	44° 46.2	· · 34.3	69° 37.7	16.5'	03° 24.1	12.2'	53.9'
16	59° 46.3	35.2	84° 13.2	16.5'	03° 36.3	12.2'	53.9'
17	74° 46.5	36.1	98° 48.6	16.4'	03° 48.5	12.2'	53.9'
18	89° 46.7	N08° 37.0	113° 24.1	16.4'	N04° 00.7	12.2'	53.9'
19	104° 46.8	37.9	127° 59.5	16.4'	04° 12.9	12.1'	53.9'
20	119° 47.0	38.8	142° 34.9	16.4'	04° 25.0	12.1'	53.9'
21	134° 47.1	· · 39.7	157° 10.4	16.4'	04° 37.2	12.1'	53.9'
22	149° 47.3	40.7	171° 45.8	16.4'	04° 49.3	12.1'	53.9'
23	164° 47.5	41.6	186° 21.2	16.4'	05° 01.4	12.1'	53.9'
SD = 16.0'			SD = 14.7'				
$d = 0.9'$							

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 47.6	N08° 42.5	200° 56.5	16.4'	N05° 13.4	12.1'	53.9'
1	194° 47.8	43.4	215° 31.9	16.3'	05° 25.5	12.0'	53.9'
2	209° 48.0	44.3	230° 07.2	16.3'	05° 37.5	12.0'	53.9'
3	224° 48.1	· · 45.2	244° 42.6	16.3'	05° 49.5	12.0'	53.9'
4	239° 48.3	46.1	259° 17.9	16.3'	06° 01.5	12.0'	53.9'
5	254° 48.4	47.0	273° 53.2	16.3'	06° 13.5	11.9'	53.9'
6	269° 48.6	N08° 48.0	288° 28.4	16.3'	N06° 25.4	11.9'	53.9'
7	284° 48.8	48.9	303° 03.7	16.2'	06° 37.3	11.9'	53.9'
8	299° 48.9	49.8	317° 38.9	16.2'	06° 49.2	11.9'	53.9'
9	314° 49.1	· · 50.7	332° 14.1	16.2'	07° 01.1	11.8'	53.9'
10	329° 49.2	51.6	346° 49.3	16.2'	07° 12.9	11.8'	53.9'
11	344° 49.4	52.5	1° 24.5	16.1'	07° 24.7	11.8'	53.9'
12	359° 49.5	N08° 53.4	15° 59.6	16.1'	N07° 36.5	11.7'	53.9'
13	14° 49.7	54.3	30° 34.7	16.1'	07° 48.2	11.7'	53.9'
14	29° 49.9	55.2	45° 09.8	16.1'	08° 00.0	11.7'	53.9'
15	44° 50.0	· · 56.1	59° 44.9	16.0'	08° 11.6	11.7'	53.9'
16	59° 50.2	57.1	74° 19.9	16.0'	08° 23.3	11.6'	53.9'
17	74° 50.3	58.0	88° 54.9	16.0'	08° 34.9	11.6'	53.9'
18	89° 50.5	N08° 58.9	103° 29.9	16.0'	N08° 46.5	11.6'	53.9'
19	104° 50.7	08° 59.8	118° 04.8	15.9'	08° 58.1	11.5'	53.9'
20	119° 50.8	09° 00.7	132° 39.8	15.9'	09° 09.6	11.5'	53.9'
21	134° 51.0	· · 01.6	147° 14.7	15.9'	09° 21.1	11.5'	53.9'
22	149° 51.1	02.5	161° 49.5	15.8'	09° 32.5	11.4'	53.9'
23	164° 51.3	03.4	176° 24.3	15.8'	09° 44.0	11.4'	53.9'
SD = 15.9'			SD = 14.7'				
$d = 0.9'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:30	04:00	20:05	21:38	—:—
N 70°	—:—	02:59	04:14	19:51	21:08	—:—
68°	01:31	03:20	04:25	19:39	20:46	22:41
66°	02:09	03:36	04:35	19:30	20:29	21:58
64°	02:35	03:49	04:42	19:22	20:15	21:30
62°	02:55	04:00	04:49	19:15	20:04	21:10
60°	03:11	04:10	04:55	19:09	19:54	20:54
N 58°	03:24	04:18	05:00	19:04	19:46	20:40
56°	03:35	04:25	05:04	18:59	19:39	20:29
54°	03:45	04:31	05:08	18:55	19:32	20:19
52°	03:53	04:36	05:12	18:51	19:27	20:11
50°	04:00	04:41	05:15	18:48	19:22	20:03
45°	04:15	04:52	05:22	18:41	19:11	19:48
N 40°	04:27	05:00	05:28	18:35	19:02	19:36
35°	04:37	05:07	05:33	18:29	18:55	19:26
30°	04:44	05:13	05:37	18:25	18:49	19:18
20°	04:57	05:23	05:45	18:17	18:39	19:06
N 10°	05:06	05:30	05:52	18:10	18:32	18:56
0°	05:13	05:37	05:58	18:04	18:25	18:49
S 10°	05:18	05:42	06:04	17:58	18:19	18:44
20°	05:22	05:48	06:10	17:52	18:14	18:40
30°	05:25	05:53	06:17	17:45	18:09	18:37
35°	05:26	05:55	06:21	17:41	18:06	18:35
40°	05:27	05:58	06:25	17:36	18:03	18:35
45°	05:27	06:01	06:30	17:31	18:00	18:34
S 50°	05:26	06:04	06:36	17:25	17:57	18:35
52°	05:26	06:05	06:39	17:22	17:56	18:35
54°	05:26	06:07	06:42	17:19	17:54	18:35
56°	05:25	06:08	06:46	17:15	17:53	18:36
58°	05:24	06:10	06:49	17:11	17:51	18:36
S 60°	05:23	06:11	06:54	17:07	17:49	18:37

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	04:12	03:51	03:29	15:24	17:12	19:03
N 70°	04:07	03:52	03:37	15:26	17:06	18:49
68°	04:03	03:53	03:43	15:28	17:02	18:37
66°	03:59	03:54	03:49	15:29	16:58	18:28
64°	03:56	03:55	03:54	15:31	16:55	18:20
62°	03:54	03:56	03:58	15:32	16:52	18:13
60°	03:51	03:56	04:01	15:33	16:50	18:08
N 58°	03:49	03:57	04:05	15:33	16:48	18:03
56°	03:47	03:57	04:07	15:34	16:46	17:58
54°	03:46	03:58	04:10	15:35	16:44	17:54
52°	03:44	03:58	04:12	15:35	16:43	17:51
50°	03:43	03:59	04:15	15:36	16:42	17:47
45°	03:40	03:59	04:19	15:37	16:39	17:40
N 40°	03:37	04:00	04:23	15:38	16:36	17:34
35°	03:35	04:01	04:27	15:39	16:34	17:29
30°	03:33	04:01	04:30	15:40	16:32	17:25
20°	03:29	04:02	04:35	15:41	16:29	17:17
N 10°	03:26	04:03	04:40	15:42	16:26	17:10
0°	03:24	04:04	04:44	15:43	16:23	17:04
S 10°	03:21	04:05	04:49	15:44	16:21	16:58
20°	03:18	04:06	04:54	15:45	16:18	16:51
30°	03:14	04:07	04:59	15:46	16:15	16:44
35°	03:12	04:07	05:02	15:47	16:13	16:39
40°	03:10	04:08	05:06	15:48	16:11	16:34
45°	03:07	04:09	05:10	15:49	16:08	16:29
S 50°	03:04	04:10	05:16	15:50	16:05	16:22
52°	03:03	04:10	05:18	15:50	16:04	16:19
54°	03:01	04:11	05:21	15:51	16:03	16:15
56°	02:59	04:12	05:23	15:51	16:01	16:12
58°	02:57	04:12	05:27	15:52	15:59	16:07
S 60°	02:55	04:13	05:30	15:52	15:57	16:03

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April 13, 14, 15 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	201°28.2	174°38.9	N10°03.3	21°59.0	N03°08.8	0°08.2	S07°16.8	161°29.3	N13°20.2	Alpheratz	357°31.5	29°15.1
1	216°30.7	189°38.5	04.5	37°02.1	09.0	15°11.0	16.7	176°31.4	20.3	Ankaa	353°04.3	-42°08.7
2	231°33.2	204°38.1	05.7	52°05.3	09.2	30°13.8	16.5	191°33.6	20.4	Schedar	349°27.5	56°41.8
3	246°35.6	219°37.6	· · 06.8	67°08.5	· · 09.4	45°16.5	· · 16.4	206°35.7	· · 20.5	Diphda	348°44.2	-17°49.6
4	261°38.1	234°37.2	08.0	82°11.6	09.5	60°19.3	16.3	221°37.9	20.6	Achernar	335°18.4	-57°05.3
5	276°40.5	249°36.7	09.2	97°14.8	09.7	75°22.0	16.2	236°40.1	20.7	Hamal	327°47.6	23°36.0
6	291°43.0	264°36.3	N10°10.3	112°17.9	N03°09.9	90°24.8	S07°16.1	251°42.2	N13°20.8	Polaris	312°35.8	89°23.2
7	306°45.5	279°35.8	11.5	127°21.1	10.1	105°27.6	16.0	266°44.4	20.9	Acamar	315°09.6	-40°11.4
8	321°47.9	294°35.4	12.7	142°24.3	10.3	120°30.3	15.8	281°46.5	21.0	Menkar	314°02.8	4°12.1
9	336°50.4	309°34.9	· · 13.8	157°27.4	· · 10.5	135°33.1	· · 15.7	296°48.7	· · 21.1	Mirfak	308°23.7	49°57.9
10	351°52.9	324°34.5	15.0	172°30.6	10.7	150°35.9	15.6	311°50.9	21.2	Aldebaran	290°35.9	16°34.0
11	6°55.3	339°34.0	16.2	187°33.7	10.9	165°38.6	15.5	326°53.0	21.3	Rigel	281°00.8	-8°10.3
12	21°57.8	354°33.6	N10°17.3	202°36.9	N03°11.0	180°41.4	S07°15.4	341°55.2	N13°21.4	Capella	280°17.1	46°01.6
13	37°00.3	9°33.1	18.5	217°40.1	11.2	195°44.2	15.2	356°57.4	21.5	Bellatrix	278°19.4	6°22.4
14	52°02.7	24°32.6	19.7	232°43.2	11.4	210°46.9	15.1	11°59.5	21.6	Elnath	277°57.7	28°37.8
15	67°05.2	39°32.2	· · 20.8	247°46.4	· · 11.6	225°49.7	· · 15.0	27°01.7	· · 21.7	Alnilam	275°34.4	-1°11.2
16	82°07.7	54°31.7	22.0	262°49.5	11.8	240°52.5	14.9	42°03.8	21.8	Betelgeuse	270°48.5	7°24.5
17	97°10.1	69°31.3	23.1	277°52.7	12.0	255°55.2	14.8	57°06.0	21.9	Canopus	263°51.0	-52°43.1
18	112°12.6	84°30.8	N10°24.3	292°55.8	N03°12.2	270°58.0	S07°14.7	72°08.2	N13°22.0	Sirius	258°23.3	-16°45.7
19	127°15.0	99°30.4	25.5	307°59.0	12.3	286°00.8	14.5	87°10.3	22.1	Adhara	255°03.2	-29°01.1
20	142°17.5	114°29.9	26.6	323°02.1	12.5	301°03.5	14.4	102°12.5	22.2	Procyon	244°47.3	5°08.7
21	157°20.0	129°29.5	· · 27.8	338°05.3	· · 12.7	316°06.3	· · 14.3	117°14.6	· · 22.3	Pollux	243°13.1	27°57.2
22	172°22.4	144°29.0	28.9	353°08.4	12.9	331°09.1	14.2	132°16.8	22.4	Avior	234°13.1	-59°36.6
23	187°24.9	159°28.6	30.1	8°11.6	13.1	346°11.8	14.1	147°19.0	22.5	Suhail	222°43.6	-43°33.5
Mer.pass. 10:32		ν -0.4' d 1.2' m -3.90		ν 3.2' d 0.2' m -1.10		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.36				

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	202°27.4	174°28.1	N10°31.3	23°14.7	N03°13.2	1°14.6	S07°14.0	162°21.1	N13°22.6
1	217°29.8	189°27.7	32.4	38°17.8	13.4	16°17.3	13.8	177°23.3	22.7
2	232°32.3	204°27.2	33.6	53°21.0	13.6	31°20.1	13.7	192°25.4	22.8
3	247°34.8	219°26.7	· · 34.7	68°24.1	· · 13.8	46°22.9	· · 13.6	207°27.6	· · 22.9
4	262°37.2	234°26.3	35.9	83°27.3	14.0	61°25.6	13.5	222°29.8	23.0
5	277°39.7	249°25.8	37.0	98°30.4	14.1	76°28.4	13.4	237°31.9	23.1
6	292°42.2	264°25.4	N10°38.2	113°33.5	N03°14.3	91°31.2	S07°13.2	252°34.1	N13°23.2
7	307°44.6	279°24.9	39.4	128°36.7	14.5	106°33.9	13.1	267°36.2	23.3
8	322°47.1	294°24.5	40.5	143°39.8	14.7	121°36.7	13.0	282°38.4	23.4
9	337°49.5	309°24.0	· · 41.7	158°43.0	· · 14.8	136°39.5	· · 12.9	297°40.6	· · 23.5
10	352°52.0	324°23.5	42.8	173°46.1	15.0	151°42.2	12.8	312°42.7	23.6
11	7°54.5	339°23.1	44.0	188°49.2	15.2	166°45.0	12.7	327°44.9	23.7
12	22°56.9	354°22.6	N10°45.1	203°52.4	N03°15.4	181°47.8	S07°12.5	342°47.0	N13°23.8
13	37°59.4	9°22.2	46.3	218°55.5	15.5	196°50.5	12.4	357°49.2	23.9
14	53°01.9	24°21.7	47.4	233°58.6	15.7	211°53.3	12.3	12°51.4	24.0
15	68°04.3	39°21.2	· · 48.6	249°01.7	· · 15.9	226°56.1	· · 12.2	27°53.5	· · 24.1
16	83°06.8	54°20.8	49.7	264°04.9	16.0	241°58.8	12.1	42°55.7	24.2
17	98°09.3	69°20.3	50.9	279°08.0	16.2	257°01.6	12.0	57°57.8	24.3
18	113°11.7	84°19.9	N10°52.0	294°11.1	N03°16.4	272°04.3	S07°11.8	73°00.0	N13°24.4
19	128°14.2	99°19.4	53.2	309°14.2	16.6	287°07.1	11.7	88°02.2	24.5
20	143°16.6	114°18.9	54.3	324°17.4	16.7	302°09.9	11.6	103°04.3	24.6
21	158°19.1	129°18.5	· · 55.5	339°20.5	· · 16.9	317°12.6	· · 11.5	118°06.5	· · 24.7
22	173°21.6	144°18.0	56.6	354°23.6	17.1	332°15.4	11.4	133°08.6	24.8
23	188°24.0	159°17.6	57.8	9°26.7	17.2	347°18.2	11.2	148°10.8	24.9
Mer.pass. 10:28		ν -0.5' d 1.2' m -3.90		ν 3.1' d 0.2' m -1.07		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.35	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	203°26.5	174°17.1	N10°58.9	24°29.8	N03°17.4	2°20.9	S07°11.1	163°13.0	N13°25.0
1	218°29.0	189°16.6	11°00.1	39°33.0	17.6	17°23.7	11.0	178°15.1	25.1
2	233°31.4	204°16.2	01.2	54°36.1	17.7	32°26.5	10.9	193°17.3	25.2
3	248°33.9	219°15.7	· · 02.4	69°39.2	· · 17.9	47°29.2	· · 10.8	208°19.4	· · 25.3
4	263°36.4	234°15.2	03.5	84°42.3	18.1	62°32.0	10.7	223°21.6	25.4
5	278°38.8	249°14.8	04.7	99°45.4	18.2	77°34.8	10.5	238°23.7	25.5
6	293°41.3	264°14.3	N11°05.8	114°48.5	N03°18.4	92°37.5	S07°10.4	253°25.9	N13°25.6
7	308°43.8	279°13.8	07.0	129°51.6	18.6	107°40.3	10.3	268°28.1	25.7
8	323°46.2	294°13.4	08.1	144°54.8	18.7	122°43.0	10.2	283°30.2	25.8
9	338°48.7	309°12.9	· · 09.2	159°57.9	· · 18.9	137°45.8	· · 10.1	298°32.4	· · 25.9
10	353°51.1	324°12.4	10.4	175°01.0	19.0	152°48.6	10.0	313°34.5	26.0
11	8°53.6	339°12.0	11.5	190°04.1	19.2	167°51.3	09.8	328°36.7	26.1
12	23°56.1	354°11.5	N11°12.7	205°07.2	N03°19.4	182°54.1	S07°09.7	343°38.9	N13°26.2
13	38°58.5	9°11.0	13.8	220°10.3	19.5	197°56.9	09.6	358°41.0	26.3
14	54°01.0	24°10.6	15.0	235°13.4	19.7	212°59.6	09.5	13°43.2	26.4
15	69°03.5	39°10.1	· · 16.1	250°16.5	· · 19.8	228°02.4	· · 09.4	28°45.3	· · 26.5
16	84°05.9	54°09.6	17.2	265°19.6	20.0	243°05.2	09.3	43°47.5	26.6
17	99°08.4	69°09.2	18.4	280°22.7	20.2	258°07.9	09.1	58°49.7	26.7
18	114°10.9	84°08.7	N11°19.5	295°25.8	N03°20.3	273°10.7	S07°09.0	73°51.8	N13°26.8
19	129°13.3	99°08.2	20.7	310°28.9	20.5	288°13.5	08.9	88°54.0	26.9
20	144°15.8	114°07.8	21.8	325°32.0	20.6	303°16.2	08.8	103°56.1	27.0
21	159°18.2	129°07.3	· · 22.9	340°35.1	· · 20.8	318°19.0	· · 08.7	118°58.3	· · 27.1
22	174°20.7	144°06.8	24.1	355°38.2	20.9	333°21.7	08.6	134°00.4	27.2
23	189°23.2	159°06.3	25.2	10°41.3	21.1	348°24.5	08.4	149°02.6	27.2
Mer.pass. 10:25		ν -0.5' d 1.1' m -3.90		ν 3.1' d 0.2' m -1.05		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.35	

Apr 13 Fri	SHA	Mer.pass
Venus	333°10.7	12:22
Mars	180°30.7	22:27
Jupiter	158°40.0	23:55
Saturn	320°01.0	13:12

Apr 14 Sat	SHA	Mer.pass
Venus	332°00.7	12:22
Mars	180°47.3	22:22
Jupiter	158°47.2	23:51
Saturn	319°53.8	13:09

Apr 15 Sun	SHA	Mer.pass
Venus	330°50.6	12:23
Mars	181°03.3	22:17
Jupiter	158°54.4	23:46
Saturn	319°46.4	13:05

Horizontal parallax	
Venus:	0.1
Mars:	0.2

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°51.4	N09°04.3	190°59.1	15.8'	N09°55.3	11.3'	53.9'
1	194°51.6	05.2	205°33.9	15.7'	10°06.7	11.3'	53.9'
2	209°51.8	06.1	220°08.6	15.7'	10°18.0	11.3'	53.9'
3	224°51.9	· · 07.0	234°43.3	15.7'	10°29.3	11.2'	53.9'
4	239°52.1	07.9	249°18.0	15.6'	10°40.5	11.2'	53.9'
5	254°52.2	08.8	263°52.6	15.6'	10°51.7	11.1'	53.9'
6	269°52.4	N09°09.7	278°27.2	15.5'	N11°02.8	11.1'	53.9'
7	284°52.5	10.7	293°01.8	15.5'	11°13.9	11.1'	53.9'
8	299°52.7	11.6	307°36.3	15.5'	11°25.0	11.0'	53.9'
9	314°52.9	· · 12.5	322°10.7	15.4'	11°36.0	11.0'	53.9'
10	329°53.0	13.4	336°45.2	15.4'	11°46.9	10.9'	53.9'
11	344°53.2	14.3	351°19.6	15.4'	11°57.9	10.9'	53.9'
12	359°53.3	N09°15.2	5°53.9	15.3'	N12°08.7	10.8'	53.9'
13	14°53.5	16.1	20°28.2	15.3'	12°19.6	10.8'	53.9'
14	29°53.6	17.0	35°02.5	15.2'	12°30.4	10.7'	53.9'
15	44°53.8	· · 17.9	49°36.7	15.2'	12°41.1	10.7'	53.9'
16	59°53.9	18.8	64°10.9	15.1'	12°51.8	10.6'	53.9'
17	74°54.1	19.7	78°45.1	15.1'	13°02.4	10.6'	53.9'
18	89°54.3	N09°20.6	93°19.2	15.1'	N13°13.0	10.5'	53.9'
19	104°54.4	21.5	107°53.2	15.0'	13°23.5	10.5'	53.9'
20	119°54.6	22.4	122°27.2	15.0'	13°34.0	10.4'	53.9'
21	134°54.7	· · 23.3	137°01.2	14.9'	13°44.4	10.4'	53.9'
22	149°54.9	24.2	151°35.1	14.9'	13°54.8	10.3'	53.9'
23	164°55.0	25.1	166°09.0	14.8'	14°05.1	10.3'	53.9'
SD = 15.9'			SD = 14.7'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°55.2	N09°26.0	180°42.8	14.8'	N14°15.4	10.2'	53.9'
1	194°55.3	26.9	195°16.6	14.7'	14°25.6	10.2'	53.9'
2	209°55.5	27.8	209°50.3	14.7'	14°35.8	10.1'	53.9'
3	224°55.6	· · 28.7	224°24.0	14.6'	14°45.9	10.0'	53.9'
4	239°55.8	29.6	238°57.7	14.6'	14°55.9	10.0'	53.9'
5	254°55.9	30.5	253°31.2	14.5'	15°05.9	9.9'	53.9'
6	269°56.1	N09°31.4	268°04.8	14.5'	N15°15.8	9.9'	53.9'
7	284°56.3	32.3	282°38.3	14.4'	15°25.6	9.8'	53.9'
8	299°56.4	33.2	297°11.7	14.4'	15°35.4	9.7'	53.9'
9	314°56.6	· · 34.1	311°45.1	14.3'	15°45.2	9.7'	53.9'
10	329°56.7	35.0	326°18.4	14.3'	15°54.8	9.6'	53.9'
11	344°56.9	35.9	340°51.7	14.2'	16°04.4	9.5'	54.0'
12	359°57.0	N09°36.8	355°24.9	14.2'	N16°14.0	9.5'	54.0'
13	14°57.2	37.7	9°58.1	14.1'	16°23.5	9.4'	54.0'
14	29°57.3	38.6	24°31.2	14.1'	16°32.9	9.3'	54.0'
15	44°57.5	· · 39.5	39°04.3	14.0'	16°42.2	9.3'	54.0'
16	59°57.6	40.4	53°37.3	14.0'	16°51.5	9.2'	54.0'
17	74°57.8	41.3	68°10.2	13.9'	17°00.7	9.1'	54.0'
18	89°57.9	N09°42.2	82°43.1	13.8'	N17°09.8	9.1'	54.0'
19	104°58.1	43.0	97°16.0	13.8'	17°18.9	9.0'	54.0'
20	119°58.2	43.9	111°48.8	13.7'	17°27.9	8.9'	54.0'
21	134°58.4	· · 44.8	126°21.5	13.7'	17°36.8	8.9'	54.0'
22	149°58.5	45.7	140°54.2	13.6'	17°45.7	8.8'	54.0'
23	164°58.7	46.6	155°26.8	13.6'	17°54.4	8.7'	54.0'
SD = 15.9'			SD = 14.7'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°58.8	N09°47.5	169°59.4	13.5'	N18°03.1	8.6'	54.0'
1	194°59.0	48.4	184°31.9	13.5'	18°11.8	8.6'	54.0'
2	209°59.1	49.3	199°04.4	13.4'	18°20.3	8.5'	54.0'
3	224°59.3	· · 50.2	213°36.8	13.3'	18°28.8	8.4'	54.1'
4	239°59.4	51.1	228°09.1	13.3'	18°37.2	8.3'	54.1'
5	254°59.6	52.0	242°41.4	13.2'	18°45.5	8.2'	54.1'
6	269°59.7	N09°52.9	257°13.6	13.2'	N18°53.8	8.2'	54.1'
7	284°59.9	53.8	271°45.8	13.1'	19°01.9	8.1'	54.1'
8	300°00.0	54.7	286°17.9	13.1'	19°10.0	8.0'	54.1'
9	315°00.2	· · 55.5	300°49.9	13.0'	19°18.0	7.9'	54.1'
10	330°00.3	56.4	315°21.9	12.9'	19°25.9	7.8'	54.1'
11	345°00.5	57.3	329°53.9	12.9'	19°33.7	7.7'	54.1'
12	0°00.6	N09°58.2	344°25.8	12.8'	N19°41.5	7.7'	54.1'
13	15°00.8	09°59.1	358°57.6	12.8'	19°49.2	7.6'	54.1'
14	30°00.9	10°00.0	13°29.3	12.7'	19°56.7	7.5'	54.2'
15	45°01.1	· · 00.9	28°01.0	12.6'	20°04.2	7.4'	54.2'
16	60°01.2	01.8	42°32.7	12.6'	20°11.6	7.3'	54.2'
17	75°01.4	02.7	57°04.3	12.5'	20°18.9	7.2'	54.2'
18	90°01.5	N10°03.6	71°35.8	12.5'	N20°26.2	7.1'	54.2'
19	105°01.7	04.4	86°07.3	12.4'	20°33.3	7.1'	54.2'
20	120°01.8	05.3	100°38.7	12.4'	20°40.4	7.0'	54.2'
21	135°01.9	· · 06.2	115°10.1	12.3'	20°47.3	6.9'	54.2'
22	150°02.1	07.1	129°41.4	12.2'	20°54.2	6.8'	54.2'
23	165°02.2	08.0	144°12.6	12.2'	21°01.0	6.7'	54.2'
SD = 15.9'			SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:05	03:44	20:21	22:03	—:—
N 70°	—:—	02:40	04:00	20:04	21:26	—:—
68°	00:52	03:04	04:12	19:51	21:00	23:27
66°	01:48	03:22	04:23	19:40	20:41	22:19
64°	02:19	03:37	04:32	19:31	20:26	21:46
62°	02:41	03:50	04:39	19:23	20:13	21:22
60°	02:59	04:00	04:46	19:16	20:03	21:04
N 58°	03:14	04:09	04:51	19:10	19:53	20:49
56°	03:26	04:17	04:57	19:05	19:45	20:37
54°	03:36	04:23	05:01	19:01	19:39	20:26
52°	03:45	04:29	05:05	18:56	19:32	20:17
50°	03:53	04:35	05:09	18:53	19:27	20:09
45°	04:09	04:46	05:17	18:44	19:15	19:52
N 40°	04:22	04:56	05:23	18:38	19:06	19:39
35°	04:32	05:03	05:29	18:32	18:58	19:29
30°	04:41	05:10	05:34	18:27	18:51	19:20
20°	04:54	05:20	05:43	18:18	18:40	19:07
N 10°	05:04	05:29	05:50	18:10	18:32	18:57
0°	05:12	05:36	05:57	18:04	18:24	18:49
S 10°	05:18	05:42	06:04	17:57	18:18	18:43
20°	05:23	05:48	06:11	17:49	18:12	18:37
30°	05:27	05:54	06:19	17:41	18:06	18:33
35°	05:28	05:58	06:23	17:37	18:02	18:32
40°	05:29	06:01	06:28	17:32	17:59	18:30
45°	05:30	06:04	06:34	17:26	17:55	18:29
S 50°	05:31	06:08	06:41	17:19	17:51	18:29
52°	05:31	06:10	06:44	17:15	17:49	18:29
54°	05:31	06:12	06:48	17:12	17:48	18:28
56°	05:31	06:14	06:52	17:08	17:45	18:28
58°	05:31	06:16	06:56	17:03	17:43	18:29
S 60°	05:30	06:18	07:01	16:58	17:41	18:29

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	03:03	02:29	□	21:06	□	□
N 70°	03:20	02:58	02:23	20:38	22:45	□
68°	03:33	03:21	03:04	20:17	22:05	· · ·
66°	03:44	03:38	03:32	20:00	21:38	23:21
64°	03:53	03:53	03:54	19:47	21:17	22:49
62°	04:01	04:05	04:11	19:36	21:00	22:25
60°	04:07	04:15	04:26	19:26	20:46	22:06
N 58°	04:13	04:24	04:38	19:18	20:35	21:51
56°	04:19	04:32	04:49	19:11	20:24	21:37
54°	04:23	04:39	04:58	19:05	20:15	21:26
52°	04:28	04:46	05:07	18:59	20:07	21:16
50°	04:32	04:51	05:15	18:54	20:00	21:07
45°	04:40	05:04	05:31	18:42	19:45	20:47
N 40°	04:48	05:14	05:44	18:33	19:32	20:32
35°	04:54	05:23	05:56	18:25	19:22	20:19
30°	04:59	05:31	06:06	18:18	19:12	20:08
20°	05:09	05:45	06:24	18:06	18:56	19:48
N 10°	05:18	05:57	06:39	17:56	18:43	19:31
0°	05:26	06:08	06:53	17:46	18:30	19:16
S 10°	05:34	06:20	07:08	17:36	18:17	19:00
20°	05:42	06:32	07:23	17:26	18:03	18:43
30°	05:52	06:46	07:41	17:14	17:47	18:24
35°	05:58	06:54	07:51	17:07	17:38	18:13
40°	06:05	07:04	08:03	17:00	17:28	18:00
45°	06:12	07:15	08:17	16:51	17:16	17:45
S 50°	06:22	07:28	08:35	16:40	17:01	17:27
52°	06:26	07:34	08:43	16:35	16:54	17:18
54°	06:31	07:41	08:52	16:30	16:47	17:09
56°	06:36	07:49	09:03	16:24	16:38	16:58
58°	06:42	07:58	09:15	16:17	16:29	16:45
S 60°	06:48	08:08	09:28	16:09	16:18	16:31

Day	Sun		Mer. Pass hh:mm	Moon		Age 0-2 1-1%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
13	00:34	00:27	12:00	11:36	23:57	
14	00:19	00:12	12:00	12:19	—:—	
15	00:05	00:02	12:00	13:04	00:41	

April 16, 17, 18 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	204°25.6	174°05.9	N11°26.3	25°44.4	N03°21.3	3°27.3	S07°08.3	164°04.8	N13°27.3	Alpheratz	357°31.5	29°15.1
1	219°28.1	189°05.4	27.5	40°47.5	21.4	18°30.0	08.2	179°06.9	27.4	Ankaa	353°04.3	-42°08.7
2	234°30.6	204°04.9	28.6	55°50.5	21.6	33°32.8	08.1	194°09.1	27.5	Schedar	349°27.5	56°41.8
3	249°33.0	219°04.5	· · 29.7	70°53.6	· · 21.7	48°35.6	· · 08.0	209°11.2	· · 27.6	Diphda	348°44.2	-17°49.6
4	264°35.5	234°04.0	30.9	85°56.7	21.9	63°38.3	07.8	224°13.4	27.7	Achernar	335°18.4	-57°05.3
5	279°38.0	249°03.5	32.0	100°59.8	22.0	78°41.1	07.7	239°15.6	27.8	Hamal	327°47.6	23°36.0
6	294°40.4	264°03.0	N11°33.2	116°02.9	N03°22.2	93°43.9	S07°07.6	254°17.7	N13°27.9	Polaris	312°36.1	89°23.2
7	309°42.9	279°02.6	34.3	131°06.0	22.3	108°46.6	07.5	269°19.9	28.0	Acamar	315°09.6	-40°11.4
8	324°45.4	294°02.1	35.4	146°09.1	22.5	123°49.4	07.4	284°22.0	28.1	Menkar	314°02.8	4°12.2
9	339°47.8	309°01.6	· · 36.5	161°12.1	· · 22.6	138°52.1	· · 07.3	299°24.2	· · 28.2	Mirfak	308°23.7	49°57.9
10	354°50.3	324°01.1	37.7	176°15.2	22.8	153°54.9	07.1	314°26.3	28.3	Aldebaran	290°35.9	16°33.9
11	9°52.7	339°00.7	38.8	191°18.3	22.9	168°57.7	07.0	329°28.5	28.4	Rigel	281°00.8	-8°10.3
12	24°55.2	354°00.2	N11°39.9	206°21.4	N03°23.1	184°00.4	S07°06.9	344°30.7	N13°28.5	Capella	280°17.1	46°01.6
13	39°57.7	8°59.7	41.1	221°24.5	23.2	199°03.2	06.8	359°32.8	28.6	Bellatrix	278°19.4	6°22.4
14	55°00.1	23°59.2	42.2	236°27.5	23.4	214°06.0	06.7	14°35.0	28.7	Elnath	277°57.7	28°37.8
15	70°02.6	38°58.8	· · 43.3	251°30.6	· · 23.5	229°08.7	· · 06.6	29°37.1	· · 28.8	Alnilam	275°34.4	-1°11.2
16	85°05.1	53°58.3	44.5	266°33.7	23.7	244°11.5	06.4	44°39.3	28.9	Betelgeuse	270°48.5	7°24.5
17	100°07.5	68°57.8	45.6	281°36.8	23.8	259°14.3	06.3	59°41.4	29.0	Canopus	263°51.1	-52°43.1
18	115°10.0	83°57.3	N11°46.7	296°39.8	N03°24.0	274°17.0	S07°06.2	74°43.6	N13°29.1	Sirius	258°23.3	-16°45.7
19	130°12.5	98°56.8	47.8	311°42.9	24.1	289°19.8	06.1	89°45.8	29.2	Adhara	255°03.2	-29°01.1
20	145°14.9	113°56.4	49.0	326°46.0	24.2	304°22.5	06.0	104°47.9	29.3	Procyon	244°47.3	5°08.7
21	160°17.4	128°55.9	· · 50.1	341°49.0	· · 24.4	319°25.3	· · 05.9	119°50.1	· · 29.4	Pollux	243°13.2	27°57.2
22	175°19.9	143°55.4	51.2	356°52.1	24.5	334°28.1	05.7	134°52.2	29.5	Avior	234°13.2	-59°36.6
23	190°22.3	158°54.9	52.4	11°55.2	24.7	349°30.8	05.6	149°54.4	29.6	Suhail	222°43.6	-43°33.5
Mer.pass. 10:21		ν -0.5' d 1.1' m -3.90		ν 3.1' d 0.2' m -1.03		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.35				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	205°24.8	173°54.5	N11°53.5	26°58.2	N03°24.8	4°33.6	S07°05.5	164°56.5	N13°29.7
1	220°27.2	188°54.0	54.6	42°01.3	25.0	19°36.4	05.4	179°58.7	29.8
2	235°29.7	203°53.5	55.7	57°04.4	25.1	34°39.1	05.3	195°00.9	29.9
3	250°32.2	218°53.0	· · 56.9	72°07.4	· · 25.2	49°41.9	· · 05.2	210°03.0	· · 30.0
4	265°34.6	233°52.5	58.0	87°10.5	25.4	64°44.6	05.1	225°05.2	30.1
5	280°37.1	248°52.0	11°59.1	102°13.6	25.5	79°47.4	04.9	240°07.3	30.2
6	295°39.6	263°51.6	N12°00.2	117°16.6	N03°25.7	94°50.2	S07°04.8	255°09.5	N13°30.3
7	310°42.0	278°51.1	01.3	132°19.7	25.8	109°52.9	04.7	270°11.6	30.4
8	325°44.5	293°50.6	02.5	147°22.7	25.9	124°55.7	04.6	285°13.8	30.5
9	340°47.0	308°50.1	· · 03.6	162°25.8	· · 26.1	139°58.5	· · 04.5	300°16.0	· · 30.6
10	355°49.4	323°49.6	04.7	177°28.8	26.2	155°01.2	04.4	315°18.1	30.7
11	10°51.9	338°49.1	05.8	192°31.9	26.3	170°04.0	04.2	330°20.3	30.8
12	25°54.4	353°48.7	N12°06.9	207°34.9	N03°26.5	185°06.7	S07°04.1	345°22.4	N13°30.9
13	40°56.8	8°48.2	06.0	222°38.0	26.6	200°09.5	04.0	0°24.6	31.0
14	55°59.3	23°47.7	09.2	237°41.0	26.7	215°12.3	03.9	15°26.7	31.1
15	71°01.7	38°47.2	· · 10.3	252°44.1	· · 26.9	230°15.0	· · 03.8	30°28.9	· · 31.2
16	86°04.2	53°46.7	11.4	267°47.1	27.0	245°17.8	03.7	45°31.1	31.3
17	101°06.7	68°46.2	12.5	282°50.2	27.1	260°20.6	03.5	60°33.2	31.4
18	116°09.1	83°45.8	N12°13.6	297°53.2	N03°27.3	275°23.3	S07°03.4	75°35.4	N13°31.5
19	131°11.6	98°45.3	14.8	312°56.3	27.4	290°26.1	03.3	90°37.5	31.6
20	146°14.1	113°44.8	15.9	327°59.3	27.5	305°28.8	03.2	105°39.7	31.7
21	161°16.5	128°44.3	· · 17.0	343°02.4	· · 27.7	320°31.6	· · 03.1	120°41.8	· · 31.8
22	176°19.0	143°43.8	18.1	358°05.4	27.8	335°34.4	03.0	135°44.0	31.9
23	191°21.5	158°43.3	19.2	13°08.4	27.9	350°37.1	02.8	150°46.1	32.0
Mer.pass. 10:17		ν -0.5' d 1.1' m -3.90		ν 3.1' d 0.1' m -1.01		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.35	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	206°23.9	173°42.8	N12°20.3	28°11.5	N03°28.1	5°39.9	S07°02.7	165°48.3	N13°32.1
1	221°26.4	188°42.3	21.4	43°14.5	28.2	20°42.7	02.6	180°50.5	32.2
2	236°28.8	203°41.8	22.6	58°17.5	28.3	35°45.4	02.5	195°52.6	32.3
3	251°31.3	218°41.4	· · 23.7	73°20.6	· · 28.4	50°48.2	· · 02.4	210°54.8	· · 32.4
4	266°33.8	233°40.9	24.8	88°23.6	28.6	65°50.9	02.3	225°56.9	32.5
5	281°36.2	248°40.4	25.9	103°26.6	28.7	80°53.7	02.2	240°59.1	32.6
6	296°38.7	263°39.9	N12°27.0	118°29.7	N03°28.8	95°56.5	S07°02.0	256°01.2	N13°32.7
7	311°41.2	278°39.4	28.1	133°32.7	28.9	110°59.2	01.9	271°03.4	32.8
8	326°43.6	293°38.9	29.2	148°35.7	29.1	126°02.0	01.8	286°05.6	32.9
9	341°46.1	308°38.4	· · 30.3	163°38.8	· · 29.2	141°04.7	· · 01.7	301°07.7	· · 33.0
10	356°48.6	323°37.9	31.4	178°41.8	29.3	156°07.5	01.6	316°09.9	33.1
11	11°51.0	338°37.4	32.5	193°44.8	29.4	171°10.3	01.5	331°12.0	33.2
12	26°53.5	353°36.9	N12°33.6	208°47.8	N03°29.6	186°13.0	S07°01.3	346°14.2	N13°33.3
13	41°56.0	8°36.4	34.8	223°50.8	29.7	201°15.8	01.2	1°16.3	33.4
14	56°58.4	23°35.9	35.9	238°53.9	29.8	216°18.5	01.1	16°18.5	33.5
15	72°00.9	38°35.4	· · 37.0	253°56.9	· · 29.9	231°21.3	· · 01.0	31°20.6	· · 33.6
16	87°03.3	53°35.0	38.1	268°59.9	30.0	246°24.1	00.9	46°22.8	33.7
17	102°05.8	68°34.5	39.2	284°02.9	30.2	261°26.8	00.8	61°25.0	33.8
18	117°08.3	83°34.0	N12°40.3	299°05.9	N03°30.3	276°29.6	S07°00.6	76°27.1	N13°33.8
19	132°10.7	98°33.5	41.4	314°09.0	30.4	291°32.4	00.5	91°29.3	33.9
20	147°13.2	113°33.0	42.5	329°12.0	30.5	306°35.1	00.4	106°31.4	34.0
21	162°15.7	128°32.5	· · 43.6	344°15.0	· · 30.6	321°37.9	· · 00.3	121°33.6	· · 34.1
22	177°18.1	143°32.0	44.7	359°18.0	30.8	336°40.6	00.2	136°35.7	34.2
23	192°20.6	158°31.5	45.8	14°21.0	30.9	351°43.4	00.1	151°37.9	34.3
Mer.pass. 10:13		ν -0.5' d 1.1' m -3.90		ν 3.0' d 0.1' m -0.98		ν 2.8' d -0.1' m -2.47		ν 2.2' d 0.1' m 0.34	

Apr 16 Mon	SHA	Mer.pass
Venus	329°40.2	12:24
Mars	181°18.7	22:12
Jupiter	159°01.6	23:42
Saturn	319°39.1	13:02

Apr 17 Tue	SHA	Mer.pass
Venus	328°29.7	12:25
Mars	181°33.5	22:08
Jupiter	159°08.8	23:37
Saturn	319°31.8	12:58

Apr 18 Wed	SHA	Mer.pass
Venus	327°18.9	12:26
Mars	181°47.5	22:03
Jupiter	159°16.0	23:33
Saturn	319°24.4	12:55

Horizontal parallax	
Venus:	0.1
Mars:	0.2

h Sun							Moon						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°02.4	N10°08.9	158°43.8	12.1'	N21°07.7	6.6'	54.3'						
1	195°02.5	09.8	173°14.9	12.1'	21°14.2	6.5'	54.3'						
2	210°02.7	10.6	187°46.0	12.0'	21°20.7	6.4'	54.3'						
3	225°02.8	11.5	202°17.0	12.0'	21°27.1	6.3'	54.3'						
4	240°03.0	12.4	216°47.9	11.9'	21°33.4	6.2'	54.3'						
5	255°03.1	13.3	231°18.8	11.8'	21°39.6	6.1'	54.3'						
6	270°03.3	N10°14.2	245°49.7	11.8'	N21°45.7	6.0'	54.3'						
7	285°03.4	15.1	260°20.4	11.7'	21°51.8	5.9'	54.3'						
8	300°03.6	16.0	274°51.2	11.7'	21°57.7	5.8'	54.4'						
9	315°03.7	16.8	289°21.8	11.6'	22°03.5	5.7'	54.4'						
10	330°03.8	17.7	303°52.4	11.6'	22°09.2	5.6'	54.4'						
11	345°04.0	18.6	318°23.0	11.5'	22°14.8	5.5'	54.4'						
12	0°04.1	N10°19.5	332°53.5	11.4'	N22°20.3	5.4'	54.4'						
13	15°04.3	20.4	347°24.0	11.4'	22°25.7	5.3'	54.4'						
14	30°04.4	21.3	1°54.3	11.3'	22°31.0	5.2'	54.4'						
15	45°04.6	22.1	16°24.7	11.3'	22°36.2	5.1'	54.4'						
16	60°04.7	23.0	30°55.0	11.2'	22°41.3	5.0'	54.5'						
17	75°04.9	23.9	45°25.2	11.2'	22°46.3	4.9'	54.5'						
18	90°05.0	N10°24.8	59°55.4	11.1'	N22°51.1	4.8'	54.5'						
19	105°05.1	25.7	74°25.5	11.1'	22°55.9	4.7'	54.5'						
20	120°05.3	26.5	88°55.6	11.0'	23°00.6	4.6'	54.5'						
21	135°05.4	27.4	103°25.6	11.0'	23°05.1	4.4'	54.5'						
22	150°05.6	28.3	117°55.6	10.9'	23°09.6	4.3'	54.5'						
23	165°05.7	29.2	132°25.5	10.9'	23°13.9	4.2'	54.6'						
SD = 15.9'							SD = 14.8'						

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°05.9	N10°30.1	146°55.4	10.8'	N23°18.1	4.1'	54.6'
1	195°06.0	30.9	161°25.2	10.8'	23°22.3	4.0'	54.6'
2	210°06.1	31.8	175°55.0	10.7'	23°26.3	3.9'	54.6'
3	225°06.3	32.7	190°24.7	10.7'	23°30.2	3.8'	54.6'
4	240°06.4	33.6	204°54.4	10.6'	23°33.9	3.7'	54.6'
5	255°06.6	34.5	219°24.0	10.6'	23°37.6	3.6'	54.6'
6	270°06.7	N10°35.3	233°53.6	10.5'	N23°41.2	3.4'	54.7'
7	285°06.9	36.2	248°23.1	10.5'	23°44.6	3.3'	54.7'
8	300°07.0	37.1	262°52.6	10.4'	23°47.9	3.2'	54.7'
9	315°07.1	38.0	277°22.0	10.4'	23°51.1	3.1'	54.7'
10	330°07.3	38.8	291°51.4	10.4'	23°54.2	3.0'	54.7'
11	345°07.4	39.7	306°20.8	10.3'	23°57.2	2.9'	54.7'
12	0°07.6	N10°40.6	320°50.1	10.3'	N24°00.0	2.7'	54.8'
13	15°07.7	41.5	335°19.4	10.2'	24°02.8	2.6'	54.8'
14	30°07.8	42.3	349°48.6	10.2'	24°05.4	2.5'	54.8'
15	45°08.0	43.2	4°17.8	10.1'	24°07.9	2.4'	54.8'
16	60°08.1	44.1	18°46.9	10.1'	24°10.3	2.3'	54.8'
17	75°08.3	45.0	33°16.1	10.1'	24°12.6	2.1'	54.9'
18	90°08.4	N10°45.8	47°45.1	10.0'	N24°14.7	2.0'	54.9'
19	105°08.5	46.7	62°14.2	10.0'	24°16.7	1.9'	54.9'
20	120°08.7	47.6	76°43.2	10.0'	24°18.6	1.8'	54.9'
21	135°08.8	48.5	91°12.1	9.9'	24°20.4	1.7'	54.9'
22	150°09.0	49.3	105°41.1	9.9'	24°22.1	1.5'	54.9'
23	165°09.1	50.2	120°09.9	9.9'	24°23.6	1.4'	55.0'
	SD = 15.9'	$d = 0.9'$			SD = 14.9'		

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°09.2	N10°51.1	134°38.8	9.8'	N24°25.0	1.3'	55.0'
1	195°09.4	52.0	149°07.6	9.8'	24°26.3	1.2'	55.0'
2	210°09.5	52.8	163°36.4	9.8'	24°27.5	1.0'	55.0'
3	225°09.7	53.7	178°05.2	9.7'	24°28.5	0.9'	55.0'
4	240°09.8	54.6	192°33.9	9.7'	24°29.4	0.8'	55.1'
5	255°09.9	55.4	207°02.6	9.7'	24°30.2	0.7'	55.1'
6	270°10.1	N10°56.3	221°31.3	9.6'	N24°30.9	0.5'	55.1'
7	285°10.2	57.2	235°59.9	9.6'	24°31.4	0.4'	55.1'
8	300°10.3	58.1	250°28.6	9.6'	24°31.8	0.3'	55.1'
9	315°10.5	58.9	264°57.2	9.6'	24°32.1	0.2'	55.2'
10	330°10.6	10°59.8	279°25.7	9.5'	24°32.3	0.0'	55.2'
11	345°10.8	11°00.7	293°54.3	9.5'	24°32.3	-0.1'	55.2'
12	0°10.9	N11°01.5	308°22.8	9.5'	N24°32.2	-0.2'	55.2'
13	15°11.0	02.4	322°51.3	9.5'	24°32.0	-0.3'	55.2'
14	30°11.2	03.3	337°19.8	9.5'	24°31.7	-0.5'	55.3'
15	45°11.3	04.1	351°48.2	9.4'	24°31.2	-0.6'	55.3'
16	60°11.4	05.0	6°16.7	9.4'	24°30.6	-0.7'	55.3'
17	75°11.6	05.9	20°45.1	9.4'	24°29.9	-0.9'	55.3'
18	90°11.7	N11°06.7	35°13.5	9.4'	N24°29.0	-1.0'	55.4'
19	105°11.8	07.6	49°41.9	9.4'	24°28.1	-1.1'	55.4'
20	120°12.0	08.5	64°10.2	9.4'	24°26.9	-1.2'	55.4'
21	135°12.1	09.3	78°38.6	9.3'	24°25.7	-1.4'	55.4'
22	150°12.2	10.2	93°06.9	9.3'	24°24.3	-1.5'	55.5'
23	165°12.4	11.1	107°35.2	9.3'	24°22.8	-1.6'	55.5'
	SD = 15.9'	$d = 0.9'$			SD = 15.0'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	01:34	03:26	20:37	22:35	—:—
N 70°	—:—	02:19	03:45	20:18	21:46	—:—
68°	—:—	02:47	03:59	20:02	21:16	—:—
66°	01:21	03:08	04:11	19:50	20:54	22:47
64°	02:01	03:25	04:21	19:40	20:37	22:03
62°	02:27	03:39	04:30	19:31	20:23	21:36
60°	02:47	03:50	04:37	19:24	20:11	21:15
N 58°	03:03	04:00	04:43	19:17	20:01	20:59
56°	03:16	04:08	04:49	19:11	19:52	20:45
54°	03:27	04:16	04:54	19:06	19:45	20:33
52°	03:37	04:22	04:59	19:02	19:38	20:23
50°	03:46	04:28	05:03	18:57	19:32	20:15
45°	04:03	04:41	05:12	18:48	19:19	19:57
N 40°	04:17	04:51	05:19	18:41	19:09	19:43
35°	04:28	04:59	05:25	18:34	19:00	19:32
30°	04:37	05:06	05:31	18:29	18:53	19:22
20°	04:51	05:18	05:40	18:19	18:41	19:08
N 10°	05:02	05:27	05:49	18:11	18:32	18:57
0°	05:11	05:35	05:56	18:03	18:24	18:48
S 10°	05:18	05:42	06:04	17:55	18:17	18:41
20°	05:23	05:49	06:12	17:47	18:10	18:35
30°	05:28	05:56	06:20	17:38	18:02	18:30
35°	05:30	06:00	06:25	17:33	17:59	18:28
40°	05:32	06:04	06:31	17:27	17:55	18:26
45°	05:34	06:08	06:38	17:21	17:50	18:24
S 50°	05:35	06:13	06:46	17:13	17:45	18:23
52°	05:36	06:15	06:49	17:09	17:43	18:22
54°	05:36	06:17	06:53	17:05	17:41	18:22
56°	05:36	06:20	06:58	17:00	17:38	18:22
58°	05:37	06:22	07:03	16:55	17:36	18:21
S 60°	05:37	06:25	07:08	16:50	17:33	18:21

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	02:30	☐	☐	00:15	☐	☐
66°	03:25	03:12	☐		01:16	☐
64°	03:58	04:09	04:37		00:20	01:37
62°	04:22	04:42	05:18	23:47		00:56
60°	04:42	05:07	05:46	23:22		00:29
N 58°	04:58	05:26	06:08	23:03		00:07
56°	05:12	05:43	06:26	22:47	23:49	
54°	05:23	05:57	06:41	22:33	23:34	
52°	05:34	06:09	06:54	22:21	23:21	
50°	05:43	06:20	07:05	22:11	23:10	
45°	06:03	06:42	07:29	21:48	22:46	23:38
N 40°	06:19	07:00	07:48	21:31	22:27	23:19
35°	06:33	07:16	08:04	21:16	22:11	23:03
30°	06:45	07:29	08:18	21:03	21:57	22:49
20°	07:06	07:52	08:42	20:41	21:34	22:26
N 10°	07:24	08:11	09:02	20:22	21:14	22:06
0°	07:40	08:30	09:21	20:04	20:55	21:47
S 10°	07:57	08:48	09:41	19:46	20:36	21:28
20°	08:15	09:08	10:01	19:27	20:15	21:07
30°	08:36	09:31	10:25	19:05	19:52	20:43
35°	08:49	09:45	10:39	18:53	19:38	20:30
40°	09:03	10:01	10:55	18:38	19:22	20:13
45°	09:20	10:19	11:14	18:21	19:03	19:54
S 50°	09:40	10:43	11:38	17:59	18:40	19:30
52°	09:50	10:54	11:50	17:49	18:28	19:18
54°	10:02	11:07	12:03	17:37	18:15	19:05
56°	10:15	11:21	12:19	17:24	18:00	18:50
58°	10:30	11:39	12:37	17:09	17:43	18:32
S 60°	10:47	12:00	12:59	16:50	17:22	18:10

April 19, 20, 21 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	207°23.1	173°31.0	N12°46.9	29°24.0	N03°31.0	6°46.2	S07°00.0	166°40.0	N13°34.4					Alpheratz	357°31.5	29°15.1	
1	222°25.5	188°30.5	48.0	44°27.0	31.1	21°48.9	06°59.8	181°42.2	34.5					Ankaa	353°04.2	-42°08.7	
2	237°28.0	203°30.0	49.1	59°30.0	31.2	36°51.7	59.7	196°44.4	34.6					Schedar	349°27.5	56°41.8	
3	252°30.5	218°29.5	· · 50.2	74°33.0	· · 31.3	51°54.4	· · 59.6	211°46.5	· · 34.7					Diphda	348°44.2	-17°49.5	
4	267°32.9	233°29.0	51.3	89°36.0	31.5	66°57.2	59.5	226°48.7	34.8					Achernar	335°18.4	-57°05.2	
5	282°35.4	248°28.5	52.4	104°39.0	31.6	82°00.0	59.4	241°50.8	34.9					Hamal	327°47.6	23°36.0	
6	297°37.8	263°28.0	N12°53.5	119°42.0	N03°31.7	97°02.7	S06°59.3	256°53.0	N13°35.0					Polaris	312°36.3	89°23.2	
7	312°40.3	278°27.5	54.6	134°45.0	31.8	112°05.5	59.2	271°55.1	35.1					Acamar	315°09.6	-40°11.4	
8	327°42.8	293°27.0	55.7	149°48.0	31.9	127°08.2	59.0	286°57.3	35.2					Menkar	314°02.8	4°12.2	
9	342°45.2	308°26.5	· · 56.8	164°51.0	· · 32.0	142°11.0	· · 58.9	301°59.4	· · 35.3					Mirfak	308°23.7	49°57.9	
10	357°47.7	323°26.0	57.9	179°54.0	32.1	157°13.8	58.8	317°01.6	35.4					Aldebaran	290°35.9	16°33.9	
11	12°50.2	338°25.5	12°59.0	194°57.0	32.2	172°16.5	58.7	332°03.8	35.5					Rigel	281°00.8	-8°10.3	
12	27°52.6	353°25.0	N13°00.1	210°00.0	N03°32.3	187°19.3	S06°58.6	347°05.9	N13°35.6					Capella	280°17.1	46°01.6	
13	42°55.1	8°24.5	01.1	225°03.0	32.5	202°22.0	58.5	2°08.1	35.7					Bellatrix	278°19.4	6°22.4	
14	57°57.6	23°24.0	02.2	240°06.0	32.6	217°24.8	58.3	17°10.2	35.8					Elnath	277°57.7	28°37.8	
15	73°00.0	38°23.5	· · 03.3	255°09.0	· · 32.7	232°27.6	· · 58.2	32°12.4	· · 35.9					Alnilam	275°34.4	-1°11.2	
16	88°02.5	53°23.0	04.4	270°12.0	32.8	247°30.3	58.1	47°14.5	36.0					Betelgeuse	270°48.5	7°24.5	
17	103°05.0	68°22.5	05.5	285°15.0	32.9	262°33.1	58.0	62°16.7	36.1					Canopus	263°51.1	-52°43.0	
18	118°07.4	83°21.9	N13°06.6	300°18.0	N03°33.0	277°35.8	S06°57.9	77°18.8	N13°36.2					Sirius	258°23.3	-16°45.7	
19	133°09.9	98°21.4	07.7	315°21.0	33.1	292°38.6	57.8	92°21.0	36.3					Adhara	255°03.2	-29°01.1	
20	148°12.3	113°20.9	08.8	330°23.9	33.2	307°41.4	57.7	107°23.1	36.4					Procyon	244°47.3	5°08.7	
21	163°14.8	128°20.4	· · 09.9	345°26.9	· · 33.3	322°44.1	· · 57.5	122°25.3	· · 36.5					Pollux	243°13.2	27°57.2	
22	178°17.3	143°19.9	11.0	0°29.9	33.4	337°46.9	57.4	137°27.5	36.6					Avior	234°13.2	-59°36.6	
23	193°19.7	158°19.4	12.1	15°32.9	33.5	352°49.6	57.3	152°29.6	36.7					Suhail	222°43.6	-43°33.5	
Mer.pass. 10:09			ν -0.5' d 1.1' m -3.90	ν 3.0' d 0.1' m -0.96	ν 2.8' d -0.1' m -2.47	ν 2.2' d 0.1' m 0.34											

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	
0	208°22.2	173°18.9	N13°13.1	30°35.9	N03°33.6	7°52.4	S06°57.2	167°31.8	N13°36.8	
1	223°24.7	188°18.4	14.2	45°38.8	33.7	22°55.2	57.1	182°33.9	36.9	
2	238°27.1	203°17.9	15.3	60°41.8	33.8	37°57.9	57.0	197°36.1	37.0	
3	253°29.6	218°17.4	· · 16.4	75°44.8	· · 33.9	53°00.7	· · 56.9	212°38.2	· · 37.1	
4	268°32.1	233°16.9	17.5	90°47.8	34.0	68°03.4	56.7	227°40.4	37.2	
5	283°34.5	248°16.4	18.6	105°50.7	34.1	83°06.2	56.6	242°42.5	37.3	
6	298°37.0	263°15.9	N13°19.7	120°53.7	N03°34.2	98°08.9	S06°56.5	257°44.7	N13°37.4	
7	313°39.4	278°15.3	20.7	135°56.7	34.3	113°11.7	56.4	272°46.8	37.5	
8	328°41.9	293°14.8	21.8	150°59.7	34.4	128°14.5	56.3	287°49.0	37.6	
9	343°44.4	308°14.3	· · 22.9	166°02.6	· · 34.5	143°17.2	· · 56.2	302°51.2	· · 37.7	
10	358°46.8	323°13.8	24.0	181°05.6	34.6	158°20.0	56.1	317°53.3	37.8	
11	13°49.3	338°13.3	25.1	196°08.6	34.7	173°22.7	55.9	332°55.5	37.9	
12	28°51.8	353°12.8	N13°26.1	211°11.5	N03°34.8	188°25.5	S06°55.8	347°57.6	N13°38.0	
13	43°54.2	8°12.3	27.2	226°14.5	34.9	203°28.3	55.7	2°59.8	38.1	
14	58°56.7	23°11.8	28.3	241°17.5	35.0	218°31.0	55.6	18°01.9	38.2	
15	73°59.2	38°11.3	· · 29.4	256°20.4	· · 35.1	233°33.8	· · 55.5	33°04.1	· · 38.3	
16	89°01.6	53°10.7	30.5	271°23.4	35.2	248°36.5	55.4	48°06.2	38.4	
17	104°04.1	68°10.2	31.5	286°26.3	35.3	263°39.3	55.3	63°08.4	38.5	
18	119°06.6	83°09.7	N13°32.6	301°29.3	N03°35.4	278°42.0	S06°55.2	78°10.5	N13°38.6	
19	134°09.0	98°09.2	33.7	316°32.3	35.5	293°44.8	55.0	93°12.7	38.7	
20	149°11.5	113°08.7	34.8	331°35.2	35.6	308°47.6	54.9	108°14.9	38.8	
21	164°13.9	128°08.2	· · 35.8	346°38.2	· · 35.6	323°50.3	· · 54.8	123°17.0	· · 38.9	
22	179°16.4	143°07.6	36.9	1°41.1	35.7	338°53.1	54.7	138°19.2	39.0	
23	194°18.9	158°07.1	38.0	16°44.1	35.8	353°55.8	54.6	153°21.3	39.1	
Mer.pass. 10:05			ν -0.5' d 1.1' m -3.90	ν 3.0' d 0.1' m -0.94	ν 2.8' d -0.1' m -2.47	ν 2.2' d 0.1' m 0.34				

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	
0	209°21.3	173°06.6	N13°39.1	31°47.0	N03°35.9	8°58.6	S06°54.5	168°23.5	N13°39.2	
1	224°23.8	188°06.1	40.1	46°50.0	36.0	24°01.3	54.4	183°25.6	39.3	
2	239°26.3	203°05.6	41.2	61°52.9	36.1	39°04.1	54.2	198°27.8	39.4	
3	254°28.7	218°05.1	· · 42.3	76°55.9	· · 36.2	54°06.9	· · 54.1	213°29.9	· · 39.5	
4	269°31.2	233°04.5	43.4	91°58.8	36.3	69°09.6	54.0	228°32.1	39.6	
5	284°33.7	248°04.0	44.4	107°01.8	36.4	84°12.4	53.9	243°34.2	39.7	
6	299°36.1	263°03.5	N13°45.5	122°04.7	N03°36.4	99°15.1	S06°53.8	258°36.4	N13°39.7	
7	314°38.6	278°03.0	46.6	137°07.6	36.5	114°17.9	53.7	273°38.5	39.8	
8	329°41.1	293°02.5	47.6	152°10.6	36.6	129°20.6	53.6	288°40.7	39.9	
9	344°43.5	308°01.9	· · 48.7	167°13.5	· · 36.7	144°23.4	· · 53.4	303°42.9	· · 40.0	
10	359°46.0	323°01.4	49.8	182°16.5	36.8	159°26.2	53.3	318°45.0	40.1	
11	14°48.4	338°00.9	50.8	197°19.4	36.9	174°28.9	53.2	333°47.2	40.2	
12	29°50.9	353°00.4	N13°51.9	212°22.3	N03°37.0	189°31.7	S06°53.1	348°49.3	N13°40.3	
13	44°53.4	7°59.9	53.0	227°25.3	37.0	204°34.4	53.0	3°51.5	40.4	
14	59°55.8	22°59.3	54.0	242°28.2	37.1	219°37.2	52.9	18°53.6	40.5	
15	74°58.3	37°58.8	· · 55.1	257°31.1	· · 37.2	234°39.9	· · 52.8	33°55.8	· · 40.6	
16	90°00.8	52°58.3	56.2	272°34.1	37.3	249°42.7	52.7	48°57.9	40.7	
17	105°03.2	67°57.8	57.2	287°37.0	37.4	264°45.5	52.5	64°00.1	40.8	
18	120°05.7	82°57.2	N13°58.3	302°39.9	N03°37.4	279°48.2	S06°52.4	79°02.2	N13°40.9	
19	135°08.2	97°56.7	13°59.4	317°42.8	37.5	294°51.0	52.3	94°04.4	41.0	
20	150°10.6	112°56.2	14°00.4	332°45.8	37.6	309°53.7	52.2	109°06.5	41.1	
21	165°13.1	127°55.7	· · 01.5	347°48.7	· · 37.7	324°56.5	· · 52.1	124°08.7	· · 41.2	
22	180°15.6	142°55.1	02.5	2°51.6	37.8	339°59.2	52.0	139°10.8	41.3	
23	195°18.0	157°54.6	03.6	17°54.5	37.8	355°02.0	51.9	154°13.0	41.4	
Mer.pass. 10:01			ν -0.5' d 1.1' m -3.90	ν 2.9' d 0.1' m -0.92	ν 2.8' d -0.1' m -2.47	ν 2.2' d 0.1' m 0.33				

Apr 19 Thu	SHA	Mer.pass
Venus	326°07.9	12:26
Mars	182°01.0	21:58
Jupiter	159°23.1	23:29
Saturn	319°17.0	12:51

Apr 20 Fri	SHA	Mer.pass
Venus	324°56.7	12:27
Mars	182°13.7	21:53
Jupiter	159°30.2	23:24
Saturn	319°09.6	12:48

Apr 21 Sat	SHA	Mer.pass
Venus	323°45.3	12:28
Mars	182°25.7	21:49
Jupiter	159°37.2	23:20
Saturn	319°02.1	12:45

Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°12.5	N11°11.9	122°03.6	9.3'	N24°21.2	-1.8'	55.5'
1	195°12.7	12.8	136°31.9	9.3'	24°19.5	-1.9'	55.5'
2	210°12.8	13.6	151°00.2	9.3'	24°17.6	-2.0'	55.5'
3	225°12.9	· · 14.5	165°28.4	9.3'	24°15.6	-2.1'	55.6'
4	240°13.1	15.4	179°56.7	9.3'	24°13.4	-2.3'	55.6'
5	255°13.2	16.2	194°25.0	9.3'	24°11.1	-2.4'	55.6'
6	270°13.3	N11°17.1	208°53.2	9.3'	N24°08.7	-2.5'	55.6'
7	285°13.5	18.0	223°21.5	9.2'	24°06.2	-2.7'	55.7'
8	300°13.6	18.8	237°49.7	9.2'	24°03.5	-2.8'	55.7'
9	315°13.7	· · 19.7	252°18.0	9.2'	24°00.8	-2.9'	55.7'
10	330°13.9	20.5	266°46.2	9.2'	23°57.8	-3.0'	55.7'
11	345°14.0	21.4	281°14.4	9.2'	23°54.8	-3.2'	55.8'
12	0°14.1	N11°22.3	295°42.7	9.2'	N23°51.6	-3.3'	55.8'
13	15°14.3	23.1	310°10.9	9.2'	23°48.3	-3.4'	55.8'
14	30°14.4	24.0	324°39.2	9.2'	23°44.9	-3.6'	55.9'
15	45°14.5	· · 24.9	339°07.4	9.2'	23°41.3	-3.7'	55.9'
16	60°14.6	25.7	353°35.6	9.2'	23°37.6	-3.8'	55.9'
17	75°14.8	26.6	8°03.9	9.2'	23°33.8	-4.0'	55.9'
18	90°14.9	N11°27.4	22°32.1	9.2'	N23°29.8	-4.1'	56.0'
19	105°15.0	28.3	37°00.3	9.2'	23°25.8	-4.2'	56.0'
20	120°15.2	29.1	51°28.6	9.3'	23°21.6	-4.3'	56.0'
21	135°15.3	· · 30.0	65°56.8	9.3'	23°17.2	-4.5'	56.0'
22	150°15.4	30.9	80°25.1	9.3'	23°12.8	-4.6'	56.1'
23	165°15.6	31.7	94°53.4	9.3'	23°08.2	-4.7'	56.1'
SD = 15.9'			SD = 15.1'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°15.7	N11°32.6	109°21.6	9.3'	N23°03.5	-4.8'	56.1'
1	195°15.8	33.4	123°49.9	9.3'	22°58.6	-5.0'	56.2'
2	210°16.0	34.3	138°18.2	9.3'	22°53.6	-5.1'	56.2'
3	225°16.1	· · 35.1	152°46.5	9.3'	22°48.5	-5.2'	56.2'
4	240°16.2	36.0	167°14.8	9.3'	22°43.3	-5.4'	56.2'
5	255°16.3	36.9	181°43.1	9.3'	22°38.0	-5.5'	56.3'
6	270°16.5	N11°37.7	196°11.5	9.3'	N22°32.5	-5.6'	56.3'
7	285°16.6	38.6	210°39.8	9.4'	22°26.9	-5.7'	56.3'
8	300°16.7	39.4	225°08.2	9.4'	22°21.1	-5.9'	56.4'
9	315°16.9	· · 40.3	239°36.5	9.4'	22°15.3	-6.0'	56.4'
10	330°17.0	41.1	254°04.9	9.4'	22°09.3	-6.1'	56.4'
11	345°17.1	42.0	268°33.3	9.4'	22°03.2	-6.2'	56.4'
12	0°17.3	N11°42.8	283°01.7	9.4'	N21°57.0	-6.3'	56.5'
13	15°17.4	43.7	297°30.2	9.4'	21°50.6	-6.5'	56.5'
14	30°17.5	44.5	311°58.6	9.5'	21°44.2	-6.6'	56.5'
15	45°17.6	· · 45.4	326°27.1	9.5'	21°37.6	-6.7'	56.6'
16	60°17.8	46.2	340°55.5	9.5'	21°30.9	-6.8'	56.6'
17	75°17.9	47.1	355°24.0	9.5'	21°24.0	-7.0'	56.6'
18	90°18.0	N11°47.9	9°52.5	9.5'	N21°17.1	-7.1'	56.7'
19	105°18.1	48.8	24°21.1	9.5'	21°10.0	-7.2'	56.7'
20	120°18.3	49.6	38°49.6	9.6'	21°02.8	-7.3'	56.7'
21	135°18.4	· · 50.5	53°18.2	9.6'	20°55.5	-7.4'	56.8'
22	150°18.5	51.3	67°46.8	9.6'	20°48.0	-7.6'	56.8'
23	165°18.7	52.2	82°15.4	9.6'	20°40.4	-7.7'	56.8'
SD = 15.9'			SD = 15.3'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°18.8	N11°53.0	96°44.0	9.6'	N20°32.8	-7.8'	56.8'
1	195°18.9	53.9	111°12.6	9.7'	20°25.0	-7.9'	56.9'
2	210°19.0	54.7	125°41.3	9.7'	20°17.1	-8.0'	56.9'
3	225°19.2	· · 55.6	140°10.0	9.7'	20°09.0	-8.1'	56.9'
4	240°19.3	56.4	154°38.7	9.7'	20°00.9	-8.3'	57.0'
5	255°19.4	57.3	169°07.4	9.7'	19°52.6	-8.4'	57.0'
6	270°19.5	N11°58.1	183°36.1	9.8'	N19°44.2	-8.5'	57.0'
7	285°19.7	59.0	198°04.9	9.8'	19°35.7	-8.6'	57.1'
8	300°19.8	11°59.8	212°33.7	9.8'	19°27.1	-8.7'	57.1'
9	315°19.9	12°00.7	227°02.5	9.8'	19°18.4	-8.8'	57.1'
10	330°20.0	01.5	241°31.3	9.8'	19°09.6	-8.9'	57.2'
11	345°20.2	02.3	256°00.2	9.9'	19°00.6	-9.1'	57.2'
12	0°20.3	N12°03.2	270°29.0	9.9'	N18°51.6	-9.2'	57.2'
13	15°20.4	04.0	284°57.9	9.9'	18°42.4	-9.3'	57.3'
14	30°20.5	04.9	299°26.8	9.9'	18°33.1	-9.4'	57.3'
15	45°20.6	· · 05.7	313°55.8	10.0'	18°23.7	-9.5'	57.3'
16	60°20.8	06.6	328°24.7	10.0'	18°14.2	-9.6'	57.4'
17	75°20.9	07.4	342°53.7	10.0'	18°04.6	-9.7'	57.4'
18	90°21.0	N12°08.2	357°22.7	10.0'	N17°54.9	-9.8'	57.5'
19	105°21.1	09.1	11°51.7	10.0'	17°45.1	-9.9'	57.5'
20	120°21.3	09.9	26°20.8	10.1'	17°35.2	-10.0'	57.5'
21	135°21.4	· · 10.8	40°49.8	10.1'	17°25.1	-10.1'	57.6'
22	150°21.5	11.6	55°18.9	10.1'	17°15.0	-10.2'	57.6'
23	165°21.6	12.5	69°48.0	10.1'	17°04.7	-10.3'	57.6'
SD = 15.9'			SD = 15.5'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	00:51	03:08	20:54	23:31	—:—
N 70°	—:—	01:55	03:29	20:32	22:10	—:—
68°	—:—	02:30	03:46	20:14	21:33	—:—
66°	00:44	02:54	04:00	20:01	21:07	23:35
64°	01:40	03:13	04:11	19:49	20:48	22:24
62°	02:12	03:28	04:20	19:39	20:32	21:50
60°	02:34	03:40	04:28	19:31	20:20	21:27
N 58°	02:52	03:51	04:35	19:24	20:09	21:08
56°	03:06	04:00	04:42	19:17	19:59	20:53
54°	03:19	04:08	04:47	19:12	19:51	20:41
52°	03:29	04:15	04:52	19:07	19:44	20:30
50°	03:39	04:22	04:57	19:02	19:37	20:21
45°	03:57	04:35	05:07	18:52	19:23	20:01
N 40°	04:12	04:46	05:15	18:44	19:12	19:47
35°	04:24	04:55	05:22	18:37	19:03	19:35
30°	04:34	05:03	05:28	18:30	18:55	19:25
20°	04:49	05:15	05:38	18:20	18:43	19:09
N 10°	05:01	05:26	05:47	18:11	18:32	18:57
0°	05:10	05:34	05:56	18:02	18:23	18:48
S 10°	05:18	05:42	06:04	17:54	18:15	18:40
20°	05:24	05:50	06:12	17:45	18:07	18:33
30°	05:30	05:58	06:22	17:35	17:59	18:27
35°	05:32	06:02	06:28	17:29	17:55	18:25
40°	05:35	06:07	06:34	17:23	17:51	18:22
45°	05:37	06:12	06:42	17:16	17:46	18:20
S 50°	05:39	06:17	06:50	17:07	17:40	18:17
52°	05:40	06:20	06:54	17:03	17:37	18:17
54°	05:41	06:22	06:59	16:58	17:35	18:16
56°	05:42	06:25	07:04	16:53	17:32	18:15
58°	05:43	06:28	07:09	16:48	17:28	18:14
S 60°	05:43	06:32	07:15	16:41	17:25	18:13

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	□	□	□	□	□	□
N 70°	□	□	06:39	□	□	05:05
68°	□	□	07:48	□	□	03:56
66°	□	06:25	08:24	□	03:28	03:19
64°	05:39	07:08	08:49	02:25	02:45	02:53
62°	06:17	07:37	09:09	01:46	02:16	02:32
60°	06:44	07:59	09:25	01:19	01:53	02:15
N 58°	07:05	08:17	09:39	00:58	01:35	02:01
56°	07:22	08:32	09:50	00:40	01:20	01:49
54°	07:37	08:45	10:01	00:26	01:07	01:38
52°	07:50	08:56	10:10	00:13	00:55	01:29
50°	08:01	09:06	10:18	00:02	00:45	01:20
45°	08:24	09:27	10:35	· · ·	00:23	01:02
N 40°	08:43	09:44	10:49	· · ·	00:06	00:47
35°	08:59	09:58	11:01	23:51	· · ·	00:34
30°	09:12	10:10	11:11	23:38	· · ·	00:23
20°	09:35	10:31	11:29	23:16	· · ·	00:04
N 10°	09:55	10:50	11:44	22:57	23:47	· · ·
0°	10:14	11:07	11:59	22:39	23:32	· · ·
S 10°	10:33	11:23	12:13	22:21	23:16	· · ·
20°	10:52	11:42	12:28	22:02	22:59	23:57
30°	11:15	12:02	12:45	21:40	22:39	23:41
35°	11:29	12:15	12:56	21:26	22:28	23:32
40°	11:44	12:28	13:07	21:11	22:14	23:21
45°	12:03	12:45	13:21	20:53	21:59	23:09
S 50°	12:26	13:05	13:37	20:31	21:39	22:54
52°	12:37	13:15	13:45	20:20	21:30	22:46
54°	12:50	13:26	13:53	20:07	21:20	22:39
56°	13:04	13:38	14:03	19:53	21:08	22:30
58°	13:21	13:52	14:14	19:37	20:54	22:19
S 60°	13:41	14:08	14:26	19:17	20:38	22:08

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12 ^h	Pass	Mer.Pass.	Lower	6-8
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	23-41%
19	00:50	00:56	11:59	16:27	04:00	
20	01:03	01:09	11:59	17:19	04:53	
21	01:15	01:21	11:59	18:11	05:45	

April 22, 23, 24 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	210°20.5	172°54.1	N14°04.7	32°57.5	N03°37.9	10°04.7	S06°51.8	169°15.2	N13°41.5	Alpheratz	357°31.4	29°15.1
1	225°22.9	187°53.6	05.7	48°00.4	38.0	25°07.5	51.6	184°17.3	41.6	Ankaa	353°04.2	-42°08.7
2	240°25.4	202°53.0	06.8	63°03.3	38.1	40°10.3	51.5	199°19.5	41.7	Schedar	349°27.4	56°41.8
3	255°27.9	217°52.5	•• 07.8	78°06.2	•• 38.1	55°13.0	•• 51.4	214°21.6	•• 41.8	Diphda	348°44.2	-17°49.5
4	270°30.3	232°52.0	08.9	93°09.1	38.2	70°15.8	51.3	229°23.8	41.9	Achernar	335°18.3	-57°05.2
5	285°32.8	247°51.4	09.9	108°12.0	38.3	85°18.5	51.2	244°25.9	42.0	Hamal	327°47.6	23°36.0
6	300°35.3	262°50.9	N14°11.0	123°15.0	N03°38.4	100°21.3	S06°51.1	259°28.1	N13°42.1	Polaris	312°36.6	89°23.2
7	315°37.7	277°50.4	12.1	138°17.9	38.4	115°24.0	51.0	274°30.2	42.2	Acamar	315°09.6	-40°11.3
8	330°40.2	292°49.9	13.1	153°20.8	38.5	130°26.8	50.9	289°32.4	42.3	Menkar	314°02.8	4°12.2
9	345°42.7	307°49.3	•• 14.2	168°23.7	•• 38.6	145°29.5	•• 50.7	304°34.5	•• 42.4	Mirfak	308°23.7	49°57.9
10	0°45.1	322°48.8	15.2	183°26.6	38.6	160°32.3	50.6	319°36.7	42.5	Aldebaran	290°35.9	16°33.9
11	15°47.6	337°48.3	16.3	198°29.5	38.7	175°35.0	50.5	334°38.8	42.6	Rigel	281°00.8	-8°10.3
12	30°50.0	352°47.7	N14°17.3	213°32.4	N03°38.8	190°37.8	S06°50.4	349°41.0	N13°42.7	Capella	280°17.1	46°01.6
13	45°52.5	7°47.2	18.4	228°35.3	38.9	205°40.6	50.3	4°43.1	42.8	Bellatrix	278°19.4	6°22.4
14	60°55.0	22°46.7	19.4	243°38.2	38.9	220°43.3	50.2	19°45.3	42.9	Elnath	277°57.7	28°37.8
15	75°57.4	37°46.1	•• 20.5	258°41.1	•• 39.0	235°46.1	•• 50.1	34°47.4	•• 43.0	Alnilam	275°34.4	-1°11.2
16	90°59.9	52°45.6	21.5	273°44.0	39.1	250°48.8	50.0	49°49.6	43.1	Betelgeuse	270°48.5	7°24.6
17	106°02.4	67°45.1	22.6	288°46.9	39.1	265°51.6	49.8	64°51.8	43.2	Canopus	263°51.1	-52°43.0
18	121°04.8	82°44.5	N14°23.6	303°49.8	N03°39.2	280°54.3	S06°49.7	79°53.9	N13°43.3	Sirius	258°23.3	-16°45.7
19	136°07.3	97°44.0	24.7	318°52.7	39.3	295°57.1	49.6	94°56.1	43.4	Adhara	255°03.3	-29°01.1
20	151°09.8	112°43.5	25.7	333°55.6	39.3	310°59.8	49.5	109°58.2	43.5	Procyon	244°47.3	5°08.8
21	166°12.2	127°42.9	•• 26.8	348°58.5	•• 39.4	326°02.6	•• 49.4	125°00.4	•• 43.6	Pollux	243°13.2	27°57.2
22	181°14.7	142°42.4	27.8	4°01.4	39.5	341°05.3	49.3	140°02.5	43.7	Avior	234°13.2	-59°36.6
23	196°17.2	157°41.8	28.9	19°04.3	39.5	356°08.1	49.2	155°04.7	43.8	Suhail	222°43.6	-43°33.5
Mer.pass. 09:57		ν -0.5' d 1.1' m -3.90		ν 2.9' d 0.1' m -0.89		ν 2.8' d -0.1' m -2.46		ν 2.2' d 0.1' m 0.33				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	211°19.6	172°41.3	N14°29.9	34°07.2	N03°39.6	11°10.8	S06°49.1	170°06.8	N13°43.9
1	226°22.1	187°40.8	31.0	49°10.1	39.7	26°13.6	49.0	185°09.0	44.0
2	241°24.5	202°40.2	32.0	64°13.0	39.7	41°16.4	48.8	200°11.1	44.1
3	256°27.0	217°39.7	•• 33.0	79°15.8	•• 39.8	56°19.1	•• 48.7	215°13.3	•• 44.2
4	271°29.5	232°39.2	34.1	94°18.7	39.8	71°21.9	48.6	230°15.4	44.3
5	286°31.9	247°38.6	35.1	109°21.6	39.9	86°24.6	48.5	245°17.6	44.4
6	301°34.4	262°38.1	N14°36.2	124°24.5	N03°40.0	101°27.4	S06°48.4	260°19.7	N13°44.5
7	316°36.9	277°37.5	37.2	139°27.4	40.0	116°30.1	48.3	275°21.9	44.6
8	331°39.3	292°37.0	38.2	154°30.3	40.1	131°32.9	48.2	290°24.0	44.7
9	346°41.8	307°36.5	•• 39.3	169°33.1	•• 40.1	146°35.6	•• 48.1	305°26.2	•• 44.8
10	1°44.3	322°35.9	40.3	184°36.0	40.2	161°38.4	48.0	320°28.3	44.8
11	16°46.7	337°35.4	41.4	199°38.9	40.3	176°41.1	47.8	335°30.5	44.9
12	31°49.2	352°34.8	N14°42.4	214°41.8	N03°40.3	191°43.9	S06°47.7	350°32.6	N13°45.0
13	46°51.7	7°34.3	43.4	229°44.7	40.4	206°46.6	47.6	5°34.8	45.1
14	61°54.1	22°33.7	44.5	244°47.5	40.4	221°49.4	47.5	20°37.0	45.2
15	76°56.6	37°33.2	•• 45.5	259°50.4	•• 40.5	236°52.1	•• 47.4	35°39.1	•• 45.3
16	91°59.0	52°32.7	46.5	274°53.3	40.5	251°54.9	47.3	50°41.3	45.4
17	107°01.5	67°32.1	47.6	289°56.1	40.6	266°57.6	47.2	65°43.4	45.5
18	122°04.0	82°31.6	N14°48.6	304°59.0	N03°40.6	282°00.4	S06°47.1	80°45.6	N13°45.6
19	137°06.4	97°31.0	49.6	320°01.9	40.7	297°03.1	47.0	95°47.7	45.7
20	152°08.9	112°30.5	50.7	335°04.7	40.7	312°05.9	46.8	110°49.9	45.8
21	167°11.4	127°29.9	•• 51.7	350°07.6	•• 40.8	327°08.6	•• 46.7	125°52.0	•• 45.9
22	182°13.8	142°29.4	52.7	5°10.5	40.8	342°11.4	46.6	140°54.2	46.0
23	197°16.3	157°28.8	53.8	20°13.3	40.9	357°14.2	46.5	155°56.3	46.1
Mer.pass. 09:53		ν -0.5' d 1.0' m -3.90		ν 2.9' d 0.1' m -0.87		ν 2.8' d -0.1' m -2.46		ν 2.2' d 0.1' m 0.32	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	212°18.8	172°28.3	N14°54.8	35°16.2	N03°40.9	12°16.9	S06°46.4	170°58.5	N13°46.2
1	227°21.2	187°27.7	55.8	50°19.0	41.0	27°19.7	46.3	186°00.6	46.3
2	242°23.7	202°27.2	56.9	65°21.9	41.0	42°22.4	46.2	201°02.8	46.4
3	257°26.1	217°26.6	•• 57.9	80°24.8	•• 41.1	57°25.2	•• 46.1	216°04.9	•• 46.5
4	272°28.6	232°26.1	58.9	95°27.6	41.1	72°27.9	46.0	231°07.1	46.6
5	287°31.1	247°25.5	14°59.9	110°30.5	41.2	87°30.7	45.8	246°09.2	46.7
6	302°33.5	262°25.0	N15°01.0	125°33.3	N03°41.2	102°33.4	S06°45.7	261°11.4	N13°46.8
7	317°36.0	277°24.4	02.0	140°36.2	41.3	117°36.2	45.6	276°13.5	46.9
8	332°38.5	292°23.9	03.0	155°39.0	41.3	132°38.9	45.5	291°15.7	47.0
9	347°40.9	307°23.3	•• 04.0	170°41.9	•• 41.4	147°41.7	•• 45.4	306°17.8	•• 47.1
10	2°43.4	322°22.8	05.1	185°44.7	41.4	162°44.4	45.3	321°20.0	47.2
11	17°45.9	337°22.2	06.1	200°47.6	41.5	177°47.2	45.2	336°22.1	47.3
12	32°48.3	352°21.7	N15°07.1	215°50.4	N03°41.5	192°49.9	S06°45.1	351°24.3	N13°47.4
13	47°50.8	7°21.1	08.1	230°53.3	41.6	207°52.7	45.0	6°26.4	47.5
14	62°53.3	22°20.6	09.1	245°56.1	41.6	222°55.4	44.9	21°28.6	47.6
15	77°55.7	37°20.0	•• 10.2	260°58.9	•• 41.6	237°58.2	•• 44.7	36°30.7	•• 47.7
16	92°58.2	52°19.5	11.2	276°01.8	41.7	253°00.9	44.6	51°32.9	47.8
17	108°00.6	67°18.9	12.2	291°04.6	41.7	268°03.7	44.5	66°35.1	47.9
18	123°03.1	82°18.4	N15°13.2	306°07.5	N03°41.8	283°06.4	S06°44.4	81°37.2	N13°48.0
19	138°05.6	97°17.8	14.2	321°10.3	41.8	298°09.2	44.3	96°39.4	48.1
20	153°08.0	112°17.2	15.2	336°13.1	41.8	313°11.9	44.2	111°41.5	48.2
21	168°10.5	127°16.7	•• 16.3	351°16.0	•• 41.9	328°14.7	•• 44.1	126°43.7	•• 48.3
22	183°13.0	142°16.1	17.3	6°18.8	41.9	343°17.4	44.0	141°45.8	48.4
23	198°15.4	157°15.6	18.3	21°21.6	42.0	358°20.2	43.9	156°48.0	48.5
Mer.pass. 09:49		ν -0.5' d 1.0' m -3.90		ν 2.9' d 0.0' m -0.85		ν 2.8' d -0.1' m -2.46		ν 2.2' d 0.1' m 0.32	

Apr 22 Sun	SHA	Mer.pass
Venus	322°33.6	12:29
Mars	182°37.0	21:44
Jupiter	159°44.3	23:15
Saturn	318°54.7	12:41

Apr 23 Mon	SHA	Mer.pass
Venus	321°21.7	12:30
Mars	182°47.6	21:39
Jupiter	159°51.2	23:11
Saturn	318°47.2	12:38

Apr 24 Tue	SHA	Mer.pass
Venus	320°09.5	12:31
Mars	182°57.4	21:35
Jupiter	159°58.1	23:07
Saturn	318°39.7	12:34

Horizontal parallax	
Venus:	0.1
Mars:	0.2

h Sun Moon						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	180°21.8	N12°13.3	84°17.1	10.1'	N16°54.4	-10.4' 57.7'
1	195°21.9	14.1	98°46.3	10.2'	16°43.9	-10.6' 57.7'
2	210°22.0	15.0	113°15.4	10.2'	16°33.4	-10.7' 57.7'
3	225°22.1	· · 15.8	127°44.6	10.2'	16°22.7	-10.8' 57.8'
4	240°22.2	16.7	142°13.8	10.2'	16°12.0	-10.9' 57.8'
5	255°22.4	17.5	156°43.1	10.2'	16°01.1	-11.0' 57.8'
6	270°22.5	N12°18.3	171°12.3	10.3'	N15°50.2	-11.0' 57.9'
7	285°22.6	19.2	185°41.6	10.3'	15°39.1	-11.1' 57.9'
8	300°22.7	20.0	200°10.8	10.3'	15°28.0	-11.2' 57.9'
9	315°22.8	· · 20.8	214°40.1	10.3'	15°16.7	-11.3' 58.0'
10	330°23.0	21.7	229°09.5	10.3'	15°05.4	-11.4' 58.0'
11	345°23.1	22.5	243°38.8	10.4'	14°54.0	-11.5' 58.1'
12	0°23.2	N12°23.3	258°08.2	10.4'	N14°42.5	-11.6' 58.1'
13	15°23.3	24.2	272°37.5	10.4'	14°30.8	-11.7' 58.1'
14	30°23.4	25.0	287°06.9	10.4'	14°19.1	-11.8' 58.2'
15	45°23.6	· · 25.9	301°36.3	10.4'	14°07.3	-11.9' 58.2'
16	60°23.7	26.7	316°05.7	10.4'	13°55.4	-12.0' 58.2'
17	75°23.8	27.5	330°35.2	10.4'	13°43.5	-12.1' 58.3'
18	90°23.9	N12°28.4	345°04.6	10.5'	N13°31.4	-12.2' 58.3'
19	105°24.0	29.2	359°34.1	10.5'	13°19.2	-12.2' 58.3'
20	120°24.1	30.0	14°03.5	10.5'	13°07.0	-12.3' 58.4'
21	135°24.3	· · 30.9	28°33.0	10.5'	12°54.7	-12.4' 58.4'
22	150°24.4	31.7	43°02.5	10.5'	12°42.2	-12.5' 58.4'
23	165°24.5	32.5	57°32.0	10.5'	12°29.7	-12.6' 58.5'
SD = 15.9' $d = 0.8'$ SD = 15.7'						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	180°24.6	N12°33.4	72°01.6	10.5'	N12°17.2	-12.7' 58.5'
1	195°24.7	34.2	86°31.1	10.5'	12°04.5	-12.7' 58.6'
2	210°24.9	35.0	101°00.6	10.6'	11°51.8	-12.8' 58.6'
3	225°25.0	· · 35.8	115°30.2	10.6'	11°39.0	-12.9' 58.6'
4	240°25.1	36.7	129°59.8	10.6'	11°26.1	-13.0' 58.7'
5	255°25.2	37.5	144°29.3	10.6'	11°13.1	-13.1' 58.7'
6	270°25.3	N12°38.3	158°58.9	10.6'	N11°00.0	-13.1' 58.7'
7	285°25.4	39.2	173°28.5	10.6'	10°46.9	-13.2' 58.8'
8	300°25.5	40.0	187°58.1	10.6'	10°33.7	-13.3' 58.8'
9	315°25.7	· · 40.8	202°27.7	10.6'	10°20.4	-13.3' 58.8'
10	330°25.8	41.6	216°57.3	10.6'	10°07.1	-13.4' 58.9'
11	345°25.9	42.5	231°26.9	10.6'	09°53.6	-13.5' 58.9'
12	0°26.0	N12°43.3	245°56.5	10.6'	N09°40.2	-13.6' 59.0'
13	15°26.1	44.1	260°26.1	10.6'	09°26.6	-13.6' 59.0'
14	30°26.2	45.0	274°55.7	10.6'	09°13.0	-13.7' 59.0'
15	45°26.4	· · 45.8	289°25.3	10.6'	08°59.3	-13.8' 59.1'
16	60°26.5	46.6	303°55.0	10.6'	08°45.5	-13.8' 59.1'
17	75°26.6	47.4	318°24.6	10.6'	08°31.7	-13.9' 59.1'
18	90°26.7	N12°48.3	332°54.2	10.6'	N08°17.8	-13.9' 59.2'
19	105°26.8	49.1	347°23.8	10.6'	08°03.9	-14.0' 59.2'
20	120°26.9	49.9	1°53.4	10.6'	07°49.9	-14.1' 59.2'
21	135°27.0	· · 50.7	16°23.0	10.6'	07°35.8	-14.1' 59.3'
22	150°27.1	51.6	30°52.6	10.6'	07°21.7	-14.2' 59.3'
23	165°27.3	52.4	45°22.2	10.6'	07°07.5	-14.2' 59.3'
SD = 15.9' $d = 0.8'$ SD = 16.0'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	180°27.4	N12°53.2	59°51.8	10.6'	N06°53.2	-14.3' 59.4'
1	195°27.5	54.0	74°21.4	10.6'	06°39.0	-14.3' 59.4'
2	210°27.6	54.8	88°50.9	10.6'	06°24.6	-14.4' 59.4'
3	225°27.7	· · 55.7	103°20.5	10.6'	06°10.2	-14.5' 59.5'
4	240°27.8	56.5	117°50.1	10.5'	05°55.8	-14.5' 59.5'
5	255°27.9	57.3	132°19.6	10.5'	05°41.3	-14.5' 59.5'
6	270°28.0	N12°58.1	146°49.1	10.5'	N05°26.7	-14.6' 59.6'
7	285°28.2	59.0	161°18.6	10.5'	05°12.1	-14.6' 59.6'
8	300°28.3	12°59.8	175°48.1	10.5'	04°57.5	-14.7' 59.6'
9	315°28.4	13°00.6	190°17.6	10.5'	04°42.8	-14.7' 59.7'
10	330°28.5	01.4	204°47.1	10.5'	04°28.1	-14.8' 59.7'
11	345°28.6	02.2	219°16.6	10.4'	04°13.3	-14.8' 59.7'
12	0°28.7	N13°03.0	233°46.0	10.4'	N03°58.5	-14.8' 59.8'
13	15°28.8	03.9	248°15.4	10.4'	03°43.6	-14.9' 59.8'
14	30°28.9	04.7	262°44.8	10.4'	03°28.7	-14.9' 59.8'
15	45°29.0	· · 05.5	277°14.2	10.4'	03°13.8	-15.0' 59.9'
16	60°29.1	06.3	291°43.6	10.3'	02°58.9	-15.0' 59.9'
17	75°29.3	07.1	306°12.9	10.3'	02°43.9	-15.0' 59.9'
18	90°29.4	N13°08.0	320°42.2	10.3'	N02°28.9	-15.1' 60.0'
19	105°29.5	08.8	335°11.5	10.3'	02°13.8	-15.1' 60.0'
20	120°29.6	09.6	349°40.8	10.2'	01°58.7	-15.1' 60.0'
21	135°29.7	· · 10.4	4°10.0	10.2'	01°43.6	-15.1' 60.1'
22	150°29.8	11.2	18°39.2	10.2'	01°28.5	-15.2' 60.1'
23	165°29.9	12.0	33°08.4	10.2'	01°13.3	-15.2' 60.1'
SD = 15.9' $d = 0.8'$ SD = 16.2'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:49	21:13	—:—	—:—
N 70°	—:—	01:27	03:14	20:46	22:39	—:—
68°	—:—	02:10	03:33	20:27	21:51	—:—
66°	—:—	02:39	03:48	20:11	21:22	—:—
64°	01:16	03:00	04:00	19:58	21:00	22:49
62°	01:55	03:17	04:11	19:48	20:42	22:06
60°	02:21	03:30	04:20	19:39	20:28	21:39
N 58°	02:41	03:42	04:27	19:31	20:16	21:19
56°	02:57	03:52	04:34	19:24	20:06	21:02
54°	03:10	04:01	04:40	19:17	19:57	20:48
52°	03:21	04:09	04:46	19:12	19:49	20:37
50°	03:31	04:16	04:51	19:07	19:42	20:27
45°	03:52	04:30	05:02	18:56	19:27	20:06
N 40°	04:07	04:42	05:10	18:47	19:15	19:50
35°	04:20	04:51	05:18	18:39	19:06	19:37
30°	04:30	05:00	05:25	18:32	18:57	19:27
20°	04:47	05:13	05:36	18:21	18:44	19:10
N 10°	04:59	05:24	05:46	18:11	18:32	18:58
0°	05:09	05:34	05:55	18:02	18:23	18:47
S 10°	05:18	05:42	06:04	17:52	18:14	18:39
20°	05:25	05:51	06:13	17:43	18:05	18:31
30°	05:32	06:00	06:24	17:32	17:57	18:25
35°	05:35	06:04	06:30	17:26	17:52	18:21
40°	05:38	06:09	06:37	17:19	17:47	18:18
45°	05:41	06:15	06:45	17:11	17:41	18:15
S 50°	05:44	06:21	06:55	17:01	17:34	18:12
52°	05:45	06:24	06:59	16:56	17:31	18:11
54°	05:46	06:27	07:04	16:51	17:28	18:10
56°	05:47	06:31	07:10	16:46	17:25	18:08
58°	05:48	06:34	07:16	16:40	17:21	18:07
S 60°	05:50	06:39	07:23	16:33	17:17	18:06

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	08:55	11:29	13:47	04:39	03:52	03:22
N 70°	09:32	11:45	13:52	04:00	03:33	03:14
68°	09:58	11:58	13:56	03:33	03:19	03:07
66°	10:17	12:08	13:59	03:12	03:07	03:01
64°	10:33	12:17	14:02	02:56	02:57	02:56
62°	10:46	12:24	14:04	02:42	02:48	02:52
60°	10:57	12:31	14:07	02:30	02:40	02:48
N 58°	11:06	12:36	14:08	02:20	02:34	02:45
56°	11:14	12:41	14:10	02:11	02:28	02:42
54°	11:22	12:46	14:12	02:03	02:22	02:40
52°	11:28	12:50	14:13	01:55	02:18	02:37
50°	11:34	12:53	14:14	01:49	02:13	02:35
45°	11:47	13:01	14:17	01:35	02:04	02:30
N 40°	11:57	13:07	14:19	01:23	01:56	02:26
35°	12:06	13:13	14:21	01:13	01:49	02:22
30°	12:14	13:18	14:23	01:04	01:43	02:19
20°	12:27	13:26	14:26	00:49	01:32	02:14
N 10°	12:39	13:33	14:28	00:36	01:23	02:09
0°	12:50	13:40	14:31	00:23	01:14	02:04
S 10°	13:01	13:47	14:33	00:10	01:05	01:59
20°	13:12	13:54	14:36	· · ·	00:55	01:54
30°	13:25	14:03	14:39	· · ·	00:44	01:49
35°	13:33	14:07	14:40	· · ·	00:38	01:45
40°	13:41	14:13	14:42	· · ·	00:30	01:41
45°	13:51	14:19	14:45	· · ·	00:22	01:37
S 50°	14:03	14:26	14:47	· · ·	00:11	01:31
52°	14:09	14:30	14:49	· · ·	00:07	01:29
54°	14:15	14:33	14:50	· · ·	00:01	01:26
56°	14:22	14:37	14:51	23:55	· · ·	01:23
58°	14:29	14:42	14:53	23:48	· · ·	01:20
S 60°	14:38	14:47	14:55	23:41	· · ·	01:16

Day	Sun		Mer. Pass	Moon		Age 9-11 52-73%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
22	01:27	01:33	11:58	19:02	06:36	
23	01:38	01:44	11:58	19:52	07:27	
24	01:49	01:55	11:58	20:43	08:17	

April 25, 26, 27 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	213° 17.9	172° 15.0	N15° 19.3	36° 24.5	N03° 42.0	13° 22.9	S06° 43.8	171° 50.1	N13° 48.6	Alpheratz	357° 31.4	29° 15.0
1	228° 20.4	187° 14.5	20.3	51° 27.3	42.0	28° 25.6	43.7	186° 52.3	48.7	Ankaa	353° 04.2	-42° 08.7
2	243° 22.8	202° 13.9	21.3	66° 30.1	42.1	43° 28.4	43.5	201° 54.4	48.8	Schedar	349° 27.4	56° 41.8
3	258° 25.3	217° 13.3	•• 22.3	81° 32.9	•• 42.1	58° 31.1	•• 43.4	216° 56.6	•• 48.9	Diphda	348° 44.2	-17° 49.5
4	273° 27.7	232° 12.8	23.4	96° 35.8	42.1	73° 33.9	43.3	231° 58.7	49.0	Achernar	335° 18.3	-57° 05.2
5	288° 30.2	247° 12.2	24.4	111° 38.6	42.2	88° 36.6	43.2	247° 00.9	49.1	Hamal	327° 47.6	23° 36.0
6	303° 32.7	262° 11.7	N15° 25.4	126° 41.4	N03° 42.2	103° 39.4	S06° 43.1	262° 03.0	N13° 49.1	Polaris	312° 36.9	89° 23.2
7	318° 35.1	277° 11.1	26.4	141° 44.2	42.2	118° 42.1	43.0	277° 05.2	49.2	Acamar	315° 09.6	-40° 11.3
8	333° 37.6	292° 10.5	27.4	156° 47.1	42.3	133° 44.9	42.9	292° 07.3	49.3	Menkar	314° 02.8	4° 12.2
9	348° 40.1	307° 10.0	•• 28.4	171° 49.9	•• 42.3	148° 47.6	•• 42.8	307° 09.5	•• 49.4	Mirfak	308° 23.7	49° 57.8
10	3° 42.5	322° 09.4	29.4	186° 52.7	42.3	163° 50.4	42.7	322° 11.6	49.5	Aldebaran	290° 35.9	16° 33.9
11	18° 45.0	337° 08.9	30.4	201° 55.5	42.4	178° 53.1	42.6	337° 13.8	49.6	Rigel	281° 00.8	-8° 10.3
12	33° 47.5	352° 08.3	N15° 31.4	216° 58.3	N03° 42.4	193° 55.9	S06° 42.5	352° 15.9	N13° 49.7	Capella	280° 17.1	46° 01.6
13	48° 49.9	7° 07.7	32.4	232° 01.1	42.4	208° 58.6	42.3	7° 18.1	49.8	Bellatrix	278° 19.4	6° 22.4
14	63° 52.4	22° 07.2	33.4	247° 03.9	42.5	224° 01.4	42.2	22° 20.2	49.9	Elnath	277° 57.7	28° 37.8
15	78° 54.9	37° 06.6	•• 34.4	262° 06.8	•• 42.5	239° 04.1	•• 42.1	37° 22.4	•• 50.0	Alnilam	275° 34.4	-1° 11.2
16	93° 57.3	52° 06.0	35.4	277° 09.6	42.5	254° 06.9	42.0	52° 24.5	50.1	Betelgeuse	270° 48.5	7° 24.6
17	108° 59.8	67° 05.5	36.4	292° 12.4	42.6	269° 09.6	41.9	67° 26.7	50.2	Canopus	263° 51.1	-52° 43.0
18	124° 02.2	82° 04.9	N15° 37.4	307° 15.2	N03° 42.6	284° 12.4	S06° 41.8	82° 28.8	N13° 50.3	Sirius	258° 23.3	-16° 45.7
19	139° 04.7	97° 04.3	38.4	322° 18.0	42.6	299° 15.1	41.7	97° 31.0	50.4	Adhara	255° 03.3	-29° 01.1
20	154° 07.2	112° 03.8	39.4	337° 20.8	42.6	314° 17.8	41.6	112° 33.1	50.5	Procyon	244° 47.3	5° 08.8
21	169° 09.6	127° 03.2	•• 40.4	352° 23.6	•• 42.7	329° 20.6	•• 41.5	127° 35.3	•• 50.6	Pollux	243° 13.2	27° 57.2
22	184° 12.1	142° 02.6	41.4	7° 26.4	42.7	344° 23.3	41.4	142° 37.4	50.7	Avior	234° 13.3	-59° 36.6
23	199° 14.6	157° 02.1	42.4	22° 29.2	42.7	359° 26.1	41.3	157° 39.6	50.8	Suhail	222° 43.6	-43° 33.5
Mer.pass. 09:45		ν -0.6' d 1.0' m -3.90		ν 2.8' d 0.0' m -0.82		ν 2.7' d -0.1' m -2.46		ν 2.2' d 0.1' m 0.31				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	214° 17.0	172° 01.5	N15° 43.4	37° 32.0	N03° 42.7	14° 28.8	S06° 41.2	172° 41.7	N13° 50.9	Alphard	217° 44.3	-8° 47.4
1	229° 19.5	187° 00.9	44.4	52° 34.8	42.8	29° 31.6	41.0	187° 43.9	51.0	Regulus	207° 30.7	11° 49.2
2	244° 22.0	202° 00.4	45.4	67° 37.6	42.8	44° 34.3	40.9	202° 46.0	51.1	Dubhe	193° 36.6	61° 35.6
3	259° 24.4	216° 59.8	•• 46.4	82° 40.4	•• 42.8	59° 37.1	•• 40.8	217° 48.2	•• 51.2	Denebola	182° 21.2	14° 24.3
4	274° 26.9	231° 59.2	47.4	97° 43.2	42.8	74° 39.8	40.7	232° 50.3	51.3	Gienah	175° 39.7	-17° 42.5
5	289° 29.4	246° 58.6	48.4	112° 46.0	42.8	89° 42.6	40.6	247° 52.5	51.4	Acrux	172° 55.4	-63° 16.0
6	304° 31.8	261° 58.1	N15° 49.4	127° 48.8	N03° 42.9	104° 45.3	S06° 40.5	262° 54.6	N13° 51.5	Gacrux	171° 47.1	-57° 16.9
7	319° 34.3	276° 57.5	50.4	142° 51.6	42.9	119° 48.1	40.4	277° 56.8	51.6	Alioth	166° 09.6	55° 48.0
8	334° 36.7	291° 56.9	51.4	157° 54.4	42.9	134° 50.8	40.3	292° 58.9	51.7	Spica	158° 18.3	-11° 19.0
9	349° 39.2	306° 56.4	•• 52.4	172° 57.1	•• 42.9	149° 53.5	•• 40.2	308° 01.1	•• 51.8	Alkaid	152° 48.9	49° 09.9
10	4° 41.7	321° 55.8	53.4	187° 59.9	42.9	164° 56.3	40.1	323° 03.2	51.9	Hadar	148° 30.3	-60° 31.0
11	19° 44.1	336° 55.2	54.4	203° 02.7	43.0	179° 59.0	40.0	338° 05.4	52.0	Menkent	147° 53.1	-36° 31.0
12	34° 46.6	351° 54.6	N15° 55.4	218° 05.5	N03° 43.0	195° 01.8	S06° 39.9	353° 07.5	N13° 52.1	Arcturus	145° 44.4	19° 01.7
13	49° 49.1	6° 54.1	56.3	233° 08.3	43.0	210° 04.5	39.7	8° 09.7	52.2	Rigel Kent.	139° 34.7	-60° 57.4
14	64° 51.5	21° 53.5	57.3	248° 11.1	43.0	225° 07.3	39.6	23° 11.8	52.3	Kochab	137° 19.0	74° 02.0
15	79° 54.0	36° 52.9	•• 58.3	263° 13.8	•• 43.0	240° 10.0	•• 39.5	38° 14.0	•• 52.4	Zuben'ubi	136° 51.8	-16° 09.9
16	94° 56.5	51° 52.3	15° 59.3	278° 16.6	43.1	255° 12.8	39.4	53° 16.1	52.5	Alphecca	126° 00.5	26° 36.8
17	109° 58.9	66° 51.8	16° 00.3	293° 19.4	43.1	270° 15.5	39.3	68° 18.3	52.6	Antares	112° 11.3	-26° 29.8
18	125° 01.4	81° 51.2	N16° 01.3	308° 22.2	N03° 43.1	285° 18.2	S06° 39.2	83° 20.4	N13° 52.7	Atria	107° 01.8	-69° 04.5
19	140° 03.8	96° 50.6	02.3	323° 25.0	43.1	300° 21.0	39.1	98° 22.6	52.8	Sabik	101° 58.5	-15° 45.6
20	155° 06.3	111° 50.0	03.2	338° 27.7	43.1	315° 23.7	39.0	113° 24.7	52.8	Shaula	96° 05.4	-37° 07.3
21	170° 08.8	126° 49.4	•• 04.2	353° 30.5	•• 43.1	330° 26.5	•• 38.9	128° 26.9	•• 52.9	Rasalhague	95° 55.1	12° 32.3
22	185° 11.2	141° 48.9	05.2	8° 33.3	43.1	345° 29.2	38.8	143° 29.0	53.0	Eltanin	90° 40.3	51° 28.9
23	200° 13.7	156° 48.3	06.2	23° 36.0	43.1	0° 32.0	38.7	158° 31.2	53.1	Kaus Aust.	83° 27.7	-34° 22.0
Mer.pass. 09:41		ν -0.6' d 1.0' m -3.90		ν 2.8' d 0.0' m -0.80		ν 2.7' d -0.1' m -2.46		ν 2.2' d 0.1' m 0.31				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	215° 16.2	171° 47.7	N16° 07.2	38° 38.8	N03° 43.2	15° 34.7	S06° 38.6	173° 33.3	N13° 53.2	Nunki	75° 43.3	-26° 15.4		
1	230° 18.6	186° 47.1	08.2	53° 41.6	43.2	30° 37.4	38.5	188° 35.5	53.3	Altair	61° 56.5	8° 56.7		
2	245° 21.1	201° 46.5	09.1	68° 44.3	43.2	45° 40.2	38.4	203° 37.6	53.4	Peacock	53° 00.4	-56° 38.0		
3	260° 23.6	216° 46.0	•• 10.1	83° 47.1	•• 43.2	60° 42.9	•• 38.3	218° 39.8	•• 53.5	Deneb	49° 23.4	45° 22.9		
4	275° 26.0	231° 45.4	11.1	98° 49.9	43.2	75° 45.7	38.1	233° 41.9	53.6	Enif	33° 35.5	10° 00.6		
5	290° 28.5	246° 44.8	12.1	113° 52.6	43.2	90° 48.4	38.0	248° 44.1	53.7	Al Na'ir	27° 28.9	-46° 48.8		
6	305° 31.0	261° 44.2	N16° 13.0	128° 55.4	N03° 43.2	105° 51.2	S06° 37.9	263° 46.2	N13° 53.8	Fomalhaut	15° 11.0	-29° 27.8		
7	320° 33.4	276° 43.6	14.0	143° 58.2	43.2	120° 53.9	37.8	278° 48.4	53.9	Scheat	13° 42.0	28° 14.4		
8	335° 35.9	291° 43.1	15.0	159° 00.9	43.2	135° 56.6	37.7	293° 50.5	54.0	Markab	13° 26.6	15° 21.7		
9	350° 38.3	306° 42.5	•• 16.0	174° 03.7	•• 43.2	150° 59.4	•• 37.6	308° 52.7	•• 54.1	Apr 25 Wed		SHA	Mer.pass	
10	5° 40.8	321° 41.9	16.9	189° 06.4	43.2	166° 02.1	37.5	323° 54.8	54.2	Venus	318° 57.1	12:31		
11	20° 43.3	336° 41.3	17.9	204° 09.2	43.3	181° 04.9	37.4	338° 57.0	54.3	Mars	183° 06.6	21:30		
12	35° 45.7	351° 40.7	N16° 18.9	219° 11.9	N03° 43.3	196° 07.6	S06° 37.3	353° 59.1	N13° 54.4	Jupiter	160° 05.0	23:02		
13	50° 48.2	6° 40.1	19.9	234° 14.7	43.3	211° 10.4	37.2	9° 01.3	54.5	Saturn				

DUT1 = UT1-UTC = +0.2091 sec $\Delta T = TT-UT1 = +68.9749$ sec

2029 April 25 to Apr. 27 UT

h Sun Moon						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	180°30.0	N13°12.8	47°37.6	10.1'	N00°58.1	-15.2' 60.1'
1	195°30.1	13.7	62°06.7	10.1'	00°42.9	-15.2' 60.2'
2	210°30.2	14.5	76°35.8	10.1'	00°27.7	-15.2' 60.2'
3	225°30.3	· · 15.3	91°04.9	10.0'	N00°12.5	-15.3' 60.2'
4	240°30.4	16.1	105°33.9	10.0'	S00°02.8	15.3' 60.3'
5	255°30.6	16.9	120°02.9	10.0'	00°18.0	15.3' 60.3'
6	270°30.7	N13°17.7	134°31.8	9.9'	S00°33.3	15.3' 60.3'
7	285°30.8	18.5	149°00.8	9.9'	00°48.6	15.3' 60.3'
8	300°30.9	19.3	163°29.7	9.8'	01°03.9	15.3' 60.4'
9	315°31.0	· · 20.1	177°58.5	9.8'	01°19.2	15.3' 60.4'
10	330°31.1	21.0	192°27.3	9.8'	01°34.6	15.3' 60.4'
11	345°31.2	21.8	206°56.1	9.7'	01°49.9	15.3' 60.5'
12	0°31.3	N13°22.6	221°24.8	9.7'	S02°05.2	15.3' 60.5'
13	15°31.4	23.4	235°53.5	9.6'	02°20.6	15.3' 60.5'
14	30°31.5	24.2	250°22.1	9.6'	02°35.9	15.3' 60.5'
15	45°31.6	· · 25.0	264°50.7	9.6'	02°51.2	15.3' 60.6'
16	60°31.7	25.8	279°19.3	9.5'	03°06.5	15.3' 60.6'
17	75°31.8	26.6	293°47.8	9.5'	03°21.9	15.3' 60.6'
18	90°31.9	N13°27.4	308°16.3	9.4'	S03°37.2	15.3' 60.6'
19	105°32.0	28.2	322°44.7	9.4'	03°52.5	15.3' 60.7'
20	120°32.1	29.0	337°13.1	9.3'	04°07.8	15.3' 60.7'
21	135°32.2	· · 29.8	351°41.4	9.3'	04°23.0	15.3' 60.7'
22	150°32.3	30.6	6°09.6	9.2'	04°38.3	15.3' 60.7'
23	165°32.4	31.5	20°37.9	9.2'	04°53.6	15.2' 60.7'
SD = 15.9' $d = 0.8'$			SD = 16.4'			
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	180°32.5	N13°32.3	35°06.0	9.1'	S05°08.8	15.2' 60.8'
1	195°32.6	33.1	49°34.1	9.1'	05°24.0	15.2' 60.8'
2	210°32.7	33.9	64°02.2	9.0'	05°39.2	15.2' 60.8'
3	225°32.9	· · 34.7	78°30.2	8.9'	05°54.4	15.1' 60.8'
4	240°33.0	35.5	92°58.1	8.9'	06°09.5	15.1' 60.8'
5	255°33.1	36.3	107°26.0	8.8'	06°24.6	15.1' 60.9'
6	270°33.2	N13°37.1	121°53.9	8.8'	S06°39.7	15.1' 60.9'
7	285°33.3	37.9	136°21.6	8.7'	06°54.8	15.0' 60.9'
8	300°33.4	38.7	150°49.3	8.7'	07°09.8	15.0' 60.9'
9	315°33.5	· · 39.5	165°17.0	8.6'	07°24.8	15.0' 60.9'
10	330°33.6	40.3	179°44.6	8.5'	07°39.8	14.9' 60.9'
11	345°33.7	41.1	194°12.1	8.5'	07°54.7	14.9' 61.0'
12	0°33.8	N13°41.9	208°39.6	8.4'	S08°09.6	14.8' 61.0'
13	15°33.9	42.7	223°07.0	8.3'	08°24.4	14.8' 61.0'
14	30°34.0	43.5	237°34.3	8.3'	08°39.2	14.7' 61.0'
15	45°34.1	· · 44.3	252°01.6	8.2'	08°53.9	14.7' 61.0'
16	60°34.2	45.1	266°28.8	8.1'	09°08.6	14.6' 61.0'
17	75°34.3	45.9	280°56.0	8.1'	09°23.3	14.6' 61.0'
18	90°34.4	N13°46.7	295°23.1	8.0'	S09°37.9	14.5' 61.1'
19	105°34.5	47.5	309°50.1	7.9'	09°52.4	14.5' 61.1'
20	120°34.6	48.3	324°17.0	7.9'	10°06.9	14.4' 61.1'
21	135°34.7	· · 49.1	338°43.9	7.8'	10°21.3	14.4' 61.1'
22	150°34.8	49.9	353°10.7	7.7'	10°35.7	14.3' 61.1'
23	165°34.9	50.7	7°37.4	7.7'	10°50.0	14.2' 61.1'
SD = 15.9' $d = 0.8'$			SD = 16.6'			
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	180°35.0	N13°51.5	22°04.1	7.6'	S11°04.2	14.2' 61.1'
1	195°35.0	52.2	36°30.7	7.5'	11°18.4	14.1' 61.1'
2	210°35.1	53.0	50°57.2	7.5'	11°32.5	14.0' 61.1'
3	225°35.2	· · 53.8	65°23.7	7.4'	11°46.5	14.0' 61.1'
4	240°35.3	54.6	79°50.1	7.3'	12°00.5	13.9' 61.2'
5	255°35.4	55.4	94°16.4	7.2'	12°14.4	13.8' 61.2'
6	270°35.5	N13°56.2	108°42.6	7.2'	S12°28.2	13.7' 61.2'
7	285°35.6	57.0	123°08.8	7.1'	12°41.9	13.7' 61.2'
8	300°35.7	57.8	137°34.9	7.0'	12°55.6	13.6' 61.2'
9	315°35.8	· · 58.6	152°00.9	6.9'	13°09.2	13.5' 61.2'
10	330°35.9	13°59.4	166°26.9	6.9'	13°22.7	13.4' 61.2'
11	345°36.0	14°00.2	180°52.7	6.8'	13°36.1	13.3' 61.2'
12	0°36.1	N14°01.0	195°18.5	6.7'	S13°49.4	13.2' 61.2'
13	15°36.2	01.8	209°44.3	6.7'	14°02.6	13.1' 61.2'
14	30°36.3	02.5	224°09.9	6.6'	14°15.8	13.0' 61.2'
15	45°36.4	· · 03.3	238°35.5	6.5'	14°28.8	12.9' 61.2'
16	60°36.5	04.1	253°01.0	6.4'	14°41.7	12.8' 61.2'
17	75°36.6	04.9	267°26.4	6.4'	14°54.6	12.8' 61.2'
18	90°36.7	N14°05.7	281°51.8	6.3'	S15°07.3	12.6' 61.2'
19	105°36.8	06.5	296°17.1	6.2'	15°20.0	12.5' 61.2'
20	120°36.9	07.3	310°42.3	6.1'	15°32.5	12.4' 61.2'
21	135°37.0	· · 08.1	325°07.4	6.1'	15°45.0	12.3' 61.2'
22	150°37.0	08.8	339°32.5	6.0'	15°57.3	12.2' 61.2'
23	165°37.1	09.6	353°57.5	5.9'	16°09.5	12.1' 61.2'
SD = 15.9' $d = 0.8'$			SD = 16.7'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:29	21:32	—:—	—:—
N 70°	—:—	00:48	02:58	21:02	23:29	—:—
68°	—:—	01:49	03:19	20:39	22:13	—:—
66°	—:—	02:23	03:36	20:22	21:37	—:—
64°	00:42	02:47	03:50	20:08	21:12	23:32
62°	01:36	03:05	04:01	19:56	20:53	22:25
60°	02:07	03:21	04:11	19:46	20:37	21:53
N 58°	02:29	03:33	04:20	19:37	20:24	21:29
56°	02:47	03:44	04:27	19:30	20:13	21:11
54°	03:01	03:54	04:34	19:23	20:03	20:56
52°	03:14	04:02	04:40	19:17	19:55	20:44
50°	03:24	04:09	04:45	19:11	19:47	20:33
45°	03:46	04:25	04:57	19:00	19:31	20:11
N 40°	04:03	04:37	05:06	18:50	19:19	19:54
35°	04:16	04:48	05:15	18:42	19:08	19:40
30°	04:27	04:57	05:22	18:34	18:59	19:29
20°	04:44	05:11	05:34	18:22	18:45	19:12
N 10°	04:58	05:23	05:45	18:11	18:33	18:58
0°	05:08	05:33	05:54	18:01	18:22	18:47
S 10°	05:18	05:43	06:04	17:51	18:13	18:38
20°	05:26	05:52	06:14	17:41	18:03	18:30
30°	05:33	06:01	06:26	17:29	17:54	18:22
35°	05:37	06:07	06:33	17:22	17:48	18:18
40°	05:40	06:12	06:40	17:15	17:43	18:15
45°	05:44	06:19	06:49	17:06	17:36	18:11
S 50°	05:48	06:26	06:59	16:55	17:29	18:07
52°	05:49	06:29	07:04	16:50	17:26	18:05
54°	05:51	06:32	07:10	16:45	17:22	18:04
56°	05:52	06:36	07:16	16:39	17:18	18:02
58°	05:54	06:40	07:22	16:32	17:14	18:00
S 60°	05:56	06:45	07:30	16:25	17:09	17:58

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	16:04	18:30	21:29	02:56	02:30	01:58
N 70°	15:59	18:13	20:44	02:56	02:38	02:17
68°	15:55	18:00	20:15	02:56	02:45	02:32
66°	15:52	17:49	19:54	02:56	02:51	02:45
64°	15:49	17:40	19:36	02:56	02:55	02:55
62°	15:47	17:33	19:22	02:56	03:00	03:04
60°	15:45	17:26	19:11	02:56	03:03	03:12
N 58°	15:43	17:20	19:01	02:56	03:07	03:19
56°	15:41	17:15	18:52	02:56	03:10	03:25
54°	15:40	17:11	18:44	02:56	03:12	03:31
52°	15:39	17:07	18:37	02:56	03:15	03:36
50°	15:37	17:03	18:31	02:56	03:17	03:40
45°	15:35	16:55	18:17	02:56	03:22	03:50
N 40°	15:33	16:49	18:06	02:55	03:26	03:59
35°	15:31	16:43	17:57	02:55	03:29	04:06
30°	15:29	16:38	17:49	02:55	03:32	04:12
20°	15:27	16:30	17:35	02:55	03:38	04:23
N 10°	15:24	16:22	17:23	02:55	03:42	04:32
0°	15:22	16:15	17:11	02:55	03:47	04:41
S 10°	15:20	16:09	17:00	02:55	03:51	04:51
20°	15:18	16:01	16:48	02:54	03:56	05:00
30°	15:15	15:53	16:35	02:54	04:02	05:11
35°	15:14	15:49	16:27	02:54	04:05	05:18
40°	15:12	15:44	16:19	02:54	04:08	05:25
45°	15:10	15:38	16:08	02:54	04:13	05:34
S 50°	15:08	15:30	15:56	02:53	04:18	05:44
52°	15:07	15:27	15:51	02:53	04:20	05:49
54°	15:06	15:24	15:44	02:53	04:22	05:54
56°	15:05	15:20	15:37	02:53	04:25	06:00
58°	15:04	15:15	15:30	02:53	04:28	06:07
S 60°	15:02	15:10	15:21	02:52	04:32	06:15

Day	Sun		Mer. Pass hh:mm	Moon		Age 12-14 83-97%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
25	02:00	02:05	11:58	21:34	09:08	
26	02:10	02:15	11:58	22:28	10:01	
27	02:20	02:24	11:58	23:25	10:56	

April 28, 29, 30 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	216°15.3	171°33.7	N16°30.5	39°44.9	N03°43.3	16°40.5	S06°36.0	174°24.9	N13°55.6	Alpheratz	357°31.4	29°15.0
1	231°17.8	186°33.1	31.5	54°47.6	43.3	31°43.2	35.9	189°27.1	55.7	Ankaa	353°04.2	-42°08.7
2	246°20.2	201°32.5	32.4	69°50.4	43.3	46°46.0	35.8	204°29.2	55.8	Schedar	349°27.4	56°41.8
3	261°22.7	216°31.9	•• 33.4	84°53.1	•• 43.3	61°48.7	•• 35.7	219°31.4	•• 55.9	Diphda	348°44.2	-17°49.5
4	276°25.2	231°31.3	34.4	99°55.8	43.3	76°51.5	35.6	234°33.5	56.0	Achernar	335°18.3	-57°05.2
5	291°27.6	246°30.7	35.3	114°58.6	43.3	91°54.2	35.5	249°35.7	56.1	Hamal	327°47.6	23°36.0
6	306°30.1	261°30.1	N16°36.3	130°01.3	N03°43.3	106°56.9	S06°35.4	264°37.8	N13°56.2	Polaris	312°36.9	89°23.2
7	321°32.6	276°29.5	37.2	145°04.0	43.3	121°59.7	35.3	279°40.0	56.2	Acamar	315°09.6	-40°11.3
8	336°35.0	291°28.9	38.2	160°06.8	43.3	137°02.4	35.2	294°42.1	56.3	Menkar	314°02.8	4°12.2
9	351°37.5	306°28.3	•• 39.1	175°09.5	•• 43.2	152°05.2	•• 35.1	309°44.3	•• 56.4	Mirfak	308°23.7	49°57.8
10	6°39.9	321°27.7	40.1	190°12.2	43.2	167°07.9	35.0	324°46.4	56.5	Aldebaran	290°35.9	16°33.9
11	21°42.4	336°27.1	41.1	205°14.9	43.2	182°10.6	34.9	339°48.6	56.6	Rigel	281°00.8	-8°10.3
12	36°44.9	351°26.5	N16°42.0	220°17.7	N03°43.2	197°13.4	S06°34.8	354°50.7	N13°56.7	Capella	280°17.1	46°01.6
13	51°47.3	6°25.9	43.0	235°20.4	43.2	212°16.1	34.7	9°52.9	56.8	Bellatrix	278°19.4	6°22.4
14	66°49.8	21°25.3	43.9	250°23.1	43.2	227°18.8	34.6	24°55.0	56.9	Elnath	277°57.8	28°37.8
15	81°52.3	36°24.7	•• 44.9	265°25.8	•• 43.2	242°21.6	•• 34.4	39°57.2	•• 57.0	Alnilam	275°34.5	-1°11.2
16	96°54.7	51°24.1	45.8	280°28.5	43.2	257°24.3	34.3	54°59.3	57.1	Betelgeuse	270°48.6	7°24.6
17	111°57.2	66°23.5	46.8	295°31.3	43.2	272°27.1	34.2	70°01.5	57.2	Canopus	263°51.1	-52°43.0
18	126°59.7	81°22.9	N16°47.7	310°34.0	N03°43.2	287°29.8	S06°34.1	85°03.6	N13°57.3	Sirius	258°23.4	-16°45.7
19	142°02.1	96°22.3	48.7	325°36.7	43.2	302°32.5	34.0	100°05.8	57.4	Adhara	255°03.3	-29°01.1
20	157°04.6	111°21.7	49.6	340°39.4	43.1	317°35.3	33.9	115°07.9	57.5	Procyon	244°47.3	5°08.8
21	172°07.1	126°21.1	•• 50.6	355°42.1	•• 43.1	332°38.0	•• 33.8	130°10.1	•• 57.6	Pollux	243°13.2	27°57.2
22	187°09.5	141°20.5	51.5	10°44.8	43.1	347°40.8	33.7	145°12.2	57.7	Avior	234°13.3	-59°36.6
23	202°12.0	156°19.9	52.5	25°47.5	43.1	2°43.5	33.6	160°14.4	57.8	Suhail	222°43.6	-43°33.5
Mer.pass. 09:33		ν -0.6' d 1.0' m -3.90		ν 2.7' d -0.0' m -0.75		ν 2.7' d -0.1' m -2.45		ν 2.1' d 0.1' m 0.29				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	217°14.4	171°19.3	N16°53.4	40°50.3	N03°43.1	17°46.2	S06°33.5	175°16.5	N13°57.9
1	232°16.9	186°18.7	54.4	55°53.0	43.1	32°49.0	33.4	190°18.7	58.0
2	247°19.4	201°18.1	55.3	70°55.7	43.1	47°51.7	33.3	205°20.8	58.1
3	262°21.8	216°17.5	•• 56.2	85°58.4	•• 43.1	62°54.4	•• 33.2	220°23.0	•• 58.2
4	277°24.3	231°16.9	57.2	101°01.1	43.0	77°57.2	33.1	235°25.1	58.3
5	292°26.8	246°16.3	58.1	116°03.8	43.0	92°59.9	33.0	250°27.3	58.4
6	307°29.2	261°15.7	N16°59.1	131°06.5	N03°43.0	108°02.6	S06°32.9	265°29.4	N13°58.5
7	322°31.7	276°15.1	17°00.0	146°09.2	43.0	123°05.4	32.8	280°31.6	58.6
8	337°34.2	291°14.5	01.0	161°11.9	43.0	138°08.1	32.7	295°33.7	58.7
9	352°36.6	306°13.9	•• 01.9	176°14.6	•• 42.9	153°10.9	•• 32.6	310°35.9	•• 58.8
10	7°39.1	321°13.3	02.8	191°17.3	42.9	168°13.6	32.5	325°38.0	58.9
11	22°41.5	336°12.7	03.8	206°20.0	42.9	183°16.3	32.4	340°40.2	59.0
12	37°44.0	351°12.1	N17°04.7	221°22.7	N03°42.9	198°19.1	S06°32.3	355°42.3	N13°59.1
13	52°46.5	6°11.5	05.6	236°25.4	42.9	213°21.8	32.2	10°44.5	59.2
14	67°48.9	21°10.9	06.6	251°28.0	42.8	228°24.5	32.1	25°46.6	59.2
15	82°51.4	36°10.2	•• 07.5	266°30.7	•• 42.8	243°27.3	•• 32.0	40°48.8	•• 59.3
16	97°53.9	51°09.6	08.4	281°33.4	42.8	258°30.0	31.9	55°50.9	59.4
17	112°56.3	66°09.0	09.4	296°36.1	42.8	273°32.7	31.7	70°53.1	59.5
18	127°58.8	81°08.4	N17°10.3	311°38.8	N03°42.8	288°35.5	S06°31.6	85°55.2	N13°59.6
19	143°01.3	96°07.8	11.2	326°41.5	42.7	303°38.2	31.5	100°57.4	59.7
20	158°03.7	111°07.2	12.2	341°44.2	42.7	318°40.9	31.4	115°59.5	59.8
21	173°06.2	126°06.6	•• 13.1	356°46.8	•• 42.7	333°43.7	•• 31.3	131°01.7	13°59.9
22	188°08.7	141°06.0	14.0	11°49.5	42.7	348°46.4	31.2	146°03.8	14°00.0
23	203°11.1	156°05.4	15.0	26°52.2	42.6	3°49.1	31.1	161°06.0	00.1
Mer.pass. 09:29		ν -0.6' d 0.9' m -3.90		ν 2.7' d -0.0' m -0.73		ν 2.7' d -0.1' m -2.45		ν 2.1' d 0.1' m 0.28	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	218°13.6	171°04.7	N17°15.9	41°54.9	N03°42.6	18°51.9	S06°31.0	176°08.1	N14°00.2
1	233°16.0	186°04.1	16.8	56°57.6	42.6	33°54.6	30.9	191°10.3	00.3
2	248°18.5	201°03.5	17.7	72°00.2	42.6	48°57.3	30.8	206°12.4	00.4
3	263°21.0	216°02.9	•• 18.7	87°02.9	•• 42.5	64°00.1	•• 30.7	221°14.6	•• 00.5
4	278°23.4	231°02.3	19.6	102°05.6	42.5	79°02.8	30.6	236°16.7	00.6
5	293°25.9	246°01.7	20.5	117°08.3	42.5	94°05.5	30.5	251°18.9	00.7
6	308°28.4	261°01.1	N17°21.4	132°10.9	N03°42.4	109°08.3	S06°30.4	266°21.0	N14°00.8
7	323°30.8	276°00.4	22.4	147°13.6	42.4	124°11.0	30.3	281°23.2	00.9
8	338°33.3	290°59.8	23.3	162°16.3	42.4	139°13.7	30.2	296°25.3	01.0
9	353°35.8	305°59.2	•• 24.2	177°18.9	•• 42.4	154°16.5	•• 30.1	311°27.5	•• 01.1
10	8°38.2	320°58.6	25.1	192°21.6	42.3	169°19.2	30.0	326°29.6	01.2
11	23°40.7	335°58.0	26.0	207°24.3	42.3	184°21.9	29.9	341°31.8	01.3
12	38°43.2	350°57.3	N17°27.0	222°26.9	N03°42.3	199°24.7	S06°29.8	356°33.9	N14°01.4
13	53°45.6	5°56.7	27.9	237°29.6	42.2	214°27.4	29.7	11°36.0	01.5
14	68°48.1	20°56.1	28.8	252°32.2	42.2	229°30.1	29.6	26°38.2	01.6
15	83°50.5	35°55.5	•• 29.7	267°34.9	•• 42.2	244°32.9	•• 29.5	41°40.3	•• 01.7
16	98°53.0	50°54.9	30.6	282°37.6	42.1	259°35.6	29.4	56°42.5	01.8
17	113°55.5	65°54.2	31.5	297°40.2	42.1	274°38.3	29.3	71°44.6	01.9
18	128°57.9	80°53.6	N17°32.5	312°42.9	N03°42.0	289°41.1	S06°29.2	86°46.8	N14°02.0
19	144°00.4	95°53.0	33.4	327°45.5	42.0	304°43.8	29.1	101°48.9	02.0
20	159°02.9	110°52.4	34.3	342°48.2	42.0	319°46.5	29.0	116°51.1	02.1
21	174°05.3	125°51.7	•• 35.2	357°50.8	•• 41.9	334°49.2	•• 28.9	131°53.2	•• 02.2
22	189°07.8	140°51.1	36.1	12°53.5	41.9	349°52.0	28.8	146°55.4	02.3
23	204°10.3	155°50.5	37.0	27°56.1	41.9	4°54.7	28.7	161°57.5	02.4
Mer.pass. 09:26		ν -0.6' d 0.9' m -3.90		ν 2.7' d -0.0' m -0.71		ν 2.7' d -0.1' m -2.44		ν 2.1' d 0.1' m 0.27	

Apr 28 Sat	SHA	Mer.pass
Venus	315°18.4	12:34
Mars	183°29.6	21:17
Jupiter	160°25.2	22:49
Saturn	318°09.6	12:21

Apr 29 Sun	SHA	Mer.pass
Venus	314°04.9	12:35
Mars	183°35.8	21:13
Jupiter	160°31.8	22:45
Saturn	318°02.1	12:17

Apr 30 Mon	SHA	Mer.pass
Venus	312°51.2	12:36
Mars	183°41.3	21:09
Jupiter	160°38.3	22:40
Saturn	317°54.5	12:14

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.2092 sec $\Delta T = TT-UT1 = +68.9748$ sec

2029 April 28 to Apr. 30 UT

h	Sun				Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°37.2	N14°10.4	8°22.4	5.8'	S16°21.6	12.0'	61.2'	
1	195°37.3	11.2	22°47.2	5.8'	16°33.6	11.9'	61.2'	
2	210°37.4	12.0	37°12.0	5.7'	16°45.5	11.8'	61.2'	
3	225°37.5	· · 12.8	51°36.7	5.6'	16°57.3	11.6'	61.2'	
4	240°37.6	13.6	66°01.3	5.6'	17°09.0	11.5'	61.2'	
5	255°37.7	14.3	80°25.9	5.5'	17°20.5	11.4'	61.2'	
6	270°37.8	N14°15.1	94°50.3	5.4'	S17°31.9	11.3'	61.1'	
7	285°37.9	15.9	109°14.8	5.3'	17°43.2	11.2'	61.1'	
8	300°38.0	16.7	123°39.1	5.3'	17°54.3	11.0'	61.1'	
9	315°38.1	· · 17.5	138°03.4	5.2'	18°05.4	10.9'	61.1'	
10	330°38.1	18.3	152°27.6	5.1'	18°16.3	10.8'	61.1'	
11	345°38.2	19.0	166°51.7	5.1'	18°27.0	10.6'	61.1'	
12	0°38.3	N14°19.8	181°15.8	5.0'	S18°37.7	10.5'	61.1'	
13	15°38.4	20.6	195°39.8	4.9'	18°48.2	10.4'	61.1'	
14	30°38.5	21.4	210°03.8	4.9'	18°58.5	10.2'	61.1'	
15	45°38.6	· · 22.2	224°27.6	4.8'	19°08.8	10.1'	61.1'	
16	60°38.7	22.9	238°51.5	4.8'	19°18.9	10.0'	61.0'	
17	75°38.8	23.7	253°15.2	4.7'	19°28.8	9.8'	61.0'	
18	90°38.9	N14°24.5	267°38.9	4.6'	S19°38.6	9.7'	61.0'	
19	105°38.9	25.3	282°02.5	4.6'	19°48.3	9.5'	61.0'	
20	120°39.0	26.0	296°26.1	4.5'	19°57.8	9.4'	61.0'	
21	135°39.1	· · 26.8	310°49.6	4.5'	20°07.2	9.2'	61.0'	
22	150°39.2	27.6	325°13.1	4.4'	20°16.4	9.1'	61.0'	
23	165°39.3	28.4	339°36.5	4.4'	20°25.5	8.9'	60.9'	
	SD = 15.9'	d = 0.8'	SD = 16.7'					
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°39.4	N14°29.2	353°59.8	4.3'	S20°34.4	8.8'	60.9'	
1	195°39.5	29.9	8°23.1	4.2'	20°43.2	8.6'	60.9'	
2	210°39.6	30.7	22°46.4	4.2'	20°51.8	8.5'	60.9'	
3	225°39.6	· · 31.5	37°09.6	4.1'	21°00.2	8.3'	60.9'	
4	240°39.7	32.3	51°32.7	4.1'	21°08.6	8.2'	60.8'	
5	255°39.8	33.0	65°55.8	4.1'	21°16.7	8.0'	60.8'	
6	270°39.9	N14°33.8	80°18.9	4.0'	S21°24.7	7.8'	60.8'	
7	285°40.0	34.6	94°41.9	4.0'	21°32.5	7.7'	60.8'	
8	300°40.1	35.3	109°04.9	3.9'	21°40.2	7.5'	60.8'	
9	315°40.2	· · 36.1	123°27.8	3.9'	21°47.7	7.3'	60.7'	
10	330°40.2	36.9	137°50.7	3.9'	21°55.1	7.2'	60.7'	
11	345°40.3	37.7	152°13.5	3.8'	22°02.2	7.0'	60.7'	
12	0°40.4	N14°38.4	166°36.4	3.8'	S22°09.3	6.9'	60.7'	
13	15°40.5	39.2	180°59.2	3.8'	22°16.1	6.7'	60.7'	
14	30°40.6	40.0	195°21.9	3.7'	22°22.8	6.5'	60.6'	
15	45°40.7	· · 40.7	209°44.6	3.7'	22°29.3	6.3'	60.6'	
16	60°40.8	41.5	224°07.3	3.7'	22°35.7	6.2'	60.6'	
17	75°40.8	42.3	238°30.0	3.6'	22°41.8	6.0'	60.6'	
18	90°40.9	N14°43.0	252°52.7	3.6'	S22°47.9	5.8'	60.5'	
19	105°41.0	43.8	267°15.3	3.6'	22°53.7	5.7'	60.5'	
20	120°41.1	44.6	281°37.9	3.6'	22°59.4	5.5'	60.5'	
21	135°41.2	· · 45.4	296°00.5	3.6'	23°04.9	5.3'	60.4'	
22	150°41.2	46.1	310°23.1	3.6'	23°10.2	5.2'	60.4'	
23	165°41.3	46.9	324°45.6	3.5'	23°15.3	5.0'	60.4'	
	SD = 15.9'	d = 0.8'	SD = 16.6'					
Mon	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°41.4	N14°47.6	339°08.2	3.5'	S23°20.3	4.8'	60.4'	
1	195°41.5	48.4	353°30.7	3.5'	23°25.1	4.6'	60.3'	
2	210°41.6	49.2	7°53.2	3.5'	23°29.8	4.5'	60.3'	
3	225°41.7	· · 49.9	22°15.8	3.5'	23°34.2	4.3'	60.3'	
4	240°41.7	50.7	36°38.3	3.5'	23°38.5	4.1'	60.2'	
5	255°41.8	51.5	51°00.8	3.5'	23°42.6	3.9'	60.2'	
6	270°41.9	N14°52.2	65°23.3	3.5'	S23°46.5	3.8'	60.2'	
7	285°42.0	53.0	79°45.9	3.5'	23°50.3	3.6'	60.2'	
8	300°42.1	53.8	94°08.4	3.5'	23°53.9	3.4'	60.1'	
9	315°42.1	· · 54.5	108°30.9	3.6'	23°57.3	3.2'	60.1'	
10	330°42.2	55.3	122°53.5	3.6'	24°00.5	3.1'	60.1'	
11	345°42.3	56.0	137°16.0	3.6'	24°03.5	2.9'	60.0'	
12	0°42.4	N14°56.8	151°38.6	3.6'	S24°06.4	2.7'	60.0'	
13	15°42.5	57.6	166°01.2	3.6'	24°09.1	2.5'	60.0'	
14	30°42.5	58.3	180°23.8	3.6'	24°11.6	2.3'	59.9'	
15	45°42.6	· · 59.1	194°46.5	3.7'	24°14.0	2.2'	59.9'	
16	60°42.7	14°59.8	209°09.1	3.7'	24°16.2	2.0'	59.9'	
17	75°42.8	15°00.6	223°31.8	3.7'	24°18.2	1.8'	59.8'	
18	90°42.8	N15°01.4	237°54.5	3.7'	S24°20.0	1.6'	59.8'	
19	105°42.9	02.1	252°17.3	3.8'	24°21.6	1.5'	59.8'	
20	120°43.0	02.9	266°40.1	3.8'	24°23.1	1.3'	59.7'	
21	135°43.1	· · 03.6	281°02.9	3.8'	24°24.4	1.1'	59.7'	
22	150°43.1	04.4	295°25.7	3.9'	24°25.5	1.0'	59.7'	
23	165°43.2	05.1	309°48.6	3.9'	24°26.5	0.8'	59.6'	
	SD = 15.9'	d = 0.8'	SD = 16.5'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:07	21:54	—:—	—:—
N 70°	—:—	—:—	02:41	21:18	—:—	—:—
68°	—:—	01:25	03:06	20:52	22:38	—:—
66°	—:—	02:06	03:24	20:33	21:53	—:—
64°	—:—	02:33	03:40	20:17	21:25	—:—
62°	01:15	02:54	03:52	20:04	21:03	22:48
60°	01:52	03:11	04:03	19:53	20:46	22:07
N 58°	02:17	03:25	04:12	19:44	20:32	21:41
56°	02:37	03:36	04:20	19:36	20:20	21:20
54°	02:53	03:46	04:27	19:28	20:10	21:04
52°	03:06	03:55	04:34	19:22	20:01	20:51
50°	03:17	04:03	04:40	19:16	19:52	20:39
45°	03:40	04:20	04:52	19:03	19:36	20:15
N 40°	03:58	04:33	05:02	18:53	19:22	19:58
35°	04:12	04:44	05:11	18:44	19:11	19:43
30°	04:24	04:54	05:19	18:36	19:01	19:32
20°	04:42	05:09	05:32	18:23	18:46	19:13
N 10°	04:56	05:22	05:43	18:11	18:33	18:59
0°	05:08	05:33	05:54	18:01	18:22	18:47
S 10°	05:18	05:43	06:04	17:50	18:12	18:37
20°	05:27	05:53	06:15	17:39	18:02	18:28
30°	05:35	06:03	06:28	17:26	17:51	18:19
35°	05:39	06:09	06:35	17:19	17:45	18:15
40°	05:43	06:15	06:43	17:11	17:39	18:11
45°	05:47	06:22	06:53	17:01	17:32	18:07
S 50°	05:52	06:30	07:04	16:50	17:24	18:02
52°	05:54	06:34	07:09	16:45	17:20	18:00
54°	05:56	06:37	07:15	16:39	17:16	17:58
56°	05:58	06:42	07:21	16:32	17:12	17:56
58°	06:00	06:46	07:29	16:25	17:07	17:54
S 60°	06:02	06:52	07:37	16:17	17:02	17:52

May 01, 02, 03 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	219°12.7	170°49.9	N17°37.9	42°58.8	N03°41.8	19°57.4	S06°28.6	176°59.7	N14°02.5					Alpheratz	357°31.4	29°15.0	
1	234°15.2	185°49.2	38.8	58°01.4	41.8	35°00.2	28.5	192°01.8	02.6					Ankaa	353°04.2	-42°08.6	
2	249°17.7	200°48.6	39.7	73°04.1	41.7	50°02.9	28.4	207°04.0	02.7					Schedar	349°27.4	56°41.8	
3	264°20.1	215°48.0	•• 40.6	88°06.7	•• 41.7	65°05.6	•• 28.3	222°06.1	•• 02.8					Diphda	348°44.1	-17°49.5	
4	279°22.6	230°47.4	41.5	103°09.4	41.7	80°08.4	28.2	237°08.3	02.9					Achernar	335°18.3	-57°05.2	
5	294°25.0	245°46.7	42.5	118°12.0	41.6	95°11.1	28.1	252°10.4	03.0					Hamal	327°47.6	23°36.0	
6	309°27.5	260°46.1	N17°43.4	133°14.7	N03°41.6	110°13.8	S06°28.0	267°12.6	N14°03.1					Polaris	312°36.6	89°23.2	
7	324°30.0	275°45.5	44.3	148°17.3	41.5	125°16.5	27.9	282°14.7	03.2					Acamar	315°09.6	-40°11.3	
8	339°32.4	290°44.9	45.2	163°19.9	41.5	140°19.3	27.8	297°16.9	03.3					Menkar	314°02.8	4°12.2	
9	354°34.9	305°44.2	•• 46.1	178°22.6	•• 41.5	155°22.0	•• 27.7	312°19.0	•• 03.4					Mirfak	308°23.7	49°57.8	
10	9°37.4	320°43.6	47.0	193°25.2	41.4	170°24.7	27.6	327°21.2	03.5					Aldebaran	290°35.9	16°33.9	
11	24°39.8	335°43.0	47.9	208°27.8	41.4	185°27.5	27.5	342°23.3	03.6					Rigel	281°00.8	-8°10.3	
12	39°42.3	350°42.3	N17°48.8	223°30.5	N03°41.3	200°30.2	S06°27.4	357°25.5	N14°03.7					Capella	280°17.1	46°01.6	
13	54°44.8	5°41.7	49.7	238°33.1	41.3	215°32.9	27.3	12°27.6	03.8					Bellatrix	278°19.4	6°22.4	
14	69°47.2	20°41.1	50.6	253°35.7	41.2	230°35.6	27.2	27°29.8	03.9					Elnath	277°57.8	28°37.8	
15	84°49.7	35°40.4	•• 51.5	268°38.4	•• 41.2	245°38.4	•• 27.1	42°31.9	•• 04.0					Alnilam	275°34.5	-1°11.2	
16	99°52.2	50°39.8	52.4	283°41.0	41.1	260°41.1	27.0	57°34.1	04.1					Betelgeuse	270°48.6	7°24.6	
17	114°54.6	65°39.2	53.3	298°43.6	41.1	275°43.8	26.9	72°36.2	04.2					Canopus	263°51.2	-52°43.0	
18	129°57.1	80°38.5	N17°54.1	313°46.2	N03°41.0	290°46.6	S06°26.8	87°38.4	N14°04.3					Sirius	258°23.4	-16°45.7	
19	144°59.5	95°37.9	55.0	328°48.9	41.0	305°49.3	26.7	102°40.5	04.4					Adhara	255°03.3	-29°01.1	
20	160°02.0	110°37.3	55.9	343°51.5	40.9	320°52.0	26.6	117°42.7	04.5					Procyon	244°47.3	5°08.8	
21	175°04.5	125°36.6	•• 56.8	358°54.1	•• 40.9	335°54.7	•• 26.5	132°44.8	•• 04.6					Pollux	243°13.2	27°57.2	
22	190°06.9	140°36.0	57.7	13°56.7	40.8	350°57.5	26.4	147°47.0	04.6					Avior	234°13.3	-59°36.6	
23	205°09.4	155°35.4	58.6	28°59.4	40.8	6°00.2	26.3	162°49.1	04.7					Suhail	222°43.7	-43°33.5	
Mer.pass. 09:22			ν -0.6' d 0.9' m -3.90	ν 2.6' d -0.0' m -0.69	ν 2.7' d -0.1' m -2.44	ν 2.1' d 0.1' m 0.26											

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	220°11.9	170°34.7	N17°59.5	44°02.0	N03°40.7	21°02.9	S06°26.2	177°51.2	N14°04.8		
1	235°14.3	185°34.1	18°00.4	59°04.6	40.7	36°05.6	26.1	192°53.4	04.9		
2	250°16.8	200°33.5	01.3	74°07.2	40.6	51°08.4	26.0	207°55.5	05.0		
3	265°19.3	215°32.8	•• 02.2	89°09.8	•• 40.6	66°11.1	•• 25.9	222°57.7	•• 05.1		
4	280°21.7	230°32.2	03.0	104°12.4	40.5	81°13.8	25.8	237°59.8	05.2		
5	295°24.2	245°31.5	03.9	119°15.0	40.5	96°16.5	25.7	253°02.0	05.3		
6	310°26.7	260°30.9	N18°04.8	134°17.7	N03°40.4	111°19.3	S06°25.6	268°04.1	N14°05.4		
7	325°29.1	275°30.3	05.7	149°20.3	40.4	126°22.0	25.5	283°06.3	05.5		
8	340°31.6	290°29.6	06.6	164°22.9	40.3	141°24.7	25.4	298°08.4	05.6		
9	355°34.0	305°29.0	•• 07.5	179°25.5	•• 40.3	156°27.4	•• 25.3	313°10.6	•• 05.7		
10	10°36.5	320°28.3	08.3	194°28.1	40.2	171°30.2	25.2	328°12.7	05.8		
11	25°39.0	335°27.7	09.2	209°30.7	40.2	186°32.9	25.1	343°14.9	05.9		
12	40°41.4	350°27.0	N18°10.1	224°33.3	N03°40.1	201°35.6	S06°25.0	358°17.0	N14°06.0		
13	55°43.9	5°26.4	11.0	239°35.9	40.0	216°38.3	24.9	13°19.2	06.1		
14	70°46.4	20°25.8	11.9	254°38.5	40.0	231°41.1	24.8	28°21.3	06.2		
15	85°48.8	35°25.1	•• 12.7	269°41.1	•• 39.9	246°43.8	•• 24.7	43°23.5	•• 06.3		
16	100°51.3	50°24.5	13.6	284°43.7	39.9	261°46.5	24.6	58°25.6	06.4		
17	115°53.8	65°23.8	14.5	299°46.3	39.8	276°49.2	24.5	73°27.8	06.5		
18	130°56.2	80°23.2	N18°15.4	314°48.9	N03°39.7	291°52.0	S06°24.4	88°29.9	N14°06.6		
19	145°58.7	95°22.5	16.2	329°51.5	39.7	306°54.7	24.3	103°32.1	06.7		
20	161°01.1	110°21.9	17.1	344°54.1	39.6	321°57.4	24.2	118°34.2	06.8		
21	176°03.6	125°21.2	•• 18.0	359°56.7	•• 39.6	337°00.1	•• 24.1	133°36.4	•• 06.9		
22	191°06.1	140°20.6	18.9	14°59.3	39.5	352°02.9	24.0	148°38.5	07.0		
23	206°08.5	155°19.9	19.7	30°01.9	39.4	7°05.6	23.9	163°40.7	07.0		
Mer.pass. 09:18			ν -0.6' d 0.9' m -3.90	ν 2.6' d -0.1' m -0.66	ν 2.7' d -0.1' m -2.44	ν 2.1' d 0.1' m 0.25					

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	221°11.0	170°19.3	N18°20.6	45°04.4	N03°39.4	22°08.3	S06°23.8	178°42.8	N14°07.1		
1	236°13.5	185°18.7	21.5	60°07.0	39.3	37°11.0	23.7	193°45.0	07.2		
2	251°15.9	200°18.0	22.3	75°09.6	39.2	52°13.7	23.6	208°47.1	07.3		
3	266°18.4	215°17.4	•• 23.2	90°12.2	•• 39.2	67°16.5	•• 23.5	223°49.2	•• 07.4		
4	281°20.9	230°16.7	24.1	105°14.8	39.1	82°19.2	23.4	238°51.4	07.5		
5	296°23.3	245°16.1	24.9	120°17.4	39.1	97°21.9	23.3	253°53.5	07.6		
6	311°25.8	260°15.4	N18°25.8	135°20.0	N03°39.0	112°24.6	S06°23.2	268°55.7	N14°07.7		
7	326°28.3	275°14.7	26.7	150°22.5	38.9	127°27.4	23.1	283°57.8	07.8		
8	341°30.7	290°14.1	27.5	165°25.1	38.9	142°30.1	23.0	299°00.0	07.9		
9	356°33.2	305°13.4	•• 28.4	180°27.7	•• 38.8	157°32.8	•• 22.9	314°02.1	•• 08.0		
10	11°35.6	320°12.8	29.3	195°30.3	38.7	172°35.5	22.8	329°04.3	08.1		
11	26°38.1	335°12.1	30.1	210°32.8	38.6	187°38.2	22.8	344°06.4	08.2		
12	41°40.6	350°11.5	N18°31.0	225°35.4	N03°38.6	202°41.0	S06°22.7	359°08.6	N14°08.3		
13	56°43.0	5°10.8	31.8	240°38.0	38.5	217°43.7	22.6	14°10.7	08.4		
14	71°45.5	20°10.2	32.7	255°40.6	38.4	232°46.4	22.5	29°12.9	08.5		
15	86°48.0	35°09.5	•• 33.6	270°43.1	•• 38.4	247°49.1	•• 22.4	44°15.0	•• 08.6		
16	101°50.4	50°08.9	34.4	285°45.7	38.3	262°51.8	22.3	59°17.2	08.7		
17	116°52.9	65°08.2	35.3	300°48.3	38.2	277°54.6	22.2	74°19.3	08.8		
18	131°55.4	80°07.5	N18°36.1	315°50.8	N03°38.2	292°57.3	S06°22.1	89°21.5	N14°08.9		
19	146°57.8	95°06.9	37.0	330°53.4	38.1	308°00.0	22.0	104°23.6	09.0		
20	162°00.3	110°06.2	37.8	345°56.0	38.0	323°02.7	21.9	119°25.8	09.1		
21	177°02.8	125°05.6	•• 38.7	0°58.5	•• 37.9	338°05.4	•• 21.8	134°27.9	•• 09.2		
22	192°05.2	140°04.9	39.5	16°01.1	37.9	353°08.2	21.7	149°30.1	09.2		
23	207°07.7	155°04.3	40.4	31°03.6	37.8	8°10.9	21.6	164°32.2	09.3		
Mer.pass. 09:14			ν -0.6' d 0.9' m -3.90	ν 2.6' d -0.1' m -0.64	ν 2.7' d -0.1' m -2.43	ν 2.1' d 0.1' m 0.24					

May 01 Tue			SHA	Mer.pass
			Venus	311°37.1 12:37
			Mars	183°46.1 21:04
			Jupiter	160°44.7 22:36
			Saturn	317°47.0 12:10

May 02 Wed			SHA	Mer.pass
			Venus	310°22.9 12:38
			Mars	183°50.1 21:00
			Jupiter	160°51.0 22:32
			Saturn	317°39.4 12:07

May 03 Thu			SHA	Mer.pass
			Venus	309°08.3 12:39
			Mars	183°53.4 20:56
			Jupiter	160°57.3 22:27
			Saturn	317°31.8 12:03

Horizontal parallax		
Venus:		0.1
Mars:		0.2

DUT1 = UT1-UTC = +0.2093 sec $\Delta T = TT-UT1 = +68.9747$ sec

2029 May 01 to May. 03 UT

h Sun Moon							
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180° 43.3	N15° 05.9	324° 11.5	4.0'	S24° 27.2	0.6'	59.6'
1	195° 43.4	06.7	338° 34.5	4.0'	24° 27.9	0.4'	59.5'
2	210° 43.5	07.4	352° 57.5	4.1'	24° 28.3	0.3'	59.5'
3	225° 43.5	· · 08.2	7° 20.6	4.1'	24° 28.6	0.1'	59.5'
4	240° 43.6	08.9	21° 43.7	4.2'	24° 28.6	-0.1'	59.4'
5	255° 43.7	09.7	36° 06.9	4.2'	24° 28.6	-0.2'	59.4'
6	270° 43.8	N15° 10.4	50° 30.1	4.3'	S24° 28.3	-0.4'	59.4'
7	285° 43.8	11.2	64° 53.4	4.3'	24° 27.9	-0.6'	59.3'
8	300° 43.9	11.9	79° 16.7	4.4'	24° 27.3	-0.8'	59.3'
9	315° 44.0	· · 12.7	93° 40.1	4.5'	24° 26.6	-0.9'	59.2'
10	330° 44.0	13.4	108° 03.6	4.5'	24° 25.7	-1.1'	59.2'
11	345° 44.1	14.2	122° 27.1	4.6'	24° 24.6	-1.2'	59.2'
12	0° 44.2	N15° 14.9	136° 50.7	4.7'	S24° 23.3	-1.4'	59.1'
13	15° 44.3	15.7	151° 14.3	4.7'	24° 21.9	-1.6'	59.1'
14	30° 44.3	16.4	165° 38.0	4.8'	24° 20.3	-1.7'	59.1'
15	45° 44.4	· · 17.2	180° 01.8	4.9'	24° 18.6	-1.9'	59.0'
16	60° 44.5	17.9	194° 25.7	4.9'	24° 16.7	-2.1'	59.0'
17	75° 44.6	18.7	208° 49.6	5.0'	24° 14.7	-2.2'	58.9'
18	90° 44.6	N15° 19.4	223° 13.6	5.1'	S24° 12.4	-2.4'	58.9'
19	105° 44.7	20.2	237° 37.7	5.2'	24° 10.1	-2.5'	58.9'
20	120° 44.8	20.9	252° 01.9	5.2'	24° 07.5	-2.7'	58.8'
21	135° 44.8	· · 21.7	266° 26.2	5.3'	24° 04.9	-2.8'	58.8'
22	150° 44.9	22.4	280° 50.5	5.4'	24° 02.0	-3.0'	58.7'
23	165° 45.0	23.2	295° 14.9	5.5'	23° 59.0	-3.1'	58.7'
SD = 15.9' $d = 0.8'$ SD = 16.2'							
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180° 45.1	N15° 23.9	309° 39.4	5.6'	S23° 55.9	-3.3'	58.7'
1	195° 45.1	24.7	324° 04.0	5.7'	23° 52.6	-3.4'	58.6'
2	210° 45.2	25.4	338° 28.6	5.8'	23° 49.2	-3.6'	58.6'
3	225° 45.3	· · 26.1	352° 53.4	5.8'	23° 45.6	-3.7'	58.5'
4	240° 45.3	26.9	7° 18.3	5.9'	23° 41.8	-3.9'	58.5'
5	255° 45.4	27.6	21° 43.2	6.0'	23° 38.0	-4.0'	58.5'
6	270° 45.5	N15° 28.4	36° 08.2	6.1'	S23° 33.9	-4.2'	58.4'
7	285° 45.5	29.1	50° 33.4	6.2'	23° 29.8	-4.3'	58.4'
8	300° 45.6	29.9	64° 58.6	6.3'	23° 25.4	-4.5'	58.3'
9	315° 45.7	· · 30.6	79° 23.9	6.4'	23° 21.0	-4.6'	58.3'
10	330° 45.7	31.3	93° 49.3	6.5'	23° 16.4	-4.7'	58.3'
11	345° 45.8	32.1	108° 14.8	6.6'	23° 11.7	-4.9'	58.2'
12	0° 45.9	N15° 32.8	122° 40.4	6.7'	S23° 06.8	-5.0'	58.2'
13	15° 45.9	33.6	137° 06.1	6.8'	23° 01.8	-5.1'	58.1'
14	30° 46.0	34.3	151° 31.9	6.9'	22° 56.7	-5.3'	58.1'
15	45° 46.1	· · 35.0	165° 57.8	7.0'	22° 51.4	-5.4'	58.1'
16	60° 46.1	35.8	180° 23.8	7.1'	22° 46.0	-5.5'	58.0'
17	75° 46.2	36.5	194° 49.9	7.2'	22° 40.5	-5.7'	58.0'
18	90° 46.3	N15° 37.2	209° 16.1	7.3'	S22° 34.8	-5.8'	57.9'
19	105° 46.3	38.0	223° 42.5	7.4'	22° 29.0	-5.9'	57.9'
20	120° 46.4	38.7	238° 08.9	7.5'	22° 23.1	-6.0'	57.9'
21	135° 46.5	· · 39.5	252° 35.4	7.6'	22° 17.1	-6.2'	57.8'
22	150° 46.5	40.2	267° 02.0	7.7'	22° 10.9	-6.3'	57.8'
23	165° 46.6	40.9	281° 28.7	7.8'	22° 04.6	-6.4'	57.7'
SD = 15.9' $d = 0.7'$ SD = 16.0'							
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180° 46.7	N15° 41.7	295° 55.6	7.9'	S21° 58.2	-6.5'	57.7'
1	195° 46.7	42.4	310° 22.5	8.0'	21° 51.7	-6.6'	57.7'
2	210° 46.8	43.1	324° 49.6	8.2'	21° 45.1	-6.8'	57.6'
3	225° 46.9	· · 43.9	339° 16.7	8.3'	21° 38.3	-6.9'	57.6'
4	240° 46.9	44.6	353° 44.0	8.4'	21° 31.4	-7.0'	57.5'
5	255° 47.0	45.3	8° 11.3	8.5'	21° 24.5	-7.1'	57.5'
6	270° 47.0	N15° 46.1	22° 38.8	8.6'	S21° 17.4	-7.2'	57.5'
7	285° 47.1	46.8	37° 06.4	8.7'	21° 10.2	-7.3'	57.4'
8	300° 47.2	47.5	51° 34.1	8.8'	21° 02.9	-7.4'	57.4'
9	315° 47.2	· · 48.3	66° 01.9	8.9'	20° 55.5	-7.5'	57.3'
10	330° 47.3	49.0	80° 29.8	9.0'	20° 47.9	-7.6'	57.3'
11	345° 47.4	49.7	94° 57.8	9.1'	20° 40.3	-7.7'	57.3'
12	0° 47.4	N15° 50.4	109° 25.9	9.2'	S20° 32.6	-7.8'	57.2'
13	15° 47.5	51.2	123° 54.1	9.3'	20° 24.7	-7.9'	57.2'
14	30° 47.5	51.9	138° 22.4	9.4'	20° 16.8	-8.0'	57.1'
15	45° 47.6	· · 52.6	152° 50.9	9.5'	20° 08.8	-8.1'	57.1'
16	60° 47.7	53.4	167° 19.4	9.6'	20° 00.7	-8.2'	57.1'
17	75° 47.7	54.1	181° 48.0	9.7'	19° 52.4	-8.3'	57.0'
18	90° 47.8	N15° 54.8	196° 16.8	9.9'	S19° 44.1	-8.4'	57.0'
19	105° 47.8	55.5	210° 45.6	10.0'	19° 35.7	-8.5'	57.0'
20	120° 47.9	56.3	225° 14.6	10.1'	19° 27.2	-8.6'	56.9'
21	135° 48.0	· · 57.0	239° 43.7	10.2'	19° 18.6	-8.7'	56.9'
22	150° 48.0	57.7	254° 12.8	10.3'	19° 09.9	-8.8'	56.8'
23	165° 48.1	58.4	268° 42.1	10.4'	19° 01.2	-8.9'	56.8'
SD = 15.9' $d = 0.7'$ SD = 15.7'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:42	22:20	—:—	—:—
N 70°	—:—	—:—	02:24	21:35	—:—	—:—
68°	—:—	00:52	02:52	21:06	23:17	—:—
66°	—:—	01:47	03:13	20:44	22:12	—:—
64°	—:—	02:19	03:29	20:27	21:38	—:—
62°	00:46	02:43	03:43	20:13	21:14	23:21
60°	01:36	03:01	03:55	20:01	20:55	22:23
N 58°	02:05	03:16	04:05	19:51	20:40	21:53
56°	02:26	03:29	04:13	19:42	20:27	21:30
54°	02:44	03:39	04:21	19:34	20:16	21:12
52°	02:58	03:49	04:28	19:27	20:06	20:58
50°	03:10	03:58	04:34	19:21	19:58	20:45
45°	03:35	04:15	04:48	19:07	19:40	20:20
N 40°	03:53	04:29	04:59	18:56	19:25	20:01
35°	04:08	04:41	05:08	18:46	19:14	19:46
30°	04:20	04:51	05:16	18:38	19:04	19:34
20°	04:40	05:07	05:30	18:24	18:47	19:14
N 10°	04:55	05:20	05:42	18:12	18:34	18:59
0°	05:07	05:32	05:54	18:00	18:22	18:47
S 10°	05:18	05:43	06:05	17:49	18:11	18:36
20°	05:27	05:54	06:16	17:37	18:00	18:26
30°	05:36	06:05	06:30	17:24	17:49	18:17
35°	05:41	06:11	06:37	17:16	17:42	18:12
40°	05:46	06:18	06:46	17:07	17:36	18:08
45°	05:50	06:25	06:56	16:57	17:28	18:03
S 50°	05:56	06:34	07:08	16:45	17:19	17:58
52°	05:58	06:38	07:14	16:39	17:15	17:55
54°	06:00	06:42	07:20	16:33	17:11	17:53
56°	06:03	06:47	07:27	16:26	17:06	17:50
58°	06:05	06:52	07:35	16:18	17:01	17:48
S 60°	06:08	06:58	07:44	16:09	16:55	17:45

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	03:44	■	■	05:19
66°	■	03:06	02:38	■	03:59	06:24
64°	01:06	01:49	02:03	03:52	05:16	06:59
62°	00:22	01:12	01:37	04:36	05:52	07:24
60°	· ·	00:46	01:16	05:05	06:18	07:44
N 58°	· ·	00:25	01:00	05:28	06:39	08:00
56°	· ·	00:08	00:45	05:46	06:56	08:14
54°	23:53	· ·	00:33	06:01	07:10	08:26
52°	23:40	· ·	00:22	06:15	07:22	08:36
50°	23:29	· ·	00:13	06:26	07:33	08:46
45°	23:06	23:52	· ·	06:51	07:56	09:05
N 40°	22:47	23:35	· ·	07:10	08:15	09:21
35°	22:31	23:21	· ·	07:26	08:30	09:35
30°	22:18	23:09	23:53	07:40	08:43	09:46
20°	21:54	22:48	23:36	08:04	09:06	10:06
N 10°	21:34	22:30	23:20	08:25	09:25	10:23
0°	21:16	22:13	23:06	08:44	09:43	10:39
S 10°	20:57	21:56	22:51	09:03	10:02	10:55
20°	20:37	21:37	22:36	09:24	10:21	11:11
30°	20:13	21:16	22:18	09:47	10:43	11:31
35°	20:00	21:04	22:08	10:01	10:56	11:42
40°	19:44	20:50	21:56	10:17	11:11	11:55
45°	19:25	20:32	21:42	10:36	11:28	12:10
S 50°	19:01	20:11	21:24	11:00	11:50	12:28
52°	18:50	20:01	21:16	11:11	12:01	12:37
54°	18:37	19:50	21:07	11:24	12:12	12:46
56°	18:22	19:37	20:57	11:39	12:26	12:57
58°	18:05	19:22	20:45	11:57	12:41	13:09
S 60°	17:44	19:04	20:31	12:18	12:59	13:24

Day	Sun			Moon		Age 18-20 91-74%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss	Mer. Pass hh:mm	Mer.Pass. Upper hh:mm	Lower hh:mm	
01	02:53	02:57	11:57	02:29	15:00	
02	03:00	03:04	11:57	03:30	15:58	
03	03:07	03:10	11:57	04:26	16:53	

May 04, 05, 06 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	222°10.1	170°03.6	N18°41.2	46°06.2	N03°37.7	23°13.6	S06°21.5	179°34.4	N14°09.4				Alpheratz	357°31.4	29°15.1		
1	237°12.6	185°02.9	42.1	61°08.8	37.6	38°16.3	21.4	194°36.5	09.5				Ankaa	353°04.2	-42°08.6		
2	252°15.1	200°02.3	42.9	76°11.3	37.6	53°19.0	21.3	209°38.6	09.6				Schedar	349°27.4	56°41.8		
3	267°17.5	215°01.6	· · 43.8	91°13.9	· · 37.5	68°21.8	· · 21.2	224°40.8	· · 09.7				Diphda	348°44.1	-17°49.5		
4	282°20.0	230°00.9	44.6	106°16.4	37.4	83°24.5	21.1	239°42.9	09.8				Achernar	335°18.3	-57°05.2		
5	297°22.5	245°00.3	45.5	121°19.0	37.3	98°27.2	21.0	254°45.1	09.9				Hamal	327°47.6	23°36.0		
6	312°24.9	259°59.6	N18°46.3	136°21.5	N03°37.3	113°29.9	S06°20.9	269°47.2	N14°10.0				Polaris	312°36.5	89°23.1		
7	327°27.4	274°59.0	47.2	151°24.1	37.2	128°32.6	20.8	284°49.4	10.1				Acamar	315°09.6	-40°11.3		
8	342°29.9	289°58.3	48.0	166°26.6	37.1	143°35.3	20.7	299°51.5	10.2				Menkar	314°02.8	4°12.2		
9	357°32.3	304°57.6	· · 48.8	181°29.2	· · 37.0	158°38.1	· · 20.6	314°53.7	· · 10.3				Mirfak	308°23.7	49°57.8		
10	12°34.8	319°57.0	49.7	196°31.7	36.9	173°40.8	20.5	329°55.8	10.4				Aldebaran	290°35.9	16°33.9		
11	27°37.3	334°56.3	50.5	211°34.3	36.9	188°43.5	20.5	344°58.0	10.5				Rigel	281°00.8	-8°10.3		
12	42°39.7	349°55.6	N18°51.4	226°36.8	N03°36.8	203°46.2	S06°20.4	0°00.1	N14°10.6				Capella	280°17.1	46°01.6		
13	57°42.2	4°55.0	52.2	241°39.4	36.7	218°48.9	20.3	15°02.3	10.7				Bellatrix	278°19.4	6°22.4		
14	72°44.6	19°54.3	53.0	256°41.9	36.6	233°51.6	20.2	30°04.4	10.8				Elnath	277°57.8	28°37.8		
15	87°47.1	34°53.6	· · 53.9	271°44.4	· · 36.5	248°54.4	· · 20.1	45°06.6	· · 10.9				Alnilam	275°34.5	-1°11.2		
16	102°49.6	49°53.0	54.7	286°47.0	36.4	263°57.1	20.0	60°08.7	11.0				Betelgeuse	270°48.6	7°24.6		
17	117°52.0	64°52.3	55.5	301°49.5	36.4	278°59.8	19.9	75°10.9	11.1				Canopus	263°51.2	-52°43.0		
18	132°54.5	79°51.6	N18°56.4	316°52.1	N03°36.3	294°02.5	S06°19.8	90°13.0	N14°11.2				Sirius	258°23.4	-16°45.7		
19	147°57.0	94°51.0	57.2	331°54.6	36.2	309°05.2	19.7	105°15.2	11.3				Adhara	255°03.3	-29°01.1		
20	162°59.4	109°50.3	58.0	346°57.1	36.1	324°07.9	19.6	120°17.3	11.3				Procyon	244°47.3	5°08.8		
21	178°01.9	124°49.6	· · 58.9	1°59.7	· · 36.0	339°10.7	· · 19.5	135°19.5	· · 11.4				Pollux	243°13.2	27°57.2		
22	193°04.4	139°48.9	18°59.7	17°02.2	35.9	354°13.4	19.4	150°21.6	11.5				Avior	234°13.3	-59°36.6		
23	208°06.8	154°48.3	19°00.5	32°04.7	35.9	9°16.1	19.3	165°23.8	11.6				Suhail	222°43.7	-43°33.5		
Mer.pass. 09:10			ν -0.7' d 0.8' m -3.90			ν 2.6' d -0.1' m -0.62			ν 2.7' d -0.1' m -2.43			ν 2.1' d 0.1' m 0.23					

Sat	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	223°09.3	169°47.6	N19°01.4	47°07.3	N03°35.8	24°18.8	S06°19.2	180°25.9						
1	238°11.7	184°46.9	02.2	62°09.8	35.7	39°21.5	19.1	195°28.0						
2	253°14.2	199°46.3	03.0	77°12.3	35.6	54°24.2	19.0	210°30.2						
3	268°16.7	214°45.6	· · 03.9	92°14.8	· · 35.5	69°26.9	· · 18.9	225°32.3						
4	283°19.1	229°44.9	04.7	107°17.4	35.4	84°29.7	18.8	240°34.5						
5	298°21.6	244°44.2	05.5	122°19.9	35.3	99°32.4	18.8	255°36.6						
6	313°24.1	259°43.6	N19°06.3	137°22.4	N03°35.2	114°35.1	S06°18.7	270°38.8						
7	328°26.5	274°42.9	07.1	152°24.9	35.1	129°37.8	18.6	285°40.9						
8	343°29.0	289°42.2	08.0	167°27.4	35.1	144°40.5	18.5	300°43.1						
9	358°31.5	304°41.5	· · 08.8	182°30.0	· · 35.0	159°43.2	· · 18.4	315°45.2						
10	13°33.9	319°40.9	09.6	197°32.5	34.9	174°45.9	18.3	330°47.4						
11	28°36.4	334°40.2	10.4	212°35.0	34.8	189°48.6	18.2	345°49.5						
12	43°38.9	349°39.5	N19°11.2	227°37.5	N03°34.7	204°51.4	S06°18.1	0°51.7						
13	58°41.3	4°38.8	12.1	242°40.0	34.6	219°54.1	18.0	15°53.8						
14	73°43.8	19°38.1	12.9	257°42.5	34.5	234°56.8	17.9	30°56.0						
15	88°46.2	34°37.5	· · 13.7	272°45.0	· · 34.4	249°59.5	· · 17.8	45°58.1						
16	103°48.7	49°36.8	14.5	287°47.6	34.3	265°02.2	17.7	61°00.3						
17	118°51.2	64°36.1	15.3	302°50.1	34.2	280°04.9	17.6	76°02.4						
18	133°53.6	79°35.4	N19°16.1	317°52.6	N03°34.1	295°07.6	S06°17.5	91°04.6						
19	148°56.1	94°34.7	17.0	332°55.1	34.0	310°10.3	17.4	106°06.7						
20	163°58.6	109°34.1	17.8	347°57.6	33.9	325°13.1	17.4	121°08.9						
21	179°01.0	124°33.4	· · 18.6	3°00.1	· · 33.8	340°15.8	· · 17.3	136°11.0						
22	194°03.5	139°32.7	19.4	18°02.6	33.7	355°18.5	17.2	151°13.1						
23	209°06.0	154°32.0	20.2	33°05.1	33.6	10°21.2	17.1	166°15.3						
Mer.pass. 09:06			ν -0.7' d 0.8' m -3.90			ν 2.5' d -0.1' m -0.60			ν 2.7' d -0.1' m -2.43			ν 2.1' d 0.1' m 0.23		

Sun	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	224°08.4	169°31.3	N19°21.0	48°07.6	N03°33.5	25°23.9	S06°17.0	181°17.4						
1	239°10.9	184°30.7	21.8	63°10.1	33.4	40°26.6	16.9	196°19.6						
2	254°13.4	199°30.0	22.6	78°12.6	33.3	55°29.3	16.8	211°21.7						
3	269°15.8	214°29.3	· · 23.4	93°15.1	· · 33.2	70°32.0	· · 16.7	226°23.9						
4	284°18.3	229°28.6	24.2	108°17.6	33.1	85°34.7	16.6	241°26.0						
5	299°20.7	244°27.9	25.0	123°20.1	33.0	100°37.4	16.5	256°28.2						
6	314°23.2	259°27.2	N19°25.8	138°22.6	N03°32.9	115°40.2	S06°16.4	271°30.3						
7	329°25.7	274°26.5	26.6	153°25.1	32.8	130°42.9	16.3	286°32.5						
8	344°28.1	289°25.8	27.4	168°27.6	32.7	145°45.6	16.3	301°34.6						
9	359°30.6	304°25.2	· · 28.2	183°30.0	· · 32.6	160°48.3	· · 16.2	316°36.8						
10	14°33.1	319°24.5	29.0	198°32.5	32.5	175°51.0	16.1	331°38.9						
11	29°35.5	334°23.8	29.8	213°35.0	32.4	190°53.7	16.0	346°41.1						
12	44°38.0	349°23.1	N19°30.6	228°37.5	N03°32.3	205°56.4	S06°15.9	1°43.2						
13	59°40.5	4°22.4	31.4	243°40.0	32.2	220°59.1	15.8	16°45.4						
14	74°42.9	19°21.7	32.2	258°42.5	32.1	236°01.8	15.7	31°47.5						
15	89°45.4	34°21.0	· · 33.0	273°45.0	· · 32.0	251°04.5	· · 15.6	46°49.7						
16	104°47.8	49°20.3	33.8	288°47.4	31.9	266°07.2	15.5	61°51.8						
17	119°50.3	64°19.6	34.6	303°49.9	31.8	281°09.9	15.4	76°53.9						
18	134°52.8	79°18.9	N19°35.4	318°52.4	N03°31.7	296°12.6	S06°15.3	91°56.1						
19	149°55.2	94°18.3	36.2	333°54.9	31.6	311°15.4	15.3	106°58.2						
20	164°57.7	109°17.6	37.0	348°57.3	31.5	326°18.1	15.2	122°00.4						
21	180°00.2	124°16.9	· · 37.8	3°59.8	· · 31.3	341°20.8	· · 15.1	137°02.5						
22	195°02.6	139°16.2	38.5	19°02.3	31.2	356°23.5	15.0	152°04.7						
23	210°05.1	154°15.5	39.3	34°04.8	31.1	11°26.2	14.9	167°06.8						
Mer.pass. 09:02			ν -0.7' d 0.8' m -3.90			ν 2.5' d -0.1' m -0.57			ν 2.7' d -0.1' m -2.42			ν 2.1' d 0.1' m 0.23		

May 04 Fri	SHA	Mer.pass
Venus	307°53.4	12:40
Mars	183°56.1	20:52
Jupiter	161°03.5	22:23
Saturn	317°24.2	12:00

May 05 Sat	SHA	Mer.pass
Venus	306°38.3	12:41
Mars	183°58.0	20:48
Jupiter	161°09.5	22:19
Saturn	317°16.6	11:57

May 06 Sun	SHA	Mer.pass
Venus	305°22.9	12:42
Mars	183°59.2	20:44
Jupiter	161°15.5	22:14
Saturn	317°09.0	11:53

Horizontal parallax		
Venus:		0.1
Mars:		0.2

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°48.1	N15°59.2	283°11.5	10.5'	S18°52.3	-8.9'	56.8'
1	195°48.2	15°59.9	297°41.0	10.6'	18°43.4	-9.0'	56.7'
2	210°48.3	16°00.6	312°10.6	10.7'	18°34.3	-9.1'	56.7'
3	225°48.3	· · 01.3	326°40.2	10.8'	18°25.2	-9.2'	56.7'
4	240°48.4	02.0	341°10.0	10.9'	18°16.0	-9.3'	56.6'
5	255°48.4	02.8	355°39.9	11.0'	18°06.8	-9.3'	56.6'
6	270°48.5	N16°03.5	10°09.9	11.1'	S17°57.4	-9.4'	56.5'
7	285°48.5	04.2	24°40.0	11.2'	17°48.0	-9.5'	56.5'
8	300°48.6	04.9	39°10.2	11.3'	17°38.5	-9.6'	56.5'
9	315°48.7	· · 05.6	53°40.5	11.4'	17°28.9	-9.7'	56.4'
10	330°48.7	06.4	68°10.9	11.5'	17°19.3	-9.7'	56.4'
11	345°48.8	07.1	82°41.3	11.6'	17°09.5	-9.8'	56.4'
12	0°48.8	N16°07.8	97°11.9	11.7'	S16°59.7	-9.9'	56.3'
13	15°48.9	08.5	111°42.6	11.8'	16°49.9	-9.9'	56.3'
14	30°48.9	09.2	126°13.4	11.9'	16°39.9	-10.0'	56.3'
15	45°49.0	· · 10.0	140°44.2	12.0'	16°29.9	-10.1'	56.2'
16	60°49.0	10.7	155°15.2	12.1'	16°19.9	-10.1'	56.2'
17	75°49.1	11.4	169°46.2	12.1'	16°09.7	-10.2'	56.2'
18	90°49.1	N16°12.1	184°17.4	12.2'	S15°59.5	-10.3'	56.1'
19	105°49.2	12.8	198°48.6	12.3'	15°49.3	-10.3'	56.1'
20	120°49.3	13.5	213°19.9	12.4'	15°39.0	-10.4'	56.1'
21	135°49.3	· · 14.2	227°51.4	12.5'	15°28.6	-10.4'	56.0'
22	150°49.4	15.0	242°22.9	12.6'	15°18.1	-10.5'	56.0'
23	165°49.4	15.7	256°54.5	12.7'	15°07.6	-10.6'	56.0'
SD = 15.9'			SD = 15.5'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°49.5	N16°16.4	271°26.1	12.8'	S14°57.1	-10.6'	55.9'
1	195°49.5	17.1	285°57.9	12.9'	14°46.5	-10.7'	55.9'
2	210°49.6	17.8	300°29.8	12.9'	14°35.8	-10.7'	55.9'
3	225°49.6	· · 18.5	315°01.7	13.0'	14°25.1	-10.8'	55.8'
4	240°49.7	19.2	329°33.7	13.1'	14°14.3	-10.8'	55.8'
5	255°49.7	19.9	344°05.8	13.2'	14°03.5	-10.9'	55.8'
6	270°49.8	N16°20.6	358°38.0	13.3'	S13°52.6	-10.9'	55.7'
7	285°49.8	21.4	13°10.3	13.3'	13°41.7	-11.0'	55.7'
8	300°49.9	22.1	27°42.6	13.4'	13°30.7	-11.0'	55.7'
9	315°49.9	· · 22.8	42°15.1	13.5'	13°19.7	-11.1'	55.7'
10	330°50.0	23.5	56°47.6	13.6'	13°08.6	-11.1'	55.6'
11	345°50.0	24.2	71°20.2	13.7'	12°57.5	-11.2'	55.6'
12	0°50.1	N16°24.9	85°52.8	13.7'	S12°46.3	-11.2'	55.6'
13	15°50.1	25.6	100°25.6	13.8'	12°35.1	-11.2'	55.5'
14	30°50.2	26.3	114°58.4	13.9'	12°23.9	-11.3'	55.5'
15	45°50.2	· · 27.0	129°31.2	14.0'	12°12.6	-11.3'	55.5'
16	60°50.3	27.7	144°04.2	14.0'	12°01.2	-11.4'	55.4'
17	75°50.3	28.4	158°37.2	14.1'	11°49.9	-11.4'	55.4'
18	90°50.4	N16°29.1	173°10.3	14.2'	S11°38.5	-11.4'	55.4'
19	105°50.4	29.8	187°43.5	14.2'	11°27.0	-11.5'	55.4'
20	120°50.5	30.5	202°16.7	14.3'	11°15.5	-11.5'	55.3'
21	135°50.5	· · 31.2	216°50.0	14.4'	11°04.0	-11.6'	55.3'
22	150°50.6	31.9	231°23.4	14.4'	10°52.4	-11.6'	55.3'
23	165°50.6	32.6	245°56.8	14.5'	10°40.9	-11.6'	55.3'
SD = 15.9'			SD = 15.3'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°50.6	N16°33.3	260°30.3	14.6'	S10°29.2	-11.7'	55.2'
1	195°50.7	34.0	275°03.9	14.6'	10°17.6	-11.7'	55.2'
2	210°50.7	34.7	289°37.5	14.7'	10°05.9	-11.7'	55.2'
3	225°50.8	· · 35.4	304°11.2	14.7'	09°54.2	-11.8'	55.1'
4	240°50.8	36.1	318°44.9	14.8'	09°42.4	-11.8'	55.1'
5	255°50.9	36.8	333°18.7	14.9'	09°30.6	-11.8'	55.1'
6	270°50.9	N16°37.5	347°52.6	14.9'	S09°18.8	-11.8'	55.1'
7	285°51.0	38.2	2°26.5	15.0'	09°07.0	-11.9'	55.0'
8	300°51.0	38.9	17°00.5	15.0'	08°55.1	-11.9'	55.0'
9	315°51.1	· · 39.6	31°34.5	15.1'	08°43.2	-11.9'	55.0'
10	330°51.1	40.3	46°08.6	15.1'	08°31.3	-11.9'	55.0'
11	345°51.1	41.0	60°42.8	15.2'	08°19.4	-12.0'	55.0'
12	0°51.2	N16°41.7	75°16.9	15.2'	S08°07.4	-12.0'	54.9'
13	15°51.2	42.4	89°51.2	15.3'	07°55.4	-12.0'	54.9'
14	30°51.3	43.1	104°25.5	15.3'	07°43.4	-12.0'	54.9'
15	45°51.3	· · 43.8	118°59.8	15.4'	07°31.4	-12.0'	54.9'
16	60°51.4	44.5	133°34.2	15.4'	07°19.4	-12.1'	54.8'
17	75°51.4	45.2	148°08.6	15.5'	07°07.3	-12.1'	54.8'
18	90°51.4	N16°45.9	162°43.1	15.5'	S06°55.2	-12.1'	54.8'
19	105°51.5	46.6	177°17.6	15.6'	06°43.1	-12.1'	54.8'
20	120°51.5	47.3	191°52.2	15.6'	06°31.0	-12.1'	54.8'
21	135°51.6	· · 48.0	206°26.8	15.6'	06°18.8	-12.2'	54.7'
22	150°51.6	48.6	221°01.5	15.7'	06°06.7	-12.2'	54.7'
23	165°51.6	49.3	235°36.1	15.7'	05°54.5	-12.2'	54.7'
SD = 15.8'			SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:11	22:54	—:—	—:—
N 70°	—:—	—:—	02:05	21:54	—:—	—:—
68°	—:—	—:—	02:37	21:20	—:—	—:—
66°	—:—	01:27	03:01	20:55	22:33	—:—
64°	—:—	02:05	03:19	20:37	21:53	—:—
62°	—:—	02:31	03:34	20:21	21:25	—:—
60°	01:17	02:51	03:47	20:08	21:05	22:42
N 58°	01:52	03:07	03:57	19:57	20:48	22:05
56°	02:16	03:21	04:07	19:48	20:34	21:40
54°	02:35	03:33	04:15	19:39	20:22	21:21
52°	02:50	03:43	04:23	19:32	20:12	21:05
50°	03:04	03:52	04:29	19:25	20:03	20:51
45°	03:29	04:11	04:43	19:11	19:44	20:25
N 40°	03:49	04:25	04:55	18:59	19:29	20:05
35°	04:05	04:38	05:05	18:49	19:16	19:49
30°	04:18	04:48	05:14	18:40	19:06	19:36
20°	04:38	05:05	05:28	18:25	18:48	19:16
N 10°	04:54	05:19	05:41	18:12	18:34	19:00
0°	05:07	05:32	05:53	18:00	18:22	18:47
S 10°	05:18	05:43	06:05	17:48	18:10	18:35
20°	05:28	05:55	06:17	17:36	17:59	18:25
30°	05:38	06:07	06:32	17:21	17:46	18:15
35°	05:43	06:13	06:40	17:13	17:40	18:10
40°	05:48	06:21	06:49	17:04	17:32	18:05
45°	05:54	06:29	07:00	16:53	17:24	17:59
S 50°	06:00	06:38	07:13	16:40	17:14	17:53
52°	06:02	06:43	07:19	16:34	17:10	17:51
54°	06:05	06:47	07:26	16:27	17:05	17:48
56°	06:08	06:52	07:33	16:20	17:00	17:45
58°	06:11	06:58	07:41	16:11	16:55	17:42
S 60°	06:14	07:04	07:51	16:01	16:48	17:39

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	03:30	02:55	■	09:02	11:09
N 70°	03:40	03:01	02:40	07:12	09:29	11:23
68°	02:58	02:40	02:27	07:53	09:49	11:33
66°	02:29	02:22	02:17	08:21	10:05	11:42
64°	02:07	02:08	02:08	08:42	10:18	11:49
62°	01:49	01:56	02:01	08:59	10:29	11:55
60°	01:35	01:46	01:54	09:13	10:39	12:01
N 58°	01:22	01:37	01:48	09:25	10:47	12:06
56°	01:11	01:29	01:43	09:35	10:54	12:10
54°	01:02	01:22	01:39	09:44	11:00	12:14
52°	00:53	01:16	01:35	09:52	11:06	12:17
50°	00:45	01:10	01:31	09:59	11:11	12:20
45°	00:29	00:58	01:23	10:15	11:22	12:27
N 40°	00:15	00:48	01:16	10:27	11:31	12:32
35°	00:03	00:39	01:10	10:38	11:39	12:37
30°	· · ·	00:31	01:04	10:47	11:45	12:41
20°	· · ·	00:18	00:55	11:03	11:57	12:48
N 10°	· · ·	00:06	00:47	11:17	12:07	12:54
0°	23:55	· · ·	00:40	11:30	12:17	13:00
S 10°	23:43	· · ·	00:32	11:43	12:26	13:06
20°	23:31	· · ·	00:24	11:56	12:36	13:12
30°	23:18	· · ·	00:14	12:12	12:47	13:19
35°	23:10	· · ·	00:09	12:21	12:54	13:23
40°	23:01	· · ·	00:03	12:31	13:01	13:27
45°	22:50	23:56	· · ·	12:43	13:09	13:33
S 50°	22:37	23:47	· · ·	12:57	13:20	13:39
52°	22:31	23:43	· · ·	13:03	13:24	13:41
54°	22:24	23:39	· · ·	13:11	13:29	13:45
56°	22:16	23:34	· · ·	13:19	13:35	13:48
58°	22:08	23:28	· · ·	13:28	13:41	13:52
S 60°	21:58	23:22	· · ·	13:39	13:49	13:56

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12 ^h	Pass	Mer.Pass.	Lower	21-23
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	64-44%
04	03:13	03:15	11:57	05:18	17:42	
05	03:18	03:20	11:57	06:06	18:28	
06	03:23	03:25	11:57	06:50	19:11	

May 07, 08, 09 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	225°07.6	169°14.8	N19°40.1	49°07.2	N03°31.0	26°28.9	S06°14.8	182°09.0	N14°16.3					Alpheratz	357°31.3	29°15.1	
1	240°10.0	184°14.1	40.9	64°09.7	30.9	41°31.6	14.7	197°11.1	16.4					Ankaa	353°04.1	-42°08.6	
2	255°12.5	199°13.4	41.7	79°12.2	30.8	56°34.3	14.6	212°13.3	16.5					Schedar	349°27.3	56°41.8	
3	270°15.0	214°12.7	· · 42.5	94°14.6	· · 30.7	71°37.0	· · 14.5	227°15.4	· · 16.6					Diphda	348°44.1	-17°49.5	
4	285°17.4	229°12.0	43.3	109°17.1	30.6	86°39.7	14.4	242°17.6	16.7					Achernar	335°18.3	-57°05.1	
5	300°19.9	244°11.3	44.0	124°19.6	30.5	101°42.4	14.3	257°19.7	16.8					Hamal	327°47.6	23°36.0	
6	315°22.3	259°10.6	N19°44.8	139°22.0	N03°30.3	116°45.1	S06°14.3	272°21.9	N14°16.9					Polaris	312°36.5	89°23.1	
7	330°24.8	274°09.9	45.6	154°24.5	30.2	131°47.8	14.2	287°24.0	16.9					Acamar	315°09.6	-40°11.3	
8	345°27.3	289°09.2	46.4	169°27.0	30.1	146°50.5	14.1	302°26.2	17.0					Menkar	314°02.8	4°12.2	
9	0°29.7	304°08.5	· · 47.2	184°29.4	· · 30.0	161°53.2	· · 14.0	317°28.3	· · 17.1					Mirfak	308°23.7	49°57.8	
10	15°32.2	319°07.8	47.9	199°31.9	29.9	176°55.9	13.9	332°30.5	17.2					Aldebaran	290°35.9	16°33.9	
11	30°34.7	334°07.1	48.7	214°34.3	29.8	191°58.6	13.8	347°32.6	17.3					Rigel	281°00.8	-8°10.3	
12	45°37.1	349°06.4	N19°49.5	229°36.8	N03°29.7	207°01.3	S06°13.7	2°34.7	N14°17.4					Capella	280°17.1	46°01.6	
13	60°39.6	4°05.7	50.3	244°39.3	29.5	222°04.0	13.6	17°36.9	17.5					Bellatrix	278°19.4	6°22.4	
14	75°42.1	19°05.0	51.0	259°41.7	29.4	237°06.7	13.5	32°39.0	17.6					Elnath	277°57.8	28°37.8	
15	90°44.5	34°04.3	· · 51.8	274°44.2	· · 29.3	252°09.5	· · 13.5	47°41.2	· · 17.7					Alnilam	275°34.5	-1°11.2	
16	105°47.0	49°03.6	52.6	289°46.6	29.2	267°12.2	13.4	62°43.3	17.8					Betelgeuse	270°48.6	7°24.6	
17	120°49.5	64°02.9	53.3	304°49.1	29.1	282°14.9	13.3	77°45.5	17.9					Canopus	263°51.2	-52°43.0	
18	135°51.9	79°02.2	N19°54.1	319°51.5	N03°29.0	297°17.6	S06°13.2	92°47.6	N14°18.0					Sirius	258°23.4	-16°45.7	
19	150°54.4	94°01.5	54.9	334°54.0	28.8	312°20.3	13.1	107°49.8	18.1					Adhara	255°03.3	-29°01.0	
20	165°56.8	109°00.8	55.6	349°56.4	28.7	327°23.0	13.0	122°51.9	18.2					Procyon	244°47.3	5°08.8	
21	180°59.3	124°00.1	· · 56.4	4°58.9	· · 28.6	342°25.7	· · 12.9	137°54.1	· · 18.3					Pollux	243°13.2	27°57.2	
22	196°01.8	138°59.4	57.2	20°01.3	28.5	357°28.4	12.8	152°56.2	18.4					Avior	234°13.4	-59°36.6	
23	211°04.2	153°58.7	57.9	35°03.8	28.4	12°31.1	12.7	167°58.4	18.5					Suhail	222°43.7	-43°33.5	
Mer.pass. 08:58			ν -0.7' d 0.8' m -3.90			ν 2.5' d -0.1' m -0.55			ν 2.7' d -0.1' m -2.42			ν 2.1' d 0.1' m 0.24					

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	226°06.7	168°58.0	N19°58.7	50°06.2	N03°28.2	27°33.8	S06°12.7	183°00.5	N14°18.5					Denebola	182°21.2	14°24.3	
1	241°09.2	183°57.3	19°59.5	65°08.6	28.1	42°36.5	12.6	198°02.7	18.6					Gienah	175°39.7	-17°42.5	
2	256°11.6	198°56.5	20°00.2	80°11.1	28.0	57°39.2	12.5	213°04.8	18.7					Acrux	172°55.4	-63°16.0	
3	271°14.1	213°55.8	· · 01.0	95°13.5	· · 27.9	72°41.9	· · 12.4	228°07.0	· · 18.8					Gacrux	171°47.1	-57°17.0	
4	286°16.6	228°55.1	01.8	110°16.0	27.7	87°44.6	12.3	243°09.1	18.9					Alioth	166°09.6	55°48.1	
5	301°19.0	243°54.4	02.5	125°18.4	27.6	102°47.3	12.2	258°11.3	19.0					Spica	158°18.3	-11°19.0	
6	316°21.5	258°53.7	N20°03.3	140°20.8	N03°27.5	117°50.0	S06°12.1	273°13.4	N14°19.1					Alkaid	152°48.9	49°10.0	
7	331°23.9	273°53.0	04.0	155°23.3	27.4	132°52.7	12.0	288°15.5	19.2					Hadar	148°30.2	-60°31.0	
8	346°26.4	288°52.3	04.8	170°25.7	27.2	147°55.4	12.0	303°17.7	19.3					Menkent	147°53.0	-36°31.0	
9	1°28.9	303°51.6	· · 05.5	185°28.1	· · 27.1	162°58.1	· · 11.9	318°19.8	· · 19.4					Arcturus	145°44.4	19°01.7	
10	16°31.3	318°50.9	06.3	200°30.6	27.0	178°00.8	11.8	333°22.0	19.5					Rigil Kent.	139°34.7	-60°57.5	
11	31°33.8	333°50.2	07.1	215°33.0	26.9	193°03.5	11.7	348°24.1	19.6					Kochab	137°19.0	74°02.0	
12	46°36.3	348°49.4	N20°07.8	230°35.4	N03°26.7	208°06.2	S06°11.6	3°26.3	N14°19.7					Zuben'ubi	136°51.8	-67°09.9	
13	61°38.7	3°48.7	08.6	245°37.8	26.6	223°08.9	11.5	18°28.4	19.8					Alphecca	126°00.5	26°36.9	
14	76°41.2	18°48.0	09.3	260°40.3	26.5	238°11.6	11.4	33°30.6	19.9					Antares	112°11.2	-26°29.8	
15	91°43.7	33°47.3	· · 10.1	275°42.7	· · 26.4	253°14.3	· · 11.3	48°32.7	· · 20.0					Atria	107°01.7	-69°04.6	
16	106°46.1	48°46.6	10.8	290°45.1	26.2	268°17.0	11.3	63°34.9	20.1					Sabik	101°58.5	-15°45.6	
17	121°48.6	63°45.9	11.6	305°47.5	26.1	283°19.7	11.2	78°37.0	20.1					Shaula	96°05.3	-37°07.3	
18	136°51.1	78°45.2	N20°12.3	320°50.0	N03°26.0	298°22.4	S06°11.1	93°39.2	N14°20.2					Rasalhague	95°55.0	12°32.3	
19	151°53.5	93°44.4	13.0	335°52.4	25.8	313°25.1	11.0	108°41.3	20.3					Eltanin	90°40.2	51°28.9	
20	166°56.0	108°43.7	13.8	350°54.8	25.7	328°27.8	10.9	123°43.5	20.4					Kaus Aust.	83°27.6	-34°22.0	
21	181°58.4	123°43.0	· · 14.5	5°57.2	· · 25.6	343°30.5	· · 10.8	138°45.6	· · 20.5					Vega	80°30.6	38°48.5	
22	197°00.9	138°42.3	15.3	20°59.6	25.4	358°33.1	10.7	153°47.8	20.6					Nunki	75°43.2	-26°15.4	
23	212°03.4	153°41.6	16.0	36°02.1	25.3	13°35.8	10.7	168°49.9	20.7					Altair	61°56.4	8°56.8	
Mer.pass. 08:54			ν -0.7' d 0.8' m -3.90			ν 2.4' d -0.1' m -0.53			ν 2.7' d -0.1' m -2.41			ν 2.1' d 0.1' m 0.25					

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	227°05.8	168°40.9	N20°16.8	51°04.5	N03°25.2	28°38.5	S06°10.6	183°52.1	N14°20.8					May 07 Mon	SHA	Mer.pass
1	242°08.3	183°40.1	17.5	66°06.9	25.0	43°41.2	10.5	198°54.2	20.9					Venus	304°07.2	12:44
2	257°10.8	198°39.4	18.2	81°09.3	24.9	58°43.9	10.4	213°56.3	21.0					Mars	183°59.7	20:40
3	272°13.2	213°38.7	· · 19.0	96°11.7	· · 24.8	73°46.6	· · 10.3	228°58.5	· · 21.1					Jupiter	161°21.3	22:10
4	287°15.7	228°38.0	19.7	111°14.1	24.6	88°49.3	10.2	244°00.6	21.2					Saturn	317°01.4	11:50
5	302°18.2	243°37.3	20.5	126°16.5	24.5	103°52.0	10.1	259°02.8	21.3							
6	317°20.6	258°36.5	N20°21.2	141°18.9	N03°24.4	118°54.7	S06°10.1	274°04.9	N14°21.4							
7	332°23.1	273°35.8	21.9	156°21.3	24.2	133°57.4	10.0	289°07.1	21.5							
8	347°25.5	288°35.1	22.7	171°23.7	24.1	149°00.1	09.9	304°09.2	21.6							
9	2°28.0	303°34.4	· · 23.4	186°26.1	· · 24.0	164°02.8	· · 09.8	319°11.4	· · 21.7							

DUT1 = UT1-UTC = +0.2095 sec ΔT = TT-UT1 = +68.9745 sec

2029 May 07 to May. 09 UT

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	180°51.7	N16°50.0	250°10.9	15.8'	S05°42.3	-12.2' 54.7'
1	195°51.7	50.7	264°45.6	15.8'	05°30.1	-12.2' 54.7'
2	210°51.8	51.4	279°20.4	15.8'	05°17.9	-12.2' 54.6'
3	225°51.8	· · 52.1	293°55.3	15.9'	05°05.7	-12.2' 54.6'
4	240°51.8	52.8	308°30.2	15.9'	04°53.5	-12.2' 54.6'
5	255°51.9	53.5	323°05.1	15.9'	04°41.2	-12.3' 54.6'
6	270°51.9	N16°54.2	337°40.0	16.0'	S04°29.0	-12.3' 54.6'
7	285°52.0	54.8	352°15.0	16.0'	04°16.7	-12.3' 54.5'
8	300°52.0	55.5	6°50.0	16.0'	04°04.4	-12.3' 54.5'
9	315°52.0	· · 56.2	21°25.0	16.1'	03°52.1	-12.3' 54.5'
10	330°52.1	56.9	36°00.1	16.1'	03°39.8	-12.3' 54.5'
11	345°52.1	57.6	50°35.2	16.1'	03°27.6	-12.3' 54.5'
12	0°52.2	N16°58.3	65°10.3	16.1'	S03°15.2	-12.3' 54.5'
13	15°52.2	58.9	79°45.4	16.2'	03°02.9	-12.3' 54.4'
14	30°52.2	16°59.6	94°20.6	16.2'	02°50.6	-12.3' 54.4'
15	45°52.3	17°00.3	108°55.8	16.2'	02°38.3	-12.3' 54.4'
16	60°52.3	01.0	123°31.0	16.2'	02°26.0	-12.3' 54.4'
17	75°52.3	01.7	138°06.3	16.3'	02°13.7	-12.3' 54.4'
18	90°52.4	N17°02.4	152°41.5	16.3'	S02°01.3	-12.3' 54.4'
19	105°52.4	03.0	167°16.8	16.3'	01°49.0	-12.3' 54.3'
20	120°52.4	03.7	181°52.1	16.3'	01°36.7	-12.3' 54.3'
21	135°52.5	· · 04.4	196°27.4	16.3'	01°24.3	-12.3' 54.3'
22	150°52.5	05.1	211°02.8	16.3'	01°12.0	-12.3' 54.3'
23	165°52.5	05.8	225°38.1	16.4'	00°59.7	-12.3' 54.3'
SD = 15.8' d = 0.7' SD = 14.9'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	180°52.6	N17°06.4	240°13.5	16.4'	S00°47.4	-12.3' 54.3'
1	195°52.6	07.1	254°48.8	16.4'	00°35.0	-12.3' 54.3'
2	210°52.6	07.8	269°24.2	16.4'	00°22.7	-12.3' 54.3'
3	225°52.7	· · 08.5	283°59.6	16.4'	S00°10.4	-12.3' 54.2'
4	240°52.7	09.1	298°35.1	16.4'	N00°01.9	12.3' 54.2'
5	255°52.7	09.8	313°10.5	16.4'	00°14.2	12.3' 54.2'
6	270°52.8	N17°10.5	327°45.9	16.4'	N00°26.5	12.3' 54.2'
7	285°52.8	11.2	342°21.3	16.4'	00°38.8	12.3' 54.2'
8	300°52.8	11.8	356°56.8	16.5'	00°51.1	12.3' 54.2'
9	315°52.9	· · 12.5	11°32.2	16.5'	01°03.4	12.3' 54.2'
10	330°52.9	13.2	26°07.7	16.5'	01°15.7	12.3' 54.2'
11	345°52.9	13.9	40°43.2	16.5'	01°27.9	12.3' 54.1'
12	0°53.0	N17°14.5	55°18.6	16.5'	N01°40.2	12.2' 54.1'
13	15°53.0	15.2	69°54.1	16.5'	01°52.4	12.2' 54.1'
14	30°53.0	15.9	84°29.6	16.5'	02°04.7	12.2' 54.1'
15	45°53.1	· · 16.5	99°05.0	16.5'	02°16.9	12.2' 54.1'
16	60°53.1	17.2	113°40.5	16.5'	02°29.1	12.2' 54.1'
17	75°53.1	17.9	128°16.0	16.5'	02°41.3	12.2' 54.1'
18	90°53.2	N17°18.5	142°51.4	16.5'	N02°53.5	12.2' 54.1'
19	105°53.2	19.2	157°26.9	16.5'	03°05.7	12.2' 54.1'
20	120°53.2	19.9	172°02.4	16.5'	03°17.8	12.1' 54.1'
21	135°53.2	· · 20.5	186°37.8	16.4'	03°30.0	12.1' 54.1'
22	150°53.3	21.2	201°13.3	16.4'	03°42.1	12.1' 54.1'
23	165°53.3	21.9	215°48.7	16.4'	03°54.2	12.1' 54.0'
SD = 15.8' d = 0.7' SD = 14.8'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	180°53.3	N17°22.5	230°24.1	16.4'	N04°06.3	12.1' 54.0'
1	195°53.4	23.2	244°59.5	16.4'	04°18.4	12.1' 54.0'
2	210°53.4	23.9	259°34.9	16.4'	04°30.5	12.0' 54.0'
3	225°53.4	· · 24.5	274°10.3	16.4'	04°42.5	12.0' 54.0'
4	240°53.4	25.2	288°45.7	16.4'	04°54.5	12.0' 54.0'
5	255°53.5	25.9	303°21.1	16.4'	05°06.5	12.0' 54.0'
6	270°53.5	N17°26.5	317°56.5	16.3'	N05°18.5	12.0' 54.0'
7	285°53.5	27.2	332°31.8	16.3'	05°30.5	11.9' 54.0'
8	300°53.5	27.9	347°07.2	16.3'	05°42.4	11.9' 54.0'
9	315°53.6	· · 28.5	1°42.5	16.3'	05°54.3	11.9' 54.0'
10	330°53.6	29.2	16°17.8	16.3'	06°06.2	11.9' 54.0'
11	345°53.6	29.8	30°53.1	16.3'	06°18.1	11.8' 54.0'
12	0°53.6	N17°30.5	45°28.3	16.2'	N06°29.9	11.8' 54.0'
13	15°53.7	31.2	60°03.6	16.2'	06°41.7	11.8' 54.0'
14	30°53.7	31.8	74°38.8	16.2'	06°53.5	11.8' 54.0'
15	45°53.7	· · 32.5	89°14.0	16.2'	07°05.3	11.7' 54.0'
16	60°53.7	33.1	103°49.2	16.2'	07°17.0	11.7' 54.0'
17	75°53.8	33.8	118°24.3	16.1'	07°28.8	11.7' 53.9'
18	90°53.8	N17°34.4	132°59.4	16.1'	N07°40.4	11.7' 53.9'
19	105°53.8	35.1	147°34.6	16.1'	07°52.1	11.6' 53.9'
20	120°53.8	35.8	162°09.6	16.1'	08°03.7	11.6' 53.9'
21	135°53.9	· · 36.4	176°44.7	16.0'	08°15.3	11.6' 53.9'
22	150°53.9	37.1	191°19.7	16.0'	08°26.9	11.5' 53.9'
23	165°53.9	37.7	205°54.7	16.0'	08°38.4	11.5' 53.9'
SD = 15.8' d = 0.7' SD = 14.7'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	00:16	—:—	—:—	—:—
N 70°	—:—	—:—	01:44	22:15	—:—	—:—
68°	—:—	—:—	02:23	21:34	—:—	—:—
66°	—:—	01:01	02:49	21:07	23:02	—:—
64°	—:—	01:49	03:09	20:46	22:08	—:—
62°	—:—	02:19	03:25	20:30	21:37	—:—
60°	00:54	02:41	03:39	20:16	21:14	23:07
N 58°	01:38	02:59	03:50	20:04	20:56	22:19
56°	02:05	03:13	04:01	19:54	20:41	21:50
54°	02:26	03:26	04:09	19:45	20:29	21:29
52°	02:43	03:37	04:17	19:37	20:17	21:12
50°	02:57	03:46	04:24	19:30	20:08	20:58
45°	03:24	04:06	04:39	19:14	19:48	20:30
N 40°	03:45	04:22	04:52	19:02	19:32	20:09
35°	04:01	04:35	05:02	18:51	19:19	19:52
30°	04:15	04:46	05:11	18:42	19:08	19:39
20°	04:36	05:04	05:27	18:26	18:50	19:17
N 10°	04:53	05:18	05:40	18:13	18:35	19:01
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:18	05:44	06:05	17:47	18:09	18:35
20°	05:29	05:56	06:19	17:34	17:57	18:24
30°	05:40	06:08	06:34	17:19	17:44	18:13
35°	05:45	06:16	06:42	17:10	17:37	18:07
40°	05:51	06:23	06:52	17:01	17:29	18:02
45°	05:57	06:32	07:03	16:49	17:20	17:56
S 50°	06:03	06:42	07:17	16:35	17:10	17:49
52°	06:06	06:47	07:24	16:29	17:05	17:46
54°	06:09	06:52	07:31	16:21	17:00	17:43
56°	06:12	06:57	07:39	16:13	16:55	17:40
58°	06:16	07:04	07:48	16:04	16:49	17:36
S 60°	06:20	07:10	07:58	15:54	16:42	17:33

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	02:30	02:08	01:47	13:02	14:50	16:39
N 70°	02:23	02:08	01:53	13:07	14:47	16:28
68°	02:17	02:07	01:58	13:10	14:44	16:19
66°	02:12	02:07	02:02	13:13	14:42	16:11
64°	02:07	02:06	02:05	13:16	14:41	16:05
62°	02:04	02:06	02:08	13:18	14:39	16:00
60°	02:00	02:06	02:11	13:20	14:38	15:55
N 58°	01:57	02:06	02:13	13:22	14:36	15:51
56°	01:55	02:05	02:16	13:23	14:35	15:47
54°	01:53	02:05	02:18	13:25	14:34	15:44
52°	01:50	02:05	02:19	13:26	14:34	15:41
50°	01:48	02:05	02:21	13:27	14:33	15:38
45°	01:44	02:04	02:24	13:29	14:31	15:32
N 40°	01:41	02:04	02:27	13:31	14:30	15:28
35°	01:38	02:04	02:30	13:33	14:28	15:23
30°	01:35	02:04	02:32	13:35	14:27	15:20
20°	01:30	02:03	02:36	13:37	14:25	15:14
N 10°	01:26	02:03	02:40	13:40	14:24	15:08
0°	01:22	02:03	02:43	13:42	14:22	15:03
S 10°	01:18	02:03	02:47	13:44	14:21	14:58
20°	01:14	02:02	02:50	13:46	14:19	14:52
30°	01:09	02:02	02:54	13:49	14:17	14:46
35°	01:06	02:02	02:57	13:50	14:16	14:42
40°	01:03	02:02	03:00	13:52	14:15	14:38
45°	00:59	02:01	03:03	13:53	14:13	14:34
S 50°	00:55	02:01	03:07	13:56	14:12	14:28
52°	00:53	02:01	03:08	13:57	14:11	14:26
54°	00:51	02:01	03:10	13:58	14:10	14:23
56°	00:48	02:01	03:13	13:59	14:09	14:20
58°	00:45	02:01	03:15	14:00	14:08	14:16
S 60°	00:42	02:01	03:18	14:02	14:07	14:12

Day	Sun		Mer. Pass	Moon		Age 24-26 35-18%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
07	03:27	03:29	11:57	07:32	19:52	
08	03:30	03:32	11:56	08:13	20:33	
09	03:33	03:35	11:56	08:53	21:13	

Aries Venus

<https://pypi.org/project/sfalmanac/>

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°53.9	N17°38.4	220°29.7	15.9'	N08°49.9	11.5'	53.9'
1	195°53.9	39.0	235°04.6	15.9'	09°01.4	11.4'	53.9'
2	210°54.0	39.7	249°39.6	15.9'	09°12.8	11.4'	53.9'
3	225°54.0	· · 40.3	264°14.4	15.9'	09°24.2	11.4'	53.9'
4	240°54.0	41.0	278°49.3	15.8'	09°35.5	11.3'	53.9'
5	255°54.0	41.6	293°24.1	15.8'	09°46.9	11.3'	53.9'
6	270°54.1	N17°42.3	307°58.9	15.7'	N09°58.2	11.2'	53.9'
7	285°54.1	42.9	322°33.6	15.7'	10°09.4	11.2'	53.9'
8	300°54.1	43.6	337°08.4	15.7'	10°20.6	11.2'	53.9'
9	315°54.1	· · 44.2	351°43.0	15.6'	10°31.8	11.1'	53.9'
10	330°54.1	44.9	6°17.7	15.6'	10°42.9	11.1'	53.9'
11	345°54.2	45.5	20°52.3	15.6'	10°54.0	11.1'	53.9'
12	0°54.2	N17°46.2	35°26.9	15.5'	N11°05.1	11.0'	53.9'
13	15°54.2	46.8	50°01.4	15.5'	11°16.1	11.0'	53.9'
14	30°54.2	47.5	64°35.9	15.5'	11°27.0	10.9'	53.9'
15	45°54.2	· · 48.1	79°10.3	15.4'	11°38.0	10.9'	53.9'
16	60°54.2	48.8	93°44.7	15.4'	11°48.8	10.8'	53.9'
17	75°54.3	49.4	108°19.1	15.3'	11°59.7	10.8'	53.9'
18	90°54.3	N17°50.1	122°53.4	15.3'	N12°10.5	10.7'	53.9'
19	105°54.3	50.7	137°27.7	15.2'	12°21.2	10.7'	53.9'
20	120°54.3	51.3	152°02.0	15.2'	12°31.9	10.7'	53.9'
21	135°54.3	· · 52.0	166°36.1	15.2'	12°42.6	10.6'	53.9'
22	150°54.3	52.6	181°10.3	15.1'	12°53.2	10.6'	54.0'
23	165°54.4	53.3	195°44.4	15.1'	13°03.7	10.5'	54.0'
SD = 15.8'			SD = 14.7'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°54.4	N17°53.9	210°18.5	15.0'	N13°14.2	10.5'	54.0'
1	195°54.4	54.5	224°52.5	15.0'	13°24.7	10.4'	54.0'
2	210°54.4	55.2	239°26.5	14.9'	13°35.1	10.3'	54.0'
3	225°54.4	· · 55.8	254°00.4	14.9'	13°45.4	10.3'	54.0'
4	240°54.4	56.5	268°34.2	14.8'	13°55.7	10.2'	54.0'
5	255°54.5	57.1	283°08.1	14.8'	14°06.0	10.2'	54.0'
6	270°54.5	N17°57.7	297°41.8	14.7'	N14°16.2	10.1'	54.0'
7	285°54.5	58.4	312°15.6	14.7'	14°26.3	10.1'	54.0'
8	300°54.5	59.0	326°49.2	14.6'	14°36.4	10.0'	54.0'
9	315°54.5	17°59.7	341°22.9	14.6'	14°46.4	10.0'	54.0'
10	330°54.5	18°00.3	355°56.4	14.5'	14°56.3	9.9'	54.0'
11	345°54.5	00.9	10°29.9	14.5'	15°06.2	9.8'	54.0'
12	0°54.6	N18°01.6	25°03.4	14.4'	N15°16.1	9.8'	54.0'
13	15°54.6	02.2	39°36.8	14.4'	15°25.9	9.7'	54.0'
14	30°54.6	02.8	54°10.2	14.3'	15°35.6	9.7'	54.0'
15	45°54.6	· · 03.5	68°43.5	14.2'	15°45.3	9.6'	54.0'
16	60°54.6	04.1	83°16.7	14.2'	15°54.9	9.5'	54.0'
17	75°54.6	04.7	97°49.9	14.1'	16°04.4	9.5'	54.0'
18	90°54.6	N18°05.4	112°23.1	14.1'	N16°13.9	9.4'	54.0'
19	105°54.6	06.0	126°56.2	14.0'	16°23.3	9.3'	54.1'
20	120°54.6	06.6	141°29.2	14.0'	16°32.6	9.3'	54.1'
21	135°54.7	· · 07.3	156°02.1	13.9'	16°41.9	9.2'	54.1'
22	150°54.7	07.9	170°35.1	13.9'	16°51.1	9.1'	54.1'
23	165°54.7	08.5	185°07.9	13.8'	17°00.3	9.1'	54.1'
SD = 15.8'			SD = 14.7'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°54.7	N18°09.1	199°40.7	13.7'	N17°09.3	9.0'	54.1'
1	195°54.7	09.8	214°13.5	13.7'	17°18.3	8.9'	54.1'
2	210°54.7	10.4	228°46.1	13.6'	17°27.3	8.9'	54.1'
3	225°54.7	· · 11.0	243°18.8	13.6'	17°36.1	8.8'	54.1'
4	240°54.7	11.6	257°51.3	13.5'	17°44.9	8.7'	54.1'
5	255°54.7	12.3	272°23.8	13.4'	17°53.6	8.6'	54.1'
6	270°54.7	N18°12.9	286°56.3	13.4'	N18°02.3	8.6'	54.1'
7	285°54.7	13.5	301°28.7	13.3'	18°10.8	8.5'	54.1'
8	300°54.8	14.2	316°01.0	13.3'	18°19.3	8.4'	54.2'
9	315°54.8	· · 14.8	330°33.3	13.2'	18°27.8	8.3'	54.2'
10	330°54.8	15.4	345°05.5	13.1'	18°36.1	8.3'	54.2'
11	345°54.8	16.0	359°37.6	13.1'	18°44.4	8.2'	54.2'
12	0°54.8	N18°16.6	14°09.7	13.0'	N18°52.5	8.1'	54.2'
13	15°54.8	17.3	28°41.7	13.0'	19°00.6	8.0'	54.2'
14	30°54.8	17.9	43°13.7	12.9'	19°08.7	7.9'	54.2'
15	45°54.8	· · 18.5	57°45.6	12.8'	19°16.6	7.9'	54.2'
16	60°54.8	19.1	72°17.4	12.8'	19°24.5	7.8'	54.2'
17	75°54.8	19.7	86°49.2	12.7'	19°32.2	7.7'	54.2'
18	90°54.8	N18°20.4	101°20.9	12.7'	N19°39.9	7.6'	54.2'
19	105°54.8	21.0	115°52.6	12.6'	19°47.5	7.5'	54.3'
20	120°54.8	21.6	130°24.2	12.5'	19°55.1	7.4'	54.3'
21	135°54.8	· · 22.2	144°55.7	12.5'	20°02.5	7.4'	54.3'
22	150°54.8	22.8	159°27.2	12.4'	20°09.9	7.3'	54.3'
23	165°54.8	23.5	173°58.6	12.3'	20°17.1	7.2'	54.3'
SD = 15.8'			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	—	—	01:21	22:40	—	—
68°	—	—	02:07	21:50	—	—
66°	—	00:18	02:37	21:19	—	—
64°	—	01:32	02:59	20:56	22:25	—
62°	—	02:07	03:17	20:38	21:49	—
60°	00:16	02:31	03:31	20:23	21:24	—
N 58°	01:23	02:50	03:44	20:10	21:04	22:35
56°	01:54	03:06	03:54	20:00	20:48	22:01
54°	02:17	03:19	04:04	19:50	20:35	21:38
52°	02:35	03:31	04:12	19:42	20:23	21:19
50°	02:50	03:41	04:20	19:34	20:13	21:04
45°	03:19	04:02	04:36	19:18	19:52	20:34
N 40°	03:41	04:18	04:49	19:05	19:35	20:13
35°	03:58	04:32	05:00	18:54	19:22	19:55
30°	04:12	04:43	05:09	18:44	19:10	19:41
20°	04:34	05:02	05:26	18:27	18:51	19:19
N 10°	04:52	05:18	05:40	18:13	18:35	19:01
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:18	05:44	06:06	17:47	18:09	18:34
20°	05:30	05:57	06:20	17:33	17:56	18:22
30°	05:41	06:10	06:35	17:17	17:42	18:11
35°	05:47	06:18	06:45	17:08	17:35	18:05
40°	05:53	06:26	06:55	16:57	17:26	17:59
45°	06:00	06:35	07:07	16:45	17:17	17:52
S 50°	06:07	06:46	07:21	16:31	17:06	17:45
52°	06:10	06:51	07:28	16:24	17:01	17:42
54°	06:13	06:57	07:36	16:16	16:56	17:39
56°	06:17	07:02	07:44	16:08	16:50	17:35
58°	06:21	07:09	07:54	15:58	16:43	17:31
S 60°	06:25	07:16	08:05	15:47	16:36	17:27

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	01:24	00:53 23:58	□	18:36	21:03	□
N 70°	01:37	01:18	00:50	18:14	20:12	□
68°	01:48	01:37	01:23	17:56	19:41	21:39
66°	01:57	01:52	01:47	17:42	19:18	20:59
64°	02:04	02:04	02:05	17:31	19:00	20:31
62°	02:11	02:15	02:21	17:21	18:45	20:10
60°	02:17	02:24	02:34	17:13	18:32	19:53
N 58°	02:22	02:32	02:45	17:06	18:22	19:38
56°	02:27	02:39	02:55	16:59	18:13	19:26
54°	02:31	02:46	03:04	16:54	18:04	19:15
52°	02:34	02:51	03:12	16:49	17:57	19:06
50°	02:38	02:57	03:19	16:44	17:51	18:57
45°	02:45	03:08	03:34	16:34	17:37	18:39
N 40°	02:51	03:17	03:46	16:26	17:25	18:25
35°	02:57	03:25	03:57	16:19	17:15	18:13
30°	03:02	03:33	04:07	16:13	17:07	18:02
20°	03:10	03:45	04:23	16:02	16:52	17:44
N 10°	03:17	03:56	04:37	15:53	16:39	17:28
0°	03:24	04:06	04:50	15:44	16:27	17:13
S 10°	03:31	04:17	05:04	15:36	16:15	16:58
20°	03:38	04:28	05:18	15:26	16:03	16:42
30°	03:47	04:41	05:35	15:16	15:48	16:24
35°	03:52	04:48	05:45	15:10	15:40	16:14
40°	03:58	04:56	05:56	15:03	15:30	16:02
45°	04:04	05:06	06:09	14:55	15:19	15:47
S 50°	04:12	05:19	06:25	14:46	15:06	15:30
52°	04:16	05:24	06:33	14:41	15:00	15:22
54°	04:20	05:30	06:41	14:37	14:53	15:13
56°	04:25	05:37	06:51	14:31	14:45	15:03
58°	04:30	05:45	07:02	14:25	14:37	14:52
S 60°	04:35	05:54	07:14	14:19	14:27	14:39

Day	Sun		Mer. Pass hh:mm	Moon		Age 27-29 11-2%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
10	03:36	03:37	11:56	09:34	21:55	
11	03:38	03:38	11:56	10:17	22:39	
12	03:39	03:39	11:56	11:02	23:25	

May 13, 14, 15 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	231°02.4	167°29.8	N21°23.4	54°50.8	N03°10.3	32°56.4	S06°02.7	187°18.2	N14°29.7	Alpheratz	357°31.3	29°15.1
1	246°04.9	182°29.0	24.0	69°53.1	10.1	47°59.1	02.6	202°20.4	29.8	Ankaa	353°04.1	-42°08.6
2	261°07.3	197°28.2	24.7	84°55.4	09.9	63°01.8	02.6	217°22.5	29.9	Schedar	349°27.3	56°41.7
3	276°09.8	212°27.5	· · 25.3	99°57.7	· · 09.7	78°04.5	· · 02.5	232°24.7	· · 30.0	Diphda	348°44.1	-17°49.5
4	291°12.2	227°26.7	25.9	115°00.0	09.6	93°07.1	02.4	247°26.8	30.1	Achernar	335°18.3	-57°05.1
5	306°14.7	242°25.9	26.6	130°02.3	09.4	108°09.8	02.3	262°29.0	30.2	Hamal	327°47.5	23°36.0
6	321°17.2	257°25.2	N21°27.2	145°04.6	N03°09.2	123°12.5	S06°02.3	277°31.1	N14°30.3	Polaris	312°35.9	89°23.1
7	336°19.6	272°24.4	27.9	160°06.9	09.0	138°15.2	02.2	292°33.2	30.4	Acamar	315°09.6	-40°11.2
8	351°22.1	287°23.6	28.5	175°09.2	08.8	153°17.8	02.1	307°35.4	30.5	Menkar	314°02.8	4°12.2
9	6°24.6	302°22.9	· · 29.1	190°11.5	· · 08.7	168°20.5	· · 02.0	322°37.5	· · 30.6	Mirfak	308°23.6	49°57.8
10	21°27.0	317°22.1	29.8	205°13.8	08.5	183°23.2	02.0	337°39.7	30.7	Aldebaran	290°35.9	16°33.9
11	36°29.5	332°21.3	30.4	220°16.1	08.3	198°25.9	01.9	352°41.8	30.8	Rigel	281°00.8	-8°10.3
12	51°32.0	347°20.6	N21°31.0	235°18.4	N03°08.1	213°28.5	S06°01.8	7°44.0	N14°30.9	Capella	280°17.1	46°01.5
13	66°34.4	2°19.8	31.7	250°20.7	07.9	228°31.2	01.7	22°46.1	30.9	Bellatrix	278°19.4	6°22.4
14	81°36.9	17°19.0	32.3	265°23.0	07.7	243°33.9	01.7	37°48.3	31.0	Elnath	277°57.8	28°37.8
15	96°39.4	32°18.3	· · 32.9	280°25.2	· · 07.6	258°36.5	· · 01.6	52°50.4	· · 31.1	Alnilam	275°34.5	-1°11.2
16	111°41.8	47°17.5	33.6	295°27.5	07.4	273°39.2	01.5	67°52.6	31.2	Betelgeuse	270°48.6	7°24.6
17	126°44.3	62°16.7	34.2	310°29.8	07.2	288°41.9	01.4	82°54.7	31.3	Canopus	263°51.2	-52°43.0
18	141°46.7	77°16.0	N21°34.8	325°32.1	N03°07.0	303°44.6	S06°01.3	97°56.9	N14°31.4	Sirius	258°23.4	-16°45.7
19	156°49.2	92°15.2	35.4	340°34.4	06.8	318°47.2	01.3	112°59.0	31.5	Adhara	255°03.4	-29°01.0
20	171°51.7	107°14.4	36.1	355°36.7	06.6	333°49.9	01.2	128°01.2	31.6	Procyon	244°47.4	5°08.8
21	186°54.1	122°13.7	· · 36.7	10°38.9	· · 06.5	348°52.6	· · 01.1	143°03.3	· · 31.7	Pollux	243°13.3	27°57.2
22	201°56.6	137°12.9	37.3	25°41.2	06.3	3°55.2	01.0	158°05.5	31.8	Avior	234°13.4	-59°36.6
23	216°59.1	152°12.1	37.9	40°43.5	06.1	18°57.9	01.0	173°07.6	31.9	Suhail	222°43.7	-43°33.5
Mer.pass. 08:34		ν -0.8' d 0.6' m -3.90		ν 2.3' d -0.2' m -0.42		ν 2.7' d -0.1' m -2.39		ν 2.1' d 0.1' m 0.29		Miaplacidus	221°37.4	-69°50.7
										Alphard	217°44.3	-8°47.4
										Regulus	207°30.7	11°49.3
										Dubhe	193°36.7	61°35.7
										Denebola	182°21.2	14°24.4
										Gienah	175°39.7	-17°42.5
										Acrux	172°55.5	-63°16.1
										Gacrux	171°47.1	-57°17.0
										Alioth	166°09.6	55°48.1
										Spica	158°18.3	-11°19.0
										Alkaid	152°48.9	49°10.0
										Hadar	148°30.2	-60°31.0
										Menkent	147°53.0	-36°31.0
										Arcturus	145°44.4	19°01.7
										Rigel Kent.	139°34.7	-60°57.5
										Kochab	137°19.0	74°02.1
										Zuben'ubi	136°51.8	-16°09.9
										Alphecca	126°00.4	26°36.9
										Antares	112°11.2	-26°29.8
										Atria	107°01.6	-69°04.6
										Sabik	101°58.4	-15°45.6
										Shaula	96°05.2	-37°07.3
										Rasalhague	95°55.0	12°32.3
										Eltanin	90°40.1	51°29.0
										Kaus Aust.	83°27.6	-34°22.0
										Vega	80°30.5	38°48.6
										Nunki	75°43.2	-26°15.4
										Altair	61°56.4	8°56.8
										Peacock	53°00.2	-56°38.0
										Deneb	49°23.2	45°23.0
										Enif	33°35.4	10°00.6
										Al Na'ir	27°28.7	-46°48.8
										Fomalhaut	15°10.9	-29°27.8
										Scheat	13°41.9	28°14.4
										Markab	13°26.5	15°21.8
										May 13 Sun	SHA	Mer.pass
										Venus	296°27.4	12:51
										Mars	183°48.4	20:18
										Jupiter	161°54.0	21:44
										Saturn	316°15.8	11:29
										May 14 Mon	SHA	Mer.pass
										Venus	295°09.8	12:52
										Mars	183°44.2	20:14
										Jupiter	161°59.1	21:40
										Saturn	316°08.2	11:26
										May 15 Tue	SHA	Mer.pass
										Venus	293°52.0	12:53
										Mars	183°39.4	20:10
										Jupiter	162°03.9	21:36
										Saturn	316°00.7	11:22

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°54.8	N18°24.1	188°29.9	12.3'	N20°24.3	7.1'	54.3'						
1	195°54.9	24.7	203°01.2	12.2'	20°31.4	7.0'	54.3'						
2	210°54.9	25.3	217°32.5	12.2'	20°38.4	6.9'	54.3'						
3	225°54.9	25.9	232°03.6	12.1'	20°45.3	6.8'	54.3'						
4	240°54.9	26.5	246°34.7	12.0'	20°52.1	6.7'	54.4'						
5	255°54.9	27.1	261°05.8	12.0'	20°58.8	6.6'	54.4'						
6	270°54.9	N18°27.8	275°36.8	11.9'	N21°05.4	6.5'	54.4'						
7	285°54.9	28.4	290°07.7	11.9'	21°12.0	6.4'	54.4'						
8	300°54.9	29.0	304°38.5	11.8'	21°18.4	6.3'	54.4'						
9	315°54.9	29.6	319°09.3	11.7'	21°24.7	6.2'	54.4'						
10	330°54.9	30.2	333°40.1	11.7'	21°31.0	6.1'	54.4'						
11	345°54.9	30.8	348°10.8	11.6'	21°37.1	6.0'	54.4'						
12	0°54.9	N18°31.4	2°41.4	11.6'	N21°43.2	5.9'	54.5'						
13	15°54.9	32.0	17°11.9	11.5'	21°49.1	5.8'	54.5'						
14	30°54.9	32.6	31°42.4	11.4'	21°55.0	5.7'	54.5'						
15	45°54.9	33.2	46°12.9	11.4'	22°00.7	5.6'	54.5'						
16	60°54.9	33.8	60°43.3	11.3'	22°06.3	5.5'	54.5'						
17	75°54.9	34.5	75°13.6	11.3'	22°11.9	5.4'	54.5'						
18	90°54.9	N18°35.1	89°43.9	11.2'	N22°17.3	5.3'	54.5'						
19	105°54.9	35.7	104°14.1	11.2'	22°22.7	5.2'	54.5'						
20	120°54.9	36.3	118°44.3	11.1'	22°27.9	5.1'	54.6'						
21	135°54.9	36.9	133°14.4	11.0'	22°33.0	5.0'	54.6'						
22	150°54.9	37.5	147°44.4	11.0'	22°38.0	4.9'	54.6'						
23	165°54.9	38.1	162°14.4	10.9'	22°43.0	4.8'	54.6'						
SD = 15.8'							SD = 14.8'						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°54.9	N18°38.7	176°44.3	10.9'	N22°47.8	4.7'	54.6'						
1	195°54.9	39.3	191°14.2	10.8'	22°52.5	4.6'	54.6'						
2	210°54.9	39.9	205°44.1	10.8'	22°57.1	4.5'	54.6'						
3	225°54.9	40.5	220°13.8	10.7'	23°01.5	4.4'	54.7'						
4	240°54.9	41.1	234°43.6	10.7'	23°05.9	4.3'	54.7'						
5	255°54.9	41.7	249°13.2	10.6'	23°10.2	4.1'	54.7'						
6	270°54.9	N18°42.3	263°42.8	10.6'	N23°14.3	4.0'	54.7'						
7	285°54.8	42.9	278°12.4	10.5'	23°18.4	3.9'	54.7'						
8	300°54.8	43.5	292°41.9	10.5'	23°22.3	3.8'	54.7'						
9	315°54.8	44.1	307°11.4	10.4'	23°26.1	3.7'	54.7'						
10	330°54.8	44.7	321°40.8	10.4'	23°29.8	3.6'	54.8'						
11	345°54.8	45.3	336°10.2	10.3'	23°33.4	3.5'	54.8'						
12	0°54.8	N18°45.9	350°39.5	10.3'	N23°36.8	3.4'	54.8'						
13	15°54.8	46.5	5°08.8	10.2'	23°40.2	3.2'	54.8'						
14	30°54.8	47.1	19°38.1	10.2'	23°43.4	3.1'	54.8'						
15	45°54.8	47.7	34°07.2	10.1'	23°46.5	3.0'	54.8'						
16	60°54.8	48.3	48°36.4	10.1'	23°49.5	2.9'	54.8'						
17	75°54.8	48.8	63°05.5	10.1'	23°52.4	2.8'	54.9'						
18	90°54.8	N18°49.4	77°34.6	10.0'	N23°55.2	2.6'	54.9'						
19	105°54.8	50.0	92°03.6	10.0'	23°57.8	2.5'	54.9'						
20	120°54.8	50.6	106°32.6	9.9'	24°00.3	2.4'	54.9'						
21	135°54.8	51.2	121°01.5	9.9'	24°02.7	2.3'	54.9'						
22	150°54.8	51.8	135°30.4	9.9'	24°05.0	2.2'	54.9'						
23	165°54.8	52.4	149°59.3	9.8'	24°07.2	2.0'	55.0'						
SD = 15.8'							SD = 14.9'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°54.7	N18°53.0	164°28.1	9.8'	N24°09.2	1.9'	55.0'						
1	195°54.7	53.6	178°56.9	9.8'	24°11.1	1.8'	55.0'						
2	210°54.7	54.2	193°25.6	9.7'	24°12.9	1.7'	55.0'						
3	225°54.7	54.7	207°54.4	9.7'	24°14.6	1.5'	55.0'						
4	240°54.7	55.3	222°23.0	9.7'	24°16.2	1.4'	55.0'						
5	255°54.7	55.9	236°51.7	9.6'	24°17.6	1.3'	55.1'						
6	270°54.7	N18°56.5	251°20.3	9.6'	N24°18.9	1.2'	55.1'						
7	285°54.7	57.1	265°48.9	9.6'	24°20.1	1.0'	55.1'						
8	300°54.7	57.7	280°17.5	9.5'	24°21.1	0.9'	55.1'						
9	315°54.7	58.3	294°46.0	9.5'	24°22.0	0.8'	55.1'						
10	330°54.7	58.8	309°14.5	9.5'	24°22.8	0.7'	55.1'						
11	345°54.6	59.4	323°43.0	9.5'	24°23.5	0.5'	55.2'						
12	0°54.6	N19°00.0	338°11.4	9.4'	N24°24.0	0.4'	55.2'						
13	15°54.6	00.6	352°39.9	9.4'	24°24.5	0.3'	55.2'						
14	30°54.6	01.2	7°08.3	9.4'	24°24.8	0.2'	55.2'						
15	45°54.6	01.8	21°36.7	9.4'	24°24.9	0.0'	55.2'						
16	60°54.6	02.3	36°05.0	9.3'	24°25.0	-0.1'	55.3'						
17	75°54.6	02.9	50°33.4	9.3'	24°24.9	-0.2'	55.3'						
18	90°54.6	N19°03.5	65°01.7	9.3'	N24°24.6	-0.3'	55.3'						
19	105°54.5	04.1	79°30.0	9.3'	24°24.3	-0.5'	55.3'						
20	120°54.5	04.7	93°58.3	9.3'	24°23.8	-0.6'	55.3'						
21	135°54.5	05.2	108°26.6	9.3'	24°23.2	-0.7'	55.3'						
22	150°54.5	05.8	122°54.8	9.2'	24°22.5	-0.9'	55.4'						
23	165°54.5	06.4	137°23.1	9.2'	24°21.6	-1.0'	55.4'						
SD = 15.8'							SD = 15.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	00:50	23:16	☐	☐
68°	☐	☐	01:51	22:06	☐	☐
66°	☐	☐	02:25	21:31	☐	☐
64°	☐	01:14	02:50	21:06	22:46	☐
62°	☐	01:54	03:09	20:46	22:02	☐
60°	☐	02:22	03:24	20:30	21:34	☐
N 58°	01:06	02:42	03:37	20:17	21:12	22:53
56°	01:43	02:59	03:49	20:05	20:55	22:13
54°	02:09	03:13	03:59	19:55	20:41	21:47
52°	02:28	03:25	04:07	19:46	20:29	21:27
50°	02:44	03:36	04:15	19:38	20:18	21:10
45°	03:15	03:58	04:32	19:21	19:56	20:39
N 40°	03:37	04:15	04:46	19:08	19:38	20:16
35°	03:55	04:29	04:57	18:56	19:24	19:58
30°	04:10	04:41	05:07	18:46	19:12	19:43
20°	04:33	05:01	05:24	18:29	18:52	19:20
N 10°	04:51	05:17	05:39	18:14	18:36	19:02
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:19	05:44	06:06	17:46	18:08	18:34
20°	05:31	05:58	06:21	17:32	17:55	18:22
30°	05:43	06:12	06:37	17:15	17:40	18:09
35°	05:49	06:20	06:47	17:06	17:32	18:03
40°	05:56	06:29	06:58	16:55	17:24	17:56
45°	06:03	06:38	07:10	16:42	17:14	17:49
S 50°	06:11	06:50	07:26	16:26	17:02	17:42
52°	06:14	06:55	07:33	16:19	16:57	17:38
54°	06:18	07:01	07:41	16:11	16:51	17:34
56°	06:22	07:07	07:50	16:02	16:45	17:30
58°	06:26	07:14	08:00	15:52	16:38	17:26
S 60°	06:30	07:22	08:12	15:40	16:30	17:22

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°						
N 70°						
68°	00:59					
66°	01:41	01:32		22:48		
64°	02:09	02:18	02:41	22:02	23:25	
62°	02:31	02:48	03:19	21:33	22:47	23:43
60°	02:49	03:11	03:46	21:10	22:20	23:16
N 58°	03:03	03:29	04:07	20:52	21:59	22:55
56°	03:16	03:45	04:24	20:37	21:42	22:37
54°	03:27	03:58	04:39	20:24	21:28	22:22
52°	03:37	04:10	04:52	20:13	21:15	22:10
50°	03:46	04:20	05:03	20:03	21:04	21:58
45°	04:05	04:42	05:27	19:41	20:40	21:35
N 40°	04:20	04:59	05:46	19:24	20:22	21:16
35°	04:33	05:14	06:01	19:10	20:06	21:00
30°	04:44	05:27	06:15	18:57	19:53	20:46
20°	05:04	05:49	06:39	18:36	19:30	20:22
N 10°	05:21	06:08	06:59	18:18	19:10	20:02
0°	05:37	06:26	07:18	18:01	18:51	19:43
S 10°	05:53	06:44	07:37	17:44	18:32	19:24
20°	06:11	07:04	07:57	17:25	18:12	19:04
30°	06:31	07:26	08:21	17:04	17:49	18:40
35°	06:42	07:39	08:35	16:52	17:36	18:26
40°	06:56	07:55	08:51	16:38	17:20	18:10
45°	07:12	08:13	09:10	16:21	17:02	17:51
S 50°	07:32	08:35	09:34	16:01	16:39	17:27
52°	07:41	08:46	09:45	15:51	16:28	17:15
54°	07:52	08:59	09:58	15:40	16:15	17:02
56°	08:04	09:13	10:13	15:27	16:01	16:47
58°	08:18	09:29	10:31	15:13	15:44	16:29
S 60°	08:34	09:49	10:53	14:56	15:24	16:07

May 16, 17, 18 UT (Wed., Thu., Fri.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	233°59.8	166°33.8	N22°07.1	57°33.8	N02°56.4	36°08.5	S05°57.4	189°52.9	N14°36.3					Alpheratz	357°31.3	29°15.1	
1	249°02.3	181°33.0	07.7	72°36.0	56.2	51°11.2	57.4	204°55.0	36.4					Ankaa	353°04.1	-42°08.6	
2	264°04.7	196°32.2	08.3	87°38.2	56.0	66°13.8	57.3	219°57.2	36.5					Schedar	349°27.2	56°41.7	
3	279°07.2	211°31.4	•• 08.8	102°40.4	•• 55.8	81°16.5	•• 57.2	234°59.3	•• 36.6					Diphda	348°44.1	-17°49.4	
4	294°09.7	226°30.6	09.4	117°42.6	55.6	96°19.1	57.1	250°01.5	36.7					Achernar	335°18.3	-57°05.1	
5	309°12.1	241°29.8	10.0	132°44.9	55.4	111°21.8	57.1	265°03.6	36.8					Hamal	327°47.5	23°36.0	
6	324°14.6	256°29.0	N22°10.5	147°47.1	N02°55.2	126°24.5	S05°57.0	280°05.8	N14°36.9					Polaris	312°35.3	89°23.1	
7	339°17.1	271°28.2	11.1	162°49.3	54.9	141°27.1	56.9	295°07.9	37.0					Acamar	315°09.6	-40°11.2	
8	354°19.5	286°27.4	11.7	177°51.5	54.7	156°29.8	56.9	310°10.1	37.1					Menkar	314°02.8	4°12.2	
9	9°22.0	301°26.6	•• 12.2	192°53.7	•• 54.5	171°32.4	•• 56.8	325°12.2	•• 37.2					Mirfak	308°23.6	49°57.8	
10	24°24.4	316°25.8	12.8	207°55.9	54.3	186°35.1	56.7	340°14.4	37.3					Aldebaran	290°35.9	16°33.9	
11	39°26.9	331°25.0	13.3	222°58.2	54.1	201°37.7	56.7	355°16.5	37.3					Rigel	281°00.8	-8°10.3	
12	54°29.4	346°24.2	N22°13.9	238°00.4	N02°53.9	216°40.4	S05°56.6	10°18.7	N14°37.4					Capella	280°17.1	46°01.5	
13	69°31.8	1°23.4	14.5	253°02.6	53.7	231°43.1	56.5	25°20.8	37.5					Bellatrix	278°19.4	6°22.4	
14	84°34.3	16°22.6	15.0	268°04.8	53.5	246°45.7	56.5	40°23.0	37.6					Elnath	277°57.8	28°37.8	
15	99°36.8	31°21.8	•• 15.6	283°07.0	•• 53.2	261°48.4	•• 56.4	55°25.1	•• 37.7					Alnilam	275°34.5	-1°11.2	
16	114°39.2	46°21.0	16.1	298°09.2	53.0	276°51.0	56.3	70°27.3	37.8					Betelgeuse	270°48.6	7°24.6	
17	129°41.7	61°20.2	16.7	313°11.4	52.8	291°53.7	56.2	85°29.4	37.9					Canopus	263°51.3	-52°43.0	
18	144°44.2	76°19.4	N22°17.2	328°13.6	N02°52.6	306°56.3	S05°56.2	100°31.6	N14°38.0					Sirius	258°23.4	-16°45.7	
19	159°46.6	91°18.6	17.8	343°15.8	52.4	321°59.0	56.1	115°33.7	38.1					Adhara	255°03.4	-29°01.0	
20	174°49.1	106°17.8	18.3	358°18.0	52.2	337°01.6	56.0	130°35.9	38.2					Procyon	244°47.4	5°08.8	
21	189°51.6	121°17.0	•• 18.9	13°20.2	•• 52.0	352°04.3	•• 56.0	145°38.0	•• 38.3					Pollux	243°13.3	27°57.2	
22	204°54.0	136°16.2	19.4	28°22.4	51.7	7°07.0	55.9	160°40.2	38.3					Avior	234°13.5	-59°36.6	
23	219°56.5	151°15.4	20.0	43°24.6	51.5	22°09.6	55.8	175°42.3	38.4					Suhail	222°43.7	-43°33.5	
Mer.pass. 08:23		ν -0.8' d 0.6' m -3.90			ν 2.2' d -0.2' m -0.36			ν 2.7' d -0.1' m -2.37			ν 2.1' d 0.1' m 0.31						

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	234°58.9	166°14.6	N22°20.5	58°26.8	N02°51.3	37°12.3	S05°55.8	190°44.5	N14°38.5				
1	250°01.4	181°13.8	21.0	73°29.0	51.1	52°14.9	55.7	205°46.6	38.6				
2	265°03.9	196°13.0	21.6	88°31.2	50.9	67°17.6	55.6	220°48.8	38.7				
3	280°06.3	211°12.2	•• 22.1	103°33.4	•• 50.6	82°20.2	•• 55.6	235°50.9	•• 38.8				
4	295°08.8	226°11.4	22.7	118°35.6	50.4	97°22.9	55.5	250°53.0	38.9				
5	310°11.3	241°10.6	23.2	133°37.8	50.2	112°25.5	55.4	265°55.2	39.0				
6	325°13.7	256°09.8	N22°23.7	148°40.0	N02°50.4	127°28.2	S05°55.4	280°57.3	N14°39.1				
7	340°16.2	271°09.0	24.3	163°42.2	49.8	142°30.8	55.3	295°59.5	39.2				
8	355°18.7	286°08.2	24.8	178°44.4	49.5	157°33.5	55.2	311°01.6	39.2				
9	10°21.1	301°07.4	•• 25.4	193°46.6	•• 49.3	172°36.1	•• 55.2	326°03.8	•• 39.3				
10	25°23.6	316°06.6	25.9	208°48.8	49.1	187°38.8	55.1	341°05.9	39.4				
11	40°26.1	331°05.8	26.4	223°50.9	48.9	202°41.4	55.0	356°08.1	39.5				
12	55°28.5	346°05.0	N22°27.0	238°53.1	N02°48.7	217°44.1	S05°55.0	11°10.2	N14°39.6				
13	70°31.0	1°04.2	27.5	253°55.3	48.4	232°46.7	54.9	26°12.4	39.7				
14	85°33.4	16°03.4	28.0	268°57.5	48.2	247°49.4	54.8	41°14.5	39.8				
15	100°35.9	31°02.6	•• 28.5	283°59.7	•• 48.0	262°52.0	•• 54.8	56°16.7	•• 39.9				
16	115°38.4	46°01.8	29.1	299°01.9	47.8	277°54.7	54.7	71°18.8	40.0				
17	130°40.8	61°00.9	29.6	314°04.0	47.5	292°57.3	54.6	86°21.0	40.1				
18	145°43.3	76°00.1	N22°30.1	329°06.2	N02°47.3	308°00.0	S05°54.6	101°23.1	N14°40.2				
19	160°45.8	90°59.3	30.6	344°08.4	47.1	323°02.6	54.5	116°25.3	40.2				
20	175°48.2	105°58.5	31.2	359°10.6	46.9	338°05.3	54.4	131°27.4	40.3				
21	190°50.7	120°57.7	•• 31.7	14°12.8	•• 46.6	353°07.9	•• 54.4	146°29.6	•• 40.4				
22	205°53.2	135°56.9	32.2	29°14.9	46.4	8°10.6	54.3	161°31.7	40.5				
23	220°55.6	150°56.1	32.7	44°17.1	46.2	23°13.2	54.2	176°33.9	40.6				
Mer.pass. 08:19		ν -0.8' d 0.5' m -3.90			ν 2.2' d -0.2' m -0.34			ν 2.7' d -0.1' m -2.37			ν 2.1' d 0.1' m 0.31		

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	235°58.1	165°55.3	N22°33.2	59°19.3	N02°46.0	38°15.9	S05°54.2	191°36.0	N14°40.7				
1	251°00.6	180°54.5	33.8	74°21.5	45.7	53°18.5	54.1	206°38.2	40.8				
2	266°03.0	195°53.7	34.3	89°23.6	45.5	68°21.2	54.1	221°40.3	40.9				
3	281°05.5	210°52.8	•• 34.8	104°25.8	•• 45.3	83°23.8	•• 54.0	236°42.5	•• 41.0				
4	296°07.9	225°52.0	35.3	119°28.0	45.0	98°26.5	53.9	251°44.6	41.1				
5	311°10.4	240°51.2	35.8	134°30.1	44.8	113°29.1	53.9	266°46.8	41.1				
6	326°12.9	255°50.4	N22°36.3	149°32.3	N02°44.6	128°31.8	S05°53.8	281°48.9	N14°41.2				
7	341°15.3	270°49.6	36.8	164°34.5	44.4	143°34.4	53.7	296°51.1	41.3				
8	356°17.8	285°48.8	37.3	179°36.6	44.1	158°37.0	53.7	311°53.2	41.4				
9	11°20.3	300°48.0	•• 37.9	194°38.8	•• 43.9	173°39.7	•• 53.6	326°55.4	•• 41.5				
10	26°22.7	315°47.1	38.4	209°41.0	43.7	188°42.3	53.5	341°57.5	41.6				
11	41°25.2	330°46.3	38.9	224°43.1	43.4	203°45.0	53.5	356°59.7	41.7				
12	56°27.7	345°45.5	N22°39.4	239°45.3	N02°43.2	218°47.6	S05°53.4	12°01.8	N14°41.8				
13	71°30.1	0°44.7	39.9	254°47.4	43.0	233°50.3	53.3	27°04.0	41.9				
14	86°32.6	15°43.9	40.4	269°49.6	42.7	248°52.9	53.3	42°06.1	42.0				
15	101°35.1	30°43.1	•• 40.9	284°51.8	•• 42.5	263°55.6	•• 53.2	57°08.3	•• 42.0				
16	116°37.5	45°42.3	41.4	299°53.9	42.3	278°58.2	53.2	72°10.4	42.1				
17	131°40.0	60°41.4	41.9	314°56.1	42.0	294°00.8	53.1	87°12.6	42.2				
18	146°42.4	75°40.6	N22°42.4	329°58.2	N02°41.8	309°03.5	S05°53.0	102°14.7	N14°42.3				
19	161°44.9	90°39.8	42.9	345°00.4	41.6	324°06.1	53.0	117°16.9	42.4				
20	176°47.4	105°39.0	43.4	0°02.5	41.3	339°08.8	52.9	132°19.0	42.5				
21	191°49.8	120°38.2	•• 43.9	15°04.7	•• 41.1	354°11.4	•• 52.8	147°21.2	•• 42.6				
22	206°52.3	135°37.3	44.4	30°06.8	40.9	9°14.1	52.8	162°23.3	42.7				
23	221°54.8	150°36.5	44.8	45°09.0	40.6	24°16.7	52.7	177°25.5	42.8				
Mer.pass. 08:15		ν -0.8' d 0.5' m -3.90			ν 2.2' d -0.2' m -0.32			ν 2.6' d -0.1' m -2.36			ν 2.1' d 0.1' m 0.31		

May 16 Wed	SHA	Mer.pass
Venus	292°34.0	12:54
Mars	183°34.0	20:07
Jupiter	162°08.7	21:32
Saturn	315°53.1	11:19

May 17 Thu	SHA	Mer.pass
Venus	291°15.7	12:56
Mars	183°27.9	20:03
Jupiter	162°13.3	21:27
Saturn	315°45.5	11:15

May 18 Fri	SHA	Mer.pass
Venus	289°57.2	12:57
Mars	183°21.2	20:00
Jupiter	162°17.8	21:23
Saturn	315°37.9	11:12

Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°54.5	N19°07.0	151°51.3	9.2'	N24°20.6	-1.1'	55.4'
1	195°54.5	07.5	166°19.5	9.2'	24°19.5	-1.2'	55.4'
2	210°54.5	08.1	180°47.7	9.2'	24°18.3	-1.4'	55.4'
3	225°54.4	· · 08.7	195°15.9	9.2'	24°16.9	-1.5'	55.5'
4	240°54.4	09.3	209°44.1	9.2'	24°15.4	-1.6'	55.5'
5	255°54.4	09.8	224°12.3	9.2'	24°13.8	-1.8'	55.5'
6	270°54.4	N19°10.4	238°40.5	9.2'	N24°12.0	-1.9'	55.5'
7	285°54.4	11.0	253°08.7	9.2'	24°10.1	-2.0'	55.5'
8	300°54.4	11.6	267°36.8	9.2'	24°08.1	-2.2'	55.6'
9	315°54.3	· · 12.1	282°05.0	9.2'	24°05.9	-2.3'	55.6'
10	330°54.3	12.7	296°33.1	9.2'	24°03.7	-2.4'	55.6'
11	345°54.3	13.3	311°01.3	9.2'	24°01.2	-2.5'	55.6'
12	0°54.3	N19°13.8	325°29.5	9.2'	N23°58.7	-2.7'	55.6'
13	15°54.3	14.4	339°57.6	9.2'	23°56.0	-2.8'	55.7'
14	30°54.3	15.0	354°25.8	9.2'	23°53.2	-2.9'	55.7'
15	45°54.2	· · 15.5	8°54.0	9.2'	23°50.3	-3.1'	55.7'
16	60°54.2	16.1	23°22.1	9.2'	23°47.3	-3.2'	55.7'
17	75°54.2	16.7	37°50.3	9.2'	23°44.1	-3.3'	55.7'
18	90°54.2	N19°17.2	52°18.5	9.2'	N23°40.8	-3.4'	55.8'
19	105°54.2	17.8	66°46.7	9.2'	23°37.3	-3.6'	55.8'
20	120°54.2	18.4	81°14.9	9.2'	23°33.8	-3.7'	55.8'
21	135°54.1	· · 18.9	95°43.1	9.2'	23°30.1	-3.8'	55.8'
22	150°54.1	19.5	110°11.3	9.2'	23°26.2	-4.0'	55.8'
23	165°54.1	20.1	124°39.5	9.2'	23°22.3	-4.1'	55.9'
SD = 15.8'			SD = 15.1'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°54.1	N19°20.6	139°07.7	9.2'	N23°18.2	-4.2'	55.9'
1	195°54.1	21.2	153°36.0	9.3'	23°14.0	-4.3'	55.9'
2	210°54.0	21.7	168°04.2	9.3'	23°09.7	-4.5'	55.9'
3	225°54.0	· · 22.3	182°32.5	9.3'	23°05.2	-4.6'	56.0'
4	240°54.0	22.9	197°00.8	9.3'	23°00.6	-4.7'	56.0'
5	255°54.0	23.4	211°29.1	9.3'	22°55.9	-4.8'	56.0'
6	270°54.0	N19°24.0	225°57.4	9.3'	N22°51.1	-5.0'	56.0'
7	285°53.9	24.5	240°25.7	9.4'	22°46.1	-5.1'	56.0'
8	300°53.9	25.1	254°54.1	9.4'	22°41.0	-5.2'	56.1'
9	315°53.9	· · 25.7	269°22.5	9.4'	22°35.8	-5.3'	56.1'
10	330°53.9	26.2	283°50.9	9.4'	22°30.5	-5.5'	56.1'
11	345°53.9	26.8	298°19.3	9.4'	22°25.0	-5.6'	56.1'
12	0°53.8	N19°27.3	312°47.7	9.5'	N22°19.5	-5.7'	56.2'
13	15°53.8	27.9	327°16.1	9.5'	22°13.8	-5.8'	56.2'
14	30°53.8	28.4	341°44.6	9.5'	22°07.9	-5.9'	56.2'
15	45°53.8	· · 29.0	356°13.1	9.5'	22°02.0	-6.1'	56.2'
16	60°53.7	29.5	10°41.6	9.5'	21°55.9	-6.2'	56.2'
17	75°53.7	30.1	25°10.2	9.6'	21°49.7	-6.3'	56.3'
18	90°53.7	N19°30.6	39°38.7	9.6'	N21°43.4	-6.4'	56.3'
19	105°53.7	31.2	54°07.3	9.6'	21°37.0	-6.5'	56.3'
20	120°53.6	31.8	68°36.0	9.6'	21°30.5	-6.7'	56.3'
21	135°53.6	· · 32.3	83°04.6	9.7'	21°23.8	-6.8'	56.4'
22	150°53.6	32.9	97°33.3	9.7'	21°17.0	-6.9'	56.4'
23	165°53.6	33.4	112°02.0	9.7'	21°10.1	-7.0'	56.4'
SD = 15.8'			SD = 15.2'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°53.5	N19°34.0	126°30.7	9.8'	N21°03.1	-7.1'	56.4'
1	195°53.5	34.5	140°59.4	9.8'	20°55.9	-7.3'	56.5'
2	210°53.5	35.0	155°28.2	9.8'	20°48.7	-7.4'	56.5'
3	225°53.5	· · 35.6	169°57.0	9.8'	20°41.3	-7.5'	56.5'
4	240°53.4	36.1	184°25.9	9.9'	20°33.9	-7.6'	56.5'
5	255°53.4	36.7	198°54.7	9.9'	20°26.3	-7.7'	56.5'
6	270°53.4	N19°37.2	213°23.6	9.9'	N20°18.5	-7.8'	56.6'
7	285°53.4	37.8	227°52.6	10.0'	20°10.7	-7.9'	56.6'
8	300°53.3	38.3	242°21.5	10.0'	20°02.8	-8.0'	56.6'
9	315°53.3	· · 38.9	256°50.5	10.0'	19°54.7	-8.2'	56.6'
10	330°53.3	39.4	271°19.5	10.0'	19°46.6	-8.3'	56.7'
11	345°53.3	39.9	285°48.6	10.1'	19°38.3	-8.4'	56.7'
12	0°53.2	N19°40.5	300°17.6	10.1'	N19°30.0	-8.5'	56.7'
13	15°53.2	41.0	314°46.8	10.1'	19°21.5	-8.6'	56.7'
14	30°53.2	41.6	329°15.9	10.2'	19°12.9	-8.7'	56.8'
15	45°53.1	· · 42.1	343°45.1	10.2'	19°04.2	-8.8'	56.8'
16	60°53.1	42.7	358°14.3	10.2'	18°55.4	-8.9'	56.8'
17	75°53.1	43.2	12°43.5	10.3'	18°46.5	-9.0'	56.8'
18	90°53.1	N19°43.7	27°12.8	10.3'	N18°37.4	-9.1'	56.9'
19	105°53.0	44.3	41°42.1	10.3'	18°28.3	-9.2'	56.9'
20	120°53.0	44.8	56°11.4	10.4'	18°19.1	-9.3'	56.9'
21	135°53.0	· · 45.3	70°40.8	10.4'	18°09.8	-9.4'	56.9'
22	150°52.9	45.9	85°10.2	10.4'	18°00.3	-9.5'	57.0'
23	165°52.9	46.4	99°39.6	10.5'	17°50.8	-9.6'	57.0'
SD = 15.8'			SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	--	--	01:34	22:24	--	--
66°	--	--	02:13	21:43	--	--
64°	--	00:50	02:40	21:15	23:12	--
62°	--	01:41	03:01	20:54	22:15	--
60°	--	02:12	03:17	20:37	21:44	--
N 58°	00:45	02:34	03:31	20:23	21:20	23:16
56°	01:32	02:52	03:43	20:11	21:02	22:25
54°	02:00	03:07	03:54	20:00	20:47	21:55
52°	02:21	03:20	04:03	19:51	20:34	21:34
50°	02:38	03:31	04:11	19:43	20:23	21:16
45°	03:10	03:54	04:29	19:25	19:59	20:44
N 40°	03:34	04:12	04:43	19:10	19:41	20:20
35°	03:52	04:27	04:55	18:58	19:27	20:01
30°	04:07	04:39	05:05	18:48	19:14	19:46
20°	04:31	04:59	05:23	18:30	18:54	19:22
N 10°	04:50	05:16	05:39	18:14	18:37	19:03
0°	05:05	05:31	05:53	18:00	18:22	18:47
S 10°	05:19	05:45	06:07	17:46	18:08	18:34
20°	05:32	05:59	06:22	17:31	17:54	18:21
30°	05:45	06:14	06:39	17:13	17:39	18:08
35°	05:51	06:22	06:49	17:03	17:31	18:01
40°	05:58	06:31	07:00	16:52	17:21	17:54
45°	06:06	06:42	07:14	16:39	17:11	17:47
S 50°	06:14	06:54	07:30	16:23	16:59	17:38
52°	06:18	06:59	07:37	16:15	16:53	17:35
54°	06:22	07:05	07:46	16:07	16:47	17:31
56°	06:26	07:12	07:55	15:57	16:40	17:26
58°	06:30	07:20	08:06	15:46	16:33	17:22
S 60°	06:35	07:28	08:18	15:34	16:24	17:17

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	05:21	☐	☐	02:15
66°	☐	04:06	06:04	☐	01:39	01:30
64°	03:32	04:55	06:33	00:23	00:50	01:01
62°	04:12	05:26	06:55	· · ·	00:19	00:38
60°	04:39	05:50	07:12	23:55	· · ·	00:20
N 58°	05:00	06:08	07:27	23:36	· · ·	00:05
56°	05:18	06:24	07:40	23:20	23:52	· · ·
54°	05:32	06:37	07:50	23:07	23:41	· · ·
52°	05:45	06:49	08:00	22:55	23:30	23:59
50°	05:56	06:59	08:09	22:44	23:21	23:52
45°	06:20	07:21	08:27	22:22	23:02	23:36
N 40°	06:39	07:38	08:42	22:04	22:47	23:24
35°	06:55	07:53	08:54	21:49	22:33	23:13
30°	07:08	08:05	09:05	21:36	22:22	23:04
20°	07:31	08:27	09:24	21:13	22:02	22:47
N 10°	07:52	08:46	09:40	20:54	21:44	22:33
0°	08:10	09:03	09:55	20:36	21:28	22:19
S 10°	08:29	09:21	10:10	20:18	21:12	22:06
20°	08:49	09:39	10:26	19:58	20:54	21:51
30°	09:13	10:01	10:44	19:35	20:34	21:34
35°	09:26	10:13	10:55	19:22	20:22	21:25
40°	09:42	10:27	11:07	19:06	20:08	21:13
45°	10:01	10:45	11:22	18:48	19:52	21:00
S 50°	10:24	11:06	11:39	18:25	19:31	20:44
52°	10:35	11:16	11:47	18:14	19:22	20:36
54°	10:48	11:27	11:56	18:01	19:11	20:27
56°	11:03	11:39	12:07	17:47	18:58	20:18
58°	11:20	11:54	12:18	17:30	18:44	20:06
S 60°	11:41	12:12	12:32	17:09	18:27	19:54

Day	Sun		Mer. Pass	Moon		Age 3-5 6-18%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
16	03:38	03:37	11:56	14:23	01:57	
17	03:36	03:35	11:56	15:16	02:49	
18	03:34	03:33	11:56	16:07	03:42	

May 19, 20, 21 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	236°57.2	165°35.7	N22°45.3	60°11.1	N02°40.4	39°19.3	S05°52.7	192°27.6	N14°42.8	Alpheratz	357°31.3	29°15.1
1	251°59.7	180°34.9	45.8	75°13.3	40.2	54°22.0	52.6	207°29.8	42.9	Ankaa	353°04.1	-42°08.6
2	267°02.2	195°34.1	46.3	90°15.4	39.9	69°24.6	52.5	222°31.9	43.0	Schedar	349°27.2	56°41.7
3	282°04.6	210°33.2	· · 46.8	105°17.6	· · 39.7	84°27.3	· · 52.5	237°34.1	· · 43.1	Diphda	348°44.0	-17°49.4
4	297°07.1	225°32.4	47.3	120°19.7	39.4	99°29.9	52.4	252°36.2	43.2	Achernar	335°18.2	-57°05.1
5	312°09.5	240°31.6	47.8	135°21.9	39.2	114°32.5	52.3	267°38.4	43.3	Hamal	327°47.5	23°36.0
6	327°12.0	255°30.8	N22°48.3	150°24.0	N02°39.0	129°35.2	S05°52.3	282°40.5	N14°43.4	Polaris	312°34.9	89°23.1
7	342°14.5	270°30.0	48.7	165°26.1	38.7	144°37.8	52.2	297°42.7	43.5	Acamar	315°09.6	-40°11.2
8	357°16.9	285°29.1	49.2	180°28.3	38.5	159°40.5	52.2	312°44.8	43.6	Menkar	314°02.8	4°12.2
9	12°19.4	300°28.3	· · 49.7	195°30.4	· · 38.2	174°43.1	· · 52.1	327°47.0	· · 43.7	Mirfak	308°23.6	49°57.8
10	27°21.9	315°27.5	50.2	210°32.6	38.0	189°45.7	52.0	342°49.1	43.7	Aldebaran	290°35.9	16°34.0
11	42°24.3	330°26.7	50.7	225°34.7	37.8	204°48.4	52.0	357°51.3	43.8	Rigel	281°00.8	-8°10.2
12	57°26.8	345°25.8	N22°51.1	240°36.8	N02°37.5	219°51.0	S05°51.9	12°53.4	N14°43.9	Capella	280°17.1	46°01.5
13	72°29.3	0°25.0	51.6	255°39.0	37.3	234°53.7	51.9	27°55.6	44.0	Bellatrix	278°19.4	6°22.4
14	87°31.7	15°24.2	52.1	270°41.1	37.0	249°56.3	51.8	42°57.7	44.1	Elnath	277°57.8	28°37.8
15	102°34.2	30°23.4	· · 52.6	285°43.2	· · 36.8	264°58.9	· · 51.7	57°59.9	· · 44.2	Alnilam	275°34.5	-1°11.2
16	117°36.7	45°22.5	53.0	300°45.4	36.5	280°01.6	51.7	73°02.0	44.3	Betelgeuse	270°48.6	7°24.6
17	132°39.1	60°21.7	53.5	315°47.5	36.3	295°04.2	51.6	88°04.2	44.4	Canopus	263°51.3	-52°43.0
18	147°41.6	75°20.9	N22°54.0	330°49.6	N02°36.1	310°06.9	S05°51.5	103°06.3	N14°44.5	Sirius	258°23.4	-16°45.6
19	162°44.0	90°20.1	54.5	345°51.8	35.8	325°09.5	51.5	118°08.5	44.5	Adhara	255°03.4	-29°01.0
20	177°46.5	105°19.2	54.9	0°53.9	35.6	340°12.1	51.4	133°10.6	44.6	Procyon	244°47.4	5°08.8
21	192°49.0	120°18.4	· · 55.4	15°56.0	· · 35.3	355°14.8	· · 51.4	148°12.8	· · 44.7	Pollux	243°13.3	27°57.2
22	207°51.4	135°17.6	55.9	30°58.1	35.1	10°17.4	51.3	163°14.9	44.8	Avior	234°13.5	-59°36.6
23	222°53.9	150°16.8	56.3	46°00.3	34.8	25°20.0	51.2	178°17.1	44.9	Suhail	222°43.8	-43°33.5
Mer.pass. 08:11		ν -0.8' d 0.5' m -3.90		ν 2.1' d -0.2' m -0.30		ν 2.6' d -0.1' m -2.36		ν 2.1' d 0.1' m 0.31				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	237°56.4	165°15.9	N22°56.8	61°02.4	N02°34.6	40°22.7	S05°51.2	193°19.2	N14°45.0
1	252°58.8	180°15.1	57.2	76°04.5	34.3	55°25.3	51.1	208°21.4	45.1
2	268°01.3	195°14.3	57.7	91°06.6	34.1	70°27.9	51.1	223°23.5	45.2
3	283°03.8	210°13.4	· · 58.2	106°08.8	· · 33.8	85°30.6	· · 51.0	238°25.7	· · 45.3
4	298°06.2	225°12.6	58.6	121°10.9	33.6	100°33.2	50.9	253°27.8	45.3
5	313°08.7	240°11.8	59.1	136°13.0	33.4	115°35.8	50.9	268°30.0	45.4
6	328°11.2	255°11.0	N22°59.5	151°15.1	N02°33.1	130°38.5	S05°50.8	283°32.1	N14°45.5
7	343°13.6	270°10.1	23°00.0	166°17.2	32.9	145°41.1	50.8	298°34.3	45.6
8	358°16.1	285°09.3	00.5	181°19.4	32.6	160°43.7	50.7	313°36.4	45.7
9	13°18.5	300°08.5	· · 00.9	196°21.5	· · 32.4	175°46.4	· · 50.7	328°38.6	· · 45.8
10	28°21.0	315°07.6	01.4	211°23.6	32.1	190°49.0	50.6	343°40.7	45.9
11	43°23.5	330°06.8	01.8	226°25.7	31.9	205°51.6	50.5	358°42.9	46.0
12	58°25.9	345°06.0	N23°02.3	241°27.8	N02°31.6	220°54.3	S05°50.5	13°45.0	N14°46.1
13	73°28.4	0°05.1	02.7	256°29.9	31.4	235°56.9	50.4	28°47.2	46.2
14	88°30.9	15°04.3	03.2	271°32.0	31.1	250°59.5	50.4	43°49.3	46.2
15	103°33.3	30°03.5	· · 03.6	286°34.1	· · 30.9	266°02.2	· · 50.3	58°51.5	· · 46.3
16	118°35.8	45°02.6	04.1	301°36.3	30.6	281°04.8	50.2	73°53.6	46.4
17	133°38.3	60°01.8	04.5	316°38.4	30.3	296°07.4	50.2	88°55.8	46.5
18	148°40.7	75°01.0	N23°04.9	331°40.5	N02°30.1	311°10.1	S05°50.1	103°57.9	N14°46.6
19	163°43.2	90°00.1	05.4	346°42.6	29.8	326°12.7	50.1	119°00.1	46.7
20	178°45.7	104°59.3	05.8	1°44.7	29.6	341°15.3	50.0	134°02.2	46.8
21	193°48.1	119°58.5	· · 06.3	16°46.8	· · 29.3	356°18.0	· · 50.0	149°04.4	· · 46.9
22	208°50.6	134°57.6	06.7	31°48.9	29.1	11°20.6	49.9	164°06.5	46.9
23	223°53.0	149°56.8	07.1	46°51.0	28.8	26°23.2	49.8	179°08.7	47.0
Mer.pass. 08:07		ν -0.8' d 0.5' m -3.90		ν 2.1' d -0.2' m -0.28		ν 2.6' d -0.1' m -2.35		ν 2.2' d 0.1' m 0.32	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	238°55.5	164°56.0	N23°07.6	61°53.1	N02°28.6	41°25.8	S05°49.8	194°10.8	N14°47.1
1	253°58.0	179°55.1	08.0	76°55.2	28.3	56°28.5	49.7	209°13.0	47.2
2	269°00.4	194°54.3	08.5	91°57.3	28.1	71°31.1	49.7	224°15.2	47.3
3	284°02.9	209°53.5	· · 08.9	106°59.4	· · 27.8	86°33.7	· · 49.6	239°17.3	· · 47.4
4	299°05.4	224°52.6	09.3	122°01.5	27.5	101°36.4	49.6	254°19.5	47.5
5	314°07.8	239°51.8	09.7	137°03.6	27.3	116°39.0	49.5	269°21.6	47.6
6	329°10.3	254°51.0	N23°10.2	152°05.7	N02°27.0	131°41.6	S05°49.4	284°23.8	N14°47.7
7	344°12.8	269°50.1	10.6	167°07.8	26.8	146°44.2	49.4	299°25.9	47.7
8	359°15.2	284°49.3	11.0	182°09.9	26.5	161°46.9	49.3	314°28.1	47.8
9	14°17.7	299°48.4	· · 11.5	197°11.9	· · 26.3	176°49.5	· · 49.3	329°30.2	· · 47.9
10	29°20.1	314°47.6	11.9	212°14.0	26.0	191°52.1	49.2	344°32.4	48.0
11	44°22.6	329°46.8	12.3	227°16.1	25.7	206°54.8	49.2	359°34.5	48.1
12	59°25.1	344°45.9	N23°12.7	242°18.2	N02°25.5	221°57.4	S05°49.1	14°36.7	N14°48.2
13	74°27.5	359°45.1	13.2	257°20.3	25.2	237°00.0	49.0	29°38.8	48.3
14	89°30.0	14°44.2	13.6	272°22.4	25.0	252°02.6	49.0	44°41.0	48.4
15	104°32.5	29°43.4	· · 14.0	287°24.5	· · 24.7	267°05.3	· · 48.9	59°43.1	· · 48.5
16	119°34.9	44°42.6	14.4	302°26.6	24.4	282°07.9	48.9	74°45.3	48.5
17	134°37.4	59°41.7	14.8	317°28.6	24.2	297°10.5	48.8	89°47.4	48.6
18	149°39.9	74°40.9	N23°15.2	332°30.7	N02°23.9	312°13.1	S05°48.8	104°49.6	N14°48.7
19	164°42.3	89°40.0	15.7	347°32.8	23.6	327°15.8	48.7	119°51.7	48.8
20	179°44.8	104°39.2	16.1	2°34.9	23.4	342°18.4	48.7	134°53.9	48.9
21	194°47.3	119°38.4	· · 16.5	17°37.0	· · 23.1	357°21.0	· · 48.6	149°56.0	· · 49.0
22	209°49.7	134°37.5	16.9	32°39.1	22.9	12°23.6	48.6	164°58.2	49.1
23	224°52.2	149°36.7	17.3	47°41.1	22.6	27°26.3	48.5	180°00.3	49.2
Mer.pass. 08:03		ν -0.8' d 0.4' m -3.90		ν 2.1' d -0.3' m -0.26		ν 2.6' d -0.1' m -2.35		ν 2.2' d 0.1' m 0.32	

May 19 Sat	SHA	Mer.pass
Venus	288°38.5	12:58
Mars	183°13.9	19:56
Jupiter	162°22.1	21:19
Saturn	315°30.4	11:09

May 20 Sun	SHA	Mer.pass
Venus	287°19.6	13:00
Mars	183°06.0	19:53
Jupiter	162°26.3	21:15
Saturn	315°22.9	11:05

May 21 Mon	SHA	Mer.pass
Venus	286°00.5	13:01
Mars	182°57.6	19:50
Jupiter	162°30.3	21:11
Saturn	315°15.3	11:02

Horizontal parallax	
Venus:	0.1
Mars:	0.2

DUT1 = UT1-UTC = +0.2098 sec ΔT = TT-UT1 = +68.9742 sec

2029 May 19 to May. 21 UT

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	HP
0	180°52.9	N19°46.9	114°09.1	10.5'	N17°41.2	-9.7'
1	195°52.8	47.5	128°38.5	10.5'	17°31.4	-9.8'
2	210°52.8	48.0	143°08.1	10.6'	17°21.6	-9.9'
3	225°52.8	· · 48.5	157°37.6	10.6'	17°11.7	-10.0'
4	240°52.7	49.1	172°07.2	10.6'	17°01.6	-10.1'
5	255°52.7	49.6	186°36.8	10.6'	16°51.5	-10.2'
6	270°52.7	N19°50.1	201°06.5	10.7'	N16°41.3	-10.3'
7	285°52.7	50.7	215°36.1	10.7'	16°31.0	-10.4'
8	300°52.6	51.2	230°05.8	10.7'	16°20.6	-10.5'
9	315°52.6	· · 51.7	244°35.6	10.8'	16°10.1	-10.6'
10	330°52.6	52.3	259°05.3	10.8'	15°59.5	-10.7'
11	345°52.5	52.8	273°35.1	10.8'	15°48.8	-10.8'
12	0°52.5	N19°53.3	288°05.0	10.9'	N15°38.0	-10.9'
13	15°52.5	53.8	302°34.8	10.9'	15°27.1	-11.0'
14	30°52.4	54.4	317°04.7	10.9'	15°16.2	-11.0'
15	45°52.4	· · 54.9	331°34.6	10.9'	15°05.1	-11.1'
16	60°52.4	55.4	346°04.5	11.0'	14°54.0	-11.2'
17	75°52.3	55.9	0°34.5	11.0'	14°42.8	-11.3'
18	90°52.3	N19°56.5	15°04.5	11.0'	N14°31.5	-11.4'
19	105°52.2	57.0	29°34.5	11.0'	14°20.1	-11.5'
20	120°52.2	57.5	44°04.6	11.1'	14°08.6	-11.6'
21	135°52.2	· · 58.0	58°34.6	11.1'	13°57.0	-11.6'
22	150°52.1	58.6	73°04.7	11.1'	13°45.4	-11.7'
23	165°52.1	59.1	87°34.8	11.1'	13°33.7	-11.8'
SD = 15.8' d = 0.5' SD = 15.6'						
Sun	GHA	Dec	GHA	ν	Dec	HP
0	180°52.1	N19°59.6	102°05.0	11.2'	N13°21.9	-11.9'
1	195°52.0	20°00.1	116°35.1	11.2'	13°10.0	-12.0'
2	210°52.0	00.6	131°05.3	11.2'	12°58.0	-12.0'
3	225°52.0	· · 01.2	145°35.5	11.2'	12°46.0	-12.1'
4	240°51.9	01.7	160°05.8	11.3'	12°33.9	-12.2'
5	255°51.9	02.2	174°36.0	11.3'	12°21.7	-12.3'
6	270°51.8	N20°02.7	189°06.3	11.3'	N12°09.5	-12.3'
7	285°51.8	03.2	203°36.6	11.3'	11°57.1	-12.4'
8	300°51.8	03.7	218°06.9	11.3'	11°44.7	-12.5'
9	315°51.7	· · 04.3	232°37.2	11.3'	11°32.2	-12.5'
10	330°51.7	04.8	247°07.6	11.4'	11°19.7	-12.6'
11	345°51.7	05.3	261°38.0	11.4'	11°07.1	-12.7'
12	0°51.6	N20°05.8	276°08.3	11.4'	N10°54.4	-12.8'
13	15°51.6	06.3	290°38.7	11.4'	10°41.6	-12.8'
14	30°51.5	06.8	305°09.2	11.4'	10°28.8	-12.9'
15	45°51.5	· · 07.3	319°39.6	11.4'	10°15.9	-13.0'
16	60°51.5	07.8	334°10.0	11.5'	10°03.0	-13.0'
17	75°51.4	08.4	348°40.5	11.5'	09°50.0	-13.1'
18	90°51.4	N20°08.9	3°10.9	11.5'	N09°36.9	-13.1'
19	105°51.3	09.4	17°41.4	11.5'	09°23.7	-13.2'
20	120°51.3	09.9	32°11.9	11.5'	09°10.5	-13.3'
21	135°51.3	· · 10.4	46°42.4	11.5'	08°57.3	-13.3'
22	150°51.2	10.9	61°12.9	11.5'	08°44.0	-13.4'
23	165°51.2	11.4	75°43.4	11.5'	08°30.6	-13.4'
SD = 15.8' d = 0.5' SD = 15.7'						
Mon	GHA	Dec	GHA	ν	Dec	HP
0	180°51.1	N20°11.9	90°13.9	11.5'	N08°17.2	-13.5'
1	195°51.1	12.4	104°44.5	11.5'	08°03.7	-13.5'
2	210°51.1	12.9	119°15.0	11.5'	07°50.1	-13.6'
3	225°51.0	· · 13.4	133°45.5	11.5'	07°36.5	-13.6'
4	240°51.0	13.9	148°16.1	11.5'	07°22.9	-13.7'
5	255°50.9	14.4	162°46.6	11.5'	07°09.2	-13.7'
6	270°50.9	N20°14.9	177°17.1	11.5'	N06°55.4	-13.8'
7	285°50.8	15.4	191°47.7	11.5'	06°41.7	-13.8'
8	300°50.8	15.9	206°18.2	11.5'	06°27.8	-13.9'
9	315°50.8	· · 16.4	220°48.8	11.5'	06°13.9	-13.9'
10	330°50.7	16.9	235°19.3	11.5'	06°00.0	-14.0'
11	345°50.7	17.4	249°49.8	11.5'	05°46.0	-14.0'
12	0°50.6	N20°17.9	264°20.3	11.5'	N05°32.0	-14.1'
13	15°50.6	18.4	278°50.9	11.5'	05°17.9	-14.1'
14	30°50.5	18.9	293°21.4	11.5'	05°03.8	-14.1'
15	45°50.5	· · 19.4	307°51.9	11.5'	04°49.7	-14.2'
16	60°50.4	19.9	322°22.4	11.5'	04°35.5	-14.2'
17	75°50.4	20.4	336°52.9	11.5'	04°21.3	-14.3'
18	90°50.4	N20°20.9	351°23.4	11.5'	N04°07.0	-14.3'
19	105°50.3	21.4	5°53.8	11.5'	03°52.7	-14.3'
20	120°50.3	21.9	20°24.3	11.4'	03°38.4	-14.4'
21	135°50.2	· · 22.4	34°54.8	11.4'	03°24.1	-14.4'
22	150°50.2	22.9	49°25.2	11.4'	03°09.7	-14.4'
23	165°50.1	23.4	63°55.6	11.4'	02°55.3	-14.4'
SD = 15.8' d = 0.5' SD = 15.9'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	—:—	—:—	01:15	22:45	—:—	—:—
66°	—:—	—:—	02:01	21:55	—:—	—:—
64°	—:—	00:03	02:31	21:25	—:—	—:—
62°	—:—	01:28	02:53	21:02	22:29	—:—
60°	—:—	02:02	03:11	20:44	21:54	—:—
N 58°	00:02	02:27	03:26	20:29	21:28	—:—
56°	01:19	02:46	03:38	20:16	21:09	22:38
54°	01:51	03:02	03:49	20:05	20:53	22:04
52°	02:14	03:15	03:59	19:55	20:39	21:41
50°	02:32	03:27	04:07	19:47	20:27	21:23
45°	03:06	03:51	04:26	19:28	20:03	20:48
N 40°	03:31	04:09	04:40	19:13	19:44	20:23
35°	03:50	04:24	04:53	19:01	19:29	20:04
30°	04:05	04:37	05:04	18:50	19:16	19:48
20°	04:30	04:58	05:22	18:31	18:55	19:23
N 10°	04:49	05:16	05:38	18:15	18:38	19:04
0°	05:05	05:31	05:53	18:00	18:22	18:48
S 10°	05:20	05:45	06:08	17:45	18:08	18:33
20°	05:33	06:00	06:23	17:30	17:53	18:20
30°	05:46	06:15	06:41	17:12	17:37	18:07
35°	05:53	06:24	06:51	17:02	17:29	18:00
40°	06:01	06:34	07:03	16:50	17:19	17:52
45°	06:08	06:44	07:17	16:36	17:08	17:44
S 50°	06:17	06:57	07:34	16:19	16:55	17:35
52°	06:21	07:03	07:42	16:11	16:49	17:31
54°	06:25	07:10	07:50	16:02	16:43	17:27
56°	06:30	07:17	08:00	15:52	16:36	17:23
58°	06:35	07:24	08:11	15:41	16:28	17:18
S 60°	06:40	07:33	08:25	15:28	16:19	17:12

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	06:12	08:56	11:11	03:12	02:13	01:42
N 70°	07:02	09:16	11:19	02:21	01:51	01:30
68°	07:33	09:32	11:26	01:48	01:33	01:21
66°	07:56	09:45	11:32	01:24	01:19	01:14
64°	08:14	09:56	11:37	01:05	01:07	01:07
62°	08:29	10:04	11:41	00:50	00:57	01:02
60°	08:41	10:12	11:44	00:37	00:48	00:57
N 58°	08:52	10:19	11:47	00:25	00:40	00:52
56°	09:01	10:25	11:50	00:15	00:34	00:48
54°	09:09	10:30	11:53	00:07	00:27	00:45
52°	09:16	10:35	11:55	· · ·	00:22	00:42
50°	09:23	10:39	11:57	· · ·	00:17	00:39
45°	09:37	10:48	12:01	· · ·	00:06	00:32
N 40°	09:48	10:56	12:05	23:57	· · ·	00:27
35°	09:58	11:03	12:08	23:49	· · ·	00:22
30°	10:06	11:08	12:11	23:42	· · ·	00:18
20°	10:21	11:18	12:16	23:30	· · ·	00:11
N 10°	10:34	11:27	12:20	23:19	· · ·	00:05
0°	10:46	11:35	12:24	23:09	23:58	· · ·
S 10°	10:58	11:43	12:28	22:59	23:52	· · ·
20°	11:10	11:52	12:32	22:48	23:46	· · ·
30°	11:25	12:02	12:37	22:36	23:38	· · ·
35°	11:33	12:07	12:40	22:29	23:34	· · ·
40°	11:42	12:14	12:43	22:20	23:29	· · ·
45°	11:53	12:21	12:46	22:11	23:23	· · ·
S 50°	12:07	12:30	12:51	21:59	23:16	· · ·
52°	12:13	12:34	12:53	21:53	23:13	· · ·
54°	12:20	12:38	12:55	21:47	23:09	· · ·
56°	12:27	12:43	12:57	21:40	23:05	· · ·
58°	12:36	12:49	13:00	21:33	23:01	· · ·
S 60°	12:45	12:55	13:03	21:24	22:56	· · ·

Day	Sun		Mer. Pass	Moon		Age 6-8 27-48%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
19	03:31	03:30	11:57	16:58	04:33	
20	03:28	03:26	11:57	17:47	05:22	
21	03:25	03:22	11:57	18:36	06:11	

May 22, 23, 24 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	239°54.6	164°35.8	N23°17.7	62°43.2	N02°22.3	42°28.9	S05°48.4	195°02.5	N14°49.2				Alpheratz	357°31.2	29°15.1		
1	254°57.1	179°35.0		77°45.3	22.1	57°31.5	48.4	210°04.6	49.3				Ankaa	353°04.0	-42°08.5		
2	269°59.6	194°34.1		92°47.4	21.8	72°34.1	48.3	225°06.8	49.4				Schedar	349°27.2	56°41.7		
3	285°02.0	209°33.3	•• 18.9	107°49.4	•• 21.5	87°36.7	•• 48.3	240°08.9	•• 49.5				Diphda	348°44.0	-17°49.4		
4	300°04.5	224°32.5		122°51.5	21.3	102°39.4	48.2	255°11.1	49.6				Achernar	335°18.2	-57°05.1		
5	315°07.0	239°31.6		137°53.6	21.0	117°42.0	48.2	270°13.2	49.7				Hamal	327°47.5	23°36.0		
6	330°09.4	254°30.8	N23°20.1	152°55.6	N02°20.7	132°44.6	S05°48.1	285°15.4	N14°49.8				Polaris	312°34.5	89°23.0		
7	345°11.9	269°29.9		167°57.7	20.5	147°47.2	48.1	300°17.5	49.9				Acamar	315°09.6	-40°11.2		
8	0°14.4	284°29.1		182°59.8	20.2	162°49.9	48.0	315°19.7	49.9				Menkar	314°02.8	4°12.2		
9	15°16.8	299°28.2	•• 21.3	198°01.9	•• 19.9	177°52.5	•• 48.0	330°21.8	•• 50.0				Mirfak	308°23.6	49°57.8		
10	30°19.3	314°27.4		213°03.9	19.7	192°55.1	47.9	345°24.0	50.1				Aldebaran	290°35.9	16°34.0		
11	45°21.7	329°26.5		228°06.0	19.4	207°57.7	47.8	0°26.2	50.2				Rigel	281°00.8	-8°10.2		
12	60°24.2	344°25.7	N23°22.5	243°08.1	N02°19.1	223°00.3	S05°47.8	15°28.3	N14°50.3				Capella	280°17.1	46°01.5		
13	75°26.7	359°24.8		258°10.1	18.9	238°03.0	47.7	30°30.5	50.4				Bellatrix	278°19.4	6°22.4		
14	90°29.1	14°24.0		273°12.2	18.6	253°05.6	47.7	45°32.6	50.5				Elnath	277°57.8	28°37.8		
15	105°31.6	29°23.2	•• 23.7	288°14.2	•• 18.3	268°08.2	•• 47.6	60°34.8	•• 50.6				Alnilam	275°34.5	-1°11.2		
16	120°34.1	44°22.3		303°16.3	18.1	283°10.8	47.6	75°36.9	50.7				Betelgeuse	270°48.6	7°24.6		
17	135°36.5	59°21.5		318°18.4	17.8	298°13.4	47.5	90°39.1	50.7				Canopus	263°51.3	-52°42.9		
18	150°39.0	74°20.6	N23°24.9	333°20.4	N02°17.5	313°16.1	S05°47.5	105°41.2	N14°50.8				Sirius	258°23.4	-16°45.6		
19	165°41.5	89°19.8		348°22.5	17.2	328°18.7	47.4	120°43.4	50.9				Adhara	255°03.4	-29°01.0		
20	180°43.9	104°18.9		3°24.5	17.0	343°21.3	47.4	135°45.5	51.0				Procyon	244°47.4	5°08.8		
21	195°46.4	119°18.1	•• 26.0	18°26.6	•• 16.7	358°23.9	•• 47.3	150°47.7	•• 51.1				Pollux	243°13.3	27°57.2		
22	210°48.9	134°17.2		33°28.7	16.4	13°26.5	47.3	165°49.8	51.2				Avior	234°13.5	-59°36.6		
23	225°51.3	149°16.4		48°30.7	16.2	28°29.1	47.2	180°52.0	51.3				Suhail	222°43.8	-43°33.5		
Mer.pass. 07:59			ν -0.8' d 0.4' m -3.90			ν 2.1' d -0.3' m -0.24			ν 2.6' d -0.1' m -2.34			ν 2.2' d 0.1' m 0.32					
Wed	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec					
0	240°53.8	164°15.5	N23°27.2	63°32.8	N02°15.9	43°31.8	S05°47.2	195°54.1	N14°51.4				Denebola	182°21.3	14°24.4		
1	255°56.2	179°14.7		78°34.8	15.6	58°34.4	47.1	210°56.3	51.4				Gienah	175°39.8	-17°42.5		
2	270°58.7	194°13.8		93°36.9	15.3	73°37.0	47.1	225°58.4	51.5				Acrux	172°55.5	-63°16.1		
3	286°01.2	209°13.0	•• 28.3	108°38.9	•• 15.1	88°39.6	•• 47.0	241°00.6	•• 51.6				Gacrux	171°47.2	-57°17.0		
4	301°03.6	224°12.1		123°41.0	14.8	103°42.2	47.0	256°02.7	51.7				Alioth	166°09.7	55°48.1		
5	316°06.1	239°11.3		138°43.0	14.5	118°44.8	46.9	271°04.9	51.8				Spica	158°18.3	-11°19.0		
6	331°08.6	254°10.4	N23°29.4	153°45.1	N02°14.2	133°47.5	S05°46.9	286°07.0	N14°51.9				Alkaid	152°49.0	49°10.1		
7	346°11.0	269°09.6		168°47.1	14.0	148°50.1	46.8	301°09.2	52.0				Hadar	148°30.2	-60°31.1		
8	1°13.5	284°08.7		183°49.2	13.7	163°52.7	46.8	316°11.3	52.1				Menkent	147°53.0	-36°31.0		
9	16°16.0	299°07.9	•• 30.6	198°51.2	•• 13.4	178°55.3	•• 46.7	331°13.5	•• 52.1				Arcturus	145°44.4	19°01.7		
10	31°18.4	314°07.0		213°53.3	13.1	193°57.9	46.7	346°15.7	52.2				Rigil Kent.	139°34.7	-60°57.6		
11	46°20.9	329°06.2		228°55.3	12.9	209°00.5	46.6	1°17.8	52.3				Kochab	137°19.1	74°02.1		
12	61°23.4	344°05.3	N23°31.7	243°57.4	N02°12.6	224°03.1	S05°46.5	16°20.0	N14°52.4				Zuben'ubi	136°51.8	-16°09.9		
13	76°25.8	359°04.4		258°59.4	12.3	239°05.8	46.5	31°22.1	52.5				Alphecca	126°00.4	26°36.9		
14	91°28.3	14°03.6		274°01.4	12.0	254°08.4	46.4	46°24.3	52.6				Antares	112°11.1	-26°29.8		
15	106°30.7	29°02.7	•• 32.8	289°03.5	•• 11.7	269°11.0	•• 46.4	61°26.4	•• 52.7				Atria	107°01.5	-69°04.6		
16	121°33.2	44°01.9		304°05.5	11.5	284°13.6	46.3	76°28.6	52.7				Sabik	101°58.4	-15°45.6		
17	136°35.7	59°01.0		319°07.6	11.2	299°16.2	46.3	91°30.7	52.8				Shaula	96°05.2	-37°07.3		
18	151°38.1	74°00.2	N23°33.8	334°09.6	N02°10.9	314°18.8	S05°46.2	106°32.9	N14°52.9				Rasalhague	95°55.0	12°32.3		
19	166°40.6	88°59.3		349°11.6	10.6	329°21.4	46.2	121°35.0	53.0				Eltanin	90°40.1	51°29.0		
20	181°43.1	103°58.5		4°13.7	10.4	344°24.0	46.1	136°37.2	53.1				Kaus Aust.	83°27.5	-34°22.0		
21	196°45.5	118°57.6	•• 34.9	19°15.7	•• 10.1	359°26.7	•• 46.1	151°39.3	•• 53.2				Vega	80°30.5	38°48.6		
22	211°48.0	133°56.8		34°17.7	09.8	14°29.3	46.0	166°41.5	53.3				Nunki	75°43.1	-26°15.4		
23	226°50.5	148°55.9		49°19.8	09.5	29°31.9	46.0	181°43.6	53.4				Altair	61°56.3	8°56.8		
Mer.pass. 07:55			ν -0.8' d 0.4' m -3.90			ν 2.1' d -0.3' m -0.22			ν 2.6' d -0.1' m -2.33			ν 2.2' d 0.1' m 0.32					
Thu	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec					
0	241°52.9	163°55.0	N23°36.0	64°21.8	N02°09.2	44°34.5	S05°46.0	196°45.8	N14°53.4				May 22 Tue	SHA	Mer.pass		
1	256°55.4	178°54.2		79°23.8	08.9	59°37.1	45.9	211°47.9	53.5				Venus	284°41.2	13:02		
2	271°57.8	193°53.3		94°25.9	08.7	74°39.7	45.9	226°50.1	53.6				Mars	182°48.6	19:46		
3	287°00.3	208°52.5	•• 37.0	109°27.9	•• 08.4	89°42.3	•• 45.8	241°52.3	•• 53.7				Jupiter	162°34.2	21:06		
4	302°02.8	223°51.6		124°29.9	08.1	104°44.9	45.8	256°54.4	53.8				Saturn	315°07.8	10:58		
5	317°05.2	238°50.8		139°31.9	07.8	119°47.5	45.7	271°56.6	53.9								
6	332°07.7	253°49.9	N23°38.1	154°34.0	N02°07.5	134°50.1	S05°45.7	286°58.7	N14°54.0								
7	347°10.2	268°49.0		169°36.0	07.2	149°52.8	45.6	302°00.9	54.1								
8	2°12.6	283°48.2		184°38.0	07.0	164°55.4	45.6	317°03.0	54.1								
9	17°15.1	298°47.3	•• 39.1	199°40.0	•• 06.7	179°58.0	•• 45.5	332°05.2	•• 54.2								
10	32°17.6	313°46.5		214°42.1	06.4	195°00.6	45.5	347°07.3	54.3								
11	47°20.0	328°45.6		229°44.1	06.1	210°03.2	45.4	2°09.5	54.4								
12	62°22.5	343°44.8	N23°40.1	244°46.1	N02°05.8	225°05.8	S05°45.4	17°11.6	N14°54.5								
13	77°25.0	358°43.9		259°48.1	05.5	240°08.4	45.3	32°13.8	54.6								
14	92°27.4	13°43.0		274°50.1	05.3	255°11.0	45.3	47°15.9	54.7								
15	107°29.9	28°42															

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°50.1	N20°23.9	78°26.0	11.4'	N02°40.8	-14.5'	59.0'
1	195°50.0	24.4	92°56.4	11.4'	02°26.3	-14.5'	59.0'
2	210°50.0	24.9	107°26.7	11.3'	02°11.8	-14.5'	59.0'
3	225°49.9	· · 25.4	121°57.1	11.3'	01°57.3	-14.5'	59.1'
4	240°49.9	25.8	136°27.4	11.3'	01°42.8	-14.6'	59.1'
5	255°49.8	26.3	150°57.7	11.3'	01°28.2	-14.6'	59.1'
6	270°49.8	N20°26.8	165°28.0	11.3'	N01°13.6	-14.6'	59.2'
7	285°49.7	27.3	179°58.2	11.2'	00°59.0	-14.6'	59.2'
8	300°49.7	27.8	194°28.5	11.2'	00°44.4	-14.6'	59.2'
9	315°49.6	· · 28.3	208°58.7	11.2'	00°29.7	-14.7'	59.2'
10	330°49.6	28.8	223°28.8	11.1'	00°15.0	-14.7'	59.3'
11	345°49.5	29.2	237°59.0	11.1'	N00°00.4	-14.7'	59.3'
12	0°49.5	N20°29.7	252°29.1	11.1'	S00°14.3	14.7'	59.3'
13	15°49.4	30.2	266°59.2	11.1'	00°29.0	14.7'	59.3'
14	30°49.4	30.7	281°29.2	11.0'	00°43.7	14.7'	59.4'
15	45°49.3	· · 31.2	295°59.3	11.0'	00°58.5	14.7'	59.4'
16	60°49.3	31.7	310°29.2	11.0'	01°13.2	14.7'	59.4'
17	75°49.2	32.1	324°59.2	10.9'	01°27.9	14.7'	59.4'
18	90°49.2	N20°32.6	339°29.1	10.9'	S01°42.7	14.7'	59.5'
19	105°49.1	33.1	353°59.0	10.8'	01°57.4	14.7'	59.5'
20	120°49.1	33.6	8°28.8	10.8'	02°12.2	14.8'	59.5'
21	135°49.0	· · 34.1	22°58.6	10.8'	02°26.9	14.8'	59.5'
22	150°49.0	34.5	37°28.4	10.7'	02°41.7	14.7'	59.6'
23	165°48.9	35.0	51°58.1	10.7'	02°56.4	14.7'	59.6'
	SD = 15.8'	$d = 0.5'$			SD = 16.1'		

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°48.9	N20°35.5	66°27.8	10.6'	S03°11.2	14.7'	59.6'
1	195°48.8	36.0	80°57.4	10.6'	03°25.9	14.7'	59.6'
2	210°48.8	36.4	95°27.0	10.5'	03°40.6	14.7'	59.7'
3	225°48.7	· · 36.9	109°56.6	10.5'	03°55.4	14.7'	59.7'
4	240°48.7	37.4	124°26.1	10.4'	04°10.1	14.7'	59.7'
5	255°48.6	37.9	138°55.5	10.4'	04°24.8	14.7'	59.7'
6	270°48.6	N20°38.3	153°24.9	10.4'	S04°39.5	14.7'	59.7'
7	285°48.5	38.8	167°54.3	10.3'	04°54.2	14.7'	59.8'
8	300°48.5	39.3	182°23.6	10.2'	05°08.8	14.6'	59.8'
9	315°48.4	· · 39.8	196°52.8	10.2'	05°23.5	14.6'	59.8'
10	330°48.4	40.2	211°22.0	10.1'	05°38.1	14.6'	59.8'
11	345°48.3	40.7	225°51.1	10.1'	05°52.7	14.6'	59.9'
12	0°48.2	N20°41.2	240°20.2	10.0'	S06°07.3	14.6'	59.9'
13	15°48.2	41.6	254°49.3	10.0'	06°21.9	14.5'	59.9'
14	30°48.1	42.1	269°18.2	9.9'	06°36.4	14.5'	59.9'
15	45°48.1	· · 42.6	283°47.1	9.9'	06°50.9	14.5'	59.9'
16	60°48.0	43.0	298°16.0	9.8'	07°05.4	14.5'	60.0'
17	75°48.0	43.5	312°44.8	9.7'	07°19.8	14.4'	60.0'
18	90°47.9	N20°44.0	327°13.5	9.7'	S07°34.3	14.4'	60.0'
19	105°47.9	44.4	341°42.2	9.6'	07°48.7	14.4'	60.0'
20	120°47.8	44.9	356°10.8	9.5'	08°03.0	14.3'	60.0'
21	135°47.7	· · 45.4	10°39.3	9.5'	08°17.3	14.3'	60.1'
22	150°47.7	45.8	25°07.8	9.4'	08°31.6	14.2'	60.1'
23	165°47.6	46.3	39°36.2	9.3'	08°45.8	14.2'	60.1'
	SD = 15.8'	$d = 0.5'$			SD = 16.3'		

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°47.6	N20°46.8	54°04.6	9.3'	S09°00.0	14.2'	60.1'
1	195°47.5	47.2	68°32.9	9.2'	09°14.2	14.1'	60.1'
2	210°47.5	47.7	83°01.1	9.1'	09°28.3	14.1'	60.2'
3	225°47.4	· · 48.1	97°29.2	9.1'	09°42.4	14.0'	60.2'
4	240°47.3	48.6	111°57.3	9.0'	09°56.4	14.0'	60.2'
5	255°47.3	49.1	126°25.3	8.9'	10°10.4	13.9'	60.2'
6	270°47.2	N20°49.5	140°53.2	8.9'	S10°24.3	13.9'	60.2'
7	285°47.2	50.0	155°21.1	8.8'	10°38.1	13.8'	60.2'
8	300°47.1	50.4	169°48.9	8.7'	10°51.9	13.7'	60.2'
9	315°47.1	· · 50.9	184°16.6	8.6'	11°05.7	13.7'	60.3'
10	330°47.0	51.3	198°44.2	8.6'	11°19.4	13.6'	60.3'
11	345°46.9	51.8	213°11.8	8.5'	11°33.0	13.6'	60.3'
12	0°46.9	N20°52.3	227°39.3	8.4'	S11°46.6	13.5'	60.3'
13	15°46.8	52.7	242°06.7	8.3'	12°00.1	13.4'	60.3'
14	30°46.8	53.2	256°34.0	8.3'	12°13.5	13.4'	60.3'
15	45°46.7	· · 53.6	271°01.3	8.2'	12°26.9	13.3'	60.3'
16	60°46.6	54.1	285°28.4	8.1'	12°40.2	13.2'	60.4'
17	75°46.6	54.5	299°55.5	8.0'	12°53.4	13.2'	60.4'
18	90°46.5	N20°55.0	314°22.6	7.9'	S13°06.5	13.1'	60.4'
19	105°46.5	55.4	328°49.5	7.9'	13°19.6	13.0'	60.4'
20	120°46.4	55.9	343°16.4	7.8'	13°32.6	12.9'	60.4'
21	135°46.3	· · 56.3	357°43.2	7.7'	13°45.5	12.8'	60.4'
22	150°46.3	56.8	12°09.9	7.6'	13°58.4	12.8'	60.4'
23	165°46.2	57.2	26°36.5	7.5'	14°11.2	12.7'	60.4'
	SD = 15.8'	$d = 0.5'$			SD = 16.4'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	—:—	—:—	00:51	23:11	—:—	—:—
66°	—:—	—:—	01:49	22:08	—:—	—:—
64°	—:—	—:—	02:22	21:34	—:—	—:—
62°	—:—	01:13	02:46	21:10	22:45	—:—
60°	—:—	01:53	03:05	20:51	22:03	—:—
N 58°	—:—	02:19	03:20	20:35	21:36	—:—
56°	01:06	02:40	03:33	20:21	21:15	22:52
54°	01:43	02:57	03:45	20:10	20:58	22:14
52°	02:07	03:11	03:55	19:59	20:44	21:48
50°	02:27	03:23	04:04	19:50	20:32	21:28
45°	03:02	03:48	04:23	19:31	20:07	20:52
N 40°	03:28	04:07	04:38	19:16	19:47	20:27
35°	03:47	04:22	04:51	19:03	19:32	20:07
30°	04:04	04:36	05:02	18:52	19:18	19:50
20°	04:29	04:57	05:21	18:32	18:56	19:25
N 10°	04:49	05:15	05:38	18:16	18:38	19:05
0°	05:05	05:31	05:53	18:00	18:22	18:48
S 10°	05:20	05:46	06:08	17:45	18:08	18:33
20°	05:34	06:01	06:24	17:29	17:52	18:19
30°	05:48	06:17	06:43	17:11	17:36	18:06
35°	05:55	06:26	06:53	17:00	17:27	17:58
40°	06:03	06:36	07:06	16:48	17:17	17:51
45°	06:11	06:47	07:20	16:33	17:06	17:42
S 50°	06:20	07:01	07:37	16:16	16:52	17:33
52°	06:25	07:07	07:46	16:08	16:46	17:28
54°	06:29	07:14	07:55	15:58	16:40	17:24
56°	06:34	07:21	08:05	15:48	16:32	17:19
58°	06:39	07:29	08:17	15:36	16:24	17:14
S 60°	06:45	07:39	08:31	15:22	16:14	17:08

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	13:22	15:37	18:10	01:16	00:52	00:24 23:44
N 70°	13:21	15:26	17:43	01:13	00:56	00:38
68°	13:20	15:17	17:22	01:11	01:00	00:49
66°	13:19	15:10	17:06	01:09	01:04	00:58
64°	13:19	15:04	16:53	01:07	01:06	01:06
62°	13:18	14:58	16:43	01:05	01:09	01:13
60°	13:18	14:54	16:33	01:04	01:11	01:19
N 58°	13:17	14:50	16:25	01:03	01:13	01:24
56°	13:17	14:46	16:18	01:02	01:15	01:29
54°	13:17	14:43	16:12	01:01	01:16	01:33
52°	13:16	14:40	16:07	01:00	01:18	01:37
50°	13:16	14:38	16:01	00:59	01:19	01:41
45°	13:16	14:32	15:51	00:57	01:22	01:48
N 40°	13:15	14:27	15:42	00:56	01:24	01:55
35°	13:15	14:23	15:34	00:54	01:26	02:00
30°	13:14	14:20	15:27	00:53	01:28	02:05
20°	13:14	14:14	15:16	00:51	01:31	02:14
N 10°	13:14	14:08	15:06	00:49	01:34	02:21
0°	13:13	14:04	14:56	00:47	01:37	02:28
S 10°	13:13	13:59	14:47	00:45	01:39	02:35
20°	13:12	13:54	14:38	00:43	01:42	02:43
30°	13:12	13:48	14:26	00:41	01:45	02:52
35°	13:12	13:45	14:20	00:40	01:47	02:57
40°	13:11	13:41	14:13	00:38	01:49	03:02
45°	13:11	13:37	14:05	00:36	01:52	03:09
S 50°	13:11	13:32	13:55	00:34	01:55	03:17
52°	13:11	13:29	13:50	00:33	01:56	03:21
54°	13:11	13:27	13:45	00:32	01:57	03:25
56°	13:10	13:24	13:40	00:31	01:59	03:29
58°	13:10	13:21	13:34	00:30	02:01	03:34
S 60°	13:10	13:18	13:27	00:28	02:03	03:40

Day	Sun		Mer. Pass hh:mm	Moon		Age 9-11 59-80%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
22	03:20	03:18	11:57	19:25	07:00	
23	03:16	03:13	11:57	20:16	07:50	
24	03:10	03:08	11:57	21:09	08:42	

May 25, 26, 27 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	242°52.1	163°34.4	N23°44.1	65°10.3	N02°02.4	45°37.1	S05°44.8	197°37.5	N14°55.5					Alpheratz	357°31.2	29°15.1	
1	257°54.5	178°33.6		44.4	80°12.3		02.1	60°39.7		44.8		212°39.6		55.6		-42°08.5	
2	272°57.0	193°32.7		44.7	95°14.3		01.8	75°42.3		44.7		227°41.8		55.7		56°41.7	
3	287°59.5	208°31.8	· ·	45.0	110°16.3	· ·	01.5	90°44.9	· ·	44.7		242°43.9	· ·	55.8		-17°49.4	
4	303°01.9	223°31.0		45.4	125°18.3		01.2	105°47.5		44.6		257°46.1		55.9		-57°05.0	
5	318°04.4	238°30.1		45.7	140°20.3		00.9	120°50.1		44.6		272°48.2		56.0		23°36.0	
6	333°06.8	253°29.3	N23°	46.0	155°22.3	N02°	00.6	135°52.7	S05°	44.5		287°50.4	N14°	56.0		89°23.0	
7	348°09.3	268°28.4		46.3	170°24.3		00.3	150°55.3		44.5		302°52.5		56.1		-40°11.2	
8	3°11.8	283°27.5		46.6	185°26.3	02°	00.0	165°57.9		44.4		317°54.7		56.2		4°12.2	
9	18°14.2	298°26.7	· ·	46.9	200°28.3	01°	59.7	181°00.5	· ·	44.4		332°56.9	· ·	56.3		49°57.8	
10	33°16.7	313°25.8		47.3	215°30.3		59.5	196°03.1		44.3		347°59.0		56.4		16°34.0	
11	48°19.2	328°24.9		47.6	230°32.3		59.2	211°05.7		44.3		3°01.2		56.5		-8°10.2	
12	63°21.6	343°24.1	N23°	47.9	245°34.3	N01°	58.9	226°08.3	S05°	44.3		18°03.3	N14°	56.6		46°01.5	
13	78°24.1	358°23.2		48.2	260°36.3		58.6	241°10.9		44.2		33°05.5		56.6		6°22.4	
14	93°26.6	13°22.3		48.5	275°38.3		58.3	256°13.5		44.2		48°07.6		56.7		28°37.8	
15	108°29.0	28°21.5	· ·	48.8	290°40.3	· ·	58.0	271°16.1	· ·	44.1		63°09.8	· ·	56.8		-1°11.2	
16	123°31.5	43°20.6		49.1	305°42.3		57.7	286°18.7		44.1		78°11.9		56.9		7°24.6	
17	138°33.9	58°19.7		49.4	320°44.3		57.4	301°21.3		44.0		93°14.1		57.0		-52°42.9	
18	153°36.4	73°18.9	N23°	49.7	335°46.3	N01°	57.1	316°23.9	S05°	44.0		108°16.2	N14°	57.1		-16°45.6	
19	168°38.9	88°18.0		50.0	350°48.3		56.8	331°26.5		43.9		123°18.4		57.2		-29°01.0	
20	183°41.3	103°17.1		50.3	5°50.3		56.5	346°29.1		43.9		138°20.6		57.2		5°08.8	
21	198°43.8	118°16.3	· ·	50.6	20°52.3	· ·	56.2	1°31.7	· ·	43.9		153°22.7	· ·	57.3		27°57.2	
22	213°46.3	133°15.4		50.9	35°54.3		55.9	16°34.3		43.8		168°24.9		57.4		-59°36.6	
23	228°48.7	148°14.5		51.2	50°56.3		55.6	31°36.9		43.8		183°27.0		57.5		-43°33.5	
Mer.pass. 07:47			ν -0.9' d 0.3' m -3.90			ν 2.0' d -0.3' m -0.18			ν 2.6' d -0.0' m -2.32			ν 2.2' d 0.1' m 0.32					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	243°51.2	163°13.7	N23°	51.5	65°58.3	N01°	55.3	46°39.5	S05°	43.7		198°29.2	N14°	57.6			
1	258°53.7	178°12.8		51.8	81°00.2		55.0	61°42.1		43.7		213°31.3		57.7		17°24.4	
2	273°56.1	193°11.9		52.1	96°02.2		54.7	76°44.7		43.6		228°33.5		57.8		-17°42.5	
3	288°58.6	208°11.1	· ·	52.4	111°04.2	· ·	54.4	91°47.3	· ·	43.6		243°35.6	· ·	57.8		-63°16.1	
4	304°01.1	223°10.2		52.7	126°06.2		54.1	106°49.9		43.5		258°37.8		57.9		-57°17.0	
5	319°03.5	238°09.3		53.0	141°08.2		53.8	121°52.5		43.5		273°39.9		58.0		55°48.1	
6	334°06.0	253°08.5	N23°	53.2	156°10.2	N01°	53.5	136°55.1	S05°	43.5		288°42.1	N14°	58.1		-11°19.0	
7	349°08.4	268°07.6		53.5	171°12.2		53.2	151°57.7		43.4		303°44.3		58.2		49°10.1	
8	4°10.9	283°06.7		53.8	186°14.1		52.9	167°00.2		43.4		318°46.4		58.3		-60°31.1	
9	19°13.4	298°05.9	· ·	54.1	201°16.1	· ·	52.6	182°02.8	· ·	43.3		333°48.6	· ·	58.4		147°53.0	
10	34°15.8	313°05.0		54.4	216°18.1		52.3	197°05.4		43.3		348°50.7		58.4		145°44.4	
11	49°18.3	328°04.1		54.7	231°20.1		52.0	212°08.0		43.2		3°52.9		58.5		139°34.7	
12	64°20.8	343°03.3	N23°	54.9	246°22.1	N01°	51.7	227°10.6	S05°	43.2		18°55.0	N14°	58.6		137°19.1	
13	79°23.2	358°02.4		55.2	261°24.0		51.4	242°13.2		43.2		33°57.2		58.7		74°02.1	
14	94°25.7	13°01.5		55.5	276°26.0		51.1	257°15.8		43.1		48°59.3		58.8		-16°09.9	
15	109°28.2	28°00.6	· ·	55.8	291°28.0	· ·	50.8	272°18.4	· ·	43.1		64°01.5	· ·	58.9		136°51.8	
16	124°30.6	42°59.8		56.1	306°30.0		50.5	287°21.0		43.0		79°03.6		59.0		126°00.4	
17	139°33.1	57°58.9		56.3	321°31.9		50.2	302°23.6		43.0		94°05.8		59.0		112°11.1	
18	154°35.6	72°58.0	N23°	56.6	336°33.9	N01°	49.9	317°26.2	S05°	43.0		109°08.0	N14°	59.1		-26°29.8	
19	169°38.0	87°57.2		56.9	351°35.9		49.6	332°28.8		42.9		124°10.1		59.2		-69°04.6	
20	184°40.5	102°56.3		57.1	6°37.8		49.3	347°31.4		42.9		139°12.3		59.3		101°58.4	
21	199°42.9	117°55.4	· ·	57.4	21°39.8	· ·	49.0	2°34.0	· ·	42.8		154°14.4	· ·	59.4		96°05.2	
22	214°45.4	132°54.6		57.7	36°41.8		48.7	17°36.6		42.8		169°16.6		59.5		-37°07.3	
23	229°47.9	147°53.7		58.0	51°43.8		48.4	32°39.2		42.7		184°18.7		59.6		95°55.0	
Mer.pass. 07:43			ν -0.9' d 0.3' m -3.90			ν 2.0' d -0.3' m -0.16			ν 2.6' d -0.0' m -2.32			ν 2.2' d 0.1' m 0.32					

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	244°50.3	162°52.8	N23°	58.2	66°45.7	N01°	48.1	47°41.7	S05°	42.7		199°20.9	N14°	59.6		
1	259°52.8	177°51.9		58.5	81°47.7		47.8	62°44.3		42.7		214°23.0		59.7		12°32.3
2	274°55.3	192°51.1		58.8	96°49.7		47.4	77°46.9		42.6		229°25.2		59.8		51°29.0
3	289°57.7	207°50.2	· ·	59.0	111°51.6	· ·	47.1	92°49.5	· ·	42.6		244°27.4	14°	59.9		-34°22.0
4	305°00.2	222°49.3		59.3	126°53.6		46.8	107°52.1		42.5		259°29.5	15°	00.0		80°30.5
5	320°02.7	237°48.4		59.5	141°55.5		46.5	122°54.7		42.5		274°31.7		00.1		38°48.6
6	335°05.1	252°47.6	N23°	59.8	156°57.5	N01°	46.2	137°57.3	S05°	42.5		289°33.8	N15°	00.2		-26°15.4
7	350°07.6	267°46.7	24°	00.1	171°59.5		45.9	152°59.9		42.4		304°36.0		00.2		61°56.3
8	5°10.1	282°45.8		00.3	187°01.4		45.6	168°02.5		42.4		319°38.1		00.3		8°56.8
9	20°12.5	297°45.0	· ·	00.6	202°03.4	· ·	45.3	183°05.0	· ·	42.3		334°40.3	· ·	00.4		53°00.0
10	35°15.0	312°44.1		00.8	217°05.4		45.0	198°07.6		42.3		349°42.4		00.5		-56°38.0
11	50°17.4	327°43.2		01.1	232°07.3		44.7	213°10.2		42.3		4°44.6		00.6		49°23.1
12	65°19.9	342°42.3	N24°	01.3	247°09.3	N01°	44.4	228°12.8	S05°	42.2		19°46.8	N15°	00.7		33°35.3
13	80°22.4	357°41.5		01.6	262°11.2		44.1	243°15.4		42.2		34°48.9		00.8		27°28.6
14	95°24.8	12°40.6		01.8	277°13.2		43.7	258°18.0		42.1		49°51.1		00.8		-46°48.7
15	110°27.3	27°39.7	· ·	02.1	292°15.1	· ·	43.4	273°20.6	· ·	42.1		64°53.2	· ·	00.9		15°10.8
16	125°29.8	42°38.8		02.3	307°17.1		43.1	288°23.2		42.1		79°55.4		01.0		-28°14.5
17	140°3															

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°46.1	N20°57.7	41°03.0	7.5'	S14°23.8	12.6'	60.5'
1	195°46.1	58.1	55°29.5	7.4'	14°36.4	12.5'	60.5'
2	210°46.0	58.6	69°55.8	7.3'	14°48.9	12.4'	60.5'
3	225°46.0	· · 59.0	84°22.1	7.2'	15°01.3	12.3'	60.5'
4	240°45.9	59.4	98°48.3	7.1'	15°13.7	12.2'	60.5'
5	255°45.8	20°59.9	113°14.5	7.0'	15°25.9	12.1'	60.5'
6	270°45.8	N21°00.3	127°40.5	7.0'	S15°38.0	12.0'	60.5'
7	285°45.7	00.8	142°06.5	6.9'	15°50.1	11.9'	60.5'
8	300°45.6	01.2	156°32.4	6.8'	16°02.0	11.8'	60.5'
9	315°45.6	· · 01.7	170°58.1	6.7'	16°13.8	11.7'	60.5'
10	330°45.5	02.1	185°23.9	6.6'	16°25.6	11.6'	60.5'
11	345°45.5	02.5	199°49.5	6.5'	16°37.2	11.5'	60.5'
12	0°45.4	N21°03.0	214°15.0	6.5'	S16°48.7	11.4'	60.5'
13	15°45.3	03.4	228°40.5	6.4'	17°00.1	11.3'	60.5'
14	30°45.3	03.9	243°05.9	6.3'	17°11.4	11.2'	60.5'
15	45°45.2	· · 04.3	257°31.2	6.2'	17°22.6	11.1'	60.5'
16	60°45.1	04.7	271°56.4	6.1'	17°33.7	11.0'	60.6'
17	75°45.1	05.2	286°21.5	6.1'	17°44.7	10.9'	60.6'
18	90°45.0	N21°05.6	300°46.6	6.0'	S17°55.5	10.7'	60.6'
19	105°44.9	06.0	315°11.6	5.9'	18°06.3	10.6'	60.6'
20	120°44.9	06.5	329°36.5	5.8'	18°16.9	10.5'	60.6'
21	135°44.8	· · 06.9	344°01.3	5.7'	18°27.4	10.4'	60.6'
22	150°44.7	07.3	358°26.0	5.7'	18°37.8	10.2'	60.6'
23	165°44.7	07.8	12°50.7	5.6'	18°48.0	10.1'	60.6'
SD = 15.8'			d = 0.4'				
SD = 15.8'			SD = 16.5'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°44.6	N21°08.2	27°15.3	5.5'	S18°58.1	10.0'	60.6'
1	195°44.5	08.6	41°39.8	5.4'	19°08.1	9.9'	60.6'
2	210°44.5	09.1	56°04.2	5.3'	19°18.0	9.7'	60.6'
3	225°44.4	· · 09.5	70°28.5	5.3'	19°27.7	9.6'	60.6'
4	240°44.3	09.9	84°52.8	5.2'	19°37.3	9.5'	60.6'
5	255°44.3	10.4	99°17.0	5.1'	19°46.7	9.3'	60.6'
6	270°44.2	N21°10.8	113°41.1	5.1'	S19°56.1	9.2'	60.6'
7	285°44.1	11.2	128°05.2	5.0'	20°05.2	9.0'	60.5'
8	300°44.1	11.6	142°29.2	4.9'	20°14.3	8.9'	60.5'
9	315°44.0	· · 12.1	156°53.1	4.8'	20°23.2	8.8'	60.5'
10	330°43.9	12.5	171°16.9	4.8'	20°31.9	8.6'	60.5'
11	345°43.8	12.9	185°40.7	4.7'	20°40.6	8.5'	60.5'
12	0°43.8	N21°13.3	200°04.4	4.6'	S20°49.0	8.3'	60.5'
13	15°43.7	13.8	214°28.1	4.6'	20°57.3	8.2'	60.5'
14	30°43.6	14.2	228°51.6	4.5'	21°05.5	8.0'	60.5'
15	45°43.6	· · 14.6	243°15.2	4.5'	21°13.5	7.9'	60.5'
16	60°43.5	15.0	257°38.6	4.4'	21°21.4	7.7'	60.5'
17	75°43.4	15.5	272°02.0	4.3'	21°29.1	7.6'	60.5'
18	90°43.4	N21°15.9	286°25.3	4.3'	S21°36.7	7.4'	60.5'
19	105°43.3	16.3	300°48.6	4.2'	21°44.1	7.3'	60.5'
20	120°43.2	16.7	315°11.8	4.2'	21°51.4	7.1'	60.5'
21	135°43.1	· · 17.1	329°35.0	4.1'	21°58.5	6.9'	60.4'
22	150°43.1	17.6	343°58.1	4.1'	22°05.4	6.8'	60.4'
23	165°43.0	18.0	358°21.2	4.0'	22°12.2	6.6'	60.4'
SD = 15.8'			d = 0.4'				
SD = 15.8'			SD = 16.5'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°42.9	N21°18.4	12°44.2	4.0'	S22°18.8	6.5'	60.4'
1	195°42.9	18.8	27°07.2	3.9'	22°25.3	6.3'	60.4'
2	210°42.8	19.2	41°30.1	3.9'	22°31.6	6.1'	60.4'
3	225°42.7	· · 19.6	55°53.0	3.8'	22°37.7	6.0'	60.4'
4	240°42.6	20.0	70°15.8	3.8'	22°43.7	5.8'	60.4'
5	255°42.6	20.5	84°38.6	3.8'	22°49.5	5.6'	60.4'
6	270°42.5	N21°20.9	99°01.4	3.7'	S22°55.1	5.5'	60.3'
7	285°42.4	21.3	113°24.1	3.7'	23°00.6	5.3'	60.3'
8	300°42.3	21.7	127°46.8	3.7'	23°05.9	5.1'	60.3'
9	315°42.3	· · 22.1	142°09.4	3.6'	23°11.0	5.0'	60.3'
10	330°42.2	22.5	156°32.1	3.6'	23°16.0	4.8'	60.3'
11	345°42.1	22.9	170°54.7	3.6'	23°20.7	4.6'	60.3'
12	0°42.1	N21°23.3	185°17.2	3.6'	S23°25.4	4.4'	60.2'
13	15°42.0	23.7	199°39.8	3.5'	23°29.8	4.3'	60.2'
14	30°41.9	24.2	214°02.3	3.5'	23°34.1	4.1'	60.2'
15	45°41.8	· · 24.6	228°24.8	3.5'	23°38.2	3.9'	60.2'
16	60°41.8	25.0	242°47.3	3.5'	23°42.1	3.8'	60.2'
17	75°41.7	25.4	257°09.8	3.5'	23°45.9	3.6'	60.2'
18	90°41.6	N21°25.8	271°32.3	3.5'	S23°49.5	3.4'	60.1'
19	105°41.5	26.2	285°54.7	3.4'	23°52.9	3.2'	60.1'
20	120°41.5	26.6	300°17.2	3.4'	23°56.1	3.1'	60.1'
21	135°41.4	· · 27.0	314°39.6	3.4'	23°59.1	2.9'	60.1'
22	150°41.3	27.4	329°02.1	3.4'	24°02.0	2.7'	60.1'
23	165°41.2	27.8	343°24.5	3.4'	24°04.7	2.5'	60.0'
SD = 15.8'			d = 0.4'				
SD = 15.8'			SD = 16.5'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	—:—	—:—	00:10	—:—	—:—	—:—
66°	—:—	—:—	01:37	22:21	—:—	—:—
64°	—:—	—:—	02:13	21:43	—:—	—:—
62°	—:—	00:57	02:39	21:17	23:03	—:—
60°	—:—	01:43	02:59	20:57	22:13	—:—
N 58°	—:—	02:12	03:15	20:40	21:44	—:—
56°	00:51	02:34	03:29	20:26	21:22	23:08
54°	01:34	02:52	03:41	20:14	21:04	22:23
52°	02:01	03:06	03:51	20:03	20:49	21:55
50°	02:21	03:19	04:01	19:54	20:36	21:34
45°	02:59	03:45	04:20	19:34	20:10	20:56
N 40°	03:25	04:05	04:36	19:18	19:50	20:30
35°	03:45	04:21	04:50	19:05	19:34	20:09
30°	04:02	04:34	05:01	18:53	19:20	19:52
20°	04:28	04:57	05:21	18:34	18:58	19:26
N 10°	04:48	05:15	05:38	18:16	18:39	19:06
0°	05:06	05:31	05:53	18:01	18:23	18:49
S 10°	05:21	05:47	06:09	17:45	18:07	18:33
20°	05:35	06:02	06:26	17:28	17:52	18:19
30°	05:49	06:19	06:45	17:09	17:35	18:05
35°	05:57	06:28	06:56	16:58	17:26	17:57
40°	06:05	06:38	07:08	16:46	17:16	17:49
45°	06:14	06:50	07:23	16:31	17:04	17:40
S 50°	06:23	07:04	07:41	16:13	16:50	17:30
52°	06:28	07:10	07:49	16:04	16:43	17:26
54°	06:33	07:17	07:59	15:55	16:36	17:21
56°	06:38	07:25	08:10	15:44	16:29	17:16
58°	06:43	07:34	08:22	15:32	16:20	17:11
S 60°	06:49	07:43	08:36	15:17	16:10	17:04

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°						
N 70°	20:31			00:13 23:24		
68°	19:43			00:35	00:14	
66°	19:12	21:28		00:53	00:46	00:35
64°	18:49	20:47	22:34	01:07	01:10	01:17
62°	18:31	20:19	21:54	01:19	01:29	01:46
60°	18:16	19:57	21:27	01:29	01:44	02:08
N 58°	18:03	19:39	21:05	01:38	01:58	02:26
56°	17:52	19:25	20:47	01:46	02:09	02:42
54°	17:43	19:12	20:33	01:54	02:19	02:55
52°	17:35	19:01	20:20	02:00	02:29	03:06
50°	17:27	18:51	20:08	02:06	02:37	03:17
45°	17:11	18:30	19:45	02:18	02:54	03:38
N 40°	16:58	18:14	19:26	02:29	03:08	03:55
35°	16:47	18:00	19:10	02:38	03:21	04:10
30°	16:37	17:47	18:56	02:46	03:31	04:23
20°	16:20	17:27	18:33	02:59	03:50	04:45
N 10°	16:06	17:09	18:13	03:11	04:06	05:04
0°	15:53	16:52	17:54	03:23	04:21	05:22
S 10°	15:39	16:36	17:36	03:34	04:36	05:39
20°	15:25	16:18	17:16	03:46	04:52	05:58
30°	15:09	15:58	16:53	04:00	05:10	06:21
35°	15:00	15:46	16:40	04:08	05:21	06:33
40°	14:50	15:33	16:24	04:18	05:34	06:48
45°	14:37	15:17	16:06	04:28	05:49	07:06
S 50°	14:23	14:58	15:43	04:42	06:07	07:28
52°	14:16	14:49	15:33	04:48	06:15	07:39
54°	14:08	14:39	15:20	04:55	06:25	07:51
56°	14:00	14:27	15:06	05:02	06:36	08:04
58°	13:50	14:14	14:50	05:11	06:49	08:20
S 60°	13:39	13:59	14:31	05:21	07:03	08:40

Day	Sun		Mer. Pass hh:mm	Moon		Age 12-14 89-99%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
25	03:05	03:02	11:57	22:07	09:38	
26	02:58	02:55	11:57	23:07	10:36	
27	02:52	02:48	11:57	—:—	11:38	

May 28, 29, 30 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	245°49.5	162°31.8	N24°04.3	67°32.7	N01°40.6	48°43.8	S05°41.8	200°12.6	N15°01.7					Alpheratz	357°31.2	29°15.1	
1	260°51.9	177°31.0		82°34.6	40.3	63°46.4	41.7	215°14.8	01.8					Ankaa	353°04.0	-42°08.5	
2	275°54.4	192°30.1		97°36.6	40.0	78°49.0	41.7	230°16.9	01.9					Schedar	349°27.1	56°41.7	
3	290°56.9	207°29.2	•• 05.0	112°38.5	•• 39.7	93°51.6	•• 41.6	245°19.1	•• 01.9					Diphda	348°44.0	-17°49.4	
4	305°59.3	222°28.3		127°40.5	39.4	108°54.2	41.6	260°21.2	02.0					Achernar	335°18.2	-57°05.0	
5	321°01.8	237°27.4		142°42.4	39.0	123°56.8	41.6	275°23.4	02.1					Hamal	327°47.4	23°36.0	
6	336°04.3	252°26.6	N24°05.6	157°44.4	N01°38.7	138°59.3	S05°41.5	290°25.6	N15°02.2					Polaris	312°32.7	89°23.0	
7	351°06.7	267°25.7		172°46.3	38.4	154°01.9	41.5	305°27.7	02.3					Acamar	315°09.6	-40°11.2	
8	6°09.2	282°24.8		187°48.2	38.1	169°04.5	41.5	320°29.9	02.4					Menkar	314°02.7	4°12.2	
9	21°11.7	297°23.9	•• 06.3	202°50.2	•• 37.8	184°07.1	•• 41.4	335°32.0	•• 02.4					Mirfak	308°23.6	49°57.8	
10	36°14.1	312°23.1		217°52.1	37.5	199°09.7	41.4	350°34.2	02.5					Aldebaran	290°35.9	16°34.0	
11	51°16.6	327°22.2		232°54.0	37.1	214°12.3	41.3	5°36.3	02.6					Rigel	281°00.8	-8°10.2	
12	66°19.0	342°21.3	N24°07.0	247°56.0	N01°36.8	229°14.8	S05°41.3	20°38.5	N15°02.7					Capella	280°17.1	46°01.5	
13	81°21.5	357°20.4		262°57.9	36.5	244°17.4	41.3	35°40.7	02.8					Bellatrix	278°19.4	6°22.4	
14	96°24.0	12°19.5		277°59.9	36.2	259°20.0	41.2	50°42.8	02.9					Elnath	277°57.8	28°37.8	
15	111°26.4	27°18.7	•• 07.7	293°01.8	•• 35.9	274°22.6	•• 41.2	65°45.0	•• 03.0					Alnilam	275°34.5	-1°11.2	
16	126°28.9	42°17.8		308°03.7	35.6	289°25.2	41.2	80°47.1	03.0					Betelgeuse	270°48.6	7°24.6	
17	141°31.4	57°16.9		323°05.7	35.2	304°27.7	41.1	95°49.3	03.1					Canopus	263°51.3	-52°42.9	
18	156°33.8	72°16.0	N24°08.3	338°07.6	N01°34.9	319°30.3	S05°41.1	110°51.4	N15°03.2					Sirius	258°23.4	-16°45.6	
19	171°36.3	87°15.2		353°09.5	34.6	334°32.9	41.0	125°53.6	03.3					Adhara	255°03.4	-29°01.0	
20	186°38.8	102°14.3		8°11.4	34.3	349°35.5	41.0	140°55.8	03.4					Procyon	244°47.4	5°08.8	
21	201°41.2	117°13.4	•• 09.0	23°13.4	•• 34.0	4°38.1	•• 41.0	155°57.9	•• 03.5					Pollux	243°13.3	27°57.2	
22	216°43.7	132°12.5		38°15.3	33.6	19°40.6	40.9	171°00.1	03.5					Avior	234°13.6	-59°36.6	
23	231°46.2	147°11.6		53°17.2	33.3	34°43.2	40.9	186°02.2	03.6					Suhail	222°43.8	-43°33.5	
Mer.pass. 07:35			ν -0.9' d 0.2' m -3.90			ν 1.9' d -0.3' m -0.13			ν 2.6' d -0.0' m -2.30			ν 2.2' d 0.1' m 0.32					
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	246°48.6	162°10.8	N24°09.6	68°19.2	N01°33.0	49°45.8	S05°40.9	201°04.4	N15°03.7					Denebola	182°21.3	14°24.4	
1	261°51.1	177°09.9		83°21.1	32.7	64°48.4	40.8	216°06.5	03.8					Gienah	175°39.8	-17°42.5	
2	276°53.5	192°09.0		98°23.0	32.4	79°51.0	40.8	231°08.7	03.9					Acrux	172°55.5	-63°16.1	
3	291°56.0	207°08.1	•• 10.2	113°24.9	•• 32.0	94°53.5	•• 40.8	246°10.9	•• 04.0					Gacrux	171°47.2	-57°17.0	
4	306°58.5	222°07.2		128°26.9	31.7	109°56.1	40.7	261°13.0	04.0					Alioth	166°09.7	55°48.1	
5	322°00.9	237°06.4		143°28.8	31.4	124°58.7	40.7	276°15.2	04.1					Spica	158°18.3	-11°19.0	
6	337°03.4	252°05.5	N24°10.8	158°30.7	N01°31.1	140°01.3	S05°40.7	291°17.3	N15°04.2					Alkaid	152°49.0	49°10.1	
7	352°05.9	267°04.6		173°32.6	30.7	155°03.8	40.6	306°19.5	04.3					Hadar	148°30.2	-60°31.1	
8	7°08.3	282°03.7		188°34.5	30.4	170°06.4	40.6	321°21.6	04.4					Menkent	147°53.0	-36°31.0	
9	22°10.8	297°02.8	•• 11.4	203°36.5	•• 30.1	185°09.0	•• 40.6	336°23.8	•• 04.5					Arcturus	145°44.4	19°01.7	
10	37°13.3	312°01.9		218°38.4	29.8	200°11.6	40.5	351°26.0	04.6					Rigil Kent.	139°34.6	-60°57.6	
11	52°15.7	327°01.1		233°40.3	29.4	215°14.1	40.5	6°28.1	04.6					Kochab	137°19.1	74°02.2	
12	67°18.2	342°00.2	N24°12.0	248°42.2	N01°29.1	230°16.7	S05°40.5	21°30.3	N15°04.7					Zuben'ubi	136°51.8	-16°09.9	
13	82°20.7	356°59.3		263°44.1	28.8	245°19.3	40.4	36°32.4	04.8					Alphecca	126°00.4	26°37.0	
14	97°23.1	11°58.4		278°46.1	28.5	260°21.9	40.4	51°34.6	04.9					Antares	112°11.1	-26°29.8	
15	112°25.6	26°57.5	•• 12.5	293°48.0	•• 28.1	275°24.4	•• 40.4	66°36.7	•• 05.0					Atria	107°01.5	-69°04.7	
16	127°28.0	41°56.7		308°49.9	27.8	290°27.0	40.3	81°38.9	05.1					Sabik	101°58.4	-15°45.6	
17	142°30.5	56°55.8		323°51.8	27.5	305°29.6	40.3	96°41.1	05.1					Shaula	96°05.2	-37°07.3	
18	157°33.0	71°54.9	N24°13.1	338°53.7	N01°27.2	320°32.2	S05°40.3	111°43.2	N15°05.2					Rasalhague	95°54.9	12°32.4	
19	172°35.4	86°54.0		353°55.6	26.8	335°34.7	40.2	126°45.4	05.3					Eltanin	90°40.0	51°29.1	
20	187°37.9	101°53.1		8°57.5	26.5	350°37.3	40.2	141°47.5	05.4					Kaus Aust.	83°27.5	-34°22.0	
21	202°40.4	116°52.2	•• 13.7	23°59.4	•• 26.2	5°39.9	•• 40.2	156°49.7	•• 05.5					Vega	80°30.4	38°48.6	
22	217°42.8	131°51.4		39°01.3	25.8	20°42.4	40.1	171°51.8	05.6					Nunki	75°43.1	-26°15.4	
23	232°45.3	146°50.5		54°03.3	25.5	35°45.0	40.1	186°54.0	05.6					Altair	61°56.3	8°56.8	
Mer.pass. 07:32			ν -0.9' d 0.2' m -3.90			ν 1.9' d -0.3' m -0.11			ν 2.6' d -0.0' m -2.30			ν 2.2' d 0.1' m 0.33					
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	247°47.8	161°49.6	N24°14.2	69°05.2	N01°25.2	50°47.6	S05°40.1	201°56.2	N15°05.7					May 28 Mon	SHA	Mer.pass	
1	262°50.2	176°48.7		84°07.1	24.9	65°50.2	40.0	216°58.3	05.8					Venus	276°42.4	13:11	
2	277°52.7	191°47.8		99°09.0	24.5	80°52.7	40.0	232°00.5	05.9					Mars	181°43.2	19:27	
3	292°55.2	206°47.0	•• 14.7	114°10.9	•• 24.2	95°55.3	•• 40.0	247°02.6	•• 06.0					Jupiter	162°54.4	20:42	
4	307°57.6	221°46.1		129°12.8	23.9	110°57.9	39.9	262°04.8	06.1					Saturn	314°23.1	10:38	
5	323°00.1	236°45.2		144°14.7	23.5	126°00.4	39.9	277°07.0	06.1								
6	338°02.5	251°44.3	N24°15.2	159°16.6	N01°23.2	141°03.0	S05°39.9	292°09.1	N15°06.2					May 29 Tue	SHA	Mer.pass	
7	353°05.0	266°43.4		174°18.5	22.9	156°05.6	39.8	307°11.3	06.3					Venus	275°22.1	13:12	
8	8°07.5	281°42.5		189°20.4	22.5	171°08.1	39.8	322°13.4	06.4					Mars	181°30.5	19:24	
9	23°09.9	296°41.6	•• 15.8	204°22.3	•• 22.2	186°10.7	•• 39.8	337°15.6	•• 06.5					Jupiter	162°57.2	20:37	
10	38°12.4	311°40.8		219°24.2	21.9	201°13.3	39.7	352°17.7	06.6					Saturn	314°15.8	10:34	
11	53°14.9	326°39.9		234°26.1	21.6	216°15.9	39.7	7°19.9	06.6								
12	68°17.3</																

h Sun							Moon						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°41.1	N21°28.2	357°46.9	3.4'	S24°07.3	2.4'	60.0'						
1	195°41.1	28.6	12°09.4	3.5'	24°09.6	2.2'	60.0'						
2	210°41.0	29.0	26°31.8	3.5'	24°11.8	2.0'	60.0'						
3	225°40.9	29.4	40°54.3	3.5'	24°13.8	1.8'	59.9'						
4	240°40.8	29.8	55°16.8	3.5'	24°15.6	1.7'	59.9'						
5	255°40.8	30.2	69°39.3	3.5'	24°17.3	1.5'	59.9'						
6	270°40.7	N21°30.6	84°01.8	3.5'	S24°18.8	1.3'	59.9'						
7	285°40.6	31.0	98°24.3	3.5'	24°20.1	1.1'	59.8'						
8	300°40.5	31.4	112°46.8	3.6'	24°21.2	0.9'	59.8'						
9	315°40.4	31.8	127°09.4	3.6'	24°22.1	0.8'	59.8'						
10	330°40.4	32.2	141°32.0	3.6'	24°22.9	0.6'	59.8'						
11	345°40.3	32.6	155°54.6	3.6'	24°23.5	0.4'	59.7'						
12	0°40.2	N21°33.0	170°17.2	3.7'	S24°23.9	0.3'	59.7'						
13	15°40.1	33.4	184°39.9	3.7'	24°24.2	0.1'	59.7'						
14	30°40.1	33.7	199°02.6	3.8'	24°24.3	-0.1'	59.7'						
15	45°40.0	34.1	213°25.4	3.8'	24°24.2	-0.3'	59.6'						
16	60°39.9	34.5	227°48.2	3.8'	24°23.9	-0.4'	59.6'						
17	75°39.8	34.9	242°11.0	3.9'	24°23.5	-0.6'	59.6'						
18	90°39.7	N21°35.3	256°33.9	3.9'	S24°22.8	-0.8'	59.6'						
19	105°39.7	35.7	270°56.8	4.0'	24°22.1	-1.0'	59.5'						
20	120°39.6	36.1	285°19.8	4.0'	24°21.1	-1.1'	59.5'						
21	135°39.5	36.5	299°42.8	4.1'	24°20.0	-1.3'	59.5'						
22	150°39.4	36.9	314°05.9	4.1'	24°18.7	-1.5'	59.4'						
23	165°39.3	37.3	328°29.0	4.2'	24°17.2	-1.6'	59.4'						
SD = 15.8' $d = 0.4'$							SD = 16.4'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°39.2	N21°37.6	342°52.2	4.2'	S24°15.6	-1.8'	59.4'						
1	195°39.2	38.0	357°15.5	4.3'	24°13.8	-2.0'	59.3'						
2	210°39.1	38.4	11°38.8	4.4'	24°11.9	-2.1'	59.3'						
3	225°39.0	38.8	26°02.1	4.4'	24°09.8	-2.3'	59.3'						
4	240°38.9	39.2	40°25.6	4.5'	24°07.5	-2.4'	59.3'						
5	255°38.8	39.6	54°49.1	4.6'	24°05.0	-2.6'	59.2'						
6	270°38.8	N21°39.9	69°12.6	4.6'	S24°02.4	-2.8'	59.2'						
7	285°38.7	40.3	83°36.3	4.7'	23°59.6	-2.9'	59.2'						
8	300°38.6	40.7	98°00.0	4.8'	23°56.7	-3.1'	59.1'						
9	315°38.5	41.1	112°23.8	4.9'	23°53.6	-3.2'	59.1'						
10	330°38.4	41.5	126°47.6	4.9'	23°50.4	-3.4'	59.1'						
11	345°38.3	41.8	141°11.6	5.0'	23°47.0	-3.6'	59.0'						
12	0°38.3	N21°42.2	155°35.6	5.1'	S23°43.4	-3.7'	59.0'						
13	15°38.2	42.6	169°59.7	5.2'	23°39.7	-3.9'	59.0'						
14	30°38.1	43.0	184°23.9	5.3'	23°35.9	-4.0'	58.9'						
15	45°38.0	43.3	198°48.2	5.4'	23°31.8	-4.2'	58.9'						
16	60°37.9	43.7	213°12.5	5.4'	23°27.7	-4.3'	58.9'						
17	75°37.8	44.1	227°36.9	5.5'	23°23.4	-4.5'	58.8'						
18	90°37.7	N21°44.5	242°01.5	5.6'	S23°18.9	-4.6'	58.8'						
19	105°37.7	44.8	256°26.1	5.7'	23°14.3	-4.8'	58.8'						
20	120°37.6	45.2	270°50.8	5.8'	23°09.5	-4.9'	58.7'						
21	135°37.5	45.6	285°15.6	5.9'	23°04.6	-5.0'	58.7'						
22	150°37.4	46.0	299°40.5	6.0'	22°59.6	-5.2'	58.7'						
23	165°37.3	46.3	314°05.5	6.1'	22°54.4	-5.3'	58.6'						
SD = 15.8' $d = 0.4'$							SD = 16.2'						
Wed	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°37.2	N21°46.7	328°30.5	6.2'	S22°49.1	-5.5'	58.6'						
1	195°37.2	47.1	342°55.7	6.3'	22°43.7	-5.6'	58.6'						
2	210°37.1	47.4	357°21.0	6.4'	22°38.1	-5.7'	58.5'						
3	225°37.0	47.8	11°46.4	6.5'	22°32.3	-5.9'	58.5'						
4	240°36.9	48.2	26°11.8	6.6'	22°26.5	-6.0'	58.4'						
5	255°36.8	48.5	40°37.4	6.7'	22°20.5	-6.1'	58.4'						
6	270°36.7	N21°48.9	55°03.1	6.8'	S22°14.3	-6.3'	58.4'						
7	285°36.6	49.3	69°28.9	6.9'	22°08.1	-6.4'	58.3'						
8	300°36.5	49.6	83°54.7	7.0'	22°01.7	-6.5'	58.3'						
9	315°36.5	50.0	98°20.7	7.1'	21°55.2	-6.6'	58.3'						
10	330°36.4	50.4	112°46.8	7.2'	21°48.5	-6.8'	58.2'						
11	345°36.3	50.7	127°13.0	7.3'	21°41.8	-6.9'	58.2'						
12	0°36.2	N21°51.1	141°39.3	7.4'	S21°34.9	-7.0'	58.2'						
13	15°36.1	51.5	156°05.7	7.5'	21°27.9	-7.1'	58.1'						
14	30°36.0	51.8	170°32.2	7.6'	21°20.8	-7.2'	58.1'						
15	45°35.9	52.2	184°58.8	7.7'	21°13.5	-7.4'	58.1'						
16	60°35.8	52.5	199°25.5	7.8'	21°06.2	-7.5'	58.0'						
17	75°35.7	52.9	213°52.4	7.9'	20°58.7	-7.6'	58.0'						
18	90°35.7	N21°53.3	228°19.3	8.0'	S20°51.1	-7.7'	57.9'						
19	105°35.6	53.6	242°46.3	8.2'	20°43.4	-7.8'	57.9'						
20	120°35.5	54.0	257°13.5	8.3'	20°35.6	-7.9'	57.9'						
21	135°35.4	54.3	271°40.7	8.4'	20°27.7	-8.0'	57.8'						
22	150°35.3	54.7	286°08.1	8.5'	20°19.7	-8.1'	57.8'						
23	165°35.2	55.0	300°35.6	8.6'	20°11.5	-8.2'	57.8'						
SD = 15.8' $d = 0.4'$							SD = 16.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	--	--	01:24	22:35	--	--
64°	--	--	02:05	21:52	--	--
62°	--	00:37	02:33	21:24	23:27	--
60°	--	01:34	02:54	21:03	22:23	--
N 58°	--	02:06	03:11	20:45	21:51	--
56°	00:33	02:29	03:25	20:31	21:27	23:29
54°	01:26	02:47	03:38	20:18	21:09	22:32
52°	01:55	03:03	03:48	20:07	20:53	22:02
50°	02:17	03:16	03:58	19:57	20:40	21:39
45°	02:55	03:42	04:18	19:37	20:13	21:00
N 40°	03:23	04:03	04:35	19:21	19:53	20:33
35°	03:44	04:19	04:48	19:07	19:36	20:12
30°	04:01	04:33	05:00	18:55	19:22	19:55
20°	04:27	04:56	05:20	18:35	18:59	19:28
N 10°	04:48	05:15	05:38	18:17	18:40	19:07
0°	05:06	05:32	05:54	18:01	18:23	18:49
S 10°	05:21	05:47	06:10	17:45	18:08	18:34
20°	05:36	06:03	06:27	17:28	17:52	18:19
30°	05:51	06:20	06:46	17:08	17:34	18:04
35°	05:59	06:30	06:58	16:57	17:25	17:56
40°	06:07	06:40	07:10	16:44	17:14	17:48
45°	06:16	06:53	07:26	16:29	17:02	17:39
S 50°	06:26	07:07	07:44	16:10	16:48	17:28
52°	06:31	07:14	07:53	16:01	16:41	17:24
54°	06:36	07:21	08:03	15:52	16:34	17:19
56°	06:41	07:29	08:14	15:40	16:26	17:13
58°	06:47	07:38	08:27	15:28	16:17	17:08
S 60°	06:53	07:48	08:42	15:13	16:07	17:01

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	██	██	██	██	██	██
N 70°	██	██	██	██	██	██
68°	██	██	██	██	██	██
66°	██	██	00:54	██	██	03:37
64°	23:42		00:07	01:41	02:43	04:23
62°	23:01	23:37	23:55	02:21	03:24	04:53
60°	22:33	23:14	23:38	02:49	03:52	05:15
N 58°	22:11	22:55	23:24	03:11	04:14	05:33
56°	21:53	22:40	23:11	03:28	04:32	05:49
54°	21:38	22:26	23:00	03:43	04:47	06:02
52°	21:25	22:14	22:51	03:56	05:00	06:13
50°	21:13	22:04	22:42	04:08	05:11	06:23
45°	20:49	21:42	22:24	04:32	05:35	06:44
N 40°	20:30	21:24	22:09	04:51	05:54	07:01
35°	20:14	21:10	21:56	05:07	06:10	07:16
30°	20:00	20:57	21:45	05:21	06:24	07:28
20°	19:36	20:34	21:26	05:45	06:47	07:49
N 10°	19:16	20:15	21:09	06:05	07:07	08:08
0°	18:57	19:57	20:54	06:24	07:26	08:25
S 10°	18:38	19:39	20:38	06:43	07:45	08:42
20°	18:17	19:20	20:21	07:04	08:05	09:00
30°	17:54	18:57	20:02	07:27	08:28	09:20
35°	17:40	18:44	19:50	07:41	08:41	09:32
40°	17:24	18:29	19:37	07:57	08:57	09:46
45°	17:05	18:11	19:22	08:16	09:15	10:02
S 50°	16:41	17:49	19:03	08:40	09:38	10:22
52°	16:29	17:38	18:54	08:51	09:49	10:32
54°	16:16	17:26	18:44	09:04	10:01	10:42
56°	16:01	17:12	18:32	09:19	10:15	10:54
58°	15:44	16:56	18:19	09:37	10:32	11:08
S 60°	15:22	16:36	18:03	09:58	10:52	11:24

May 31, 01, 02 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	248°46.9	161°28.4	N24°18.1	69°50.7	N01°17.2	51°49.2	S05°39.3	202°48.0	N15°07.7						Alpheratz	357°31.2	29°15.1
1	263°49.4	176°27.5	18.3	84°52.6	16.9	66°51.8	39.3	217°50.1	07.8						Ankaa	353°04.0	-42°08.5
2	278°51.8	191°26.6	18.4	99°54.5	16.5	81°54.3	39.2	232°52.3	07.9						Schedar	349°27.1	56°41.7
3	293°54.3	206°25.7	•• 18.6	114°56.3	•• 16.2	96°56.9	•• 39.2	247°54.4	•• 08.0						Diphda	348°44.0	-17°49.4
4	308°56.8	221°24.8	18.7	129°58.2	15.9	111°59.5	39.2	262°56.6	08.1						Achernar	335°18.1	-57°05.0
5	323°59.2	236°23.9	18.8	145°00.1	15.5	127°02.0	39.2	277°58.8	08.1						Hamal	327°47.4	23°36.0
6	339°01.7	251°23.1	N24°19.0	160°02.0	N01°15.2	142°04.6	S05°39.1	293°00.9	N15°08.2						Polaris	312°31.9	89°23.0
7	354°04.2	266°22.2	19.1	175°03.9	14.8	157°07.2	39.1	308°03.1	08.3						Acamar	315°09.5	-40°11.1
8	9°06.6	281°21.3	19.3	190°05.8	14.5	172°09.7	39.1	323°05.2	08.4						Menkar	314°02.7	4°12.2
9	24°09.1	296°20.4	•• 19.4	205°07.6	•• 14.2	187°12.3	•• 39.0	338°07.4	•• 08.5						Mirfak	308°23.5	49°57.8
10	39°11.5	311°19.5	19.5	220°09.5	13.8	202°14.8	39.0	353°09.6	08.6						Aldebaran	290°35.9	16°34.0
11	54°14.0	326°18.6	19.7	235°11.4	13.5	217°17.4	39.0	8°11.7	08.6						Rigel	281°00.8	-8°10.2
12	69°16.5	341°17.7	N24°19.8	250°13.3	N01°13.1	232°20.0	S05°39.0	23°13.9	N15°08.7						Capella	280°17.1	46°01.5
13	84°18.9	356°16.9	19.9	265°15.1	12.8	247°22.5	38.9	38°16.0	08.8						Bellatrix	278°19.4	6°22.4
14	99°21.4	11°16.0	20.1	280°17.0	12.5	262°25.1	38.9	53°18.2	08.9						Elnath	277°57.8	28°37.8
15	114°23.9	26°15.1	•• 20.2	295°18.9	•• 12.1	277°27.7	•• 38.9	68°20.4	•• 09.0						Alnilam	275°34.5	-1°11.2
16	129°26.3	41°14.2	20.3	310°20.8	11.8	292°30.2	38.8	83°22.5	09.0						Betelgeuse	270°48.6	7°24.6
17	144°28.8	56°13.3	20.5	325°22.7	11.4	307°32.8	38.8	98°24.7	09.1						Canopus	263°51.3	-52°42.9
18	159°31.3	71°12.4	N24°20.6	340°24.5	N01°11.1	322°35.3	S05°38.8	113°26.8	N15°09.2						Sirius	258°23.4	-16°45.6
19	174°33.7	86°11.5	20.7	355°26.4	10.8	337°37.9	38.8	128°29.0	09.3						Adhara	255°03.4	-29°01.0
20	189°36.2	101°10.7	20.8	10°28.3	10.4	352°40.4	38.7	143°31.2	09.4						Procyon	244°47.4	5°08.8
21	204°38.6	116°09.8	•• 21.0	25°30.1	•• 10.1	7°43.0	•• 38.7	158°33.3	•• 09.5						Pollux	243°13.3	27°57.2
22	219°41.1	131°08.9	21.1	40°32.0	09.7	22°45.6	38.7	173°35.5	09.5						Avior	234°13.6	-59°36.6
23	234°43.6	146°08.0	21.2	55°33.9	09.4	37°48.1	38.7	188°37.6	09.6						Suhail	222°43.8	-43°33.5
Mer.pass. 07:24			ν -0.9' d 0.1' m -3.90			ν 1.9' d -0.3' m -0.07		ν 2.6' d -0.0' m -2.28		ν 2.2' d 0.1' m 0.33					Miaplacidus	221°37.7	-69°50.7
															Alphard	217°44.4	-8°47.4
															Regulus	207°30.8	11°49.3
															Dubhe	193°36.9	61°35.7
															Denebola	182°21.3	14°24.4
															Gienah	175°39.8	-17°42.6
															Acrux	172°55.6	-63°16.1
															Gacrux	171°47.2	-57°17.0
															Alioth	166°09.7	55°48.1
															Spica	158°18.3	-11°19.0
															Alkaid	152°49.0	49°10.1
															Hadar	148°30.2	-60°31.1
															Menkent	147°53.0	-36°31.1
															Arcturus	145°44.4	19°01.8
															Rigel Kent.	139°34.6	-60°57.6
															Kochab	137°19.2	74°02.2
															Zuben'ubi	136°51.8	-16°09.9
															Alphecca	126°00.4	26°37.0
															Antares	112°11.1	-26°29.8
															Atria	107°01.4	-69°04.7
															Sabik	101°58.4	-15°45.6
															Shaula	96°05.1	-37°07.3
															Rasalhague	95°54.9	12°32.4
															Eltanin	90°40.0	51°29.1
															Kaus Aust.	83°27.5	-34°22.0
															Vega	80°30.4	38°48.6
															Nunki	75°43.1	-26°15.4
															Altair	61°56.3	8°56.8
															Peacock	52°59.9	-56°38.0
															Deneb	49°23.1	45°23.0
															Enif	33°35.2	10°00.7
															Al Na'ir	27°28.5	-46°48.7
															Fomalhaut	15°10.7	-29°27.7
															Scheat	13°41.7	28°14.5
															Markab	13°26.4	15°21.8
															May 31 Thu	SHA	Mer.pass
															Venus	272°41.5	13:15
															Mars	181°03.8	19:18
															Jupiter	163°02.3	20:29
															Saturn	314°01.1	10:27
															Jun 01 Fri	SHA	Mer.pass
															Venus	271°21.1	13:16
															Mars	180°49.7	19:15
															Jupiter	163°04.6	20:25
															Saturn	313°53.7	10:24
															Jun 02 Sat	SHA	Mer.pass
															Venus	270°00.6	13:18
															Mars	180°35.2	19:12
															Jupiter	163°06.8	20:21
															Saturn	313°46.5	10:20
															Horizontal parallax		
															Venus:	0.1	
															Mars:	0.2	
Mer.pass. 07:16			ν -0.9' d 0.1' m -3.89			ν 1.8' d -0.4' m -0.04		ν 2.6' d -0.0' m -2.27		ν 2.2' d 0.1' m 0.33							

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°35.1	N21°55.4	315°03.2	8.7'	S20°03.3	-8.3'	57.7'
1	195°35.0	55.7	329°30.9	8.8'	19°55.0	-8.4'	57.7'
2	210°34.9	56.1	343°58.7	8.9'	19°46.5	-8.5'	57.6'
3	225°34.8	· · 56.5	358°26.6	9.0'	19°38.0	-8.6'	57.6'
4	240°34.8	56.8	12°54.6	9.1'	19°29.4	-8.7'	57.6'
5	255°34.7	57.2	27°22.7	9.2'	19°20.6	-8.8'	57.5'
6	270°34.6	N21°57.5	41°50.9	9.3'	S19°11.8	-8.9'	57.5'
7	285°34.5	57.9	56°19.3	9.4'	19°02.9	-9.0'	57.5'
8	300°34.4	58.2	70°47.7	9.6'	18°53.9	-9.1'	57.4'
9	315°34.3	· · 58.6	85°16.3	9.7'	18°44.8	-9.2'	57.4'
10	330°34.2	58.9	99°45.0	9.8'	18°35.6	-9.3'	57.4'
11	345°34.1	59.2	114°13.7	9.9'	18°26.3	-9.4'	57.3'
12	0°34.0	N21°59.6	128°42.6	10.0'	S18°16.9	-9.4'	57.3'
13	15°33.9	21°59.9	143°11.6	10.1'	18°07.5	-9.5'	57.2'
14	30°33.8	22°00.3	157°40.7	10.2'	17°58.0	-9.6'	57.2'
15	45°33.7	· · 00.6	172°09.9	10.3'	17°48.3	-9.7'	57.2'
16	60°33.6	01.0	186°39.2	10.4'	17°38.6	-9.8'	57.1'
17	75°33.6	01.3	201°08.6	10.5'	17°28.9	-9.9'	57.1'
18	90°33.5	N22°01.7	215°38.1	10.6'	S17°19.0	-9.9'	57.1'
19	105°33.4	02.0	230°07.7	10.7'	17°09.1	-10.0'	57.0'
20	120°33.3	02.3	244°37.4	10.8'	16°59.1	-10.1'	57.0'
21	135°33.2	· · 02.7	259°07.2	10.9'	16°49.0	-10.1'	57.0'
22	150°33.1	03.0	273°37.2	11.0'	16°38.9	-10.2'	56.9'
23	165°33.0	03.4	288°07.2	11.1'	16°28.6	-10.3'	56.9'
SD = 15.8'			SD = 15.7'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°32.9	N22°03.7	302°37.3	11.2'	S16°18.3	-10.4'	56.8'
1	195°32.8	04.0	317°07.5	11.3'	16°08.0	-10.4'	56.8'
2	210°32.7	04.4	331°37.8	11.4'	15°57.6	-10.5'	56.8'
3	225°32.6	· · 04.7	346°08.3	11.5'	15°47.1	-10.6'	56.7'
4	240°32.5	05.0	0°38.8	11.6'	15°36.5	-10.6'	56.7'
5	255°32.4	05.4	15°09.4	11.7'	15°25.9	-10.7'	56.7'
6	270°32.3	N22°05.7	29°40.1	11.8'	S15°15.2	-10.7'	56.6'
7	285°32.2	06.0	44°10.9	11.9'	15°04.5	-10.8'	56.6'
8	300°32.1	06.4	58°41.8	12.0'	14°53.7	-10.9'	56.6'
9	315°32.0	· · 06.7	73°12.8	12.1'	14°42.8	-10.9'	56.5'
10	330°31.9	07.0	87°43.9	12.2'	14°31.9	-11.0'	56.5'
11	345°31.8	07.4	102°15.0	12.3'	14°21.0	-11.0'	56.5'
12	0°31.7	N22°07.7	116°46.3	12.4'	S14°09.9	-11.1'	56.4'
13	15°31.6	08.0	131°17.7	12.4'	13°58.9	-11.1'	56.4'
14	30°31.5	08.4	145°49.1	12.5'	13°47.7	-11.2'	56.4'
15	45°31.4	· · 08.7	160°20.6	12.6'	13°36.5	-11.2'	56.3'
16	60°31.4	09.0	174°52.3	12.7'	13°25.3	-11.3'	56.3'
17	75°31.3	09.4	189°24.0	12.8'	13°14.0	-11.3'	56.3'
18	90°31.2	N22°09.7	203°55.8	12.9'	S13°02.7	-11.4'	56.2'
19	105°31.1	10.0	218°27.7	13.0'	12°51.4	-11.4'	56.2'
20	120°31.0	10.3	232°59.6	13.0'	12°39.9	-11.5'	56.2'
21	135°30.9	· · 10.7	247°31.7	13.1'	12°28.5	-11.5'	56.1'
22	150°30.8	11.0	262°03.8	13.2'	12°17.0	-11.5'	56.1'
23	165°30.7	11.3	276°36.0	13.3'	12°05.4	-11.6'	56.1'
SD = 15.8'			SD = 15.5'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°30.6	N22°11.6	291°08.3	13.4'	S11°53.9	-11.6'	56.0'
1	195°30.5	11.9	305°40.7	13.4'	11°42.2	-11.7'	56.0'
2	210°30.4	12.3	320°13.1	13.5'	11°30.6	-11.7'	56.0'
3	225°30.3	· · 12.6	334°45.6	13.6'	11°18.9	-11.7'	55.9'
4	240°30.2	12.9	349°18.2	13.7'	11°07.2	-11.8'	55.9'
5	255°30.1	13.2	3°50.9	13.8'	10°55.4	-11.8'	55.9'
6	270°30.0	N22°13.5	18°23.7	13.8'	S10°43.6	-11.8'	55.8'
7	285°29.9	13.9	32°56.5	13.9'	10°31.8	-11.9'	55.8'
8	300°29.8	14.2	47°29.4	14.0'	10°19.9	-11.9'	55.8'
9	315°29.7	· · 14.5	62°02.4	14.0'	10°08.0	-11.9'	55.7'
10	330°29.6	14.8	76°35.4	14.1'	09°56.1	-12.0'	55.7'
11	345°29.5	15.1	91°08.5	14.2'	09°44.1	-12.0'	55.7'
12	0°29.4	N22°15.4	105°41.7	14.2'	S09°32.1	-12.0'	55.7'
13	15°29.3	15.8	120°14.9	14.3'	09°20.1	-12.0'	55.6'
14	30°29.2	16.1	134°48.2	14.4'	09°08.1	-12.1'	55.6'
15	45°29.1	· · 16.4	149°21.6	14.4'	08°56.0	-12.1'	55.6'
16	60°29.0	16.7	163°55.0	14.5'	08°43.9	-12.1'	55.5'
17	75°28.9	17.0	178°28.5	14.6'	08°31.8	-12.1'	55.5'
18	90°28.8	N22°17.3	193°02.1	14.6'	S08°19.6	-12.2'	55.5'
19	105°28.6	17.6	207°35.7	14.7'	08°07.5	-12.2'	55.5'
20	120°28.5	17.9	222°09.4	14.7'	07°55.3	-12.2'	55.4'
21	135°28.4	· · 18.2	236°43.1	14.8'	07°43.1	-12.2'	55.4'
22	150°28.3	18.5	251°16.9	14.8'	07°30.9	-12.2'	55.4'
23	165°28.2	18.9	265°50.7	14.9'	07°18.6	-12.3'	55.3'
SD = 15.8'			SD = 15.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	—:—	—:—	01:11	22:49	—:—	—:—
64°	—:—	—:—	01:57	22:01	—:—	—:—
62°	—:—	—:—	02:27	21:31	—:—	—:—
60°	—:—	01:25	02:49	21:08	22:33	—:—
N 58°	—:—	02:00	03:07	20:50	21:58	—:—
56°	—:—	02:24	03:22	20:35	21:33	—:—
54°	01:18	02:43	03:35	20:22	21:13	22:40
52°	01:50	02:59	03:46	20:11	20:57	22:08
50°	02:12	03:13	03:56	20:01	20:44	21:44
45°	02:53	03:40	04:16	19:40	20:16	21:04
N 40°	03:21	04:01	04:33	19:23	19:55	20:36
35°	03:42	04:18	04:47	19:09	19:38	20:14
30°	04:00	04:32	04:59	18:57	19:24	19:56
20°	04:27	04:56	05:20	18:36	19:00	19:29
N 10°	04:48	05:15	05:38	18:18	18:41	19:08
0°	05:06	05:32	05:54	18:02	18:24	18:50
S 10°	05:22	05:48	06:11	17:45	18:08	18:34
20°	05:37	06:04	06:28	17:28	17:51	18:19
30°	05:52	06:22	06:48	17:08	17:34	18:03
35°	06:00	06:32	06:59	16:56	17:24	17:55
40°	06:09	06:42	07:13	16:43	17:13	17:47
45°	06:18	06:55	07:28	16:27	17:01	17:37
S 50°	06:29	07:10	07:47	16:08	16:46	17:27
52°	06:34	07:17	07:56	15:59	16:39	17:22
54°	06:39	07:24	08:07	15:49	16:31	17:17
56°	06:44	07:32	08:18	15:37	16:23	17:11
58°	06:50	07:42	08:31	15:24	16:14	17:05
S 60°	06:57	07:52	08:47	15:09	16:03	16:58

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	02:03	01:17	■	06:11	08:35
N 70°	02:34	01:24	00:58	03:54	06:49	08:52
68°	01:20	00:56	00:42	05:07	07:15	09:06
66°	00:42	00:35	00:30	05:43	07:35	09:17
64°	00:16	00:18	00:19	06:09	07:51	09:26
62°	· ·	00:04	00:10	06:29	08:04	09:34
60°	23:52	· ·	00:02	06:46	08:15	09:41
N 58°	23:42	23:55	· ·	06:59	08:25	09:47
56°	23:33	23:49	· ·	07:11	08:33	09:52
54°	23:25	23:43	23:59	07:21	08:40	09:57
52°	23:18	23:39	23:56	07:30	08:47	10:01
50°	23:11	23:34	23:53	07:38	08:53	10:05
45°	22:57	23:24	23:47	07:56	09:06	10:13
N 40°	22:45	23:16	23:43	08:10	09:16	10:20
35°	22:35	23:09	23:38	08:22	09:25	10:26
30°	22:27	23:03	23:35	08:32	09:33	10:31
20°	22:11	22:52	23:28	08:50	09:46	10:40
N 10°	21:58	22:42	23:23	09:05	09:58	10:48
0°	21:46	22:33	23:18	09:19	10:09	10:55
S 10°	21:33	22:24	23:12	09:33	10:20	11:02
20°	21:20	22:15	23:07	09:48	10:31	11:10
30°	21:04	22:04	23:00	10:05	10:44	11:18
35°	20:55	21:57	22:56	10:15	10:51	11:23
40°	20:45	21:50	22:52	10:27	11:00	11:28
45°	20:33	21:41	22:47	10:40	11:10	11:35
S 50°	20:18	21:31	22:41	10:56	11:22	11:42
52°	20:11	21:26	22:38	11:03	11:27	11:46
54°	20:03	21:21	22:35	11:12	11:33	11:50
56°	19:55	21:15	22:32	11:21	11:40	11:54
58°	19:45	21:08	22:28	11:31	11:47	11:59
S 60°	19:33	21:01	22:24	11:43	11:56	12:04

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	18-20
	mm:ss	mm:ss	mm:ss	hh:mm	hh:mm	87-70%
31	02:20	02:16	11:58	03:06	15:32	
01	02:12	02:07	11:58	03:57	16:21	
02	02:02	01:57	11:58	04:44	17:06	

June 03, 04, 05 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	251°44.3	160°24.5	N24°25.6	72°04.5	N00°52.2	54°53.1	S05°37.5	205°23.5	N15°13.6
1	266°46.8	175°23.6	25.6	87°06.4	51.9	69°55.7	37.5	220°25.7	13.7
2	281°49.2	190°22.7	25.7	102°08.2	51.5	84°58.2	37.4	235°27.8	13.8
3	296°51.7	205°21.8	•• 25.7	117°10.0	•• 51.1	100°00.8	•• 37.4	250°30.0	•• 13.9
4	311°54.2	220°20.9	25.8	132°11.8	50.8	115°03.3	37.4	265°32.2	14.0
5	326°56.6	235°20.0	25.8	147°13.7	50.4	130°05.8	37.4	280°34.3	14.0
6	341°59.1	250°19.1	N24°25.9	162°15.5	N00°50.1	145°08.4	S05°37.3	295°36.5	N15°14.1
7	357°01.6	265°18.2	25.9	177°17.3	49.7	160°10.9	37.3	310°38.7	14.2
8	12°04.0	280°17.3	26.0	192°19.1	49.3	175°13.5	37.3	325°40.8	14.3
9	27°06.5	295°16.4	•• 26.0	207°21.0	•• 49.0	190°16.0	•• 37.3	340°43.0	•• 14.4
10	42°09.0	310°15.6	26.1	222°22.8	48.6	205°18.6	37.3	355°45.1	14.4
11	57°11.4	325°14.7	26.1	237°24.6	48.3	220°21.1	37.2	10°47.3	14.5
12	72°13.9	340°13.8	N24°26.2	252°26.4	N00°47.9	235°23.6	S05°37.2	25°49.5	N15°14.6
13	87°16.4	355°12.9	26.2	267°28.3	47.5	250°26.2	37.2	40°51.6	14.7
14	102°18.8	10°12.0	26.3	282°30.1	47.2	265°28.7	37.2	55°53.8	14.8
15	117°21.3	25°11.1	•• 26.3	297°31.9	•• 46.8	280°31.3	•• 37.2	70°56.0	•• 14.8
16	132°23.7	40°10.2	26.3	312°33.7	46.5	295°33.8	37.1	85°58.1	14.9
17	147°26.2	55°09.3	26.4	327°35.5	46.1	310°36.3	37.1	101°00.3	15.0
18	162°28.7	70°08.4	N24°26.4	342°37.4	N00°45.7	325°38.9	S05°37.1	116°02.4	N15°15.1
19	177°31.1	85°07.6	26.5	357°39.2	45.4	340°41.4	37.1	131°04.6	15.2
20	192°33.6	100°06.7	26.5	12°41.0	45.0	355°44.0	37.1	146°06.8	15.2
21	207°36.1	115°05.8	•• 26.5	27°42.8	•• 44.6	10°46.5	•• 37.1	161°08.9	•• 15.3
22	222°38.5	130°04.9	26.6	42°44.6	44.3	25°49.0	37.0	176°11.1	15.4
23	237°41.0	145°04.0	26.6	57°46.4	43.9	40°51.6	37.0	191°13.3	15.5
Mer.pass. 07:12		ν -0.9' d 0.1' m -3.89		ν 1.8' d -0.4' m -0.02		ν 2.5' d -0.0' m -2.26		ν 2.2' d 0.1' m 0.33	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	252°43.5	160°03.1	N24°26.6	72°48.3	N00°43.6	55°54.1	S05°37.0	206°15.4	N15°15.6
1	267°45.9	175°02.2	26.6	87°50.1	43.2	70°56.6	37.0	221°17.6	15.6
2	282°48.4	190°01.3	26.7	102°51.9	42.8	85°59.2	37.0	236°19.7	15.7
3	297°50.8	205°00.4	•• 26.7	117°53.7	•• 42.5	101°01.7	•• 36.9	251°21.9	•• 15.8
4	312°53.3	219°59.5	26.7	132°55.5	42.1	116°04.3	36.9	266°24.1	15.9
5	327°55.8	234°58.7	26.7	147°57.3	41.7	131°06.8	36.9	281°26.2	16.0
6	342°58.2	249°57.8	N24°26.8	162°59.1	N00°41.4	146°09.3	S05°36.9	296°28.4	N15°16.1
7	358°00.7	264°56.9	26.8	178°00.9	41.0	161°11.9	36.9	311°30.6	16.1
8	13°03.2	279°56.0	26.8	193°02.7	40.6	176°14.4	36.9	326°32.7	16.2
9	28°05.6	294°55.1	•• 26.8	208°04.5	•• 40.3	191°16.9	•• 36.8	341°34.9	•• 16.3
10	43°08.1	309°54.2	26.8	223°06.3	39.9	206°19.5	36.8	356°37.1	16.4
11	58°10.6	324°53.3	26.9	238°08.1	39.5	221°22.0	36.8	11°39.2	16.5
12	73°13.0	339°52.4	N24°26.9	253°10.0	N00°39.2	236°24.5	S05°36.8	26°41.4	N15°16.5
13	88°15.5	354°51.5	26.9	268°11.8	38.8	251°27.1	36.8	41°43.5	16.6
14	103°18.0	9°50.7	26.9	283°13.6	38.4	266°29.6	36.8	56°45.7	16.7
15	118°20.4	24°49.8	•• 26.9	298°15.4	•• 38.1	281°32.1	•• 36.7	71°47.9	•• 16.8
16	133°22.9	39°48.9	26.9	313°17.2	37.7	296°34.7	36.7	86°50.0	16.9
17	148°25.3	54°48.0	26.9	328°19.0	37.3	311°37.2	36.7	101°52.2	16.9
18	163°27.8	69°47.1	N24°26.9	343°20.8	N00°37.0	326°39.7	S05°36.7	116°54.4	N15°17.0
19	178°30.3	84°46.2	26.9	358°22.6	36.6	341°42.3	36.7	131°56.5	17.1
20	193°32.7	99°45.3	26.9	13°24.4	36.2	356°44.8	36.7	146°58.7	17.2
21	208°35.2	114°44.4	•• 26.9	28°26.2	•• 35.8	11°47.3	•• 36.6	162°00.9	•• 17.3
22	223°37.7	129°43.5	26.9	43°28.0	35.5	26°49.9	36.6	177°03.0	17.3
23	238°40.1	144°42.7	26.9	58°29.8	35.1	41°52.4	36.6	192°05.2	17.4
Mer.pass. 07:08		ν -0.9' d 0.0' m -3.89		ν 1.8' d -0.4' m -0.00		ν 2.5' d -0.0' m -2.26		ν 2.2' d 0.1' m 0.33	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	253°42.6	159°41.8	N24°26.9	73°31.5	N00°34.7	56°54.9	S05°36.6	207°07.4	N15°17.5
1	268°45.1	174°40.9	26.9	88°33.3	34.4	71°57.5	36.6	222°09.5	17.6
2	283°47.5	189°40.0	26.9	103°35.1	34.0	87°00.0	36.6	237°11.7	17.6
3	298°50.0	204°39.1	•• 26.9	118°36.9	•• 33.6	102°02.5	•• 36.5	252°13.8	•• 17.7
4	313°52.5	219°38.2	26.9	133°38.7	33.2	117°05.0	36.5	267°16.0	17.8
5	328°54.9	234°37.3	26.9	148°40.5	32.9	132°07.6	36.5	282°18.2	17.9
6	343°57.4	249°36.4	N24°26.9	163°42.3	N00°32.5	147°10.1	S05°36.5	297°20.3	N15°18.0
7	358°59.8	264°35.5	26.9	178°44.1	32.1	162°12.6	36.5	312°22.5	18.0
8	14°02.3	279°34.7	26.9	193°45.9	31.8	177°15.2	36.5	327°24.7	18.1
9	29°04.8	294°33.8	•• 26.9	208°47.7	•• 31.4	192°17.7	•• 36.5	342°26.8	•• 18.2
10	44°07.2	309°32.9	26.9	223°49.5	31.0	207°20.2	36.4	357°29.0	18.3
11	59°09.7	324°32.0	26.9	238°51.2	30.6	222°22.7	36.4	12°31.2	18.4
12	74°12.2	339°31.1	N24°26.8	253°53.0	N00°30.3	237°25.3	S05°36.4	27°33.3	N15°18.4
13	89°14.6	354°30.2	26.8	268°54.8	29.9	252°27.8	36.4	42°35.5	18.5
14	104°17.1	9°29.3	26.8	283°56.6	29.5	267°30.3	36.4	57°37.7	18.6
15	119°19.6	24°28.4	•• 26.8	298°58.4	•• 29.1	282°32.8	•• 36.4	72°39.8	•• 18.7
16	134°22.0	39°27.6	26.8	314°00.2	28.8	297°35.4	36.4	87°42.0	18.8
17	149°24.5	54°26.7	26.7	329°02.0	28.4	312°37.9	36.3	102°44.2	18.8
18	164°26.9	69°25.8	N24°26.7	344°03.7	N00°28.0	327°40.4	S05°36.3	117°46.3	N15°18.9
19	179°29.4	84°24.9	26.7	359°05.5	27.6	342°42.9	36.3	132°48.5	19.0
20	194°31.9	99°24.0	26.7	14°07.3	27.3	357°45.5	36.3	147°50.7	19.1
21	209°34.3	114°23.1	•• 26.6	29°09.1	•• 26.9	12°48.0	•• 36.3	162°52.8	•• 19.2
22	224°36.8	129°22.2	26.6	44°10.9	26.5	27°50.5	36.3	177°55.0	19.2
23	239°39.3	144°21.3	26.6	59°12.6	26.1	42°53.0	36.3	192°57.1	19.3
Mer.pass. 07:04		ν -0.9' d -0.0' m -3.89		ν 1.8' d -0.4' m 0.01		ν 2.5' d -0.0' m -2.25		ν 2.2' d 0.1' m 0.33	

Stars		
	SHA	Dec
Alpheratz	357°31.1	29°15.1
Ankaa	353°03.9	-42°08.5
Schedar	349°27.0	56°41.7
Diphda	348°43.9	-17°49.4
Achernar	335°18.1	-57°05.0
Hamal	327°47.4	23°36.0
Polaris	312°31.2	89°23.0
Acamar	315°09.5	-40°11.1
Menkar	314°02.7	4°12.2
Mirfak	308°23.5	49°57.7
Aldebaran	290°35.9	16°34.0
Rigel	281°00.8	-8°10.2
Capella	280°17.1	46°01.5
Bellatrix	278°19.4	6°22.4
Elnath	277°57.7	28°37.8
Alnilam	275°34.5	-1°11.1
Betelgeuse	270°48.6	7°24.6
Canopus	263°51.3	-52°42.9
Sirius	258°23.4	-16°45.6
Adhara	255°03.4	-29°01.0
Procyon	244°47.4	5°08.8
Pollux	243°13.3	27°57.2
Avior	234°13.6	-59°36.6
Suhail	222°43.8	-43°33.5
Miaplacidus	221°37.7	-69°50.7
Alphard	217°44.4	-8°47.4
Regulus	207°30.8	11°49.3
Dubhe	193°36.9	61°35.7
Denebola	182°21.3	14°24.4
Gienah	175°39.8	-17°42.5
Acrux	172°55.6	-63°16.1
Gacrux	171°47.2	-57°17.0
Alioth	166°09.8	55°48.1
Spica	158°18.3	-11°19.0
Alkaid	152°49.0	49°10.1
Hadar	148°30.2	-60°31.1
Menkent	147°53.0	-36°31.1
Arcturus	145°44.4	19°01.8
Rigel Kent.	139°34.7	-60°57.6
Kochab	137°19.2	74°02.2
Zuben'ubi	136°51.8	-16°09.9
Alphecca	126°00.4	26°37.0
Antares	112°11.1	-26°29.8
Atria	107°01.4	-69°04.7
Sabik	101°58.4	-15°45.6
Shaula	96°05.1	-37°07.4
Rasalhague	95°54.9	12°32.4
Eltanin	90°40.0	51°29.1
Kaus Aust.	83°27.4	-34°22.0
Vega	80°30.4	38°48.7
Nunki	75°43.1	-26°15.4
Altair	61°56.2	8°56.9
Peacock	52°59.9	-56°38.0
Deneb	49°23.0	45°23.0
Enif	33°35.2	10°00.7
Al Na'ir	27°28.5	-46°48.7
Fomalhaut	15°10.7	-29°27.7
Scheat	13°41.7	28°14.5
Markab	13°26.3	15°21.8

Jun 03 Sun	SHA	Mer.pass
Venus	268°40.1	13:19
Mars	180°20.2	19:09
Jupiter	163°08.8	20:17
Saturn	313°39.2	10:17

Jun 04 Mon	SHA	Mer.pass
Venus	267°19.7	13:21
Mars	180°04.8	19:06
Jupiter	163°10.7	20:13
Saturn	313°32.0	10:13

Jun 05 Tue	SHA	Mer.pass
Venus	265°59.2	13:22
Mars	179°49.0	19:04
Jupiter	163°12.3	20:09
Saturn	313°24.8	10:10

Horizontal parallax		
Venus:		0.1

DUT1 = UT1-UTC = +0.2102 sec $\Delta T = TT-UT1 = +68.9738$ sec

2029 June 03 to Jun. 05 UT

h Sun Moon						
Sun	GHA	Dec	GHA	ν	Dec	HP
0	180°28.1	N22°19.2	280°24.6	15.0'	S07°06.4	-12.3'
1	195°28.0	19.5	294°58.6	15.0'	06°54.1	-12.3'
2	210°27.9	19.8	309°32.6	15.1'	06°41.8	-12.3'
3	225°27.8	· · 20.1	324°06.7	15.1'	06°29.5	-12.3'
4	240°27.7	20.4	338°40.8	15.2'	06°17.2	-12.3'
5	255°27.6	20.7	353°14.9	15.2'	06°04.9	-12.3'
6	270°27.5	N22°21.0	7°49.1	15.3'	S05°52.5	-12.4'
7	285°27.4	21.3	22°23.4	15.3'	05°40.2	-12.4'
8	300°27.3	21.6	36°57.7	15.3'	05°27.8	-12.4'
9	315°27.2	· · 21.9	51°32.1	15.4'	05°15.4	-12.4'
10	330°27.1	22.2	66°06.4	15.4'	05°03.0	-12.4'
11	345°27.0	22.5	80°40.9	15.5'	04°50.6	-12.4'
12	0°26.9	N22°22.8	95°15.4	15.5'	S04°38.2	-12.4'
13	15°26.8	23.1	109°49.9	15.6'	04°25.8	-12.4'
14	30°26.7	23.4	124°24.4	15.6'	04°13.4	-12.4'
15	45°26.6	· · 23.7	138°59.0	15.6'	04°00.9	-12.4'
16	60°26.5	24.0	153°33.7	15.7'	03°48.5	-12.4'
17	75°26.4	24.3	168°08.3	15.7'	03°36.1	-12.4'
18	90°26.3	N22°24.6	182°43.0	15.7'	S03°23.6	-12.4'
19	105°26.1	24.8	197°17.8	15.8'	03°11.2	-12.4'
20	120°26.0	25.1	211°52.5	15.8'	02°58.7	-12.5'
21	135°25.9	· · 25.4	226°27.3	15.8'	02°46.3	-12.5'
22	150°25.8	25.7	241°02.2	15.9'	02°33.8	-12.5'
23	165°25.7	26.0	255°37.0	15.9'	02°21.4	-12.5'
SD = 15.8' d = 0.3' SD = 15.1'						
Mon	GHA	Dec	GHA	ν	Dec	HP
0	180°25.6	N22°26.3	270°11.9	15.9'	S02°08.9	-12.5'
1	195°25.5	26.6	284°46.8	15.9'	01°56.5	-12.4'
2	210°25.4	26.9	299°21.8	16.0'	01°44.0	-12.4'
3	225°25.3	· · 27.2	313°56.8	16.0'	01°31.6	-12.4'
4	240°25.2	27.5	328°31.8	16.0'	01°19.1	-12.4'
5	255°25.1	27.7	343°06.8	16.0'	01°06.7	-12.4'
6	270°25.0	N22°28.0	357°41.8	16.1'	S00°54.3	-12.4'
7	285°24.9	28.3	12°16.9	16.1'	00°41.8	-12.4'
8	300°24.8	28.6	26°52.0	16.1'	00°29.4	-12.4'
9	315°24.6	· · 28.9	41°27.1	16.1'	00°17.0	-12.4'
10	330°24.5	29.2	56°02.2	16.1'	S00°04.6	-12.4'
11	345°24.4	29.4	70°37.3	16.2'	N00°07.8	12.4'
12	0°24.3	N22°29.7	85°12.5	16.2'	N00°20.2	12.4'
13	15°24.2	30.0	99°47.7	16.2'	00°32.6	12.4'
14	30°24.1	30.3	114°22.8	16.2'	00°45.0	12.4'
15	45°24.0	· · 30.6	128°58.0	16.2'	00°57.4	12.4'
16	60°23.9	30.8	143°33.3	16.2'	01°09.7	12.3'
17	75°23.8	31.1	158°08.5	16.2'	01°22.1	12.3'
18	90°23.7	N22°31.4	172°43.7	16.2'	N01°34.4	12.3'
19	105°23.6	31.7	187°18.9	16.2'	01°46.7	12.3'
20	120°23.4	32.0	201°54.2	16.3'	01°59.1	12.3'
21	135°23.3	· · 32.2	216°29.4	16.3'	02°11.4	12.3'
22	150°23.2	32.5	231°04.7	16.3'	02°23.6	12.3'
23	165°23.1	32.8	245°40.0	16.3'	02°35.9	12.3'
SD = 15.8' d = 0.3' SD = 14.9'						
Tue	GHA	Dec	GHA	ν	Dec	HP
0	180°23.0	N22°33.1	260°15.2	16.3'	N02°48.2	12.2'
1	195°22.9	33.3	274°50.5	16.3'	03°00.4	12.2'
2	210°22.8	33.6	289°25.8	16.3'	03°12.6	12.2'
3	225°22.7	· · 33.9	304°01.0	16.3'	03°24.8	12.2'
4	240°22.6	34.1	318°36.3	16.3'	03°37.0	12.2'
5	255°22.4	34.4	333°11.6	16.3'	03°49.2	12.1'
6	270°22.3	N22°34.7	347°46.9	16.3'	N04°01.3	12.1'
7	285°22.2	34.9	2°22.1	16.3'	04°13.4	12.1'
8	300°22.1	35.2	16°57.4	16.3'	04°25.6	12.1'
9	315°22.0	· · 35.5	31°32.6	16.3'	04°37.6	12.1'
10	330°21.9	35.8	46°07.9	16.2'	04°49.7	12.0'
11	345°21.8	36.0	60°43.1	16.2'	05°01.7	12.0'
12	0°21.7	N22°36.3	75°18.4	16.2'	N05°13.8	12.0'
13	15°21.5	36.5	89°53.6	16.2'	05°25.8	12.0'
14	30°21.4	36.8	104°28.8	16.2'	05°37.7	11.9'
15	45°21.3	· · 37.1	119°04.0	16.2'	05°49.7	11.9'
16	60°21.2	37.3	133°39.2	16.2'	06°01.6	11.9'
17	75°21.1	37.6	148°14.4	16.2'	06°13.5	11.9'
18	90°21.0	N22°37.9	162°49.6	16.2'	N06°25.4	11.8'
19	105°20.9	38.1	177°24.7	16.1'	06°37.2	11.8'
20	120°20.8	38.4	191°59.8	16.1'	06°49.0	11.8'
21	135°20.6	· · 38.6	206°35.0	16.1'	07°00.8	11.8'
22	150°20.5	38.9	221°10.1	16.1'	07°12.6	11.7'
23	165°20.4	39.2	235°45.2	16.1'	07°24.3	11.7'
SD = 15.8' d = 0.3' SD = 14.8'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	--	--	00:58	23:03	--	--
64°	--	--	01:50	22:08	--	--
62°	--	--	02:22	21:36	--	--
60°	--	01:17	02:45	21:13	22:42	--
N 58°	--	01:54	03:03	20:54	22:04	--
56°	--	02:20	03:19	20:39	21:38	--
54°	01:10	02:40	03:32	20:25	21:18	22:49
52°	01:45	02:57	03:44	20:14	21:01	22:13
50°	02:09	03:11	03:54	20:03	20:47	21:49
45°	02:50	03:38	04:15	19:42	20:19	21:07
N 40°	03:19	04:00	04:32	19:25	19:57	20:38
35°	03:41	04:17	04:46	19:11	19:40	20:16
30°	03:59	04:32	04:59	18:58	19:25	19:58
20°	04:27	04:55	05:20	18:37	19:01	19:30
N 10°	04:48	05:15	05:38	18:19	18:42	19:09
0°	05:06	05:32	05:55	18:02	18:24	18:50
S 10°	05:23	05:49	06:11	17:45	18:08	18:34
20°	05:38	06:05	06:29	17:28	17:51	18:19
30°	05:54	06:23	06:49	17:07	17:33	18:03
35°	06:02	06:33	07:01	16:55	17:23	17:55
40°	06:11	06:44	07:15	16:42	17:12	17:46
45°	06:20	06:57	07:31	16:26	16:59	17:36
S 50°	06:31	07:12	07:50	16:06	16:44	17:25
52°	06:36	07:19	07:59	15:57	16:37	17:20
54°	06:41	07:27	08:10	15:47	16:30	17:15
56°	06:47	07:35	08:22	15:35	16:21	17:09
58°	06:53	07:45	08:35	15:21	16:11	17:03
S 60°	07:00	07:56	08:51	15:05	16:01	16:56

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	00:49	00:27	00:05 23:43	10:34	12:24	14:12
N 70°	00:39	00:24	00:09 23:54	10:41	12:23	14:04
68°	00:31	00:21	00:12	10:47	12:23	13:57
66°	00:24	00:19	00:15	10:52	12:23	13:52
64°	00:19	00:18	00:17	10:56	12:22	13:47
62°	00:14	00:16	00:19	11:00	12:22	13:43
60°	00:09	00:15	00:20	11:03	12:22	13:40
N 58°	00:05	00:14	00:22	11:06	12:22	13:37
56°	00:02	00:13	00:23	11:08	12:22	13:34
54°	· · ·	00:12	00:24	11:10	12:21	13:31
52°	· · ·	00:11	00:25	11:12	12:21	13:29
50°	· · ·	00:10	00:26	11:14	12:21	13:27
45°	· · ·	00:08	00:29	11:18	12:21	13:23
N 40°	· · ·	00:07	00:31	11:21	12:21	13:19
35°	· · ·	00:06	00:32	11:24	12:20	13:16
30°	· · ·	00:05	00:34	11:27	12:20	13:13
20°	· · ·	00:03	00:36	11:31	12:20	13:08
N 10°	· · ·	00:01	00:38	11:34	12:20	13:04
0°	· · ·	00:00	00:40	11:38	12:19	13:00
S 10°	23:58	· · ·	00:43	11:41	12:19	12:56
20°	23:56	· · ·	00:45	11:45	12:19	12:52
30°	23:54	· · ·	00:48	11:49	12:18	12:47
35°	23:53	· · ·	00:49	11:51	12:18	12:44
40°	23:52	· · ·	00:51	11:54	12:18	12:41
45°	23:51	· · ·	00:53	11:57	12:18	12:38
S 50°	23:49	· · ·	00:55	12:01	12:17	12:34
52°	23:48	· · ·	00:56	12:02	12:17	12:32
54°	23:47	· · ·	00:58	12:04	12:17	12:29
56°	23:46	· · ·	00:59	12:06	12:17	12:27
58°	23:45	· · ·	01:01	12:08	12:16	12:25
S 60°	23:44	· · ·	01:02	12:11	12:16	12:22

Day	Sun		Mer. Pass hh:mm	Moon		Age 21-23 60-41%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
03	01:53	01:48	11:58	05:28	17:49	
04	01:42	01:37	11:58	06:09	18:30	
05	01:32	01:27	11:59	06:50	19:11	

June 06, 07, 08 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	254° 41.7	159° 20.5	N24° 26.6	74° 14.4	N00° 25.7	57° 55.6	S05° 36.3	207° 59.3	N15° 19.4	Alpheratz	357° 31.1	29° 15.1
1	269° 44.2	174° 19.6	26.5	89° 16.2	25.4	72° 58.1	36.2	223° 01.5	19.5	Ankaa	353° 03.9	-42° 08.5
2	284° 46.7	189° 18.7	26.5	104° 18.0	25.0	88° 00.6	36.2	238° 03.6	19.6	Schedar	349° 27.0	56° 41.7
3	299° 49.1	204° 17.8	•• 26.5	119° 19.8	•• 24.6	103° 03.1	•• 36.2	253° 05.8	•• 19.6	Diphda	348° 43.9	-17° 49.4
4	314° 51.6	219° 16.9	26.4	134° 21.5	24.2	118° 05.7	36.2	268° 08.0	19.7	Achernar	335° 18.1	-57° 05.0
5	329° 54.1	234° 16.0	26.4	149° 23.3	23.9	133° 08.2	36.2	283° 10.1	19.8	Hamal	327° 47.4	23° 36.0
6	344° 56.5	249° 15.1	N24° 26.3	164° 25.1	N00° 23.5	148° 10.7	S05° 36.2	298° 12.3	N15° 19.9	Polaris	312° 30.4	89° 23.0
7	359° 59.0	264° 14.2	26.3	179° 26.8	23.1	163° 13.2	36.2	313° 14.5	19.9	Acamar	315° 09.5	-40° 11.1
8	15° 01.4	279° 13.4	26.3	194° 28.6	22.7	178° 15.8	36.2	328° 16.6	20.0	Menkar	314° 02.7	4° 12.2
9	30° 03.9	294° 12.5	•• 26.2	209° 30.4	•• 22.3	193° 18.3	•• 36.1	343° 18.8	•• 20.1	Mirfak	308° 23.5	49° 57.7
10	45° 06.4	309° 11.6	26.2	224° 32.2	22.0	208° 20.8	36.1	358° 21.0	20.2	Aldebaran	290° 35.9	16° 34.0
11	60° 08.8	324° 10.7	26.1	239° 33.9	21.6	223° 23.3	36.1	13° 23.1	20.3	Rigel	281° 00.8	-8° 10.2
12	75° 11.3	339° 09.8	N24° 26.1	254° 35.7	N00° 21.2	238° 25.8	S05° 36.1	28° 25.3	N15° 20.3	Capella	280° 17.1	46° 01.5
13	90° 13.8	354° 08.9	26.0	269° 37.5	20.8	253° 28.4	36.1	43° 27.5	20.4	Bellatrix	278° 19.4	6° 22.4
14	105° 16.2	9° 08.0	26.0	284° 39.2	20.4	268° 30.9	36.1	58° 29.6	20.5	Elnath	277° 57.7	28° 37.8
15	120° 18.7	24° 07.1	•• 25.9	299° 41.0	•• 20.1	283° 33.4	•• 36.1	73° 31.8	•• 20.6	Alnilam	275° 34.5	-1° 11.1
16	135° 21.2	39° 06.3	25.9	314° 42.8	19.7	298° 35.9	36.1	88° 34.0	20.7	Betelgeuse	270° 48.6	7° 24.6
17	150° 23.6	54° 05.4	25.8	329° 44.5	19.3	313° 38.4	36.1	103° 36.1	20.7	Canopus	263° 51.3	-52° 42.9
18	165° 26.1	69° 04.5	N24° 25.8	344° 46.3	N00° 18.9	328° 40.9	S05° 36.0	118° 38.3	N15° 20.8	Sirius	258° 23.4	-16° 45.6
19	180° 28.6	84° 03.6	25.7	359° 48.1	18.5	343° 43.5	36.0	133° 40.5	20.9	Adhara	255° 03.4	-29° 01.0
20	195° 31.0	99° 02.7	25.7	14° 49.8	18.1	358° 46.0	36.0	148° 42.6	21.0	Procyon	244° 47.4	5° 08.8
21	210° 33.5	114° 01.8	•• 25.6	29° 51.6	•• 17.8	13° 48.5	•• 36.0	163° 44.8	•• 21.0	Pollux	243° 13.3	27° 57.2
22	225° 35.9	129° 00.9	25.6	44° 53.4	17.4	28° 51.0	36.0	178° 47.0	21.1	Avior	234° 13.6	-59° 36.6
23	240° 38.4	144° 00.1	25.5	59° 55.1	17.0	43° 53.5	36.0	193° 49.1	21.2	Suhail	222° 43.9	-43° 33.4
Mer.pass. 07:00		ν -0.9' d -0.0' m -3.89		ν 1.8' d -0.4' m 0.03		ν 2.5' d -0.0' m -2.24		ν 2.2' d 0.1' m 0.33				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	255° 40.9	158° 59.2	N24° 25.4	74° 56.9	N00° 16.6	58° 56.1	S05° 36.0	208° 51.3	N15° 21.3	Alphard	217° 44.4	-8° 47.4
1	270° 43.3	173° 58.3	25.4	89° 58.6	16.2	73° 58.6	36.0	223° 53.5	21.4	Regulus	207° 30.8	11° 49.3
2	285° 45.8	188° 57.4	25.3	105° 00.4	15.8	89° 01.1	36.0	238° 55.6	21.4	Dubhe	193° 36.9	61° 35.7
3	300° 48.3	203° 56.5	•• 25.3	120° 02.2	•• 15.5	104° 03.6	•• 36.0	253° 57.8	•• 21.5	Denebola	182° 21.3	14° 24.4
4	315° 50.7	218° 55.6	25.2	135° 03.9	15.1	119° 06.1	35.9	269° 00.0	21.6	Gienah	175° 39.8	-17° 42.5
5	330° 53.2	233° 54.7	25.1	150° 05.7	14.7	134° 08.6	35.9	284° 02.1	21.7	Acrux	172° 55.6	-63° 16.1
6	345° 55.7	248° 53.9	N24° 25.1	165° 07.4	N00° 14.3	149° 11.1	S05° 35.9	299° 04.3	N15° 21.8	Gacrux	171° 47.3	-57° 17.0
7	0° 58.1	263° 53.0	25.0	180° 09.2	13.9	164° 13.7	35.9	314° 06.5	21.8	Alioth	166° 09.8	55° 48.2
8	16° 00.6	278° 52.1	24.9	195° 10.9	13.5	179° 16.2	35.9	329° 08.6	21.9	Spica	158° 18.3	-11° 19.0
9	31° 03.0	293° 51.2	•• 24.8	210° 12.7	•• 13.1	194° 18.7	•• 35.9	344° 10.8	•• 22.0	Alkaid	152° 49.0	49° 10.1
10	46° 05.5	308° 50.3	24.8	225° 14.4	12.8	209° 21.2	35.9	359° 13.0	22.1	Hadar	148° 30.3	-60° 31.1
11	61° 08.0	323° 49.4	24.7	240° 16.2	12.4	224° 23.7	35.9	14° 15.1	22.1	Menkent	147° 53.0	-36° 31.1
12	76° 10.4	338° 48.6	N24° 24.6	255° 18.0	N00° 12.0	239° 26.2	S05° 35.9	29° 17.3	N15° 22.2	Arcturus	145° 44.4	19° 01.8
13	91° 12.9	353° 47.7	24.5	270° 19.7	11.6	254° 28.7	35.9	44° 19.5	22.3	Rigil Kent.	139° 34.7	-60° 57.6
14	106° 15.4	8° 46.8	24.5	285° 21.5	11.2	269° 31.3	35.9	59° 21.6	22.4	Kochab	137° 19.2	74° 02.2
15	121° 17.8	23° 45.9	•• 24.4	300° 23.2	•• 10.8	284° 33.8	•• 35.9	74° 23.8	•• 22.5	Zuben'ubi	136° 51.8	-16° 09.9
16	136° 20.3	38° 45.0	24.3	315° 25.0	10.4	299° 36.3	35.8	89° 26.0	22.5	Alphecca	126° 00.4	26° 37.0
17	151° 22.8	53° 44.1	24.2	330° 26.7	10.0	314° 38.8	35.8	104° 28.2	22.6	Antares	112° 11.1	-26° 29.8
18	166° 25.2	68° 43.2	N24° 24.1	345° 28.5	N00° 09.7	329° 41.3	S05° 35.8	119° 30.3	N15° 22.7	Atria	107° 01.4	-69° 04.7
19	181° 27.7	83° 42.4	24.1	0° 30.2	09.3	344° 43.8	35.8	134° 32.5	22.8	Sabik	101° 58.3	-15° 45.6
20	196° 30.2	98° 41.5	24.0	15° 31.9	08.9	359° 46.3	35.8	149° 34.7	22.8	Shaula	96° 05.1	-37° 07.4
21	211° 32.6	113° 40.6	•• 23.9	30° 33.7	•• 08.5	14° 48.8	•• 35.8	164° 36.8	•• 22.9	Rasalhague	95° 54.9	12° 32.4
22	226° 35.1	128° 39.7	23.8	45° 35.4	08.1	29° 51.3	35.8	179° 39.0	23.0	Eltanin	90° 40.0	51° 29.1
23	241° 37.5	143° 38.8	23.7	60° 37.2	07.7	44° 53.9	35.8	194° 41.2	23.1	Kaus Aust.	83° 27.4	-34° 22.0
Mer.pass. 06:56		ν -0.9' d -0.1' m -3.89		ν 1.8' d -0.4' m 0.04		ν 2.5' d -0.0' m -2.23		ν 2.2' d 0.1' m 0.33				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	256° 40.0	158° 37.9	N24° 23.6	75° 38.9	N00° 07.3	59° 56.4	S05° 35.8	209° 43.3	N15° 23.2	Nunki	75° 43.0	-26° 15.4
1	271° 42.5	173° 37.1	23.5	90° 40.7	06.9	74° 58.9	35.8	224° 45.5	23.2	Altair	61° 56.2	8° 56.9
2	286° 44.9	188° 36.2	23.4	105° 42.4	06.5	90° 01.4	35.8	239° 47.7	23.3	Peacock	52° 59.9	-56° 38.0
3	301° 47.4	203° 35.3	•• 23.3	120° 44.2	•• 06.1	105° 03.9	•• 35.8	254° 49.8	•• 23.4	Deneb	49° 23.0	45° 23.1
4	316° 49.9	218° 34.4	23.2	135° 45.9	05.8	120° 06.4	35.8	269° 52.0	23.5	Enif	33° 35.2	10° 00.7
5	331° 52.3	233° 33.5	23.1	150° 47.6	05.4	135° 08.9	35.8	284° 54.2	23.5	Al Na'ir	27° 28.5	-46° 48.7
6	346° 54.8	248° 32.7	N24° 23.0	165° 49.4	N00° 05.0	150° 11.4	S05° 35.7	299° 56.3	N15° 23.6	Fomalhaut	15° 10.7	-29° 27.7
7	1° 57.3	263° 31.8	22.9	180° 51.1	04.6	165° 13.9	35.7	314° 58.5	23.7	Scheat	13° 41.7	28° 14.5
8	16° 59.7	278° 30.9	22.8	195° 52.9	04.2	180° 16.4	35.7	330° 00.7	23.8	Markab	13° 26.3	15° 21.8
9	32° 02.2	293° 30.0	•• 22.7	210° 54.6	•• 03.8	195° 18.9	•• 35.7	345° 02.8	•• 23.9			
10	47° 04.7	308° 29.1	22.6	225° 56.3	03.4	210° 21.5	35.7	0° 05.0	23.9			
11	62° 07.1	323° 28.2	22.5	240° 58.1	03.0	225° 24.0	35.7	15° 07.2	24.0			
12	77° 09.6	338° 27.4	N24° 22.4	255° 59.8	N00° 02.6	240° 26.5	S05° 35.7	30° 09.4	N15° 24.1			
13	92° 12.0	353° 26.5	22.3	271° 01.5	02.2	255° 29.0	35.7	45° 11.5	24.2			
14	107° 14.5	8° 25.6	22.2	286° 03.3	01.8	270° 31.5	35.7	60° 13.7	24.2			
15	122° 17.0	23° 24.7	•• 22.1	301° 05.0	•• 01.4	285° 34.0	•• 35.7	75° 15.9	•• 24.3			
16	137° 19.4	38° 23.8	22.0	316° 06.7	01.0	300° 36.5	35.7	90° 18.0	24.4			
17	152° 21.9	53° 23										

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°20.3	N22°39.4	250°20.2	16.0'	N07°36.0	11.7'	54.1'
1	195°20.2	39.7	264°55.3	16.0'	07°47.7	11.6'	54.1'
2	210°20.1	39.9	279°30.3	16.0'	07°59.3	11.6'	54.1'
3	225°20.0	· · 40.2	294°05.3	16.0'	08°10.9	11.6'	54.1'
4	240°19.8	40.4	308°40.3	16.0'	08°22.5	11.5'	54.1'
5	255°19.7	40.7	323°15.2	15.9'	08°34.0	11.5'	54.1'
6	270°19.6	N22°40.9	337°50.1	15.9'	N08°45.6	11.5'	54.1'
7	285°19.5	41.2	352°25.1	15.9'	08°57.0	11.4'	54.1'
8	300°19.4	41.4	6°59.9	15.8'	09°08.5	11.4'	54.1'
9	315°19.3	· · 41.7	21°34.8	15.8'	09°19.9	11.4'	54.1'
10	330°19.2	41.9	36°09.6	15.8'	09°31.2	11.3'	54.1'
11	345°19.0	42.2	50°44.4	15.8'	09°42.6	11.3'	54.1'
12	0°18.9	N22°42.4	65°19.2	15.7'	N09°53.9	11.3'	54.1'
13	15°18.8	42.7	79°53.9	15.7'	10°05.1	11.2'	54.1'
14	30°18.7	42.9	94°28.6	15.7'	10°16.3	11.2'	54.1'
15	45°18.6	· · 43.2	109°03.2	15.6'	10°27.5	11.1'	54.1'
16	60°18.5	43.4	123°37.9	15.6'	10°38.6	11.1'	54.1'
17	75°18.3	43.7	138°12.5	15.6'	10°49.7	11.1'	54.1'
18	90°18.2	N22°43.9	152°47.0	15.5'	N11°00.8	11.0'	54.1'
19	105°18.1	44.2	167°21.6	15.5'	11°11.8	11.0'	54.1'
20	120°18.0	44.4	181°56.1	15.5'	11°22.7	10.9'	54.1'
21	135°17.9	· · 44.6	196°30.5	15.4'	11°33.7	10.9'	54.1'
22	150°17.8	44.9	211°04.9	15.4'	11°44.5	10.8'	54.1'
23	165°17.6	45.1	225°39.3	15.3'	11°55.4	10.8'	54.1'
SD = 15.8'			SD = 14.8'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°17.5	N22°45.4	240°13.7	15.3'	N12°06.2	10.7'	54.1'
1	195°17.4	45.6	254°47.9	15.3'	12°16.9	10.7'	54.1'
2	210°17.3	45.8	269°22.2	15.2'	12°27.6	10.6'	54.1'
3	225°17.2	· · 46.1	283°56.4	15.2'	12°38.3	10.6'	54.1'
4	240°17.0	46.3	298°30.6	15.1'	12°48.9	10.5'	54.1'
5	255°16.9	46.6	313°04.7	15.1'	12°59.4	10.5'	54.1'
6	270°16.8	N22°46.8	327°38.8	15.0'	N13°09.9	10.4'	54.1'
7	285°16.7	47.0	342°12.8	15.0'	13°20.4	10.4'	54.1'
8	300°16.6	47.3	356°46.8	14.9'	13°30.8	10.3'	54.1'
9	315°16.5	· · 47.5	11°20.8	14.9'	13°41.1	10.3'	54.1'
10	330°16.3	47.7	25°54.7	14.9'	13°51.4	10.2'	54.1'
11	345°16.2	48.0	40°28.5	14.8'	14°01.6	10.2'	54.1'
12	0°16.1	N22°48.2	55°02.3	14.8'	N14°11.8	10.1'	54.1'
13	15°16.0	48.4	69°36.1	14.7'	14°21.9	10.1'	54.1'
14	30°15.9	48.7	84°09.8	14.7'	14°32.0	10.0'	54.1'
15	45°15.7	· · 48.9	98°43.4	14.6'	14°42.0	10.0'	54.1'
16	60°15.6	49.1	113°17.0	14.6'	14°52.0	9.9'	54.1'
17	75°15.5	49.4	127°50.6	14.5'	15°01.9	9.8'	54.1'
18	90°15.4	N22°49.6	142°24.1	14.4'	N15°11.8	9.8'	54.1'
19	105°15.3	49.8	156°57.5	14.4'	15°21.5	9.7'	54.1'
20	120°15.1	50.0	171°30.9	14.3'	15°31.3	9.7'	54.1'
21	135°15.0	· · 50.3	186°04.3	14.3'	15°40.9	9.6'	54.1'
22	150°14.9	50.5	200°37.6	14.2'	15°50.5	9.5'	54.1'
23	165°14.8	50.7	215°10.8	14.2'	16°00.1	9.5'	54.1'
SD = 15.8'			SD = 14.7'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°14.7	N22°50.9	229°44.0	14.1'	N16°09.6	9.4'	54.1'
1	195°14.5	51.2	244°17.1	14.1'	16°19.0	9.4'	54.2'
2	210°14.4	51.4	258°50.1	14.0'	16°28.3	9.3'	54.2'
3	225°14.3	· · 51.6	273°23.1	13.9'	16°37.6	9.2'	54.2'
4	240°14.2	51.8	287°56.1	13.9'	16°46.8	9.2'	54.2'
5	255°14.1	52.0	302°29.0	13.8'	16°56.0	9.1'	54.2'
6	270°13.9	N22°52.3	317°01.8	13.8'	N17°05.1	9.0'	54.2'
7	285°13.8	52.5	331°34.6	13.7'	17°14.1	8.9'	54.2'
8	300°13.7	52.7	346°07.3	13.7'	17°23.0	8.9'	54.2'
9	315°13.6	· · 52.9	0°39.9	13.6'	17°31.9	8.8'	54.2'
10	330°13.5	53.1	15°12.5	13.5'	17°40.7	8.7'	54.2'
11	345°13.3	53.3	29°45.1	13.5'	17°49.4	8.7'	54.2'
12	0°13.2	N22°53.6	44°17.5	13.4'	N17°58.1	8.6'	54.2'
13	15°13.1	53.8	58°49.9	13.3'	18°06.7	8.5'	54.3'
14	30°13.0	54.0	73°22.3	13.3'	18°15.2	8.4'	54.3'
15	45°12.8	· · 54.2	87°54.6	13.2'	18°23.6	8.4'	54.3'
16	60°12.7	54.4	102°26.8	13.2'	18°32.0	8.3'	54.3'
17	75°12.6	54.6	116°59.0	13.1'	18°40.3	8.2'	54.3'
18	90°12.5	N22°54.8	131°31.1	13.0'	N18°48.5	8.1'	54.3'
19	105°12.4	55.0	146°03.1	13.0'	18°56.6	8.0'	54.3'
20	120°12.2	55.3	160°35.1	12.9'	19°04.7	8.0'	54.3'
21	135°12.1	· · 55.5	175°07.0	12.8'	19°12.6	7.9'	54.3'
22	150°12.0	55.7	189°38.8	12.8'	19°20.5	7.8'	54.3'
23	165°11.9	55.9	204°10.6	12.7'	19°28.3	7.7'	54.4'
SD = 15.8'			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	--	--	00:44	23:19	--	--
64°	--	--	01:44	22:15	--	--
62°	--	--	02:17	21:42	--	--
60°	--	01:09	02:42	21:17	22:51	--
N 58°	--	01:50	03:01	20:58	22:10	--
56°	--	02:17	03:17	20:42	21:42	--
54°	01:03	02:37	03:30	20:28	21:21	22:57
52°	01:41	02:54	03:42	20:16	21:04	22:19
50°	02:06	03:09	03:52	20:06	20:50	21:53
45°	02:48	03:37	04:14	19:44	20:21	21:10
N 40°	03:18	03:59	04:31	19:27	19:59	20:40
35°	03:40	04:16	04:46	19:12	19:42	20:18
30°	03:58	04:31	04:58	19:00	19:27	20:00
20°	04:26	04:55	05:20	18:38	19:03	19:32
N 10°	04:48	05:15	05:38	18:20	18:43	19:10
0°	05:07	05:33	05:55	18:03	18:25	18:51
S 10°	05:23	05:49	06:12	17:46	18:08	18:35
20°	05:39	06:06	06:30	17:28	17:52	18:19
30°	05:55	06:25	06:51	17:07	17:33	18:03
35°	06:03	06:35	07:03	16:55	17:23	17:55
40°	06:12	06:46	07:16	16:41	17:12	17:45
45°	06:22	06:59	07:33	16:25	16:59	17:36
S 50°	06:33	07:15	07:53	16:05	16:43	17:24
52°	06:38	07:22	08:02	15:56	16:36	17:19
54°	06:44	07:30	08:13	15:45	16:28	17:14
56°	06:50	07:38	08:25	15:33	16:19	17:08
58°	06:56	07:48	08:39	15:19	16:10	17:02
S 60°	07:03	07:59	08:55	15:02	15:58	16:54

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	23:16	22:35	☐	16:06	18:17	☐
N 70°	23:36	23:13	22:25	15:48	17:40	20:02
68°	00:02 23:52	23:39	23:21	15:34	17:15	19:07
66°	00:10	00:05	00:00 23:54	15:22	16:56	18:34
64°	00:16	00:16	00:16	15:13	16:40	18:11
62°	00:21	00:25	00:30	15:04	16:27	17:52
60°	00:26	00:33	00:42	14:57	16:16	17:36
N 58°	00:30	00:40	00:52	14:51	16:07	17:23
56°	00:34	00:46	01:01	14:46	15:59	17:12
54°	00:37	00:52	01:09	14:41	15:52	17:02
52°	00:40	00:57	01:16	14:37	15:45	16:54
50°	00:43	01:01	01:22	14:33	15:39	16:46
45°	00:49	01:11	01:36	14:24	15:27	16:29
N 40°	00:54	01:20	01:47	14:17	15:16	16:16
35°	00:59	01:27	01:57	14:11	15:07	16:04
30°	01:03	01:33	02:06	14:06	14:59	15:54
20°	01:09	01:44	02:21	13:57	14:46	15:37
N 10°	01:16	01:54	02:34	13:49	14:35	15:22
0°	01:21	02:03	02:46	13:41	14:24	15:08
S 10°	01:27	02:12	02:59	13:34	14:13	14:54
20°	01:33	02:22	03:12	13:26	14:01	14:40
30°	01:40	02:33	03:28	13:17	13:48	14:23
35°	01:44	02:40	03:37	13:12	13:41	14:13
40°	01:49	02:48	03:47	13:06	13:32	14:02
45°	01:55	02:56	03:59	12:59	13:22	13:49
S 50°	02:01	03:07	04:14	12:51	13:10	13:33
52°	02:04	03:12	04:21	12:47	13:04	13:25
54°	02:07	03:18	04:28	12:43	12:58	13:17
56°	02:11	03:24	04:37	12:38	12:52	13:08
58°	02:15	03:31	04:47	12:33	12:44	12:58
S 60°	02:20	03:38	04:58	12:28	12:35	12:46

Day	Sun		Mer. Pass hh:mm	Moon		Age 24-26 32-16%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
06	01:21	01:16	11:59	07:31	19:52	
07	01:10	01:04	11:59	08:13	20:35	
08	00:59	00:53	11:59	08:57	21:20	

June 09, 10, 11 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	257°39.1	158°16.8	N24°21.1	76°20.6	S00°02.1	60°56.5	S05°35.7	210°35.4	N15°25.0	Alpheratz	357°31.1	29°15.1
1	272°41.6	173°15.9		91°22.3	02.5	75°59.0	35.7	225°37.6	25.1	Ankaa	353°03.9	-42°08.5
2	287°44.1	188°15.0		106°24.0	02.9	91°01.5	35.7	240°39.7	25.2	Schedar	349°27.0	56°41.7
3	302°46.5	203°14.2	•• 20.7	121°25.7	•• 03.3	106°04.0	•• 35.6	255°41.9	•• 25.2	Diphda	348°43.9	-17°49.4
4	317°49.0	218°13.3		136°27.5	03.7	121°06.5	35.6	270°44.1	25.3	Achernar	335°18.1	-57°05.0
5	332°51.5	233°12.4		151°29.2	04.1	136°09.0	35.6	285°46.2	25.4	Hamal	327°47.4	23°36.0
6	347°53.9	248°11.5	N24°20.3	166°30.9	S00°04.5	151°11.5	S05°35.6	300°48.4	N15°25.5	Polaris	312°29.3	89°23.0
7	2°56.4	263°10.6		181°32.6	04.9	166°14.0	35.6	315°50.6	25.6	Acamar	315°09.5	-40°11.1
8	17°58.9	278°09.8		196°34.4	05.3	181°16.5	35.6	330°52.7	25.6	Menkar	314°02.7	4°12.2
9	33°01.3	293°08.9	•• 19.9	211°36.1	•• 05.7	196°19.0	•• 35.6	345°54.9	•• 25.7	Mirfak	308°23.5	49°57.7
10	48°03.8	308°08.0		226°37.8	06.1	211°21.5	35.6	0°57.1	25.8	Aldebaran	290°35.9	16°34.0
11	63°06.3	323°07.1		241°39.5	06.5	226°24.0	35.6	15°59.3	25.9	Rigel	281°00.8	-8°10.2
12	78°08.7	338°06.2	N24°19.5	256°41.2	S00°06.9	241°26.5	S05°35.6	31°01.4	N15°25.9	Capella	280°17.1	46°01.5
13	93°11.2	353°05.4		271°43.0	07.3	256°29.0	35.6	46°03.6	26.0	Bellatrix	278°19.4	6°22.4
14	108°13.6	8°04.5		286°44.7	07.7	271°31.5	35.6	61°05.8	26.1	Elnath	277°57.7	28°37.8
15	123°16.1	23°03.6	•• 19.1	301°46.4	•• 08.1	286°34.0	•• 35.6	76°07.9	•• 26.2	Alnilam	275°34.5	-1°11.1
16	138°18.6	38°02.7		316°48.1	08.5	301°36.5	35.6	91°10.1	26.2	Betelgeuse	270°48.6	7°24.6
17	153°21.0	53°01.9		331°49.8	08.9	316°39.0	35.6	106°12.3	26.3	Canopus	263°51.3	-52°42.9
18	168°23.5	68°01.0	N24°18.7	346°51.5	S00°09.3	331°41.5	S05°35.6	121°14.4	N15°26.4	Sirius	258°23.4	-16°45.6
19	183°26.0	83°00.1		1°53.3	09.7	346°44.0	35.6	136°16.6	26.5	Adhara	255°03.4	-29°00.9
20	198°28.4	97°59.2		16°55.0	10.1	1°46.5	35.6	151°18.8	26.5	Procyon	244°47.4	5°08.8
21	213°30.9	112°58.3	•• 18.3	31°56.7	•• 10.5	16°49.0	•• 35.6	166°21.0	•• 26.6	Pollux	243°13.3	27°57.2
22	228°33.4	127°57.5		46°58.4	10.9	31°51.5	35.6	181°23.1	26.7	Avior	234°13.6	-59°36.6
23	243°35.8	142°56.6		62°00.1	11.3	46°54.0	35.6	196°25.3	26.8	Suhail	222°43.9	-43°33.4
Mer.pass. 06:48		ν -0.9' d -0.1' m -3.89		ν 1.7' d 0.4' m 0.08		ν 2.5' d -0.0' m -2.22		ν 2.2' d 0.1' m 0.33		Miaplacidus	221°37.8	-69°50.7
										Alphard	217°44.4	-8°47.4
										Regulus	207°30.8	11°49.3
										Dubhe	193°37.0	61°35.7
										Denebola	182°21.3	14°24.4
										Gienah	175°39.8	-17°42.5
										Acrux	172°55.6	-63°16.1
										Gacrux	171°47.3	-57°17.0
										Alioth	166°09.8	55°48.2
										Spica	158°18.3	-11°19.0
										Alkaid	152°49.0	49°10.1
										Hadar	148°30.3	-60°31.1
										Menkent	147°53.0	-36°31.1
										Arcturus	145°44.4	19°01.8
										Rigel Kent.	139°34.7	-60°57.6
										Kochab	137°19.3	74°02.2
										Zuben'ubi	136°51.8	-16°09.9
										Alphecca	126°00.4	26°37.0
										Antares	112°11.1	-26°29.8
										Atria	107°01.4	-69°04.7
										Sabik	101°58.3	-15°45.6
										Shaula	96°05.1	-37°07.4
										Rasalhague	95°54.9	12°32.4
										Eltanin	90°40.0	51°29.1
										Kaus Aust.	83°27.4	-34°22.0
										Vega	80°30.4	38°48.7
										Nunki	75°43.0	-26°15.4
										Altair	61°56.2	8°56.9
										Peacock	52°59.8	-56°38.0
										Deneb	49°23.0	45°23.1
										Enif	33°35.1	10°00.7
										Al Na'ir	27°28.4	-46°48.7
										Fomalhaut	15°10.6	-29°27.7
										Scheat	13°41.7	28°14.5
										Markab	13°26.3	15°21.8
										Jun 09 Sat	SHA	Mer.pass
										Venus	260°37.6	13:28
										Mars	178°41.4	18:52
										Jupiter	163°17.4	19:53
										Saturn	312°56.2	09:56
										Jun 10 Sun	SHA	Mer.pass
										Venus	259°17.4	13:29
										Mars	178°23.5	18:50
										Jupiter	163°18.2	19:49
										Saturn	312°49.2	09:53
										Jun 11 Mon	SHA	Mer.pass
										Venus	257°57.3	13:30
										Mars	178°05.3	18:47
										Jupiter	163°18.9	19:45
										Saturn	312°42.2	09:49
								</				

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°11.7	N22°56.1	218°42.3	12.6'	N19°36.1	7.6'	54.4'
1	195°11.6	56.3	233°13.9	12.6'	19°43.7	7.6'	54.4'
2	210°11.5	56.5	247°45.5	12.5'	19°51.3	7.5'	54.4'
3	225°11.4	· · 56.7	262°17.0	12.5'	19°58.7	7.4'	54.4'
4	240°11.2	56.9	276°48.5	12.4'	20°06.1	7.3'	54.4'
5	255°11.1	57.1	291°19.9	12.3'	20°13.4	7.2'	54.4'
6	270°11.0	N22°57.3	305°51.2	12.3'	N20°20.6	7.1'	54.4'
7	285°10.9	57.5	320°22.5	12.2'	20°27.7	7.0'	54.4'
8	300°10.8	57.7	334°53.7	12.1'	20°34.8	6.9'	54.5'
9	315°10.6	· · 57.9	349°24.8	12.1'	20°41.7	6.9'	54.5'
10	330°10.5	58.1	3°55.9	12.0'	20°48.6	6.8'	54.5'
11	345°10.4	58.3	18°26.9	11.9'	20°55.3	6.7'	54.5'
12	0°10.3	N22°58.5	32°57.8	11.9'	N21°02.0	6.6'	54.5'
13	15°10.1	58.7	47°28.7	11.8'	21°08.6	6.5'	54.5'
14	30°10.0	58.9	61°59.5	11.7'	21°15.1	6.4'	54.5'
15	45°09.9	· · 59.1	76°30.2	11.7'	21°21.5	6.3'	54.6'
16	60°09.8	59.3	91°00.9	11.6'	21°27.7	6.2'	54.6'
17	75°09.6	59.5	105°31.5	11.6'	21°33.9	6.1'	54.6'
18	90°09.5	N22°59.7	120°02.1	11.5'	N21°40.0	6.0'	54.6'
19	105°09.4	22°59.9	134°32.6	11.4'	21°46.0	5.9'	54.6'
20	120°09.3	23°00.1	149°03.0	11.4'	21°51.9	5.8'	54.6'
21	135°09.1	· · 00.3	163°33.3	11.3'	21°57.7	5.7'	54.6'
22	150°09.0	00.5	178°03.6	11.2'	22°03.4	5.6'	54.7'
23	165°08.9	00.7	192°33.9	11.2'	22°09.0	5.5'	54.7'
SD = 15.7'			SD = 14.8'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°08.8	N23°00.8	207°04.0	11.1'	N22°14.5	5.4'	54.7'
1	195°08.6	01.0	221°34.2	11.0'	22°19.9	5.3'	54.7'
2	210°08.5	01.2	236°04.2	11.0'	22°25.2	5.2'	54.7'
3	225°08.4	· · 01.4	250°34.2	10.9'	22°30.4	5.1'	54.7'
4	240°08.2	01.6	265°04.1	10.9'	22°35.4	5.0'	54.7'
5	255°08.1	01.8	279°34.0	10.8'	22°40.4	4.9'	54.8'
6	270°08.0	N23°02.0	294°03.8	10.7'	N22°45.3	4.8'	54.8'
7	285°07.9	02.2	308°33.5	10.7'	22°50.0	4.6'	54.8'
8	300°07.7	02.3	323°03.2	10.6'	22°54.7	4.5'	54.8'
9	315°07.6	· · 02.5	337°32.8	10.6'	22°59.2	4.4'	54.8'
10	330°07.5	02.7	352°02.4	10.5'	23°03.6	4.3'	54.8'
11	345°07.4	02.9	6°31.9	10.5'	23°08.0	4.2'	54.9'
12	0°07.2	N23°03.1	21°01.4	10.4'	N23°12.2	4.1'	54.9'
13	15°07.1	03.2	35°30.8	10.3'	23°16.3	4.0'	54.9'
14	30°07.0	03.4	50°00.1	10.3'	23°20.2	3.9'	54.9'
15	45°06.9	· · 03.6	64°29.4	10.2'	23°24.1	3.8'	54.9'
16	60°06.7	03.8	78°58.7	10.2'	23°27.8	3.6'	54.9'
17	75°06.6	04.0	93°27.8	10.1'	23°31.5	3.5'	55.0'
18	90°06.5	N23°04.1	107°57.0	10.1'	N23°35.0	3.4'	55.0'
19	105°06.3	04.3	122°26.0	10.0'	23°38.4	3.3'	55.0'
20	120°06.2	04.5	136°55.1	10.0'	23°41.7	3.2'	55.0'
21	135°06.1	· · 04.7	151°24.0	9.9'	23°44.9	3.0'	55.0'
22	150°06.0	04.8	165°53.0	9.9'	23°47.9	2.9'	55.1'
23	165°05.8	05.0	180°21.8	9.8'	23°50.8	2.8'	55.1'
SD = 15.7'			SD = 14.9'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°05.7	N23°05.2	194°50.7	9.8'	N23°53.7	2.7'	55.1'
1	195°05.6	05.4	209°19.4	9.7'	23°56.3	2.6'	55.1'
2	210°05.4	05.5	223°48.2	9.7'	23°58.9	2.4'	55.1'
3	225°05.3	· · 05.7	238°16.9	9.6'	24°01.4	2.3'	55.1'
4	240°05.2	05.9	252°45.5	9.6'	24°03.7	2.2'	55.2'
5	255°05.1	06.0	267°14.1	9.6'	24°05.9	2.1'	55.2'
6	270°04.9	N23°06.2	281°42.7	9.5'	N24°08.0	2.0'	55.2'
7	285°04.8	06.4	296°11.2	9.5'	24°09.9	1.8'	55.2'
8	300°04.7	06.5	310°39.7	9.4'	24°11.8	1.7'	55.2'
9	315°04.6	· · 06.7	325°08.1	9.4'	24°13.5	1.6'	55.3'
10	330°04.4	06.9	339°36.5	9.4'	24°15.1	1.5'	55.3'
11	345°04.3	07.0	354°04.8	9.3'	24°16.5	1.3'	55.3'
12	0°04.2	N23°07.2	8°33.2	9.3'	N24°17.8	1.2'	55.3'
13	15°04.0	07.4	23°01.4	9.2'	24°19.1	1.1'	55.3'
14	30°03.9	07.5	37°29.7	9.2'	24°20.1	1.0'	55.4'
15	45°03.8	· · 07.7	51°57.9	9.2'	24°21.1	0.8'	55.4'
16	60°03.6	07.9	66°26.1	9.2'	24°21.9	0.7'	55.4'
17	75°03.5	08.0	80°54.2	9.1'	24°22.6	0.6'	55.4'
18	90°03.4	N23°08.2	95°22.4	9.1'	N24°23.2	0.4'	55.4'
19	105°03.3	08.3	109°50.4	9.1'	24°23.6	0.3'	55.4'
20	120°03.1	08.5	124°18.5	9.0'	24°23.9	0.2'	55.5'
21	135°03.0	· · 08.7	138°46.5	9.0'	24°24.1	0.0'	55.5'
22	150°02.9	08.8	153°14.6	9.0'	24°24.1	-0.1'	55.5'
23	165°02.7	09.0	167°42.5	9.0'	24°24.1	-0.2'	55.5'
SD = 15.7'			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	--	--	00:27	23:39	--	--
64°	--	--	01:39	22:21	--	--
62°	--	--	02:14	21:46	--	--
60°	--	01:02	02:39	21:21	22:59	--
N 58°	--	01:46	02:59	21:01	22:14	--
56°	--	02:14	03:15	20:45	21:46	--
54°	00:57	02:35	03:29	20:31	21:25	23:04
52°	01:37	02:52	03:41	20:19	21:07	22:23
50°	02:03	03:07	03:51	20:08	20:52	21:57
45°	02:47	03:36	04:13	19:46	20:23	21:12
N 40°	03:17	03:58	04:31	19:28	20:01	20:42
35°	03:40	04:16	04:46	19:14	19:43	20:20
30°	03:58	04:31	04:58	19:01	19:28	20:01
20°	04:26	04:55	05:20	18:39	19:04	19:33
N 10°	04:49	05:16	05:39	18:20	18:43	19:10
0°	05:07	05:33	05:56	18:03	18:26	18:52
S 10°	05:24	05:50	06:13	17:46	18:09	18:35
20°	05:40	06:07	06:31	17:28	17:52	18:19
30°	05:56	06:26	06:52	17:07	17:33	18:03
35°	06:04	06:36	07:04	16:55	17:23	17:54
40°	06:14	06:48	07:18	16:41	17:11	17:45
45°	06:24	07:01	07:35	16:24	16:58	17:35
S 50°	06:35	07:17	07:55	16:04	16:42	17:24
52°	06:40	07:24	08:04	15:54	16:35	17:19
54°	06:46	07:32	08:15	15:44	16:27	17:13
56°	06:52	07:41	08:28	15:31	16:18	17:07
58°	06:58	07:51	08:42	15:17	16:08	17:00
S 60°	07:06	08:02	08:59	15:00	15:57	16:53

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	23:47	23:33	☐	20:20	22:20	☐
64°	00:19	00:26	00:43	19:43	21:10	22:19
62°	00:38	00:53	01:18	19:16	20:35	21:39
60°	00:54	01:14	01:44	18:56	20:09	21:11
N 58°	01:08	01:31	02:05	18:39	19:49	20:49
56°	01:20	01:46	02:21	18:24	19:32	20:32
54°	01:30	01:58	02:36	18:12	19:18	20:17
52°	01:39	02:09	02:48	18:01	19:06	20:04
50°	01:47	02:19	02:59	17:52	18:55	19:53
45°	02:05	02:40	03:22	17:32	18:32	19:29
N 40°	02:19	02:57	03:41	17:15	18:14	19:10
35°	02:32	03:11	03:57	17:02	17:59	18:54
30°	02:42	03:24	04:10	16:50	17:46	18:40
20°	03:01	03:45	04:33	16:29	17:23	18:17
N 10°	03:17	04:03	04:53	16:12	17:03	17:56
0°	03:32	04:21	05:12	15:55	16:45	17:37
S 10°	03:48	04:38	05:31	15:39	16:27	17:18
20°	04:04	04:57	05:51	15:21	16:07	16:58
30°	04:23	05:19	06:14	15:01	15:45	16:34
35°	04:34	05:32	06:28	14:50	15:32	16:20
40°	04:47	05:46	06:44	14:36	15:17	16:04
45°	05:02	06:04	07:03	14:20	14:59	15:45
S 50°	05:20	06:26	07:26	14:01	14:36	15:21
52°	05:29	06:36	07:38	13:52	14:26	15:10
54°	05:39	06:48	07:51	13:41	14:14	14:57
56°	05:50	07:01	08:06	13:30	14:00	14:42
58°	06:03	07:17	08:23	13:16	13:44	14:24
S 60°	06:18	07:36	08:45	13:01	13:25	14:02

Day	Sun		Mer. Pass	Moon		Age 27-29 10-1%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
09	00:47	00:41	11:59	09:44	22:08	
10	00:35	00:29	12:00	10:33	22:58	
11	00:23	00:17	12:00	11:25	23:51	

June 12, 13, 14 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	260°36.6	157°13.9	N24°09.1	78°23.2	S00°31.3	63°56.0	S05°35.7	213°11.8	N15°30.5	Alpheratz	357°31.1	29°15.1
1	275°39.0	172°13.0	08.9	93°24.8	31.7	78°58.4	35.7	228°13.9	30.6	Ankaa	353°03.9	-42°08.4
2	290°41.5	187°12.2	08.7	108°26.5	32.2	94°00.9	35.7	243°16.1	30.6	Schedar	349°26.9	56°41.7
3	305°44.0	202°11.3	•• 08.5	123°28.2	•• 32.6	109°03.4	•• 35.7	258°18.3	•• 30.7	Diphda	348°43.9	-17°49.3
4	320°46.4	217°10.4	08.3	138°29.9	33.0	124°05.9	35.7	273°20.5	30.8	Achernar	335°18.0	-57°04.9
5	335°48.9	232°09.6	08.1	153°31.6	33.4	139°08.4	35.7	288°22.6	30.9	Hamal	327°47.3	23°36.0
6	350°51.4	247°08.7	N24°07.9	168°33.2	S00°33.8	154°10.8	S05°35.7	303°24.8	N15°30.9	Polaris	312°28.0	89°23.0
7	5°53.8	262°07.8	07.6	183°34.9	34.2	169°13.3	35.7	318°27.0	31.0	Acamar	315°09.5	-40°11.1
8	20°56.3	277°07.0	07.4	198°36.6	34.6	184°15.8	35.7	333°29.2	31.1	Menkar	314°02.7	4°12.3
9	35°58.7	292°06.1	•• 07.2	213°38.2	•• 35.1	199°18.3	•• 35.7	348°31.3	•• 31.2	Mirfak	308°23.5	49°57.7
10	51°01.2	307°05.2	07.0	228°39.9	35.5	214°20.8	35.8	3°33.5	31.2	Aldebaran	290°35.8	16°34.0
11	66°03.7	322°04.4	06.8	243°41.6	35.9	229°23.2	35.8	18°35.7	31.3	Rigel	281°00.8	-8°10.2
12	81°06.1	337°03.5	N24°06.5	258°43.3	S00°36.3	244°25.7	S05°35.8	33°37.9	N15°31.4	Capella	280°17.1	46°01.5
13	96°08.6	352°02.6	06.3	273°44.9	36.7	259°28.2	35.8	48°40.0	31.5	Bellatrix	278°19.4	6°22.4
14	111°11.1	7°01.8	06.1	288°46.6	37.1	274°30.7	35.8	63°42.2	31.5	Elnath	277°57.7	28°37.8
15	126°13.5	22°00.9	•• 05.9	303°48.3	•• 37.6	289°33.1	•• 35.8	78°44.4	•• 31.6	Alnilam	275°34.4	-1°11.1
16	141°16.0	37°00.1	05.6	318°49.9	38.0	304°35.6	35.8	93°46.6	31.7	Betelgeuse	270°48.6	7°24.6
17	156°18.5	51°59.2	05.4	333°51.6	38.4	319°38.1	35.8	108°48.7	31.8	Canopus	263°51.3	-52°42.8
18	171°20.9	66°58.3	N24°05.2	348°53.3	S00°38.8	334°40.6	S05°35.8	123°50.9	N15°31.8	Sirius	258°23.4	-16°45.6
19	186°23.4	81°57.5	04.9	3°54.9	39.2	349°43.1	35.8	138°53.1	31.9	Adhara	255°03.4	-29°00.9
20	201°25.9	96°56.6	04.7	18°56.6	39.7	4°45.5	35.8	153°55.3	32.0	Procyon	244°47.4	5°08.8
21	216°28.3	111°55.7	•• 04.5	33°58.3	•• 40.1	19°48.0	•• 35.8	168°57.4	•• 32.0	Pollux	243°13.3	27°57.2
22	231°30.8	126°54.9	04.2	48°59.9	40.5	34°50.5	35.8	183°59.6	32.1	Avior	234°13.7	-59°36.6
23	246°33.2	141°54.0	04.0	64°01.6	40.9	49°53.0	35.8	199°01.8	32.2	Suhail	222°43.9	-43°33.4
Mer.pass. 06:36		ν -0.9' d -0.2' m -3.89			ν 1.7' d 0.4' m 0.12		ν 2.5' d 0.0' m -2.20		ν 2.2' d 0.1' m 0.33			

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	261°35.7	156°53.2	N24°03.8	79°03.3	S00°41.3	64°55.4	S05°35.8	214°04.0	N15°32.3	Alphard	217°44.4	-8°47.4
1	276°38.2	171°52.3	03.5	94°04.9	41.8	79°57.9	35.9	229°06.1	32.3	Regulus	207°30.8	11°49.3
2	291°40.6	186°51.4	03.3	109°06.6	42.2	95°00.4	35.9	244°08.3	32.4	Dubhe	193°37.0	61°35.7
3	306°43.1	201°50.6	•• 03.0	124°08.3	•• 42.6	110°02.9	•• 35.9	259°10.5	•• 32.5	Denebola	182°21.3	14°24.4
4	321°45.6	216°49.7	02.8	139°09.9	43.0	125°05.3	35.9	274°12.7	32.6	Gienah	175°39.8	-17°42.5
5	336°48.0	231°48.9	02.5	154°11.6	43.4	140°07.8	35.9	289°14.8	32.6	Acrux	172°55.6	-63°16.1
6	351°50.5	246°48.0	N24°02.3	169°13.2	S00°43.9	155°10.3	S05°35.9	304°17.0	N15°32.7	Gacrux	171°47.3	-57°17.1
7	6°53.0	261°47.1	02.1	184°14.9	44.3	170°12.7	35.9	319°19.2	32.8	Alioth	166°09.8	55°48.2
8	21°55.4	276°46.3	01.8	199°16.6	44.7	185°15.2	35.9	334°21.4	32.9	Spica	158°18.3	-11°19.0
9	36°57.9	291°45.4	•• 01.6	214°18.2	•• 45.1	200°17.7	•• 35.9	349°23.5	•• 32.9	Alkaid	152°49.0	49°10.1
10	52°00.4	306°44.6	01.3	229°19.9	45.5	215°20.2	35.9	4°25.7	33.0	Hadar	148°30.3	-60°31.1
11	67°02.8	321°43.7	01.1	244°21.5	46.0	230°22.6	35.9	19°27.9	33.1	Menkent	147°53.0	-36°31.1
12	82°05.3	336°42.9	N24°00.8	259°23.2	S00°46.4	245°25.1	S05°36.0	34°30.1	N15°33.2	Arcturus	145°44.4	19°01.8
13	97°07.7	351°42.0	00.5	274°24.8	46.8	260°27.6	36.0	49°32.2	33.2	Rigel Kent.	139°34.7	-60°57.6
14	112°10.2	6°41.1	00.3	289°26.5	47.2	275°30.0	36.0	64°34.4	33.3	Kochab	137°19.3	74°02.2
15	127°12.7	21°40.3	24°00.0	304°28.1	•• 47.7	290°32.5	•• 36.0	79°36.6	•• 33.4	Zuben'ubi	136°51.8	-16°09.9
16	142°15.1	36°39.4	23°59.8	319°29.8	48.1	305°35.0	36.0	94°38.8	33.5	Alphecca	126°00.4	26°37.0
17	157°17.6	51°38.6	59.5	334°31.5	48.5	320°37.5	36.0	109°40.9	33.5	Antares	112°11.1	-26°29.8
18	172°20.1	66°37.7	N23°59.3	349°33.1	S00°48.9	335°39.9	S05°36.0	124°43.1	N15°33.6	Atria	107°01.4	-69°04.7
19	187°22.5	81°36.9	59.0	4°34.8	49.4	350°42.4	36.0	139°45.3	33.7	Sabik	101°58.3	-15°45.6
20	202°25.0	96°36.0	58.7	19°36.4	49.8	5°44.9	36.0	154°47.5	33.7	Shaula	96°05.1	-37°07.4
21	217°27.5	111°35.1	•• 58.5	34°38.1	•• 50.2	20°47.3	•• 36.0	169°49.7	•• 33.8	Rasalhague	95°54.9	12°32.4
22	232°29.9	126°34.3	58.2	49°39.7	50.6	35°49.8	36.0	184°51.8	33.9	Eltanin	90°40.0	51°29.1
23	247°32.4	141°33.4	57.9	64°41.4	51.1	50°52.3	36.1	199°54.0	34.0	Kaus Aust.	83°27.4	-34°22.0
Mer.pass. 06:33		ν -0.9' d -0.2' m -3.89			ν 1.7' d 0.4' m 0.14		ν 2.5' d 0.0' m -2.19		ν 2.2' d 0.1' m 0.32			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	262°34.9	156°32.6	N23°57.7	79°43.0	S00°51.5	65°54.7	S05°36.1	214°56.2	N15°34.0	Nunki	75°43.0	-26°15.4
1	277°37.3	171°31.7	57.4	94°44.7	51.9	80°57.2	36.1	229°58.4	34.1	Altair	61°56.2	8°56.9
2	292°39.8	186°30.9	57.1	109°46.3	52.3	95°59.7	36.1	245°00.5	34.2	Peacock	52°59.8	-56°38.0
3	307°42.2	201°30.0	•• 56.9	124°47.9	•• 52.8	111°02.1	•• 36.1	260°02.7	•• 34.3	Deneb	49°23.0	45°23.1
4	322°44.7	216°29.2	56.6	139°49.6	53.2	126°04.6	36.1	275°04.9	34.3	Enif	33°35.1	10°00.7
5	337°47.2	231°28.3	56.3	154°51.2	53.6	141°07.1	36.1	290°07.1	34.4	Al Na'ir	27°28.4	-46°48.7
6	352°49.6	246°27.5	N23°56.0	169°52.9	S00°54.0	156°09.5	S05°36.1	305°09.3	N15°34.5	Fomalhaut	15°10.6	-29°27.7
7	7°52.1	261°26.6	55.8	184°54.5	54.5	171°12.0	36.1	320°11.4	34.6	Scheat	13°41.6	28°14.5
8	22°54.6	276°25.8	55.5	199°56.2	54.9	186°14.5	36.2	335°13.6	34.6	Markab	13°26.3	25°21.9
9	37°57.0	291°24.9	•• 55.2	214°57.8	•• 55.3	201°16.9	•• 36.2	350°15.8	•• 34.7			
10	52°59.5	306°24.1	54.9	229°59.5	55.7	216°19.4	36.2	5°18.0	34.8	Jun 12 Tue	SHA	Mer.pass
11	68°02.0	321°23.2	54.6	245°01.1	56.2	231°21.9	36.2	20°20.1	34.8	Venus	256°37.3	13:32
12	83°04.4	336°22.4	N23°54.4	260°02.7	S00°56.6	246°24.3	S05°36.2	35°22.3	N15°34.9	Mars	177°46.6	18:44
13	98°06.9	351°21.5	54.1	275°04.4	57.0	261°26.8	36.2	50°24.5	35.0	Jupiter	163°19.4	19:41
14	113°09.3	6°20.7	53.8	290°06.0	57.5	276°29.3	36.2	65°26.7	35.1	Saturn	312°35.2	09:46
15	128°11.8	21°19.8	•• 53.5	305°07.7	•• 57.9	291°31.7	•• 36.2	80°28.9	•• 35.1			
16	143°14.3	36°19.0	53.2	320°09.3	58.3	306°34.2	36.3	95°31.0	35.2	Jun 13 Wed	SHA	Mer.pass
17	158°16.7	51°18.1	52.9	335°10.9	58.7	321°36.6	36.3	110°33.2	35.3	Venus	255°17.5	13:33
18	173°19.2	66°17.3	N23°52.6	350°12.6	S00°59.2	336°39.1	S05°36.3	125°35.4	N15°35.4	Mars	177°27.6	18:42
19	188°21.7	81°16.4	52.3	5°14.2	00°59.6	351°41.6	36.3	140°37.6	35.4	Jupiter	163°19.7	19:37
20	203°24.1	96°15.6	52.1	20°15.8	01°00.0	6°44.0	36.3	155°39.7	35.5	Saturn	312°28.2	09:42

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°02.6	N23°09.1	182°10.5	8.9'	N24°23.9	-0.3'	55.5'
1	195°02.5	09.3	196°38.4	8.9'	24°23.5	-0.5'	55.6'
2	210°02.3	09.4	211°06.4	8.9'	24°23.0	-0.6'	55.6'
3	225°02.2	· · 09.6	225°34.3	8.9'	24°22.4	-0.7'	55.6'
4	240°02.1	09.7	240°02.1	8.9'	24°21.7	-0.9'	55.6'
5	255°02.0	09.9	254°30.0	8.8'	24°20.8	-1.0'	55.6'
6	270°01.8	N23°10.1	268°57.8	8.8'	N24°19.8	-1.1'	55.7'
7	285°01.7	10.2	283°25.7	8.8'	24°18.7	-1.3'	55.7'
8	300°01.6	10.4	297°53.5	8.8'	24°17.4	-1.4'	55.7'
9	315°01.4	· · 10.5	312°21.3	8.8'	24°16.0	-1.5'	55.7'
10	330°01.3	10.7	326°49.1	8.8'	24°14.5	-1.7'	55.7'
11	345°01.2	10.8	341°16.8	8.8'	24°12.9	-1.8'	55.8'
12	0°01.0	N23°10.9	355°44.6	8.8'	N24°11.1	-1.9'	55.8'
13	15°00.9	11.1	10°12.4	8.8'	24°09.1	-2.1'	55.8'
14	30°00.8	11.2	24°40.1	8.7'	24°07.1	-2.2'	55.8'
15	45°00.6	· · 11.4	39°07.9	8.7'	24°04.9	-2.3'	55.9'
16	60°00.5	11.5	53°35.6	8.7'	24°02.6	-2.5'	55.9'
17	75°00.4	11.7	68°03.4	8.7'	24°00.1	-2.6'	55.9'
18	90°00.3	N23°11.8	82°31.1	8.7'	N23°57.5	-2.7'	55.9'
19	105°00.1	12.0	96°58.8	8.7'	23°54.8	-2.8'	55.9'
20	120°00.0	12.1	111°26.6	8.7'	23°52.0	-3.0'	56.0'
21	134°59.9	· · 12.2	125°54.3	8.7'	23°49.0	-3.1'	56.0'
22	149°59.7	12.4	140°22.0	8.7'	23°45.9	-3.2'	56.0'
23	164°59.6	12.5	154°49.8	8.7'	23°42.6	-3.4'	56.0'
SD = 15.7'			SD = 15.1'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179°59.5	N23°12.7	169°17.5	8.8'	N23°39.3	-3.5'	56.0'
1	194°59.3	12.8	183°45.3	8.8'	23°35.7	-3.6'	56.1'
2	209°59.2	12.9	198°13.0	8.8'	23°32.1	-3.8'	56.1'
3	224°59.1	· · 13.1	212°40.8	8.8'	23°28.3	-3.9'	56.1'
4	239°58.9	13.2	227°08.6	8.8'	23°24.4	-4.0'	56.1'
5	254°58.8	13.3	241°36.3	8.8'	23°20.4	-4.2'	56.1'
6	269°58.7	N23°13.5	256°04.1	8.8'	N23°16.2	-4.3'	56.2'
7	284°58.5	13.6	270°31.9	8.8'	23°12.0	-4.4'	56.2'
8	299°58.4	13.8	284°59.8	8.8'	23°07.5	-4.6'	56.2'
9	314°58.3	· · 13.9	299°27.6	8.8'	23°03.0	-4.7'	56.2'
10	329°58.1	14.0	313°55.4	8.9'	22°58.3	-4.8'	56.2'
11	344°58.0	14.1	328°23.3	8.9'	22°53.5	-4.9'	56.3'
12	359°57.9	N23°14.3	342°51.2	8.9'	N22°48.6	-5.1'	56.3'
13	14°57.7	14.4	357°19.1	8.9'	22°43.5	-5.2'	56.3'
14	29°57.6	14.5	11°47.0	8.9'	22°38.3	-5.3'	56.3'
15	44°57.5	· · 14.7	26°14.9	9.0'	22°33.0	-5.4'	56.4'
16	59°57.3	14.8	40°42.9	9.0'	22°27.5	-5.6'	56.4'
17	74°57.2	14.9	55°10.9	9.0'	22°22.0	-5.7'	56.4'
18	89°57.1	N23°15.0	69°38.9	9.0'	N22°16.3	-5.8'	56.4'
19	104°57.0	15.2	84°06.9	9.0'	22°10.4	-5.9'	56.4'
20	119°56.8	15.3	98°35.0	9.1'	22°04.5	-6.1'	56.5'
21	134°56.7	· · 15.4	113°03.0	9.1'	21°58.4	-6.2'	56.5'
22	149°56.6	15.5	127°31.1	9.1'	21°52.2	-6.3'	56.5'
23	164°56.4	15.7	141°59.2	9.2'	21°45.9	-6.4'	56.5'
SD = 15.7'			SD = 15.3'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°56.3	N23°15.8	156°27.4	9.2'	N21°39.5	-6.6'	56.5'
1	194°56.2	15.9	170°55.6	9.2'	21°32.9	-6.7'	56.6'
2	209°56.0	16.0	185°23.8	9.2'	21°26.2	-6.8'	56.6'
3	224°55.9	· · 16.1	199°52.0	9.3'	21°19.4	-6.9'	56.6'
4	239°55.8	16.3	214°20.3	9.3'	21°12.5	-7.0'	56.6'
5	254°55.6	16.4	228°48.6	9.3'	21°05.4	-7.2'	56.6'
6	269°55.5	N23°16.5	243°16.9	9.4'	N20°58.3	-7.3'	56.7'
7	284°55.4	16.6	257°45.3	9.4'	20°51.0	-7.4'	56.7'
8	299°55.2	16.7	272°13.7	9.4'	20°43.6	-7.5'	56.7'
9	314°55.1	· · 16.9	286°42.1	9.5'	20°36.1	-7.6'	56.7'
10	329°54.9	17.0	301°10.6	9.5'	20°28.4	-7.7'	56.8'
11	344°54.8	17.1	315°39.0	9.5'	20°20.7	-7.9'	56.8'
12	359°54.7	N23°17.2	330°07.6	9.6'	N20°12.8	-8.0'	56.8'
13	14°54.5	17.3	344°36.1	9.6'	20°04.9	-8.1'	56.8'
14	29°54.4	17.4	359°04.7	9.6'	19°56.8	-8.2'	56.8'
15	44°54.3	· · 17.5	13°33.4	9.7'	19°48.6	-8.3'	56.9'
16	59°54.1	17.6	28°02.0	9.7'	19°40.2	-8.4'	56.9'
17	74°54.0	17.8	42°30.7	9.7'	19°31.8	-8.5'	56.9'
18	89°53.9	N23°17.9	56°59.5	9.8'	N19°23.3	-8.6'	56.9'
19	104°53.7	18.0	71°28.2	9.8'	19°14.6	-8.8'	56.9'
20	119°53.6	18.1	85°57.1	9.9'	19°05.9	-8.9'	57.0'
21	134°53.5	· · 18.2	100°25.9	9.9'	18°57.0	-9.0'	57.0'
22	149°53.3	18.3	114°54.8	9.9'	18°48.1	-9.1'	57.0'
23	164°53.2	18.4	129°23.7	10.0'	18°39.0	-9.2'	57.0'
SD = 15.7'			SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	--	--	01:35	22:26	--	--
62°	--	--	02:11	21:50	--	--
60°	--	00:56	02:37	21:24	23:06	--
N 58°	--	01:43	02:57	21:04	22:18	--
56°	--	02:12	03:14	20:47	21:49	--
54°	00:51	02:34	03:28	20:33	21:27	23:11
52°	01:34	02:51	03:40	20:21	21:09	22:27
50°	02:01	03:06	03:50	20:10	20:54	21:59
45°	02:46	03:35	04:13	19:48	20:25	21:14
N 40°	03:16	03:58	04:31	19:30	20:03	20:44
35°	03:39	04:16	04:45	19:15	19:45	20:21
30°	03:58	04:31	04:58	19:02	19:29	20:03
20°	04:27	04:56	05:20	18:40	19:05	19:34
N 10°	04:49	05:16	05:39	18:21	18:44	19:11
0°	05:08	05:34	05:56	18:04	18:26	18:53
S 10°	05:25	05:51	06:14	17:46	18:09	18:36
20°	05:41	06:08	06:32	17:28	17:52	18:20
30°	05:57	06:27	06:53	17:07	17:33	18:03
35°	06:06	06:37	07:05	16:55	17:23	17:55
40°	06:15	06:49	07:19	16:41	17:11	17:45
45°	06:25	07:02	07:36	16:24	16:58	17:35
S 50°	06:37	07:18	07:57	16:03	16:42	17:23
52°	06:42	07:26	08:06	15:54	16:35	17:18
54°	06:48	07:34	08:17	15:43	16:26	17:13
56°	06:54	07:43	08:30	15:30	16:17	17:06
58°	07:00	07:53	08:44	15:16	16:07	17:00
S 60°	07:08	08:04	09:02	14:58	15:56	16:52

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°						
N 70°						
68°			02:43			00:42
66°		01:39	03:41	23:55	23:44	23:37
64°	01:24	02:39	04:15	22:55	23:10	23:16
62°	02:04	03:13	04:39	22:21	22:45	22:59
60°	02:31	03:37	04:58	21:56	22:25	22:44
N 58°	02:53	03:57	05:14	21:36	22:09	22:32
56°	03:10	04:13	05:28	21:19	21:55	22:21
54°	03:25	04:27	05:39	21:05	21:43	22:11
52°	03:38	04:40	05:50	20:53	21:32	22:03
50°	03:50	04:50	05:59	20:42	21:22	21:55
45°	04:13	05:13	06:18	20:19	21:02	21:39
N 40°	04:32	05:31	06:34	20:01	20:46	21:25
35°	04:48	05:46	06:47	19:45	20:32	21:14
30°	05:02	05:59	06:59	19:32	20:20	21:03
20°	05:26	06:21	07:18	19:09	19:59	20:46
N 10°	05:46	06:40	07:35	18:49	19:41	20:31
0°	06:05	06:58	07:51	18:31	19:24	20:16
S 10°	06:24	07:16	08:07	18:12	19:07	20:02
20°	06:44	07:35	08:24	17:52	18:48	19:46
30°	07:08	07:58	08:43	17:29	18:27	19:28
35°	07:21	08:11	08:55	17:15	18:15	19:18
40°	07:37	08:25	09:08	16:59	18:00	19:06
45°	07:56	08:43	09:23	16:40	17:43	18:51
S 50°	08:20	09:05	09:41	16:17	17:22	18:34
52°	08:32	09:16	09:50	16:06	17:12	18:25
54°	08:45	09:27	10:00	15:53	17:00	18:16
56°	08:59	09:41	10:11	15:38	16:47	18:05
58°	09:17	09:56	10:24	15:20	16:32	17:53
S 60°	09:39	10:15	10:38	14:59	16:14	17:39

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	0-2 0-4%
12	00:10	00:04	12:00	12:18	--	
13	00:02	00:08	12:00	13:11	00:44	
14	00:15	00:21	12:00	14:04	01:38	

June 15, 16, 17 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	SHA	Dec				
0	263°34.0	156°12.2	N23°50.9	80°22.4	S01°01.8	66°53.9	S05°36.4	215°48.5	N15°35.8	Alpheratz	357°31.0	29°15.1											
1	278°36.5	171°11.3	50.6	95°24.0	02.2	81°56.3	36.4	230°50.6	35.9	Ankaa	353°03.8	-42°08.4											
2	293°38.9	186°10.5	50.3	110°25.6	02.6	96°58.8	36.4	245°52.8	35.9	Schedar	349°26.9	56°41.7											
3	308°41.4	201°09.6	•• 50.0	125°27.3	•• 03.0	112°01.3	•• 36.4	260°55.0	•• 36.0	Diphda	348°43.8	-17°49.3											
4	323°43.8	216°08.8	49.7	140°28.9	03.5	127°03.7	36.4	275°57.2	36.1	Achernar	335°18.0	-57°04.9											
5	338°46.3	231°07.9	49.4	155°30.5	03.9	142°06.2	36.4	290°59.4	36.2	Hamal	327°47.3	23°36.0											
6	353°48.8	246°07.1	N23°49.1	170°32.2	S01°04.3	157°08.6	S05°36.4	306°01.5	N15°36.2	Polaris	312°26.9	89°22.9											
7	8°51.2	261°06.2	48.8	185°33.8	04.8	172°11.1	36.5	321°03.7	36.3	Acamar	315°09.5	-40°11.1											
8	23°53.7	276°05.4	48.5	200°35.4	05.2	187°13.6	36.5	336°05.9	36.4	Menkar	314°02.7	4°12.3											
9	38°56.2	291°04.6	•• 48.1	215°37.1	•• 05.6	202°16.0	•• 36.5	351°08.1	•• 36.4	Mirfak	308°23.4	49°57.7											
10	53°58.6	306°03.7	47.8	230°38.7	06.1	217°18.5	36.5	6°10.3	36.5	Aldebaran	290°35.8	16°34.0											
11	69°01.1	321°02.9	47.5	245°40.3	06.5	232°20.9	36.5	21°12.4	36.6	Rigel	281°00.8	-8°10.2											
12	84°03.6	336°02.0	N23°47.2	260°41.9	S01°06.9	247°23.4	S05°36.5	36°14.6	N15°36.7	Capella	280°17.0	46°01.5											
13	99°06.0	351°01.2	46.9	275°43.6	07.4	262°25.8	36.5	51°16.8	36.7	Bellatrix	278°19.4	6°22.4											
14	114°08.5	6°00.3	46.6	290°45.2	07.8	277°28.3	36.6	66°19.0	36.8	Elnath	277°57.7	28°37.8											
15	129°11.0	20°59.5	•• 46.3	305°46.8	•• 08.2	292°30.8	•• 36.6	81°21.2	•• 36.9	Alnilam	275°34.4	-1°11.1											
16	144°13.4	35°58.6	46.0	320°48.4	08.7	307°33.2	36.6	96°23.3	37.0	Betelgeuse	270°48.5	7°24.6											
17	159°15.9	50°57.8	45.6	335°50.1	09.1	322°35.7	36.6	111°25.5	37.0	Canopus	263°51.3	-52°42.8											
18	174°18.3	65°57.0	N23°45.3	350°51.7	S01°09.5	337°38.1	S05°36.6	126°27.7	N15°37.1	Sirius	258°23.4	-16°45.6											
19	189°20.8	80°56.1	45.0	5°53.3	10.0	352°40.6	36.6	141°29.9	37.2	Adhara	255°03.4	-29°00.9											
20	204°23.3	95°55.3	44.7	20°54.9	10.4	7°43.0	36.7	156°32.1	37.2	Procyon	244°47.4	5°08.8											
21	219°25.7	110°54.4	•• 44.4	35°56.5	•• 10.9	22°45.5	•• 36.7	171°34.2	•• 37.3	Pollux	243°13.3	27°57.2											
22	234°28.2	125°53.6	44.0	50°58.2	11.3	37°47.9	36.7	186°36.4	37.4	Avior	234°13.7	-59°36.6											
23	249°30.7	140°52.8	43.7	65°59.8	11.7	52°50.4	36.7	201°38.6	37.5	Suhail	222°43.9	-43°33.4											
Mer.pass. 06:25				ν -0.8' d -0.3' m -3.89				ν 1.6' d 0.4' m 0.17				ν 2.5' d 0.0' m -2.18				ν 2.2' d 0.1' m 0.32							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	264°33.1	155°51.9	N23°43.4	81°01.4	S01°12.2	67°52.8	S05°36.7	216°40.8	N15°37.5										
1	279°35.6	170°51.1	43.1	96°03.0	12.6	82°55.3	36.7	231°43.0	37.6										
2	294°38.1	185°50.2	42.7	111°04.6	13.0	97°57.8	36.8	246°45.1	37.7										
3	309°40.5	200°49.4	•• 42.4	126°06.3	•• 13.5	113°00.2	•• 36.8	261°47.3	•• 37.7										
4	324°43.0	215°48.6	42.1	141°07.9	13.9	128°02.7	36.8	276°49.5	37.8										
5	339°45.5	230°47.7	41.7	156°09.5	14.3	143°05.1	36.8	291°51.7	37.9										
6	354°47.9	245°46.9	N23°41.4	171°11.1	S01°14.8	158°07.6	S05°36.8	306°53.9	N15°38.0										
7	9°50.4	260°46.0	41.1	186°12.7	15.2	173°10.0	36.8	321°56.0	38.0										
8	24°52.8	275°45.2	40.7	201°14.3	15.7	188°12.5	36.9	336°58.2	38.1										
9	39°55.3	290°44.4	•• 40.4	216°15.9	•• 16.1	203°14.9	•• 36.9	352°00.4	•• 38.2										
10	54°57.8	305°43.5	40.1	231°17.6	16.5	218°17.4	36.9	7°02.6	38.2										
11	70°00.2	320°42.7	39.7	246°19.2	17.0	233°19.8	36.9	22°04.8	38.3										
12	85°02.7	335°41.9	N23°39.4	261°20.8	S01°17.4	248°22.3	S05°36.9	37°07.0	N15°38.4										
13	100°05.2	350°41.0	39.0	276°22.4	17.8	263°24.7	36.9	52°09.1	38.5										
14	115°07.6	5°40.2	38.7	291°24.0	18.3	278°27.2	37.0	67°11.3	38.5										
15	130°10.1	20°39.4	•• 38.4	306°25.6	•• 18.7	293°29.6	•• 37.0	82°13.5	•• 38.6										
16	145°12.6	35°38.5	38.0	321°27.2	19.2	308°32.1	37.0	97°15.7	38.7										
17	160°15.0	50°37.7	37.7	336°28.8	19.6	323°34.5	37.0	112°17.9	38.7										
18	175°17.5	65°36.9	N23°37.3	351°30.4	S01°20.0	338°37.0	S05°37.0	127°20.0	N15°38.8										
19	190°19.9	80°36.0	37.0	6°32.0	20.5	353°39.4	37.1	142°22.2	38.9										
20	205°22.4	95°35.2	36.6	21°33.6	20.9	8°41.9	37.1	157°24.4	39.0										
21	220°24.9	110°34.4	•• 36.3	36°35.3	•• 21.4	23°44.3	•• 37.1	172°26.6	•• 39.0										
22	235°27.3	125°33.5	35.9	51°36.9	21.8	38°46.8	37.1	187°28.8	39.1										
23	250°29.8	140°32.7	35.6	66°38.5	22.2	53°49.2	37.1	202°31.0	39.2										
Mer.pass. 06:21				ν -0.8' d -0.3' m -3.89				ν 1.6' d 0.4' m 0.18				ν 2.5' d 0.0' m -2.17				ν 2.2' d 0.1' m 0.32			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	265°32.3	155°31.9	N23°35.2	81°40.1	S01°22.7	68°51.7	S05°37.2	217°33.1	N15°39.2										
1	280°34.7	170°31.0	34.9	96°41.7	23.1	83°54.1	37.2	232°35.3	39.3										
2	295°37.2	185°30.2	34.5	111°43.3	23.6	98°56.5	37.2	247°37.5	39.4										
3	310°39.7	200°29.4	•• 34.1	126°44.9	•• 24.0	113°59.0	•• 37.2	262°39.7	•• 39.5										
4	325°42.1	215°28.5	33.8	141°46.5	24.5	129°01.4	37.2	277°41.9	39.5										
5	340°44.6	230°27.7	33.4	156°48.1	24.9	144°03.9	37.2	292°44.1	39.6										
6	355°47.1	245°26.9	N23°33.1	171°49.7	S01°25.3	159°06.3	S05°37.3	307°46.2	N15°39.7										
7	10°49.5	260°26.0	32.7	186°51.3	25.8	174°08.8	37.3	322°48.4	39.7										
8	25°52.0	275°25.2	32.3	201°52.9	26.2	189°11.2	37.3	337°50.6	39.8										
9	40°54.4	290°24.4	•• 32.0	216°54.5	•• 26.7	204°13.7	•• 37.3	352°52.8	•• 39.9										
10	55°56.9	305°23.6	31.6	231°56.1	27.1	219°16.1	37.4	7°55.0	40.0										
11	70°59.4	320°22.7	31.2	246°57.7	27.6	234°18.5	37.4	22°57.2	40.0										
12	86°01.8	335°21.9	N23°30.9	261°59.3	S01°28.0	249°21.0	S05°37.4	37°59.3	N15°40.1										
13	101°04.3	350°21.1	30.5	277°00.9	28.4	264°23.4	37.4	53°01.5	40.2										
14	116°06.8	5°20.3	30.1	292°02.5	28.9	279°25.9	37.4	68°03.7	40.2										
15	131°09.2	20°19.4	•• 29.8	307°04.1	•• 29.3	294°28.3	•• 37.5	83°05.9	•• 40.3										
16	146°11.7	35°18.6	29.4	322°05.7	29.8	309°30.8	37.5	98°08.1	40.4										
17	161°14.2	50°17.8	29.0	337°07.3	30.2	324°33.2	37.5	113°10.3	40.4										
18	176°16.6	65°17.0	N23°28.6	352°08.8	S01°30.7	339°35.6	S05°37.5	128°12.4	N15°40.5										
19	191°19.1	80°16.1	28.3	7°10.4	31.1	354°38.1	37.5	143°14.6	40.6										
20	206°21.6	95°15.3	27.9	22°12.0	31.6	9°40.5	37.6	158°16.8	40.7										
21	221°24.0	110°14.5	•• 27.5	37°13.6	•• 32.0	24°43.0	•• 37.6	173°19.0	•• 40.7										
22	236°26.5	125°13.7	27.1	52°15.2	32.4	39°45.4	37.6	188°21.2	40.8										
23	251°28.9	140°12.8	26.7	67°16.8	32.9	54°47.8	37.6	203°23.4	40.9										
Mer.pass. 06:17				ν -0.8' d -0.4' m -3.89				ν 1.6' d 0.4' m 0.20				ν 2.4' d 0.0' m -2.16				ν 2.2' d 0.1' m 0.32			

Stars		
Alphard	217°44.4	-8°47.4
Regulus	207°30.8	11°49.3
Dubhe	193°37.0	61°35.7
Denebola	182°21.3	14°24.4
Gienah	175°39.8	-17°42.5
Acrux	172°55.7	-63°16.1
Gacrux	171°47.3	-57°17.1
Alioth	166°09.8	55°48.2
Spica	158°18.3	-11°19.0
Alkaid	152°49.0	49°10.1
Hadar	148°30.3	-60°31.2
Menkent	147°53.0	-36°31.1
Arcturus	145°44.5	19°01.8
Rigel Kent.	139°34.7	-60°57.6
Kochab	137°19.3	74°02.2
Zuben'ubi	136°51.8	-16°09.9
Alphecca	126°00.4	26°37.0
Antares	112°11.0	-26°29.8
Atria	107°01.4	-69°04.7
Sabik	101°58.3	-15°45.6
Shaula	96°05.1	-37°07.4
Rasalhague	95°54.9	12°32.4
Eltanin	90°40.0	51°29.2
Kaus Aust.	83°27.4	-34°22.0
Vega	80°30.4	38°48.7
Nunki	75°43.0	-26°15.4
Altair	61°56.2	8°56.9
Peacock	52°59.8	-56°38.0
Deneb	49°22.9	45°23.1
Enif	33°35.1	10°00.7
Al Na'ir	27°28.4	-46°48.7
Fomalhaut	15°10.6	-29°27.7
Scheat	13°41.6	28°14.5
Markab	13°26.2	15°21.9
Jun 15 Fri		
Venus	252°38.2	13:36
Mars	176°48.4	18:36
Jupiter	163°19.9	19:29
Saturn	312°14.5	09:35
Jun 16 Sat		
Venus	251°18.8	13:37
Mars	176°28.3	18:34
Jupiter	163°19.7	19:25
Saturn	312°07.7	09:32
Jun 17 Sun		
Venus	249°59.6	13:

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°53.1	N23°18.5	143°52.7	10.0'	N18°29.8	-9.3'	57.0'
1	194°52.9	18.6	158°21.7	10.0'	18°20.5	-9.4'	57.1'
2	209°52.8	18.7	172°50.7	10.1'	18°11.1	-9.5'	57.1'
3	224°52.7	· · 18.8	187°19.8	10.1'	18°01.6	-9.6'	57.1'
4	239°52.5	18.9	201°48.9	10.2'	17°52.1	-9.7'	57.1'
5	254°52.4	19.0	216°18.1	10.2'	17°42.4	-9.8'	57.2'
6	269°52.3	N23°19.1	230°47.3	10.2'	N17°32.6	-9.9'	57.2'
7	284°52.1	19.2	245°16.5	10.3'	17°22.7	-10.0'	57.2'
8	299°52.0	19.3	259°45.8	10.3'	17°12.7	-10.1'	57.2'
9	314°51.9	· · 19.4	274°15.1	10.3'	17°02.6	-10.2'	57.2'
10	329°51.7	19.5	288°44.4	10.4'	16°52.4	-10.3'	57.3'
11	344°51.6	19.6	303°13.8	10.4'	16°42.2	-10.4'	57.3'
12	359°51.5	N23°19.7	317°43.2	10.5'	N16°31.8	-10.5'	57.3'
13	14°51.3	19.8	332°12.7	10.5'	16°21.3	-10.6'	57.3'
14	29°51.2	19.9	346°42.2	10.5'	16°10.8	-10.6'	57.3'
15	44°51.1	· · 20.0	1°11.7	10.6'	16°00.1	-10.7'	57.4'
16	59°50.9	20.1	15°41.3	10.6'	15°49.4	-10.8'	57.4'
17	74°50.8	20.2	30°10.9	10.7'	15°38.6	-10.9'	57.4'
18	89°50.6	N23°20.3	44°40.6	10.7'	N15°27.6	-11.0'	57.4'
19	104°50.5	20.4	59°10.3	10.7'	15°16.6	-11.1'	57.4'
20	119°50.4	20.4	73°40.0	10.8'	15°05.5	-11.2'	57.5'
21	134°50.2	· · 20.5	88°09.7	10.8'	14°54.4	-11.3'	57.5'
22	149°50.1	20.6	102°39.5	10.8'	14°43.1	-11.3'	57.5'
23	164°50.0	20.7	117°09.4	10.9'	14°31.8	-11.4'	57.5'
SD = 15.7'			SD = 15.6'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°49.8	N23°20.8	131°39.2	10.9'	N14°20.3	-11.5'	57.5'
1	194°49.7	20.9	146°09.1	10.9'	14°08.8	-11.6'	57.6'
2	209°49.6	21.0	160°39.1	11.0'	13°57.2	-11.7'	57.6'
3	224°49.4	· · 21.1	175°09.0	11.0'	13°45.6	-11.7'	57.6'
4	239°49.3	21.1	189°39.1	11.0'	13°33.8	-11.8'	57.6'
5	254°49.2	21.2	204°09.1	11.1'	13°22.0	-11.9'	57.6'
6	269°49.0	N23°21.3	218°39.2	11.1'	N13°10.1	-12.0'	57.7'
7	284°48.9	21.4	233°09.3	11.1'	12°58.1	-12.0'	57.7'
8	299°48.8	21.5	247°39.4	11.2'	12°46.1	-12.1'	57.7'
9	314°48.6	· · 21.6	262°09.6	11.2'	12°34.0	-12.2'	57.7'
10	329°48.5	21.6	276°39.8	11.2'	12°21.8	-12.3'	57.7'
11	344°48.3	21.7	291°10.0	11.3'	12°09.5	-12.3'	57.8'
12	359°48.2	N23°21.8	305°40.3	11.3'	N11°57.2	-12.4'	57.8'
13	14°48.1	21.9	320°10.5	11.3'	11°44.8	-12.5'	57.8'
14	29°47.9	22.0	334°40.9	11.3'	11°32.3	-12.5'	57.8'
15	44°47.8	· · 22.0	349°11.2	11.4'	11°19.8	-12.6'	57.8'
16	59°47.7	22.1	3°41.6	11.4'	11°07.2	-12.7'	57.9'
17	74°47.5	22.2	18°12.0	11.4'	10°54.5	-12.7'	57.9'
18	89°47.4	N23°22.3	32°42.4	11.5'	N10°41.8	-12.8'	57.9'
19	104°47.3	22.3	47°12.9	11.5'	10°29.0	-12.9'	57.9'
20	119°47.1	22.4	61°43.3	11.5'	10°16.1	-12.9'	57.9'
21	134°47.0	· · 22.5	76°13.8	11.5'	10°03.2	-13.0'	58.0'
22	149°46.9	22.6	90°44.3	11.5'	09°50.3	-13.0'	58.0'
23	164°46.7	22.6	105°14.9	11.6'	09°37.2	-13.1'	58.0'
SD = 15.7'			SD = 15.7'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°46.6	N23°22.7	119°45.5	11.6'	N09°24.2	-13.1'	58.0'
1	194°46.4	22.8	134°16.0	11.6'	09°11.0	-13.2'	58.0'
2	209°46.3	22.8	148°46.7	11.6'	08°57.8	-13.2'	58.1'
3	224°46.2	· · 22.9	163°17.3	11.6'	08°44.6	-13.3'	58.1'
4	239°46.0	23.0	177°47.9	11.7'	08°31.3	-13.4'	58.1'
5	254°45.9	23.0	192°18.6	11.7'	08°17.9	-13.4'	58.1'
6	269°45.8	N23°23.1	206°49.3	11.7'	N08°04.5	-13.5'	58.1'
7	284°45.6	23.2	221°20.0	11.7'	07°51.1	-13.5'	58.2'
8	299°45.5	23.2	235°50.7	11.7'	07°37.6	-13.5'	58.2'
9	314°45.4	· · 23.3	250°21.4	11.7'	07°24.0	-13.6'	58.2'
10	329°45.2	23.4	264°52.1	11.8'	07°10.4	-13.6'	58.2'
11	344°45.1	23.4	279°22.9	11.8'	06°56.8	-13.7'	58.2'
12	359°44.9	N23°23.5	293°53.7	11.8'	N06°43.1	-13.7'	58.2'
13	14°44.8	23.5	308°24.4	11.8'	06°29.4	-13.8'	58.3'
14	29°44.7	23.6	322°55.2	11.8'	06°15.6	-13.8'	58.3'
15	44°44.5	· · 23.7	337°26.0	11.8'	06°01.8	-13.8'	58.3'
16	59°44.4	23.7	351°56.8	11.8'	05°48.0	-13.9'	58.3'
17	74°44.3	23.8	6°27.6	11.8'	05°34.1	-13.9'	58.3'
18	89°44.1	N23°23.8	20°58.5	11.8'	N05°20.2	-13.9'	58.4'
19	104°44.0	23.9	35°29.3	11.8'	05°06.2	-14.0'	58.4'
20	119°43.9	24.0	50°00.1	11.8'	04°52.3	-14.0'	58.4'
21	134°43.7	· · 24.0	64°30.9	11.8'	04°38.2	-14.0'	58.4'
22	149°43.6	24.1	79°01.8	11.8'	04°24.2	-14.1'	58.4'
23	164°43.5	24.1	93°32.6	11.8'	04°10.1	-14.1'	58.4'
SD = 15.7'			SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	--	--	01:32	22:30	--	--
62°	--	--	02:10	21:52	--	--
60°	--	00:52	02:36	21:26	23:11	--
N 58°	--	01:41	02:56	21:06	22:21	--
56°	--	02:11	03:13	20:49	21:51	--
54°	00:47	02:33	03:27	20:35	21:29	23:15
52°	01:33	02:51	03:39	20:22	21:11	22:29
50°	02:00	03:06	03:50	20:11	20:56	22:02
45°	02:46	03:35	04:13	19:49	20:26	21:16
N 40°	03:16	03:58	04:31	19:31	20:04	20:46
35°	03:39	04:16	04:46	19:16	19:46	20:22
30°	03:58	04:31	04:59	19:03	19:30	20:04
20°	04:27	04:56	05:21	18:41	19:06	19:35
N 10°	04:49	05:17	05:40	18:22	18:45	19:12
0°	05:08	05:35	05:57	18:04	18:27	18:53
S 10°	05:25	05:52	06:15	17:47	18:10	18:36
20°	05:41	06:09	06:33	17:29	17:53	18:20
30°	05:58	06:28	06:54	17:07	17:34	18:04
35°	06:07	06:38	07:07	16:55	17:23	17:55
40°	06:16	06:50	07:21	16:41	17:11	17:45
45°	06:26	07:04	07:37	16:24	16:58	17:35
S 50°	06:38	07:20	07:58	16:03	16:42	17:23
52°	06:43	07:27	08:08	15:54	16:34	17:18
54°	06:49	07:35	08:19	15:42	16:26	17:12
56°	06:55	07:44	08:32	15:30	16:17	17:06
58°	07:02	07:55	08:46	15:15	16:07	17:00
S 60°	07:09	08:06	09:04	14:58	15:55	16:52

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°		06:26	08:45		00:37	00:01 23:35
N 70°	04:30	06:52	08:56	00:46	00:09 23:48	23:30
68°	05:09	07:11	09:06	00:06 23:49	23:36	23:26
66°	05:36	07:26	09:13	23:32	23:27	23:22
64°	05:57	07:39	09:19	23:18	23:19	23:19
62°	06:13	07:49	09:25	23:07	23:12	23:16
60°	06:27	07:58	09:29	22:57	23:06	23:14
N 58°	06:39	08:06	09:34	22:48	23:01	23:12
56°	06:49	08:13	09:37	22:40	22:56	23:10
54°	06:58	08:19	09:41	22:34	22:52	23:08
52°	07:06	08:24	09:43	22:27	22:48	23:06
50°	07:13	08:29	09:46	22:22	22:45	23:05
45°	07:28	08:40	09:52	22:10	22:37	23:02
N 40°	07:41	08:48	09:57	22:00	22:30	22:59
35°	07:51	08:56	10:01	21:51	22:25	22:57
30°	08:00	09:03	10:05	21:43	22:20	22:55
20°	08:16	09:14	10:11	21:30	22:11	22:51
N 10°	08:30	09:24	10:17	21:18	22:04	22:48
0°	08:43	09:33	10:22	21:07	21:56	22:45
S 10°	08:56	09:42	10:27	20:56	21:49	22:42
20°	09:09	09:52	10:32	20:44	21:41	22:38
30°	09:25	10:03	10:39	20:30	21:32	22:34
35°	09:34	10:10	10:42	20:22	21:27	22:32
40°	09:44	10:17	10:46	20:13	21:21	22:30
45°	09:56	10:25	10:51	20:02	21:14	22:27
S 50°	10:11	10:35	10:57	19:49	21:06	22:23
52°	10:18	10:40	10:59	19:43	21:02	22:21
54°	10:25	10:45	11:02	19:36	20:57	22:20
56°	10:33	10:51	11:05	19:28	20:53	22:18
58°	10:43	10:57	11:09	19:20	20:47	22:15
S 60°	10:54	11:04	11:12	19:10	20:41	22:13

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Upper hh:mm	Lower hh:mm	3-5 9-24%
15	00:28	00:34	12:01	14:55	02:30	
16	00:41	00:47	12:01	15:45	03:20	
17	00:54	01:00	12:01	16:33	04:09	

June 18, 19, 20 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	266°31.4	155°12.0	N23°26.4	82°18.4	S01°33.3	69°50.3	S05°37.6	218°25.5	N15°40.9	Alpheratz	357°31.0	29°15.1
1	281°33.9	170°11.2	26.0	97°20.0	33.8	84°52.7	37.7	233°27.7	41.0	Ankaa	353°03.8	-42°08.4
2	296°36.3	185°10.4	25.6	112°21.6	34.2	99°55.2	37.7	248°29.9	41.1	Schedar	349°26.9	56°41.7
3	311°38.8	200°09.5	•• 25.2	127°23.2	•• 34.7	114°57.6	•• 37.7	263°32.1	•• 41.2	Diphda	348°43.8	-17°49.3
4	326°41.3	215°08.7	24.8	142°24.8	35.1	130°00.0	37.7	278°34.3	41.2	Achernar	335°18.0	-57°04.9
5	341°43.7	230°07.9	24.4	157°26.3	35.6	145°02.5	37.8	293°36.5	41.3	Hamal	327°47.3	23°36.0
6	356°46.2	245°07.1	N23°24.0	172°27.9	S01°36.0	160°04.9	S05°37.8	308°38.6	N15°41.4	Polaris	312°25.9	89°22.9
7	11°48.7	260°06.3	23.6	187°29.5	36.5	175°07.4	37.8	323°40.8	41.4	Acamar	315°09.4	-40°11.0
8	26°51.1	275°05.4	23.3	202°31.1	36.9	190°09.8	37.8	338°43.0	41.5	Menkar	314°02.6	4°12.3
9	41°53.6	290°04.6	•• 22.9	217°32.7	•• 37.4	205°12.2	•• 37.9	353°45.2	•• 41.6	Mirfak	308°23.4	49°57.7
10	56°56.0	305°03.8	22.5	232°34.3	37.8	220°14.7	37.9	8°47.4	41.6	Aldebaran	290°35.8	16°34.0
11	71°58.5	320°03.0	22.1	247°35.8	38.3	235°17.1	37.9	23°49.6	41.7	Rigel	281°00.8	-8°10.2
12	87°01.0	335°02.2	N23°21.7	262°37.4	S01°38.7	250°19.5	S05°37.9	38°51.8	N15°41.8	Capella	280°17.0	46°01.5
13	102°03.4	350°01.3	21.3	277°39.0	39.2	265°22.0	37.9	53°53.9	41.9	Bellatrix	278°19.4	6°22.4
14	117°05.9	5°00.5	20.9	292°40.6	39.6	280°24.4	38.0	68°56.1	41.9	Elnath	277°57.7	28°37.8
15	132°08.4	19°59.7	•• 20.5	307°42.2	•• 40.0	295°26.9	•• 38.0	83°58.3	•• 42.0	Alnilam	275°34.4	-1°11.1
16	147°10.8	34°58.9	20.1	322°43.8	40.5	310°29.3	38.0	99°00.5	42.1	Betelgeuse	270°48.5	7°24.6
17	162°13.3	49°58.1	19.7	337°45.3	40.9	325°31.7	38.0	114°02.7	42.1	Canopus	263°51.3	-52°42.8
18	177°15.8	64°57.3	N23°19.3	352°46.9	S01°41.4	340°34.2	S05°38.1	129°04.9	N15°42.2	Sirius	258°23.4	-16°45.6
19	192°18.2	79°56.4	18.9	7°48.5	41.8	355°36.6	38.1	144°07.1	42.3	Adhara	255°03.4	-29°00.9
20	207°20.7	94°55.6	18.5	22°50.1	42.3	10°39.0	38.1	159°09.2	42.3	Procyon	244°47.4	5°08.8
21	222°23.2	109°54.8	•• 18.1	37°51.7	•• 42.7	25°41.5	•• 38.1	174°11.4	•• 42.4	Pollux	243°13.3	27°57.2
22	237°25.6	124°54.0	17.6	52°53.2	43.2	40°43.9	38.2	189°13.6	42.5	Avior	234°13.7	-59°36.5
23	252°28.1	139°53.2	17.2	67°54.8	43.6	55°46.3	38.2	204°15.8	42.5	Suhail	222°43.9	-43°33.4
Mer.pass. 06:13		ν -0.8' d -0.4' m -3.89		ν 1.6' d 0.4' m 0.21		ν 2.4' d 0.0' m -2.16		ν 2.2' d 0.1' m 0.32				
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	267°30.5	154°52.4	N23°16.8	82°56.4	S01°44.1	70°48.8	S05°38.2	219°18.0	N15°42.6	Alphard	217°44.4	-8°47.4
1	282°33.0	169°51.6	16.4	97°58.0	44.6	85°51.2	38.2	234°20.2	42.7	Regulus	207°30.8	11°49.3
2	297°35.5	184°50.7	16.0	112°59.5	45.0	100°53.6	38.3	249°22.4	42.8	Dubhe	193°37.0	61°35.7
3	312°37.9	199°49.9	•• 15.6	128°01.1	•• 45.5	115°56.1	•• 38.3	264°24.5	•• 42.8	Denebola	182°21.3	14°24.4
4	327°40.4	214°49.1	15.2	143°02.7	45.9	130°58.5	38.3	279°26.7	42.9	Gienah	175°39.8	-17°42.5
5	342°42.9	229°48.3	14.8	158°04.3	46.4	146°00.9	38.3	294°28.9	43.0	Acrux	172°55.7	-63°16.1
6	357°45.3	244°47.5	N23°14.3	173°05.8	S01°46.8	161°03.3	S05°38.4	309°31.1	N15°43.0	Gacrux	171°47.3	-57°17.1
7	12°47.8	259°46.7	13.9	188°07.4	47.3	176°05.8	38.4	324°33.3	43.1	Alioth	166°09.9	55°48.2
8	27°50.3	274°45.9	13.5	203°09.0	47.7	191°08.2	38.4	339°35.5	43.2	Spica	158°18.4	-11°19.0
9	42°52.7	289°45.1	•• 13.1	218°10.5	•• 48.2	206°10.6	•• 38.4	354°37.7	•• 43.2	Alkaid	152°49.1	-49°10.1
10	57°55.2	304°44.2	12.7	233°12.1	48.6	221°13.1	38.5	9°39.9	43.3	Hadar	148°30.3	-60°31.2
11	72°57.7	319°43.4	12.2	248°13.7	49.1	236°15.5	38.5	24°42.0	43.4	Menkent	147°53.1	-36°31.1
12	88°00.1	334°42.6	N23°11.8	263°15.3	S01°49.5	251°17.9	S05°38.5	39°44.2	N15°43.5	Arcturus	145°44.5	19°01.8
13	103°02.6	349°41.8	11.4	278°16.8	50.0	266°20.4	38.6	54°46.4	43.5	Rigel Kent.	139°34.7	-60°57.6
14	118°05.0	4°41.0	11.0	293°18.4	50.4	281°22.8	38.6	69°48.6	43.6	Kochab	137°19.4	74°02.2
15	133°07.5	19°40.2	•• 10.5	308°20.0	•• 50.9	296°25.2	•• 38.6	84°50.8	•• 43.7	Zuben'ubi	136°51.8	-16°09.9
16	148°10.0	34°39.4	10.1	323°21.5	51.3	311°27.6	38.6	99°53.0	43.7	Alphecca	126°00.4	26°37.0
17	163°12.4	49°38.6	09.7	338°23.1	51.8	326°30.1	38.7	114°55.2	43.8	Antares	112°11.0	-26°29.8
18	178°14.9	64°37.8	N23°09.2	353°24.7	S01°52.2	341°32.5	S05°38.7	129°57.4	N15°43.9	Atria	107°01.4	-69°04.8
19	193°17.4	79°37.0	08.8	8°26.2	52.7	356°34.9	38.7	144°59.5	43.9	Sabik	101°58.3	-15°45.6
20	208°19.8	94°36.2	08.4	23°27.8	53.2	11°37.4	38.7	160°01.7	44.0	Shaula	96°05.1	-37°07.4
21	223°22.3	109°35.4	•• 07.9	38°29.4	•• 53.6	26°39.8	•• 38.8	175°03.9	•• 44.1	Rasalhague	95°54.9	12°32.4
22	238°24.8	124°34.6	07.5	53°30.9	54.1	41°42.2	38.8	190°06.1	44.1	Eltanin	90°40.0	51°29.2
23	253°27.2	139°33.8	07.1	68°32.5	54.5	56°44.6	38.8	205°08.3	44.2	Kaus Aust.	83°27.4	-34°22.0
Mer.pass. 06:09		ν -0.8' d -0.4' m -3.89		ν 1.6' d 0.5' m 0.23		ν 2.4' d 0.0' m -2.15		ν 2.2' d 0.1' m 0.32				
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	268°29.7	154°33.0	N23°06.6	83°34.1	S01°55.0	71°47.1	S05°38.8	220°10.5	N15°44.3	Nunki	75°43.0	-26°15.4
1	283°32.1	169°32.1	06.2	98°35.6	55.4	86°49.5	38.9	235°12.7	44.3	Altair	61°56.2	8°56.9
2	298°34.6	184°31.3	05.8	113°37.2	55.9	101°51.9	38.9	250°14.9	44.4	Peacock	52°59.7	-56°38.0
3	313°37.1	199°30.5	•• 05.3	128°38.7	•• 56.3	116°54.3	•• 38.9	265°17.0	•• 44.5	Deneb	49°22.9	45°23.1
4	328°39.5	214°29.7	04.9	143°40.3	56.8	131°56.8	39.0	280°19.2	44.6	Enif	33°35.1	10°00.7
5	343°42.0	229°28.9	04.4	158°41.9	57.3	146°59.2	39.0	295°21.4	44.6	Al Na'ir	27°28.3	-46°48.7
6	358°44.5	244°28.1	N23°04.0	173°43.4	S01°57.7	162°01.6	S05°39.0	310°23.6	N15°44.7	Fomalhaut	15°10.6	-29°27.7
7	13°46.9	259°27.3	03.5	188°45.0	58.2	177°04.0	39.0	325°25.8	44.8	Scheat	13°41.6	28°14.5
8	28°49.4	274°26.5	03.1	203°46.5	58.6	192°06.5	39.1	340°28.0	44.8	Markab	13°26.2	15°21.9
9	43°51.9	289°25.7	•• 02.6	218°48.1	•• 59.1	207°08.9	•• 39.1	355°30.2	•• 44.9			
10	58°54.3	304°24.9	02.2	233°49.6	01°59.5	222°11.3	39.1	10°32.4	45.0	Jun 18 Mon	SHA	Mer.pass
11	73°56.8	319°24.1	01.7	248°51.2	02°00.0	237°13.7	39.2	25°34.6	45.0	Venus	248°40.6	13:40
12	88°59.3	334°23.3	N23°01.3	263°52.8	S02°00.5	252°16.1	S05°39.2	40°36.7	N15°45.1	Mars	175°47.0	18:29
13	104°01.7	349°22.5	00.8	278°54.3	00.9	267°18.6	39.2	55°38.9	45.2	Jupiter	163°18.9	19:18
14	119°04.2	4°21.7	23°00.4	293°55.9	01.4	282°21.0	39.2	70°41.1	45.2	Saturn	311°54.1	09:25
15	134°06.6	19°20.9	22°59.9	308°57.4	•• 01.8	297°23.4	•• 39.3	85°43.3	•• 45.3			
16	149°09.1	34°20.1	59.5	323°59.0	02.3	312°25.8	39.3	100°45.5	45.4	Jun 19 Tue	SHA	Mer.pass
17	164°11.6	49°19.3	59.0	339°00.5	02.8	327°28.3	39.3	115°47.7	45.4	Venus	247°21.8	13:41
18	179°14.0	64°18.5	N22°58.5	354°02.1	S02°03.2	342°30.7	S05°39.4	130°49.9	N15°45.5	Mars	175°25.8	18:26
19	194°16.5	79°17.8	58.1	9°03.6	03.7	357°33.1	39.4	145°52.1	45.6	Jupiter	163°18.2	19:14
20	209°19.0	94°17.0	57.6	24°05.2	04.1	12°35.5	39.4	160°54.3	45.6	Saturn	311°47.4	09:

DUT1 = UT1-UTC = +0.2106 sec ΔT = TT-UT1 = +68.9734 sec

2029 June 18 to Jun. 20 UT

h		Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°43.3	N23°24.2	108°03.4	11.8'	N03°56.0	-14.1'	58.5'	
1	194°43.2	24.2	122°34.3	11.8'	03°41.9	-14.2'	58.5'	
2	209°43.0	24.3	137°05.1	11.8'	03°27.7	-14.2'	58.5'	
3	224°42.9	· · 24.3	151°35.9	11.8'	03°13.5	-14.2'	58.5'	
4	239°42.8	24.4	166°06.8	11.8'	02°59.3	-14.2'	58.5'	
5	254°42.6	24.4	180°37.6	11.8'	02°45.1	-14.3'	58.6'	
6	269°42.5	N23°24.5	195°08.4	11.8'	N02°30.8	-14.3'	58.6'	
7	284°42.4	24.5	209°39.2	11.8'	02°16.5	-14.3'	58.6'	
8	299°42.2	24.6	224°10.0	11.8'	02°02.2	-14.3'	58.6'	
9	314°42.1	· · 24.6	238°40.8	11.8'	01°47.9	-14.3'	58.6'	
10	329°42.0	24.7	253°11.6	11.8'	01°33.6	-14.4'	58.6'	
11	344°41.8	24.7	267°42.4	11.8'	01°19.2	-14.4'	58.7'	
12	359°41.7	N23°24.8	282°13.1	11.7'	N01°04.9	-14.4'	58.7'	
13	14°41.5	24.8	296°43.9	11.7'	00°50.5	-14.4'	58.7'	
14	29°41.4	24.8	311°14.6	11.7'	00°36.1	-14.4'	58.7'	
15	44°41.3	· · 24.9	325°45.4	11.7'	00°21.7	-14.4'	58.7'	
16	59°41.1	24.9	340°16.1	11.7'	N00°07.3	-14.4'	58.8'	
17	74°41.0	25.0	354°46.7	11.7'	S00°07.1	14.4'	58.8'	
18	89°40.9	N23°25.0	9°17.4	11.7'	S00°21.6	14.4'	58.8'	
19	104°40.7	25.0	23°48.1	11.6'	00°36.0	14.4'	58.8'	
20	119°40.6	25.1	38°18.7	11.6'	00°50.5	14.4'	58.8'	
21	134°40.5	· · 25.1	52°49.3	11.6'	01°04.9	14.4'	58.8'	
22	149°40.3	25.2	67°19.9	11.6'	01°19.3	14.4'	58.9'	
23	164°40.2	25.2	81°50.4	11.5'	01°33.8	14.4'	58.9'	
SD = 15.7'			d = 0.1'	SD = 15.9'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°40.0	N23°25.2	96°21.0	11.5'	S01°48.2	14.4'	58.9'	
1	194°39.9	25.3	110°51.5	11.5'	02°02.7	14.4'	58.9'	
2	209°39.8	25.3	125°22.0	11.5'	02°17.1	14.4'	58.9'	
3	224°39.6	· · 25.3	139°52.4	11.4'	02°31.6	14.4'	58.9'	
4	239°39.5	25.4	154°22.9	11.4'	02°46.0	14.4'	58.9'	
5	254°39.4	25.4	168°53.3	11.4'	03°00.4	14.4'	59.0'	
6	269°39.2	N23°25.4	183°23.6	11.3'	S03°14.8	14.4'	59.0'	
7	284°39.1	25.5	197°54.0	11.3'	03°29.3	14.4'	59.0'	
8	299°39.0	25.5	212°24.3	11.3'	03°43.7	14.4'	59.0'	
9	314°38.8	· · 25.5	226°54.5	11.2'	03°58.0	14.4'	59.0'	
10	329°38.7	25.6	241°24.8	11.2'	04°12.4	14.4'	59.0'	
11	344°38.5	25.6	255°55.0	11.2'	04°26.8	14.3'	59.1'	
12	359°38.4	N23°25.6	270°25.1	11.1'	S04°41.1	14.3'	59.1'	
13	14°38.3	25.6	284°55.2	11.1'	04°55.4	14.3'	59.1'	
14	29°38.1	25.7	299°25.3	11.0'	05°09.7	14.3'	59.1'	
15	44°38.0	· · 25.7	313°55.3	11.0'	05°24.0	14.3'	59.1'	
16	59°37.9	25.7	328°25.3	11.0'	05°38.3	14.2'	59.1'	
17	74°37.7	25.7	342°55.3	10.9'	05°52.5	14.2'	59.1'	
18	89°37.6	N23°25.8	357°25.2	10.9'	S06°06.8	14.2'	59.2'	
19	104°37.5	25.8	11°55.1	10.8'	06°21.0	14.2'	59.2'	
20	119°37.3	25.8	26°24.9	10.8'	06°35.1	14.1'	59.2'	
21	134°37.2	· · 25.8	40°54.6	10.7'	06°49.3	14.1'	59.2'	
22	149°37.0	25.8	55°24.4	10.7'	07°03.4	14.1'	59.2'	
23	164°36.9	25.9	69°54.0	10.6'	07°17.4	14.0'	59.2'	
SD = 15.7'			d = 0.0'	SD = 16.1'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°36.8	N23°25.9	84°23.7	10.6'	S07°31.5	14.0'	59.2'	
1	194°36.6	25.9	98°53.2	10.5'	07°45.5	14.0'	59.3'	
2	209°36.5	25.9	113°22.7	10.5'	07°59.5	13.9'	59.3'	
3	224°36.4	· · 25.9	127°52.2	10.4'	08°13.4	13.9'	59.3'	
4	239°36.2	26.0	142°21.6	10.4'	08°27.3	13.9'	59.3'	
5	254°36.1	26.0	156°51.0	10.3'	08°41.2	13.8'	59.3'	
6	269°36.0	N23°26.0	171°20.3	10.2'	S08°55.0	13.8'	59.3'	
7	284°35.8	26.0	185°49.5	10.2'	09°08.8	13.7'	59.3'	
8	299°35.7	26.0	200°18.7	10.1'	09°22.5	13.7'	59.4'	
9	314°35.6	· · 26.0	214°47.8	10.1'	09°36.2	13.6'	59.4'	
10	329°35.4	26.0	229°16.9	10.0'	09°49.8	13.6'	59.4'	
11	344°35.3	26.0	243°45.9	9.9'	10°03.4	13.5'	59.4'	
12	359°35.1	N23°26.1	258°14.8	9.9'	S10°16.9	13.5'	59.4'	
13	14°35.0	26.1	272°43.7	9.8'	10°30.4	13.4'	59.4'	
14	29°34.9	26.1	287°12.5	9.8'	10°43.9	13.4'	59.4'	
15	44°34.7	· · 26.1	301°41.3	9.7'	10°57.2	13.3'	59.4'	
16	59°34.6	26.1	316°10.0	9.6'	11°10.6	13.3'	59.4'	
17	74°34.5	26.1	330°38.6	9.6'	11°23.8	13.2'	59.5'	
18	89°34.3	N23°26.1	345°07.2	9.5'	S11°37.0	13.1'	59.5'	
19	104°34.2	26.1	359°35.7	9.4'	11°50.2	13.1'	59.5'	
20	119°34.1	26.1	14°04.1	9.4'	12°03.3	13.0'	59.5'	
21	134°33.9	· · 26.1	28°32.4	9.3'	12°16.3	13.0'	59.5'	
22	149°33.8	26.1	43°00.7	9.2'	12°29.2	12.9'	59.5'	
23	164°33.7	26.1	57°28.9	9.1'	12°42.1	12.8'	59.5'	
SD = 15.7'			d = 0.0'	SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	—:—	—:—	01:31	22:32	—:—	—:—
62°	—:—	—:—	02:09	21:54	—:—	—:—
60°	—:—	00:49	02:36	21:27	23:14	—:—
N 58°	—:—	01:40	02:56	21:07	22:23	—:—
56°	—:—	02:10	03:13	20:50	21:53	—:—
54°	00:45	02:33	03:27	20:36	21:30	23:18
52°	01:32	02:51	03:40	20:23	21:12	22:31
50°	02:00	03:06	03:50	20:13	20:57	22:03
45°	02:46	03:36	04:13	19:50	20:27	21:17
N 40°	03:16	03:58	04:31	19:32	20:05	20:47
35°	03:40	04:16	04:46	19:17	19:47	20:23
30°	03:58	04:32	04:59	19:04	19:31	20:05
20°	04:27	04:57	05:21	18:42	19:06	19:36
N 10°	04:50	05:17	05:40	18:23	18:46	19:13
0°	05:09	05:35	05:58	18:05	18:28	18:54
S 10°	05:26	05:52	06:15	17:48	18:10	18:37
20°	05:42	06:10	06:34	17:29	17:53	18:21
30°	05:59	06:29	06:55	17:08	17:34	18:04
35°	06:08	06:39	07:07	16:55	17:24	17:55
40°	06:17	06:51	07:22	16:41	17:12	17:46
45°	06:27	07:05	07:38	16:24	16:58	17:36
S 50°	06:39	07:21	07:59	16:04	16:42	17:24
52°	06:44	07:28	08:09	15:54	16:35	17:18
54°	06:50	07:36	08:20	15:43	16:26	17:13
56°	06:56	07:45	08:33	15:30	16:17	17:07
58°	07:03	07:56	08:48	15:15	16:07	17:00
S 60°	07:11	08:07	09:05	14:58	15:55	16:52

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	10:54	13:04	15:24	23:11	22:46	22:13
N 70°	10:56	12:57	15:04	23:14	22:56	22:35
68°	10:57	12:50	14:49	23:15	23:05	22:52
66°	10:59	12:45	14:36	23:17	23:12	23:07
64°	11:00	12:41	14:26	23:18	23:18	23:18
62°	11:00	12:37	14:17	23:20	23:23	23:28
60°	11:01	12:34	14:10	23:21	23:28	23:37
N 58°	11:02	12:31	14:03	23:22	23:32	23:45
56°	11:03	12:29	13:57	23:23	23:36	23:51
54°	11:03	12:27	13:52	23:23	23:39	23:57
52°	11:04	12:25	13:48	23:24	23:42	· · ·
50°	11:04	12:23	13:43	23:25	23:45	· · ·
45°	11:05	12:19	13:34	23:26	23:51	· · ·
N 40°	11:06	12:16	13:27	23:27	23:56	· · ·
35°	11:06	12:13	13:21	23:28	· · ·	00:00
30°	11:07	12:10	13:15	23:29	· · ·	00:04
20°	11:08	12:06	13:05	23:31	· · ·	00:11
N 10°	11:09	12:02	12:57	23:32	· · ·	00:17
0°	11:10	11:59	12:49	23:33	· · ·	00:23
S 10°	11:11	11:55	12:42	23:34	· · ·	00:28
20°	11:12	11:52	12:34	23:36	· · ·	00:34
30°	11:13	11:48	12:24	23:37	· · ·	00:41
35°	11:14	11:46	12:19	23:38	· · ·	00:45
40°	11:15	11:43	12:13	23:39	· · ·	00:49
45°	11:16	11:40	12:06	23:40	· · ·	00:55
S 50°	11:17	11:37	11:58	23:41	· · ·	01:01
52°	11:17	11:35	11:54	23:42	· · ·	01:04
54°	11:18	11:33	11:50	23:43	· · ·	01:07
56°	11:18	11:31	11:46	23:43	· · ·	01:11
58°	11:19	11:29	11:41	23:44	· · ·	01:15
S 60°	11:20	11:27	11:35	23:45	· · ·	01:19

Day	Sun			Moon		
	Eqn.of Time	Mer.	Mer.Pass.	Upper	Lower	Age
	00 ^h mm:ss	12 ^h mm:ss	Pass hh:mm	hh:mm	hh:mm	6-8 34-57%
18	01:07	01:13	12:01	17:22	04:57	
19	01:20	01:26	12:01	18:11	05:46	
20	01:33	01:39	12:02	19:02	06:36	

June 21, 22, 23 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	269°28.8	154°13.8	N22°55.8	84°11.4	S02°06.0	72°45.2	S05°39.6	221°03.0	N15°45.9	Alpheratz 357°31.0 29°15.1 Ankaa 353°03.8 -42°08.4 Schedar 349°26.8 56°41.7 Diphda 348°43.8 -17°49.3 Achernar 335°17.9 -57°04.9 Hamal 327°47.3 23°36.0 Polaris 312°24.7 89°22.9 Acamar 315°09.4 -40°11.0 Menkar 314°02.6 4°12.3 Mirfak 308°23.4 49°57.7 Aldebaran 290°35.8 16°34.0 Rigel 281°00.8 -8°10.2 Capella 280°17.0 46°01.5 Bellatrix 278°19.4 6°22.5 Elnath 277°57.7 28°37.8 Alnilam 275°34.4 -1°11.1 Betelgeuse 270°48.5 7°24.6 Canopus 263°51.3 -52°42.8 Sirius 258°23.4 -16°45.6 Adhara 255°03.4 -29°00.9 Procyon 244°47.4 5°08.8 Pollux 243°13.3 27°57.2 Avior 234°13.7 -59°36.5 Suhail 222°43.9 -43°33.4 Miaplacidus 221°37.9 -69°50.6 Alphard 217°44.4 -8°47.4 Regulus 207°30.8 11°49.3 Dubhe 193°37.1 61°35.7 Denebola 182°21.3 14°24.4 Gienah 175°39.8 -17°42.5 Acrux 172°55.7 -63°16.1 Gacrux 171°47.3 -57°17.1 Alioth 166°09.9 55°48.2 Spica 158°18.4 -11°19.0 Alkaid 152°49.1 49°10.1 Hadar 148°30.3 -60°31.2 Menkent 147°53.1 -36°31.1 Arcturus 145°44.5 19°01.8 Rigel Kent. 139°34.7 -60°57.6 Kochab 137°19.4 74°02.3 Zuben'ubi 136°51.8 -16°09.9 Alphecca 126°00.4 26°37.0 Antares 112°11.0 -26°29.8 Atria 107°01.4 -69°04.8 Sabik 101°58.3 -15°45.6 Shaula 96°05.1 -37°07.4 Rasalhague 95°54.9 12°32.4 Eltanin 90°40.0 51°29.2 Kaus Aust. 83°27.4 -34°22.0 Vega 80°30.3 38°48.7 Nunki 75°43.0 -26°15.4 Altair 61°56.1 8°56.9 Peacock 52°59.7 -56°38.0 Deneb 49°22.9 45°23.1 Enif 33°35.1 10°00.7 Al Na'ir 27°28.3 -46°48.7 Fomalhaut 15°10.5 -29°27.7 Scheat 13°41.6 28°14.5 Markab 13°26.2 15°21.9 Jun 21 ThuSHAMer.pass Venus244°45.013:44 Mars174°42.618:21 Jupiter163°16.419:06 Saturn311°34.209:14 Jun 22 FriSHAMer.pass Venus243°26.913:45 Mars174°20.518:19 Jupiter163°15.219:02 Saturn311°27.609:11 Jun 23 SatSHAMer.pass Venus242°09.113:46 Mars173°58.018:16 Jupiter163°13.918:58 Saturn311°21.109:07 Horizontal parallax Venus:0.1 Mars:0.1							
1	284°31.3	169°13.0	55.3	99°12.9	06.4	87°47.6	39.6	236°05.2	46.0								
2	299°33.8	184°12.2	54.8	114°14.5	06.9	102°50.0	39.6	251°07.4	46.1								
3	314°36.2	199°11.4	•• 54.4	129°16.0	•• 07.4	117°52.5	•• 39.6	266°09.6	•• 46.1								
4	329°38.7	214°10.6	53.9	144°17.6	07.8	132°54.9	39.7	281°11.8	46.2								
5	344°41.1	229°09.8	53.4	159°19.1	08.3	147°57.3	39.7	296°14.0	46.3								
6	359°43.6	244°09.0	N22°53.0	174°20.7	S02°08.7	162°59.7	S05°39.7	311°16.2	N15°46.3								
7	14°46.1	259°08.2	52.5	189°22.2	09.2	178°02.1	39.8	326°18.3	46.4								
8	29°48.5	274°07.4	52.0	204°23.8	09.7	193°04.5	39.8	341°20.5	46.5								
9	44°51.0	289°06.6	•• 51.5	219°25.3	•• 10.1	208°07.0	•• 39.8	356°22.7	•• 46.5								
10	59°53.5	304°05.9	51.1	234°26.9	10.6	223°09.4	39.9	11°24.9	46.6								
11	74°55.9	319°05.1	50.6	249°28.4	11.0	238°11.8	39.9	26°27.1	46.7								
12	89°58.4	334°04.3	N22°50.1	264°29.9	S02°11.5	253°14.2	S05°39.9	41°29.3	N15°46.7								
13	105°00.9	349°03.5	49.6	279°31.5	12.0	268°16.6	40.0	56°31.5	46.8								
14	120°03.3	4°02.7	49.1	294°33.0	12.4	283°19.0	40.0	71°33.7	46.9								
15	135°05.8	19°01.9	•• 48.6	309°34.6	•• 12.9	298°21.4	•• 40.0	86°35.9	•• 46.9								
16	150°08.2	34°01.1	48.2	324°36.1	13.4	313°23.9	40.1	101°38.1	47.0								
17	165°10.7	49°00.3	47.7	339°37.7	13.8	328°26.3	40.1	116°40.3	47.1								
18	180°13.2	63°59.6	N22°47.2	354°39.2	S02°14.3	343°28.7	S05°40.1	131°42.4	N15°47.1								
19	195°15.6	78°58.8	46.7	9°40.7	14.7	358°31.1	40.2	146°44.6	47.2								
20	210°18.1	93°58.0	46.2	24°42.3	15.2	13°33.5	40.2	161°46.8	47.3								
21	225°20.6	108°57.2	•• 45.7	39°43.8	•• 15.7	28°35.9	•• 40.2	176°49.0	•• 47.3								
22	240°23.0	123°56.4	45.2	54°45.3	16.1	43°38.3	40.3	191°51.2	47.4								
23	255°25.5	138°55.6	44.7	69°46.9	16.6	58°40.8	40.3	206°53.4	47.5								
Mer.pass. 06:01		ν -0.8' d -0.5' m -3.89			ν 1.5' d 0.5' m 0.25			ν 2.4' d 0.0' m -2.14			ν 2.2' d 0.1' m 0.32						
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	270°28.0	153°54.8	N22°44.3	84°48.4	S02°17.1	73°43.2	S05°40.3	221°55.6	N15°47.5								
1	285°30.4	168°54.1	43.8	99°50.0	17.5	88°45.6	40.4	236°57.8	47.6								
2	300°32.9	183°53.3	43.3	114°51.5	18.0	103°48.0	40.4	252°00.0	47.7								
3	315°35.4	198°52.5	•• 42.8	129°53.0	•• 18.5	118°50.4	•• 40.4	267°02.2	•• 47.7								
4	330°37.8	213°51.7	42.3	144°54.6	18.9	133°52.8	40.5	282°04.4	47.8								
5	345°40.3	228°50.9	41.8	159°56.1	19.4	148°55.2	40.5	297°06.6	47.9								
6	0°42.7	243°50.2	N22°41.3	174°57.6	S02°19.9	163°57.6	S05°40.5	312°08.8	N15°47.9								
7	15°45.2	258°49.4	40.8	189°59.2	20.3	179°00.0	40.6	327°10.9	48.0								
8	30°47.7	273°48.6	40.3	205°00.7	20.8	194°02.5	40.6	342°13.1	48.1								
9	45°50.1	288°47.8	•• 39.8	220°02.2	•• 21.3	209°04.9	•• 40.6	357°15.3	•• 48.1								
10	60°52.6	303°47.0	39.3	235°03.8	21.7	224°07.3	40.7	12°17.5	48.2								
11	75°55.1	318°46.3	38.8	250°05.3	22.2	239°09.7	40.7	27°19.7	48.3								
12	90°57.5	333°45.5	N22°38.3	265°06.8	S02°22.7	254°12.1	S05°40.7	42°21.9	N15°48.3								
13	106°00.0	348°44.7	37.7	280°08.3	23.1	269°14.5	40.8	57°24.1	48.4								
14	121°02.5	3°43.9	37.2	295°09.9	23.6	284°16.9	40.8	72°26.3	48.5								
15	136°04.9	18°43.2	•• 36.7	310°11.4	•• 24.1	299°19.3	•• 40.8	87°28.5	•• 48.5								
16	151°07.4	33°42.4	36.2	325°12.9	24.5	314°21.7	40.9	102°30.7	48.6								
17	166°09.9	48°41.6	35.7	340°14.5	25.0	329°24.1	40.9	117°32.9	48.7								
18	181°12.3	63°40.8	N22°35.2	355°16.0	S02°25.5	344°26.5	S05°40.9	132°35.1	N15°48.7								
19	196°14.8	78°40.1	34.7	10°17.5	25.9	359°28.9	41.0	147°37.3	48.8								
20	211°17.2	93°39.3	34.2	25°19.0	26.4	14°31.4	41.0	162°39.5	48.9								
21	226°19.7	108°38.5	•• 33.7	40°20.6	•• 26.9	29°33.8	•• 41.0	177°41.6	•• 48.9								
22	241°22.2	123°37.7	33.1	55°22.1	27.3	44°36.2	41.1	192°43.8	49.0								
23	256°24.6	138°37.0	32.6	70°23.6	27.8	59°38.6	41.1	207°46.0	49.1								
Mer.pass. 05:57		ν -0.8' d -0.5' m -3.89			ν 1.5' d 0.5' m 0.27			ν 2.4' d 0.0' m -2.13			ν 2.2' d 0.1' m 0.32						
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	271°27.1	153°36.2	N22°32.1	85°25.1	S02°28.3	74°41.0	S05°41.2	222°48.2	N15°49.1								
1	286°29.6	168°35.4	31.6	100°26.7	28.7	89°43.4	41.2	237°50.4	49.2								
2	301°32.0	183°34.6	31.1	115°28.2	29.2	104°45.8	41.2	252°52.6	49.3								
3	316°34.5	198°33.9	•• 30.5	130°29.7	•• 29.7	119°48.2	•• 41.3	267°54.8	•• 49.3								
4	331°37.0	213°33.1	30.0	145°31.2	30.2	134°50.6	41.3	282°57.0	49.4								
5	346°39.4	228°32.3	29.5	160°32.7	30.6	149°53.0	41.3	297°59.2	49.5								
6	1°41.9	243°31.6	N22°29.0	175°34.3	S02°31.1	164°55.4	S05°41.4	313°01.4	N15°49.5								
7	16°44.4	258°30.8	28.4	190°35.8	31.6	179°57.8	41.4	328°03.6	49.6								
8	31°46.8	273°30.0	27.9	205°37.3	32.0	195°00.2	41.4	343°05.8	49.7								
9	46°49.3	288°29.3	•• 27.4	220°38.8	•• 32.5	210°02.6	•• 41.5	358°08.0	•• 49.7								
10	61°51.7	303°28.5	26.8	235°40.3	33.0	225°05.0	41.5	13°10.2	49.8								
11	76°54.2	318°27.7	26.3	250°41.9	33.4	240°07.4	41.6	28°12.4	49.9								
12	91°56.7	333°26.9	N22°25.8	265°43.4	S02°33.9	255°09.8	S05°41.6	43°14.6	N15°49.9								
13	106°59.1	348°26.2	25.2	280°44.9	34.4	270°12.2	41.6	58°16.8	50.0								
14	122°01.6	3°25.4	24.7	295°46.4	34.9	285°14.6	41.7	73°19.0	50.1								
15	137°04.1	18°24.7	•• 24.2	310°47.9	•• 35.3	300°17.0	•• 41.7	88°21.2	•• 50.1								
16	152°06.5	33°23.9	23.6	325°49.4	35.8	315°19.4	41.7	103°23.3	50.2								
17	167°09.0	48°23.1	23.1	340°51.0	36.3	330°21.8	41.8	118°25.5	50.3								
18	182°11.5	63°22.4	N22°22.6	355°52.5	S02°36.7	345°24.2	S05°41.8	133°27.7	N15°50.3								
19	197°13.9	78°21.6	22.0	10°54.0	37.2	0°26.6	41.9	148°29.9	50.4								
20	212°16.4	93°20.8	21.5	25°55.5	37.7	15°29.0	41.9	163°32.1	50.5								
21	227°18.8	108°20.1	•• 20.9	40°57.0	•• 38.2	30°31.4	•• 41.9	178°34.3	•• 50.5								
22	242°21.3	123°19.3	20.4	55°58.5	38.6	45°33.8	42.0	193°36.5	50.6								
23	257°23.8	138°18.5	19.8	71°00.0	39.1	60°36.2	42.0	208°38.7	50.7								
Mer.pass. 05:53		ν -0.8' d -0.5' m -3.89			ν 1.5' d 0.5' m 0.28			ν 2.4' d 0.0' m -2.12			ν 2.2' d 0.1' m 0.32						

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°33.5	N23°26.1	71°57.1	9.1'	S12°54.9	12.7'	59.5'
1	194°33.4	26.1	86°25.1	9.0'	13°07.7	12.7'	59.5'
2	209°33.2	26.1	100°53.1	8.9'	13°20.4	12.6'	59.5'
3	224°33.1	· · 26.1	115°21.1	8.9'	13°33.0	12.5'	59.6'
4	239°33.0	26.1	129°48.9	8.8'	13°45.5	12.5'	59.6'
5	254°32.8	26.1	144°16.7	8.7'	13°57.9	12.4'	59.6'
6	269°32.7	N23°26.1	158°44.4	8.6'	S14°10.3	12.3'	59.6'
7	284°32.6	26.1	173°12.0	8.6'	14°22.6	12.2'	59.6'
8	299°32.4	26.1	187°39.6	8.5'	14°34.8	12.1'	59.6'
9	314°32.3	· · 26.1	202°07.1	8.4'	14°47.0	12.0'	59.6'
10	329°32.2	26.1	216°34.5	8.3'	14°59.0	12.0'	59.6'
11	344°32.0	26.1	231°01.8	8.3'	15°11.0	11.9'	59.6'
12	359°31.9	N23°26.1	245°29.1	8.2'	S15°22.8	11.8'	59.6'
13	14°31.8	26.1	259°56.2	8.1'	15°34.6	11.7'	59.6'
14	29°31.6	26.1	274°23.3	8.0'	15°46.3	11.6'	59.6'
15	44°31.5	· · 26.1	288°50.4	7.9'	15°57.9	11.5'	59.7'
16	59°31.3	26.1	303°17.3	7.9'	16°09.4	11.4'	59.7'
17	74°31.2	26.0	317°44.2	7.8'	16°20.8	11.3'	59.7'
18	89°31.1	N23°26.0	332°10.9	7.7'	S16°32.2	11.2'	59.7'
19	104°30.9	26.0	346°37.7	7.6'	16°43.4	11.1'	59.7'
20	119°30.8	26.0	1°04.3	7.5'	16°54.5	11.0'	59.7'
21	134°30.7	· · 26.0	15°30.8	7.5'	17°05.5	10.9'	59.7'
22	149°30.5	26.0	29°57.3	7.4'	17°16.4	10.8'	59.7'
23	164°30.4	26.0	44°23.7	7.3'	17°27.2	10.7'	59.7'
SD = 15.7'			SD = 16.2'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°30.3	N23°25.9	58°50.0	7.2'	S17°37.9	10.6'	59.7'
1	194°30.1	25.9	73°16.2	7.2'	17°48.5	10.5'	59.7'
2	209°30.0	25.9	87°42.4	7.1'	17°59.0	10.4'	59.7'
3	224°29.9	· · 25.9	102°08.5	7.0'	18°09.4	10.3'	59.7'
4	239°29.7	25.9	116°34.5	6.9'	18°19.6	10.1'	59.7'
5	254°29.6	25.9	131°00.4	6.8'	18°29.8	10.0'	59.7'
6	269°29.5	N23°25.8	145°25.2	6.8'	S18°39.8	9.9'	59.7'
7	284°29.3	25.8	159°52.0	6.7'	18°49.7	9.8'	59.7'
8	299°29.2	25.8	174°17.7	6.6'	18°59.5	9.7'	59.7'
9	314°29.1	· · 25.8	188°43.3	6.5'	19°09.2	9.6'	59.7'
10	329°28.9	25.8	203°08.8	6.5'	19°18.8	9.4'	59.7'
11	344°28.8	25.7	217°34.3	6.4'	19°28.2	9.3'	59.7'
12	359°28.6	N23°25.7	231°59.7	6.3'	S19°37.5	9.2'	59.7'
13	14°28.5	25.7	246°25.0	6.2'	19°46.7	9.1'	59.7'
14	29°28.4	25.7	260°50.2	6.2'	19°55.8	8.9'	59.7'
15	44°28.2	· · 25.6	275°15.4	6.1'	20°04.7	8.8'	59.7'
16	59°28.1	25.6	289°40.4	6.0'	20°13.5	8.7'	59.7'
17	74°28.0	25.6	304°05.5	5.9'	20°22.1	8.5'	59.7'
18	89°27.8	N23°25.5	318°30.4	5.9'	S20°30.7	8.4'	59.7'
19	104°27.7	25.5	332°55.3	5.8'	20°39.1	8.3'	59.7'
20	119°27.6	25.5	347°20.1	5.7'	20°47.3	8.1'	59.7'
21	134°27.4	· · 25.5	1°44.8	5.7'	20°55.5	8.0'	59.7'
22	149°27.3	25.4	16°09.5	5.6'	21°03.4	7.8'	59.7'
23	164°27.2	25.4	30°34.1	5.5'	21°11.3	7.7'	59.7'
SD = 15.7'			SD = 16.3'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°27.0	N23°25.4	44°58.6	5.5'	S21°19.0	7.6'	59.7'
1	194°26.9	25.3	59°23.0	5.4'	21°26.6	7.4'	59.7'
2	209°26.8	25.3	73°47.4	5.3'	21°34.0	7.3'	59.7'
3	224°26.6	· · 25.3	88°11.8	5.3'	21°41.3	7.1'	59.7'
4	239°26.5	25.2	102°36.1	5.2'	21°48.4	7.0'	59.7'
5	254°26.4	25.2	117°00.3	5.2'	21°55.4	6.8'	59.7'
6	269°26.2	N23°25.1	131°24.4	5.1'	S22°02.2	6.7'	59.7'
7	284°26.1	25.1	145°48.5	5.0'	22°08.9	6.5'	59.7'
8	299°26.0	25.1	160°12.6	5.0'	22°15.4	6.4'	59.7'
9	314°25.8	· · 25.0	174°36.6	4.9'	22°21.8	6.2'	59.7'
10	329°25.7	25.0	189°00.5	4.9'	22°28.0	6.1'	59.7'
11	344°25.6	25.0	203°24.4	4.8'	22°34.1	5.9'	59.7'
12	359°25.4	N23°24.9	217°48.2	4.8'	S22°40.0	5.8'	59.7'
13	14°25.3	24.9	232°12.0	4.7'	22°45.8	5.6'	59.7'
14	29°25.2	24.8	246°35.7	4.7'	22°51.4	5.5'	59.7'
15	44°25.0	· · 24.8	260°59.4	4.6'	22°56.9	5.3'	59.7'
16	59°24.9	24.7	275°23.0	4.6'	23°02.2	5.1'	59.6'
17	74°24.8	24.7	289°46.6	4.6'	23°07.3	5.0'	59.6'
18	89°24.6	N23°24.6	304°10.2	4.5'	S23°12.3	4.8'	59.6'
19	104°24.5	24.6	318°33.7	4.5'	23°17.1	4.7'	59.6'
20	119°24.4	24.6	332°57.2	4.4'	23°21.8	4.5'	59.6'
21	134°24.2	· · 24.5	347°20.6	4.4'	23°26.2	4.3'	59.6'
22	149°24.1	24.5	1°44.0	4.4'	23°30.6	4.2'	59.6'
23	164°24.0	24.4	16°07.4	4.3'	23°34.7	4.0'	59.6'
SD = 15.7'			SD = 16.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	--	--	01:32	22:32	--	--
62°	--	--	02:10	21:54	--	--
60°	--	00:50	02:36	21:28	23:14	--
N 58°	--	01:41	02:57	21:07	22:23	--
56°	--	02:11	03:13	20:51	21:53	--
54°	00:46	02:33	03:28	20:36	21:31	23:18
52°	01:33	02:51	03:40	20:24	21:13	22:31
50°	02:01	03:06	03:51	20:13	20:58	22:03
45°	02:46	03:36	04:14	19:51	20:28	21:18
N 40°	03:17	03:59	04:32	19:33	20:05	20:47
35°	03:40	04:17	04:47	19:17	19:47	20:24
30°	03:59	04:32	05:00	19:04	19:32	20:05
20°	04:28	04:57	05:22	18:42	19:07	19:36
N 10°	04:51	05:18	05:41	18:23	18:46	19:14
0°	05:10	05:36	05:58	18:06	18:28	18:55
S 10°	05:27	05:53	06:16	17:48	18:11	18:37
20°	05:43	06:10	06:34	17:30	17:54	18:21
30°	05:59	06:29	06:56	17:08	17:35	18:05
35°	06:08	06:40	07:08	16:56	17:24	17:56
40°	06:18	06:52	07:22	16:42	17:12	17:47
45°	06:28	07:05	07:39	16:25	16:59	17:36
S 50°	06:40	07:21	08:00	16:04	16:43	17:24
52°	06:45	07:29	08:10	15:54	16:35	17:19
54°	06:51	07:37	08:21	15:43	16:27	17:13
56°	06:57	07:46	08:34	15:31	16:18	17:07
58°	07:04	07:56	08:48	15:16	16:08	17:00
S 60°	07:11	08:08	09:06	14:58	15:56	16:53

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	18:24			21:06		
N 70°	17:30			22:02		
68°	16:58	19:33		22:36	22:00	
66°	16:34	18:41	21:06	23:00	22:52	22:33
64°	16:16	18:09	20:01	23:20	23:25	23:39
62°	16:01	17:46	19:26	23:36	23:49	· ·
60°	15:48	17:27	19:01	23:49	· ·	00:08
N 58°	15:37	17:12	18:41	· ·	00:01	00:24
56°	15:28	16:59	18:24	· ·	00:11	00:38
54°	15:20	16:47	18:10	· ·	00:20	00:50
52°	15:12	16:37	17:58	00:03	00:28	01:00
50°	15:06	16:28	17:47	00:08	00:35	01:10
45°	14:52	16:09	17:24	00:19	00:51	01:30
N 40°	14:40	15:54	17:06	00:28	01:04	01:46
35°	14:30	15:41	16:51	00:35	01:15	02:00
30°	14:22	15:30	16:38	00:42	01:24	02:12
20°	14:07	15:11	16:15	00:54	01:41	02:32
N 10°	13:54	14:54	15:56	01:04	01:55	02:50
0°	13:42	14:39	15:38	01:14	02:09	03:07
S 10°	13:31	14:24	15:20	01:24	02:23	03:24
20°	13:18	14:07	15:01	01:35	02:37	03:42
30°	13:04	13:49	14:40	01:47	02:54	04:02
35°	12:56	13:38	14:27	01:54	03:04	04:14
40°	12:47	13:26	14:12	02:02	03:15	04:28
45°	12:36	13:11	13:55	02:11	03:28	04:45
S 50°	12:23	12:54	13:33	02:22	03:45	05:06
52°	12:17	12:46	13:23	02:28	03:52	05:16
54°	12:10	12:36	13:12	02:33	04:01	05:27
56°	12:03	12:26	12:59	02:40	04:11	05:39
58°	11:55	12:15	12:44	02:47	04:22	05:54
S 60°	11:46	12:01	12:26	02:56	04:34	06:11

Day	Sun		Mer. Pass	Moon		Age 9-11 68-87%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer. Pass. hh:mm	Lower Mer. Pass. hh:mm	
21	01:46	01:52	12:02	19:56	07:28	
22	01:59	02:05	12:02	20:53	08:24	
23	02:12	02:18	12:02	21:53	09:22	

June 24, 25, 26 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	272°26.2	153°17.8	N22°19.3	86°01.6	S02°39.6	75°38.6	S05°42.1	223°40.9	N15°50.7	Alpheratz	357°31.0	29°15.1
1	287°28.7	168°17.0	18.7	101°03.1	40.1	90°41.0	42.1	238°43.1	50.8	Ankaa	353°03.7	-42°08.4
2	302°31.2	183°16.3	18.2	116°04.6	40.5	105°43.4	42.1	253°45.3	50.9	Schedar	349°26.8	56°41.7
3	317°33.6	198°15.5	•• 17.7	131°06.1	•• 41.0	120°45.8	•• 42.2	268°47.5	•• 50.9	Diphda	348°43.8	-17°49.3
4	332°36.1	213°14.7	17.1	146°07.6	41.5	135°48.2	42.2	283°49.7	51.0	Achernar	335°17.9	-57°04.9
5	347°38.6	228°14.0	16.6	161°09.1	41.9	150°50.6	42.3	298°51.9	51.0	Hamal	327°47.2	23°36.0
6	2°41.0	243°13.2	N22°16.0	176°10.6	S02°42.4	165°53.0	S05°42.3	313°54.1	N15°51.1	Polaris	312°23.1	89°22.9
7	17°43.5	258°12.5	15.4	191°12.1	42.9	180°55.4	42.3	328°56.3	51.2	Acamar	315°09.4	-40°11.0
8	32°46.0	273°11.7	14.9	206°13.6	43.4	195°57.8	42.4	343°58.5	51.2	Menkar	314°02.6	4°12.3
9	47°48.4	288°11.0	•• 14.3	221°15.1	•• 43.8	211°00.2	•• 42.4	359°00.7	•• 51.3	Mirfak	308°23.4	49°57.7
10	62°50.9	303°10.2	13.8	236°16.6	44.3	226°02.6	42.5	14°02.9	51.4	Aldebaran	290°35.8	16°34.0
11	77°53.3	318°09.4	13.2	251°18.1	44.8	241°05.0	42.5	29°05.1	51.4	Rigel	281°00.7	-8°10.1
12	92°55.8	333°08.7	N22°12.7	266°19.6	S02°45.3	256°07.4	S05°42.5	44°07.3	N15°51.5	Capella	280°17.0	46°01.4
13	107°58.3	348°07.9	12.1	281°21.1	45.7	271°09.8	42.6	59°09.5	51.6	Bellatrix	278°19.3	6°22.5
14	123°00.7	3°07.2	11.5	296°22.7	46.2	286°12.2	42.6	74°11.7	51.6	Elnath	277°57.7	28°37.8
15	138°03.2	18°06.4	•• 11.0	311°24.2	•• 46.7	301°14.6	•• 42.7	89°13.9	•• 51.7	Alnilam	275°34.4	-1°11.1
16	153°05.7	33°05.7	10.4	326°25.7	47.2	316°17.0	42.7	104°16.1	51.8	Betelgeuse	270°48.5	7°24.6
17	168°08.1	48°04.9	09.9	341°27.2	47.6	331°19.4	42.7	119°18.3	51.8	Canopus	263°51.3	-52°42.8
18	183°10.6	63°04.2	N22°09.3	356°28.7	S02°48.1	346°21.8	S05°42.8	134°20.5	N15°51.9	Sirius	258°23.4	-16°45.5
19	198°13.1	78°03.4	08.7	11°30.2	48.6	1°24.1	42.8	149°22.7	52.0	Adhara	255°03.4	-29°00.9
20	213°15.5	93°02.7	08.2	26°31.7	49.1	16°26.5	42.9	164°24.9	52.0	Procyon	244°47.4	5°08.8
21	228°18.0	108°01.9	•• 07.6	41°33.2	•• 49.6	31°28.9	•• 42.9	179°27.0	•• 52.1	Pollux	243°13.3	27°57.2
22	243°20.5	123°01.2	07.0	56°34.7	50.0	46°31.3	42.9	194°29.2	52.2	Avior	234°13.7	-59°36.5
23	258°22.9	138°00.4	06.4	71°36.2	50.5	61°33.7	43.0	209°31.4	52.2	Suhail	222°43.9	-43°33.4
Mer.pass. 05:49		ν -0.8' d -0.5' m -3.89		ν 1.5' d 0.5' m 0.29		ν 2.4' d 0.0' m -2.12		ν 2.2' d 0.1' m 0.32				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	273°25.4	152°59.7	N22°05.9	86°37.7	S02°51.0	76°36.1	S05°43.0	224°33.6	N15°52.3	Alphard	217°44.4	-8°47.4
1	288°27.8	167°58.9	05.3	101°39.2	51.5	91°38.5	43.1	239°35.8	52.3	Regulus	207°30.8	11°49.3
2	303°30.3	182°58.2	04.7	116°40.7	51.9	106°40.9	43.1	254°38.0	52.4	Dubhe	193°37.1	61°35.7
3	318°32.8	197°57.4	•• 04.1	131°42.2	•• 52.4	121°43.3	•• 43.2	269°40.2	•• 52.5	Denebola	182°21.4	14°24.4
4	333°35.2	212°56.7	03.6	146°43.7	52.9	136°45.7	43.2	284°42.4	52.5	Gienah	175°39.8	-17°42.5
5	348°37.7	227°55.9	03.0	161°45.2	53.4	151°48.1	43.2	299°44.6	52.6	Acrux	172°55.7	-63°16.1
6	3°40.2	242°55.2	N22°02.4	176°46.6	S02°53.8	166°50.5	S05°43.3	314°46.8	N15°52.7	Gacrux	171°47.4	-57°17.1
7	18°42.6	257°54.4	01.8	191°48.1	54.3	181°52.8	43.3	329°49.0	52.7	Alioth	166°09.9	55°48.2
8	33°45.1	272°53.7	01.3	206°49.6	54.8	196°55.2	43.4	344°51.2	52.8	Spica	158°18.4	-11°19.0
9	48°47.6	287°52.9	•• 00.7	221°51.1	•• 55.3	211°57.6	•• 43.4	359°53.4	•• 52.9	Alkaid	152°49.1	49°10.2
10	63°50.0	302°52.2	22°00.1	236°52.6	55.8	227°00.0	43.4	14°55.6	52.9	Hadar	148°30.3	-60°31.2
11	78°52.5	317°51.5	21°59.5	251°54.1	56.2	242°02.4	43.5	29°57.8	53.0	Menkent	147°53.1	-36°31.1
12	93°55.0	332°50.7	N21°58.9	266°55.6	S02°56.7	257°04.8	S05°43.5	45°00.0	N15°53.1	Arcturus	145°44.5	19°01.8
13	108°57.4	347°50.0	58.3	281°57.1	57.2	272°07.2	43.6	60°02.2	53.1	Rigel Kent.	139°34.7	-60°57.7
14	123°59.9	2°49.2	57.7	296°58.6	57.7	287°09.6	43.6	75°04.4	53.2	Kochab	137°19.5	74°02.3
15	139°02.3	17°48.5	•• 57.2	312°00.1	•• 58.2	302°12.0	•• 43.7	90°06.6	•• 53.2	Zuben'ubi	136°51.8	-16°09.9
16	154°04.8	32°47.7	56.6	327°01.6	58.6	317°14.3	43.7	105°08.8	53.3	Alphecca	126°00.4	26°37.1
17	169°07.3	47°47.0	56.0	342°03.1	59.1	332°16.7	43.8	120°11.0	53.4	Antares	112°11.0	-26°29.8
18	184°09.7	62°46.3	N21°55.4	357°04.6	S02°59.6	347°19.1	S05°43.8	135°13.2	N15°53.4	Atria	107°01.4	-69°04.8
19	199°12.2	77°45.5	54.8	12°06.1	03°00.1	2°21.5	43.8	150°15.4	53.5	Sabik	101°58.3	-15°45.6
20	214°14.7	92°44.8	54.2	27°07.5	00.6	17°23.9	43.9	165°17.6	53.6	Shaula	96°05.0	-37°07.4
21	229°17.1	107°44.0	•• 53.6	42°09.0	•• 01.0	32°26.3	•• 43.9	180°19.8	•• 53.6	Rasalhague	95°54.9	12°32.5
22	244°19.6	122°43.3	53.0	57°10.5	01.5	47°28.7	44.0	195°22.0	53.7	Eltanin	90°40.0	51°29.2
23	259°22.1	137°42.6	52.4	72°12.0	02.0	62°31.0	44.0	210°24.2	53.8	Kaus Aust.	83°27.3	-34°22.0
Mer.pass. 05:45		ν -0.7' d -0.6' m -3.89		ν 1.5' d 0.5' m 0.31		ν 2.4' d 0.0' m -2.11		ν 2.2' d 0.1' m 0.31		Vega	80°30.3	38°48.8

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	274°24.5	152°41.8	N21°51.8	87°13.5	S03°02.5	77°33.4	S05°44.1	225°26.4	N15°53.8	Nunki	75°42.9	-26°15.4
1	289°27.0	167°41.1	51.2	102°15.0	03.0	92°35.8	44.1	240°28.6	53.9	Altair	61°56.1	8°56.9
2	304°29.5	182°40.4	50.6	117°16.5	03.4	107°38.2	44.1	255°30.8	54.0	Peacock	52°59.7	-56°38.0
3	319°31.9	197°39.6	•• 50.0	132°17.9	•• 03.9	122°40.6	•• 44.2	270°33.0	•• 54.0	Deneb	49°22.9	45°23.1
4	334°34.4	212°38.9	49.4	147°19.4	04.4	137°43.0	44.2	285°35.2	54.1	Enif	33°35.0	10°00.7
5	349°36.8	227°38.1	48.8	162°20.9	04.9	152°45.4	44.3	300°37.4	54.1	Al Na'ir	27°28.3	-46°48.7
6	4°39.3	242°37.4	N21°48.2	177°22.4	S03°05.4	167°47.7	S05°44.3	315°39.6	N15°54.2	Fomalhaut	15°10.5	-29°27.6
7	19°41.8	257°36.7	47.6	192°23.9	05.9	182°50.1	44.4	330°41.8	54.3	Scheat	13°41.5	28°14.6
8	34°44.2	272°35.9	47.0	207°25.4	06.3	197°52.5	44.4	345°44.0	54.3	Markab	13°26.2	15°21.9
9	49°46.7	287°35.2	•• 46.4	222°26.8	•• 06.8	212°54.9	•• 44.5	0°46.2	•• 54.4			
10	64°49.2	302°34.5	45.8	237°28.3	07.3	227°57.3	44.5	15°48.4	54.5	Jun 24 Sun	SHA	Mer.pass
11	79°51.6	317°33.8	45.2	252°29.8	07.8	242°59.6	44.6	30°50.6	54.5	Venus	240°51.5	13:48
12	94°54.1	332°33.0	N21°44.6	267°31.3	S03°08.3	258°02.0	S05°44.6	45°52.8	N15°54.6	Mars	173°35.3	18:14
13	109°56.6	347°32.3	44.0	282°32.8	08.7	273°04.4	44.6	60°55.0	54.7	Jupiter	163°12.4	18:54
14	124°59.0	2°31.6	43.3	297°34.3	09.2	288°06.8	44.7	75°57.2	54.7	Saturn	311°14.7	09:04
15	140°01.5	17°30.8	•• 42.7	312°35.7	•• 09.7	303°09.2	•• 44.7	90°59.4	•• 54.8			
16	155°04.0	32°30.1	42.1	327°37.2	10.2	318°11.6	44.8	106°01.6	54.8	Jun 25 Mon	SHA	Mer.pass
17	170°06.4	47°29.4	41.5	342°38.7	10.7	333°13.9	44.8	121°03.8	54.9	Venus	239°34.3	13:49
18	185°08.9	62°28.6	N21°40.9	357°40.2	S03°11.2	348°16.3	S05°44.9	136°06.0	N15°55.0	Mars	173°12.3	18:12
19	200°11.3	77°27.9	40.3	12°41.6	11.6	3°18.7	44.9	151°08.2	55.0	Jupiter	163°10.7	18:51
20	215°13.8	92°27.2	39.6	27°43.1	12.1	18°21.1	45.0	166°10.4	55.1	Saturn	311°08.3	

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°23.8	N23°24.4	30°30.7	4.3'	S23°38.7	3.8'	59.6'						
1	194°23.7	24.3	44°54.0	4.3'	23°42.6	3.7'	59.6'						
2	209°23.6	24.3	59°17.3	4.3'	23°46.3	3.5'	59.6'						
3	224°23.4	· · 24.2	73°40.6	4.2'	23°49.8	3.3'	59.5'						
4	239°23.3	24.2	88°03.8	4.2'	23°53.1	3.2'	59.5'						
5	254°23.2	24.1	102°27.1	4.2'	23°56.3	3.0'	59.5'						
6	269°23.0	N23°24.0	116°50.3	4.2'	S23°59.3	2.8'	59.5'						
7	284°22.9	24.0	131°13.5	4.2'	24°02.1	2.7'	59.5'						
8	299°22.8	23.9	145°36.6	4.2'	24°04.8	2.5'	59.5'						
9	314°22.6	· · 23.9	159°59.8	4.2'	24°07.3	2.3'	59.5'						
10	329°22.5	23.8	174°23.0	4.1'	24°09.6	2.2'	59.5'						
11	344°22.4	23.8	188°46.1	4.1'	24°11.8	2.0'	59.4'						
12	359°22.2	N23°23.7	203°09.2	4.1'	S24°13.8	1.8'	59.4'						
13	14°22.1	23.6	217°32.4	4.1'	24°15.6	1.7'	59.4'						
14	29°22.0	23.6	231°55.5	4.1'	24°17.3	1.5'	59.4'						
15	44°21.8	· · 23.5	246°18.7	4.1'	24°18.8	1.3'	59.4'						
16	59°21.7	23.5	260°41.8	4.1'	24°20.1	1.1'	59.4'						
17	74°21.6	23.4	275°04.9	4.2'	24°21.2	1.0'	59.4'						
18	89°21.4	N23°23.3	289°28.1	4.2'	S24°22.2	0.8'	59.3'						
19	104°21.3	23.3	303°51.3	4.2'	24°23.0	0.6'	59.3'						
20	119°21.2	23.2	318°14.4	4.2'	24°23.7	0.5'	59.3'						
21	134°21.0	· · 23.1	332°37.6	4.2'	24°24.1	0.3'	59.3'						
22	149°20.9	23.1	347°00.8	4.2'	24°24.4	0.1'	59.3'						
23	164°20.8	23.0	1°24.0	4.2'	24°24.6	-0.0'	59.3'						
SD = 15.7' d = -0.1'							SD = 16.2'						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°20.6	N23°22.9	15°47.3	4.3'	S24°24.5	-0.2'	59.2'						
1	194°20.5	22.9	30°10.6	4.3'	24°24.3	-0.4'	59.2'						
2	209°20.4	22.8	44°33.8	4.3'	24°24.0	-0.5'	59.2'						
3	224°20.3	· · 22.7	58°57.2	4.3'	24°23.4	-0.7'	59.2'						
4	239°20.1	22.7	73°20.5	4.4'	24°22.7	-0.9'	59.2'						
5	254°20.0	22.6	87°43.9	4.4'	24°21.8	-1.0'	59.2'						
6	269°19.9	N23°22.5	102°07.3	4.4'	S24°20.8	-1.2'	59.1'						
7	284°19.7	22.5	116°30.8	4.5'	24°19.6	-1.4'	59.1'						
8	299°19.6	22.4	130°54.2	4.5'	24°18.2	-1.5'	59.1'						
9	314°19.5	· · 22.3	145°17.8	4.6'	24°16.7	-1.7'	59.1'						
10	329°19.3	22.2	159°41.3	4.6'	24°15.0	-1.9'	59.1'						
11	344°19.2	22.2	174°04.9	4.7'	24°13.1	-2.0'	59.0'						
12	359°19.1	N23°22.1	188°28.6	4.7'	S24°11.1	-2.2'	59.0'						
13	14°18.9	22.0	202°52.3	4.8'	24°08.9	-2.3'	59.0'						
14	29°18.8	21.9	217°16.1	4.8'	24°06.6	-2.5'	59.0'						
15	44°18.7	· · 21.9	231°39.9	4.9'	24°04.1	-2.7'	59.0'						
16	59°18.5	21.8	246°03.7	4.9'	24°01.4	-2.8'	58.9'						
17	74°18.4	21.7	260°27.7	5.0'	23°58.6	-3.0'	58.9'						
18	89°18.3	N23°21.6	274°51.6	5.0'	S23°55.6	-3.1'	58.9'						
19	104°18.2	21.5	289°15.7	5.1'	23°52.5	-3.3'	58.9'						
20	119°18.0	21.5	303°39.8	5.2'	23°49.2	-3.5'	58.8'						
21	134°17.9	· · 21.4	318°03.9	5.2'	23°45.7	-3.6'	58.8'						
22	149°17.8	21.3	332°28.1	5.3'	23°42.1	-3.8'	58.8'						
23	164°17.6	21.2	346°52.4	5.4'	23°38.3	-3.9'	58.8'						
SD = 15.7' d = -0.1'							SD = 16.2'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°17.5	N23°21.1	1°16.8	5.4'	S23°34.4	-4.1'	58.8'						
1	194°17.4	21.0	15°41.2	5.5'	23°30.3	-4.2'	58.7'						
2	209°17.2	21.0	30°05.7	5.6'	23°26.1	-4.4'	58.7'						
3	224°17.1	· · 20.9	44°30.3	5.6'	23°21.8	-4.5'	58.7'						
4	239°17.0	20.8	58°54.9	5.7'	23°17.3	-4.7'	58.7'						
5	254°16.9	20.7	73°19.6	5.8'	23°12.6	-4.8'	58.6'						
6	269°16.7	N23°20.6	87°44.4	5.9'	S23°07.8	-5.0'	58.6'						
7	284°16.6	20.5	102°09.3	6.0'	23°02.8	-5.1'	58.6'						
8	299°16.5	20.4	116°34.3	6.0'	22°57.7	-5.2'	58.6'						
9	314°16.3	· · 20.3	130°59.3	6.1'	22°52.5	-5.4'	58.5'						
10	329°16.2	20.2	145°24.4	6.2'	22°47.1	-5.5'	58.5'						
11	344°16.1	20.2	159°49.6	6.3'	22°41.6	-5.7'	58.5'						
12	359°15.9	N23°20.1	174°14.9	6.4'	S22°36.0	-5.8'	58.4'						
13	14°15.8	20.0	188°40.3	6.5'	22°30.2	-5.9'	58.4'						
14	29°15.7	19.9	203°05.7	6.5'	22°24.2	-6.1'	58.4'						
15	44°15.6	· · 19.8	217°31.3	6.6'	22°18.2	-6.2'	58.4'						
16	59°15.4	19.7	231°56.9	6.7'	22°12.0	-6.3'	58.3'						
17	74°15.3	19.6	246°22.6	6.8'	22°05.7	-6.5'	58.3'						
18	89°15.2	N23°19.5	260°48.4	6.9'	S21°59.2	-6.6'	58.3'						
19	104°15.0	19.4	275°14.3	7.0'	21°52.6	-6.7'	58.3'						
20	119°14.9	19.3	289°40.3	7.1'	21°45.9	-6.8'	58.2'						
21	134°14.8	· · 19.2	304°06.4	7.2'	21°39.1	-7.0'	58.2'						
22	149°14.7	19.1	318°32.6	7.3'	21°32.1	-7.1'	58.2'						
23	164°14.5	19.0	332°58.9	7.4'	21°25.0	-7.2'	58.1'						
SD = 15.7' d = -0.1'							SD = 16.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	—:—	—:—	01:33	22:31	—:—	—:—
62°	—:—	—:—	02:11	21:54	—:—	—:—
60°	—:—	00:52	02:37	21:28	23:12	—:—
N 58°	—:—	01:42	02:58	21:07	22:23	—:—
56°	—:—	02:12	03:15	20:51	21:53	—:—
54°	00:48	02:34	03:29	20:36	21:31	23:16
52°	01:34	02:52	03:41	20:24	21:13	22:31
50°	02:02	03:07	03:52	20:13	20:58	22:03
45°	02:47	03:37	04:14	19:51	20:28	21:18
N 40°	03:18	04:00	04:32	19:33	20:06	20:47
35°	03:41	04:18	04:48	19:18	19:48	20:24
30°	04:00	04:33	05:01	19:05	19:32	20:06
20°	04:29	04:58	05:23	18:43	19:07	19:37
N 10°	04:51	05:18	05:41	18:24	18:47	19:14
0°	05:10	05:37	05:59	18:06	18:29	18:55
S 10°	05:27	05:54	06:16	17:49	18:12	18:38
20°	05:43	06:11	06:35	17:30	17:55	18:22
30°	06:00	06:30	06:56	17:09	17:36	18:06
35°	06:09	06:40	07:09	16:57	17:25	17:57
40°	06:18	06:52	07:23	16:43	17:13	17:47
45°	06:28	07:06	07:39	16:26	17:00	17:37
S 50°	06:40	07:22	08:00	16:05	16:44	17:25
52°	06:45	07:29	08:10	15:55	16:36	17:20
54°	06:51	07:37	08:21	15:44	16:28	17:14
56°	06:57	07:46	08:34	15:32	16:19	17:08
58°	07:04	07:57	08:48	15:17	16:09	17:02
S 60°	07:11	08:08	09:06	15:00	15:57	16:54

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	23:56	■	■	■
66°	■	23:19	22:58	■	■	00:36
64°	21:29	22:10	22:24		00:19	01:45
62°	20:46	21:35	21:59	00:14	01:02	02:20
60°	20:17	21:09	21:40	00:40	01:31	02:46
N 58°	19:55	20:48	21:23	01:00	01:53	03:06
56°	19:37	20:31	21:09	01:17	02:12	03:22
54°	19:21	20:17	20:57	01:31	02:27	03:37
52°	19:08	20:04	20:46	01:44	02:40	03:49
50°	18:56	19:53	20:37	01:55	02:52	04:00
45°	18:32	19:30	20:17	02:18	03:16	04:22
N 40°	18:13	19:11	20:00	02:36	03:35	04:41
35°	17:57	18:56	19:47	02:52	03:51	04:56
30°	17:43	18:42	19:34	03:06	04:05	05:09
20°	17:19	18:19	19:14	03:29	04:29	05:31
N 10°	16:59	17:59	18:56	03:49	04:50	05:51
0°	16:40	17:40	18:39	04:07	05:09	06:09
S 10°	16:20	17:22	18:22	04:26	05:28	06:27
20°	16:00	17:02	18:04	04:46	05:48	06:46
30°	15:37	16:38	17:43	05:09	06:12	07:08
35°	15:23	16:25	17:31	05:23	06:26	07:21
40°	15:07	16:09	17:16	05:38	06:41	07:35
45°	14:48	15:50	17:00	05:57	07:00	07:53
S 50°	14:24	15:27	16:39	06:20	07:24	08:14
52°	14:13	15:16	16:29	06:32	07:35	08:25
54°	14:00	15:03	16:18	06:44	07:48	08:36
56°	13:45	14:48	16:05	06:59	08:03	08:49
58°	13:28	14:31	15:50	07:16	08:20	09:04
S 60°	13:07	14:10	15:32	07:37	08:42	09:23

June 27, 28, 29 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	275°23.7	152°24.3	N21°37.2	87°49.0	S03°14.1	78°30.6	S05°45.2	226°19.3	N15°55.3	Alpheratz	357°30.9	29°15.2
1	290°26.1	167°23.6	36.5	102°50.5	14.6	93°33.0	45.2	241°21.5	55.4	Ankaa	353°03.7	-42°08.4
2	305°28.6	182°22.8	35.9	117°52.0	15.0	108°35.3	45.3	256°23.7	55.5	Schedar	349°26.7	56°41.8
3	320°31.1	197°22.1	•• 35.3	132°53.4	•• 15.5	123°37.7	•• 45.3	271°25.9	•• 55.5	Diphda	348°43.7	-17°49.3
4	335°33.5	212°21.4	34.7	147°54.9	16.0	138°40.1	45.3	286°28.1	55.6	Achernar	335°17.9	-57°04.9
5	350°36.0	227°20.7	34.0	162°56.4	16.5	153°42.5	45.4	301°30.3	55.7	Hamal	327°47.2	23°36.0
6	5°38.5	242°19.9	N21°33.4	177°57.9	S03°17.0	168°44.9	S05°45.4	316°32.5	N15°55.7	Polaris	312°21.7	89°22.9
7	20°40.9	257°19.2	32.8	192°59.3	17.5	183°47.2	45.5	331°34.7	55.8	Acamar	315°09.4	-40°11.0
8	35°43.4	272°18.5	32.1	208°00.8	18.0	198°49.6	45.5	346°36.9	55.9	Menkar	314°02.6	4°12.3
9	50°45.8	287°17.8	•• 31.5	223°02.3	•• 18.4	213°52.0	•• 45.6	1°39.1	•• 55.9	Mirfak	308°23.3	49°57.7
10	65°48.3	302°17.1	30.9	238°03.7	18.9	228°54.4	45.6	16°41.3	56.0	Aldebaran	290°35.8	16°34.0
11	80°50.8	317°16.3	30.2	253°05.2	19.4	243°56.7	45.7	31°43.5	56.0	Rigel	281°00.7	-8°10.1
12	95°53.2	332°15.6	N21°29.6	268°06.7	S03°19.9	258°59.1	S05°45.7	46°45.7	N15°56.1	Capella	280°17.0	46°01.4
13	110°55.7	347°14.9	29.0	283°08.2	20.4	274°01.5	45.8	61°47.9	56.2	Bellatrix	278°19.3	6°22.5
14	125°58.2	2°14.2	28.3	298°09.6	20.9	289°03.9	45.8	76°50.1	56.2	Elnath	277°57.7	28°37.8
15	141°00.6	17°13.5	•• 27.7	313°11.1	•• 21.4	304°06.2	•• 45.9	91°52.3	•• 56.3	Alnilam	275°34.4	-1°11.1
16	156°03.1	32°12.8	27.1	328°12.6	21.8	319°08.6	45.9	106°54.5	56.4	Betelgeuse	270°48.5	7°24.6
17	171°05.6	47°12.0	26.4	343°14.0	22.3	334°11.0	46.0	121°56.7	56.4	Canopus	263°51.3	-52°42.8
18	186°08.0	62°11.3	N21°25.8	358°15.5	S03°22.8	349°13.4	S05°46.0	136°58.9	N15°56.5	Sirius	258°23.4	-16°45.5
19	201°10.5	77°10.6	25.1	13°17.0	23.3	4°15.7	46.1	152°01.1	56.5	Adhara	255°03.4	-29°00.9
20	216°12.9	92°09.9	24.5	28°18.4	23.8	19°18.1	46.1	167°03.3	56.6	Procyon	244°47.4	5°08.8
21	231°15.4	107°09.2	•• 23.8	43°19.9	•• 24.3	34°20.5	•• 46.2	182°05.5	•• 56.7	Pollux	243°13.3	27°57.2
22	246°17.9	122°08.5	23.2	58°21.3	24.8	49°22.9	46.2	197°07.7	56.7	Avior	234°13.7	-59°36.5
23	261°20.3	137°07.8	22.5	73°22.8	25.3	64°25.2	46.3	212°09.9	56.8	Suhail	222°43.9	-43°33.4
Mer.pass. 05:37		ν -0.7' d -0.6' m -3.89		ν 1.5' d 0.5' m 0.33		ν 2.4' d 0.0' m -2.10		ν 2.2' d 0.1' m 0.31				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	276°22.8	152°07.0	N21°21.9	88°24.3	S03°25.8	79°27.6	S05°46.3	227°12.1	N15°56.9
1	291°25.3	167°06.3	21.3	103°25.7	26.2	94°30.0	46.4	242°14.3	56.9
2	306°27.7	182°05.6	20.6	118°27.2	26.7	109°32.3	46.4	257°16.6	57.0
3	321°30.2	197°04.9	•• 20.0	133°28.7	•• 27.2	124°34.7	•• 46.5	272°18.8	•• 57.0
4	336°32.7	212°04.2	19.3	148°30.1	27.7	139°37.1	46.5	287°21.0	57.1
5	351°35.1	227°03.5	18.6	163°31.6	28.2	154°39.5	46.6	302°23.2	57.2
6	6°37.6	242°02.8	N21°18.0	178°33.0	S03°28.7	169°41.8	S05°46.6	317°25.4	N15°57.2
7	21°40.1	257°02.1	17.3	193°34.5	29.2	184°44.2	46.7	332°27.6	57.3
8	36°42.5	272°01.4	16.7	208°36.0	29.7	199°46.6	46.7	347°29.8	57.3
9	51°45.0	287°00.7	•• 16.0	223°37.4	•• 30.2	214°48.9	•• 46.8	2°32.0	•• 57.4
10	66°47.4	301°59.9	15.4	238°38.9	30.6	229°51.3	46.8	17°34.2	57.5
11	81°49.9	316°59.2	14.7	253°40.3	31.1	244°53.7	46.9	32°36.4	57.5
12	96°52.4	331°58.5	N21°14.0	268°41.8	S03°31.6	259°56.0	S05°46.9	47°38.6	N15°57.6
13	111°54.8	346°57.8	13.4	283°43.3	32.1	274°58.4	47.0	62°40.8	57.7
14	126°57.3	1°57.1	12.7	298°44.7	32.6	290°00.8	47.0	77°43.0	57.7
15	141°59.8	16°56.4	•• 12.1	313°46.2	•• 33.1	305°03.1	•• 47.1	92°45.2	•• 57.8
16	157°02.2	31°55.7	11.4	328°47.6	33.6	320°05.5	47.1	107°47.4	57.8
17	172°04.7	46°55.0	10.7	343°49.1	34.1	335°07.9	47.2	122°49.6	57.9
18	187°07.2	61°54.3	N21°10.1	358°50.5	S03°34.6	350°10.2	S05°47.2	137°51.8	N15°58.0
19	202°09.6	76°53.6	09.4	13°52.0	35.1	5°12.6	47.3	152°54.1	58.0
20	217°12.1	91°52.9	08.7	28°53.4	35.6	20°15.0	47.3	167°56.3	58.1
21	232°14.6	106°52.2	•• 08.1	43°54.9	•• 36.0	35°17.3	•• 47.4	182°58.5	•• 58.1
22	247°17.0	121°51.5	07.4	58°56.3	36.5	50°19.7	47.4	198°00.7	58.2
23	262°19.5	136°50.8	06.7	73°57.8	37.0	65°22.1	47.5	213°02.9	58.3
Mer.pass. 05:34		ν -0.7' d -0.6' m -3.89		ν 1.5' d 0.5' m 0.34		ν 2.4' d 0.0' m -2.09		ν 2.2' d 0.1' m 0.31	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	277°21.9	151°50.1	N21°06.0	88°59.2	S03°37.5	80°24.4	S05°47.5	228°05.1	N15°58.3
1	292°24.4	166°49.4	05.4	104°00.7	38.0	95°26.8	47.6	243°07.3	58.4
2	307°26.9	181°48.7	04.7	119°02.1	38.5	110°29.2	47.7	258°09.5	58.5
3	322°29.3	196°48.0	•• 04.0	134°03.6	•• 39.0	125°31.5	•• 47.7	273°11.7	•• 58.5
4	337°31.8	211°47.3	03.3	149°05.0	39.5	140°33.9	47.8	288°13.9	58.6
5	352°34.3	226°46.6	02.7	164°06.5	40.0	155°36.3	47.8	303°16.1	58.6
6	7°36.7	241°45.9	N21°02.0	179°07.9	S03°40.5	170°38.6	S05°47.9	318°18.3	N15°58.7
7	22°39.2	256°45.2	01.3	194°09.4	41.0	185°41.0	47.9	333°20.5	58.8
8	37°41.7	271°44.5	00.6	209°10.8	41.5	200°43.4	48.0	348°22.7	58.8
9	52°44.1	286°43.8	21°00.0	224°12.3	•• 42.0	215°45.7	•• 48.0	3°25.0	•• 58.9
10	67°46.6	301°43.1	20°59.3	239°13.7	42.5	230°48.1	48.1	18°27.2	58.9
11	82°49.1	316°42.4	58.6	254°15.2	42.9	245°50.4	48.1	33°29.4	59.0
12	97°51.5	331°41.7	N20°57.9	269°16.6	S03°43.4	260°52.8	S05°48.2	48°31.6	N15°59.1
13	112°54.0	346°41.1	57.2	284°18.1	43.9	275°55.2	48.2	63°33.8	59.1
14	127°56.4	1°40.4	56.5	299°19.5	44.4	290°57.5	48.3	78°36.0	59.2
15	142°58.9	16°39.7	•• 55.8	314°21.0	•• 44.9	305°59.9	•• 48.3	93°38.2	•• 59.2
16	158°01.4	31°39.0	55.2	329°22.4	45.4	321°02.2	48.4	108°40.4	59.3
17	173°03.8	46°38.3	54.5	344°23.9	45.9	336°04.6	48.5	123°42.6	59.4
18	188°06.3	61°37.6	N20°53.8	359°25.3	S03°46.4	351°07.0	S05°48.5	138°44.8	N15°59.4
19	203°08.8	76°36.9	53.1	14°26.7	46.9	6°09.3	48.6	153°47.0	59.5
20	218°11.2	91°36.2	52.4	29°28.2	47.4	21°11.7	48.6	168°49.2	59.6
21	233°13.7	106°35.5	•• 51.7	44°29.6	•• 47.9	36°14.0	•• 48.7	183°51.5	•• 59.6
22	248°16.2	121°34.8	51.0	59°31.1	48.4	51°16.4	48.7	198°53.7	59.7
23	263°18.6	136°34.2	50.3	74°32.5	48.9	66°18.8	48.8	213°55.9	59.7
Mer.pass. 05:30		ν -0.7' d -0.7' m -3.89		ν 1.5' d 0.5' m 0.36		ν 2.4' d 0.1' m -2.08		ν 2.2' d 0.1' m 0.31	

Jun 27 Wed	SHA	Mer.pass
Venus	237°00.6	13:51
Mars	172°25.4	18:07
Jupiter	163°06.9	18:43
Saturn	310°55.6	08:53

Jun 28 Thu	SHA	Mer.pass
Venus	235°44.2	13:52
Mars	172°01.5	18:05
Jupiter	163°04.8	18:39
Saturn	310°49.3	08:50

Jun 29 Fri	SHA	Mer.pass
Venus	234°28.2	13:53
Mars	171°37.3	18:02
Jupiter	163°02.5	18:35
Saturn	310°43.1	08:46

Horizontal parallax	
Venus:	0.1
Mars:	0.1

DUT1 = UT1-UTC = +0.2109 sec ΔT = TT-UT1 = +68.9731 sec

2029 June 27 to Jun. 29 UT

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179°14.4	N23°18.9	347°25.2	7.5'	S21°17.8	-7.3'	58.1'
1	194°14.3	18.8	1°51.7	7.6'	21°10.5	-7.4'	58.1'
2	209°14.1	18.7	16°18.3	7.7'	21°03.0	-7.6'	58.1'
3	224°14.0	•• 18.6	30°44.9	7.8'	20°55.5	-7.7'	58.0'
4	239°13.9	18.5	45°11.7	7.9'	20°47.8	-7.8'	58.0'
5	254°13.8	18.4	59°38.6	8.0'	20°40.0	-7.9'	58.0'
6	269°13.6	N23°18.3	74°05.5	8.1'	S20°32.1	-8.0'	57.9'
7	284°13.5	18.2	88°32.6	8.2'	20°24.0	-8.1'	57.9'
8	299°13.4	18.1	102°59.7	8.3'	20°15.9	-8.2'	57.9'
9	314°13.2	•• 18.0	117°27.0	8.4'	20°07.7	-8.3'	57.9'
10	329°13.1	17.8	131°54.3	8.5'	19°59.3	-8.5'	57.8'
11	344°13.0	17.7	146°21.8	8.6'	19°50.9	-8.6'	57.8'
12	359°12.9	N23°17.6	160°49.3	8.7'	S19°42.3	-8.7'	57.8'
13	14°12.7	17.5	175°17.0	8.8'	19°33.6	-8.8'	57.7'
14	29°12.6	17.4	189°44.8	8.9'	19°24.9	-8.9'	57.7'
15	44°12.5	•• 17.3	204°12.6	9.0'	19°16.0	-9.0'	57.7'
16	59°12.3	17.2	218°40.6	9.1'	19°07.1	-9.1'	57.6'
17	74°12.2	17.1	233°08.6	9.2'	18°58.0	-9.2'	57.6'
18	89°12.1	N23°17.0	247°36.8	9.3'	S18°48.8	-9.2'	57.6'
19	104°12.0	16.8	262°05.0	9.4'	18°39.6	-9.3'	57.6'
20	119°11.8	16.7	276°33.4	9.5'	18°30.3	-9.4'	57.5'
21	134°11.7	•• 16.6	291°01.9	9.6'	18°20.8	-9.5'	57.5'
22	149°11.6	16.5	305°30.4	9.7'	18°11.3	-9.6'	57.5'
23	164°11.5	16.4	319°59.1	9.8'	18°01.7	-9.7'	57.4'
SD = 15.7' d = -0.1'			SD = 15.9'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°11.3	N23°16.3	334°27.8	9.9'	S17°52.0	-9.8'	57.4'
1	194°11.2	16.1	348°56.7	10.0'	17°42.2	-9.9'	57.4'
2	209°11.1	16.0	3°25.7	10.1'	17°32.3	-9.9'	57.3'
3	224°10.9	•• 15.9	17°54.7	10.2'	17°22.4	-10.0'	57.3'
4	239°10.8	15.8	32°23.9	10.3'	17°12.4	-10.1'	57.3'
5	254°10.7	15.7	46°53.1	10.4'	17°02.3	-10.2'	57.2'
6	269°10.6	N23°15.5	61°22.5	10.5'	S16°52.1	-10.3'	57.2'
7	284°10.4	15.4	75°51.9	10.5'	16°41.8	-10.3'	57.2'
8	299°10.3	15.3	90°21.5	10.6'	16°31.5	-10.4'	57.1'
9	314°10.2	•• 15.2	104°51.1	10.7'	16°21.1	-10.5'	57.1'
10	329°10.1	15.0	119°20.9	10.8'	16°10.6	-10.6'	57.1'
11	344°09.9	14.9	133°50.7	10.9'	16°00.0	-10.6'	57.1'
12	359°09.8	N23°14.8	148°20.6	11.0'	S15°49.4	-10.7'	57.0'
13	14°09.7	14.7	162°50.7	11.1'	15°38.7	-10.8'	57.0'
14	29°09.6	14.5	177°20.8	11.2'	15°28.0	-10.8'	57.0'
15	44°09.4	•• 14.4	191°51.0	11.3'	15°17.1	-10.9'	56.9'
16	59°09.3	14.3	206°21.3	11.4'	15°06.3	-10.9'	56.9'
17	74°09.2	14.1	220°51.7	11.5'	14°55.3	-11.0'	56.9'
18	89°09.1	N23°14.0	235°22.2	11.6'	S14°44.3	-11.1'	56.8'
19	104°08.9	13.9	249°52.8	11.7'	14°33.2	-11.1'	56.8'
20	119°08.8	13.7	264°23.5	11.8'	14°22.1	-11.2'	56.8'
21	134°08.7	•• 13.6	278°54.2	11.9'	14°10.9	-11.2'	56.7'
22	149°08.6	13.5	293°25.1	11.9'	13°59.7	-11.3'	56.7'
23	164°08.4	13.3	307°56.0	12.0'	13°48.4	-11.4'	56.7'
SD = 15.7' d = -0.1'			SD = 15.7'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°08.3	N23°13.2	322°27.1	12.1'	S13°37.0	-11.4'	56.7'
1	194°08.2	13.1	336°58.2	12.2'	13°25.6	-11.5'	56.6'
2	209°08.1	12.9	351°29.4	12.3'	13°14.2	-11.5'	56.6'
3	224°07.9	•• 12.8	6°00.7	12.4'	13°02.7	-11.6'	56.6'
4	239°07.8	12.7	20°32.1	12.5'	12°51.1	-11.6'	56.5'
5	254°07.7	12.5	35°03.5	12.5'	12°39.5	-11.6'	56.5'
6	269°07.6	N23°12.4	49°35.1	12.6'	S12°27.9	-11.7'	56.5'
7	284°07.4	12.2	64°06.7	12.7'	12°16.2	-11.7'	56.4'
8	299°07.3	12.1	78°38.4	12.8'	12°04.4	-11.8'	56.4'
9	314°07.2	•• 12.0	93°10.2	12.9'	11°52.7	-11.8'	56.4'
10	329°07.1	11.8	107°42.0	12.9'	11°40.8	-11.9'	56.3'
11	344°06.9	11.7	122°14.0	13.0'	11°29.0	-11.9'	56.3'
12	359°06.8	N23°11.5	136°46.0	13.1'	S11°17.1	-11.9'	56.3'
13	14°06.7	11.4	151°18.1	13.2'	11°05.1	-12.0'	56.3'
14	29°06.6	11.2	165°50.3	13.2'	10°53.2	-12.0'	56.2'
15	44°06.4	•• 11.1	180°22.5	13.3'	10°41.2	-12.0'	56.2'
16	59°06.3	10.9	194°54.8	13.4'	10°29.1	-12.1'	56.2'
17	74°06.2	10.8	209°27.2	13.5'	10°17.0	-12.1'	56.1'
18	89°06.1	N23°10.7	223°59.7	13.5'	S10°04.9	-12.1'	56.1'
19	104°05.9	10.5	238°32.2	13.6'	09°52.8	-12.2'	56.1'
20	119°05.8	10.4	253°04.9	13.7'	09°40.6	-12.2'	56.0'
21	134°05.7	•• 10.2	267°37.5	13.7'	09°28.4	-12.2'	56.0'
22	149°05.6	10.1	282°10.3	13.8'	09°16.2	-12.3'	56.0'
23	164°05.5	09.9	296°43.1	13.9'	09°03.9	-12.3'	56.0'
SD = 15.7' d = -0.1'			SD = 15.5'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	--	--	01:37	22:29	--	--
62°	--	--	02:14	21:52	--	--
60°	--	00:57	02:40	21:27	23:08	--
N 58°	--	01:45	03:00	21:07	22:21	--
56°	--	02:14	03:16	20:50	21:52	--
54°	00:53	02:36	03:30	20:36	21:30	23:12
52°	01:37	02:54	03:43	20:24	21:12	22:29
50°	02:04	03:09	03:53	20:13	20:57	22:02
45°	02:49	03:38	04:16	19:51	20:28	21:17
N 40°	03:19	04:01	04:34	19:33	20:06	20:47
35°	03:42	04:19	04:49	19:18	19:48	20:24
30°	04:01	04:34	05:01	19:05	19:33	20:06
20°	04:30	04:59	05:23	18:43	19:08	19:37
N 10°	04:52	05:19	05:42	18:24	18:47	19:15
0°	05:11	05:37	06:00	18:07	18:30	18:56
S 10°	05:28	05:54	06:17	17:50	18:13	18:39
20°	05:44	06:11	06:35	17:31	17:55	18:23
30°	06:00	06:30	06:56	17:10	17:37	18:06
35°	06:09	06:41	07:09	16:58	17:26	17:58
40°	06:18	06:52	07:23	16:44	17:14	17:49
45°	06:29	07:06	07:39	16:27	17:01	17:38
S 50°	06:40	07:22	08:00	16:07	16:45	17:27
52°	06:45	07:29	08:10	15:57	16:38	17:21
54°	06:51	07:37	08:21	15:46	16:30	17:16
56°	06:57	07:46	08:33	15:33	16:21	17:10
58°	07:04	07:56	08:48	15:19	16:11	17:03
S 60°	07:11	08:08	09:05	15:02	15:59	16:56

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°		01:10 23:46	23:12		02:40	05:47
N 70°	23:55	23:20	22:59		03:54	06:11
68°	23:17	23:00	22:48	02:00	04:31	06:30
66°	22:50	22:44	22:38	02:57	04:56	06:44
64°	22:29	22:31	22:31	03:30	05:16	06:56
62°	22:12	22:19	22:24	03:55	05:32	07:06
60°	21:58	22:10	22:18	04:14	05:46	07:15
N 58°	21:46	22:01	22:13	04:30	05:57	07:22
56°	21:35	21:54	22:08	04:43	06:07	07:29
54°	21:26	21:47	22:04	04:55	06:16	07:35
52°	21:18	21:41	22:00	05:05	06:23	07:40
50°	21:10	21:36	21:57	05:14	06:30	07:45
45°	20:54	21:24	21:49	05:33	06:45	07:55
N 40°	20:41	21:14	21:43	05:49	06:57	08:04
35°	20:29	21:06	21:38	06:02	07:08	08:11
30°	20:20	20:58	21:33	06:14	07:17	08:17
20°	20:02	20:46	21:25	06:33	07:32	08:28
N 10°	19:47	20:34	21:17	06:50	07:46	08:38
0°	19:33	20:24	21:10	07:05	07:58	08:46
S 10°	19:19	20:13	21:03	07:21	08:10	08:55
20°	19:04	20:02	20:56	07:38	08:23	09:04
30°	18:47	19:49	20:47	07:56	08:38	09:15
35°	18:37	19:41	20:43	08:07	08:47	09:21
40°	18:25	19:32	20:37	08:20	08:57	09:28
45°	18:11	19:22	20:30	08:35	09:08	09:36
S 50°	17:54	19:10	20:23	08:53	09:22	09:45
52°	17:46	19:04	20:19	09:01	09:28	09:50
54°	17:38	18:58	20:15	09:10	09:35	09:54
56°	17:28	18:50	20:10	09:21	09:43	10:00
58°	17:16	18:42	20:06	09:33	09:52	10:06
S 60°	17:03	18:33	20:00	09:47	10:02	10:12

Day	Sun		Mer. Pass hh:mm	Moon		Age 15-17 99-91%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
27	03:02	03:09	12:03	00:52	13:20	
28	03:15	03:21	12:03	01:46	14:11	
29	03:27	03:33	12:04	02:35	14:58	

June 30, 01, 02 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	278°21.1	151°33.5	N20°49.6	89°33.9	S03°49.4	81°21.1	S05°48.8	228°58.1	N15°59.8	Alpheratz	357°30.9	29°15.2
1	293°23.5	166°32.8	48.9	104°35.4	49.9	96°23.5	48.9	244°00.3	59.9	Ankaa	353°03.7	-42°08.4
2	308°26.0	181°32.1	48.2	119°36.8	50.4	111°25.8	48.9	259°02.5	15°59.9	Schedar	349°26.7	56°41.8
3	323°28.5	196°31.4	· · 47.5	134°38.3	· · 50.9	126°28.2	· · 49.0	274°04.7	16°00.0	Diphda	348°43.7	-17°49.3
4	338°30.9	211°30.7	46.8	149°39.7	51.4	141°30.6	49.1	289°06.9	00.0	Achernar	335°17.8	-57°04.9
5	353°33.4	226°30.0	46.1	164°41.1	51.9	156°32.9	49.1	304°09.1	00.1	Hamal	327°47.2	23°36.0
6	8°35.9	241°29.4	N20°45.4	179°42.6	S03°52.4	171°35.3	S05°49.2	319°11.3	N16°00.2	Polaris	312°20.5	89°22.9
7	23°38.3	256°28.7	44.7	194°44.0	52.9	186°37.6	49.2	334°13.5	00.2	Acamar	315°09.4	-40°11.0
8	38°40.8	271°28.0	44.0	209°45.4	53.3	201°40.0	49.3	349°15.8	00.3	Menkar	314°02.6	4°12.3
9	53°43.3	286°27.3	· · 43.3	224°46.9	· · 53.8	216°42.3	· · 49.3	4°18.0	· · 00.3	Mirfak	308°23.3	49°57.7
10	68°45.7	301°26.6	42.6	239°48.3	54.3	231°44.7	49.4	19°20.2	00.4	Aldebaran	290°35.7	16°34.0
11	83°48.2	316°26.0	41.9	254°49.8	54.8	246°47.1	49.4	34°22.4	00.5	Rigel	281°00.7	-8°10.1
12	98°50.7	331°25.3	N20°41.2	269°51.2	S03°55.3	261°49.4	S05°49.5	49°24.6	N16°00.5	Capella	280°17.0	46°01.4
13	113°53.1	346°24.6	40.5	284°52.6	55.8	276°51.8	49.6	64°26.8	00.6	Bellatrix	278°19.3	6°22.5
14	128°55.6	1°23.9	39.8	299°54.1	56.3	291°54.1	49.6	79°29.0	00.6	Elnath	277°57.6	28°37.8
15	143°58.0	16°23.2	· · 39.0	314°55.5	· · 56.8	306°56.5	· · 49.7	94°31.2	· · 00.7	Alnilam	275°34.4	-1°11.1
16	159°00.5	31°22.6	38.3	329°56.9	57.3	321°58.8	49.7	109°33.4	00.8	Betelgeuse	270°48.5	7°24.6
17	174°03.0	46°21.9	37.6	344°58.3	57.8	337°01.2	49.8	124°35.7	00.8	Canopus	263°51.3	-52°42.8
18	189°05.4	61°21.2	N20°36.9	359°59.8	S03°58.3	352°03.5	S05°49.8	139°37.9	N16°00.9	Sirius	258°23.4	-16°45.5
19	204°07.9	76°20.5	36.2	15°01.2	58.8	7°05.9	49.9	154°40.1	00.9	Adhara	255°03.4	-29°00.9
20	219°10.4	91°19.8	35.5	30°02.6	59.3	22°08.2	50.0	169°42.3	01.0	Procyon	244°47.4	5°08.8
21	234°12.8	106°19.2	· · 34.8	45°04.1	03°59.8	37°10.6	· · 50.0	184°44.5	· · 01.1	Pollux	243°13.3	27°57.2
22	249°15.3	121°18.5	34.0	60°05.5	04°00.3	52°13.0	50.1	199°46.7	01.1	Avior	234°13.7	-59°36.5
23	264°17.8	136°17.8	33.3	75°06.9	00.8	67°15.3	50.1	214°48.9	01.2	Suhail	222°43.9	-43°33.4
Mer.pass. 05:26		ν -0.7' d -0.7' m -3.89		ν 1.4' d 0.5' m 0.37		ν 2.4' d 0.1' m -2.07		ν 2.2' d 0.1' m 0.31		Miaplacidus	221°38.0	-69°50.6
										Alphard	217°44.4	-8°47.4
										Regulus	207°30.8	11°49.3
										Dubhe	193°37.1	61°35.6
										Denebola	182°21.4	14°24.4
										Gienah	175°39.8	-17°42.5
										Acrux	172°55.8	-63°16.1
										Gacrux	171°47.4	-57°17.1
										Alioth	166°09.9	55°48.2
										Spica	158°18.4	-11°19.0
										Alkaid	152°49.1	49°10.2
										Hadar	148°30.4	-60°31.2
										Menkent	147°53.1	-36°31.1
										Arcturus	145°44.5	19°01.8
										Rigel Kent.	139°34.8	-60°57.7
										Kochab	137°19.5	74°02.3
										Zuben'ubi	136°51.8	-16°09.9
										Alphecca	126°00.4	26°37.1
										Antares	112°11.0	-26°29.8
										Atria	107°01.4	-69°04.8
										Sabik	101°58.3	-15°45.6
										Shaula	96°05.0	-37°07.4
										Rasalhague	95°54.9	12°32.5
										Eltanin	90°40.0	51°29.2
										Kaus Aust.	83°27.3	-34°22.0
										Vega	80°30.3	38°48.8
										Nunki	75°42.9	-26°15.4
										Altair	61°56.1	8°57.0
										Peacock	52°59.6	-56°38.0
										Deneb	49°22.8	45°23.2
										Enif	33°35.0	10°00.8
										Al Na'ir	27°28.2	-46°48.7
										Fomalhaut	15°10.4	-29°27.6
										Scheat	13°41.5	28°14.6
										Markab	13°26.1	15°21.9
										Jun 30 Sat	SHA	Mer.pass
										Venus	233°12.4	13:54
										Mars	171°12.9	18:00
										Jupiter	163°00.0	18:32
										Saturn	310°37.0	08:43
										Jul 01 Sun	SHA	Mer.pass
										Venus	231°56.9	13:55
										Mars	170°48.1	17:58
										Jupiter	162°57.4	18:28
										Saturn	310°30.9	08:39
										Jul 02 Mon	SHA	Mer.pass
										Venus	230°41.8	13:57
										Mars	170°23.1	17:55
										Jupiter	162°54.7	18:24
										Saturn	310°24.9	08:36

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°05.3	N23°09.7	311°16.0	13.9'	S08°51.6	-12.3'	55.9'
1	194°05.2	09.6	325°48.9	14.0'	08°39.3	-12.3'	55.9'
2	209°05.1	09.4	340°21.9	14.1'	08°27.0	-12.4'	55.9'
3	224°05.0	· · 09.3	354°55.0	14.1'	08°14.7	-12.4'	55.8'
4	239°04.8	09.1	9°28.2	14.2'	08°02.3	-12.4'	55.8'
5	254°04.7	09.0	24°01.3	14.3'	07°49.9	-12.4'	55.8'
6	269°04.6	N23°08.8	38°34.6	14.3'	S07°37.5	-12.4'	55.8'
7	284°04.5	08.7	53°07.9	14.4'	07°25.0	-12.5'	55.7'
8	299°04.4	08.5	67°41.3	14.4'	07°12.6	-12.5'	55.7'
9	314°04.2	· · 08.3	82°14.7	14.5'	07°00.1	-12.5'	55.7'
10	329°04.1	08.2	96°48.2	14.5'	06°47.6	-12.5'	55.7'
11	344°04.0	08.0	111°21.8	14.6'	06°35.1	-12.5'	55.6'
12	359°03.9	N23°07.9	125°55.4	14.7'	S06°22.6	-12.5'	55.6'
13	14°03.7	07.7	140°29.0	14.7'	06°10.1	-12.5'	55.6'
14	29°03.6	07.5	155°02.7	14.8'	05°57.5	-12.6'	55.5'
15	44°03.5	· · 07.4	169°36.5	14.8'	05°45.0	-12.6'	55.5'
16	59°03.4	07.2	184°10.3	14.9'	05°32.4	-12.6'	55.5'
17	74°03.3	07.1	198°44.2	14.9'	05°19.8	-12.6'	55.5'
18	89°03.1	N23°06.9	213°18.1	15.0'	S05°07.2	-12.6'	55.4'
19	104°03.0	06.7	227°52.0	15.0'	04°54.6	-12.6'	55.4'
20	119°02.9	06.6	242°26.0	15.0'	04°42.0	-12.6'	55.4'
21	134°02.8	· · 06.4	257°00.1	15.1'	04°29.4	-12.6'	55.4'
22	149°02.7	06.2	271°34.2	15.1'	04°16.8	-12.6'	55.3'
23	164°02.5	06.1	286°08.3	15.2'	04°04.2	-12.6'	55.3'
SD = 15.7'			SD = 15.3'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°02.4	N23°05.9	300°42.5	15.2'	S03°51.5	-12.6'	55.3'
1	194°02.3	05.7	315°16.7	15.3'	03°38.9	-12.6'	55.3'
2	209°02.2	05.5	329°50.9	15.3'	03°26.3	-12.6'	55.2'
3	224°02.1	· · 05.4	344°25.2	15.3'	03°13.6	-12.6'	55.2'
4	239°01.9	05.2	358°59.6	15.4'	03°01.0	-12.6'	55.2'
5	254°01.8	05.0	13°33.9	15.4'	02°48.4	-12.6'	55.2'
6	269°01.7	N23°04.9	28°08.3	15.4'	S02°35.7	-12.6'	55.2'
7	284°01.6	04.7	42°42.8	15.5'	02°23.1	-12.6'	55.1'
8	299°01.5	04.5	57°17.2	15.5'	02°10.5	-12.6'	55.1'
9	314°01.3	· · 04.3	71°51.7	15.5'	01°57.8	-12.6'	55.1'
10	329°01.2	04.2	86°26.3	15.6'	01°45.2	-12.6'	55.1'
11	344°01.1	04.0	101°00.8	15.6'	01°32.6	-12.6'	55.0'
12	359°01.0	N23°03.8	115°35.4	15.6'	S01°20.0	-12.6'	55.0'
13	14°00.9	03.6	130°10.1	15.7'	01°07.4	-12.6'	55.0'
14	29°00.7	03.4	144°44.7	15.7'	00°54.8	-12.6'	55.0'
15	44°00.6	· · 03.3	159°19.4	15.7'	00°42.2	-12.6'	55.0'
16	59°00.5	03.1	173°54.1	15.7'	00°29.6	-12.6'	54.9'
17	74°00.4	02.9	188°28.8	15.7'	00°17.0	-12.6'	54.9'
18	89°00.3	N23°02.7	203°03.6	15.8'	S00°04.4	-12.6'	54.9'
19	104°00.2	02.5	217°38.3	15.8'	N00°08.2	12.6'	54.9'
20	119°00.0	02.4	232°13.1	15.8'	00°20.7	12.5'	54.9'
21	133°59.9	· · 02.2	246°47.9	15.8'	00°33.3	12.5'	54.8'
22	148°59.8	02.0	261°22.8	15.9'	00°45.8	12.5'	54.8'
23	163°59.7	01.8	275°57.6	15.9'	00°58.3	12.5'	54.8'
SD = 15.7'			SD = 15.1'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°59.6	N23°01.6	290°32.5	15.9'	N01°10.8	12.5'	54.8'
1	193°59.4	01.4	305°07.4	15.9'	01°23.3	12.5'	54.8'
2	208°59.3	01.2	319°42.3	15.9'	01°35.8	12.5'	54.7'
3	223°59.2	· · 01.1	334°17.2	15.9'	01°48.3	12.4'	54.7'
4	238°59.1	00.9	348°52.1	15.9'	02°00.7	12.4'	54.7'
5	253°59.0	00.7	3°27.1	15.9'	02°13.1	12.4'	54.7'
6	268°58.9	N23°00.5	18°02.0	16.0'	N02°25.6	12.4'	54.7'
7	283°58.7	00.3	32°37.0	16.0'	02°37.9	12.4'	54.7'
8	298°58.6	23°00.1	47°11.9	16.0'	02°50.3	12.4'	54.6'
9	313°58.5	22°59.9	61°46.9	16.0'	03°02.7	12.3'	54.6'
10	328°58.4	59.7	76°21.9	16.0'	03°15.0	12.3'	54.6'
11	343°58.3	59.5	90°56.9	16.0'	03°27.3	12.3'	54.6'
12	358°58.2	N22°59.3	105°31.9	16.0'	N03°39.6	12.3'	54.6'
13	13°58.0	59.1	120°06.9	16.0'	03°51.9	12.3'	54.6'
14	28°57.9	58.9	134°41.9	16.0'	04°04.2	12.2'	54.5'
15	43°57.8	· · 58.7	149°16.9	16.0'	04°16.4	12.2'	54.5'
16	58°57.7	58.6	163°51.9	16.0'	04°28.6	12.2'	54.5'
17	73°57.6	58.4	178°26.9	16.0'	04°40.8	12.2'	54.5'
18	88°57.5	N22°58.2	193°01.9	16.0'	N04°53.0	12.1'	54.5'
19	103°57.4	58.0	207°36.9	16.0'	05°05.1	12.1'	54.5'
20	118°57.2	57.8	222°11.9	16.0'	05°17.2	12.1'	54.5'
21	133°57.1	· · 57.6	236°46.9	16.0'	05°29.3	12.1'	54.4'
22	148°57.0	57.4	251°21.8	16.0'	05°41.4	12.0'	54.4'
23	163°56.9	57.2	265°56.8	16.0'	05°53.4	12.0'	54.4'
SD = 15.7'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	--	--	00:23	23:38	--	--
64°	--	--	01:42	22:25	--	--
62°	--	--	02:17	21:50	--	--
60°	--	01:04	02:42	21:25	23:02	--
N 58°	--	01:49	03:02	21:05	22:18	--
56°	--	02:17	03:19	20:49	21:50	--
54°	00:59	02:39	03:32	20:35	21:28	23:07
52°	01:40	02:56	03:45	20:23	21:11	22:27
50°	02:07	03:11	03:55	20:12	20:56	22:00
45°	02:51	03:40	04:17	19:50	20:27	21:17
N 40°	03:21	04:02	04:35	19:33	20:05	20:47
35°	03:44	04:20	04:50	19:18	19:47	20:24
30°	04:02	04:35	05:03	19:05	19:32	20:06
20°	04:31	05:00	05:24	18:44	19:08	19:37
N 10°	04:53	05:20	05:43	18:25	18:48	19:15
0°	05:12	05:38	06:00	18:08	18:30	18:56
S 10°	05:28	05:55	06:17	17:50	18:13	18:40
20°	05:44	06:12	06:36	17:32	17:56	18:24
30°	06:00	06:30	06:57	17:11	17:38	18:07
35°	06:09	06:41	07:09	16:59	17:27	17:59
40°	06:18	06:52	07:23	16:45	17:16	17:50
45°	06:28	07:05	07:39	16:29	17:03	17:40
S 50°	06:40	07:21	07:59	16:09	16:47	17:28
52°	06:45	07:29	08:09	15:59	16:40	17:23
54°	06:51	07:37	08:20	15:48	16:32	17:18
56°	06:57	07:45	08:32	15:36	16:23	17:12
58°	07:03	07:55	08:47	15:21	16:13	17:05
S 60°	07:10	08:07	09:04	15:04	16:01	16:58

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	22:48	22:26	22:04	07:56	09:51	11:41
N 70°	22:42	22:27	22:12	08:07	09:53	11:36
68°	22:37	22:28	22:18	08:16	09:56	11:32
66°	22:33	22:28	22:24	08:24	09:57	11:28
64°	22:30	22:29	22:28	08:30	09:59	11:25
62°	22:27	22:30	22:32	08:35	10:00	11:23
60°	22:24	22:30	22:36	08:40	10:01	11:21
N 58°	22:22	22:30	22:39	08:44	10:02	11:19
56°	22:20	22:31	22:42	08:48	10:03	11:17
54°	22:18	22:31	22:44	08:51	10:04	11:15
52°	22:17	22:32	22:46	08:54	10:05	11:14
50°	22:15	22:32	22:48	08:57	10:06	11:13
45°	22:12	22:32	22:53	09:02	10:07	11:10
N 40°	22:09	22:33	22:57	09:07	10:08	11:08
35°	22:06	22:33	23:00	09:11	10:09	11:06
30°	22:04	22:34	23:03	09:15	10:10	11:04
20°	22:00	22:35	23:08	09:21	10:12	11:01
N 10°	21:57	22:35	23:13	09:26	10:13	10:58
0°	21:54	22:36	23:17	09:32	10:14	10:56
S 10°	21:51	22:37	23:21	09:37	10:15	10:53
20°	21:47	22:37	23:26	09:42	10:17	10:51
30°	21:44	22:38	23:31	09:48	10:18	10:48
35°	21:41	22:39	23:35	09:51	10:19	10:46
40°	21:39	22:39	23:38	09:55	10:20	10:44
45°	21:36	22:40	23:42	10:00	10:21	10:42
S 50°	21:33	22:41	23:47	10:05	10:22	10:39
52°	21:31	22:41	23:49	10:07	10:23	10:38
54°	21:29	22:41	23:52	10:10	10:23	10:36
56°	21:27	22:42	23:55	10:13	10:24	10:35
58°	21:25	22:42	23:58	10:16	10:25	10:33
S 60°	21:23	22:43	· ·	10:20	10:26	10:31

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer.Pass. Upper hh:mm	Lower hh:mm	18-20 84-67%
30	03:39	03:45	12:04	03:21	15:43	
01	03:50	03:56	12:04	04:04	16:25	
02	04:02	04:07	12:04	04:46	17:06	

July 03, 04, 05 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	281°18.5	150°45.5	N19°56.9	91°16.4	S04°25.4	84°10.2	S05°53.1	231°37.4	N16°04.1	Alpheratz	357°30.9	29°15.2	
1	296°21.0	165°44.8	56.1	106°17.8	25.9	99°12.6	53.1	246°39.6	04.1	Ankaa	353°03.7	-42°08.4	
2	311°23.4	180°44.2	55.4	121°19.2	26.4	114°14.9	53.2	261°41.8	04.2	Schedar	349°26.7	56°41.8	
3	326°25.9	195°43.5	•• 54.6	136°20.6	•• 26.9	129°17.3	•• 53.3	276°44.1	•• 04.2	Diphda	348°43.7	-17°49.3	
4	341°28.4	210°42.9	53.8	151°22.0	27.5	144°19.6	53.3	291°46.3	04.3	Achernar	335°17.8	-57°04.9	
5	356°30.8	225°42.3	53.1	166°23.4	28.0	159°21.9	53.4	306°48.5	04.3	Hamal	327°47.2	23°36.0	
6	11°33.3	240°41.6	N19°52.3	181°24.8	S04°28.5	174°24.3	S05°53.5	321°50.7	N16°04.4	Polaris	312°19.2	89°22.9	
7	26°35.7	255°41.0	51.5	196°26.2	29.0	189°26.6	53.5	336°52.9	04.5	Acamar	315°09.3	-40°11.0	
8	41°38.2	270°40.3	50.7	211°27.6	29.5	204°29.0	53.6	351°55.1	04.5	Menkar	314°02.5	4°12.3	
9	56°40.7	285°39.7	•• 50.0	226°29.0	•• 30.0	219°31.3	•• 53.7	6°57.4	•• 04.6	Mirfak	308°23.3	49°57.7	
10	71°43.1	300°39.0	49.2	241°30.4	30.5	234°33.6	53.7	21°59.6	04.6	Aldebaran	290°35.7	16°34.0	
11	86°45.6	315°38.4	48.4	256°31.8	31.0	249°36.0	53.8	37°01.8	04.7	Rigel	281°00.7	-8°10.1	
12	101°48.1	330°37.8	N19°47.6	271°33.2	S04°31.5	264°38.3	S05°53.8	52°04.0	N16°04.7	Capella	280°17.0	46°01.4	
13	116°50.5	345°37.1	46.9	286°34.6	32.0	279°40.6	53.9	67°06.2	04.8	Bellatrix	278°19.3	6°22.5	
14	131°53.0	0°36.5	46.1	301°36.0	32.5	294°43.0	54.0	82°08.4	04.9	Elnath	277°57.6	28°37.8	
15	146°55.5	15°35.8	•• 45.3	316°37.4	•• 33.0	309°45.3	•• 54.0	97°10.7	•• 04.9	Alnilam	275°34.4	-1°11.1	
16	161°57.9	30°35.2	44.5	331°38.8	33.5	324°47.6	54.1	112°12.9	05.0	Betelgeuse	270°48.5	7°24.6	
17	177°00.4	45°34.6	43.7	346°40.2	34.0	339°50.0	54.2	127°15.1	05.0	Canopus	263°51.3	-52°42.7	
18	192°02.9	60°33.9	N19°43.0	1°41.6	S04°34.6	354°52.3	S05°54.2	142°17.3	N16°05.1	Sirius	258°23.4	-16°45.5	
19	207°05.3	75°33.3	42.2	16°43.0	35.1	9°54.6	54.3	157°19.5	05.1	Adhara	255°03.4	-29°00.9	
20	222°07.8	90°32.7	41.4	31°44.4	35.6	24°57.0	54.4	172°21.8	05.2	Procyon	244°47.4	5°08.8	
21	237°10.2	105°32.0	•• 40.6	46°45.8	•• 36.1	39°59.3	•• 54.4	187°24.0	•• 05.3	Pollux	243°13.3	27°57.2	
22	252°12.7	120°31.4	39.8	61°47.2	36.6	55°01.6	54.5	202°26.2	05.3	Avior	234°13.8	-59°36.5	
23	267°15.2	135°30.8	39.0	76°48.6	37.1	70°04.0	54.6	217°28.4	05.4	Suhail	222°43.9	-43°33.4	
Mer.pass. 05:14		ν -0.6' d -0.8' m -3.89		ν 1.4' d 0.5' m 0.40		ν 2.3' d 0.1' m -2.05		ν 2.2' d 0.1' m 0.30		Miaplacidus		221°38.1	-69°50.6
										Alphard		217°44.4	-8°47.4
										Regulus		207°30.8	11°49.3
										Dubhe		193°37.1	61°35.6
										Denebola		182°21.4	14°24.4
										Gienah		175°39.9	-17°42.5
										Acrux		172°55.8	-63°16.1
										Gacrux		171°47.4	-57°17.1
										Alioth		166°10.0	55°48.2
										Spica		158°18.4	-11°19.0
										Alkaid		152°49.1	49°10.2
										Hadar		148°30.4	-60°31.2
										Menkent		147°53.1	-36°31.1
										Arcturus		145°44.5	19°01.8
										Rigel Kent.		139°34.8	-60°57.7
										Kochab		137°19.6	74°02.3
										Zuben'ubi		136°51.8	-16°09.9
										Alphecca		126°00.5	26°37.1
										Antares		112°11.0	-26°29.8
										Atria		107°01.4	-69°04.8
										Sabik		101°58.3	-15°45.6
										Shaula		96°05.0	-37°07.4
										Rasalhague		95°54.9	12°32.5
										Eltanin		90°40.0	51°29.3
										Kaus Aust.		83°27.3	-34°22.0
										Vega		80°30.3	38°48.8
										Nunki		75°42.9	-26°15.4
										Altair		61°56.1	8°57.0
										Peacock		52°59.6	-56°38.1
										Deneb		49°22.8	45°23.2
										Enif		33°35.0	10°00.8
										Al Na'ir		27°28.2	-46°48.7
										Fomalhaut		15°10.4	-29°27.6
										Scheat		13°41.5	28°14.6
										Markab		13°26.1	15°21.9
										Jul 03 Tue		SHA	Mer.pass
										Venus		229°27.0	13:58
										Mars		169°57.9	17:53
										Jupiter		162°51.7	18:20
										Saturn		310°18.9	08:32
										Jul 04 Wed		SHA	Mer.pass
										Venus		228°12.5	13:59
										Mars		169°32.4	17:51
										Jupiter		162°48.7	18:17
										Saturn		310°13.0	08:29
										Jul 05 Thu		SHA	Mer.pass
										Venus		226°58.3	14:00
										Mars		169°06.6	17:49
										Jupiter		162°45.4	18:13
										Saturn		310°07.1	08:25
										Horizontal parallax			
										Venus:		0.1	
										Mars:		0.1	

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	178°56.8	N22°56.9	280°31.8	16.0'	N06°05.4	12.0' 54.4'
1	193°56.7	56.7	295°06.8	16.0'	06°17.4	11.9' 54.4'
2	208°56.5	56.5	309°41.7	15.9'	06°29.3	11.9' 54.4'
3	223°56.4	· · 56.3	324°16.7	15.9'	06°41.2	11.9' 54.4'
4	238°56.3	56.1	338°51.6	15.9'	06°53.1	11.9' 54.4'
5	253°56.2	55.9	353°26.5	15.9'	07°05.0	11.8' 54.4'
6	268°56.1	N22°55.7	8°01.5	15.9'	N07°16.8	11.8' 54.3'
7	283°56.0	55.5	22°36.4	15.9'	07°28.6	11.8' 54.3'
8	298°55.9	55.3	37°11.2	15.9'	07°40.3	11.7' 54.3'
9	313°55.8	· · 55.1	51°46.1	15.9'	07°52.1	11.7' 54.3'
10	328°55.6	54.9	66°21.0	15.8'	08°03.7	11.7' 54.3'
11	343°55.5	54.7	80°55.8	15.8'	08°15.4	11.6' 54.3'
12	358°55.4	N22°54.5	95°30.6	15.8'	N08°27.0	11.6' 54.3'
13	13°55.3	54.3	110°05.4	15.8'	08°38.6	11.5' 54.3'
14	28°55.2	54.0	124°40.2	15.8'	08°50.1	11.5' 54.3'
15	43°55.1	· · 53.8	139°15.0	15.7'	09°01.7	11.5' 54.3'
16	58°55.0	53.6	153°49.7	15.7'	09°13.1	11.4' 54.3'
17	73°54.8	53.4	168°24.4	15.7'	09°24.6	11.4' 54.3'
18	88°54.7	N22°53.2	182°59.1	15.7'	N09°35.9	11.4' 54.2'
19	103°54.6	53.0	197°33.8	15.6'	09°47.3	11.3' 54.2'
20	118°54.5	52.8	212°08.4	15.6'	09°58.6	11.3' 54.2'
21	133°54.4	· · 52.5	226°43.1	15.6'	10°09.9	11.2' 54.2'
22	148°54.3	52.3	241°17.7	15.6'	10°21.1	11.2' 54.2'
23	163°54.2	52.1	255°52.2	15.5'	10°32.3	11.1' 54.2'
SD = 15.7' d = -0.2' SD = 14.8'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	178°54.1	N22°51.9	270°26.7	15.5'	N10°43.4	11.1' 54.2'
1	193°54.0	51.7	285°01.3	15.5'	10°54.5	11.1' 54.2'
2	208°53.8	51.4	299°35.7	15.4'	11°05.6	11.0' 54.2'
3	223°53.7	· · 51.2	314°10.2	15.4'	11°16.6	11.0' 54.2'
4	238°53.6	51.0	328°44.6	15.4'	11°27.6	10.9' 54.2'
5	253°53.5	50.8	343°19.0	15.3'	11°38.5	10.9' 54.2'
6	268°53.4	N22°50.5	357°53.3	15.3'	N11°49.4	10.8' 54.2'
7	283°53.3	50.3	12°27.6	15.3'	12°00.2	10.8' 54.2'
8	298°53.2	50.1	27°01.9	15.2'	12°11.0	10.7' 54.2'
9	313°53.1	· · 49.9	41°36.1	15.2'	12°21.7	10.7' 54.2'
10	328°53.0	49.6	56°10.3	15.2'	12°32.4	10.6' 54.2'
11	343°52.8	49.4	70°44.5	15.1'	12°43.0	10.6' 54.2'
12	358°52.7	N22°49.2	85°18.6	15.1'	N12°53.6	10.5' 54.2'
13	13°52.6	49.0	99°52.7	15.0'	13°04.1	10.5' 54.2'
14	28°52.5	48.7	114°26.8	15.0'	13°14.6	10.4' 54.2'
15	43°52.4	· · 48.5	129°00.8	15.0'	13°25.1	10.4' 54.2'
16	58°52.3	48.3	143°34.7	14.9'	13°35.4	10.3' 54.2'
17	73°52.2	48.0	158°08.6	14.9'	13°45.7	10.3' 54.2'
18	88°52.1	N22°47.8	172°42.5	14.8'	N13°56.0	10.2' 54.2'
19	103°52.0	47.6	187°16.4	14.8'	14°06.2	10.2' 54.2'
20	118°51.9	47.3	201°50.1	14.7'	14°16.4	10.1' 54.2'
21	133°51.7	· · 47.1	216°23.9	14.7'	14°26.5	10.0' 54.2'
22	148°51.6	46.9	230°57.6	14.7'	14°36.6	10.0' 54.2'
23	163°51.5	46.6	245°31.2	14.6'	14°46.5	9.9' 54.2'
SD = 15.7' d = -0.2' SD = 14.8'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	178°51.4	N22°46.4	260°04.8	14.6'	N14°56.4	9.9' 54.2'
1	193°51.3	46.2	274°38.4	14.5'	15°06.3	9.8' 54.2'
2	208°51.2	45.9	289°11.9	14.5'	15°16.1	9.7' 54.2'
3	223°51.1	· · 45.7	303°45.4	14.4'	15°25.8	9.7' 54.2'
4	238°51.0	45.5	318°18.8	14.4'	15°35.5	9.6' 54.2'
5	253°50.9	45.2	332°52.1	14.3'	15°45.1	9.6' 54.2'
6	268°50.8	N22°45.0	347°25.4	14.3'	N15°54.7	9.5' 54.2'
7	283°50.7	44.7	1°58.7	14.2'	16°04.2	9.4' 54.2'
8	298°50.6	44.5	16°31.9	14.2'	16°13.6	9.4' 54.2'
9	313°50.5	· · 44.3	31°05.1	14.1'	16°23.0	9.3' 54.2'
10	328°50.3	44.0	45°38.2	14.0'	16°32.3	9.2' 54.2'
11	343°50.2	43.8	60°11.2	14.0'	16°41.5	9.2' 54.2'
12	358°50.1	N22°43.5	74°44.2	13.9'	N16°50.7	9.1' 54.2'
13	13°50.0	43.3	89°17.1	13.9'	16°59.8	9.0' 54.2'
14	28°49.9	43.0	103°50.0	13.8'	17°08.8	9.0' 54.2'
15	43°49.8	· · 42.8	118°22.8	13.8'	17°17.8	8.9' 54.2'
16	58°49.7	42.5	132°55.6	13.7'	17°26.7	8.8' 54.3'
17	73°49.6	42.3	147°28.3	13.6'	17°35.5	8.7' 54.3'
18	88°49.5	N22°42.0	162°00.9	13.6'	N17°44.3	8.7' 54.3'
19	103°49.4	41.8	176°33.5	13.5'	17°52.9	8.6' 54.3'
20	118°49.3	41.5	191°06.1	13.5'	18°01.5	8.5' 54.3'
21	133°49.2	· · 41.3	205°38.5	13.4'	18°10.1	8.5' 54.3'
22	148°49.1	41.0	220°10.9	13.4'	18°18.5	8.4' 54.3'
23	163°49.0	40.8	234°43.3	13.3'	18°26.9	8.3' 54.3'
SD = 15.7' d = -0.2' SD = 14.8'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	--	--	00:43	23:21	--	--
64°	--	--	01:48	22:20	--	--
62°	--	--	02:22	21:46	--	--
60°	--	01:12	02:46	21:22	22:55	--
N 58°	--	01:54	03:05	21:03	22:14	--
56°	--	02:21	03:21	20:47	21:47	--
54°	01:06	02:42	03:35	20:33	21:26	23:01
52°	01:45	02:59	03:47	20:22	21:09	22:23
50°	02:10	03:13	03:57	20:11	20:55	21:58
45°	02:53	03:42	04:19	19:50	20:26	21:15
N 40°	03:23	04:04	04:37	19:32	20:05	20:46
35°	03:45	04:22	04:51	19:18	19:47	20:23
30°	04:03	04:37	05:04	19:05	19:32	20:05
20°	04:32	05:01	05:25	18:44	19:08	19:37
N 10°	04:54	05:21	05:44	18:25	18:48	19:15
0°	05:12	05:38	06:01	18:08	18:31	18:57
S 10°	05:29	05:55	06:18	17:51	18:14	18:40
20°	05:45	06:12	06:36	17:33	17:57	18:24
30°	06:00	06:30	06:56	17:13	17:39	18:09
35°	06:09	06:40	07:08	17:01	17:29	18:00
40°	06:18	06:52	07:22	16:47	17:17	17:51
45°	06:28	07:05	07:38	16:31	17:04	17:41
S 50°	06:39	07:20	07:59	16:11	16:49	17:30
52°	06:44	07:28	08:08	16:01	16:42	17:25
54°	06:50	07:35	08:19	15:51	16:34	17:20
56°	06:56	07:44	08:31	15:38	16:25	17:14
58°	07:02	07:54	08:45	15:24	16:15	17:07
S 60°	07:09	08:05	09:01	15:08	16:04	17:00

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	21:40	21:06	19:41	13:32	15:34	18:31
N 70°	21:56	21:35	21:02	13:18	15:07	17:11
68°	22:08	21:57	21:41	13:07	14:47	16:33
66°	22:19	22:14	22:09	12:58	14:31	16:07
64°	22:28	22:28	22:30	12:51	14:18	15:47
62°	22:35	22:40	22:47	12:44	14:07	15:31
60°	22:42	22:50	23:01	12:39	13:58	15:17
N 58°	22:48	22:59	23:13	12:34	13:49	15:05
56°	22:53	23:07	23:24	12:30	13:42	14:55
54°	22:58	23:14	23:33	12:26	13:36	14:47
52°	23:02	23:20	23:41	12:22	13:30	14:39
50°	23:06	23:26	23:49	12:19	13:25	14:32
45°	23:14	23:38	· ·	12:12	13:14	14:17
N 40°	23:21	23:48	· ·	12:06	13:05	14:04
35°	23:28	23:57	· ·	12:01	12:57	13:54
30°	23:33	· ·	00:05	11:57	12:51	13:45
20°	23:42	· ·	00:18	11:50	12:39	13:29
N 10°	23:51	· ·	00:30	11:43	12:29	13:15
0°	23:59	· ·	00:41	11:37	12:19	13:03
S 10°	· ·	00:06	00:52	11:31	12:09	12:50
20°	· ·	00:15	01:05	11:24	11:59	12:36
30°	· ·	00:25	01:18	11:17	11:48	12:21
35°	· ·	00:30	01:26	11:13	11:41	12:12
40°	· ·	00:37	01:36	11:08	11:33	12:02
45°	· ·	00:44	01:47	11:02	11:25	11:50
S 50°	· ·	00:53	02:00	10:56	11:14	11:35
52°	· ·	00:58	02:06	10:53	11:09	11:29
54°	· ·	01:02	02:13	10:49	11:04	11:21
56°	· ·	01:07	02:20	10:46	10:58	11:13
58°	· ·	01:13	02:29	10:42	10:52	11:04
S 60°	00:01	01:20	02:39	10:37	10:44	10:53

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
03	04:13	04:18	12:04	05:27	17:48	
04	04:24	04:29	12:04	06:09	18:30	
05	04:34	04:39	12:05	06:52	19:14	

July 06, 07, 08 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn	
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	284°15.9	150°00.4	N18°59.3	92°56.5	S05°02.2	86°58.0	S05°57.9	234°17.3	N16°08.1
1	299°18.4	164°59.8	58.5	107°57.8	02.7	102°00.3	58.0	249°19.5	08.2
2	314°20.8	179°59.2	57.7	122°59.2	03.2	117°02.6	58.0	264°21.7	08.2
3	329°23.3	194°58.6	•• 56.8	138°00.6	•• 03.7	132°04.9	•• 58.1	279°23.9	•• 08.3
4	344°25.8	209°58.0	56.0	153°01.9	04.2	147°07.3	58.2	294°26.2	08.3
5	359°28.2	224°57.4	55.2	168°03.3	04.7	162°09.6	58.2	309°28.4	08.4
6	14°30.7	239°56.8	N18°54.3	183°04.7	S05°05.3	177°11.9	S05°58.3	324°30.6	N16°08.4
7	29°33.2	254°56.2	53.5	198°06.1	05.8	192°14.2	58.4	339°32.8	08.5
8	44°35.6	269°55.6	52.6	213°07.4	06.3	207°16.5	58.5	354°35.1	08.6
9	59°38.1	284°55.0	•• 51.8	228°08.8	•• 06.8	222°18.8	•• 58.5	9°37.3	•• 08.6
10	74°40.6	299°54.4	51.0	243°10.2	07.3	237°21.2	58.6	24°39.5	08.7
11	89°43.0	314°53.8	50.1	258°11.5	07.8	252°23.5	58.7	39°41.7	08.7
12	104°45.5	329°53.2	N18°49.3	273°12.9	S05°08.4	267°25.8	S05°58.7	54°44.0	N16°08.8
13	119°47.9	344°52.6	48.4	288°14.3	08.9	282°28.1	58.8	69°46.2	08.8
14	134°50.4	359°52.0	47.6	303°15.6	09.4	297°30.4	58.9	84°48.4	08.9
15	149°52.9	14°51.4	•• 46.7	318°17.0	•• 09.9	312°32.7	•• 59.0	99°50.6	•• 08.9
16	164°55.3	29°50.8	45.9	333°18.4	10.4	327°35.1	59.0	114°52.9	09.0
17	179°57.8	44°50.2	45.1	348°19.7	10.9	342°37.4	59.1	129°55.1	09.1
18	195°00.3	59°49.6	N18°44.2	3°21.1	S05°11.5	357°39.7	S05°59.2	144°57.3	N16°09.1
19	210°02.7	74°49.0	43.4	18°22.5	12.0	12°42.0	59.2	159°59.5	09.2
20	225°05.2	89°48.4	42.5	33°23.8	12.5	27°44.3	59.3	175°01.8	09.2
21	240°07.7	104°47.8	•• 41.7	48°25.2	•• 13.0	42°46.6	•• 59.4	190°04.0	•• 09.3
22	255°10.1	119°47.2	40.8	63°26.6	13.5	57°48.9	59.5	205°06.2	09.3
23	270°12.6	134°46.6	40.0	78°27.9	14.0	72°51.3	59.5	220°08.5	09.4
Mer.pass. 05:02		ν -0.6' d -0.8' m -3.89		ν 1.4' d 0.5' m 0.44		ν 2.3' d 0.1' m -2.04		ν 2.2' d 0.1' m 0.30	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	285°15.1	149°46.0	N18°39.1	93°29.3	S05°14.6	87°53.6	S05°59.6	235°10.7	N16°09.4
1	300°17.5	164°45.5	38.2	108°30.7	15.1	102°55.9	59.7	250°12.9	09.5
2	315°20.0	179°44.9	37.4	123°32.0	15.6	117°58.2	59.8	265°15.1	09.5
3	330°22.4	194°44.3	•• 36.5	138°33.4	•• 16.1	133°00.5	•• 59.8	280°17.4	•• 09.6
4	345°24.9	209°43.7	35.7	153°34.7	16.6	148°02.8	05°59.9	295°19.6	09.6
5	0°27.4	224°43.1	34.8	168°36.1	17.2	163°05.1	06°00.0	310°21.8	09.7
6	15°29.8	239°42.5	N18°34.0	183°37.5	S05°17.7	178°07.5	S06°00.0	325°24.0	N16°09.8
7	30°32.3	254°41.9	33.1	198°38.8	18.2	193°09.8	00.1	340°26.3	09.8
8	45°34.8	269°41.3	32.2	213°40.2	18.7	208°12.1	00.2	355°28.5	09.9
9	60°37.2	284°40.7	•• 31.4	228°41.5	•• 19.2	223°14.4	•• 00.3	10°30.7	•• 09.9
10	75°39.7	299°40.2	30.5	243°42.9	19.7	238°16.7	00.3	25°33.0	10.0
11	90°42.2	314°39.6	29.7	258°44.3	20.3	253°19.0	00.4	40°35.2	10.0
12	105°44.6	329°39.0	N18°28.8	273°45.6	S05°20.8	268°21.3	S06°00.5	55°37.4	N16°10.1
13	120°47.1	344°38.4	27.9	288°47.0	21.3	283°23.6	00.6	70°39.6	10.1
14	135°49.6	359°37.8	27.1	303°48.3	21.8	298°25.9	00.6	85°41.9	10.2
15	150°52.0	14°37.2	•• 26.2	318°49.7	•• 22.3	313°28.3	•• 00.7	100°44.1	•• 10.2
16	165°54.5	29°36.7	25.3	333°51.0	22.9	328°30.6	00.8	115°46.3	10.3
17	180°56.9	44°36.1	24.5	348°52.4	23.4	343°32.9	00.9	130°48.6	10.3
18	195°59.4	59°35.5	N18°23.6	3°53.8	S05°23.9	358°35.2	S06°00.9	145°50.8	N16°10.4
19	211°01.9	74°34.9	22.7	18°55.1	24.4	13°37.5	01.0	160°53.0	10.5
20	226°04.3	89°34.3	21.9	33°56.5	24.9	28°39.8	01.1	175°55.2	10.5
21	241°06.8	104°33.7	•• 21.0	48°57.8	•• 25.5	43°42.1	•• 01.2	190°57.5	•• 10.6
22	256°09.3	119°33.2	20.1	63°59.2	26.0	58°44.4	01.2	205°59.7	10.6
23	271°11.7	134°32.6	19.3	79°00.5	26.5	73°46.7	01.3	221°01.9	10.7
Mer.pass. 04:58		ν -0.6' d -0.9' m -3.89		ν 1.4' d 0.5' m 0.45		ν 2.3' d 0.1' m -2.03		ν 2.2' d 0.1' m 0.29	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	286°14.2	149°32.0	N18°18.4	94°01.9	S05°27.0	88°49.0	S06°01.4	236°04.2	N16°10.7
1	301°16.7	164°31.4	17.5	109°03.2	27.5	103°51.3	01.5	251°06.4	10.8
2	316°19.1	179°30.9	16.6	124°04.6	28.1	118°53.6	01.5	266°08.6	10.8
3	331°21.6	194°30.3	•• 15.8	139°05.9	•• 28.6	133°56.0	•• 01.6	281°10.8	•• 10.9
4	346°24.0	209°29.7	14.9	154°07.3	29.1	148°58.3	01.7	296°13.1	10.9
5	1°26.5	224°29.1	14.0	169°08.6	29.6	164°00.6	01.8	311°15.3	11.0
6	16°29.0	239°28.6	N18°13.1	184°10.0	S05°30.1	179°02.9	S06°01.8	326°17.5	N16°11.0
7	31°31.4	254°28.0	12.2	199°11.3	30.7	194°05.2	01.9	341°19.8	11.1
8	46°33.9	269°27.4	11.4	214°12.7	31.2	209°07.5	02.0	356°22.0	11.1
9	61°36.4	284°26.8	•• 10.5	229°14.0	•• 31.7	224°09.8	•• 02.1	11°24.2	•• 11.2
10	76°38.8	299°26.3	09.6	244°15.4	32.2	239°12.1	02.2	26°26.5	11.3
11	91°41.3	314°25.7	08.7	259°16.7	32.7	254°14.4	02.2	41°28.7	11.3
12	106°43.8	329°25.1	N18°07.8	274°18.1	S05°33.3	269°16.7	S06°02.3	56°30.9	N16°11.4
13	121°46.2	344°24.5	06.9	289°19.4	33.8	284°19.0	02.4	71°33.1	11.4
14	136°48.7	359°24.0	06.1	304°20.8	34.3	299°21.3	02.5	86°35.4	11.5
15	151°51.2	14°23.4	•• 05.2	319°22.1	•• 34.8	314°23.6	•• 02.5	101°37.6	•• 11.5
16	166°53.6	29°22.8	04.3	334°23.5	35.4	329°25.9	02.6	116°39.8	11.6
17	181°56.1	44°22.3	03.4	349°24.8	35.9	344°28.2	02.7	131°42.1	11.6
18	196°58.5	59°21.7	N18°02.5	4°26.1	S05°36.4	359°30.5	S06°02.8	146°44.3	N16°11.7
19	212°01.0	74°21.1	01.6	19°27.5	36.9	14°32.8	02.8	161°46.5	11.7
20	227°03.5	89°20.6	18°00.7	34°28.8	37.4	29°35.1	02.9	176°48.8	11.8
21	242°05.9	104°20.0	17°59.8	49°30.2	•• 38.0	44°37.4	•• 03.0	191°51.0	•• 11.8
22	257°08.4	119°19.4	59.0	64°31.5	38.5	59°39.7	03.1	206°53.2	11.9
23	272°10.9	134°18.9	58.1	79°32.9	39.0	74°42.0	03.2	221°55.5	11.9
Mer.pass. 04:54		ν -0.6' d -0.9' m -3.90		ν 1.4' d 0.5' m 0.46		ν 2.3' d 0.1' m -2.02		ν 2.2' d 0.1' m 0.29	

Stars			SHA	Dec
Alpheratz	357°30.9	29°15.2		
Ankaa	353°03.6	-42°08.4		
Schedar	349°26.6	56°41.8		
Diphda	348°43.7	-17°49.3		
Achernar	335°17.8	-57°04.9		
Hamal	327°47.2	23°36.0		
Polaris	312°17.7	89°22.9		
Acamar	315°09.3	-40°11.0		
Menkar	314°02.5	4°12.3		
Mirfak	308°23.2	49°57.7		
Aldebaran	290°35.7	16°34.0		
Rigel	281°00.7	-8°10.1		
Capella	280°16.9	46°01.4		
Bellatrix	278°19.3	6°22.5		
Elnath	277°57.6	28°37.8		
Alnilam	275°34.4	-1°11.1		
Betelgeuse	270°48.5	7°24.6		
Canopus	263°51.3	-52°42.7		
Sirius	258°23.4	-16°45.5		
Adhara	255°03.4	-29°00.8		
Procyon	244°47.4	5°08.8		
Pollux	243°13.3	27°57.2		
Avior	234°13.8	-59°36.5		
Suhail	222°43.9	-43°33.4		
Miaplacidus	221°38.1	-69°50.6		
Alphard	217°44.4	-8°47.4		
Regulus	207°30.8	11°49.3		
Dubhe	193°37.2	61°35.6		
Denebola	182°21.4	14°24.4		
Gienah	175°39.9	-17°42.5		
Acrux	172°55.8	-63°16.1		
Gacrux	171°47.4	-57°17.1		
Alioth	166°10.0	55°48.2		
Spica	158°18.4	-11°19.0		
Alkaid	152°49.2	49°10.2		
Hadar	148°30.4	-60°31.2		
Menkent	147°53.1	-36°31.1		
Arcturus	145°44.5	19°01.8		
Rigel Kent.	139°34.8	-60°57.7		
Kochab	137°19.7	74°02.3		
Zuben'ubi	136°51.8	-16°09.9		
Alphecca	126°00.5	26°37.1		
Antares	112°11.0	-26°29.8		
Atria	107°01.4	-69°04.8		
Sabik	101°58.3	-15°45.6		
Shaula	96°05.0	-37°07.4		
Rasalhague	95°54.9	12°32.5		
Eltanin	90°40.0	51°29.3		
Kaus Aust.	83°27.3	-34°22.0		
Vega	80°30.3	38°48.8		
Nunki	75°42.9	-26°15.4		
Altair	61°56.1	8°57.0		
Peacock	52°59.6	-56°38.1		
Deneb	49°22.8	45°23.2		
Enif	33°35.0	10°00.8		
Al Na'ir	27°28.2	-46°48.7		
Fomalhaut	15°10.4	-29°27.6		
Scheat	13°41.4	28°14.6		
Markab	13°26.1	15°21.9		

Jul 06 Fri	SHA	Mer.pass
Venus	225°44.5	14:01
Mars	168°40.5	17:47
Jupiter	162°42.1	18:09
Saturn	310°01.4	08:22

Jul 07 Sat	SHA
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h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°48.9	N22°40.5	249°15.6	13.2'	N18°35.2	8.2'	54.3'
1	193°48.8	40.3	263°47.8	13.2'	18°43.4	8.2'	54.3'
2	208°48.7	40.0	278°20.0	13.1'	18°51.6	8.1'	54.3'
3	223°48.5	· · 39.8	292°52.1	13.0'	18°59.7	8.0'	54.3'
4	238°48.4	39.5	307°24.1	13.0'	19°07.7	7.9'	54.4'
5	253°48.3	39.3	321°56.1	12.9'	19°15.6	7.8'	54.4'
6	268°48.2	N22°39.0	336°28.0	12.9'	N19°23.4	7.8'	54.4'
7	283°48.1	38.8	350°59.9	12.8'	19°31.2	7.7'	54.4'
8	298°48.0	38.5	5°31.7	12.7'	19°38.8	7.6'	54.4'
9	313°47.9	· · 38.2	20°03.4	12.7'	19°46.4	7.5'	54.4'
10	328°47.8	38.0	34°35.1	12.6'	19°53.9	7.4'	54.4'
11	343°47.7	37.7	49°06.7	12.5'	20°01.3	7.3'	54.4'
12	358°47.6	N22°37.5	63°38.2	12.5'	N20°08.7	7.2'	54.4'
13	13°47.5	37.2	78°09.7	12.4'	20°15.9	7.2'	54.5'
14	28°47.4	36.9	92°41.1	12.3'	20°23.1	7.1'	54.5'
15	43°47.3	· · 36.7	107°12.5	12.3'	20°30.1	7.0'	54.5'
16	58°47.2	36.4	121°43.7	12.2'	20°37.1	6.9'	54.5'
17	73°47.1	36.1	136°15.0	12.2'	20°44.0	6.8'	54.5'
18	88°47.0	N22°35.9	150°46.1	12.1'	N20°50.8	6.7'	54.5'
19	103°46.9	35.6	165°17.2	12.0'	20°57.5	6.6'	54.5'
20	118°46.8	35.4	179°48.2	12.0'	21°04.1	6.5'	54.6'
21	133°46.7	· · 35.1	194°19.2	11.9'	21°10.7	6.4'	54.6'
22	148°46.6	34.8	208°50.1	11.8'	21°17.1	6.3'	54.6'
23	163°46.5	34.6	223°20.9	11.8'	21°23.4	6.2'	54.6'
SD = 15.7' d = -0.3' SD = 14.8'							
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°46.4	N22°34.3	237°51.7	11.7'	N21°29.7	6.1'	54.6'
1	193°46.3	34.0	252°22.4	11.6'	21°35.8	6.1'	54.6'
2	208°46.2	33.7	266°53.0	11.6'	21°41.9	6.0'	54.6'
3	223°46.1	· · 33.5	281°23.6	11.5'	21°47.8	5.9'	54.7'
4	238°46.0	33.2	295°54.1	11.4'	21°53.7	5.8'	54.7'
5	253°45.9	32.9	310°24.5	11.4'	21°59.4	5.7'	54.7'
6	268°45.8	N22°32.7	324°54.9	11.3'	N22°05.1	5.6'	54.7'
7	283°45.7	32.4	339°25.2	11.2'	22°10.6	5.5'	54.7'
8	298°45.6	32.1	353°55.4	11.2'	22°16.1	5.4'	54.7'
9	313°45.5	· · 31.8	8°25.6	11.1'	22°21.4	5.2'	54.8'
10	328°45.4	31.6	22°55.7	11.1'	22°26.7	5.1'	54.8'
11	343°45.3	31.3	37°25.8	11.0'	22°31.8	5.0'	54.8'
12	358°45.2	N22°31.0	51°55.8	10.9'	N22°36.9	4.9'	54.8'
13	13°45.1	30.7	66°25.7	10.9'	22°41.8	4.8'	54.8'
14	28°45.0	30.4	80°55.5	10.8'	22°46.6	4.7'	54.8'
15	43°44.9	· · 30.2	95°25.3	10.7'	22°51.4	4.6'	54.9'
16	58°44.8	29.9	109°55.1	10.7'	22°56.0	4.5'	54.9'
17	73°44.7	29.6	124°24.8	10.6'	23°00.5	4.4'	54.9'
18	88°44.6	N22°29.3	138°54.4	10.6'	N23°04.9	4.3'	54.9'
19	103°44.5	29.0	153°23.9	10.5'	23°09.2	4.2'	54.9'
20	118°44.4	28.8	167°53.4	10.4'	23°13.3	4.1'	55.0'
21	133°44.3	· · 28.5	182°22.9	10.4'	23°17.4	4.0'	55.0'
22	148°44.2	28.2	196°52.2	10.3'	23°21.3	3.8'	55.0'
23	163°44.1	27.9	211°21.6	10.3'	23°25.2	3.7'	55.0'
SD = 15.7' d = -0.3' SD = 14.9'							
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°44.0	N22°27.6	225°50.8	10.2'	N23°28.9	3.6'	55.0'
1	193°43.9	27.3	240°20.0	10.1'	23°32.5	3.5'	55.1'
2	208°43.8	27.1	254°49.2	10.1'	23°36.0	3.4'	55.1'
3	223°43.7	· · 26.8	269°18.3	10.0'	23°39.4	3.3'	55.1'
4	238°43.6	26.5	283°47.3	10.0'	23°42.7	3.1'	55.1'
5	253°43.5	26.2	298°16.3	9.9'	23°45.8	3.0'	55.1'
6	268°43.4	N22°25.9	312°45.2	9.9'	N23°48.8	2.9'	55.2'
7	283°43.3	25.6	327°14.1	9.8'	23°51.7	2.8'	55.2'
8	298°43.2	25.3	341°42.9	9.8'	23°54.5	2.7'	55.2'
9	313°43.1	· · 25.0	356°11.7	9.7'	23°57.2	2.5'	55.2'
10	328°43.0	24.7	10°40.4	9.7'	23°59.7	2.4'	55.2'
11	343°42.9	24.5	25°09.0	9.6'	24°02.2	2.3'	55.3'
12	358°42.8	N22°24.2	39°37.6	9.6'	N24°04.5	2.2'	55.3'
13	13°42.8	23.9	54°06.2	9.5'	24°06.7	2.1'	55.3'
14	28°42.7	23.6	68°34.7	9.5'	24°08.7	1.9'	55.3'
15	43°42.6	· · 23.3	83°03.2	9.4'	24°10.7	1.8'	55.3'
16	58°42.5	23.0	97°31.6	9.4'	24°12.5	1.7'	55.4'
17	73°42.4	22.7	112°00.0	9.3'	24°14.2	1.6'	55.4'
18	88°42.3	N22°22.4	126°28.3	9.3'	N24°15.7	1.4'	55.4'
19	103°42.2	22.1	140°56.6	9.2'	24°17.2	1.3'	55.4'
20	118°42.1	21.8	155°24.8	9.2'	24°18.5	1.2'	55.5'
21	133°42.0	· · 21.5	169°53.0	9.2'	24°19.7	1.1'	55.5'
22	148°41.9	21.2	184°21.2	9.1'	24°20.7	0.9'	55.5'
23	163°41.8	20.9	198°49.3	9.1'	24°21.7	0.8'	55.5'
SD = 15.7' d = -0.3' SD = 15.0'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	--	--	00:59	23:06	--	--
64°	--	--	01:55	22:13	--	--
62°	--	--	02:27	21:42	--	--
60°	--	01:21	02:50	21:19	22:47	--
N 58°	--	01:59	03:09	21:00	22:09	--
56°	--	02:25	03:25	20:45	21:44	--
54°	01:14	02:46	03:38	20:31	21:24	22:54
52°	01:50	03:02	03:50	20:20	21:07	22:19
50°	02:14	03:16	04:00	20:10	20:53	21:55
45°	02:56	03:44	04:21	19:48	20:25	21:13
N 40°	03:25	04:06	04:38	19:31	20:04	20:44
35°	03:47	04:23	04:53	19:17	19:46	20:22
30°	04:05	04:38	05:05	19:05	19:32	20:05
20°	04:33	05:02	05:26	18:44	19:08	19:37
N 10°	04:55	05:22	05:44	18:25	18:48	19:15
0°	05:13	05:39	06:01	18:09	18:31	18:57
S 10°	05:29	05:55	06:18	17:52	18:15	18:41
20°	05:45	06:12	06:36	17:34	17:58	18:25
30°	06:00	06:30	06:56	17:14	17:40	18:10
35°	06:09	06:40	07:08	17:02	17:30	18:02
40°	06:17	06:51	07:21	16:49	17:19	17:53
45°	06:27	07:04	07:37	16:33	17:06	17:43
S 50°	06:38	07:19	07:57	16:13	16:51	17:32
52°	06:43	07:26	08:06	16:04	16:44	17:27
54°	06:48	07:34	08:17	15:53	16:36	17:22
56°	06:54	07:43	08:29	15:42	16:28	17:16
58°	07:00	07:52	08:42	15:28	16:18	17:10
S 60°	07:07	08:03	08:59	15:12	16:07	17:03

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	21:12	☐	☐	18:39	☐	☐
66°	22:02	21:53	☐	17:49	19:41	☐
64°	22:34	22:46	23:15	17:18	18:49	20:07
62°	22:58	23:18	23:55	16:55	18:17	19:28
60°	23:17	23:42	· ·	16:37	17:53	19:00
N 58°	23:33	· ·	00:02	16:21	17:34	18:39
56°	23:46	· ·	00:18	16:08	17:18	18:22
54°	23:58	· ·	00:31	15:57	17:05	18:07
52°	· ·	00:08	00:43	15:47	16:53	17:54
50°	· ·	00:18	00:54	15:38	16:42	17:43
45°	00:05	00:37	01:17	15:19	16:21	17:19
N 40°	00:18	00:53	01:35	15:04	16:03	17:00
35°	00:30	01:07	01:50	14:51	15:48	16:44
30°	00:40	01:19	02:03	14:40	15:35	16:31
20°	00:57	01:39	02:25	14:21	15:14	16:07
N 10°	01:12	01:57	02:45	14:04	14:55	15:47
0°	01:26	02:13	03:03	13:48	14:37	15:28
S 10°	01:40	02:30	03:22	13:33	14:20	15:09
20°	01:55	02:48	03:41	13:16	14:01	14:49
30°	02:13	03:09	04:04	12:58	13:39	14:26
35°	02:23	03:21	04:18	12:47	13:26	14:12
40°	02:35	03:35	04:33	12:34	13:12	13:57
45°	02:49	03:51	04:52	12:19	12:55	13:38
S 50°	03:06	04:12	05:15	12:01	12:33	13:14
52°	03:14	04:22	05:26	11:53	12:23	13:03
54°	03:23	04:33	05:39	11:43	12:12	12:50
56°	03:34	04:46	05:53	11:32	11:59	12:36
58°	03:45	05:00	06:10	11:20	11:44	12:18
S 60°	03:59	05:18	06:31	11:06	11:26	11:57

Day	Sun		Mer. Pass hh:mm	Moon		Age 24-26 29-14%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer.Pass. hh:mm	Lower Mer.Pass. hh:mm	
06	04:45	04:50	12:05	07:37	20:01	
07	04:54	04:59	12:05	08:25	20:50	
08	05:04	05:09	12:05	09:16	21:42	

July 09, 10, 11 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	287°13.3	149°18.3	N17°57.2	94°34.2	S05°39.5	89°44.3	S06°03.2	236°57.7	N16°12.0	Alpheratz 357°30.8 29°15.2 Ankaa 353°03.6 -42°08.4 Schedar 349°26.6 56°41.8 Diphda 348°43.7 -17°49.2 Achernar 335°17.7 -57°04.8 Hamal 327°47.1 23°36.0 Polaris 312°16.0 89°22.9 Acamar 315°09.3 -40°11.0 Menkar 314°02.5 4°12.3 Mirfak 308°23.2 49°57.7 Aldebaran 290°35.7 16°34.0 Rigel 281°00.7 -8°10.1 Capella 280°16.9 46°01.4 Bellatrix 278°19.3 6°22.5 Elnath 277°57.6 28°37.8 Alnilam 275°34.3 -1°11.1 Betelgeuse 270°48.5 7°24.6 Canopus 263°51.3 -52°42.7 Sirius 258°23.4 -16°45.5 Adhara 255°03.4 -29°00.8 Procyon 244°47.4 5°08.8 Pollux 243°13.3 27°57.2 Avior 234°13.8 -59°36.5 Suhail 222°43.9 -43°33.3 Miaplacidus 221°38.1 -69°50.6 Alphard 217°44.4 -8°47.4 Regulus 207°30.8 11°49.3 Dubhe 193°37.2 61°35.6 Denebola 182°21.4 14°24.4 Gienah 175°39.9 -17°42.5 Acrux 172°55.9 -63°16.1 Gacrux 171°47.5 -57°17.1 Alioth 166°10.0 55°48.2 Spica 158°18.4 -11°19.0 Alkaid 152°49.2 49°10.2 Hadar 148°30.4 -60°31.2 Menkent 147°53.1 -36°31.1 Arcturus 145°44.5 19°01.8 Rigel Kent. 139°34.8 -60°57.7 Kochab 137°19.7 74°02.3 Zuben'ubi 136°51.8 -16°09.9 Alphecca 126°00.5 26°37.1 Antares 112°11.0 -26°29.8 Atria 107°01.4 -69°04.8 Sabik 101°58.3 -15°45.6 Shaula 96°05.0 -37°07.4 Rasalhague 95°54.9 12°32.5 Eltanin 90°40.0 51°29.3 Kaus Aust. 83°27.3 -34°22.0 Vega 80°30.3 38°48.8 Nunki 75°42.9 -26°15.4 Altair 61°56.1 8°57.0 Peacock 52°59.6 -56°38.1 Deneb 49°22.8 45°23.2 Enif 33°34.9 10°00.8 Al Na'ir 27°28.1 -46°48.7 Fomalhaut 15°10.4 -29°27.6 Scheat 13°41.4 28°14.6 Markab 13°26.1 15°22.0							
1	302°15.8	164°17.7	56.3	109°35.5	40.1	104°46.6	03.3	251°59.9	12.0								
2	317°18.3	179°17.2	55.4	124°36.9	40.6	119°48.9	03.4	267°02.2	12.1								
3	332°20.7	194°16.6	•• 54.5	139°38.2	•• 41.1	134°51.2	•• 03.5	282°04.4	•• 12.1								
4	347°23.2	209°16.1	53.6	154°39.6	41.6	149°53.5	03.5	297°06.6	12.2								
5	2°25.7	224°15.5	52.7	169°40.9	42.2	164°55.8	03.6	312°08.9	12.3								
6	17°28.1	239°14.9	N17°51.8	184°42.2	S05°42.7	179°58.1	S06°03.7	327°11.1	N16°12.3								
7	32°30.6	254°14.4	50.9	199°43.6	43.2	195°00.4	03.8	342°13.3	12.4								
8	47°33.0	269°13.8	50.0	214°44.9	43.7	210°02.7	03.9	357°15.6	12.4								
9	62°35.5	284°13.3	•• 49.1	229°46.3	•• 44.3	225°05.0	•• 03.9	12°17.8	•• 12.5								
10	77°38.0	299°12.7	48.2	244°47.6	44.8	240°07.3	04.0	27°20.0	12.5								
11	92°40.4	314°12.1	47.3	259°48.9	45.3	255°09.6	04.1	42°22.3	12.6								
12	107°42.9	329°11.6	N17°46.4	274°50.3	S05°45.8	270°11.9	S06°04.2	57°24.5	N16°12.6								
13	122°45.4	344°11.0	45.5	289°51.6	46.3	285°14.2	04.3	72°26.7	12.7								
14	137°47.8	359°10.5	44.6	304°52.9	46.9	300°16.5	04.3	87°29.0	12.7								
15	152°50.3	14°09.9	•• 43.7	319°54.3	•• 47.4	315°18.8	•• 04.4	102°31.2	•• 12.8								
16	167°52.8	29°09.4	42.8	334°55.6	47.9	330°21.1	04.5	117°33.4	12.8								
17	182°55.2	44°08.8	41.9	349°56.9	48.4	345°23.4	04.6	132°35.7	12.9								
18	197°57.7	59°08.3	N17°41.0	4°58.3	S05°49.0	0°25.7	S06°04.7	147°37.9	N16°12.9								
19	213°00.2	74°07.7	40.0	19°59.6	49.5	15°28.0	04.7	162°40.1	13.0								
20	228°02.6	89°07.1	39.1	35°00.9	50.0	30°30.3	04.8	177°42.4	13.0								
21	243°05.1	104°06.6	•• 38.2	50°02.3	•• 50.5	45°32.6	•• 04.9	192°44.6	•• 13.1								
22	258°07.5	119°06.0	37.3	65°03.6	51.1	60°34.9	05.0	207°46.8	13.1								
23	273°10.0	134°05.5	36.4	80°04.9	51.6	75°37.2	05.1	222°49.1	13.2								
Mer.pass. 04:50		ν -0.6' d -0.9' m -3.90			ν 1.3' d 0.5' m 0.47			ν 2.3' d 0.1' m -2.02			ν 2.2' d 0.1' m 0.29						

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	288°12.5	149°04.9	N17°35.5	95°06.3	S05°52.1	90°39.5	S06°05.1	237°51.3	N16°13.2				
1	303°14.9	164°04.4	34.6	110°07.6	52.7	105°41.8	05.2	252°53.5	13.3				
2	318°17.4	179°03.8	33.7	125°08.9	53.2	120°44.1	05.3	267°55.8	13.3				
3	333°19.9	194°03.3	•• 32.7	140°10.3	•• 53.7	135°46.4	•• 05.4	282°58.0	•• 13.4				
4	348°22.3	209°02.7	31.8	155°11.6	54.2	150°48.7	05.5	298°00.2	13.4				
5	3°24.8	224°02.2	30.9	170°12.9	54.8	165°51.0	05.5	313°02.5	13.5				
6	18°27.3	239°01.6	N17°30.0	185°14.3	S05°55.3	180°53.2	S06°05.6	328°04.7	N16°13.5				
7	33°29.7	254°01.1	29.1	200°15.6	55.8	195°55.5	05.7	343°06.9	13.6				
8	48°32.2	269°00.6	28.2	215°16.9	56.3	210°57.8	05.8	358°09.2	13.6				
9	63°34.7	284°00.0	•• 27.2	230°18.2	•• 56.9	226°00.1	•• 05.9	13°11.4	•• 13.7				
10	78°37.1	298°59.5	26.3	245°19.6	57.4	241°02.4	05.9	28°13.7	13.8				
11	93°39.6	313°58.9	25.4	260°20.9	57.9	256°04.7	06.0	43°15.9	13.8				
12	108°42.0	328°58.4	N17°24.5	275°22.2	S05°58.4	271°07.0	S06°06.1	58°18.1	N16°13.9				
13	123°44.5	343°57.8	23.6	290°23.5	59.0	286°09.3	06.2	73°20.4	13.9				
14	138°47.0	358°57.3	22.6	305°24.9	05°59.5	301°11.6	06.3	88°22.6	14.0				
15	153°49.4	13°56.8	•• 21.7	320°26.2	06°00.0	316°13.9	•• 06.4	103°24.8	•• 14.0				
16	168°51.9	28°56.2	20.8	335°27.5	00.5	331°16.2	06.4	118°27.1	14.1				
17	183°54.4	43°55.7	19.9	350°28.8	01.1	346°18.5	06.5	133°29.3	14.1				
18	198°56.8	58°55.1	N17°18.9	5°30.2	S06°01.6	1°20.8	S06°06.6	148°31.6	N16°14.2				
19	213°59.3	73°54.6	18.0	20°31.5	02.1	16°23.0	06.7	163°33.8	14.2				
20	229°01.8	88°54.1	17.1	35°32.8	02.7	31°25.3	06.8	178°36.0	14.3				
21	244°04.2	103°53.5	•• 16.1	50°34.1	•• 03.2	46°27.6	•• 06.8	193°38.3	•• 14.3				
22	259°06.7	118°53.0	15.2	65°35.5	03.7	61°29.9	06.9	208°40.5	14.4				
23	274°09.2	133°52.4	14.3	80°36.8	04.2	76°32.2	07.0	223°42.7	14.4				
Mer.pass. 04:46		ν -0.5' d -0.9' m -3.90			ν 1.3' d 0.5' m 0.48			ν 2.3' d 0.1' m -2.01			ν 2.2' d 0.1' m 0.29		

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	289°11.6	148°51.9	N17°13.3	95°38.1	S06°04.8	91°34.5	S06°07.1	238°45.0	N16°14.5				
1	304°14.1	163°51.4	12.4	110°39.4	05.3	106°36.8	07.2	253°47.2	14.5				
2	319°16.5	178°50.8	11.5	125°40.7	05.8	121°39.1	07.3	268°49.5	14.6				
3	334°19.0	193°50.3	•• 10.5	140°42.1	•• 06.4	136°41.4	•• 07.3	283°51.7	•• 14.6				
4	349°21.5	208°49.8	09.6	155°43.4	06.9	151°43.6	07.4	298°53.9	14.7				
5	4°23.9	223°49.2	08.7	170°44.7	07.4	166°45.9	07.5	313°56.2	14.7				
6	19°26.4	238°48.7	N17°07.7	185°46.0	S06°07.9	181°48.2	S06°07.6	328°58.4	N16°14.8				
7	34°28.9	253°48.2	06.8	200°47.3	08.5	196°50.5	07.7	344°00.6	14.8				
8	49°31.3	268°47.6	05.9	215°48.7	09.0	211°52.8	07.8	359°02.9	14.9				
9	64°33.8	283°47.1	•• 04.9	230°50.0	•• 09.5	226°55.1	•• 07.8	14°05.1	•• 14.9				
10	79°36.3	298°46.6	04.0	245°51.3	10.1	241°57.4	07.9	29°07.4	15.0				
11	94°38.7	313°46.0	03.0	260°52.6	10.6	256°59.7	08.0	44°09.6	15.0				
12	109°41.2	328°45.5	N17°02.1	275°53.9	S06°11.1	272°01.9	S06°08.1	59°11.8	N16°15.1				
13	124°43.6	343°45.0	01.2	290°55.2	11.6	287°04.3	08.2	74°14.1	15.1				
14	139°46.1	358°44.4	17°00.2	305°56.6	12.2	302°06.5	08.3	89°16.3	15.2				
15	154°48.6	13°43.9	16°59.3	320°57.9	•• 12.7	317°08.8	•• 08.3	104°18.6	•• 15.2				
16	169°51.0	28°43.4	58.3	335°59.2	13.2	332°11.1	08.4	119°20.8	15.3				
17	184°53.5	43°42.9	57.4	351°00.5	13.8	347°13.4	08.5	134°23.0	15.3				
18	199°56.0	58°42.3	N16°56.4	6°01.8	S06°14.3	2°15.6	S06°08.6	149°25.3	N16°15.4				
19	214°58.4	73°41.8	55.5	21°03.1	14.8	17°17.9	08.7	164°27.5	15.4				
20	230°00.9	88°41.3	54.6	36°04.4	15.3	32°20.2	08.8	179°29.8	15.5				
21	245°03.4	103°40.8	•• 53.6	51°05.7	•• 15.9	47°22.5	•• 08.9	194°32.0	•• 15.5				
22	260°05.8	118°40.2	52.7	66°07.1	16.4	62°24.8	08.9	209°34.2	15.6				
23	275°08.3	133°39.7	51.7	81°08.4	16.9	77°27.1	09.0	224°36.5	15.6				
Mer.pass. 04:42		ν -0.5' d -0.9' m -3.90			ν 1.3' d 0.5' m 0.49			ν 2.3' d 0.1' m -2.00			ν 2.2' d 0.1' m 0.29		

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	HP
0	178°41.7	N22°20.6	213°17.4	9.0'	N24°22.5	0.7' 55.6'
1	193°41.6	20.3	227°45.4	9.0'	24°23.1	0.5' 55.6'
2	208°41.5	20.0	242°13.4	9.0'	24°23.7	0.4' 55.6'
3	223°41.4	· · 19.7	256°41.4	8.9'	24°24.1	0.3' 55.6'
4	238°41.3	19.4	271°09.3	8.9'	24°24.4	0.2' 55.6'
5	253°41.3	19.1	285°37.2	8.9'	24°24.5	0.0' 55.7'
6	268°41.2	N22°18.8	300°05.1	8.8'	N24°24.6	-0.1' 55.7'
7	283°41.1	18.5	314°32.9	8.8'	24°24.4	-0.2' 55.7'
8	298°41.0	18.2	329°00.7	8.8'	24°24.2	-0.4' 55.7'
9	313°40.9	· · 17.8	343°28.5	8.7'	24°23.8	-0.5' 55.8'
10	328°40.8	17.5	357°56.3	8.7'	24°23.3	-0.6' 55.8'
11	343°40.7	17.2	12°24.0	8.7'	24°22.7	-0.8' 55.8'
12	358°40.6	N22°16.9	26°51.7	8.7'	N24°21.9	-0.9' 55.8'
13	13°40.5	16.6	41°19.3	8.6'	24°21.0	-1.0' 55.9'
14	28°40.4	16.3	55°47.0	8.6'	24°20.0	-1.2' 55.9'
15	43°40.3	· · 16.0	70°14.6	8.6'	24°18.8	-1.3' 55.9'
16	58°40.2	15.7	84°42.2	8.6'	24°17.5	-1.4' 55.9'
17	73°40.2	15.4	99°09.8	8.6'	24°16.1	-1.6' 56.0'
18	88°40.1	N22°15.1	113°37.3	8.5'	N24°14.5	-1.7' 56.0'
19	103°40.0	14.7	128°04.9	8.5'	24°12.8	-1.8' 56.0'
20	118°39.9	14.4	142°32.4	8.5'	24°11.0	-2.0' 56.0'
21	133°39.8	· · 14.1	156°59.9	8.5'	24°09.0	-2.1' 56.1'
22	148°39.7	13.8	171°27.4	8.5'	24°06.9	-2.2' 56.1'
23	163°39.6	13.5	185°54.9	8.5'	24°04.7	-2.4' 56.1'
SD = 15.7' d = -0.3' SD = 15.2'						
Tue	GHA	Dec	GHA	ν	Dec	HP
0	178°39.5	N22°13.2	200°22.3	8.5'	N24°02.3	-2.5' 56.1'
1	193°39.4	12.8	214°49.8	8.4'	23°59.8	-2.6' 56.2'
2	208°39.4	12.5	229°17.3	8.4'	23°57.1	-2.8' 56.2'
3	223°39.3	· · 12.2	243°44.7	8.4'	23°54.4	-2.9' 56.2'
4	238°39.2	11.9	258°12.1	8.4'	23°51.4	-3.1' 56.2'
5	253°39.1	11.6	272°39.5	8.4'	23°48.4	-3.2' 56.3'
6	268°39.0	N22°11.2	287°07.0	8.4'	N23°45.2	-3.3' 56.3'
7	283°38.9	10.9	301°34.4	8.4'	23°41.9	-3.5' 56.3'
8	298°38.8	10.6	316°01.8	8.4'	23°38.4	-3.6' 56.3'
9	313°38.7	· · 10.3	330°29.2	8.4'	23°34.8	-3.7' 56.4'
10	328°38.7	10.0	344°56.6	8.4'	23°31.1	-3.9' 56.4'
11	343°38.6	09.6	359°24.0	8.4'	23°27.3	-4.0' 56.4'
12	358°38.5	N22°09.3	13°51.4	8.4'	N23°23.3	-4.1' 56.4'
13	13°38.4	09.0	28°18.8	8.4'	23°19.1	-4.3' 56.5'
14	28°38.3	08.7	42°46.3	8.4'	23°14.9	-4.4' 56.5'
15	43°38.2	· · 08.3	57°13.7	8.4'	23°10.5	-4.5' 56.5'
16	58°38.1	08.0	71°41.1	8.4'	23°06.0	-4.7' 56.5'
17	73°38.0	07.7	86°08.6	8.4'	23°01.3	-4.8' 56.6'
18	88°38.0	N22°07.3	100°36.0	8.5'	N22°56.5	-4.9' 56.6'
19	103°37.9	07.0	115°03.5	8.5'	22°51.6	-5.1' 56.6'
20	118°37.8	06.7	129°30.9	8.5'	22°46.5	-5.2' 56.6'
21	133°37.7	· · 06.3	143°58.4	8.5'	22°41.3	-5.3' 56.7'
22	148°37.6	06.0	158°25.9	8.5'	22°36.0	-5.5' 56.7'
23	163°37.5	05.7	172°53.4	8.5'	22°30.6	-5.6' 56.7'
SD = 15.7' d = -0.3' SD = 15.3'						
Wed	GHA	Dec	GHA	ν	Dec	HP
0	178°37.5	N22°05.4	187°20.9	8.5'	N22°25.0	-5.7' 56.7'
1	193°37.4	05.0	201°48.4	8.5'	22°19.3	-5.8' 56.8'
2	208°37.3	04.7	216°16.0	8.6'	22°13.4	-6.0' 56.8'
3	223°37.2	· · 04.3	230°43.6	8.6'	22°07.5	-6.1' 56.8'
4	238°37.1	04.0	245°11.1	8.6'	22°01.4	-6.2' 56.9'
5	253°37.0	03.7	259°38.7	8.6'	21°55.2	-6.4' 56.9'
6	268°36.9	N22°03.3	274°06.4	8.6'	N21°48.8	-6.5' 56.9'
7	283°36.9	03.0	288°34.0	8.7'	21°42.3	-6.6' 56.9'
8	298°36.8	02.7	303°01.7	8.7'	21°35.7	-6.7' 57.0'
9	313°36.7	· · 02.3	317°29.4	8.7'	21°29.0	-6.9' 57.0'
10	328°36.6	02.0	331°57.1	8.7'	21°22.1	-7.0' 57.0'
11	343°36.5	01.6	346°24.8	8.8'	21°15.1	-7.1' 57.0'
12	358°36.5	N22°01.3	0°52.6	8.8'	N21°08.0	-7.2' 57.0'
13	13°36.4	01.0	15°20.4	8.8'	21°00.8	-7.4' 57.1'
14	28°36.3	00.6	29°48.2	8.8'	20°53.4	-7.5' 57.1'
15	43°36.2	22°00.3	44°16.0	8.9'	20°45.9	-7.6' 57.1'
16	58°36.1	21°59.9	58°43.9	8.9'	20°38.3	-7.7' 57.1'
17	73°36.0	59.6	73°11.8	8.9'	20°30.6	-7.8' 57.2'
18	88°36.0	N21°59.2	87°39.7	9.0'	N20°22.8	-8.0' 57.2'
19	103°35.9	58.9	102°07.7	9.0'	20°14.8	-8.1' 57.2'
20	118°35.8	58.6	116°35.7	9.0'	20°06.7	-8.2' 57.2'
21	133°35.7	· · 58.2	131°03.7	9.0'	19°58.5	-8.3' 57.3'
22	148°35.6	57.9	145°31.7	9.1'	19°50.2	-8.4' 57.3'
23	163°35.6	57.5	159°59.8	9.1'	19°41.8	-8.6' 57.3'
SD = 15.7' d = -0.3' SD = 15.5'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	--	--	01:14	22:53	--	--
64°	--	--	02:02	22:07	--	--
62°	--	--	02:32	21:37	--	--
60°	--	01:30	02:55	21:15	22:38	--
N 58°	--	02:05	03:13	20:57	22:04	--
56°	--	02:30	03:28	20:42	21:39	--
54°	01:22	02:50	03:41	20:29	21:20	22:46
52°	01:55	03:06	03:53	20:18	21:04	22:14
50°	02:19	03:20	04:03	20:08	20:51	21:51
45°	02:59	03:47	04:23	19:47	20:23	21:11
N 40°	03:28	04:08	04:40	19:30	20:02	20:43
35°	03:49	04:25	04:54	19:16	19:45	20:21
30°	04:07	04:40	05:07	19:04	19:31	20:04
20°	04:34	05:03	05:27	18:43	19:08	19:36
N 10°	04:56	05:22	05:45	18:26	18:48	19:15
0°	05:13	05:39	06:02	18:09	18:31	18:57
S 10°	05:29	05:56	06:18	17:53	18:15	18:41
20°	05:45	06:12	06:36	17:35	17:59	18:26
30°	06:00	06:30	06:56	17:15	17:41	18:11
35°	06:08	06:39	07:07	17:04	17:32	18:03
40°	06:17	06:50	07:20	16:51	17:21	17:54
45°	06:26	07:03	07:36	16:35	17:08	17:45
S 50°	06:37	07:18	07:55	16:16	16:53	17:34
52°	06:42	07:25	08:04	16:07	16:47	17:30
54°	06:47	07:32	08:15	15:57	16:39	17:24
56°	06:52	07:40	08:26	15:45	16:31	17:19
58°	06:58	07:50	08:39	15:32	16:22	17:13
S 60°	07:05	08:00	08:55	15:16	16:11	17:06

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°						
N 70°						23:28
68°						22:27
66°			01:08		22:00	21:52
64°	· · ·	00:18	01:49	20:56	21:18	21:27
62°	· · ·	00:55	02:17	20:19	20:50	21:07
60°	00:22	01:21	02:38	19:52	20:28	20:50
N 58°	00:44	01:42	02:56	19:31	20:10	20:36
56°	01:01	01:59	03:11	19:14	19:55	20:24
54°	01:16	02:14	03:23	19:00	19:42	20:14
52°	01:29	02:26	03:34	18:47	19:30	20:05
50°	01:40	02:37	03:44	18:36	19:20	19:56
45°	02:04	03:01	04:05	18:12	18:59	19:38
N 40°	02:23	03:19	04:22	17:53	18:41	19:23
35°	02:39	03:35	04:36	17:38	18:27	19:11
30°	02:53	03:48	04:48	17:24	18:14	19:00
20°	03:16	04:11	05:09	17:01	17:52	18:41
N 10°	03:37	04:31	05:27	16:40	17:33	18:25
0°	03:56	04:49	05:44	16:21	17:15	18:09
S 10°	04:15	05:08	06:00	16:03	16:58	17:54
20°	04:35	05:28	06:18	15:42	16:38	17:37
30°	04:59	05:51	06:39	15:19	16:16	17:17
35°	05:13	06:04	06:51	15:05	16:03	17:06
40°	05:29	06:20	07:05	14:49	15:48	16:53
45°	05:48	06:38	07:21	14:30	15:30	16:37
S 50°	06:12	07:01	07:41	14:06	15:08	16:18
52°	06:23	07:12	07:50	13:54	14:57	16:09
54°	06:37	07:24	08:01	13:41	14:45	15:59
56°	06:52	07:38	08:13	13:26	14:31	15:47
58°	07:10	07:55	08:27	13:08	14:15	15:34
S 60°	07:32	08:15	08:43	12:46	13:55	15:18

Day	Sun		Mer. Pass	Moon		Age 27-0 8-0%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
09	05:13	05:18	12:05	10:09	22:35	
10	05:22	05:26	12:05	11:02	23:30	
11	05:30	05:34	12:06	11:56	--	

July 12, 13, 14 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	290°10.8	148°39.2	N16°50.8	96°09.7	S06°17.5	92°29.3	S06°09.1	239°38.7	N16°15.7					Alpheratz	357°30.8	29°15.2	
1	305°13.2	163°38.7	49.8	111°11.0	18.0	107°31.6	09.2	254°41.0	15.7					Ankaa	353°03.6	-42°08.4	
2	320°15.7	178°38.1	48.9	126°12.3	18.5	122°33.9	09.3	269°43.2	15.8					Schedar	349°26.6	56°41.8	
3	335°18.1	193°37.6	· · 47.9	141°13.6	· · 19.1	137°36.2	· · 09.4	284°45.4	· · 15.8					Diphda	348°43.6	-17°49.2	
4	350°20.6	208°37.1	46.9	156°14.9	19.6	152°38.5	09.5	299°47.7	15.9					Achernar	335°17.7	-57°04.8	
5	5°23.1	223°36.6	46.0	171°16.2	20.1	167°40.8	09.5	314°49.9	15.9					Hamal	327°47.1	23°36.1	
6	20°25.5	238°36.1	N16°45.0	186°17.5	S06°20.7	182°43.0	S06°09.6	329°52.2	N16°16.0					Polaris	312°14.5	89°22.9	
7	35°28.0	253°35.5	44.1	201°18.8	21.2	197°45.3	09.7	344°54.4	16.0					Acamar	315°09.3	-40°10.9	
8	50°30.5	268°35.0	43.1	216°20.2	21.7	212°47.6	09.8	359°56.6	16.1					Menkar	314°02.5	4°12.3	
9	65°32.9	283°34.5	· · 42.2	231°21.5	· · 22.2	227°49.9	· · 09.9	14°58.9	· · 16.1					Mirfak	308°23.2	49°57.7	
10	80°35.4	298°34.0	41.2	246°22.8	22.8	242°52.2	10.0	30°01.1	16.2					Aldebaran	290°35.7	16°34.0	
11	95°37.9	313°33.5	40.3	261°24.1	23.3	257°54.4	10.1	45°03.4	16.2					Rigel	281°00.7	-8°10.1	
12	110°40.3	328°33.0	N16°39.3	276°25.4	S06°23.8	272°56.7	S06°10.1	60°05.6	N16°16.3					Capella	280°16.9	46°01.4	
13	125°42.8	343°32.4	38.3	291°26.7	24.4	287°59.0	10.2	75°07.9	16.3					Bellatrix	278°19.3	6°22.5	
14	140°45.3	358°31.9	37.4	306°28.0	24.9	303°01.3	10.3	90°10.1	16.4					Elnath	277°57.6	28°37.8	
15	155°47.7	13°31.4	· · 36.4	321°29.3	· · 25.4	318°03.6	· · 10.4	105°12.3	· · 16.4					Alnilam	275°34.3	-1°11.1	
16	170°50.2	28°30.9	35.4	336°30.6	26.0	333°05.8	10.5	120°14.6	16.5					Betelgeuse	270°48.4	7°24.6	
17	185°52.6	43°30.4	34.5	351°31.9	26.5	348°08.1	10.6	135°16.8	16.5					Canopus	263°51.3	-52°42.7	
18	200°55.1	58°29.9	N16°33.5	6°33.2	S06°27.0	3°10.4	S06°10.7	150°19.1	N16°16.6					Sirius	258°23.4	-16°45.5	
19	215°57.6	73°29.4	32.6	21°34.5	27.6	18°12.7	10.7	165°21.3	16.6					Adhara	255°03.4	-29°00.8	
20	231°00.0	88°28.9	31.6	36°35.8	28.1	33°15.0	10.8	180°23.6	16.7					Procyon	244°47.4	5°08.8	
21	246°02.5	103°28.3	· · 30.6	51°37.1	· · 28.6	48°17.2	· · 10.9	195°25.8	· · 16.7					Pollux	243°13.3	27°57.2	
22	261°05.0	118°27.8	29.7	66°38.4	29.2	63°19.5	11.0	210°28.0	16.7					Avior	234°13.8	-59°36.4	
23	276°07.4	133°27.3	28.7	81°39.7	29.7	78°21.8	11.1	225°30.3	16.8					Suhail	222°43.9	-43°33.3	
Mer.pass. 04:39			ν -0.5' d -1.0' m -3.90			ν 1.3' d 0.5' m 0.50			ν 2.3' d 0.1' m -2.00			ν 2.2' d 0.0' m 0.28					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	291°09.9	148°26.8	N16°27.7	96°41.0	S06°30.2	93°24.1	S06°11.2	240°32.5	N16°16.8					
1	306°12.4	163°26.3	26.8	111°42.3	30.8	108°26.3	11.3	255°34.8	16.9					
2	321°14.8	178°25.8	25.8	126°43.6	31.3	123°28.6	11.4	270°37.0	16.9					
3	336°17.3	193°25.3	· · 24.8	141°44.9	· · 31.8	138°30.9	· · 11.4	285°39.3	· · 17.0					
4	351°19.8	208°24.8	23.8	156°46.2	32.4	153°33.2	11.5	300°41.5	17.0					
5	6°22.2	223°24.3	22.9	171°47.5	32.9	168°35.4	11.6	315°43.8	17.1					
6	21°24.7	238°23.8	N16°21.9	186°48.8	S06°33.4	183°37.7	S06°11.7	330°46.0	N16°17.1					
7	36°27.1	253°23.3	20.9	201°50.1	34.0	198°40.0	11.8	345°48.2	17.2					
8	51°29.6	268°22.8	20.0	216°51.4	34.5	213°42.3	11.9	0°50.5	17.2					
9	66°32.1	283°22.3	· · 19.0	231°52.7	· · 35.0	228°44.5	· · 12.0	15°52.7	· · 17.3					
10	81°34.5	298°21.8	18.0	246°54.0	35.6	243°46.8	12.1	30°55.0	17.3					
11	96°37.0	313°21.3	17.0	261°55.3	36.1	258°49.1	12.2	45°57.2	17.4					
12	111°39.5	328°20.7	N16°16.0	276°56.6	S06°36.6	273°51.4	S06°12.2	60°59.5	N16°17.4					
13	126°41.9	343°20.2	15.1	291°57.9	37.2	288°53.6	12.3	76°01.7	17.5					
14	141°44.4	358°19.7	14.1	306°59.2	37.7	303°55.9	12.4	91°04.0	17.5					
15	156°46.9	13°19.2	· · 13.1	322°00.5	· · 38.2	318°58.2	· · 12.5	106°06.2	· · 17.6					
16	171°49.3	28°18.7	12.1	337°01.8	38.8	334°00.5	12.6	121°08.4	17.6					
17	186°51.8	43°18.2	11.2	352°03.1	39.3	349°02.7	12.7	136°10.7	17.7					
18	201°54.3	58°17.7	N16°10.2	7°04.4	S06°39.8	4°05.0	S06°12.8	151°12.9	N16°17.7					
19	216°56.7	73°17.2	09.2	22°05.7	40.4	19°07.3	12.9	166°15.2	17.8					
20	231°59.2	88°16.8	08.2	37°06.9	40.9	34°09.5	13.0	181°17.4	17.8					
21	247°01.6	103°16.3	· · 07.2	52°08.2	· · 41.4	49°11.8	· · 13.0	196°19.7	· · 17.9					
22	262°04.1	118°15.8	06.2	67°09.5	42.0	64°14.1	13.1	211°21.9	17.9					
23	277°06.6	133°15.3	05.3	82°10.8	42.5	79°16.4	13.2	226°24.2	18.0					
Mer.pass. 04:35			ν -0.5' d -1.0' m -3.90			ν 1.3' d 0.5' m 0.51			ν 2.3' d 0.1' m -1.99			ν 2.2' d 0.0' m 0.28		

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	292°09.0	148°14.8	N16°04.3	97°12.1	S06°43.0	94°18.6	S06°13.3	241°26.4	N16°18.0					
1	307°11.5	163°14.3	03.3	112°13.4	43.6	109°20.9	13.4	256°28.7	18.1					
2	322°14.0	178°13.8	02.3	127°14.7	44.1	124°23.2	13.5	271°30.9	18.1					
3	337°16.4	193°13.3	· · 01.3	142°16.0	· · 44.6	139°25.4	· · 13.6	286°33.2	· · 18.1					
4	352°18.9	208°12.8	16°00.3	157°17.3	45.2	154°27.7	13.7	301°35.4	18.2					
5	7°21.4	223°12.3	15°59.3	172°18.6	45.7	169°30.0	13.8	316°37.7	18.2					
6	22°23.8	238°11.8	N15°58.3	187°19.8	S06°46.2	184°32.2	S06°13.9	331°39.9	N16°18.3					
7	37°26.3	253°11.3	50.2	202°21.1	46.8	199°34.5	13.9	346°42.1	18.3					
8	52°28.7	268°10.8	56.4	217°22.4	47.3	214°36.8	14.0	1°44.4	18.4					
9	67°31.2	283°10.3	· · 55.4	232°23.7	· · 47.8	229°39.1	· · 14.1	16°46.6	· · 18.4					
10	82°33.7	298°09.8	54.4	247°25.0	48.4	244°41.3	14.2	31°48.9	18.5					
11	97°36.1	313°09.3	53.4	262°26.3	48.9	259°43.6	14.3	46°51.1	18.5					
12	112°38.6	328°08.9	N15°52.4	277°27.6	S06°49.5	274°45.9	S06°14.4	61°53.4	N16°18.6					
13	127°41.1	343°08.4	51.4	292°28.9	50.0	289°48.1	14.5	76°55.6	18.6					
14	142°43.5	358°07.9	50.4	307°30.1	50.5	304°50.4	14.6	91°57.9	18.7					
15	157°46.0	13°07.4	· · 49.4	322°31.4	· · 51.1	319°52.7	· · 14.7	107°00.1	· · 18.7					
16	172°48.5	28°06.9	48.4	337°32.7	51.6	334°54.9	14.8	122°02.4	18.8					
17	187°50.9	43°06.4	47.4	352°34.0	52.1	349°57.2	14.9	137°04.6	18.8					
18	202°53.4	58°05.9	N15°46.4	7°35.3	S06°52.7	4°59.5	S06°14.9	152°06.9	N16°18.9					
19	217°55.9	73°05.5	45.4	22°36.6	53.2	20°01.7	15.0	167°09.1	18.9					
20	232°58.3	88°05.0	44.4	37°37.8	53.7	35°04.0	15.1	182°11.4	19.0					
21	248°00.8	103°04.5	· · 43.4	52°39.1	· · 54.3	50°06.3	· · 15.2	197°13.6	· · 19.0					
22	263°03.2	118°04.0	42.4	67°40.4	54.8	65°08.5	15.3	212°15.9	19.0					
23	278°05.7	133°03.5	41.4	82°41.7	55.4	80°10.8	15.4	227°18.1	19.1					
Mer.pass. 04:31			ν -0.5' d -1.0' m -3.90			ν 1.3' d 0.5' m 0.52			ν 2.3' d 0.1' m -1.99			ν 2.2' d 0.0' m 0.28		

Jul 12 Thu	SHA	Mer.pass
Venus	218°28.4	14:06
Mars	165°58.9	17:34
Jupiter	162°18.6	17:47
Saturn	309°28.0	08:00

Jul 13 Fri	SHA	Mer.pass
Venus	217°16.9	14:07
Mars	165°31.1	17:32
Jupiter	162°14.2	17:44
Saturn	309°22.6	07:57

Jul 14 Sat	SHA	Mer.pass
Venus	216°05.7	14:07
Mars	165°03.1	17:30
Jupiter	162°09.6	17:40
Saturn	309°17.4	07:53

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h Sun Moon						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	178°35.5	N21°57.2	174°27.9	9.1'	N19°33.2	-8.7' 57.3'
1	193°35.4	56.8	188°56.1	9.2'	19°24.5	-8.8' 57.4'
2	208°35.3	56.5	203°24.2	9.2'	19°15.8	-8.9' 57.4'
3	223°35.2	· · 56.1	217°52.4	9.2'	19°06.9	-9.0' 57.4'
4	238°35.2	55.8	232°20.7	9.3'	18°57.9	-9.1' 57.4'
5	253°35.1	55.4	246°49.0	9.3'	18°48.7	-9.2' 57.5'
6	268°35.0	N21°55.1	261°17.3	9.4'	N18°39.5	-9.3' 57.5'
7	283°34.9	54.7	275°45.6	9.4'	18°30.2	-9.4' 57.5'
8	298°34.9	54.3	290°14.0	9.4'	18°20.7	-9.6' 57.5'
9	313°34.8	· · 54.0	304°42.4	9.5'	18°11.2	-9.7' 57.6'
10	328°34.7	53.6	319°10.9	9.5'	18°01.5	-9.8' 57.6'
11	343°34.6	53.3	333°39.4	9.5'	17°51.7	-9.9' 57.6'
12	358°34.5	N21°52.9	348°07.9	9.6'	N17°41.9	-10.0' 57.6'
13	13°34.5	52.6	2°36.5	9.6'	17°31.9	-10.1' 57.6'
14	28°34.4	52.2	17°05.1	9.6'	17°21.8	-10.2' 57.7'
15	43°34.3	· · 51.9	31°33.7	9.7'	17°11.6	-10.3' 57.7'
16	58°34.2	51.5	46°02.4	9.7'	17°01.3	-10.4' 57.7'
17	73°34.2	51.1	60°31.1	9.8'	16°50.9	-10.5' 57.7'
18	88°34.1	N21°50.8	74°59.9	9.8'	N16°40.5	-10.6' 57.8'
19	103°34.0	50.4	89°28.7	9.8'	16°29.9	-10.7' 57.8'
20	118°33.9	50.1	103°57.5	9.9'	16°19.2	-10.8' 57.8'
21	133°33.9	· · 49.7	118°26.4	9.9'	16°08.4	-10.9' 57.8'
22	148°33.8	49.3	132°55.3	9.9'	15°57.5	-11.0' 57.8'
23	163°33.7	49.0	147°24.2	10.0'	15°46.6	-11.1' 57.9'
SD = 15.7' d = -0.3' SD = 15.6'						
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	178°33.6	N21°48.6	161°53.2	10.0'	N15°35.5	-11.2' 57.9'
1	193°33.6	48.2	176°22.2	10.1'	15°24.4	-11.2' 57.9'
2	208°33.5	47.9	190°51.3	10.1'	15°13.1	-11.3' 57.9'
3	223°33.4	· · 47.5	205°20.4	10.1'	15°01.8	-11.4' 57.9'
4	238°33.3	47.1	219°49.5	10.2'	14°50.4	-11.5' 58.0'
5	253°33.3	46.8	234°18.7	10.2'	14°38.9	-11.6' 58.0'
6	268°33.2	N21°46.4	248°47.9	10.2'	N14°27.3	-11.7' 58.0'
7	283°33.1	46.0	263°17.1	10.3'	14°15.6	-11.8' 58.0'
8	298°33.0	45.7	277°46.4	10.3'	14°03.8	-11.8' 58.0'
9	313°33.0	· · 45.3	292°15.7	10.3'	13°52.0	-11.9' 58.1'
10	328°32.9	44.9	306°45.0	10.4'	13°40.1	-12.0' 58.1'
11	343°32.8	44.5	321°14.4	10.4'	13°28.0	-12.1' 58.1'
12	358°32.8	N21°44.2	335°43.8	10.5'	N13°16.0	-12.2' 58.1'
13	13°32.7	43.8	350°13.3	10.5'	13°03.8	-12.2' 58.1'
14	28°32.6	43.4	4°42.8	10.5'	12°51.6	-12.3' 58.2'
15	43°32.5	· · 43.1	19°12.3	10.5'	12°39.2	-12.4' 58.2'
16	58°32.5	42.7	33°41.9	10.6'	12°26.8	-12.5' 58.2'
17	73°32.4	42.3	48°11.4	10.6'	12°14.4	-12.5' 58.2'
18	88°32.3	N21°41.9	62°41.1	10.7'	N12°01.8	-12.6' 58.2'
19	103°32.3	41.6	77°10.7	10.7'	11°49.2	-12.7' 58.3'
20	118°32.2	41.2	91°40.4	10.7'	11°36.5	-12.7' 58.3'
21	133°32.1	· · 40.8	106°10.1	10.7'	11°23.8	-12.8' 58.3'
22	148°32.0	40.4	120°39.9	10.8'	11°11.0	-12.9' 58.3'
23	163°32.0	40.0	135°09.6	10.8'	10°58.1	-12.9' 58.3'
SD = 15.7' d = -0.4' SD = 15.8'						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	178°31.9	N21°39.7	149°39.4	10.8'	N10°45.2	-13.0' 58.3'
1	193°31.8	39.3	164°09.3	10.9'	10°32.2	-13.1' 58.4'
2	208°31.8	38.9	178°39.1	10.9'	10°19.1	-13.1' 58.4'
3	223°31.7	· · 38.5	193°09.0	10.9'	10°05.9	-13.2' 58.4'
4	238°31.6	38.1	207°39.0	10.9'	09°52.8	-13.3' 58.4'
5	253°31.6	37.8	222°08.9	11.0'	09°39.5	-13.3' 58.4'
6	268°31.5	N21°37.4	236°38.9	11.0'	N09°26.2	-13.4' 58.4'
7	283°31.4	37.0	251°08.9	11.0'	09°12.8	-13.4' 58.5'
8	298°31.4	36.6	265°38.9	11.1'	08°59.4	-13.5' 58.5'
9	313°31.3	· · 36.2	280°09.0	11.1'	08°45.9	-13.5' 58.5'
10	328°31.2	35.8	294°39.0	11.1'	08°32.4	-13.6' 58.5'
11	343°31.1	35.4	309°09.1	11.1'	08°18.8	-13.6' 58.5'
12	358°31.1	N21°35.1	323°39.3	11.1'	N08°05.2	-13.7' 58.5'
13	13°31.0	34.7	338°09.4	11.2'	07°51.5	-13.7' 58.6'
14	28°30.9	34.3	352°39.6	11.2'	07°37.8	-13.8' 58.6'
15	43°30.9	· · 33.9	7°09.7	11.2'	07°24.1	-13.8' 58.6'
16	58°30.8	33.5	21°39.9	11.2'	07°10.2	-13.9' 58.6'
17	73°30.8	33.1	36°10.2	11.2'	06°56.4	-13.9' 58.6'
18	88°30.7	N21°32.7	50°40.4	11.3'	N06°42.5	-13.9' 58.6'
19	103°30.6	32.3	65°10.7	11.3'	06°28.5	-14.0' 58.6'
20	118°30.6	31.9	79°40.9	11.3'	06°14.6	-14.0' 58.7'
21	133°30.5	· · 31.5	94°11.2	11.3'	06°00.6	-14.1' 58.7'
22	148°30.4	31.1	108°41.5	11.3'	05°46.5	-14.1' 58.7'
23	163°30.4	30.7	123°11.9	11.3'	05°32.4	-14.1' 58.7'
SD = 15.7' d = -0.4' SD = 15.9'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	--	--	01:28	22:40	--	--
64°	--	--	02:10	21:59	--	--
62°	--	00:34	02:39	21:31	23:28	--
60°	--	01:39	03:00	21:10	22:30	--
N 58°	--	02:12	03:18	20:53	21:58	--
56°	00:32	02:36	03:32	20:38	21:35	23:32
54°	01:31	02:54	03:45	20:26	21:16	22:38
52°	02:01	03:10	03:56	20:15	21:01	22:09
50°	02:23	03:23	04:06	20:05	20:48	21:47
45°	03:03	03:50	04:26	19:45	20:21	21:08
N 40°	03:30	04:10	04:42	19:29	20:01	20:41
35°	03:51	04:27	04:56	19:15	19:44	20:20
30°	04:09	04:41	05:08	19:03	19:30	20:03
20°	04:36	05:04	05:28	18:43	19:07	19:36
N 10°	04:56	05:23	05:46	18:26	18:48	19:15
0°	05:14	05:40	06:02	18:09	18:32	18:58
S 10°	05:30	05:56	06:18	17:53	18:16	18:42
20°	05:44	06:12	06:35	17:36	18:00	18:27
30°	05:59	06:29	06:55	17:17	17:43	18:12
35°	06:07	06:39	07:06	17:06	17:33	18:05
40°	06:16	06:49	07:19	16:53	17:23	17:56
45°	06:25	07:01	07:34	16:37	17:11	17:47
S 50°	06:35	07:16	07:53	16:19	16:56	17:37
52°	06:40	07:22	08:02	16:10	16:50	17:32
54°	06:45	07:30	08:12	16:00	16:42	17:27
56°	06:50	07:38	08:23	15:49	16:34	17:22
58°	06:56	07:47	08:36	15:36	16:25	17:16
S 60°	07:02	07:57	08:51	15:21	16:15	17:10

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°		03:43	06:14	23:08	22:24	21:56
N 70°	01:33	04:18	06:29	22:32	22:07	21:48
68°	02:33	04:42	06:41	22:06	21:53	21:42
66°	03:07	05:01	06:51	21:46	21:41	21:36
64°	03:31	05:16	06:59	21:30	21:32	21:32
62°	03:50	05:28	07:06	21:17	21:23	21:28
60°	04:06	05:39	07:12	21:05	21:16	21:24
N 58°	04:19	05:48	07:17	20:55	21:10	21:21
56°	04:31	05:56	07:22	20:47	21:04	21:18
54°	04:41	06:03	07:26	20:39	20:59	21:16
52°	04:50	06:09	07:30	20:32	20:54	21:13
50°	04:58	06:15	07:33	20:25	20:50	21:11
45°	05:15	06:27	07:41	20:12	20:41	21:07
N 40°	05:28	06:37	07:47	20:00	20:33	21:03
35°	05:40	06:46	07:52	19:51	20:26	20:59
30°	05:50	06:53	07:57	19:42	20:20	20:56
20°	06:07	07:06	08:05	19:27	20:10	20:51
N 10°	06:23	07:18	08:12	19:14	20:01	20:46
0°	06:37	07:28	08:19	19:02	19:52	20:42
S 10°	06:51	07:39	08:25	18:49	19:44	20:38
20°	07:06	07:50	08:32	18:36	19:35	20:33
30°	07:23	08:03	08:40	18:20	19:24	20:27
35°	07:33	08:10	08:45	18:11	19:18	20:24
40°	07:44	08:19	08:50	18:01	19:11	20:20
45°	07:57	08:28	08:56	17:49	19:02	20:16
S 50°	08:13	08:40	09:03	17:34	18:52	20:11
52°	08:21	08:45	09:06	17:27	18:48	20:09
54°	08:29	08:51	09:10	17:19	18:42	20:06
56°	08:39	08:58	09:13	17:11	18:37	20:03
58°	08:49	09:05	09:18	17:01	18:30	20:00
S 60°	09:01	09:14	09:23	16:49	18:23	19:56

Day	Sun		Mer. Pass	Moon		Age 1-3 0-7%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
12	05:38	05:42	12:06	12:49	00:23	
13	05:45	05:49	12:06	13:40	01:15	
14	05:52	05:56	12:06	14:30	02:06	

July 15, 16, 17 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	293°08.2	148°03.0	N15°40.4	97°43.0	S06°55.9	95°13.1	S06°15.5	242°20.4	N16°19.1	Alpheratz	357°30.8	29°15.2
1	308°10.6	163°02.6		39.4	112°44.2		56.4	110°15.3		15.6	257°22.6	19.2
2	323°13.1	178°02.1		38.4	127°45.5		57.0	125°17.6		15.7	272°24.9	19.2
3	338°15.6	193°01.6	· ·	37.4	142°46.8	· ·	57.5	140°19.8	· ·	15.8	287°27.1	· · 19.3
4	353°18.0	208°01.1		36.4	157°48.1		58.0	155°22.1		15.9	302°29.4	19.3
5	8°20.5	223°00.6		35.4	172°49.4		58.6	170°24.4		16.0	317°31.6	19.4
6	23°23.0	238°00.2	N15°34.4	187°50.6	S06°59.1	185°26.6	S06°16.1	332°33.9	N16°19.4			
7	38°25.4	252°59.7		33.4	202°51.9		06°59.6	200°28.9		16.1	347°36.1	19.5
8	53°27.9	267°59.2		32.4	217°53.2		07°00.2	215°31.2		16.2	2°38.4	19.5
9	68°30.4	282°58.7	· ·	31.3	232°54.5	· ·	00.7	230°33.4	· ·	16.3	17°40.6	· · 19.6
10	83°32.8	297°58.2		30.3	247°55.8		01.3	245°35.7		16.4	32°42.9	19.6
11	98°35.3	312°57.8		29.3	262°57.0		01.8	260°37.9		16.5	47°45.1	19.6
12	113°37.7	327°57.3	N15°28.3	277°58.3	S07°02.3	275°40.2	S06°16.6	62°47.4	N16°19.7			
13	128°40.2	342°56.8		27.3	292°59.6		02.9	290°42.5		16.7	77°49.6	19.7
14	143°42.7	357°56.3		26.3	308°00.9		03.4	305°44.7		16.8	92°51.9	19.8
15	158°45.1	12°55.9	· ·	25.3	323°02.1	· ·	03.9	320°47.0	· ·	16.9	107°54.1	· · 19.8
16	173°47.6	27°55.4		24.3	338°03.4		04.5	335°49.3		17.0	122°56.4	19.9
17	188°50.1	42°54.9		23.3	353°04.7		05.0	350°51.5		17.1	137°58.6	19.9
18	203°52.5	57°54.5	N15°22.2	8°06.0	S07°05.6	5°53.8	S06°17.2	153°00.9	N16°20.0			
19	218°55.0	72°54.0		21.2	23°07.2		06.1	20°56.0		17.3	168°03.1	20.0
20	233°57.5	87°53.5		20.2	38°08.5		06.6	35°58.3		17.4	183°05.4	20.1
21	248°59.9	102°53.0	· ·	19.2	53°09.8	· ·	07.2	51°00.6	· ·	17.5	198°07.6	· · 20.1
22	264°02.4	117°52.6		18.2	68°11.0		07.7	66°02.8		17.6	213°09.9	20.2
23	279°04.8	132°52.1		17.2	83°12.3		08.3	81°05.1		17.6	228°12.1	20.2
Mer.pass. 04:27		ν -0.5' d -1.0' m -3.90		ν 1.3' d 0.5' m 0.53		ν 2.3' d 0.1' m -1.98		ν 2.2' d 0.0' m 0.28				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	294°07.3	147°51.6	N15°16.1	98°13.6	S07°08.8	96°07.3	S06°17.7	243°14.4	N16°20.2			
1	309°09.8	162°51.2		15.1	113°14.9		09.3	111°09.6		17.8	258°16.7	20.3
2	324°12.2	177°50.7		14.1	128°16.1		09.9	126°11.9		17.9	273°18.9	20.3
3	339°14.7	192°50.2	· ·	13.1	143°17.4	· ·	10.4	141°14.1	· ·	18.0	288°21.2	· · 20.4
4	354°17.2	207°49.8		12.1	158°18.7		10.9	156°16.4		18.1	303°23.4	20.4
5	9°19.6	222°49.3		11.0	173°19.9		11.5	171°18.6		18.2	318°25.7	20.5
6	24°22.1	237°48.8	N15°10.0	188°21.2	S07°12.0	186°20.9	S06°18.3	333°27.9	N16°20.5			
7	39°24.6	252°48.4		09.0	203°22.5		12.6	201°23.1		18.4	348°30.2	20.6
8	54°27.0	267°47.9		08.0	218°23.7		13.1	216°25.4		18.5	3°32.4	20.6
9	69°29.5	282°47.4	· ·	06.9	233°25.0	· ·	13.6	231°27.7	· ·	18.6	18°34.7	· · 20.7
10	84°32.0	297°47.0		05.9	248°26.3		14.2	246°29.9		18.7	33°36.9	20.7
11	99°34.4	312°46.5		04.9	263°27.5		14.7	261°32.2		18.8	48°39.2	20.7
12	114°36.9	327°46.1	N15°03.9	278°28.8	S07°15.3	276°34.4	S06°18.9	63°41.4	N16°20.8			
13	129°39.3	342°45.6		02.8	293°30.1		15.8	291°36.7		19.0	78°43.7	20.8
14	144°41.8	357°45.1		01.8	308°31.3		16.3	306°38.9		19.1	93°46.0	20.9
15	159°44.3	12°44.7	15°00.8	323°32.6	· ·	16.9		321°41.2	· ·	19.2	108°48.2	· · 20.9
16	174°46.7	27°44.2	14°59.7	338°33.9		17.4		336°43.4		19.3	123°50.5	21.0
17	189°49.2	42°43.8		58.7	353°35.1		18.0	351°45.7		19.4	138°52.7	21.0
18	204°51.7	57°43.3	N14°57.7	8°36.4	S07°18.5	6°48.0	S06°19.5	153°55.0	N16°21.1			
19	219°54.1	72°42.8		56.6	23°37.7		19.0	21°50.2		19.6	168°57.2	21.1
20	234°56.6	87°42.4		55.6	38°38.9		19.6	36°52.5		19.6	183°59.5	21.2
21	249°59.1	102°41.9	· ·	54.6	53°40.2	· ·	20.1	51°54.7	· ·	19.7	199°01.7	· · 21.2
22	265°01.5	117°41.5		53.5	68°41.4		20.7	66°57.0		19.8	214°04.0	21.2
23	280°04.0	132°41.0		52.5	83°42.7		21.2	81°59.2		19.9	229°06.2	21.3
Mer.pass. 04:23		ν -0.5' d -1.0' m -3.90		ν 1.3' d 0.5' m 0.54		ν 2.3' d 0.1' m -1.97		ν 2.3' d 0.0' m 0.27				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	295°06.4	147°40.6	N14°51.5	98°44.0	S07°21.7	97°01.5	S06°20.0	244°08.5	N16°21.3			
1	310°08.9	162°40.1		50.4	113°45.2		22.3	112°03.7		20.1	259°10.8	21.4
2	325°11.4	177°39.6		49.4	128°46.5		22.8	127°06.0		20.2	274°13.0	21.4
3	340°13.8	192°39.2	· ·	48.4	143°47.7	· ·	23.4	142°08.2	· ·	20.3	289°15.3	· · 21.5
4	355°16.3	207°38.7		47.3	158°49.0		23.9	157°10.5		20.4	304°17.5	21.5
5	10°18.8	222°38.3		46.3	173°50.3		24.4	172°12.7		20.5	319°19.8	21.6
6	25°21.2	237°37.8	N14°45.3	188°51.5	S07°25.0	187°15.0	S06°20.6	334°22.0	N16°21.6			
7	40°23.7	252°37.4		44.2	203°52.8		25.5	202°17.2		20.7	349°24.3	21.6
8	55°26.2	267°36.9		43.2	218°54.0		26.1	217°19.5		20.8	4°26.6	21.7
9	70°28.6	282°36.5	· ·	42.1	233°55.3	· ·	26.6	232°21.7	· ·	20.9	19°28.8	· · 21.7
10	85°31.1	297°36.0		41.1	248°56.6		27.1	247°24.0		21.0	34°31.1	21.8
11	100°33.6	312°35.6		40.0	263°57.8		27.7	262°26.2		21.1	49°33.3	21.8
12	115°36.0	327°35.1	N14°39.0	278°59.1	S07°28.2	277°28.5	S06°21.2	64°35.6	N16°21.9			
13	130°38.5	342°34.7		38.0	294°00.3		28.8	292°30.7		21.3	79°37.8	21.9
14	145°40.9	357°34.2		36.9	309°01.6		29.3	307°33.0		21.4	94°40.1	22.0
15	160°43.4	12°33.8	· ·	35.9	324°02.8	· ·	29.8	322°35.2	· ·	21.5	109°42.4	· · 22.0
16	175°45.9	27°33.3		34.8	339°04.1		30.4	337°37.5		21.6	124°44.6	22.0
17	190°48.3	42°32.9		33.8	354°05.3		30.9	352°39.7		21.7	139°46.9	22.1
18	205°50.8	57°32.5	N14°32.7	9°06.6	S07°31.5	7°42.0	S06°21.8	154°49.1	N16°22.1			
19	220°53.3	72°32.0		31.7	24°07.9		32.0	22°44.2		21.9	169°51.4	22.2
20	235°55.7	87°31.6		30.6	39°09.1		32.6	37°46.5		22.0	184°53.6	22.2
21	250°58.2	102°31.1	· ·	29.6	54°10.4	· ·	33.1	52°48.7	· ·	22.1	199°55.9	· · 22.3
22	266°00.7	117°30.7		28.5	69°11.6		33.6	67°51.0		22.2	214°58.2	22.3
23	281°03.1	132°30.2		27.5	84°12.9		34.2	82°53.2		22.3	230°00.4	22.4
Mer.pass. 04:19		ν -0.5' d -1.0' m -3.90		ν 1.3' d 0.5' m 0.55		ν 2.3' d 0.1' m -1.97		ν 2.3' d 0.0' m 0.27				

Jul 15 Sun	SHA	Mer.pass
Venus	214°54.9	14:08
Mars	164°34.8	17:28
Jupiter	162°04.9	17:36
Saturn	309°12.2	07:49

Jul 16 Mon	SHA	Mer.pass
Venus	213°44.3	14:09
Mars	164°06.3	17:26
Jupiter	162°00.0	17:33
Saturn	309°07.1	07:46

Jul 17 Tue	SHA	Mer.pass
Venus	212°34.1	14:10
Mars	163°37.5	17:24
Jupiter	161°55.0	17:29
Saturn	309°02.1	07:42

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h Sun Moon						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	178°30.3	N21°30.4	137°42.2	11.3'	N05°18.3	-14.2' 58.7'
1	193°30.2	30.0	152°12.5	11.4'	05°04.1	-14.2' 58.7'
2	208°30.2	29.6	166°42.9	11.4'	04°50.0	-14.2' 58.7'
3	223°30.1	· · 29.2	181°13.2	11.4'	04°35.7	-14.2' 58.7'
4	238°30.0	28.8	195°43.6	11.4'	04°21.5	-14.3' 58.8'
5	253°30.0	28.4	210°14.0	11.4'	04°07.2	-14.3' 58.8'
6	268°29.9	N21°28.0	224°44.4	11.4'	N03°52.9	-14.3' 58.8'
7	283°29.9	27.6	239°14.8	11.4'	03°38.6	-14.3' 58.8'
8	298°29.8	27.2	253°45.2	11.4'	03°24.2	-14.4' 58.8'
9	313°29.7	· · 26.8	268°15.6	11.4'	03°09.9	-14.4' 58.8'
10	328°29.7	26.4	282°46.0	11.4'	02°55.5	-14.4' 58.8'
11	343°29.6	26.0	297°16.5	11.4'	02°41.1	-14.4' 58.8'
12	358°29.5	N21°25.6	311°46.9	11.4'	N02°26.6	-14.4' 58.9'
13	13°29.5	25.2	326°17.3	11.4'	02°12.2	-14.5' 58.9'
14	28°29.4	24.8	340°47.7	11.4'	01°57.7	-14.5' 58.9'
15	43°29.4	· · 24.4	355°18.1	11.4'	01°43.3	-14.5' 58.9'
16	58°29.3	23.9	9°48.6	11.4'	01°28.8	-14.5' 58.9'
17	73°29.2	23.5	24°19.0	11.4'	01°14.3	-14.5' 58.9'
18	88°29.2	N21°23.1	38°49.4	11.4'	N00°59.8	-14.5' 58.9'
19	103°29.1	22.7	53°19.8	11.4'	00°45.2	-14.5' 58.9'
20	118°29.1	22.3	67°50.2	11.4'	00°30.7	-14.5' 58.9'
21	133°29.0	· · 21.9	82°20.6	11.4'	00°16.2	-14.5' 58.9'
22	148°28.9	21.5	96°51.0	11.4'	N00°01.6	-14.5' 59.0'
23	163°28.9	21.1	111°21.4	11.4'	S00°12.9	14.5' 59.0'
SD = 15.7' d = -0.4' SD = 16.0'						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	178°28.8	N21°20.7	125°51.8	11.4'	S00°27.5	14.5' 59.0'
1	193°28.8	20.3	140°22.1	11.4'	00°42.0	14.5' 59.0'
2	208°28.7	19.9	154°52.5	11.3'	00°56.6	14.5' 59.0'
3	223°28.6	· · 19.4	169°22.8	11.3'	01°11.1	14.5' 59.0'
4	238°28.6	19.0	183°53.2	11.3'	01°25.7	14.5' 59.0'
5	253°28.5	18.6	198°23.5	11.3'	01°40.2	14.5' 59.0'
6	268°28.5	N21°18.2	212°53.8	11.3'	S01°54.7	14.5' 59.0'
7	283°28.4	17.8	227°24.1	11.3'	02°09.2	14.5' 59.0'
8	298°28.4	17.4	241°54.4	11.3'	02°23.8	14.5' 59.0'
9	313°28.3	· · 17.0	256°24.6	11.2'	02°38.3	14.5' 59.0'
10	328°28.2	16.5	270°54.8	11.2'	02°52.8	14.5' 59.1'
11	343°28.2	16.1	285°25.0	11.2'	03°07.3	14.5' 59.1'
12	358°28.1	N21°15.7	299°55.2	11.2'	S03°21.7	14.5' 59.1'
13	13°28.1	15.3	314°25.4	11.1'	03°36.2	14.4' 59.1'
14	28°28.0	14.9	328°55.6	11.1'	03°50.6	14.4' 59.1'
15	43°28.0	· · 14.5	343°25.7	11.1'	04°05.0	14.4' 59.1'
16	58°27.9	14.0	357°55.8	11.1'	04°19.4	14.4' 59.1'
17	73°27.9	13.6	12°25.8	11.0'	04°33.8	14.4' 59.1'
18	88°27.8	N21°13.2	26°55.9	11.0'	S04°48.2	14.3' 59.1'
19	103°27.7	12.8	41°25.9	11.0'	05°02.5	14.3' 59.1'
20	118°27.7	12.3	55°55.9	11.0'	05°16.8	14.3' 59.1'
21	133°27.6	· · 11.9	70°25.9	10.9'	05°31.1	14.3' 59.1'
22	148°27.6	11.5	84°55.8	10.9'	05°45.4	14.2' 59.1'
23	163°27.5	11.1	99°25.7	10.9'	05°59.6	14.2' 59.1'
SD = 15.7' d = -0.4' SD = 16.1'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	178°27.5	N21°10.6	113°55.5	10.8'	S06°13.8	14.2' 59.1'
1	193°27.4	10.2	128°25.4	10.8'	06°28.0	14.1' 59.1'
2	208°27.4	09.8	142°55.2	10.8'	06°42.1	14.1' 59.2'
3	223°27.3	· · 09.4	157°24.9	10.7'	06°56.2	14.1' 59.2'
4	238°27.3	08.9	171°54.7	10.7'	07°10.3	14.0' 59.2'
5	253°27.2	08.5	186°24.3	10.6'	07°24.3	14.0' 59.2'
6	268°27.2	N21°08.1	200°54.0	10.6'	S07°38.3	13.9' 59.2'
7	283°27.1	07.7	215°23.6	10.6'	07°52.2	13.9' 59.2'
8	298°27.1	07.2	229°53.2	10.5'	08°06.1	13.9' 59.2'
9	313°27.0	· · 06.8	244°22.7	10.5'	08°20.0	13.8' 59.2'
10	328°27.0	06.4	258°52.2	10.4'	08°33.8	13.8' 59.2'
11	343°26.9	05.9	273°21.6	10.4'	08°47.6	13.7' 59.2'
12	358°26.9	N21°05.5	287°51.0	10.4'	S09°01.3	13.7' 59.2'
13	13°26.8	05.1	302°20.4	10.3'	09°15.0	13.6' 59.2'
14	28°26.8	04.6	316°49.7	10.3'	09°28.6	13.6' 59.2'
15	43°26.7	· · 04.2	331°18.9	10.2'	09°42.2	13.5' 59.2'
16	58°26.7	03.8	345°48.2	10.2'	09°55.7	13.5' 59.2'
17	73°26.6	03.3	0°17.3	10.1'	10°09.2	13.4' 59.2'
18	88°26.6	N21°02.9	14°46.4	10.1'	S10°22.6	13.4' 59.2'
19	103°26.5	02.5	29°15.5	10.0'	10°36.0	13.3' 59.2'
20	118°26.5	02.0	43°44.5	10.0'	10°49.3	13.2' 59.2'
21	133°26.4	· · 01.6	58°13.5	9.9'	11°02.5	13.2' 59.2'
22	148°26.4	01.1	72°42.4	9.9'	11°15.7	13.1' 59.2'
23	163°26.3	00.7	87°11.3	9.8'	11°28.8	13.1' 59.2'
SD = 15.7' d = -0.4' SD = 16.1'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	--	--	01:41	22:28	--	--
64°	--	--	02:19	21:51	--	--
62°	--	00:59	02:46	21:25	23:08	--
60°	--	01:49	03:06	21:05	22:20	--
N 58°	--	02:19	03:23	20:48	21:51	--
56°	00:54	02:41	03:37	20:34	21:29	23:13
54°	01:40	02:59	03:49	20:22	21:12	22:30
52°	02:08	03:14	03:59	20:12	20:57	22:03
50°	02:29	03:27	04:09	20:03	20:44	21:42
45°	03:06	03:53	04:29	19:43	20:19	21:05
N 40°	03:33	04:13	04:45	19:27	19:59	20:38
35°	03:54	04:29	04:58	19:14	19:43	20:18
30°	04:11	04:43	05:10	19:02	19:29	20:01
20°	04:37	05:05	05:30	18:42	19:07	19:35
N 10°	04:57	05:24	05:47	18:25	18:48	19:15
0°	05:15	05:40	06:03	18:10	18:32	18:58
S 10°	05:30	05:56	06:18	17:54	18:17	18:43
20°	05:44	06:11	06:35	17:37	18:01	18:28
30°	05:59	06:28	06:54	17:19	17:44	18:14
35°	06:06	06:37	07:05	17:08	17:35	18:06
40°	06:14	06:48	07:18	16:55	17:25	17:58
45°	06:23	07:00	07:32	16:40	17:13	17:49
S 50°	06:33	07:14	07:51	16:22	16:59	17:40
52°	06:38	07:20	07:59	16:13	16:53	17:35
54°	06:42	07:27	08:09	16:04	16:46	17:30
56°	06:47	07:35	08:20	15:53	16:38	17:25
58°	06:53	07:43	08:32	15:41	16:29	17:20
S 60°	06:59	07:53	08:46	15:26	16:19	17:14

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	08:27	10:37	12:53	21:32	21:07	20:38
N 70°	08:32	10:32	12:37	21:32	21:15	20:55
68°	08:35	10:29	12:25	21:31	21:21	21:10
66°	08:38	10:26	12:15	21:31	21:27	21:21
64°	08:41	10:23	12:06	21:31	21:31	21:31
62°	08:43	10:21	11:59	21:31	21:35	21:40
60°	08:45	10:19	11:53	21:31	21:39	21:47
N 58°	08:47	10:17	11:48	21:31	21:42	21:54
56°	08:49	10:15	11:43	21:31	21:45	21:59
54°	08:50	10:14	11:39	21:31	21:47	22:05
52°	08:51	10:13	11:35	21:31	21:50	22:09
50°	08:52	10:12	11:32	21:31	21:52	22:13
45°	08:55	10:09	11:24	21:31	21:56	22:23
N 40°	08:57	10:07	11:18	21:31	22:00	22:31
35°	08:59	10:05	11:13	21:31	22:03	22:37
30°	09:00	10:04	11:08	21:31	22:06	22:43
20°	09:03	10:01	11:00	21:31	22:12	22:53
N 10°	09:06	09:59	10:53	21:31	22:16	23:02
0°	09:08	09:57	10:47	21:31	22:20	23:11
S 10°	09:10	09:55	10:40	21:31	22:25	23:20
20°	09:13	09:53	10:34	21:31	22:29	23:29
30°	09:15	09:50	10:26	21:31	22:34	23:39
35°	09:17	09:49	10:22	21:30	22:37	23:45
40°	09:19	09:47	10:17	21:30	22:41	23:52
45°	09:21	09:45	10:11	21:30	22:45	· · ·
S 50°	09:23	09:43	10:04	21:30	22:50	· · ·
52°	09:24	09:42	10:01	21:30	22:52	· · ·
54°	09:26	09:41	09:58	21:30	22:54	· · ·
56°	09:27	09:40	09:54	21:30	22:57	· · ·
58°	09:29	09:39	09:50	21:30	23:00	· · ·
S 60°	09:30	09:37	09:45	21:29	23:03	· · ·

Day	Sun		Mer. Pass	Moon		Age 4-6 13-32%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
15	05:59	06:02	12:06	15:19	02:55	
16	06:05	06:07	12:06	16:09	03:44	
17	06:10	06:13	12:06	16:59	04:33	

July 18, 19, 20 UT (Wed., Thu., Fri.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Wed	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	SHA	Dec		
0	296°05.6	147°29.8	N14°26.4	99°14.1	S07°34.7	97°55.5	S06°22.4	245°02.7	N16°22.4					Alpheratz	357°30.8 29°15.2		
1	311°08.1	162°29.3	25.4	114°15.4	35.3	112°57.7	22.5	260°04.9	22.4					Ankaa	353°03.5 -42°08.4		
2	326°10.5	177°28.9	24.3	129°16.6	35.8	128°00.0	22.6	275°07.2	22.5					Schedar	349°26.5 56°41.8		
3	341°13.0	192°28.5	•• 23.3	144°17.9	•• 36.3	143°02.2	•• 22.7	290°09.5	•• 22.5					Diphda	348°43.6 -17°49.2		
4	356°15.4	207°28.0	22.2	159°19.1	36.9	158°04.5	22.8	305°11.7	22.6					Achernar	335°17.6 -57°04.8		
5	11°17.9	222°27.6	21.2	174°20.4	37.4	173°06.7	22.9	320°14.0	22.6					Hamal	327°47.1 23°36.1		
6	26°20.4	237°27.1	N14°20.1	189°21.6	S07°38.0	188°09.0	S06°23.0	335°16.2	N16°22.7					Polaris	312°11.5 89°22.9		
7	41°22.8	252°26.7	19.1	204°22.9	38.5	203°11.2	23.1	350°18.5	22.7					Acamar	315°09.2 -40°10.9		
8	56°25.3	267°26.3	18.0	219°24.1	39.1	218°13.4	23.2	5°20.8	22.7					Menkar	314°02.4 4°12.4		
9	71°27.8	282°25.8	•• 17.0	234°25.4	•• 39.6	233°15.7	•• 23.3	20°23.0	•• 22.8					Mirfak	308°23.1 49°57.7		
10	86°30.2	297°25.4	15.9	249°26.6	40.1	248°17.9	23.4	35°25.3	22.8					Aldebaran	290°35.6 16°34.0		
11	101°32.7	312°25.0	14.8	264°27.9	40.7	263°20.2	23.5	50°27.5	22.9					Rigel	281°00.6 -8°10.1		
12	116°35.2	327°24.5	N14°13.8	279°29.1	S07°41.2	278°22.4	S06°23.6	65°29.8	N16°22.9					Capella	280°16.8 46°01.4		
13	131°37.6	342°24.1	12.7	294°30.4	41.8	293°24.7	23.7	80°32.1	23.0					Bellatrix	278°19.2 6°22.5		
14	146°40.1	357°23.7	11.7	309°31.6	42.3	308°26.9	23.8	95°34.3	23.0					Elnath	277°57.5 28°37.8		
15	161°42.5	12°23.2	•• 10.6	324°32.8	•• 42.9	323°29.2	•• 23.9	110°36.6	•• 23.1					Alnilam	275°34.3 -1°11.0		
16	176°45.0	27°22.8	09.5	339°34.1	43.4	338°31.4	24.0	125°38.8	23.1					Betelgeuse	270°48.4 7°24.7		
17	191°47.5	42°22.4	08.5	354°35.3	43.9	353°33.6	24.1	140°41.1	23.1					Canopus	263°51.3 -52°42.7		
18	206°49.9	57°21.9	N14°07.4	9°36.6	S07°44.5	8°35.9	S06°24.2	155°43.4	N16°23.2					Sirius	258°23.3 -16°45.5		
19	221°52.4	72°21.5	06.4	24°37.8	45.0	23°38.1	24.3	170°45.6	23.2					Adhara	255°03.3 -29°00.8		
20	236°54.9	87°21.1	05.3	39°39.1	45.6	38°40.4	24.4	185°47.9	23.3					Procyon	244°47.3 5°08.8		
21	251°57.3	102°20.6	•• 04.2	54°40.3	•• 46.1	53°42.6	•• 24.5	200°50.2	•• 23.3					Pollux	243°13.2 27°57.1		
22	266°59.8	117°20.2	03.2	69°41.6	46.7	68°44.9	24.6	215°52.4	23.4					Avior	234°13.8 -59°36.4		
23	282°02.3	132°19.8	02.1	84°42.8	47.2	83°47.1	24.7	230°54.7	23.4					Suhail	222°44.0 -43°33.3		
Mer.pass. 04:15			ν -0.4' d -1.1' m -3.90			ν 1.3' d 0.5' m 0.56			ν 2.2' d 0.1' m -1.96			ν 2.3' d 0.0' m 0.27					

Thu	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	
0	297°04.7	147°19.3	N14°01.0	99°44.0	S07°47.7	98°49.3	S06°24.8	245°56.9	N16°23.4							
1	312°07.2	162°18.9	14°00.0	114°45.3	48.3	113°51.6	24.9	260°59.2	23.5							
2	327°09.7	177°18.5	13°58.9	129°46.5	48.8	128°53.8	25.0	276°01.5	23.5							
3	342°12.1	192°18.1	•• 57.8	144°47.8	•• 49.4	143°56.1	•• 25.1	291°03.7	•• 23.6							
4	357°14.6	207°17.6	56.8	159°49.0	49.9	158°58.3	25.2	306°06.0	23.6							
5	12°17.0	222°17.2	55.7	174°50.2	50.5	174°00.5	25.3	321°08.3	23.7							
6	27°19.5	237°16.8	N13°54.6	189°51.5	S07°51.0	189°02.8	S06°25.4	336°10.5	N16°23.7							
7	42°22.0	252°16.3	53.6	204°52.7	51.5	204°05.0	25.5	351°12.8	23.7							
8	57°24.4	267°15.9	52.5	219°54.0	52.1	219°07.3	25.6	6°15.0	23.8							
9	72°26.9	282°15.5	•• 51.4	234°55.2	•• 52.6	234°09.5	•• 25.7	21°17.3	•• 23.8							
10	87°29.4	297°15.1	50.4	249°56.4	53.2	249°11.7	25.8	36°19.6	23.9							
11	102°31.8	312°14.7	49.3	264°57.7	53.7	264°14.0	25.9	51°21.8	23.9							
12	117°34.3	327°14.2	N13°48.2	279°58.9	S07°54.3	279°16.2	S06°26.0	66°24.1	N16°23.9							
13	132°36.8	342°13.8	47.1	295°00.2	54.8	294°18.5	26.1	81°26.4	24.0							
14	147°39.2	357°13.4	46.1	310°01.4	55.4	309°20.7	26.2	96°28.6	24.0							
15	162°41.7	12°13.0	•• 45.0	325°02.6	•• 55.9	324°22.9	•• 26.3	111°30.9	•• 24.1							
16	177°44.2	27°12.5	43.9	340°03.9	56.4	339°25.2	26.4	126°33.2	24.1							
17	192°46.6	42°12.1	42.8	355°05.1	57.0	354°27.4	26.5	141°35.4	24.2							
18	207°49.1	57°11.7	N13°41.8	10°06.3	S07°57.5	9°29.7	S06°26.6	156°37.7	N16°24.2							
19	222°51.5	72°11.3	40.7	25°07.6	58.1	24°31.9	26.7	171°39.9	24.2							
20	237°54.0	87°10.9	39.6	40°08.8	58.6	39°34.1	26.8	186°42.2	24.3							
21	252°56.5	102°10.4	•• 38.5	55°10.0	•• 59.2	54°36.4	•• 26.9	201°44.5	•• 24.3							
22	267°58.9	117°10.0	37.5	70°11.3	07°59.7	69°38.6	27.0	216°46.7	24.4							
23	283°01.4	132°09.6	36.4	85°12.5	08°00.3	84°40.8	27.1	231°49.0	24.4							
Mer.pass. 04:11			ν -0.4' d -1.1' m -3.91			ν 1.2' d 0.5' m 0.57			ν 2.2' d 0.1' m -1.96			ν 2.3' d 0.0' m 0.27				

Fri	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec
0	298°03.9	147°09.2	N13°35.3	100°13.7	S08°00.8	99°43.1	S06°27.2	246°51.3	N16°24.5						
1	313°06.3	162°08.8	34.2	115°15.0	01.3	114°45.3	27.3	261°53.5	24.5						
2	328°08.8	177°08.4	33.1	130°16.2	01.9	129°47.5	27.4	276°55.8	24.5						
3	343°11.3	192°07.9	•• 32.0	145°17.4	•• 02.4	144°49.8	•• 27.5	291°58.1	•• 24.6						
4	358°13.7	207°07.5	31.0	160°18.7	03.0	159°52.0	27.6	307°00.3	24.6						
5	13°16.2	222°07.1	29.9	175°19.9	03.5	174°54.2	27.7	322°02.6	24.7						
6	28°18.6	237°06.7	N13°28.8	190°21.1	S08°04.1	189°56.5	S06°27.8	337°04.9	N16°24.7						
7	43°21.1	252°06.3	27.7	205°22.4	04.6	204°58.7	28.0	352°07.1	24.7						
8	58°23.6	267°05.9	26.6	220°23.6	05.2	220°01.0	28.1	7°09.4	24.8						
9	73°26.0	282°05.5	•• 25.5	235°24.8	•• 05.7	235°03.2	•• 28.2	22°11.7	•• 24.8						
10	88°28.5	297°05.1	24.5	250°26.0	06.3	250°05.4	28.3	37°13.9	24.9						
11	103°31.0	312°04.6	23.4	265°27.3	06.8	265°07.7	28.4	52°16.2	24.9						
12	118°33.4	327°04.2	N13°22.3	280°28.5	S08°07.3	280°09.9	S06°28.5	67°18.5	N16°25.0						
13	133°35.9	342°03.8	21.2	295°29.7	07.9	295°12.1	28.6	82°20.7	25.0						
14	148°38.4	357°03.4	20.1	310°31.0	08.4	310°14.4	28.7	97°23.0	25.0						
15	163°40.8	12°03.0	•• 19.0	325°32.2	•• 09.0	325°16.6	•• 28.8	112°25.3	•• 25.1						
16	178°43.3	27°02.6	17.9	3403											

h		Sun		Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°26.3	N21°00.3	101°40.1	9.8'	S11°41.9	13.0'	59.2'
1	193°26.2	20°59.8	116°08.8	9.7'	11°54.9	12.9'	59.2'
2	208°26.2	59.4	130°37.5	9.6'	12°07.8	12.9'	59.2'
3	223°26.1	· · 58.9	145°06.2	9.6'	12°20.7	12.8'	59.2'
4	238°26.1	58.5	159°34.8	9.5'	12°33.5	12.7'	59.2'
5	253°26.0	58.0	174°03.3	9.5'	12°46.2	12.6'	59.2'
6	268°26.0	N20°57.6	188°31.8	9.4'	S12°58.8	12.6'	59.2'
7	283°26.0	57.2	203°00.2	9.4'	13°11.4	12.5'	59.2'
8	298°25.9	56.7	217°28.6	9.3'	13°23.9	12.4'	59.2'
9	313°25.9	· · 56.3	231°56.9	9.2'	13°36.3	12.3'	59.2'
10	328°25.8	55.8	246°25.1	9.2'	13°48.7	12.3'	59.2'
11	343°25.8	55.4	260°53.3	9.1'	14°00.9	12.2'	59.2'
12	358°25.7	N20°54.9	275°21.4	9.1'	S14°13.1	12.1'	59.2'
13	13°25.7	54.5	289°49.5	9.0'	14°25.2	12.0'	59.2'
14	28°25.6	54.0	304°17.5	8.9'	14°37.2	11.9'	59.2'
15	43°25.6	· · 53.6	318°45.4	8.9'	14°49.2	11.8'	59.2'
16	58°25.6	53.1	333°13.3	8.8'	15°01.0	11.8'	59.2'
17	73°25.5	52.7	347°41.1	8.7'	15°12.8	11.7'	59.2'
18	88°25.5	N20°52.2	2°08.8	8.7'	S15°24.4	11.6'	59.2'
19	103°25.4	51.8	16°36.5	8.6'	15°36.0	11.5'	59.2'
20	118°25.4	51.3	31°04.1	8.6'	15°47.5	11.4'	59.2'
21	133°25.3	· · 50.9	45°31.7	8.5'	15°58.9	11.3'	59.2'
22	148°25.3	50.4	59°59.1	8.4'	16°10.2	11.2'	59.2'
23	163°25.3	50.0	74°26.6	8.4'	16°21.4	11.1'	59.2'
SD = 15.7'		$d = -0.4'$	SD = 16.2'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°25.2	N20°49.5	88°53.9	8.3'	S16°32.5	11.0'	59.2'
1	193°25.2	49.1	103°21.2	8.2'	16°43.5	10.9'	59.2'
2	208°25.1	48.6	117°48.4	8.2'	16°54.4	10.8'	59.2'
3	223°25.1	· · 48.1	132°15.6	8.1'	17°05.2	10.7'	59.2'
4	238°25.1	47.7	146°42.7	8.0'	17°15.9	10.6'	59.2'
5	253°25.0	47.2	161°09.7	8.0'	17°26.5	10.5'	59.2'
6	268°25.0	N20°46.8	175°36.7	7.9'	S17°37.0	10.4'	59.2'
7	283°24.9	46.3	190°03.6	7.8'	17°47.4	10.3'	59.2'
8	298°24.9	45.9	204°30.4	7.8'	17°57.7	10.2'	59.2'
9	313°24.9	· · 45.4	218°57.2	7.7'	18°07.9	10.1'	59.2'
10	328°24.8	44.9	233°23.9	7.6'	18°17.9	9.9'	59.2'
11	343°24.8	44.5	247°50.5	7.6'	18°27.9	9.8'	59.2'
12	358°24.7	N20°44.0	262°17.1	7.5'	S18°37.7	9.7'	59.2'
13	13°24.7	43.5	276°43.6	7.4'	18°47.5	9.6'	59.2'
14	28°24.7	43.1	291°10.0	7.4'	18°57.1	9.5'	59.2'
15	43°24.6	· · 42.6	305°36.4	7.3'	19°06.5	9.4'	59.2'
16	58°24.6	42.2	320°02.7	7.2'	19°15.9	9.3'	59.2'
17	73°24.5	41.7	334°28.9	7.2'	19°25.2	9.1'	59.2'
18	88°24.5	N20°41.2	348°55.1	7.1'	S19°34.3	9.0'	59.2'
19	103°24.5	40.8	3°21.2	7.0'	19°43.3	8.9'	59.2'
20	118°24.4	40.3	17°47.2	7.0'	19°52.2	8.8'	59.2'
21	133°24.4	· · 39.8	32°13.2	6.9'	20°01.0	8.6'	59.2'
22	148°24.4	39.4	46°39.1	6.9'	20°09.6	8.5'	59.1'
23	163°24.3	38.9	61°05.0	6.8'	20°18.1	8.4'	59.1'
SD = 15.7'		$d = -0.5'$	SD = 16.2'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°24.3	N20°38.4	75°30.8	6.7'	S20°26.5	8.3'	59.1'
1	193°24.3	37.9	89°56.5	6.7'	20°34.7	8.1'	59.1'
2	208°24.2	37.5	104°22.2	6.6'	20°42.8	8.0'	59.1'
3	223°24.2	· · 37.0	118°47.8	6.6'	20°50.8	7.9'	59.1'
4	238°24.2	36.5	133°13.4	6.5'	20°58.7	7.7'	59.1'
5	253°24.1	36.1	147°38.9	6.4'	21°06.4	7.6'	59.1'
6	268°24.1	N20°35.6	162°04.3	6.4'	S21°14.0	7.5'	59.1'
7	283°24.1	35.1	176°29.7	6.3'	21°21.5	7.3'	59.1'
8	298°24.0	34.6	190°55.0	6.3'	21°28.8	7.2'	59.1'
9	313°24.0	· · 34.2	205°20.3	6.2'	21°35.9	7.0'	59.1'
10	328°24.0	33.7	219°45.5	6.2'	21°43.0	6.9'	59.1'
11	343°23.9	33.2	234°10.7	6.1'	21°49.9	6.8'	59.1'
12	358°23.9	N20°32.7	248°35.8	6.1'	S21°56.6	6.6'	59.1'
13	13°23.9	32.3	263°00.9	6.0'	22°03.2	6.5'	59.1'
14	28°23.8	31.8	277°25.9	6.0'	22°09.7	6.3'	59.0'
15	43°23.8	· · 31.3	291°50.8	5.9'	22°16.0	6.2'	59.0'
16	58°23.8	30.8	306°15.7	5.9'	22°22.2	6.0'	59.0'
17	73°23.7	30.4	320°40.6	5.8'	22°28.3	5.9'	59.0'
18	88°23.7	N20°29.9	335°05.4	5.8'	S22°34.1	5.7'	59.0'
19	103°23.7	29.4	349°30.2	5.7'	22°39.9	5.6'	59.0'
20	118°23.6	28.9	3°54.9	5.7'	22°45.5	5.4'	59.0'
21	133°23.6	· · 28.4	18°19.6	5.6'	22°50.9	5.3'	59.0'
22	148°23.6	27.9	32°44.3	5.6'	22°56.2	5.1'	59.0'
23	163°23.5	27.5	47°08.9	5.6'	23°01.4	5.0'	59.0'
SD = 15.7'		$d = -0.5'$	SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	—:—	—:—	00:49	23:14	—:—	—:—
66°	—:—	—:—	01:54	22:15	—:—	—:—
64°	—:—	—:—	02:28	21:42	—:—	—:—
62°	—:—	01:17	02:53	21:18	22:51	—:—
60°	—:—	01:59	03:12	20:59	22:11	—:—
N 58°	—:—	02:26	03:28	20:43	21:44	—:—
56°	01:10	02:47	03:41	20:30	21:24	22:58
54°	01:49	03:04	03:53	20:19	21:07	22:21
52°	02:15	03:19	04:03	20:09	20:53	21:56
50°	02:34	03:31	04:12	19:59	20:41	21:37
45°	03:10	03:56	04:32	19:40	20:16	21:01
N 40°	03:36	04:16	04:47	19:25	19:57	20:36
35°	03:56	04:31	05:00	19:12	19:41	20:16
30°	04:13	04:45	05:11	19:01	19:28	20:00
20°	04:38	05:07	05:31	18:42	19:06	19:34
N 10°	04:58	05:25	05:47	18:25	18:48	19:14
0°	05:15	05:41	06:03	18:10	18:32	18:58
S 10°	05:30	05:56	06:18	17:55	18:17	18:43
20°	05:44	06:11	06:34	17:39	18:02	18:29
30°	05:58	06:27	06:53	17:20	17:46	18:15
35°	06:05	06:36	07:03	17:09	17:37	18:08
40°	06:13	06:46	07:16	16:57	17:27	18:00
45°	06:21	06:57	07:30	16:43	17:16	17:52
S 50°	06:31	07:11	07:48	16:25	17:02	17:42
52°	06:35	07:17	07:56	16:17	16:56	17:38
54°	06:39	07:24	08:05	16:08	16:49	17:34
56°	06:44	07:31	08:16	15:58	16:42	17:29
58°	06:50	07:40	08:27	15:46	16:34	17:24
S 60°	06:55	07:49	08:41	15:32	16:24	17:18

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	15:30			19:50		
N 70°	14:54	18:06		20:28	19:11	
68°	14:28	16:47		20:55	20:30	
66°	14:09	16:10	18:22	21:16	21:09	20:57
64°	13:53	15:44	17:34	21:32	21:36	21:45
62°	13:41	15:24	17:04	21:46	21:57	22:16
60°	13:30	15:07	16:41	21:58	22:14	22:39
N 58°	13:20	14:53	16:23	22:08	22:28	22:58
56°	13:12	14:41	16:07	22:17	22:41	23:14
54°	13:05	14:31	15:54	22:25	22:52	23:28
52°	12:59	14:22	15:42	22:32	23:01	23:40
50°	12:53	14:14	15:32	22:39	23:10	23:50
45°	12:40	13:56	15:11	22:53	23:29	· ·
N 40°	12:30	13:42	14:54	23:04	23:44	· ·
35°	12:21	13:30	14:39	23:14	23:57	· ·
30°	12:14	13:20	14:27	23:23	· ·	00:08
20°	12:00	13:02	14:05	23:38	· ·	00:27
N 10°	11:49	12:47	13:47	23:51	· ·	00:44
0°	11:38	12:33	13:30	· ·	00:04	01:00
S 10°	11:28	12:19	13:13	· ·	00:16	01:15
20°	11:17	12:03	12:55	· ·	00:30	01:32
30°	11:04	11:46	12:34	· ·	00:45	01:52
35°	10:57	11:36	12:22	· ·	00:54	02:03
40°	10:49	11:25	12:08	· ·	01:04	02:16
45°	10:39	11:12	11:51	00:00	01:16	02:32
S 50°	10:28	10:56	11:31	00:10	01:31	02:51
52°	10:22	10:48	11:22	00:15	01:38	03:00
54°	10:17	10:40	11:11	00:20	01:46	03:11
56°	10:10	10:31	10:59	00:25	01:54	03:22
58°	10:03	10:20	10:45	00:31	02:04	03:36
S 60°	09:55	10:08	10:29	00:38	02:15	03:51

Day	Sun		Mer. Pass	Moon		Age 7-9 43-66%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
18	06:15	06:17	12:06	17:51	05:25	
19	06:19	06:21	12:06	18:46	06:18	
20	06:23	06:24	12:06	19:44	07:15	

Aries Venus

<https://pypi.org/project/sfalmanac/> Page 136

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	178°23.5	N20°27.0	61°33.4	5.5'	S23°06.4	4.8' 59.0'
1	193°23.5	26.5	75°58.0	5.5'	23°11.2	4.7' 59.0'
2	208°23.5	26.0	90°22.5	5.5'	23°15.9	4.5' 58.9'
3	223°23.4	· · 25.5	104°46.9	5.4'	23°20.4	4.4' 58.9'
4	238°23.4	25.0	119°11.4	5.4'	23°24.8	4.2' 58.9'
5	253°23.4	24.6	133°35.8	5.4'	23°29.0	4.1' 58.9'
6	268°23.3	N20°24.1	148°00.1	5.3'	S23°33.1	3.9' 58.9'
7	283°23.3	23.6	162°24.5	5.3'	23°37.0	3.8' 58.9'
8	298°23.3	23.1	176°48.8	5.3'	23°40.8	3.6' 58.9'
9	313°23.3	· · 22.6	191°13.1	5.3'	23°44.4	3.4' 58.9'
10	328°23.2	22.1	205°37.3	5.2'	23°47.8	3.3' 58.9'
11	343°23.2	21.6	220°01.6	5.2'	23°51.1	3.1' 58.9'
12	358°23.2	N20°21.1	234°25.8	5.2'	S23°54.2	3.0' 58.8'
13	13°23.2	20.6	248°50.0	5.2'	23°57.2	2.8' 58.8'
14	28°23.1	20.1	263°14.2	5.2'	24°00.0	2.7' 58.8'
15	43°23.1	· · 19.7	277°38.4	5.2'	24°02.7	2.5' 58.8'
16	58°23.1	19.2	292°02.5	5.1'	24°05.2	2.3' 58.8'
17	73°23.1	18.7	306°26.7	5.1'	24°07.5	2.2' 58.8'
18	88°23.0	N20°18.2	320°50.8	5.1'	S24°09.7	2.0' 58.8'
19	103°23.0	17.7	335°15.0	5.1'	24°11.7	1.9' 58.8'
20	118°23.0	17.2	349°39.1	5.1'	24°13.5	1.7' 58.8'
21	133°23.0	· · 16.7	4°03.2	5.1'	24°15.2	1.5' 58.7'
22	148°22.9	16.2	18°27.3	5.1'	24°16.8	1.4' 58.7'
23	163°22.9	15.7	32°51.4	5.1'	24°18.1	1.2' 58.7'
SD = 15.7' d = -0.5'			SD = 16.1'			
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	178°22.9	N20°15.2	47°15.5	5.1'	S24°19.4	1.0' 58.7'
1	193°22.9	14.7	61°39.7	5.1'	24°20.4	0.9' 58.7'
2	208°22.8	14.2	76°03.8	5.1'	24°21.3	0.7' 58.7'
3	223°22.8	· · 13.7	90°27.9	5.1'	24°22.0	0.6' 58.7'
4	238°22.8	13.2	104°52.1	5.1'	24°22.6	0.4' 58.7'
5	253°22.8	12.7	119°16.2	5.2'	24°23.0	0.2' 58.6'
6	268°22.8	N20°12.2	133°40.4	5.2'	S24°23.2	0.1' 58.6'
7	283°22.7	11.7	148°04.5	5.2'	24°23.3	-0.1' 58.6'
8	298°22.7	11.2	162°28.7	5.2'	24°23.3	-0.2' 58.6'
9	313°22.7	· · 10.7	176°52.9	5.2'	24°23.0	-0.4' 58.6'
10	328°22.7	10.2	191°17.2	5.2'	24°22.6	-0.6' 58.6'
11	343°22.6	09.7	205°41.4	5.3'	24°22.1	-0.7' 58.6'
12	358°22.6	N20°09.2	220°05.7	5.3'	S24°21.4	-0.9' 58.5'
13	13°22.6	08.7	234°30.0	5.3'	24°20.5	-1.0' 58.5'
14	28°22.6	08.2	248°54.3	5.4'	24°19.5	-1.2' 58.5'
15	43°22.6	· · 07.7	263°18.7	5.4'	24°18.3	-1.3' 58.5'
16	58°22.6	07.2	277°43.0	5.4'	24°16.9	-1.5' 58.5'
17	73°22.5	06.6	292°07.5	5.4'	24°15.4	-1.7' 58.5'
18	88°22.5	N20°06.1	306°31.9	5.5'	S24°13.8	-1.8' 58.4'
19	103°22.5	05.6	320°56.4	5.5'	24°12.0	-2.0' 58.4'
20	118°22.5	05.1	335°20.9	5.6'	24°10.0	-2.1' 58.4'
21	133°22.5	· · 04.6	349°45.5	5.6'	24°07.9	-2.3' 58.4'
22	148°22.4	04.1	4°10.1	5.6'	24°05.6	-2.4' 58.4'
23	163°22.4	03.6	18°34.7	5.7'	24°03.1	-2.6' 58.4'
SD = 15.7' d = -0.5'			SD = 16.0'			
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	178°22.4	N20°03.1	32°59.4	5.7'	S24°00.5	-2.7' 58.3'
1	193°22.4	02.6	47°24.1	5.8'	23°57.8	-2.9' 58.3'
2	208°22.4	02.1	61°48.9	5.8'	23°54.9	-3.0' 58.3'
3	223°22.4	· · 01.5	76°13.7	5.9'	23°51.9	-3.2' 58.3'
4	238°22.3	01.0	90°38.5	5.9'	23°48.7	-3.3' 58.3'
5	253°22.3	00.5	105°03.5	6.0'	23°45.3	-3.5' 58.3'
6	268°22.3	N20°00.0	119°28.5	6.0'	S23°41.8	-3.6' 58.2'
7	283°22.3	19°59.5	133°53.6	6.1'	23°38.2	-3.8' 58.2'
8	298°22.3	59.0	148°18.6	6.2'	23°34.4	-3.9' 58.2'
9	313°22.3	· · 58.4	162°43.8	6.2'	23°30.5	-4.1' 58.2'
10	328°22.2	57.9	177°09.0	6.3'	23°26.4	-4.2' 58.2'
11	343°22.2	57.4	191°34.3	6.3'	23°22.2	-4.4' 58.2'
12	358°22.2	N19°56.9	205°59.6	6.4'	S23°17.8	-4.5' 58.1'
13	13°22.2	56.4	220°25.0	6.5'	23°13.3	-4.7' 58.1'
14	28°22.2	55.9	234°50.5	6.5'	23°08.6	-4.8' 58.1'
15	43°22.2	· · 55.3	249°16.0	6.6'	23°03.9	-4.9' 58.1'
16	58°22.2	54.8	263°41.6	6.7'	22°58.9	-5.1' 58.1'
17	73°22.2	54.3	278°07.2	6.7'	22°53.9	-5.2' 58.0'
18	88°22.1	N19°53.8	292°33.0	6.8'	S22°48.6	-5.3' 58.0'
19	103°22.1	53.2	306°58.8	6.9'	22°43.3	-5.5' 58.0'
20	118°22.1	52.7	321°24.6	6.9'	22°37.8	-5.6' 58.0'
21	133°22.1	· · 52.2	335°50.6	7.0'	22°32.2	-5.7' 58.0'
22	148°22.1	51.7	350°16.6	7.1'	22°26.5	-5.9' 57.9'
23	163°22.1	51.2	4°42.7	7.2'	22°20.6	-6.0' 57.9'
SD = 15.7' d = -0.5'			SD = 15.9'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	--	--	01:16	22:50	--	--
66°	--	--	02:06	22:03	--	--
64°	--	--	02:37	21:33	--	--
62°	--	01:32	03:00	21:11	22:36	--
60°	--	02:09	03:18	20:53	22:02	--
N 58°	--	02:34	03:34	20:38	21:37	--
56°	01:24	02:54	03:46	20:25	21:18	22:45
54°	01:58	03:10	03:58	20:14	21:02	22:13
52°	02:21	03:24	04:07	20:05	20:48	21:50
50°	02:40	03:35	04:16	19:56	20:36	21:31
45°	03:15	04:00	04:35	19:38	20:13	20:57
N 40°	03:39	04:18	04:50	19:23	19:54	20:33
35°	03:59	04:34	05:02	19:10	19:39	20:13
30°	04:15	04:47	05:13	18:59	19:26	19:58
20°	04:40	05:08	05:32	18:41	19:05	19:33
N 10°	04:59	05:25	05:48	18:25	18:47	19:14
0°	05:15	05:41	06:03	18:10	18:32	18:58
S 10°	05:30	05:55	06:18	17:55	18:18	18:43
20°	05:43	06:10	06:33	17:40	18:03	18:30
30°	05:57	06:26	06:51	17:22	17:47	18:17
35°	06:04	06:34	07:02	17:12	17:39	18:10
40°	06:11	06:44	07:14	17:00	17:29	18:02
45°	06:19	06:55	07:27	16:46	17:18	17:54
S 50°	06:28	07:08	07:44	16:29	17:05	17:45
52°	06:32	07:14	07:52	16:21	17:00	17:41
54°	06:36	07:20	08:01	16:12	16:53	17:37
56°	06:41	07:27	08:11	16:02	16:46	17:33
58°	06:46	07:35	08:22	15:51	16:38	17:28
S 60°	06:51	07:44	08:36	15:38	16:29	17:22

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°						
N 70°						
68°						
66°			21:15			
64°	19:12	20:10	20:32	22:12	23:18	· ·
62°	18:31	19:30	20:03	22:53	23:58	· ·
60°	18:03	19:03	19:41	23:21	· ·	00:25
N 58°	17:41	18:41	19:22	23:43	· ·	00:47
56°	17:23	18:24	19:07	· ·	00:01	01:04
54°	17:08	18:09	18:54	· ·	00:16	01:19
52°	16:55	17:56	18:42	· ·	00:29	01:32
50°	16:44	17:44	18:32	· ·	00:41	01:43
45°	16:20	17:20	18:11	00:12	01:05	02:07
N 40°	16:01	17:01	17:53	00:30	01:24	02:26
35°	15:45	16:45	17:38	00:45	01:40	02:41
30°	15:31	16:31	17:26	00:58	01:54	02:55
20°	15:08	16:08	17:04	01:20	02:18	03:18
N 10°	14:48	15:48	16:45	01:40	02:38	03:38
0°	14:29	15:29	16:27	01:58	02:57	03:57
S 10°	14:10	15:10	16:09	02:16	03:16	04:15
20°	13:50	14:49	15:50	02:35	03:37	04:35
30°	13:27	14:26	15:28	02:58	04:00	04:58
35°	13:13	14:12	15:15	03:11	04:14	05:11
40°	12:58	13:56	15:00	03:26	04:30	05:27
45°	12:39	13:37	14:42	03:44	04:49	05:45
S 50°	12:16	13:13	14:20	04:07	05:13	06:07
52°	12:05	13:02	14:10	04:17	05:24	06:18
54°	11:53	12:49	13:58	04:30	05:37	06:30
56°	11:39	12:34	13:44	04:44	05:52	06:44
58°	11:22	12:16	13:28	05:00	06:10	07:01
S 60°	11:02	11:55	13:09	05:20	06:32	07:20

Day	Sun		Mer. Pass hh:mm	Moon		Age 10-12 76-92%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
21	06:26	06:27	12:06	20:43	08:13	
22	06:28	06:29	12:06	21:43	09:13	
23	06:30	06:31	12:07	22:40	10:12	

July 24, 25, 26 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	302°00.4	146°31.6	N11°49.0	102°10.3	S08°53.3	103°16.7	S06°37.5	250°29.4	N16°28.3	Alpheratz	357°30.7	29°15.3
1	317°02.9	161°31.2	47.9	117°11.5	53.9	118°18.9	37.7	265°31.7	28.3	Ankaa	353°03.5	-42°08.4
2	332°05.4	176°30.9	46.7	132°12.7	54.4	133°21.1	37.8	280°33.9	28.4	Schedar	349°26.4	56°41.8
3	347°07.8	191°30.5	· · 45.6	147°13.9	· · 55.0	148°23.3	· · 37.9	295°36.2	· · 28.4	Diphda	348°43.5	-17°49.2
4	2°10.3	206°30.1	44.5	162°15.1	55.5	163°25.5	38.0	310°38.5	28.4	Achernar	335°17.6	-57°04.8
5	17°12.7	221°29.8	43.3	177°16.3	56.1	178°27.7	38.1	325°40.8	28.5	Hamal	327°47.0	23°36.1
6	32°15.2	236°29.4	N11°42.2	192°17.5	S08°56.6	193°29.9	S06°38.2	340°43.1	N16°28.5	Polaris	312°07.9	89°22.9
7	47°17.7	251°29.0	41.1	207°18.7	57.2	208°32.2	38.3	355°45.3	28.6	Acamar	315°09.2	-40°10.9
8	62°20.1	266°28.7	39.9	222°19.9	57.7	223°34.4	38.4	10°47.6	28.6	Menkar	314°02.4	4°12.4
9	77°22.6	281°28.3	· · 38.8	237°21.1	· · 58.3	238°36.6	· · 38.5	25°49.9	· · 28.6	Mirfak	308°23.0	49°57.7
10	92°25.1	296°27.9	37.7	252°22.3	58.8	253°38.8	38.7	40°52.2	28.7	Aldebaran	290°35.6	16°34.0
11	107°27.5	311°27.6	36.5	267°23.4	59.4	268°41.0	38.8	55°54.5	28.7	Rigel	281°00.6	-8°10.1
12	122°30.0	326°27.2	N11°35.4	282°24.6	S08°59.9	283°43.2	S06°38.9	70°56.7	N16°28.7	Capella	280°16.8	46°01.4
13	137°32.5	341°26.9	34.2	297°25.8	09°00.5	298°45.4	39.0	85°59.0	28.8	Bellatrix	278°19.2	6°22.5
14	152°34.9	356°26.5	33.1	312°27.0	01.0	313°47.6	39.1	101°01.3	28.8	Elnath	277°57.5	28°37.8
15	167°37.4	11°26.1	· · 32.0	327°28.2	· · 01.6	328°49.8	· · 39.2	116°03.6	· · 28.9	Alnilam	275°34.3	-1°11.0
16	182°39.9	26°25.8	30.8	342°29.4	02.1	343°52.1	39.3	131°05.9	28.9	Betelgeuse	270°48.4	7°24.7
17	197°42.3	41°25.4	29.7	357°30.6	02.7	358°54.3	39.4	146°08.1	28.9	Canopus	263°51.2	-52°42.6
18	212°44.8	56°25.1	N11°28.5	12°31.8	S09°03.2	13°56.5	S06°39.6	161°10.4	N16°29.0	Sirius	258°23.3	-16°45.4
19	227°47.2	71°24.7	27.4	27°33.0	03.8	28°58.7	39.7	176°12.7	29.0	Adhara	255°03.3	-29°00.8
20	242°49.7	86°24.3	26.3	42°34.2	04.3	44°00.9	39.8	191°15.0	29.0	Procyon	244°47.3	5°08.8
21	257°52.2	101°24.0	· · 25.1	57°35.3	· · 04.9	59°03.1	· · 39.9	206°17.3	· · 29.1	Pollux	243°13.2	27°57.1
22	272°54.6	116°23.6	24.0	72°36.5	05.4	74°05.3	40.0	221°19.5	29.1	Avior	234°13.8	-59°36.4
23	287°57.1	131°23.3	22.8	87°37.7	06.0	89°07.5	40.1	236°21.8	29.2	Suhail	222°44.0	-43°33.3
Mer.pass. 03:51		ν -0.4' d -1.1' m -3.91		ν 1.2' d 0.5' m 0.62		ν 2.2' d 0.1' m -1.93		ν 2.3' d 0.0' m 0.25		Miaplacidus	221°38.2	-69°50.5
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	302°59.6	146°22.9	N11°21.7	102°38.9	S09°06.5	104°09.7	S06°40.2	251°24.1	N16°29.2	Alphard	217°44.4	-8°47.3
1	318°02.0	161°22.6	20.5	117°40.1	07.1	119°11.9	40.4	266°26.4	29.2	Regulus	207°30.8	11°49.3
2	333°04.5	176°22.2	19.4	132°41.3	07.6	134°14.1	40.5	281°28.7	29.3	Dubhe	193°37.3	61°35.6
3	348°07.0	191°21.8	· · 18.2	147°42.5	· · 08.2	149°16.4	· · 40.6	296°31.0	· · 29.3	Denebola	182°21.4	14°24.4
4	3°09.4	206°21.5	17.1	162°43.7	08.7	164°18.6	40.7	311°33.2	29.3	Gienah	175°39.9	-17°42.5
5	18°11.9	221°21.1	15.9	177°44.8	09.3	179°20.8	40.8	326°35.5	29.4	Acrux	172°56.0	-63°16.1
6	33°14.4	236°20.8	N11°14.8	192°46.0	S09°09.8	194°23.0	S06°40.9	341°37.8	N16°29.4	Gacrux	171°47.6	-57°17.0
7	48°16.8	251°20.4	13.7	207°47.2	10.4	209°25.2	41.0	356°40.1	29.5	Alioth	166°10.1	55°48.2
8	63°19.3	266°20.1	12.5	222°48.4	10.9	224°27.4	41.2	11°42.4	29.5	Spica	158°18.4	-11°19.0
9	78°21.7	281°19.7	· · 11.4	237°49.6	· · 11.5	239°29.6	· · 41.3	26°44.7	· · 29.5	Alkaid	152°49.3	49°10.2
10	93°24.2	296°19.4	10.2	252°50.8	12.0	254°31.8	41.4	41°46.9	29.6	Hadar	148°30.5	-60°31.2
11	108°26.7	311°19.0	09.1	267°51.9	12.6	269°34.0	41.5	56°49.2	29.6	Menkent	147°53.2	-36°31.1
12	123°29.1	326°18.7	N11°07.9	282°53.1	S09°13.1	284°36.2	S06°41.6	71°51.5	N16°29.6	Arcturus	145°44.6	19°01.9
13	138°31.6	341°18.3	06.8	297°54.3	13.7	299°38.4	41.7	86°53.8	29.7	Rigel Kent.	139°34.9	-60°57.7
14	153°34.1	356°18.0	05.6	312°55.5	14.2	314°40.6	41.8	101°56.1	29.7	Kochab	137°20.0	74°02.3
15	168°36.5	11°17.6	· · 04.5	327°56.7	· · 14.8	329°42.8	· · 42.0	116°58.4	· · 29.7	Zuben'ubi	136°51.8	-16°09.9
16	183°39.0	26°17.3	03.3	342°57.9	15.3	344°45.0	42.1	132°00.6	29.8	Alphecca	126°00.5	26°37.1
17	198°41.5	41°16.9	02.1	357°59.0	15.9	359°47.2	42.2	147°02.9	29.8	Antares	112°11.1	-26°29.8
18	213°43.9	56°16.6	N11°01.0	13°00.2	S09°16.4	14°49.4	S06°42.3	162°05.2	N16°29.9	Atria	107°01.5	-69°04.9
19	228°46.4	71°16.2	10°59.8	28°01.4	17.0	29°51.7	42.4	177°07.5	29.9	Sabik	101°58.3	-15°45.6
20	243°48.9	86°15.9	58.7	43°02.6	17.5	44°53.9	42.5	192°09.8	29.9	Shaula	96°05.0	-37°07.4
21	258°51.3	101°15.5	· · 57.5	58°03.8	· · 18.1	59°56.1	· · 42.6	207°12.1	· · 30.0	Rasalhague	95°54.9	12°32.5
22	273°53.8	116°15.2	56.4	73°04.9	18.6	74°58.3	42.8	222°14.4	30.0	Eltanin	90°40.0	51°29.4
23	288°56.2	131°14.8	55.2	88°06.1	19.2	90°00.5	42.9	237°16.6	30.0	Kaus Aust.	83°27.3	-34°22.1
Mer.pass. 03:47		ν -0.4' d -1.1' m -3.91		ν 1.2' d 0.5' m 0.63		ν 2.2' d 0.1' m -1.92		ν 2.3' d 0.0' m 0.25		Vega	80°30.3	38°48.9
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	303°58.7	146°14.5	N10°54.1	103°07.3	S09°19.7	105°02.7	S06°43.0	252°18.9	N16°30.1	Nunki	75°42.9	-26°15.4
1	319°01.2	161°14.2	52.9	118°08.5	20.3	120°04.9	43.1	267°21.2	30.1	Altair	61°56.0	8°57.0
2	334°03.6	176°13.8	51.8	133°09.6	20.8	135°07.1	43.2	282°23.5	30.1	Peacock	52°59.5	-56°38.1
3	349°06.1	191°13.5	· · 50.6	148°10.8	· · 21.4	150°09.3	· · 43.3	297°25.8	· · 30.2	Deneb	49°22.7	45°23.3
4	4°08.6	206°13.1	49.4	163°12.0	21.9	165°11.5	43.4	312°28.1	30.2	Enif	33°34.9	10°00.9
5	19°11.0	221°12.8	48.3	178°13.2	22.5	180°13.7	43.6	327°30.4	30.2	Al Na'ir	27°28.0	-46°48.7
6	34°13.5	236°12.4	N10°47.1	193°14.4	S09°23.0	195°15.9	S06°43.7	342°32.6	N16°30.3	Fomalhaut	15°10.3	-29°27.6
7	49°16.0	251°12.1	46.0	208°15.5	23.6	210°18.1	43.8	357°34.9	30.3	Scheat	13°41.3	28°14.7
8	64°18.4	266°11.8	44.8	223°16.7	24.1	225°20.3	43.9	12°37.2	30.4	Markab	13°26.0	15°22.0
9	79°20.9	281°11.4	· · 43.6	238°17.9	· · 24.7	240°22.5	· · 44.0	27°39.5	· · 30.4			
10	94°23.3	296°11.1	42.5	253°19.1	25.2	255°24.7	44.1	42°41.8	30.4	Jul 24 Tue	SHA	Mer.pass
11	109°25.8	311°10.7	41.3	268°20.2	25.8	270°26.9	44.3	57°44.1	30.5	Venus	204°31.2	14:14
12	124°28.3	326°10.4	N10°40.2	283°21.4	S09°26.3	285°29.1	S06°44.4	72°46.4	N16°30.5	Mars	160°09.9	17:10
13	139°30.7	341°10.0	39.0	298°22.6	26.9	300°31.3	44.5	87°48.6	30.5	Jupiter	161°16.2	17:04
14	154°33.2	356°09.7	37.8	313°23.8	27.4	315°33.5	44.6	102°50.9	30.6	Saturn	308°29.0	07:17
15	169°35.7	11°09.4	· · 36.7	328°24.9	· · 28.0	330°35.7	· · 44.7	117°53.2	· · 30.6			
16	184°38.1	26°09.0	35.5	343°26.1	28.5	345°37.9	44.8	132°55.5	30.6	Jul 25 Wed	SHA	Mer.pass
17	199°40.6	41°08.7	34.3	358°27.3	29.1	0°40.1	45.0	147°57.8	30.7	Venus	203°23.3	14:15
18	214°43.1	56°08.4	N10°33.2	13°28.4	S09°29.6	15°42.3	S06°45.1	163°00.1	N16°30.7	Mars	159°39.3	17:08
19	229°45.5	71°08.0	32.0	28°29.6	30.2	30°44.5	45.2	178°02.4	30.7	Jupiter	161°10.2	17:01
20	244°48.0	86°07.7	30.9	43°30.8	30.7	45°46.7	45.3	193°0°				

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.1	N19°50.6	19°08.8	7.2'	S22°14.6	-6.1'	57.9'
1	193°22.1	50.1	33°35.1	7.3'	22°08.4	-6.3'	57.9'
2	208°22.1	49.6	48°01.4	7.4'	22°02.2	-6.4'	57.9'
3	223°22.0	· · 49.0	62°27.8	7.5'	21°55.8	-6.5'	57.8'
4	238°22.0	48.5	76°54.3	7.6'	21°49.2	-6.6'	57.8'
5	253°22.0	48.0	91°20.8	7.6'	21°42.6	-6.8'	57.8'
6	268°22.0	N19°47.5	105°47.5	7.7'	S21°35.8	-6.9'	57.8'
7	283°22.0	46.9	120°14.2	7.8'	21°28.9	-7.0'	57.8'
8	298°22.0	46.4	134°41.0	7.9'	21°21.9	-7.1'	57.7'
9	313°22.0	· · 45.9	149°07.9	8.0'	21°14.8	-7.3'	57.7'
10	328°22.0	45.3	163°34.8	8.1'	21°07.5	-7.4'	57.7'
11	343°22.0	44.8	178°01.9	8.1'	21°00.1	-7.5'	57.7'
12	358°22.0	N19°44.3	192°29.0	8.2'	S20°52.7	-7.6'	57.6'
13	13°22.0	43.7	206°56.2	8.3'	20°45.1	-7.7'	57.6'
14	28°21.9	43.2	221°23.6	8.4'	20°37.3	-7.8'	57.6'
15	43°21.9	· · 42.7	235°50.9	8.5'	20°29.5	-7.9'	57.6'
16	58°21.9	42.1	250°18.4	8.6'	20°21.6	-8.0'	57.6'
17	73°21.9	41.6	264°46.0	8.7'	20°13.5	-8.2'	57.5'
18	88°21.9	N19°41.1	279°13.6	8.7'	S20°05.4	-8.3'	57.5'
19	103°21.9	40.5	293°41.4	8.8'	19°57.1	-8.4'	57.5'
20	118°21.9	40.0	308°09.2	8.9'	19°48.7	-8.5'	57.5'
21	133°21.9	· · 39.5	322°37.1	9.0'	19°40.2	-8.6'	57.4'
22	148°21.9	38.9	337°05.1	9.1'	19°31.7	-8.7'	57.4'
23	163°21.9	38.4	351°33.2	9.2'	19°23.0	-8.8'	57.4'
SD = 15.7'			SD = 15.8'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.9	N19°37.8	6°01.4	9.3'	S19°14.2	-8.9'	57.4'
1	193°21.9	37.3	20°29.6	9.4'	19°05.3	-9.0'	57.3'
2	208°21.9	36.8	34°58.0	9.4'	18°56.4	-9.1'	57.3'
3	223°21.9	· · 36.2	49°26.4	9.5'	18°47.3	-9.2'	57.3'
4	238°21.9	35.7	63°55.0	9.6'	18°38.1	-9.3'	57.3'
5	253°21.9	35.1	78°23.6	9.7'	18°28.9	-9.4'	57.3'
6	268°21.9	N19°34.6	92°52.3	9.8'	S18°19.5	-9.4'	57.2'
7	283°21.9	34.1	107°21.1	9.9'	18°10.1	-9.5'	57.2'
8	298°21.9	33.5	121°50.0	10.0'	18°00.5	-9.6'	57.2'
9	313°21.9	· · 33.0	136°19.0	10.1'	17°50.9	-9.7'	57.2'
10	328°21.9	32.4	150°48.0	10.2'	17°41.2	-9.8'	57.1'
11	343°21.9	31.9	165°17.2	10.2'	17°31.4	-9.9'	57.1'
12	358°21.8	N19°31.3	179°46.4	10.3'	S17°21.5	-10.0'	57.1'
13	13°21.8	30.8	194°15.8	10.4'	17°11.6	-10.0'	57.1'
14	28°21.8	30.2	208°45.2	10.5'	17°01.5	-10.1'	57.0'
15	43°21.8	· · 29.7	223°14.7	10.6'	16°51.4	-10.2'	57.0'
16	58°21.8	29.1	237°44.3	10.7'	16°41.2	-10.3'	57.0'
17	73°21.8	28.6	252°13.9	10.8'	16°30.9	-10.4'	57.0'
18	88°21.8	N19°28.0	266°43.7	10.9'	S16°20.6	-10.4'	56.9'
19	103°21.8	27.5	281°13.6	10.9'	16°10.2	-10.5'	56.9'
20	118°21.8	26.9	295°43.5	11.0'	15°59.7	-10.6'	56.9'
21	133°21.8	· · 26.4	310°13.5	11.1'	15°49.1	-10.6'	56.9'
22	148°21.8	25.8	324°43.6	11.2'	15°38.5	-10.7'	56.8'
23	163°21.8	25.3	339°13.8	11.3'	15°27.7	-10.8'	56.8'
SD = 15.7'			SD = 15.6'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.8	N19°24.7	353°44.1	11.4'	S15°17.0	-10.8'	56.8'
1	193°21.9	24.2	8°14.5	11.4'	15°06.1	-10.9'	56.8'
2	208°21.9	23.6	22°44.9	11.5'	14°55.2	-11.0'	56.7'
3	223°21.9	· · 23.1	37°15.4	11.6'	14°44.2	-11.0'	56.7'
4	238°21.9	22.5	51°46.0	11.7'	14°33.2	-11.1'	56.7'
5	253°21.9	22.0	66°16.7	11.8'	14°22.1	-11.2'	56.7'
6	268°21.9	N19°21.4	80°47.5	11.9'	S14°10.9	-11.2'	56.6'
7	283°21.9	20.9	95°18.3	11.9'	13°59.7	-11.3'	56.6'
8	298°21.9	20.3	109°49.3	12.0'	13°48.5	-11.3'	56.6'
9	313°21.9	· · 19.7	124°20.3	12.1'	13°37.1	-11.4'	56.6'
10	328°21.9	19.2	138°51.4	12.2'	13°25.7	-11.4'	56.5'
11	343°21.9	18.6	153°22.6	12.2'	13°14.3	-11.5'	56.5'
12	358°21.9	N19°18.1	167°53.8	12.3'	S13°02.8	-11.5'	56.5'
13	13°21.9	17.5	182°25.1	12.4'	12°51.3	-11.6'	56.5'
14	28°21.9	16.9	196°56.5	12.5'	12°39.7	-11.6'	56.4'
15	43°21.9	· · 16.4	211°28.0	12.6'	12°28.0	-11.7'	56.4'
16	58°21.9	15.8	225°59.6	12.6'	12°16.3	-11.7'	56.4'
17	73°21.9	15.3	240°31.2	12.7'	12°04.6	-11.8'	56.4'
18	88°21.9	N19°14.7	255°02.9	12.8'	S11°52.8	-11.8'	56.3'
19	103°21.9	14.1	269°34.7	12.8'	11°41.0	-11.9'	56.3'
20	118°21.9	13.6	284°06.5	12.9'	11°29.1	-11.9'	56.3'
21	133°21.9	· · 13.0	298°38.4	13.0'	11°17.2	-12.0'	56.3'
22	148°21.9	12.4	313°10.4	13.1'	11°05.3	-12.0'	56.2'
23	163°22.0	11.9	327°42.5	13.1'	10°53.3	-12.0'	56.2'
SD = 15.7'			SD = 15.5'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	—	—	01:37	22:31	—	—
66°	—	—	02:19	21:51	—	—
64°	—	00:47	02:47	21:24	23:16	—
62°	—	01:46	03:08	21:03	22:23	—
60°	—	02:18	03:25	20:46	21:52	—
N 58°	00:43	02:42	03:39	20:32	21:29	23:21
56°	01:37	03:00	03:52	20:20	21:11	22:33
54°	02:07	03:16	04:02	20:10	20:56	22:04
52°	02:28	03:29	04:12	20:00	20:43	21:43
50°	02:46	03:40	04:20	19:52	20:32	21:26
45°	03:19	04:03	04:38	19:35	20:09	20:53
N 40°	03:43	04:21	04:52	19:20	19:51	20:29
35°	04:02	04:36	05:04	19:08	19:37	20:11
30°	04:17	04:49	05:15	18:58	19:24	19:56
20°	04:41	05:09	05:33	18:40	19:04	19:32
N 10°	05:00	05:26	05:49	18:24	18:47	19:13
0°	05:16	05:41	06:03	18:10	18:32	18:57
S 10°	05:29	05:55	06:17	17:56	18:18	18:44
20°	05:42	06:09	06:32	17:41	18:04	18:31
30°	05:55	06:24	06:50	17:24	17:49	18:18
35°	06:02	06:33	07:00	17:14	17:41	18:12
40°	06:09	06:42	07:11	17:02	17:32	18:05
45°	06:17	06:52	07:24	16:49	17:21	17:57
S 50°	06:25	07:05	07:41	16:33	17:09	17:49
52°	06:29	07:10	07:48	16:25	17:03	17:45
54°	06:33	07:16	07:57	16:17	16:57	17:41
56°	06:37	07:23	08:06	16:07	16:50	17:37
58°	06:42	07:31	08:17	15:57	16:43	17:32
S 60°	06:47	07:39	08:30	15:44	16:34	17:27

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	22:21	21:38	■	■	02:51
N 70°	22:43	21:45	21:20	■	00:42	03:26
68°	21:41	21:19	21:05	■	01:43	03:51
66°	21:05	20:59	20:53	00:14	02:18	04:09
64°	20:40	20:43	20:43	00:57	02:42	04:25
62°	20:20	20:29	20:35	01:26	03:02	04:37
60°	20:03	20:17	20:27	01:48	03:18	04:48
N 58°	19:49	20:07	20:21	02:05	03:31	04:57
56°	19:37	19:59	20:15	02:20	03:42	05:05
54°	19:27	19:51	20:09	02:33	03:52	05:12
52°	19:17	19:44	20:05	02:44	04:01	05:18
50°	19:09	19:37	20:00	02:54	04:09	05:24
45°	18:51	19:24	19:51	03:15	04:26	05:36
N 40°	18:36	19:12	19:43	03:32	04:40	05:47
35°	18:24	19:03	19:36	03:46	04:51	05:55
30°	18:13	18:54	19:30	03:58	05:01	06:03
20°	17:54	18:39	19:20	04:19	05:19	06:16
N 10°	17:38	18:26	19:11	04:37	05:34	06:27
0°	17:22	18:14	19:02	04:54	05:48	06:38
S 10°	17:07	18:02	18:54	05:10	06:01	06:48
20°	16:50	17:49	18:44	05:28	06:16	06:59
30°	16:31	17:34	18:34	05:49	06:33	07:12
35°	16:20	17:25	18:28	06:00	06:43	07:19
40°	16:07	17:15	18:21	06:14	06:54	07:27
45°	15:52	17:03	18:12	06:30	07:07	07:36
S 50°	15:33	16:49	18:03	06:50	07:22	07:48
52°	15:25	16:42	17:58	06:59	07:30	07:53
54°	15:15	16:34	17:53	07:09	07:38	07:59
56°	15:03	16:26	17:47	07:21	07:47	08:05
58°	14:50	16:16	17:41	07:34	07:57	08:12
S 60°	14:35	16:05	17:34	07:50	08:09	08:21

Day	Sun		Mer. Pass hh:mm	Moon		Age 13-15 97-100%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
24	06:32	06:32	12:07	23:35	11:08	
25	06:32	06:33	12:07	—	12:01	
26	06:33	06:32	12:07	00:26	12:50	

July 27, 28, 29 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	304°57.8	146°06.3	N10°26.2	103°35.5	S09°32.9	105°55.5	S06°45.8	253°13.8	N16°30.9	Alpheratz	357°30.7	29°15.3					
1	320°00.3	161°06.0	25.0	118°36.6	33.5	120°57.7	45.9	268°16.1	31.0	Ankaa	353°03.4	-42°08.3					
2	335°02.8	176°05.7	23.9	133°37.8	34.0	135°59.9	46.0	283°18.4	31.0	Schedar	349°26.4	56°41.9					
3	350°05.2	191°05.3	•• 22.7	148°39.0	•• 34.6	151°02.1	•• 46.1	298°20.7	•• 31.0	Diphda	348°43.5	-17°49.2					
4	5°07.7	206°05.0	21.5	163°40.1	35.2	166°04.3	46.3	313°23.0	31.1	Achernar	335°17.5	-57°04.8					
5	20°10.2	221°04.7	20.3	178°41.3	35.7	181°06.5	46.4	328°25.3	31.1	Hamal	327°47.0	23°36.1					
6	35°12.6	236°04.3	N10°19.2	193°42.5	S09°36.3	196°08.7	S06°46.5	343°27.6	N16°31.1	Polaris	312°06.5	89°22.9					
7	50°15.1	251°04.0	18.0	208°43.6	36.8	211°10.9	46.6	358°29.8	31.2	Acamar	315°09.1	-40°10.9					
8	65°17.6	266°03.7	16.8	223°44.8	37.4	226°13.1	46.7	13°32.1	31.2	Menkar	314°02.3	4°12.4					
9	80°20.0	281°03.4	•• 15.7	238°46.0	•• 37.9	241°15.3	•• 46.8	28°34.4	•• 31.2	Mirfak	308°23.0	49°57.7					
10	95°22.5	296°03.0	14.5	253°47.1	38.5	256°17.5	47.0	43°36.7	31.3	Aldebaran	290°35.6	16°34.0					
11	110°25.0	311°02.7	13.3	268°48.3	39.0	271°19.7	47.1	58°39.0	31.3	Rigel	281°00.6	-8°10.0					
12	125°27.4	326°02.4	N10°12.1	283°49.5	S09°39.6	286°21.9	S06°47.2	73°41.3	N16°31.3	Capella	280°16.7	46°01.4					
13	140°29.9	341°02.0	11.0	298°50.6	40.1	301°24.0	47.3	88°43.6	31.4	Bellatrix	278°19.2	6°22.5					
14	155°32.3	356°01.7	09.8	313°51.8	40.7	316°26.2	47.4	103°45.9	31.4	Elnath	277°57.5	28°37.8					
15	170°34.8	11°01.4	•• 08.6	328°53.0	•• 41.2	331°28.4	•• 47.6	118°48.2	•• 31.4	Alnilam	275°34.2	-1°11.0					
16	185°37.3	26°01.1	07.5	343°54.1	41.8	346°30.6	47.7	133°50.5	31.5	Betelgeuse	270°48.4	7°24.7					
17	200°39.7	41°00.7	06.3	358°55.3	42.3	1°32.8	47.8	148°52.8	31.5	Canopus	263°51.2	-52°42.6					
18	215°42.2	56°00.4	N10°05.1	13°56.4	S09°42.9	16°35.0	S06°47.9	163°55.0	N16°31.5	Sirius	258°23.3	-16°45.4					
19	230°44.7	71°00.1	03.9	28°57.6	43.4	31°37.2	48.0	178°57.3	31.6	Adhara	255°03.3	-29°00.8					
20	245°47.1	85°59.7	02.8	43°58.8	44.0	46°39.4	48.1	193°59.6	31.6	Procyon	244°47.3	5°08.8					
21	260°49.6	100°59.4	•• 01.6	58°59.9	•• 44.5	61°41.6	•• 48.3	209°01.9	•• 31.6	Pollux	243°13.2	27°57.1					
22	275°52.1	115°59.1	10°00.4	74°01.1	45.1	76°43.8	48.4	224°04.2	31.7	Avior	234°13.8	-59°36.4					
23	290°54.5	130°58.8	09°59.2	89°02.3	45.6	91°46.0	48.5	239°06.5	31.7	Suhail	222°44.0	-43°33.3					
Mer.pass. 03:40			ν -0.3' d -1.2' m -3.92			ν 1.2' d 0.6' m 0.64			ν 2.2' d 0.1' m -1.91			ν 2.3' d 0.0' m 0.24					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	305°57.0	145°58.5	N09°58.0	104°03.4	S09°46.2	106°48.2	S06°48.6	254°08.8	N16°31.7					
1	320°59.4	160°58.1	56.9	119°04.6	46.7	121°50.4	48.7	269°11.1	31.8					
2	336°01.9	175°57.8	55.7	134°05.7	47.3	136°52.6	48.9	284°13.4	31.8					
3	351°04.4	190°57.5	•• 54.5	149°06.9	•• 47.8	151°54.8	•• 49.0	299°15.7	•• 31.9					
4	6°06.8	205°57.2	53.3	164°08.1	48.4	166°57.0	49.1	314°18.0	31.9					
5	21°09.3	220°56.8	52.2	179°09.2	48.9	181°59.1	49.2	329°20.3	31.9					
6	36°11.8	235°56.5	N09°51.0	194°10.4	S09°49.5	197°01.3	S06°49.3	344°22.6	N16°32.0					
7	51°14.2	250°56.2	49.8	209°11.5	50.0	212°03.5	49.5	359°24.9	32.0					
8	66°16.7	265°55.9	48.6	224°12.7	50.6	227°05.7	49.6	14°27.1	32.0					
9	81°19.2	280°55.6	•• 47.4	239°13.8	•• 51.2	242°07.9	•• 49.7	29°29.4	•• 32.1					
10	96°21.6	295°55.2	46.2	254°15.0	51.7	257°10.1	49.8	44°31.7	32.1					
11	111°24.1	310°54.9	45.1	269°16.2	52.3	272°12.3	49.9	59°34.0	32.1					
12	126°26.6	325°54.6	N09°43.9	284°17.3	S09°52.8	287°14.5	S06°50.1	74°36.3	N16°32.2					
13	141°29.0	340°54.3	42.7	299°18.5	53.4	302°16.7	50.2	89°38.6	32.2					
14	156°31.5	355°54.0	41.5	314°19.6	53.9	317°18.9	50.3	104°40.9	32.2					
15	171°33.9	10°53.6	•• 40.3	329°20.8	•• 54.5	332°21.1	•• 50.4	119°43.2	•• 32.3					
16	186°36.4	25°53.3	39.1	344°21.9	55.0	347°23.2	50.5	134°45.5	32.3					
17	201°38.9	40°53.0	38.0	359°23.1	55.6	2°25.4	50.7	149°47.8	32.3					
18	216°41.3	55°52.7	N09°36.8	14°24.2	S09°56.1	17°27.6	S06°50.8	164°50.1	N16°32.4					
19	231°43.8	70°52.4	35.6	29°25.4	56.7	32°29.8	50.9	179°52.4	32.4					
20	246°46.3	85°52.1	34.4	44°26.5	57.2	47°32.0	51.0	194°54.7	32.4					
21	261°48.7	100°51.7	•• 33.2	59°27.7	•• 57.8	62°34.2	•• 51.1	209°57.0	•• 32.5					
22	276°51.2	115°51.4	32.0	74°28.9	58.3	77°36.4	51.3	224°59.3	32.5					
23	291°53.7	130°51.1	30.8	89°30.0	58.9	92°38.6	51.4	240°01.6	32.5					
Mer.pass. 03:36			ν -0.3' d -1.2' m -3.92			ν 1.2' d 0.6' m 0.65			ν 2.2' d 0.1' m -1.91			ν 2.3' d 0.0' m 0.24		

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	306°56.1	145°50.8	N09°29.7	104°31.2	S09°59.4	107°40.8	S06°51.5	255°03.9	N16°32.6					
1	321°58.6	160°50.5	28.5	119°32.3	10°00.0	122°42.9	51.6	270°06.2	32.6					
2	337°01.0	175°50.2	27.3	134°33.5	00.5	137°45.1	51.7	285°08.5	32.6					
3	352°03.5	190°49.9	•• 26.1	149°34.6	•• 01.1	152°47.3	•• 51.9	300°10.8	•• 32.7					
4	7°06.0	205°49.6	24.9	164°35.8	01.6	167°49.5	52.0	315°13.0	32.7					
5	22°08.4	220°49.2	23.7	179°36.9	02.2	182°51.7	52.1	330°15.3	32.7					
6	37°10.9	235°48.9	N09°22.5	194°38.1	S10°02.7	197°53.9	S06°52.2	345°17.6	N16°32.7					
7	52°13.4	250°48.6	21.3	209°39.2	03.3	212°56.1	52.4	0°19.9	32.8					
8	67°15.8	265°48.3	20.1	224°40.4	03.9	227°58.3	52.5	15°22.2	32.8					
9	82°18.3	280°48.0	•• 19.0	239°41.5	•• 04.4	243°00.4	•• 52.6	30°24.5	•• 32.8					
10	97°20.8	295°47.7	17.8	254°42.7	05.0	258°02.6	52.7	45°26.8	32.9					
11	112°23.2	310°47.4	16.6	269°43.8	05.5	273°04.8	52.8	60°29.1	32.9					
12	127°25.7	325°47.1	N09°15.4	284°44.9	S10°06.1	288°07.0	S06°53.0	75°31.4	N16°32.9					
13	142°28.2	340°46.8	14.2	299°46.1	06.6	303°09.2	53.1	90°33.7	33.0					
14	157°30.6	355°46.5	13.0	314°47.2	07.2	318°11.4	53.2	105°36.0	33.0					
15	172°33.1	10°46.1	•• 11.8	329°48.4	•• 07.7	333°13.5	•• 53.3	120°38.3	•• 33.0					
16	187°35.5	25°45.8	10.6	344°49.5	08.3	348°15.7	53.5	135°40.6	33.1					
17	202°38.0	40°45.5	09.4	359°50.7	08.8	3°17.9	53.6	150°42.9	33.1					
18	217°40.5	55°45.2	N09°08.2	14°51.8	S10°09.4	18°20.1	S06°53.7	165°45.2	N16°33.1					
19	232°42.9	70°44.9	07.0	29°53.0	09.9	33°22.3	53.8	180°47.5	33.2					
20	247°45.4	85°44.6	05.8	44°54.1	10.5	48°24.5	53.9	195°49.8	33.2					
21	262°47.9	100°44.3	•• 04.6	59°55.3	•• 11.0	63°26.7	•• 54.1	210°52.1	•• 33.2					
22	277°50.3	115°44.0	03.4	74°56.4	11.6	78°28.8	54.2	225°54.4	33.3					
23	292°52.8	130°43.7	02.2	89°57.5	12.1	93°31.0	54.3	240°56.7	33.3					
Mer.pass. 03:32			ν -0.3' d -1.2' m -3.92			ν 1.2' d 0.6' m 0.66			ν 2.2' d 0.1' m -1.90			ν 2.3' d 0.0' m 0.24		

Jul 27 Fri	SHA	Mer.pass
Venus	201°08.5	14:16
Mars	158°37.6	17:04
Jupiter	160°57.6	16:54
Saturn	308°16.0	07:06

Jul 28 Sat	SHA	Mer.pass
Venus	200°01.5	14:16
Mars	158°06.4	17:02
Jupiter	160°51.2	16:50
Saturn	308°11.8	07:02

Jul 29 Sun	SHA	Mer.pass
Venus	198°54.7	14:17
Mars	157°35.0	17:01
Jupiter	160°44.6	16:47
Saturn	308°07.7	06:59

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.0	N19°11.3	342°14.6	13.2'	S10°41.2	-12.1'	56.2'
1	193°22.0	10.7	356°46.8	13.3'	10°29.2	-12.1'	56.2'
2	208°22.0	10.2	11°19.0	13.3'	10°17.1	-12.1'	56.1'
3	223°22.0	· · 09.6	25°51.4	13.4'	10°04.9	-12.2'	56.1'
4	238°22.0	09.0	40°23.8	13.5'	09°52.7	-12.2'	56.1'
5	253°22.0	08.5	54°56.2	13.5'	09°40.5	-12.2'	56.1'
6	268°22.0	N19°07.9	69°28.8	13.6'	S09°28.3	-12.3'	56.0'
7	283°22.0	07.3	84°01.3	13.6'	09°16.0	-12.3'	56.0'
8	298°22.0	06.8	98°34.0	13.7'	09°03.7	-12.3'	56.0'
9	313°22.0	· · 06.2	113°06.7	13.8'	08°51.4	-12.4'	56.0'
10	328°22.1	05.6	127°39.5	13.8'	08°39.1	-12.4'	55.9'
11	343°22.1	05.1	142°12.3	13.9'	08°26.7	-12.4'	55.9'
12	358°22.1	N19°04.5	156°45.2	14.0'	S08°14.3	-12.4'	55.9'
13	13°22.1	03.9	171°18.1	14.0'	08°01.9	-12.5'	55.9'
14	28°22.1	03.3	185°51.2	14.1'	07°49.4	-12.5'	55.8'
15	43°22.1	· · 02.8	200°24.2	14.1'	07°36.9	-12.5'	55.8'
16	58°22.1	02.2	214°57.3	14.2'	07°24.4	-12.5'	55.8'
17	73°22.1	01.6	229°30.5	14.2'	07°11.9	-12.5'	55.8'
18	88°22.1	N19°01.0	244°03.7	14.3'	S06°59.4	-12.6'	55.7'
19	103°22.2	19°00.5	258°37.0	14.3'	06°46.8	-12.6'	55.7'
20	118°22.2	18°59.9	273°10.4	14.4'	06°34.3	-12.6'	55.7'
21	133°22.2	· · 59.3	287°43.7	14.4'	06°21.7	-12.6'	55.7'
22	148°22.2	58.7	302°17.2	14.5'	06°09.1	-12.6'	55.7'
23	163°22.2	58.1	316°50.7	14.5'	05°56.5	-12.6'	55.6'
SD = 15.7'			SD = 15.3'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.2	N18°57.6	331°24.2	14.6'	S05°43.8	-12.6'	55.6'
1	193°22.2	57.0	345°57.8	14.6'	05°31.2	-12.7'	55.6'
2	208°22.3	56.4	0°31.4	14.7'	05°18.5	-12.7'	55.6'
3	223°22.3	· · 55.8	15°05.1	14.7'	05°05.9	-12.7'	55.5'
4	238°22.3	55.2	29°38.8	14.8'	04°53.2	-12.7'	55.5'
5	253°22.3	54.7	44°12.6	14.8'	04°40.5	-12.7'	55.5'
6	268°22.3	N18°54.1	58°46.4	14.8'	S04°27.8	-12.7'	55.5'
7	283°22.3	53.5	73°20.2	14.9'	04°15.1	-12.7'	55.4'
8	298°22.3	52.9	87°54.1	14.9'	04°02.4	-12.7'	55.4'
9	313°22.4	· · 52.3	102°28.0	15.0'	03°49.7	-12.7'	55.4'
10	328°22.4	51.7	117°02.0	15.0'	03°37.0	-12.7'	55.4'
11	343°22.4	51.2	131°36.0	15.0'	03°24.3	-12.7'	55.4'
12	358°22.4	N18°50.6	146°10.0	15.1'	S03°11.6	-12.7'	55.3'
13	13°22.4	50.0	160°44.1	15.1'	02°58.8	-12.7'	55.3'
14	28°22.4	49.4	175°18.2	15.1'	02°46.1	-12.7'	55.3'
15	43°22.5	· · 48.8	189°52.4	15.2'	02°33.4	-12.7'	55.3'
16	58°22.5	48.2	204°26.6	15.2'	02°20.7	-12.7'	55.2'
17	73°22.5	47.6	219°00.8	15.2'	02°08.0	-12.7'	55.2'
18	88°22.5	N18°47.1	233°35.0	15.3'	S01°55.2	-12.7'	55.2'
19	103°22.5	46.5	248°09.3	15.3'	01°42.5	-12.7'	55.2'
20	118°22.6	45.9	262°43.6	15.3'	01°29.8	-12.7'	55.2'
21	133°22.6	· · 45.3	277°17.9	15.4'	01°17.1	-12.7'	55.1'
22	148°22.6	44.7	291°52.3	15.4'	01°04.4	-12.7'	55.1'
23	163°22.6	44.1	306°26.7	15.4'	00°51.7	-12.7'	55.1'
SD = 15.7'			SD = 15.2'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.6	N18°43.5	321°01.1	15.4'	S00°39.0	-12.7'	55.1'
1	193°22.7	42.9	335°35.6	15.5'	00°26.4	-12.7'	55.1'
2	208°22.7	42.3	350°10.0	15.5'	00°13.7	-12.7'	55.0'
3	223°22.7	· · 41.7	4°44.5	15.5'	S00°01.0	-12.7'	55.0'
4	238°22.7	41.1	19°19.0	15.5'	N00°11.6	12.6'	55.0'
5	253°22.7	40.5	33°53.6	15.6'	00°24.3	12.6'	55.0'
6	268°22.8	N18°39.9	48°28.1	15.6'	N00°36.9	12.6'	55.0'
7	283°22.8	39.3	63°02.7	15.6'	00°49.5	12.6'	54.9'
8	298°22.8	38.8	77°37.3	15.6'	01°02.1	12.6'	54.9'
9	313°22.8	· · 38.2	92°11.9	15.6'	01°14.7	12.6'	54.9'
10	328°22.8	37.6	106°46.5	15.6'	01°27.3	12.6'	54.9'
11	343°22.9	37.0	121°21.2	15.7'	01°39.8	12.5'	54.9'
12	358°22.9	N18°36.4	135°55.8	15.7'	N01°52.4	12.5'	54.9'
13	13°22.9	35.8	150°30.5	15.7'	02°04.9	12.5'	54.8'
14	28°22.9	35.2	165°05.2	15.7'	02°17.4	12.5'	54.8'
15	43°23.0	· · 34.6	179°39.9	15.7'	02°29.9	12.5'	54.8'
16	58°23.0	34.0	194°14.6	15.7'	02°42.4	12.5'	54.8'
17	73°23.0	33.4	208°49.3	15.7'	02°54.8	12.4'	54.8'
18	88°23.0	N18°32.8	223°24.0	15.7'	N03°07.2	12.4'	54.7'
19	103°23.1	32.2	237°58.7	15.7'	03°19.7	12.4'	54.7'
20	118°23.1	31.6	252°33.5	15.7'	03°32.0	12.4'	54.7'
21	133°23.1	· · 31.0	267°08.2	15.8'	03°44.4	12.3'	54.7'
22	148°23.1	30.4	281°43.0	15.8'	03°56.8	12.3'	54.7'
23	163°23.2	29.7	296°17.7	15.8'	04°09.1	12.3'	54.7'
SD = 15.7'			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	—:—	—:—	00:42	23:17	—:—	—:—
68°	—:—	—:—	01:55	22:13	—:—	—:—
66°	—:—	—:—	02:31	21:39	—:—	—:—
64°	—:—	01:15	02:56	21:14	22:52	—:—
62°	—:—	02:00	03:16	20:55	22:10	—:—
60°	—:—	02:28	03:32	20:39	21:42	—:—
N 58°	01:08	02:50	03:45	20:26	21:21	22:59
56°	01:49	03:07	03:57	20:15	21:04	22:21
54°	02:15	03:21	04:07	20:05	20:50	21:55
52°	02:36	03:34	04:16	19:56	20:38	21:36
50°	02:52	03:45	04:24	19:48	20:27	21:20
45°	03:23	04:07	04:41	19:31	20:05	20:49
N 40°	03:46	04:24	04:55	19:18	19:48	20:26
35°	04:04	04:38	05:07	19:06	19:34	20:08
30°	04:19	04:51	05:17	18:56	19:22	19:53
20°	04:42	05:10	05:34	18:39	19:02	19:30
N 10°	05:01	05:27	05:49	18:24	18:46	19:12
0°	05:16	05:41	06:03	18:10	18:32	18:57
S 10°	05:29	05:55	06:17	17:56	18:18	18:44
20°	05:41	06:08	06:31	17:42	18:05	18:32
30°	05:54	06:23	06:48	17:25	17:51	18:20
35°	06:00	06:31	06:58	17:16	17:43	18:13
40°	06:07	06:39	07:08	17:05	17:34	18:07
45°	06:14	06:49	07:21	16:52	17:24	18:00
S 50°	06:22	07:01	07:37	16:37	17:12	17:52
52°	06:25	07:06	07:44	16:29	17:07	17:48
54°	06:29	07:12	07:52	16:21	17:01	17:45
56°	06:33	07:19	08:01	16:12	16:55	17:41
58°	06:37	07:26	08:11	16:02	16:48	17:37
S 60°	06:42	07:34	08:23	15:51	16:40	17:32

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	21:11	20:48	20:27	05:13	07:13	09:06
N 70°	21:02	20:46	20:31	05:29	07:20	09:04
68°	20:54	20:45	20:35	05:42	07:25	09:03
66°	20:48	20:43	20:39	05:52	07:29	09:02
64°	20:43	20:42	20:42	06:01	07:33	09:01
62°	20:38	20:41	20:44	06:08	07:36	09:00
60°	20:34	20:40	20:46	06:15	07:38	08:59
N 58°	20:31	20:40	20:48	06:20	07:41	08:58
56°	20:28	20:39	20:50	06:25	07:43	08:58
54°	20:25	20:38	20:51	06:30	07:45	08:57
52°	20:22	20:38	20:53	06:34	07:46	08:57
50°	20:20	20:37	20:54	06:37	07:48	08:56
45°	20:15	20:36	20:57	06:45	07:51	08:55
N 40°	20:10	20:35	20:59	06:51	07:54	08:55
35°	20:06	20:34	21:01	06:57	07:56	08:54
30°	20:03	20:34	21:03	07:02	07:58	08:53
20°	19:57	20:32	21:07	07:10	08:02	08:52
N 10°	19:52	20:31	21:09	07:17	08:05	08:51
0°	19:47	20:30	21:12	07:24	08:08	08:51
S 10°	19:43	20:29	21:15	07:31	08:11	08:50
20°	19:37	20:28	21:18	07:38	08:14	08:49
30°	19:31	20:27	21:21	07:46	08:18	08:48
35°	19:28	20:26	21:23	07:51	08:20	08:47
40°	19:24	20:26	21:26	07:56	08:22	08:46
45°	19:20	20:25	21:28	08:02	08:24	08:45
S 50°	19:14	20:24	21:31	08:09	08:27	08:44
52°	19:12	20:23	21:33	08:12	08:29	08:44
54°	19:09	20:23	21:34	08:16	08:30	08:44
56°	19:06	20:22	21:36	08:20	08:32	08:43
58°	19:03	20:21	21:38	08:24	08:34	08:42
S 60°	18:59	20:21	21:40	08:29	08:36	08:42

Day	Sun		Mer. Pass hh:mm	Moon		Age 16-18 98-88%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
27	06:32	06:32	12:07	01:13	13:36	
28	06:31	06:30	12:07	01:58	14:19	
29	06:29	06:28	12:06	02:40	15:01	

July 30, 31, 01 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	307°55.3	145°43.4	N09°01.0	104°58.7	S10°12.7	108°33.2	S06°54.4	255°59.0	N16°33.3	Alpheratz	357°30.7	29°15.3	
1	322°57.7	160°43.1	08°59.8	119°59.8	13.2	123°35.4	54.6	271°01.3	33.4	Ankaa	353°03.4	-42°08.3	
2	338°00.2	175°42.8		135°01.0	13.8	138°37.6	54.7	286°03.6	33.4	Schedar	349°26.4	56°41.9	
3	353°02.7	190°42.5	•• 57.4	150°02.1	•• 14.3	153°39.7	•• 54.8	301°05.9	•• 33.4	Diphda	348°43.5	-17°49.2	
4	8°05.1	205°42.2		165°03.3	14.9	168°41.9	54.9	316°08.2	33.5	Achernar	335°17.5	-57°04.8	
5	23°07.6	220°41.9		180°04.4	15.5	183°44.1	55.1	331°10.5	33.5	Hamal	327°46.9	23°36.1	
6	38°10.0	235°41.6	N08°53.9	195°05.5	S10°16.0	198°46.3	S06°55.2	346°12.8	N16°33.5	Polaris	312°05.0	89°22.9	
7	53°12.5	250°41.3		210°06.7	16.6	213°48.5	55.3	1°15.1	33.6	Acamar	315°09.1	-40°10.9	
8	68°15.0	265°41.0		225°07.8	17.1	228°50.7	55.4	16°17.4	33.6	Menkar	314°02.3	4°12.4	
9	83°17.4	280°40.7	•• 50.2	240°09.0	•• 17.7	243°52.8	•• 55.5	31°19.7	•• 33.6	Mirfak	308°23.0	49°57.7	
10	98°19.9	295°40.4		255°10.1	18.2	258°55.0	55.7	46°22.0	33.6	Aldebaran	290°35.6	16°34.0	
11	113°22.4	310°40.1		270°11.2	18.8	273°57.2	55.8	61°24.3	33.7	Rigel	281°00.5	-8°10.0	
12	128°24.8	325°39.8	N08°46.6	285°12.4	S10°19.3	288°59.4	S06°55.9	76°26.6	N16°33.7	Capella	280°16.7	46°01.4	
13	143°27.3	340°39.5		300°13.5	19.9	304°01.6	56.0	91°28.9	33.7	Bellatrix	278°19.2	6°22.5	
14	158°29.8	355°39.2		315°14.6	20.4	319°03.7	56.2	106°31.2	33.8	Elnath	277°57.5	28°37.8	
15	173°32.2	10°38.9	•• 43.0	330°15.8	•• 21.0	334°05.9	•• 56.3	121°33.5	•• 33.8	Alnilam	275°34.2	-1°11.0	
16	188°34.7	25°38.6		345°16.9	21.5	349°08.1	56.4	136°35.8	33.8	Betelgeuse	270°48.4	7°24.7	
17	203°37.1	40°38.3		0°18.1	22.1	4°10.3	56.5	151°38.1	33.9	Canopus	263°51.2	-52°42.6	
18	218°39.6	55°38.0	N08°39.4	15°19.2	S10°22.6	19°12.5	S06°56.7	166°40.4	N16°33.9	Sirius	258°23.3	-16°45.4	
19	233°42.1	70°37.7		30°20.3	23.2	34°14.6	56.8	181°42.7	33.9	Adhara	255°03.3	-29°00.7	
20	248°44.5	85°37.4		45°21.5	23.7	49°16.8	56.9	196°45.0	34.0	Procyon	244°47.3	5°08.8	
21	263°47.0	100°37.1	•• 35.8	60°22.6	•• 24.3	64°19.0	•• 57.0	211°47.3	•• 34.0	Pollux	243°13.2	27°57.1	
22	278°49.5	115°36.8		75°23.7	24.8	79°21.2	57.2	226°49.6	34.0	Avior	234°13.8	-59°36.3	
23	293°51.9	130°36.5		90°24.9	25.4	94°23.3	57.3	241°51.9	34.1	Suhail	222°44.0	-43°33.3	
Mer.pass. 03:28		ν -0.3' d -1.2' m -3.92		ν 1.1' d 0.6' m 0.67		ν 2.2' d 0.1' m -1.90		ν 2.3' d 0.0' m 0.23		Miaplacidus	221°38.2	-69°50.5	
										Alphard	217°44.4	-8°47.3	
										Regulus	207°30.8	11°49.3	
										Dubhe	193°37.3	61°35.6	
										Denebola	182°21.4	14°24.4	
										Gienah	175°39.9	-17°42.5	
										Acrux	172°56.1	-63°16.1	
										Gacrux	171°47.6	-57°17.0	
										Alioth	166°10.1	55°48.2	
										Spica	158°18.5	-11°19.0	
										Alkaid	152°49.3	49°10.2	
										Hadar	148°30.6	-60°31.2	
										Menkent	147°53.2	-36°31.1	
										Arcturus	145°44.6	19°01.9	
										Rigil Kent.	139°35.0	-60°57.7	
										Kochab	137°20.1	74°02.3	
										Zuben'ubi	136°51.9	-16°09.9	
										Alphecca	126°00.5	26°37.1	
										Antares	112°11.1	-26°29.8	
										Atria	107°01.5	-69°04.9	
										Sabik	101°58.3	-15°45.6	
										Shaula	96°05.0	-37°07.4	
										Rasalhague	95°54.9	12°32.6	
										Eltanin	90°40.0	51°29.4	
										Kaus Aust.	83°27.3	-34°22.1	
										Vega	80°30.3	38°48.9	
										Nunki	75°42.9	-26°15.4	
										Altair	61°56.0	8°57.0	
										Peacock	52°59.5	-56°38.1	
										Deneb	49°22.7	45°23.3	
										Enif	33°34.8	10°00.9	
										Al Na'ir	27°28.0	-46°48.7	
										Fomalhaut	15°10.2	-29°27.6	
										Scheat	13°41.3	28°14.7	
										Markab	13°25.9	15°22.0	
										Jul 30 Mon		SHA	Mer.pass
										Venus		197°48.1	14:17
										Mars		157°03.4	16:59
										Jupiter		160°37.9	16:43
										Saturn		308°03.8	06:55
										Jul 31 Tue		SHA	Mer.pass
										Venus		196°41.8	14:18
										Mars		156°31.6	16:57
										Jupiter		160°31.1	16:40
										Saturn		307°59.9	06:51
										Aug 01 Wed		SHA	Mer.pass
										Venus		195°35.7	14:18
										Mars		155°59.6	16:55
										Jupiter		160°24.2	16:36
										Saturn		307°56.0	06:48
										Horizontal parallax			
										Venus:		0.1	
										Mars:		0.1	
Mer.pass. 03:20		ν -0.3' d -1.2' m -3.93		ν 1.1' d 0.6' m 0.68		ν 2.2' d 0.1' m -1.89		ν 2.3' d 0.0' m 0.23					

DUT1 = UT1-UTC = +0.2116 sec ΔT = TT-UT1 = +68.9724 sec

2029 July 30 to Aug. 01 UT

Sun			Moon			
h	GHA	Dec	GHA	ν	Dec	d
Mon	HP					
0	178°23.2	N18°29.1	310°52.5	15.8'	N04°21.4	12.3'
1	193°23.2	28.5	325°27.3	15.8'	04°33.6	12.2'
2	208°23.2	27.9	340°02.0	15.8'	04°45.9	12.2'
3	223°23.3	· · 27.3	354°36.8	15.8'	04°58.1	12.2'
4	238°23.3	26.7	9°11.6	15.8'	05°10.3	12.2'
5	253°23.3	26.1	23°46.4	15.8'	05°22.5	12.1'
6	268°23.4	N18°25.5	38°21.1	15.8'	N05°34.6	12.1'
7	283°23.4	24.9	52°55.9	15.8'	05°46.7	12.1'
8	298°23.4	24.3	67°30.6	15.8'	05°58.8	12.0'
9	313°23.4	· · 23.7	82°05.4	15.8'	06°10.8	12.0'
10	328°23.5	23.1	96°40.1	15.7'	06°22.8	12.0'
11	343°23.5	22.5	111°14.9	15.7'	06°34.8	12.0'
12	358°23.5	N18°21.8	125°49.6	15.7'	N06°46.8	11.9'
13	13°23.6	21.2	140°24.4	15.7'	06°58.7	11.9'
14	28°23.6	20.6	154°59.1	15.7'	07°10.6	11.9'
15	43°23.6	· · 20.0	169°33.8	15.7'	07°22.4	11.8'
16	58°23.6	19.4	184°08.5	15.7'	07°34.2	11.8'
17	73°23.7	18.8	198°43.2	15.7'	07°46.0	11.7'
18	88°23.7	N18°18.2	213°17.9	15.7'	N07°57.8	11.7'
19	103°23.7	17.6	227°52.5	15.7'	08°09.5	11.7'
20	118°23.8	16.9	242°27.2	15.6'	08°21.1	11.6'
21	133°23.8	· · 16.3	257°01.8	15.6'	08°32.8	11.6'
22	148°23.8	15.7	271°36.4	15.6'	08°44.4	11.6'
23	163°23.9	15.1	286°11.0	15.6'	08°55.9	11.5'
SD = 15.7'			SD = 14.9'			

Tue	GHA	Dec	GHA	ν	Dec	d
HP						
0	178°23.9	N18°14.5	300°45.6	15.6'	N09°07.4	11.5'
1	193°23.9	13.9	315°20.2	15.6'	09°18.9	11.4'
2	208°24.0	13.2	329°54.7	15.5'	09°30.3	11.4'
3	223°24.0	· · 12.6	344°29.3	15.5'	09°41.7	11.3'
4	238°24.0	12.0	359°03.8	15.5'	09°53.1	11.3'
5	253°24.1	11.4	13°38.3	15.5'	10°04.4	11.3'
6	268°24.1	N18°10.8	28°12.7	15.4'	N10°15.6	11.2'
7	283°24.1	10.1	42°47.2	15.4'	10°26.9	11.2'
8	298°24.2	09.5	57°21.6	15.4'	10°38.0	11.1'
9	313°24.2	· · 08.9	71°56.0	15.4'	10°49.2	11.1'
10	328°24.2	08.3	86°30.4	15.3'	11°00.3	11.0'
11	343°24.3	07.6	101°04.7	15.3'	11°11.3	11.0'
12	358°24.3	N18°07.0	115°39.0	15.3'	N11°22.3	10.9'
13	13°24.3	06.4	130°13.3	15.3'	11°33.2	10.9'
14	28°24.4	05.8	144°47.6	15.2'	11°44.1	10.8'
15	43°24.4	· · 05.1	159°21.8	15.2'	11°54.9	10.8'
16	58°24.5	04.5	173°56.0	15.2'	12°05.7	10.7'
17	73°24.5	03.9	188°30.2	15.1'	12°16.5	10.7'
18	88°24.5	N18°03.3	203°04.3	15.1'	N12°27.2	10.6'
19	103°24.6	02.6	217°38.4	15.1'	12°37.8	10.6'
20	118°24.6	02.0	232°12.5	15.0'	12°48.4	10.5'
21	133°24.6	· · 01.4	246°46.5	15.0'	12°58.9	10.5'
22	148°24.7	00.8	261°20.5	15.0'	13°09.4	10.4'
23	163°24.7	00.1	275°54.5	14.9'	13°19.8	10.4'
SD = 15.7'			SD = 14.8'			

Wed	GHA	Dec	GHA	ν	Dec	d
HP						
0	178°24.8	N17°59.5	290°28.4	14.9'	N13°30.2	10.3'
1	193°24.8	58.9	305°02.3	14.9'	13°40.5	10.3'
2	208°24.8	58.2	319°36.2	14.8'	13°50.8	10.2'
3	223°24.9	· · 57.6	334°10.0	14.8'	14°01.0	10.1'
4	238°24.9	57.0	348°43.8	14.7'	14°11.1	10.1'
5	253°25.0	56.3	3°17.5	14.7'	14°21.2	10.0'
6	268°25.0	N17°55.7	17°51.3	14.7'	N14°31.3	10.0'
7	283°25.0	55.1	32°24.9	14.6'	14°41.2	9.9'
8	298°25.1	54.4	46°58.5	14.6'	14°51.1	9.8'
9	313°25.1	· · 53.8	61°32.1	14.5'	15°01.0	9.8'
10	328°25.2	53.2	76°05.6	14.5'	15°10.8	9.7'
11	343°25.2	52.5	90°39.1	14.4'	15°20.5	9.7'
12	358°25.2	N17°51.9	105°12.6	14.4'	N15°30.1	9.6'
13	13°25.3	51.3	119°46.0	14.4'	15°39.7	9.5'
14	28°25.3	50.6	134°19.4	14.3'	15°49.3	9.5'
15	43°25.4	· · 50.0	148°52.7	14.3'	15°58.8	9.4'
16	58°25.4	49.4	163°25.9	14.2'	16°08.2	9.3'
17	73°25.5	48.7	177°59.1	14.2'	16°17.5	9.3'
18	88°25.5	N17°48.1	192°32.3	14.1'	N16°26.8	9.2'
19	103°25.5	47.4	207°05.4	14.1'	16°36.0	9.1'
20	118°25.6	46.8	221°38.5	14.0'	16°45.1	9.1'
21	133°25.6	· · 46.2	236°11.5	14.0'	16°54.2	9.0'
22	148°25.7	45.5	250°44.5	13.9'	17°03.2	8.9'
23	163°25.7	44.9	265°17.4	13.9'	17°12.1	8.9'
SD = 15.8'			SD = 14.8'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	—:—	—:—	01:20	22:45	—:—	—:—
68°	—:—	—:—	02:11	21:57	—:—	—:—
66°	—:—	—:—	02:43	21:27	23:59	—:—
64°	—:—	01:35	03:06	21:05	22:32	—:—
62°	—:—	02:12	03:24	20:47	21:57	—:—
60°	—:—	02:38	03:39	20:32	21:33	—:—
N 58°	01:26	02:57	03:52	20:20	21:13	22:42
56°	02:00	03:14	04:02	20:09	20:57	22:10
54°	02:24	03:27	04:12	20:00	20:44	21:46
52°	02:43	03:39	04:21	19:51	20:32	21:28
50°	02:58	03:49	04:28	19:44	20:22	21:13
45°	03:28	04:11	04:44	19:28	20:01	20:44
N 40°	03:50	04:27	04:58	19:15	19:45	20:22
35°	04:07	04:41	05:09	19:03	19:31	20:05
30°	04:21	04:53	05:19	18:54	19:20	19:51
20°	04:44	05:12	05:35	18:37	19:01	19:29
N 10°	05:01	05:27	05:50	18:23	18:45	19:11
0°	05:16	05:41	06:03	18:10	18:32	18:57
S 10°	05:29	05:54	06:16	17:57	18:19	18:44
20°	05:40	06:07	06:30	17:43	18:06	18:33
30°	05:52	06:21	06:46	17:27	17:52	18:21
35°	05:58	06:28	06:55	17:18	17:45	18:15
40°	06:04	06:37	07:06	17:08	17:36	18:09
45°	06:11	06:46	07:18	16:55	17:27	18:03
S 50°	06:18	06:57	07:33	16:41	17:16	17:55
52°	06:21	07:02	07:39	16:34	17:11	17:52
54°	06:25	07:08	07:47	16:26	17:06	17:49
56°	06:28	07:14	07:56	16:18	17:00	17:45
58°	06:32	07:20	08:05	16:08	16:53	17:41
S 60°	06:36	07:28	08:16	15:57	16:46	17:37

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	20:04	19:36	18:49	10:57	12:53	15:10
N 70°	20:16	19:58	19:33	10:47	12:33	14:28
68°	20:26	20:15	20:02	10:39	12:17	13:59
66°	20:34	20:29	20:24	10:32	12:04	13:38
64°	20:41	20:41	20:42	10:27	11:54	13:22
62°	20:47	20:51	20:57	10:22	11:45	13:08
60°	20:52	21:00	21:09	10:18	11:37	12:56
N 58°	20:57	21:07	21:20	10:15	11:30	12:46
56°	21:01	21:14	21:29	10:11	11:24	12:37
54°	21:05	21:20	21:38	10:08	11:19	12:30
52°	21:08	21:25	21:45	10:06	11:14	12:23
50°	21:11	21:30	21:52	10:04	11:10	12:16
45°	21:18	21:41	22:06	09:58	11:01	12:03
N 40°	21:24	21:50	22:18	09:54	10:53	11:52
35°	21:29	21:57	22:29	09:50	10:47	11:43
30°	21:33	22:04	22:38	09:47	10:41	11:35
20°	21:41	22:16	22:53	09:42	10:31	11:21
N 10°	21:47	22:26	23:07	09:37	10:22	11:08
0°	21:54	22:36	23:20	09:32	10:14	10:57
S 10°	22:00	22:46	23:33	09:28	10:06	10:46
20°	22:07	22:56	23:47	09:23	09:57	10:33
30°	22:15	23:09	· · ·	09:17	09:47	10:20
35°	22:19	23:16	· · ·	09:14	09:42	10:12
40°	22:25	23:24	· · ·	09:10	09:35	10:03
45°	22:31	23:33	· · ·	09:06	09:28	09:52
S 50°	22:38	23:45	· · ·	09:01	09:19	09:39
52°	22:41	23:50	· · ·	08:59	09:15	09:33
54°	22:45	23:56	· · ·	08:57	09:11	09:27
56°	22:49	· · ·	00:02	08:54	09:06	09:20
58°	22:54	· · ·	00:10	08:51	09:00	09:11
S 60°	22:59	· · ·	00:18	08:48	08:54	09:02

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	19-21
				hh:mm	hh:mm	81-64%
30	06:27	06:26	12:06	03:22	15:43	
31	06:24	06:23	12:06	04:04	16:25	
01	06:21	06:19	12:06	04:46	17:08	

August 02, 03, 04 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	310°52.7	145°22.5	N07°33.9	106°20.0	S10°52.5	111°09.8	S07°03.5	258°45.0	N16°35.5					Alpheratz	357°30.6	29°15.3	
1	325°55.1	160°22.3		121°21.1	53.0	126°12.0	03.6	273°47.3	35.5					Ankaa	353°03.4	-42°08.3	
2	340°57.6	175°22.0		136°22.2	53.6	141°14.2	03.7	288°49.6	35.6					Schedar	349°26.3	56°41.9	
3	356°00.1	190°21.7	•• 30.2	151°23.3	•• 54.2	156°16.3	•• 03.9	303°51.9	•• 35.6					Diphda	348°43.5	-17°49.2	
4	11°02.5	205°21.4		166°24.4	54.7	171°18.5	04.0	318°54.2	35.6					Achernar	335°17.5	-57°04.8	
5	26°05.0	220°21.1		181°25.6	55.3	186°20.7	04.1	333°56.5	35.7					Hamal	327°46.9	23°36.1	
6	41°07.5	235°20.9	N07°26.5	196°26.7	S10°55.8	201°22.8	S07°04.3	348°58.9	N16°35.7					Polaris	312°03.2	89°22.9	
7	56°09.9	250°20.6		211°27.8	56.4	216°25.0	04.4	4°01.2	35.7					Acamar	315°09.1	-40°10.9	
8	71°12.4	265°20.3		226°28.9	56.9	231°27.2	04.5	19°03.5	35.7					Menkar	314°02.3	4°12.4	
9	86°14.8	280°20.1	•• 22.9	241°30.0	•• 57.5	246°29.3	•• 04.6	34°05.8	•• 35.8					Mirfak	308°23.0	49°57.7	
10	101°17.3	295°19.8		256°31.1	58.0	261°31.5	04.8	49°08.1	35.8					Aldebaran	290°35.5	16°34.0	
11	116°19.8	310°19.5		271°32.2	58.6	276°33.7	04.9	64°10.4	35.8					Rigel	281°00.5	-8°10.0	
12	131°22.2	325°19.2	N07°19.2	286°33.4	S10°59.1	291°35.8	S07°05.0	79°12.7	N16°35.9					Capella	280°16.7	46°01.4	
13	146°24.7	340°19.0		301°34.5	10°59.7	306°38.0	05.2	94°15.0	35.9					Bellatrix	278°19.1	6°22.5	
14	161°27.2	355°18.7		316°35.6	11°00.2	321°40.1	05.3	109°17.4	35.9					Elnath	277°57.4	28°37.8	
15	176°29.6	10°18.4	•• 15.5	331°36.7	•• 00.8	336°42.3	•• 05.4	124°19.7	•• 35.9					Alnilam	275°34.2	-1°11.0	
16	191°32.1	25°18.1		346°37.8	01.3	351°44.5	05.6	139°22.0	36.0					Betelgeuse	270°48.3	7°24.7	
17	206°34.6	40°17.9		1°38.9	01.9	6°46.6	05.7	154°24.3	36.0					Canopus	263°51.2	-52°42.6	
18	221°37.0	55°17.6	N07°11.8	16°40.0	S11°02.4	21°48.8	S07°05.8	169°26.6	N16°36.0					Sirius	258°23.3	-16°45.4	
19	236°39.5	70°17.3		31°41.1	03.0	36°51.0	05.9	184°28.9	36.1					Adhara	255°03.3	-29°00.7	
20	251°42.0	85°17.1		46°42.2	03.6	51°53.1	06.1	199°31.2	36.1					Procyon	244°47.3	5°08.8	
21	266°44.4	100°16.8	•• 08.1	61°43.3	•• 04.1	66°55.3	•• 06.2	214°33.6	•• 36.1					Pollux	243°13.2	27°57.1	
22	281°46.9	115°16.5		76°44.4	04.7	81°57.5	06.3	229°35.9	36.1					Avior	234°13.8	-59°36.3	
23	296°49.3	130°16.3		91°45.6	05.2	96°59.6	06.5	244°38.2	36.2					Suhail	222°44.0	-43°33.2	
Mer.pass. 03:16			ν -0.3' d -1.2' m -3.93			ν 1.1' d 0.6' m 0.69			ν 2.2' d 0.1' m -1.88			ν 2.3' d 0.0' m 0.22					
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	311°51.8	145°16.0	N07°04.4	106°46.7	S11°05.8	112°01.8	S07°06.6	259°40.5	N16°36.2					Dubhe	193°37.3	61°35.6	
1	326°54.3	160°15.7		121°47.8	06.3	127°03.9	06.7	274°42.8	36.2					Denebola	182°21.4	14°24.4	
2	341°56.7	175°15.5		136°48.9	06.9	142°06.1	06.9	289°45.1	36.3					Gienah	175°39.9	-17°42.5	
3	356°59.2	190°15.2	07°00.7	151°50.0	•• 07.4	157°08.3	•• 07.0	304°47.4	•• 36.3					Acrux	172°56.1	-63°16.1	
4	12°01.7	205°14.9	06°59.5	166°51.1	08.0	172°10.4	07.1	319°49.8	36.3					Gacrux	171°47.6	-57°17.0	
5	27°04.1	220°14.7		181°52.2	08.5	187°12.6	07.2	334°52.1	36.3					Alioth	166°10.2	55°48.2	
6	42°06.6	235°14.4	N06°57.0	196°53.3	S11°09.1	202°14.8	S07°07.4	349°54.4	N16°36.4					Spica	158°18.5	-11°19.0	
7	57°09.1	250°14.1		211°54.4	09.6	217°16.9	07.5	4°56.7	36.4					Alkaid	152°49.3	49°10.2	
8	72°11.5	265°13.9		226°55.5	10.2	232°19.1	07.6	19°59.0	36.4					Hadar	148°30.6	-60°31.2	
9	87°14.0	280°13.6	•• 53.3	241°56.6	•• 10.7	247°21.2	•• 07.8	35°01.3	•• 36.4					Menkent	147°53.2	-36°31.1	
10	102°16.5	295°13.3		256°57.7	11.3	262°23.4	07.9	50°03.6	36.5					Arcturus	145°44.6	19°01.9	
11	117°18.9	310°13.1		271°58.8	11.8	277°25.6	08.0	65°06.0	36.5					Rigel Kent.	139°35.0	-60°57.7	
12	132°21.4	325°12.8	N06°49.6	286°59.9	S11°12.4	292°27.7	S07°08.2	80°08.3	N16°36.5					Kochab	137°20.2	74°02.3	
13	147°23.8	340°12.5		302°01.0	13.0	307°29.9	08.3	95°10.6	36.6					Zuben'ubi	136°51.9	-16°09.9	
14	162°26.3	355°12.3		317°02.1	13.5	322°32.0	08.4	110°12.9	36.6					Alphecca	126°00.6	26°37.1	
15	177°28.8	10°12.0	•• 45.9	332°03.2	•• 14.1	337°34.2	•• 08.6	125°15.2	•• 36.6					Antares	112°11.1	-26°29.8	
16	192°31.2	25°11.7		347°04.3	14.6	352°36.4	08.7	140°17.5	36.6					Atria	107°01.5	-69°04.9	
17	207°33.7	40°11.5		2°05.4	15.2	7°38.5	08.8	155°19.9	36.7					Sabik	101°58.3	-15°45.5	
18	222°36.2	55°11.2	N06°42.2	17°06.5	S11°15.7	22°40.7	S07°08.9	170°22.2	N16°36.7					Shaula	96°05.0	-37°07.4	
19	237°38.6	70°11.0		32°07.6	16.3	37°42.8	09.1	185°24.5	36.7					Rasalhague	95°54.9	12°32.6	
20	252°41.1	85°10.7		47°08.7	16.8	52°45.0	09.2	200°26.8	36.7					Eltanin	90°40.1	51°29.4	
21	267°43.6	100°10.4	•• 38.5	62°09.8	•• 17.4	67°47.2	•• 09.3	215°29.1	•• 36.8					Kaus Aust.	83°27.3	-34°22.1	
22	282°46.0	115°10.2		77°10.9	17.9	82°49.3	09.5	230°31.5	36.8					Vega	80°30.3	38°49.0	
23	297°48.5	130°09.9		92°12.0	18.5	97°51.5	09.6	245°33.8	36.8					Nunki	75°42.9	-26°15.4	
Mer.pass. 03:12			ν -0.3' d -1.2' m -3.93			ν 1.1' d 0.6' m 0.70			ν 2.2' d 0.1' m -1.88			ν 2.3' d 0.0' m 0.22					
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	312°51.0	145°09.7	N06°34.8	107°13.1	S11°19.0	112°53.6	S07°09.7	260°36.1	N16°36.9					Altair	61°56.0	8°57.1	
1	327°53.4	160°09.4		122°14.2	19.6	127°55.8	09.9	275°38.4	36.9					Peacock	52°59.5	-56°38.1	
2	342°55.9	175°09.1		137°15.3	20.1	142°58.0	10.0	290°40.7	36.9					Deneb	49°22.7	45°23.4	
3	357°58.3	190°08.9	•• 31.1	152°16.4	•• 20.7	158°00.1	•• 10.1	305°43.0	•• 36.9					Enif	33°34.8	10°00.9	
4	13°00.8	205°08.6		167°17.5	21.2	173°02.3	10.3	320°45.4	37.0					Al Na'ir	27°28.0	-46°48.7	
5	28°03.3	220°08.4		182°18.6	21.8	188°04.4	10.4	335°47.7	37.0					Fomalhaut	15°10.2	-29°27.6	
6	43°05.7	235°08.1	N06°27.4	197°19.7	S11°22.3	203°06.6	S07°10.5	350°50.0	N16°37.0					Scheat	13°41.3	28°14.7	
7	58°08.2	250°07.8		212°20.8	22.9	218°08.7	10.7	5°52.3	37.0					Markab	13°25.9	15°22.0	
8	73°10.7	265°07.6		227°21.9	23.5	233°10.9	10.8	20°54.6	37.1					Aug 02 Thu			
9	88°13.1	280°07.3	•• 23.7	242°23.0	•• 24.0	248°13.1	•• 10.9	35°57.0	•• 37.1					Venus	194°29.9	14:19	
10	103°15.6	295°07.1		257°24.1	24.6	263°15.2	11.1	50°59.3	37.1					Mars	155°27.3	16:53	
11	118°18.1																

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°25.8	N17°44.2	279°50.3	13.8'	N17°21.0	8.8'	54.2'
1	193°25.8	43.6	294°23.1	13.8'	17°29.7	8.7'	54.2'
2	208°25.8	42.9	308°55.9	13.7'	17°38.5	8.6'	54.2'
3	223°25.9	· · 42.3	323°28.6	13.7'	17°47.1	8.6'	54.2'
4	238°25.9	41.7	338°01.3	13.6'	17°55.7	8.5'	54.2'
5	253°26.0	41.0	352°33.9	13.6'	18°04.2	8.4'	54.2'
6	268°26.0	N17°40.4	7°06.5	13.5'	N18°12.6	8.3'	54.2'
7	283°26.1	39.7	21°39.0	13.4'	18°20.9	8.3'	54.2'
8	298°26.1	39.1	36°11.4	13.4'	18°29.2	8.2'	54.2'
9	313°26.2	· · 38.4	50°43.8	13.3'	18°37.4	8.1'	54.2'
10	328°26.2	37.8	65°16.2	13.3'	18°45.5	8.0'	54.3'
11	343°26.3	37.1	79°48.4	13.2'	18°53.5	8.0'	54.3'
12	358°26.3	N17°36.5	94°20.7	13.2'	N19°01.5	7.9'	54.3'
13	13°26.4	35.8	108°52.8	13.1'	19°09.3	7.8'	54.3'
14	28°26.4	35.2	123°25.0	13.1'	19°17.1	7.7'	54.3'
15	43°26.5	· · 34.5	137°57.0	13.0'	19°24.8	7.6'	54.3'
16	58°26.5	33.9	152°29.0	12.9'	19°32.5	7.5'	54.3'
17	73°26.6	33.2	167°00.9	12.9'	19°40.0	7.5'	54.3'
18	88°26.6	N17°32.6	181°32.8	12.8'	N19°47.5	7.4'	54.3'
19	103°26.7	31.9	196°04.7	12.8'	19°54.9	7.3'	54.3'
20	118°26.7	31.3	210°36.4	12.7'	20°02.1	7.2'	54.3'
21	133°26.8	· · 30.6	225°08.1	12.6'	20°09.4	7.1'	54.3'
22	148°26.8	30.0	239°39.8	12.6'	20°16.5	7.0'	54.4'
23	163°26.9	29.3	254°11.4	12.5'	20°23.5	6.9'	54.4'
SD = 15.8'			SD = 14.8'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°26.9	N17°28.7	268°42.9	12.5'	N20°30.5	6.9'	54.4'
1	193°27.0	28.0	283°14.4	12.4'	20°37.3	6.8'	54.4'
2	208°27.0	27.4	297°45.8	12.4'	20°44.1	6.7'	54.4'
3	223°27.1	· · 26.7	312°17.1	12.3'	20°50.8	6.6'	54.4'
4	238°27.1	26.1	326°48.4	12.2'	20°57.4	6.5'	54.4'
5	253°27.2	25.4	341°19.7	12.2'	21°03.9	6.4'	54.4'
6	268°27.2	N17°24.7	355°50.8	12.1'	N21°10.3	6.3'	54.5'
7	283°27.3	24.1	10°21.9	12.0'	21°16.6	6.2'	54.5'
8	298°27.3	23.4	24°53.0	12.0'	21°22.8	6.1'	54.5'
9	313°27.4	· · 22.8	39°24.0	11.9'	21°28.9	6.0'	54.5'
10	328°27.4	22.1	53°54.9	11.9'	21°34.9	5.9'	54.5'
11	343°27.5	21.5	68°25.8	11.8'	21°40.9	5.8'	54.5'
12	358°27.5	N17°20.8	82°56.6	11.7'	N21°46.7	5.7'	54.5'
13	13°27.6	20.1	97°27.3	11.7'	21°52.4	5.6'	54.5'
14	28°27.6	19.5	111°58.0	11.6'	21°58.1	5.5'	54.6'
15	43°27.7	· · 18.8	126°28.6	11.6'	22°03.6	5.4'	54.6'
16	58°27.8	18.1	140°59.2	11.5'	22°09.1	5.3'	54.6'
17	73°27.8	17.5	155°29.7	11.4'	22°14.4	5.2'	54.6'
18	88°27.9	N17°16.8	170°00.1	11.4'	N22°19.7	5.1'	54.6'
19	103°27.9	16.2	184°30.5	11.3'	22°24.8	5.0'	54.6'
20	118°28.0	15.5	199°00.8	11.3'	22°29.8	4.9'	54.7'
21	133°28.0	· · 14.8	213°31.1	11.2'	22°34.8	4.8'	54.7'
22	148°28.1	14.2	228°01.3	11.1'	22°39.6	4.7'	54.7'
23	163°28.2	13.5	242°31.5	11.1'	22°44.3	4.6'	54.7'
SD = 15.8'			SD = 14.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°28.2	N17°12.8	257°01.5	11.0'	N22°48.9	4.5'	54.7'
1	193°28.3	12.2	271°31.6	11.0'	22°53.5	4.4'	54.7'
2	208°28.3	11.5	286°01.5	10.9'	22°57.9	4.3'	54.8'
3	223°28.4	· · 10.8	300°31.4	10.9'	23°02.2	4.2'	54.8'
4	238°28.4	10.2	315°01.3	10.8'	23°06.4	4.1'	54.8'
5	253°28.5	09.5	329°31.1	10.7'	23°10.4	4.0'	54.8'
6	268°28.6	N17°08.8	344°00.8	10.7'	N23°14.4	3.9'	54.8'
7	283°28.6	08.2	358°30.5	10.6'	23°18.3	3.8'	54.9'
8	298°28.7	07.5	13°00.1	10.6'	23°22.0	3.6'	54.9'
9	313°28.7	· · 06.8	27°29.7	10.5'	23°25.7	3.5'	54.9'
10	328°28.8	06.2	41°59.2	10.5'	23°29.2	3.4'	54.9'
11	343°28.9	05.5	56°28.7	10.4'	23°32.6	3.3'	54.9'
12	358°28.9	N17°04.8	70°58.1	10.3'	N23°35.9	3.2'	55.0'
13	13°29.0	04.1	85°27.4	10.3'	23°39.1	3.1'	55.0'
14	28°29.0	03.5	99°56.7	10.2'	23°42.2	3.0'	55.0'
15	43°29.1	· · 02.8	114°25.9	10.2'	23°45.2	2.8'	55.0'
16	58°29.2	02.1	128°55.1	10.1'	23°48.0	2.7'	55.0'
17	73°29.2	01.4	143°24.3	10.1'	23°50.7	2.6'	55.1'
18	88°29.3	N17°00.8	157°53.3	10.0'	N23°53.3	2.5'	55.1'
19	103°29.3	17°00.1	172°22.4	10.0'	23°55.8	2.4'	55.1'
20	118°29.4	16°59.4	186°51.3	9.9'	23°58.2	2.3'	55.1'
21	133°29.5	· · 58.7	201°20.3	9.9'	24°00.5	2.1'	55.2'
22	148°29.5	58.1	215°49.2	9.8'	24°02.6	2.0'	55.2'
23	163°29.6	57.4	230°18.0	9.8'	24°04.6	1.9'	55.2'
SD = 15.8'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°	--	--	01:45	22:21	--	--
68°	--	--	02:26	21:42	--	--
66°	--	00:56	02:54	21:15	23:06	--
64°	--	01:52	03:15	20:55	22:15	--
62°	--	02:24	03:32	20:38	21:45	--
60°	00:51	02:47	03:46	20:25	21:23	23:12
N 58°	01:42	03:05	03:58	20:13	21:05	22:27
56°	02:11	03:20	04:08	20:03	20:50	21:59
54°	02:32	03:33	04:17	19:54	20:38	21:38
52°	02:50	03:44	04:25	19:46	20:27	21:21
50°	03:04	03:54	04:32	19:39	20:17	21:07
45°	03:32	04:14	04:48	19:24	19:57	20:39
N 40°	03:53	04:30	05:00	19:11	19:41	20:18
35°	04:10	04:43	05:11	19:01	19:28	20:02
30°	04:24	04:55	05:20	18:52	19:17	19:48
20°	04:45	05:13	05:36	18:36	18:59	19:27
N 10°	05:02	05:28	05:50	18:22	18:44	19:10
0°	05:16	05:41	06:03	18:10	18:31	18:56
S 10°	05:28	05:53	06:15	17:57	18:19	18:44
20°	05:39	06:06	06:29	17:44	18:07	18:33
30°	05:50	06:19	06:44	17:29	17:54	18:23
35°	05:56	06:26	06:53	17:20	17:47	18:17
40°	06:01	06:34	07:02	17:10	17:39	18:11
45°	06:07	06:43	07:14	16:59	17:30	18:05
S 50°	06:14	06:53	07:28	16:45	17:20	17:59
52°	06:17	06:58	07:35	16:38	17:15	17:56
54°	06:20	07:03	07:42	16:31	17:10	17:53
56°	06:23	07:09	07:50	16:23	17:04	17:50
58°	06:27	07:15	07:59	16:14	16:58	17:46
S 60°	06:31	07:22	08:09	16:04	16:51	17:42

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°						
N 70°	18:31			17:03		
68°	19:42			15:53		
66°	20:19	20:13	19:58	15:17	17:02	19:02
64°	20:45	20:54	21:14	14:52	16:22	17:46
62°	21:06	21:22	21:50	14:32	15:54	17:10
60°	21:23	21:44	22:16	14:15	15:33	16:44
N 58°	21:37	22:01	22:37	14:02	15:16	16:24
56°	21:49	22:16	22:54	13:50	15:01	16:07
54°	22:00	22:29	23:08	13:40	14:48	15:53
52°	22:09	22:40	23:21	13:31	14:37	15:40
50°	22:18	22:51	23:32	13:23	14:28	15:29
45°	22:36	23:12	23:55	13:05	14:07	15:06
N 40°	22:51	23:29	· · ·	12:51	13:50	14:48
35°	23:03	23:43	· · ·	12:40	13:36	14:32
30°	23:15	23:56	· · ·	12:29	13:24	14:19
20°	23:34	· · ·	00:18	12:11	13:03	13:56
N 10°	23:50	· · ·	00:36	11:56	12:46	13:37
0°	· · ·	00:06	00:54	11:42	12:29	13:18
S 10°	· · ·	00:21	01:12	11:27	12:12	13:00
20°	· · ·	00:38	01:31	11:12	11:54	12:40
30°	00:03	00:58	01:53	10:55	11:34	12:18
35°	00:12	01:09	02:06	10:44	11:22	12:04
40°	00:23	01:22	02:21	10:33	11:08	11:49
45°	00:35	01:38	02:38	10:19	10:52	11:31
S 50°	00:51	01:57	03:00	10:03	10:32	11:08
52°	00:58	02:06	03:11	09:55	10:22	10:58
54°	01:06	02:16	03:23	09:47	10:12	10:45
56°	01:15	02:28	03:37	09:37	10:00	10:31
58°	01:26	02:41	03:53	09:26	09:46	10:15
S 60°	01:38	02:57	04:12	09:13	09:30	09:55

Day	Sun		Mer. Pass hh:mm	Moon		Age 22-24 55-36%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Pass. hh:mm	Lower Pass. hh:mm	
02	06:17	06:15	12:06	05:31	17:54	
03	06:12	06:10	12:06	06:17	18:41	
04	06:07	06:04	12:06	07:06	19:32	

August 05, 06, 07 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	313°50.1	145°03.5	N06°05.0	107°39.4	S11°32.3	113°45.4	S07°12.9	261°31.8	N16°37.5	Alpheratz	357°30.6	29°15.3
1	328°52.6	160°03.2	03.8	122°40.5	32.8	128°47.5	13.1	276°34.1	37.5	Ankaa	353°03.4	-42°08.3
2	343°55.0	175°03.0	02.5	137°41.6	33.4	143°49.7	13.2	291°36.4	37.5	Schedar	349°26.3	56°41.9
3	358°57.5	190°02.7	•• 01.3	152°42.6	•• 34.0	158°51.8	•• 13.3	306°38.7	•• 37.6	Diphda	348°43.5	-17°49.2
4	13°59.9	205°02.5	06°00.0	167°43.7	34.5	173°54.0	13.5	321°41.1	37.6	Achernar	335°17.4	-57°04.8
5	29°02.4	220°02.2	05°58.8	182°44.8	35.1	188°56.1	13.6	336°43.4	37.6	Hamal	327°46.9	23°36.1
6	44°04.9	235°02.0	N05°57.5	197°45.9	S11°35.6	203°58.3	S07°13.7	351°45.7	N16°37.6	Polaris	312°01.4	89°22.9
7	59°07.3	250°01.7	56.3	212°47.0	36.2	219°00.4	13.9	6°48.0	37.7	Acamar	315°09.1	-40°10.9
8	74°09.8	265°01.5	55.1	227°48.1	36.7	234°02.6	14.0	21°50.4	37.7	Menkar	314°02.3	4°12.4
9	89°12.3	280°01.2	•• 53.8	242°49.2	•• 37.3	249°04.7	•• 14.1	36°52.7	•• 37.7	Mirfak	308°22.9	49°57.7
10	104°14.7	295°01.0	52.6	257°50.2	37.8	264°06.9	14.3	51°55.0	37.7	Aldebaran	290°35.5	16°34.0
11	119°17.2	310°00.7	51.3	272°51.3	38.4	279°09.0	14.4	66°57.3	37.8	Rigel	281°00.5	-8°10.0
12	134°19.7	325°00.5	N05°50.1	287°52.4	S11°38.9	294°11.2	S07°14.5	81°59.7	N16°37.8	Capella	280°16.7	46°01.4
13	149°22.1	340°00.2	48.8	302°53.5	39.5	309°13.3	14.7	97°02.0	37.8	Bellatrix	278°19.1	6°22.5
14	164°24.6	355°00.0	47.6	317°54.6	40.0	324°15.5	14.8	112°04.3	37.8	Elnath	277°57.4	28°37.8
15	179°27.1	9°59.7	•• 46.3	332°55.7	•• 40.6	339°17.6	•• 14.9	127°06.6	•• 37.9	Alnilam	275°34.2	-1°11.0
16	194°29.5	24°59.5	45.1	347°56.8	41.1	354°19.8	15.1	142°09.0	37.9	Betelgeuse	270°48.3	7°24.7
17	209°32.0	39°59.3	43.8	2°57.8	41.7	9°22.0	15.2	157°11.3	37.9	Canopus	263°51.2	-52°42.6
18	224°34.4	54°59.0	N05°42.6	17°58.9	S11°42.2	24°24.1	S07°15.3	172°13.6	N16°37.9	Sirius	258°23.3	-16°45.4
19	239°36.9	69°58.8	41.3	33°00.0	42.8	39°26.2	15.5	187°15.9	38.0	Adhara	255°03.3	-29°00.7
20	254°39.4	84°58.5	40.1	48°01.1	43.3	54°28.4	15.6	202°18.3	38.0	Procyon	244°47.3	5°08.8
21	269°41.8	99°58.3	•• 38.8	63°02.2	•• 43.9	69°30.5	•• 15.8	217°20.6	•• 38.0	Pollux	243°13.2	27°57.1
22	284°44.3	114°58.0	37.6	78°03.2	44.4	84°32.7	15.9	232°22.9	38.0	Avior	234°13.8	-59°36.3
23	299°46.8	129°57.8	36.3	93°04.3	45.0	99°34.8	16.0	247°25.2	38.1	Suhail	222°44.0	-43°33.2
Mer.pass. 03:04		ν -0.3' d -1.2' m -3.94		ν 1.1' d 0.6' m 0.71		ν 2.2' d 0.1' m -1.87		ν 2.3' d 0.0' m 0.21		Miaplacidus	221°38.2	-69°50.5
										Alphard	217°44.4	-8°47.3
										Regulus	207°30.8	11°49.3
										Dubhe	193°37.3	61°35.5
										Denebola	182°21.4	14°24.4
										Gienah	175°39.9	-17°42.5
										Acrux	172°56.1	-63°16.1
										Gacrux	171°47.6	-57°17.0
										Alioth	166°10.2	55°48.1
										Spica	158°18.5	-11°19.0
										Alkaid	152°49.3	49°10.2
										Hadar	148°30.6	-60°31.2
										Menkent	147°53.2	-36°31.1
										Arcturus	145°44.6	19°01.9
										Rigil Kent.	139°35.0	-60°57.7
										Kochab	137°20.2	74°02.3
										Zuben'ubi	136°51.9	-16°09.9
										Alphecca	126°00.6	26°37.2
										Antares	112°11.1	-26°29.8
										Atria	107°01.6	-69°04.9
										Sabik	101°58.3	-15°45.5
										Shaula	96°05.0	-37°07.4
										Rasalhague	95°54.9	12°32.6
										Eltanin	90°40.1	51°29.4
										Kaus Aust.	83°27.3	-34°22.1
										Vega	80°30.3	38°49.0
										Nunki	75°42.9	-26°15.4
										Altair	61°56.0	8°57.1
										Peacock	52°59.4	-56°38.1
										Deneb	49°22.7	45°23.4
										Enif	33°34.8	10°00.9
										Al Na'ir	27°28.0	-46°48.7
										Fomalhaut	15°10.2	-29°27.6
										Scheat	13°41.2	28°14.7
										Markab	13°25.9	15°22.1
										Aug 05 Sun		
										Venus	191°13.4	14:20
										Mars	153°49.3	16:48
										Jupiter	159°55.3	16:23
										Saturn	307°41.7	06:33
										Aug 06 Mon		
										Venus	190°08.3	14:20
										Mars	153°16.2	16:46
										Jupiter	159°47.8	16:19
										Saturn	307°38.3	06:29
										Aug 07 Tue		
										Venus	189°03.4	14:21
										Mars	152°42.9	16:45
										Jupiter	159°40.1	16:16
										Saturn	307°35.1	06:25
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.1	
Mer.pass. 02:56		ν -0.2' d -1.3' m -3.94		ν 1.1' d 0.6' m 0.73		ν 2.1' d 0.1' m -1.86		ν 2.3' d 0.0' m 0.21				

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°29.7	N16°56.7	244°46.8	9.7'	N24°06.5	1.8'	55.2'
1	193°29.7	56.0	259°15.5	9.7'	24°08.3	1.7'	55.3'
2	208°29.8	55.4	273°44.2	9.6'	24°09.9	1.5'	55.3'
3	223°29.8	· · 54.7	288°12.8	9.6'	24°11.5	1.4'	55.3'
4	238°29.9	54.0	302°41.4	9.6'	24°12.9	1.3'	55.3'
5	253°30.0	53.3	317°10.0	9.5'	24°14.2	1.2'	55.4'
6	268°30.0	N16°52.6	331°38.5	9.5'	N24°15.3	1.0'	55.4'
7	283°30.1	52.0	346°07.0	9.4'	24°16.3	0.9'	55.4'
8	298°30.2	51.3	0°35.4	9.4'	24°17.3	0.8'	55.4'
9	313°30.2	· · 50.6	15°03.8	9.3'	24°18.0	0.7'	55.5'
10	328°30.3	49.9	29°32.1	9.3'	24°18.7	0.5'	55.5'
11	343°30.4	49.2	44°00.4	9.3'	24°19.2	0.4'	55.5'
12	358°30.4	N16°48.6	58°28.7	9.2'	N24°19.6	0.3'	55.5'
13	13°30.5	47.9	72°56.9	9.2'	24°19.9	0.1'	55.6'
14	28°30.6	47.2	87°25.1	9.2'	24°20.1	0.0'	55.6'
15	43°30.6	· · 46.5	101°53.2	9.1'	24°20.1	-0.1'	55.6'
16	58°30.7	45.8	116°21.3	9.1'	24°20.0	-0.2'	55.6'
17	73°30.8	45.1	130°49.4	9.0'	24°19.7	-0.4'	55.7'
18	88°30.8	N16°44.4	145°17.5	9.0'	N24°19.4	-0.5'	55.7'
19	103°30.9	43.8	159°45.5	9.0'	24°18.9	-0.6'	55.7'
20	118°31.0	43.1	174°13.5	9.0'	24°18.2	-0.8'	55.7'
21	133°31.0	· · 42.4	188°41.4	8.9'	24°17.5	-0.9'	55.8'
22	148°31.1	41.7	203°09.3	8.9'	24°16.6	-1.0'	55.8'
23	163°31.2	41.0	217°37.2	8.9'	24°15.6	-1.2'	55.8'
SD = 15.8'			SD = 15.1'				
$d = -0.7'$							

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°31.2	N16°40.3	232°05.1	8.8'	N24°14.4	-1.3'	55.9'
1	193°31.3	39.6	246°32.9	8.8'	24°13.1	-1.4'	55.9'
2	208°31.4	38.9	261°00.7	8.8'	24°11.7	-1.5'	55.9'
3	223°31.5	· · 38.3	275°28.5	8.8'	24°10.2	-1.7'	55.9'
4	238°31.5	37.6	289°56.3	8.7'	24°08.5	-1.8'	56.0'
5	253°31.6	36.9	304°24.0	8.7'	24°06.7	-1.9'	56.0'
6	268°31.7	N16°36.2	318°51.7	8.7'	N24°04.7	-2.1'	56.0'
7	283°31.7	35.5	333°19.4	8.7'	24°02.7	-2.2'	56.1'
8	298°31.8	34.8	347°47.1	8.7'	24°00.5	-2.3'	56.1'
9	313°31.9	· · 34.1	2°14.8	8.6'	23°58.1	-2.5'	56.1'
10	328°31.9	33.4	16°42.4	8.6'	23°55.6	-2.6'	56.2'
11	343°32.0	32.7	31°10.0	8.6'	23°53.0	-2.7'	56.2'
12	358°32.1	N16°32.0	45°37.7	8.6'	N23°50.3	-2.9'	56.2'
13	13°32.2	31.3	60°05.2	8.6'	23°47.4	-3.0'	56.2'
14	28°32.2	30.6	74°32.8	8.6'	23°44.4	-3.1'	56.3'
15	43°32.3	· · 29.9	89°00.4	8.6'	23°41.3	-3.3'	56.3'
16	58°32.4	29.2	103°28.0	8.5'	23°38.0	-3.4'	56.3'
17	73°32.5	28.5	117°55.5	8.5'	23°34.6	-3.5'	56.4'
18	88°32.5	N16°27.8	132°23.0	8.5'	N23°31.0	-3.7'	56.4'
19	103°32.6	27.1	146°50.6	8.5'	23°27.3	-3.8'	56.4'
20	118°32.7	26.4	161°18.1	8.5'	23°23.5	-3.9'	56.5'
21	133°32.8	· · 25.8	175°45.6	8.5'	23°19.6	-4.1'	56.5'
22	148°32.8	25.1	190°13.1	8.5'	23°15.5	-4.2'	56.5'
23	163°32.9	24.4	204°40.6	8.5'	23°11.3	-4.3'	56.5'
SD = 15.8'			SD = 15.2'				
$d = -0.7'$							

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°33.0	N16°23.7	219°08.1	8.5'	N23°06.9	-4.5'	56.6'
1	193°33.1	23.0	233°35.6	8.5'	23°02.4	-4.6'	56.6'
2	208°33.1	22.3	248°03.1	8.5'	22°57.8	-4.7'	56.6'
3	223°33.2	· · 21.6	262°30.6	8.5'	22°53.1	-4.9'	56.7'
4	238°33.3	20.8	276°58.1	8.5'	22°48.2	-5.0'	56.7'
5	253°33.4	20.1	291°25.5	8.5'	22°43.2	-5.1'	56.7'
6	268°33.4	N16°19.4	305°53.0	8.5'	N22°38.0	-5.3'	56.8'
7	283°33.5	18.7	320°20.5	8.5'	22°32.8	-5.4'	56.8'
8	298°33.6	18.0	334°48.0	8.5'	22°27.4	-5.5'	56.8'
9	313°33.7	· · 17.3	349°15.5	8.5'	22°21.8	-5.7'	56.9'
10	328°33.7	16.6	3°43.1	8.5'	22°16.1	-5.8'	56.9'
11	343°33.8	15.9	18°10.6	8.5'	22°10.3	-5.9'	56.9'
12	358°33.9	N16°15.2	32°38.1	8.5'	N22°04.4	-6.1'	57.0'
13	13°34.0	14.5	47°05.6	8.5'	21°58.3	-6.2'	57.0'
14	28°34.1	13.8	61°33.2	8.6'	21°52.1	-6.3'	57.0'
15	43°34.1	· · 13.1	76°00.7	8.6'	21°45.8	-6.5'	57.0'
16	58°34.2	12.4	90°28.3	8.6'	21°39.3	-6.6'	57.1'
17	73°34.3	11.7	104°55.9	8.6'	21°32.8	-6.7'	57.1'
18	88°34.4	N16°11.0	119°23.5	8.6'	N21°26.1	-6.8'	57.1'
19	103°34.5	10.3	133°51.1	8.6'	21°19.2	-7.0'	57.2'
20	118°34.5	09.6	148°18.7	8.6'	21°12.2	-7.1'	57.2'
21	133°34.6	· · 08.9	162°46.3	8.6'	21°05.1	-7.2'	57.2'
22	148°34.7	08.1	177°13.9	8.7'	20°57.9	-7.3'	57.3'
23	163°34.8	07.4	191°41.6	8.7'	20°50.6	-7.5'	57.3'
SD = 15.8'			SD = 15.4'				
$d = -0.7'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:02	22:57	—:—	—:—
N 70°	—:—	—:—	02:06	22:00	—:—	—:—
68°	—:—	—:—	02:41	21:27	—:—	—:—
66°	—:—	01:26	03:05	21:03	22:39	—:—
64°	—:—	02:08	03:25	20:45	22:00	—:—
62°	—:—	02:35	03:40	20:30	21:33	—:—
60°	01:17	02:56	03:53	20:17	21:13	22:48
N 58°	01:55	03:13	04:04	20:06	20:56	22:13
56°	02:21	03:27	04:14	19:57	20:43	21:48
54°	02:41	03:39	04:22	19:48	20:31	21:29
52°	02:57	03:50	04:30	19:41	20:21	21:13
50°	03:10	03:59	04:37	19:34	20:11	21:00
45°	03:37	04:18	04:51	19:20	19:53	20:34
N 40°	03:57	04:33	05:03	19:08	19:38	20:14
35°	04:13	04:46	05:13	18:58	19:25	19:58
30°	04:26	04:56	05:22	18:49	19:15	19:45
20°	04:47	05:14	05:37	18:34	18:57	19:25
N 10°	05:03	05:28	05:50	18:21	18:43	19:09
0°	05:16	05:41	06:02	18:09	18:31	18:56
S 10°	05:27	05:53	06:14	17:57	18:19	18:45
20°	05:38	06:04	06:27	17:45	18:08	18:34
30°	05:48	06:16	06:41	17:31	17:56	18:24
35°	05:53	06:23	06:50	17:22	17:49	18:19
40°	05:58	06:31	06:59	17:13	17:42	18:14
45°	06:04	06:39	07:10	17:02	17:33	18:08
S 50°	06:10	06:49	07:23	16:49	17:24	18:02
52°	06:13	06:53	07:29	16:43	17:19	18:00
54°	06:15	06:58	07:36	16:36	17:15	17:57
56°	06:18	07:03	07:44	16:29	17:09	17:54
58°	06:21	07:09	07:52	16:20	17:04	17:51
S 60°	06:25	07:15	08:02	16:10	16:57	17:48

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	23:40	☐	☐	20:55
66°	☐	22:26	· ·	☐	20:15	20:07
64°	22:00	23:19	· ·	18:49	19:22	19:35
62°	22:39	23:51	· ·	18:10	18:49	19:12
60°	23:06	· ·	00:15	17:43	18:25	18:53
N 58°	23:27	· ·	00:34	17:22	18:06	18:37
56°	23:45	· ·	00:50	17:04	17:50	18:24
54°	■	☐	01:04	16:49	17:36	18:12
52°	· ·	00:12	01:15	16:36	17:24	18:02
50°	· ·	00:24	01:26	16:25	17:13	17:53
45°	· ·	00:47	01:48	16:01	16:51	17:33
N 40°	00:14	01:06	02:05	15:42	16:33	17:17
35°	00:29	01:22	02:20	15:27	16:17	17:04
30°	00:43	01:35	02:33	15:13	16:04	16:52
20°	01:06	01:59	02:55	14:49	15:42	16:32
N 10°	01:26	02:19	03:14	14:29	15:22	16:14
0°	01:45	02:38	03:32	14:10	15:04	15:57
S 10°	02:04	02:57	03:49	13:51	14:45	15:41
20°	02:24	03:17	04:08	13:31	14:25	15:23
30°	02:47	03:40	04:30	13:07	14:02	15:02
35°	03:01	03:54	04:43	12:54	13:49	14:50
40°	03:17	04:10	04:57	12:38	13:33	14:36
45°	03:36	04:28	05:15	12:18	13:15	14:19
S 50°	04:00	04:52	05:36	11:55	12:52	13:58
52°	04:11	05:03	05:46	11:43	12:40	13:49
54°	04:24	05:16	05:58	11:30	12:28	13:37
56°	04:39	05:31	06:11	11:15	12:13	13:25
58°	04:57	05:48	06:26	10:57	11:56	13:10
S 60°	05:18	06:09	06:44	10:36	11:35	12:52

Day	Sun		Mer. Pass hh:mm	Moon		Age 25-27 27-11%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
05	06:01	05:58	12:06	07:58	20:24	
06	05:55	05:52	12:06	08:51	21:18	
07	05:48	05:44	12:06	09:45	22:11	

August 08, 09, 10 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	316°47.5	144°46.1	N04°34.8	108°56.8	S12°12.0	116°19.9	S07°22.7	264°19.4	N16°39.2	Alpheratz	357°30.6	29°15.3	
1	331°50.0	159°45.8		123°57.9	12.6	131°22.0	22.9	279°21.7	39.3	Ankaa	353°03.3	-42°08.3	
2	346°52.4	174°45.6		138°59.0	13.1	146°24.2	23.0	294°24.1	39.3	Schedar	349°26.3	56°41.9	
3	1°54.9	189°45.4	•• 31.0	154°00.0	•• 13.7	161°26.3	•• 23.1	309°26.4	•• 39.3	Diphda	348°43.4	-17°49.2	
4	16°57.4	204°45.2		169°01.1	14.2	176°28.5	23.3	324°28.7	39.3	Achernar	335°17.4	-57°04.8	
5	31°59.8	219°44.9		184°02.1	14.8	191°30.6	23.4	339°31.1	39.3	Hamal	327°46.9	23°36.1	
6	47°02.3	234°44.7	N04°27.2	199°03.2	S12°15.3	206°32.7	S07°23.6	354°33.4	N16°39.4	Polaris	311°59.7	89°22.9	
7	62°04.8	249°44.5		214°04.3	15.9	221°34.9	23.7	9°35.7	39.4	Acamar	315°09.0	-40°10.9	
8	77°07.2	264°44.2		229°05.3	16.4	236°37.0	23.8	24°38.1	39.4	Menkar	314°02.3	4°12.4	
9	92°09.7	279°44.0	•• 23.4	244°06.4	•• 17.0	251°39.1	•• 24.0	39°40.4	•• 39.4	Mirfak	308°22.9	49°57.7	
10	107°12.2	294°43.8		259°07.4	17.5	266°41.3	24.1	54°42.8	39.5	Aldebaran	290°35.5	16°34.0	
11	122°14.6	309°43.5		274°08.5	18.1	281°43.4	24.3	69°45.1	39.5	Rigel	281°00.5	-8°10.0	
12	137°17.1	324°43.3	N04°19.6	289°09.6	S12°18.6	296°45.6	S07°24.4	84°47.4	N16°39.5	Capella	280°16.6	46°01.4	
13	152°19.5	339°43.1		304°10.6	19.2	311°47.7	24.5	99°49.8	39.5	Bellatrix	278°19.1	6°22.5	
14	167°22.0	354°42.9		319°11.7	19.7	326°49.8	24.7	114°52.1	39.5	Elnath	277°57.4	28°37.8	
15	182°24.5	9°42.6	•• 15.8	334°12.7	•• 20.3	341°52.0	•• 24.8	129°54.4	•• 39.6	Alnilam	275°34.2	-1°11.0	
16	197°26.9	24°42.4		349°13.8	20.8	356°54.1	25.0	144°56.8	39.6	Betelgeuse	270°48.3	7°24.7	
17	212°29.4	39°42.2		4°14.9	21.4	11°56.2	25.1	159°59.1	39.6	Canopus	263°51.2	-52°42.6	
18	227°31.9	54°41.9	N04°12.0	19°15.9	S12°21.9	26°58.4	S07°25.2	175°01.4	N16°39.6	Sirius	258°23.2	-16°45.4	
19	242°34.3	69°41.7		34°17.0	22.5	42°00.5	25.4	190°03.8	39.7	Adhara	255°03.3	-29°00.7	
20	257°36.8	84°41.5		49°18.0	23.0	57°02.6	25.5	205°06.1	39.7	Procyon	244°47.3	5°08.9	
21	272°39.3	99°41.3	•• 08.2	64°19.1	•• 23.6	72°04.8	•• 25.7	220°08.5	•• 39.7	Pollux	243°13.2	27°57.1	
22	287°41.7	114°41.0		79°20.1	24.1	87°06.9	25.8	235°10.8	39.7	Avior	234°13.7	-59°36.3	
23	302°44.2	129°40.8		94°21.2	24.7	102°09.0	25.9	250°13.1	39.7	Suhail	222°43.9	-43°33.2	
Mer.pass. 02:52		ν -0.2' d -1.3' m -3.94		ν 1.1' d 0.6' m 0.73		ν 2.1' d 0.1' m -1.85		ν 2.3' d 0.0' m 0.20		Miaplacidus	221°38.2	-69°50.4	
										Alphard	217°44.4	-8°47.3	
										Regulus	207°30.8	11°49.3	
										Dubhe	193°37.3	61°35.5	
										Denebola	182°21.4	14°24.4	
										Gienah	175°39.9	-17°42.5	
										Acrux	172°56.1	-63°16.1	
										Gacrux	171°47.6	-57°17.0	
										Alioth	166°10.2	55°48.1	
										Spica	158°18.5	-11°19.0	
										Alkaid	152°49.4	49°10.2	
										Hadar	148°30.7	-60°31.2	
										Menkent	147°53.2	-36°31.1	
										Arcturus	145°44.6	19°01.9	
										Rigil Kent.	139°35.1	-60°57.7	
										Kochab	137°20.3	74°02.3	
										Zuben'ubi	136°51.9	-16°09.9	
										Alphecca	126°00.6	26°37.2	
										Antares	112°11.1	-26°29.8	
										Atria	107°01.6	-69°04.9	
										Sabik	101°58.3	-15°45.5	
										Shaula	96°05.1	-37°07.4	
										Rasalhague	95°54.9	12°32.6	
										Eltanin	90°40.1	51°29.4	
										Kaus Aust.	83°27.3	-34°22.1	
										Vega	80°30.4	38°49.0	
										Nunki	75°42.9	-26°15.4	
										Altair	61°56.0	8°57.1	
										Peacock	52°59.4	-56°38.2	
										Deneb	49°22.7	45°23.4	
										Enif	33°34.8	10°00.9	
										Al Na'ir	27°27.9	-46°48.7	
										Fomalhaut	15°10.2	-29°27.6	
										Scheat	13°41.2	28°14.8	
										Markab	13°25.9	15°22.1	
										Aug 08 Wed		SHA	Mer.pass
										Venus	187°58.6	14:21	
										Mars	152°09.3	16:43	
										Jupiter	159°32.4	16:12	
										Saturn	307°31.9	06:22	
										Aug 09 Thu		SHA	Mer.pass
										Venus	186°53.9	14:22	
										Mars	151°35.6	16:41	
										Jupiter	159°24.5	16:09	
										Saturn	307°28.8	06:18	
										Aug 10 Fri		SHA	Mer.pass
										Venus	185°49.4	14:22	
										Mars	151°01.6	16:40	
										Jupiter	159°16.6	16:06	
										Saturn	307°25.8	06:14	
										Horizontal parallax			
										Venus:	0.1		
										Mars:	0.1		
Mer.pass. 02:44		ν -0.2' d -1.3' m -3.95		ν 1.0' d 0.5' m 0.75		ν 2.1' d 0.1' m -1.84		ν 2.3' d 0.0' m 0.19					

DUT1 = UT1-UTC = +0.2118 sec ΔT = TT-UT1 = +68.9722 sec

2029 August 08 to Aug. 10 UT

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°34.9	N16°06.7	206°09.3	8.7'	N20°43.1	-7.6'	57.3'
1	193°34.9	06.0	220°37.0	8.7'	20°35.5	-7.7'	57.4'
2	208°35.0	05.3	235°04.7	8.7'	20°27.8	-7.8'	57.4'
3	223°35.1	04.6	249°32.4	8.7'	20°19.9	-8.0'	57.4'
4	238°35.2	03.9	264°00.2	8.8'	20°12.0	-8.1'	57.5'
5	253°35.3	03.2	278°27.9	8.8'	20°03.9	-8.2'	57.5'
6	268°35.4	N16°02.4	292°55.7	8.8'	N19°55.7	-8.3'	57.5'
7	283°35.4	01.7	307°23.5	8.8'	19°47.3	-8.5'	57.6'
8	298°35.5	01.0	321°51.4	8.9'	19°38.9	-8.6'	57.6'
9	313°35.6	16°00.3	336°19.2	8.9'	19°30.3	-8.7'	57.6'
10	328°35.7	15°59.6	350°47.1	8.9'	19°21.6	-8.8'	57.6'
11	343°35.8	58.9	5°15.0	8.9'	19°12.8	-8.9'	57.7'
12	358°35.9	N15°58.2	19°42.9	8.9'	N19°03.9	-9.0'	57.7'
13	13°35.9	57.4	34°10.9	9.0'	18°54.8	-9.2'	57.7'
14	28°36.0	56.7	48°38.8	9.0'	18°45.6	-9.3'	57.8'
15	43°36.1	56.0	63°06.8	9.0'	18°36.4	-9.4'	57.8'
16	58°36.2	55.3	77°34.8	9.0'	18°27.0	-9.5'	57.8'
17	73°36.3	54.6	92°02.9	9.1'	18°17.5	-9.6'	57.9'
18	88°36.4	N15°53.9	106°31.0	9.1'	N18°07.9	-9.7'	57.9'
19	103°36.4	53.1	120°59.0	9.1'	17°58.1	-9.8'	57.9'
20	118°36.5	52.4	135°27.2	9.2'	17°48.3	-10.0'	57.9'
21	133°36.6	51.7	149°55.3	9.2'	17°38.3	-10.1'	58.0'
22	148°36.7	51.0	164°23.5	9.2'	17°28.3	-10.2'	58.0'
23	163°36.8	50.3	178°51.7	9.2'	17°18.1	-10.3'	58.0'
SD = 15.8' d = -0.7'			SD = 15.6'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°36.9	N15°49.5	193°19.9	9.3'	N17°07.8	-10.4'	58.1'
1	193°37.0	48.8	207°48.2	9.3'	16°57.4	-10.5'	58.1'
2	208°37.1	48.1	222°16.5	9.3'	16°46.9	-10.6'	58.1'
3	223°37.1	47.4	236°44.8	9.3'	16°36.4	-10.7'	58.2'
4	238°37.2	46.6	251°13.1	9.4'	16°25.7	-10.8'	58.2'
5	253°37.3	45.9	265°41.5	9.4'	16°14.9	-10.9'	58.2'
6	268°37.4	N15°45.2	280°09.9	9.4'	N16°04.0	-11.0'	58.2'
7	283°37.5	44.5	294°38.3	9.5'	15°53.0	-11.1'	58.3'
8	298°37.6	43.7	309°06.8	9.5'	15°41.9	-11.2'	58.3'
9	313°37.7	43.0	323°35.3	9.5'	15°30.7	-11.3'	58.3'
10	328°37.8	42.3	338°03.8	9.5'	15°19.4	-11.4'	58.4'
11	343°37.9	41.6	352°32.3	9.6'	15°08.0	-11.5'	58.4'
12	358°37.9	N15°40.8	7°00.9	9.6'	N14°56.5	-11.6'	58.4'
13	13°38.0	40.1	21°29.5	9.6'	14°44.9	-11.7'	58.4'
14	28°38.1	39.4	35°58.1	9.7'	14°33.2	-11.8'	58.5'
15	43°38.2	38.7	50°26.8	9.7'	14°21.5	-11.9'	58.5'
16	58°38.3	37.9	64°55.4	9.7'	14°09.6	-11.9'	58.5'
17	73°38.4	37.2	79°24.1	9.7'	13°57.7	-12.0'	58.5'
18	88°38.5	N15°36.5	93°52.9	9.8'	N13°45.6	-12.1'	58.6'
19	103°38.6	35.7	108°21.6	9.8'	13°33.5	-12.2'	58.6'
20	118°38.7	35.0	122°50.4	9.8'	13°21.3	-12.3'	58.6'
21	133°38.8	34.3	137°19.3	9.8'	13°09.0	-12.4'	58.6'
22	148°38.9	33.6	151°48.1	9.9'	12°56.6	-12.5'	58.7'
23	163°39.0	32.8	166°17.0	9.9'	12°44.2	-12.5'	58.7'
SD = 15.8' d = -0.7'			SD = 15.8'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°39.1	N15°32.1	180°45.9	9.9'	N12°31.6	-12.6'	58.7'
1	193°39.1	31.4	195°14.8	9.9'	12°19.0	-12.7'	58.7'
2	208°39.2	30.6	209°43.7	10.0'	12°06.3	-12.8'	58.8'
3	223°39.3	29.9	224°12.7	10.0'	11°53.6	-12.8'	58.8'
4	238°39.4	29.2	238°41.7	10.0'	11°40.7	-12.9'	58.8'
5	253°39.5	28.4	253°10.7	10.0'	11°27.8	-13.0'	58.8'
6	268°39.6	N15°27.7	267°39.8	10.1'	N11°14.8	-13.1'	58.9'
7	283°39.7	27.0	282°08.9	10.1'	11°01.7	-13.1'	58.9'
8	298°39.8	26.2	296°37.9	10.1'	10°48.6	-13.2'	58.9'
9	313°39.9	25.5	311°07.1	10.1'	10°35.4	-13.3'	58.9'
10	328°40.0	24.7	325°36.2	10.2'	10°22.1	-13.3'	59.0'
11	343°40.1	24.0	340°05.4	10.2'	10°08.8	-13.4'	59.0'
12	358°40.2	N15°23.3	354°34.6	10.2'	N09°55.3	-13.5'	59.0'
13	13°40.3	22.5	9°03.8	10.2'	09°41.9	-13.5'	59.0'
14	28°40.4	21.8	23°33.0	10.2'	09°28.3	-13.6'	59.0'
15	43°40.5	21.1	38°02.2	10.3'	09°14.7	-13.7'	59.1'
16	58°40.6	20.3	52°31.5	10.3'	09°01.1	-13.7'	59.1'
17	73°40.7	19.6	67°00.8	10.3'	08°47.4	-13.8'	59.1'
18	88°40.8	N15°18.8	81°30.1	10.3'	N08°33.6	-13.8'	59.1'
19	103°40.9	18.1	95°59.4	10.3'	08°19.8	-13.9'	59.1'
20	118°41.0	17.4	110°28.8	10.4'	08°05.9	-13.9'	59.2'
21	133°41.1	16.6	124°58.1	10.4'	07°51.9	-14.0'	59.2'
22	148°41.2	15.9	139°27.5	10.4'	07°37.9	-14.0'	59.2'
23	163°41.3	15.1	153°56.9	10.4'	07°23.9	-14.1'	59.2'
SD = 15.8' d = -0.7'			SD = 16.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:38	22:25	—:—	—:—
N 70°	—:—	—:—	02:25	21:41	—:—	—:—
68°	—:—	00:36	02:55	21:13	23:17	—:—
66°	—:—	01:47	03:17	20:52	22:18	—:—
64°	—:—	02:22	03:34	20:35	21:45	—:—
62°	00:33	02:47	03:48	20:21	21:22	23:23
60°	01:36	03:06	04:00	20:09	21:03	22:29
N 58°	02:08	03:21	04:11	19:59	20:48	21:59
56°	02:31	03:34	04:20	19:50	20:35	21:37
54°	02:49	03:45	04:28	19:42	20:24	21:20
52°	03:04	03:55	04:35	19:35	20:14	21:05
50°	03:16	04:04	04:41	19:29	20:06	20:53
45°	03:41	04:22	04:55	19:15	19:48	20:28
N 40°	04:00	04:36	05:06	19:04	19:34	20:10
35°	04:16	04:48	05:16	18:55	19:22	19:55
30°	04:28	04:58	05:24	18:47	19:12	19:42
20°	04:48	05:15	05:38	18:32	18:56	19:23
N 10°	05:03	05:29	05:51	18:20	18:42	19:08
0°	05:16	05:41	06:02	18:09	18:30	18:55
S 10°	05:26	05:52	06:13	17:58	18:19	18:45
20°	05:36	06:03	06:25	17:46	18:09	18:35
30°	05:46	06:14	06:39	17:32	17:57	18:26
35°	05:50	06:20	06:47	17:25	17:51	18:21
40°	05:55	06:27	06:56	17:16	17:44	18:16
45°	06:00	06:35	07:06	17:06	17:36	18:11
S 50°	06:06	06:44	07:18	16:53	17:28	18:06
52°	06:08	06:48	07:24	16:48	17:24	18:04
54°	06:10	06:52	07:30	16:41	17:19	18:01
56°	06:13	06:57	07:37	16:34	17:15	17:59
58°	06:16	07:03	07:45	16:26	17:09	17:56
S 60°	06:18	07:08	07:54	16:17	17:03	17:53

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	□	00:30	03:27	21:58	20:52	20:19
N 70°	□	01:27	03:48	21:00	20:29	20:08
68°	□	02:00	04:04	20:26	20:11	19:59
66°	□	02:24	04:17	20:01	19:56	19:51
64°	□	02:43	04:28	19:41	19:44	19:44
62°	□	02:58	04:38	19:25	19:33	19:39
60°	□	03:11	04:45	19:12	19:24	19:34
N 58°	□	03:21	04:52	19:00	19:16	19:29
56°	□	03:31	04:58	18:50	19:09	19:25
54°	□	03:39	05:04	18:41	19:03	19:22
52°	□	03:47	05:09	18:33	18:57	19:18
50°	□	03:53	05:13	18:25	18:52	19:16
45°	□	04:08	05:22	18:10	18:41	19:09
N 40°	□	04:20	05:30	17:57	18:32	19:04
35°	□	04:29	05:37	17:46	18:24	18:59
30°	□	04:38	05:43	17:36	18:17	18:55
20°	□	04:53	05:53	17:19	18:04	18:47
N 10°	□	05:06	06:02	17:05	17:54	18:41
0°	□	05:18	06:10	16:51	17:43	18:35
S 10°	□	05:31	06:19	16:37	17:33	18:28
20°	□	05:44	06:27	16:22	17:22	18:22
30°	□	05:58	06:37	16:05	17:09	18:14
35°	□	06:07	06:43	15:55	17:02	18:09
40°	□	06:16	06:50	15:43	16:53	18:04
45°	□	06:28	06:57	15:29	16:43	17:58
S 50°	□	06:41	07:06	15:13	16:31	17:51
52°	□	06:48	07:10	15:05	16:25	17:48
54°	□	06:55	07:15	14:56	16:19	17:44
56°	□	07:02	07:20	14:46	16:12	17:40
58°	□	07:11	07:26	14:34	16:04	17:36
S 60°	□	07:21	07:32	14:21	15:55	17:30

Day	Sun		Mer.	Moon		Age
	Eqn. of Time	12h	Pass	Mer. Pass.	Lower	28-0
	00h	12h	hh:mm	Upper	hh:mm	5-0%
08	05:41	05:37	12:06	10:38	23:05	
09	05:32	05:28	12:05	11:31	23:57	
10	05:24	05:19	12:05	12:22	—:—	

August 11, 12, 13 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	319°44.9	144°30.0	N03°03.4	110°12.4	S12°51.6	118°53.4	S07°32.9	267°07.9	N16°40.8	Alpheratz	357°30.6	29°15.3
1	334°47.4	159°29.8	02.1	125°13.4	52.2	133°55.6	33.0	282°10.2	40.8	Ankaa	353°03.3	-42°08.4
2	349°49.9	174°29.6	03°00.8	140°14.5	52.7	148°57.7	33.2	297°12.6	40.8	Schedar	349°26.2	56°41.9
3	4°52.3	189°29.4	02°59.6	155°15.5	53.3	163°59.8	33.3	312°14.9	40.8	Diphda	348°43.4	-17°49.2
4	19°54.8	204°29.2	58.3	170°16.5	53.8	179°01.9	33.5	327°17.3	40.8	Achernar	335°17.4	-57°04.8
5	34°57.2	219°28.9	57.0	185°17.6	54.3	194°04.1	33.6	342°19.6	40.8	Hamal	327°46.8	23°36.1
6	49°59.7	234°28.7	N02°55.7	200°18.6	S12°54.9	209°06.2	S07°33.7	357°22.0	N16°40.9	Polaris	311°58.2	89°22.9
7	65°02.2	249°28.5	54.5	215°19.6	55.4	224°08.3	33.9	12°24.3	40.9	Acamar	315°09.0	-40°10.9
8	80°04.6	264°28.3	53.2	230°20.7	56.0	239°10.4	34.0	27°26.7	40.9	Menkar	314°02.2	4°12.4
9	95°07.1	279°28.1	51.9	245°21.7	56.5	254°12.6	34.2	42°29.0	40.9	Mirfak	308°22.8	49°57.7
10	110°09.6	294°27.9	50.6	260°22.7	57.1	269°14.7	34.3	57°31.4	40.9	Aldebaran	290°35.5	16°34.0
11	125°12.0	309°27.7	49.4	275°23.8	57.6	284°16.8	34.5	72°33.7	41.0	Rigel	281°00.5	-8°10.0
12	140°14.5	324°27.5	N02°48.1	290°24.8	S12°58.2	299°18.9	S07°34.6	87°36.1	N16°41.0	Capella	280°16.6	46°01.4
13	155°17.0	339°27.2	46.8	305°25.8	58.7	314°21.0	34.8	102°38.4	41.0	Bellatrix	278°19.1	6°22.5
14	170°19.4	354°27.0	45.5	320°26.9	59.3	329°23.2	34.9	117°40.8	41.0	Elnath	277°57.4	28°37.8
15	185°21.9	9°26.8	44.3	335°27.9	12°59.8	344°25.3	35.0	132°43.1	41.0	Alnilam	275°34.1	-1°11.0
16	200°24.4	24°26.6	43.0	350°28.9	13°00.4	359°27.4	35.2	147°45.5	41.1	Betelgeuse	270°48.3	7°24.7
17	215°26.8	39°26.4	41.7	5°29.9	00.9	14°29.5	35.3	162°47.8	41.1	Canopus	263°51.1	-52°42.5
18	230°29.3	54°26.2	N02°40.4	20°31.0	S13°01.5	29°31.7	S07°35.5	177°50.2	N16°41.1	Sirius	258°23.2	-16°45.4
19	245°31.7	69°26.0	39.1	35°32.0	02.0	44°33.8	35.6	192°52.5	41.1	Adhara	255°03.2	-29°00.7
20	260°34.2	84°25.8	37.9	50°33.0	02.6	59°35.9	35.8	207°54.9	41.1	Procyon	244°47.3	5°08.9
21	275°36.7	99°25.6	36.6	65°34.1	03.1	74°38.0	35.9	222°57.2	41.2	Pollux	243°13.1	27°57.1
22	290°39.1	114°25.3	35.3	80°35.1	03.7	89°40.1	36.0	237°59.6	41.2	Avior	234°13.7	-59°36.3
23	305°41.6	129°25.1	34.0	95°36.1	04.2	104°42.3	36.2	253°01.9	41.2	Suhail	222°43.9	-43°33.2
Mer.pass. 02:41		ν -0.2' d -1.3' m -3.95		ν 1.0' d 0.5' m 0.75		ν 2.1' d 0.1' m -1.84		ν 2.3' d 0.0' m 0.19		Miaplacidus	221°38.2	-69°50.4
										Alphard	217°44.4	-8°47.3
										Regulus	207°30.8	11°49.3
										Dubhe	193°37.3	61°35.5
										Denebola	182°21.5	14°24.4
										Gienah	175°39.9	-17°42.5
										Acrux	172°56.1	-63°16.1
										Gacrux	171°47.7	-57°17.0
										Alioth	166°10.2	55°48.1
										Spica	158°18.5	-11°19.0
										Alkaid	152°49.4	49°10.2
										Hadar	148°30.7	-60°31.2
										Menkent	147°53.2	-36°31.1
										Arcturus	145°44.6	19°01.9
										Rigil Kent.	139°35.1	-60°57.7
										Kochab	137°20.3	74°02.3
										Zuben'ubi	136°51.9	-16°09.9
										Alphecca	126°00.6	26°37.2
										Antares	112°11.1	-26°29.8
										Atria	107°01.6	-69°04.9
										Sabik	101°58.3	-15°45.5
										Shaula	96°05.1	-37°07.4
										Rasalhague	95°54.9	12°32.6
										Eltanin	90°40.1	51°29.4
										Kaus Aust.	83°27.3	-34°22.1
										Vega	80°30.4	38°49.0
										Nunki	75°42.9	-26°15.4
										Altair	61°56.0	8°57.1
										Peacock	52°59.4	-56°38.2
										Deneb	49°22.7	45°23.4
										Enif	33°34.8	10°00.9
										Al Na'ir	27°27.9	-46°48.7
										Fomalhaut	15°10.2	-29°27.6
										Scheat	13°41.2	28°14.8
										Markab	13°25.9	15°22.1
										Aug 11 Sat	SHA	Mer.pass
										Venus	184°45.1	14:22
										Mars	150°27.5	16:38
										Jupiter	159°08.5	16:02
										Saturn	307°23.0	06:11
										Aug 12 Sun	SHA	Mer.pass
										Venus	183°40.9	14:23
										Mars	149°53.1	16:36
										Jupiter	159°00.3	15:59
										Saturn	307°20.2	06:07
										Aug 13 Mon	SHA	Mer.pass
										Venus	182°36.7	14:23
										Mars	149°18.5	16:35
										Jupiter	158°52.0	15:55
										Saturn	307°17.5	06:03
										Horizontal parallax		
										Venus:		0.1
										Mars:		0.1

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°41.4	N15°14.4	168°26.3	10.4'	N07°09.8	-14.1'	59.2'
1	193°41.5	13.7	182°55.7	10.4'	06°55.7	-14.2'	59.3'
2	208°41.6	12.9	197°25.1	10.4'	06°41.5	-14.2'	59.3'
3	223°41.7	12.2	211°54.6	10.5'	06°27.3	-14.3'	59.3'
4	238°41.8	11.4	226°24.0	10.5'	06°13.0	-14.3'	59.3'
5	253°41.9	10.7	240°53.5	10.5'	05°58.7	-14.3'	59.3'
6	268°42.0	N15°09.9	255°23.0	10.5'	N05°44.4	-14.4'	59.3'
7	283°42.1	09.2	269°52.5	10.5'	05°30.0	-14.4'	59.4'
8	298°42.2	08.4	284°22.0	10.5'	05°15.5	-14.5'	59.4'
9	313°42.3	07.7	298°51.5	10.5'	05°01.1	-14.5'	59.4'
10	328°42.4	07.0	313°21.0	10.5'	04°46.6	-14.5'	59.4'
11	343°42.5	06.2	327°50.5	10.5'	04°32.1	-14.6'	59.4'
12	358°42.6	N15°05.5	342°20.1	10.5'	N04°17.5	-14.6'	59.4'
13	13°42.7	04.7	356°49.6	10.5'	04°02.9	-14.6'	59.5'
14	28°42.8	04.0	11°19.2	10.5'	03°48.3	-14.6'	59.5'
15	43°42.9	03.2	25°48.7	10.6'	03°33.6	-14.7'	59.5'
16	58°43.0	02.5	40°18.3	10.6'	03°19.0	-14.7'	59.5'
17	73°43.1	01.7	54°47.8	10.6'	03°04.3	-14.7'	59.5'
18	88°43.2	N15°01.0	69°17.4	10.6'	N02°49.6	-14.7'	59.5'
19	103°43.3	15°00.2	83°46.9	10.6'	02°34.8	-14.8'	59.5'
20	118°43.4	14°59.5	98°16.5	10.6'	02°20.1	-14.8'	59.5'
21	133°43.5	58.7	112°46.1	10.6'	02°05.3	-14.8'	59.6'
22	148°43.6	58.0	127°15.6	10.6'	01°50.5	-14.8'	59.6'
23	163°43.7	57.2	141°45.2	10.6'	01°35.7	-14.8'	59.6'
SD = 15.8'			SD = 16.2'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°43.8	N14°56.5	156°14.7	10.6'	N01°20.9	-14.8'	59.6'
1	193°43.9	55.7	170°44.3	10.5'	01°06.1	-14.8'	59.6'
2	208°44.0	55.0	185°13.8	10.5'	00°51.2	-14.8'	59.6'
3	223°44.1	54.2	199°43.4	10.5'	00°36.4	-14.9'	59.6'
4	238°44.2	53.5	214°12.9	10.5'	00°21.5	-14.9'	59.6'
5	253°44.3	52.7	228°42.4	10.5'	N00°06.6	-14.9'	59.6'
6	268°44.5	N14°51.9	243°11.9	10.5'	S00°08.2	14.9'	59.6'
7	283°44.6	51.2	257°41.5	10.5'	00°23.1	14.9'	59.7'
8	298°44.7	50.4	272°11.0	10.5'	00°38.0	14.9'	59.7'
9	313°44.8	49.7	286°40.4	10.5'	00°52.8	14.9'	59.7'
10	328°44.9	48.9	301°09.9	10.5'	01°07.7	14.9'	59.7'
11	343°45.0	48.2	315°39.4	10.5'	01°22.6	14.9'	59.7'
12	358°45.1	N14°47.4	330°08.9	10.4'	S01°37.4	14.9'	59.7'
13	13°45.2	46.6	344°38.3	10.4'	01°52.3	14.8'	59.7'
14	28°45.3	45.9	359°07.7	10.4'	02°07.1	14.8'	59.7'
15	43°45.4	45.1	13°37.1	10.4'	02°21.9	14.8'	59.7'
16	58°45.5	44.4	28°06.5	10.4'	02°36.8	14.8'	59.7'
17	73°45.6	43.6	42°35.9	10.4'	02°51.6	14.8'	59.7'
18	88°45.8	N14°42.9	57°05.3	10.3'	S03°06.4	14.8'	59.7'
19	103°45.9	42.1	71°34.6	10.3'	03°21.1	14.8'	59.7'
20	118°46.0	41.3	86°03.9	10.3'	03°35.9	14.7'	59.7'
21	133°46.1	40.6	100°33.2	10.3'	03°50.6	14.7'	59.7'
22	148°46.2	39.8	115°02.5	10.3'	04°05.4	14.7'	59.7'
23	163°46.3	39.1	129°31.8	10.2'	04°20.1	14.7'	59.8'
SD = 15.8'			SD = 16.3'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°46.4	N14°38.3	144°01.0	10.2'	S04°34.7	14.6'	59.8'
1	193°46.5	37.5	158°30.2	10.2'	04°49.4	14.6'	59.8'
2	208°46.6	36.8	172°59.4	10.2'	05°04.0	14.6'	59.8'
3	223°46.7	36.0	187°28.6	10.1'	05°18.6	14.6'	59.8'
4	238°46.9	35.2	201°57.7	10.1'	05°33.2	14.5'	59.8'
5	253°47.0	34.5	216°26.8	10.1'	05°47.7	14.5'	59.8'
6	268°47.1	N14°33.7	230°55.9	10.1'	S06°02.2	14.5'	59.8'
7	283°47.2	32.9	245°24.9	10.0'	06°16.7	14.4'	59.8'
8	298°47.3	32.2	259°54.0	10.0'	06°31.1	14.4'	59.8'
9	313°47.4	31.4	274°22.9	10.0'	06°45.5	14.4'	59.8'
10	328°47.5	30.7	288°51.9	9.9'	06°59.8	14.3'	59.8'
11	343°47.6	29.9	303°20.8	9.9'	07°14.1	14.3'	59.8'
12	358°47.8	N14°29.1	317°49.7	9.9'	S07°28.4	14.2'	59.8'
13	13°47.9	28.4	332°18.6	9.8'	07°42.6	14.2'	59.8'
14	28°48.0	27.6	346°47.4	9.8'	07°56.8	14.1'	59.8'
15	43°48.1	26.8	1°16.2	9.8'	08°10.9	14.1'	59.8'
16	58°48.2	26.0	15°45.0	9.7'	08°25.0	14.0'	59.8'
17	73°48.3	25.3	30°13.7	9.7'	08°39.1	14.0'	59.8'
18	88°48.5	N14°24.5	44°42.4	9.6'	S08°53.0	13.9'	59.8'
19	103°48.6	23.7	59°11.0	9.6'	09°07.0	13.9'	59.8'
20	118°48.7	23.0	73°39.6	9.6'	09°20.8	13.8'	59.8'
21	133°48.8	22.2	88°08.2	9.5'	09°34.7	13.8'	59.8'
22	148°48.9	21.4	102°36.7	9.5'	09°48.4	13.7'	59.7'
23	163°49.0	20.7	117°05.2	9.4'	10°02.1	13.6'	59.7'
SD = 15.8'			SD = 16.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:04	21:59	—:—	—:—
N 70°	—:—	—:—	02:42	21:24	—:—	—:—
68°	—:—	01:19	03:08	20:59	22:42	—:—
66°	—:—	02:06	03:28	20:40	21:59	—:—
64°	—:—	02:35	03:43	20:24	21:31	—:—
62°	01:10	02:57	03:56	20:12	21:10	22:52
60°	01:53	03:15	04:08	20:01	20:53	22:13
N 58°	02:20	03:29	04:17	19:51	20:39	21:47
56°	02:40	03:41	04:25	19:43	20:27	21:27
54°	02:57	03:52	04:33	19:36	20:17	21:11
52°	03:11	04:01	04:40	19:29	20:08	20:58
50°	03:22	04:09	04:46	19:23	20:00	20:46
45°	03:46	04:26	04:58	19:11	19:43	20:23
N 40°	04:04	04:40	05:09	19:00	19:30	20:05
35°	04:18	04:51	05:18	18:52	19:19	19:51
30°	04:30	05:00	05:26	18:44	19:09	19:39
20°	04:49	05:16	05:39	18:31	18:54	19:21
N 10°	05:04	05:29	05:51	18:19	18:41	19:06
0°	05:15	05:40	06:02	18:08	18:30	18:55
S 10°	05:26	05:51	06:12	17:58	18:19	18:45
20°	05:35	06:01	06:24	17:47	18:09	18:36
30°	05:43	06:12	06:36	17:34	17:59	18:27
35°	05:47	06:17	06:44	17:27	17:53	18:23
40°	05:52	06:24	06:52	17:19	17:47	18:19
45°	05:56	06:31	07:02	17:09	17:40	18:14
S 50°	06:01	06:39	07:13	16:58	17:32	18:10
52°	06:03	06:43	07:18	16:52	17:28	18:08
54°	06:05	06:47	07:24	16:46	17:24	18:06
56°	06:07	06:51	07:31	16:40	17:20	18:04
58°	06:09	06:56	07:38	16:33	17:15	18:02
S 60°	06:12	07:01	07:47	16:24	17:10	17:59

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	05:47	08:01	10:17	19:54	19:30	19:02
N 70°	05:56	08:00	10:06	19:51	19:34	19:16
68°	06:03	07:59	09:57	19:48	19:38	19:28
66°	06:08	07:58	09:49	19:46	19:42	19:37
64°	06:13	07:58	09:43	19:45	19:45	19:45
62°	06:17	07:57	09:38	19:43	19:47	19:52
60°	06:21	07:57	09:33	19:42	19:49	19:58
N 58°	06:24	07:56	09:29	19:41	19:51	20:03
56°	06:27	07:56	09:26	19:39	19:53	20:08
54°	06:29	07:56	09:22	19:38	19:55	20:12
52°	06:32	07:55	09:19	19:38	19:56	20:16
50°	06:34	07:55	09:17	19:37	19:58	20:19
45°	06:38	07:54	09:11	19:35	20:00	20:27
N 40°	06:42	07:54	09:06	19:33	20:03	20:33
35°	06:45	07:54	09:02	19:32	20:05	20:39
30°	06:48	07:53	08:59	19:31	20:07	20:44
20°	06:53	07:53	08:53	19:29	20:10	20:52
N 10°	06:57	07:52	08:47	19:27	20:13	21:00
0°	07:01	07:52	08:43	19:25	20:16	21:07
S 10°	07:05	07:51	08:38	19:23	20:18	21:14
20°	07:09	07:51	08:32	19:21	20:21	21:22
30°	07:14	07:50	08:27	19:19	20:24	21:30
35°	07:17	07:50	08:23	19:18	20:26	21:35
40°	07:20	07:50	08:20	19:16	20:28	21:41
45°	07:24	07:49	08:15	19:14	20:31	21:48
S 50°	07:28	07:49	08:10	19:12	20:34	21:56
52°	07:30	07:49	08:08	19:11	20:35	22:00
54°	07:32	07:49	08:05	19:10	20:37	22:04
56°	07:35	07:49	08:03	19:09	20:38	22:08
58°	07:37	07:48	07:59	19:08	20:40	22:13
S 60°	07:40	07:48	07:56	19:06	20:42	22:19

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12h	Pass	Mer.Pass.	Lower	1-3
	00h	mm:ss	hh:mm	Upper	hh:mm	1-11%
11	05:15	05:10	12:05	13:13	00:48	
12	05:05	05:00	12:05	14:04	01:38	
13	04:54	04:49	12:05	14:55	02:29	

August 14, 15, 16 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars				
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	322°42.3	144°15.1	N01°31.3	111°26.0	S13°31.0	121°26.0	S07°43.4	269°57.3	N16°42.0	Alpheratz	357°30.6	29°15.3		
1	337°44.8	159°14.9		126°27.0	31.5	136°28.1	43.5	284°59.6	42.1	Ankaa	353°03.3	-42°08.4		
2	352°47.3	174°14.7		141°28.0	32.1	151°30.2	43.7	300°02.0	42.1	Schedar	349°26.2	56°41.9		
3	7°49.7	189°14.5	•• 27.4	156°29.1	•• 32.6	166°32.3	•• 43.8	315°04.3	•• 42.1	Diphda	348°43.4	-17°49.2		
4	22°52.2	204°14.3		171°30.1	33.1	181°34.4	43.9	330°06.7	42.1	Achernar	335°17.3	-57°04.8		
5	37°54.7	219°14.1		186°31.1	33.7	196°36.6	44.1	345°09.1	42.1	Hamal	327°46.8	23°36.1		
6	52°57.1	234°13.9	N01°23.6	201°32.1	S13°34.2	211°38.7	S07°44.2	0°11.4	N16°42.1	Polaris	311°56.5	89°22.9		
7	67°59.6	249°13.7		216°33.1	34.8	226°40.8	44.4	15°13.8	42.2	Acamar	315°09.0	-40°10.9		
8	83°02.1	264°13.5		231°34.1	35.3	241°42.9	44.5	30°16.1	42.2	Menkar	314°02.2	4°12.4		
9	98°04.5	279°13.3	•• 19.7	246°35.1	•• 35.9	256°45.0	•• 44.7	45°18.5	•• 42.2	Mirfak	308°22.8	49°57.7		
10	113°07.0	294°13.1		261°36.1	36.4	271°47.1	44.8	60°20.9	42.2	Aldebaran	290°35.4	16°34.0		
11	128°09.4	309°12.9		276°37.1	37.0	286°49.2	45.0	75°23.2	42.2	Rigel	281°00.4	-8°10.0		
12	143°11.9	324°12.7	N01°15.9	291°38.1	S13°37.5	301°51.3	S07°45.1	90°25.6	N16°42.2	Capella	280°16.6	46°01.4		
13	158°14.4	339°12.5		306°39.1	38.0	316°53.4	45.3	105°27.9	42.2	Bellatrix	278°19.1	6°22.5		
14	173°16.8	354°12.3		321°40.1	38.6	331°55.6	45.4	120°30.3	42.3	Elnath	277°57.3	28°37.8		
15	188°19.3	9°12.1	•• 12.0	336°41.1	•• 39.1	346°57.7	•• 45.6	135°32.7	•• 42.3	Alnilam	275°34.1	-1°11.0		
16	203°21.8	24°11.9		351°42.1	39.7	1°59.8	45.7	150°35.0	42.3	Betelgeuse	270°48.3	7°24.7		
17	218°24.2	39°11.7		6°43.1	40.2	17°01.9	45.9	165°37.4	42.3	Canopus	263°51.1	-52°42.5		
18	233°26.7	54°11.5	N01°08.1	21°44.1	S13°40.8	32°04.0	S07°46.0	180°39.8	N16°42.3	Sirius	258°23.2	-16°45.4		
19	248°29.2	69°11.3		36°45.1	41.3	47°06.1	46.2	195°42.1	42.3	Adhara	255°03.2	-29°00.7		
20	263°31.6	84°11.1		51°46.1	41.8	62°08.2	46.3	210°44.5	42.4	Procyon	244°47.2	5°08.9		
21	278°34.1	99°10.9	•• 04.3	66°47.1	•• 42.4	77°10.3	•• 46.5	225°46.8	•• 42.4	Pollux	243°13.1	27°57.1		
22	293°36.5	114°10.7		81°48.1	42.9	92°12.4	46.6	240°49.2	42.4	Avior	234°13.7	-59°36.3		
23	308°39.0	129°10.5		96°49.1	43.5	107°14.5	46.8	255°51.6	42.4	Suhail	222°43.9	-43°33.2		
Mer.pass. 02:29		ν -0.2' d -1.3' m -3.96		ν 1.0' d 0.5' m 0.77		ν 2.1' d 0.1' m -1.83		ν 2.4' d 0.0' m 0.18		Miaplacidus	221°38.2	-69°50.4		
												Alphard	217°44.4	-8°47.3
												Regulus	207°30.8	11°49.3
												Dubhe	193°37.3	61°35.5
												Denebola	182°21.5	14°24.4
												Gienah	175°40.0	-17°42.5
												Acrux	172°56.2	-63°16.1
												Gacrux	171°47.7	-57°17.0
												Alioth	166°10.2	55°48.1
												Spica	158°18.5	-11°19.0
												Alkaid	152°49.4	49°10.2
												Hadar	148°30.7	-60°31.2
												Menkent	147°53.3	-36°31.1
												Arcturus	145°44.6	19°01.9
												Rigil Kent.	139°35.1	-60°57.7
												Kochab	137°20.4	74°02.3
												Zuben'ubi	136°51.9	-16°09.9
												Alphecca	126°00.6	26°37.2
												Antares	112°11.1	-26°29.8
												Atria	107°01.7	-69°04.9
												Sabik	101°58.3	-15°45.5
												Shaula	96°05.1	-37°07.4
												Rasalhague	95°54.9	12°32.6
												Eltanin	90°40.1	51°29.4
												Kaus Aust.	83°27.3	-34°22.1
												Vega	80°30.4	38°49.0
												Nunki	75°42.9	-26°15.4
												Altair	61°56.0	8°57.1
												Peacock	52°59.4	-56°38.2
												Deneb	49°22.7	45°23.4
												Enif	33°34.8	10°00.9
												Al Na'ir	27°27.9	-46°48.8
												Fomalhaut	15°10.2	-29°27.6
												Scheat	13°41.2	28°14.8
												Markab	13°25.9	15°22.1
												Aug 14 Tue	SHA	Mer.pass
												Venus	181°32.7	14:23
												Mars	148°43.7	16:33
												Jupiter	158°43.7	15:52
												Saturn	307°14.9	05:59
												Aug 15 Wed	SHA	Mer.pass
												Venus	180°28.8	14:23
												Mars	148°08.7	16:32
												Jupiter	158°35.2	15:49
												Saturn	307°12.5	05:55
												Aug 16 Thu	SHA	Mer.pass
												Venus	179°25.0	14:24
												Mars	147°33.4	16:30
												Jupiter	158°26.6	15:45
												Saturn	307°10.1	05:52
												Horizontal parallax		
												Venus:	0.1	
												Mars:	0.1	
Mer.pass. 02:21		ν -0.2' d -1.3' m -3.97		ν 1.0' d 0.5' m 0.79		ν 2.1' d 0.2' m -1.82		ν 2.4' d 0.0' m 0.17						

DUT1 = UT1-UTC = +0.2120 sec $\Delta T = TT-UT1 = +68.9720$ sec

2029 August 14 to Aug. 16 UT

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	178°49.1	N14°19.9	131°33.6	9.4'	S10°15.8	13.6' 59.7'
1	193°49.3	19.1	146°02.0	9.4'	10°29.4	13.5' 59.7'
2	208°49.4	18.3	160°30.4	9.3'	10°42.9	13.5' 59.7'
3	223°49.5	· · 17.6	174°58.7	9.3'	10°56.3	13.4' 59.7'
4	238°49.6	16.8	189°27.0	9.2'	11°09.7	13.3' 59.7'
5	253°49.7	16.0	203°55.2	9.2'	11°23.0	13.2' 59.7'
6	268°49.9	N14°15.3	218°23.4	9.1'	S11°36.3	13.2' 59.7'
7	283°50.0	14.5	232°51.5	9.1'	11°49.4	13.1' 59.7'
8	298°50.1	13.7	247°19.6	9.0'	12°02.6	13.0' 59.7'
9	313°50.2	· · 12.9	261°47.6	9.0'	12°15.6	13.0' 59.7'
10	328°50.3	12.2	276°15.6	8.9'	12°28.5	12.9' 59.7'
11	343°50.5	11.4	290°43.6	8.9'	12°41.4	12.8' 59.7'
12	358°50.6	N14°10.6	305°11.5	8.8'	S12°54.2	12.7' 59.7'
13	13°50.7	09.8	319°39.3	8.8'	13°06.9	12.6' 59.7'
14	28°50.8	09.0	334°07.1	8.8'	13°19.6	12.6' 59.7'
15	43°50.9	· · 08.3	348°34.9	8.7'	13°32.1	12.5' 59.7'
16	58°51.1	07.5	3°02.6	8.7'	13°44.6	12.4' 59.6'
17	73°51.2	06.7	17°30.2	8.6'	13°57.0	12.3' 59.6'
18	88°51.3	N14°05.9	31°57.8	8.5'	S14°09.3	12.2' 59.6'
19	103°51.4	05.2	46°25.4	8.5'	14°21.5	12.1' 59.6'
20	118°51.5	04.4	60°52.9	8.4'	14°33.6	12.0' 59.6'
21	133°51.7	· · 03.6	75°20.3	8.4'	14°45.7	11.9' 59.6'
22	148°51.8	02.8	89°47.7	8.3'	14°57.6	11.8' 59.6'
23	163°51.9	02.0	104°15.1	8.3'	15°09.4	11.8' 59.6'
SD = 15.8' $d = -0.8'$ SD = 16.3'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	178°52.0	N14°01.3	118°42.3	8.2'	S15°21.2	11.7' 59.6'
1	193°52.1	14°00.5	133°09.6	8.2'	15°32.9	11.6' 59.6'
2	208°52.3	13°59.7	147°36.8	8.1'	15°44.4	11.5' 59.6'
3	223°52.4	· · 58.9	162°03.9	8.1'	15°55.9	11.4' 59.5'
4	238°52.5	58.1	176°31.0	8.0'	16°07.2	11.3' 59.5'
5	253°52.6	57.4	190°58.0	8.0'	16°18.5	11.2' 59.5'
6	268°52.8	N13°56.6	205°25.0	7.9'	S16°29.6	11.0' 59.5'
7	283°52.9	55.8	219°51.9	7.9'	16°40.7	10.9' 59.5'
8	298°53.0	55.0	234°18.8	7.8'	16°51.6	10.8' 59.5'
9	313°53.1	· · 54.2	248°45.6	7.8'	17°02.4	10.7' 59.5'
10	328°53.3	53.4	263°12.3	7.7'	17°13.2	10.6' 59.5'
11	343°53.4	52.6	277°39.0	7.7'	17°23.8	10.5' 59.5'
12	358°53.5	N13°51.9	292°05.7	7.6'	S17°34.3	10.4' 59.4'
13	13°53.6	51.1	306°32.3	7.5'	17°44.7	10.3' 59.4'
14	28°53.8	50.3	320°58.8	7.5'	17°55.0	10.2' 59.4'
15	43°53.9	· · 49.5	335°25.3	7.4'	18°05.1	10.0' 59.4'
16	58°54.0	48.7	349°51.8	7.4'	18°15.2	9.9' 59.4'
17	73°54.1	47.9	4°18.1	7.3'	18°25.1	9.8' 59.4'
18	88°54.3	N13°47.1	18°44.5	7.3'	S18°34.9	9.7' 59.4'
19	103°54.4	46.4	33°10.8	7.2'	18°44.6	9.6' 59.4'
20	118°54.5	45.6	47°37.0	7.2'	18°54.2	9.5' 59.3'
21	133°54.7	· · 44.8	62°03.2	7.1'	19°03.6	9.3' 59.3'
22	148°54.8	44.0	76°29.3	7.1'	19°13.0	9.2' 59.3'
23	163°54.9	43.2	90°55.4	7.0'	19°22.2	9.1' 59.3'
SD = 15.8' $d = -0.8'$ SD = 16.2'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	178°55.0	N13°42.4	105°21.4	7.0'	S19°31.2	9.0' 59.3'
1	193°55.2	41.6	119°47.4	6.9'	19°40.2	8.8' 59.3'
2	208°55.3	40.8	134°13.3	6.9'	19°49.0	8.7' 59.3'
3	223°55.4	· · 40.0	148°39.2	6.8'	19°57.7	8.6' 59.2'
4	238°55.6	39.2	163°05.0	6.8'	20°06.3	8.4' 59.2'
5	253°55.7	38.5	177°30.8	6.7'	20°14.7	8.3' 59.2'
6	268°55.8	N13°37.7	191°56.5	6.7'	S20°23.0	8.2' 59.2'
7	283°55.9	36.9	206°22.2	6.6'	20°31.2	8.0' 59.2'
8	298°56.1	36.1	220°47.9	6.6'	20°39.2	7.9' 59.2'
9	313°56.2	· · 35.3	235°13.5	6.6'	20°47.1	7.8' 59.2'
10	328°56.3	34.5	249°39.0	6.5'	20°54.9	7.6' 59.1'
11	343°56.5	33.7	264°04.5	6.5'	21°02.5	7.5' 59.1'
12	358°56.6	N13°32.9	278°30.0	6.4'	S21°10.0	7.4' 59.1'
13	13°56.7	32.1	292°55.4	6.4'	21°17.4	7.2' 59.1'
14	28°56.9	31.3	307°20.8	6.3'	21°24.6	7.1' 59.1'
15	43°57.0	· · 30.5	321°46.1	6.3'	21°31.7	6.9' 59.1'
16	58°57.1	29.7	336°11.4	6.3'	21°38.6	6.8' 59.1'
17	73°57.3	28.9	350°36.7	6.2'	21°45.4	6.7' 59.0'
18	88°57.4	N13°28.1	5°01.9	6.2'	S21°52.1	6.5' 59.0'
19	103°57.5	27.3	19°27.1	6.2'	21°58.6	6.4' 59.0'
20	118°57.7	26.5	33°52.3	6.1'	22°04.9	6.2' 59.0'
21	133°57.8	· · 25.7	48°17.4	6.1'	22°11.2	6.1' 59.0'
22	148°57.9	24.9	62°42.5	6.1'	22°17.2	5.9' 59.0'
23	163°58.1	24.1	77°07.5	6.0'	22°23.2	5.8' 58.9'
SD = 15.8' $d = -0.8'$ SD = 16.2'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:26	21:37	—:—	—:—
N 70°	—:—	—:—	02:58	21:07	23:25	—:—
68°	—:—	01:45	03:21	20:45	22:17	—:—
66°	—:—	02:22	03:38	20:28	21:42	—:—
64°	—:—	02:48	03:53	20:14	21:18	23:29
62°	01:34	03:08	04:05	20:02	20:59	22:29
60°	02:07	03:23	04:15	19:52	20:43	21:58
N 58°	02:31	03:37	04:24	19:44	20:30	21:35
56°	02:49	03:48	04:31	19:36	20:19	21:17
54°	03:04	03:58	04:38	19:29	20:10	21:02
52°	03:17	04:06	04:44	19:23	20:01	20:50
50°	03:28	04:14	04:50	19:18	19:54	20:39
45°	03:51	04:30	05:02	19:06	19:38	20:17
N 40°	04:08	04:43	05:12	18:56	19:25	20:00
35°	04:21	04:53	05:20	18:48	19:15	19:47
30°	04:32	05:02	05:28	18:41	19:06	19:36
20°	04:50	05:17	05:40	18:29	18:51	19:18
N 10°	05:04	05:29	05:51	18:18	18:39	19:05
0°	05:15	05:40	06:01	18:08	18:29	18:54
S 10°	05:24	05:49	06:11	17:58	18:20	18:44
20°	05:33	05:59	06:22	17:48	18:10	18:36
30°	05:41	06:09	06:33	17:36	18:00	18:29
35°	05:44	06:14	06:40	17:29	17:55	18:25
40°	05:48	06:20	06:48	17:21	17:49	18:21
45°	05:52	06:26	06:57	17:13	17:43	18:18
S 50°	05:56	06:34	07:08	17:02	17:36	18:14
52°	05:58	06:37	07:13	16:57	17:32	18:12
54°	05:59	06:41	07:18	16:52	17:29	18:10
56°	06:01	06:45	07:24	16:45	17:25	18:09
58°	06:03	06:49	07:31	16:39	17:21	18:07
S 60°	06:05	06:54	07:39	16:31	17:16	18:05

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	12:46	■	■	18:24	■	■
N 70°	12:19	14:58	■	18:53	18:10	■
68°	11:59	14:13	17:21	19:15	18:56	17:47
66°	11:44	13:43	15:50	19:32	19:26	19:18
64°	11:31	13:21	15:12	19:46	19:49	19:57
62°	11:20	13:04	14:46	19:58	20:07	20:24
60°	11:11	12:49	14:25	20:08	20:23	20:45
N 58°	11:03	12:37	14:08	20:17	20:36	21:02
56°	10:56	12:26	13:54	20:25	20:47	21:17
54°	10:50	12:17	13:42	20:32	20:57	21:30
52°	10:44	12:09	13:31	20:38	21:06	21:41
50°	10:39	12:01	13:21	20:44	21:14	21:51
45°	10:28	11:46	13:01	20:57	21:31	22:12
N 40°	10:20	11:33	12:45	21:07	21:45	22:29
35°	10:12	11:22	12:31	21:16	21:57	22:43
30°	10:05	11:12	12:19	21:24	22:07	22:56
20°	09:54	10:56	11:59	21:37	22:25	23:17
N 10°	09:44	10:42	11:42	21:49	22:41	23:36
0°	09:35	10:29	11:25	22:00	22:55	23:53
S 10°	09:26	10:16	11:09	22:11	23:10	· ·
20°	09:16	10:02	10:52	22:23	23:26	· ·
30°	09:05	09:46	10:32	22:37	23:44	· ·
35°	08:59	09:37	10:21	22:45	23:55	· ·
40°	08:51	09:27	10:08	22:54	· ·	00:07
45°	08:43	09:15	09:52	23:05	· ·	00:22
S 50°	08:33	09:00	09:33	23:18	· ·	00:40
52°	08:29	08:54	09:25	23:24	· ·	00:48
54°	08:24	08:46	09:15	23:31	· ·	00:57
56°	08:18	08:38	09:04	23:39	· ·	01:08
58°	08:12	08:28	08:51	23:47	· ·	01:20
S 60°	08:05	08:18	08:36	23:57	· ·	01:34

Day	Sun		Mer. Pass	Moon		Age 4-6 20-41%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
14	04:43	04:38	12:05	15:47	03:21	
15	04:32	04:26	12:04	16:42	04:14	
16	04:20	04:14	12:04	17:39	05:10	

August 17, 18, 19 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	325°39.8	144°01.0	S00°01.3	112°37.8	S14°10.0	123°57.6	S07°54.1	272°47.6	N16°43.1	Alpheratz	357°30.5	29°15.4											
1	340°42.2	159°00.9	02.6	127°38.8	10.6	138°59.7	54.3	287°49.9	43.1	Ankaa	353°03.3	-42°08.4											
2	355°44.7	174°00.7	03.9	142°39.7	11.1	154°01.8	54.4	302°52.3	43.1	Schedar	349°26.2	56°41.9											
3	10°47.1	189°00.5	•• 05.2	157°40.7	•• 11.6	169°03.9	•• 54.6	317°54.7	•• 43.1	Diphda	348°43.4	-17°49.2											
4	25°49.6	204°00.3	06.5	172°41.7	12.2	184°06.0	54.7	332°57.0	43.1	Achernar	335°17.3	-57°04.8											
5	40°52.1	219°00.1	07.8	187°42.7	12.7	199°08.1	54.9	347°59.4	43.2	Hamal	327°46.8	23°36.2											
6	55°54.5	233°59.9	S00°09.1	202°43.7	S14°13.2	214°10.2	S07°55.0	3°01.8	N16°43.2	Polaris	311°54.7	89°22.9											
7	70°57.0	248°59.7	10.4	217°44.6	13.8	229°12.3	55.2	18°04.2	43.2	Acamar	315°09.0	-40°10.9											
8	85°59.5	263°59.5	11.7	232°45.6	14.3	244°14.4	55.4	33°06.5	43.2	Menkar	314°02.2	4°12.4											
9	101°01.9	278°59.3	•• 12.9	247°46.6	•• 14.9	259°16.5	•• 55.5	48°08.9	•• 43.2	Mirfak	308°22.8	49°57.7											
10	116°04.4	293°59.2	14.2	262°47.6	15.4	274°18.6	55.7	63°11.3	43.2	Aldebaran	290°35.4	16°34.0											
11	131°06.9	308°59.0	15.5	277°48.6	15.9	289°20.7	55.8	78°13.6	43.2	Rigel	281°00.4	-8°10.0											
12	146°09.3	323°58.8	S00°16.8	292°49.5	S14°16.5	304°22.8	S07°56.0	93°16.0	N16°43.3	Capella	280°16.6	46°01.4											
13	161°11.8	338°58.6	18.1	307°50.5	17.0	319°24.9	56.1	108°18.4	43.3	Bellatrix	278°19.0	6°22.5											
14	176°14.3	353°58.4	19.4	322°51.5	17.6	334°27.0	56.3	123°20.8	43.3	Elnath	277°57.3	28°37.8											
15	191°16.7	8°58.2	•• 20.7	337°52.5	•• 18.1	349°29.1	•• 56.4	138°23.1	•• 43.3	Alnilam	275°34.1	-1°11.0											
16	206°19.2	23°58.0	22.0	352°53.5	18.6	4°31.2	56.6	153°25.5	43.3	Betelgeuse	270°48.2	7°24.7											
17	221°21.6	38°57.8	23.2	7°54.4	19.2	19°33.3	56.7	168°27.9	43.3	Canopus	263°51.1	-52°42.5											
18	236°24.1	53°57.6	S00°24.5	22°55.4	S14°19.7	34°35.4	S07°56.9	183°30.3	N16°43.3	Sirius	258°23.2	-16°45.4											
19	251°26.6	68°57.5	25.8	37°56.4	20.3	49°37.5	57.0	198°32.6	43.3	Adhara	255°03.2	-29°00.7											
20	266°29.0	83°57.3	27.1	52°57.4	20.8	64°39.6	57.2	213°35.0	43.4	Procyon	244°47.2	5°08.9											
21	281°31.5	98°57.1	•• 28.4	67°58.3	•• 21.3	79°41.7	•• 57.3	228°37.4	•• 43.4	Pollux	243°13.1	27°57.1											
22	296°34.0	113°56.9	29.7	82°59.3	21.9	94°43.8	57.5	243°39.8	43.4	Avior	234°13.7	-59°36.3											
23	311°36.4	128°56.7	31.0	98°00.3	22.4	109°45.9	57.6	258°42.1	43.4	Suhail	222°43.9	-43°33.2											
Mer.pass. 02:17				ν -0.2' d 1.3' m -3.97				ν 1.0' d 0.5' m 0.79				ν 2.1' d 0.2' m -1.81				ν 2.4' d 0.0' m 0.17							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	326°38.9	143°56.5	S00°32.3	113°01.3	S14°22.9	124°48.0	S07°57.8	273°44.5	N16°43.4										
1	341°41.4	158°56.3	33.6	128°02.2	23.5	139°50.1	57.9	288°46.9	43.4										
2	356°43.8	173°56.1	34.8	143°03.2	24.0	154°52.2	58.1	303°49.3	43.4										
3	11°46.3	188°56.0	•• 36.1	158°04.2	•• 24.6	169°54.3	•• 58.3	318°51.6	•• 43.4										
4	26°48.8	203°55.8	37.4	173°05.2	25.1	184°56.4	58.4	333°54.0	43.4										
5	41°51.2	218°55.6	38.7	188°06.1	25.6	199°58.5	58.6	348°56.4	43.5										
6	56°53.7	233°55.4	S00°40.0	203°07.1	S14°26.2	215°00.6	S07°58.7	3°58.8	N16°43.5										
7	71°56.1	248°55.2	41.3	218°08.1	26.7	230°02.7	58.9	19°01.1	43.5										
8	86°58.6	263°55.0	42.6	233°09.1	27.2	245°04.8	59.0	34°03.5	43.5										
9	102°01.1	278°54.8	•• 43.9	248°10.0	•• 27.8	260°06.9	•• 59.2	49°05.9	•• 43.5										
10	117°03.5	293°54.7	45.2	263°11.0	28.3	275°09.0	59.3	64°08.3	43.5										
11	132°06.0	308°54.5	46.4	278°12.0	28.9	290°11.0	59.5	79°10.7	43.5										
12	147°08.5	323°54.3	S00°47.7	293°12.9	S14°29.4	305°13.1	S07°59.6	94°13.0	N16°43.5										
13	162°10.9	338°54.1	49.0	308°13.9	29.9	320°15.2	59.8	109°15.4	43.6										
14	177°13.4	353°53.9	50.3	323°14.9	30.5	335°17.3	07°59.9	124°17.8	43.6										
15	192°15.9	8°53.7	•• 51.6	338°15.9	•• 31.0	350°19.4	08°00.1	139°20.2	•• 43.6										
16	207°18.3	23°53.5	52.9	353°16.8	31.5	5°21.5	00.2	154°22.5	43.6										
17	222°20.8	38°53.4	54.2	8°17.8	32.1	20°23.6	00.4	169°24.9	43.6										
18	237°23.3	53°53.2	S00°55.5	23°18.8	S14°32.6	35°25.7	S08°00.6	184°27.3	N16°43.6										
19	252°25.7	68°53.0	56.8	38°19.7	33.2	50°27.8	00.7	199°29.7	43.6										
20	267°28.2	83°52.8	58.0	53°20.7	33.7	65°29.9	00.9	214°32.1	43.6										
21	282°30.6	98°52.6	00°59.3	68°21.7	•• 34.2	80°32.0	•• 01.0	229°34.4	•• 43.6										
22	297°33.1	113°52.4	01°00.6	83°22.6	34.8	95°34.1	01.2	244°36.8	43.7										
23	312°35.6	128°52.3	01.9	98°23.6	35.3	110°36.2	01.3	259°39.2	43.7										
Mer.pass. 02:13				ν -0.2' d 1.3' m -3.97				ν 1.0' d 0.5' m 0.80				ν 2.1' d 0.2' m -1.81				ν 2.4' d 0.0' m 0.16			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	327°38.0	143°52.1	S01°03.2	113°24.6	S14°35.8	125°38.3	S08°01.5	274°41.6	N16°43.7										
1	342°40.5	158°51.9	04.5	128°25.5	36.4	140°40.4	01.6	289°44.0	43.7										
2	357°43.0	173°51.7	05.8	143°26.5	36.9	155°42.4	01.8	304°46.3	43.7										
3	12°45.4	188°51.5	•• 07.1	158°27.5	•• 37.4	170°44.5	•• 01.9	319°48.7	•• 43.7										
4	27°47.9	203°51.3	08.3	173°28.4	38.0	185°46.6	02.1	334°51.1	43.7										
5	42°50.4	218°51.1	09.6	188°29.4	38.5	200°48.7	02.3	349°53.5	43.7										
6	57°52.8	233°51.0	S01°10.9	203°30.4	S14°39.0	215°50.8	S08°02.4	4°55.9	N16°43.7										
7	72°55.3	248°50.8	12.2	218°31.3	39.6	230°52.9	02.6	19°58.2	43.8										
8	87°57.7	263°50.6	13.5	233°32.3	40.1	245°55.0	02.7	35°00.6	43.8										
9	103°00.2	278°50.4	•• 14.8	248°33.2	•• 40.7	260°57.1	•• 02.9	50°03.0	•• 43.8										
10	118°02.7	293°50.2	16.1	263°34.2	41.2	275°59.2	03.0	65°05.4	43.8										
11	133°05.1	308°50.0	17.4	278°35.2	41.7	291°01.3	03.2	80°07.8	43.8										
12	148°07.6	323°49.9	S01°18.7	293°36.1	S14°42.3	306°03.4	S08°03.3	95°10.2	N16°43.8										
13	163°10.1	338°49.7	19.9	308°37.1	42.8	321°05.4	03.5	110°12.5	43.8										
14	178°12.5	353°49.5	21.2	323°38.1	43.3	336°07.5	03.6	125°14.9	43.8										
15	193°15.0	8°49.3	•• 22.5	338°39.0	•• 43.9	351°09.6	•• 03.8	140°17.3	•• 43.8										
16	208°17.5	23°49.1	23.8	353°40.0	44.4	6°11.7	04.0	155°19.7	43.8										
17	223°19.9	38°48.9	25.1	8°40.9	44.9	21°13.8	04.1	170°22.1	43.9										
18	238°22.4	53°48.8	S01°26.4	23°41.9	S14°45.5	36°15.9	S08°04.3	185°24.4	N16°43.9										
19	253°24.9	68°48.6	27.7	38°42.9	46.0	51°18.0	04.4	200°26.8	43.9										
20	268°27.3	83°48.4	29.0	53°43.8	46.5	66°20.1	04.6	215°29.2	43.9										
21	283°29.8	98°48.2	•• 30.3	68°44.8	•• 47.1	81°22.2	•• 04.7	230°31.6	•• 43.9										
22	298°32.2	113°48.0	31.5	83°45.7	47.6	96°24.2	04.9	245°34.0	43.9										
23	313°34.7	128°47.9	32.8	98°46.7	48.1	111°26.3	05.0	260°36.4	43.9										
Mer.pass. 02:09				ν -0.2' d 1.3' m -3.98				ν 1.0' d 0.5' m 0.80				ν 2.1' d 0.2' m -1.81				ν 2.4' d 0.0' m 0.16			

Aug 17 Fri	SHA	Mer.pass
Venus	178°21.3	14:24
Mars	146°58.0	16:28
Jupiter	158°17.9	15:42
Saturn	307°07.8	05:48

Aug 18 Sat	SHA	Mer.pass
Venus	177°17.6	14:24
Mars	146°22.4	16:27
Jupiter	158°09.1	15:39
Saturn	307°05.6	05:44

Aug 19 Sun	SHA	Mer.pass
Venus	176°14.0	14:25
Mars	145°46.5	16:25
Jupiter	158°00.2	15:35
Saturn	307°03.5	05:40

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°58.2	N13°23.3	91°32.6	6.0'	S22°28.9	5.6'	58.9'
1	193°58.3	22.5	105°57.5	6.0'	22°34.6	5.5'	58.9'
2	208°58.5	21.7	120°22.5	5.9'	22°40.1	5.3'	58.9'
3	223°58.6	· · 20.9	134°47.4	5.9'	22°45.4	5.2'	58.9'
4	238°58.7	20.1	149°12.3	5.9'	22°50.6	5.0'	58.9'
5	253°58.9	19.3	163°37.2	5.9'	22°55.6	4.9'	58.8'
6	268°59.0	N13°18.5	178°02.1	5.8'	S23°00.5	4.7'	58.8'
7	283°59.1	17.7	192°26.9	5.8'	23°05.3	4.6'	58.8'
8	298°59.3	16.9	206°51.7	5.8'	23°09.9	4.4'	58.8'
9	313°59.4	· · 16.1	221°16.5	5.8'	23°14.3	4.3'	58.8'
10	328°59.5	15.3	235°41.3	5.8'	23°18.6	4.1'	58.7'
11	343°59.7	14.5	250°06.0	5.7'	23°22.7	4.0'	58.7'
12	358°59.8	N13°13.7	264°30.8	5.7'	S23°26.7	3.8'	58.7'
13	13°59.9	12.9	278°55.5	5.7'	23°30.5	3.7'	58.7'
14	29°00.1	12.1	293°20.2	5.7'	23°34.2	3.5'	58.7'
15	44°00.2	· · 11.3	307°44.9	5.7'	23°37.7	3.4'	58.7'
16	59°00.4	10.5	322°09.6	5.7'	23°41.0	3.2'	58.6'
17	74°00.5	09.7	336°34.2	5.7'	23°44.3	3.1'	58.6'
18	89°00.6	N13°08.9	350°58.9	5.7'	S23°47.3	2.9'	58.6'
19	104°00.8	08.1	5°23.6	5.7'	23°50.2	2.7'	58.6'
20	119°00.9	07.3	19°48.2	5.6'	23°52.9	2.6'	58.6'
21	134°01.1	· · 06.5	34°12.9	5.6'	23°55.5	2.4'	58.6'
22	149°01.2	05.7	48°37.5	5.6'	23°58.0	2.3'	58.5'
23	164°01.3	04.9	63°02.2	5.6'	24°00.2	2.1'	58.5'
	SD = 15.8'	d = -0.8'	SD = 16.1'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°01.5	N13°04.1	77°26.8	5.7'	S24°02.4	2.0'	58.5'
1	194°01.6	03.3	91°51.5	5.7'	24°04.3	1.8'	58.5'
2	209°01.7	02.5	106°16.1	5.7'	24°06.1	1.7'	58.5'
3	224°01.9	· · 01.6	120°40.8	5.7'	24°07.8	1.5'	58.4'
4	239°02.0	00.8	135°05.4	5.7'	24°09.3	1.3'	58.4'
5	254°02.2	13°00.0	149°30.1	5.7'	24°10.6	1.2'	58.4'
6	269°02.3	N12°59.2	163°54.8	5.7'	S24°11.8	1.0'	58.4'
7	284°02.5	58.4	178°19.5	5.7'	24°12.8	0.9'	58.4'
8	299°02.6	57.6	192°44.2	5.7'	24°13.7	0.7'	58.3'
9	314°02.7	· · 56.8	207°08.9	5.7'	24°14.4	0.6'	58.3'
10	329°02.9	56.0	221°33.7	5.8'	24°15.0	0.4'	58.3'
11	344°03.0	55.2	235°58.4	5.8'	24°15.4	0.3'	58.3'
12	359°03.2	N12°54.4	250°23.2	5.8'	S24°15.6	0.1'	58.3'
13	14°03.3	53.5	264°48.0	5.8'	24°15.7	-0.1'	58.3'
14	29°03.4	52.7	279°12.8	5.8'	24°15.7	-0.2'	58.2'
15	44°03.6	· · 51.9	293°37.7	5.9'	24°15.5	-0.4'	58.2'
16	59°03.7	51.1	308°02.5	5.9'	24°15.1	-0.5'	58.2'
17	74°03.9	50.3	322°27.4	5.9'	24°14.6	-0.7'	58.2'
18	89°04.0	N12°49.5	336°52.4	6.0'	S24°13.9	-0.8'	58.2'
19	104°04.2	48.7	351°17.3	6.0'	24°13.1	-1.0'	58.1'
20	119°04.3	47.8	5°42.3	6.0'	24°12.1	-1.1'	58.1'
21	134°04.4	· · 47.0	20°07.4	6.1'	24°11.0	-1.3'	58.1'
22	149°04.6	46.2	34°32.4	6.1'	24°09.7	-1.4'	58.1'
23	164°04.7	45.4	48°57.5	6.1'	24°08.3	-1.6'	58.1'
	SD = 15.8'	d = -0.8'	SD = 16.0'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°04.9	N12°44.6	63°22.6	6.2'	S24°06.7	-1.7'	58.0'
1	194°05.0	43.8	77°47.8	6.2'	24°05.0	-1.9'	58.0'
2	209°05.2	43.0	92°13.0	6.3'	24°03.1	-2.0'	58.0'
3	224°05.3	· · 42.1	106°38.3	6.3'	24°01.1	-2.2'	58.0'
4	239°05.5	41.3	121°03.6	6.3'	23°58.9	-2.3'	58.0'
5	254°05.6	40.5	135°28.9	6.4'	23°56.6	-2.5'	57.9'
6	269°05.8	N12°39.7	149°54.3	6.4'	S23°54.1	-2.6'	57.9'
7	284°05.9	38.9	164°19.7	6.5'	23°51.5	-2.8'	57.9'
8	299°06.0	38.0	178°45.2	6.5'	23°48.7	-2.9'	57.9'
9	314°06.2	· · 37.2	193°10.7	6.6'	23°45.8	-3.1'	57.9'
10	329°06.3	36.4	207°36.3	6.6'	23°42.8	-3.2'	57.8'
11	344°06.5	35.6	222°02.0	6.7'	23°39.6	-3.3'	57.8'
12	359°06.6	N12°34.8	236°27.6	6.7'	S23°36.2	-3.5'	57.8'
13	14°06.8	33.9	250°53.4	6.8'	23°32.7	-3.6'	57.8'
14	29°06.9	33.1	265°19.2	6.9'	23°29.1	-3.8'	57.8'
15	44°07.1	· · 32.3	279°45.0	6.9'	23°25.4	-3.9'	57.7'
16	59°07.2	31.5	294°11.0	7.0'	23°21.5	-4.0'	57.7'
17	74°07.4	30.7	308°36.9	7.0'	23°17.4	-4.2'	57.7'
18	89°07.5	N12°29.8	323°03.0	7.1'	S23°13.3	-4.3'	57.7'
19	104°07.7	29.0	337°29.1	7.2'	23°08.9	-4.4'	57.7'
20	119°07.8	28.2	351°55.2	7.2'	23°04.5	-4.6'	57.6'
21	134°08.0	· · 27.4	6°21.4	7.3'	22°59.9	-4.7'	57.6'
22	149°08.1	26.6	20°47.7	7.4'	22°55.2	-4.9'	57.6'
23	164°08.3	25.7	35°14.1	7.4'	22°50.3	-5.0'	57.6'
	SD = 15.8'	d = -0.8'	SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:46	21:17	—:—	—:—
N 70°	—:—	01:16	03:13	20:51	22:41	—:—
68°	—:—	02:07	03:33	20:32	21:55	—:—
66°	—:—	02:38	03:49	20:16	21:26	—:—
64°	01:07	03:00	04:02	20:04	21:05	22:51
62°	01:52	03:18	04:13	19:53	20:47	22:10
60°	02:20	03:32	04:22	19:44	20:33	21:44
N 58°	02:41	03:44	04:30	19:36	20:21	21:23
56°	02:58	03:55	04:37	19:29	20:11	21:07
54°	03:12	04:04	04:44	19:23	20:02	20:54
52°	03:24	04:12	04:49	19:17	19:55	20:42
50°	03:34	04:19	04:54	19:12	19:47	20:32
45°	03:55	04:34	05:05	19:01	19:33	20:11
N 40°	04:11	04:46	05:15	18:52	19:21	19:56
35°	04:24	04:56	05:22	18:45	19:11	19:43
30°	04:35	05:04	05:29	18:38	19:03	19:32
20°	04:51	05:18	05:41	18:26	18:49	19:16
N 10°	05:04	05:29	05:51	18:16	18:38	19:03
0°	05:15	05:39	06:00	18:07	18:28	18:53
S 10°	05:23	05:48	06:10	17:58	18:19	18:44
20°	05:31	05:57	06:19	17:48	18:11	18:37
30°	05:38	06:06	06:30	17:37	18:02	18:30
35°	05:41	06:11	06:37	17:31	17:57	18:27
40°	05:44	06:16	06:44	17:24	17:52	18:24
45°	05:48	06:22	06:52	17:16	17:46	18:21
S 50°	05:51	06:29	07:02	17:06	17:40	18:18
52°	05:52	06:32	07:07	17:02	17:37	18:16
54°	05:53	06:35	07:12	16:57	17:34	18:15
56°	05:55	06:38	07:17	16:51	17:30	18:14
58°	05:56	06:42	07:23	16:45	17:26	18:12
S 60°	05:58	06:46	07:30	16:38	17:22	18:11

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	18:21	■	19:30	18:50	■	21:44
64°	16:54	18:06	18:37	20:17	21:08	22:36
62°	16:17	17:25	18:05	20:54	21:49	23:08
60°	15:51	16:57	17:41	21:21	22:17	23:32
N 58°	15:30	16:35	17:22	21:42	22:38	23:51
56°	15:13	16:18	17:06	21:59	22:56	· · ·
54°	14:58	16:03	16:52	22:14	23:11	· · ·
52°	14:46	15:49	16:40	22:26	23:24	· · ·
50°	14:35	15:38	16:29	22:38	23:35	· · ·
45°	14:12	15:14	16:07	23:01	23:59	· · ·
N 40°	13:53	14:55	15:49	23:20	· · ·	00:18
35°	13:38	14:39	15:34	23:36	· · ·	00:34
30°	13:24	14:25	15:20	23:49	· · ·	00:48
20°	13:01	14:02	14:58	· · ·	00:13	01:11
N 10°	12:42	13:41	14:38	· · ·	00:33	01:31
0°	12:24	13:22	14:20	· · ·	00:52	01:50
S 10°	12:05	13:03	14:02	00:10	01:10	02:09
20°	11:46	12:43	13:42	00:29	01:31	02:29
30°	11:23	12:20	13:20	00:51	01:54	02:52
35°	11:10	12:06	13:06	01:03	02:07	03:06
40°	10:55	11:50	12:51	01:18	02:23	03:21
45°	10:37	11:31	12:33	01:35	02:42	03:40
S 50°	10:15	11:08	12:10	01:57	03:05	04:03
52°	10:05	10:56	11:59	02:07	03:17	04:14
54°	09:53	10:43	11:47	02:18	03:29	04:26
56°	09:39	10:29	11:33	02:32	03:44	04:41
58°	09:24	10:11	11:16	02:47	04:01	04:58
S 60°	09:05	09:50	10:56	03:06	04:23	05:18

Day	Sun		Mer. Pass	Moon		Age 7-9 53-74%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer. Pass. hh:mm	Lower Mer. Pass. hh:mm	
17	04:07	04:01	12:04	18:38	06:08	
18	03:54	03:47	12:04	19:36	07:07	
19	03:40	03:33	12:04	20:34	08:05	

August 20, 21, 22 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars			
Mon	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	SHA	Dec		
0	328°37.2	143°47.7	S01°34.1	113°47.7	S14°48.7	126°28.4	S08°05.2	275°38.8	N16°43.9	275°38.8	N16°43.9	275°38.8	Alpheratz	357°30.5	29°15.4	Alpheratz	357°30.5	29°15.4
1	343°39.6	158°47.5	35.4	128°48.6	49.2	141°30.5	05.4	290°41.1	43.9	290°41.1	43.9	290°41.1	Ankaa	353°03.3	-42°08.4	Ankaa	353°03.3	-42°08.4
2	358°42.1	173°47.3	36.7	143°49.6	49.7	156°32.6	05.5	305°43.5	43.9	305°43.5	43.9	305°43.5	Schedar	349°26.1	56°42.0	Schedar	349°26.1	56°42.0
3	13°44.6	188°47.1	· · 38.0	158°50.5	· · 50.3	171°34.7	· · 05.7	320°45.9	· · 44.0	320°45.9	· · 44.0	320°45.9	Diphda	348°43.4	-17°49.2	Diphda	348°43.4	-17°49.2
4	28°47.0	203°46.9	39.3	173°51.5	50.8	186°36.8	05.8	335°48.3	44.0	335°48.3	44.0	335°48.3	Achernar	335°17.3	-57°04.8	Achernar	335°17.3	-57°04.8
5	43°49.5	218°46.8	40.6	188°52.4	51.3	201°38.9	06.0	350°50.7	44.0	350°50.7	44.0	350°50.7	Hamal	327°46.8	23°36.2	Hamal	327°46.8	23°36.2
6	58°52.0	233°46.6	S01°41.9	203°53.4	S14°51.9	216°40.9	S08°06.1	5°53.1	N16°44.0	5°53.1	N16°44.0	5°53.1	Polaris	311°52.9	89°22.9	Polaris	311°52.9	89°22.9
7	73°54.4	248°46.4	43.1	218°54.3	52.4	231°43.0	06.3	20°55.4	44.0	20°55.4	44.0	20°55.4	Acamar	315°08.9	-40°10.9	Acamar	315°08.9	-40°10.9
8	88°56.9	263°46.2	44.4	233°55.3	52.9	246°45.1	06.4	35°57.8	44.0	35°57.8	44.0	35°57.8	Menkar	314°02.2	4°12.4	Menkar	314°02.2	4°12.4
9	103°59.4	278°46.0	· · 45.7	248°56.3	· · 53.5	261°47.2	· · 06.6	51°00.2	· · 44.0	51°00.2	· · 44.0	51°00.2	Mirfak	308°22.7	49°57.8	Mirfak	308°22.7	49°57.8
10	119°01.8	293°45.9	47.0	263°57.2	54.0	276°49.3	06.8	66°02.6	44.0	66°02.6	44.0	66°02.6	Aldebaran	290°35.4	16°34.1	Aldebaran	290°35.4	16°34.1
11	134°04.3	308°45.7	48.3	278°58.2	54.5	291°51.4	06.9	81°05.0	44.0	81°05.0	44.0	81°05.0	Rigel	281°00.4	-8°10.0	Rigel	281°00.4	-8°10.0
12	149°06.7	323°45.5	S01°49.6	293°59.1	S14°55.1	306°53.5	S08°07.1	96°07.4	N16°44.0	96°07.4	N16°44.0	96°07.4	Capella	280°16.5	46°01.4	Capella	280°16.5	46°01.4
13	164°09.2	338°45.3	50.9	309°00.1	55.6	321°55.6	07.2	111°09.8	44.1	111°09.8	44.1	111°09.8	Bellatrix	278°19.0	6°22.6	Bellatrix	278°19.0	6°22.6
14	179°11.7	353°45.1	52.2	324°01.0	56.1	336°57.6	07.4	126°12.1	44.1	126°12.1	44.1	126°12.1	Elnath	277°57.3	28°37.8	Elnath	277°57.3	28°37.8
15	194°14.1	8°44.9	· · 53.4	339°02.0	· · 56.7	351°59.7	· · 07.5	141°14.5	· · 44.1	141°14.5	· · 44.1	141°14.5	Alnilam	275°34.1	-1°11.0	Alnilam	275°34.1	-1°11.0
16	209°16.6	23°44.8	54.7	354°02.9	57.2	7°01.8	07.7	156°16.9	44.1	156°16.9	44.1	156°16.9	Betelgeuse	270°48.2	7°24.7	Betelgeuse	270°48.2	7°24.7
17	224°19.1	38°44.6	56.0	9°03.9	57.7	22°03.9	07.9	171°19.3	44.1	171°19.3	44.1	171°19.3	Canopus	263°51.1	-52°42.5	Canopus	263°51.1	-52°42.5
18	239°21.5	53°44.4	S01°57.3	24°04.8	S14°58.3	37°06.0	S08°08.0	186°21.7	N16°44.1	186°21.7	N16°44.1	186°21.7	Sirius	258°23.2	-16°45.4	Sirius	258°23.2	-16°45.4
19	254°24.0	68°44.2	58.6	39°05.8	58.8	52°08.1	08.2	201°24.1	44.1	201°24.1	44.1	201°24.1	Adhara	255°03.2	-29°00.7	Adhara	255°03.2	-29°00.7
20	269°26.5	83°44.0	01°59.9	54°06.7	59.3	67°10.2	08.3	216°26.5	44.1	216°26.5	44.1	216°26.5	Procyon	244°47.2	5°08.9	Procyon	244°47.2	5°08.9
21	284°28.9	98°43.9	02°01.2	69°07.7	14°59.9	82°12.2	· · 08.5	231°28.9	· · 44.1	231°28.9	· · 44.1	231°28.9	Pollux	243°13.1	27°57.1	Pollux	243°13.1	27°57.1
22	299°31.4	113°43.7	02.5	84°08.6	15°00.4	97°14.3	08.6	246°31.2	44.1	246°31.2	44.1	246°31.2	Avior	234°13.7	-59°36.2	Avior	234°13.7	-59°36.2
23	314°33.9	128°43.5	03.7	99°09.6	00.9	112°16.4	08.8	261°33.6	44.1	261°33.6	44.1	261°33.6	Suhail	222°43.9	-43°33.2	Suhail	222°43.9	-43°33.2
Mer.pass. 02:05			ν -0.2' d 1.3' m -3.98			ν 1.0' d 0.5' m 0.81			ν 2.1' d 0.2' m -1.80			ν 2.4' d 0.0' m 0.15						

Tue	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	329°36.3	143°43.3	S02°05.0	114°10.5	S15°01.5	127°18.5	S08°09.0	276°36.0	N16°44.2					
1	344°38.8	158°43.1	06.3	129°11.5	02.0	142°20.6	09.1	291°38.4	44.2					
2	359°41.2	173°43.0	07.6	144°12.4	02.5	157°22.7	09.3	306°40.8	44.2					
3	14°43.7	188°42.8	· · 08.9	159°13.4	· · 03.1	172°24.7	· · 09.4	321°43.2	· · 44.2					
4	29°46.2	203°42.6	10.2	174°14.3	03.6	187°26.8	09.6	336°45.6	44.2					
5	44°48.6	218°42.4	11.5	189°15.3	04.1	202°28.9	09.7	351°48.0	44.2					
6	59°51.1	233°42.2	S02°12.8	204°16.2	S15°04.6	217°31.0	S08°09.9	6°50.4	N16°44.2					
7	74°53.6	248°42.1	14.0	219°17.2	05.2	232°33.1	10.1	21°52.7	44.2					
8	89°56.0	263°41.9	15.3	234°18.1	05.7	247°35.2	10.2	36°55.1	44.2					
9	104°58.5	278°41.7	· · 16.6	249°19.1	· · 06.2	262°37.2	· · 10.4	51°57.5	· · 44.2					
10	120°01.0	293°41.5	17.9	264°20.0	06.8	277°39.3	10.5	66°59.9	44.2					
11	135°03.4	308°41.3	19.2	279°21.0	07.3	292°41.4	10.7	82°02.3	44.3					
12	150°05.9	323°41.2	S02°20.5	294°21.9	S15°07.8	307°43.5	S08°10.8	97°04.7	N16°44.3					
13	165°08.3	338°41.0	21.8	309°22.8	08.4	322°45.6	11.0	112°07.1	44.3					
14	180°10.8	353°40.8	23.1	324°23.8	08.9	337°47.7	11.2	127°09.5	44.3					
15	195°13.3	8°40.6	· · 24.3	339°24.7	· · 09.4	352°49.7	· · 11.3	142°11.9	· · 44.3					
16	210°15.7	23°40.4	25.6	354°25.7	10.0	7°51.8	11.5	157°14.3	44.3					
17	225°18.2	38°40.3	26.9	9°26.6	10.5	22°53.9	11.6	172°16.7	44.3					
18	240°20.7	53°40.1	S02°28.2	24°27.6	S15°11.0	37°56.0	S08°11.8	187°19.0	N16°44.3					
19	255°23.1	68°39.9	29.5	39°28.5	11.5	52°58.1	11.9	202°21.4	44.3					
20	270°25.6	83°39.7	30.8	54°29.4	12.1	68°00.1	12.1	217°23.8	44.3					
21	285°28.1	98°39.5	· · 32.1	69°30.4	· · 12.6	83°02.2	· · 12.3	232°26.2	· · 44.3					
22	300°30.5	113°39.4	33.3	84°31.3	13.1	98°04.3	12.4	247°28.6	44.3					
23	315°33.0	128°39.2	34.6	99°32.3	13.7	113°06.4	12.6	262°31.0	44.4					
Mer.pass. 02:01			ν -0.2' d 1.3' m -3.99			ν 0.9' d 0.5' m 0.81			ν 2.1' d 0.2' m -1.80			ν 2.4' d 0.0' m 0.15		

Wed	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	
0	330°35.5	143°39.0	S02°35.9	114°33.2	S15°14.2	128°08.5	S08°12.7	277°33.4	N16°44.4
1	345°37.9	158°38.8	37.2	129°34.2	14.7	143°10.5	12.9	292°35.8	44.4
2	0°40.4	173°38.7	38.5	144°35.1	15.2	158°12.6	13.0	307°38.2	44.4
3	15°42.8	188°38.5	· · 39.8	159°36.0	· · 15.8	173°14.7	· · 13.2	322°40.6	· · 44.4
4	30°45.3	203°38.3	41.1	174°37.0	16.3	188°16.8	13.4	337°43.0	44.4
5	45°47.8	218°38.1	42.3	189°37.9	16.8	203°18.9	13.5	352°45.4	44.4
6	60°50.2	233°37.9	S02°43.6	204°38.8	S15°17.4	218°20.9	S08°13.7	7°47.8	N16°44.4
7	75°52.7	248°37.8	44.9	219°39.8	17.9	233°23.0	13.8	22°50.2	44.4
8	90°55.2	263°37.6	46.2	234°40.7	18.4	248°25.1	14.0	37°52.6	44.4
9	105°57.6	278°37.4	· · 47.5	249°41.7	· · 19.0	263°27.2	· · 14.2	52°54.9	· · 44.4
10	121°00.1	293°37.2	48.8	264°42.6	19.5	278°29.3	14.3	67°57.3	44.4
11	136°02.6	308°37.0	50.1	279°43.5	20.0	293°31.3	14.5	82°59.7	44.4
12	151°05.0	323°36.9	S02°51.3	294°44.5	S15°20.5	308°33.4	S08°14.6	98°02.1	N16°44.5
13	166°07.5	338°36.7	52.6	309°45.4	21.1	323°35.5	14.8	113°04.5	44.5
14	181°10.0	353°36.5	53.9	324°46.3	21.6	338°37.6	14.9	128°06.9	44.5
15	196°12.4	8°36.3	· · 55.2	339°47.3	· · 22.1	353°39.7	· · 15.1	143°09.3	· · 44

DUT1 = UT1-UTC = +0.2121 sec ΔT = TT-UT1 = +68.9719 sec

2029 August 20 to Aug. 22 UT

h	Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°08.4	N12°24.9	49°40.5	7.5'	S22°45.4	-5.1'	57.6'
1	194°08.6	24.1	64°07.0	7.6'	22°40.2	-5.2'	57.5'
2	209°08.7	23.3	78°33.5	7.6'	22°35.0	-5.4'	57.5'
3	224°08.9	· · 22.4	93°00.2	7.7'	22°29.6	-5.5'	57.5'
4	239°09.0	21.6	107°26.8	7.8'	22°24.1	-5.6'	57.5'
5	254°09.2	20.8	121°53.6	7.8'	22°18.5	-5.8'	57.5'
6	269°09.3	N12°20.0	136°20.4	7.9'	S22°12.7	-5.9'	57.4'
7	284°09.5	19.1	150°47.3	8.0'	22°06.9	-6.0'	57.4'
8	299°09.6	18.3	165°14.3	8.0'	22°00.9	-6.1'	57.4'
9	314°09.8	· · 17.5	179°41.4	8.1'	21°54.7	-6.2'	57.4'
10	329°09.9	16.6	194°08.5	8.2'	21°48.5	-6.4'	57.4'
11	344°10.1	15.8	208°35.7	8.3'	21°42.1	-6.5'	57.3'
12	359°10.2	N12°15.0	223°03.0	8.3'	S21°35.6	-6.6'	57.3'
13	14°10.4	14.2	237°30.3	8.4'	21°29.0	-6.7'	57.3'
14	29°10.5	13.3	251°57.7	8.5'	21°22.3	-6.8'	57.3'
15	44°10.7	· · 12.5	266°25.2	8.6'	21°15.5	-7.0'	57.3'
16	59°10.8	11.7	280°52.8	8.7'	21°08.5	-7.1'	57.2'
17	74°11.0	10.8	295°20.5	8.7'	21°01.4	-7.2'	57.2'
18	89°11.1	N12°10.0	309°48.2	8.8'	S20°54.2	-7.3'	57.2'
19	104°11.3	09.2	324°16.0	8.9'	20°47.0	-7.4'	57.2'
20	119°11.5	08.4	338°43.9	9.0'	20°39.5	-7.5'	57.1'
21	134°11.6	· · 07.5	353°11.9	9.0'	20°32.0	-7.6'	57.1'
22	149°11.8	06.7	7°39.9	9.1'	20°24.4	-7.7'	57.1'
23	164°11.9	05.9	22°08.0	9.2'	20°16.7	-7.8'	57.1'
SD = 15.8'			SD = 15.7'				
$d = -0.8'$							

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179°12.1	N12°05.0	36°36.2	9.3'	S20°08.8	-7.9'	57.1'
1	194°12.2	04.2	51°04.5	9.4'	20°00.9	-8.0'	57.0'
2	209°12.4	03.4	65°32.9	9.4'	19°52.9	-8.1'	57.0'
3	224°12.5	· · 02.5	80°01.3	9.5'	19°44.7	-8.2'	57.0'
4	239°12.7	01.7	94°29.9	9.6'	19°36.5	-8.3'	57.0'
5	254°12.9	00.9	108°58.5	9.7'	19°28.1	-8.4'	57.0'
6	269°13.0	N12°00.0	123°27.2	9.8'	S19°19.7	-8.5'	56.9'
7	284°13.2	11°59.2	137°55.9	9.9'	19°11.1	-8.6'	56.9'
8	299°13.3	58.4	152°24.8	9.9'	19°02.5	-8.7'	56.9'
9	314°13.5	· · 57.5	166°53.7	10.0'	18°53.8	-8.8'	56.9'
10	329°13.6	56.7	181°22.7	10.1'	18°44.9	-8.9'	56.9'
11	344°13.8	55.9	195°51.8	10.2'	18°36.0	-9.0'	56.8'
12	359°13.9	N11°55.0	210°21.0	10.3'	S18°27.0	-9.1'	56.8'
13	14°14.1	54.2	224°50.2	10.3'	18°17.9	-9.2'	56.8'
14	29°14.3	53.3	239°19.6	10.4'	18°08.7	-9.3'	56.8'
15	44°14.4	· · 52.5	253°49.0	10.5'	17°59.4	-9.4'	56.8'
16	59°14.6	51.7	268°18.5	10.6'	17°50.1	-9.5'	56.7'
17	74°14.7	50.8	282°48.1	10.7'	17°40.6	-9.5'	56.7'
18	89°14.9	N11°50.0	297°17.7	10.7'	S17°31.1	-9.6'	56.7'
19	104°15.1	49.2	311°47.5	10.8'	17°21.5	-9.7'	56.7'
20	119°15.2	48.3	326°17.3	10.9'	17°11.8	-9.8'	56.7'
21	134°15.4	· · 47.5	340°47.2	11.0'	17°02.0	-9.9'	56.6'
22	149°15.5	46.6	355°17.1	11.1'	16°52.1	-9.9'	56.6'
23	164°15.7	45.8	9°47.2	11.1'	16°42.2	-10.0'	56.6'
SD = 15.8'			SD = 15.6'				
$d = -0.8'$							

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179°15.9	N11°45.0	24°17.3	11.2'	S16°32.2	-10.1'	56.6'
1	194°16.0	44.1	38°47.5	11.3'	16°22.1	-10.2'	56.5'
2	209°16.2	43.3	53°17.8	11.4'	16°11.9	-10.2'	56.5'
3	224°16.3	· · 42.4	67°48.2	11.4'	16°01.7	-10.3'	56.5'
4	239°16.5	41.6	82°18.6	11.5'	15°51.4	-10.4'	56.5'
5	254°16.7	40.8	96°49.1	11.6'	15°41.0	-10.5'	56.5'
6	269°16.8	N11°39.9	111°19.7	11.7'	S15°30.5	-10.5'	56.4'
7	284°17.0	39.1	125°50.4	11.7'	15°20.0	-10.6'	56.4'
8	299°17.1	38.2	140°21.1	11.8'	15°09.4	-10.7'	56.4'
9	314°17.3	· · 37.4	154°51.9	11.9'	14°58.8	-10.7'	56.4'
10	329°17.5	36.5	169°22.8	12.0'	14°48.1	-10.8'	56.4'
11	344°17.6	35.7	183°53.8	12.0'	14°37.3	-10.8'	56.3'
12	359°17.8	N11°34.9	198°24.8	12.1'	S14°26.4	-10.9'	56.3'
13	14°17.9	34.0	212°55.9	12.2'	14°15.5	-11.0'	56.3'
14	29°18.1	33.2	227°27.1	12.3'	14°04.6	-11.0'	56.3'
15	44°18.3	· · 32.3	241°58.4	12.3'	13°53.5	-11.1'	56.3'
16	59°18.4	31.5	256°29.7	12.4'	13°42.5	-11.1'	56.2'
17	74°18.6	30.6	271°01.1	12.5'	13°31.3	-11.2'	56.2'
18	89°18.8	N11°29.8	285°32.6	12.5'	S13°20.1	-11.2'	56.2'
19	104°18.9	28.9	300°04.1	12.6'	13°08.9	-11.3'	56.2'
20	119°19.1	28.1	314°35.7	12.7'	12°57.6	-11.4'	56.2'
21	134°19.3	· · 27.3	329°07.4	12.7'	12°46.2	-11.4'	56.1'
22	149°19.4	26.4	343°39.1	12.8'	12°34.8	-11.5'	56.1'
23	164°19.6	25.6	358°10.9	12.9'	12°23.4	-11.5'	56.1'
SD = 15.8'			SD = 15.4'				
$d = -0.8'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	03:04	20:58	23:24	—:—
N 70°	—:—	01:47	03:27	20:35	22:12	—:—
68°	—:—	02:25	03:45	20:18	21:36	—:—
66°	—:—	02:52	03:59	20:05	21:11	23:30
64°	01:34	03:11	04:11	19:53	20:52	22:26
62°	02:08	03:27	04:21	19:44	20:36	21:53
60°	02:33	03:41	04:29	19:35	20:23	21:30
N 58°	02:51	03:52	04:37	19:28	20:13	21:12
56°	03:07	04:01	04:43	19:22	20:03	20:57
54°	03:19	04:10	04:49	19:16	19:55	20:45
52°	03:30	04:17	04:54	19:11	19:48	20:34
50°	03:40	04:24	04:59	19:06	19:41	20:25
45°	04:00	04:38	05:09	18:56	19:27	20:05
N 40°	04:15	04:49	05:17	18:48	19:16	19:51
35°	04:27	04:58	05:25	18:41	19:07	19:39
30°	04:37	05:06	05:31	18:35	19:00	19:29
20°	04:52	05:19	05:42	18:24	18:47	19:13
N 10°	05:04	05:30	05:51	18:15	18:36	19:02
0°	05:14	05:39	06:00	18:06	18:28	18:52
S 10°	05:22	05:47	06:08	17:58	18:19	18:44
20°	05:29	05:55	06:17	17:49	18:12	18:37
30°	05:35	06:03	06:27	17:39	18:04	18:32
35°	05:38	06:07	06:33	17:33	17:59	18:29
40°	05:40	06:12	06:40	17:27	17:55	18:26
45°	05:43	06:17	06:47	17:19	17:50	18:24
S 50°	05:45	06:23	06:56	17:11	17:44	18:22
52°	05:46	06:26	07:00	17:06	17:41	18:21
54°	05:47	06:29	07:05	17:02	17:38	18:20
56°	05:48	06:32	07:10	16:57	17:35	18:19
58°	05:49	06:35	07:16	16:51	17:32	18:18
S 60°	05:50	06:39	07:22	16:45	17:29	18:17

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	■	21:04	20:04	■	23:54	· · ·
N 70°	■	20:10	19:41	■	■	00:46
68°	20:05	19:37	19:22	23:04	· · ·	01:18
66°	19:19	19:13	19:08	23:48	· · ·	01:41
64°	18:49	18:54	18:55	· · ·	00:18	01:59
62°	18:26	18:38	18:45	· · ·	00:40	02:14
60°	18:08	18:25	18:36	· · ·	00:58	02:26
N 58°	17:53	18:13	18:28	· · ·	01:13	02:37
56°	17:40	18:03	18:21	00:07	01:25	02:46
54°	17:28	17:55	18:15	00:20	01:36	02:54
52°	17:18	17:47	18:09	00:32	01:46	03:02
50°	17:09	17:40	18:04	00:42	01:55	03:08
45°	16:50	17:24	17:53	01:04	02:13	03:22
N 40°	16:34	17:12	17:44	01:22	02:28	03:34
35°	16:21	17:01	17:36	01:36	02:40	03:44
30°	16:09	16:52	17:29	01:49	02:51	03:52
20°	15:49	16:35	17:17	02:11	03:10	04:07
N 10°	15:32	16:21	17:06	02:29	03:26	04:19
0°	15:15	16:08	16:56	02:47	03:41	04:31
S 10°	14:59	15:54	16:46	03:04	03:56	04:43
20°	14:42	15:40	16:35	03:23	04:12	04:56
30°	14:21	15:23	16:23	03:44	04:30	05:10
35°	14:10	15:13	16:16	03:56	04:40	05:18
40°	13:56	15:02	16:08	04:11	04:52	05:27
45°	13:40	14:49	15:58	04:28	05:06	05:38
S 50°	13:20	14:33	15:47	04:48	05:23	05:51
52°	13:11	14:26	15:41	04:58	05:31	05:57
54°	13:00	14:17	15:35	05:09	05:40	06:04
56°	12:48	14:08	15:28	05:22	05:50	06:11
58°	12:34	13:57	15:21	05:36	06:02	06:19
S 60°	12:17	13:45	15:12	05:53	06:15	06:29

Day	Sun			Moon			Age 10-12 83-95%
	Eqn. of Time		Mer.	Mer. Pass.			
	00 ^h	12 ^h	Pass	Upper	Lower		
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm		
20	03:26	03:19	12:03	21:28	09:01		
21	03:12	03:04	12:03	22:20	09:54		
22	02:57	02:49	12:03	23:08	10:44		

August 23, 24, 25 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	331°34.6	143°34.7	S03°06.8	114°55.7	S15°26.9	128°58.3	S08°16.5	278°30.9	N16°44.5	Alpheratz	357°30.5	29°15.4					
1	346°37.1	158°34.5	08.1	129°56.6	27.4	144°00.4	16.7	293°33.3	44.5	Ankaa	353°03.2	-42°08.4					
2	1°39.5	173°34.4	09.3	144°57.5	27.9	159°02.5	16.9	308°35.7	44.5	Schedar	349°26.1	56°42.0					
3	16°42.0	188°34.2	•• 10.6	159°58.5	•• 28.4	174°04.6	•• 17.0	323°38.1	•• 44.6	Diphda	348°43.3	-17°49.2					
4	31°44.4	203°34.0	11.9	174°59.4	29.0	189°06.7	17.2	338°40.5	44.6	Achernar	335°17.2	-57°04.8					
5	46°46.9	218°33.8	13.2	190°00.3	29.5	204°08.7	17.3	353°42.9	44.6	Hamal	327°46.8	23°36.2					
6	61°49.4	233°33.7	S03°14.5	205°01.3	S15°30.0	219°10.8	S08°17.5	8°45.3	N16°44.6	Polaris	311°51.4	89°22.9					
7	76°51.8	248°33.5	15.8	220°02.2	30.5	234°12.9	17.6	23°47.7	44.6	Acamar	315°08.9	-40°10.8					
8	91°54.3	263°33.3	17.0	235°03.1	31.1	249°15.0	17.8	38°50.1	44.6	Menkar	314°02.1	4°12.4					
9	106°56.8	278°33.1	•• 18.3	250°04.1	•• 31.6	264°17.0	•• 18.0	53°52.5	•• 44.6	Mirfak	308°22.7	49°57.8					
10	121°59.2	293°32.9	19.6	265°05.0	32.1	279°19.1	18.1	68°54.9	44.6	Aldebaran	290°35.4	16°34.1					
11	137°01.7	308°32.8	20.9	280°05.9	32.7	294°21.2	18.3	83°57.3	44.6	Rigel	281°00.4	-8°10.0					
12	152°04.2	323°32.6	S03°22.2	295°06.8	S15°33.2	309°23.3	S08°18.4	98°59.7	N16°44.6	Capella	280°16.5	46°01.4					
13	167°06.6	338°32.4	23.5	310°07.8	33.7	324°25.3	18.6	114°02.1	44.6	Bellatrix	278°19.0	6°22.6					
14	182°09.1	353°32.2	24.7	325°08.7	34.2	339°27.4	18.8	129°04.5	44.6	Elnath	277°57.3	28°37.8					
15	197°11.6	8°32.1	•• 26.0	340°09.6	•• 34.8	354°29.5	•• 18.9	144°06.9	•• 44.6	Alnilam	275°34.1	-1°11.0					
16	212°14.0	23°31.9	27.3	355°10.6	35.3	9°31.6	19.1	159°09.3	44.6	Betelgeuse	270°48.2	7°24.7					
17	227°16.5	38°31.7	28.6	10°11.5	35.8	24°33.6	19.2	174°11.7	44.6	Canopus	263°51.0	-52°42.5					
18	242°18.9	53°31.5	S03°29.9	25°12.4	S15°36.3	39°35.7	S08°19.4	189°14.1	N16°44.7	Sirius	258°23.2	-16°45.4					
19	257°21.4	68°31.3	31.2	40°13.3	36.9	54°37.8	19.6	204°16.5	44.7	Adhara	255°03.2	-29°00.7					
20	272°23.9	83°31.2	32.4	55°14.3	37.4	69°39.9	19.7	219°18.9	44.7	Procyon	244°47.2	5°08.9					
21	287°26.3	98°31.0	•• 33.7	70°15.2	•• 37.9	84°41.9	•• 19.9	234°21.3	•• 44.7	Pollux	243°13.1	27°57.1					
22	302°28.8	113°30.8	35.0	85°16.1	38.4	99°44.0	20.0	249°23.7	44.7	Avior	234°13.7	-59°36.2					
23	317°31.3	128°30.6	36.3	100°17.0	39.0	114°46.1	20.2	264°26.1	44.7	Suhail	222°43.9	-43°33.2					
Mer.pass. 01:53			ν -0.2' d 1.3' m -3.99			ν 0.9' d 0.5' m 0.83			ν 2.1' d 0.2' m -1.79			ν 2.4' d 0.0' m 0.14					

Fri	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	332°33.7	143°30.5	S03°37.6	115°18.0	S15°39.5	129°48.1	S08°20.4	279°28.5	N16°44.7					
1	347°36.2	158°30.3	38.9	130°18.9	40.0	144°50.2	20.5	294°30.9	44.7					
2	2°38.7	173°30.1	40.1	145°19.8	40.5	159°52.3	20.7	309°33.3	44.7					
3	17°41.1	188°29.9	•• 41.4	160°20.7	•• 41.0	174°54.4	•• 20.8	324°35.7	•• 44.7					
4	32°43.6	203°29.8	42.7	175°21.6	41.6	189°56.4	21.0	339°38.1	44.7					
5	47°46.1	218°29.6	44.0	190°22.6	42.1	204°58.5	21.2	354°40.5	44.7					
6	62°48.5	233°29.4	S03°45.3	205°23.5	S15°42.6	220°00.6	S08°21.3	9°42.9	N16°44.7					
7	77°51.0	248°29.2	46.5	220°24.4	43.1	235°02.7	21.5	24°45.3	44.7					
8	92°53.4	263°29.0	47.8	235°25.3	43.7	250°04.7	21.7	39°47.7	44.7					
9	107°55.9	278°28.9	•• 49.1	250°26.2	•• 44.2	265°06.8	•• 21.8	54°50.1	•• 44.7					
10	122°58.4	293°28.7	50.4	265°27.2	44.7	280°08.9	22.0	69°52.5	44.7					
11	138°00.8	308°28.5	51.7	280°28.1	45.2	295°10.9	22.1	84°54.9	44.7					
12	153°03.3	323°28.3	S03°53.0	295°29.0	S15°45.8	310°13.0	S08°22.3	99°57.3	N16°44.8					
13	168°05.8	338°28.2	54.2	310°29.9	46.3	325°15.1	22.5	114°59.7	44.8					
14	183°08.2	353°28.0	55.5	325°30.8	46.8	340°17.2	22.6	130°02.1	44.8					
15	198°10.7	8°27.8	•• 56.8	340°31.8	•• 47.3	355°19.2	•• 22.8	145°04.5	•• 44.8					
16	213°13.2	23°27.6	58.1	355°32.7	47.8	10°21.3	22.9	160°06.9	44.8					
17	228°15.6	38°27.4	03°59.4	10°33.6	48.4	25°23.4	23.1	175°09.3	44.8					
18	243°18.1	53°27.3	S04°00.6	25°34.5	S15°48.9	40°25.4	S08°23.3	190°11.7	N16°44.8					
19	258°20.5	68°27.1	01.9	40°35.4	49.4	55°27.5	23.4	205°14.1	44.8					
20	273°23.0	83°26.9	03.2	55°36.4	49.9	70°29.6	23.6	220°16.5	44.8					
21	288°25.5	98°26.7	•• 04.5	70°37.3	•• 50.5	85°31.6	•• 23.7	235°18.9	•• 44.8					
22	303°27.9	113°26.6	05.8	85°38.2	51.0	100°33.7	23.9	250°21.4	44.8					
23	318°30.4	128°26.4	07.0	100°39.1	51.5	115°35.8	24.1	265°23.8	44.8					
Mer.pass. 01:49			ν -0.2' d 1.3' m -4.00			ν 0.9' d 0.5' m 0.83			ν 2.1' d 0.2' m -1.79			ν 2.4' d 0.0' m 0.13		

Sat	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	333°32.9	143°26.2	S04°08.3	115°40.0	S15°52.0	130°37.9	S08°24.2	280°26.2	N16°44.8					
1	348°35.3	158°26.0	09.6	130°40.9	52.5	145°39.9	24.4	295°28.6	44.8					
2	3°37.8	173°25.9	10.9	145°41.8	53.1	160°42.0	24.6	310°31.0	44.8					
3	18°40.3	188°25.7	•• 12.2	160°42.8	•• 53.6	175°44.1	•• 24.7	325°33.4	•• 44.8					
4	33°42.7	203°25.5	13.4	175°43.7	54.1	190°46.1	24.9	340°35.8	44.8					
5	48°45.2	218°25.3	14.7	190°44.6	54.6	205°48.2	25.0	355°38.2	44.8					
6	63°47.7	233°25.1	S04°16.0	205°45.5	S15°55.1	220°50.3	S08°25.2	10°40.6	N16°44.8					
7	78°50.1	248°25.0	17.3	220°46.4	55.7	235°52.3	25.4	25°43.0	44.8					
8	93°52.6	263°24.8	18.5	235°47.3	56.2	250°54.4	25.5	40°45.4	44.8					
9	108°55.0	278°24.6	•• 19.8	250°48.2	•• 56.7	265°56.5	•• 25.7	55°47.8	•• 44.9					
10	123°57.5	293°24.4	21.1	265°49.1	57.2	280°58.5	25.8	70°50.2	44.9					
11	139°00.0	308°24.3	22.4	280°50.1	57.7	296°00.6	26.0	85°52.6	44.9					
12	154°02.4	323°24.1	S04°23.7	295°51.0	S15°58.3	311°02.7	S08°26.2	100°55.1	N16°44.9					
13	169°04.9	338°23.9	24.9	310°51.9	58.8	326°04.7	26.3	115°57.5	44.9					
14	184°07.4	353°23.7	26.2	325°52.8	59.3	341°06.8	26.5	130°59.9	44.9					
15	199°09.8	8°23.5	•• 27.5	340°53.7	15°59.8	356°08.9	•• 26.7	146°02.3	•• 44.9					
16	214°12.3	23°23.4	28.8	355°54.6	16°00.3	11°10.9	26.8	161°04.7	44.9					
17	229°14.8	38°23.2	30.1	10°55.5	00.9	26°13.0	27.0	176°07.1	44.9					
18	244°17.2	53°23.0	S04°31.3	25°56.4	S16°01.4	41°15.1	S08°27.1	191°09.5	N16°44.9					
19	259°19.7	68°22.8	32.6	40°57.3	01.9	56°17.1	27.3	206°11.9	44.9					
20	274°22.1	83°22.7	33.9	55°58.2	02.4	71°19.2	27.5	221°14.3	44.9					
21	289°24.6	98°22.5	•• 35.2	70°59.1	•• 02.9	86°21.3	•• 27.6	236°16.7	•• 44.9					
22	304°27.1	113°22.3	36.4	86°00.1	03.5	101°23.3	27.8	251°19.1	44.9					
23	319°29.5	128°22.1	37.7	101°01.0	04.0	116°25.4	28.0	266°21.6	44.9					
Mer.pass. 01:46			ν -0.2' d 1.3' m -4.00			ν 0.9' d 0.5' m 0.84			ν 2.1' d 0.2' m -1.79			ν 2.4' d 0.0' m 0.13		

Aug 23 Thu	SHA	Mer.pass
Venus	172°00.1	14:26
Mars	143°21.1	16:19
Jupiter	157°23.8	15:22
Saturn	306°56.3	05:25

Aug 24 Fri	SHA	Mer.pass
Venus	170°56.7	14:26
Mars	142°44.2	16:18
Jupiter	157°14.4	15:19
Saturn	306°54.7	05:21

Aug 25 Sat	SHA	Mer.pass
Venus	169°53.3	14:26
Mars	142°07.1	16:16
Jupiter	157°05.0	15:15
Saturn	306°53.3	05:17

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°19.7	N11°24.7	12°42.8	12.9'	S12°11.9	-11.6'	56.1'
1	194°19.9	23.9	27°14.7	13.0'	12°00.3	-11.6'	56.0'
2	209°20.1	23.0	41°46.7	13.1'	11°48.7	-11.6'	56.0'
3	224°20.2	· · 22.2	56°18.8	13.1'	11°37.1	-11.7'	56.0'
4	239°20.4	21.3	70°50.9	13.2'	11°25.4	-11.7'	56.0'
5	254°20.6	20.5	85°23.1	13.3'	11°13.7	-11.8'	56.0'
6	269°20.7	N11°19.6	99°55.4	13.3'	S11°01.9	-11.8'	55.9'
7	284°20.9	18.8	114°27.7	13.4'	10°50.1	-11.9'	55.9'
8	299°21.1	17.9	129°00.1	13.4'	10°38.2	-11.9'	55.9'
9	314°21.2	· · 17.1	143°32.5	13.5'	10°26.3	-11.9'	55.9'
10	329°21.4	16.2	158°05.0	13.6'	10°14.4	-12.0'	55.9'
11	344°21.6	15.4	172°37.6	13.6'	10°02.4	-12.0'	55.8'
12	359°21.7	N11°14.5	187°10.2	13.7'	S09°50.4	-12.0'	55.8'
13	14°21.9	13.7	201°42.9	13.7'	09°38.4	-12.1'	55.8'
14	29°22.1	12.8	216°15.6	13.8'	09°26.3	-12.1'	55.8'
15	44°22.2	· · 12.0	230°48.4	13.8'	09°14.2	-12.1'	55.8'
16	59°22.4	11.1	245°21.2	13.9'	09°02.1	-12.2'	55.7'
17	74°22.6	10.3	259°54.1	13.9'	08°49.9	-12.2'	55.7'
18	89°22.7	N11°09.4	274°27.1	14.0'	S08°37.7	-12.2'	55.7'
19	104°22.9	08.6	289°00.1	14.1'	08°25.5	-12.3'	55.7'
20	119°23.1	07.7	303°33.1	14.1'	08°13.3	-12.3'	55.7'
21	134°23.2	· · 06.8	318°06.2	14.2'	08°01.0	-12.3'	55.6'
22	149°23.4	06.0	332°39.4	14.2'	07°48.7	-12.3'	55.6'
23	164°23.6	05.1	347°12.6	14.3'	07°36.3	-12.4'	55.6'
SD = 15.8'			SD = 15.3'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°23.7	N11°04.3	1°45.9	14.3'	S07°24.0	-12.4'	55.6'
1	194°23.9	03.4	16°19.2	14.3'	07°11.6	-12.4'	55.6'
2	209°24.1	02.6	30°52.5	14.4'	06°59.2	-12.4'	55.5'
3	224°24.3	· · 01.7	45°25.9	14.4'	06°46.8	-12.4'	55.5'
4	239°24.4	00.9	59°59.3	14.5'	06°34.4	-12.5'	55.5'
5	254°24.6	11°00.0	74°32.8	14.5'	06°21.9	-12.5'	55.5'
6	269°24.8	N10°59.1	89°06.4	14.6'	S06°09.5	-12.5'	55.5'
7	284°24.9	58.3	103°39.9	14.6'	05°57.0	-12.5'	55.4'
8	299°25.1	57.4	118°13.5	14.7'	05°44.5	-12.5'	55.4'
9	314°25.3	· · 56.6	132°47.2	14.7'	05°31.9	-12.5'	55.4'
10	329°25.4	55.7	147°20.9	14.7'	05°19.4	-12.5'	55.4'
11	344°25.6	54.9	161°54.6	14.8'	05°06.9	-12.6'	55.4'
12	359°25.8	N10°54.0	176°28.4	14.8'	S04°54.3	-12.6'	55.3'
13	14°26.0	53.1	191°02.2	14.8'	04°41.7	-12.6'	55.3'
14	29°26.1	52.3	205°36.0	14.9'	04°29.2	-12.6'	55.3'
15	44°26.3	· · 51.4	220°09.9	14.9'	04°16.6	-12.6'	55.3'
16	59°26.5	50.6	234°43.9	15.0'	04°04.0	-12.6'	55.3'
17	74°26.6	49.7	249°17.8	15.0'	03°51.4	-12.6'	55.3'
18	89°26.8	N10°48.8	263°51.8	15.0'	S03°38.8	-12.6'	55.2'
19	104°27.0	48.0	278°25.8	15.1'	03°26.1	-12.6'	55.2'
20	119°27.2	47.1	292°59.9	15.1'	03°13.5	-12.6'	55.2'
21	134°27.3	· · 46.3	307°34.0	15.1'	03°00.9	-12.6'	55.2'
22	149°27.5	45.4	322°08.1	15.1'	02°48.3	-12.6'	55.2'
23	164°27.7	44.5	336°42.2	15.2'	02°35.6	-12.6'	55.1'
SD = 15.8'			SD = 15.2'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°27.9	N10°43.7	351°16.4	15.2'	S02°23.0	-12.6'	55.1'
1	194°28.0	42.8	5°50.6	15.2'	02°10.4	-12.6'	55.1'
2	209°28.2	42.0	20°24.8	15.3'	01°57.7	-12.6'	55.1'
3	224°28.4	· · 41.1	34°59.1	15.3'	01°45.1	-12.6'	55.1'
4	239°28.5	40.2	49°33.4	15.3'	01°32.5	-12.6'	55.1'
5	254°28.7	39.4	64°07.7	15.3'	01°19.8	-12.6'	55.0'
6	269°28.9	N10°38.5	78°42.0	15.4'	S01°07.2	-12.6'	55.0'
7	284°29.1	37.6	93°16.3	15.4'	00°54.6	-12.6'	55.0'
8	299°29.2	36.8	107°50.7	15.4'	00°42.0	-12.6'	55.0'
9	314°29.4	· · 35.9	122°25.1	15.4'	00°29.4	-12.6'	55.0'
10	329°29.6	35.0	136°59.5	15.4'	00°16.8	-12.6'	54.9'
11	344°29.8	34.2	151°34.0	15.5'	S00°04.2	-12.6'	54.9'
12	359°29.9	N10°33.3	166°08.4	15.5'	N00°08.4	12.6'	54.9'
13	14°30.1	32.4	180°42.9	15.5'	00°21.0	12.6'	54.9'
14	29°30.3	31.6	195°17.4	15.5'	00°33.6	12.6'	54.9'
15	44°30.5	· · 30.7	209°51.9	15.5'	00°46.1	12.5'	54.9'
16	59°30.6	29.8	224°26.4	15.5'	00°58.7	12.5'	54.8'
17	74°30.8	29.0	239°00.9	15.5'	01°11.2	12.5'	54.8'
18	89°31.0	N10°28.1	253°35.5	15.6'	N01°23.7	12.5'	54.8'
19	104°31.2	27.2	268°10.0	15.6'	01°36.2	12.5'	54.8'
20	119°31.4	26.4	282°44.6	15.6'	01°48.7	12.5'	54.8'
21	134°31.5	· · 25.5	297°19.2	15.6'	02°01.2	12.5'	54.8'
22	149°31.7	24.6	311°53.8	15.6'	02°13.6	12.4'	54.8'
23	164°31.9	23.8	326°28.4	15.6'	02°26.1	12.4'	54.7'
SD = 15.8'			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	01:18	03:21	20:39	22:35	—:—
N 70°	—:—	02:11	03:41	20:20	21:48	—:—
68°	—:—	02:42	03:57	20:05	21:18	—:—
66°	01:08	03:05	04:09	19:53	20:56	22:47
64°	01:54	03:23	04:20	19:43	20:39	22:05
62°	02:23	03:37	04:29	19:34	20:25	21:38
60°	02:44	03:49	04:36	19:27	20:14	21:17
N 58°	03:01	03:59	04:43	19:20	20:04	21:01
56°	03:15	04:08	04:49	19:14	19:55	20:48
54°	03:27	04:16	04:54	19:09	19:47	20:36
52°	03:37	04:23	04:59	19:04	19:41	20:26
50°	03:46	04:29	05:03	19:00	19:35	20:17
45°	04:04	04:42	05:13	18:51	19:22	19:59
N 40°	04:18	04:52	05:20	18:44	19:12	19:46
35°	04:29	05:01	05:27	18:37	19:03	19:34
30°	04:39	05:08	05:33	18:31	18:56	19:25
20°	04:53	05:20	05:43	18:22	18:44	19:11
N 10°	05:05	05:30	05:51	18:13	18:35	19:00
0°	05:13	05:38	05:59	18:06	18:27	18:51
S 10°	05:21	05:45	06:07	17:58	18:19	18:44
20°	05:27	05:53	06:15	17:50	18:12	18:38
30°	05:32	06:00	06:24	17:41	18:05	18:33
35°	05:34	06:04	06:29	17:36	18:01	18:31
40°	05:36	06:08	06:35	17:30	17:57	18:29
45°	05:38	06:12	06:42	17:23	17:53	18:27
S 50°	05:40	06:17	06:50	17:15	17:48	18:26
52°	05:40	06:20	06:54	17:11	17:46	18:25
54°	05:41	06:22	06:58	17:07	17:43	18:25
56°	05:42	06:25	07:03	17:03	17:41	18:24
58°	05:42	06:28	07:08	16:58	17:38	18:24
S 60°	05:42	06:31	07:14	16:52	17:35	18:23

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	19:33	19:10	18:49	02:34	04:38	06:32
N 70°	19:21	19:05	18:51	02:55	04:48	06:34
68°	19:11	19:02	18:53	03:12	04:56	06:35
66°	19:03	18:58	18:54	03:25	05:03	06:36
64°	18:56	18:55	18:55	03:36	05:08	06:37
62°	18:50	18:53	18:56	03:45	05:13	06:38
60°	18:44	18:51	18:57	03:53	05:17	06:39
N 58°	18:40	18:49	18:58	04:00	05:21	06:39
56°	18:35	18:47	18:58	04:06	05:24	06:40
54°	18:31	18:46	18:59	04:12	05:27	06:40
52°	18:28	18:44	19:00	04:17	05:30	06:41
50°	18:25	18:43	19:00	04:21	05:32	06:41
45°	18:18	18:40	19:01	04:31	05:37	06:42
N 40°	18:12	18:38	19:02	04:39	05:41	06:43
35°	18:07	18:36	19:03	04:45	05:45	06:43
30°	18:03	18:34	19:04	04:51	05:48	06:44
20°	17:55	18:31	19:06	05:01	05:54	06:45
N 10°	17:48	18:28	19:07	05:10	05:59	06:45
0°	17:42	18:26	19:08	05:19	06:03	06:46
S 10°	17:36	18:23	19:09	05:27	06:08	06:47
20°	17:29	18:20	19:10	05:35	06:12	06:48
30°	17:21	18:17	19:12	05:45	06:18	06:48
35°	17:17	18:15	19:13	05:51	06:21	06:49
40°	17:11	18:13	19:14	05:57	06:24	06:49
45°	17:05	18:11	19:15	06:05	06:28	06:50
S 50°	16:58	18:08	19:16	06:14	06:33	06:51
52°	16:55	18:07	19:17	06:18	06:35	06:51
54°	16:51	18:05	19:18	06:22	06:37	06:51
56°	16:47	18:04	19:18	06:27	06:40	06:52
58°	16:43	18:02	19:19	06:32	06:43	06:52
S 60°	16:38	18:00	19:20	06:39	06:46	06:52

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h			Upper	Lower	13-15
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	99-99%
23	02:41	02:33	12:03	23:53	11:30	
24	02:25	02:17	12:02	—:—	12:15	
25	02:09	02:00	12:02	00:36	12:57	

August 26, 27, 28 UT (Sun., Mon., Tue.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	334°32.0	143°22.0	S04°39.0	116°01.9	S16°04.5	131°27.5	S08°28.1	281°24.0	N16°44.9	Alpheratz	357°30.5	29°15.4											
1	349°34.5	158°21.8	40.3	131°02.8	05.0	146°29.5	28.3	296°26.4	44.9	Ankaa	353°03.2	-42°08.4											
2	4°36.9	173°21.6	41.5	146°03.7	05.5	161°31.6	28.4	311°28.8	44.9	Schedar	349°26.1	56°42.0											
3	19°39.4	188°21.4	· · 42.8	161°04.6	· · 06.1	176°33.7	· · 28.6	326°31.2	· · 44.9	Diphda	348°43.3	-17°49.2											
4	34°41.9	203°21.2	44.1	176°05.5	06.6	191°35.7	28.8	341°33.6	44.9	Achernar	335°17.2	-57°04.8											
5	49°44.3	218°21.1	45.4	191°06.4	07.1	206°37.8	28.9	356°36.0	44.9	Hamal	327°46.7	23°36.2											
6	64°46.8	233°20.9	S04°46.7	206°07.3	S16°07.6	221°39.9	S08°29.1	11°38.4	N16°44.9	Polaris	311°49.9	89°22.9											
7	79°49.3	248°20.7	47.9	221°08.2	08.1	236°41.9	29.3	26°40.8	44.9	Acamar	315°08.9	-40°10.8											
8	94°51.7	263°20.5	49.2	236°09.1	08.6	251°44.0	29.4	41°43.3	44.9	Menkar	314°02.1	4°12.4											
9	109°54.2	278°20.4	· · 50.5	251°10.0	· · 09.2	266°46.1	· · 29.6	56°45.7	· · 44.9	Mirfak	308°22.7	49°57.8											
10	124°56.6	293°20.2	51.8	266°10.9	09.7	281°48.1	29.7	71°48.1	44.9	Aldebaran	290°35.3	16°34.1											
11	139°59.1	308°20.0	53.0	281°11.8	10.2	296°50.2	29.9	86°50.5	44.9	Rigel	281°00.4	-8°10.0											
12	155°01.6	323°19.8	S04°54.3	296°12.7	S16°10.7	311°52.2	S08°30.1	101°52.9	N16°44.9	Capella	280°16.5	46°01.4											
13	170°04.0	338°19.6	55.6	311°13.6	11.2	326°54.3	30.2	116°55.3	45.0	Bellatrix	278°19.0	6°22.6											
14	185°06.5	353°19.5	56.9	326°14.5	11.7	341°56.4	30.4	131°57.7	45.0	Elnath	277°57.2	28°37.8											
15	200°09.0	8°19.3	· · 58.1	341°15.4	· · 12.3	356°58.4	· · 30.6	147°00.1	· · 45.0	Alnilam	275°34.0	-1°11.0											
16	215°11.4	23°19.1	04°59.4	356°16.3	12.8	12°00.5	30.7	162°02.6	45.0	Betelgeuse	270°48.2	7°24.7											
17	230°13.9	38°18.9	05°00.7	11°17.2	13.3	27°02.6	30.9	177°05.0	45.0	Canopus	263°51.0	-52°42.5											
18	245°16.4	53°18.8	S05°02.0	26°18.1	S16°13.8	42°04.6	S08°31.0	192°07.4	N16°45.0	Sirius	258°23.1	-16°45.4											
19	260°18.8	68°18.6	03.2	41°19.0	14.3	57°06.7	31.2	207°09.8	45.0	Adhara	255°03.2	-29°00.7											
20	275°21.3	83°18.4	04.5	56°19.9	14.8	72°08.8	31.4	222°12.2	45.0	Procyon	244°47.2	5°08.9											
21	290°23.7	98°18.2	· · 05.8	71°20.8	· · 15.4	87°10.8	· · 31.5	237°14.6	· · 45.0	Pollux	243°13.1	27°57.1											
22	305°26.2	113°18.0	07.0	86°21.7	15.9	102°12.9	31.7	252°17.0	45.0	Avior	234°13.7	-59°36.2											
23	320°28.7	128°17.9	08.3	101°22.6	16.4	117°14.9	31.9	267°19.5	45.0	Suhail	222°43.9	-43°33.1											
Mer.pass. 01:42				ν -0.2' d 1.3' m -4.01				ν 0.9' d 0.5' m 0.84				ν 2.1' d 0.2' m -1.78				ν 2.4' d 0.0' m 0.13							

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	335°31.1	143°17.7	S05°09.6	116°23.5	S16°16.9	132°17.0	S08°32.0	282°21.9	N16°45.0	Denebola	182°21.5	14°24.4							
1	350°33.6	158°17.5	10.9	131°24.4	17.4	147°19.1	32.2	297°24.3	45.0	Gienah	175°40.0	-17°42.4							
2	5°36.1	173°17.3	12.1	146°25.3	17.9	162°21.1	32.4	312°26.7	45.0	Acrux	172°56.2	-63°16.0							
3	20°38.5	188°17.2	· · 13.4	161°26.2	· · 18.5	177°23.2	· · 32.5	327°29.1	· · 45.0	Gacrux	171°47.7	-57°16.9							
4	35°41.0	203°17.0	14.7	176°27.1	19.0	192°25.3	32.7	342°31.5	45.0	Alioth	166°10.3	55°48.1							
5	50°43.5	218°16.8	16.0	191°28.0	19.5	207°27.3	32.8	357°33.9	45.0	Spica	158°18.5	-11°19.0							
6	65°45.9	233°16.6	S05°17.2	206°28.9	S16°20.0	222°29.4	S08°33.0	12°36.4	N16°45.0	Alkaid	152°49.5	-49°10.1							
7	80°48.4	248°16.4	18.5	221°29.8	20.5	237°31.4	33.2	27°38.8	45.0	Hadar	148°30.8	-60°31.1							
8	95°50.9	263°16.3	19.8	236°30.7	21.0	252°33.5	33.3	42°41.2	45.0	Menkent	147°53.3	-36°31.0							
9	110°53.3	278°16.1	· · 21.0	251°31.6	· · 21.5	267°35.6	· · 33.5	57°43.6	· · 45.0	Arcturus	145°44.7	19°01.9							
10	125°55.8	293°15.9	22.3	266°32.5	22.1	282°37.6	33.7	72°46.0	45.0	Rigel Kent.	139°35.2	-60°57.7							
11	140°58.2	308°15.7	23.6	281°33.4	22.6	297°39.7	33.8	87°48.4	45.0	Kochab	137°20.6	74°02.3							
12	156°00.7	323°15.6	S05°24.9	296°34.3	S16°23.1	312°41.7	S08°34.0	102°50.9	N16°45.0	Zuben'ubi	136°51.9	-16°09.9							
13	171°03.2	338°15.4	26.1	311°35.2	23.6	327°43.8	34.2	117°53.3	45.0	Alphecca	126°00.7	26°37.2							
14	186°05.6	353°15.2	27.4	326°36.1	24.1	342°45.9	34.3	132°55.7	45.0	Antares	112°11.2	-26°29.8							
15	201°08.1	8°15.0	· · 28.7	341°36.9	· · 24.6	357°47.9	· · 34.5	147°58.1	· · 45.0	Atria	107°01.8	-69°04.9							
16	216°10.6	23°14.8	29.9	356°37.8	25.1	12°50.0	34.6	163°00.5	45.0	Sabik	101°58.4	-15°45.5							
17	231°13.0	38°14.7	31.2	11°38.7	25.7	27°52.0	34.8	178°02.9	45.0	Shaula	96°05.1	-37°07.4							
18	246°15.5	53°14.5	S05°32.5	26°39.6	S16°26.2	42°54.1	S08°35.0	193°05.4	N16°45.0	Rasalhague	95°55.0	12°32.6							
19	261°18.0	68°14.3	33.8	41°40.5	26.7	57°56.2	35.1	208°07.8	45.0	Eltanin	90°40.2	51°29.5							
20	276°20.4	83°14.1	35.0	56°41.4	27.2	72°58.2	35.3	223°10.2	45.0	Kaus Aust.	83°27.3	-34°22.1							
21	291°22.9	98°13.9	· · 36.3	71°42.3	· · 27.7	88°00.3	· · 35.5	238°12.6	· · 45.0	Vega	80°30.4	38°49.0							
22	306°25.4	113°13.8	37.6	86°43.2	28.2	103°02.3	35.6	253°15.0	45.0	Nunki	75°42.9	-26°15.4							
23	321°27.8	128°13.6	38.8	101°44.1	28.7	118°04.4	35.8	268°17.5	45.0	Altair	61°56.0	8°57.1							
Mer.pass. 01:38				ν -0.2' d 1.3' m -4.01				ν 0.9' d 0.5' m 0.85				ν 2.1' d 0.2' m -1.78				ν 2.4' d 0.0' m 0.12			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	336°30.3	143°13.4	S05°40.1	116°45.0	S16°29.2	133°06.5	S08°36.0	283°19.9	N16°45.0	Aug 26 Sun	SHA	Mer.pass							
1	351°32.7	158°13.2	41.4	131°45.9	29.8	148°08.5	36.1	298°22.3	45.0	Venus	168°49.9	14:27							
2	6°35.2	173°13.1	42.6	146°46.7	30.3	163°10.6	36.3	313°24.7	45.0	Mars	141°29.9	16:15							
3	21°37.7	188°12.9	· · 43.9	161°47.6	· · 30.8	178°12.6	· · 36.5	328°27.1	· · 45.0	Jupiter	156°55.5	15:12							
4	36°40.1	203°12.7	45.2	176°48.5	31.3	193°14.7	36.6	343°29.6	45.0	Saturn	306°52.0	05:14							
5	51°42.6	218°12.5	46.4	191°49.4	31.8	208°16.7	36.8	358°32.0	45.0										
6	66°45.1	233°12.3	S05°47.7	206°50.3	S16°32.3	223°18.8	S08°36.9	13°34.4	N16°45.0										
7	81°47.5	248°12.2	49.0	221°51.2	32.8	238°20.9	37.1	28°36.8	45.0										
8	96°50.0	263°12.0	50.3	236°52.1	33.3	253°22.9	37.3	43°39.2	45.0										
9	111°52.5	278°11.8	· · 51.5	251°53.0	· · 33.9	268°25.0	· · 37.4	58°41.7	· · 45.0										
10	126°54.9	293°11.6	52.8	266°53.8	34.4	283°27.0	37.6	73°44.1	45.0										
11	141°57.4	308°11.4	54.1	281°54.7	34.9	298°29.1	37.8	88°46.5	45.0										
12	156°59.8	323°11.3	S05°55.3	296°55.6	S16°35.4	313°31.1	S08°37.9	103°48.9	N16°45.0										
13	172°02.3	338°11.1	56.6	311°56.5	35.9	328°33.2	38.1	118°51.3	45.0										
14	187°04.8	353°10.9	57.9	326°57.4	36.4	343°35.3	38.3	133°53.8	45.0										
15	202°07.2	8°10.7	05°59.1	341°58.3	· · 36.9	358°37.3	· · 38.4	148°56.2	· · 45.0										
16	217°09.7	23°10.5	06°00.4	356°59.1	37.4	13°39.4	38.6	163°58.6	45.0										
17	232°12.2	38°10.4	01.7	12°00.0	37.9	28°41.4	38.8	179°01.0	45.1										
18	247°14.6	53°10.2	S06°02.9	27°00.9	S16°38.5	43°43.5	S08°38.9	194°03.5	N16°45.1										
19	262°17.1	68°10.0	04.2	42°01.8	39.0	58°45.5	39.1	209°05.9	45.1										
20	277°19.6	83°09.8	05.5	57°02.7	39.5	73°47.6	39.3	224°08.3	45.1										
21	292°22.0	98°09.6	· · 06.7	72°03.6	· · 40.0	88°49.6	· · 39.4	239°10.7	· · 45.1										
22	307°24.5	113°09.5	08.0	87°04.4	40.5	103°51.7	39.6	254°13.2	45.1										
23	322°27.0	128°09.3	09.3	102°05.3	41.0	118°53.8	39.8	269°15.6	45.1										
Mer.pass. 01:34				ν -0.2' d 1.3' m -4.02				ν 0.9' d 0.5' m 0.85				ν 2.1' d 0.2' m -1.78				ν 2.4' d 0.0' m 0.12			

Aug 26 Sun			SHA	Mer.pass
Venus	168°49.9		14:27	
Mars	141°29.9		16:15	
Jupiter	156°55.5		15:12	
Saturn	306°52.0		05:14	

Aug 27 Mon			SHA	Mer.pass
Venus	167°46.6		14:27	
Mars	140°52.4		16:13	
Jupiter	156°45.9		15:09	
Saturn	306°50.7		05:10	

Aug 28 Tue			SHA	Mer.pass
Venus	166°43.1		14:27	
Mars	140°14.7		16:12	
Jupiter	156°36.2		15:05	
Saturn	306°49.6		05:06	

Horizontal parallax			
Venus:		0.1	
M			

DUT1 = UT1-UTC = +0.2122 sec ΔT = TT-UT1 = +68.9718 sec

2029 August 26 to Aug. 28 UT

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°32.1	N10°22.9	341°03.0	15.6'	N02°38.5	12.4'	54.7'
1	194°32.2	22.0	355°37.6	15.6'	02°50.9	12.4'	54.7'
2	209°32.4	21.2	10°12.3	15.6'	03°03.3	12.4'	54.7'
3	224°32.6	· · 20.3	24°46.9	15.6'	03°15.7	12.3'	54.7'
4	239°32.8	19.4	39°21.5	15.6'	03°28.0	12.3'	54.7'
5	254°32.9	18.6	53°56.2	15.6'	03°40.3	12.3'	54.6'
6	269°33.1	N10°17.7	68°30.8	15.7'	N03°52.6	12.3'	54.6'
7	284°33.3	16.8	83°05.5	15.7'	04°04.9	12.3'	54.6'
8	299°33.5	15.9	97°40.1	15.7'	04°17.2	12.2'	54.6'
9	314°33.7	· · 15.1	112°14.8	15.7'	04°29.4	12.2'	54.6'
10	329°33.8	14.2	126°49.4	15.7'	04°41.6	12.2'	54.6'
11	344°34.0	13.3	141°24.1	15.7'	04°53.8	12.2'	54.6'
12	359°34.2	N10°12.5	155°58.7	15.7'	N05°05.9	12.1'	54.5'
13	14°34.4	11.6	170°33.4	15.6'	05°18.1	12.1'	54.5'
14	29°34.6	10.7	185°08.0	15.6'	05°30.2	12.1'	54.5'
15	44°34.7	· · 09.8	199°42.7	15.6'	05°42.2	12.0'	54.5'
16	59°34.9	09.0	214°17.3	15.6'	05°54.3	12.0'	54.5'
17	74°35.1	08.1	228°52.0	15.6'	06°06.3	12.0'	54.5'
18	89°35.3	N10°07.2	243°26.6	15.6'	N06°18.3	11.9'	54.5'
19	104°35.5	06.3	258°01.2	15.6'	06°30.2	11.9'	54.5'
20	119°35.6	05.5	272°35.8	15.6'	06°42.1	11.9'	54.4'
21	134°35.8	· · 04.6	287°10.4	15.6'	06°54.0	11.9'	54.4'
22	149°36.0	03.7	301°45.1	15.6'	07°05.9	11.8'	54.4'
23	164°36.2	02.8	316°19.6	15.6'	07°17.7	11.8'	54.4'
SD = 15.8'			SD = 14.9'				
$d = -0.9'$							

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°36.4	N10°02.0	330°54.2	15.6'	N07°29.5	11.7'	54.4'
1	194°36.5	01.1	345°28.8	15.6'	07°41.2	11.7'	54.4'
2	209°36.7	10°00.2	0°03.3	15.5'	07°52.9	11.7'	54.4'
3	224°36.9	09°59.3	14°37.9	15.5'	08°04.6	11.6'	54.4'
4	239°37.1	58.5	29°12.4	15.5'	08°16.2	11.6'	54.4'
5	254°37.3	57.6	43°46.9	15.5'	08°27.8	11.6'	54.3'
6	269°37.5	N09°56.7	58°21.4	15.5'	N08°39.4	11.5'	54.3'
7	284°37.6	55.8	72°55.9	15.5'	08°50.9	11.5'	54.3'
8	299°37.8	54.9	87°30.4	15.5'	09°02.4	11.4'	54.3'
9	314°38.0	· · 54.1	102°04.9	15.4'	09°13.8	11.4'	54.3'
10	329°38.2	53.2	116°39.3	15.4'	09°25.2	11.4'	54.3'
11	344°38.4	52.3	131°13.7	15.4'	09°36.6	11.3'	54.3'
12	359°38.6	N09°51.4	145°48.1	15.4'	N09°47.9	11.3'	54.3'
13	14°38.7	50.6	160°22.5	15.4'	09°59.2	11.2'	54.3'
14	29°38.9	49.7	174°56.9	15.3'	10°10.4	11.2'	54.3'
15	44°39.1	· · 48.8	189°31.2	15.3'	10°21.6	11.1'	54.2'
16	59°39.3	47.9	204°05.5	15.3'	10°32.7	11.1'	54.2'
17	74°39.5	47.0	218°39.8	15.3'	10°43.8	11.0'	54.2'
18	89°39.7	N09°46.1	233°14.1	15.2'	N10°54.8	11.0'	54.2'
19	104°39.8	45.3	247°48.3	15.2'	11°05.8	10.9'	54.2'
20	119°40.0	44.4	262°22.5	15.2'	11°16.8	10.9'	54.2'
21	134°40.2	· · 43.5	276°56.7	15.2'	11°27.6	10.8'	54.2'
22	149°40.4	42.6	291°30.9	15.1'	11°38.5	10.8'	54.2'
23	164°40.6	41.7	306°05.0	15.1'	11°49.3	10.7'	54.2'
SD = 15.8'			SD = 14.8'				
$d = -0.9'$							

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179°40.8	N09°40.9	320°39.2	15.1'	N12°00.0	10.7'	54.2'
1	194°40.9	40.0	335°13.2	15.1'	12°10.7	10.6'	54.2'
2	209°41.1	39.1	349°47.3	15.0'	12°21.4	10.6'	54.2'
3	224°41.3	· · 38.2	4°21.3	15.0'	12°32.0	10.5'	54.2'
4	239°41.5	37.3	18°55.3	15.0'	12°42.5	10.5'	54.1'
5	254°41.7	36.4	33°29.3	14.9'	12°53.0	10.4'	54.1'
6	269°41.9	N09°35.6	48°03.2	14.9'	N13°03.4	10.4'	54.1'
7	284°42.1	34.7	62°37.1	14.9'	13°13.8	10.3'	54.1'
8	299°42.2	33.8	77°11.0	14.8'	13°24.1	10.3'	54.1'
9	314°42.4	· · 32.9	91°44.8	14.8'	13°34.3	10.2'	54.1'
10	329°42.6	32.0	106°18.7	14.8'	13°44.5	10.1'	54.1'
11	344°42.8	31.1	120°52.4	14.7'	13°54.7	10.1'	54.1'
12	359°43.0	N09°30.2	135°26.2	14.7'	N14°04.8	10.0'	54.1'
13	14°43.2	29.4	149°59.9	14.7'	14°14.8	10.0'	54.1'
14	29°43.4	28.5	164°33.5	14.6'	14°24.7	9.9'	54.1'
15	44°43.6	· · 27.6	179°07.2	14.6'	14°34.7	9.8'	54.1'
16	59°43.7	26.7	193°40.8	14.6'	14°44.5	9.8'	54.1'
17	74°43.9	25.8	208°14.3	14.5'	14°54.3	9.7'	54.1'
18	89°44.1	N09°24.9	222°47.8	14.5'	N15°04.0	9.7'	54.1'
19	104°44.3	24.0	237°21.3	14.4'	15°13.6	9.6'	54.1'
20	119°44.5	23.2	251°54.7	14.4'	15°23.2	9.5'	54.1'
21	134°44.7	· · 22.3	266°28.1	14.4'	15°32.8	9.5'	54.1'
22	149°44.9	21.4	281°01.5	14.3'	15°42.2	9.4'	54.1'
23	164°45.1	20.5	295°34.8	14.3'	15°51.6	9.3'	54.1'
SD = 15.8'			SD = 14.8'				
$d = -0.9'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	01:52	03:37	20:22	22:03	—:—
N 70°	—:—	02:31	03:55	20:05	21:27	—:—
68°	—:—	02:58	04:08	19:52	21:01	23:22
66°	01:37	03:17	04:19	19:41	20:42	22:19
64°	02:12	03:33	04:29	19:32	20:27	21:47
62°	02:36	03:46	04:37	19:24	20:14	21:23
60°	02:55	03:57	04:44	19:18	20:04	21:05
N 58°	03:10	04:06	04:50	19:12	19:55	20:50
56°	03:23	04:14	04:55	19:07	19:47	20:38
54°	03:34	04:22	05:00	19:02	19:40	20:27
52°	03:43	04:28	05:04	18:58	19:34	20:18
50°	03:51	04:34	05:08	18:54	19:28	20:10
45°	04:08	04:45	05:16	18:46	19:16	19:53
N 40°	04:21	04:55	05:23	18:39	19:07	19:41
35°	04:32	05:03	05:29	18:33	18:59	19:30
30°	04:41	05:10	05:34	18:28	18:53	19:22
20°	04:54	05:21	05:43	18:19	18:42	19:08
N 10°	05:05	05:30	05:51	18:12	18:33	18:58
0°	05:13	05:37	05:58	18:05	18:26	18:50
S 10°	05:19	05:44	06:05	17:58	18:19	18:44
20°	05:25	05:50	06:13	17:51	18:13	18:39
30°	05:29	05:57	06:21	17:42	18:07	18:35
35°	05:31	06:00	06:26	17:38	18:03	18:33
40°	05:32	06:03	06:31	17:33	18:00	18:31
45°	05:33	06:07	06:37	17:27	17:56	18:30
S 50°	05:34	06:12	06:44	17:19	17:52	18:30
52°	05:34	06:13	06:48	17:16	17:50	18:29
54°	05:34	06:15	06:51	17:12	17:48	18:29
56°	05:35	06:18	06:55	17:08	17:46	18:29
58°	05:35	06:20	07:00	17:04	17:44	18:30
S 60°	05:34	06:22	07:05	16:59	17:42	18:30

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	18:28	18:03	17:28	08:23	10:16	12:20
N 70°	18:36	18:20	18:00	08:17	10:01	11:50
68°	18:43	18:34	18:23	08:12	09:49	11:28
66°	18:50	18:45	18:41	08:08	09:39	11:11
64°	18:55	18:55	18:56	08:04	09:30	10:57
62°	18:59	19:03	19:08	08:01	09:23	10:46
60°	19:03	19:10	19:19	07:58	09:17	10:36
N 58°	19:07	19:16	19:28	07:56	09:12	10:28
56°	19:10	19:22	19:36	07:54	09:07	10:20
54°	19:13	19:27	19:44	07:52	09:03	10:13
52°	19:15	19:32	19:50	07:50	08:59	10:07
50°	19:17	19:36	19:56	07:49	08:56	10:02
45°	19:23	19:45	20:09	07:45	08:48	09:50
N 40°	19:27	19:52	20:20	07:43	08:42	09:41
35°	19:31	19:59	20:29	07:40	08:36	09:33
30°	19:34	20:05	20:37	07:38	08:32	09:26
20°	19:40	20:15	20:51	07:34	08:24	09:13
N 10°	19:45	20:23	21:03	07:31	08:17	09:02
0°	19:50	20:32	21:15	07:28	08:10	08:52
S 10°	19:55	20:40	21:27	07:25	08:03	08:42
20°	20:00	20:49	21:39	07:22	07:56	08:32
30°	20:06	21:00	21:53	07:18	07:48	08:20
35°	20:09	21:06	22:02	07:16	07:44	08:13
40°	20:13	21:12	22:11	07:14	07:38	08:05
45°	20:18	21:20	22:23	07:11	07:32	07:56
S 50°	20:23	21:30	22:36	07:08	07:25	07:44
52°	20:26	21:35	22:43	07:06	07:22	07:39
54°	20:29	21:40	22:50	07:04	07:18	07:34
56°	20:32	21:45	22:58	07:03	07:14	07:28
58°	20:35	21:51	23:07	07:01	07:10	07:21
S 60°	20:39	21:58	23:17	06:58	07:05	07:13

Day	Sun		Mer. Pass hh:mm	Moon		Age 16-18 96-86%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
26	01:52	01:43	12:02	01:18	13:39	
27	01:35	01:26	12:01	02:00	14:21	
28	01:17	01:08	12:01	02:42	15:04	

August 29, 30, 31 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	337°29.4	143°09.1	S06°10.5	117°06.2	S16°41.5	133°55.8	S08°39.9	284°18.0	N16°45.1	Alpheratz	357°30.5 29°15.4
1	352°31.9	158°08.9	11.8	132°07.1	42.0	148°57.9	40.1	299°20.4	45.1	Ankaa	353°03.2 -42°08.4
2	7°34.3	173°08.7	13.0	147°08.0	42.5	163°59.9	40.3	314°22.8	45.1	Schedar	349°26.1 56°42.0
3	22°36.8	188°08.6	· · 14.3	162°08.8	· · 43.0	179°02.0	· · 40.4	329°25.3	· · 45.1	Diphda	348°43.3 -17°49.2
4	37°39.3	203°08.4	15.6	177°09.7	43.5	194°04.0	40.6	344°27.7	45.1	Achernar	335°17.2 -57°04.8
5	52°41.7	218°08.2	16.8	192°10.6	44.1	209°06.1	40.8	359°30.1	45.1	Hamal	327°46.7 23°36.2
6	67°44.2	233°08.0	S06°18.1	207°11.5	S16°44.6	224°08.1	S08°40.9	14°32.5	N16°45.1	Polaris	311°48.2 89°22.9
7	82°46.7	248°07.8	19.4	222°12.4	45.1	239°10.2	41.1	29°35.0	45.1	Acamar	315°08.9 -40°10.9
8	97°49.1	263°07.7	20.6	237°13.2	45.6	254°12.3	41.2	44°37.4	45.1	Menkar	314°02.1 4°12.4
9	112°51.6	278°07.5	· · 21.9	252°14.1	· · 46.1	269°14.3	· · 41.4	59°39.8	· · 45.1	Mirfak	308°22.7 49°57.8
10	127°54.1	293°07.3	23.2	267°15.0	46.6	284°16.4	41.6	74°42.2	45.1	Aldebaran	290°35.3 16°34.1
11	142°56.5	308°07.1	24.4	282°15.9	47.1	299°18.4	41.7	89°44.7	45.1	Rigel	281°00.3 -8°10.0
12	157°59.0	323°06.9	S06°25.7	297°16.7	S16°47.6	314°20.5	S08°41.9	104°47.1	N16°45.1	Capella	280°16.4 46°01.4
13	173°01.4	338°06.8	26.9	312°17.6	48.1	329°22.5	42.1	119°49.5	45.1	Bellatrix	278°18.9 6°22.6
14	188°03.9	353°06.6	28.2	327°18.5	48.6	344°24.6	42.2	134°52.0	45.1	Elnath	277°57.2 28°37.8
15	203°06.4	8°06.4	· · 29.5	342°19.4	· · 49.1	359°26.6	· · 42.4	149°54.4	· · 45.1	Alnilam	275°34.0 -1°11.0
16	218°08.8	23°06.2	30.7	357°20.2	49.6	14°28.7	42.6	164°56.8	45.1	Betelgeuse	270°48.2 7°24.7
17	233°11.3	38°06.0	32.0	12°21.1	50.1	29°30.7	42.7	179°59.2	45.1	Canopus	263°51.0 -52°42.5
18	248°13.8	53°05.9	S06°33.3	27°22.0	S16°50.7	44°32.8	S08°42.9	195°01.7	N16°45.1	Sirius	258°23.1 -16°45.4
19	263°16.2	68°05.7	34.5	42°22.9	51.2	59°34.8	43.1	210°04.1	45.1	Adhara	255°03.1 -29°00.6
20	278°18.7	83°05.5	35.8	57°23.7	51.7	74°36.9	43.2	225°06.5	45.1	Procyon	244°47.2 5°08.9
21	293°21.2	98°05.3	· · 37.0	72°24.6	· · 52.2	89°38.9	· · 43.4	240°08.9	· · 45.1	Pollux	243°13.0 27°57.1
22	308°23.6	113°05.1	38.3	87°25.5	52.7	104°41.0	43.6	255°11.4	45.1	Avior	234°13.6 -59°36.2
23	323°26.1	128°04.9	39.6	102°26.4	53.2	119°43.0	43.7	270°13.8	45.1	Suhail	222°43.9 -43°33.1
Mer.pass. 01:30		ν -0.2' d 1.3' m -4.02		ν 0.9' d 0.5' m 0.86		ν 2.1' d 0.2' m -1.77		ν 2.4' d 0.0' m 0.11			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	338°28.6	143°04.8	S06°40.8	117°27.2	S16°53.7	134°45.1	S08°43.9	285°16.2	N16°45.1
1	353°31.0	158°04.6	42.1	132°28.1	54.2	149°47.1	44.1	300°18.7	45.1
2	8°33.5	173°04.4	43.3	147°29.0	54.7	164°49.2	44.2	315°21.1	45.0
3	23°35.9	188°04.2	· · 44.6	162°29.8	· · 55.2	179°51.2	· · 44.4	330°23.5	· · 45.0
4	38°38.4	203°04.0	45.9	177°30.7	55.7	194°53.3	44.6	345°25.9	45.0
5	53°40.9	218°03.9	47.1	192°31.6	56.2	209°55.4	44.7	0°28.4	45.0
6	68°43.3	233°03.7	S06°48.4	207°32.5	S16°56.7	224°57.4	S08°44.9	15°30.8	N16°45.0
7	83°45.8	248°03.5	49.6	222°33.3	57.2	239°59.5	45.1	30°33.2	45.0
8	98°48.3	263°03.3	50.9	237°34.2	57.7	255°01.5	45.2	45°35.7	45.0
9	113°50.7	278°03.1	· · 52.1	252°35.1	· · 58.2	270°03.6	· · 45.4	60°38.1	· · 45.0
10	128°53.2	293°02.9	53.4	267°35.9	58.7	285°05.6	45.6	75°40.5	45.0
11	143°55.7	308°02.8	54.7	282°36.8	59.2	300°07.7	45.7	90°43.0	45.0
12	158°58.1	323°02.6	S06°55.9	297°37.7	S16°59.8	315°09.7	S08°45.9	105°45.4	N16°45.0
13	174°00.6	338°02.4	57.2	312°38.5	17°00.3	330°11.8	46.1	120°47.8	45.0
14	189°03.1	353°02.2	58.4	327°39.4	00.8	345°13.8	46.2	135°50.2	45.0
15	204°05.5	8°02.0	06°59.7	342°40.3	· · 01.3	0°15.9	· · 46.4	150°52.7	· · 45.0
16	219°08.0	23°01.8	07°01.0	357°41.1	01.8	15°17.9	46.6	165°55.1	45.0
17	234°10.4	38°01.7	02.2	12°42.0	02.3	30°20.0	46.7	180°57.5	45.0
18	249°12.9	53°01.5	S07°03.5	27°42.9	S17°02.8	45°22.0	S08°46.9	196°00.0	N16°45.0
19	264°15.4	68°01.3	04.7	42°43.7	03.3	60°24.0	47.1	211°02.4	45.0
20	279°17.8	83°01.1	06.0	57°44.6	03.8	75°26.1	47.2	226°04.8	45.0
21	294°20.3	98°00.9	· · 07.2	72°45.5	· · 04.3	90°28.1	· · 47.4	241°07.3	· · 45.0
22	309°22.8	113°00.7	08.5	87°46.3	04.8	105°30.2	47.6	256°09.7	45.0
23	324°25.2	128°00.6	09.7	102°47.2	05.3	120°32.2	47.7	271°12.1	45.0
Mer.pass. 01:26		ν -0.2' d 1.3' m -4.02		ν 0.9' d 0.5' m 0.86		ν 2.1' d 0.2' m -1.77		ν 2.4' d -0.0' m 0.11	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	339°27.7	143°00.4	S07°11.0	117°48.0	S17°05.8	135°34.3	S08°47.9	286°14.6	N16°45.0
1	354°30.2	158°00.2	12.3	132°48.9	06.3	150°36.3	48.1	301°17.0	45.0
2	9°32.6	173°00.0	13.5	147°49.8	06.8	165°38.4	48.3	316°19.4	45.0
3	24°35.1	187°59.8	· · 14.8	162°50.6	· · 07.3	180°40.4	· · 48.4	331°21.9	· · 45.0
4	39°37.5	202°59.6	16.0	177°51.5	07.8	195°42.5	48.6	346°24.3	45.0
5	54°40.0	217°59.5	17.3	192°52.4	08.3	210°44.5	48.8	1°26.7	45.0
6	69°42.5	232°59.3	S07°18.5	207°53.2	S17°08.8	225°46.6	S08°48.9	16°29.2	N16°45.0
7	84°44.9	247°59.1	19.8	222°54.1	09.3	240°48.6	49.1	31°31.6	45.0
8	99°47.4	262°58.9	21.0	237°54.9	09.8	255°50.7	49.3	46°34.0	45.0
9	114°49.9	277°58.7	· · 22.3	252°55.8	· · 10.3	270°52.7	· · 49.4	61°36.5	· · 45.0
10	129°52.3	292°58.5	23.5	267°56.7	10.8	285°54.8	49.6	76°38.9	45.0
11	144°54.8	307°58.4	24.8	282°57.5	11.3	300°56.8	49.8	91°41.3	45.0
12	159°57.3	322°58.2	S07°26.0	297°58.4	S17°11.8	315°58.9	S08°49.9	106°43.8	N16°45.0
13	174°59.7	337°58.0	27.3	312°59.2	12.3	331°00.9	50.1	121°46.2	45.0
14	190°02.2	352°57.8	28.5	328°00.1	12.8	346°03.0	50.3	136°48.6	45.0
15	205°04.7	7°57.6	· · 29.8	343°01.0	· · 13.3	1°05.0	· · 50.4	151°51.1	· · 45.0
16	220°07.1	22°57.4	31.0	358°01.8	13.8	16°07.0	50.6	166°53.5	45.0
17	235°09.6	37°57.2	32.3	13°02.7	14.3	31°09.1	50.8	181°55.9	45.0
18	250°12.0	52°57.1	S07°33.5	28°03.5	S17°14.8	46°11.1	S08°50.9	196°58.4	N16°45.0
19	265°14.5	67°56.9	34.8	43°04.4	15.3	61°13.2	51.1	212°00.8	45.0
20	280°17.0	82°56.7	36.0	58°05.2	15.8	76°15.2	51.3	227°03.3	45.0
21	295°19.4	97°56.5	· · 37.3	73°06.1	· · 16.3	91°17.3	· · 51.4	242°05.7	· · 45.0
22	310°21.9	112°56.3	38.5	88°07.0	16.8	106°19.3	51.6	257°08.1	45.0
23	325°24.4	127°56.1	39.8	103°07.8	17.3	121°21.4	51.8	272°10.6	45.0
Mer.pass. 01:22		ν -0.2' d 1.3' m -4.03		ν 0.9' d 0.5' m 0.87		ν 2.0' d 0.2' m -1.77		ν 2.4' d -0.0' m 0.10	

Aug 29 Wed	SHA	Mer.pass
Venus	165°39.7	14:28
Mars	139°36.8	16:11
Jupiter	156°26.4	15:02
Saturn	306°48.6	05:02

Aug 30 Thu	SHA	Mer.pass
Venus	164°36.2	14:28
Mars	138°58.7	16:09
Jupiter	156°16.5	14:59
Saturn	306°47.7	04:58

Aug 31 Fri	SHA	Mer.pass
Venus	163°32.7	14:28
Mars	138°20.4	16:08
Jupiter	156°06.6	14:56
Saturn	306°46.9	04:54

Horizontal parallax	
Venus:	0.1
Mars:	0.1

DUT1 = UT1-UTC = +0.2123 sec $\Delta T = TT-UT1 = +68.9717$ sec

2029 August 29 to Aug. 31 UT

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	d HP
Wed							
0	179°45.2	N09°19.6		310°08.1	14.2'	N16°00.9	9.3' 54.1'
1	194°45.4	18.7		324°41.3	14.2'	16°10.2	9.2' 54.1'
2	209°45.6	17.8		339°14.5	14.1'	16°19.4	9.1' 54.1'
3	224°45.8	•• 16.9		353°47.7	14.1'	16°28.5	9.1' 54.1'
4	239°46.0	16.0		8°20.8	14.1'	16°37.6	9.0' 54.1'
5	254°46.2	15.2		22°53.8	14.0'	16°46.6	8.9' 54.1'
6	269°46.4	N09°14.3		37°26.9	14.0'	N16°55.5	8.8' 54.1'
7	284°46.6	13.4		51°59.8	13.9'	17°04.3	8.8' 54.1'
8	299°46.8	12.5		66°32.8	13.9'	17°13.1	8.7' 54.1'
9	314°47.0	•• 11.6		81°05.6	13.8'	17°21.8	8.6' 54.1'
10	329°47.1	10.7		95°38.5	13.8'	17°30.4	8.6' 54.1'
11	344°47.3	09.8		110°11.3	13.7'	17°39.0	8.5' 54.1'
12	359°47.5	N09°08.9		124°44.0	13.7'	N17°47.4	8.4' 54.1'
13	14°47.7	08.0		139°16.7	13.7'	17°55.8	8.3' 54.1'
14	29°47.9	07.1		153°49.4	13.6'	18°04.2	8.3' 54.1'
15	44°48.1	•• 06.2		168°22.0	13.6'	18°12.4	8.2' 54.1'
16	59°48.3	05.3		182°54.5	13.5'	18°20.6	8.1' 54.1'
17	74°48.5	04.5		197°27.0	13.5'	18°28.7	8.0' 54.1'
18	89°48.7	N09°03.6		211°59.5	13.4'	N18°36.7	7.9' 54.1'
19	104°48.9	02.7		226°31.9	13.4'	18°44.6	7.9' 54.1'
20	119°49.0	01.8		241°04.3	13.3'	18°52.5	7.8' 54.1'
21	134°49.2	•• 00.9		255°36.6	13.3'	19°00.3	7.7' 54.1'
22	149°49.4	00°00.0		270°08.9	13.2'	19°08.0	7.6' 54.1'
23	164°49.6	08°59.1		284°41.1	13.2'	19°15.6	7.5' 54.1'
SD = 15.8'				SD = 14.7'			
$d = -0.9'$							

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 49.8	N08° 58.2	299° 13.2	13.1'	N19° 23.1	7.5'	54.1'
1	194° 50.0	57.3	313° 45.3	13.1'	19° 30.6	7.4'	54.1'
2	209° 50.2	56.4	328° 17.4	13.0'	19° 38.0	7.3'	54.1'
3	224° 50.4	• • 55.5	342° 49.4	13.0'	19° 45.2	7.2'	54.1'
4	239° 50.6	54.6	357° 21.4	12.9'	19° 52.4	7.1'	54.1'
5	254° 50.8	53.7	11° 53.3	12.9'	19° 59.5	7.0'	54.1'
6	269° 51.0	N08° 52.8	26° 25.1	12.8'	N20° 06.6	6.9'	54.2'
7	284° 51.2	51.9	40° 56.9	12.8'	20° 13.5	6.9'	54.2'
8	299° 51.4	51.0	55° 28.7	12.7'	20° 20.4	6.8'	54.2'
9	314° 51.6	• • 50.1	70° 00.4	12.6'	20° 27.1	6.7'	54.2'
10	329° 51.7	49.2	84° 32.1	12.6'	20° 33.8	6.6'	54.2'
11	344° 51.9	48.3	99° 03.7	12.5'	20° 40.4	6.5'	54.2'
12	359° 52.1	N08° 47.4	113° 35.2	12.5'	N20° 46.9	6.4'	54.2'
13	14° 52.3	46.5	128° 06.7	12.4'	20° 53.3	6.3'	54.2'
14	29° 52.5	45.6	142° 38.1	12.4'	20° 59.6	6.2'	54.2'
15	44° 52.7	• • 44.7	157° 09.5	12.3'	21° 05.8	6.1'	54.2'
16	59° 52.9	43.8	171° 40.9	12.3'	21° 11.9	6.0'	54.2'
17	74° 53.1	42.9	186° 12.1	12.2'	21° 18.0	5.9'	54.2'
18	89° 53.3	N08° 42.0	200° 43.4	12.2'	N21° 23.9	5.8'	54.3'
19	104° 53.5	41.1	215° 14.5	12.1'	21° 29.8	5.8'	54.3'
20	119° 53.7	40.2	229° 45.7	12.1'	21° 35.5	5.7'	54.3'
21	134° 53.9	• • 39.3	244° 16.7	12.0'	21° 41.2	5.6'	54.3'
22	149° 54.1	38.4	258° 47.8	12.0'	21° 46.7	5.5'	54.3'
23	164° 54.3	37.5	273° 18.7	11.9'	21° 52.2	5.4'	54.3'
	SD = 15.8'	$d = -0.9'$			SD = 14.8'		

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°54.5	N08°36.6	287°49.6	11.9'	N21°57.5	5.3'	54.3'
1	194°54.7	35.7	302°20.5	11.8'	22°02.8	5.2'	54.3'
2	209°54.9	34.8	316°51.3	11.8'	22°08.0	5.1'	54.4'
3	224°55.1	• • 33.9	331°22.1	11.7'	22°13.0	5.0'	54.4'
4	239°55.2	33.0	345°52.8	11.7'	22°18.0	4.9'	54.4'
5	254°55.4	32.1	0°23.4	11.6'	22°22.9	4.8'	54.4'
6	269°55.6	N08°31.2	14°54.0	11.6'	N22°27.6	4.7'	54.4'
7	284°55.8	30.3	29°24.6	11.5'	22°32.3	4.6'	54.4'
8	299°56.0	29.4	43°55.1	11.5'	22°36.9	4.5'	54.4'
9	314°56.2	• • 28.5	58°25.6	11.4'	22°41.3	4.3'	54.4'
10	329°56.4	27.6	72°56.0	11.3'	22°45.7	4.2'	54.5'
11	344°56.6	26.7	87°26.3	11.3'	22°49.9	4.1'	54.5'
12	359°56.8	N08°25.8	101°56.6	11.2'	N22°54.0	4.0'	54.5'
13	14°57.0	24.9	116°26.9	11.2'	22°58.1	3.9'	54.5'
14	29°57.2	24.0	130°57.1	11.1'	23°02.0	3.8'	54.5'
15	44°57.4	• • 23.1	145°27.2	11.1'	23°05.8	3.7'	54.5'
16	59°57.6	22.2	159°57.3	11.1'	23°09.5	3.6'	54.6'
17	74°57.8	21.3	174°27.4	11.0'	23°13.1	3.5'	54.6'
18	89°58.0	N08°20.4	188°57.4	11.0'	N23°16.6	3.4'	54.6'
19	104°58.2	19.5	203°27.3	10.9'	23°20.0	3.3'	54.6'
20	119°58.4	18.6	217°57.2	10.9'	23°23.3	3.2'	54.6'
21	134°58.6	• • 17.7	232°27.1	10.8'	23°26.5	3.1'	54.6'
22	149°58.8	16.8	246°56.9	10.8'	23°29.5	2.9'	54.7'
23	164°59.0	15.9	261°26.7	10.7'	23°32.5	2.8'	54.7'
	SD = 15.8'	$d = -0.9'$			SD = 14.8'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:18	03:53	20:05	21:37	—:—
N 70°	—:—	02:49	04:08	19:50	21:07	—:—
68°	01:12	03:12	04:20	19:39	20:45	22:39
66°	01:58	03:30	04:29	19:29	20:28	21:57
64°	02:27	03:44	04:38	19:22	20:15	21:30
62°	02:48	03:55	04:45	19:15	20:04	21:09
60°	03:05	04:05	04:51	19:09	19:54	20:53
N 58°	03:19	04:14	04:56	19:04	19:46	20:40
56°	03:31	04:21	05:01	18:59	19:39	20:28
54°	03:41	04:27	05:05	18:55	19:32	20:19
52°	03:49	04:33	05:09	18:51	19:27	20:10
50°	03:57	04:38	05:12	18:48	19:21	20:03
45°	04:13	04:49	05:20	18:40	19:11	19:47
N 40°	04:25	04:58	05:26	18:34	19:02	19:35
35°	04:35	05:05	05:31	18:29	18:55	19:26
30°	04:43	05:12	05:36	18:25	18:49	19:18
20°	04:55	05:22	05:44	18:17	18:39	19:05
N 10°	05:05	05:30	05:51	18:10	18:31	18:56
0°	05:12	05:36	05:57	18:04	18:25	18:49
S 10°	05:18	05:42	06:03	17:58	18:19	18:43
20°	05:22	05:48	06:10	17:51	18:13	18:39
30°	05:25	05:53	06:17	17:44	18:08	18:36
35°	05:27	05:56	06:22	17:40	18:05	18:35
40°	05:28	05:59	06:26	17:35	18:03	18:34
45°	05:28	06:02	06:32	17:30	18:00	18:34
S 50°	05:28	06:06	06:38	17:24	17:56	18:34
52°	05:28	06:07	06:41	17:21	17:55	18:34
54°	05:28	06:09	06:44	17:18	17:53	18:34
56°	05:27	06:10	06:48	17:14	17:52	18:35
58°	05:27	06:12	06:52	17:10	17:50	18:35
S 60°	05:26	06:14	06:56	17:06	17:48	18:36

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	☐	☐	☐	☐	☐	☐
N 70°	17:25	☐	☐	13:57	☐	☐
68°	18:08	17:38	☐	13:15	15:21	☐
66°	18:37	18:32	18:28	12:47	14:28	16:13
64°	18:59	19:05	19:20	12:26	13:55	15:21
62°	19:16	19:29	19:52	12:09	13:32	14:50
60°	19:31	19:49	20:16	11:55	13:13	14:26
N 58°	19:44	20:05	20:35	11:43	12:57	14:07
56°	19:55	20:18	20:51	11:33	12:44	13:51
54°	20:04	20:30	21:05	11:24	12:32	13:38
52°	20:13	20:41	21:17	11:15	12:22	13:26
50°	20:21	20:50	21:27	11:08	12:13	13:15
45°	20:37	21:10	21:50	10:53	11:54	12:54
N 40°	20:51	21:26	22:07	10:40	11:39	12:36
35°	21:02	21:40	22:23	10:29	11:26	12:21
30°	21:12	21:52	22:36	10:20	11:14	12:09
20°	21:30	22:12	22:58	10:04	10:55	11:47
N 10°	21:45	22:30	23:18	09:49	10:38	11:28
0°	22:00	22:47	23:36	09:36	10:22	11:10
S 10°	22:14	23:04	23:54	09:23	10:06	10:53
20°	22:30	23:22	•• ••	09:09	09:50	10:34
30°	22:48	23:42	•• ••	08:53	09:31	10:12
35°	22:58	23:55	•• ••	08:44	09:19	09:59
40°	23:10	•• ••	00:09	08:34	09:07	09:45
45°	23:25	•• ••	00:25	08:21	08:52	09:28
S 50°	23:42	•• ••	00:46	08:07	08:33	09:06
52°	23:50	•• ••	00:56	08:00	08:25	08:56
54°	•• ••	00:00	01:07	07:52	08:15	08:45
56°	•• ••	00:10	01:20	07:43	08:04	08:32
58°	•• ••	00:22	01:35	07:34	07:52	08:16
S 60°	•• ••	00:36	01:52	07:23	07:37	07:58

Day	Sun			Moon		
	Eqn.of Time		Mer.	Mer.Pass.		Age
	00 ^h	12 ^h	Pass	Upper	Lower	19-21
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	79-61%
29	00:59	00:50	12:01	03:26	15:48	
30	00:41	00:31	12:01	04:11	16:34	
31	00:22	00:13	12:00	04:58	17:23	

September 01, 02, 03 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	340°26.8	142°55.9	S07°41.0	118°08.7	S17°17.8	136°23.4	S08°51.9	287°13.0	N16°45.0	Alpheratz	357°30.5	29°15.4	
1	355°29.3	157°55.8	42.3	133°09.5	18.3	151°25.5	52.1	302°15.4	45.0	Ankaa	353°03.2	-42°08.4	
2	10°31.8	172°55.6	43.5	148°10.4	18.8	166°27.5	52.3	317°17.9	45.0	Schedar	349°26.1	56°42.0	
3	25°34.2	187°55.4	· · 44.8	163°11.2	· · 19.3	181°29.5	· · 52.5	332°20.3	· · 45.0	Diphda	348°43.3	-17°49.2	
4	40°36.7	202°55.2	46.0	178°12.1	19.8	196°31.6	52.6	347°22.8	45.0	Achernar	335°17.2	-57°04.8	
5	55°39.2	217°55.0	47.3	193°12.9	20.3	211°33.6	52.8	2°25.2	45.0	Hamal	327°46.7	23°36.2	
6	70°41.6	232°54.8	S07°48.5	208°13.8	S17°20.8	226°35.7	S08°53.0	17°27.6	N16°45.0	Polaris	311°46.5	89°22.9	
7	85°44.1	247°54.6	49.8	223°14.6	21.3	241°37.7	53.1	32°30.1	44.9	Acamar	315°08.8	-40°10.9	
8	100°46.5	262°54.4	51.0	238°15.5	21.8	256°39.8	53.3	47°32.5	44.9	Menkar	314°02.1	4°12.5	
9	115°49.0	277°54.3	· · 52.3	253°16.3	· · 22.3	271°41.8	· · 53.5	62°35.0	· · 44.9	Mirfak	308°22.6	49°57.8	
10	130°51.5	292°54.1	53.5	268°17.2	22.8	286°43.9	53.6	77°37.4	44.9	Aldebaran	290°35.3	16°34.1	
11	145°53.9	307°53.9	54.8	283°18.0	23.3	301°45.9	53.8	92°39.8	44.9	Rigel	281°00.3	-8°10.0	
12	160°56.4	322°53.7	S07°56.0	298°18.9	S17°23.8	316°47.9	S08°54.0	107°42.3	N16°44.9	Capella	280°16.4	46°01.4	
13	175°58.9	337°53.5	57.3	313°19.7	24.3	331°50.0	54.1	122°44.7	44.9	Bellatrix	278°18.9	6°22.6	
14	191°01.3	352°53.3	58.5	328°20.6	24.8	346°52.0	54.3	137°47.1	44.9	Elnath	277°57.2	28°37.8	
15	206°03.8	7°53.1	07°59.8	343°21.4	· · 25.3	1°54.1	· · 54.5	152°49.6	· · 44.9	Alnilam	275°34.0	-1°11.0	
16	221°06.3	22°52.9	08°01.0	358°22.3	25.8	16°56.1	54.6	167°52.0	44.9	Betelgeuse	270°48.1	7°24.7	
17	236°08.7	37°52.8	02.2	13°23.1	26.3	31°58.2	54.8	182°54.5	44.9	Canopus	263°51.0	-52°42.5	
18	251°11.2	52°52.6	S08°03.5	28°24.0	S17°26.8	47°00.2	S08°55.0	197°56.9	N16°44.9	Sirius	258°23.1	-16°45.4	
19	266°13.7	67°52.4	04.7	43°24.8	27.3	62°02.2	55.2	212°59.4	44.9	Adhara	255°03.1	-29°00.6	
20	281°16.1	82°52.2	06.0	58°25.7	27.8	77°04.3	55.3	228°01.8	44.9	Procyon	244°47.1	5°08.9	
21	296°18.6	97°52.0	· · 07.2	73°26.5	· · 28.3	92°06.3	· · 55.5	243°04.2	· · 44.9	Pollux	243°13.0	27°57.1	
22	311°21.0	112°51.8	08.5	88°27.4	28.7	107°08.4	55.7	258°06.7	44.9	Avior	234°13.6	-59°36.2	
23	326°23.5	127°51.6	09.7	103°28.2	29.2	122°10.4	55.8	273°09.1	44.9	Suhail	222°43.9	-43°33.1	
Mer.pass. 01:18		ν -0.2' d 1.2' m -4.03		ν 0.9' d 0.5' m 0.87		ν 2.0' d 0.2' m -1.76		ν 2.4' d -0.0' m 0.10					
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	341°26.0	142°51.4	S08°11.0	118°29.1	S17°29.7	137°12.4	S08°56.0	288°11.6	N16°44.9	Dubhe	193°37.3	61°35.4	
1	356°28.4	157°51.3	12.2	133°29.9	30.2	152°14.5	56.2	303°14.0	44.9	Denebola	182°21.5	14°24.4	
2	11°30.9	172°51.1	13.4	148°30.8	30.7	167°16.5	56.3	318°16.4	44.9	Gienah	175°40.0	-17°42.4	
3	26°33.4	187°50.9	· · 14.7	163°31.6	· · 31.2	182°18.6	· · 56.5	333°18.9	· · 44.9	Acrux	172°56.3	-63°16.0	
4	41°35.8	202°50.7	15.9	178°32.4	31.7	197°20.6	56.7	348°21.3	44.9	Gacrux	171°47.8	-57°16.9	
5	56°38.3	217°50.5	17.2	193°33.3	32.2	212°22.7	56.9	3°23.8	44.9	Alioth	166°10.3	55°48.1	
6	71°40.8	232°50.3	S08°18.4	208°34.1	S17°32.7	227°24.7	S08°57.0	18°26.2	N16°44.9	Spica	158°18.6	-11°18.9	
7	86°43.2	247°50.1	19.6	223°35.0	33.2	242°26.7	57.2	33°28.7	44.9	Alkaid	152°49.5	49°10.1	
8	101°45.7	262°49.9	20.9	238°35.8	33.7	257°28.8	57.4	48°31.1	44.9	Hadar	148°30.8	-60°31.1	
9	116°48.1	277°49.7	· · 22.1	253°36.7	· · 34.2	272°30.8	· · 57.5	63°33.5	· · 44.9	Menkent	147°53.3	-36°31.0	
10	131°50.6	292°49.5	23.4	268°37.5	34.7	287°32.9	57.7	78°36.0	44.9	Arcturus	145°44.7	19°01.8	
11	146°53.1	307°49.4	24.6	283°38.3	35.2	302°34.9	57.9	93°38.4	44.8	Rigil Kent.	139°35.3	-60°57.6	
12	161°55.5	322°49.2	S08°25.9	298°39.2	S17°35.7	317°36.9	S08°58.0	108°40.9	N16°44.8	Kochab	137°20.7	74°02.3	
13	176°58.0	337°49.0	27.1	313°40.0	36.2	332°39.0	58.2	123°43.3	44.8	Zuben'ubi	136°52.0	-16°09.8	
14	192°00.5	352°48.8	28.3	328°40.9	36.7	347°41.0	58.4	138°45.8	44.8	Alphecca	126°00.7	26°37.2	
15	207°02.9	7°48.6	· · 29.6	343°41.7	· · 37.2	2°43.1	· · 58.5	153°48.2	· · 44.8	Antares	112°11.2	-26°29.8	
16	222°05.4	22°48.4	30.8	358°42.5	37.6	17°45.1	58.7	168°50.7	44.8	Atria	107°01.9	-69°04.9	
17	237°07.9	37°48.2	32.0	13°43.4	38.1	32°47.1	58.9	183°53.1	44.8	Sabik	101°58.4	-15°45.5	
18	252°10.3	52°48.0	S08°33.3	28°44.2	S17°38.6	47°49.2	S08°59.1	198°55.5	N16°44.8	Shaula	96°05.2	-37°07.4	
19	267°12.8	67°47.8	34.5	43°45.1	39.1	62°51.2	59.2	213°58.0	44.8	Rasalhague	95°55.0	12°32.6	
20	282°15.3	82°47.6	35.8	58°45.9	39.6	77°53.2	59.4	229°00.4	44.8	Eltanin	90°40.3	51°29.5	
21	297°17.7	97°47.4	· · 37.0	73°46.7	· · 40.1	92°55.3	· · 59.6	244°02.9	· · 44.8	Kaus Aust.	83°27.4	-34°22.1	
22	312°20.2	112°47.3	38.2	88°47.6	40.6	107°57.3	59.7	259°05.3	44.8	Vega	80°30.5	38°49.1	
23	327°22.6	127°47.1	39.5	103°48.4	41.1	122°59.4	59.9	274°07.8	44.8	Nunki	75°42.9	-26°15.4	
Mer.pass. 01:14		ν -0.2' d 1.2' m -4.04		ν 0.8' d 0.5' m 0.88		ν 2.0' d 0.2' m -1.76		ν 2.4' d -0.0' m 0.09		Altair	61°56.1	8°57.1	
										Peacock	52°59.5	-56°38.2	
										Deneb	49°22.8	45°23.5	
										Enif	33°34.8	10°01.0	
										Al Na'ir	27°27.9	-46°48.8	
										Fomalhaut	15°10.1	-29°27.6	
										Scheat	13°41.1	28°14.9	
										Markab	13°25.8	15°22.1	
											Sep 01 Sat	SHA	Mer.pass
											Venus	162°29.1	14:28
											Mars	137°41.8	16:07
											Jupiter	155°56.6	14:52
											Saturn	306°46.2	04:50
											Sep 02 Sun	SHA	Mer.pass
											Venus	161°25.5	14:29
											Mars	137°03.1	16:05
											Jupiter	155°46.5	14:49
											Saturn	306°45.6	04:46
											Sep 03 Mon	SHA	Mer.pass
											Venus	160°21.8	14:29
											Mars	136°24.1	16:04
											Jupiter	155°36.3	14:46
											Saturn	306°45.1	04:43
											Horizontal parallax		
											Venus:	0.1	
											Mars:	0.1	
Mer.pass. 01:10		ν -0.2' d 1.2' m -4.04		ν 0.8' d 0.5' m 0.88		ν 2.0' d 0.2' m -1.76		ν 2.4' d -0.0' m 0.09					

DUT1 = UT1-UTC = +0.2123 sec ΔT = TT-UT1 = +68.9717 sec

2029 September 01 to Sep. 03 UT

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°59.2	N08°14.9	275°56.4	10.7'	N23°35.3	2.7'	54.7'
1	194°59.4	14.0	290°26.0	10.6'	23°38.0	2.6'	54.7'
2	209°59.6	13.1	304°55.7	10.6'	23°40.6	2.5'	54.7'
3	224°59.8	· · 12.2	319°25.3	10.5'	23°43.1	2.4'	54.8'
4	240°00.0	11.3	333°54.8	10.5'	23°45.5	2.3'	54.8'
5	255°00.2	10.4	348°24.3	10.5'	23°47.7	2.1'	54.8'
6	270°00.4	N08°09.5	2°53.7	10.4'	N23°49.9	2.0'	54.8'
7	285°00.6	08.6	17°23.2	10.4'	23°51.9	1.9'	54.8'
8	300°00.8	07.7	31°52.5	10.3'	23°53.8	1.8'	54.9'
9	315°01.0	· · 06.8	46°21.8	10.3'	23°55.6	1.7'	54.9'
10	330°01.2	05.9	60°51.1	10.2'	23°57.3	1.6'	54.9'
11	345°01.4	05.0	75°20.4	10.2'	23°58.9	1.4'	54.9'
12	0°01.6	N08°04.1	89°49.6	10.2'	N24°00.3	1.3'	55.0'
13	15°01.8	03.1	104°18.7	10.1'	24°01.7	1.2'	55.0'
14	30°02.0	02.2	118°47.9	10.1'	24°02.9	1.1'	55.0'
15	45°02.2	· · 01.3	133°16.9	10.0'	24°04.0	1.0'	55.0'
16	60°02.4	08°00.4	147°46.0	10.0'	24°04.9	0.8'	55.0'
17	75°02.6	07°59.5	162°15.0	10.0'	24°05.8	0.7'	55.1'
18	90°02.8	N07°58.6	176°44.0	9.9'	N24°06.5	0.6'	55.1'
19	105°03.0	57.7	191°12.9	9.9'	24°07.1	0.5'	55.1'
20	120°03.2	56.8	205°41.8	9.9'	24°07.6	0.4'	55.1'
21	135°03.4	· · 55.9	220°10.7	9.8'	24°08.0	0.2'	55.2'
22	150°03.6	54.9	234°39.5	9.8'	24°08.2	0.1'	55.2'
23	165°03.8	54.0	249°08.3	9.8'	24°08.3	-0.0'	55.2'
SD = 15.8'			SD = 14.9'				
$d = -0.9'$							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°04.0	N07°53.1	263°37.1	9.7'	N24°08.3	-0.1'	55.2'
1	195°04.2	52.2	278°05.8	9.7'	24°08.2	-0.3'	55.3'
2	210°04.4	51.3	292°34.5	9.7'	24°07.9	-0.4'	55.3'
3	225°04.6	· · 50.4	307°03.2	9.6'	24°07.6	-0.5'	55.3'
4	240°04.8	49.5	321°31.9	9.6'	24°07.1	-0.6'	55.3'
5	255°05.0	48.6	336°00.5	9.6'	24°06.4	-0.7'	55.4'
6	270°05.2	N07°47.6	350°29.1	9.6'	N24°05.7	-0.9'	55.4'
7	285°05.4	46.7	4°57.6	9.5'	24°04.8	-1.0'	55.4'
8	300°05.6	45.8	19°26.2	9.5'	24°03.8	-1.1'	55.5'
9	315°05.8	· · 44.9	33°54.7	9.5'	24°02.7	-1.3'	55.5'
10	330°06.0	44.0	48°23.2	9.5'	24°01.4	-1.4'	55.5'
11	345°06.2	43.1	62°51.6	9.4'	24°00.1	-1.5'	55.5'
12	0°06.4	N07°42.2	77°20.1	9.4'	N23°58.6	-1.6'	55.6'
13	15°06.6	41.2	91°48.5	9.4'	23°56.9	-1.8'	55.6'
14	30°06.8	40.3	106°16.9	9.4'	23°55.2	-1.9'	55.6'
15	45°07.0	· · 39.4	120°45.3	9.4'	23°53.3	-2.0'	55.7'
16	60°07.2	38.5	135°13.6	9.3'	23°51.3	-2.1'	55.7'
17	75°07.4	37.6	149°41.9	9.3'	23°49.1	-2.3'	55.7'
18	90°07.6	N07°36.7	164°10.3	9.3'	N23°46.9	-2.4'	55.7'
19	105°07.8	35.8	178°38.5	9.3'	23°44.5	-2.5'	55.8'
20	120°08.0	34.8	193°06.8	9.3'	23°41.9	-2.7'	55.8'
21	135°08.2	· · 33.9	207°35.1	9.2'	23°39.3	-2.8'	55.8'
22	150°08.4	33.0	222°03.3	9.2'	23°36.5	-2.9'	55.9'
23	165°08.6	32.1	236°31.6	9.2'	23°33.6	-3.0'	55.9'
SD = 15.8'			SD = 15.1'				
$d = -0.9'$							

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°08.8	N07°31.2	250°59.8	9.2'	N23°30.6	-3.2'	55.9'
1	195°09.0	30.3	265°28.0	9.2'	23°27.4	-3.3'	56.0'
2	210°09.2	29.3	279°56.2	9.2'	23°24.1	-3.4'	56.0'
3	225°09.4	· · 28.4	294°24.3	9.2'	23°20.7	-3.5'	56.0'
4	240°09.7	27.5	308°52.5	9.2'	23°17.1	-3.7'	56.1'
5	255°09.9	26.6	323°20.7	9.1'	23°13.5	-3.8'	56.1'
6	270°10.1	N07°25.7	337°48.8	9.1'	N23°09.7	-3.9'	56.1'
7	285°10.3	24.7	352°16.9	9.1'	23°05.7	-4.1'	56.2'
8	300°10.5	23.8	6°45.1	9.1'	23°01.7	-4.2'	56.2'
9	315°10.7	· · 22.9	21°13.2	9.1'	22°57.5	-4.3'	56.2'
10	330°10.9	22.0	35°41.3	9.1'	22°53.1	-4.4'	56.3'
11	345°11.1	21.1	50°09.4	9.1'	22°48.7	-4.6'	56.3'
12	0°11.3	N07°20.1	64°37.5	9.1'	N22°44.1	-4.7'	56.3'
13	15°11.5	19.2	79°05.6	9.1'	22°39.4	-4.8'	56.4'
14	30°11.7	18.3	93°33.7	9.1'	22°34.6	-5.0'	56.4'
15	45°11.9	· · 17.4	108°01.8	9.1'	22°29.6	-5.1'	56.4'
16	60°12.1	16.5	122°29.9	9.1'	22°24.5	-5.2'	56.5'
17	75°12.3	15.5	136°58.0	9.1'	22°19.3	-5.3'	56.5'
18	90°12.5	N07°14.6	151°26.1	9.1'	N22°14.0	-5.5'	56.5'
19	105°12.7	13.7	165°54.2	9.1'	22°08.5	-5.6'	56.6'
20	120°12.9	12.8	180°22.3	9.1'	22°02.9	-5.7'	56.6'
21	135°13.1	· · 11.9	194°50.3	9.1'	21°57.2	-5.9'	56.6'
22	150°13.3	10.9	209°18.4	9.1'	21°51.3	-6.0'	56.7'
23	165°13.6	10.0	223°46.5	9.1'	21°45.4	-6.1'	56.7'
SD = 15.8'			SD = 15.3'				
$d = -0.9'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:40	04:07	19:48	21:14	—:—
N 70°	00:25	03:06	04:20	19:36	20:49	23:09
68°	01:42	03:26	04:31	19:26	20:30	22:10
66°	02:17	03:41	04:39	19:18	20:15	21:37
64°	02:41	03:54	04:46	19:11	20:03	21:14
62°	03:00	04:04	04:52	19:05	19:53	20:56
60°	03:15	04:13	04:58	19:00	19:44	20:41
N 58°	03:28	04:21	05:02	18:55	19:37	20:29
56°	03:38	04:27	05:07	18:51	19:30	20:19
54°	03:47	04:33	05:10	18:48	19:25	20:10
52°	03:55	04:38	05:14	18:44	19:19	20:02
50°	04:02	04:43	05:17	18:41	19:15	19:55
45°	04:17	04:53	05:23	18:35	19:05	19:41
N 40°	04:28	05:01	05:29	18:30	18:57	19:30
35°	04:37	05:08	05:34	18:25	18:51	19:21
30°	04:45	05:13	05:38	18:21	18:45	19:14
20°	04:56	05:22	05:45	18:14	18:36	19:03
N 10°	05:05	05:29	05:51	18:08	18:29	18:54
0°	05:11	05:35	05:56	18:03	18:24	18:48
S 10°	05:16	05:41	06:02	17:57	18:19	18:43
20°	05:20	05:45	06:07	17:52	18:14	18:40
30°	05:22	05:50	06:14	17:46	18:10	18:37
35°	05:23	05:52	06:18	17:42	18:08	18:37
40°	05:23	05:54	06:22	17:38	18:05	18:37
45°	05:23	05:57	06:26	17:34	18:03	18:37
S 50°	05:22	05:59	06:32	17:28	18:01	18:38
52°	05:22	06:01	06:34	17:26	18:00	18:39
54°	05:21	06:02	06:37	17:23	17:58	18:39
56°	05:20	06:03	06:40	17:20	17:57	18:40
58°	05:19	06:04	06:44	17:16	17:56	18:41
S 60°	05:18	06:06	06:48	17:13	17:55	18:43

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	20:23	☐	☐	19:43
66°	☐	19:52	21:48	☐	18:23	18:18
64°	19:53	20:56	22:25	16:33	17:18	17:40
62°	20:30	21:31	22:52	15:56	16:43	17:13
60°	20:57	21:56	23:12	15:29	16:18	16:52
N 58°	21:18	22:16	23:29	15:08	15:58	16:35
56°	21:35	22:33	23:43	14:51	15:41	16:20
54°	21:50	22:47	23:55	14:37	15:27	16:07
52°	22:02	22:59	· · ·	14:24	15:15	15:56
50°	22:14	23:10	· · ·	14:13	15:04	15:46
45°	22:37	23:33	· · ·	13:50	14:41	15:26
N 40°	22:56	23:51	· · ·	13:31	14:22	15:09
35°	23:11	· · ·	00:06	13:15	14:07	14:54
30°	23:25	· · ·	00:19	13:02	13:53	14:42
20°	23:48	· · ·	00:42	12:39	13:31	14:21
N 10°	· · ·	00:08	01:01	12:19	13:11	14:02
0°	· · ·	00:27	01:19	12:00	12:52	13:45
S 10°	· · ·	00:46	01:37	11:42	12:33	13:27
20°	00:14	01:06	01:57	11:22	12:13	13:08
30°	00:36	01:29	02:19	10:59	11:50	12:47
35°	00:50	01:43	02:32	10:45	11:37	12:34
40°	01:05	01:59	02:48	10:29	11:21	12:19
45°	01:24	02:17	03:06	10:11	11:02	12:02
S 50°	01:47	02:41	03:28	09:48	10:39	11:40
52°	01:58	02:52	03:39	09:36	10:27	11:29
54°	02:10	03:05	03:51	09:24	10:15	11:18
56°	02:24	03:20	04:04	09:09	10:00	11:04
58°	02:41	03:37	04:20	08:52	09:43	10:48
S 60°	03:02	03:59	04:40	08:32	09:21	10:29

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	22-24
		mm:ss		hh:mm	hh:mm	52-33%
01	00:03	00:06	12:00	05:48	18:14	
02	00:16	00:26	12:00	06:39	19:06	
03	00:35	00:45	11:59	07:32	19:58	

September 04, 05, 06 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	343°24.3	142°42.2	S09°10.3	119°09.2	S17°53.3	138°50.3	S09°04.2	290°09.0	N16°44.7	Alpheratz	357°30.5	29°15.4
1	358°26.7	157°42.0	11.5	134°10.1	53.8	153°52.3	04.3	305°11.4	44.7	Ankaa	353°03.2	-42°08.4
2	13°29.2	172°41.8	12.8	149°10.9	54.3	168°54.4	04.5	320°13.9	44.6	Schedar	349°26.0	56°42.0
3	28°31.6	187°41.6	•• 14.0	164°11.7	•• 54.8	183°56.4	•• 04.7	335°16.3	•• 44.6	Diphda	348°43.3	-17°49.2
4	43°34.1	202°41.4	15.2	179°12.5	55.3	198°58.4	04.9	350°18.8	44.6	Achernar	335°17.1	-57°04.8
5	58°36.6	217°41.2	16.4	194°13.4	55.8	214°00.5	05.0	5°21.3	44.6	Hamal	327°46.7	23°36.2
6	73°39.0	232°41.0	S09°17.7	209°14.2	S17°56.2	229°02.5	S09°05.2	20°23.7	N16°44.6	Polaris	311°44.8	89°22.9
7	88°41.5	247°40.8	18.9	224°15.0	56.7	244°04.5	05.4	35°26.2	44.6	Acamar	315°08.8	-40°10.9
8	103°44.0	262°40.7	20.1	239°15.9	57.2	259°06.6	05.5	50°28.6	44.6	Menkar	314°02.1	4°12.5
9	118°46.4	277°40.5	•• 21.3	254°16.7	•• 57.7	274°08.6	•• 05.7	65°31.1	•• 44.6	Mirfak	308°22.6	49°57.8
10	133°48.9	292°40.3	22.6	269°17.5	58.2	289°10.6	05.9	80°33.5	44.6	Aldebaran	290°35.3	16°34.1
11	148°51.4	307°40.1	23.8	284°18.3	58.7	304°12.7	06.1	95°36.0	44.6	Rigel	281°00.3	-8°10.0
12	163°53.8	322°39.9	S09°25.0	299°19.1	S17°59.2	319°14.7	S09°06.2	110°38.4	N16°44.6	Capella	280°16.4	46°01.4
13	178°56.3	337°39.7	26.3	314°20.0	17°59.6	334°16.7	06.4	125°40.9	44.6	Bellatrix	278°18.9	6°22.6
14	193°58.7	352°39.5	27.5	329°20.8	18°00.1	349°18.8	06.6	140°43.3	44.6	Elnath	277°57.2	28°37.8
15	209°01.2	7°39.3	•• 28.7	344°21.6	•• 00.6	4°20.8	•• 06.7	155°45.8	•• 44.6	Alnilam	275°34.0	-1°11.0
16	224°03.7	22°39.1	29.9	359°22.4	01.1	19°22.8	06.9	170°48.2	44.6	Betelgeuse	270°48.1	7°24.7
17	239°06.1	37°38.9	31.2	14°23.3	01.6	34°24.9	07.1	185°50.7	44.6	Canopus	263°50.9	-52°42.5
18	254°08.6	52°38.7	S09°32.4	29°24.1	S18°02.1	49°26.9	S09°07.3	200°53.1	N16°44.5	Sirius	258°23.1	-16°45.3
19	269°11.1	67°38.5	33.6	44°24.9	02.6	64°28.9	07.4	215°55.6	44.5	Adhara	255°03.1	-29°00.6
20	284°13.5	82°38.3	34.8	59°25.7	03.0	79°31.0	07.6	230°58.1	44.5	Procyon	244°47.1	5°08.9
21	299°16.0	97°38.1	•• 36.0	74°26.5	•• 03.5	94°33.0	•• 07.8	246°00.5	•• 44.5	Pollux	243°13.0	27°57.1
22	314°18.5	112°37.9	37.3	89°27.4	04.0	109°35.0	07.9	261°03.0	44.5	Avior	234°13.6	-59°36.2
23	329°20.9	127°37.7	38.5	104°28.2	04.5	124°37.1	08.1	276°05.4	44.5	Suhail	222°43.9	-43°33.1
Mer.pass. 01:06		ν -0.2' d 1.2' m -4.05		ν 0.8' d 0.5' m 0.89		ν 2.0' d 0.2' m -1.75		ν 2.5' d -0.0' m 0.08		Miaplacidus	221°38.1	-69°50.3
Alphard 217°44.4 -8°47.3 Regulus 207°30.8 11°49.3 Dubhe 193°37.3 61°35.4 Denebola 182°21.5 14°24.4 Gienah 175°40.0 -17°42.4 Acrux 172°56.3 -63°16.0 Gacrux 171°47.8 -57°16.9 Alioth 166°10.3 55°48.0 Spica 158°18.6 -11°18.9 Alkaid 152°49.5 49°10.1 Hadar 148°30.9 -60°31.1 Menkent 147°53.3 -36°31.0 Arcturus 145°44.7 19°01.8 Rigil Kent. 139°35.3 -60°57.6 Kochab 137°20.8 74°02.3 Zuben'ubi 136°52.0 -16°09.8 Alphecca 126°00.7 26°37.2 Antares 112°11.2 -26°29.8 Atria 107°01.9 -69°04.9 Sabik 101°58.4 -15°45.5 Shaula 96°05.2 -37°07.4 Rasalhague 95°55.0 12°32.6 Eltanin 90°40.3 51°29.5 Kaus Aust. 83°27.4 -34°22.1 Vega 80°30.5 38°49.1 Nunki 75°42.9 -26°15.4 Altair 61°56.1 8°57.1 Peacock 52°59.5 -56°38.2 Deneb 49°22.8 45°23.5 Enif 33°34.8 10°01.0 Al Na'ir 27°27.9 -46°48.8 Fomalhaut 15°10.1 -29°27.6 Scheat 13°41.1 28°14.9 Markab 13°25.8 15°22.2												
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	344°23.4	142°37.5	S09°39.7	119°29.0	S18°05.0	139°39.1	S09°08.3	291°07.9	N16°44.5	Sep 04 Tue	SHA	
1	359°25.9	157°37.3	40.9	134°29.8	05.5	154°41.1	08.5	306°10.3	44.5	Venus	159°18.0	14:29
2	14°28.3	172°37.1	42.2	149°30.6	05.9	169°43.1	08.6	321°12.8	44.5	Mars	135°45.0	16:02
3	29°30.8	187°36.9	•• 43.4	164°31.5	•• 06.4	184°45.2	•• 08.8	336°15.2	•• 44.5	Jupiter	155°26.0	14:43
4	44°33.2	202°36.7	44.6	179°32.3	06.9	199°47.2	09.0	351°17.7	44.5	Saturn	306°44.7	04:39
5	59°35.7	217°36.5	45.8	194°33.1	07.4	214°49.2	09.2	6°20.2	44.5			
6	74°38.2	232°36.3	S09°47.0	209°33.9	S18°07.9	229°51.3	S09°09.3	21°22.6	N16°44.5	Sep 05 Wed	SHA	Mer.pass
7	89°40.6	247°36.1	48.3	224°34.7	08.4	244°53.3	09.5	36°25.1	44.5	Venus	158°14.1	14:30
8	104°43.1	262°35.9	49.5	239°35.6	08.8	259°55.3	09.7	51°27.5	44.4	Mars	135°05.6	16:01
9	119°45.6	277°35.7	•• 50.7	254°36.4	•• 09.3	274°57.4	•• 09.8	66°30.0	•• 44.4	Jupiter	155°15.7	14:39
10	134°48.0	292°35.5	51.9	269°37.2	09.8	289°59.4	10.0	81°32.4	44.4	Saturn	306°44.5	04:35
11	149°50.5	307°35.3	53.1	284°38.0	10.3	305°01.4	10.2	96°34.9	44.4			
12	164°53.0	322°35.1	S09°54.4	299°38.8	S18°10.8	320°03.5	S09°10.4	111°37.4	N16°44.4	Sep 06 Thu	SHA	Mer.pass
13	179°55.4	337°34.9	55.6	314°39.6	11.2	335°05.5	10.5	126°39.8	44.4	Venus	157°10.1	14:30
14	194°57.9	352°34.7	56.8	329°40.4	11.7	350°07.5	10.7	141°42.3	44.4	Mars	134°26.0	16:00
15	210°00.4	7°34.5	•• 58.0	344°41.3	•• 12.2	5°09.5	•• 10.9	156°44.7	•• 44.4	Jupiter	155°05.3	14:36
16	225°02.8	22°34.3	09°59.2	359°42.1	12.7	20°11.6	11.0	171°47.2	44.4	Saturn	306°44.3	04:31
17	240°05.3	37°34.1	10°00.4	14°42.9	13.2	35°13.6	11.2	186°49.7	44.4			
18	255°07.7	52°33.9	S10°01.7	29°43.7	S18°13.6	50°15.6	S09°11.4	201°52.1	N16°44.4	Horizontal parallax	Venus:	0.1
19	270°10.2	67°33.7	02.9	44°44.5	14.1	65°17.7	11.6	216°54.6	44.4		Mars:	0.1
20	285°12.7	82°33.5	04.1	59°45.3	14.6	80°19.7	11.7	231°57.0	44.4			
21	300°15.1	97°33.3	•• 05.3	74°46.1	•• 15.1	95°21.7	•• 11.9	246°59.5	•• 44.4			
22	315°17.6	112°33.1	06.5	89°46.9	15.6	110°23.8	12.1	262°01.9	44.3			
23	330°20.1	127°32.9	07.7	104°47.8	16.0	125°25.8	12.3	277°04.4	44.3			
Mer.pass. 01:02		ν -0.2' d 1.2' m -4.06		ν 0.8' d 0.5' m 0.89		ν 2.0' d 0.2' m -1.75		ν 2.5' d -0.0' m 0.08				
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	345°22.5	142°32.7	S10°08.9	119°48.6	S18°16.5	140°27.8	S09°12.4	292°06.9	N16°44.3			
1	0°25.0	157°32.5	10.2	134°49.4	17.0	155°29.8	12.6	307°09.3	44.3			
2	15°27.5	172°32.3	11.4	149°50.2	17.5	170°31.9	12.8	322°11.8	44.3			
3	30°29.9	187°32.1	•• 12.6	164°51.0	•• 18.0	185°33.9	•• 12.9	337°14.3	•• 44.3			
4	45°32.4	202°31.8	13.8	179°51.8	18.4	200°35.9	13.1	352°16.7	44.3			
5	60°34.8	217°31.6	15.0	194°52.6	18.9	215°38.0	13.3	7°19.2	44.3			
6	75°37.3	232°31.4	S10°16.2	209°53.4	S18°19.4	230°40.0	S09°13.5	22°21.6	N16°44.3			
7	90°39.8	247°31.2	17.4	224°54.2	19.9	245°42.0	13.6	37°24.1	44.3			
8	105°42.2	262°31.0	18.6	239°55.0	20.4	260°44.0	13.8	52°26.6	44.3			
9	120°44.7	277°30.8	•• 19.9	254°55.8	•• 20.8	275°46.1	•• 14.0	67°29.0	•• 44.3			
10	135°47.2	292°30.6	21.1	269°56.7	21.3	290°48.1	14.2	82°31.5	44.2			
11	150°49.6	307°30.4	22.3	284°57.5	21.8	305°50.1	14.3	97°33.9	44.2			
12	165°52.1	322°30.2	S10°23.5	299°58.3	S18°22.3	320°52.1	S09°14.5	112°36.4	N16°44.2			
13	180°54.6	337°30.0	24.7	314°59.1	22.7	335°54.2	14.7	127°38.9	44.2			
14	195°57.0	352°29.8	25.9	329°59.9	23.2	350°56.2	14.9	142°41.3	44.2			
15	210°59.5	7°29.6	•• 27.1	345°00.7	•• 23.7	5°58.2	•• 15.0	157°43.8	•• 44.2			
16	226°02.0	22°29.4	28.3	0°01.5	24.2	21°00.3	15.2	172°46.3	44.2			
17	241°04.4	37°29.2	29.5	15°02.3	24.6	36°02.3	15.4	187°48.7	44.2			
18	256°06.9	52°29.0	S10°30.7	30°03.1	S18°25.1	51°04.3	S09°15.5	202°51.2	N16°44.2			
19	271°09.3	67°28.8	31.9	45°03.9	25.6	66°06.3	15.7	217°53.6	44.2			
20	286°11.8	82°28.6	33.1	60°04.7	26.1	81°08.4	15.9	232°56.1	44.2			
21	301°14.3	97°28.4	•• 34.									

DUT1 = UT1-UTC = +0.2124 sec ΔT = TT-UT1 = +68.9716 sec

2029 September 04 to Sep. 06 UT

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180° 13.8	N07° 09.1	238° 14.6	9.1'	N21° 39.3	-6.2'	56.7'
1	195° 14.0	08.2	252° 42.7	9.1'	21° 33.0	-6.4'	56.8'
2	210° 14.2	07.2	267° 10.8	9.1'	21° 26.7	-6.5'	56.8'
3	225° 14.4	· · 06.3	281° 38.9	9.1'	21° 20.2	-6.6'	56.8'
4	240° 14.6	05.4	296° 07.0	9.1'	21° 13.6	-6.7'	56.9'
5	255° 14.8	04.5	310° 35.2	9.1'	21° 06.9	-6.9'	56.9'
6	270° 15.0	N07° 03.6	325° 03.3	9.1'	N21° 00.0	-7.0'	57.0'
7	285° 15.2	02.6	339° 31.4	9.1'	20° 53.0	-7.1'	57.0'
8	300° 15.4	01.7	353° 59.6	9.1'	20° 45.9	-7.2'	57.0'
9	315° 15.6	07° 00.8	8° 27.7	9.2'	20° 38.7	-7.3'	57.1'
10	330° 15.8	06° 59.9	22° 55.9	9.2'	20° 31.4	-7.5'	57.1'
11	345° 16.0	58.9	37° 24.0	9.2'	20° 23.9	-7.6'	57.1'
12	0° 16.2	N06° 58.0	51° 52.2	9.2'	N20° 16.3	-7.7'	57.2'
13	15° 16.4	57.1	66° 20.4	9.2'	20° 08.6	-7.8'	57.2'
14	30° 16.7	56.2	80° 48.6	9.2'	20° 00.8	-8.0'	57.2'
15	45° 16.9	· · 55.2	95° 16.8	9.2'	19° 52.8	-8.1'	57.3'
16	60° 17.1	54.3	109° 45.0	9.2'	19° 44.7	-8.2'	57.3'
17	75° 17.3	53.4	124° 13.2	9.2'	19° 36.5	-8.3'	57.4'
18	90° 17.5	N06° 52.5	138° 41.4	9.2'	N19° 28.2	-8.4'	57.4'
19	105° 17.7	51.5	153° 09.7	9.3'	19° 19.8	-8.5'	57.4'
20	120° 17.9	50.6	167° 38.0	9.3'	19° 11.3	-8.7'	57.5'
21	135° 18.1	· · 49.7	182° 06.2	9.3'	19° 02.6	-8.8'	57.5'
22	150° 18.3	48.7	196° 34.5	9.3'	18° 53.8	-8.9'	57.5'
23	165° 18.5	47.8	211° 02.8	9.3'	18° 44.9	-9.0'	57.6'
	SD = 15.9'	$d = -0.9'$			SD = 15.5'		

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180° 18.7	N06° 46.9	225° 31.1	9.3'	N18° 35.9	-9.1'	57.6'
1	195° 18.9	46.0	239° 59.5	9.3'	18° 26.8	-9.2'	57.7'
2	210° 19.2	45.0	254° 27.8	9.4'	18° 17.5	-9.4'	57.7'
3	225° 19.4	· · 44.1	268° 56.1	9.4'	18° 08.2	-9.5'	57.7'
4	240° 19.6	43.2	283° 24.5	9.4'	17° 58.7	-9.6'	57.8'
5	255° 19.8	42.3	297° 52.9	9.4'	17° 49.1	-9.7'	57.8'
6	270° 20.0	N06° 41.3	312° 21.3	9.4'	N17° 39.5	-9.8'	57.8'
7	285° 20.2	40.4	326° 49.7	9.4'	17° 29.7	-9.9'	57.9'
8	300° 20.4	39.5	341° 18.1	9.4'	17° 19.7	-10.0'	57.9'
9	315° 20.6	· · 38.5	355° 46.6	9.5'	17° 09.7	-10.1'	57.9'
10	330° 20.8	37.6	10° 15.1	9.5'	16° 59.6	-10.2'	58.0'
11	345° 21.0	36.7	24° 43.5	9.5'	16° 49.4	-10.3'	58.0'
12	0° 21.2	N06° 35.8	39° 12.0	9.5'	N16° 39.0	-10.4'	58.1'
13	15° 21.5	34.8	53° 40.5	9.5'	16° 28.6	-10.6'	58.1'
14	30° 21.7	33.9	68° 09.0	9.5'	16° 18.0	-10.7'	58.1'
15	45° 21.9	· · 33.0	82° 37.6	9.6'	16° 07.4	-10.8'	58.2'
16	60° 22.1	32.0	97° 06.1	9.6'	15° 56.6	-10.9'	58.2'
17	75° 22.3	31.1	111° 34.7	9.6'	15° 45.8	-11.0'	58.2'
18	90° 22.5	N06° 30.2	126° 03.3	9.6'	N15° 34.8	-11.1'	58.3'
19	105° 22.7	29.2	140° 31.9	9.6'	15° 23.7	-11.2'	58.3'
20	120° 22.9	28.3	155° 00.5	9.6'	15° 12.6	-11.3'	58.4'
21	135° 23.1	· · 27.4	169° 29.1	9.6'	15° 01.3	-11.4'	58.4'
22	150° 23.3	26.4	183° 57.8	9.7'	14° 50.0	-11.5'	58.4'
23	165° 23.6	25.5	198° 26.4	9.7'	14° 38.5	-11.6'	58.5'
	SD = 15.9'	$d = -0.9'$			SD = 15.7'		

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180° 23.8	N06° 24.6	212° 55.1	9.7'	N14° 26.9	-11.6'	58.5'
1	195° 24.0	23.7	227° 23.8	9.7'	14° 15.3	-11.7'	58.5'
2	210° 24.2	22.7	241° 52.5	9.7'	14° 03.6	-11.8'	58.6'
3	225° 24.4	· · 21.8	256° 21.2	9.7'	13° 51.7	-11.9'	58.6'
4	240° 24.6	20.9	270° 50.0	9.7'	13° 39.8	-12.0'	58.6'
5	255° 24.8	19.9	285° 18.7	9.8'	13° 27.8	-12.1'	58.7'
6	270° 25.0	N06° 19.0	299° 47.5	9.8'	N13° 15.7	-12.2'	58.7'
7	285° 25.2	18.1	314° 16.3	9.8'	13° 03.5	-12.3'	58.7'
8	300° 25.5	17.1	328° 45.0	9.8'	12° 51.2	-12.4'	58.8'
9	315° 25.7	· · 16.2	343° 13.8	9.8'	12° 38.8	-12.5'	58.8'
10	330° 25.9	15.3	357° 42.7	9.8'	12° 26.4	-12.5'	58.8'
11	345° 26.1	14.3	12° 11.5	9.8'	12° 13.8	-12.6'	58.9'
12	0° 26.3	N06° 13.4	26° 40.3	9.9'	N12° 01.2	-12.7'	58.9'
13	15° 26.5	12.5	41° 09.2	9.9'	11° 48.5	-12.8'	59.0'
14	30° 26.7	11.5	55° 38.0	9.9'	11° 35.7	-12.9'	59.0'
15	45° 26.9	· · 10.6	70° 06.9	9.9'	11° 22.9	-12.9'	59.0'
16	60° 27.1	09.7	84° 35.8	9.9'	11° 09.9	-13.0'	59.1'
17	75° 27.4	08.7	99° 04.7	9.9'	10° 56.9	-13.1'	59.1'
18	90° 27.6	N06° 07.8	113° 33.6	9.9'	N10° 43.8	-13.2'	59.1'
19	105° 27.8	06.8	128° 02.5	9.9'	10° 30.6	-13.2'	59.1'
20	120° 28.0	05.9	142° 31.4	9.9'	10° 17.4	-13.3'	59.2'
21	135° 28.2	· · 05.0	157° 00.4	9.9'	10° 04.1	-13.4'	59.2'
22	150° 28.4	04.0	171° 29.3	9.9'	09° 50.7	-13.5'	59.2'
23	165° 28.6	03.1	185° 58.3	10.0'	09° 37.2	-13.5'	59.3'
	SD = 15.9'	$d = -0.9'$			SD = 16.0'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	03:00	04:22	19:32	20:53	—:—
N 70°	01:20	03:22	04:33	19:22	20:31	22:27
68°	02:05	03:39	04:42	19:13	20:15	21:46
66°	02:34	03:52	04:49	19:06	20:02	21:19
64°	02:55	04:04	04:55	19:00	19:51	20:59
62°	03:11	04:13	05:00	18:55	19:42	20:43
60°	03:25	04:21	05:05	18:51	19:35	20:30
N 58°	03:36	04:28	05:09	18:47	19:28	20:19
56°	03:45	04:34	05:12	18:43	19:22	20:10
54°	03:54	04:39	05:16	18:40	19:17	20:02
52°	04:01	04:44	05:19	18:37	19:12	19:55
50°	04:08	04:48	05:21	18:35	19:08	19:48
45°	04:21	04:57	05:27	18:29	18:59	19:35
N 40°	04:31	05:04	05:32	18:25	18:52	19:25
35°	04:40	05:10	05:36	18:21	18:47	19:17
30°	04:47	05:15	05:39	18:17	18:42	19:10
20°	04:57	05:23	05:45	18:12	18:34	19:00
N 10°	05:05	05:29	05:51	18:06	18:28	18:52
0°	05:10	05:35	05:55	18:02	18:23	18:47
S 10°	05:14	05:39	06:00	17:57	18:18	18:43
20°	05:17	05:43	06:05	17:53	18:15	18:40
30°	05:19	05:46	06:10	17:47	18:11	18:39
35°	05:19	05:48	06:13	17:44	18:10	18:39
40°	05:18	05:50	06:17	17:41	18:08	18:39
45°	05:18	05:51	06:21	17:37	18:06	18:40
S 50°	05:16	05:53	06:26	17:32	18:05	18:42
52°	05:15	05:54	06:28	17:30	18:04	18:43
54°	05:14	05:55	06:30	17:28	18:04	18:44
56°	05:13	05:55	06:33	17:26	18:03	18:46
58°	05:11	05:56	06:35	17:23	18:02	18:48
S 60°	05:09	05:57	06:39	17:20	18:01	18:50

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	☐	☐	00:24	☐	19:25	18:45
N 70°	22:20	· ·	00:55	19:37	18:52	18:29
68°	23:11	· ·	01:18	18:46	18:28	18:16
66°	23:42	· ·	01:36	18:13	18:09	18:05
64°	· ·	00:05	01:50	17:49	17:54	17:56
62°	· ·	00:24	02:02	17:30	17:41	17:48
60°	· ·	00:39	02:12	17:14	17:30	17:41
N 58°	· ·	00:52	02:20	17:01	17:20	17:35
56°	· ·	01:03	02:28	16:49	17:12	17:30
54°	· ·	01:12	02:35	16:39	17:04	17:25
52°	00:06	01:21	02:41	16:30	16:57	17:20
50°	00:16	01:29	02:46	16:22	16:51	17:16
45°	00:36	01:45	02:58	16:04	16:38	17:08
N 40°	00:52	01:58	03:08	15:50	16:27	17:00
35°	01:06	02:10	03:16	15:38	16:17	16:54
30°	01:18	02:20	03:24	15:27	16:09	16:48
20°	01:38	02:37	03:36	15:09	15:55	16:39
N 10°	01:56	02:51	03:47	14:53	15:42	16:30
0°	02:12	03:05	03:57	14:38	15:30	16:22
S 10°	02:29	03:19	04:07	14:22	15:18	16:14
20°	02:46	03:33	04:18	14:06	15:05	16:05
30°	03:06	03:50	04:30	13:47	14:50	15:54
35°	03:18	04:00	04:38	13:36	14:41	15:49
40°	03:32	04:11	04:46	13:23	14:31	15:42
45°	03:47	04:23	04:55	13:08	14:19	15:34
S 50°	04:07	04:39	05:06	12:50	14:05	15:24
52°	04:16	04:46	05:11	12:41	13:58	15:20
54°	04:26	04:54	05:17	12:31	13:51	15:15
56°	04:38	05:03	05:23	12:20	13:43	15:09
58°	04:51	05:14	05:30	12:07	13:33	15:03
S 60°	05:07	05:25	05:38	11:52	13:22	14:56

Day	Sun		Mer. Pass hh:mm	Moon		Age 25-27 23-8%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
04	00:55	01:05	11:59	08:25	20:51	
05	01:15	01:25	11:59	09:18	21:44	
06	01:35	01:45	11:58	10:09	22:35	

September 07, 08, 09 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	346°21.7	142°27.7	S10°38.0	120°07.9	S18°28.0	141°16.5	S09°16.6	293°06.0	N16°44.1	Alpheratz	357°30.5	29°15.4					
1	1°24.1	157°27.5	39.2	135°08.7	28.4	156°18.5	16.8	308°08.4	44.1	Ankaa	353°03.2	-42°08.4					
2	16°26.6	172°27.3	40.4	150°09.5	28.9	171°20.5	16.9	323°10.9	44.1	Schedar	349°26.0	56°42.1					
3	31°29.1	187°27.1	•• 41.6	165°10.3	•• 29.4	186°22.5	•• 17.1	338°13.4	•• 44.1	Diphda	348°43.3	-17°49.2					
4	46°31.5	202°26.9	42.8	180°11.1	29.9	201°24.6	17.3	353°15.8	44.1	Achernar	335°17.1	-57°04.9					
5	61°34.0	217°26.7	44.0	195°11.9	30.3	216°26.6	17.5	8°18.3	44.1	Hamal	327°46.7	23°36.2					
6	76°36.5	232°26.5	S10°45.2	210°12.7	S18°30.8	231°28.6	S09°17.6	23°20.8	N16°44.1	Polaris	311°43.4	89°22.9					
7	91°38.9	247°26.3	46.4	225°13.5	31.3	246°30.6	17.8	38°23.2	44.1	Acamar	315°08.8	-40°10.9					
8	106°41.4	262°26.1	47.6	240°14.3	31.8	261°32.7	18.0	53°25.7	44.1	Menkar	314°02.0	4°12.5					
9	121°43.8	277°25.9	•• 48.8	255°15.1	•• 32.2	276°34.7	•• 18.1	68°28.2	•• 44.0	Mirfak	308°22.6	49°57.8					
10	136°46.3	292°25.7	50.0	270°15.9	32.7	291°36.7	18.3	83°30.6	44.0	Aldebaran	290°35.2	16°34.1					
11	151°48.8	307°25.4	51.2	285°16.7	33.2	306°38.7	18.5	98°33.1	44.0	Rigel	281°00.3	-8°10.0					
12	166°51.2	322°25.2	S10°52.4	300°17.5	S18°33.7	321°40.8	S09°18.7	113°35.6	N16°44.0	Capella	280°16.3	46°01.4					
13	181°53.7	337°25.0	53.6	315°18.3	34.1	336°42.8	18.8	128°38.0	44.0	Bellatrix	278°18.9	6°22.6					
14	196°56.2	352°24.8	54.8	330°19.1	34.6	351°44.8	19.0	143°40.5	44.0	Elnath	277°57.1	28°37.8					
15	211°58.6	7°24.6	•• 56.0	345°19.9	•• 35.1	6°46.8	•• 19.2	158°43.0	•• 44.0	Alnilam	275°34.0	-1°11.0					
16	227°01.1	22°24.4	57.2	0°20.7	35.5	21°48.9	19.4	173°45.4	44.0	Betelgeuse	270°48.1	7°24.7					
17	242°03.6	37°24.2	58.4	15°21.5	36.0	36°50.9	19.5	188°47.9	44.0	Canopus	263°50.9	-52°42.5					
18	257°06.0	52°24.0	S10°59.6	30°22.3	S18°36.5	51°52.9	S09°19.7	203°50.4	N16°44.0	Sirius	258°23.1	-16°45.3					
19	272°08.5	67°23.8	11°00.8	45°23.1	37.0	66°54.9	19.9	218°52.8	43.9	Adhara	255°03.1	-29°00.6					
20	287°10.9	82°23.5	02.0	60°23.9	37.4	81°57.0	20.1	233°55.3	43.9	Procyon	244°47.1	5°08.9					
21	302°13.4	97°23.3	•• 03.2	75°24.7	•• 37.9	96°59.0	•• 20.2	248°57.8	•• 43.9	Pollux	243°13.0	27°57.1					
22	317°15.9	112°23.1	04.4	90°25.5	38.4	112°01.0	20.4	264°00.2	43.9	Avior	234°13.6	-59°36.2					
23	332°18.3	127°22.9	05.6	105°26.3	38.8	127°03.0	20.6	279°02.7	43.9	Suhail	222°43.8	-43°33.1					
Mer.pass. 00:54			ν -0.2' d 1.2' m -4.07			ν 0.8' d 0.5' m 0.90			ν 2.0' d 0.2' m -1.75			ν 2.5' d -0.0' m 0.07					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	347°20.8	142°22.7	S11°06.8	120°27.1	S18°39.3	142°05.0	S09°20.8	294°05.2	N16°43.9					
1	2°23.3	157°22.5	08.0	135°27.9	39.8	157°07.1	20.9	309°07.7	43.9					
2	17°25.7	172°22.3	09.2	150°28.6	40.3	172°09.1	21.1	324°10.1	43.9					
3	32°28.2	187°22.1	•• 10.4	165°29.4	•• 40.7	187°11.1	•• 21.3	339°12.6	•• 43.9					
4	47°30.7	202°21.9	11.6	180°30.2	41.2	202°13.1	21.5	354°15.1	43.9					
5	62°33.1	217°21.6	12.8	195°31.0	41.7	217°15.2	21.6	9°17.5	43.8					
6	77°35.6	232°21.4	S11°14.0	210°31.8	S18°42.1	232°17.2	S09°21.8	24°20.0	N16°43.8					
7	92°38.1	247°21.2	15.2	225°32.6	42.6	247°19.2	22.0	39°22.5	43.8					
8	107°40.5	262°21.0	16.4	240°33.4	43.1	262°21.2	22.2	54°24.9	43.8					
9	122°43.0	277°20.8	•• 17.5	255°34.2	•• 43.5	277°23.2	•• 22.3	69°27.4	•• 43.8					
10	137°45.4	292°20.6	18.7	270°35.0	44.0	292°25.3	22.5	84°29.9	43.8					
11	152°47.9	307°20.4	19.9	285°35.8	44.5	307°27.3	22.7	99°32.4	43.8					
12	167°50.4	322°20.1	S11°21.1	300°36.5	S18°44.9	322°29.3	S09°22.9	114°34.8	N16°43.8					
13	182°52.8	337°19.9	22.3	315°37.3	45.4	337°31.3	23.0	129°37.3	43.8					
14	197°55.3	352°19.7	23.5	330°38.1	45.9	352°33.4	23.2	144°39.8	43.7					
15	212°57.8	7°19.5	•• 24.7	345°38.9	•• 46.3	7°35.4	•• 23.4	159°42.2	•• 43.7					
16	228°00.2	22°19.3	25.9	0°39.7	46.8	22°37.4	23.5	174°44.7	43.7					
17	243°02.7	37°19.1	27.1	15°40.5	47.3	37°39.4	23.7	189°47.2	43.7					
18	258°05.2	52°18.9	S11°28.3	30°41.3	S18°47.7	52°41.4	S09°23.9	204°49.7	N16°43.7					
19	273°07.6	67°18.6	29.5	45°42.1	48.2	67°43.5	24.1	219°52.1	43.7					
20	288°10.1	82°18.4	30.6	60°42.8	48.7	82°45.5	24.2	234°54.6	43.7					
21	303°12.5	97°18.2	•• 31.8	75°43.6	•• 49.1	97°47.5	•• 24.4	249°57.1	•• 43.7					
22	318°15.0	112°18.0	33.0	90°44.4	49.6	112°49.5	24.6	264°59.6	43.7					
23	333°17.5	127°17.8	34.2	105°45.2	50.1	127°51.5	24.8	280°02.0	43.7					
Mer.pass. 00:50			ν -0.2' d 1.2' m -4.07			ν 0.8' d 0.5' m 0.90			ν 2.0' d 0.2' m -1.74			ν 2.5' d -0.0' m 0.06		

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	348°19.9	142°17.6	S11°35.4	120°46.0	S18°50.5	142°53.6	S09°24.9	295°04.5	N16°43.6					
1	3°22.4	157°17.3	36.6	135°46.8	51.0	157°55.6	25.1	310°07.0	43.6					
2	18°24.9	172°17.1	37.8	150°47.6	51.5	172°57.6	25.3	325°09.5	43.6					
3	33°27.3	187°16.9	•• 39.0	165°48.3	•• 51.9	187°59.6	•• 25.5	340°11.9	•• 43.6					
4	48°29.8	202°16.7	40.1	180°49.1	52.4	203°01.6	25.6	355°14.4	43.6					
5	63°32.3	217°16.5	41.3	195°49.9	52.9	218°03.7	25.8	10°16.9	43.6					
6	78°34.7	232°16.3	S11°42.5	210°50.7	S18°53.3	233°05.7	S09°26.0	25°19.4	N16°43.6					
7	93°37.2	247°16.0	43.7	225°51.5	53.8	248°07.7	26.2	40°21.8	43.6					
8	108°39.7	262°15.8	44.9	240°52.2	54.3	263°09.7	26.3	55°24.3	43.6					
9	123°42.1	277°15.6	•• 46.1	255°53.0	•• 54.7	278°11.7	•• 26.5	70°26.8	•• 43.5					
10	138°44.6	292°15.4	47.2	270°53.8	55.2	293°13.7	26.7	85°29.3	43.5					
11	153°47.0	307°15.2	48.4	285°54.6	55.6	308°15.8	26.9	100°31.7	43.5					
12	168°49.5	322°14.9	S11°49.6	300°55.4	S18°56.1	323°17.8	S09°27.0	115°34.2	N16°43.5					
13	183°52.0	337°14.7	50.8	315°56.2	56.6	338°19.8	27.2	130°36.7	43.5					
14	198°54.4	352°14.5	52.0	330°56.9	57.0	353°21.8	27.4	145°39.2	43.5					
15	213°56.9	7°14.3	•• 53.2	345°57.7	•• 57.5	8°23.8	•• 27.6	160°41.6	•• 43.5					
16	228°59.4	22°14.1	54.3	0°58.5	58.0	23°25.9	27.8	175°44.1	43.5					
17	244°01.8	37°13.8	55.5	15°59.3	58.4	38°27.9	27.9	190°46.6	43.4					
18	259°04.3	52°13.6	S11°56.7	31°00.0	S18°58.9	53°29.9	S09°28.1	205°49.1	N16°43.4					
19	274°06.8	67°13.4	57.9	46°00.8	59.4	68°31.9	28.3	220°51.5	43.4					
20	289°09.2	82°13.2	11°59.1	61°01.6	18°59.8	83°33.9	28.5	235°54.0	43.4					
21	304°11.7	97°13.0	12°00.2	76°02.4	19°00.3	98°35.9	•• 28.6	250°56.5	•• 43.4					
22	319°14.1	112°12.7	01.4	91°03.2	00.7	113°38.0	28.8	265°59.0	43.4					
23	334°16.6	127°12.5	02.6	106°03.9	01.2	128°40.0	29.0	281°01.5	43.4					
Mer.pass. 00:47			ν -0.2' d 1.2' m -4.08			ν 0.8' d 0.5' m 0.91			ν 2.0' d 0.2' m -1.74			ν 2.5' d -0.0' m 0.06		

Sep 07 Fri	SHA	Mer.pass
Venus	156°06.1	14:30
Mars	133°46.3	15:59
Jupiter	154°54.8	14:33
Saturn	306°44.3	04:27

Sep 08 Sat	SHA	Mer.pass
Venus	155°01.9	14:31
Mars	133°06.3	15:57
Jupiter	154°44.2	14:30
Saturn	306°44.4	04:23

Sep 09 Sun	SHA	Mer.pass
Venus	153°57.6	14:31
Mars	132°26.0	15:56
Jupiter	154°33.6	14:26
Saturn	306°44.6	04:19

Horizontal parallax		
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h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°28.8	N06°02.2	200°27.2	10.0'	N09°23.7	-13.6'	59.3'
1	195°29.1	01.2	214°56.2	10.0'	09°10.1	-13.7'	59.3'
2	210°29.3	06°00.3	229°25.1	10.0'	08°56.5	-13.7'	59.4'
3	225°29.5	05°59.4	243°54.1	10.0'	08°42.7	-13.8'	59.4'
4	240°29.7	58.4	258°23.1	10.0'	08°28.9	-13.8'	59.4'
5	255°29.9	57.5	272°52.1	10.0'	08°15.1	-13.9'	59.5'
6	270°30.1	N05°56.6	287°21.1	10.0'	N08°01.2	-14.0'	59.5'
7	285°30.3	55.6	301°50.1	10.0'	07°47.2	-14.0'	59.5'
8	300°30.5	54.7	316°19.0	10.0'	07°33.2	-14.1'	59.6'
9	315°30.8	· · 53.7	330°48.0	10.0'	07°19.1	-14.1'	59.6'
10	330°31.0	52.8	345°17.0	10.0'	07°05.0	-14.2'	59.6'
11	345°31.2	51.9	359°46.0	10.0'	06°50.8	-14.2'	59.6'
12	0°31.4	N05°50.9	14°15.0	10.0'	N06°36.6	-14.3'	59.7'
13	15°31.6	50.0	28°44.0	10.0'	06°22.3	-14.3'	59.7'
14	30°31.8	49.0	43°13.0	10.0'	06°07.9	-14.4'	59.7'
15	45°32.0	· · 48.1	57°42.0	10.0'	05°53.5	-14.4'	59.7'
16	60°32.3	47.2	72°11.0	10.0'	05°39.1	-14.5'	59.8'
17	75°32.5	46.2	86°40.0	10.0'	05°24.6	-14.5'	59.8'
18	90°32.7	N05°45.3	101°09.0	10.0'	N05°10.1	-14.6'	59.8'
19	105°32.9	44.4	115°38.0	10.0'	04°55.5	-14.6'	59.8'
20	120°33.1	43.4	130°07.0	10.0'	04°40.9	-14.6'	59.9'
21	135°33.3	· · 42.5	144°35.9	10.0'	04°26.3	-14.7'	59.9'
22	150°33.5	41.5	159°04.9	10.0'	04°11.6	-14.7'	59.9'
23	165°33.8	40.6	173°33.9	10.0'	03°56.9	-14.8'	59.9'
SD = 15.9'			SD = 16.2'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°34.0	N05°39.7	188°02.8	9.9'	N03°42.1	-14.8'	60.0'
1	195°34.2	38.7	202°31.8	9.9'	03°27.3	-14.8'	60.0'
2	210°34.4	37.8	217°00.7	9.9'	03°12.5	-14.8'	60.0'
3	225°34.6	· · 36.8	231°29.6	9.9'	02°57.7	-14.9'	60.0'
4	240°34.8	35.9	245°58.5	9.9'	02°42.8	-14.9'	60.1'
5	255°35.0	35.0	260°27.4	9.9'	02°27.9	-14.9'	60.1'
6	270°35.3	N05°34.0	274°56.3	9.9'	N02°13.0	-14.9'	60.1'
7	285°35.5	33.1	289°25.2	9.9'	01°58.1	-15.0'	60.1'
8	300°35.7	32.1	303°54.1	9.9'	01°43.1	-15.0'	60.1'
9	315°35.9	· · 31.2	318°22.9	9.8'	01°28.1	-15.0'	60.2'
10	330°36.1	30.2	332°51.8	9.8'	01°13.1	-15.0'	60.2'
11	345°36.3	29.3	347°20.6	9.8'	00°58.1	-15.0'	60.2'
12	0°36.5	N05°28.4	1°49.4	9.8'	N00°43.1	-15.0'	60.2'
13	15°36.8	27.4	16°18.2	9.8'	00°28.0	-15.1'	60.2'
14	30°37.0	26.5	30°47.0	9.8'	N00°12.9	-15.1'	60.3'
15	45°37.2	· · 25.5	45°15.7	9.7'	S00°02.1	15.1'	60.3'
16	60°37.4	24.6	59°44.4	9.7'	00°17.2	15.1'	60.3'
17	75°37.6	23.7	74°13.1	9.7'	00°32.3	15.1'	60.3'
18	90°37.8	N05°22.7	88°41.8	9.7'	S00°47.4	15.1'	60.3'
19	105°38.1	21.8	103°10.5	9.6'	01°02.5	15.1'	60.3'
20	120°38.3	20.8	117°39.1	9.6'	01°17.5	15.1'	60.4'
21	135°38.5	· · 19.9	132°07.8	9.6'	01°32.6	15.1'	60.4'
22	150°38.7	18.9	146°36.4	9.6'	01°47.7	15.1'	60.4'
23	165°38.9	18.0	161°04.9	9.5'	02°02.8	15.1'	60.4'
SD = 15.9'			SD = 16.4'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°39.1	N05°17.0	175°33.5	9.5'	S02°17.9	15.1'	60.4'
1	195°39.4	16.1	190°02.0	9.5'	02°33.0	15.1'	60.4'
2	210°39.6	15.2	204°30.5	9.5'	02°48.0	15.1'	60.4'
3	225°39.8	· · 14.2	218°59.0	9.4'	03°03.1	15.0'	60.5'
4	240°40.0	13.3	233°27.4	9.4'	03°18.1	15.0'	60.5'
5	255°40.2	12.3	247°55.8	9.4'	03°33.1	15.0'	60.5'
6	270°40.4	N05°11.4	262°24.2	9.3'	S03°48.1	15.0'	60.5'
7	285°40.7	10.4	276°52.5	9.3'	04°03.1	15.0'	60.5'
8	300°40.9	09.5	291°20.9	9.3'	04°18.1	14.9'	60.5'
9	315°41.1	· · 08.5	305°49.1	9.2'	04°33.0	14.9'	60.5'
10	330°41.3	07.6	320°17.4	9.2'	04°48.0	14.9'	60.5'
11	345°41.5	06.7	334°45.6	9.2'	05°02.9	14.9'	60.5'
12	0°41.7	N05°05.7	349°13.8	9.1'	S05°17.7	14.8'	60.5'
13	15°42.0	04.8	3°41.9	9.1'	05°32.6	14.8'	60.5'
14	30°42.2	03.8	18°10.0	9.1'	05°47.4	14.8'	60.6'
15	45°42.4	· · 02.9	32°38.1	9.0'	06°02.2	14.7'	60.6'
16	60°42.6	01.9	47°06.1	9.0'	06°16.9	14.7'	60.6'
17	75°42.8	01.0	61°34.1	9.0'	06°31.6	14.7'	60.6'
18	90°43.0	N05°00.0	76°02.1	8.9'	S06°46.3	14.6'	60.6'
19	105°43.3	04°59.1	90°30.0	8.9'	07°01.0	14.6'	60.6'
20	120°43.5	58.1	104°57.9	8.8'	07°15.5	14.6'	60.6'
21	135°43.7	· · 57.2	119°25.7	8.8'	07°30.1	14.5'	60.6'
22	150°43.9	56.2	133°53.5	8.8'	07°44.6	14.5'	60.6'
23	165°44.1	55.3	148°21.3	8.7'	07°59.1	14.4'	60.6'
SD = 15.9'			SD = 16.5'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	00:43	03:18	04:36	19:16	20:33	22:53
N 70°	01:50	03:37	04:45	19:07	20:15	21:57
68°	02:25	03:52	04:52	19:00	20:01	21:25
66°	02:49	04:03	04:58	18:55	19:49	21:03
64°	03:07	04:13	05:04	18:50	19:40	20:45
62°	03:22	04:21	05:08	18:45	19:32	20:31
60°	03:34	04:28	05:12	18:42	19:25	20:19
N 58°	03:44	04:35	05:15	18:38	19:19	20:09
56°	03:53	04:40	05:18	18:36	19:14	20:01
54°	04:00	04:45	05:21	18:33	19:09	19:53
52°	04:07	04:49	05:23	18:31	19:05	19:47
50°	04:13	04:53	05:26	18:28	19:01	19:41
45°	04:25	05:01	05:30	18:24	18:54	19:29
N 40°	04:35	05:07	05:34	18:20	18:47	19:20
35°	04:42	05:12	05:38	18:17	18:42	19:12
30°	04:48	05:17	05:41	18:14	18:38	19:06
20°	04:58	05:24	05:46	18:09	18:31	18:57
N 10°	05:05	05:29	05:50	18:05	18:26	18:50
0°	05:09	05:34	05:54	18:01	18:22	18:46
S 10°	05:13	05:37	05:58	17:57	18:18	18:43
20°	05:15	05:40	06:02	17:53	18:15	18:41
30°	05:15	05:43	06:07	17:49	18:13	18:41
35°	05:15	05:44	06:09	17:46	18:12	18:41
40°	05:14	05:45	06:12	17:44	18:11	18:42
45°	05:12	05:46	06:15	17:41	18:10	18:44
S 50°	05:10	05:47	06:19	17:37	18:09	18:47
52°	05:08	05:47	06:21	17:35	18:09	18:48
54°	05:07	05:47	06:23	17:33	18:09	18:50
56°	05:05	05:48	06:25	17:31	18:09	18:52
58°	05:03	05:48	06:27	17:29	18:08	18:54
S 60°	05:00	05:48	06:30	17:27	18:08	18:57

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	02:53	05:11	07:27	18:18	17:53	17:28
N 70°	03:07	05:14	07:21	18:11	17:55	17:38
68°	03:18	05:17	07:16	18:05	17:56	17:46
66°	03:27	05:19	07:12	18:01	17:57	17:52
64°	03:35	05:21	07:08	17:57	17:57	17:58
62°	03:41	05:22	07:05	17:53	17:58	18:03
60°	03:47	05:24	07:02	17:50	17:58	18:07
N 58°	03:52	05:25	07:00	17:47	17:59	18:11
56°	03:56	05:26	06:58	17:45	17:59	18:14
54°	04:00	05:27	06:56	17:43	18:00	18:17
52°	04:04	05:28	06:54	17:41	18:00	18:20
50°	04:07	05:29	06:52	17:39	18:00	18:23
45°	04:14	05:31	06:49	17:35	18:01	18:28
N 40°	04:19	05:32	06:46	17:31	18:02	18:33
35°	04:24	05:34	06:44	17:28	18:02	18:37
30°	04:29	05:35	06:42	17:26	18:03	18:40
20°	04:36	05:37	06:38	17:21	18:03	18:47
N 10°	04:43	05:39	06:35	17:17	18:04	18:52
0°	04:49	05:40	06:32	17:13	18:05	18:57
S 10°	04:55	05:42	06:29	17:09	18:05	19:02
20°	05:01	05:44	06:26	17:05	18:06	19:08
30°	05:09	05:46	06:23	17:00	18:07	19:14
35°	05:13	05:47	06:21	16:57	18:07	19:18
40°	05:18	05:48	06:19	16:54	18:07	19:22
45°	05:23	05:50	06:16	16:50	18:08	19:27
S 50°	05:30	05:52	06:13	16:46	18:08	19:32
52°	05:33	05:53	06:12	16:44	18:09	19:35
54°	05:36	05:53	06:11	16:41	18:09	19:38
56°	05:40	05:54	06:09	16:39	18:09	19:41
58°	05:44	05:56	06:07	16:36	18:10	19:45
S 60°	05:48	05:57	06:05	16:33	18:10	19:49

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	28-1 3-1%
07	01:55	02:06	11:58	11:01	23:27	
08	02:16	02:26	11:58	11:52	—:—	
09	02:37	02:47	11:57	12:45	00:18	

September 10, 11, 12 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	349°19.1	142°12.3	S12°03.8	121°04.7	S19°01.7	143°42.0	S09°29.2	296°03.9	N16°43.4			296°03.9	N16°43.4	Alpheratz	357°30.5	29°15.4	
1	4°21.5	157°12.1	04.9	136°05.5	02.1	158°44.0	29.3	311°06.4	43.3			311°06.4	43.3	Ankaa	353°03.1	-42°08.4	
2	19°24.0	172°11.9	06.1	151°06.3	02.6	173°46.0	29.5	326°08.9	43.3			326°08.9	43.3	Schedar	349°26.0	56°42.1	
3	34°26.5	187°11.6	•• 07.3	166°07.0	•• 03.0	188°48.0	•• 29.7	341°11.4	•• 43.3			341°11.4	•• 43.3	Diphda	348°43.3	-17°49.2	
4	49°28.9	202°11.4	08.5	181°07.8	03.5	203°50.1	29.9	356°13.9	43.3			356°13.9	43.3	Achernar	335°17.1	-57°04.9	
5	64°31.4	217°11.2	09.7	196°08.6	04.0	218°52.1	30.0	11°16.3	43.3			11°16.3	43.3	Hamal	327°46.6	23°36.2	
6	79°33.9	232°11.0	S12°10.8	211°09.4	S19°04.4	233°54.1	S09°30.2	26°18.8	N16°43.3			26°18.8	N16°43.3	Polaris	311°42.0	89°22.9	
7	94°36.3	247°10.7	12.0	226°10.1	04.9	248°56.1	30.4	41°21.3	43.3			41°21.3	43.3	Acamar	315°08.8	-40°10.9	
8	109°38.8	262°10.5	13.2	241°10.9	05.3	263°58.1	30.6	56°23.8	43.3			56°23.8	43.3	Menkar	314°02.0	4°12.5	
9	124°41.3	277°10.3	•• 14.3	256°11.7	•• 05.8	279°00.1	•• 30.7	71°26.3	•• 43.2			71°26.3	•• 43.2	Mirfak	308°22.5	49°57.8	
10	139°43.7	292°10.1	15.5	271°12.4	06.3	294°02.2	30.9	86°28.7	43.2			86°28.7	43.2	Aldebaran	290°35.2	16°34.1	
11	154°46.2	307°09.8	16.7	286°13.2	06.7	309°04.2	31.1	101°31.2	43.2			101°31.2	43.2	Rigel	281°00.3	-8°10.0	
12	169°48.6	322°09.6	S12°17.9	301°14.0	S19°07.2	324°06.2	S09°31.3	116°33.7	N16°43.2			116°33.7	N16°43.2	Capella	280°16.3	46°01.4	
13	184°51.1	337°09.4	19.0	316°14.8	07.6	339°08.2	31.4	131°36.2	43.2			131°36.2	43.2	Bellatrix	278°18.9	6°22.6	
14	199°53.6	352°09.2	20.2	331°15.5	08.1	354°10.2	31.6	146°38.7	43.2			146°38.7	43.2	Elnath	277°57.1	28°37.8	
15	214°56.0	7°08.9	•• 21.4	346°16.3	•• 08.5	9°12.2	•• 31.8	161°41.1	•• 43.2			161°41.1	•• 43.2	Alnilam	275°33.9	-1°11.0	
16	229°58.5	22°08.7	22.6	1°17.1	09.0	24°14.2	32.0	176°43.6	43.2			176°43.6	43.2	Betelgeuse	270°48.1	7°24.7	
17	245°01.0	37°08.5	23.7	16°17.8	09.5	39°16.3	32.1	191°46.1	43.1			191°46.1	43.1	Canopus	263°50.9	-52°42.5	
18	260°03.4	52°08.3	S12°24.9	31°18.6	S19°09.9	54°18.3	S09°32.3	206°48.6	N16°43.1			206°48.6	N16°43.1	Sirius	258°23.0	-16°45.3	
19	275°05.9	67°08.0	26.1	46°19.4	10.4	69°20.3	32.5	221°51.1	43.1			221°51.1	43.1	Adhara	255°03.1	-29°00.6	
20	290°08.4	82°07.8	27.2	61°20.1	10.8	84°22.3	32.7	236°53.6	43.1			236°53.6	43.1	Procyon	244°47.1	5°08.9	
21	305°10.8	97°07.6	•• 28.4	76°20.9	•• 11.3	99°24.3	•• 32.8	251°56.0	•• 43.1			251°56.0	•• 43.1	Pollux	243°13.0	27°57.1	
22	320°13.3	112°07.4	29.6	91°21.7	11.7	114°26.3	33.0	266°58.5	43.1			266°58.5	43.1	Avior	234°13.6	-59°36.2	
23	335°15.8	127°07.1	30.7	106°22.4	12.2	129°28.3	33.2	282°01.0	43.1			282°01.0	43.1	Suhail	222°43.8	-43°33.1	
Mer.pass. 00:43			ν -0.2' d 1.2' m -4.08			ν 0.8' d 0.5' m 0.91			ν 2.0' d 0.2' m -1.74			ν 2.5' d -0.0' m 0.05					
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	350°18.2	142°06.9	S12°31.9	121°23.2	S19°12.7	144°30.4	S09°33.4	297°03.5	N16°43.1			297°03.5	N16°43.1	Alpheratz	357°30.5	29°15.4	
1	5°20.7	157°06.7	33.1	136°24.0	13.1	159°32.4	33.6	312°06.0	43.0			312°06.0	43.0	Ankaa	353°03.1	-42°08.4	
2	20°23.1	172°06.5	34.2	151°24.7	13.6	174°34.4	33.7	327°08.4	43.0			327°08.4	43.0	Schedar	349°26.0	56°42.1	
3	35°25.6	187°06.2	•• 35.4	166°25.5	•• 14.0	189°36.4	•• 33.9	342°10.9	•• 43.0			342°10.9	•• 43.0	Diphda	348°43.3	-17°49.2	
4	50°28.1	202°06.0	36.6	181°26.3	14.5	204°38.4	34.1	357°13.4	43.0			357°13.4	43.0	Achernar	335°17.1	-57°04.9	
5	65°30.5	217°05.8	37.7	196°27.0	14.9	219°40.4	34.3	12°15.9	43.0			12°15.9	43.0	Hamal	327°46.6	23°36.2	
6	80°33.0	232°05.6	S12°38.9	211°27.8	S19°15.4	234°42.4	S09°34.4	27°18.4	N16°43.0			27°18.4	N16°43.0	Polaris	311°42.0	89°22.9	
7	95°35.5	247°05.3	40.1	226°28.6	15.8	249°44.5	34.6	42°20.9	43.0			42°20.9	43.0	Acamar	315°08.8	-40°10.9	
8	110°37.9	262°05.1	41.2	241°29.3	16.3	264°46.5	34.8	57°23.4	42.9			57°23.4	42.9	Menkar	314°02.0	4°12.5	
9	125°40.4	277°04.9	•• 42.4	256°30.1	•• 16.7	279°48.5	•• 35.0	72°25.8	•• 42.9			72°25.8	•• 42.9	Mirfak	308°22.5	49°57.8	
10	140°42.9	292°04.6	43.5	271°30.9	17.2	294°50.5	35.1	87°28.3	42.9			87°28.3	42.9	Aldebaran	290°35.2	16°34.1	
11	155°45.3	307°04.4	44.7	286°31.6	17.7	309°52.5	35.3	102°30.8	42.9			102°30.8	42.9	Rigel	281°00.3	-8°10.0	
12	170°47.8	322°04.2	S12°45.9	301°32.4	S19°18.1	324°54.5	S09°35.5	117°33.3	N16°42.9			117°33.3	N16°42.9	Capella	280°16.3	46°01.4	
13	185°50.2	337°03.9	47.0	316°33.2	18.6	339°56.5	35.7	132°35.8	42.9			132°35.8	42.9	Bellatrix	278°18.9	6°22.6	
14	200°52.7	352°03.7	48.2	331°33.9	19.0	354°58.5	35.8	147°38.3	42.9			147°38.3	42.9	Elnath	277°57.1	28°37.8	
15	215°55.2	7°03.5	•• 49.4	346°34.7	•• 19.5	10°00.6	•• 36.0	162°40.8	•• 42.9			162°40.8	•• 42.9	Alnilam	275°33.9	-1°11.0	
16	230°57.6	22°03.3	50.5	1°35.4	19.9	25°02.6	36.2	177°43.2	42.8			177°43.2	42.8	Betelgeuse	270°48.1	7°24.7	
17	246°00.1	37°03.0	51.7	16°36.2	20.4	40°04.6	36.4	192°45.7	42.8			192°45.7	42.8	Canopus	263°50.9	-52°42.5	
18	261°02.6	52°02.8	S12°52.8	31°37.0	S19°20.8	55°06.6	S09°36.6	207°48.2	N16°42.8			207°48.2	N16°42.8	Sirius	258°23.0	-16°45.3	
19	276°05.0	67°02.6	54.0	46°37.7	21.3	70°08.6	36.7	222°50.7	42.8			222°50.7	42.8	Adhara	255°03.1	-29°00.6	
20	291°07.5	82°02.3	55.1	61°38.5	21.7	85°10.6	36.9	237°53.2	42.8			237°53.2	42.8	Procyon	244°47.1	5°08.9	
21	306°10.0	97°02.1	•• 56.3	76°39.2	•• 22.2	100°12.6	•• 37.1	252°55.7	•• 42.8			252°55.7	•• 42.8	Pollux	243°13.0	27°57.1	
22	321°12.4	112°01.9	57.5	91°40.0	22.6	115°14.6	37.3	267°58.2	42.8			267°58.2	42.8	Avior	234°13.6	-59°36.2	
23	336°14.9	127°01.6	58.6	106°40.8	23.1	130°16.6	37.4	283°00.6	42.7			283°00.6	42.7	Suhail	222°43.8	-43°33.1	
Mer.pass. 00:39			ν -0.2' d 1.2' m -4.09			ν 0.8' d 0.5' m 0.92			ν 2.0' d 0.2' m -1.74			ν 2.5' d -0.0' m 0.05					
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	351°17.4	142°01.4	S12°59.8	121°41.5	S19°23.5	145°18.7	S09°37.6	298°03.1	N16°42.7			298°03.1	N16°42.7	Alpheratz	357°30.5	29°15.4	
1	6°19.8	157°01.2	13°00.9	136°42.3	24.0	160°20.7	37.8	313°05.6	42.7			313°05.6	42.7	Ankaa	353°03.1	-42°08.4	
2	21°22.3	172°00.9	02.1	151°43.0	24.4	175°22.7	38.0	328°08.1	42.7			328°08.1	42.7	Schedar	349°26.0	56°42.1	
3	36°24.7	187°00.7	•• 03.2	166°43.8	•• 24.9	190°24.7	•• 38.1	343°10.6	•• 42.7			343°10.6	•• 42.7	Diphda	348°43.3	-17°49.2	
4	51°27.2	202°00.5	04.4	181°44.5	25.3	205°26.7	38.3	358°13.1	42.7			358°13.1	42.7	Achernar	335°17.1	-57°04.9	
5	66°29.7</																

h	Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°44.3	N04°54.4	162°49.0	8.7'	S08°13.5	14.4'	60.6'
1	195°44.6	53.4	177°16.7	8.6'	08°27.8	14.3'	60.6'
2	210°44.8	52.5	191°44.3	8.6'	08°42.2	14.3'	60.6'
3	225°45.0	· · 51.5	206°11.9	8.5'	08°56.4	14.2'	60.6'
4	240°45.2	50.6	220°39.4	8.5'	09°10.6	14.1'	60.6'
5	255°45.4	49.6	235°06.9	8.4'	09°24.8	14.1'	60.6'
6	270°45.7	N04°48.7	249°34.4	8.4'	S09°38.9	14.0'	60.6'
7	285°45.9	47.7	264°01.8	8.4'	09°52.9	14.0'	60.6'
8	300°46.1	46.8	278°29.1	8.3'	10°06.8	13.9'	60.6'
9	315°46.3	· · 45.8	292°56.4	8.3'	10°20.7	13.8'	60.6'
10	330°46.5	44.9	307°23.7	8.2'	10°34.6	13.8'	60.6'
11	345°46.7	43.9	321°50.9	8.2'	10°48.3	13.7'	60.6'
12	0°47.0	N04°43.0	336°18.1	8.1'	S11°02.0	13.6'	60.6'
13	15°47.2	42.0	350°45.2	8.1'	11°15.7	13.6'	60.6'
14	30°47.4	41.1	5°12.3	8.0'	11°29.2	13.5'	60.6'
15	45°47.6	· · 40.1	19°39.3	8.0'	11°42.7	13.4'	60.6'
16	60°47.8	39.2	34°06.2	7.9'	11°56.1	13.3'	60.6'
17	75°48.1	38.2	48°33.2	7.9'	12°09.4	13.2'	60.6'
18	90°48.3	N04°37.3	63°00.0	7.8'	S12°22.7	13.2'	60.6'
19	105°48.5	36.3	77°26.8	7.8'	12°35.9	13.1'	60.6'
20	120°48.7	35.4	91°53.6	7.7'	12°48.9	13.0'	60.5'
21	135°48.9	· · 34.4	106°20.3	7.7'	13°01.9	12.9'	60.5'
22	150°49.1	33.5	120°47.0	7.6'	13°14.8	12.8'	60.5'
23	165°49.4	32.5	135°13.6	7.6'	13°27.7	12.7'	60.5'
SD = 15.9'			SD = 16.5'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°49.6	N04°31.6	149°40.1	7.5'	S13°40.4	12.6'	60.5'
1	195°49.8	30.6	164°06.7	7.5'	13°53.1	12.6'	60.5'
2	210°50.0	29.7	178°33.1	7.4'	14°05.6	12.5'	60.5'
3	225°50.2	· · 28.7	192°59.5	7.3'	14°18.1	12.4'	60.5'
4	240°50.5	27.8	207°25.9	7.3'	14°30.4	12.3'	60.5'
5	255°50.7	26.8	221°52.2	7.2'	14°42.7	12.2'	60.5'
6	270°50.9	N04°25.9	236°18.4	7.2'	S14°54.9	12.1'	60.5'
7	285°51.1	24.9	250°44.6	7.1'	15°06.9	12.0'	60.4'
8	300°51.3	24.0	265°10.7	7.1'	15°18.9	11.9'	60.4'
9	315°51.6	· · 23.0	279°36.8	7.0'	15°30.7	11.8'	60.4'
10	330°51.8	22.1	294°02.8	7.0'	15°42.5	11.6'	60.4'
11	345°52.0	21.1	308°28.8	6.9'	15°54.1	11.5'	60.4'
12	0°52.2	N04°20.2	322°54.8	6.9'	S16°05.7	11.4'	60.4'
13	15°52.4	19.2	337°20.6	6.8'	16°17.1	11.3'	60.4'
14	30°52.7	18.2	351°46.5	6.8'	16°28.4	11.2'	60.4'
15	45°52.9	· · 17.3	6°12.2	6.7'	16°39.6	11.1'	60.3'
16	60°53.1	16.3	20°38.0	6.7'	16°50.7	11.0'	60.3'
17	75°53.3	15.4	35°03.6	6.6'	17°01.7	10.9'	60.3'
18	90°53.5	N04°14.4	49°29.3	6.6'	S17°12.6	10.7'	60.3'
19	105°53.8	13.5	63°54.8	6.5'	17°23.3	10.6'	60.3'
20	120°54.0	12.5	78°20.3	6.5'	17°33.9	10.5'	60.3'
21	135°54.2	· · 11.6	92°45.8	6.4'	17°44.4	10.4'	60.2'
22	150°54.4	10.6	107°11.2	6.4'	17°54.8	10.3'	60.2'
23	165°54.6	09.7	121°36.6	6.3'	18°05.0	10.1'	60.2'
SD = 15.9'			SD = 16.5'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°54.9	N04°08.7	136°01.9	6.3'	S18°15.2	10.0'	60.2'
1	195°55.1	07.8	150°27.2	6.2'	18°25.2	9.9'	60.2'
2	210°55.3	06.8	164°52.5	6.2'	18°35.0	9.7'	60.2'
3	225°55.5	· · 05.9	179°17.6	6.1'	18°44.8	9.6'	60.1'
4	240°55.7	04.9	193°42.8	6.1'	18°54.4	9.5'	60.1'
5	255°56.0	03.9	208°07.9	6.0'	19°03.9	9.4'	60.1'
6	270°56.2	N04°03.0	222°32.9	6.0'	S19°13.2	9.2'	60.1'
7	285°56.4	02.0	236°57.9	6.0'	19°22.5	9.1'	60.1'
8	300°56.6	01.1	251°22.9	5.9'	19°31.5	8.9'	60.0'
9	315°56.8	04°00.1	265°47.8	5.9'	19°40.5	8.8'	60.0'
10	330°57.1	03°59.2	280°12.7	5.8'	19°49.3	8.7'	60.0'
11	345°57.3	58.2	294°37.5	5.8'	19°58.0	8.5'	60.0'
12	0°57.5	N03°57.3	309°02.3	5.8'	S20°06.5	8.4'	60.0'
13	15°57.7	56.3	323°27.1	5.7'	20°14.9	8.3'	59.9'
14	30°57.9	55.3	337°51.8	5.7'	20°23.1	8.1'	59.9'
15	45°58.2	· · 54.4	352°16.5	5.6'	20°31.3	8.0'	59.9'
16	60°58.4	53.4	6°41.1	5.6'	20°39.2	7.8'	59.9'
17	75°58.6	52.5	21°05.7	5.6'	20°47.0	7.7'	59.9'
18	90°58.8	N03°51.5	35°30.3	5.5'	S20°54.7	7.5'	59.8'
19	105°59.1	50.6	49°54.9	5.5'	21°02.3	7.4'	59.8'
20	120°59.3	49.6	64°19.4	5.5'	21°09.6	7.2'	59.8'
21	135°59.5	· · 48.7	78°43.8	5.4'	21°16.9	7.1'	59.8'
22	150°59.7	47.7	93°08.3	5.4'	21°24.0	6.9'	59.7'
23	165°59.9	46.7	107°32.7	5.4'	21°30.9	6.8'	59.7'
SD = 15.9'			SD = 16.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:31	03:35	04:50	19:00	20:14	22:13
N 70°	02:14	03:51	04:57	18:53	19:59	21:33
68°	02:42	04:04	05:03	18:48	19:46	21:07
66°	03:03	04:14	05:08	18:43	19:37	20:47
64°	03:19	04:23	05:12	18:39	19:28	20:31
62°	03:32	04:30	05:16	18:36	19:21	20:19
60°	03:43	04:36	05:19	18:33	19:15	20:08
N 58°	03:52	04:41	05:22	18:30	19:10	19:59
56°	04:00	04:46	05:24	18:28	19:05	19:52
54°	04:07	04:50	05:26	18:26	19:01	19:45
52°	04:13	04:54	05:28	18:24	18:58	19:39
50°	04:18	04:57	05:30	18:22	18:55	19:34
45°	04:29	05:04	05:34	18:18	18:48	19:23
N 40°	04:38	05:10	05:37	18:15	18:42	19:14
35°	04:45	05:15	05:40	18:12	18:38	19:08
30°	04:50	05:18	05:42	18:10	18:34	19:02
20°	04:59	05:24	05:47	18:06	18:28	18:54
N 10°	05:05	05:29	05:50	18:03	18:24	18:48
0°	05:08	05:33	05:53	18:00	18:20	18:45
S 10°	05:11	05:35	05:56	17:57	18:18	18:42
20°	05:12	05:37	06:00	17:54	18:16	18:41
30°	05:11	05:39	06:03	17:50	18:14	18:42
35°	05:10	05:40	06:05	17:49	18:14	18:43
40°	05:09	05:40	06:07	17:47	18:14	18:45
45°	05:06	05:40	06:10	17:44	18:13	18:47
S 50°	05:03	05:40	06:13	17:41	18:14	18:51
52°	05:01	05:40	06:14	17:40	18:14	18:53
54°	04:59	05:40	06:15	17:39	18:14	18:55
56°	04:57	05:40	06:17	17:37	18:14	18:57
58°	04:54	05:40	06:19	17:35	18:15	19:00
S 60°	04:51	05:39	06:21	17:34	18:15	19:04

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	09:53	12:51	■	16:57	15:57	■
N 70°	09:34	12:02	■	17:18	16:47	■
68°	09:19	11:31	14:01	17:34	17:20	16:51
66°	09:07	11:08	13:14	17:48	17:43	17:38
64°	08:58	10:50	12:44	17:59	18:02	18:09
62°	08:49	10:36	12:21	18:09	18:18	18:33
60°	08:42	10:24	12:03	18:17	18:31	18:51
N 58°	08:36	10:13	11:48	18:25	18:42	19:07
56°	08:31	10:04	11:35	18:31	18:52	19:20
54°	08:26	09:56	11:24	18:37	19:01	19:32
52°	08:21	09:49	11:14	18:42	19:09	19:42
50°	08:17	09:42	11:06	18:47	19:16	19:51
45°	08:09	09:29	10:47	18:58	19:31	20:11
N 40°	08:01	09:17	10:32	19:06	19:44	20:27
35°	07:55	09:08	10:19	19:14	19:54	20:40
30°	07:50	08:59	10:08	19:20	20:04	20:52
20°	07:41	08:45	09:50	19:32	20:20	21:12
N 10°	07:33	08:32	09:33	19:42	20:34	21:30
0°	07:25	08:21	09:18	19:51	20:48	21:46
S 10°	07:18	08:09	09:03	20:01	21:01	22:03
20°	07:10	07:57	08:47	20:11	21:16	22:20
30°	07:02	07:43	08:29	20:23	21:32	22:41
35°	06:57	07:35	08:18	20:30	21:42	22:53
40°	06:51	07:26	08:06	20:37	21:53	23:06
45°	06:44	07:16	07:52	20:46	22:06	23:23
S 50°	06:37	07:03	07:35	20:57	22:22	23:43
52°	06:33	06:57	07:27	21:03	22:29	23:53
54°	06:29	06:51	07:18	21:08	22:38	· · ·
56°	06:25	06:44	07:08	21:15	22:47	· · ·
58°	06:20	06:36	06:57	21:22	22:58	· · ·
S 60°	06:15	06:27	06:44	21:30	23:10	· · ·

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	00 ^h	Mer. Pass	Upper	Lower	2-4
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	4-18%
10	02:57	03:08	11:57	13:38	01:11	
11	03:18	03:29	11:57	14:34	02:06	
12	03:39	03:50	11:56	15:32	03:03	

September 13, 14, 15 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	352°16.5	141°55.8	S13°27.4	121°59.6	S19°34.3	146°06.9	S09°41.9	299°02.9	N16°42.4	Alpheratz	357°30.4	29°15.5											
1	7°19.0	156°55.5	28.5	137°00.4	34.7	161°08.9	42.0	314°05.4	42.4	Ankaa	353°03.1	-42°08.4											
2	22°21.4	171°55.3	29.7	152°01.1	35.2	176°10.9	42.2	329°07.9	42.3	Schedar	349°26.0	56°42.1											
3	37°23.9	186°55.1	· · 30.8	167°01.9	· · 35.6	191°12.9	· · 42.4	344°10.4	· · 42.3	Diphda	348°43.2	-17°49.2											
4	52°26.3	201°54.8	32.0	182°02.6	36.1	206°14.9	42.6	359°12.9	42.3	Achernar	335°17.1	-57°04.9											
5	67°28.8	216°54.6	33.1	197°03.4	36.5	221°16.9	42.8	14°15.4	42.3	Hamal	327°46.6	23°36.2											
6	82°31.3	231°54.3	S13°34.2	212°04.1	S19°37.0	236°18.9	S09°42.9	29°17.9	N16°42.3	Polaris	311°40.3	89°23.0											
7	97°33.7	246°54.1	35.4	227°04.8	37.4	251°20.9	43.1	44°20.3	42.3	Acamar	315°08.7	-40°10.9											
8	112°36.2	261°53.9	36.5	242°05.6	37.8	266°23.0	43.3	59°22.8	42.3	Menkar	314°02.0	4°12.5											
9	127°38.7	276°53.6	· · 37.7	257°06.3	· · 38.3	281°25.0	· · 43.5	74°25.3	· · 42.2	Mirfak	308°22.5	49°57.8											
10	142°41.1	291°53.4	38.8	272°07.1	38.7	296°27.0	43.6	89°27.8	42.2	Aldebaran	290°35.2	16°34.1											
11	157°43.6	306°53.1	39.9	287°07.8	39.2	311°29.0	43.8	104°30.3	42.2	Rigel	281°00.2	-8°10.0											
12	172°46.1	321°52.9	S13°41.1	302°08.6	S19°39.6	326°31.0	S09°44.0	119°32.8	N16°42.2	Capella	280°16.3	46°01.4											
13	187°48.5	336°52.7	42.2	317°09.3	40.1	341°33.0	44.2	134°35.3	42.2	Bellatrix	278°18.8	6°22.6											
14	202°51.0	351°52.4	43.4	332°10.1	40.5	356°35.0	44.4	149°37.8	42.2	Elnath	277°57.1	28°37.8											
15	217°53.5	6°52.2	· · 44.5	347°10.8	· · 41.0	11°37.0	· · 44.5	164°40.3	· · 42.1	Alnilam	275°33.9	-1°11.0											
16	232°55.9	21°51.9	45.6	2°11.6	41.4	26°39.0	44.7	179°42.8	42.1	Betelgeuse	270°48.0	7°24.7											
17	247°58.4	36°51.7	46.8	17°12.3	41.8	41°41.0	44.9	194°45.3	42.1	Canopus	263°50.8	-52°42.5											
18	263°00.8	51°51.5	S13°47.9	32°13.0	S19°42.3	56°43.0	S09°45.1	209°47.8	N16°42.1	Sirius	258°23.0	-16°45.3											
19	278°03.3	66°51.2	49.0	47°13.8	42.7	71°45.0	45.2	224°50.3	42.1	Adhara	255°03.0	-29°00.6											
20	293°05.8	81°51.0	50.2	62°14.5	43.2	86°47.0	45.4	239°52.8	42.1	Procyon	244°47.1	5°08.9											
21	308°08.2	96°50.7	· · 51.3	77°15.3	· · 43.6	101°49.0	· · 45.6	254°55.3	· · 42.0	Pollux	243°12.9	27°57.1											
22	323°10.7	111°50.5	52.5	92°16.0	44.0	116°51.0	45.8	269°57.8	42.0	Avior	234°13.5	-59°36.1											
23	338°13.2	126°50.2	53.6	107°16.8	44.5	131°53.1	46.0	285°00.3	42.0	Suhail	222°43.8	-43°33.1											
Mer.pass. 00:31				ν -0.2' d 1.1' m -4.10				ν 0.7' d 0.4' m 0.92				ν 2.0' d 0.2' m -1.73				ν 2.5' d -0.0' m 0.04							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	353°15.6	141°50.0	S13°54.7	122°17.5	S19°44.9	146°55.1	S09°46.1	300°02.8	N16°42.0										
1	8°18.1	156°49.8	55.9	137°18.2	45.4	161°57.1	46.3	315°05.3	42.0										
2	23°20.6	171°49.5	57.0	152°19.0	45.8	176°59.1	46.5	330°07.8	42.0										
3	38°23.0	186°49.3	· · 58.1	167°19.7	· · 46.2	192°01.1	· · 46.7	345°10.3	· · 41.9										
4	53°25.5	201°49.0	13°59.2	182°20.5	46.7	207°03.1	46.8	0°12.8	41.9										
5	68°28.0	216°48.8	14°00.4	197°21.2	47.1	222°05.1	47.0	15°15.3	41.9										
6	83°30.4	231°48.5	S14°01.5	212°21.9	S19°47.6	237°07.1	S09°47.2	30°17.8	N16°41.9										
7	98°32.9	246°48.3	02.6	227°22.7	48.0	252°09.1	47.4	45°20.3	41.9										
8	113°35.3	261°48.0	03.8	242°23.4	48.4	267°11.1	47.6	60°22.7	41.9										
9	128°37.8	276°47.8	· · 04.9	257°24.2	· · 48.9	282°13.1	· · 47.7	75°25.2	· · 41.9										
10	143°40.3	291°47.6	06.0	272°24.9	49.3	297°15.1	47.9	90°27.7	41.8										
11	158°42.7	306°47.3	07.2	287°25.6	49.8	312°17.1	48.1	105°30.2	41.8										
12	173°45.2	321°47.1	S14°08.3	302°26.4	S19°50.2	327°19.1	S09°48.3	120°32.7	N16°41.8										
13	188°47.7	336°46.8	09.4	317°27.1	50.6	342°21.1	48.5	135°35.2	41.8										
14	203°50.1	351°46.6	10.5	332°27.8	51.1	357°23.1	48.6	150°37.7	41.8										
15	218°52.6	6°46.3	· · 11.7	347°28.6	· · 51.5	12°25.1	· · 48.8	165°40.2	· · 41.8										
16	233°55.1	21°46.1	12.8	2°29.3	51.9	27°27.1	49.0	180°42.7	41.7										
17	248°57.5	36°45.8	13.9	17°30.0	52.4	42°29.1	49.2	195°45.2	41.7										
18	264°00.0	51°45.6	S14°15.0	32°30.8	S19°52.8	57°31.1	S09°49.3	210°47.7	N16°41.7										
19	279°02.4	66°45.3	16.2	47°31.5	53.2	72°33.1	49.5	225°50.2	41.7										
20	294°04.9	81°45.1	17.3	62°32.3	53.7	87°35.2	49.7	240°52.7	41.7										
21	309°07.4	96°44.8	· · 18.4	77°33.0	· · 54.1	102°37.2	· · 49.9	255°55.2	· · 41.6										
22	324°09.8	111°44.6	19.5	92°33.7	54.6	117°39.2	50.1	270°57.7	41.6										
23	339°12.3	126°44.3	20.6	107°34.5	55.0	132°41.2	50.2	286°00.2	41.6										
Mer.pass. 00:27				ν -0.2' d 1.1' m -4.11				ν 0.7' d 0.4' m 0.93				ν 2.0' d 0.2' m -1.73				ν 2.5' d -0.0' m 0.03			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	354°14.8	141°44.1	S14°21.8	122°35.2	S19°55.4	147°43.2	S09°50.4	301°02.7	N16°41.6										
1	9°17.2	156°43.8	22.9	137°35.9	55.9	162°45.2	50.6	316°05.2	41.6										
2	24°19.7	171°43.6	24.0	152°36.7	56.3	177°47.2	50.8	331°07.7	41.6										
3	39°22.2	186°43.3	· · 25.1	167°37.4	· · 56.7	192°49.2	· · 51.0	346°10.3	· · 41.5										
4	54°24.6	201°43.1	26.2	182°38.1	57.2	207°51.2	51.1	1°12.8	41.5										
5	69°27.1	216°42.8	27.4	197°38.8	57.6	222°53.2	51.3	16°15.3	41.5										
6	84°29.6	231°42.6	S14°28.5	212°39.6	S19°58.0	237°55.2	S09°51.5	31°17.8	N16°41.5										
7	99°32.0	246°42.3	29.6	227°40.3	58.5	252°57.2	51.7	46°20.3	41.5										
8	114°34.5	261°42.1	30.7	242°41.0	58.9	267°59.2	51.8	61°22.8	41.5										
9	129°36.9	276°41.8	· · 31.8	257°41.8	· · 59.3	283°01.2	· · 52.0	76°25.3	· · 41.4										
10	144°39.4	291°41.6	33.0	272°42.5	19°59.8	298°03.2	52.2	91°27.8	41.4										
11	159°41.9	306°41.3	34.1	287°43.2	20°00.2	313°05.2	52.4	106°30.3	41.4										
12	174°44.3	321°41.1	S14°35.2	302°44.0	S20°00.6	328°07.2	S09°52.6	121°32.8	N16°41.4										
13	189°46.8	336°40.8	36.3	317°44.7	01.1	343°09.2	52.7	136°35.3	41.4										
14	204°49.3	351°40.6	37.4	332°45.4	01.5	358°11.2	52.9	151°37.8	41.4										
15	219°51.7	6°40.3	· · 38.5	347°46.1	· · 01.9	13°13.2	· · 53.1	166°40.3	· · 41.3										
16	234°54.2	21°40.1	39.6	2°46.9	02.3	28°15.2	53.3	181°42.8	41.3										
17	249°56.7	36°39.8	40.7	17°47.6	02.8	43°17.2	53.5	196°45.3	41.3										
18	264°59.1	51°39.6	S14°41.9	32°48.3	S20°03.2	58°19.2	S09°53.6	211°47.8	N16°41.3										
19	280°01.6	66°39.3	43.0	47°49.0	03.6	73°21.2	53.8	226°50.3	41.3										
20	295°04.1	81°39.1	44.1	62°49.8	04.1	88°23.2	54.0	241°52.8	41.2										
21	310°06.5	96°38.8	· · 45.2	77°50.5	· · 04.5	103°25.2	· · 54.2	256°55.3	· · 41.2										
22	325°09.0	111°38.6	46.3	92°51.2	04.9	118°27.2	54.4	271°57.8	41.2										
23	340°11.4	126°38.3	47.4	107°51.9	05.4	133°29.2	54.5	287°00.3	41.2										
Mer.pass. 00:23				ν -0.2' d 1.1' m -4.12				ν 0.7' d 0.4' m 0.93				ν 2.0' d 0.2' m -1.73				ν 2.5' d -0.0' m 0.03			

Sep 13 Thu	SHA	Mer.pass
Venus	149°39.3	14:33
Mars	129°43.1	15:51
Jupiter	153°50.4	14:14
Saturn	306°46.4	04:03

Sep 14 Fri	SHA	Mer.pass
Venus	148°34.4	14:33
Mars	129°01.9	15:50
Jupiter	153°39.4	14:10
Saturn	306°47.1	03:59

Sep 15 Sat	SHA	Mer.pass
Venus	147°29.3	14:33
Mars	128°20.4	15:49
Jupiter	153°28.4	14:07
Saturn	306°48.0	03:55

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	181°00.2	N03°45.8	121°57.1	5.4'	S21°37.7	6.6'	59.7'
1	196°00.4	44.8	136°21.5	5.3'	21°44.3	6.5'	59.7'
2	211°00.6	43.9	150°45.8	5.3'	21°50.8	6.3'	59.6'
3	226°00.8	· · 42.9	165°10.1	5.3'	21°57.2	6.2'	59.6'
4	241°01.0	42.0	179°34.4	5.3'	22°03.4	6.0'	59.6'
5	256°01.3	41.0	193°58.7	5.3'	22°09.4	5.9'	59.6'
6	271°01.5	N03°40.0	208°23.0	5.2'	S22°15.3	5.7'	59.5'
7	286°01.7	39.1	222°47.2	5.2'	22°21.0	5.6'	59.5'
8	301°01.9	38.1	237°11.4	5.2'	22°26.6	5.4'	59.5'
9	316°02.2	· · 37.2	251°35.6	5.2'	22°32.0	5.3'	59.5'
10	331°02.4	36.2	265°59.8	5.2'	22°37.2	5.1'	59.4'
11	346°02.6	35.3	280°24.0	5.2'	22°42.3	4.9'	59.4'
12	1°02.8	N03°34.3	294°48.1	5.1'	S22°47.3	4.8'	59.4'
13	16°03.0	33.3	309°12.3	5.1'	22°52.1	4.6'	59.4'
14	31°03.3	32.4	323°36.4	5.1'	22°56.7	4.5'	59.3'
15	46°03.5	· · 31.4	338°00.5	5.1'	23°01.2	4.3'	59.3'
16	61°03.7	30.5	352°24.7	5.1'	23°05.5	4.2'	59.3'
17	76°03.9	29.5	6°48.8	5.1'	23°09.6	4.0'	59.3'
18	91°04.2	N03°28.5	21°12.9	5.1'	S23°13.6	3.8'	59.2'
19	106°04.4	27.6	35°37.0	5.1'	23°17.5	3.7'	59.2'
20	121°04.6	26.6	50°01.2	5.1'	23°21.2	3.5'	59.2'
21	136°04.8	· · 25.7	64°25.3	5.1'	23°24.7	3.4'	59.2'
22	151°05.0	24.7	78°49.4	5.1'	23°28.0	3.2'	59.1'
23	166°05.3	23.8	93°13.5	5.1'	23°31.2	3.0'	59.1'
SD = 15.9'			SD = 16.3'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	181°05.5	N03°22.8	107°37.7	5.1'	S23°34.3	2.9'	59.1'
1	196°05.7	21.8	122°01.8	5.2'	23°37.2	2.7'	59.0'
2	211°05.9	20.9	136°26.0	5.2'	23°39.9	2.6'	59.0'
3	226°06.2	· · 19.9	150°50.1	5.2'	23°42.5	2.4'	59.0'
4	241°06.4	19.0	165°14.3	5.2'	23°44.9	2.2'	59.0'
5	256°06.6	18.0	179°38.5	5.2'	23°47.1	2.1'	58.9'
6	271°06.8	N03°17.0	194°02.7	5.2'	S23°49.2	1.9'	58.9'
7	286°07.0	16.1	208°26.9	5.2'	23°51.1	1.8'	58.9'
8	301°07.3	15.1	222°51.2	5.3'	23°52.9	1.6'	58.9'
9	316°07.5	· · 14.2	237°15.4	5.3'	23°54.5	1.4'	58.8'
10	331°07.7	13.2	251°39.7	5.3'	23°55.9	1.3'	58.8'
11	346°07.9	12.2	266°04.0	5.3'	23°57.2	1.1'	58.8'
12	1°08.2	N03°11.3	280°28.3	5.4'	S23°58.4	1.0'	58.7'
13	16°08.4	10.3	294°52.7	5.4'	23°59.3	0.8'	58.7'
14	31°08.6	09.4	309°17.1	5.4'	24°00.2	0.7'	58.7'
15	46°08.8	· · 08.4	323°41.5	5.5'	24°00.8	0.5'	58.7'
16	61°09.0	07.4	338°06.0	5.5'	24°01.3	0.3'	58.6'
17	76°09.3	06.5	352°30.5	5.5'	24°01.7	0.2'	58.6'
18	91°09.5	N03°05.5	6°55.0	5.6'	S24°01.9	0.0'	58.6'
19	106°09.7	04.6	21°19.5	5.6'	24°01.9	-0.1'	58.5'
20	121°09.9	03.6	35°44.1	5.6'	24°01.8	-0.3'	58.5'
21	136°10.2	· · 02.6	50°08.8	5.7'	24°01.5	-0.4'	58.5'
22	151°10.4	01.7	64°33.4	5.7'	24°01.1	-0.6'	58.5'
23	166°10.6	00.7	78°58.1	5.8'	24°00.5	-0.7'	58.4'
SD = 15.9'			SD = 16.1'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	181°10.8	N02°59.7	93°22.9	5.8'	S23°59.7	-0.9'	58.4'
1	196°11.0	58.8	107°47.7	5.8'	23°58.9	-1.0'	58.4'
2	211°11.3	57.8	122°12.5	5.9'	23°57.8	-1.2'	58.3'
3	226°11.5	· · 56.9	136°37.4	5.9'	23°56.6	-1.3'	58.3'
4	241°11.7	55.9	151°02.4	6.0'	23°55.3	-1.5'	58.3'
5	256°11.9	54.9	165°27.4	6.0'	23°53.8	-1.6'	58.3'
6	271°12.2	N02°54.0	179°52.4	6.1'	S23°52.1	-1.8'	58.2'
7	286°12.4	53.0	194°17.5	6.2'	23°50.3	-1.9'	58.2'
8	301°12.6	52.0	208°42.7	6.2'	23°48.4	-2.1'	58.2'
9	316°12.8	· · 51.1	223°07.9	6.3'	23°46.3	-2.2'	58.1'
10	331°13.1	50.1	237°33.2	6.3'	23°44.1	-2.4'	58.1'
11	346°13.3	49.2	251°58.5	6.4'	23°41.7	-2.5'	58.1'
12	1°13.5	N02°48.2	266°23.9	6.4'	S23°39.2	-2.7'	58.1'
13	16°13.7	47.2	280°49.3	6.5'	23°36.5	-2.8'	58.0'
14	31°13.9	46.3	295°14.8	6.6'	23°33.7	-3.0'	58.0'
15	46°14.2	· · 45.3	309°40.4	6.6'	23°30.7	-3.1'	58.0'
16	61°14.4	44.3	324°06.0	6.7'	23°27.6	-3.2'	57.9'
17	76°14.6	43.4	338°31.7	6.8'	23°24.4	-3.4'	57.9'
18	91°14.8	N02°42.4	352°57.5	6.8'	S23°21.0	-3.5'	57.9'
19	106°15.1	41.5	7°23.3	6.9'	23°17.5	-3.7'	57.9'
20	121°15.3	40.5	21°49.2	7.0'	23°13.8	-3.8'	57.8'
21	136°15.5	· · 39.5	36°15.2	7.0'	23°10.0	-3.9'	57.8'
22	151°15.7	38.6	50°41.2	7.1'	23°06.1	-4.1'	57.8'
23	166°16.0	37.6	65°07.3	7.2'	23°02.1	-4.2'	57.7'
SD = 15.9'			SD = 15.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:01	03:51	05:04	18:44	19:56	21:42
N 70°	02:35	04:05	05:09	18:39	19:43	21:11
68°	02:58	04:16	05:14	18:35	19:33	20:49
66°	03:16	04:24	05:18	18:31	19:24	20:32
64°	03:30	04:32	05:21	18:28	19:17	20:18
62°	03:41	04:38	05:24	18:26	19:11	20:07
60°	03:51	04:43	05:26	18:23	19:06	19:58
N 58°	03:59	04:48	05:28	18:21	19:01	19:50
56°	04:06	04:52	05:30	18:20	18:57	19:43
54°	04:13	04:56	05:32	18:18	18:54	19:37
52°	04:18	04:59	05:33	18:17	18:51	19:31
50°	04:23	05:02	05:35	18:15	18:48	19:27
45°	04:33	05:08	05:38	18:12	18:42	19:19
N 40°	04:41	05:13	05:40	18:10	18:37	19:09
35°	04:47	05:17	05:42	18:08	18:33	19:03
30°	04:52	05:20	05:44	18:06	18:30	18:58
20°	04:59	05:25	05:47	18:03	18:25	18:51
N 10°	05:04	05:29	05:50	18:01	18:22	18:46
0°	05:07	05:32	05:52	17:59	18:19	18:43
S 10°	05:09	05:33	05:54	17:57	18:18	18:42
20°	05:09	05:35	05:57	17:54	18:16	18:42
30°	05:08	05:35	05:59	17:52	18:16	18:44
35°	05:06	05:35	06:01	17:51	18:16	18:45
40°	05:04	05:35	06:02	17:49	18:16	18:48
45°	05:01	05:35	06:04	17:48	18:17	18:51
S 50°	04:56	05:34	06:06	17:46	18:18	18:56
52°	04:54	05:33	06:07	17:45	18:19	18:58
54°	04:52	05:33	06:08	17:44	18:19	19:00
56°	04:49	05:32	06:09	17:43	18:20	19:03
58°	04:46	05:31	06:10	17:42	18:21	19:07
S 60°	04:42	05:30	06:12	17:41	18:22	19:11

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	15:26	■	17:38	17:31	■	19:26
64°	14:32	15:55	16:39	18:26	19:07	20:25
62°	13:59	15:16	16:05	18:59	19:46	20:58
60°	13:35	14:49	15:40	19:23	20:13	21:23
N 58°	13:15	14:27	15:20	19:43	20:35	21:43
56°	12:59	14:10	15:03	19:59	20:52	21:59
54°	12:46	13:55	14:49	20:13	21:07	22:13
52°	12:34	13:42	14:37	20:25	21:20	22:25
50°	12:23	13:31	14:26	20:36	21:31	22:36
45°	12:01	13:07	14:03	20:59	21:55	22:58
N 40°	11:44	12:49	13:45	21:17	22:14	23:16
35°	11:29	12:33	13:30	21:32	22:29	23:31
30°	11:16	12:19	13:16	21:45	22:43	23:44
20°	10:54	11:56	12:54	22:08	23:06	· · ·
N 10°	10:35	11:36	12:34	22:28	23:27	· · ·
0°	10:17	11:17	12:16	22:46	23:45	· · ·
S 10°	10:00	10:58	11:57	23:04	· · ·	00:04
20°	09:41	10:38	11:37	23:24	· · ·	00:24
30°	09:20	10:15	11:15	23:47	· · ·	00:47
35°	09:07	10:02	11:01	· · ·	00:00	01:01
40°	08:53	09:46	10:46	· · ·	00:15	01:16
45°	08:36	09:28	10:28	· · ·	00:33	01:35
S 50°	08:15	09:05	10:05	· · ·	00:56	01:58
52°	08:05	08:54	09:54	· · ·	01:07	02:09
54°	07:54	08:41	09:41	00:03	01:19	02:21
56°	07:41	08:27	09:27	00:16	01:34	02:36
58°	07:27	08:10	09:10	00:30	01:50	02:53
S 60°	07:09	07:50	08:50	00:47	02:10	03:13

Day	Sun		Mer.	Moon		Age
	Eqn. of Time		Pass	Upper	Lower	5-7
	00 ^h	12 ^h	hh:mm	hh:mm	hh:mm	27-49%
13	04:01	04:11	11:56	16:32	04:02	
14	04:22	04:33	11:55	17:31	05:01	
15	04:43	04:54	11:55	18:29	06:01	

September 16, 17, 18 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	355°13.9	141°38.0	S14°48.5	122°52.7	S20°05.8	148°31.2	S09°54.7	302°02.8	N16°41.2	Alpheratz	357°30.4 29°15.5
1	10°16.4	156°37.8	49.6	137°53.4	06.2	163°33.2	54.9	317°05.3	41.2	Ankaa	353°03.1 -42°08.4
2	25°18.8	171°37.5	50.7	152°54.1	06.6	178°35.2	55.1	332°07.8	41.1	Schedar	349°26.0 56°42.1
3	40°21.3	186°37.3	· · 51.8	167°54.8	· · 07.1	193°37.2	· · 55.2	347°10.4	· · 41.1	Diphda	348°43.2 -17°49.2
4	55°23.8	201°37.0	52.9	182°55.6	07.5	208°39.2	55.4	2°12.9	41.1	Achernar	335°17.1 -57°04.9
5	70°26.2	216°36.8	54.1	197°56.3	07.9	223°41.2	55.6	17°15.4	41.1	Hamal	327°46.6 23°36.2
6	85°28.7	231°36.5	S14°55.2	212°57.0	S20°08.4	238°43.2	S09°55.8	32°17.9	N16°41.1	Polaris	311°38.7 89°23.0
7	100°31.2	246°36.3	56.3	227°57.7	08.8	253°45.2	56.0	47°20.4	41.0	Acamar	315°08.7 -40°10.9
8	115°33.6	261°36.0	57.4	242°58.5	09.2	268°47.2	56.1	62°22.9	41.0	Menkar	314°02.0 4°12.5
9	130°36.1	276°35.7	· · 58.5	257°59.2	· · 09.6	283°49.2	· · 56.3	77°25.4	· · 41.0	Mirfak	308°22.5 49°57.8
10	145°38.6	291°35.5	14°59.6	272°59.9	10.1	298°51.2	56.5	92°27.9	41.0	Aldebaran	290°35.2 16°34.1
11	160°41.0	306°35.2	15°00.7	288°00.6	10.5	313°53.2	56.7	107°30.4	41.0	Rigel	281°00.2 -8°10.0
12	175°43.5	321°35.0	S15°01.8	303°01.3	S20°10.9	328°55.2	S09°56.9	122°32.9	N16°41.0	Capella	280°16.2 46°01.4
13	190°45.9	336°34.7	02.9	318°02.1	11.3	343°57.2	57.0	137°35.4	40.9	Bellatrix	278°18.8 6°22.6
14	205°48.4	351°34.5	04.0	333°02.8	11.8	358°59.2	57.2	152°37.9	40.9	Elnath	277°57.1 28°37.8
15	220°50.9	6°34.2	· · 05.1	348°03.5	· · 12.2	14°01.2	· · 57.4	167°40.4	· · 40.9	Alnilam	275°33.9 -1°11.0
16	235°53.3	21°33.9	06.2	3°04.2	12.6	29°03.2	57.6	182°43.0	40.9	Betelgeuse	270°48.0 7°24.7
17	250°55.8	36°33.7	07.3	18°04.9	13.0	44°05.2	57.8	197°45.5	40.9	Canopus	263°50.8 -52°42.5
18	265°58.3	51°33.4	S15°08.4	33°05.7	S20°13.5	59°07.2	S09°57.9	212°48.0	N16°40.8	Sirius	258°23.0 -16°45.3
19	281°00.7	66°33.2	09.5	48°06.4	13.9	74°09.2	58.1	227°50.5	40.8	Adhara	255°03.0 -29°00.6
20	296°03.2	81°32.9	10.6	63°07.1	14.3	89°11.2	58.3	242°53.0	40.8	Procyon	244°47.0 5°08.9
21	311°05.7	96°32.6	· · 11.7	78°07.8	· · 14.7	104°13.2	· · 58.5	257°55.5	· · 40.8	Pollux	243°12.9 27°57.1
22	326°08.1	111°32.4	12.8	93°08.5	15.2	119°15.2	58.7	272°58.0	40.8	Avior	234°13.5 -59°36.1
23	341°10.6	126°32.1	13.9	108°09.2	15.6	134°17.2	58.8	288°00.5	40.7	Suhail	222°43.8 -43°33.1
Mer.pass. 00:19		ν -0.3' d 1.1' m -4.12		ν 0.7' d 0.4' m 0.94		ν 2.0' d 0.2' m -1.72		ν 2.5' d -0.0' m 0.02		Miaplacidus	221°38.0 -69°50.3
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	Alphard	217°44.3 -8°47.3
0	356°13.1	141°31.9	S15°15.0	123°10.0	S20°16.0	149°19.2	S09°59.0	303°03.0	N16°40.7	Regulus	207°30.7 11°49.3
1	11°15.5	156°31.6	16.1	138°10.7	16.4	164°21.2	59.2	318°05.5	40.7	Dubhe	193°37.3 61°35.3
2	26°18.0	171°31.3	17.1	153°11.4	16.9	179°23.2	59.4	333°08.0	40.7	Denebola	182°21.5 14°24.4
3	41°20.4	186°31.1	· · 18.2	168°12.1	· · 17.3	194°25.2	· · 59.6	348°10.6	· · 40.7	Gienah	175°40.0 -17°42.4
4	56°22.9	201°30.8	19.3	183°12.8	17.7	209°27.2	59.7	3°13.1	40.6	Acrux	172°56.3 -63°15.9
5	71°25.4	216°30.6	20.4	198°13.5	18.1	224°29.2	09°59.9	18°15.6	40.6	Gacrux	171°47.8 -57°16.9
6	86°27.8	231°30.3	S15°21.5	213°14.2	S20°18.6	239°31.2	S10°00.1	33°18.1	N16°40.6	Alioth	166°10.4 55°48.0
7	101°30.3	246°30.0	22.6	228°15.0	19.0	254°33.2	00.3	48°20.6	40.6	Spica	158°18.6 -11°18.9
8	116°32.8	261°29.8	23.7	243°15.7	19.4	269°35.2	00.5	63°23.1	40.6	Alkaid	152°49.6 49°10.1
9	131°35.2	276°29.5	· · 24.8	258°16.4	· · 19.8	284°37.2	· · 00.6	78°25.6	· · 40.6	Hadar	148°30.9 -60°31.1
10	146°37.7	291°29.2	25.9	273°17.1	20.2	299°39.2	00.8	93°28.1	40.5	Menkent	147°53.4 -36°31.0
11	161°40.2	306°29.0	27.0	288°17.8	20.7	314°41.2	01.0	108°30.7	40.5	Arcturus	145°44.7 19°01.8
12	176°42.6	321°28.7	S15°28.1	303°18.5	S20°21.1	329°43.2	S10°01.2	123°33.2	N16°40.5	Rigel Kent.	139°35.4 -60°57.6
13	191°45.1	336°28.4	29.1	318°19.2	21.5	344°45.2	01.4	138°35.7	40.5	Kochab	137°21.0 74°02.2
14	206°47.5	351°28.2	30.2	333°19.9	21.9	359°47.2	01.5	153°38.2	40.5	Zuben'ubi	136°52.0 -16°09.8
15	221°50.0	6°27.9	· · 31.3	348°20.7	· · 22.3	14°49.1	· · 01.7	168°40.7	· · 40.4	Alphecca	126°00.8 26°37.1
16	236°52.5	21°27.7	32.4	3°21.4	22.8	29°51.1	01.9	183°43.2	40.4	Antares	112°11.3 -26°29.8
17	251°54.9	36°27.4	33.5	18°22.1	23.2	44°53.1	02.1	198°45.7	40.4	Atria	107°02.1 -69°04.9
18	266°57.4	51°27.1	S15°34.6	33°22.8	S20°23.6	59°55.1	S10°02.3	213°48.2	N16°40.4	Sabik	101°58.5 -15°45.5
19	281°59.9	66°26.9	35.7	48°23.5	24.0	74°57.1	02.4	228°50.8	40.4	Shaula	96°05.2 -37°07.4
20	297°02.3	81°26.6	36.7	63°24.2	24.4	89°59.1	02.6	243°53.3	40.3	Rasalhague	95°55.0 12°32.6
21	312°04.8	96°26.3	· · 37.8	78°24.9	· · 24.9	105°01.1	· · 02.8	258°55.8	· · 40.3	Eltanin	90°40.4 51°29.5
22	327°07.3	111°26.1	38.9	93°25.6	25.3	120°03.1	03.0	273°58.3	40.3	Kaus Aust.	83°27.4 -34°22.1
23	342°09.7	126°25.8	40.0	108°26.3	25.7	135°05.1	03.2	289°00.8	40.3	Vega	80°30.5 38°49.1
Mer.pass. 00:15		ν -0.3' d 1.1' m -4.13		ν 0.7' d 0.4' m 0.94		ν 2.0' d 0.2' m -1.72		ν 2.5' d -0.0' m 0.02		Nunki	75°43.0 -26°15.4
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	Altair	61°56.1 8°57.1
0	357°12.2	141°25.5	S15°41.1	123°27.0	S20°26.1	150°07.1	S10°03.3	304°03.3	N16°40.3	Peacock	52°59.5 -56°38.3
1	12°14.7	156°25.3	42.2	138°27.7	26.5	165°09.1	03.5	319°05.8	40.2	Deneb	49°22.8 45°23.6
2	27°17.1	171°25.0	43.2	153°28.5	26.9	180°11.1	03.7	334°08.4	40.2	Enif	33°34.8 10°01.0
3	42°19.6	186°24.7	· · 44.3	168°29.2	· · 27.4	195°13.1	· · 03.9	349°10.9	· · 40.2	Al Na'ir	27°27.9 -46°48.8
4	57°22.0	201°24.5	45.4	183°29.9	27.8	210°15.1	04.1	4°13.4	40.2	Fomalhaut	15°10.1 -29°27.6
5	72°24.5	216°24.2	46.5	198°30.6	28.2	225°17.1	04.2	19°15.9	40.2	Scheat	13°41.1 28°14.9
6	87°27.0	231°23.9	S15°47.6	213°31.3	S20°28.6	240°19.1	S10°04.4	34°18.4	N16°40.1	Markab	13°25.8 15°22.2
7	102°29.4	246°23.7	48.6	228°32.0	29.0	255°21.1	04.6	49°20.9	40.1	Sep 16 Sun	SHA Mer.pass
8	117°31.9	261°23.4	49.7	243°32.7	29.4	270°23.1	04.8	64°23.5	40.1	Venus	146°24.1 14:34
9	132°34.4	276°23.1	· · 50.8	258°33.4	· · 29.9	285°25.1	· · 05.0	79°26.0	· · 40.1	Mars	127°38.8 15:48
10	147°36.8	291°22.8	51.9	273°34.1	30.3	300°27.1	05.1	94°28.5	40.1	Jupiter	153°17.3 14:04
11	162°39.3	306°22.6	52.9	288°34.8	30.7	315°29.1	05.3	109°31.0	40.0	Saturn	306°48.9 03:51
12	177°41.8	321°22.3	S15°54.0	303°35.5	S20°31.1	330°31.0	S10°05.5	124°33.5	N16°40.0	Sep 17 Mon	SHA Mer.pass
13	192°44.2	336°22.0	55.1	318°36.2	31.5	345°33.0	05.7	139°36.0	40.0	Venus	145°18.8 14:34
14	207°46.7	351°21.8	56.2	333°36.9	31.9	0°35.0	05.9	154°38.6	40.0	Mars	126°56.9 15:47
15	222°49.2	6°21.5	· · 57.2	348°37.6	· · 32.3	15°37.0	· · 06.0	169°41.1	· · 39.9	Jupiter	153°06.1 14:01
16	237°51.6	21°21.2	58.3	3°38.3	32.8	30°39.0	06.2	184°43.6	39.9	Saturn	306°50.0 03:47
17	252°54.1	36°21.0	15°59.4	18°39.0	33.2	45°41.0	06.4	199°46.1	39.9	Sep 18 Tue	SHA Mer.pass
18	267°56.5	51°20.7	S16°00.4	33°39.7	S20°33.6	60°43.0	S10°06.6	214°48.6	N16°39.9	Venus	144°13.3 14:35
19	282°59.0	66°20.4	01.5	48°40.4	34.0	75°45.0	06.8	229°51.1	39.9	Mars	126°14.9 15:45
20	298°01.5	81°20.1	02.6	63°41.1	34.4	90°47.0	06.9	244°53.7	39.8	Jupiter	152°54.9 13:58
21	313°03.9	96°19.9	· · 03.7	78°41.8	· · 34.8	105°49.0	· · 07.1	259°56.2	· · 39.8	Saturn	306°51.1 03:43
22	328°06.4	111°19.6	04.7	93°42.5	35.2	120°51.0	07.3	274°58.7	39.8	Horizontal parallax	
23	343°08.9	126°19.3	05.8	108°43.2	35.6	135°53.0	07.5	290°01.2	39.8	Venus:	0.2
Mer.pass. 00:11		ν -0.3' d 1.1' m -4.14		ν 0.7' d 0.4' m 0.94		ν 2.0' d 0.2' m -1.72		ν 2.5' d -0.0' m 0.01		Mars:	0.1

DUT1 = UT1-UTC = +0.2126 sec ΔT = TT-UT1 = +68.9714 sec

2029 September 16 to Sep. 18 UT

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	181°16.2	N02°36.6	79°33.5	7.2'	S22°57.9	-4.3'	57.7'
1	196°16.4	35.7	93°59.7	7.3'	22°53.5	-4.5'	57.7'
2	211°16.6	34.7	108°26.0	7.4'	22°49.1	-4.6'	57.7'
3	226°16.9	· · 33.7	122°52.4	7.5'	22°44.5	-4.7'	57.6'
4	241°17.1	32.8	137°18.9	7.5'	22°39.7	-4.9'	57.6'
5	256°17.3	31.8	151°45.4	7.6'	22°34.9	-5.0'	57.6'
6	271°17.5	N02°30.9	166°12.1	7.7'	S22°29.9	-5.1'	57.6'
7	286°17.7	29.9	180°38.8	7.8'	22°24.8	-5.2'	57.5'
8	301°18.0	28.9	195°05.5	7.9'	22°19.6	-5.4'	57.5'
9	316°18.2	· · 28.0	209°32.4	7.9'	22°14.2	-5.5'	57.5'
10	331°18.4	27.0	223°59.3	8.0'	22°08.7	-5.6'	57.4'
11	346°18.6	26.0	238°26.3	8.1'	22°03.1	-5.7'	57.4'
12	1°18.9	N02°25.1	252°53.4	8.2'	S21°57.4	-5.8'	57.4'
13	16°19.1	24.1	267°20.6	8.2'	21°51.5	-6.0'	57.4'
14	31°19.3	23.1	281°47.8	8.3'	21°45.6	-6.1'	57.3'
15	46°19.5	· · 22.2	296°15.2	8.4'	21°39.5	-6.2'	57.3'
16	61°19.8	21.2	310°42.6	8.5'	21°33.3	-6.3'	57.3'
17	76°20.0	20.2	325°10.1	8.6'	21°27.0	-6.4'	57.3'
18	91°20.2	N02°19.3	339°37.6	8.7'	S21°20.5	-6.5'	57.2'
19	106°20.4	18.3	354°05.3	8.7'	21°14.0	-6.7'	57.2'
20	121°20.7	17.3	8°33.0	8.8'	21°07.3	-6.8'	57.2'
21	136°20.9	· · 16.4	23°00.9	8.9'	21°00.5	-6.9'	57.1'
22	151°21.1	15.4	37°28.8	9.0'	20°53.7	-7.0'	57.1'
23	166°21.3	14.4	51°56.7	9.1'	20°46.7	-7.1'	57.1'
SD = 15.9'			SD = 15.7'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	181°21.5	N02°13.5	66°24.8	9.2'	S20°39.6	-7.2'	57.1'
1	196°21.8	12.5	80°53.0	9.2'	20°32.3	-7.3'	57.0'
2	211°22.0	11.5	95°21.2	9.3'	20°25.0	-7.4'	57.0'
3	226°22.2	· · 10.6	109°49.5	9.4'	20°17.6	-7.5'	57.0'
4	241°22.4	09.6	124°17.9	9.5'	20°10.1	-7.6'	57.0'
5	256°22.7	08.7	138°46.4	9.6'	20°02.5	-7.7'	56.9'
6	271°22.9	N02°07.7	153°15.0	9.7'	S19°54.7	-7.8'	56.9'
7	286°23.1	06.7	167°43.6	9.7'	19°46.9	-7.9'	56.9'
8	301°23.3	05.8	182°12.4	9.8'	19°39.0	-8.0'	56.9'
9	316°23.6	· · 04.8	196°41.2	9.9'	19°31.0	-8.1'	56.8'
10	331°23.8	03.8	211°10.1	10.0'	19°22.9	-8.2'	56.8'
11	346°24.0	02.9	225°39.1	10.1'	19°14.6	-8.3'	56.8'
12	1°24.2	N02°01.9	240°08.2	10.2'	S19°06.3	-8.4'	56.8'
13	16°24.5	00.9	254°37.4	10.2'	18°57.9	-8.5'	56.7'
14	31°24.7	02°00.0	269°06.6	10.3'	18°49.4	-8.6'	56.7'
15	46°24.9	01°59.0	283°35.9	10.4'	18°40.9	-8.7'	56.7'
16	61°25.1	58.0	298°05.3	10.5'	18°32.2	-8.8'	56.7'
17	76°25.3	57.1	312°34.8	10.6'	18°23.4	-8.8'	56.6'
18	91°25.6	N01°56.1	327°04.4	10.7'	S18°14.6	-8.9'	56.6'
19	106°25.8	55.1	341°34.0	10.7'	18°05.6	-9.0'	56.6'
20	121°26.0	54.2	356°03.8	10.8'	17°56.6	-9.1'	56.6'
21	136°26.2	· · 53.2	10°33.6	10.9'	17°47.5	-9.2'	56.5'
22	151°26.5	52.2	25°03.5	11.0'	17°38.3	-9.3'	56.5'
23	166°26.7	51.2	39°33.5	11.1'	17°29.1	-9.3'	56.5'
SD = 15.9'			SD = 15.6'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	181°26.9	N01°50.3	54°03.5	11.1'	S17°19.7	-9.4'	56.5'
1	196°27.1	49.3	68°33.7	11.2'	17°10.3	-9.5'	56.4'
2	211°27.4	48.3	83°03.9	11.3'	17°00.8	-9.6'	56.4'
3	226°27.6	· · 47.4	97°34.2	11.4'	16°51.2	-9.7'	56.4'
4	241°27.8	46.4	112°04.6	11.5'	16°41.5	-9.7'	56.4'
5	256°28.0	45.4	126°35.0	11.5'	16°31.8	-9.8'	56.3'
6	271°28.3	N01°44.5	141°05.6	11.6'	S16°22.0	-9.9'	56.3'
7	286°28.5	43.5	155°36.2	11.7'	16°12.1	-9.9'	56.3'
8	301°28.7	42.5	170°06.9	11.8'	16°02.2	-10.0'	56.3'
9	316°28.9	· · 41.6	184°37.7	11.8'	15°52.2	-10.1'	56.3'
10	331°29.2	40.6	199°08.5	11.9'	15°42.1	-10.2'	56.2'
11	346°29.4	39.6	213°39.4	12.0'	15°31.9	-10.2'	56.2'
12	1°29.6	N01°38.7	228°10.4	12.1'	S15°21.7	-10.3'	56.2'
13	16°29.8	37.7	242°41.5	12.1'	15°11.4	-10.4'	56.2'
14	31°30.0	36.7	257°12.6	12.2'	15°01.1	-10.4'	56.1'
15	46°30.3	· · 35.8	271°43.8	12.3'	14°50.7	-10.5'	56.1'
16	61°30.5	34.8	286°15.1	12.4'	14°40.2	-10.5'	56.1'
17	76°30.7	33.8	300°46.5	12.4'	14°29.6	-10.6'	56.1'
18	91°30.9	N01°32.9	315°17.9	12.5'	S14°19.0	-10.7'	56.0'
19	106°31.2	31.9	329°49.4	12.6'	14°08.4	-10.7'	56.0'
20	121°31.4	30.9	344°21.0	12.6'	13°57.7	-10.8'	56.0'
21	136°31.6	· · 30.0	358°52.6	12.7'	13°46.9	-10.8'	56.0'
22	151°31.8	29.0	13°24.3	12.8'	13°36.1	-10.9'	56.0'
23	166°32.1	28.0	27°56.1	12.8'	13°25.2	-10.9'	55.9'
SD = 15.9'			SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:26	04:06	05:17	18:29	19:39	21:17
N 70°	02:53	04:18	05:21	18:25	19:28	20:52
68°	03:13	04:27	05:24	18:22	19:19	20:33
66°	03:28	04:35	05:27	18:20	19:12	20:18
64°	03:40	04:41	05:29	18:18	19:06	20:06
62°	03:51	04:46	05:31	18:16	19:01	19:56
60°	03:59	04:51	05:33	18:14	18:56	19:47
N 58°	04:07	04:55	05:35	18:13	18:52	19:40
56°	04:13	04:58	05:36	18:12	18:49	19:34
54°	04:19	05:01	05:37	18:11	18:46	19:29
52°	04:24	05:04	05:38	18:10	18:43	19:24
50°	04:28	05:07	05:39	18:09	18:41	19:20
45°	04:37	05:12	05:41	18:07	18:36	19:11
N 40°	04:44	05:16	05:43	18:05	18:32	19:04
35°	04:49	05:19	05:44	18:04	18:29	18:59
30°	04:54	05:22	05:46	18:03	18:27	18:54
20°	05:00	05:26	05:48	18:01	18:23	18:48
N 10°	05:04	05:29	05:50	17:59	18:20	18:44
0°	05:06	05:30	05:51	17:58	18:18	18:42
S 10°	05:07	05:32	05:53	17:56	18:17	18:42
20°	05:07	05:32	05:54	17:55	18:17	18:43
30°	05:04	05:32	05:56	17:54	18:18	18:45
35°	05:02	05:31	05:56	17:53	18:18	18:48
40°	04:59	05:30	05:57	17:52	18:19	18:51
45°	04:55	05:29	05:58	17:51	18:21	18:55
S 50°	04:50	05:27	05:59	17:50	18:22	19:00
52°	04:47	05:26	06:00	17:50	18:23	19:03
54°	04:44	05:25	06:01	17:49	18:25	19:06
56°	04:41	05:24	06:01	17:49	18:26	19:09
58°	04:37	05:23	06:02	17:48	18:27	19:14
S 60°	04:32	05:21	06:03	17:48	18:29	19:18

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	18:27	■	■	■
N 70°	■	18:33	17:59	■	22:15	· ·
68°	18:27	17:53	17:37	20:33	22:54	· ·
66°	17:30	17:25	17:20	21:29	23:21	· ·
64°	16:56	17:03	17:06	22:02	23:42	· ·
62°	16:31	16:46	16:54	22:26	23:59	· ·
60°	16:12	16:31	16:44	22:46	· ·	00:12
N 58°	15:55	16:19	16:35	23:01	· ·	00:24
56°	15:41	16:08	16:28	23:15	· ·	00:34
54°	15:29	15:59	16:21	23:27	· ·	00:43
52°	15:19	15:50	16:14	23:37	· ·	00:51
50°	15:09	15:42	16:09	23:46	· ·	00:58
45°	14:49	15:26	15:56	· ·	00:05	01:14
N 40°	14:33	15:13	15:46	· ·	00:21	01:26
35°	14:19	15:01	15:37	· ·	00:34	01:37
30°	14:07	14:51	15:30	· ·	00:45	01:46
20°	13:46	14:34	15:16	00:06	01:05	02:02
N 10°	13:28	14:18	15:04	00:25	01:22	02:15
0°	13:12	14:04	14:53	00:43	01:37	02:28
S 10°	12:55	13:50	14:42	01:01	01:53	02:41
20°	12:37	13:35	14:30	01:19	02:09	02:54
30°	12:16	13:17	14:17	01:41	02:28	03:10
35°	12:04	13:07	14:09	01:54	02:39	03:19
40°	11:50	12:55	14:00	02:08	02:52	03:29
45°	11:33	12:41	13:49	02:26	03:07	03:40
S 50°	11:12	12:24	13:36	02:47	03:25	03:55
52°	11:02	12:16	13:30	02:57	03:33	04:01
54°	10:51	12:07	13:23	03:08	03:43	04:08
56°	10:39	11:57	13:16	03:21	03:54	04:17
58°	10:24	11:45	13:08	03:36	04:06	04:26
S 60°	10:07	11:32	12:58	03:54	04:20	04:36

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12 ^h	Pass	Mer.Pass.	Lower	8-10
	00 ^h	mm:ss	hh:mm	Upper	hh:mm	60-80%
16	05:05	05:15	11:55	19:25	06:57	
17	05:26	05:37	11:54	20:16	07:51	
18	05:48	05:58	11:54	21:05	08:41	

September 19, 20, 21 UT (Wed., Thu., Fri.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	358°11.3	141°19.0	S16°06.9	123°43.9	S20°36.1	150°55.0	S10°07.7	305°03.7	N16°39.8	Alpheratz	357°30.4	29°15.5											
1	13°13.8	156°18.8	07.9	138°44.6	36.5	165°57.0	07.8	320°06.3	39.7	Ankaa	353°03.1	-42°08.4											
2	28°16.3	171°18.5	09.0	153°45.3	36.9	180°59.0	08.0	335°08.8	39.7	Schedar	349°26.0	56°42.1											
3	43°18.7	186°18.2	•• 10.1	168°46.0	•• 37.3	196°01.0	•• 08.2	350°11.3	•• 39.7	Diphda	348°43.2	-17°49.2											
4	58°21.2	201°18.0	11.1	183°46.7	37.7	211°02.9	08.4	5°13.8	39.7	Achernar	335°17.0	-57°04.9											
5	73°23.6	216°17.7	12.2	198°47.4	38.1	226°04.9	08.6	20°16.3	39.7	Hamal	327°46.6	23°36.2											
6	88°26.1	231°17.4	S16°13.3	213°48.1	S20°38.5	241°06.9	S10°08.7	35°18.9	N16°39.6	Polaris	311°37.4	89°23.0											
7	103°28.6	246°17.1	14.3	228°48.8	38.9	256°08.9	08.9	50°21.4	39.6	Acamar	315°08.7	-40°10.9											
8	118°31.0	261°16.9	15.4	243°49.5	39.3	271°10.9	09.1	65°23.9	39.6	Menkar	314°02.0	4°12.5											
9	133°33.5	276°16.6	•• 16.4	258°50.2	•• 39.7	286°12.9	•• 09.3	80°26.4	•• 39.6	Mirfak	308°22.4	49°57.8											
10	148°36.0	291°16.3	17.5	273°50.9	40.2	301°14.9	09.5	95°28.9	39.5	Aldebaran	290°35.2	16°34.1											
11	163°38.4	306°16.0	18.6	288°51.6	40.6	316°16.9	09.7	110°31.5	39.5	Rigel	281°00.2	-8°10.0											
12	178°40.9	321°15.8	S16°19.6	303°52.3	S20°41.0	331°18.9	S10°09.8	125°34.0	N16°39.5	Capella	280°16.2	46°01.4											
13	193°43.4	336°15.5	20.7	318°53.0	41.4	346°20.9	10.0	140°36.5	39.5	Bellatrix	278°18.8	6°22.6											
14	208°45.8	351°15.2	21.7	333°53.7	41.8	1°22.9	10.2	155°39.0	39.5	Elnath	277°57.0	28°37.8											
15	223°48.3	6°14.9	•• 22.8	348°54.4	•• 42.2	16°24.9	•• 10.4	170°41.5	•• 39.4	Alnilam	275°33.9	-1°11.0											
16	238°50.8	21°14.6	23.9	3°55.1	42.6	31°26.8	10.6	185°44.1	39.4	Betelgeuse	270°48.0	7°24.7											
17	253°53.2	36°14.4	24.9	18°55.8	43.0	46°28.8	10.7	200°46.6	39.4	Canopus	263°50.8	-52°42.5											
18	268°55.7	51°14.1	S16°26.0	33°56.5	S20°43.4	61°30.8	S10°10.9	215°49.1	N16°39.4	Sirius	258°23.0	-16°45.3											
19	283°58.1	66°13.8	27.0	48°57.2	43.8	76°32.8	11.1	230°51.6	39.4	Adhara	255°03.0	-29°00.6											
20	299°00.6	81°13.5	28.1	63°57.8	44.2	91°34.8	11.3	245°54.2	39.3	Procyon	244°47.0	5°08.9											
21	314°03.1	96°13.3	•• 29.1	78°58.5	•• 44.6	106°36.8	•• 11.5	260°56.7	•• 39.3	Pollux	243°12.9	27°57.1											
22	329°05.5	111°13.0	30.2	93°59.2	45.0	121°38.8	11.6	275°59.2	39.3	Avior	234°13.5	-59°36.1											
23	344°08.0	126°12.7	31.3	108°59.9	45.4	136°40.8	11.8	291°01.7	39.3	Suhail	222°43.8	-43°33.1											
Mer.pass. 00:07				ν -0.3' d 1.1' m -4.14				ν 0.7' d 0.4' m 0.95				ν 2.0' d 0.2' m -1.72				ν 2.5' d -0.0' m 0.01							

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	359°10.5	141°12.4	S16°32.3	124°00.6	S20°45.9	151°42.8	S10°12.0	306°04.2	N16°39.2										
1	14°12.9	156°12.1	33.4	139°01.3	46.3	166°44.8	12.2	321°06.8	39.2										
2	29°15.4	171°11.9	34.4	154°02.0	46.7	181°46.8	12.4	336°09.3	39.2										
3	44°17.9	186°11.6	•• 35.5	169°02.7	•• 47.1	196°48.7	•• 12.5	351°11.8	•• 39.2										
4	59°20.3	201°11.3	36.5	184°03.4	47.5	211°50.7	12.7	6°14.3	39.2										
5	74°22.8	216°11.0	37.6	199°04.1	47.9	226°52.7	12.9	21°16.9	39.1										
6	89°25.2	231°10.7	S16°38.6	214°04.8	S20°48.3	241°54.7	S10°13.1	36°19.4	N16°39.1										
7	104°27.7	246°10.5	39.7	229°05.4	48.7	256°56.7	13.3	51°21.9	39.1										
8	119°30.2	261°10.2	40.7	244°06.1	49.1	271°58.7	13.4	66°24.4	39.1										
9	134°32.6	276°09.9	•• 41.8	259°06.8	•• 49.5	287°00.7	•• 13.6	81°27.0	•• 39.0										
10	149°35.1	291°09.6	42.8	274°07.5	49.9	302°02.7	13.8	96°29.5	39.0										
11	164°37.6	306°09.3	43.9	289°08.2	50.3	317°04.7	14.0	111°32.0	39.0										
12	179°40.0	321°09.1	S16°44.9	304°08.9	S20°50.7	332°06.7	S10°14.2	126°34.5	N16°39.0										
13	194°42.5	336°08.8	45.9	319°09.6	51.1	347°08.6	14.4	141°37.1	39.0										
14	209°45.0	351°08.5	47.0	334°10.3	51.5	2°10.6	14.5	156°39.6	38.9										
15	224°47.4	6°08.2	•• 48.0	349°10.9	•• 51.9	17°12.6	•• 14.7	171°42.1	•• 38.9										
16	239°49.9	21°07.9	49.1	4°11.6	52.3	32°14.6	14.9	186°44.6	38.9										
17	254°52.4	36°07.6	50.1	19°12.3	52.7	47°16.6	15.1	201°47.2	38.9										
18	269°54.8	51°07.4	S16°51.2	34°13.0	S20°53.1	62°18.6	S10°15.3	216°49.7	N16°38.8										
19	284°57.3	66°07.1	52.2	49°13.7	53.5	77°20.6	15.4	231°52.2	38.8										
20	299°59.7	81°06.8	53.2	64°14.4	53.9	92°22.6	15.6	246°54.8	38.8										
21	315°02.2	96°06.5	•• 54.3	79°15.1	•• 54.3	107°24.6	•• 15.8	261°57.3	•• 38.8										
22	330°04.7	111°06.2	55.3	94°15.7	54.7	122°26.5	16.0	276°59.8	38.7										
23	345°07.1	126°05.9	56.4	109°16.4	55.1	137°28.5	16.2	292°02.3	38.7										
Mer.pass. 23:59				ν -0.3' d 1.1' m -4.15				ν 0.7' d 0.4' m 0.95				ν 2.0' d 0.2' m -1.72				ν 2.5' d -0.0' m 0.00			

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	0°09.6	141°05.6	S16°57.4	124°17.1	S20°55.5	152°30.5	S10°16.3	307°04.9	N16°38.7										
1	15°12.1	156°05.4	58.4	139°17.8	55.9	167°32.5	16.5	322°07.4	38.7										
2	30°14.5	171°05.1	16°59.5	154°18.5	56.3	182°34.5	16.7	337°09.9	38.7										
3	45°17.0	186°04.8	17°00.5	169°19.1	•• 56.7	197°36.5	•• 16.9	352°12.5	•• 38.6										
4	60°19.5	201°04.5	01.5	184°19.8	57.1	212°38.5	17.1	7°15.0	38.6										
5	75°21.9	216°04.2	02.6	199°20.5	57.5	227°40.5	17.3	22°17.5	38.6										
6	90°24.4	231°03.9	S17°03.6	214°21.2	S20°57.9	242°42.5	S10°17.4	37°20.0	N16°38.6										
7	105°26.8	246°03.6	04.7	229°21.9	58.3	257°44.4	17.6	52°22.6	38.5										
8	120°29.3	261°03.4	05.7	244°22.6	58.7	272°46.4	17.8	67°25.1	38.5										
9	135°31.8	276°03.1	•• 06.7	259°23.2	•• 59.1	287°48.4	•• 18.0	82°27.6	•• 38.5										
10	150°34.2	291°02.8	07.7	274°23.9	59.5	302°50.4	18.2	97°30.2	38.5										
11	165°36.7	306°02.5	08.8	289°24.6	20°59.9	317°52.4	18.3	112°32.7	38.4										
12	180°39.2	321°02.2	S17°09.8	304°25.3	S21°00.3	332°54.4	S10°18.5	127°35.2	N16°38.4										
13	195°41.6	336°01.9	10.8	319°26.0	00.7	347°56.4	18.7	142°37.7	38.4										
14	210°44.1	351°01.6	11.9	334°26.6	01.1	2°58.4	18.9	157°40.3	38.4										
15	225°46.6	6°01.3	•• 12.9	349°27.3	•• 01.5	18°00.3	•• 19.1	172°42.8	•• 38.4										
16	240°49.0	21°01.0	13.9	4°28.0	01.9	33°02.1	19.3	187°45.3	38.3										
17	255°51.5	36°00.8	15.0	19°28.7	02.2	48°04.3	19.4	202°47.9	38.3										
18	270°54.0	51°00.5	S17°16.0	34°29.3	S21°02.6	63°06.3	S10°19.6	217°50.4	N16°38.3										
19	285°56.4	66°00.2	17.0	49°30.0	03.0	78°08.3	19.8	232°52.9	38.3										
20	300°58.9	80°59.9	18.0	64°30.7	03.4	93°10.3	20.0	247°55.5	38.2										
21	316°01.3	95°59.6	•• 19.1	79°31.4	•• 03.8	108°12.3	•• 20.2	262°58.0	•• 38.2										
22	331°03.8	110°59.3	20.1	94°32.0	04.2	123°14.2	20.3	278°00.5	38.2										
23	346°06.3	125°59.0	21.1	109°32.7	04.6	138°16.2	20.5	293°03.1	38.2										
Mer.pass. 23:55				ν -0.3' d 1.0' m -4.16				ν 0.7' d 0.4' m 0.95				ν 2.0' d 0.2' m -1.71				ν 2.5' d -0.0' m -0.00			

Sep 19 Wed	SHA	Mer.pass
Venus	143°07.7	14:35
Mars	125°32.6	15:44
Jupiter	152°43.6	13:54
Saturn	306°52.4	03:39

Sep 20 Thu	SHA	Mer.pass
Venus	142°02.0	14:35
Mars	124°50.1	15:43
Jupiter	152°32.3	13:51
Saturn	306°53.8	03:35

Sep 21 Fri	SHA	Mer.pass
Venus	140°56.0	14:36
Mars	124°07.5	15:42
Jupiter	152°20.9	13:48
Saturn	306°55.3	03:31

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

DUT1 = UT1-UTC = +0.2127 sec ΔT = TT-UT1 = +68.9713 sec

2029 September 19 to Sep. 21 UT

h Sun Moon						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	181°32.3	N01°27.0	42°28.0	12.9'	S13°14.2	-11.0' 55.9'
1	196°32.5	26.1	56°59.9	13.0'	13°03.2	-11.0' 55.9'
2	211°32.7	25.1	71°31.9	13.0'	12°52.2	-11.1' 55.9'
3	226°33.0	· · 24.1	86°03.9	13.1'	12°41.1	-11.1' 55.9'
4	241°33.2	23.2	100°36.0	13.2'	12°30.0	-11.2' 55.8'
5	256°33.4	22.2	115°08.2	13.2'	12°18.8	-11.2' 55.8'
6	271°33.6	N01°21.2	129°40.4	13.3'	S12°07.5	-11.3' 55.8'
7	286°33.8	20.3	144°12.7	13.4'	11°56.3	-11.3' 55.8'
8	301°34.1	19.3	158°45.1	13.4'	11°44.9	-11.4' 55.7'
9	316°34.3	· · 18.3	173°17.5	13.5'	11°33.6	-11.4' 55.7'
10	331°34.5	17.4	187°50.0	13.5'	11°22.1	-11.5' 55.7'
11	346°34.7	16.4	202°22.6	13.6'	11°10.7	-11.5' 55.7'
12	1°35.0	N01°15.4	216°55.2	13.7'	S10°59.2	-11.5' 55.7'
13	16°35.2	14.4	231°27.8	13.7'	10°47.6	-11.6' 55.6'
14	31°35.4	13.5	246°00.5	13.8'	10°36.0	-11.6' 55.6'
15	46°35.6	· · 12.5	260°33.3	13.8'	10°24.4	-11.7' 55.6'
16	61°35.9	11.5	275°06.1	13.9'	10°12.8	-11.7' 55.6'
17	76°36.1	10.6	289°39.0	13.9'	10°01.1	-11.7' 55.6'
18	91°36.3	N01°09.6	304°12.0	14.0'	S09°49.3	-11.8' 55.5'
19	106°36.5	08.6	318°45.0	14.0'	09°37.6	-11.8' 55.5'
20	121°36.7	07.7	333°18.0	14.1'	09°25.8	-11.8' 55.5'
21	136°37.0	· · 06.7	347°51.1	14.1'	09°13.9	-11.9' 55.5'
22	151°37.2	05.7	2°24.2	14.2'	09°02.1	-11.9' 55.5'
23	166°37.4	04.7	16°57.4	14.2'	08°50.2	-11.9' 55.5'
SD = 15.9' d = -1.0' SD = 15.3'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	181°37.6	N01°03.8	31°30.7	14.3'	S08°38.2	-12.0' 55.4'
1	196°37.9	02.8	46°04.0	14.3'	08°26.3	-12.0' 55.4'
2	211°38.1	01.8	60°37.3	14.4'	08°14.3	-12.0' 55.4'
3	226°38.3	01°00.9	75°10.7	14.4'	08°02.3	-12.0' 55.4'
4	241°38.5	00°59.9	89°44.2	14.5'	07°50.3	-12.1' 55.4'
5	256°38.8	58.9	104°17.7	14.5'	07°38.2	-12.1' 55.3'
6	271°39.0	N00°58.0	118°51.2	14.6'	S07°26.1	-12.1' 55.3'
7	286°39.2	57.0	133°24.8	14.6'	07°14.0	-12.1' 55.3'
8	301°39.4	56.0	147°58.4	14.7'	07°01.9	-12.2' 55.3'
9	316°39.6	· · 55.0	162°32.0	14.7'	06°49.7	-12.2' 55.3'
10	331°39.9	54.1	177°05.7	14.7'	06°37.5	-12.2' 55.2'
11	346°40.1	53.1	191°39.5	14.8'	06°25.3	-12.2' 55.2'
12	1°40.3	N00°52.1	206°13.2	14.8'	S06°13.1	-12.2' 55.2'
13	16°40.5	51.2	220°47.1	14.9'	06°00.9	-12.3' 55.2'
14	31°40.8	50.2	235°20.9	14.9'	05°48.6	-12.3' 55.2'
15	46°41.0	· · 49.2	249°54.8	14.9'	05°36.4	-12.3' 55.2'
16	61°41.2	48.2	264°28.7	15.0'	05°24.1	-12.3' 55.1'
17	76°41.4	47.3	279°02.7	15.0'	05°11.8	-12.3' 55.1'
18	91°41.6	N00°46.3	293°36.7	15.0'	S04°59.5	-12.3' 55.1'
19	106°41.9	45.3	308°10.7	15.1'	04°47.2	-12.3' 55.1'
20	121°42.1	44.4	322°44.8	15.1'	04°34.8	-12.3' 55.1'
21	136°42.3	· · 43.4	337°18.9	15.1'	04°22.5	-12.4' 55.1'
22	151°42.5	42.4	351°53.0	15.2'	04°10.1	-12.4' 55.0'
23	166°42.8	41.5	6°27.2	15.2'	03°57.7	-12.4' 55.0'
SD = 15.9' d = -1.0' SD = 15.1'						
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	181°43.0	N00°40.5	21°01.4	15.2'	S03°45.4	-12.4' 55.0'
1	196°43.2	39.5	35°35.6	15.2'	03°33.0	-12.4' 55.0'
2	211°43.4	38.5	50°09.8	15.3'	03°20.6	-12.4' 55.0'
3	226°43.6	· · 37.6	64°44.1	15.3'	03°08.2	-12.4' 55.0'
4	241°43.9	36.6	79°18.4	15.3'	02°55.8	-12.4' 54.9'
5	256°44.1	35.6	93°52.7	15.3'	02°43.4	-12.4' 54.9'
6	271°44.3	N00°34.7	108°27.1	15.4'	S02°31.0	-12.4' 54.9'
7	286°44.5	33.7	123°01.4	15.4'	02°18.6	-12.4' 54.9'
8	301°44.8	32.7	137°35.8	15.4'	02°06.1	-12.4' 54.9'
9	316°45.0	· · 31.7	152°10.2	15.4'	01°53.7	-12.4' 54.9'
10	331°45.2	30.8	166°44.7	15.5'	01°41.3	-12.4' 54.8'
11	346°45.4	29.8	181°19.1	15.5'	01°28.9	-12.4' 54.8'
12	1°45.6	N00°28.8	195°53.6	15.5'	S01°16.5	-12.4' 54.8'
13	16°45.9	27.9	210°28.1	15.5'	01°04.0	-12.4' 54.8'
14	31°46.1	26.9	225°02.6	15.5'	00°51.6	-12.4' 54.8'
15	46°46.3	· · 25.9	239°37.1	15.5'	00°39.2	-12.4' 54.8'
16	61°46.5	24.9	254°11.7	15.6'	00°26.8	-12.4' 54.8'
17	76°46.8	24.0	268°46.3	15.6'	00°14.4	-12.4' 54.7'
18	91°47.0	N00°23.0	283°20.8	15.6'	S00°02.0	-12.4' 54.7'
19	106°47.2	22.0	297°55.4	15.6'	N00°10.4	12.4' 54.7'
20	121°47.4	21.0	312°30.0	15.6'	00°22.8	12.4' 54.7'
21	136°47.6	· · 20.1	327°04.7	15.6'	00°35.1	12.4' 54.7'
22	151°47.9	19.1	341°39.3	15.6'	00°47.5	12.4' 54.7'
23	166°48.1	18.1	356°13.9	15.6'	00°59.9	12.3' 54.7'
SD = 15.9' d = -1.0' SD = 15.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:47	04:21	05:31	18:13	19:22	20:55
N 70°	03:09	04:31	05:33	18:11	19:13	20:33
68°	03:26	04:38	05:35	18:10	19:06	20:17
66°	03:40	04:45	05:36	18:08	19:00	20:04
64°	03:51	04:50	05:38	18:07	18:55	19:53
62°	04:00	04:54	05:39	18:06	18:51	19:45
60°	04:07	04:58	05:40	18:05	18:47	19:37
N 58°	04:14	05:01	05:41	18:04	18:44	19:31
56°	04:20	05:04	05:42	18:04	18:41	19:25
54°	04:25	05:07	05:42	18:03	18:38	19:20
52°	04:29	05:09	05:43	18:03	18:36	19:16
50°	04:33	05:11	05:44	18:02	18:34	19:12
45°	04:41	05:15	05:45	18:01	18:30	19:05
N 40°	04:47	05:19	05:46	18:00	18:27	18:59
35°	04:52	05:21	05:47	17:59	18:25	18:54
30°	04:56	05:23	05:47	17:59	18:23	18:51
20°	05:01	05:26	05:48	17:58	18:20	18:45
N 10°	05:04	05:28	05:49	17:57	18:18	18:43
0°	05:05	05:29	05:50	17:57	18:17	18:41
S 10°	05:05	05:30	05:51	17:56	18:17	18:41
20°	05:04	05:29	05:51	17:56	18:18	18:43
30°	05:00	05:28	05:52	17:55	18:19	18:47
35°	04:57	05:27	05:52	17:55	18:20	18:50
40°	04:54	05:25	05:52	17:55	18:22	18:54
45°	04:49	05:23	05:53	17:55	18:24	18:58
S 50°	04:43	05:21	05:53	17:55	18:27	19:05
52°	04:40	05:19	05:53	17:55	18:28	19:08
54°	04:36	05:18	05:53	17:55	18:30	19:12
56°	04:32	05:16	05:53	17:55	18:32	19:16
58°	04:28	05:14	05:53	17:55	18:34	19:20
S 60°	04:23	05:12	05:53	17:55	18:36	19:26

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	17:54	17:30	17:09	00:03	02:11	04:06
N 70°	17:38	17:23	17:09	00:30	02:24	04:10
68°	17:26	17:17	17:08	00:50	02:35	04:13
66°	17:16	17:12	17:08	01:06	02:43	04:16
64°	17:07	17:08	17:08	01:18	02:50	04:19
62°	17:00	17:04	17:08	01:29	02:57	04:21
60°	16:54	17:01	17:07	01:38	03:02	04:23
N 58°	16:48	16:58	17:07	01:46	03:07	04:24
56°	16:43	16:56	17:07	01:53	03:11	04:26
54°	16:38	16:53	17:07	02:00	03:14	04:27
52°	16:34	16:51	17:07	02:05	03:18	04:28
50°	16:30	16:49	17:07	02:10	03:21	04:29
45°	16:22	16:45	17:07	02:21	03:27	04:32
N 40°	16:15	16:42	17:06	02:30	03:33	04:34
35°	16:09	16:39	17:06	02:38	03:38	04:35
30°	16:04	16:36	17:06	02:45	03:42	04:37
20°	15:55	16:31	17:06	02:56	03:49	04:39
N 10°	15:47	16:27	17:06	03:06	03:55	04:42
0°	15:39	16:23	17:05	03:16	04:01	04:44
S 10°	15:32	16:19	17:05	03:25	04:06	04:46
20°	15:24	16:15	17:05	03:35	04:12	04:48
30°	15:14	16:10	17:05	03:46	04:19	04:50
35°	15:09	16:08	17:05	03:53	04:23	04:51
40°	15:03	16:05	17:05	04:00	04:28	04:53
45°	14:56	16:01	17:05	04:08	04:33	04:55
S 50°	14:47	15:57	17:05	04:18	04:39	04:57
52°	14:43	15:55	17:05	04:23	04:41	04:58
54°	14:39	15:53	17:04	04:28	04:44	04:59
56°	14:34	15:50	17:04	04:34	04:48	05:00
58°	14:29	15:47	17:04	04:40	04:52	05:01
S 60°	14:22	15:44	17:04	04:47	04:56	05:03

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
19	06:09	06:20	11:54	21:50	09:28	
20	06:31	06:41	11:53	22:33	10:12	
21	06:52	07:03	11:53	23:16	10:55	

September 22, 23, 24 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	1°08.7	140°58.7	S17°22.1	124°33.4	S21°05.0	153°18.2	S10°20.7	308°05.6	N16°38.1	Alpheratz	357°30.4 29°15.5
1	16°11.2	155°58.4	23.2	139°34.1	05.4	168°20.2	20.9	323°08.1	38.1	Ankaa	353°03.1 -42°08.4
2	31°13.7	170°58.1	24.2	154°34.7	05.8	183°22.2	21.1	338°10.7	38.1	Schedar	349°26.0 56°42.1
3	46°16.1	185°57.8	· · 25.2	169°35.4	· · 06.2	198°24.2	· · 21.3	353°13.2	· · 38.1	Diphda	348°43.2 -17°49.2
4	61°18.6	200°57.5	26.2	184°36.1	06.6	213°26.2	21.4	8°15.7	38.0	Achernar	335°17.0 -57°04.9
5	76°21.1	215°57.3	27.2	199°36.8	07.0	228°28.1	21.6	23°18.3	38.0	Hamal	327°46.6 23°36.3
6	91°23.5	230°57.0	S17°28.3	214°37.4	S21°07.3	243°30.1	S10°21.8	38°20.8	N16°38.0	Polaris	311°36.2 89°23.0
7	106°26.0	245°56.7	29.3	229°38.1	07.7	258°32.1	22.0	53°23.3	38.0	Acamar	315°08.7 -40°10.9
8	121°28.5	260°56.4	30.3	244°38.8	08.1	273°34.1	22.2	68°25.9	37.9	Menkar	314°02.0 4°12.5
9	136°30.9	275°56.1	· · 31.3	259°39.5	· · 08.5	288°36.1	· · 22.3	83°28.4	· · 37.9	Mirfak	308°22.4 49°57.8
10	151°33.4	290°55.8	32.3	274°40.1	08.9	303°38.1	22.5	98°30.9	37.9	Aldebaran	290°35.1 16°34.1
11	166°35.8	305°55.5	33.3	289°40.8	09.3	318°40.1	22.7	113°33.5	37.9	Rigel	281°00.2 -8°10.0
12	181°38.3	320°55.2	S17°34.4	304°41.5	S21°09.7	333°42.0	S10°22.9	128°36.0	N16°37.8	Capella	280°16.2 46°01.4
13	196°40.8	335°54.9	35.4	319°42.1	10.1	348°44.0	23.1	143°38.5	37.8	Bellatrix	278°18.8 6°22.6
14	211°43.2	350°54.6	36.4	334°42.8	10.5	3°46.0	23.3	158°41.1	37.8	Elnath	277°57.0 28°37.8
15	226°45.7	5°54.3	· · 37.4	349°43.5	· · 10.9	18°48.0	· · 23.4	173°43.6	· · 37.8	Alnilam	275°33.9 -1°11.0
16	241°48.2	20°54.0	38.4	4°44.2	11.2	33°50.0	23.6	188°46.1	37.8	Betelgeuse	270°48.0 7°24.7
17	256°50.6	35°53.7	39.4	19°44.8	11.6	48°52.0	23.8	203°48.7	37.7	Canopus	263°50.7 -52°42.4
18	271°53.1	50°53.4	S17°40.4	34°45.5	S21°12.0	63°54.0	S10°24.0	218°51.2	N16°37.7	Sirius	258°23.0 -16°45.3
19	286°55.6	65°53.1	41.4	49°46.2	12.4	78°55.9	24.2	233°53.7	37.7	Adhara	255°03.0 -29°00.6
20	301°58.0	80°52.8	42.5	64°46.8	12.8	93°57.9	24.3	248°56.3	37.7	Procyon	244°47.0 5°08.9
21	317°00.5	95°52.5	· · 43.5	79°47.5	· · 13.2	108°59.9	· · 24.5	263°58.8	· · 37.6	Pollux	243°12.9 27°57.1
22	332°02.9	110°52.2	44.5	94°48.2	13.6	124°01.9	24.7	279°01.3	37.6	Avior	234°13.4 -59°36.1
23	347°05.4	125°51.9	45.5	109°48.8	14.0	139°03.9	24.9	294°03.9	37.6	Suhail	222°43.8 -43°33.1
Mer.pass. 23:51		ν -0.3' d 1.0' m -4.17		ν 0.7' d 0.4' m 0.96		ν 2.0' d 0.2' m -1.71		ν 2.5' d -0.0' m -0.01		Miaplacidus 221°37.9 -69°50.2	
Alphard 217°44.3 -8°47.2											
Regulus 207°30.7 11°49.3											
Dubhe 193°37.3 61°35.3											
Denebola 182°21.5 14°24.4											
Gienah 175°40.0 -17°42.4											
Acrux 172°56.3 -63°15.9											
Gacrux 171°47.8 -57°16.8											
Alioth 166°10.4 55°48.0											
Spica 158°18.6 -11°18.9											
Alkaid 152°49.6 49°10.0											
Hadar 148°31.0 -60°31.1											
Menkent 147°53.4 -36°31.0											
Arcturus 145°44.8 19°01.8											
Rigel Kent. 139°35.4 -60°57.6											
Kochab 137°21.1 74°02.2											
Zuben'ubi 136°52.0 -16°09.8											
Alphecca 126°00.8 26°37.1											
Antares 112°11.3 -26°29.8											
Atria 107°02.1 -69°04.9											
Sabik 101°58.5 -15°45.5											
Shaula 96°05.3 -37°07.4											
Rasalhague 95°55.1 12°32.6											
Eltanin 90°40.4 51°29.5											
Kaus Aust. 83°27.5 -34°22.1											
Vega 80°30.6 38°49.1											
Nunki 75°43.0 -26°15.4											
Altair 61°56.1 8°57.2											
Peacock 52°59.6 -56°38.3											
Deneb 49°22.9 45°23.6											
Enif 33°34.8 10°01.0											
Al Na'ir 27°27.9 -46°48.9											
Fomalhaut 15°10.1 -29°27.7											
Scheat 13°41.1 28°14.9											
Markab 13°25.8 15°22.2											
Sep 22 Sat SHA Mer.pass											
Venus 139°50.0 14:36											
Mars 123°24.7 15:41											
Jupiter 152°09.5 13:45											
Saturn 306°56.9 03:27											
Sep 23 Sun SHA Mer.pass											
Venus 138°43.8 14:37											
Mars 122°41.6 15:40											
Jupiter 151°58.0 13:42											
Saturn 306°58.5 03:23											
Sep 24 Mon SHA Mer.pass											
Venus 137°37.4 14:37											
Mars 121°58.4 15:39											
Jupiter 151°46.4 13:39											
Saturn 307°00.3 03:19											
Horizontal parallax											
Venus: 0.2											
Mars: 0.1											
Mer.pass. 23:44		ν -0.3' d 1.0' m -4.18		ν 0.7' d 0.4' m 0.96		ν 2.0' d 0.2' m -1.71		ν 2.5' d -0.0' m -0.02			

DUT1 = UT1-UTC = +0.2127 sec ΔT = TT-UT1 = +68.9713 sec

2029 September 22 to Sep. 24 UT

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	181°48.3	N00°17.2	10°48.6	15.7'	N01°12.2	12.3'	54.6'
1	196°48.5	16.2	25°23.2	15.7'	01°24.5	12.3'	54.6'
2	211°48.7	15.2	39°57.9	15.7'	01°36.9	12.3'	54.6'
3	226°49.0	· · 14.2	54°32.6	15.7'	01°49.2	12.3'	54.6'
4	241°49.2	13.3	69°07.3	15.7'	02°01.5	12.3'	54.6'
5	256°49.4	12.3	83°41.9	15.7'	02°13.8	12.3'	54.6'
6	271°49.6	N00°11.3	98°16.6	15.7'	N02°26.0	12.3'	54.6'
7	286°49.9	10.4	112°51.3	15.7'	02°38.3	12.2'	54.5'
8	301°50.1	09.4	127°26.0	15.7'	02°50.5	12.2'	54.5'
9	316°50.3	· · 08.4	142°00.7	15.7'	03°02.7	12.2'	54.5'
10	331°50.5	07.4	156°35.4	15.7'	03°14.9	12.2'	54.5'
11	346°50.7	06.5	171°10.1	15.7'	03°27.1	12.2'	54.5'
12	1°51.0	N00°05.5	185°44.9	15.7'	N03°39.3	12.1'	54.5'
13	16°51.2	04.5	200°19.6	15.7'	03°51.4	12.1'	54.5'
14	31°51.4	03.5	214°54.3	15.7'	04°03.5	12.1'	54.5'
15	46°51.6	· · 02.6	229°29.0	15.7'	04°15.6	12.1'	54.4'
16	61°51.8	01.6	244°03.7	15.7'	04°27.7	12.1'	54.4'
17	76°52.1	N00°00.6	258°38.4	15.7'	04°39.8	12.0'	54.4'
18	91°52.3	S00°00.3	273°13.1	15.7'	N04°51.8	12.0'	54.4'
19	106°52.5	01.3	287°47.8	15.7'	05°03.8	12.0'	54.4'
20	121°52.7	02.3	302°22.5	15.7'	05°15.8	12.0'	54.4'
21	136°52.9	· · 03.3	316°57.1	15.7'	05°27.7	11.9'	54.4'
22	151°53.2	04.2	331°31.8	15.7'	05°39.7	11.9'	54.4'
23	166°53.4	05.2	346°06.5	15.7'	05°51.6	11.9'	54.4'
SD = 15.9'			SD = 14.9'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	181°53.6	S00°06.2	0°41.1	15.6'	N06°03.4	11.8'	54.3'
1	196°53.8	07.1	15°15.8	15.6'	06°15.3	11.8'	54.3'
2	211°54.0	08.1	29°50.4	15.6'	06°27.1	11.8'	54.3'
3	226°54.3	· · 09.1	44°25.0	15.6'	06°38.9	11.8'	54.3'
4	241°54.5	10.1	58°59.7	15.6'	06°50.6	11.7'	54.3'
5	256°54.7	11.0	73°34.3	15.6'	07°02.3	11.7'	54.3'
6	271°54.9	S00°12.0	88°08.9	15.6'	N07°14.0	11.7'	54.3'
7	286°55.1	13.0	102°43.4	15.6'	07°25.7	11.6'	54.3'
8	301°55.4	14.0	117°18.0	15.6'	07°37.3	11.6'	54.3'
9	316°55.6	· · 14.9	131°52.6	15.5'	07°48.9	11.5'	54.2'
10	331°55.8	15.9	146°27.1	15.5'	08°00.4	11.5'	54.2'
11	346°56.0	16.9	161°01.6	15.5'	08°11.9	11.5'	54.2'
12	1°56.2	S00°17.8	175°36.1	15.5'	N08°23.4	11.4'	54.2'
13	16°56.5	18.8	190°10.6	15.5'	08°34.9	11.4'	54.2'
14	31°56.7	19.8	204°45.1	15.4'	08°46.3	11.4'	54.2'
15	46°56.9	· · 20.8	219°19.5	15.4'	08°57.6	11.3'	54.2'
16	61°57.1	21.7	233°53.9	15.4'	09°09.0	11.3'	54.2'
17	76°57.3	22.7	248°28.4	15.4'	09°20.2	11.2'	54.2'
18	91°57.6	S00°23.7	263°02.7	15.4'	N09°31.5	11.2'	54.2'
19	106°57.8	24.7	277°37.1	15.3'	09°42.7	11.2'	54.2'
20	121°58.0	25.6	292°11.5	15.3'	09°53.8	11.1'	54.1'
21	136°58.2	· · 26.6	306°45.8	15.3'	10°04.9	11.1'	54.1'
22	151°58.4	27.6	321°20.1	15.3'	10°16.0	11.0'	54.1'
23	166°58.7	28.6	335°54.4	15.3'	10°27.0	11.0'	54.1'
SD = 15.9'			SD = 14.8'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	181°58.9	S00°29.5	350°28.6	15.2'	N10°38.0	10.9'	54.1'
1	196°59.1	30.5	5°02.9	15.2'	10°49.0	10.9'	54.1'
2	211°59.3	31.5	19°37.1	15.2'	10°59.8	10.8'	54.1'
3	226°59.5	· · 32.4	34°11.2	15.2'	11°10.7	10.8'	54.1'
4	241°59.7	33.4	48°45.4	15.1'	11°21.5	10.7'	54.1'
5	257°00.0	34.4	63°19.5	15.1'	11°32.2	10.7'	54.1'
6	272°00.2	S00°35.4	77°53.6	15.1'	N11°42.9	10.6'	54.1'
7	287°00.4	36.3	92°27.7	15.0'	11°53.5	10.6'	54.1'
8	302°00.6	37.3	107°01.7	15.0'	12°04.1	10.5'	54.1'
9	317°00.8	· · 38.3	121°35.7	15.0'	12°14.7	10.5'	54.1'
10	332°01.1	39.3	136°09.7	15.0'	12°25.2	10.4'	54.0'
11	347°01.3	40.2	150°43.7	14.9'	12°35.6	10.4'	54.0'
12	2°01.5	S00°41.2	165°17.6	14.9'	N12°46.0	10.3'	54.0'
13	17°01.7	42.2	179°51.5	14.9'	12°56.3	10.3'	54.0'
14	32°01.9	43.1	194°25.3	14.8'	13°06.6	10.2'	54.0'
15	47°02.1	· · 44.1	208°59.2	14.8'	13°16.8	10.2'	54.0'
16	62°02.4	45.1	223°32.9	14.8'	13°27.0	10.1'	54.0'
17	77°02.6	46.1	238°06.7	14.7'	13°37.1	10.0'	54.0'
18	92°02.8	S00°47.0	252°40.4	14.7'	N13°47.1	10.0'	54.0'
19	107°03.0	48.0	267°14.1	14.7'	13°57.1	9.9'	54.0'
20	122°03.2	49.0	281°47.8	14.6'	14°07.0	9.9'	54.0'
21	137°03.5	· · 50.0	296°21.4	14.6'	14°16.9	9.8'	54.0'
22	152°03.7	50.9	310°55.0	14.5'	14°26.7	9.7'	54.0'
23	167°03.9	51.9	325°28.5	14.5'	14°36.4	9.7'	54.0'
SD = 15.9'			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:06	04:36	05:44	17:58	19:06	20:34
N 70°	03:25	04:43	05:45	17:58	18:59	20:16
68°	03:39	04:49	05:45	17:57	18:53	20:02
66°	03:51	04:55	05:46	17:57	18:48	19:51
64°	04:01	04:59	05:46	17:57	18:44	19:41
62°	04:09	05:02	05:47	17:56	18:40	19:34
60°	04:15	05:05	05:47	17:56	18:38	19:27
N 58°	04:21	05:08	05:47	17:56	18:35	19:22
56°	04:26	05:10	05:48	17:56	18:33	19:17
54°	04:31	05:12	05:48	17:56	18:31	19:13
52°	04:34	05:14	05:48	17:56	18:29	19:09
50°	04:38	05:16	05:48	17:55	18:28	19:06
45°	04:45	05:19	05:48	17:55	18:25	18:59
N 40°	04:50	05:22	05:49	17:55	18:22	18:54
35°	04:54	05:24	05:49	17:55	18:20	18:50
30°	04:57	05:25	05:49	17:55	18:19	18:47
20°	05:02	05:27	05:49	17:55	18:17	18:43
N 10°	05:04	05:28	05:49	17:55	18:16	18:41
0°	05:04	05:28	05:49	17:55	18:16	18:40
S 10°	05:03	05:28	05:49	17:56	18:17	18:41
20°	05:01	05:27	05:49	17:56	18:18	18:44
30°	04:56	05:24	05:48	17:57	18:21	18:49
35°	04:53	05:22	05:48	17:57	18:23	18:52
40°	04:49	05:20	05:47	17:58	18:25	18:57
45°	04:43	05:18	05:47	17:58	18:28	19:02
S 50°	04:36	05:14	05:46	17:59	18:32	19:10
52°	04:32	05:12	05:46	18:00	18:33	19:13
54°	04:28	05:10	05:46	18:00	18:35	19:17
56°	04:24	05:08	05:45	18:00	18:38	19:22
58°	04:19	05:06	05:45	18:01	18:40	19:27
S 60°	04:13	05:03	05:44	18:02	18:43	19:34

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	16:49	16:26	15:58	05:55	07:46	09:43
N 70°	16:55	16:40	16:23	05:52	07:34	09:20
68°	17:00	16:51	16:42	05:49	07:25	09:03
66°	17:04	17:00	16:57	05:47	07:17	08:49
64°	17:08	17:08	17:09	05:45	07:11	08:37
62°	17:11	17:15	17:20	05:44	07:05	08:27
60°	17:14	17:21	17:29	05:42	07:01	08:19
N 58°	17:16	17:26	17:37	05:41	06:56	08:12
56°	17:18	17:31	17:45	05:40	06:53	08:05
54°	17:20	17:35	17:51	05:39	06:49	08:00
52°	17:22	17:39	17:57	05:38	06:46	07:55
50°	17:24	17:42	18:02	05:37	06:44	07:50
45°	17:28	17:50	18:13	05:35	06:38	07:40
N 40°	17:31	17:56	18:23	05:34	06:33	07:32
35°	17:33	18:01	18:31	05:32	06:28	07:25
30°	17:36	18:06	18:38	05:31	06:25	07:18
20°	17:40	18:15	18:50	05:29	06:18	07:08
N 10°	17:44	18:22	19:01	05:27	06:13	06:58
0°	17:47	18:29	19:12	05:26	06:07	06:49
S 10°	17:51	18:36	19:22	05:24	06:02	06:41
20°	17:54	18:44	19:33	05:22	05:56	06:31
30°	17:59	18:52	19:46	05:20	05:50	06:21
35°	18:01	18:57	19:54	05:19	05:46	06:15
40°	18:04	19:03	20:02	05:18	05:42	06:08
45°	18:07	19:10	20:12	05:16	05:37	06:00
S 50°	18:11	19:18	20:24	05:14	05:32	05:50
52°	18:13	19:22	20:30	05:13	05:29	05:46
54°	18:15	19:26	20:36	05:12	05:26	05:41
56°	18:18	19:30	20:43	05:11	05:23	05:36
58°	18:20	19:36	20:51	05:10	05:20	05:30
S 60°	18:23	19:41	21:00	05:09	05:16	05:23

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	14-16 99-98%
22	07:13	07:24	11:53	23:57	11:36	
23	07:34	07:45	11:52	—:—	12:18	
24	07:55	08:06	11:52	00:39	13:01	

September 25, 26, 27 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	4°06.1	140°37.0	S18°34.0	125°21.1	S21°32.5	155°41.0	S10°33.8	311°08.4	N16°36.3	Alpheratz	357°30.4 29°15.5
1	19°08.6	155°36.7	35.0	140°21.8	32.9	170°43.0	34.0	326°10.9	36.3	Ankaa	353°03.1 -42°08.5
2	34°11.1	170°36.4	36.0	155°22.4	33.3	185°44.9	34.2	341°13.5	36.3	Schedar	349°25.9 56°42.2
3	49°13.5	185°36.1	· · 36.9	170°23.1	· · 33.7	200°46.9	· · 34.4	356°16.0	· · 36.2	Diphda	348°43.2 -17°49.2
4	64°16.0	200°35.8	37.9	185°23.7	34.0	215°48.9	34.6	11°18.6	36.2	Achernar	335°17.0 -57°04.9
5	79°18.5	215°35.5	38.9	200°24.4	34.4	230°50.9	34.7	26°21.1	36.2	Hamal	327°46.6 23°36.3
6	94°20.9	230°35.1	S18°39.9	215°25.0	S21°34.8	245°52.8	S10°34.9	41°23.7	N16°36.2	Polaris	311°34.8 89°23.0
7	109°23.4	245°34.8	40.8	230°25.7	35.1	260°54.8	35.1	56°26.2	36.1	Acamar	315°08.7 -40°10.9
8	124°25.9	260°34.5	41.8	245°26.3	35.5	275°56.8	35.3	71°28.7	36.1	Menkar	314°01.9 4°12.5
9	139°28.3	275°34.2	· · 42.8	260°27.0	· · 35.9	290°58.8	· · 35.5	86°31.3	· · 36.1	Mirfak	308°22.4 49°57.8
10	154°30.8	290°33.9	43.7	275°27.6	36.2	306°00.8	35.7	101°33.8	36.0	Aldebaran	290°35.1 16°34.1
11	169°33.3	305°33.6	44.7	290°28.2	36.6	321°02.7	35.8	116°36.4	36.0	Rigel	281°00.1 -8°10.0
12	184°35.7	320°33.3	S18°45.7	305°28.9	S21°37.0	336°04.7	S10°36.0	131°38.9	N16°36.0	Capella	280°16.2 46°01.4
13	199°38.2	335°32.9	46.6	320°29.5	37.4	351°06.7	36.2	146°41.5	36.0	Bellatrix	278°18.7 6°22.6
14	214°40.6	350°32.6	47.6	335°30.2	37.7	6°08.7	36.4	161°44.0	35.9	Elnath	277°57.0 28°37.8
15	229°43.1	5°32.3	· · 48.6	350°30.8	· · 38.1	21°10.7	· · 36.6	176°46.6	· · 35.9	Alnilam	275°33.8 -1°11.0
16	244°45.6	20°32.0	49.5	5°31.5	38.5	36°12.6	36.7	191°49.1	35.9	Betelgeuse	270°48.0 7°24.7
17	259°48.0	35°31.7	50.5	20°32.1	38.8	51°14.6	36.9	206°51.7	35.9	Canopus	263°50.7 -52°42.4
18	274°50.5	50°31.4	S18°51.4	35°32.8	S21°39.2	66°16.6	S10°37.1	221°54.2	N16°35.8	Sirius	258°22.9 -16°45.3
19	289°53.0	65°31.1	52.4	50°33.4	39.6	81°18.6	37.3	236°56.8	35.8	Adhara	255°02.9 -29°00.6
20	304°55.4	80°30.7	53.4	65°34.1	39.9	96°20.5	37.5	251°59.3	35.8	Procyon	244°47.0 5°08.9
21	319°57.9	95°30.4	· · 54.3	80°34.7	· · 40.3	111°22.5	· · 37.7	267°01.9	· · 35.7	Pollux	243°12.9 27°57.1
22	335°00.4	110°30.1	55.3	95°35.3	40.7	126°24.5	37.8	282°04.4	35.7	Avior	234°13.4 -59°36.1
23	350°02.8	125°29.8	56.2	110°36.0	41.0	141°26.5	38.0	297°07.0	35.7	Suhail	222°43.7 -43°33.0
Mer.pass. 23:40		ν -0.3' d 1.0' m -4.19		ν 0.7' d 0.4' m 0.97		ν 2.0' d 0.2' m -1.71		ν 2.5' d -0.0' m -0.02		Miaplacidus	221°37.9 -69°50.2
Alphard 217°44.3 -8°47.2 Regulus 207°30.7 11°49.2 Dubhe 193°37.3 61°35.3 Denebola 182°21.4 14°24.4 Gienah 175°40.0 -17°42.4 Acrux 172°56.3 -63°15.9 Gacrux 171°47.8 -57°16.8 Alioth 166°10.4 55°47.9 Spica 158°18.6 -11°18.9 Alkaid 152°49.6 49°10.0 Hadar 148°31.0 -60°31.0 Menkent 147°53.4 -36°31.0 Arcturus 145°44.8 19°01.8 Rigel Kent. 139°35.4 -60°57.6 Kochab 137°21.1 74°02.2 Zuben'ubi 136°52.0 -16°09.8 Alphecca 126°00.8 26°37.1 Antares 112°11.3 -26°29.8 Atria 107°02.2 -69°04.9 Sabik 101°58.5 -15°45.5 Shaula 96°05.3 -37°07.4 Rasalhague 95°55.1 12°32.6 Eltanin 90°40.5 51°29.5 Kaus Aust. 83°27.5 -34°22.1 Vega 80°30.6 38°49.1 Nunki 75°43.0 -26°15.4 Altair 61°56.1 8°57.2 Peacock 52°59.6 -56°38.3 Deneb 49°22.9 45°23.6 Enif 33°34.8 10°01.0 Al Na'ir 27°27.9 -46°48.9 Fomalhaut 15°10.1 -29°27.7 Scheat 13°41.1 28°14.9 Markab 13°25.8 15°22.2											
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	5°05.3	140°29.5	S18°57.2	125°36.6	S21°41.4	156°28.5	S10°38.2	312°09.5	N16°35.7		
1	20°07.8	155°29.2	58.2	140°37.3	41.8	171°30.4	38.4	327°12.1	35.6		
2	35°10.2	170°28.8	18°59.1	155°37.9	42.1	186°32.4	38.6	342°14.6	35.6		
3	50°12.7	185°28.5	19°00.1	170°38.6	· · 42.5	201°34.4	· · 38.8	357°17.2	· · 35.6		
4	65°15.1	200°28.2	01.0	185°39.2	42.8	216°36.4	38.9	12°19.7	35.5		
5	80°17.6	215°27.9	02.0	200°39.8	43.2	231°38.3	39.1	27°22.3	35.5		
6	95°20.1	230°27.6	S19°02.9	215°40.5	S21°43.6	246°40.3	S10°39.3	42°24.8	N16°35.5		
7	110°22.5	245°27.2	03.9	230°41.1	43.9	261°42.3	39.5	57°27.4	35.5		
8	125°25.0	260°26.9	04.8	245°41.8	44.3	276°44.3	39.7	72°29.9	35.4		
9	140°27.5	275°26.6	· · 05.8	260°42.4	· · 44.7	291°46.2	· · 39.9	87°32.5	· · 35.4		
10	155°29.9	290°26.3	06.7	275°43.0	45.0	306°48.2	40.0	102°35.0	35.4		
11	170°32.4	305°26.0	07.7	290°43.7	45.4	321°50.2	40.2	117°37.6	35.4		
12	185°34.9	320°25.6	S19°08.6	305°44.3	S21°45.8	336°52.2	S10°40.4	132°40.1	N16°35.3		
13	200°37.3	335°25.3	09.6	320°45.0	46.1	351°54.2	40.6	147°42.7	35.3		
14	215°39.8	350°25.0	10.5	335°45.6	46.5	6°56.1	40.8	162°45.2	35.3		
15	230°42.2	5°24.7	· · 11.5	350°46.2	· · 46.8	21°58.1	· · 41.0	177°47.8	· · 35.2		
16	245°44.7	20°24.4	12.4	5°46.9	47.2	37°00.1	41.1	192°50.3	35.2		
17	260°47.2	35°24.0	13.4	20°47.5	47.6	52°02.1	41.3	207°52.9	35.2		
18	275°49.6	50°23.7	S19°14.3	35°48.1	S21°47.9	67°04.0	S10°41.5	222°55.4	N16°35.2		
19	290°52.1	65°23.4	15.2	50°48.8	48.3	82°06.0	41.7	237°58.0	35.1		
20	305°54.6	80°23.1	16.2	65°49.4	48.6	97°08.0	41.9	253°00.5	35.1		
21	320°57.0	95°22.8	· · 17.1	80°50.0	· · 49.0	112°10.0	· · 42.1	268°03.1	· · 35.1		
22	335°59.5	110°22.4	18.1	95°50.7	49.4	127°11.9	42.2	283°05.6	35.0		
23	351°02.0	125°22.1	19.0	110°51.3	49.7	142°13.9	42.4	298°08.2	35.0		
Mer.pass. 23:36		ν -0.3' d 1.0' m -4.20		ν 0.6' d 0.4' m 0.97		ν 2.0' d 0.2' m -1.71		ν 2.5' d -0.0' m -0.03			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	6°04.4	140°21.8	S19°20.0	125°52.0	S21°50.1	157°15.9	S10°42.6	313°10.7	N16°35.0		
1	21°06.9	155°21.5	20.9	140°52.6	50.4	172°17.9	42.8	328°13.3	35.0		
2	36°09.4	170°21.1	21.8	155°53.2	50.8	187°19.8	43.0	343°15.9	34.9		
3	51°11.8	185°20.8	· · 22.8	170°53.9	· · 51.1	202°21.8	· · 43.2	358°18.4	· · 34.9		
4	66°14.3	200°20.5	23.7	185°54.5	51.5	217°23.8	43.3	13°21.0	34.9		
5	81°16.7	215°20.2	24.6	200°55.1	51.9	232°25.8	43.5	28°23.5	34.8		
6	96°19.2	230°19.8	S19°25.6	215°55.8	S21°52.2	247°27.7	S10°43.7	43°26.1	N16°34.8		
7	111°21.7	245°19.5	26.5	230°56.4	52.6	262°29.7	43.9	58°28.6	34.8		
8	126°24.1	260°19.2	27.4	245°57.0	52.9	277°31.7	44.1	73°31.2	34.8		
9	141°26.6	275°18.9	· · 28.4	260°57.7	· · 53.3	292°33.7	· · 44.3	88°33.7	· · 34.7		
10	156°29.1	290°18.5	29.3	275°58.3	53.6	307°35.6	44.4	103°36.3	34.7		
11	171°31.5	305°18.2	30.2	290°58.9	54.0	322°37.6	44.6	118°38.8	34.7		
12	186°34.0	320°17.9	S19°31.2	305°59.5	S21°54.3	337°39.6	S10°44.8	133°41.4	N16°34.6		
13	201°36.5	335°17.6	32.1	321°00.2	54.7	352°41.6	45.0	148°44.0	34.6		
14	216°38.9	350°17.2	33.0	336°00.8	55.1	7°43.5	45.2	163°46.5	34.6		
15	231°41.4	5°16.9	· · 34.0	351°01.4	· · 55.4	22°45.5	· · 45.4	178°49.1	· · 34.6		
16	246°43.9	20°16.6	34.9	6°02.1	55.8	37°47.5	45.5	193°51.6	34.5		
17	261°46.3	35°16.2	35.8	21°02.7	56.1	52°49.4	45.7	208°54.2	34.5		
18	276°48.8	50°15.9	S19°36.7	36°03.3	S21°56.5	67°51.4	S10°45.9	223°56.7	N16°34.5		
19	291°51.2	65°15.6	37.7	51°04.0	56.8	82°53.4	46.1	238°59.3	34.4		
20	306°53.7	80°15.3	38.6	66°04.6	57.2	97°55.4	46.3	254°01.9	34.4		
21	321°56.2	95°14.9	· · 39.5	81°05.2	· · 57.5	112°57.3	· · 46.5	269°04.4	· · 34.4		
22	336°58.6	110°14.6	40.4	96°05.8	57.9	127°59.3	46.6	284°07.0	34.3		
23	352°01.1	125°14.3	41.4	111°06.5	58.2	143°01.3	46.8	299°09.5	34.3		
Mer.pass. 23:32		ν -0.3' d 0.9' m -4.20		ν 0.6' d 0.4' m 0.97		ν 2.0' d 0.2' m -1.70		ν 2.6' d -0.0' m -0.03			
Sep 25 Tue SHA Mer.pass Venus 136°30.9 14:38 Mars 121°15.0 15:38 Jupiter 151°34.8 13:35 Saturn 307°02.2 03:15 Sep 26 Wed SHA Mer.pass Venus 135°24.2 14:38 Mars 120°31.3 15:37 Jupiter 151°23.2 13:32 Saturn 307°04.2 03:11 Sep 27 Thu SHA Mer.pass Venus 134°17.4 14:39 Mars 119°47.5 15:36 Jupiter 151°11.5 13:29 Saturn 307°06.3 03:07 Horizontal parallax Venus: 0.2 Mars: 0.1											

DUT1 = UT1-UTC = +0.2128 sec $\Delta T = TT-UT1 = +68.9712$ sec

2029 September 25 to Sep. 27 UT

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	182°04.1	S00°52.9	340°02.0	14.5'	N14°46.1	9.6' 54.0'
1	197°04.3	53.9	354°35.5	14.4'	14°55.8	9.6' 54.0'
2	212°04.5	54.8	9°08.9	14.4'	15°05.3	9.5' 54.0'
3	227°04.8	55.8	23°42.3	14.4'	15°14.8	9.4' 54.0'
4	242°05.0	56.8	38°15.7	14.3'	15°24.2	9.4' 54.0'
5	257°05.2	57.7	52°49.0	14.3'	15°33.6	9.3' 54.0'
6	272°05.4	S00°58.7	67°22.3	14.2'	N15°42.9	9.2' 54.0'
7	287°05.6	00°59.7	81°55.5	14.2'	15°52.1	9.2' 54.0'
8	302°05.8	01°00.7	96°28.7	14.2'	16°01.3	9.1' 54.0'
9	317°06.1	01.6	111°01.9	14.1'	16°10.4	9.0' 54.0'
10	332°06.3	02.6	125°35.0	14.1'	16°19.4	9.0' 54.0'
11	347°06.5	03.6	140°08.1	14.0'	16°28.4	8.9' 54.0'
12	2°06.7	S01°04.6	154°41.2	14.0'	N16°37.3	8.8' 53.9'
13	17°06.9	05.5	169°14.2	14.0'	16°46.1	8.7' 53.9'
14	32°07.1	06.5	183°47.1	13.9'	16°54.8	8.7' 53.9'
15	47°07.4	07.5	198°20.0	13.9'	17°03.5	8.6' 53.9'
16	62°07.6	08.5	212°52.9	13.8'	17°12.1	8.5' 53.9'
17	77°07.8	09.4	227°25.7	13.8'	17°20.6	8.5' 53.9'
18	92°08.0	S01°10.4	241°58.5	13.7'	N17°29.1	8.4' 53.9'
19	107°08.2	11.4	256°31.3	13.7'	17°37.5	8.3' 53.9'
20	122°08.4	12.3	271°04.0	13.7'	17°45.8	8.2' 53.9'
21	137°08.6	13.3	285°36.6	13.6'	17°54.0	8.2' 53.9'
22	152°08.9	14.3	300°09.2	13.6'	18°02.2	8.1' 53.9'
23	167°09.1	15.3	314°41.8	13.5'	18°10.2	8.0' 53.9'
SD = 15.9' $d = 1.0'$ SD = 14.7'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	182°09.3	S01°16.2	329°14.3	13.5'	N18°18.2	7.9' 53.9'
1	197°09.5	17.2	343°46.8	13.4'	18°26.2	7.8' 53.9'
2	212°09.7	18.2	358°19.2	13.4'	18°34.0	7.8' 53.9'
3	227°09.9	19.2	12°51.6	13.3'	18°41.8	7.7' 53.9'
4	242°10.1	20.1	27°24.0	13.3'	18°49.4	7.6' 54.0'
5	257°10.4	21.1	41°56.3	13.3'	18°57.0	7.5' 54.0'
6	272°10.6	S01°22.1	56°28.5	13.2'	N19°04.5	7.4' 54.0'
7	287°10.8	23.0	71°00.7	13.2'	19°12.0	7.3' 54.0'
8	302°11.0	24.0	85°32.9	13.1'	19°19.3	7.3' 54.0'
9	317°11.2	25.0	100°05.0	13.1'	19°26.6	7.2' 54.0'
10	332°11.4	26.0	114°37.1	13.0'	19°33.8	7.1' 54.0'
11	347°11.7	26.9	129°09.1	13.0'	19°40.9	7.0' 54.0'
12	2°11.9	S01°27.9	143°41.1	12.9'	N19°47.9	6.9' 54.0'
13	17°12.1	28.9	158°13.0	12.9'	19°54.8	6.8' 54.0'
14	32°12.3	29.9	172°44.9	12.8'	20°01.6	6.8' 54.0'
15	47°12.5	30.8	187°16.8	12.8'	20°08.4	6.7' 54.0'
16	62°12.7	31.8	201°48.6	12.8'	20°15.1	6.6' 54.0'
17	77°12.9	32.8	216°20.3	12.7'	20°21.6	6.5' 54.0'
18	92°13.1	S01°33.8	230°52.0	12.7'	N20°28.1	6.4' 54.0'
19	107°13.4	34.7	245°23.7	12.6'	20°34.5	6.3' 54.0'
20	122°13.6	35.7	259°55.3	12.6'	20°40.8	6.2' 54.0'
21	137°13.8	36.7	274°26.9	12.5'	20°47.0	6.1' 54.0'
22	152°14.0	37.6	288°58.4	12.5'	20°53.1	6.0' 54.0'
23	167°14.2	38.6	303°29.9	12.4'	20°59.2	5.9' 54.0'
SD = 15.9' $d = 1.0'$ SD = 14.7'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	182°14.4	S01°39.6	318°01.3	12.4'	N21°05.1	5.8' 54.0'
1	197°14.6	40.6	332°32.7	12.3'	21°10.9	5.7' 54.0'
2	212°14.9	41.5	347°04.0	12.3'	21°16.7	5.7' 54.0'
3	227°15.1	42.5	1°35.3	12.3'	21°22.3	5.6' 54.1'
4	242°15.3	43.5	16°06.6	12.2'	21°27.9	5.5' 54.1'
5	257°15.5	44.5	30°37.8	12.2'	21°33.4	5.4' 54.1'
6	272°15.7	S01°45.4	45°09.0	12.1'	N21°38.7	5.3' 54.1'
7	287°15.9	46.4	59°40.1	12.1'	21°44.0	5.2' 54.1'
8	302°16.1	47.4	74°11.1	12.0'	21°49.2	5.1' 54.1'
9	317°16.3	48.3	88°42.2	12.0'	21°54.2	5.0' 54.1'
10	332°16.6	49.3	103°13.2	11.9'	21°59.2	4.9' 54.1'
11	347°16.8	50.3	117°44.1	11.9'	22°04.1	4.8' 54.1'
12	2°17.0	S01°51.3	132°15.0	11.9'	N22°08.8	4.7' 54.1'
13	17°17.2	52.2	146°45.8	11.8'	22°13.5	4.6' 54.1'
14	32°17.4	53.2	161°16.7	11.8'	22°18.1	4.5' 54.1'
15	47°17.6	54.2	175°47.4	11.7'	22°22.6	4.4' 54.2'
16	62°17.8	55.2	190°18.1	11.7'	22°26.9	4.3' 54.2'
17	77°18.0	56.1	204°48.8	11.6'	22°31.2	4.2' 54.2'
18	92°18.2	S01°57.1	219°19.5	11.6'	N22°35.4	4.1' 54.2'
19	107°18.5	58.1	233°50.1	11.6'	22°39.4	4.0' 54.2'
20	122°18.7	01°59.0	248°20.6	11.5'	22°43.4	3.8' 54.2'
21	137°18.9	02°00.0	262°51.1	11.5'	22°47.2	3.7' 54.2'
22	152°19.1	01.0	277°21.6	11.4'	22°51.0	3.6' 54.2'
23	167°19.3	02.0	291°52.1	11.4'	22°54.6	3.5' 54.2'
SD = 15.9' $d = 1.0'$ SD = 14.7'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:23	04:50	05:57	17:43	18:50	20:15
N 70°	03:39	04:56	05:57	17:44	18:44	20:00
68°	03:52	05:00	05:56	17:45	18:40	19:47
66°	04:02	05:04	05:55	17:45	18:36	19:38
64°	04:10	05:08	05:55	17:46	18:33	19:30
62°	04:17	05:10	05:54	17:46	18:31	19:23
60°	04:23	05:13	05:54	17:47	18:28	19:17
N 58°	04:28	05:15	05:54	17:47	18:26	19:13
56°	04:33	05:16	05:53	17:48	18:25	19:08
54°	04:36	05:18	05:53	17:48	18:23	19:05
52°	04:40	05:19	05:53	17:49	18:22	19:02
50°	04:43	05:20	05:53	17:49	18:21	18:59
45°	04:49	05:23	05:52	17:50	18:19	18:53
N 40°	04:53	05:25	05:52	17:50	18:17	18:49
35°	04:56	05:26	05:51	17:51	18:16	18:45
30°	04:59	05:27	05:51	17:51	18:15	18:43
20°	05:02	05:28	05:50	17:52	18:14	18:40
N 10°	05:04	05:28	05:49	17:53	18:14	18:39
0°	05:03	05:27	05:48	17:54	18:15	18:39
S 10°	05:02	05:26	05:47	17:56	18:17	18:41
20°	04:58	05:24	05:46	17:57	18:19	18:45
30°	04:53	05:20	05:44	17:59	18:22	18:50
35°	04:49	05:18	05:43	18:00	18:25	18:55
40°	04:44	05:15	05:42	18:01	18:28	19:00
45°	04:37	05:12	05:41	18:02	18:31	19:06
S 50°	04:29	05:07	05:40	18:04	18:36	19:15
52°	04:25	05:05	05:39	18:05	18:38	19:19
54°	04:20	05:03	05:38	18:05	18:41	19:23
56°	04:15	05:00	05:37	18:06	18:44	19:29
58°	04:10	04:57	05:36	18:07	18:47	19:35
S 60°	04:03	04:53	05:35	18:09	18:51	19:42

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	15:06	□	□	12:06	□	□
N 70°	15:57	□	□	11:16	□	□
68°	16:30	16:11	□	10:45	12:38	□
66°	16:53	16:50	16:49	10:22	12:00	13:40
64°	17:12	17:18	17:30	10:05	11:33	12:59
62°	17:28	17:39	17:58	09:50	11:12	12:31
60°	17:41	17:56	18:20	09:38	10:55	12:10
N 58°	17:52	18:11	18:37	09:27	10:41	11:52
56°	18:01	18:23	18:52	09:18	10:29	11:38
54°	18:10	18:34	19:05	09:10	10:19	11:25
52°	18:18	18:44	19:17	09:03	10:09	11:14
50°	18:25	18:53	19:27	08:56	10:01	11:04
45°	18:40	19:11	19:48	08:42	09:44	10:43
N 40°	18:52	19:26	20:05	08:31	09:29	10:27
35°	19:03	19:39	20:19	08:21	09:17	10:13
30°	19:12	19:50	20:32	08:12	09:06	10:01
20°	19:29	20:09	20:54	07:58	08:48	09:40
N 10°	19:43	20:26	21:12	07:45	08:33	09:22
0°	19:56	20:42	21:30	07:33	08:18	09:05
S 10°	20:09	20:58	21:48	07:21	08:03	08:48
20°	20:24	21:15	22:06	07:08	07:48	08:30
30°	20:40	21:35	22:28	06:54	07:30	08:10
35°	20:50	21:46	22:41	06:46	07:20	07:58
40°	21:01	21:59	22:56	06:36	07:08	07:44
45°	21:14	22:15	23:14	06:25	06:54	07:27
S 50°	21:30	22:34	23:36	06:12	06:37	07:07
52°	21:38	22:44	23:46	06:06	06:29	06:58
54°	21:46	22:54	23:58	05:59	06:20	06:47
56°	21:55	23:06	□	05:51	06:10	06:35
58°	22:06	23:19	□	05:43	05:59	06:21
S 60°	22:19	23:35	□	05:33	05:46	06:05

Day	Sun		Mer. Pass	Moon		Age 17-19 95-84%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Upper Mer. Pass hh:mm	Lower Mer. Pass hh:mm	
25	08:16	08:27	11:52	01:22	13:44	
26	08:37	08:47	11:51	02:07	14:30	
27	08:58	09:08	11:51	02:53	15:17	

September 28, 29, 30 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	7°03.6	140°13.9	S19°42.3	126°07.1	S21°58.6	158°03.3	S10°47.0	314°12.1	N16°34.3			Alpheratz	357°30.4	29°15.5			
1	22°06.0	155°13.6	43.2	141°07.7	58.9	173°05.2	47.2	329°14.6	34.3			Ankaa	353°03.1	-42°08.5			
2	37°08.5	170°13.3	44.1	156°08.3	59.3	188°07.2	47.4	344°17.2	34.2			Schedar	349°25.9	56°42.2			
3	52°11.0	185°12.9	· · 45.0	171°09.0	21°59.6	203°09.2	· · 47.6	359°19.8	· · 34.2			Diphda	348°43.2	-17°49.2			
4	67°13.4	200°12.6	46.0	186°09.6	22°00.0	218°11.2	47.7	14°22.3	34.2			Achernar	335°17.0	-57°04.9			
5	82°15.9	215°12.3	46.9	201°10.2	00.3	233°13.1	47.9	29°24.9	34.1			Hamal	327°46.5	23°36.3			
6	97°18.3	230°12.0	S19°47.8	216°10.8	S22°00.7	248°15.1	S10°48.1	44°27.4	N16°34.1			Polaris	311°33.4	89°23.0			
7	112°20.8	245°11.6	48.7	231°11.5	01.0	263°17.1	48.3	59°30.0	34.1			Acamar	315°08.6	-40°10.9			
8	127°23.3	260°11.3	49.6	246°12.1	01.4	278°19.0	48.5	74°32.6	34.0			Menkar	314°01.9	4°12.5			
9	142°25.7	275°11.0	· · 50.5	261°12.7	· · 01.7	293°21.0	· · 48.7	89°35.1	· · 34.0			Mirfak	308°22.4	49°57.9			
10	157°28.2	290°10.6	51.5	276°13.3	02.1	308°23.0	48.8	104°37.7	34.0			Aldebaran	290°35.1	16°34.1			
11	172°30.7	305°10.3	52.4	291°14.0	02.4	323°25.0	49.0	119°40.2	34.0			Rigel	281°00.1	-8°10.0			
12	187°33.1	320°10.0	S19°53.3	306°14.6	S22°02.8	338°26.9	S10°49.2	134°42.8	N16°33.9			Capella	280°16.1	46°01.4			
13	202°35.6	335°09.6	54.2	321°15.2	03.1	353°28.9	49.4	149°45.3	33.9			Bellatrix	278°18.7	6°22.6			
14	217°38.1	350°09.3	55.1	336°15.8	03.5	8°30.9	49.6	164°47.9	33.9			Elnath	277°57.0	28°37.8			
15	232°40.5	5°09.0	· · 56.0	351°16.4	· · 03.8	23°32.9	· · 49.8	179°50.5	· · 33.8			Alnilam	275°33.8	-1°11.0			
16	247°43.0	20°08.6	56.9	6°17.1	04.2	38°34.8	49.9	194°53.0	33.8			Betelgeuse	270°47.9	7°24.7			
17	262°45.5	35°08.3	57.8	21°17.7	04.5	53°36.8	50.1	209°55.6	33.8			Canopus	263°50.7	-52°42.5			
18	277°47.9	50°08.0	S19°58.7	36°18.3	S22°04.9	68°38.8	S10°50.3	224°58.2	N16°33.8			Sirius	258°22.9	-16°45.3			
19	292°50.4	65°07.6	19°59.7	51°18.9	05.2	83°40.7	50.5	240°00.7	33.7			Adhara	255°02.9	-29°00.6			
20	307°52.8	80°07.3	20°00.6	66°19.6	05.5	98°42.7	50.7	255°03.3	33.7			Procyon	244°47.0	5°08.9			
21	322°55.3	95°07.0	· · 01.5	81°20.2	· · 05.9	113°44.7	· · 50.9	270°05.8	· · 33.7			Pollux	243°12.8	27°57.0			
22	337°57.8	110°06.6	02.4	96°20.8	06.2	128°46.7	51.0	285°08.4	33.6			Avior	234°13.4	-59°36.1			
23	353°00.2	125°06.3	03.3	111°21.4	06.6	143°48.6	51.2	300°11.0	33.6			Suhail	222°43.7	-43°33.0			
Mer.pass. 23:28			ν -0.3' d 0.9' m -4.21	ν 0.6' d 0.4' m 0.98			ν 2.0' d 0.2' m -1.70	ν 2.6' d -0.0' m -0.04									

Sat	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	8°02.7	140°06.0	S20°04.2	126°22.0	S22°06.9	158°50.6	S10°51.4	315°13.5	N16°33.6		
1	23°05.2	155°05.6	05.1	141°22.6	07.3	173°52.6	51.6	330°16.1	33.5		
2	38°07.6	170°05.3	06.0	156°23.3	07.6	188°54.5	51.8	345°18.6	33.5		
3	53°10.1	185°04.9	· · 06.9	171°23.9	· · 08.0	203°56.5	· · 52.0	0°21.2	· · 33.5		
4	68°12.6	200°04.6	07.8	186°24.5	08.3	218°58.5	52.1	15°23.8	33.4		
5	83°15.0	215°04.3	08.7	201°25.1	08.6	234°00.4	52.3	30°26.3	33.4		
6	98°17.5	230°03.9	S20°09.6	216°25.7	S22°09.0	249°02.4	S10°52.5	45°28.9	N16°33.4		
7	113°20.0	245°03.6	10.5	231°26.4	09.3	264°04.4	52.7	60°31.5	33.4		
8	128°22.4	260°03.3	11.4	246°27.0	09.7	279°06.4	52.9	75°34.0	33.3		
9	143°24.9	275°02.9	· · 12.3	261°27.6	· · 10.0	294°08.3	· · 53.1	90°36.6	· · 33.3		
10	158°27.3	290°02.6	13.2	276°28.2	10.3	309°10.3	53.2	105°39.1	33.3		
11	173°29.8	305°02.2	14.1	291°28.8	10.7	324°12.3	53.4	120°41.7	33.2		
12	188°32.3	320°01.9	S20°15.0	306°29.4	S22°11.0	339°14.2	S10°53.6	135°44.3	N16°33.2		
13	203°34.7	335°01.6	15.9	321°30.0	11.4	354°16.2	53.8	150°46.8	33.2		
14	218°37.2	350°01.2	16.8	336°30.7	11.7	9°18.2	54.0	165°49.4	33.1		
15	233°39.7	5°00.9	· · 17.6	351°31.3	· · 12.0	24°20.2	· · 54.2	180°52.0	· · 33.1		
16	248°42.1	20°00.5	18.5	6°31.9	12.4	39°22.1	54.4	195°54.5	33.1		
17	263°44.6	35°00.2	19.4	21°32.5	12.7	54°24.1	54.5	210°57.1	33.0		
18	278°47.1	49°59.9	S20°20.3	36°33.1	S22°13.1	69°26.1	S10°54.7	225°59.7	N16°33.0		
19	293°49.5	64°59.5	21.2	51°33.7	13.4	84°28.0	54.9	241°02.2	33.0		
20	308°52.0	79°59.2	22.1	66°34.3	13.7	99°30.0	55.1	256°04.8	33.0		
21	323°54.4	94°58.8	· · 23.0	81°35.0	· · 14.1	114°32.0	· · 55.3	271°07.4	· · 32.9		
22	338°56.9	109°58.5	23.9	96°35.6	14.4	129°33.9	55.5	286°09.9	32.9		
23	353°59.4	124°58.2	24.7	111°36.2	14.7	144°35.9	55.6	301°12.5	32.9		
Mer.pass. 23:24			ν -0.3' d 0.9' m -4.22	ν 0.6' d 0.3' m 0.98			ν 2.0' d 0.2' m -1.70	ν 2.6' d -0.0' m -0.05			

Sun	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	9°01.8	139°57.8	S20°25.6	126°36.8	S22°15.1	159°37.9	S10°55.8	316°15.1	N16°32.8		
1	24°04.3	154°57.5	26.5	141°37.4	15.4	174°39.8	56.0	331°17.6	32.8		
2	39°06.8	169°57.1	27.4	156°38.0	15.8	189°41.8	56.2	346°20.2	32.8		
3	54°09.2	184°56.8	· · 28.3	171°38.6	· · 16.1	204°43.8	· · 56.4	1°22.8	· · 32.7		
4	69°11.7	199°56.4	29.2	186°39.2	16.4	219°45.8	56.6	16°25.3	32.7		
5	84°14.2	214°56.1	30.0	201°39.8	16.8	234°47.7	56.7	31°27.9	32.7		
6	99°16.6	229°55.8	S20°30.9	216°40.4	S22°17.1	249°49.7	S10°56.9	46°30.5	N16°32.6		
7	114°19.1	244°55.4	31.8	231°41.1	17.4	264°51.7	57.1	61°33.0	32.6		
8	129°21.6	259°55.1	32.7	246°41.7	17.8	279°53.6	57.3	76°35.6	32.6		
9	144°24.0	274°54.7	· · 33.6	261°42.3	· · 18.1	294°55.6	· · 57.5	91°38.2	· · 32.5		
10	159°26.5	289°54.4	34.4	276°42.9	18.4	309°57.6	57.7	106°40.7	32.5		
11	174°28.9	304°54.0	35.3	291°43.5	18.8	324°59.5	57.8	121°43.3	32.5		
12	189°31.4	319°53.7	S20°36.2	306°44.1	S22°19.1	340°01.5	S10°58.0	136°45.9	N16°32.5		
13	204°33.9	334°53.4	37.1	321°44.7	19.4	355°03.5	58.2	151°48.4	32.4		
14	219°36.3	349°53.0	37.9	336°45.3	19.8	10°05.4	58.4	166°51.0	32.4		
15	234°38.8	4°52.7	· · 38.8	351°45.9	· · 20.1	25°07.4	· · 58.6	181°53.6	· · 32.4		
16	249°41.3	19°52.3	39.7	6°46.5	20.4	40°09.4	58.8	196°56.1	32.3		
17	264°43.7	34°52.0	40.6	21°47.1	20.7	55°11.3	58.9	211°58.7	32.3		
18	279°46.2	49°51.6	S20°41.4	36°47.7	S22°21.1	70°13.3	S10°59.1	227°01.3	N16°32.3		
19	294°48.7	64°51.3	42.3	51°48.3	21.4	85°15.3	59.3	242°03.8	32.2		
20	309°51.1	79°50.9	43.2	66°48.9	21.7	100°17.2	59.5	257°06.4	32.2		
21	324°53.6	94°50.6	· · 44.0	81°49.5	· · 22.1	115°19.2	· · 59.7	272°09.0	· · 32.2		
22	339°56.1	109°50.2	44.9	96°50.1	22.4	130°21.2	10°59.9	287°11.5	32.1		
23	354°58.5	124°49.9	45.8	111°50.7	22.7	145°23.1	11°00.1	302°14.1	32.1		
Mer.pass. 23:20			ν -0.3' d 0.9' m -4.23	ν 0.6' d 0.3' m 0.98			ν 2.0' d 0.2' m -1.70	ν 2.6' d -0.0' m -0.05			

Sep 28 Fri	SHA	Mer.pass
Venus	133°10.4	14:39
Mars	119°03.5	15:35
Jupiter	150°59.7	13:26
Saturn	307°08.5	03:03

Sep 29 Sat	SHA	Mer.pass
Venus	132°03.2	14:40
Mars	118°19.3	15:34
Jupiter	150°47.9	13:23
Saturn	307°10.8	02:59

Sep 30 Sun	SHA	Mer.pass
Venus	130°56.0	14:40
Mars	117°34.9	15:33
Jupiter	150°36.0	13:20
Saturn	307°13.2	02:54

Horizontal parallax	
Venus:	0.2
Mars:	0.1

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	182°19.5	S02°02.9	306°22.4	11.4'	N22°58.1	3.4'	54.3'
1	197°19.7	03.9	320°52.8	11.3'	23°01.6	3.3'	54.3'
2	212°19.9	04.9	335°23.1	11.3'	23°04.9	3.2'	54.3'
3	227°20.1	· · 05.9	349°53.4	11.2'	23°08.1	3.1'	54.3'
4	242°20.4	06.8	4°23.6	11.2'	23°11.2	3.0'	54.3'
5	257°20.6	07.8	18°53.8	11.2'	23°14.2	2.9'	54.3'
6	272°20.8	S02°08.8	33°24.0	11.1'	N23°17.1	2.8'	54.3'
7	287°21.0	09.7	47°54.1	11.1'	23°19.9	2.7'	54.3'
8	302°21.2	10.7	62°24.2	11.1'	23°22.5	2.6'	54.4'
9	317°21.4	· · 11.7	76°54.3	11.0'	23°25.1	2.4'	54.4'
10	332°21.6	12.7	91°24.3	11.0'	23°27.5	2.3'	54.4'
11	347°21.8	13.6	105°54.3	11.0'	23°29.9	2.2'	54.4'
12	2°22.0	S02°14.6	120°24.2	10.9'	N23°32.1	2.1'	54.4'
13	17°22.2	15.6	134°54.2	10.9'	23°34.2	2.0'	54.4'
14	32°22.4	16.5	149°24.0	10.9'	23°36.2	1.9'	54.4'
15	47°22.7	· · 17.5	163°53.9	10.8'	23°38.1	1.8'	54.5'
16	62°22.9	18.5	178°23.7	10.8'	23°39.9	1.7'	54.5'
17	77°23.1	19.5	192°53.5	10.8'	23°41.5	1.5'	54.5'
18	92°23.3	S02°20.4	207°23.3	10.7'	N23°43.1	1.4'	54.5'
19	107°23.5	21.4	221°53.0	10.7'	23°44.5	1.3'	54.5'
20	122°23.7	22.4	236°22.7	10.7'	23°45.8	1.2'	54.5'
21	137°23.9	· · 23.3	250°52.3	10.6'	23°47.0	1.1'	54.6'
22	152°24.1	24.3	265°22.0	10.6'	23°48.1	1.0'	54.6'
23	167°24.3	25.3	279°51.6	10.6'	23°49.1	0.9'	54.6'
SD = 16.0'			SD = 14.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	182°24.5	S02°26.3	294°21.2	10.6'	N23°50.0	0.7'	54.6'
1	197°24.7	27.2	308°50.7	10.5'	23°50.7	0.6'	54.6'
2	212°24.9	28.2	323°20.3	10.5'	23°51.3	0.5'	54.7'
3	227°25.2	· · 29.2	337°49.8	10.5'	23°51.8	0.4'	54.7'
4	242°25.4	30.1	352°19.2	10.5'	23°52.2	0.3'	54.7'
5	257°25.6	31.1	6°48.7	10.4'	23°52.5	0.2'	54.7'
6	272°25.8	S02°32.1	21°18.1	10.4'	N23°52.7	0.0'	54.7'
7	287°26.0	33.1	35°47.5	10.4'	23°52.7	-0.1'	54.8'
8	302°26.2	34.0	50°16.9	10.4'	23°52.6	-0.2'	54.8'
9	317°26.4	· · 35.0	64°46.2	10.3'	23°52.4	-0.3'	54.8'
10	332°26.6	36.0	79°15.6	10.3'	23°52.1	-0.4'	54.8'
11	347°26.8	36.9	93°44.9	10.3'	23°51.7	-0.6'	54.8'
12	2°27.0	S02°37.9	108°14.2	10.3'	N23°51.1	-0.7'	54.9'
13	17°27.2	38.9	122°43.5	10.3'	23°50.5	-0.8'	54.9'
14	32°27.4	39.9	137°12.7	10.2'	23°49.7	-0.9'	54.9'
15	47°27.6	· · 40.8	151°42.0	10.2'	23°48.8	-1.0'	54.9'
16	62°27.8	41.8	166°11.2	10.2'	23°47.7	-1.1'	55.0'
17	77°28.1	42.8	180°40.4	10.2'	23°46.6	-1.3'	55.0'
18	92°28.3	S02°43.7	195°09.6	10.2'	N23°45.3	-1.4'	55.0'
19	107°28.5	44.7	209°38.7	10.2'	23°43.9	-1.5'	55.0'
20	122°28.7	45.7	224°07.9	10.1'	23°42.4	-1.6'	55.0'
21	137°28.9	· · 46.7	238°37.0	10.1'	23°40.8	-1.7'	55.1'
22	152°29.1	47.6	253°06.2	10.1'	23°39.0	-1.9'	55.1'
23	167°29.3	48.6	267°35.3	10.1'	23°37.2	-2.0'	55.1'
SD = 16.0'			SD = 14.9'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	182°29.5	S02°49.6	282°04.4	10.1'	N23°35.2	-2.1'	55.1'
1	197°29.7	50.5	296°33.5	10.1'	23°33.1	-2.2'	55.2'
2	212°29.9	51.5	311°02.5	10.1'	23°30.8	-2.4'	55.2'
3	227°30.1	· · 52.5	325°31.6	10.1'	23°28.5	-2.5'	55.2'
4	242°30.3	53.5	340°00.7	10.0'	23°26.0	-2.6'	55.2'
5	257°30.5	54.4	354°29.7	10.0'	23°23.4	-2.7'	55.3'
6	272°30.7	S02°55.4	8°58.8	10.0'	N23°20.7	-2.8'	55.3'
7	287°30.9	56.4	23°27.8	10.0'	23°17.9	-3.0'	55.3'
8	302°31.1	57.3	37°56.8	10.0'	23°14.9	-3.1'	55.4'
9	317°31.3	· · 58.3	52°25.8	10.0'	23°11.8	-3.2'	55.4'
10	332°31.5	02°59.3	66°54.8	10.0'	23°08.6	-3.3'	55.4'
11	347°31.7	03°00.2	81°23.8	10.0'	23°05.3	-3.4'	55.4'
12	2°31.9	S03°01.2	95°52.8	10.0'	N23°01.9	-3.6'	55.5'
13	17°32.1	02.2	110°21.8	10.0'	22°58.3	-3.7'	55.5'
14	32°32.3	03.2	124°50.8	10.0'	22°54.6	-3.8'	55.5'
15	47°32.6	· · 04.1	139°19.8	10.0'	22°50.8	-3.9'	55.5'
16	62°32.8	05.1	153°48.8	10.0'	22°46.9	-4.0'	55.6'
17	77°33.0	06.1	168°17.8	10.0'	22°42.8	-4.2'	55.6'
18	92°33.2	S03°07.0	182°46.7	10.0'	N22°38.7	-4.3'	55.6'
19	107°33.4	08.0	197°15.7	10.0'	22°34.4	-4.4'	55.7'
20	122°33.6	09.0	211°44.7	10.0'	22°30.0	-4.5'	55.7'
21	137°33.8	· · 09.9	226°13.7	10.0'	22°25.5	-4.6'	55.7'
22	152°34.0	10.9	240°42.7	10.0'	22°20.8	-4.8'	55.8'
23	167°34.2	11.9	255°11.6	10.0'	22°16.1	-4.9'	55.8'
SD = 16.0'			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:40	05:03	06:11	17:27	18:34	19:57
N 70°	03:53	05:08	06:08	17:30	18:30	19:44
68°	04:04	05:11	06:07	17:32	18:27	19:34
66°	04:13	05:14	06:05	17:34	18:25	19:25
64°	04:20	05:16	06:04	17:35	18:22	19:18
62°	04:26	05:18	06:02	17:37	18:21	19:13
60°	04:31	05:20	06:01	17:38	18:19	19:08
N 58°	04:35	05:21	06:00	17:39	18:18	19:04
56°	04:39	05:22	05:59	17:40	18:17	19:00
54°	04:42	05:23	05:59	17:41	18:16	18:57
52°	04:45	05:24	05:58	17:42	18:15	18:54
50°	04:47	05:25	05:57	17:42	18:14	18:52
45°	04:52	05:26	05:56	17:44	18:13	18:47
N 40°	04:56	05:27	05:54	17:45	18:12	18:44
35°	04:59	05:28	05:53	17:47	18:12	18:41
30°	05:01	05:28	05:52	17:48	18:12	18:39
20°	05:03	05:28	05:50	17:50	18:12	18:37
N 10°	05:03	05:28	05:49	17:52	18:13	18:37
0°	05:02	05:26	05:47	17:53	18:14	18:38
S 10°	05:00	05:24	05:45	17:55	18:16	18:41
20°	04:55	05:21	05:43	17:58	18:20	18:45
30°	04:49	05:17	05:41	18:00	18:24	18:52
35°	04:44	05:14	05:39	18:02	18:27	18:57
40°	04:38	05:10	05:37	18:04	18:31	19:03
45°	04:31	05:06	05:35	18:06	18:35	19:10
S 50°	04:22	05:01	05:33	18:08	18:41	19:20
52°	04:17	04:58	05:32	18:10	18:44	19:24
54°	04:12	04:55	05:31	18:11	18:47	19:29
56°	04:07	04:52	05:29	18:12	18:50	19:35
58°	04:00	04:48	05:28	18:14	18:54	19:42
S 60°	03:52	04:44	05:26	18:16	18:58	19:50

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	16:53	17:39	19:19	15:18	16:17	16:24
64°	17:55	18:46	20:03	14:16	15:10	15:40
62°	18:30	19:21	20:32	13:41	14:35	15:11
60°	18:55	19:46	20:54	13:16	14:10	14:49
N 58°	19:15	20:06	21:11	12:56	13:50	14:30
56°	19:32	20:23	21:26	12:40	13:33	14:15
54°	19:46	20:37	21:39	12:26	13:19	14:02
52°	19:58	20:49	21:51	12:14	13:06	13:50
50°	20:09	21:00	22:01	12:03	12:55	13:40
45°	20:32	21:23	22:22	11:40	12:32	13:19
N 40°	20:50	21:41	22:39	11:22	12:14	13:01
35°	21:05	21:57	22:53	11:07	11:59	12:47
30°	21:19	22:10	23:05	10:54	11:45	12:34
20°	21:41	22:33	23:26	10:31	11:22	12:12
N 10°	22:01	22:52	23:45	10:12	11:03	11:53
0°	22:20	23:10	· · ·	09:54	10:44	11:35
S 10°	22:38	23:29	· · ·	09:36	10:25	11:17
20°	22:58	23:48	· · ·	09:16	10:06	10:58
30°	23:21	· · ·	00:11	08:54	09:43	10:36
35°	23:34	· · ·	00:24	08:41	09:29	10:23
40°	23:50	· · ·	00:40	08:26	09:14	10:08
45°	· · ·	00:08	00:58	08:08	08:55	09:50
S 50°	· · ·	00:32	01:20	07:45	08:32	09:28
52°	· · ·	00:43	01:31	07:35	08:21	09:17
54°	· · ·	00:55	01:43	07:22	08:08	09:05
56°	00:12	01:10	01:57	07:09	07:54	08:51
58°	00:28	01:27	02:14	06:53	07:37	08:35
S 60°	00:47	01:47	02:34	06:33	07:16	08:15

Day	Sun		Mer. Pass hh:mm	Moon		Age 20-22 77-59%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
28	09:18	09:28	11:51	03:42	16:07	
29	09:38	09:48	11:50	04:32	16:57	
30	09:58	10:08	11:50	05:23	17:48	

October 01, 02, 03 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	10°01.0	139°49.5	S20°46.6	126°51.4	S22°23.1	160°25.1	S11°00.2	317°16.7	N16°32.1	Alpheratz	357°30.4	29°15.5
1	25°03.4	154°49.2	47.5	141°52.0	23.4	175°27.1	00.4	332°19.3	32.0	Ankaa	353°03.1	-42°08.5
2	40°05.9	169°48.8	48.4	156°52.6	23.7	190°29.0	00.6	347°21.8	32.0	Schedar	349°25.9	56°42.2
3	55°08.4	184°48.5	· · 49.2	171°53.2	· · 24.0	205°31.0	· · 00.8	2°24.4	· · 32.0	Diphda	348°43.2	-17°49.2
4	70°10.8	199°48.1	50.1	186°53.8	24.4	220°33.0	01.0	17°27.0	31.9	Achernar	335°17.0	-57°04.9
5	85°13.3	214°47.8	50.9	201°54.4	24.7	235°34.9	01.2	32°29.5	31.9	Hamal	327°46.5	23°36.3
6	100°15.8	229°47.5	S20°51.8	216°55.0	S22°25.0	250°36.9	S11°01.3	47°32.1	N16°31.9	Polaris	311°32.1	89°23.0
7	115°18.2	244°47.1	52.7	231°55.6	25.3	265°38.9	01.5	62°34.7	31.8	Acamar	315°08.6	-40°10.9
8	130°20.7	259°46.8	53.5	246°56.2	25.7	280°40.8	01.7	77°37.2	31.8	Menkar	314°01.9	4°12.5
9	145°23.2	274°46.4	· · 54.4	261°56.8	· · 26.0	295°42.8	· · 01.9	92°39.8	· · 31.8	Mirfak	308°22.3	49°57.9
10	160°25.6	289°46.1	55.2	276°57.4	26.3	310°44.8	02.1	107°42.4	31.8	Aldebaran	290°35.1	16°34.1
11	175°28.1	304°45.7	56.1	291°58.0	26.7	325°46.7	02.3	122°45.0	31.7	Rigel	281°00.1	-8°10.0
12	190°30.6	319°45.4	S20°57.0	306°58.6	S22°27.0	340°48.7	S11°02.4	137°47.5	N16°31.7	Capella	280°16.1	46°01.4
13	205°33.0	334°45.0	57.8	321°59.2	27.3	355°50.7	02.6	152°50.1	31.7	Bellatrix	278°18.7	6°22.6
14	220°35.5	349°44.7	58.7	336°59.8	27.6	10°52.6	02.8	167°52.7	31.6	Elnath	277°56.9	28°37.8
15	235°37.9	4°44.3	20°59.5	352°00.4	· · 27.9	25°54.6	· · 03.0	182°55.3	· · 31.6	Alnilam	275°33.8	-1°11.0
16	250°40.4	19°43.9	21°00.4	7°01.0	28.3	40°56.6	03.2	197°57.8	31.6	Betelgeuse	270°47.9	7°24.7
17	265°42.9	34°43.6	01.2	22°01.6	28.6	55°58.5	03.4	213°00.4	31.5	Canopus	263°50.7	-52°42.5
18	280°45.3	49°43.2	S21°02.1	37°02.2	S22°28.9	71°00.5	S11°03.6	228°03.0	N16°31.5	Sirius	258°22.9	-16°45.3
19	295°47.8	64°42.9	02.9	52°02.8	29.2	86°02.5	03.7	243°05.5	31.5	Adhara	255°02.9	-29°00.6
20	310°50.3	79°42.5	03.8	67°03.3	29.6	101°04.4	03.9	258°08.1	31.4	Procyon	244°46.9	5°08.9
21	325°52.7	94°42.2	· · 04.6	82°03.9	· · 29.9	116°06.4	· · 04.1	273°10.7	· · 31.4	Pollux	243°12.8	27°57.0
22	340°55.2	109°41.8	05.5	97°04.5	30.2	131°08.4	04.3	288°13.3	31.4	Avior	234°13.3	-59°36.1
23	355°57.7	124°41.5	06.3	112°05.1	30.5	146°10.3	04.5	303°15.8	31.3	Suhail	222°43.7	-43°33.0
Mer.pass. 23:16		ν -0.3' d 0.9' m -4.24		ν 0.6' d 0.3' m 0.99		ν 2.0' d 0.2' m -1.70		ν 2.6' d -0.0' m -0.06		Miaplacidus	221°37.8	-69°50.2
										Alphard	217°44.2	-8°47.3
										Regulus	207°30.7	11°49.2
										Dubhe	193°37.2	61°35.2
										Denebola	182°21.4	14°24.3
										Gienah	175°40.0	-17°42.4
										Acrux	172°56.3	-63°15.9
										Gacrux	171°47.8	-57°16.8
										Alioth	166°10.4	55°47.9
										Spica	158°18.6	-11°18.9
										Alkaid	152°49.6	49°10.0
										Hadar	148°31.0	-60°31.0
										Menkent	147°53.4	-36°30.9
										Arcturus	145°44.8	19°01.8
										Rigel Kent.	139°35.4	-60°57.5
										Kochab	137°21.2	74°02.1
										Zuben'ubi	136°52.1	-16°09.8
										Alphecca	126°00.8	26°37.1
										Antares	112°11.3	-26°29.8
										Atria	107°02.2	-69°04.9
										Sabik	101°58.5	-15°45.5
										Shaula	96°05.3	-37°07.4
										Rasalhague	95°55.1	12°32.6
										Eltanin	90°40.5	51°29.5
										Kaus Aust.	83°27.5	-34°22.1
										Vega	80°30.6	38°49.1
										Nunki	75°43.0	-26°15.4
										Altair	61°56.2	8°57.2
										Peacock	52°59.6	-56°38.3
										Deneb	49°22.9	45°23.6
										Enif	33°34.8	10°01.0
										Al Na'ir	27°27.9	-46°48.9
										Fomalhaut	15°10.1	-29°27.7
										Scheat	13°41.1	28°15.0
										Markab	13°25.8	15°22.2
										Oct 01 Mon	SHA	Mer.pass
										Venus	129°48.6	14:41
										Mars	116°50.4	15:32
										Jupiter	150°24.1	13:17
										Saturn	307°15.7	02:50
										Oct 02 Tue	SHA	Mer.pass
										Venus	128°41.0	14:42
										Mars	116°05.6	15:31
										Jupiter	150°12.2	13:13
										Saturn	307°18.3	02:46
										Oct 03 Wed	SHA	Mer.pass
										Venus	127°33.3	14:42
										Mars	115°20.7	15:30
										Jupiter	150°00.2	13:10
										Saturn	307°21.0	02:42
										Horizontal parallax		
										Venus:	0.2	
										Mars:	0.1	
0	11°00.1	139°41.1	S21°07.2	127°05.7	S22°30.9	161°12.3	S11°04.7	318°18.4	N16°31.3			
1	26°02.6	154°40.8	08.0	142°06.3	31.2	176°14.3	04.8	333°21.0	31.3			
2	41°05.0	169°40.4	08.9	157°06.9	31.5	191°16.2	05.0	348°23.6	31.2			
3	56°07.5	184°40.1	· · 09.7	172°07.5	· · 31.8	206°18.2	· · 05.2	3°26.1	· · 31.2			
4	71°10.0	199°39.7	10.5	187°08.1	32.1	221°20.2	05.4	18°28.7	31.2			
5	86°12.4	214°39.4	11.4	202°08.7	32.5	236°22.1	05.6	33°31.3	31.1			
6	101°14.9	229°39.0	S21°12.2	217°09.3	S22°32.8	251°24.1	S11°05.8	48°33.9	N16°31.1			
7	116°17.4	244°38.6	13.1	232°09.9	33.1	266°26.1	05.9	63°36.4	31.1			
8	131°19.8	259°38.3	13.9	247°10.5	33.4	281°28.0	06.1	78°39.0	31.0			
9	146°22.3	274°37.9	· · 14.7	262°11.1	· · 33.7	296°30.0	· · 06.3	93°41.6	· · 31.0			
10	161°24.8	289°37.6	15.6	277°11.7	34.0	311°32.0	06.5	108°44.2	31.0			
11	176°27.2	304°37.2	16.4	292°12.3	34.4	326°33.9	06.7	123°46.7	30.9			
12	191°29.7	319°36.9	S21°17.3	307°12.9	S22°34.7	341°35.9	S11°06.9	138°49.3	N16°30.9			
13	206°32.2	334°36.5	18.1	322°13.4	35.0	356°37.8	07.1	153°51.9	30.9			
14	221°34.6	349°36.2	18.9	337°14.0	35.3	11°39.8	07.2	168°54.5	30.8			
15	236°37.1	4°35.8	· · 19.8	352°14.6	· · 35.6	26°41.8	· · 07.4	183°57.0	· · 30.8			
16	251°39.5	19°35.4	20.6	7°15.2	35.9	41°43.7	07.6	198°59.6	30.8			
17	266°42.0	34°35.1	21.4	22°15.8	36.3	56°45.7	07.8	214°02.2	30.7			
18	281°44.5	49°34.7	S21°22.3	37°16.4	S22°36.6	71°47.7	S11°08.0	229°04.8	N16°30.7			
19	296°46.9	64°34.4	23.1	52°17.0	36.9	86°49.6	08.2	244°07.3	30.7			
20	311°49.4	79°34.0	23.9	67°17.6	37.2	101°51.6	08.3	259°09.9	30.6			
21	326°51.9	94°33.7	· · 24.7	82°18.2	· · 37.5	116°53.6	· · 08.5	274°12.5	· · 30.6			
22	341°54.3	109°33.3	25.6	97°18.8	37.8	131°55.5	08.7	289°15.1	30.6			
23	356°56.8	124°32.9	26.4	112°19.3	38.1	146°57.5	08.9	304°17.7	30.5			
Mer.pass. 23:12		ν -0.4' d 0.8' m -4.25		ν 0.6' d 0.3' m 0.99		ν 2.0' d 0.2' m -1.70		ν 2.6' d -0.0' m -0.06				
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	11°59.3	139°32.6	S21°27.2	127°19.9	S22°38.5	161°59.5	S11°09.1	319°20.2	N16°30.5			
1	27°01.7	154°32.2	28.0	142°20.5	38.8	177°01.4	09.3	334°22.8	30.5			
2	42°04.2	169°31.9	28.9	157°21.1	39.1	192°03.4	09.4	349°25.4	30.4			
3	57°06.7	184°31.5	· · 29.7	172°21.7	· · 39.4	207°05.3	· · 09.6	4°28.0	· · 30.4			
4	72°09.1	199°31.1	30.5	187°22.3	39.7	222°07.3	09.8	19°30.5	30.4			
5	87°11.6	214°30.8	31.3	202°22.9	40.0	237°09.3	10.0	34°33.1	30.3			
6	102°14.0	229°30.4	S21°32.2	217°23.4	S22°40.3	252°11.2	S11°10.2	49°35.7	N16°30.3			
7	117°16.5	244°30.1	32.0	232°24.0	40.6	267°13.2	10.4	64°38.3	30.3			
8	132°19.0	259°29.7	33.8	247°24.6	40.9	282°15.2	10.6	79°40.9	30.2			
9	147°21.4	274°29.3	· · 34.6	262°25.2	· · 41.3	297°17.1	· · 10.7	94°43.4	· · 30.2			
10	162°23.9	289°29.0	35.4	277°25.8	41.6	312°19.1	10.9	109°46.0	30.2			
11	177°26.4	304°28.6	36.3	292°26.4	41.9	327°21.0	11.1	124°48.6	30.1			
12	192°28.8	319°28.3	S21°37.1	307°27.0	S22°42.2	342°23.0	S11°11.3	139°51.2	N16°30.1			
13	207°31.3	334°27.9	37.9	322°27.5	42.5	357°25.0	11.5	154°53.8	30.1			
14	222°33.8	349°27.5	38.7	337°28.1	42.8	12°26.9	11.7	169°56.3	30.0			
15	237°36.2	4°27.2	· · 39.5	352°28.7	· · 43.1							

DUT1 = UT1-UTC = +0.2129 sec ΔT = TT-UT1 = +68.9711 sec

2029 October 01 to Oct. 03 UT

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	182°34.4	S03°12.8	269°40.6	10.0'	N22°11.2	-5.0' 55.8'
1	197°34.6	13.8	284°09.6	10.0'	22°06.2	-5.1' 55.9'
2	212°34.8	14.8	298°38.6	10.0'	22°01.0	-5.2' 55.9'
3	227°35.0	· · 15.8	313°07.6	10.0'	21°55.8	-5.4' 55.9'
4	242°35.2	16.7	327°36.5	10.0'	21°50.4	-5.5' 55.9'
5	257°35.4	17.7	342°05.5	10.0'	21°44.9	-5.6' 56.0'
6	272°35.6	S03°18.7	356°34.5	10.0'	N21°39.3	-5.7' 56.0'
7	287°35.8	19.6	11°03.5	10.0'	21°33.6	-5.8' 56.0'
8	302°36.0	20.6	25°32.5	10.0'	21°27.8	-6.0' 56.1'
9	317°36.2	· · 21.6	40°01.6	10.0'	21°21.8	-6.1' 56.1'
10	332°36.4	22.5	54°30.6	10.0'	21°15.8	-6.2' 56.1'
11	347°36.6	23.5	68°59.6	10.0'	21°09.6	-6.3' 56.2'
12	2°36.8	S03°24.5	83°28.6	10.0'	N21°03.3	-6.4' 56.2'
13	17°37.0	25.4	97°57.7	10.0'	20°56.8	-6.5' 56.2'
14	32°37.2	26.4	112°26.7	10.0'	20°50.3	-6.7' 56.3'
15	47°37.4	· · 27.4	126°55.7	10.1'	20°43.6	-6.8' 56.3'
16	62°37.6	28.4	141°24.8	10.1'	20°36.9	-6.9' 56.3'
17	77°37.8	29.3	155°53.9	10.1'	20°30.0	-7.0' 56.4'
18	92°38.0	S03°30.3	170°22.9	10.1'	N20°23.0	-7.1' 56.4'
19	107°38.2	31.3	184°52.0	10.1'	20°15.9	-7.2' 56.5'
20	122°38.4	32.2	199°21.1	10.1'	20°08.6	-7.3' 56.5'
21	137°38.6	· · 33.2	213°50.2	10.1'	20°01.3	-7.5' 56.5'
22	152°38.8	34.2	228°19.3	10.1'	19°53.8	-7.6' 56.6'
23	167°39.0	35.1	242°48.4	10.1'	19°46.3	-7.7' 56.6'
SD = 16.0' d = 1.0' SD = 15.2'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	182°39.2	S03°36.1	257°17.6	10.1'	N19°38.6	-7.8' 56.6'
1	197°39.4	37.1	271°46.7	10.1'	19°30.8	-7.9' 56.7'
2	212°39.6	38.0	286°15.8	10.2'	19°22.9	-8.0' 56.7'
3	227°39.8	· · 39.0	300°45.0	10.2'	19°14.8	-8.1' 56.7'
4	242°40.0	40.0	315°14.2	10.2'	19°06.7	-8.2' 56.8'
5	257°40.2	40.9	329°43.3	10.2'	18°58.5	-8.4' 56.8'
6	272°40.4	S03°41.9	344°12.5	10.2'	N18°50.1	-8.5' 56.9'
7	287°40.6	42.9	358°41.7	10.2'	18°41.7	-8.6' 56.9'
8	302°40.8	43.8	13°10.9	10.2'	18°33.1	-8.7' 56.9'
9	317°41.0	· · 44.8	27°40.1	10.2'	18°24.4	-8.8' 57.0'
10	332°41.2	45.8	42°09.4	10.2'	18°15.6	-8.9' 57.0'
11	347°41.4	46.7	56°38.6	10.2'	18°06.7	-9.0' 57.0'
12	2°41.6	S03°47.7	71°07.8	10.3'	N17°57.7	-9.1' 57.1'
13	17°41.8	48.7	85°37.1	10.3'	17°48.6	-9.2' 57.1'
14	32°42.0	49.6	100°06.4	10.3'	17°39.4	-9.3' 57.2'
15	47°42.1	· · 50.6	114°35.7	10.3'	17°30.1	-9.4' 57.2'
16	62°42.3	51.6	129°04.9	10.3'	17°20.6	-9.5' 57.2'
17	77°42.5	52.5	143°34.2	10.3'	17°11.1	-9.6' 57.3'
18	92°42.7	S03°53.5	158°03.6	10.3'	N17°01.5	-9.7' 57.3'
19	107°42.9	54.5	172°32.9	10.3'	16°51.7	-9.8' 57.4'
20	122°43.1	55.4	187°02.2	10.3'	16°41.9	-9.9' 57.4'
21	137°43.3	· · 56.4	201°31.6	10.4'	16°31.9	-10.0' 57.4'
22	152°43.5	57.4	216°00.9	10.4'	16°21.9	-10.1' 57.5'
23	167°43.7	58.3	230°30.3	10.4'	16°11.8	-10.2' 57.5'
SD = 16.0' d = 1.0' SD = 15.4'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	182°43.9	S03°59.3	244°59.6	10.4'	N16°01.5	-10.3' 57.5'
1	197°44.1	04°00.3	259°29.0	10.4'	15°51.2	-10.4' 57.6'
2	212°44.3	01.2	273°58.4	10.4'	15°40.7	-10.5' 57.6'
3	227°44.5	· · 02.2	288°27.8	10.4'	15°30.2	-10.6' 57.7'
4	242°44.7	03.2	302°57.2	10.4'	15°19.6	-10.7' 57.7'
5	257°44.9	04.1	317°26.6	10.4'	15°08.8	-10.8' 57.7'
6	272°45.1	S04°05.1	331°56.0	10.4'	N14°58.0	-10.9' 57.8'
7	287°45.3	06.1	346°25.5	10.4'	14°47.1	-11.0' 57.8'
8	302°45.5	07.0	0°54.9	10.4'	14°36.1	-11.1' 57.9'
9	317°45.7	· · 08.0	15°24.4	10.5'	14°25.0	-11.2' 57.9'
10	332°45.9	09.0	29°53.8	10.5'	14°13.8	-11.3' 57.9'
11	347°46.1	09.9	44°23.3	10.5'	14°02.5	-11.4' 58.0'
12	2°46.2	S04°10.9	58°52.7	10.5'	N13°51.1	-11.5' 58.0'
13	17°46.4	11.9	73°22.2	10.5'	13°39.6	-11.6' 58.1'
14	32°46.6	12.8	87°51.7	10.5'	13°28.0	-11.7' 58.1'
15	47°46.8	· · 13.8	102°21.2	10.5'	13°16.4	-11.7' 58.1'
16	62°47.0	14.8	116°50.7	10.5'	13°04.6	-11.8' 58.2'
17	77°47.2	15.7	131°20.1	10.5'	12°52.8	-11.9' 58.2'
18	92°47.4	S04°16.7	145°49.6	10.5'	N12°40.9	-12.0' 58.3'
19	107°47.6	17.6	160°19.1	10.5'	12°28.9	-12.1' 58.3'
20	122°47.8	18.6	174°48.6	10.5'	12°16.8	-12.2' 58.3'
21	137°48.0	· · 19.6	189°18.1	10.5'	12°04.7	-12.2' 58.4'
22	152°48.2	20.5	203°47.7	10.5'	11°52.4	-12.3' 58.4'
23	167°48.4	21.5	218°17.2	10.5'	11°40.1	-12.4' 58.5'
SD = 16.0' d = 1.0' SD = 15.7'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:55	05:17	06:24	17:12	18:19	19:40
N 70°	04:07	05:20	06:20	17:16	18:17	19:29
68°	04:16	05:22	06:17	17:19	18:15	19:20
66°	04:23	05:23	06:14	17:22	18:13	19:13
64°	04:29	05:25	06:12	17:25	18:12	19:07
62°	04:34	05:26	06:10	17:27	18:11	19:03
60°	04:38	05:27	06:08	17:29	18:10	18:59
N 58°	04:42	05:28	06:07	17:31	18:09	18:55
56°	04:45	05:28	06:05	17:32	18:09	18:52
54°	04:48	05:29	06:04	17:33	18:09	18:50
52°	04:50	05:29	06:03	17:35	18:08	18:47
50°	04:52	05:30	06:02	17:36	18:08	18:45
45°	04:56	05:30	05:59	17:38	18:08	18:41
N 40°	04:59	05:30	05:57	17:40	18:07	18:39
35°	05:01	05:30	05:56	17:42	18:08	18:37
30°	05:02	05:30	05:54	17:44	18:08	18:36
20°	05:04	05:29	05:51	17:47	18:09	18:35
N 10°	05:03	05:28	05:49	17:50	18:11	18:35
0°	05:01	05:25	05:46	17:52	18:13	18:37
S 10°	04:58	05:22	05:43	17:55	18:16	18:41
20°	04:53	05:18	05:40	17:58	18:20	18:46
30°	04:45	05:13	05:37	18:02	18:26	18:54
35°	04:40	05:09	05:35	18:04	18:30	18:59
40°	04:33	05:05	05:33	18:07	18:34	19:06
45°	04:25	05:00	05:30	18:09	18:39	19:14
S 50°	04:15	04:54	05:26	18:13	18:46	19:25
52°	04:10	04:51	05:25	18:15	18:49	19:30
54°	04:04	04:47	05:23	18:16	18:52	19:36
56°	03:58	04:44	05:21	18:18	18:56	19:42
58°	03:51	04:39	05:19	18:21	19:01	19:50
S 60°	03:42	04:35	05:17	18:23	19:06	19:59

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	☐	21:16	23:57	☐	18:04	17:11
N 70°	☐	22:04	· · ·	☐	17:15	16:48
68°	20:27	22:34	· · ·	17:04	16:43	16:31
66°	21:08	22:57	· · ·	16:23	16:20	16:17
64°	21:35	23:14	· · ·	15:54	16:01	16:05
62°	21:56	23:29	· · ·	15:33	15:46	15:55
60°	22:13	23:41	· · ·	15:15	15:33	15:46
N 58°	22:28	23:51	· · ·	15:00	15:22	15:39
56°	22:40	· · ·	00:00	14:47	15:12	15:32
54°	22:51	· · ·	00:08	14:36	15:04	15:26
52°	23:00	· · ·	00:15	14:27	14:56	15:20
50°	23:09	· · ·	00:22	14:18	14:49	15:15
45°	23:26	· · ·	00:36	13:59	14:34	15:04
N 40°	23:41	· · ·	00:47	13:44	14:21	14:55
35°	23:53	· · ·	00:57	13:31	14:11	14:48
30°	· · ·	00:04	01:05	13:19	14:01	14:41
20°	· · ·	00:22	01:20	13:00	13:45	14:29
N 10°	· · ·	00:38	01:32	12:43	13:31	14:18
0°	00:02	00:53	01:44	12:27	13:18	14:08
S 10°	00:19	01:08	01:56	12:10	13:04	13:58
20°	00:37	01:24	02:08	11:53	12:50	13:47
30°	00:58	01:42	02:23	11:33	12:33	13:35
35°	01:10	01:52	02:31	11:21	12:23	13:28
40°	01:24	02:04	02:40	11:08	12:12	13:20
45°	01:41	02:19	02:51	10:52	11:59	13:10
S 50°	02:02	02:36	03:04	10:32	11:43	12:58
52°	02:11	02:44	03:10	10:23	11:35	12:53
54°	02:22	02:53	03:17	10:12	11:27	12:47
56°	02:35	03:03	03:25	10:00	11:17	12:40
58°	02:49	03:14	03:33	09:46	11:06	12:32
S 60°	03:06	03:27	03:43	09:30	10:54	12:24

Day	Sun		Mer. Pass	Moon		Age 23-25 49-29%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
01	10:18	10:27	11:50	06:14	18:40	
02	10:37	10:46	11:49	07:05	19:31	
03	10:56	11:05	11:49	07:56	20:21	

October 04, 05, 06 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	12°58.4	139°23.9	S21°46.8	127°33.9	S22°45.9	162°46.6	S11°13.5	320°22.1	N16°29.7	Alpheratz	357°30.4	29°15.5											
1	28°00.9	154°23.5	47.6	142°34.5	46.2	177°48.5	13.7	335°24.7	29.6	Ankaa	353°03.1	-42°08.5											
2	43°03.3	169°23.2	48.4	157°35.1	46.5	192°50.5	13.9	350°27.3	29.6	Schedar	349°25.9	56°42.2											
3	58°05.8	184°22.8	· · 49.2	172°35.7	· · 46.8	207°52.4	· · 14.1	5°29.9	· · 29.6	Diphda	348°43.2	-17°49.2											
4	73°08.3	199°22.5	50.0	187°36.3	47.1	222°54.4	14.2	20°32.5	29.5	Achernar	335°17.0	-57°05.0											
5	88°10.7	214°22.1	50.8	202°36.8	47.4	237°56.4	14.4	35°35.1	29.5	Hamal	327°46.5	23°36.3											
6	103°13.2	229°21.7	S21°51.6	217°37.4	S22°47.7	252°58.3	S11°14.6	50°37.6	N16°29.5	Polaris	311°31.0	89°23.0											
7	118°15.6	244°21.4	52.4	232°38.0	48.0	268°00.3	14.8	65°40.2	29.4	Acamar	315°08.6	-40°10.9											
8	133°18.1	259°21.0	53.2	247°38.6	48.3	283°02.2	15.0	80°42.8	29.4	Menkar	314°01.9	4°12.5											
9	148°20.6	274°20.6	· · 54.0	262°39.1	· · 48.6	298°04.2	· · 15.2	95°45.4	· · 29.4	Mirfak	308°22.3	49°57.9											
10	163°23.0	289°20.3	54.8	277°39.7	48.9	313°06.2	15.3	110°48.0	29.3	Aldebaran	290°35.0	16°34.1											
11	178°25.5	304°19.9	55.6	292°40.3	49.2	328°08.1	15.5	125°50.6	29.3	Rigel	281°00.1	-8°10.0											
12	193°28.0	319°19.5	S21°56.4	307°40.9	S22°49.5	343°10.1	S11°15.7	140°53.1	N16°29.3	Capella	280°16.1	46°01.4											
13	208°30.4	334°19.2	57.2	322°41.4	49.8	358°12.1	15.9	155°55.7	29.2	Bellatrix	278°18.7	6°22.6											
14	223°32.9	349°18.8	58.0	337°42.0	50.1	13°14.0	16.1	170°58.3	29.2	Elnath	277°56.9	28°37.8											
15	238°35.4	4°18.4	· · 58.8	352°42.6	· · 50.4	28°16.0	· · 16.3	186°00.9	· · 29.2	Alnilam	275°33.8	-1°11.0											
16	253°37.8	19°18.1	21°59.6	7°43.2	50.7	43°17.9	16.5	201°03.5	29.1	Betelgeuse	270°47.9	7°24.7											
17	268°40.3	34°17.7	22°00.4	22°43.7	51.0	58°19.9	16.6	216°06.1	29.1	Canopus	263°50.6	-52°42.5											
18	283°42.8	49°17.3	S22°01.2	37°44.3	S22°51.3	73°21.9	S11°16.8	231°08.6	N16°29.0	Sirius	258°22.9	-16°45.3											
19	298°45.2	64°17.0	01.9	52°44.9	51.6	88°23.8	17.0	246°11.2	29.0	Adhara	255°02.9	-29°00.6											
20	313°47.7	79°16.6	02.7	67°45.5	51.9	103°25.8	17.2	261°13.8	29.0	Procyon	244°46.9	5°08.8											
21	328°50.1	94°16.2	· · 03.5	82°46.0	· · 52.2	118°27.7	· · 17.4	276°16.4	· · 28.9	Pollux	243°12.8	27°57.0											
22	343°52.6	109°15.9	04.3	97°46.6	52.5	133°29.7	17.6	291°19.0	28.9	Avior	234°13.3	-59°36.1											
23	358°55.1	124°15.5	05.1	112°47.2	52.8	148°31.7	17.7	306°21.6	28.9	Suhail	222°43.7	-43°33.0											
Mer.pass. 23:04				ν -0.4' d 0.8' m -4.26				ν 0.6' d 0.3' m 0.99				ν 2.0' d 0.2' m -1.70				ν 2.6' d -0.0' m -0.07							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	13°57.5	139°15.1	S22°05.9	127°47.8	S22°53.1	163°33.6	S11°17.9	321°24.2	N16°28.8	Dubhe	193°37.2	61°35.2							
1	29°00.0	154°14.8	06.7	142°48.3	53.4	178°35.6	18.1	336°26.7	28.8	Denebola	182°21.4	14°24.3							
2	44°02.5	169°14.4	07.5	157°48.9	53.7	193°37.5	18.3	351°29.3	28.8	Gienah	175°40.0	-17°42.4							
3	59°04.9	184°14.0	· · 08.2	172°49.5	· · 54.0	208°39.5	· · 18.5	6°31.9	· · 28.7	Acrux	172°56.3	-63°15.8							
4	74°07.4	199°13.6	09.0	187°50.1	54.3	223°41.5	18.7	21°34.5	28.7	Gacrux	171°47.8	-57°16.8							
5	89°09.9	214°13.3	09.8	202°50.6	54.6	238°43.4	18.9	36°37.1	28.7	Alioth	166°10.4	55°47.9							
6	104°12.3	229°12.9	S22°10.6	217°51.2	S22°54.9	253°45.4	S11°19.0	51°39.7	N16°28.6	Spica	158°18.6	-11°18.9							
7	119°14.8	244°12.5	11.4	232°51.8	55.2	268°47.3	19.2	66°42.3	28.6	Alkaid	152°49.6	49°10.0							
8	134°17.2	259°12.2	12.1	247°52.3	55.5	283°49.3	19.4	81°44.9	28.6	Hadar	148°31.0	-60°31.0							
9	149°19.7	274°11.8	· · 12.9	262°52.9	· · 55.7	298°51.3	· · 19.6	96°47.4	· · 28.5	Menkent	147°53.4	-36°30.9							
10	164°22.2	289°11.4	13.7	277°53.5	56.0	313°53.2	19.8	111°50.0	28.5	Arcturus	145°44.8	19°01.8							
11	179°24.6	304°11.1	14.5	292°54.0	56.3	328°55.2	20.0	126°52.6	28.4	Rigel Kent.	139°35.5	-60°57.5							
12	194°27.1	319°10.7	S22°15.2	307°54.6	S22°56.6	343°57.1	S11°20.1	141°55.2	N16°28.4	Kochab	137°21.2	74°02.1							
13	209°29.6	334°10.3	16.0	322°55.2	56.9	358°59.1	20.3	156°57.8	28.4	Zuben'ubi	136°52.1	-16°09.8							
14	224°32.0	349°09.9	16.8	337°55.7	57.2	14°01.0	20.5	172°00.4	28.3	Alphecca	126°00.8	26°37.1							
15	239°34.5	4°09.6	· · 17.6	352°56.3	· · 57.5	29°03.0	· · 20.7	187°03.0	· · 28.3	Antares	112°11.3	-26°29.8							
16	254°37.0	19°09.2	18.3	7°56.9	57.8	44°05.0	20.9	202°05.6	28.3	Atria	107°02.3	-69°04.9							
17	269°39.4	34°08.8	19.1	22°57.4	58.1	59°06.9	21.1	217°08.1	28.2	Sabik	101°58.6	-15°45.5							
18	284°41.9	49°08.5	S22°19.9	37°58.0	S22°58.4	74°08.9	S11°21.2	232°10.7	N16°28.2	Shaula	96°05.3	-37°07.4							
19	299°44.4	64°08.1	20.6	52°58.6	58.7	89°10.8	21.4	247°13.3	28.2	Rasalhague	95°55.1	12°32.6							
20	314°46.8	79°07.7	21.4	67°59.1	59.0	104°12.8	21.6	262°15.9	28.1	Eltanin	90°40.5	51°29.5							
21	329°49.3	94°07.3	· · 22.2	82°59.7	· · 59.2	119°14.8	· · 21.8	277°18.5	· · 28.1	Kaus Aust.	83°27.5	-34°22.1							
22	344°51.7	109°07.0	22.9	98°00.3	59.5	134°16.7	22.0	292°21.1	28.1	Vega	80°30.7	38°49.1							
23	359°54.2	124°06.6	23.7	113°00.8	59.8	149°18.7	22.2	307°23.7	28.0	Nunki	75°43.1	-26°15.4							
Mer.pass. 23:00				ν -0.4' d 0.8' m -4.27				ν 0.6' d 0.3' m 1.00				ν 2.0' d 0.2' m -1.69				ν 2.6' d -0.0' m -0.08			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	14°56.7	139°06.2	S22°24.5	128°01.4	S23°00.1	164°20.6	S11°22.4	322°26.3	N16°28.0	Oct 04 Thu	SHA	Mer.pass							
1	29°59.1	154°05.8	25.2	143°02.0	00.4	179°22.6	22.5	337°28.9	27.9	Venus	126°25.5	14:43							
2	45°01.6	169°05.5	26.0	158°02.5	00.7	194°24.6	22.7	352°31.4	27.9	Mars	114°35.5	15:29							
3	60°04.1	184°05.1	· · 26.8	173°03.1	· · 01.0	209°26.5	· · 22.9	7°34.0	· · 27.9	Jupiter	149°48.2	13:07							
4	75°06.5	199°04.7	27.5	188°03.7	01.3	224°28.5	23.1	22°36.6	27.8	Saturn	307°23.8	02:38							
5	90°09.0	214°04.4	28.3	203°04.2	01.6	239°30.4	23.3	37°39.2	27.8										
6	105°11.5	229°04.0	S22°29.0	218°04.8	S23°01.8	254°32.4	S11°23.5	52°41.8	N16°27.8										
7	120°13.9	244°03.6	29.8	233°05.4	02.1	269°34.3	23.6	67°44.4	27.7										
8	135°16.4	259°03.2	30.5	248°05.9	02.4	284°36.3	23.8	82°47.0	27.7										
9	150°18.8	274°02.9	· · 31.3	263°06.5	· · 02.7	299°38.3	· · 24.0	97°49.6	· · 27.7										
10	165°21.3	289°02.5	32.1	278°07.0	03.0	314°40.2	24.2	112°52.2	27.6										
11	180°23.8	304°02.1	32.8	293°07.6	03.3	329°42.2	24.4	127°54.8	27.6										
12	195°26.2	319°01.7	S22°33.6	308°08.2	S23°03.6	344°44.1	S11°24.6	142°57.3	N16°27.5										
13	210°28.7	334°01.4	34.3	323°08.7	03.8	359°46.1	24.8	157°59.9	27.5										
14	225°31.2	349°01.0	35.1	338°09.3	04.1	14°48.0	24.9	173°02.5	27.5										
15	240°33.6	4°00.6	· · 35.8	353°09.8	· · 04.4	29°50.0	· · 25.1	188°05.1	· · 27.4										
16	255°36.1	19°00.2	36.6	8°10.4	04.7	44°52.0	25.3	203°07.7	27.4										
17	270°38.6	33°59.9	37.3	23°11.0	05.0	59°53.9	25.5	218°10.3	27.4										
18	285°41.0	48°59.5	S22°38.1	38°11.5	S23°05.3	74°55.9	S11°25.7	233°12.9	N16°27.3										
19	300°43.5	63°59.1	38.8	53°12.1	05.5	89°57.8	25.9	248°15.5	27.3										
20	315°46.0	78°58.7	39.6	68°12.6	05.8	104°59.8	26.0	263°18.1	27.3										
21	330°48.4	93°58.3	· · 40.3	83°13.2	· · 06.1	120°01.7	· · 26.2	278°20.7	· · 27.2										
22	345°50.9	108°58.0	41.1	98°13.8	06.4	135°03.7	26.4	293°23.3	27.2										
23	0°53.3	123°57.6	41.8	113°14.3	06.7	150°05.7	26.6	308°25.9	27.1										
Mer.pass. 22:56				ν -0.4' d 0.8' m -4.28				ν 0.6' d 0.3' m 1.00				ν 2.0' d 0.2' m -1.69				ν 2.6' d -0.0' m -0.08			

Oct 04 Thu	SHA	Mer.pass
Venus	126°25.5	14:43
Mars	114°35.5	15:29
Jupiter	149°48.2	13:07
Saturn	307°23.8	02:38

Oct 05 Fri	SHA	Mer.pass
Venus	125°17.6	14:43
Mars	113°50.2	15:28
Jupiter	149°36.1	13:04
Saturn	307°26.6	02:34

Oct 06 Sat	SHA	Mer.pass
Venus	124°09.5	14:44
Mars	113°04.7	15:27
Jupiter	149°24.0	13:01
Saturn	307°29.6	02:30

Horizontal parallax		
Venus:		0.2
Mars:		0.1

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	182°48.6	S04°22.5	232°46.7	10.5'	N11°27.7	-12.5'	58.5'
1	197°48.7	23.4	247°16.2	10.5'	11°15.2	-12.6'	58.5'
2	212°48.9	24.4	261°45.7	10.5'	11°02.6	-12.6'	58.6'
3	227°49.1	· · 25.4	276°15.2	10.5'	10°50.0	-12.7'	58.6'
4	242°49.3	26.3	290°44.7	10.5'	10°37.3	-12.8'	58.7'
5	257°49.5	27.3	305°14.2	10.5'	10°24.5	-12.9'	58.7'
6	272°49.7	S04°28.2	319°43.7	10.5'	N10°11.6	-12.9'	58.7'
7	287°49.9	29.2	334°13.2	10.5'	09°58.7	-13.0'	58.8'
8	302°50.1	30.2	348°42.7	10.5'	09°45.7	-13.1'	58.8'
9	317°50.3	· · 31.1	3°12.2	10.5'	09°32.6	-13.2'	58.9'
10	332°50.5	32.1	17°41.7	10.5'	09°19.4	-13.2'	58.9'
11	347°50.6	33.1	32°11.2	10.5'	09°06.2	-13.3'	58.9'
12	2°50.8	S04°34.0	46°40.7	10.5'	N08°52.9	-13.4'	59.0'
13	17°51.0	35.0	61°10.2	10.5'	08°39.6	-13.4'	59.0'
14	32°51.2	36.0	75°39.6	10.5'	08°26.1	-13.5'	59.1'
15	47°51.4	· · 36.9	90°09.1	10.5'	08°12.6	-13.6'	59.1'
16	62°51.6	37.9	104°38.5	10.4'	07°59.1	-13.6'	59.1'
17	77°51.8	38.8	119°08.0	10.4'	07°45.5	-13.7'	59.2'
18	92°52.0	S04°39.8	133°37.4	10.4'	N07°31.8	-13.7'	59.2'
19	107°52.2	40.8	148°06.8	10.4'	07°18.1	-13.8'	59.3'
20	122°52.4	41.7	162°36.2	10.4'	07°04.3	-13.8'	59.3'
21	137°52.5	· · 42.7	177°05.6	10.4'	06°50.4	-13.9'	59.3'
22	152°52.7	43.7	191°35.0	10.4'	06°36.5	-14.0'	59.4'
23	167°52.9	44.6	206°04.4	10.4'	06°22.6	-14.0'	59.4'
SD = 16.0' $d = 1.0'$ SD = 16.0'							

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	182°53.1	S04°45.6	220°33.7	10.3'	N06°08.6	-14.1'	59.4'
1	197°53.3	46.5	235°03.1	10.3'	05°54.5	-14.1'	59.5'
2	212°53.5	47.5	249°32.4	10.3'	05°40.4	-14.2'	59.5'
3	227°53.7	· · 48.5	264°01.7	10.3'	05°26.2	-14.2'	59.6'
4	242°53.9	49.4	278°31.0	10.3'	05°12.0	-14.3'	59.6'
5	257°54.0	50.4	293°00.3	10.3'	04°57.7	-14.3'	59.6'
6	272°54.2	S04°51.3	307°29.6	10.2'	N04°43.4	-14.4'	59.7'
7	287°54.4	52.3	321°58.8	10.2'	04°29.1	-14.4'	59.7'
8	302°54.6	53.3	336°28.0	10.2'	04°14.7	-14.4'	59.7'
9	317°54.8	· · 54.2	350°57.2	10.2'	04°00.3	-14.5'	59.8'
10	332°55.0	55.2	5°26.4	10.2'	03°45.8	-14.5'	59.8'
11	347°55.2	56.2	19°55.5	10.1'	03°31.3	-14.6'	59.8'
12	2°55.3	S04°57.1	34°24.7	10.1'	N03°16.7	-14.6'	59.9'
13	17°55.5	58.1	48°53.8	10.1'	03°02.1	-14.6'	59.9'
14	32°55.7	04°59.0	63°22.9	10.1'	02°47.5	-14.7'	59.9'
15	47°55.9	05°00.0	77°51.9	10.0'	02°32.9	-14.7'	60.0'
16	62°56.1	01.0	92°20.9	10.0'	02°18.2	-14.7'	60.0'
17	77°56.3	01.9	106°49.9	10.0'	02°03.5	-14.7'	60.0'
18	92°56.5	S05°02.9	121°18.9	9.9'	N01°48.7	-14.8'	60.1'
19	107°56.6	03.8	135°47.9	9.9'	01°33.9	-14.8'	60.1'
20	122°56.8	04.8	150°16.8	9.9'	01°19.2	-14.8'	60.1'
21	137°57.0	· · 05.8	164°45.7	9.9'	01°04.3	-14.8'	60.2'
22	152°57.2	06.7	179°14.5	9.8'	00°49.5	-14.9'	60.2'
23	167°57.4	07.7	193°43.3	9.8'	00°34.6	-14.9'	60.2'
SD = 16.0' $d = 1.0'$ SD = 16.2'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	182°57.6	S05°08.6	208°12.1	9.8'	N00°19.8	-14.9'	60.3'
1	197°57.7	09.6	222°40.9	9.7'	N00°04.9	-14.9'	60.3'
2	212°57.9	10.5	237°09.6	9.7'	S00°10.1	14.9'	60.3'
3	227°58.1	· · 11.5	251°38.3	9.6'	00°25.0	14.9'	60.3'
4	242°58.3	12.5	266°06.9	9.6'	00°39.9	14.9'	60.4'
5	257°58.5	13.4	280°35.5	9.6'	00°54.9	15.0'	60.4'
6	272°58.7	S05°14.4	295°04.1	9.5'	S01°09.8	15.0'	60.4'
7	287°58.8	15.3	309°32.6	9.5'	01°24.8	15.0'	60.5'
8	302°59.0	16.3	324°01.1	9.4'	01°39.8	15.0'	60.5'
9	317°59.2	· · 17.3	338°29.5	9.4'	01°54.7	15.0'	60.5'
10	332°59.4	18.2	352°57.9	9.4'	02°09.7	15.0'	60.5'
11	347°59.6	19.2	7°26.3	9.3'	02°24.7	15.0'	60.6'
12	2°59.7	S05°20.1	21°54.6	9.3'	S02°39.7	15.0'	60.6'
13	17°59.9	21.1	36°22.9	9.2'	02°54.6	15.0'	60.6'
14	33°00.1	22.0	50°51.1	9.2'	03°09.6	15.0'	60.6'
15	48°00.3	· · 23.0	65°19.3	9.1'	03°24.6	15.0'	60.7'
16	63°00.5	24.0	79°47.5	9.1'	03°39.5	14.9'	60.7'
17	78°00.6	24.9	94°15.6	9.0'	03°54.5	14.9'	60.7'
18	93°00.8	S05°25.9	108°43.6	9.0'	S04°09.4	14.9'	60.7'
19	108°01.0	26.8	123°11.6	8.9'	04°24.3	14.9'	60.8'
20	123°01.2	27.8	137°39.5	8.9'	04°39.2	14.9'	60.8'
21	138°01.4	· · 28.7	152°07.4	8.8'	04°54.1	14.9'	60.8'
22	153°01.5	29.7	166°35.3	8.8'	05°09.0	14.8'	60.8'
23	168°01.7	30.7	181°03.1	8.7'	05°23.8	14.8'	60.8'
SD = 16.0' $d = 1.0'$ SD = 16.4'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:10	05:30	06:38	16:57	18:04	19:23
N 70°	04:20	05:32	06:32	17:02	18:03	19:14
68°	04:27	05:32	06:28	17:07	18:02	19:07
66°	04:33	05:33	06:24	17:11	18:02	19:01
64°	04:38	05:33	06:21	17:14	18:02	18:57
62°	04:42	05:34	06:18	17:17	18:01	18:53
60°	04:46	05:34	06:16	17:20	18:01	18:49
N 58°	04:49	05:34	06:13	17:22	18:01	18:47
56°	04:51	05:34	06:11	17:24	18:01	18:44
54°	04:53	05:34	06:10	17:26	18:01	18:42
52°	04:55	05:34	06:08	17:28	18:01	18:40
50°	04:57	05:34	06:06	17:29	18:02	18:39
45°	05:00	05:34	06:03	17:33	18:02	18:36
N 40°	05:02	05:33	06:00	17:36	18:03	18:34
35°	05:03	05:33	05:58	17:38	18:03	18:33
30°	05:04	05:32	05:56	17:40	18:04	18:32
20°	05:04	05:30	05:52	17:44	18:06	18:32
N 10°	05:03	05:27	05:48	17:48	18:09	18:34
0°	05:00	05:24	05:45	17:52	18:12	18:36
S 10°	04:56	05:20	05:42	17:55	18:16	18:41
20°	04:50	05:16	05:38	17:59	18:21	18:47
30°	04:41	05:09	05:33	18:04	18:28	18:56
35°	04:35	05:05	05:31	18:07	18:32	19:02
40°	04:28	05:00	05:28	18:10	18:37	19:09
45°	04:19	04:55	05:24	18:13	18:43	19:19
S 50°	04:08	04:47	05:20	18:18	18:51	19:30
52°	04:02	04:44	05:18	18:20	18:54	19:36
54°	03:56	04:40	05:16	18:22	18:58	19:42
56°	03:49	04:35	05:14	18:24	19:03	19:50
58°	03:41	04:31	05:11	18:27	19:08	19:58
S 60°	03:31	04:25	05:08	18:30	19:13	20:08

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	· · ·	02:14	04:28	16:41	16:17	15:53
N 70°	00:17	02:22	04:27	16:30	16:14	15:58
68°	00:33	02:29	04:26	16:21	16:12	16:03
66°	00:45	02:34	04:25	16:13	16:10	16:06
64°	00:56	02:39	04:24	16:07	16:08	16:09
62°	01:05	02:43	04:23	16:01	16:07	16:12
60°	01:12	02:46	04:23	15:56	16:05	16:14
N 58°	01:19	02:49	04:22	15:52	16:04	16:16
56°	01:25	02:52	04:22	15:48	16:03	16:18
54°	01:30	02:55	04:22	15:45	16:02	16:20
52°	01:35	02:57	04:21	15:42	16:01	16:21
50°	01:39	02:59	04:21	15:39	16:01	16:23
45°	01:48	03:03	04:20	15:32	15:59	16:26
N 40°	01:56	03:07	04:20	15:27	15:57	16:28
35°	02:02	03:10	04:19	15:22	15:56	16:30
30°	02:08	03:13	04:19	15:18	15:55	16:32
20°	02:18	03:17	04:18	15:11	15:53	16:36
N 10°	02:27	03:21	04:17	15:05	15:51	16:39
0°	02:35	03:25	04:17	14:59	15:50	16:42
S 10°	02:43	03:29	04:16	14:53	15:48	16:44
20°	02:51	03:33	04:16	14:46	15:46	16:47
30°	03:01	03:38	04:15	14:39	15:44	16:51
35°	03:07	03:41	04:15	14:34	15:43	16:53
40°	03:13	03:44	04:14	14:29	15:41	16:55
45°	03:20	03:47	04:14	14:24	15:39	16:58
S 50°	03:29	03:52	04:14	14:17	15:37	17:01
52°	03:33	03:54	04:13	14:13	15:36	17:02
54°	03:38	03:56	04:13	14:10	15:35	17:04
56°	03:42	03:58	04:13	14:06	15:34	17:05
58°	03:48	04:01	04:13	14:01	15:33	17:07
S 60°	03:54	04:03	04:12	13:56	15:32	17:09

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	26-28
	mm:ss	mm:ss	mm:ss	hh:mm	hh:mm	19-5%
04	11:14	11:23	11:49	08:47	21:12	
05	11:32	11:41	11:48	09:37	22:03	
06	11:50	11:59	11:48	10:29	22:56	

October 07, 08, 09 UT (Sun., Mon., Tue.)

Aries			Venus		Mars		Jupiter		Saturn		Stars			
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	15°55.8	138°57.2	S22°42.5	128°14.9	S23°06.9	165°07.6	S11°26.8	323°28.5	N16°27.1			Alpheratz	357°30.4	29°15.5
1	30°58.3	153°56.8		43.3	143°15.4		07.2	180°09.6		27.0		Ankaa	353°03.1	-42°08.5
2	46°00.7	168°56.5		44.0	158°16.0		07.5	195°11.5		27.2		Schedar	349°25.9	56°42.2
3	61°03.2	183°56.1	· ·	44.8	173°16.5	· ·	07.8	210°13.5	· ·	27.3		Diphda	348°43.2	-17°49.2
4	76°05.7	198°55.7		45.5	188°17.1		08.1	225°15.4		27.5		Achernar	335°17.0	-57°05.0
5	91°08.1	213°55.3		46.2	203°17.7		08.3	240°17.4		27.7		Hamal	327°46.5	23°36.3
6	106°10.6	228°55.0	S22°47.0	218°18.2	S23°08.6	255°19.4	S11°27.9	53°44.0	N16°26.9			Polaris	311°29.9	89°23.1
7	121°13.1	243°54.6		47.7	233°18.8		08.9	270°21.3		28.1		Acamar	315°08.6	-40°10.9
8	136°15.5	258°54.2		48.4	248°19.3		09.2	285°23.3		28.3		Menkar	314°01.9	4°12.5
9	151°18.0	273°53.8	· ·	49.2	263°19.9	· ·	09.4	300°25.2	· ·	28.4		Mirfak	308°22.3	49°57.9
10	166°20.4	288°53.4		49.9	278°20.4		09.7	315°27.2		28.6		Aldebaran	290°35.0	16°34.1
11	181°22.9	303°53.1		50.7	293°21.0		10.0	330°29.1		28.8		Rigel	281°00.1	-8°10.0
12	196°25.4	318°52.7	S22°51.4	308°21.5	S23°10.3	345°31.1	S11°29.0	143°59.6	N16°26.7			Capella	280°16.0	46°01.4
13	211°27.8	333°52.3		52.1	323°22.1		10.6	0°33.0		29.2		Bellatrix	278°18.7	6°22.6
14	226°30.3	348°51.9		52.8	338°22.7		10.8	15°35.0		29.4		Elnath	277°56.9	28°37.8
15	241°32.8	3°51.5	· ·	53.6	353°23.2	· ·	11.1	30°37.0	· ·	29.5		Alnilam	275°33.7	-1°11.0
16	256°35.2	18°51.2		54.3	8°23.8		11.4	45°38.9		29.7		Betelgeuse	270°47.9	7°24.7
17	271°37.7	33°50.8		55.0	23°24.3		11.7	60°40.9		29.9		Canopus	263°50.6	-52°42.5
18	286°40.2	48°50.4	S22°55.8	38°24.9	S23°11.9	75°42.8	S11°30.1	234°15.2	N16°26.4			Sirius	258°22.8	-16°45.3
19	301°42.6	63°50.0		56.5	53°25.4		12.2	90°44.8		30.3		Adhara	255°02.8	-29°00.6
20	316°45.1	78°49.6		57.2	68°26.0		12.5	105°46.7		30.5		Procyon	244°46.9	5°08.8
21	331°47.6	93°49.3	· ·	57.9	83°26.5	· ·	12.7	120°48.7	· ·	30.7		Pollux	243°12.8	27°57.0
22	346°50.0	108°48.9		58.7	98°27.1		13.0	135°50.6		30.8		Avior	234°13.3	-59°36.1
23	1°52.5	123°48.5		59.4	113°27.6		13.3	150°52.6		31.0		Suhail	222°43.7	-43°33.0
Mer.pass. 22:53		ν -0.4' d 0.7' m -4.29			ν 0.6' d 0.3' m 1.00		ν 2.0' d 0.2' m -1.69		ν 2.6' d -0.0' m -0.09					

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	
0	16°54.9	138°48.1	S23°00.1	128°28.2	S23°13.6	165°54.6	S11°31.2	324°30.7	N16°26.2	
1	31°57.4	153°47.7		00.8	143°28.7		13.8	339°33.3	26.2	
2	46°59.9	168°47.4		01.5	158°29.3		14.1	354°35.9	26.1	
3	62°02.3	183°47.0	· ·	02.3	173°29.8	· ·	14.4	9°38.5	· · 26.1	
4	77°04.8	198°46.6		03.0	188°30.4		14.6	24°41.1	26.1	
5	92°07.3	213°46.2		03.7	203°30.9		14.9	39°43.7	26.0	
6	107°09.7	228°45.8	S23°04.4	218°31.5	S23°15.2	256°06.3	S11°32.3	54°46.3	N16°26.0	
7	122°12.2	243°45.5		05.1	233°32.0		15.5	69°48.9	25.9	
8	137°14.7	258°45.1		05.8	248°32.6		15.7	84°51.5	25.9	
9	152°17.1	273°44.7	· ·	06.5	263°33.1	· ·	16.0	99°54.1	· · 25.9	
10	167°19.6	288°44.3		07.3	278°33.7		16.3	114°56.7	25.8	
11	182°22.0	303°43.9		08.0	293°34.2		16.5	129°59.3	25.8	
12	197°24.5	318°43.5	S23°08.7	308°34.8	S23°16.8	346°18.0	S11°33.4	145°01.9	N16°25.8	
13	212°27.0	333°43.2		09.4	323°35.3		17.1	1°20.0	33.6	
14	227°29.4	348°42.8		10.1	338°35.8		17.3	16°21.9	33.8	
15	242°31.9	3°42.4	· ·	10.8	353°36.4	· ·	17.6	31°23.9	· · 34.0	
16	257°34.4	18°42.0		11.5	8°36.9		17.9	46°25.8	34.2	
17	272°36.8	33°41.6		12.2	23°37.5		18.1	61°27.8	34.3	
18	287°39.3	48°41.2	S23°12.9	38°38.0	S23°18.4	76°29.7	S11°34.5	235°17.5	N16°25.5	
19	302°41.8	63°40.9		13.6	53°38.6		18.7	250°20.1	25.5	
20	317°44.2	78°40.5		14.3	68°39.1		18.9	265°22.7	25.4	
21	332°46.7	93°40.1	· ·	15.0	83°39.7	· ·	19.2	121°35.6	· · 35.1	
22	347°49.2	108°39.7		15.7	98°40.2		19.4	136°37.6	35.3	
23	2°51.6	123°39.3		16.4	113°40.7		19.7	151°39.5	35.5	
Mer.pass. 22:49		ν -0.4' d 0.7' m -4.30			ν 0.6' d 0.3' m 1.01		ν 2.0' d 0.2' m -1.69		ν 2.6' d -0.0' m -0.09	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	
0	17°54.1	138°38.9	S23°17.1	128°41.3	S23°20.0	166°41.5	S11°35.6	325°33.1	N16°25.3	
1	32°56.5	153°38.6		17.8	143°41.8		20.2	340°35.7	25.3	
2	47°59.0	168°38.2		18.5	158°42.4		20.5	355°38.3	25.2	
3	63°01.5	183°37.8	· ·	19.2	173°42.9	· ·	20.8	10°40.9	· · 25.2	
4	78°03.9	198°37.4		19.9	188°43.5		21.0	25°43.5	25.1	
5	93°06.4	213°37.0		20.6	203°44.0		21.3	40°46.1	25.1	
6	108°08.9	228°36.6	S23°21.3	218°44.5	S23°21.5	256°53.2	S11°36.7	55°48.7	N16°25.1	
7	123°11.3	243°36.3		22.0	233°45.1		21.8	70°51.3	25.0	
8	138°13.8	258°35.9		22.7	248°45.6		22.1	85°53.9	25.0	
9	153°16.3	273°35.5	· ·	23.4	263°46.2	· ·	22.3	100°56.5	· · 24.9	
10	168°18.7	288°35.1		24.1	278°46.7		22.6	115°59.1	24.9	
11	183°21.2	303°34.7		24.7	293°47.2		22.8	131°01.7	24.9	
12	198°23.7	318°34.3	S23°25.4	308°47.8	S23°23.1	347°04.9	S11°37.8	146°04.3	N16°24.8	
13	213°26.1	333°33.9		26.1	323°48.3		23.4	161°06.9	24.8	
14	228°28.6	348°33.6		26.8	338°48.9		23.6	176°09.5	24.8	
15	243°31.0	3°33.2	· ·	27.5	353°49.4	· ·	23.9	191°12.1	· · 24.7	
16	258°33.5	18°32.8		28.2	8°49.9		24.1	206°14.7	24.7	
17	273°36.0	33°32.4		28.9	23°50.5		24.4	221°17.3	24.6	
18	288°38.4	48°32.0	S23°29.5	38°51.0	S23°24.6	77°16.6	S11°39.0	236°19.9	N16°24.6	
19	303°40.9	63°31.6		30.2	53°51.6		24.9	251°22.5	24.6	
20	318°43.4	78°31.2		30.9	68°52.1		25.2	266°25.1	24.5	
21	333°45.8	93°30.9	· ·	31.6	83°52.6	· ·	25.4	281°27.7	· · 24.5	
22	348°48.3	108°30.5		32.3	98°53.2		25.7	296°30.3	24.4	
23	3°50.8	123°30.1		32.9	113°53.7		25.9	311°32.9	24.4	
Mer.pass. 22:45		ν -0.4' d 0.7' m -4.31			ν 0.5' d 0.3' m 1.01		ν 2.0' d 0.2' m -1.69		ν 2.6' d -0.0' m -0.10	

Oct 07 Sun	SHA	Mer.pass
Venus	123°01.4	14:45
Mars	112°19.1	15:26
Jupiter	149°11.8	12:58
Saturn	307°32.6	02:26
Oct 08 Mon	SHA	Mer.pass
Venus	121°53.2	14:45
Mars	111°33.2	15:26
Jupiter	148°59.6	12:55
Saturn	307°35.8	02:22
Oct 09 Tue	SHA	Mer.pass
Venus	120°44.9	14:46
Mars	110°47.2	15:25
Jupiter	148°47.4	12:52
Saturn	307°39.0	02:17
Horizontal parallax		
Venus:		0.2
Mars:		0.1

DUT1 = UT1-UTC = +0.2130 sec $\Delta T = TT-UT1 = +68.9710$ sec

2029 October 07 to Oct. 09 UT

h							Moon						
Sun							Moon						
Sun	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°01.9	S05°31.6	195°30.8	8.7'	S05°38.7	14.8'	60.9'						
1	198°02.1	32.6	209°58.5	8.6'	05°53.5	14.8'	60.9'						
2	213°02.3	33.5	224°26.1	8.6'	06°08.2	14.7'	60.9'						
3	228°02.4	· · 34.5	238°53.7	8.5'	06°23.0	14.7'	60.9'						
4	243°02.6	35.4	253°21.2	8.5'	06°37.7	14.7'	60.9'						
5	258°02.8	36.4	267°48.7	8.4'	06°52.4	14.6'	61.0'						
6	273°03.0	S05°37.4	282°16.1	8.4'	S07°07.0	14.6'	61.0'						
7	288°03.2	38.3	296°43.5	8.3'	07°21.6	14.6'	61.0'						
8	303°03.3	39.3	311°10.8	8.2'	07°36.2	14.5'	61.0'						
9	318°03.5	· · 40.2	325°38.0	8.2'	07°50.7	14.5'	61.0'						
10	333°03.7	41.2	340°05.2	8.1'	08°05.2	14.4'	61.0'						
11	348°03.9	42.1	354°32.3	8.1'	08°19.6	14.4'	61.0'						
12	3°04.0	S05°43.1	8°59.4	8.0'	S08°34.0	14.4'	61.1'						
13	18°04.2	44.0	23°26.4	7.9'	08°48.4	14.3'	61.1'						
14	33°04.4	45.0	37°53.3	7.9'	09°02.7	14.2'	61.1'						
15	48°04.6	· · 46.0	52°20.2	7.8'	09°16.9	14.2'	61.1'						
16	63°04.8	46.9	66°47.0	7.8'	09°31.1	14.1'	61.1'						
17	78°04.9	47.9	81°13.7	7.7'	09°45.3	14.1'	61.1'						
18	93°05.1	S05°48.8	95°40.4	7.6'	S09°59.4	14.0'	61.1'						
19	108°05.3	49.8	110°07.1	7.6'	10°13.4	14.0'	61.1'						
20	123°05.5	50.7	124°33.6	7.5'	10°27.4	13.9'	61.1'						
21	138°05.6	· · 51.7	139°00.1	7.4'	10°41.3	13.8'	61.2'						
22	153°05.8	52.6	153°26.5	7.4'	10°55.1	13.8'	61.2'						
23	168°06.0	53.6	167°52.9	7.3'	11°08.9	13.7'	61.2'						
SD = 16.0' d = 1.0'							SD = 16.6'						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°06.2	S05°54.5	182°19.2	7.2'	S11°22.5	13.6'	61.2'						
1	198°06.3	55.5	196°45.4	7.2'	11°36.2	13.6'	61.2'						
2	213°06.5	56.4	211°11.6	7.1'	11°49.7	13.5'	61.2'						
3	228°06.7	· · 57.4	225°37.7	7.0'	12°03.2	13.4'	61.2'						
4	243°06.9	58.4	240°03.8	7.0'	12°16.6	13.3'	61.2'						
5	258°07.0	05°59.3	254°29.7	6.9'	12°29.9	13.2'	61.2'						
6	273°07.2	S06°00.3	268°55.6	6.8'	S12°43.2	13.2'	61.2'						
7	288°07.4	01.2	283°21.5	6.8'	12°56.3	13.1'	61.2'						
8	303°07.6	02.2	297°47.2	6.7'	13°09.4	13.0'	61.2'						
9	318°07.7	· · 03.1	312°12.9	6.6'	13°22.4	12.9'	61.2'						
10	333°07.9	04.1	326°38.6	6.6'	13°35.3	12.8'	61.2'						
11	348°08.1	05.0	341°04.1	6.5'	13°48.1	12.7'	61.2'						
12	3°08.2	S06°06.0	355°29.7	6.4'	S14°00.8	12.6'	61.2'						
13	18°08.4	06.9	9°55.1	6.4'	14°13.4	12.5'	61.2'						
14	33°08.6	07.9	24°20.5	6.3'	14°25.9	12.4'	61.2'						
15	48°08.8	· · 08.8	38°45.8	6.2'	14°38.4	12.3'	61.2'						
16	63°08.9	09.8	53°11.0	6.2'	14°50.7	12.2'	61.2'						
17	78°09.1	10.7	67°36.2	6.1'	15°02.9	12.1'	61.2'						
18	93°09.3	S06°11.7	82°01.3	6.0'	S15°15.0	12.0'	61.2'						
19	108°09.4	12.6	96°26.3	6.0'	15°27.0	11.9'	61.2'						
20	123°09.6	13.6	110°51.3	5.9'	15°39.0	11.8'	61.2'						
21	138°09.8	· · 14.5	125°16.2	5.8'	15°50.8	11.7'	61.2'						
22	153°10.0	15.5	139°41.0	5.8'	16°02.4	11.6'	61.2'						
23	168°10.1	16.4	154°05.8	5.7'	16°14.0	11.5'	61.2'						
SD = 16.0' d = 1.0'							SD = 16.7'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°10.3	S06°17.4	168°30.5	5.6'	S16°25.5	11.3'	61.2'						
1	198°10.5	18.3	182°55.1	5.6'	16°36.8	11.2'	61.2'						
2	213°10.6	19.3	197°19.7	5.5'	16°48.0	11.1'	61.2'						
3	228°10.8	· · 20.2	211°44.2	5.5'	16°59.2	11.0'	61.1'						
4	243°11.0	21.2	226°08.7	5.4'	17°10.1	10.9'	61.1'						
5	258°11.1	22.1	240°33.1	5.3'	17°21.0	10.7'	61.1'						
6	273°11.3	S06°23.1	254°57.4	5.3'	S17°31.7	10.6'	61.1'						
7	288°11.5	24.0	269°21.7	5.2'	17°42.3	10.5'	61.1'						
8	303°11.7	25.0	283°45.9	5.2'	17°52.8	10.4'	61.1'						
9	318°11.8	· · 25.9	298°10.1	5.1'	18°03.2	10.2'	61.1'						
10	333°12.0	26.9	312°34.2	5.0'	18°13.4	10.1'	61.1'						
11	348°12.2	27.8	326°58.2	5.0'	18°23.5	10.0'	61.1'						
12	3°12.3	S06°28.8	341°22.2	4.9'	S18°33.5	9.8'	61.0'						
13	18°12.5	29.7	355°46.1	4.9'	18°43.3	9.7'	61.0'						
14	33°12.7	30.7	10°10.0	4.8'	18°53.0	9.5'	61.0'						
15	48°12.8	· · 31.6	24°33.8	4.8'	19°02.5	9.4'	61.0'						
16	63°13.0	32.6	38°57.6	4.7'	19°11.9	9.3'	61.0'						
17	78°13.2	33.5	53°21.3	4.7'	19°21.2	9.1'	61.0'						
18	93°13.3	S06°34.5	67°45.0	4.6'	S19°30.3	9.0'	61.0'						
19	108°13.5	35.4	82°08.6	4.6'	19°39.3	8.8'	60.9'						
20	123°13.7	36.3	96°32.2	4.5'	19°48.1	8.7'	60.9'						
21	138°13.8	· · 37.3	110°55.7	4.5'	19°56.8	8.5'	60.9'						
22	153°14.0	38.2	125°19.2	4.4'	20°05.4	8.4'	60.9'						
23	168°14.2	39.2	139°42.6	4.4'	20°13.7	8.2'	60.9'						
SD = 16.0' d = 0.9'							SD = 16.7'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:25	05:44	06:52	16:41	17:49	19:07
N 70°	04:32	05:43	06:45	16:48	17:49	19:00
68°	04:38	05:43	06:39	16:54	17:50	18:55
66°	04:43	05:42	06:34	17:00	17:51	18:50
64°	04:47	05:42	06:30	17:04	17:51	18:46
62°	04:50	05:42	06:26	17:08	17:52	18:43
60°	04:53	05:41	06:23	17:11	17:52	18:41
N 58°	04:55	05:41	06:20	17:14	17:53	18:38
56°	04:57	05:40	06:17	17:16	17:54	18:37
54°	04:59	05:40	06:15	17:19	17:54	18:35
52°	05:00	05:39	06:13	17:21	17:55	18:34
50°	05:01	05:39	06:11	17:23	17:55	18:33
45°	05:04	05:38	06:07	17:27	17:57	18:30
N 40°	05:05	05:36	06:03	17:31	17:58	18:29
35°	05:06	05:35	06:00	17:34	17:59	18:29
30°	05:06	05:34	05:58	17:37	18:01	18:29
20°	05:05	05:31	05:53	17:42	18:04	18:30
N 10°	05:03	05:27	05:48	17:46	18:07	18:32
0°	04:59	05:23	05:44	17:51	18:12	18:36
S 10°	04:54	05:19	05:40	17:55	18:16	18:41
20°	04:47	05:13	05:35	18:00	18:22	18:48
30°	04:37	05:06	05:30	18:06	18:30	18:58
35°	04:31	05:01	05:27	18:09	18:35	19:05
40°	04:23	04:55	05:23	18:13	18:40	19:13
45°	04:13	04:49	05:19	18:17	18:47	19:23
S 50°	04:01	04:40	05:14	18:22	18:56	19:36
52°	03:55	04:37	05:11	18:25	19:00	19:42
54°	03:48	04:32	05:09	18:28	19:04	19:49
56°	03:40	04:27	05:06	18:31	19:09	19:57
58°	03:31	04:22	05:02	18:34	19:15	20:06
S 60°	03:20	04:15	04:59	18:38	19:21	20:17

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	06:48	09:25	■	15:27	14:47	■
N 70°	06:36	08:57	11:48	15:41	15:18	14:31
68°	06:27	08:36	10:57	15:53	15:41	15:23
66°	06:19	08:19	10:25	16:02	15:59	15:55
64°	06:13	08:06	10:02	16:11	16:14	16:20
62°	06:07	07:54	09:44	16:18	16:26	16:39
60°	06:02	07:45	09:28	16:24	16:37	16:55
N 58°	05:58	07:37	09:16	16:30	16:46	17:08
56°	05:55	07:29	09:05	16:34	16:54	17:20
54°	05:51	07:23	08:55	16:39	17:01	17:30
52°	05:48	07:17	08:46	16:43	17:08	17:39
50°	05:45	07:12	08:39	16:46	17:14	17:48
45°	05:40	07:01	08:23	16:54	17:27	18:05
N 40°	05:35	06:52	08:09	17:01	17:37	18:19
35°	05:30	06:44	07:58	17:07	17:47	18:32
30°	05:27	06:37	07:48	17:12	17:55	18:42
20°	05:20	06:25	07:31	17:21	18:09	19:01
N 10°	05:15	06:15	07:17	17:28	18:21	19:17
0°	05:10	06:05	07:03	17:36	18:32	19:32
S 10°	05:05	05:56	06:50	17:43	18:44	19:47
20°	04:59	05:46	06:36	17:51	18:56	20:03
30°	04:53	05:34	06:20	18:00	19:11	20:22
35°	04:50	05:28	06:10	18:05	19:19	20:33
40°	04:46	05:21	06:00	18:11	19:28	20:45
45°	04:42	05:12	05:47	18:18	19:39	21:00
S 50°	04:36	05:02	05:32	18:26	19:53	21:19
52°	04:34	04:57	05:25	18:30	19:59	21:27
54°	04:31	04:52	05:18	18:34	20:06	21:37
56°	04:28	04:46	05:09	18:39	20:14	21:48
58°	04:25	04:40	04:59	18:44	20:23	22:00
S 60°	04:22	04:33	04:48	18:50	20:33	22:15

October 10, 11, 12 UT (Wed., Thu., Fri.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	18°53.2	138°29.7	S23°33.6	128°54.2	S23°26.2	167°28.3	S11°40.1	326°35.6	N16°24.4					Alpheratz	357°30.4	29°15.5	
1	33°55.7	153°29.3	34.3	143°54.8	26.4	182°30.3	40.2	341°38.2	24.3					Ankaa	353°03.1	-42°08.5	
2	48°58.1	168°28.9	35.0	158°55.3	26.7	197°32.2	40.4	356°40.8	24.3					Schedar	349°25.9	56°42.2	
3	64°00.6	183°28.5	· · 35.6	173°55.8	· · 26.9	212°34.2	· · 40.6	11°43.4	· · 24.2					Diphda	348°43.2	-17°49.2	
4	79°03.1	198°28.1	36.3	188°56.4	27.2	227°36.1	40.8	26°46.0	24.2					Achernar	335°17.0	-57°05.0	
5	94°05.5	213°27.8	37.0	203°56.9	27.4	242°38.1	41.0	41°48.6	24.2					Hamal	327°46.5	23°36.3	
6	109°08.0	228°27.4	S23°37.7	218°57.4	S23°27.7	257°40.0	S11°41.2	56°51.2	N16°24.1					Polaris	311°28.7	89°23.1	
7	124°10.5	243°27.0	38.3	233°58.0	27.9	272°42.0	41.4	71°53.8	24.1					Acamar	315°08.6	-40°10.9	
8	139°12.9	258°26.6	39.0	248°58.5	28.2	287°43.9	41.5	86°56.4	24.0					Menkar	314°01.9	4°12.5	
9	154°15.4	273°26.2	· · 39.7	263°59.0	· · 28.5	302°45.9	· · 41.7	101°59.0	· · 24.0					Mirfak	308°22.3	49°57.9	
10	169°17.9	288°25.8	40.3	278°59.6	28.7	317°47.8	41.9	117°01.6	24.0					Aldebaran	290°35.0	16°34.1	
11	184°20.3	303°25.4	41.0	294°00.1	29.0	332°49.8	42.1	132°04.2	23.9					Rigel	281°00.0	-8°10.0	
12	199°22.8	318°25.1	S23°41.7	309°00.6	S23°29.2	347°51.7	S11°42.3	147°06.8	N16°23.9					Capella	280°16.0	46°01.4	
13	214°25.3	333°24.7	42.3	324°01.2	29.5	2°53.7	42.5	162°09.4	23.9					Bellatrix	278°18.6	6°22.6	
14	229°27.7	348°24.3	43.0	339°01.7	29.7	17°55.6	42.6	177°12.0	23.8					Elnath	277°56.9	28°37.8	
15	244°30.2	3°23.9	· · 43.6	354°02.2	· · 30.0	32°57.6	· · 42.8	192°14.6	· · 23.8					Alnilam	275°33.7	-1°11.0	
16	259°32.6	18°23.5	44.3	9°02.8	30.2	47°59.6	43.0	207°17.2	23.7					Betelgeuse	270°47.8	7°24.7	
17	274°35.1	33°23.1	45.0	24°03.3	30.4	63°01.5	43.2	222°19.8	23.7					Canopus	263°50.6	-52°42.5	
18	289°37.6	48°22.7	S23°45.6	39°03.8	S23°30.7	78°03.5	S11°43.4	237°22.4	N16°23.7					Sirius	258°22.8	-16°45.4	
19	304°40.0	63°22.3	46.3	54°04.4	30.9	93°05.4	43.6	252°25.1	23.6					Adhara	255°02.8	-29°00.6	
20	319°42.5	78°21.9	46.9	69°04.9	31.2	108°07.4	43.7	267°27.7	23.6					Procyon	244°46.9	5°08.8	
21	334°45.0	93°21.6	· · 47.6	84°05.4	· · 31.4	123°09.3	· · 43.9	282°30.3	· · 23.5					Pollux	243°12.7	27°57.0	
22	349°47.4	108°21.2	48.3	99°06.0	31.7	138°11.3	44.1	297°32.9	23.5					Avior	234°13.2	-59°36.1	
23	4°49.9	123°20.8	48.9	114°06.5	31.9	153°13.2	44.3	312°35.5	23.5					Suhail	222°43.6	-43°33.0	
Mer.pass. 22:41			ν -0.4' d 0.7' m -4.32	ν 0.5' d 0.3' m 1.01	ν 2.0' d 0.2' m -1.69	ν 2.6' d -0.0' m -0.10											

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	19°52.4	138°20.4	S23°49.6	129°07.0	S23°32.2	168°15.2	S11°44.5	327°38.1	N16°23.4		
1	34°54.8	153°20.0	50.2	144°07.5	32.4	183°17.1	44.7	342°40.7	23.4		
2	49°57.3	168°19.6	50.9	159°08.1	32.7	198°19.1	44.9	357°43.3	23.3		
3	64°59.8	183°19.2	· · 51.5	174°08.6	· · 32.9	213°21.0	· · 45.0	12°45.9	· · 23.3		
4	80°02.2	198°18.8	52.2	189°09.1	33.2	228°23.0	45.2	27°48.5	23.3		
5	95°04.7	213°18.5	52.8	204°09.7	33.4	243°24.9	45.4	42°51.1	23.2		
6	110°07.1	228°18.1	S23°53.5	219°10.2	S23°33.6	258°26.9	S11°45.6	57°53.7	N16°23.2		
7	125°09.6	243°17.7	54.1	234°10.7	33.9	273°28.8	45.8	72°56.3	23.1		
8	140°12.1	258°17.3	54.8	249°11.2	34.1	288°30.8	46.0	87°59.0	23.1		
9	155°14.5	273°16.9	· · 55.4	264°11.8	· · 34.4	303°32.7	· · 46.1	103°01.6	· · 23.1		
10	170°17.0	288°16.5	56.1	279°12.3	34.6	318°34.7	46.3	118°04.2	23.0		
11	185°19.5	303°16.1	56.7	294°12.8	34.8	333°36.6	46.5	133°06.8	23.0		
12	200°21.9	318°15.7	S23°57.3	309°13.3	S23°35.1	348°38.6	S11°46.7	148°09.4	N16°22.9		
13	215°24.4	333°15.3	58.0	324°13.9	35.3	3°40.5	46.9	163°12.0	22.9		
14	230°26.9	348°15.0	58.6	339°14.4	35.6	18°42.5	47.1	178°14.6	22.9		
15	245°29.3	3°14.6	· · 59.3	354°14.9	· · 35.8	33°44.4	· · 47.3	193°17.2	· · 22.8		
16	260°31.8	18°14.2	23°59.9	9°15.4	36.0	48°46.4	47.4	208°19.8	22.8		
17	275°34.3	33°13.8	24°00.5	24°16.0	36.3	63°48.3	47.6	223°22.4	22.7		
18	290°36.7	48°13.4	S24°01.2	39°16.5	S23°36.5	78°50.3	S11°47.8	238°25.0	N16°22.7		
19	305°39.2	63°13.0	01.8	54°17.0	36.8	93°52.2	48.0	253°27.7	22.7		
20	320°41.6	78°12.6	02.4	69°17.5	37.0	108°54.2	48.2	268°30.3	22.6		
21	335°44.1	93°12.2	· · 03.1	84°18.1	· · 37.2	123°56.1	· · 48.4	283°32.9	· · 22.6		
22	350°46.6	108°11.8	03.7	99°18.6	37.5	138°58.1	48.5	298°35.5	22.5		
23	5°49.0	123°11.4	04.3	114°19.1	37.7	154°00.0	48.7	313°38.1	22.5		
Mer.pass. 22:37			ν -0.4' d 0.7' m -4.33	ν 0.5' d 0.2' m 1.01	ν 2.0' d 0.2' m -1.69	ν 2.6' d -0.0' m -0.11					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	20°51.5	138°11.1	S24°05.0	129°19.6	S23°38.0	169°02.0	S11°48.9	328°40.7	N16°22.5		
1	35°54.0	153°10.7	05.6	144°20.2	38.2	184°03.9	49.1	343°43.3	22.4		
2	50°56.4	168°10.3	06.2	159°20.7	38.4	199°05.9	49.3	358°45.9	22.4		
3	65°58.9	183°09.9	· · 06.9	174°21.2	· · 38.7	214°07.8	· · 49.5	13°48.5	· · 22.3		
4	81°01.4	198°09.5	07.5	189°21.7	38.9	229°09.8	49.6	28°51.1	22.3		
5	96°03.8	213°09.1	08.1	204°22.2	39.1	244°11.7	49.8	43°53.8	22.2		
6	111°06.3	228°08.7	S24°08.7	219°22.8	S23°39.4	259°13.7	S11°50.0	58°56.4	N16°22.2		
7	126°08.8	243°08.3	09.4	234°23.3	39.6	274°15.6	50.2	73°59.0	22.2		
8	141°11.2	258°07.9	10.0	249°23.8	39.8	289°17.5	50.4	89°01.6	22.1		
9	156°13.7	273°07.5	· · 10.6	264°24.3	· · 40.1	304°19.5	· · 50.6	104°04.2	· · 22.1		
10	171°16.1	288°07.2	11.2	279°24.8	40.3	319°21.4	50.8	119°06.8	22.0		
11	186°18.6	303°06.8	11.8	294°25.4	40.5	334°23.4	50.9	134°09.4	22.0		
12	201°21.1	318°06.4	S24°12.5	309°25.9	S23°40.8	349°25.3	S11°51.1	149°12.0	N16°22.0		
13	216°23.5	333°06.0	13.1	324°26.4	41.0	4°27.3	51.3	164°14.7	21.9		
14	231°26.0	348°05.6	13.7	339°26.9	41.2	19°29.2	51.5	179°17.3	21.9		
15	246°28.5	3°05.2	· · 14.3	354°27.4	· · 41.5	34°31.2	· · 51.7	194°19.9	· · 21.8		
16	261°30.9	18°04.8	14.9	9°28.0	41.7	49°33.1	51.9	209°22.5	21.8		
17	276°33.4	33°04.4	15.5	24°28.5	41.9	64°35.1	52.0	224°25.1	21.8		
18	291°35.9	48°04.0	S24°16.2	39°29.0	S23°42.1	79°37.0	S11°52.2	239°27.7	N16°21.7		
19	306°38.3	63°03.6	16.8	54°29.5	42.4	94°39.0	52.4	254°30.3	21.7		
20	321°40.8	78°03.3	17.4	69°30.0	42.6	109°40.9	52.6	269°32.9	21.6		
21	336°43.2	93°02.9	· · 18.0	84°30.5	· · 42.8	124°42.9	· · 52.8	284°35.6	· · 21.6		
22	351°45.7	108°02.5	18.6	99°31.1	43.1	139°44.8	53.0	299°38.2	21.6		
23	6°48.2	123°02.1	19.2	114°31.6	43.3	154°46.8	53.1	314°40.8	21.5		
Mer.pass. 22:33			ν -0.4' d 0.6' m -4.34	ν 0.5' d 0.2' m 1.02	ν 1.9' d 0.2' m -1.69	ν 2.6' d -0.0' m -0.11					

Oct 10 Wed	SHA	Mer.pass
Venus	119°36.5	14:46
Mars	110°01.0	15:24
Jupiter	148°35.1	12:48
Saturn	307°42.3	02:13

Oct 11 Thu	SHA	Mer.pass
Venus	118°28.0	14:47
Mars	109°14.7	15:23
Jupiter	148°22.8	12:45
Saturn	307°45.7	02:09

Oct 12 Fri	SHA	Mer.pass
Venus	117°19.6	14:48
Mars	108°28.1	15:22
Jupiter	148°10.5	12:42
Saturn	307°49.2	02:05

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

DUT1 = UT1-UTC = +0.2130 sec $\Delta T = TT-UT1 = +68.9710$ sec

2029 October 10 to Oct. 12 UT

h		Sun			Moon		
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	183°14.3	S06°40.1	154°06.0	4.3'	S20°22.0	8.1'	60.8'
1	198°14.5	41.1	168°29.4	4.3'	20°30.1	7.9'	60.8'
2	213°14.7	42.0	182°52.7	4.3'	20°38.0	7.8'	60.8'
3	228°14.8	· · 43.0	197°15.9	4.2'	20°45.8	7.6'	60.8'
4	243°15.0	43.9	211°39.2	4.2'	20°53.4	7.5'	60.8'
5	258°15.2	44.9	226°02.4	4.2'	21°00.9	7.3'	60.7'
6	273°15.3	S06°45.8	240°25.5	4.1'	S21°08.2	7.2'	60.7'
7	288°15.5	46.8	254°48.6	4.1'	21°15.4	7.0'	60.7'
8	303°15.7	47.7	269°11.7	4.1'	21°22.4	6.8'	60.7'
9	318°15.8	· · 48.6	283°34.8	4.0'	21°29.2	6.7'	60.7'
10	333°16.0	49.6	297°57.8	4.0'	21°35.9	6.5'	60.6'
11	348°16.1	50.5	312°20.8	4.0'	21°42.4	6.4'	60.6'
12	3°16.3	S06°51.5	326°43.8	4.0'	S21°48.8	6.2'	60.6'
13	18°16.5	52.4	341°06.8	3.9'	21°55.0	6.0'	60.6'
14	33°16.6	53.4	355°29.7	3.9'	22°01.0	5.9'	60.5'
15	48°16.8	· · 54.3	9°52.7	3.9'	22°06.9	5.7'	60.5'
16	63°17.0	55.3	24°15.6	3.9'	22°12.6	5.5'	60.5'
17	78°17.1	56.2	38°38.4	3.9'	22°18.2	5.4'	60.5'
18	93°17.3	S06°57.1	53°01.3	3.9'	S22°23.6	5.2'	60.4'
19	108°17.4	58.1	67°24.2	3.8'	22°28.8	5.0'	60.4'
20	123°17.6	06°59.0	81°47.0	3.8'	22°33.8	4.9'	60.4'
21	138°17.8	07°00.0	96°09.9	3.8'	22°38.7	4.7'	60.3'
22	153°17.9	00.9	110°32.7	3.8'	22°43.4	4.5'	60.3'
23	168°18.1	01.9	124°55.5	3.8'	22°47.9	4.4'	60.3'
SD = 16.0'		d = 0.9'	SD = 16.6'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	183°18.2	S07°02.8	139°18.4	3.8'	S22°52.3	4.2'	60.3'
1	198°18.4	03.7	153°41.2	3.8'	22°56.5	4.0'	60.2'
2	213°18.6	04.7	168°04.0	3.8'	23°00.6	3.9'	60.2'
3	228°18.7	· · 05.6	182°26.8	3.8'	23°04.4	3.7'	60.2'
4	243°18.9	06.6	196°49.7	3.8'	23°08.1	3.5'	60.1'
5	258°19.1	07.5	211°12.5	3.8'	23°11.7	3.4'	60.1'
6	273°19.2	S07°08.5	225°35.4	3.9'	S23°15.0	3.2'	60.1'
7	288°19.4	09.4	239°58.2	3.9'	23°18.2	3.0'	60.1'
8	303°19.5	10.3	254°21.1	3.9'	23°21.2	2.9'	60.0'
9	318°19.7	· · 11.3	268°44.0	3.9'	23°24.1	2.7'	60.0'
10	333°19.8	12.2	283°06.9	3.9'	23°26.8	2.5'	60.0'
11	348°20.0	13.2	297°29.8	3.9'	23°29.3	2.3'	59.9'
12	3°20.2	S07°14.1	311°52.7	4.0'	S23°31.6	2.2'	59.9'
13	18°20.3	15.0	326°15.7	4.0'	23°33.8	2.0'	59.9'
14	33°20.5	16.0	340°38.7	4.0'	23°35.8	1.8'	59.8'
15	48°20.6	· · 16.9	355°01.7	4.0'	23°37.7	1.7'	59.8'
16	63°20.8	17.9	9°24.7	4.1'	23°39.3	1.5'	59.8'
17	78°21.0	18.8	23°47.8	4.1'	23°40.8	1.3'	59.7'
18	93°21.1	S07°19.7	38°10.9	4.1'	S23°42.2	1.2'	59.7'
19	108°21.3	20.7	52°34.0	4.2'	23°43.3	1.0'	59.7'
20	123°21.4	21.6	66°57.2	4.2'	23°44.3	0.8'	59.6'
21	138°21.6	· · 22.6	81°20.4	4.3'	23°45.2	0.7'	59.6'
22	153°21.7	23.5	95°43.7	4.3'	23°45.8	0.5'	59.6'
23	168°21.9	24.4	110°07.0	4.3'	23°46.3	0.3'	59.5'
SD = 16.0'		d = 0.9'	SD = 16.4'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°22.0	S07°25.4	124°30.3	4.4'	S23°46.7	0.2'	59.5'
1	198°22.2	26.3	138°53.7	4.4'	23°46.9	0.0'	59.5'
2	213°22.4	27.2	153°17.1	4.5'	23°46.9	-0.2'	59.4'
3	228°22.5	· · 28.2	167°40.6	4.5'	23°46.7	-0.3'	59.4'
4	243°22.7	29.1	182°04.1	4.6'	23°46.4	-0.5'	59.4'
5	258°22.8	30.1	196°27.7	4.6'	23°45.9	-0.6'	59.3'
6	273°23.0	S07°31.0	210°51.3	4.7'	S23°45.3	-0.8'	59.3'
7	288°23.1	31.9	225°15.0	4.7'	23°44.5	-1.0'	59.3'
8	303°23.3	32.9	239°38.8	4.8'	23°43.5	-1.1'	59.2'
9	318°23.4	· · 33.8	254°02.6	4.9'	23°42.4	-1.3'	59.2'
10	333°23.6	34.7	268°26.4	4.9'	23°41.1	-1.4'	59.1'
11	348°23.7	35.7	282°50.4	5.0'	23°39.7	-1.6'	59.1'
12	3°23.9	S07°36.6	297°14.4	5.1'	S23°38.1	-1.8'	59.1'
13	18°24.1	37.6	311°38.4	5.1'	23°36.3	-1.9'	59.0'
14	33°24.2	38.5	326°02.5	5.2'	23°34.4	-2.1'	59.0'
15	48°24.4	· · 39.4	340°26.7	5.3'	23°32.3	-2.2'	59.0'
16	63°24.5	40.4	354°51.0	5.3'	23°30.1	-2.4'	58.9'
17	78°24.7	41.3	9°15.3	5.4'	23°27.8	-2.5'	58.9'
18	93°24.8	S07°42.2	23°39.7	5.5'	S23°25.2	-2.7'	58.9'
19	108°25.0	43.2	38°04.2	5.6'	23°22.6	-2.8'	58.8'
20	123°25.1	44.1	52°28.8	5.6'	23°19.8	-3.0'	58.8'
21	138°25.3	· · 45.0	66°53.4	5.7'	23°16.8	-3.1'	58.8'
22	153°25.4	46.0	81°18.1	5.8'	23°13.7	-3.3'	58.7'
23	168°25.6	46.9	95°42.9	5.9'	23°10.4	-3.4'	58.7'
SD = 16.0'		d = 0.9'	SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:38	05:57	07:06	16:25	17:34	18:52
N 70°	04:44	05:55	06:57	16:34	17:36	18:47
68°	04:49	05:53	06:50	16:42	17:38	18:42
66°	04:53	05:52	06:44	16:48	17:40	18:39
64°	04:56	05:51	06:38	16:53	17:41	18:36
62°	04:58	05:49	06:34	16:58	17:43	18:34
60°	05:00	05:48	06:30	17:02	17:44	18:32
N 58°	05:02	05:47	06:27	17:06	17:45	18:30
56°	05:03	05:46	06:23	17:09	17:46	18:29
54°	05:04	05:45	06:21	17:12	17:47	18:28
52°	05:05	05:44	06:18	17:14	17:48	18:27
50°	05:06	05:43	06:16	17:17	17:49	18:26
45°	05:07	05:41	06:11	17:22	17:51	18:25
N 40°	05:08	05:39	06:06	17:26	17:53	18:25
35°	05:08	05:37	06:03	17:30	17:55	18:25
30°	05:08	05:35	05:59	17:33	17:57	18:25
20°	05:06	05:31	05:54	17:39	18:02	18:27
N 10°	05:03	05:27	05:48	17:45	18:06	18:30
0°	04:58	05:23	05:43	17:50	18:11	18:35
S 10°	04:52	05:17	05:38	17:55	18:16	18:41
20°	04:44	05:10	05:33	18:01	18:23	18:49
30°	04:34	05:02	05:26	18:08	18:32	19:00
35°	04:26	04:57	05:23	18:11	18:37	19:08
40°	04:18	04:51	05:18	18:16	18:43	19:16
45°	04:07	04:43	05:13	18:21	18:51	19:27
S 50°	03:53	04:34	05:07	18:27	19:01	19:41
52°	03:47	04:29	05:04	18:30	19:05	19:48
54°	03:39	04:25	05:01	18:33	19:10	19:56
56°	03:31	04:19	04:58	18:37	19:16	20:05
58°	03:21	04:13	04:54	18:41	19:22	20:15
S 60°	03:09	04:06	04:50	18:45	19:30	20:27

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	12:36	14:46	15:39	15:52	15:53	17:08
64°	11:56	13:34	14:34	16:33	17:05	18:12
62°	11:29	12:58	13:59	17:01	17:41	18:47
60°	11:07	12:32	13:33	17:23	18:07	19:13
N 58°	10:50	12:12	13:13	17:41	18:28	19:33
56°	10:36	11:55	12:56	17:56	18:45	19:49
54°	10:23	11:40	12:42	18:08	18:59	20:03
52°	10:12	11:28	12:30	18:20	19:12	20:16
50°	10:02	11:17	12:19	18:30	19:23	20:26
45°	09:42	10:54	11:56	18:51	19:46	20:49
N 40°	09:25	10:35	11:37	19:08	20:05	21:07
35°	09:11	10:20	11:22	19:23	20:20	21:22
30°	08:59	10:07	11:08	19:36	20:34	21:36
20°	08:38	09:44	10:45	19:57	20:57	21:58
N 10°	08:21	09:24	10:26	20:16	21:17	22:17
0°	08:04	09:06	10:07	20:34	21:35	22:36
S 10°	07:48	08:48	09:49	20:51	21:54	22:54
20°	07:30	08:28	09:29	21:10	22:14	23:13
30°	07:10	08:06	09:06	21:32	22:37	23:35
35°	06:58	07:53	08:53	21:44	22:50	23:48
40°	06:45	07:38	08:37	21:59	23:06	· · ·
45°	06:29	07:20	08:19	22:17	23:24	· · ·
S 50°	06:10	06:58	07:56	22:38	23:47	· · ·
52°	06:01	06:47	07:45	22:49	23:58	· · ·
54°	05:51	06:35	07:33	23:00	· · ·	00:10
56°	05:39	06:22	07:19	23:14	· · ·	00:25
58°	05:26	06:06	07:02	23:29	· · ·	00:41
S 60°	05:11	05:47	06:42	23:48	· · ·	01:01

Day	Sun			Moon		Age 3-5 7-24%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	Mer. Pass hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	
10	12:57	13:05	11:47	14:19	01:48	
11	13:13	13:21	11:47	15:21	02:50	
12	13:28	13:36	11:46	16:21	03:51	

Aries		Venus		Mars		Jupiter		Saturn	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	21°50.6	138°01.7	S24°19.8	129°32.1	S23°43.5	169°48.7	S11°53.3	329°43.4	N16°21.5
1	36°53.1	153°01.3	20.4	144°32.6	43.7	184°50.7	53.5	344°46.0	21.4
2	51°55.6	168°00.9	21.0	159°33.1	44.0	199°52.6	53.7	359°48.6	21.4
3	66°58.0	183°00.5	· · 21.6	174°33.6	· · 44.2	214°54.6	· · 53.9	14°51.2	· · 21.3
4	82°00.5	198°00.1	22.2	189°34.2	44.4	229°56.5	54.1	29°53.9	21.3
5	97°03.0	212°59.7	22.8	204°34.7	44.6	244°58.5	54.2	44°56.5	21.3
6	112°05.4	227°59.4	S24°23.4	219°35.2	S23°44.9	260°00.4	S11°54.4	59°59.1	N16°21.2
7	127°07.9	242°59.0	24.0	234°35.7	45.1	275°02.4	54.6	75°01.7	21.2
8	142°10.4	257°58.6	24.6	249°36.2	45.3	290°04.3	54.8	90°04.3	21.1
9	157°12.8	272°58.2	· · 25.2	264°36.7	· · 45.5	305°06.3	· · 55.0	105°06.9	· · 21.1
10	172°15.3	287°57.8	25.8	279°37.2	45.8	320°08.2	55.2	120°09.5	21.1
11	187°17.7	302°57.4	26.4	294°37.8	46.0	335°10.1	55.4	135°12.2	21.0
12	202°20.2	317°57.0	S24°27.0	309°38.3	S23°46.2	350°12.1	S11°55.5	150°14.8	N16°21.0
13	217°22.7	332°56.6	27.6	324°38.8	46.4	5°14.0	55.7	165°17.4	20.9
14	232°25.1	347°56.2	28.2	339°39.3	46.7	20°16.0	55.9	180°20.0	20.9
15	247°27.6	2°55.8	· · 28.8	354°39.8	· · 46.9	35°17.9	· · 56.1	195°22.6	· · 20.9
16	262°30.1	17°55.4	29.4	9°40.3	47.1	50°19.9	56.3	210°25.2	20.8
17	277°32.5	32°55.1	30.0	24°40.8	47.3	65°21.8	56.5	225°27.9	20.8
18	292°35.0	47°54.7	S24°30.6	39°41.3	S23°47.5	80°23.8	S11°56.6	240°30.5	N16°20.7
19	307°37.5	62°54.3	31.2	54°41.8	47.8	95°25.7	56.8	255°33.1	20.7
20	322°39.9	77°53.9	31.8	69°42.4	48.0	110°27.7	57.0	270°35.7	20.6
21	337°42.4	92°53.5	· · 32.3	84°42.9	· · 48.2	125°29.6	· · 57.2	285°38.3	· · 20.6
22	352°44.9	107°53.1	32.9	99°43.4	48.4	140°31.6	57.4	300°40.9	20.6
23	7°47.3	122°52.7	33.5	114°43.9	48.6	155°33.5	57.6	315°43.6	20.5
Mer.pass. 22:29		ν -0.4' d 0.6' m -4.35		ν 0.5' d 0.2' m 1.02		ν 1.9' d 0.2' m -1.69		ν 2.6' d -0.0' m -0.12	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	22° 49.8	137° 52.3	S24° 34.1	129° 44.4	S23° 48.9	170° 35.5	S11° 57.7	330° 46.2	N16° 20.5
1	37° 52.2	152° 51.9	34.7	144° 44.9	49.1	185° 37.4	57.9	345° 48.8	20.4
2	52° 54.7	167° 51.5	35.3	159° 45.4	49.3	200° 39.4	58.1	0° 51.4	20.4
3	67° 57.2	182° 51.2	· · 35.8	174° 45.9	· · 49.5	215° 41.3	· · 58.3	15° 54.0	· · 20.4
4	82° 59.6	197° 50.8	36.4	189° 46.4	49.7	230° 43.2	58.5	30° 56.6	20.3
5	98° 02.1	212° 50.4	37.0	204° 46.9	49.9	245° 45.2	58.7	45° 59.3	20.3
6	113° 04.6	227° 50.0	S24° 37.6	219° 47.4	S23° 50.2	260° 47.1	S11° 58.8	61° 01.9	N16° 20.2
7	128° 07.0	242° 49.6	38.2	234° 48.0	50.4	275° 49.1	59.0	76° 04.5	20.2
8	143° 09.5	257° 49.2	38.7	249° 48.5	50.6	290° 51.0	59.2	91° 07.1	20.1
9	158° 12.0	272° 48.8	· · 39.3	264° 49.0	· · 50.8	305° 53.0	· · 59.4	106° 09.7	· · 20.1
10	173° 14.4	287° 48.4	39.9	279° 49.5	51.0	320° 54.9	59.6	121° 12.4	20.1
11	188° 16.9	302° 48.0	40.5	294° 50.0	51.2	335° 56.9	11° 59.8	136° 15.0	20.0
12	203° 19.4	317° 47.6	S24° 41.0	309° 50.5	S23° 51.4	350° 58.8	S12° 00.0	151° 17.6	N16° 20.0
13	218° 21.8	332° 47.3	41.6	324° 51.0	51.7	6° 00.8	00.1	166° 20.2	19.9
14	233° 24.3	347° 46.9	42.2	339° 51.5	51.9	21° 02.7	00.3	181° 22.8	19.9
15	248° 26.7	2° 46.5	· · 42.7	354° 52.0	· · 52.1	36° 04.7	· · 00.5	196° 25.4	· · 19.8
16	263° 29.2	17° 46.1	43.3	9° 52.5	52.3	51° 06.6	00.7	211° 28.1	19.8
17	278° 31.7	32° 45.7	43.9	24° 53.0	52.5	66° 08.5	00.9	226° 30.7	19.8
18	293° 34.1	47° 45.3	S24° 44.4	39° 53.5	S23° 52.7	81° 10.5	S12° 01.1	241° 33.3	N16° 19.7
19	308° 36.6	62° 44.9	45.0	54° 54.0	52.9	96° 12.4	01.2	256° 35.9	19.7
20	323° 39.1	77° 44.5	45.6	69° 54.5	53.1	111° 14.4	01.4	271° 38.5	19.6
21	338° 41.5	92° 44.1	· · 46.1	84° 55.0	· · 53.3	126° 16.3	· · 01.6	286° 41.2	· · 19.6
22	353° 44.0	107° 43.7	46.7	99° 55.5	53.6	141° 18.3	01.8	301° 43.8	19.6
23	8° 46.5	122° 43.4	47.3	114° 56.0	53.8	156° 20.2	02.0	316° 46.4	19.5
Mer.pass. 22:25		ν -0.4' d 0.6' m -4.36		ν 0.5' d 0.2' m 1.02		ν 1.9' d 0.2' m -1.69		ν 2.6' d -0.0' m -0.12	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	23°48.9	137°43.0	S24°47.8	129°56.5	S23°54.0	171°22.2	S12°02.2	331°49.0	N16°19.5
1	38°51.4	152°42.6	48.4	144°57.0	54.2	186°24.1	02.3	346°51.6	19.4
2	53°53.8	167°42.2	48.9	159°57.6	54.4	201°26.1	02.5	1°54.3	19.4
3	68°56.3	182°41.8	· · 49.5	174°58.1	· · 54.6	216°28.0	· · 02.7	16°56.9	· · 19.3
4	83°58.8	197°41.4	50.0	189°58.6	54.8	231°30.0	02.9	31°59.5	19.3
5	99°01.2	212°41.0	50.6	204°59.1	55.0	246°31.9	03.1	47°02.1	19.3
6	114°03.7	227°40.6	S24°51.2	219°59.6	S23°55.2	261°33.8	S12°03.3	62°04.7	N16°19.2
7	129°06.2	242°40.2	51.7	235°00.1	55.4	276°35.8	03.4	77°07.4	19.2
8	144°08.6	257°39.9	52.3	250°00.6	55.6	291°37.7	03.6	92°10.0	19.1
9	159°11.1	272°39.5	· · 52.8	265°01.1	· · 55.8	306°39.7	· · 03.8	107°12.6	· · 19.1
10	174°13.6	287°39.1	53.4	280°01.6	56.0	321°41.6	04.0	122°15.2	19.0
11	189°16.0	302°38.7	53.9	295°02.1	56.2	336°43.6	04.2	137°17.9	19.0
12	204°18.5	317°38.3	S24°54.5	310°02.6	S23°56.5	351°45.5	S12°04.4	152°20.5	N16°19.0
13	219°21.0	332°37.9	55.0	325°03.1	56.7	6°47.5	04.5	167°23.1	18.9
14	234°23.4	347°37.5	55.5	340°03.6	56.9	21°49.4	04.7	182°25.7	18.9
15	249°25.9	2°37.1	· · 56.1	355°04.1	· · 57.1	36°51.3	· · 04.9	197°28.3	· · 18.8
16	264°28.3	17°36.7	56.6	10°04.6	57.3	51°53.3	05.1	212°31.0	18.8
17	279°30.8	32°36.4	57.2	25°05.1	57.5	66°55.2	05.3	227°33.6	18.7
18	294°33.3	47°36.0	S24°57.7	40°05.6	S23°57.7	81°57.2	S12°05.5	242°36.2	N16°18.7
19	309°35.7	62°35.6	58.3	55°06.1	57.9	96°59.1	05.6	257°38.8	18.7
20	324°38.2	77°35.2	58.8	70°06.6	58.1	112°01.1	05.8	272°41.5	18.6
21	339°40.7	92°34.8	· · 59.3	85°07.1	· · 58.3	127°03.0	· · 06.0	287°44.1	· · 18.6
22	354°43.1	107°34.4	24°59.9	100°07.6	58.5	142°05.0	06.2	302°46.7	18.5
23	9°45.6	122°34.0	25°00.4	115°08.0	58.7	157°06.9	06.4	317°49.3	18.5
Mer.pass. 22:21		$\nu -0.4' d \ 0.6'$ m -4.37		$\nu \ 0.5' d \ 0.2'$ m 1.02		$\nu \ 1.9' d \ 0.2'$ m -1.69		$\nu \ 2.6' d \ -0.0'$ m -0.13	

	Stars	
	SHA	Dec
Alpheratz	357° 30.4	29° 15.6
Ankaa	353° 03.1	-42° 08.5
Schedar	340° 25.9	56° 42.2
Diphda	348° 43.2	-17° 49.2
Achernar	335° 16.9	-57° 05.0
Hamal	327° 46.5	23° 36.3
Polaris	311° 27.4	89° 23.1
Acamar	315° 08.6	-40° 10.9
Menkar	314° 01.8	4° 12.5
Mirfak	308° 22.2	49° 57.9
Aldebaran	290° 35.0	16° 34.1
Rigel	281° 00.0	-8° 10.0
Capella	280° 16.0	46° 01.4
Bellatrix	278° 18.6	6° 22.6
Elnath	277° 56.8	28° 37.8
Anilam	275° 33.7	-1° 11.0
Betelgeuse	270° 47.8	7° 24.7
Canopus	263° 50.5	-52° 42.5
Sirius	258° 22.8	-16° 45.4
Adhara	255° 02.8	-29° 00.6
Procyon	244° 46.9	5° 08.8
Pollux	243° 12.7	27° 57.0
Avior	234° 13.2	-59° 36.1
Suhail	222° 43.6	-43° 33.0
Miaplacidus	221° 37.6	-69° 50.2
Alphard	217° 44.2	-8° 47.3
Regulus	207° 30.6	11° 49.2
Dubhe	193° 37.1	61° 35.2
Denebola	182° 21.4	14° 24.3
Gienah	175° 39.9	-17° 42.4
Acrux	172° 56.2	-63° 15.8
Gacrux	171° 47.7	-57° 16.7
Alioth	166° 10.4	55° 47.8
Spica	158° 18.6	-11° 18.9
Alkaid	152° 49.6	49° 09.9
Hadar	148° 31.0	-60° 31.0
Menkent	147° 53.4	-36° 30.9
Arcturus	145° 44.8	19° 01.8
Rigel Kent.	139° 35.5	-60° 57.5
Kochab	137° 21.3	74° 02.1
Zuben'ubi	136° 52.1	-16° 09.8
Alphecca	126° 00.9	26° 37.1
Antares	112° 11.4	-26° 29.8
Atria	107° 02.4	-69° 04.9
Sabik	101° 58.6	-15° 45.5
Shaula	96° 05.4	-37° 07.4
Rasalhague	95° 55.2	12° 32.6
Eltanin	90° 40.6	51° 29.5
Kaus Aust.	83° 27.6	-34° 22.1
Vega	80° 30.7	38° 49.1
Nunki	75° 43.1	-26° 15.4
Altair	61° 56.2	8° 57.2
Peacock	52° 59.7	-56° 38.3
Deneb	49° 23.0	45° 23.7
Enif	33° 34.9	10° 01.0
Al Na'ir	27° 28.0	-46° 48.9
Fomalhaut	15° 10.1	-29° 27.7
Scheat	13° 41.1	28° 15.0
Markab	13° 25.8	15° 22.2

Oct 13 Sat	SHA	Mer.pass
Venus	116°11.1	14:48
Mars	107°41.5	15:21
Jupiter	147°58.1	12:39
Saturn	307°52.8	02:01

Oct 14 Sun	SHA	Mer.pass
Venus	115°02.5	14:49
Mars	106°54.6	15:21
Jupiter	147°45.7	12:36
Saturn	307°56.4	01:57

Oct 15 Mon	SHA	Mer.pass
Venus	113°54.1	14:50
Mars	106°07.6	15:20
Jupiter	147°33.2	12:33
Saturn	308°00.1	01:52

Horizontal parallax	
Venus:	0.2
Mars:	0.1

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°25.7	S07°47.8	110°07.7	5.9'	S23°07.0	-3.5'	58.6'
1	198°25.9	48.8	124°32.7	6.0'	23°03.5	-3.7'	58.6'
2	213°26.0	49.7	138°57.7	6.1'	22°59.8	-3.8'	58.6'
3	228°26.2	· · 50.6	153°22.8	6.2'	22°56.0	-4.0'	58.5'
4	243°26.3	51.6	167°48.0	6.3'	22°52.0	-4.1'	58.5'
5	258°26.5	52.5	182°13.3	6.4'	22°47.9	-4.2'	58.5'
6	273°26.6	S07°53.4	196°38.7	6.5'	S22°43.6	-4.4'	58.4'
7	288°26.8	54.4	211°04.1	6.5'	22°39.2	-4.5'	58.4'
8	303°26.9	55.3	225°29.7	6.6'	22°34.7	-4.7'	58.3'
9	318°27.1	· · 56.2	239°55.3	6.7'	22°30.1	-4.8'	58.3'
10	333°27.2	57.2	254°21.0	6.8'	22°25.3	-4.9'	58.3'
11	348°27.4	58.1	268°46.8	6.9'	22°20.4	-5.1'	58.2'
12	3°27.5	S07°59.0	283°12.7	7.0'	S22°15.3	-5.2'	58.2'
13	18°27.7	08°00.0	297°38.7	7.1'	22°10.1	-5.3'	58.2'
14	33°27.8	00.9	312°04.8	7.2'	22°04.8	-5.4'	58.1'
15	48°28.0	· · 01.8	326°31.0	7.3'	21°59.4	-5.6'	58.1'
16	63°28.1	02.8	340°57.3	7.4'	21°53.8	-5.7'	58.1'
17	78°28.3	03.7	355°23.6	7.5'	21°48.1	-5.8'	58.0'
18	93°28.4	S08°04.6	9°50.1	7.6'	S21°42.3	-5.9'	58.0'
19	108°28.5	05.5	24°16.6	7.6'	21°36.4	-6.1'	57.9'
20	123°28.7	06.5	38°43.3	7.7'	21°30.3	-6.2'	57.9'
21	138°28.8	· · 07.4	53°10.0	7.8'	21°24.2	-6.3'	57.9'
22	153°29.0	08.3	67°36.8	7.9'	21°17.9	-6.4'	57.8'
23	168°29.1	09.3	82°03.8	8.0'	21°11.5	-6.5'	57.8'
SD = 16.0'			SD = 16.0'				
$d = 0.9'$							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	183°29.3	S08°10.2	96°30.8	8.1'	S21°05.0	-6.6'	57.8'
1	198°29.4	11.1	110°57.9	8.2'	20°58.3	-6.7'	57.7'
2	213°29.6	12.1	125°25.2	8.3'	20°51.6	-6.9'	57.7'
3	228°29.7	· · 13.0	139°55.2	8.4'	20°44.7	-7.0'	57.7'
4	243°29.9	13.9	154°19.9	8.5'	20°37.7	-7.1'	57.6'
5	258°30.0	14.8	168°47.4	8.6'	20°30.7	-7.2'	57.6'
6	273°30.1	S08°15.8	183°15.0	8.7'	S20°23.5	-7.3'	57.6'
7	288°30.3	16.7	197°42.7	8.8'	20°16.2	-7.4'	57.5'
8	303°30.4	17.6	212°10.5	8.9'	20°08.8	-7.5'	57.5'
9	318°30.6	· · 18.6	226°38.5	9.0'	20°01.3	-7.6'	57.5'
10	333°30.7	19.5	241°06.5	9.1'	19°53.7	-7.7'	57.4'
11	348°30.9	20.4	255°34.6	9.2'	19°46.0	-7.8'	57.4'
12	3°31.0	S08°21.3	270°02.7	9.3'	S19°38.2	-7.9'	57.3'
13	18°31.1	22.3	284°31.0	9.4'	19°30.3	-8.0'	57.3'
14	33°31.3	23.2	298°59.4	9.5'	19°22.3	-8.1'	57.3'
15	48°31.4	· · 24.1	313°27.9	9.6'	19°14.2	-8.2'	57.2'
16	63°31.6	25.0	327°56.5	9.7'	19°06.0	-8.3'	57.2'
17	78°31.7	26.0	342°25.2	9.8'	18°57.7	-8.4'	57.2'
18	93°31.9	S08°26.9	356°54.0	9.9'	S18°49.4	-8.5'	57.1'
19	108°32.0	27.8	11°22.8	10.0'	18°40.9	-8.6'	57.1'
20	123°32.1	28.7	25°51.8	10.1'	18°32.3	-8.6'	57.1'
21	138°32.3	· · 29.7	40°20.8	10.2'	18°23.7	-8.7'	57.0'
22	153°32.4	30.6	54°50.0	10.2'	18°15.0	-8.8'	57.0'
23	168°32.6	31.5	69°19.3	10.3'	18°06.2	-8.9'	57.0'
SD = 16.0'			SD = 15.8'				
$d = 0.9'$							

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	183°32.7	S08°32.4	83°48.6	10.4'	S17°57.3	-9.0'	56.9'
1	198°32.8	33.4	98°18.0	10.5'	17°48.3	-9.1'	56.9'
2	213°33.0	34.3	112°47.6	10.6'	17°39.2	-9.1'	56.9'
3	228°33.1	· · 35.2	127°17.2	10.7'	17°30.1	-9.2'	56.8'
4	243°33.4	36.1	141°46.9	10.8'	17°20.8	-9.3'	56.8'
5	258°33.3	37.1	156°16.7	10.9'	17°11.5	-9.4'	56.8'
6	273°33.5	S08°38.0	170°46.6	11.0'	S17°02.2	-9.5'	56.7'
7	288°33.7	38.9	185°16.6	11.1'	16°52.7	-9.5'	56.7'
8	303°33.8	39.8	199°46.7	11.2'	16°43.2	-9.6'	56.7'
9	318°33.9	· · 40.7	214°16.8	11.3'	16°33.6	-9.7'	56.7'
10	333°34.1	41.7	228°47.1	11.3'	16°23.9	-9.7'	56.6'
11	348°34.2	42.6	243°17.4	11.4'	16°14.2	-9.8'	56.6'
12	3°34.4	S08°43.5	257°47.9	11.5'	S16°04.3	-9.9'	56.6'
13	18°34.5	44.4	272°18.4	11.6'	15°54.5	-10.0'	56.5'
14	33°34.6	45.4	286°49.0	11.7'	15°44.5	-10.0'	56.5'
15	48°34.8	· · 46.3	301°19.7	11.8'	15°34.5	-10.1'	56.5'
16	63°34.9	47.2	315°50.4	11.9'	15°24.4	-10.1'	56.4'
17	78°35.0	48.1	330°21.3	11.9'	15°14.3	-10.2'	56.4'
18	93°35.2	S08°49.0	344°52.2	12.0'	S15°04.1	-10.3'	56.4'
19	108°35.3	50.0	359°23.3	12.1'	14°53.8	-10.3'	56.3'
20	123°35.4	50.9	13°54.4	12.2'	14°43.4	-10.4'	56.3'
21	138°35.6	· · 51.8	28°25.6	12.3'	14°33.1	-10.4'	56.3'
22	153°35.7	52.7	42°56.8	12.3'	14°22.6	-10.5'	56.3'
23	168°35.8	53.6	57°28.2	12.4'	14°12.1	-10.6'	56.2'
SD = 16.0'			SD = 15.5'				
$d = 0.9'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:52	06:10	07:20	16:10	17:20	18:37
N 70°	04:56	06:07	07:10	16:21	17:23	18:33
68°	04:59	06:04	07:01	16:29	17:26	18:30
66°	05:02	06:01	06:54	16:37	17:29	18:28
64°	05:04	05:59	06:47	16:43	17:31	18:26
62°	05:06	05:57	06:42	16:49	17:33	18:25
60°	05:07	05:55	06:37	16:53	17:35	18:23
N 58°	05:08	05:54	06:33	16:57	17:37	18:22
56°	05:09	05:52	06:30	17:01	17:39	18:22
54°	05:10	05:51	06:26	17:05	17:40	18:21
52°	05:10	05:49	06:23	17:08	17:42	18:21
50°	05:11	05:48	06:21	17:10	17:43	18:20
45°	05:11	05:45	06:15	17:17	17:46	18:20
N 40°	05:11	05:42	06:10	17:22	17:49	18:20
35°	05:10	05:40	06:05	17:26	17:52	18:21
30°	05:09	05:37	06:01	17:30	17:54	18:22
20°	05:07	05:32	05:55	17:37	17:59	18:25
N 10°	05:03	05:27	05:48	17:43	18:04	18:29
0°	04:57	05:22	05:43	17:49	18:10	18:34
S 10°	04:51	05:15	05:37	17:55	18:17	18:41
20°	04:42	05:08	05:30	18:02	18:24	18:50
30°	04:30	04:59	05:23	18:09	18:34	19:03
35°	04:22	04:53	05:19	18:14	18:40	19:10
40°	04:13	04:46	05:14	18:19	18:47	19:20
45°	04:01	04:38	05:08	18:25	18:55	19:32
S 50°	03:46	04:27	05:01	18:32	19:06	19:47
52°	03:39	04:22	04:58	18:35	19:11	19:54
54°	03:31	04:17	04:54	18:39	19:16	20:03
56°	03:21	04:11	04:50	18:43	19:23	20:12
58°	03:11	04:04	04:46	18:48	19:30	20:24
S 60°	02:58	03:56	04:41	18:53	19:38	20:37

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	■	16:50	■	■	21:35
N 70°	■	16:58	16:15	■	19:42	22:08
68°	17:00	16:07	15:50	17:47	20:32	22:31
66°	15:38	15:35	15:31	19:09	21:04	22:49
64°	15:00	15:11	15:15	19:46	21:27	23:04
62°	14:33	14:52	15:02	20:13	21:45	23:16
60°	14:12	14:36	14:51	20:33	22:00	23:27
N 58°	13:55	14:23	14:41	20:50	22:13	23:36
56°	13:40	14:11	14:33	21:04	22:24	23:43
54°	13:28	14:01	14:25	21:16	22:33	23:50
52°	13:17	13:52	14:18	21:27	22:42	23:56
50°	13:07	13:44	14:12	21:37	22:50	· ·
45°	12:46	13:26	13:59	21:57	23:06	· ·
N 40°	12:29	13:12	13:48	22:13	23:19	· ·
35°	12:15	13:00	13:38	22:27	23:30	· ·
30°	12:03	12:49	13:30	22:38	23:40	· ·
20°	11:41	12:31	13:16	22:59	23:57	· ·
N 10°	11:23	12:15	13:03	23:16	· ·	00:12
0°	11:06	12:00	12:51	23:32	· ·	00:25
S 10°	10:48	11:45	12:39	23:49	· ·	00:39
20°	10:30	11:29	12:26	· ·	00:06	00:53
30°	10:09	11:11	12:12	· ·	00:26	01:09
35°	09:56	11:00	12:03	· ·	00:37	01:19
40°	09:42	10:48	11:53	00:03	00:50	01:29
45°	09:24	10:33	11:42	00:20	01:06	01:42
S 50°	09:03	10:15	11:28	00:42	01:24	01:57
52°	08:53	10:07	11:21	00:52	01:33	02:04
54°	08:42	09:57	11:14	01:04	01:43	02:12
56°	08:29	09:46	11:06	01:17	01:55	02:21
58°	08:13	09:34	10:57	01:33	02:07	02:31
S 60°	07:55	09:19	10:46	01:51	02:22	02:42

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		6-8
	00 ^h	mm:ss	hh:mm	Upper	Lower	34-56%
13	13:43	13:50	11:46	17:19	04:51	
14	13:57	14:04	11:46	18:13	05:47	
15	14:11	14:17	11:46	19:03	06:38	

October 16, 17, 18 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	24°48.1	137°33.6	S25°01.0	130°08.5	S23°58.9	172°08.8	S12°06.6	332°51.9	N16°18.4	Alpheratz	357°30.4	29°15.6					
1	39°50.5	152°33.3	01.5	145°09.0	59.1	187°10.8	06.7	347°54.6	18.4	Ankaa	353°03.1	-42°08.5					
2	54°53.0	167°32.9	02.0	160°09.5	59.3	202°12.7	06.9	2°57.2	18.4	Schedar	349°25.9	56°42.3					
3	69°55.5	182°32.5	•• 02.6	175°10.0	•• 59.5	217°14.7	•• 07.1	17°59.8	•• 18.3	Diphda	348°43.2	-17°49.2					
4	84°57.9	197°32.1	03.1	190°10.5	59.7	232°16.6	07.3	33°02.4	18.3	Achernar	335°16.9	-57°05.0					
5	100°00.4	212°31.7	03.6	205°11.0	23°59.9	247°18.6	07.5	48°05.1	18.2	Hamal	327°46.5	23°36.3					
6	115°02.8	227°31.3	S25°04.1	220°11.5	S24°00.1	262°20.5	S12°07.7	63°07.7	N16°18.2	Polaris	311°26.6	89°23.1					
7	130°05.3	242°30.9	04.7	235°12.0	00.3	277°22.5	07.8	78°10.3	18.1	Acamar	315°08.6	-40°11.0					
8	145°07.8	257°30.5	05.2	250°12.5	00.5	292°24.4	08.0	93°12.9	18.1	Menkar	314°01.8	4°12.5					
9	160°10.2	272°30.2	•• 05.7	265°13.0	•• 00.6	307°26.3	•• 08.2	108°15.6	•• 18.1	Mirfak	308°22.2	49°57.9					
10	175°12.7	287°29.8	06.3	280°13.5	00.8	322°28.3	08.4	123°18.2	18.0	Aldebaran	290°35.0	16°34.1					
11	190°15.2	302°29.4	06.8	295°14.0	01.0	337°30.2	08.6	138°20.8	18.0	Rigel	281°00.0	-8°10.0					
12	205°17.6	317°29.0	S25°07.3	310°14.5	S24°01.2	352°32.2	S12°08.8	153°23.4	N16°17.9	Capella	280°15.9	46°01.4					
13	220°20.1	332°28.6	07.8	325°15.0	01.4	7°34.1	08.9	168°26.1	17.9	Bellatrix	278°18.6	6°22.6					
14	235°22.6	347°28.2	08.3	340°15.5	01.6	22°36.1	09.1	183°28.7	17.8	Elnath	277°56.8	28°37.8					
15	250°25.0	2°27.8	•• 08.9	355°16.0	•• 01.8	37°38.0	•• 09.3	198°31.3	•• 17.8	Alnilam	275°33.7	-1°11.0					
16	265°27.5	17°27.5	09.4	10°16.5	02.0	52°40.0	09.5	213°33.9	17.8	Betelgeuse	270°47.8	7°24.7					
17	280°29.9	32°27.1	09.9	25°17.0	02.2	67°41.9	09.7	228°36.6	17.7	Canopus	263°50.5	-52°42.5					
18	295°32.4	47°26.7	S25°10.4	40°17.5	S24°02.4	82°43.8	S12°09.9	243°39.2	N16°17.7	Sirius	258°22.8	-16°45.4					
19	310°34.9	62°26.3	10.9	55°17.9	02.6	97°45.8	10.0	258°41.8	17.6	Adhara	255°02.8	-29°00.6					
20	325°37.3	77°25.9	11.5	70°18.4	02.8	112°47.7	10.2	273°44.4	17.6	Procyon	244°46.8	5°08.8					
21	340°39.8	92°25.5	•• 12.0	85°18.9	•• 03.0	127°49.7	•• 10.4	288°47.1	•• 17.5	Pollux	243°12.7	27°57.0					
22	355°42.3	107°25.1	12.5	100°19.4	03.2	142°51.6	10.6	303°49.7	17.5	Avior	234°13.2	-59°36.1					
23	10°44.7	122°24.7	13.0	115°19.9	03.3	157°53.6	10.8	318°52.3	17.4	Suhail	222°43.6	-43°33.0					
Mer.pass. 22:17			ν -0.4' d 0.5' m -4.38			ν 0.5' d 0.2' m 1.03			ν 1.9' d 0.2' m -1.68			ν 2.6' d -0.0' m -0.13					

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	25°47.2	137°24.4	S25°13.5	130°20.4	S24°03.5	172°55.5	S12°11.0	333°54.9	N16°17.4					
1	40°49.7	152°24.0	14.0	145°20.9	03.7	187°57.4	11.1	348°57.6	17.4					
2	55°52.1	167°23.6	14.5	160°21.4	03.9	202°59.4	11.3	4°00.2	17.3					
3	70°54.6	182°23.2	•• 15.0	175°21.9	•• 04.1	218°01.3	•• 11.5	19°02.8	•• 17.3					
4	85°57.1	197°22.8	15.5	190°22.4	04.3	233°03.3	11.7	34°05.4	17.2					
5	100°59.5	212°22.4	16.1	205°22.9	04.5	248°05.2	11.9	49°08.1	17.2					
6	116°02.0	227°22.1	S25°16.6	220°23.3	S24°04.7	263°07.2	S12°12.1	64°10.7	N16°17.1					
7	131°04.4	242°21.7	17.1	235°23.8	04.9	278°09.1	12.2	79°13.3	17.1					
8	146°06.9	257°21.3	17.6	250°24.3	05.0	293°11.0	12.4	94°16.0	17.1					
9	161°09.4	272°20.9	•• 18.1	265°24.8	•• 05.2	308°13.0	•• 12.6	109°18.6	•• 17.0					
10	176°11.8	287°20.5	18.6	280°25.3	05.4	323°14.9	12.8	124°21.2	17.0					
11	191°14.3	302°20.1	19.1	295°25.8	05.6	338°16.9	13.0	139°23.8	16.9					
12	206°16.8	317°19.7	S25°19.6	310°26.3	S24°05.8	353°18.8	S12°13.2	154°26.5	N16°16.9					
13	221°19.2	332°19.4	20.1	325°26.8	06.0	8°20.8	13.3	169°29.1	16.8					
14	236°21.7	347°19.0	20.6	340°27.3	06.2	23°22.7	13.5	184°31.7	16.8					
15	251°24.2	2°18.6	•• 21.1	355°27.7	•• 06.3	38°24.6	•• 13.7	199°34.4	•• 16.7					
16	266°26.6	17°18.2	21.6	10°28.2	06.5	53°26.6	13.9	214°37.0	16.7					
17	281°29.1	32°17.8	22.1	25°28.7	06.7	68°28.5	14.1	229°39.6	16.7					
18	296°31.6	47°17.4	S25°22.5	40°29.2	S24°06.9	83°30.5	S12°14.3	244°42.2	N16°16.6					
19	311°34.0	62°17.1	23.0	55°29.7	07.1	98°32.4	14.4	259°44.9	16.6					
20	326°36.5	77°16.7	23.5	70°30.2	07.2	113°34.4	14.6	274°47.5	16.5					
21	341°38.9	92°16.3	•• 24.0	85°30.7	•• 07.4	128°36.3	•• 14.8	289°50.1	•• 16.5					
22	356°41.4	107°15.9	24.5	100°31.1	07.6	143°38.2	15.0	304°52.7	16.4					
23	11°43.9	122°15.5	25.0	115°31.6	07.8	158°40.2	15.2	319°55.4	16.4					
Mer.pass. 22:13			ν -0.4' d 0.5' m -4.39			ν 0.5' d 0.2' m 1.03			ν 1.9' d 0.2' m -1.68			ν 2.6' d -0.0' m -0.14		

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	26°46.3	137°15.2	S25°25.5	130°32.1	S24°08.0	173°42.1	S12°15.4	334°58.0	N16°16.3					
1	41°48.8	152°14.8	26.0	145°32.6	08.2	188°44.1	15.5	350°00.6	16.3					
2	56°51.3	167°14.4	26.5	160°33.1	08.3	203°46.0	15.7	5°03.3	16.3					
3	71°53.7	182°14.0	•• 26.9	175°33.6	•• 08.5	218°48.0	•• 15.9	20°05.9	•• 16.2					
4	86°56.2	197°13.6	27.4	190°34.1	08.7	233°49.9	16.1	35°08.5	16.2					
5	101°58.7	212°13.2	27.9	205°34.5	08.9	248°51.8	16.3	50°11.2	16.1					
6	117°01.1	227°12.9	S25°28.4	220°35.0	S24°09.0	263°53.8	S12°16.5	65°13.8	N16°16.1					
7	132°03.6	242°12.5	28.9	235°35.5	09.2	278°55.7	16.6	80°16.4	16.0					
8	147°06.0	257°12.1	29.3	250°36.0	09.4	293°57.7	16.8	95°19.0	16.0					
9	162°08.5	272°11.7	•• 29.8	265°36.5	•• 09.6	308°59.6	•• 17.0	110°21.7	•• 16.0					
10	177°11.0	287°11.3	30.3	280°37.0	09.7	324°01.6	17.2	125°24.3	15.9					
11	192°13.4	302°11.0	30.8	295°37.4	09.9	339°03.5	17.4	140°26.9	15.9					
12	207°15.9	317°10.6	S25°31.3	310°37.9	S24°10.1	354°05.4	S12°17.6	155°29.6	N16°15.8					
13	222°18.4	332°10.2	31.7	325°38.4	10.3	9°07.4	17.7	170°32.2	15.8					
14	237°20.8	347°09.8	32.2	340°38.9	10.4	24°09.3	17.9	185°34.8	15.7					
15	252°23.3	2°09.4	•• 32.7	355°39.4	•• 10.6	39°11.3	•• 18.1	200°37.5	•• 15.7					
16	267°25.8	17°09.1	33.1	10°39.8	10.8	54°13.2	18.3	215°40.1	15.6					
17	282°28.2	32°08.7	33.6	25°40.3	11.0	69°15.1	18.5	230°42.7	15.6					
18	297°30.7	47°08.3	S25°34.1	40°40.8	S24°11.1	84°17.1	S12°18.7	245°45.4	N16°15.6					
19	312°33.2	62°07.9	34.5	55°41.3	11.3	99°19.0	18.8	260°48.0	15.5					
20	327°35.6	77°07.5	35.0	70°41.8	11.5	114°21.0	19.0	275°50.6	15.5					
21	342°38.1	92°07.2	•• 35.5	85°42.3	•• 11.7	129°22.9	•• 19.2	290°53.2	•• 15.4					
22	357°40.5	107°06.8	35.9	100°42.7	11.8	144°24.8	19.4	305°55.9	15.4					
23	12°43.0	122°06.4	36.4	115°43.2	12.0	159°26.8	19.6	320°58.5	15.3					
Mer.pass. 22:09			ν -0.4' d 0.5' m -4.40			ν 0.5' d 0.2' m 1.03			ν 1.9' d 0.2' m -1.68			ν 2.6' d -0.0' m -0.14		

Oct 16 Tue	SHA	Mer.pass
Venus	112°45.6	14:50
Mars	105°20.5	15:19
Jupiter	147°20.8	12:30
Saturn	308°03.9	01:48

Oct 17 Wed	SHA	Mer.pass
Venus	111°37.2	14:51
Mars	104°33.2	15:18
Jupiter	147°08.3	12:27
Saturn	308°07.7	01:44

Oct 18 Thu	SHA	Mer.pass
Venus	110°28.8	14:51
Mars	103°45.8	15:17
Jupiter	146°55.8	12:24
Saturn	308°11.7	01:40

Horizontal parallax		
Venus:		0.2
Mars:		0.1

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	183°36.0	S08°54.6	71°59.6	12.5'	S14°01.5	-10.6'	56.2'
1	198°36.1	55.5	86°31.1	12.6'	13°50.9	-10.7'	56.2'
2	213°36.2	56.4	101°02.7	12.7'	13°40.2	-10.7'	56.1'
3	228°36.4	· · 57.3	115°34.3	12.7'	13°29.5	-10.8'	56.1'
4	243°36.5	58.2	130°06.1	12.8'	13°18.7	-10.8'	56.1'
5	258°36.6	08°59.2	144°37.9	12.9'	13°07.9	-10.9'	56.1'
6	273°36.8	S09°00.1	159°09.8	13.0'	S12°57.0	-10.9'	56.0'
7	288°36.9	01.0	173°41.7	13.0'	12°46.1	-11.0'	56.0'
8	303°37.0	01.9	188°13.8	13.1'	12°35.1	-11.0'	56.0'
9	318°37.2	· · 02.8	202°45.8	13.2'	12°24.1	-11.1'	55.9'
10	333°37.3	03.7	217°18.0	13.2'	12°13.0	-11.1'	55.9'
11	348°37.4	04.7	231°50.3	13.3'	12°01.9	-11.2'	55.9'
12	3°37.6	S09°05.6	246°22.6	13.4'	S11°50.8	-11.2'	55.9'
13	18°37.7	06.5	260°54.9	13.4'	11°39.6	-11.2'	55.8'
14	33°37.8	07.4	275°27.4	13.5'	11°28.3	-11.3'	55.8'
15	48°38.0	· · 08.3	289°59.9	13.6'	11°17.0	-11.3'	55.8'
16	63°38.1	09.2	304°32.5	13.6'	11°05.7	-11.4'	55.8'
17	78°38.2	10.1	319°05.1	13.7'	10°54.4	-11.4'	55.7'
18	93°38.4	S09°11.1	333°37.8	13.8'	S10°43.0	-11.4'	55.7'
19	108°38.5	12.0	348°10.6	13.8'	10°31.5	-11.5'	55.7'
20	123°38.6	12.9	2°43.4	13.9'	10°20.0	-11.5'	55.7'
21	138°38.7	· · 13.8	17°16.3	13.9'	10°08.5	-11.5'	55.6'
22	153°38.9	14.7	31°49.2	14.0'	09°57.0	-11.6'	55.6'
23	168°39.0	15.6	46°22.2	14.1'	09°45.4	-11.6'	55.6'
SD = 16.0'			SD = 15.3'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	183°39.1	S09°16.6	60°55.3	14.1'	S09°33.8	-11.6'	55.6'
1	198°39.3	17.5	75°28.4	14.2'	09°22.2	-11.7'	55.5'
2	213°39.4	18.4	90°01.6	14.2'	09°10.5	-11.7'	55.5'
3	228°39.5	· · 19.3	104°34.8	14.3'	08°58.8	-11.7'	55.5'
4	243°39.6	20.2	119°08.1	14.3'	08°47.1	-11.8'	55.5'
5	258°39.8	21.1	133°41.5	14.4'	08°35.3	-11.8'	55.4'
6	273°39.9	S09°22.0	148°14.8	14.4'	S08°23.5	-11.8'	55.4'
7	288°40.0	22.9	162°48.3	14.5'	08°11.7	-11.8'	55.4'
8	303°40.1	23.9	177°21.8	14.5'	07°59.9	-11.9'	55.4'
9	318°40.3	· · 24.8	191°55.3	14.6'	07°48.0	-11.9'	55.3'
10	333°40.4	25.7	206°28.9	14.6'	07°36.1	-11.9'	55.3'
11	348°40.5	26.6	221°02.5	14.7'	07°24.2	-11.9'	55.3'
12	3°40.6	S09°27.5	235°36.2	14.7'	S07°12.3	-12.0'	55.3'
13	18°40.8	28.4	250°10.0	14.8'	07°00.3	-12.0'	55.3'
14	33°40.9	29.3	264°43.7	14.8'	06°48.4	-12.0'	55.2'
15	48°41.0	· · 30.2	279°17.6	14.9'	06°36.4	-12.0'	55.2'
16	63°41.1	31.1	293°51.4	14.9'	06°24.4	-12.0'	55.2'
17	78°41.3	32.0	308°25.3	14.9'	06°12.3	-12.0'	55.2'
18	93°41.4	S09°33.0	322°59.3	15.0'	S06°00.3	-12.1'	55.2'
19	108°41.5	33.9	337°33.3	15.0'	05°48.2	-12.1'	55.1'
20	123°41.6	34.8	352°07.3	15.1'	05°36.2	-12.1'	55.1'
21	138°41.8	· · 35.7	6°41.3	15.1'	05°24.1	-12.1'	55.1'
22	153°41.9	36.6	21°15.5	15.1'	05°12.0	-12.1'	55.1'
23	168°42.0	37.5	35°49.6	15.2'	04°59.9	-12.1'	55.0'
SD = 16.0'			SD = 15.2'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	183°42.1	S09°38.4	50°23.8	15.2'	S04°47.7	-12.1'	55.0'
1	198°42.2	39.3	64°58.0	15.2'	04°35.6	-12.2'	55.0'
2	213°42.4	40.2	79°32.2	15.3'	04°23.4	-12.2'	55.0'
3	228°42.5	· · 41.1	94°06.5	15.3'	04°11.3	-12.2'	55.0'
4	243°42.6	42.0	108°40.8	15.3'	03°59.1	-12.2'	55.0'
5	258°42.7	42.9	123°15.1	15.4'	03°46.9	-12.2'	54.9'
6	273°42.9	S09°43.8	137°49.5	15.4'	S03°34.7	-12.2'	54.9'
7	288°43.0	44.8	152°23.9	15.4'	03°22.5	-12.2'	54.9'
8	303°43.1	45.7	166°58.3	15.4'	03°10.3	-12.2'	54.9'
9	318°43.2	· · 46.6	181°32.8	15.5'	02°58.1	-12.2'	54.9'
10	333°43.3	47.5	196°07.2	15.5'	02°45.9	-12.2'	54.8'
11	348°43.5	48.4	210°41.7	15.5'	02°33.7	-12.2'	54.8'
12	3°43.6	S09°49.3	225°16.3	15.5'	S02°21.5	-12.2'	54.8'
13	18°43.7	50.2	239°50.8	15.6'	02°09.3	-12.2'	54.8'
14	33°43.8	51.1	254°25.4	15.6'	01°57.1	-12.2'	54.8'
15	48°43.9	· · 52.0	269°00.0	15.6'	01°44.8	-12.2'	54.8'
16	63°44.0	52.9	283°34.6	15.6'	01°32.6	-12.2'	54.7'
17	78°44.2	53.8	298°09.2	15.7'	01°20.4	-12.2'	54.7'
18	93°44.3	S09°54.7	312°43.9	15.7'	S01°08.2	-12.2'	54.7'
19	108°44.4	55.6	327°18.5	15.7'	00°56.0	-12.2'	54.7'
20	123°44.5	56.5	341°53.2	15.7'	00°43.7	-12.2'	54.7'
21	138°44.6	· · 57.4	356°27.9	15.7'	00°31.5	-12.2'	54.7'
22	153°44.7	58.3	11°02.7	15.7'	00°19.3	-12.2'	54.6'
23	168°44.9	59.2	25°37.4	15.7'	00°07.1	-12.2'	54.6'
SD = 16.0'			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:05	06:23	07:35	15:54	17:05	18:23
N 70°	05:08	06:18	07:22	16:06	17:10	18:21
68°	05:10	06:14	07:12	16:17	17:15	18:19
66°	05:11	06:11	07:04	16:26	17:18	18:17
64°	05:13	06:08	06:56	16:33	17:22	18:16
62°	05:13	06:05	06:50	16:39	17:24	18:16
60°	05:14	06:02	06:45	16:45	17:27	18:15
N 58°	05:15	06:00	06:40	16:49	17:29	18:15
56°	05:15	05:58	06:36	16:54	17:32	18:15
54°	05:15	05:56	06:32	16:58	17:34	18:14
52°	05:15	05:54	06:29	17:01	17:35	18:14
50°	05:15	05:53	06:25	17:04	17:37	18:15
45°	05:15	05:49	06:19	17:11	17:41	18:15
N 40°	05:14	05:45	06:13	17:17	17:45	18:16
35°	05:13	05:42	06:08	17:22	17:48	18:17
30°	05:11	05:39	06:03	17:27	17:51	18:19
20°	05:08	05:33	05:56	17:35	17:57	18:23
N 10°	05:03	05:27	05:49	17:42	18:03	18:28
0°	04:57	05:21	05:42	17:49	18:10	18:34
S 10°	04:49	05:14	05:35	17:55	18:17	18:42
20°	04:39	05:06	05:28	18:03	18:25	18:52
30°	04:26	04:55	05:20	18:11	18:36	19:05
35°	04:18	04:49	05:15	18:16	18:42	19:13
40°	04:08	04:41	05:09	18:22	18:50	19:24
45°	03:55	04:32	05:03	18:29	18:59	19:37
S 50°	03:39	04:21	04:55	18:37	19:11	19:53
52°	03:31	04:15	04:51	18:41	19:17	20:01
54°	03:22	04:09	04:47	18:45	19:23	20:10
56°	03:12	04:03	04:43	18:49	19:30	20:21
58°	03:00	03:55	04:38	18:54	19:37	20:33
S 60°	02:46	03:46	04:32	19:00	19:46	20:48

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	16:12	15:47	15:26	23:49	· · ·	01:45
N 70°	15:54	15:38	15:24	· · ·	00:05	01:51
68°	15:39	15:30	15:22	· · ·	00:18	01:57
66°	15:28	15:24	15:21	· · ·	00:28	02:01
64°	15:18	15:19	15:19	· · ·	00:37	02:05
62°	15:09	15:14	15:18	· · ·	00:44	02:08
60°	15:02	15:10	15:17	· · ·	00:50	02:11
N 58°	14:55	15:06	15:16	· · ·	00:56	02:14
56°	14:50	15:03	15:15	· · ·	01:01	02:16
54°	14:44	15:00	15:14	· · ·	01:05	02:18
52°	14:40	14:58	15:14	· · ·	01:09	02:20
50°	14:35	14:55	15:13	00:02	01:13	02:21
45°	14:26	14:50	15:12	00:14	01:20	02:25
N 40°	14:18	14:45	15:11	00:24	01:27	02:28
35°	14:12	14:42	15:10	00:32	01:32	02:30
30°	14:06	14:38	15:09	00:40	01:37	02:32
20°	13:55	14:32	15:07	00:53	01:45	02:36
N 10°	13:46	14:27	15:06	01:04	01:53	02:40
0°	13:38	14:22	15:05	01:14	01:59	02:43
S 10°	13:29	14:17	15:03	01:24	02:06	02:46
20°	13:20	14:12	15:02	01:35	02:13	02:49
30°	13:10	14:06	15:00	01:47	02:21	02:53
35°	13:04	14:03	15:00	01:54	02:26	02:55
40°	12:57	13:59	14:59	02:02	02:31	02:57
45°	12:49	13:54	14:57	02:12	02:37	03:00
S 50°	12:39	13:48	14:56	02:23	02:44	03:03
52°	12:35	13:46	14:55	02:28	02:47	03:04
54°	12:30	13:43	14:55	02:34	02:51	03:06
56°	12:24	13:40	14:54	02:40	02:55	03:08
58°	12:18	13:37	14:53	02:47	02:59	03:10
S 60°	12:11	13:33	14:52	02:55	03:04	03:12

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	9-11 66-83%
16	14:24	14:30	11:45	19:49	07:26	
17	14:37	14:43	11:45	20:32	08:11	
18	14:49	14:54	11:45	21:15	08:54	

October 19, 20, 21 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	27°45.5	137°06.0	S25°36.9	130°43.7	S24°12.2	174°28.7	S12°19.8	336°01.1	N16°15.3					Alpheratz	357°30.4	29°15.6	
1	42°47.9	152°05.6	37.3	145°44.2	12.3	189°30.7	19.9	351°03.8	15.2					Ankaa	353°03.1	-42°08.5	
2	57°50.4	167°05.3	37.8	160°44.7	12.5	204°32.6	20.1	6°06.4	15.2					Schedar	349°25.9	56°42.3	
3	72°52.9	182°04.9	•• 38.3	175°45.1	•• 12.7	219°34.6	•• 20.3	21°09.0	•• 15.1					Diphda	348°43.2	-17°49.2	
4	87°55.3	197°04.5	38.7	190°45.6	12.8	234°36.5	20.5	36°11.7	15.1					Achernar	335°16.9	-57°05.0	
5	102°57.8	212°04.1	39.2	205°46.1	13.0	249°38.4	20.7	51°14.3	15.1					Hamal	327°46.5	23°36.3	
6	118°00.3	227°03.8	S25°39.6	220°46.6	S24°13.2	264°40.4	S12°20.8	66°16.9	N16°15.0					Polaris	311°25.8	89°23.1	
7	133°02.7	242°03.4	40.1	235°47.0	13.4	279°42.3	21.0	81°19.6	15.0					Acamar	315°08.5	-40°11.0	
8	148°05.2	257°03.0	40.5	250°47.5	13.5	294°44.3	21.2	96°22.2	14.9					Menkar	314°01.8	4°12.5	
9	163°07.6	272°02.6	•• 41.0	265°48.0	•• 13.7	309°46.2	•• 21.4	111°24.8	•• 14.9					Mirfak	308°22.2	49°57.9	
10	178°10.1	287°02.3	41.4	280°48.5	13.9	324°48.1	21.6	126°27.5	14.8					Aldebaran	290°35.0	16°34.1	
11	193°12.6	302°01.9	41.9	295°49.0	14.0	339°50.1	21.8	141°30.1	14.8					Rigel	281°00.0	-8°10.0	
12	208°15.0	317°01.5	S25°42.3	310°49.4	S24°14.2	354°52.0	S12°21.9	156°32.7	N16°14.7					Capella	280°15.9	46°01.4	
13	223°17.5	332°01.1	42.8	325°49.9	14.3	9°54.0	22.1	171°35.4	14.7					Bellatrix	278°18.6	6°22.6	
14	238°20.0	347°00.7	43.2	340°50.4	14.5	24°55.9	22.3	186°38.0	14.7					Elnath	277°56.8	28°37.8	
15	253°22.4	2°00.4	•• 43.7	355°50.9	•• 14.7	39°57.8	•• 22.5	201°40.6	•• 14.6					Alnilam	275°33.7	-1°11.0	
16	268°24.9	17°00.0	44.1	10°51.3	14.8	54°59.8	22.7	216°43.3	14.6					Betelgeuse	270°47.8	7°24.7	
17	283°27.4	31°59.6	44.6	25°51.8	15.0	70°01.7	22.9	231°45.9	14.5					Canopus	263°50.5	-52°42.5	
18	298°29.8	46°59.2	S25°45.0	40°52.3	S24°15.2	85°03.7	S12°23.0	246°48.5	N16°14.5					Sirius	258°22.8	-16°45.4	
19	313°32.3	61°58.9	45.5	55°52.8	15.3	100°05.6	23.2	261°51.2	14.4					Adhara	255°02.8	-29°00.6	
20	328°34.8	76°58.5	45.9	70°53.2	15.5	115°07.5	23.4	276°53.8	14.4					Procyon	244°46.8	5°08.8	
21	343°37.2	91°58.1	•• 46.3	85°53.7	•• 15.7	130°09.5	•• 23.6	291°56.4	•• 14.3					Pollux	243°12.7	27°57.0	
22	358°39.7	106°57.8	46.8	100°54.2	15.8	145°11.4	23.8	306°59.1	14.3					Avior	234°13.1	-59°36.1	
23	13°42.1	121°57.4	47.2	115°54.7	16.0	160°13.4	24.0	322°01.7	14.2					Suhail	222°43.5	-43°33.0	
Mer.pass. 22:05			ν -0.4' d 0.5' m -4.41			ν 0.5' d 0.2' m 1.03			ν 1.9' d 0.2' m -1.68			ν 2.6' d -0.0' m -0.14					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	28°44.6	136°57.0	S25°47.7	130°55.1	S24°16.1	175°15.3	S12°24.1	337°04.3	N16°14.2					
1	43°47.1	151°56.6	48.1	145°55.6	16.3	190°17.3	24.3	352°07.0	14.2					
2	58°49.5	166°56.3	48.5	160°56.1	16.5	205°19.2	24.5	7°09.6	14.1					
3	73°52.0	181°55.9	•• 49.0	175°56.6	•• 16.6	220°21.1	•• 24.7	22°12.2	•• 14.1					
4	88°54.5	196°55.5	49.4	190°57.0	16.8	235°23.1	24.9	37°14.9	14.0					
5	103°56.9	211°55.1	49.8	205°57.5	16.9	250°25.0	25.0	52°17.5	14.0					
6	118°59.4	226°54.8	S25°50.3	220°58.0	S24°17.1	265°27.0	S12°25.2	67°20.1	N16°13.9					
7	134°01.9	241°54.4	50.7	235°58.4	17.3	280°28.9	25.4	82°22.8	13.9					
8	149°04.3	256°54.0	51.1	250°58.9	17.4	295°30.8	25.6	97°25.4	13.8					
9	164°06.8	271°53.7	•• 51.6	265°59.4	•• 17.6	310°32.8	•• 25.8	112°28.1	•• 13.8					
10	179°09.2	286°53.3	52.0	280°59.9	17.7	325°34.7	26.0	127°30.7	13.7					
11	194°11.7	301°52.9	52.4	296°00.3	17.9	340°36.7	26.1	142°33.3	13.7					
12	209°14.2	316°52.5	S25°52.8	311°00.8	S24°18.0	355°38.6	S12°26.3	157°36.0	N16°13.7					
13	224°16.6	331°52.2	53.3	326°01.3	18.2	10°40.5	26.5	172°38.6	13.6					
14	239°19.1	346°51.8	53.7	341°01.7	18.3	25°42.5	26.7	187°41.2	13.6					
15	254°21.6	1°51.4	•• 54.1	356°02.2	•• 18.5	40°44.4	•• 26.9	202°43.9	•• 13.5					
16	269°24.0	16°51.1	54.5	11°02.7	18.7	55°46.3	27.1	217°46.5	13.5					
17	284°26.5	31°50.7	54.9	26°03.2	18.8	70°48.3	27.2	232°49.1	13.4					
18	299°29.0	46°50.3	S25°55.4	41°03.6	S24°19.0	85°50.2	S12°27.4	247°51.8	N16°13.4					
19	314°31.4	61°50.0	55.8	56°04.1	19.1	100°52.2	27.6	262°54.4	13.3					
20	329°33.9	76°49.6	56.2	71°04.6	19.3	115°54.1	27.8	277°57.1	13.3					
21	344°36.4	91°49.2	•• 56.6	86°05.0	•• 19.4	130°56.0	•• 28.0	292°59.7	•• 13.2					
22	359°38.8	106°48.9	57.0	101°05.5	19.6	145°58.0	28.1	308°02.3	13.2					
23	14°41.3	121°48.5	57.4	116°06.0	19.7	160°59.9	28.3	323°05.0	13.2					
Mer.pass. 22:01			ν -0.4' d 0.4' m -4.42			ν 0.5' d 0.2' m 1.03			ν 1.9' d 0.2' m -1.68			ν 2.6' d -0.0' m -0.15		

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	29°43.7	136°48.1	S25°57.9	131°06.4	S24°19.9	176°01.9	S12°28.5	338°07.6	N16°13.1					
1	44°46.2	151°47.7	58.3	146°06.9	20.0	191°03.8	28.7	353°10.2	13.1					
2	59°48.7	166°47.4	58.7	161°07.4	20.2	206°05.7	28.9	8°12.9	13.0					
3	74°51.1	181°47.0	•• 59.1	176°07.8	•• 20.3	221°07.7	•• 29.1	23°15.5	•• 13.0					
4	89°53.6	196°46.6	59.5	191°08.3	20.5	236°09.6	29.2	38°18.1	12.9					
5	104°56.1	211°46.3	25°59.9	206°08.8	20.6	251°11.6	29.4	53°20.8	12.9					
6	119°58.5	226°45.9	S26°00.3	221°09.2	S24°20.8	266°13.5	S12°29.6	68°23.4	N16°12.8					
7	135°01.0	241°45.5	00.7	236°09.7	20.9	281°15.4	29.8	83°26.1	12.8					
8	150°03.5	256°45.2	01.1	251°10.2	21.1	296°17.4	30.0	98°28.7	12.7					
9	165°05.9	271°44.8	•• 01.5	266°10.6	•• 21.2	311°19.3	•• 30.1	113°31.3	•• 12.7					
10	180°08.4	286°44.5	01.9	281°11.1	21.3	326°21.3	30.3	128°34.0	12.7					
11	195°10.8	301°44.1	02.3	296°11.6	21.5	341°23.2	30.5	143°36.6	12.6					
12	210°13.3	316°43.7	S26°02.7	311°12.0	S24°21.6	356°25.1	S12°30.7	158°39.3	N16°12.6					
13	225°15.8	331°43.4	03.1	326°12.5	21.8	11°27.1	30.9	173°41.9	12.5					
14	240°18.2	346°43.0	03.5	341°13.0	21.9	26°29.0	31.1	188°44.5	12.5					
15	255°20.7	1°42.6	•• 03.9	356°13.4	•• 22.1	41°31.0	•• 31.2	203°47.2	•• 12.4					
16	270°23.2	16°42.3	04.3	11°13.9	22.2	56°32.9	31.4	218°49.8	12.4					
17	285°25.6	31°41.9	04.7	26°14.4	22.4	71°34.8	31.6	233°52.4	12.3					
18	300°28.1	46°41.5	S26°05.1	41°14.8	S24°22.5	86°36.8	S12°31.8	248°55.1	N16°12.3					
19	315°30.6	61°41.2	05.5	56°15.3	22.6	101°38.7	32.0	263°57.7	12.2					
20	330°33.0	76°40.8	05.9	71°15.8	22.8	116°40.6	32.1	279°00.4	12.2					
21	345°35.5	91°40.5	•• 06.3	86°16.2	•• 22.9	131°42.6	•• 32.3	294°03.0	•• 12.1					
22	0°38.0	106°40.1	06.7	101°16.7	23.1	146°44.5	32.5	309°05.6	12.1					
23	15°40.4	121°39.7	07.1	116°17.2	23.2	161°46.5	32.7	324°08.3	12.1					
Mer.pass. 21:57			ν -0.4' d 0.4' m -4.43			ν 0.5' d 0.2' m 1.04			ν 1.9' d 0.2' m -1.68			ν 2.6' d -0.0' m -0.15		

Oct 19 Fri	SHA	Mer.pass
Venus	109°20.6	14:52
Mars	102°58.2	15:17
Jupiter	146°43.3	12:20
Saturn	308°15.7	01:36

Oct 20 Sat	SHA	Mer.pass
Venus	108°12.4	14:5

DUT1 = UT1-UTC = +0.2132 sec $\Delta T = TT-UT1 = +68.9708$ sec

2029 October 19 to Oct. 21 UT

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°45.0	S10°00.1	40°12.1	15.8'	N00°05.1	12.2'	54.6'
1	198°45.1	01.0	54°46.9	15.8'	00°17.3	12.2'	54.6'
2	213°45.2	01.9	69°21.7	15.8'	00°29.4	12.2'	54.6'
3	228°45.3	· · 02.8	83°56.5	15.8'	00°41.6	12.2'	54.6'
4	243°45.4	03.7	98°31.2	15.8'	00°53.8	12.2'	54.5'
5	258°45.6	04.6	113°06.0	15.8'	01°05.9	12.1'	54.5'
6	273°45.7	S10°05.5	127°40.9	15.8'	N01°18.1	12.1'	54.5'
7	288°45.8	06.4	142°15.7	15.8'	01°30.2	12.1'	54.5'
8	303°45.9	07.3	156°50.5	15.8'	01°42.3	12.1'	54.5'
9	318°46.0	· · 08.2	171°25.3	15.8'	01°54.4	12.1'	54.5'
10	333°46.1	09.1	186°00.2	15.8'	02°06.5	12.1'	54.5'
11	348°46.2	10.0	200°35.0	15.8'	02°18.6	12.1'	54.5'
12	3°46.3	S10°10.9	215°09.9	15.8'	N02°30.7	12.1'	54.4'
13	18°46.5	11.8	229°44.7	15.8'	02°42.8	12.0'	54.4'
14	33°46.6	12.7	244°19.6	15.9'	02°54.8	12.0'	54.4'
15	48°46.7	· · 13.6	258°54.4	15.9'	03°06.8	12.0'	54.4'
16	63°46.8	14.5	273°29.3	15.8'	03°18.8	12.0'	54.4'
17	78°46.9	15.4	288°04.1	15.8'	03°30.8	12.0'	54.4'
18	93°47.0	S10°16.3	302°39.0	15.8'	N03°42.8	12.0'	54.4'
19	108°47.1	17.2	317°13.8	15.8'	03°54.7	11.9'	54.4'
20	123°47.2	18.1	331°48.6	15.8'	04°06.7	11.9'	54.3'
21	138°47.3	· · 19.0	346°23.5	15.8'	04°18.6	11.9'	54.3'
22	153°47.5	19.9	0°58.3	15.8'	04°30.5	11.9'	54.3'
23	168°47.6	20.8	15°33.2	15.8'	04°42.3	11.8'	54.3'
SD = 16.1'			SD = 14.9'				
$d = 0.9'$							
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°47.7	S10°21.7	30°08.0	15.8'	N04°54.2	11.8'	54.3'
1	198°47.8	22.6	44°42.8	15.8'	05°06.0	11.8'	54.3'
2	213°47.9	23.5	59°17.6	15.8'	05°17.8	11.8'	54.3'
3	228°48.0	· · 24.4	73°52.4	15.8'	05°29.6	11.7'	54.3'
4	243°48.1	25.3	88°27.2	15.8'	05°41.3	11.7'	54.2'
5	258°48.2	26.2	103°02.0	15.8'	05°53.0	11.7'	54.2'
6	273°48.3	S10°27.1	117°36.7	15.8'	N06°04.7	11.7'	54.2'
7	288°48.4	28.0	132°11.5	15.7'	06°16.4	11.6'	54.2'
8	303°48.5	28.8	146°46.2	15.7'	06°28.0	11.6'	54.2'
9	318°48.6	· · 29.7	161°21.0	15.7'	06°39.7	11.6'	54.2'
10	333°48.8	30.6	175°55.7	15.7'	06°51.2	11.6'	54.2'
11	348°48.9	31.5	190°30.4	15.7'	07°02.8	11.5'	54.2'
12	3°49.0	S10°32.4	205°05.1	15.7'	N07°14.3	11.5'	54.2'
13	18°49.1	33.3	219°39.8	15.7'	07°25.8	11.5'	54.2'
14	33°49.2	34.2	234°14.4	15.6'	07°37.3	11.4'	54.2'
15	48°49.3	· · 35.1	248°49.1	15.6'	07°48.7	11.4'	54.1'
16	63°49.4	36.0	263°23.7	15.6'	08°00.1	11.4'	54.1'
17	78°49.5	36.9	277°58.3	15.6'	08°11.4	11.3'	54.1'
18	93°49.6	S10°37.8	292°32.9	15.6'	N08°22.7	11.3'	54.1'
19	108°49.7	38.7	307°07.4	15.5'	08°34.0	11.2'	54.1'
20	123°49.8	39.6	321°42.0	15.5'	08°45.3	11.2'	54.1'
21	138°49.9	· · 40.4	336°16.5	15.5'	08°56.5	11.2'	54.1'
22	153°50.0	41.3	350°51.0	15.5'	09°07.6	11.1'	54.1'
23	168°50.1	42.2	5°25.5	15.5'	09°18.8	11.1'	54.1'
SD = 16.1'			SD = 14.8'				
$d = 0.9'$							
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	183°50.2	S10°43.1	19°59.9	15.4'	N09°29.9	11.1'	54.1'
1	198°50.3	44.0	34°34.4	15.4'	09°40.9	11.0'	54.1'
2	213°50.4	44.9	49°08.8	15.4'	09°51.9	11.0'	54.1'
3	228°50.5	· · 45.8	63°43.2	15.4'	10°02.9	10.9'	54.0'
4	243°50.6	46.7	78°17.5	15.3'	10°13.8	10.9'	54.0'
5	258°50.7	47.6	92°51.8	15.3'	10°24.7	10.8'	54.0'
6	273°50.8	S10°48.4	107°26.1	15.3'	N10°35.6	10.8'	54.0'
7	288°50.9	49.3	122°00.4	15.2'	10°46.4	10.8'	54.0'
8	303°51.0	50.2	136°34.7	15.2'	10°57.1	10.7'	54.0'
9	318°51.1	· · 51.1	151°08.9	15.2'	11°07.8	10.7'	54.0'
10	333°51.2	52.0	165°43.1	15.2'	11°18.5	10.6'	54.0'
11	348°51.3	52.9	180°17.2	15.1'	11°29.1	10.6'	54.0'
12	3°51.4	S10°53.8	194°51.4	15.1'	N11°39.7	10.5'	54.0'
13	18°51.5	54.6	209°25.5	15.1'	11°50.2	10.5'	54.0'
14	33°51.6	55.5	223°59.5	15.0'	12°00.6	10.4'	54.0'
15	48°51.7	· · 56.4	238°33.6	15.0'	12°11.1	10.4'	54.0'
16	63°51.8	57.3	253°07.6	15.0'	12°21.4	10.3'	54.0'
17	78°51.9	58.2	267°41.5	14.9'	12°31.7	10.3'	54.0'
18	93°52.0	S10°59.1	282°15.5	14.9'	N12°42.0	10.2'	54.0'
19	108°52.1	11°00.0	296°49.4	14.9'	12°52.2	10.2'	54.0'
20	123°52.2	00.8	311°23.2	14.8'	13°02.4	10.1'	54.0'
21	138°52.3	· · 01.7	325°57.1	14.8'	13°12.5	10.0'	53.9'
22	153°52.4	02.6	340°30.9	14.8'	13°22.5	10.0'	53.9'
23	168°52.5	03.5	355°04.6	14.7'	13°32.5	9.9'	53.9'
SD = 16.1'			SD = 14.7'				
$d = 0.9'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:18	06:37	07:50	15:37	16:51	18:09
N 70°	05:19	06:30	07:36	15:52	16:58	18:08
68°	05:20	06:25	07:24	16:04	17:03	18:08
66°	05:20	06:20	07:14	16:14	17:08	18:07
64°	05:21	06:16	07:06	16:23	17:12	18:07
62°	05:21	06:13	06:58	16:30	17:16	18:07
60°	05:21	06:09	06:52	16:36	17:19	18:07
N 58°	05:21	06:07	06:47	16:42	17:22	18:07
56°	05:21	06:04	06:42	16:46	17:25	18:08
54°	05:20	06:02	06:38	16:51	17:27	18:08
52°	05:20	05:59	06:34	16:55	17:29	18:08
50°	05:20	05:57	06:30	16:58	17:31	18:09
45°	05:18	05:53	06:22	17:06	17:36	18:10
N 40°	05:17	05:48	06:16	17:13	17:40	18:12
35°	05:15	05:45	06:10	17:19	17:44	18:14
30°	05:13	05:41	06:05	17:24	17:48	18:16
20°	05:08	05:34	05:57	17:33	17:55	18:21
N 10°	05:03	05:27	05:49	17:41	18:02	18:27
0°	04:56	05:20	05:41	17:48	18:09	18:34
S 10°	04:48	05:13	05:34	17:56	18:17	18:42
20°	04:37	05:03	05:26	18:04	18:27	18:53
30°	04:23	04:52	05:16	18:14	18:38	19:07
35°	04:14	04:45	05:11	18:19	18:45	19:16
40°	04:03	04:37	05:05	18:25	18:54	19:28
45°	03:49	04:27	04:57	18:33	19:04	19:41
S 50°	03:32	04:14	04:49	18:42	19:16	19:59
52°	03:23	04:08	04:45	18:46	19:22	20:08
54°	03:14	04:02	04:40	18:51	19:29	20:18
56°	03:03	03:55	04:35	18:56	19:37	20:29
58°	02:49	03:46	04:30	19:01	19:45	20:43
S 60°	02:34	03:37	04:23	19:08	19:55	20:59

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	15:07	14:46	14:21	03:34	05:22	07:15
N 70°	15:11	14:57	14:42	03:33	05:13	06:57
68°	15:14	15:06	14:58	03:32	05:06	06:42
66°	15:17	15:14	15:11	03:31	05:01	06:30
64°	15:20	15:20	15:22	03:31	04:56	06:21
62°	15:22	15:26	15:31	03:30	04:51	06:12
60°	15:24	15:31	15:39	03:30	04:48	06:05
N 58°	15:25	15:35	15:46	03:29	04:44	05:59
56°	15:27	15:39	15:52	03:29	04:41	05:54
54°	15:28	15:42	15:58	03:29	04:39	05:49
52°	15:29	15:45	16:03	03:28	04:37	05:44
50°	15:30	15:48	16:08	03:28	04:34	05:40
45°	15:33	15:55	16:18	03:28	04:30	05:32
N 40°	15:35	16:00	16:26	03:27	04:26	05:25
35°	15:37	16:04	16:33	03:27	04:23	05:18
30°	15:38	16:08	16:40	03:26	04:20	05:13
20°	15:41	16:15	16:51	03:26	04:15	05:04
N 10°	15:44	16:22	17:01	03:25	04:10	04:55
0°	15:46	16:28	17:10	03:25	04:06	04:48
S 10°	15:48	16:33	17:19	03:24	04:02	04:40
20°	15:51	16:40	17:29	03:23	03:57	04:32
30°	15:54	16:47	17:41	03:23	03:52	04:23
35°	15:56	16:51	17:47	03:22	03:50	04:18
40°	15:58	16:56	17:55	03:22	03:46	04:12
45°	16:00	17:02	18:04	03:21	03:43	04:05
S 50°	16:03	17:09	18:14	03:21	03:38	03:57
52°	16:04	17:12	18:20	03:20	03:36	03:53
54°	16:05	17:15	18:25	03:20	03:34	03:49
56°	16:07	17:19	18:31	03:20	03:31	03:44
58°	16:08	17:23	18:38	03:19	03:29	03:39
S 60°	16:10	17:28	18:46	03:19	03:26	03:33

Day	Sun			Moon		Age 12-14 90-98%
	Eqn. of Time		Mer.	Mer.Pass.		
	00 ^h mm:ss	12 ^h mm:ss	Pass hh:mm	Upper hh:mm	Lower hh:mm	
19	15:00	15:05	11:45	21:56	09:35	
20	15:11	15:16	11:45	22:38	10:17	
21	15:21	15:26	11:45	23:20	10:59	

October 22, 23, 24 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	30°42.9	136°39.4	S26°07.5	131°17.6	S24°23.4	176°48.4	S12°32.9	339°10.9	N16°12.0	Alpheratz	357°30.4	29°15.6
1	45°45.3	151°39.0	07.8	146°18.1	23.5	191°50.3	33.1	354°13.6	12.0	Ankaa	353°03.1	-42°08.6
2	60°47.8	166°38.7	08.2	161°18.5	23.6	206°52.3	33.2	9°16.2	11.9	Schedar	349°25.9	56°42.3
3	75°50.3	181°38.3	· · 08.6	176°19.0	· · 23.8	221°54.2	· · 33.4	24°18.8	· · 11.9	Diphda	348°43.2	-17°49.2
4	90°52.7	196°37.9	09.0	191°19.5	23.9	236°56.2	33.6	39°21.5	11.8	Achernar	335°16.9	-57°05.1
5	105°55.2	211°37.6	09.4	206°19.9	24.1	251°58.1	33.8	54°24.1	11.8	Hamal	327°46.5	23°36.3
6	120°57.7	226°37.2	S26°09.8	221°20.4	S24°24.2	267°00.0	S12°34.0	69°26.8	N16°11.7	Polaris	311°25.0	89°23.1
7	136°00.1	241°36.9	10.1	236°20.9	24.3	282°02.0	34.1	84°29.4	11.7	Acamar	315°08.5	-40°11.0
8	151°02.6	256°36.5	10.5	251°21.3	24.5	297°03.9	34.3	99°32.0	11.6	Menkar	314°01.8	4°12.5
9	166°05.1	271°36.1	· · 10.9	266°21.8	· · 24.6	312°05.8	· · 34.5	114°34.7	· · 11.6	Mirfak	308°22.2	49°57.9
10	181°07.5	286°35.8	11.3	281°22.2	24.7	327°07.8	34.7	129°37.3	11.5	Aldebaran	290°34.9	16°34.1
11	196°10.0	301°35.4	11.7	296°22.7	24.9	342°09.7	34.9	144°40.0	11.5	Rigel	281°00.0	-8°10.0
12	211°12.5	316°35.1	S26°12.0	311°23.2	S24°25.0	357°11.7	S12°35.1	159°42.6	N16°11.4	Capella	280°15.9	46°01.4
13	226°14.9	331°34.7	12.4	326°23.6	25.1	12°13.6	35.2	174°45.2	11.4	Bellatrix	278°18.6	6°22.6
14	241°17.4	346°34.4	12.8	341°24.1	25.3	27°15.5	35.4	189°47.9	11.4	Elnath	277°56.8	28°37.8
15	256°19.8	1°34.0	· · 13.2	356°24.5	· · 25.4	42°17.5	· · 35.6	204°50.5	· · 11.3	Alnilam	275°33.6	-1°11.0
16	271°22.3	16°33.6	13.5	11°25.0	25.5	57°19.4	35.8	219°53.2	11.3	Betelgeuse	270°47.8	7°24.7
17	286°24.8	31°33.3	13.9	26°25.5	25.7	72°21.3	36.0	234°55.8	11.2	Canopus	263°50.4	-52°42.5
18	301°27.2	46°32.9	S26°14.3	41°25.9	S24°25.8	87°23.3	S12°36.1	249°58.5	N16°11.2	Sirius	258°22.7	-16°45.4
19	316°29.7	61°32.6	14.6	56°26.4	25.9	102°25.2	36.3	265°01.1	11.1	Adhara	255°02.7	-29°00.6
20	331°32.2	76°32.2	15.0	71°26.8	26.1	117°27.2	36.5	280°03.7	11.1	Procyon	244°46.8	5°08.8
21	346°34.6	91°31.9	· · 15.4	86°27.3	· · 26.2	132°29.1	· · 36.7	295°06.4	· · 11.0	Pollux	243°12.6	27°57.0
22	1°37.1	106°31.5	15.7	101°27.8	26.3	147°31.0	36.9	310°09.0	11.0	Avior	234°13.1	-59°36.1
23	16°39.6	121°31.2	16.1	116°28.2	26.5	162°33.0	37.1	325°11.7	10.9	Suhail	222°43.5	-43°33.0
Mer.pass. 21:54		ν -0.4' d 0.4' m -4.44		ν 0.5' d 0.1' m 1.04		ν 1.9' d 0.2' m -1.68		ν 2.6' d -0.0' m -0.16		Miaplacidus 221°37.5 -69°50.2		
Alphard 217°44.1 -8°47.3												
Regulus 207°30.6 11°49.2												
Dubhe 193°37.1 61°35.1												
Denebola 182°21.4 14°24.3												
Gienah 175°39.9 -17°42.4												
Acrux 172°56.2 -63°15.8												
Gacrux 171°47.7 -57°16.7												
Alioth 166°10.4 55°47.8												
Spica 158°18.6 -11°18.9												
Alkaid 152°49.6 49°09.9												
Hadar 148°31.0 -60°30.9												
Menkent 147°53.4 -36°30.9												
Arcturus 145°44.8 19°01.7												
Rigel Kent. 139°35.5 -60°57.5												
Kochab 137°21.4 74°02.0												
Zuben'ubi 136°52.1 -16°09.8												
Alphecca 126°00.9 26°37.1												
Antares 112°11.4 -26°29.7												
Atria 107°02.5 -69°04.8												
Sabik 101°58.6 -15°45.5												
Shaula 96°05.4 -37°07.4												
Rasalhague 95°55.2 12°32.6												
Eltanin 90°40.7 51°29.5												
Kaus Aust. 83°27.6 -34°22.1												
Vega 80°30.8 38°49.1												
Nunki 75°43.2 -26°15.4												
Altair 61°56.2 8°57.2												
Peacock 52°59.8 -56°38.3												
Deneb 49°23.0 45°23.7												
Enif 33°34.9 10°01.0												
Al Na'ir 27°28.0 -46°49.0												
Fomalhaut 15°10.1 -29°27.7												
Scheat 13°41.2 28°15.0												
Markab 13°25.8 15°22.2												
Oct 22 Mon SHA Mer.pass												
Venus 105°56.5 14:54												
Mars 100°34.7 15:14												
Jupiter 146°05.5 12:11												
Saturn 308°28.0 01:23												
Oct 23 Tue SHA Mer.pass												
Venus 104°48.8 14:54												
Mars 99°46.7 15:14												
Jupiter 145°52.9 12:08												
Saturn 308°32.3 01:19												
Oct 24 Wed SHA Mer.pass												
Venus 103°41.3 14:55												
Mars 98°58.4 15:13												
Jupiter 145°40.2 12:05												
Saturn 308°36.6 01:15												
Horizontal parallax												
Venus: 0.2												
Mars: 0.1												
Mer.pass. 21:46 ν -0.3' d 0.3' m -4.47 ν 0.5' d 0.1' m 1.04 ν 1.9' d 0.2' m -1.68 ν 2.6' d -0.0' m -0.17												

DUT1 = UT1-UTC = +0.2132 sec ΔT = TT-UT1 = +68.9708 sec

2029 October 22 to Oct. 24 UT

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	183°52.6	S11°04.4	9°38.3	14.7'	N13°42.4	9.9' 53.9'
1	198°52.7	05.3	24°12.0	14.6'	13°52.3	9.8' 53.9'
2	213°52.8	06.1	38°45.7	14.6'	14°02.1	9.8' 53.9'
3	228°52.9	· · 07.0	53°19.3	14.6'	14°11.9	9.7' 53.9'
4	243°53.0	07.9	67°52.9	14.5'	14°21.6	9.6' 53.9'
5	258°53.1	08.8	82°26.4	14.5'	14°31.3	9.6' 53.9'
6	273°53.2	S11°09.7	96°59.9	14.5'	N14°40.8	9.5' 53.9'
7	288°53.2	10.5	111°33.3	14.4'	14°50.4	9.5' 53.9'
8	303°53.3	11.4	126°06.8	14.4'	14°59.8	9.4' 53.9'
9	318°53.4	· · 12.3	140°40.1	14.3'	15°09.2	9.3' 53.9'
10	333°53.5	13.2	155°13.5	14.3'	15°18.6	9.3' 53.9'
11	348°53.6	14.1	169°46.8	14.3'	15°27.8	9.2' 53.9'
12	3°53.7	S11°14.9	184°20.0	14.2'	N15°37.0	9.1' 53.9'
13	18°53.8	15.8	198°53.2	14.2'	15°46.2	9.1' 53.9'
14	33°53.9	16.7	213°26.4	14.1'	15°55.2	9.0' 53.9'
15	48°54.0	· · 17.6	227°59.5	14.1'	16°04.2	8.9' 53.9'
16	63°54.1	18.4	242°26.4	14.0'	16°13.2	8.9' 53.9'
17	78°54.2	19.3	257°05.7	14.0'	16°22.1	8.8' 53.9'
18	93°54.3	S11°20.2	271°38.7	14.0'	N16°30.9	8.7' 53.9'
19	108°54.3	21.1	286°11.6	13.9'	16°39.6	8.7' 53.9'
20	123°54.4	22.0	300°44.6	13.9'	16°48.2	8.6' 53.9'
21	138°54.5	· · 22.8	315°17.4	13.8'	16°56.8	8.5' 53.9'
22	153°54.6	23.7	329°50.3	13.8'	17°05.4	8.4' 53.9'
23	168°54.7	24.6	344°23.0	13.7'	17°13.8	8.4' 53.9'
SD = 16.1' d = 0.9' SD = 14.7'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	183°54.8	S11°25.5	358°55.8	13.7'	N17°22.2	8.3' 53.9'
1	198°54.9	26.3	13°28.5	13.7'	17°30.5	8.2' 53.9'
2	213°55.0	27.2	28°01.1	13.6'	17°38.7	8.1' 53.9'
3	228°55.1	· · 28.1	42°33.7	13.6'	17°46.8	8.1' 53.9'
4	243°55.1	29.0	57°06.3	13.5'	17°54.9	8.0' 53.9'
5	258°55.2	29.8	71°38.8	13.5'	18°02.9	7.9' 53.9'
6	273°55.3	S11°30.7	86°11.3	13.4'	N18°10.8	7.8' 53.9'
7	288°55.4	31.6	100°43.7	13.4'	18°18.7	7.8' 53.9'
8	303°55.5	32.5	115°16.1	13.3'	18°26.4	7.7' 53.9'
9	318°55.6	· · 33.3	129°48.4	13.3'	18°34.1	7.6' 53.9'
10	333°55.7	34.2	144°20.7	13.2'	18°41.7	7.5' 53.9'
11	348°55.7	35.1	158°53.0	13.2'	18°49.2	7.4' 53.9'
12	3°55.8	S11°35.9	173°25.2	13.2'	N18°56.7	7.4' 53.9'
13	18°55.9	36.8	187°57.3	13.1'	19°04.0	7.3' 53.9'
14	33°56.0	37.7	202°29.4	13.1'	19°11.3	7.2' 53.9'
15	48°56.1	· · 38.6	217°01.5	13.0'	19°18.5	7.1' 53.9'
16	63°56.2	39.4	231°33.5	13.0'	19°25.6	7.0' 53.9'
17	78°56.3	40.3	246°05.5	12.9'	19°32.6	6.9' 53.9'
18	93°56.3	S11°41.2	260°37.4	12.9'	N19°39.5	6.8' 53.9'
19	108°56.4	42.0	275°09.3	12.8'	19°46.4	6.8' 53.9'
20	123°56.5	42.9	289°41.1	12.8'	19°53.2	6.7' 53.9'
21	138°56.6	· · 43.8	304°12.9	12.7'	19°59.8	6.6' 53.9'
22	153°56.7	44.6	318°44.7	12.7'	20°06.4	6.5' 53.9'
23	168°56.7	45.5	333°16.4	12.7'	20°12.9	6.4' 53.9'
SD = 16.1' d = 0.9' SD = 14.7'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	183°56.8	S11°46.4	347°48.0	12.6'	N20°19.3	6.3' 53.9'
1	198°56.9	47.3	2°19.7	12.6'	20°25.6	6.2' 53.9'
2	213°57.0	48.1	16°51.2	12.5'	20°31.9	6.1' 53.9'
3	228°57.1	· · 49.0	31°22.8	12.5'	20°38.0	6.0' 53.9'
4	243°57.1	49.9	45°54.2	12.4'	20°44.1	6.0' 53.9'
5	258°57.2	50.7	60°25.7	12.4'	20°50.0	5.9' 53.9'
6	273°57.3	S11°51.6	74°57.1	12.3'	N20°55.9	5.8' 53.9'
7	288°57.4	52.5	89°28.4	12.3'	21°01.6	5.7' 54.0'
8	303°57.5	53.3	103°59.7	12.3'	21°07.3	5.6' 54.0'
9	318°57.5	· · 54.2	118°31.0	12.2'	21°12.9	5.5' 54.0'
10	333°57.6	55.0	133°02.2	12.2'	21°18.4	5.4' 54.0'
11	348°57.7	55.9	147°33.4	12.1'	21°23.8	5.3' 54.0'
12	3°57.8	S11°56.8	162°04.5	12.1'	N21°29.1	5.2' 54.0'
13	18°57.9	57.6	176°35.6	12.0'	21°34.2	5.1' 54.0'
14	33°57.9	58.5	191°06.6	12.0'	21°39.3	5.0' 54.0'
15	48°58.0	11°59.4	205°37.7	12.0'	21°44.3	4.9' 54.0'
16	63°58.1	12°00.2	220°08.6	11.9'	21°49.2	4.8' 54.0'
17	78°58.2	01.1	234°39.5	11.9'	21°54.0	4.7' 54.0'
18	93°58.2	S12°02.0	249°10.4	11.8'	N21°58.7	4.6' 54.0'
19	108°58.3	02.8	263°41.3	11.8'	22°03.3	4.5' 54.0'
20	123°58.4	03.7	278°12.1	11.8'	22°07.8	4.4' 54.0'
21	138°58.5	· · 04.5	292°42.9	11.7'	22°12.2	4.3' 54.0'
22	153°58.5	05.4	307°13.6	11.7'	22°16.5	4.2' 54.0'
23	168°58.6	06.3	321°44.3	11.7'	22°20.7	4.1' 54.0'
SD = 16.1' d = 0.9' SD = 14.7'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:31	06:50	08:06	15:21	16:37	17:56
N 70°	05:30	06:42	07:49	15:38	16:45	17:56
68°	05:30	06:35	07:35	15:52	16:52	17:57
66°	05:30	06:29	07:24	16:03	16:58	17:57
64°	05:29	06:25	07:15	16:13	17:03	17:58
62°	05:28	06:20	07:07	16:21	17:07	17:59
60°	05:28	06:16	07:00	16:28	17:11	17:59
N 58°	05:27	06:13	06:54	16:34	17:15	18:00
56°	05:27	06:10	06:48	16:39	17:18	18:01
54°	05:26	06:07	06:43	16:44	17:21	18:02
52°	05:25	06:04	06:39	16:49	17:23	18:03
50°	05:24	06:02	06:35	16:53	17:26	18:03
45°	05:22	05:56	06:26	17:01	17:31	18:06
N 40°	05:20	05:52	06:19	17:09	17:36	18:08
35°	05:18	05:47	06:13	17:15	17:41	18:10
30°	05:15	05:43	06:07	17:21	17:45	18:13
20°	05:09	05:35	05:58	17:31	17:53	18:19
N 10°	05:03	05:28	05:49	17:39	18:01	18:25
0°	04:55	05:20	05:41	17:48	18:09	18:33
S 10°	04:46	05:11	05:33	17:56	18:18	18:43
20°	04:35	05:01	05:24	18:05	18:28	18:54
30°	04:19	04:49	05:13	18:16	18:40	19:10
35°	04:10	04:41	05:07	18:22	18:48	19:20
40°	03:58	04:32	05:01	18:29	18:57	19:31
45°	03:44	04:21	04:52	18:37	19:08	19:46
S 50°	03:25	04:08	04:43	18:47	19:22	20:05
52°	03:16	04:02	04:38	18:51	19:28	20:15
54°	03:05	03:55	04:33	18:57	19:36	20:25
56°	02:53	03:47	04:28	19:02	19:44	20:38
58°	02:39	03:37	04:22	19:08	19:53	20:53
S 60°	02:21	03:27	04:15	19:16	20:04	21:11

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	13:44	□	□	09:23	□	□
N 70°	14:21	13:43	□	08:47	10:59	□
68°	14:48	14:34	14:03	08:22	10:09	12:17
66°	15:08	15:06	15:05	08:03	09:38	11:15
64°	15:24	15:30	15:40	07:47	09:15	10:41
62°	15:38	15:48	16:05	07:34	08:56	10:16
60°	15:50	16:04	16:25	07:23	08:41	09:57
N 58°	16:00	16:17	16:42	07:14	08:29	09:41
56°	16:08	16:29	16:56	07:06	08:18	09:27
54°	16:16	16:39	17:08	06:59	08:08	09:15
52°	16:23	16:48	17:19	06:52	07:59	09:05
50°	16:30	16:56	17:28	06:46	07:52	08:55
45°	16:44	17:13	17:48	06:34	07:35	08:36
N 40°	16:55	17:27	18:05	06:23	07:22	08:20
35°	17:05	17:40	18:19	06:14	07:11	08:07
30°	17:13	17:50	18:31	06:07	07:01	07:55
20°	17:28	18:08	18:51	05:53	06:44	07:35
N 10°	17:41	18:24	19:10	05:42	06:29	07:18
0°	17:54	18:39	19:26	05:31	06:15	07:02
S 10°	18:06	18:54	19:43	05:20	06:02	06:46
20°	18:19	19:10	20:02	05:08	05:47	06:29
30°	18:35	19:29	20:23	04:55	05:30	06:09
35°	18:43	19:40	20:35	04:48	05:21	05:58
40°	18:54	19:52	20:49	04:39	05:10	05:44
45°	19:06	20:07	21:06	04:29	04:57	05:29
S 50°	19:20	20:25	21:27	04:17	04:41	05:10
52°	19:27	20:34	21:37	04:12	04:34	05:01
54°	19:35	20:43	21:49	04:05	04:26	04:51
56°	19:43	20:54	22:02	03:59	04:17	04:40
58°	19:53	21:07	22:17	03:51	04:06	04:27
S 60°	20:04	21:21	22:35	03:42	03:54	04:12

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
22	15:30	15:35	11:44	—:—	11:42	
23	15:39	15:43	11:44	00:04	12:27	
24	15:47	15:51	11:44	00:50	13:14	

October 25, 26, 27 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	33°40.3	136°14.3	S26°32.6	131°50.4	S24°32.4	179°07.9	S12°45.9	342°21.2	N16°08.6	Alpheratz	357°30.4	29°15.6											
1	48°42.8	151°13.9	32.9	146°50.9	32.5	194°09.8	46.1	357°23.9	08.6	Ankaa	353°03.1	-42°08.6											
2	63°45.2	166°13.6	33.3	161°51.3	32.6	209°11.8	46.3	12°26.5	08.5	Schedar	349°25.9	56°42.3											
3	78°47.7	181°13.3	•• 33.6	176°51.8	•• 32.7	224°13.7	•• 46.5	27°29.2	•• 08.5	Diphda	348°43.2	-17°49.2											
4	93°50.2	196°12.9	33.9	191°52.2	32.8	239°15.6	46.7	42°31.8	08.4	Achernar	335°16.9	-57°05.1											
5	108°52.6	211°12.6	34.2	206°52.7	32.9	254°17.6	46.8	57°34.5	08.4	Hamal	327°46.5	23°36.3											
6	123°55.1	226°12.3	S26°34.5	221°53.1	S24°33.0	269°19.5	S12°47.0	72°37.1	N16°08.3	Polaris	311°24.1	89°23.2											
7	138°57.5	241°11.9	34.8	236°53.5	33.1	284°21.4	47.2	87°39.8	08.3	Acamar	315°08.5	-40°11.0											
8	154°00.0	256°11.6	35.1	251°54.0	33.2	299°23.4	47.4	102°42.4	08.2	Menkar	314°01.8	4°12.5											
9	169°02.5	271°11.3	•• 35.4	266°54.4	•• 33.3	314°25.3	•• 47.6	117°45.1	•• 08.2	Mirfak	308°22.2	49°57.9											
10	184°04.9	286°10.9	35.7	281°54.9	33.4	329°27.2	47.7	132°47.7	08.2	Aldebaran	290°34.9	16°34.1											
11	199°07.4	301°10.6	36.0	296°55.3	33.5	344°29.2	47.9	147°50.4	08.1	Rigel	280°59.9	-8°10.0											
12	214°09.9	316°10.3	S26°36.3	311°55.8	S24°33.6	359°31.1	S12°48.1	162°53.0	N16°08.1	Capella	280°15.9	46°01.4											
13	229°12.3	331°10.0	36.6	326°56.2	33.7	14°33.0	48.3	177°55.6	08.0	Bellatrix	278°18.5	6°22.6											
14	244°14.8	346°09.6	36.9	341°56.7	33.9	29°35.0	48.5	192°58.3	08.0	Elnath	277°56.8	28°37.8											
15	259°17.3	1°09.3	•• 37.2	356°57.1	•• 34.0	44°36.9	•• 48.6	208°00.9	•• 07.9	Alnilam	275°33.6	-1°11.0											
16	274°19.7	16°09.0	37.5	11°57.6	34.1	59°38.9	48.8	223°03.6	07.9	Betelgeuse	270°47.7	7°24.7											
17	289°22.2	31°08.6	37.8	26°58.0	34.2	74°40.8	49.0	238°06.2	07.8	Canopus	263°50.4	-52°42.5											
18	304°24.7	46°08.3	S26°38.1	41°58.4	S24°34.3	89°42.7	S12°49.2	253°08.9	N16°07.8	Sirius	258°22.7	-16°45.4											
19	319°27.1	61°08.0	38.4	56°58.9	34.4	104°44.7	49.4	268°11.5	07.7	Adhara	255°02.7	-29°00.6											
20	334°29.6	76°07.7	38.7	71°59.3	34.5	119°46.6	49.5	283°14.2	07.7	Procyon	244°46.8	5°08.8											
21	349°32.0	91°07.3	•• 39.0	86°59.8	•• 34.6	134°48.5	•• 49.7	298°16.8	•• 07.6	Pollux	243°12.6	27°57.0											
22	4°34.5	106°07.0	39.2	102°00.2	34.7	149°50.5	49.9	313°19.5	07.6	Avior	234°13.1	-59°36.1											
23	19°37.0	121°06.7	39.5	117°00.7	34.8	164°52.4	50.1	328°22.1	07.5	Suhail	222°43.5	-43°33.0											
Mer.pass. 21:42				ν -0.3' d 0.3' m -4.48				ν 0.4' d 0.1' m 1.05				ν 1.9' d 0.2' m -1.68				ν 2.6' d -0.0' m -0.17							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	34°39.4	136°06.4	S26°39.8	132°01.1	S24°34.9	179°54.3	S12°50.3	343°24.8	N16°07.5	Alpheratz	357°30.4	29°15.6							
1	49°41.9	151°06.0	40.1	147°01.6	35.0	194°56.3	50.4	358°27.4	07.4	Ankaa	353°03.1	-42°08.6							
2	64°44.4	166°05.7	40.4	162°02.0	35.1	209°58.2	50.6	13°30.1	07.4	Schedar	349°25.9	56°42.3							
3	79°46.8	181°05.4	•• 40.7	177°02.4	•• 35.2	225°00.1	•• 50.8	28°32.7	•• 07.3	Diphda	348°43.2	-17°49.2							
4	94°49.3	196°05.1	41.0	192°02.9	35.3	240°02.1	51.0	43°35.4	07.3	Achernar	335°16.9	-57°05.1							
5	109°51.8	211°04.7	41.2	207°03.3	35.4	255°04.0	51.2	58°38.0	07.2	Hamal	327°46.5	23°36.3							
6	124°54.2	226°04.4	S26°41.5	222°03.8	S24°35.5	270°05.9	S12°51.3	73°40.7	N16°07.2	Polaris	311°24.1	89°23.2							
7	139°56.7	241°04.1	41.8	237°04.2	35.5	285°07.9	51.5	88°43.3	07.1	Acamar	315°08.5	-40°11.0							
8	154°59.2	256°03.8	42.1	252°04.7	35.6	300°09.8	51.7	103°46.0	07.1	Menkar	314°01.8	4°12.5							
9	170°01.6	271°03.5	•• 42.4	267°05.1	•• 35.7	315°11.8	•• 51.9	118°48.6	•• 07.0	Mirfak	308°22.2	49°57.9							
10	185°04.1	286°03.1	42.6	282°05.5	35.8	330°13.7	52.1	133°51.3	07.0	Aldebaran	290°34.9	16°34.1							
11	200°06.5	301°02.8	42.9	297°06.0	35.9	345°15.6	52.2	148°53.9	07.0	Rigel	280°59.9	-8°10.0							
12	215°09.0	316°02.5	S26°43.2	312°06.4	S24°36.0	0°17.6	S12°52.4	163°56.6	N16°06.9	Capella	280°15.9	46°01.4							
13	230°11.5	331°02.2	43.5	327°06.9	36.1	15°19.5	52.6	178°59.2	06.9	Bellatrix	278°18.5	6°22.6							
14	245°13.9	346°01.9	43.7	342°07.3	36.2	30°21.4	52.8	194°01.9	06.8	Elnath	277°56.8	28°37.8							
15	260°16.4	1°01.5	•• 44.0	357°07.7	•• 36.3	45°23.4	•• 53.0	209°04.5	•• 06.8	Alnilam	275°33.6	-1°11.0							
16	275°18.9	16°01.2	44.3	12°08.2	36.4	60°25.3	53.1	224°07.2	06.7	Betelgeuse	270°47.7	7°24.7							
17	290°21.3	31°00.9	44.5	27°08.6	36.5	75°27.2	53.3	239°09.8	06.7	Canopus	263°50.4	-52°42.5							
18	305°23.8	46°00.6	S26°44.8	42°09.1	S24°36.6	90°29.2	S12°53.5	254°12.5	N16°06.6	Sirius	258°22.7	-16°45.4							
19	320°26.3	61°00.3	45.1	57°09.5	36.7	105°31.1	53.7	269°15.1	06.6	Adhara	255°02.7	-29°00.6							
20	335°28.7	76°00.0	45.3	72°09.9	36.8	120°33.0	53.9	284°17.8	06.5	Procyon	244°46.8	5°08.8							
21	350°31.2	90°59.7	•• 45.6	87°10.4	•• 36.9	135°35.0	•• 54.0	299°20.4	•• 06.5	Pollux	243°12.6	27°57.0							
22	5°33.6	105°59.3	45.9	102°10.8	36.9	150°36.9	54.2	314°23.1	06.4	Avior	234°13.1	-56°36.1							
23	20°36.1	120°59.0	46.1	117°11.3	37.0	165°38.8	54.4	329°25.7	06.4	Suhail	222°43.5	-43°33.0							
Mer.pass. 21:38				ν -0.3' d 0.3' m -4.49				ν 0.4' d 0.1' m 1.05				ν 1.9' d 0.2' m -1.68				ν 2.6' d -0.0' m -0.18			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	35°38.6	135°58.7	S26°46.4	132°11.7	S24°37.1	180°40.8	S12°54.6	344°28.4	N16°06.3	Alpheratz	357°30.4	29°15.6	
1	50°41.0	150°58.4	46.7	147°12.1	37.2	195°42.7	54.8	359°31.0	06.3	Ankaa	353°03.1	-42°08.6	
2	65°43.5	165°58.1	46.9	162°12.6	37.3	210°44.6	54.9	14°33.7	06.2	Schedar	349°25.9	56°42.3	
3	80°46.0	180°57.8	•• 47.2	177°13.0	•• 37.4	225°46.6	•• 55.1	29°36.3	•• 06.2	Diphda	348°43.2	-17°49.2	
4	95°48.4	195°57.5	47.4	192°13.5	37.5	240°48.5	55.3	44°39.0	06.1	Achernar	335°16.9	-57°05.1	
5	110°50.9	210°57.2	47.7	207°13.9	37.6	255°50.4	55.5	59°41.6	06.1	Hamal	327°46.5	23°36.3	
6	125°53.4	225°56.8	S26°47.9	222°14.3	S24°37.6	270°52.4	S12°55.7	74°44.3	N16°06.0	Polaris	311°24.1	89°23.2	
7	140°55.8	240°56.5	48.2	237°14.8	37.7	285°54.3	55.8	89°46.9	06.0	Acamar	315°08.5	-40°11.0	
8	155°58.3	255°56.2	48.5	252°15.2	37.8	300°56.3	56.0	104°49.6	05.9	Menkar	314°01.8	4°12.5	
9	171°00.8	270°55.9	•• 48.7	267°15.6	•• 37.9	315°58.2	•• 56.2	119°52.2	•• 05.9	Mirfak	308°22.2	49°57.9	
10	186°03.2	285°55.6	49.0	282°16.1	38.0	331°00.1	56.4	134°54.9	05.8	Aldebaran	290°34.9	16°34.1	
11	201°05.7	300°55.3	49.2	297°16.5	38.1	346°02.1	56.6	149°57.5	05.8	Rigel	280°59.9	-8°10.0	
12	216°08.1	315°55.0	S26°49.5	312°17.0	S24°38.2	1°04.0	S12°56.7	165°00.2	N16°05.7	Capella	280°15.9	46°01.4	
13	231°10.6	330°54.7	49.7	327°17.4	38.2	16°05.9	56.9	180°02.8	05.7	Bellatrix	278°18.5	6°22.6	
14	246°13.1	345°54.4	50.0	342°17.8	38.3	31°07.9	57.1	195°05.5	05.6	Elnath	277°56.8	28°37.8	
15	261°15.5	0°54.1	•• 50.2	357°18.3	•• 38.4	46°09.8	•• 57.3	210°08.1	•• 05.6	Alnilam	275°33.1	-1°11.0	
16	276°18.0	15°53.8	50.4	12°18.7	38.5	61°11.7	57.5	225°10.8	05.6	Betelgeuse	270°47.7	7°24.7	
17	291°20.5	30°53.5	50.7	27°19.1	38.6	76°13.7	57.6	240°13.5	05.5	Canopus	263°50.4	-52°42.5	
18	306°22.9	45°53.2	S26°50.9	42°19.6	S24°38.7	91°15.6	S12°57.8	255°16.1	N16°05.5	Sirius	258°22.7	-16°45.4	
19	321°25.4	60°52.9	51.2	57°20.0	38.								

DUT1 = UT1-UTC = +0.2133 sec ΔT = TT-UT1 = +68.9707 sec

2029 October 25 to Oct. 27 UT

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	183°58.7	S12°07.1	336°14.9	11.6'	N22°24.8	4.0'	54.1'
1	198°58.8	08.0	350°45.5	11.6'	22°28.8	3.9'	54.1'
2	213°58.8	08.8	5°16.1	11.5'	22°32.7	3.8'	54.1'
3	228°58.9	· · 09.7	19°46.7	11.5'	22°36.5	3.7'	54.1'
4	243°59.0	10.6	34°17.2	11.5'	22°40.1	3.6'	54.1'
5	258°59.0	11.4	48°47.6	11.4'	22°43.7	3.5'	54.1'
6	273°59.1	S12°12.3	63°18.1	11.4'	N22°47.2	3.4'	54.1'
7	288°59.2	13.1	77°48.5	11.4'	22°50.5	3.2'	54.1'
8	303°59.3	14.0	92°18.8	11.3'	22°53.8	3.1'	54.1'
9	318°59.3	· · 14.9	106°49.2	11.3'	22°56.9	3.0'	54.1'
10	333°59.4	15.7	121°19.5	11.3'	22°59.9	2.9'	54.1'
11	348°59.5	16.6	135°49.7	11.2'	23°02.9	2.8'	54.1'
12	3°59.5	S12°17.4	150°20.0	11.2'	N23°05.7	2.7'	54.2'
13	18°59.6	18.3	164°50.2	11.2'	23°08.4	2.6'	54.2'
14	33°59.7	19.1	179°20.4	11.1'	23°11.0	2.5'	54.2'
15	48°59.8	· · 20.0	193°50.5	11.1'	23°13.5	2.4'	54.2'
16	63°59.8	20.9	208°20.6	11.1'	23°15.9	2.3'	54.2'
17	78°59.9	21.7	222°50.7	11.1'	23°18.1	2.2'	54.2'
18	94°00.0	S12°22.6	237°20.8	11.0'	N23°20.3	2.1'	54.2'
19	109°00.0	23.4	251°50.8	11.0'	23°22.4	1.9'	54.2'
20	124°00.1	24.3	266°20.8	11.0'	23°24.3	1.8'	54.2'
21	139°00.2	· · 25.1	280°50.8	11.0'	23°26.1	1.7'	54.3'
22	154°00.2	26.0	295°20.7	10.9'	23°27.8	1.6'	54.3'
23	169°00.3	26.8	309°50.7	10.9'	23°29.4	1.5'	54.3'
SD = 16.1'			SD = 14.7'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	184°00.4	S12°27.7	324°20.6	10.9'	N23°30.9	1.4'	54.3'
1	199°00.4	28.5	338°50.5	10.9'	23°32.3	1.3'	54.3'
2	214°00.5	29.4	353°20.3	10.8'	23°33.6	1.2'	54.3'
3	229°00.6	· · 30.2	7°50.2	10.8'	23°34.7	1.0'	54.3'
4	244°00.6	31.1	22°20.0	10.8'	23°35.8	0.9'	54.3'
5	259°00.7	31.9	36°49.8	10.8'	23°36.7	0.8'	54.4'
6	274°00.7	S12°32.8	51°19.5	10.8'	N23°37.5	0.7'	54.4'
7	289°00.8	33.6	65°49.3	10.7'	23°38.2	0.6'	54.4'
8	304°00.9	34.5	80°19.0	10.7'	23°38.8	0.5'	54.4'
9	319°00.9	· · 35.3	94°48.7	10.7'	23°39.3	0.4'	54.4'
10	334°01.0	36.2	109°18.4	10.7'	23°39.6	0.2'	54.4'
11	349°01.1	37.0	123°48.1	10.7'	23°39.9	0.1'	54.4'
12	4°01.1	S12°37.9	138°17.8	10.7'	N23°40.0	0.0'	54.5'
13	19°01.2	38.7	152°47.4	10.6'	23°40.0	-0.1'	54.5'
14	34°01.3	39.6	167°17.1	10.6'	23°39.9	-0.2'	54.5'
15	49°01.3	· · 40.4	181°46.7	10.6'	23°39.7	-0.3'	54.5'
16	64°01.4	41.3	196°16.3	10.6'	23°39.3	-0.5'	54.5'
17	79°01.4	42.1	210°45.9	10.6'	23°38.9	-0.6'	54.5'
18	94°01.5	S12°43.0	225°15.5	10.6'	N23°38.3	-0.7'	54.5'
19	109°01.6	43.8	239°45.1	10.6'	23°37.6	-0.8'	54.6'
20	124°01.6	44.7	254°14.6	10.6'	23°36.8	-0.9'	54.6'
21	139°01.7	· · 45.5	268°44.2	10.5'	23°35.9	-1.0'	54.6'
22	154°01.7	46.4	283°13.8	10.5'	23°34.8	-1.2'	54.6'
23	169°01.8	47.2	297°43.3	10.5'	23°33.7	-1.3'	54.6'
SD = 16.1'			SD = 14.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	184°01.8	S12°48.1	312°12.8	10.5'	N23°32.4	-1.4'	54.6'
1	199°01.9	48.9	326°42.4	10.5'	23°31.0	-1.5'	54.7'
2	214°02.0	49.7	341°11.9	10.5'	23°29.5	-1.6'	54.7'
3	229°02.0	· · 50.6	355°41.4	10.5'	23°27.9	-1.7'	54.7'
4	244°02.1	51.4	10°10.9	10.5'	23°26.2	-1.9'	54.7'
5	259°02.1	52.3	24°40.4	10.5'	23°24.3	-2.0'	54.7'
6	274°02.2	S12°53.1	39°09.9	10.5'	N23°22.4	-2.1'	54.7'
7	289°02.2	54.0	53°39.4	10.5'	23°20.3	-2.2'	54.8'
8	304°02.3	54.8	68°08.9	10.5'	23°18.1	-2.3'	54.8'
9	319°02.4	· · 55.6	82°38.4	10.5'	23°15.8	-2.4'	54.8'
10	334°02.4	56.5	97°07.9	10.5'	23°13.3	-2.5'	54.8'
11	349°02.5	57.3	111°37.4	10.5'	23°10.8	-2.7'	54.8'
12	4°02.5	S12°58.2	126°06.9	10.5'	N23°08.1	-2.8'	54.9'
13	19°02.6	59.0	140°36.4	10.5'	23°05.3	-2.9'	54.9'
14	34°02.6	12°59.8	155°05.9	10.5'	23°02.4	-3.0'	54.9'
15	49°02.7	13°00.7	169°35.4	10.5'	22°59.4	-3.1'	54.9'
16	64°02.7	01.5	184°04.9	10.5'	22°56.3	-3.2'	54.9'
17	79°02.8	02.4	198°34.4	10.5'	22°53.0	-3.4'	55.0'
18	94°02.8	S13°03.2	213°03.9	10.5'	N22°49.7	-3.5'	55.0'
19	109°02.9	04.0	227°33.4	10.5'	22°46.2	-3.6'	55.0'
20	124°02.9	04.9	242°03.0	10.5'	22°42.6	-3.7'	55.0'
21	139°03.0	· · 05.7	256°32.5	10.5'	22°38.9	-3.8'	55.0'
22	154°03.0	06.6	271°02.0	10.5'	22°35.1	-3.9'	55.1'
23	169°03.1	07.4	285°31.6	10.6'	22°31.1	-4.1'	55.1'
SD = 16.1'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:43	07:03	08:23	15:04	16:23	17:43
N 70°	05:41	06:54	08:03	15:24	16:33	17:45
68°	05:40	06:46	07:47	15:39	16:41	17:46
66°	05:38	06:39	07:35	15:52	16:48	17:48
64°	05:37	06:33	07:24	16:03	16:54	17:49
62°	05:36	06:28	07:15	16:12	16:59	17:51
60°	05:35	06:23	07:07	16:19	17:03	17:52
N 58°	05:33	06:19	07:01	16:26	17:07	17:53
56°	05:32	06:16	06:55	16:32	17:11	17:55
54°	05:31	06:13	06:49	16:38	17:14	17:56
52°	05:30	06:09	06:44	16:43	17:18	17:57
50°	05:29	06:07	06:40	16:47	17:20	17:58
45°	05:26	06:00	06:30	16:57	17:27	18:01
N 40°	05:23	05:55	06:22	17:05	17:33	18:04
35°	05:20	05:50	06:16	17:12	17:38	18:07
30°	05:17	05:45	06:10	17:18	17:42	18:10
20°	05:10	05:36	05:59	17:29	17:51	18:17
N 10°	05:03	05:28	05:49	17:38	18:00	18:25
0°	04:55	05:19	05:41	17:47	18:08	18:33
S 10°	04:45	05:10	05:31	17:56	18:18	18:43
20°	04:32	04:59	05:22	18:06	18:29	18:56
30°	04:16	04:46	05:10	18:18	18:43	19:12
35°	04:06	04:37	05:04	18:24	18:51	19:23
40°	03:53	04:28	04:56	18:32	19:01	19:35
45°	03:38	04:16	04:48	18:41	19:13	19:51
S 50°	03:18	04:02	04:37	18:52	19:27	20:12
52°	03:08	03:55	04:32	18:57	19:34	20:22
54°	02:56	03:47	04:27	19:02	19:42	20:33
56°	02:43	03:39	04:21	19:09	19:51	20:47
58°	02:27	03:29	04:14	19:16	20:01	21:03
S 60°	02:08	03:17	04:06	19:24	20:13	21:24

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	15:10	15:41	17:03	12:52	14:05	14:28
64°	16:01	16:43	17:51	12:01	13:03	13:40
62°	16:33	17:17	18:21	11:29	12:28	13:09
60°	16:57	17:42	18:44	11:06	12:03	12:46
N 58°	17:16	18:02	19:02	10:47	11:43	12:28
56°	17:32	18:19	19:18	10:31	11:27	12:12
54°	17:45	18:33	19:31	10:18	11:13	11:59
52°	17:57	18:45	19:42	10:06	11:00	11:47
50°	18:08	18:56	19:53	09:55	10:50	11:37
45°	18:30	19:19	20:14	09:34	10:27	11:15
N 40°	18:48	19:37	20:31	09:16	10:09	10:57
35°	19:03	19:52	20:46	09:01	09:53	10:42
30°	19:16	20:05	20:59	08:48	09:40	10:29
20°	19:38	20:28	21:20	08:27	09:18	10:07
N 10°	19:57	20:47	21:39	08:08	08:58	09:48
0°	20:16	21:06	21:56	07:50	08:40	09:30
S 10°	20:34	21:24	22:14	07:32	08:21	09:12
20°	20:53	21:44	22:32	07:13	08:02	08:53
30°	21:16	22:06	22:54	06:52	07:39	08:30
35°	21:29	22:19	23:06	06:39	07:26	08:17
40°	21:44	22:35	23:21	06:24	07:10	08:02
45°	22:02	22:53	23:38	06:07	06:52	07:44
S 50°	22:25	23:16	23:59	05:46	06:29	07:22
52°	22:36	23:26	· · ·	05:35	06:18	07:11
54°	22:48	23:39	· · ·	05:24	06:06	06:59
56°	23:02	23:53	· · ·	05:11	05:52	06:45
58°	23:19	· · ·	00:09	04:55	05:35	06:28
S 60°	23:39	· · ·	00:29	04:37	05:15	06:09

Day	Sun		Mer. Pass	Moon		Age 18-20 94-82%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
25	15:55	15:58	11:44	01:38	14:03	
26	16:01	16:05	11:44	02:28	14:53	
27	16:07	16:10	11:44	03:18	15:43	

October 28, 29, 30 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	36°37.7	135°51.4	S26°52.4	132°22.2	S24°39.1	181°27.2	S12°58.9	345°32.0	N16°05.2	Alpheratz	357°30.4	29°15.6
1	51°40.2	150°51.1	52.6	147°22.6	39.2	196°29.1	59.1	0°34.7	05.1	Ankaa	353°03.1	-42°08.6
2	66°42.6	165°50.8	52.8	162°23.0	39.3	211°31.1	59.3	15°37.3	05.1	Schedar	349°25.9	56°42.3
3	81°45.1	180°50.5	· · 53.1	177°23.5	· · 39.4	226°33.0	· · 59.4	30°40.0	· · 05.0	Diphda	348°43.2	-17°49.2
4	96°47.6	195°50.2	53.3	192°23.9	39.4	241°34.9	59.6	45°42.6	05.0	Achernar	335°16.9	-57°05.1
5	111°50.0	210°49.9	53.5	207°24.3	39.5	256°36.9	12°59.8	60°45.3	04.9	Hamal	327°46.4	23°36.3
6	126°52.5	225°49.6	S26°53.8	222°24.8	S24°39.6	271°38.8	S13°00.0	75°47.9	N16°04.9	Polaris	311°23.3	89°23.2
7	141°55.0	240°49.3	54.0	237°25.2	39.7	286°40.7	00.2	90°50.6	04.8	Acamar	315°08.5	-40°11.0
8	156°57.4	255°49.0	54.2	252°25.6	39.7	301°42.7	00.3	105°53.2	04.8	Menkar	314°01.8	4°12.5
9	171°59.9	270°48.7	· · 54.5	267°26.1	· · 39.8	316°44.6	· · 00.5	120°55.9	· · 04.7	Mirfak	308°22.1	49°58.0
10	187°02.4	285°48.4	54.7	282°26.5	39.9	331°46.5	00.7	135°58.6	04.7	Aldebaran	290°34.9	16°34.1
11	202°04.8	300°48.1	54.9	297°26.9	40.0	346°48.5	00.9	151°01.2	04.6	Rigel	280°59.9	-8°10.0
12	217°07.3	315°47.8	S26°55.1	312°27.4	S24°40.0	1°50.4	S13°01.1	166°03.9	N16°04.6	Capella	280°15.8	46°01.4
13	232°09.8	330°47.5	55.4	327°27.8	40.1	16°52.3	01.2	181°06.5	04.5	Bellatrix	278°18.5	6°22.5
14	247°12.2	345°47.2	55.6	342°28.2	40.2	31°54.3	01.4	196°09.2	04.5	Elnath	277°56.7	28°37.8
15	262°14.7	0°46.9	· · 55.8	357°28.7	· · 40.3	46°56.2	· · 01.6	211°11.8	· · 04.4	Alnilam	275°33.6	-1°11.0
16	277°17.1	15°46.6	56.0	12°29.1	40.3	61°58.1	01.8	226°14.5	04.4	Betelgeuse	270°47.7	7°24.7
17	292°19.6	30°46.4	56.2	27°29.5	40.4	77°00.1	02.0	241°17.1	04.3	Canopus	263°50.4	-52°42.5
18	307°22.1	45°46.1	S26°56.5	42°30.0	S24°40.5	92°02.0	S13°02.1	256°19.8	N16°04.3	Sirius	258°22.7	-16°45.4
19	322°24.5	60°45.8	56.7	57°30.4	40.5	107°03.9	02.3	271°22.4	04.2	Adhara	255°02.7	-29°00.7
20	337°27.0	75°45.5	56.9	72°30.8	40.6	122°05.9	02.5	286°25.1	04.2	Procyon	244°46.7	5°08.8
21	352°29.5	90°45.2	· · 57.1	87°31.3	· · 40.7	137°07.8	· · 02.7	301°27.7	· · 04.1	Pollux	243°12.6	27°57.0
22	7°31.9	105°44.9	57.3	102°31.7	40.7	152°09.7	02.8	316°30.4	04.1	Avior	234°13.0	-59°36.1
23	22°34.4	120°44.6	57.5	117°32.1	40.8	167°11.7	03.0	331°33.1	04.0	Suhail	222°43.5	-43°33.0
Mer.pass. 21:30		ν -0.3' d 0.2' m -4.51		ν 0.4' d 0.1' m 1.05		ν 1.9' d 0.2' m -1.68		ν 2.7' d -0.0' m -0.19		Miaplacidus	221°37.4	-69°50.2
										Alphard	217°44.0	-8°47.3
										Regulus	207°30.5	11°49.2
										Dubhe	193°37.0	61°35.1
										Denebola	182°21.3	14°24.3
										Gienah	175°39.9	-17°42.4
										Acrux	172°56.1	-63°15.7
										Gacrux	171°47.7	-57°16.7
										Alioth	166°10.3	55°47.7
										Spica	158°18.5	-11°18.9
										Alkaid	152°49.6	49°09.8
										Hadar	148°31.0	-60°30.9
										Menkent	147°53.4	-36°30.9
										Arcturus	145°44.8	19°01.7
										Rigel Kent.	139°35.5	-60°57.4
										Kochab	137°21.4	74°02.0
										Zuben'ubi	136°52.1	-16°09.8
										Alphecca	126°00.9	26°37.0
										Antares	112°11.4	-26°29.7
										Atria	107°02.5	-69°04.8
										Sabik	101°58.6	-15°45.5
										Shaula	96°05.4	-37°07.4
										Rasalhague	95°55.2	12°32.6
										Eltanin	90°40.7	51°29.5
										Kaus Aust.	83°27.6	-34°22.1
										Vega	80°30.8	38°49.1
										Nunki	75°43.2	-26°15.4
										Altair	61°56.3	8°57.2
										Peacock	52°59.9	-56°38.3
										Deneb	49°23.1	45°23.7
										Enif	33°34.9	10°01.0
										Al Na'ir	27°28.1	-46°49.0
										Fomalhaut	15°10.1	-29°27.7
										Scheat	13°41.2	28°15.0
										Markab	13°25.8	15°22.3
Oct 28 Sun		SHA		Mer.pass								
		Venus		99°13.6		14:57						
		Mars		95°44.5		15:10						
		Jupiter		144°49.5		11:53						
		Saturn		308°54.3		00:58						
Oct 29 Mon		SHA		Mer.pass								
		Venus		98°07.5		14:57						
		Mars		94°55.7		15:09						
		Jupiter		144°36.8		11:50						
		Saturn		308°58.9		00:53						
Oct 30 Tue		SHA		Mer.pass								
		Venus		97°01.7		14:58						
		Mars		94°06.8		15:09						
		Jupiter		144°24.0		11:46						
		Saturn		309°03.4		00:49						
Horizontal parallax												
		Venus:		0.2								
		Mars:		0.1								
Mer.pass. 21:22		ν -0.3' d 0.2' m -4.54		ν 0.4' d 0.1' m 1.06		ν 1.9' d 0.2' m -1.68		ν 2.7' d -0.0' m -0.20				

DUT1 = UT1-UTC = +0.2133 sec ΔT = TT-UT1 = +68.9707 sec

2029 October 28 to Oct. 30 UT

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	184°03.1	S13°08.2	300°01.1	10.6'	N22°27.1	-4.2'	55.1'
1	199°03.2	09.1	314°30.7	10.6'	22°22.9	-4.3'	55.1'
2	214°03.2	09.9	329°00.3	10.6'	22°18.6	-4.4'	55.2'
3	229°03.3	· · 10.7	343°29.8	10.6'	22°14.2	-4.5'	55.2'
4	244°03.3	11.6	357°59.4	10.6'	22°09.7	-4.6'	55.2'
5	259°03.4	12.4	12°29.0	10.6'	22°05.1	-4.7'	55.2'
6	274°03.4	S13°13.2	26°58.6	10.6'	N22°00.4	-4.8'	55.3'
7	289°03.5	14.1	41°28.2	10.6'	21°55.5	-5.0'	55.3'
8	304°03.5	14.9	55°57.9	10.6'	21°50.6	-5.1'	55.3'
9	319°03.6	· · 15.7	70°27.5	10.7'	21°45.5	-5.2'	55.3'
10	334°03.6	16.6	84°57.2	10.7'	21°40.3	-5.3'	55.4'
11	349°03.7	17.4	99°26.8	10.7'	21°35.0	-5.4'	55.4'
12	4°03.7	S13°18.2	113°56.5	10.7'	N21°29.6	-5.5'	55.4'
13	19°03.8	19.1	128°26.2	10.7'	21°24.1	-5.6'	55.4'
14	34°03.8	19.9	142°55.9	10.7'	21°18.5	-5.7'	55.5'
15	49°03.9	· · 20.7	157°25.6	10.7'	21°12.7	-5.9'	55.5'
16	64°03.9	21.6	171°55.4	10.7'	21°06.9	-6.0'	55.5'
17	79°04.0	22.4	186°25.1	10.8'	21°00.9	-6.1'	55.5'
18	94°04.0	S13°23.2	200°54.9	10.8'	N20°54.8	-6.2'	55.6'
19	109°04.0	24.1	215°24.7	10.8'	20°48.6	-6.3'	55.6'
20	124°04.1	24.9	229°54.4	10.8'	20°42.4	-6.4'	55.6'
21	139°04.1	· · 25.7	244°24.3	10.8'	20°36.0	-6.5'	55.6'
22	154°04.2	26.5	258°54.1	10.8'	20°29.4	-6.6'	55.7'
23	169°04.2	27.4	273°23.9	10.9'	20°22.8	-6.7'	55.7'
SD = 16.1'			SD = 15.0'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	184°04.3	S13°28.2	287°53.8	10.9'	N20°16.1	-6.8'	55.7'
1	199°04.3	29.0	302°23.7	10.9'	20°09.3	-6.9'	55.7'
2	214°04.3	29.9	316°53.5	10.9'	20°02.3	-7.0'	55.8'
3	229°04.4	· · 30.7	331°23.5	10.9'	19°55.3	-7.1'	55.8'
4	244°04.4	31.5	345°53.4	10.9'	19°48.2	-7.3'	55.8'
5	259°04.5	32.3	0°23.3	11.0'	19°40.9	-7.4'	55.9'
6	274°04.5	S13°33.2	14°53.3	11.0'	N19°33.5	-7.5'	55.9'
7	289°04.5	34.0	29°23.3	11.0'	19°26.1	-7.6'	55.9'
8	304°04.6	34.8	43°53.2	11.0'	19°18.5	-7.7'	56.0'
9	319°04.6	· · 35.6	58°23.3	11.0'	19°10.8	-7.8'	56.0'
10	334°04.7	36.5	72°53.3	11.0'	19°03.1	-7.9'	56.0'
11	349°04.7	37.3	87°23.3	11.1'	18°55.2	-8.0'	56.0'
12	4°04.7	S13°38.1	101°53.4	11.1'	N18°47.2	-8.1'	56.1'
13	19°04.8	38.9	116°23.5	11.1'	18°39.1	-8.2'	56.1'
14	34°04.8	39.7	130°53.6	11.1'	18°31.0	-8.3'	56.1'
15	49°04.8	· · 40.6	145°23.7	11.1'	18°22.7	-8.4'	56.2'
16	64°04.9	41.4	159°53.8	11.1'	18°14.3	-8.5'	56.2'
17	79°04.9	42.2	174°24.0	11.2'	18°05.8	-8.6'	56.2'
18	94°05.0	S13°43.0	188°54.1	11.2'	N17°57.2	-8.7'	56.3'
19	109°05.0	43.9	203°24.3	11.2'	17°48.5	-8.8'	56.3'
20	124°05.0	44.7	217°54.5	11.2'	17°39.8	-8.9'	56.3'
21	139°05.1	· · 45.5	232°24.7	11.2'	17°30.9	-9.0'	56.4'
22	154°05.1	46.3	246°55.0	11.2'	17°21.9	-9.1'	56.4'
23	169°05.1	47.1	261°25.2	11.3'	17°12.9	-9.2'	56.4'
SD = 16.1'			SD = 15.2'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	184°05.2	S13°48.0	275°55.5	11.3'	N17°03.7	-9.3'	56.5'
1	199°05.2	48.8	290°25.7	11.3'	16°54.4	-9.4'	56.5'
2	214°05.2	49.6	304°56.0	11.3'	16°45.1	-9.4'	56.5'
3	229°05.3	· · 50.4	319°26.4	11.3'	16°35.6	-9.5'	56.6'
4	244°05.3	51.2	333°56.7	11.3'	16°26.1	-9.6'	56.6'
5	259°05.3	52.0	348°27.0	11.4'	16°16.5	-9.7'	56.6'
6	274°05.4	S13°52.9	2°57.4	11.4'	N16°06.7	-9.8'	56.7'
7	289°05.4	53.7	17°27.7	11.4'	15°56.9	-9.9'	56.7'
8	304°05.4	54.5	31°58.1	11.4'	15°47.0	-10.0'	56.7'
9	319°05.5	· · 55.3	46°28.5	11.4'	15°37.0	-10.1'	56.8'
10	334°05.5	56.1	60°58.9	11.4'	15°26.9	-10.2'	56.8'
11	349°05.5	56.9	75°29.4	11.4'	15°16.7	-10.3'	56.8'
12	4°05.5	S13°57.8	89°59.8	11.4'	N15°06.5	-10.4'	56.9'
13	19°05.6	58.6	104°30.2	11.5'	14°56.1	-10.4'	56.9'
14	34°05.6	13°59.4	119°00.7	11.5'	14°45.7	-10.5'	56.9'
15	49°05.6	14°00.2	133°31.2	11.5'	14°35.2	-10.6'	57.0'
16	64°05.7	01.0	148°01.7	11.5'	14°24.5	-10.7'	57.0'
17	79°05.7	01.8	162°32.2	11.5'	14°13.8	-10.8'	57.0'
18	94°05.7	S14°02.6	177°02.7	11.5'	N14°03.1	-10.9'	57.1'
19	109°05.7	03.4	191°33.2	11.5'	13°52.2	-10.9'	57.1'
20	124°05.8	04.3	206°03.7	11.5'	13°41.3	-11.0'	57.1'
21	139°05.8	· · 05.1	220°34.2	11.5'	13°30.2	-11.1'	57.2'
22	154°05.8	05.9	235°04.8	11.5'	13°19.1	-11.2'	57.2'
23	169°05.9	06.7	249°35.3	11.6'	13°07.9	-11.3'	57.3'
SD = 16.1'			SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:55	07:17	08:40	14:46	16:09	17:30
N 70°	05:52	07:05	08:17	15:09	16:20	17:33
68°	05:50	06:56	07:59	15:27	16:30	17:36
66°	05:47	06:48	07:45	15:41	16:38	17:39
64°	05:45	06:41	07:34	15:53	16:45	17:41
62°	05:43	06:36	07:24	16:03	16:51	17:43
60°	05:41	06:30	07:15	16:11	16:56	17:45
N 58°	05:40	06:26	07:08	16:19	17:00	17:47
56°	05:38	06:22	07:01	16:25	17:05	17:48
54°	05:36	06:18	06:55	16:31	17:09	17:50
52°	05:35	06:15	06:50	16:37	17:12	17:52
50°	05:33	06:11	06:45	16:42	17:15	17:53
45°	05:30	06:04	06:35	16:52	17:23	17:57
N 40°	05:26	05:58	06:26	17:01	17:29	18:01
35°	05:23	05:52	06:18	17:09	17:35	18:04
30°	05:19	05:47	06:12	17:15	17:40	18:08
20°	05:12	05:38	06:00	17:27	17:50	18:16
N 10°	05:03	05:28	05:50	17:37	17:59	18:24
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:44	05:09	05:30	17:57	18:19	18:44
20°	04:30	04:57	05:20	18:08	18:31	18:58
30°	04:13	04:43	05:08	18:20	18:45	19:15
35°	04:02	04:34	05:01	18:27	18:54	19:26
40°	03:49	04:24	04:53	18:36	19:05	19:40
45°	03:32	04:11	04:43	18:45	19:17	19:56
S 50°	03:11	03:56	04:31	18:57	19:33	20:18
52°	03:00	03:48	04:26	19:02	19:40	20:29
54°	02:48	03:40	04:20	19:08	19:49	20:42
56°	02:33	03:31	04:14	19:15	19:58	20:57
58°	02:16	03:20	04:06	19:23	20:10	21:15
S 60°	01:54	03:07	03:58	19:31	20:23	21:38

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	□	17:55	21:11	□	17:08	15:35
N 70°	□	19:24	21:38	□	15:37	15:06
68°	17:55	20:03	21:58	15:21	14:57	14:45
66°	18:45	20:29	22:14	14:31	14:30	14:28
64°	19:16	20:50	22:27	13:59	14:08	14:13
62°	19:39	21:06	22:37	13:35	13:51	14:02
60°	19:58	21:20	22:47	13:16	13:37	13:51
N 58°	20:13	21:32	22:55	13:01	13:25	13:42
56°	20:26	21:42	23:02	12:47	13:14	13:35
54°	20:38	21:51	23:08	12:35	13:04	13:28
52°	20:48	21:59	23:14	12:25	12:56	13:21
50°	20:57	22:06	23:19	12:16	12:48	13:16
45°	21:15	22:21	23:30	11:56	12:32	13:03
N 40°	21:31	22:34	23:39	11:40	12:19	12:53
35°	21:44	22:44	23:47	11:27	12:07	12:44
30°	21:55	22:53	23:53	11:15	11:57	12:36
20°	22:14	23:09	· ·	10:55	11:40	12:23
N 10°	22:31	23:23	· ·	10:37	11:25	12:11
0°	22:46	23:36	· ·	10:20	11:10	11:59
S 10°	23:02	23:49	· ·	10:04	10:56	11:48
20°	23:19	· ·	00:03	09:46	10:41	11:36
30°	23:38	· ·	00:18	09:25	10:23	11:22
35°	23:49	· ·	00:27	09:13	10:12	11:14
40°	· ·	00:01	00:38	08:59	10:00	11:05
45°	· ·	00:16	00:50	08:43	09:46	10:54
S 50°	· ·	00:34	01:04	08:22	09:29	10:40
52°	00:09	00:43	01:11	08:12	09:21	10:34
54°	00:20	00:53	01:18	08:01	09:12	10:27
56°	00:33	01:03	01:27	07:49	09:01	10:19
58°	00:48	01:16	01:36	07:34	08:50	10:11
S 60°	01:05	01:30	01:47	07:17	08:36	10:01

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	21-23
				hh:mm	hh:mm	74-55%
28	16:13	16:15	11:44	04:08	16:33	
29	16:17	16:19	11:44	04:58	17:23	
30	16:21	16:22	11:44	05:48	18:12	

October 31, 01, 02 UT (Wed., Thu., Fri.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Wed	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	SHA	Dec	
0	39°35.1	135°31.3	S27°06.7	132°53.0	S24°43.6	183°46.4	S13°11.8	348°43.2	N16°01.6	01.6	348°43.2	N16°01.6	Alpheratz	357°30.4	29°15.6		
1	54°37.6	150°31.1	06.8	147°53.4	43.7	198°48.3	11.9	3°45.9	01.6	01.6	353°03.1	-42°08.6	Ankaa	353°03.1	-42°08.6		
2	69°40.1	165°30.8	07.0	162°53.8	43.7	213°50.3	12.1	18°48.5	01.5	01.5	349°25.9	56°42.3	Schedar	349°25.9	56°42.3		
3	84°42.5	180°30.6	· · 07.1	177°54.3	· · 43.8	228°52.2	· · 12.3	33°51.2	· · 01.5	01.5	348°43.2	-17°49.2	Diphda	348°43.2	-17°49.2		
4	99°45.0	195°30.3	07.3	192°54.7	43.8	243°54.1	12.5	48°53.9	01.4	01.4	335°16.9	-57°05.1	Achernar	335°16.9	-57°05.1		
5	114°47.5	210°30.1	07.5	207°55.1	43.8	258°56.1	12.7	63°56.5	01.4	01.4	327°46.4	23°36.3	Hamal	327°46.4	23°36.3		
6	129°49.9	225°29.8	S27°07.6	222°55.5	S24°43.9	273°58.0	S13°12.8	78°59.2	N16°01.3	01.3	311°22.7	89°23.2	Polaris	311°22.7	89°23.2		
7	144°52.4	240°29.6	07.8	237°55.9	43.9	288°59.9	13.0	94°01.8	01.3	01.3	315°08.5	-40°11.0	Acamar	315°08.5	-40°11.0		
8	159°54.8	255°29.3	07.9	252°56.4	44.0	304°01.9	13.2	109°04.5	01.2	01.2	314°01.8	4°12.5	Menkar	314°01.8	4°12.5		
9	174°57.3	270°29.1	· · 08.1	267°56.8	· · 44.0	319°03.8	· · 13.4	124°07.1	· · 01.2	01.2	308°22.1	49°58.0	Mirfak	308°22.1	49°58.0		
10	189°59.8	285°28.8	08.2	282°57.2	44.1	334°05.7	13.5	139°09.8	01.1	01.1	290°34.9	16°34.1	Aldebaran	290°34.9	16°34.1		
11	205°02.2	300°28.6	08.4	297°57.6	44.1	349°07.7	13.7	154°12.5	01.1	01.1	280°59.9	-8°10.0	Rigel	280°59.9	-8°10.0		
12	220°04.7	315°28.3	S27°08.5	312°58.0	S24°44.1	4°09.6	S13°13.9	169°15.1	N16°01.0	01.0	280°15.8	46°01.4	Capella	280°15.8	46°01.4		
13	235°07.2	330°28.1	08.7	327°58.5	44.2	19°11.5	14.1	184°17.8	01.0	01.0	278°18.5	6°22.5	Bellatrix	278°18.5	6°22.5		
14	250°09.6	345°27.9	08.8	342°58.9	44.2	34°13.5	14.3	199°20.4	00.9	00.9	277°56.7	28°37.8	Elnath	277°56.7	28°37.8		
15	265°12.1	0°27.6	· · 09.0	357°59.3	· · 44.3	49°15.4	· · 14.4	214°23.1	· · 00.9	00.9	275°33.6	-1°11.0	Alnilam	275°33.6	-1°11.0		
16	280°14.6	15°27.4	09.1	12°59.7	44.3	64°17.3	14.6	229°25.8	00.8	00.8	270°47.7	7°24.7	Betelgeuse	270°47.7	7°24.7		
17	295°17.0	30°27.1	09.3	28°00.1	44.3	79°19.2	14.8	244°28.4	00.8	00.8	263°50.4	-52°42.5	Canopus	263°50.4	-52°42.5		
18	310°19.5	45°26.9	S27°09.4	43°00.6	S24°44.4	94°21.2	S13°15.0	259°31.1	N16°00.7	00.7	258°22.7	-16°45.4	Sirius	258°22.7	-16°45.4		
19	325°22.0	60°26.6	09.5	58°01.0	44.4	109°23.1	15.1	274°33.7	00.7	00.7	255°02.7	-29°00.7	Adhara	255°02.7	-29°00.7		
20	340°24.4	75°26.4	09.7	73°01.4	44.4	124°25.0	15.3	289°36.4	00.6	00.6	244°46.7	5°08.8	Procyon	244°46.7	5°08.8		
21	355°26.9	90°26.2	· · 09.8	88°01.8	· · 44.5	139°27.0	· · 15.5	304°39.1	· · 00.6	00.6	243°12.6	27°57.0	Pollux	243°12.6	27°57.0		
22	10°29.3	105°25.9	10.0	103°02.2	44.5	154°28.9	15.7	319°41.7	00.5	00.5	234°13.0	-59°36.1	Avior	234°13.0	-59°36.1		
23	25°31.8	120°25.7	10.1	118°02.7	44.6	169°30.8	15.9	334°44.4	00.5	00.5	222°43.4	-43°33.0	Suhail	222°43.4	-43°33.0		
Mer.pass. 21:18			ν -0.3' d 0.2' m -4.55			ν 0.4' d 0.0' m 1.06			ν 1.9' d 0.2' m -1.68			ν 2.7' d -0.0' m -0.20					

Thu	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec									
0	40°34.3	135°25.4	S27°10.2	133°03.1	S24°44.6	184°32.8	S13°16.0	349°47.0	N16°00.4								
1	55°36.7	150°25.2	10.4	148°03.5	44.6	199°34.7	16.2	4°49.7	00.4								
2	70°39.2	165°25.0	10.5	163°03.9	44.7	214°36.6	16.4	19°52.4	00.3								
3	85°41.7	180°24.7	· · 10.6	178°04.3	· · 44.7	229°38.6	· · 16.6	34°55.0	· · 00.3								
4	100°44.1	195°24.5	10.8	193°04.7	44.7	244°40.5	16.7	49°57.7	00.2								
5	115°46.6	210°24.3	10.9	208°05.2	44.8	259°42.4	16.9	65°00.3	00.2								
6	130°49.1	225°24.0	S27°11.0	223°05.6	S24°44.8	274°44.4	S13°17.1	80°03.0	N16°00.1								
7	145°51.5	240°23.8	11.2	238°06.0	44.8	289°46.3	17.3	95°05.7	00.1								
8	160°54.0	255°23.6	11.3	253°06.4	44.9	304°48.2	17.5	110°08.3	00.0								
9	175°56.4	270°23.3	· · 11.4	268°06.8	· · 44.9	319°50.2	· · 17.6	125°11.0	16°00.0								
10	190°58.9	285°23.1	11.5	283°07.2	44.9	334°52.1	17.8	140°13.6	15°59.9								
11	206°01.4	300°22.9	11.7	298°07.7	45.0	349°54.0	18.0	155°16.3	59.9								
12	221°03.8	315°22.7	S27°11.8	313°08.1	S24°45.0	4°56.0	S13°18.2	170°19.0	N15°59.8								
13	236°06.3	330°22.4	11.9	328°08.5	45.0	19°57.9	18.3	185°21.6	59.8								
14	251°08.8	345°22.2	12.0	343°08.9	45.0	34°59.8	18.5	200°24.3	59.7								
15	266°11.2	0°22.0	· · 12.2	358°09.3	· · 45.1	50°01.8	· · 18.7	215°26.9	· · 59.7								
16	281°13.7	15°21.7	12.3	13°09.7	45.1	65°03.7	18.9	230°29.6	59.6								
17	296°16.2	30°21.5	12.4	28°10.2	45.1	80°05.6	19.1	245°32.3	59.6								
18	311°18.6	45°21.3	S27°12.5	43°10.6	S24°45.2	95°07.6	S13°19.2	260°34.9	N15°59.5								
19	326°21.1	60°21.1	12.6	58°11.0	45.2	110°09.5	19.4	275°37.6	59.5								
20	341°23.6	75°20.9	12.8	73°11.4	45.2	125°11.4	19.6	290°40.2	59.4								
21	356°26.0	90°20.6	· · 12.9	88°11.8	· · 45.2	140°13.3	· · 19.8	305°42.9	· · 59.4								
22	11°28.5	105°20.4	13.0	103°12.2	45.3	155°15.3	19.9	320°45.6	59.3								
23	26°30.9	120°20.2	13.1	118°12.6	45.3	170°17.2	20.1	335°48.2	59.3								
Mer.pass. 21:14			ν -0.2' d 0.1' m -4.56			ν 0.4' d 0.0' m 1.06			ν 1.9' d 0.2' m -1.68			ν 2.7' d -0.0' m -0.21					

Fri	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec									
0	41°33.4	135°20.0	S27°13.2	133°13.1	S24°45.3	185°19.1	S13°20.3	350°50.9	N15°59.2								
1	56°35.9	150°19.8	13.3	148°13.5	45.3	200°21.1	20.5	5°53.5	59.2								
2	71°38.3	165°19.5	13.4	163°13.9	45.4	215°23.0	20.6	20°56.2	59.1								
3	86°40.8	180°19.3	· · 13.5	178°14.3	· · 45.4	230°24.9	· · 20.8	35°58.9	· · 59.1								
4	101°43.3	195°19.1	13.6	193°14.7	45.4	245°26.9	21.0	51°01.5	59.0								
5	116°45.7	210°18.9	13.8	208°15.1	45.4	260°28.8	21.2	66°04.2	59.0								
6	131°48.2	225°18.7	S27°13.9	223°15.5	S24°45.5	275°30.7	S13°21.4	81°06.8	N15°58.9								
7	146°50.7	240°18.5	14.0	238°16.0	45.5	290°32.7	21.5	96°09.5	58.9								
8	161°53.1	255°18.3	14.1	253°16.4	45.5	305°34.6	21.7	111°12.2	58.8								
9	176°55.6	270°18.0	· · 14.2	268°16.8	· · 45.5	320°36.5	· · 21.9	126°14.8	· · 58.8								
10	191°58.1	285°17.8	14.3	283°17.2	45.5	335°38.5	22.1	141°17.5	58.7								
11	207°00.5	300°17.6	14.4	298°17.6	45.6	350°40.4	22.2	156°20.2	58.7								
12	222°03.0	315°17.4	S27°14.5	313°18.0	S24°45.6	5°42.3	S13°22.4	171°22.8	N15°58.6								
13	237°05.4	330°17.2	14.6	328°18.4	45.6	20°44.3	22.6	186°25.5	58.6								
14	252°07.9	345°17.0	14.7	343°18.8	45.6	35°46.2	22.8	201°28.1	58.5								
15	267°10.4	0°16.8	· · 14.8	358°19.3	· · 45.6	50°48.1	· · 22.9	216°30.8	· · 58.5								
16	282°12.8	15°16.6	14.9	13°19.7	45.6	65°50.0	23.1	231°33.5	58.4								
17	297°15.3	30°16.4	15.0	28°20.1	45.7	80°52.0	23.3	246°36.1	58.4								
18	312°17.8	45°16.2	S27°15.0	43°20.5	S24°45.7	95°53.9	S13°23.5	261°38.8	N15°58.3								
19	327°20.2	60°16.0	15.1	58°20.9	45.7	110°55.8	23.7	276°41.5	58.3								
20	342°22.7	75°15.8	15.2	73°21.3	45.7	125°57.8	23.8	291°44.1	58.2								
21	357°25.2	90°15.6	· · 15.3	88°21.7	· · 45.7	140°59.7	· · 24.0	306°46.8	· · 58.2								
22	12°27.6	105°15.4	15.4	103°22.1	45.7	156°01.6	24.2	321°49.4	58.1								
23	27°30.1	120°15.2	15.5	118°22.6	45.8	171°03.6	24.4	336°52.1	58.1								
Mer.pass. 21:10			ν -0.2' d 0.1' m -4.57			ν 0.4' d 0.0' m 1.06			ν 1.9' d 0.2' m -1.68			ν 2.7' d -0.0' m -0.21					

Oct 31 Wed	SHA	Mer.pass
Venus	95°56.2	14:58
Mars	93°17.9	15:08
Jupiter	144°11.3	11:43
Saturn	309°08.1	00:45

Nov 01 Thu	SHA	Mer.pass
Venus	94°51.2	14:59
Mars	92°28.8	15:07
Jupiter	143°58.5	11:40
Saturn	309°12.8	00:41

Nov 02 Fri	SHA
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h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	184°05.9	S14°07.5	264°05.9	11.6'	N12°56.6	-11.4'	57.3'
1	199°05.9	08.3	278°36.4	11.6'	12°45.3	-11.4'	57.3'
2	214°05.9	09.1	293°07.0	11.6'	12°33.9	-11.5'	57.4'
3	229°06.0	· · 09.9	307°37.6	11.6'	12°22.3	-11.6'	57.4'
4	244°06.0	10.7	322°08.2	11.6'	12°10.8	-11.7'	57.4'
5	259°06.0	11.5	336°38.7	11.6'	11°59.1	-11.7'	57.5'
6	274°06.0	S14°12.3	351°09.3	11.6'	N11°47.4	-11.8'	57.5'
7	289°06.0	13.2	5°39.9	11.6'	11°35.5	-11.9'	57.6'
8	304°06.1	14.0	20°10.5	11.6'	11°23.6	-12.0'	57.6'
9	319°06.1	· · 14.8	34°41.1	11.6'	11°11.7	-12.0'	57.6'
10	334°06.1	15.6	49°11.7	11.6'	10°59.7	-12.1'	57.7'
11	349°06.1	16.4	63°42.3	11.6'	10°47.5	-12.2'	57.7'
12	4°06.2	S14°17.2	78°12.9	11.6'	N10°35.4	-12.2'	57.7'
13	19°06.2	18.0	92°43.5	11.6'	10°23.1	-12.3'	57.8'
14	34°06.2	18.8	107°14.1	11.6'	10°10.8	-12.4'	57.8'
15	49°06.2	· · 19.6	121°44.6	11.6'	09°58.4	-12.5'	57.9'
16	64°06.2	20.4	136°15.2	11.6'	09°46.0	-12.5'	57.9'
17	79°06.3	21.2	150°45.8	11.6'	09°33.5	-12.6'	57.9'
18	94°06.3	S14°22.0	165°16.4	11.6'	N09°20.9	-12.6'	58.0'
19	109°06.3	22.8	179°47.0	11.6'	09°08.2	-12.7'	58.0'
20	124°06.3	23.6	194°17.5	11.6'	08°55.5	-12.8'	58.1'
21	139°06.3	· · 24.4	208°48.1	11.5'	08°42.7	-12.8'	58.1'
22	154°06.4	25.2	223°18.6	11.5'	08°29.9	-12.9'	58.1'
23	169°06.4	26.0	237°49.2	11.5'	08°17.0	-13.0'	58.2'
SD = 16.1'			SD = 15.6'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	184°06.4	S14°26.8	252°19.7	11.5'	N08°04.1	-13.0'	58.2'
1	199°06.4	27.6	266°50.2	11.5'	07°51.0	-13.1'	58.3'
2	214°06.4	28.4	281°20.7	11.5'	07°38.0	-13.1'	58.3'
3	229°06.4	· · 29.2	295°51.2	11.5'	07°24.8	-13.2'	58.3'
4	244°06.5	30.0	310°21.7	11.5'	07°11.6	-13.2'	58.4'
5	259°06.5	30.8	324°52.2	11.5'	06°58.4	-13.3'	58.4'
6	274°06.5	S14°31.6	339°22.7	11.4'	N06°45.1	-13.4'	58.4'
7	289°06.5	32.4	353°53.1	11.4'	06°31.8	-13.4'	58.5'
8	304°06.5	33.2	8°23.5	11.4'	06°18.4	-13.5'	58.5'
9	319°06.5	· · 34.0	22°53.9	11.4'	06°04.9	-13.5'	58.6'
10	334°06.5	34.8	37°24.3	11.4'	05°51.4	-13.6'	58.6'
11	349°06.6	35.6	51°54.7	11.4'	05°37.8	-13.6'	58.6'
12	4°06.6	S14°36.4	66°25.1	11.3'	N05°24.2	-13.6'	58.7'
13	19°06.6	37.2	80°55.4	11.3'	05°10.6	-13.7'	58.7'
14	34°06.6	38.0	95°25.7	11.3'	04°56.9	-13.7'	58.8'
15	49°06.6	· · 38.8	109°56.0	11.3'	04°43.2	-13.8'	58.8'
16	64°06.6	39.6	124°26.2	11.2'	04°29.4	-13.8'	58.8'
17	79°06.6	40.4	138°56.5	11.2'	04°15.6	-13.9'	58.9'
18	94°06.6	S14°41.1	153°26.7	11.2'	N04°01.7	-13.9'	58.9'
19	109°06.6	41.9	167°56.9	11.2'	03°47.8	-13.9'	59.0'
20	124°06.7	42.7	182°27.1	11.1'	03°33.8	-14.0'	59.0'
21	139°06.7	· · 43.5	196°57.2	11.1'	03°19.9	-14.0'	59.0'
22	154°06.7	44.3	211°27.3	11.1'	03°05.8	-14.1'	59.1'
23	169°06.7	45.1	225°57.4	11.0'	02°51.8	-14.1'	59.1'
SD = 16.1'			SD = 15.9'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	184°06.7	S14°45.9	240°27.4	11.0'	N02°37.7	-14.1'	59.1'
1	199°06.7	46.7	254°57.4	11.0'	02°23.6	-14.2'	59.2'
2	214°06.7	47.5	269°27.4	10.9'	02°09.4	-14.2'	59.2'
3	229°06.7	· · 48.3	283°57.4	10.9'	01°55.2	-14.2'	59.3'
4	244°06.7	49.1	298°27.3	10.9'	01°41.0	-14.2'	59.3'
5	259°06.7	49.8	312°57.1	10.8'	01°26.8	-14.3'	59.3'
6	274°06.7	S14°50.6	327°27.0	10.8'	N01°12.5	-14.3'	59.4'
7	289°06.7	51.4	341°56.8	10.8'	00°58.2	-14.3'	59.4'
8	304°06.7	52.2	356°26.5	10.7'	00°43.9	-14.3'	59.4'
9	319°06.8	· · 53.0	10°56.3	10.7'	00°29.5	-14.4'	59.5'
10	334°06.8	53.8	25°25.9	10.6'	00°15.2	-14.4'	59.5'
11	349°06.8	54.6	39°55.6	10.6'	N00°00.8	-14.4'	59.6'
12	4°06.8	S14°55.3	54°25.2	10.6'	S00°13.6	14.4'	59.6'
13	19°06.8	56.1	68°54.7	10.5'	00°28.0	14.4'	59.6'
14	34°06.8	56.9	83°24.2	10.5'	00°42.5	14.4'	59.7'
15	49°06.8	· · 57.7	97°53.7	10.4'	00°56.9	14.5'	59.7'
16	64°06.8	58.5	112°23.1	10.4'	01°11.4	14.5'	59.7'
17	79°06.8	14°59.3	126°52.5	10.3'	01°25.8	14.5'	59.8'
18	94°06.8	S15°00.1	141°21.8	10.3'	S01°40.3	14.5'	59.8'
19	109°06.8	00.8	155°51.0	10.2'	01°54.8	14.5'	59.8'
20	124°06.8	01.6	170°20.3	10.2'	02°09.3	14.5'	59.9'
21	139°06.8	· · 02.4	184°49.4	10.1'	02°23.8	14.5'	59.9'
22	154°06.8	03.2	199°18.5	10.1'	02°38.3	14.5'	59.9'
23	169°06.8	04.0	213°47.6	10.0'	02°52.8	14.5'	60.0'
SD = 16.1'			SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:07	07:30	08:58	14:27	15:55	17:18
N 70°	06:03	07:17	08:32	14:54	16:08	17:23
68°	05:59	07:06	08:12	15:14	16:19	17:26
66°	05:56	06:58	07:56	15:30	16:28	17:30
64°	05:53	06:50	07:43	15:43	16:36	17:33
62°	05:50	06:43	07:32	15:54	16:43	17:36
60°	05:48	06:38	07:23	16:03	16:49	17:38
N 58°	05:46	06:32	07:15	16:12	16:54	17:40
56°	05:44	06:28	07:07	16:19	16:59	17:43
54°	05:42	06:23	07:01	16:25	17:03	17:45
52°	05:40	06:20	06:55	16:31	17:07	17:47
50°	05:38	06:16	06:50	16:36	17:10	17:49
45°	05:33	06:08	06:39	16:48	17:18	17:53
N 40°	05:29	06:01	06:29	16:57	17:25	17:57
35°	05:25	05:55	06:21	17:06	17:32	18:02
30°	05:21	05:49	06:14	17:13	17:37	18:06
20°	05:13	05:39	06:02	17:25	17:48	18:14
N 10°	05:04	05:29	05:51	17:36	17:58	18:23
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:42	05:08	05:30	17:58	18:19	18:45
20°	04:28	04:55	05:18	18:09	18:32	18:59
30°	04:10	04:40	05:05	18:22	18:48	19:18
35°	03:58	04:31	04:58	18:30	18:57	19:30
40°	03:44	04:20	04:49	18:39	19:08	19:44
45°	03:27	04:06	04:39	18:49	19:22	20:02
S 50°	03:04	03:50	04:26	19:02	19:39	20:25
52°	02:52	03:42	04:20	19:08	19:47	20:37
54°	02:39	03:33	04:14	19:15	19:56	20:50
56°	02:23	03:23	04:07	19:22	20:06	21:06
58°	02:04	03:11	03:59	19:30	20:18	21:26
S 60°	01:39	02:57	03:49	19:40	20:32	21:53

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	23:26	· · ·	01:35	15:02	14:38	14:15
N 70°	23:39	· · ·	01:38	14:47	14:31	14:17
68°	23:49	· · ·	01:41	14:35	14:26	14:18
66°	23:58	· · ·	01:43	14:25	14:22	14:18
64°	· · ·	00:05	01:45	14:16	14:18	14:19
62°	· · ·	00:11	01:46	14:09	14:15	14:20
60°	· · ·	00:16	01:48	14:02	14:12	14:20
N 58°	· · ·	00:20	01:49	13:57	14:09	14:21
56°	· · ·	00:25	01:50	13:52	14:07	14:21
54°	· · ·	00:28	01:51	13:47	14:05	14:22
52°	· · ·	00:31	01:52	13:43	14:03	14:22
50°	· · ·	00:34	01:52	13:39	14:01	14:22
45°	· · ·	00:41	01:54	13:31	13:57	14:23
N 40°	· · ·	00:46	01:55	13:24	13:54	14:24
35°	· · ·	00:51	01:57	13:18	13:51	14:24
30°	· · ·	00:55	01:58	13:13	13:49	14:25
20°	00:05	01:02	02:00	13:04	13:44	14:25
N 10°	00:15	01:08	02:01	12:56	13:41	14:26
0°	00:25	01:14	02:03	12:48	13:37	14:27
S 10°	00:34	01:19	02:04	12:40	13:33	14:27
20°	00:45	01:25	02:06	12:32	13:29	14:28
30°	00:56	01:32	02:08	12:23	13:25	14:29
35°	01:03	01:36	02:09	12:17	13:22	14:29
40°	01:10	01:41	02:10	12:11	13:19	14:29
45°	01:19	01:46	02:12	12:03	13:15	14:30
S 50°	01:29	01:52	02:14	11:55	13:11	14:30
52°	01:34	01:55	02:14	11:50	13:09	14:31
54°	01:40	01:58	02:15	11:46	13:07	14:31
56°	01:46	02:01	02:16	11:41	13:05	14:31
58°	01:52	02:05	02:17	11:35	13:02	14:32
S 60°	01:59	02:09	02:18	11:29	12:59	14:32

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Upper	Lower	24-26
	00 ^h	mm:ss	hh:mm	hh:mm	hh:mm	45-24%
31	16:24	16:25	11:44	06:37	19:01	
01	16:26	16:26	11:44	07:25	19:50	
02	16:27	16:27	11:44	08:15	20:40	

November 03, 04, 05 UT (Sat., Sun., Mon.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	42°32.5	135°15.0	S27°15.6	133°23.0	S24°45.8	186°05.5	S13°24.5	351°54.8	N15°58.0					Alpheratz	357°30.4	29°15.6	
1	57°35.0	150°14.8	15.7	148°23.4	45.8	201°07.4	24.7	6°57.4	58.0					Ankaa	353°03.1	-42°08.6	
2	72°37.5	165°14.6	15.8	163°23.8	45.8	216°09.4	24.9	22°00.1	57.9					Schedar	349°25.9	56°42.3	
3	87°39.9	180°14.4	•• 15.8	178°24.2	•• 45.8	231°11.3	•• 25.1	37°02.8	•• 57.9					Diphda	348°43.2	-17°49.3	
4	102°42.4	195°14.2	15.9	193°24.6	45.8	246°13.2	25.2	52°05.4	57.8					Achernar	335°16.9	-57°05.1	
5	117°44.9	210°14.0	16.0	208°25.0	45.8	261°15.2	25.4	67°08.1	57.8					Hamal	327°46.4	23°36.3	
6	132°47.3	225°13.8	S27°16.1	223°25.4	S24°45.8	276°17.1	S13°25.6	82°10.7	N15°57.7					Polaris	311°22.3	89°23.2	
7	147°49.8	240°13.6	16.2	238°25.8	45.9	291°19.0	25.8	97°13.4	57.7					Acamar	315°08.5	-40°11.0	
8	162°52.3	255°13.4	16.2	253°26.2	45.9	306°20.9	25.9	112°16.1	57.6					Menkar	314°01.8	4°12.5	
9	177°54.7	270°13.2	•• 16.3	268°26.7	•• 45.9	321°22.9	•• 26.1	127°18.7	•• 57.6					Mirfak	308°22.1	49°58.0	
10	192°57.2	285°13.0	16.4	283°27.1	45.9	336°24.8	26.3	142°21.4	57.5					Aldebaran	290°34.9	16°34.1	
11	207°59.7	300°12.8	16.5	298°27.5	45.9	351°26.7	26.5	157°24.1	57.5					Rigel	280°59.9	-8°10.0	
12	223°02.1	315°12.6	S27°16.6	313°27.9	S24°45.9	6°28.7	S13°26.7	172°26.7	N15°57.4					Capella	280°15.8	46°01.4	
13	238°04.6	330°12.5	16.6	328°28.3	45.9	21°30.6	26.8	187°29.4	57.4					Bellatrix	278°18.5	6°22.5	
14	253°07.0	345°12.3	16.7	343°28.7	45.9	36°32.5	27.0	202°32.0	57.3					Elnath	277°56.7	28°37.8	
15	268°09.5	0°12.1	•• 16.8	358°29.1	•• 45.9	51°34.5	•• 27.2	217°34.7	•• 57.3					Alnilam	275°33.6	-1°11.0	
16	283°12.0	15°11.9	16.8	13°29.5	45.9	66°36.4	27.4	232°37.4	57.2					Betelgeuse	270°47.7	7°24.7	
17	298°14.4	30°11.7	16.9	28°29.9	45.9	81°38.3	27.5	247°40.0	57.2					Canopus	263°50.3	-52°42.5	
18	313°16.9	45°11.5	S27°17.0	43°30.3	S24°45.9	96°40.3	S13°27.7	262°42.7	N15°57.1					Sirius	258°22.6	-16°45.4	
19	328°19.4	60°11.4	17.1	58°30.7	45.9	111°42.2	27.9	277°45.4	57.1					Adhara	255°02.6	-29°00.7	
20	343°21.8	75°11.2	17.1	73°31.2	45.9	126°44.1	28.1	292°48.0	57.0					Procyon	244°46.7	5°08.8	
21	358°24.3	90°11.0	•• 17.2	88°31.6	•• 46.0	141°46.1	•• 28.2	307°50.7	•• 57.0					Pollux	243°12.5	27°57.0	
22	13°26.8	105°10.8	17.2	103°32.0	46.0	156°48.0	28.4	322°53.3	56.9					Avior	234°12.9	-59°36.1	
23	28°29.2	120°10.6	17.3	118°32.4	46.0	171°49.9	28.6	337°56.0	56.9					Suhail	222°43.4	-43°33.0	
Mer.pass. 21:06			ν -0.2' d 0.1' m -4.58	ν 0.4' d 0.0' m 1.06	ν 1.9' d 0.2' m -1.68	ν 2.7' d -0.1' m -0.22											

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	43°31.7	135°10.5	S27°17.4	133°32.8	S24°46.0	186°51.8	S13°28.8	352°58.7	N15°56.8
1	58°34.1	150°10.3	17.4	148°33.2	46.0	201°53.8	28.9	8°01.3	56.8
2	73°36.6	165°10.1	17.5	163°33.6	46.0	216°55.7	29.1	23°04.0	56.7
3	88°39.1	180°09.9	•• 17.6	178°34.0	•• 46.0	231°57.6	•• 29.3	38°06.7	•• 56.7
4	103°41.5	195°09.8	17.6	193°34.4	46.0	246°59.6	29.5	53°09.3	56.6
5	118°44.0	210°09.6	17.7	208°34.8	46.0	262°01.5	29.6	68°12.8	56.6
6	133°46.5	225°09.4	S27°17.7	223°35.2	S24°46.0	277°03.4	S13°29.8	83°14.7	N15°56.5
7	148°48.9	240°09.2	17.8	238°35.6	46.0	292°05.4	30.0	98°17.3	56.5
8	163°51.4	255°09.1	17.8	253°36.0	46.0	307°07.3	30.2	113°20.0	56.4
9	178°53.9	270°08.9	•• 17.9	268°36.4	•• 46.0	322°09.2	•• 30.3	128°22.7	•• 56.4
10	193°56.3	285°08.7	17.9	283°36.9	46.0	337°11.2	30.5	143°25.3	56.3
11	208°58.8	300°08.6	18.0	298°37.3	46.0	352°13.1	30.7	158°28.0	56.3
12	224°01.3	315°08.4	S27°18.1	313°37.7	S24°46.0	7°15.0	S13°30.9	173°30.6	N15°56.2
13	239°03.7	330°08.2	18.1	328°38.1	46.0	22°17.0	31.0	188°33.3	56.2
14	254°06.2	345°08.1	18.1	343°38.5	46.0	37°18.9	31.2	203°36.0	56.1
15	269°08.6	0°07.9	•• 18.2	358°38.9	•• 45.9	52°20.8	•• 31.4	218°38.6	•• 56.1
16	284°11.1	15°07.7	18.2	13°39.3	45.9	67°22.7	31.6	233°41.3	56.0
17	299°13.6	30°07.6	18.3	28°39.7	45.9	82°24.7	31.7	248°44.0	56.0
18	314°16.0	45°07.4	S27°18.3	43°40.1	S24°45.9	97°26.6	S13°31.9	263°46.6	N15°55.9
19	329°18.5	60°07.3	18.4	58°40.5	45.9	112°28.5	32.1	278°49.3	55.9
20	344°21.0	75°07.1	18.4	73°40.9	45.9	127°30.5	32.3	293°52.0	55.8
21	359°23.4	90°06.9	•• 18.5	88°41.3	•• 45.9	142°32.4	•• 32.4	308°54.6	•• 55.8
22	14°25.9	105°06.8	18.5	103°41.7	45.9	157°34.3	32.6	323°57.3	55.7
23	29°28.4	120°06.6	18.5	118°42.1	45.9	172°36.3	32.8	339°00.0	55.7
Mer.pass. 21:02			ν -0.2' d 0.1' m -4.60	ν 0.4' d 0.0' m 1.07	ν 1.9' d 0.2' m -1.68	ν 2.7' d -0.1' m -0.22			

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	44°30.8	135°06.5	S27°18.6	133°42.5	S24°45.9	187°38.2	S13°33.0	354°02.6	N15°55.6
1	59°33.3	150°06.3	18.6	148°42.9	45.9	202°40.1	33.1	9°05.3	55.6
2	74°35.8	165°06.2	18.7	163°43.3	45.9	217°42.1	33.3	24°07.9	55.5
3	89°38.2	180°06.0	•• 18.7	178°43.7	•• 45.9	232°44.0	•• 33.5	39°10.6	•• 55.5
4	104°40.7	195°05.9	18.7	193°44.1	45.9	247°45.9	33.7	54°13.3	55.4
5	119°43.1	210°05.7	18.8	208°44.5	45.8	262°47.8	33.8	69°15.9	55.4
6	134°45.6	225°05.6	S27°18.8	223°44.9	S24°45.8	277°49.8	S13°34.0	84°18.6	N15°55.3
7	149°48.1	240°05.4	18.8	238°45.4	45.8	292°51.7	34.2	99°21.3	55.3
8	164°50.5	255°05.3	18.9	253°45.8	45.8	307°53.6	34.4	114°23.9	55.2
9	179°53.0	270°05.1	•• 18.9	268°46.2	•• 45.8	322°55.6	•• 34.5	129°26.6	•• 55.2
10	194°55.5	285°05.0	18.9	283°46.6	45.8	337°57.5	34.7	144°29.3	55.1
11	209°57.9	300°04.8	18.9	298°47.0	45.8	352°59.4	34.9	159°31.9	55.1
12	225°00.4	315°04.7	S27°19.0	313°47.4	S24°45.8	8°01.4	S13°35.1	174°34.6	N15°55.0
13	240°02.9	330°04.5	19.0	328°47.8	45.7	23°03.3	35.2	189°37.3	55.0
14	255°05.3	345°04.4	19.0	343°48.2	45.7	38°05.2	35.4	204°39.9	54.9
15	270°07.8	0°04.3	•• 19.0	358°48.6	•• 45.7	53°07.2	•• 35.6	219°42.6	•• 54.9
16	285°10.2	15°04.1	19.1	13°49.0	45.7	68°09.1	35.8	234°45.3	54.8
17	300°12.7	30°04.0	19.1	28°49.4	45.7	83°11.0	35.9	249°47.9	54.8
18	315°15.2	45°03.8	S27°19.1	43°49.8	S24°45.7	98°12.9	S13°36.1	264°50.6	N15°54.7
19	330°17.6	60°03.7	19.1	58°50.2	45.6	113°14.9	36.3	279°53.3	54.7
20	345°20.1	75°03.6	19.1	73°50.6	45.6	128°16.8	36.5	294°55.9	54.6
21	0°22.6	90°03.4	•• 19.2	88°51.0	•• 45.6	143°18.7	•• 36.6	309°58.6	•• 54.6
22	15°25.0	105°03.3	19.2	103°51.4	45.6	158°20.7	36.8	325°01.2	54.5
23	30°27.5	120°03.2	19.2	118°51.8	45.6	173°22.6	37.0	340°03.9	54.5
Mer.pass. 20:58			ν -0.2' d 0.0' m -4.61	ν 0.4' d -0.0' m 1.07	ν 1.9' d 0.2' m -1.68	ν 2.7' d -0.1' m -0.23			

Nov 03 Sat	SHA	Mer.pass
Venus	92°42.4	14:59
Mars	90°50.4	15:06
Jupiter	143°33.0	11:34
Saturn	309°22.2	00:32

Nov 04 Sun	SHA	Mer.pass
Venus	91°38.8	14:59
Mars	90°01.1	15:05
Jupiter	143°20.2	11:31
Saturn	309°27.0	00:28

Nov 05 Mon	SHA	Mer.pass
Venus	90°35.6	15:00
Mars	89°11.7	15:05
Jupiter	143°07.4	11:28
Saturn	309°31.8	00:24

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	184°06.8	S15°04.7	228°16.6	9.9'	S03°07.4	14.5' 60.0'
1	199°06.8	05.5	242°45.5	9.9'	03°21.9	14.5' 60.0'
2	214°06.8	06.3	257°14.4	9.8'	03°36.4	14.5' 60.1'
3	229°06.8	· · 07.1	271°43.3	9.8'	03°50.9	14.5' 60.1'
4	244°06.8	07.9	286°12.0	9.7'	04°05.4	14.5' 60.1'
5	259°06.8	08.6	300°40.8	9.7'	04°19.9	14.5' 60.2'
6	274°06.8	S15°09.4	315°09.4	9.6'	S04°34.4	14.5' 60.2'
7	289°06.8	10.2	329°38.0	9.5'	04°48.8	14.5' 60.2'
8	304°06.8	11.0	344°06.5	9.5'	05°03.3	14.4' 60.3'
9	319°06.8	· · 11.7	358°35.0	9.4'	05°17.8	14.4' 60.3'
10	334°06.8	12.5	13°03.4	9.3'	05°32.2	14.4' 60.3'
11	349°06.8	13.3	27°31.7	9.3'	05°46.6	14.4' 60.4'
12	4°06.8	S15°14.1	42°00.0	9.2'	S06°01.0	14.4' 60.4'
13	19°06.8	14.8	56°28.2	9.1'	06°15.4	14.4' 60.4'
14	34°06.8	15.6	70°56.4	9.1'	06°29.7	14.3' 60.5'
15	49°06.7	· · 16.4	85°24.4	9.0'	06°44.1	14.3' 60.5'
16	64°06.7	17.2	99°52.4	8.9'	06°58.4	14.3' 60.5'
17	79°06.7	17.9	114°20.4	8.9'	07°12.7	14.3' 60.5'
18	94°06.7	S15°18.7	128°48.2	8.8'	S07°26.9	14.2' 60.6'
19	109°06.7	19.5	143°16.0	8.7'	07°41.1	14.2' 60.6'
20	124°06.7	20.3	157°43.7	8.6'	07°55.3	14.2' 60.6'
21	139°06.7	· · 21.0	172°11.3	8.6'	08°09.5	14.1' 60.7'
22	154°06.7	21.8	186°38.9	8.5'	08°23.6	14.1' 60.7'
23	169°06.7	22.6	201°06.4	8.4'	08°37.7	14.0' 60.7'
SD = 16.1' $d = 0.8'$ SD = 16.4'						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	184°06.7	S15°23.3	215°33.8	8.3'	S08°51.7	14.0' 60.7'
1	199°06.7	24.1	230°01.2	8.3'	09°05.7	14.0' 60.8'
2	214°06.7	24.9	244°28.4	8.2'	09°19.6	13.9' 60.8'
3	229°06.7	· · 25.6	258°55.6	8.1'	09°33.5	13.9' 60.8'
4	244°06.6	26.4	273°22.7	8.0'	09°47.4	13.8' 60.8'
5	259°06.6	27.2	287°49.7	7.9'	10°01.2	13.8' 60.9'
6	274°06.6	S15°27.9	302°16.7	7.9'	S10°15.0	13.7' 60.9'
7	289°06.6	28.7	316°43.6	7.8'	10°28.7	13.6' 60.9'
8	304°06.6	29.5	331°10.3	7.7'	10°42.3	13.6' 60.9'
9	319°06.6	· · 30.2	345°37.0	7.6'	10°55.9	13.5' 60.9'
10	334°06.6	31.0	0°03.7	7.5'	11°09.4	13.5' 61.0'
11	349°06.6	31.8	14°30.2	7.5'	11°22.9	13.4' 61.0'
12	4°06.5	S15°32.5	28°56.7	7.4'	S11°36.3	13.3' 61.0'
13	19°06.5	33.3	43°23.0	7.3'	11°49.6	13.3' 61.0'
14	34°06.5	34.1	57°49.3	7.2'	12°02.9	13.2' 61.0'
15	49°06.5	· · 34.8	72°15.5	7.1'	12°16.1	13.1' 61.1'
16	64°06.5	35.6	86°41.7	7.0'	12°29.2	13.1' 61.1'
17	79°06.5	36.4	101°07.7	7.0'	12°42.3	13.0' 61.1'
18	94°06.5	S15°37.1	115°33.7	6.9'	S12°55.3	12.9' 61.1'
19	109°06.4	37.9	129°59.5	6.8'	13°08.2	12.8' 61.1'
20	124°06.4	38.6	144°25.3	6.7'	13°21.0	12.7' 61.1'
21	139°06.4	· · 39.4	158°51.0	6.6'	13°33.8	12.7' 61.2'
22	154°06.4	40.2	173°16.6	6.5'	13°46.4	12.6' 61.2'
23	169°06.4	40.9	187°42.2	6.4'	13°59.0	12.5' 61.2'
SD = 16.1' $d = 0.8'$ SD = 16.6'						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	184°06.4	S15°41.7	202°07.6	6.4'	S14°11.5	12.4' 61.2'
1	199°06.3	42.4	216°33.0	6.3'	14°23.9	12.3' 61.2'
2	214°06.3	43.2	230°58.2	6.2'	14°36.2	12.2' 61.2'
3	229°06.3	· · 44.0	245°23.4	6.1'	14°48.4	12.1' 61.2'
4	244°06.3	44.7	259°48.5	6.0'	15°00.5	12.0' 61.3'
5	259°06.3	45.5	274°13.5	5.9'	15°12.5	11.9' 61.3'
6	274°06.3	S15°46.2	288°38.5	5.8'	S15°24.5	11.8' 61.3'
7	289°06.2	47.0	303°03.3	5.8'	15°36.3	11.7' 61.3'
8	304°06.2	47.7	317°28.1	5.7'	15°48.0	11.6' 61.3'
9	319°06.2	· · 48.5	331°52.7	5.6'	15°59.6	11.5' 61.3'
10	334°06.2	49.3	346°17.3	5.5'	16°11.1	11.4' 61.3'
11	349°06.1	50.0	0°41.8	5.4'	16°22.5	11.3' 61.3'
12	4°06.1	S15°50.8	15°06.2	5.3'	S16°33.8	11.2' 61.3'
13	19°06.1	51.5	29°30.6	5.3'	16°45.0	11.1' 61.3'
14	34°06.1	52.3	43°54.8	5.2'	16°56.0	10.9' 61.3'
15	49°06.1	· · 53.0	58°19.0	5.1'	17°07.0	10.8' 61.3'
16	64°06.0	53.8	72°43.1	5.0'	17°17.8	10.7' 61.3'
17	79°06.0	54.5	87°07.1	4.9'	17°28.5	10.6' 61.4'
18	94°06.0	S15°55.3	101°31.0	4.8'	S17°39.1	10.5' 61.4'
19	109°06.0	56.0	115°54.9	4.8'	17°49.5	10.3' 61.4'
20	124°05.9	56.8	130°18.7	4.7'	17°59.9	10.2' 61.4'
21	139°05.9	· · 57.5	144°42.4	4.6'	18°10.1	10.1' 61.4'
22	154°05.9	58.3	159°06.0	4.5'	18°20.1	9.9' 61.4'
23	169°05.9	59.0	173°29.5	4.5'	18°30.1	9.8' 61.4'
SD = 16.1' $d = 0.8'$ SD = 16.7'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:19	07:44	09:18	14:08	15:42	17:06
N 70°	06:13	07:29	08:47	14:39	15:57	17:12
68°	06:09	07:17	08:25	15:01	16:09	17:17
66°	06:04	07:07	08:07	15:19	16:19	17:21
64°	06:01	06:58	07:53	15:33	16:28	17:25
62°	05:58	06:51	07:41	15:45	16:35	17:28
60°	05:55	06:44	07:30	15:56	16:42	17:32
N 58°	05:52	06:39	07:22	16:05	16:47	17:34
56°	05:49	06:34	07:14	16:13	16:53	17:37
54°	05:47	06:29	07:07	16:20	16:57	17:40
52°	05:44	06:25	07:01	16:26	17:02	17:42
50°	05:42	06:21	06:55	16:31	17:06	17:44
45°	05:37	06:12	06:43	16:44	17:15	17:49
N 40°	05:32	06:04	06:33	16:54	17:22	17:54
35°	05:28	05:58	06:24	17:03	17:29	17:59
30°	05:23	05:51	06:16	17:10	17:35	18:04
20°	05:14	05:40	06:03	17:24	17:47	18:13
N 10°	05:04	05:29	05:51	17:36	17:58	18:23
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:41	05:07	05:29	17:58	18:20	18:46
20°	04:26	04:54	05:17	18:11	18:34	19:01
30°	04:07	04:37	05:03	18:25	18:50	19:21
35°	03:55	04:27	04:55	18:33	19:00	19:33
40°	03:40	04:16	04:45	18:43	19:12	19:48
45°	03:21	04:02	04:34	18:54	19:26	20:07
S 50°	02:57	03:44	04:21	19:07	19:44	20:32
52°	02:45	03:36	04:15	19:14	19:53	20:44
54°	02:30	03:26	04:08	19:21	20:03	20:59
56°	02:13	03:15	04:00	19:28	20:14	21:17
58°	01:52	03:03	03:51	19:37	20:27	21:39
S 60°	01:22	02:48	03:41	19:48	20:42	22:10

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	03:47	06:10	09:06	13:52	13:22	12:27
N 70°	03:41	05:52	08:20	14:01	13:43	13:15
68°	03:36	05:37	07:50	14:09	13:59	13:46
66°	03:32	05:26	07:28	14:15	14:12	14:09
64°	03:28	05:16	07:10	14:21	14:23	14:28
62°	03:25	05:08	06:56	14:25	14:33	14:43
60°	03:23	05:02	06:44	14:30	14:41	14:56
N 58°	03:20	04:55	06:34	14:33	14:48	15:07
56°	03:18	04:50	06:25	14:37	14:54	15:17
54°	03:16	04:45	06:17	14:40	15:00	15:26
52°	03:15	04:41	06:10	14:42	15:05	15:33
50°	03:13	04:37	06:04	14:45	15:10	15:40
45°	03:10	04:29	05:50	14:50	15:20	15:55
N 40°	03:07	04:22	05:39	14:55	15:29	16:08
35°	03:05	04:16	05:29	14:59	15:36	16:18
30°	03:03	04:11	05:21	15:02	15:42	16:28
20°	02:59	04:02	05:07	15:08	15:54	16:44
N 10°	02:56	03:54	04:55	15:13	16:04	16:58
0°	02:54	03:47	04:43	15:18	16:13	17:11
S 10°	02:51	03:40	04:32	15:23	16:22	17:25
20°	02:48	03:32	04:20	15:29	16:32	17:39
30°	02:45	03:23	04:06	15:35	16:44	17:55
35°	02:43	03:19	03:58	15:38	16:50	18:05
40°	02:41	03:13	03:49	15:42	16:58	18:16
45°	02:38	03:07	03:39	15:47	17:07	18:29
S 50°	02:35	02:59	03:27	15:53	17:18	18:44
52°	02:34	02:56	03:21	15:55	17:23	18:52
54°	02:33	02:52	03:15	15:58	17:28	19:00
56°	02:31	02:48	03:08	16:01	17:34	19:09
58°	02:29	02:43	03:00	16:05	17:41	19:20
S 60°	02:27	02:38	02:51	16:09	17:49	19:32

Day	Sun		Mer. Pass hh:mm	Moon		Age 27-29 15-2%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
03	16:27	16:27	11:44	09:06	21:32	
04	16:27	16:26	11:44	10:00	22:28	
05	16:25	16:25	11:44	10:57	23:27	

November 06, 07, 08 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	45°30.0	135°03.0	S27°19.2	133°52.2	S24°45.6	188°24.5	S13°37.2	355°06.6	N15°54.4	Alpheratz	357°30.4	29°15.6
1	60°32.4	150°02.9	19.2	148°52.6	45.5	203°26.5	37.3	10°09.2	54.4	Ankaa	353°03.1	-42°08.6
2	75°34.9	165°02.8	19.2	163°53.0	45.5	218°28.4	37.5	25°11.9	54.3	Schedar	349°25.9	56°42.4
3	90°37.4	180°02.7	•• 19.2	178°53.4	•• 45.5	233°30.3	•• 37.7	40°14.6	•• 54.3	Diphda	348°43.2	-17°49.3
4	105°39.8	195°02.5	19.3	193°53.8	45.5	248°32.3	37.9	55°17.2	54.2	Achernar	335°17.0	-57°05.1
5	120°42.3	210°02.4	19.3	208°54.2	45.4	263°34.2	38.0	70°19.9	54.2	Hamal	327°46.4	23°36.3
6	135°44.7	225°02.3	S27°19.3	223°54.6	S24°45.4	278°36.1	S13°38.2	85°22.6	N15°54.1	Polaris	311°21.7	89°23.2
7	150°47.2	240°02.2	19.3	238°55.0	45.4	293°38.0	38.4	100°25.2	54.1	Acamar	315°08.5	-40°11.1
8	165°49.7	255°02.0	19.3	253°55.4	45.4	308°40.0	38.6	115°27.9	54.0	Menkar	314°01.8	4°12.5
9	180°52.1	270°01.9	•• 19.3	268°55.8	•• 45.4	323°41.9	•• 38.7	130°30.6	•• 54.0	Mirfak	308°22.1	49°58.0
10	195°54.6	285°01.8	19.3	283°56.2	45.3	338°43.8	38.9	145°33.2	53.9	Aldebaran	290°34.9	16°34.1
11	210°57.1	300°01.7	19.3	298°56.6	45.3	353°45.8	39.1	160°35.9	53.9	Rigel	280°59.9	-8°10.0
12	225°59.5	315°01.6	S27°19.3	313°57.0	S24°45.3	8°47.7	S13°39.3	175°38.6	N15°53.8	Capella	280°15.8	46°01.4
13	241°02.0	330°01.4	19.3	328°57.4	45.3	23°49.6	39.4	190°41.2	53.8	Bellatrix	278°18.5	6°22.5
14	256°04.5	345°01.3	19.3	343°57.8	45.2	38°51.6	39.6	205°43.9	53.7	Elnath	277°56.7	28°37.8
15	271°06.9	0°01.2	•• 19.3	358°58.2	•• 45.2	53°53.5	•• 39.8	220°46.6	•• 53.7	Alnilam	275°33.6	-1°11.0
16	286°09.4	15°01.1	19.3	13°58.6	45.2	68°55.4	40.0	235°49.2	53.6	Betelgeuse	270°47.7	7°24.7
17	301°11.9	30°01.0	19.3	28°59.0	45.2	83°57.4	40.1	250°51.9	53.6	Canopus	263°50.3	-52°42.5
18	316°14.3	45°00.9	S27°19.3	43°59.4	S24°45.1	98°59.3	S13°40.3	265°54.6	N15°53.5	Sirius	258°22.6	-16°45.4
19	331°16.8	60°00.8	19.3	58°59.8	45.1	114°01.2	40.5	280°57.2	53.5	Adhara	255°02.6	-29°00.7
20	346°19.2	75°00.7	19.3	74°00.2	45.1	129°03.1	40.7	295°59.9	53.4	Procyon	244°46.7	5°08.8
21	1°21.7	90°00.5	•• 19.3	89°00.6	•• 45.0	144°05.1	•• 40.8	311°02.6	•• 53.4	Pollux	243°12.5	27°57.0
22	16°24.2	105°00.4	19.3	104°01.0	45.0	159°07.0	41.0	326°05.2	53.3	Avior	234°12.9	-59°36.1
23	31°26.6	120°00.3	19.3	119°01.4	45.0	174°08.9	41.2	341°07.9	53.3	Suhail	222°43.4	-43°33.0
Mer.pass. 20:55		ν -0.1' d 0.0' m -4.62		ν 0.4' d -0.0' m 1.07		ν 1.9' d 0.2' m -1.68		ν 2.7' d -0.1' m -0.24		Miaplacidus	221°37.2	-69°50.2
Alphard 217°44.0 -8°47.3 Regulus 207°30.5 11°49.1 Dubhe 193°36.9 61°35.1 Denebola 182°21.3 14°24.2 Gienah 175°39.8 -17°42.4 Acrux 172°56.0 -63°15.7 Gacrux 171°47.6 -57°16.7 Alioth 166°10.3 55°47.7 Spica 158°18.5 -11°18.9 Alkaid 152°49.6 49°09.8 Hadar 148°30.9 -60°30.9 Menkent 147°53.4 -36°30.9 Arcturus 145°44.8 19°01.7 Rigil Kent. 139°35.5 -60°57.4 Kochab 137°21.5 74°01.9 Zuben'ubi 136°52.0 -16°09.8 Alphecca 126°00.9 26°37.0 Antares 112°11.4 -26°29.7 Atria 107°02.5 -69°04.8 Sabik 101°58.6 -15°45.5 Shaula 96°05.5 -37°07.4 Rasalhague 95°55.2 12°32.6 Eltanin 90°40.8 51°29.4 Kaus Aust. 83°27.7 -34°22.1 Vega 80°30.9 38°49.0 Nunki 75°43.2 -26°15.4 Altair 61°56.3 8°57.1 Peacock 52°59.9 -56°38.3 Deneb 49°23.1 45°23.7 Enif 33°34.9 10°01.0 Al Na'ir 27°28.1 -46°49.0 Fomalhaut 15°10.2 -29°27.8 Scheat 13°41.2 28°15.0 Markab 13°25.9 15°22.3												
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	46°29.1	135°00.2	S27°19.3	134°01.8	S24°44.9	189°10.9	S13°41.4	356°10.6	N15°53.2			
1	61°31.6	150°00.1	19.3	149°02.2	44.9	204°12.8	41.5	11°13.2	53.2			
2	76°34.0	165°00.0	19.2	164°02.6	44.9	219°14.7	41.7	26°15.9	53.1			
3	91°36.5	179°59.9	•• 19.2	179°03.0	•• 44.9	234°16.7	•• 41.9	41°18.6	•• 53.1			
4	106°39.0	194°59.8	19.2	194°03.4	44.8	249°18.6	42.0	56°21.2	53.0			
5	121°41.4	209°59.7	19.2	209°03.8	44.8	264°20.5	42.2	71°23.9	53.0			
6	136°43.9	224°59.6	S27°19.2	224°04.2	S24°44.8	279°22.5	S13°42.4	86°26.6	N15°52.9			
7	151°46.3	239°59.5	19.2	239°04.6	44.7	294°24.4	42.6	101°29.2	52.8			
8	166°48.8	254°59.4	19.2	254°05.0	44.7	309°26.3	42.7	116°31.9	52.8			
9	181°51.3	269°59.3	•• 19.1	269°05.4	•• 44.7	324°28.2	•• 42.9	131°34.6	•• 52.7			
10	196°53.7	284°59.2	19.1	284°05.8	44.6	339°30.2	43.1	146°37.2	52.7			
11	211°56.2	299°59.1	19.1	299°06.2	44.6	354°32.1	43.3	161°39.9	52.6			
12	226°58.7	314°59.0	S27°19.1	314°06.6	S24°44.5	9°34.0	S13°43.4	176°42.6	N15°52.6			
13	242°01.1	329°59.0	19.1	329°07.0	44.5	24°36.0	43.6	191°45.2	52.5			
14	257°03.6	344°58.9	19.0	344°07.4	44.5	39°37.9	43.8	206°47.9	52.5			
15	272°06.1	359°58.8	•• 19.0	359°07.8	•• 44.4	54°39.8	•• 44.0	221°50.6	•• 52.4			
16	287°08.5	14°58.7	19.0	14°08.2	44.4	69°41.8	44.1	236°53.2	52.4			
17	302°11.0	29°58.6	19.0	29°08.6	44.4	84°43.7	44.3	251°55.9	52.3			
18	317°13.5	44°58.5	S27°18.9	44°09.0	S24°44.3	99°45.6	S13°44.5	266°58.6	N15°52.3			
19	332°15.9	59°58.4	18.9	59°09.4	44.3	114°47.6	44.7	282°01.2	52.2			
20	347°18.4	74°58.4	18.9	74°09.7	44.2	129°49.5	44.8	297°03.9	52.2			
21	2°20.8	89°58.3	•• 18.9	89°10.1	•• 44.2	144°51.4	•• 45.0	312°06.6	•• 52.1			
22	17°23.3	104°58.2	18.8	104°10.5	44.2	159°53.3	45.2	327°09.2	52.1			
23	32°25.8	119°58.1	18.8	119°10.9	44.1	174°55.3	45.3	342°11.9	52.0			
Mer.pass. 20:51		ν -0.1' d -0.0' m -4.63		ν 0.4' d -0.0' m 1.07		ν 1.9' d 0.2' m -1.68		ν 2.7' d -0.1' m -0.24				
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	47°28.2	134°58.0	S27°18.8	134°11.3	S24°44.1	189°57.2	S13°45.5	357°14.6	N15°52.0			
1	62°30.7	149°58.0	18.7	149°11.7	44.0	204°59.1	45.7	12°17.2	51.9			
2	77°33.2	164°57.9	18.7	164°12.1	44.0	220°01.1	45.9	27°19.9	51.9			
3	92°35.6	179°57.8	•• 18.7	179°12.5	•• 44.0	235°03.0	•• 46.0	42°22.6	•• 51.8			
4	107°38.1	194°57.7	18.6	194°12.9	43.9	250°04.9	46.2	57°25.2	51.8			
5	122°40.6	209°57.7	18.6	209°13.3	43.9	265°06.9	46.4	72°27.9	51.7			
6	137°43.0	224°57.6	S27°18.6	224°13.7	S24°43.8	280°08.8	S13°46.6	87°30.6	N15°51.7			
7	152°45.5	239°57.5	18.5	239°14.1	43.8	295°10.7	46.7	102°33.2	51.6			
8	167°48.0	254°57.5	18.5	254°14.5	43.7	310°12.7	46.9	117°35.9	51.6			
9	182°50.4	269°57.4	•• 18.4	269°14.9	•• 43.7	325°14.6	•• 47.1	132°38.6	•• 51.5			
10	197°52.9	284°57.3	18.4	284°15.3	43.6	340°16.5	47.3	147°41.2	51.5			
11	212°55.3	299°57.3	18.3	299°15.7	43.6	355°18.4	47.4	162°43.9	51.4			
12	227°57.8	314°57.2	S27°18.3	314°16.1	S24°43.5	10°20.4	S13°47.6	177°46.6	N15°51.4			
13	243°00.3	329°57.1	18.3	329°16.5	43.5	25°22.3	47.8	192°49.3	51.3			
14	258°02.7	344°57.1	18.2	344°16.9	43.4	40°24.2	47.9	207°51.9	51.3			
15	273°05.2	359°57.0	•• 18.2	359°17.3	•• 43.4	55°26.2	•• 48.1	222°54.6	•• 51.2			
16	288°07.7	14°56.9	18.1	14°17.7	43.3	70°28.1	48.3	237°57.3	51.2			
17	303°10.1	29°56.9	18.1	29°18.1	43.3	85°30.0	48.5	252°59.9	51.1			
18	318°12.6	44°56.8	S27°18.0	44°18.4	S24°43.2	100°32.0	S13°48.6	268°02.6	N15°51.1			
19	333°15.1	59°56.8	18.0	59°18.8	43.2	115°33.9	48.8	283°05.3	51.0			
20	348°17.5	74°56.7	17.9	74°19.2	43.1	130°35.8	49.0	298°07.9	51.0			
21	3°20.0	89°56.7	•• 17.9	89°19.6	•• 43.1	145°37.8	•• 49.2	313°10.6	•• 50.9			
22	18°22.5	104°56.6	17.8	104°2								

DUT1 = UT1-UTC = +0.2135 sec $\Delta T = TT-UT1 = +68.9705$ sec

2029 November 06 to Nov. 08 UT

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	184°05.8	S15°59.8	187°53.0	4.4'	S18°39.9	9.7' 61.4'
1	199°05.8	16°00.5	202°16.4	4.3'	18°49.6	9.5' 61.4'
2	214°05.8	01.3	216°39.7	4.2'	18°59.1	9.4' 61.4'
3	229°05.8	· · 02.0	231°02.9	4.2'	19°08.5	9.3' 61.4'
4	244°05.7	02.8	245°26.1	4.1'	19°17.8	9.1' 61.4'
5	259°05.7	03.5	259°49.2	4.0'	19°26.9	9.0' 61.4'
6	274°05.7	S16°04.3	274°12.2	4.0'	S19°35.9	8.8' 61.3'
7	289°05.6	05.0	288°35.2	3.9'	19°44.7	8.7' 61.3'
8	304°05.6	05.7	302°58.1	3.8'	19°53.4	8.5' 61.3'
9	319°05.6	· · 06.5	317°20.9	3.8'	20°01.9	8.4' 61.3'
10	334°05.6	07.2	331°43.7	3.7'	20°10.3	8.2' 61.3'
11	349°05.5	08.0	346°06.4	3.6'	20°18.5	8.1' 61.3'
12	4°05.5	S16°08.7	0°29.1	3.6'	S20°26.6	7.9' 61.3'
13	19°05.5	09.5	14°51.7	3.5'	20°34.5	7.8' 61.3'
14	34°05.4	10.2	29°14.2	3.5'	20°42.3	7.6' 61.3'
15	49°05.4	· · 10.9	43°36.7	3.4'	20°49.9	7.5' 61.3'
16	64°05.4	11.7	57°59.1	3.4'	20°57.4	7.3' 61.3'
17	79°05.3	12.4	72°21.5	3.3'	21°04.7	7.1' 61.3'
18	94°05.3	S16°13.2	86°43.8	3.3'	S21°11.8	7.0' 61.3'
19	109°05.3	13.9	101°06.1	3.2'	21°18.8	6.8' 61.2'
20	124°05.2	14.6	115°28.3	3.2'	21°25.6	6.7' 61.2'
21	139°05.2	· · 15.4	129°50.5	3.1'	21°32.3	6.5' 61.2'
22	154°05.2	16.1	144°12.6	3.1'	21°38.8	6.3' 61.2'
23	169°05.1	16.9	158°34.7	3.1'	21°45.1	6.2' 61.2'
SD = 16.1' d = 0.7' SD = 16.7'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	184°05.1	S16°17.6	172°56.8	3.0'	S21°51.3	6.0' 61.2'
1	199°05.1	18.3	187°18.8	3.0'	21°57.3	5.8' 61.2'
2	214°05.0	19.1	201°40.8	3.0'	22°03.1	5.7' 61.1'
3	229°05.0	· · 19.8	216°02.7	2.9'	22°08.7	5.5' 61.1'
4	244°05.0	20.5	230°24.6	2.9'	22°14.2	5.3' 61.1'
5	259°04.9	21.3	244°46.5	2.9'	22°19.5	5.1' 61.1'
6	274°04.9	S16°22.0	259°08.4	2.8'	S22°24.7	5.0' 61.1'
7	289°04.8	22.7	273°30.3	2.8'	22°29.6	4.8' 61.1'
8	304°04.8	23.5	287°52.1	2.8'	22°34.4	4.6' 61.0'
9	319°04.8	· · 24.2	302°13.9	2.8'	22°39.0	4.4' 61.0'
10	334°04.7	24.9	316°35.7	2.8'	22°43.5	4.3' 61.0'
11	349°04.7	25.7	330°57.4	2.8'	22°47.8	4.1' 61.0'
12	4°04.7	S16°26.4	345°19.2	2.7'	S22°51.9	3.9' 61.0'
13	19°04.6	27.1	359°40.9	2.7'	22°55.8	3.7' 60.9'
14	34°04.6	27.9	14°02.7	2.7'	22°59.5	3.6' 60.9'
15	49°04.5	· · 28.6	28°24.4	2.7'	23°03.1	3.4' 60.9'
16	64°04.5	29.3	42°46.1	2.7'	23°06.5	3.2' 60.9'
17	79°04.5	30.1	57°07.9	2.7'	23°09.7	3.0' 60.8'
18	94°04.4	S16°30.8	71°29.6	2.7'	S23°12.7	2.9' 60.8'
19	109°04.4	31.5	85°51.4	2.7'	23°15.6	2.7' 60.8'
20	124°04.3	32.2	100°13.1	2.8'	23°18.3	2.5' 60.8'
21	139°04.3	· · 33.0	114°34.8	2.8'	23°20.8	2.3' 60.8'
22	154°04.2	33.7	128°56.6	2.8'	23°23.1	2.1' 60.7'
23	169°04.2	34.4	143°18.4	2.8'	23°25.2	2.0' 60.7'
SD = 16.1' d = 0.7' SD = 16.7'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	184°04.2	S16°35.1	157°40.2	2.8'	S23°27.2	1.8' 60.7'
1	199°04.1	35.9	172°02.0	2.8'	23°29.0	1.6' 60.6'
2	214°04.1	36.6	186°23.8	2.9'	23°30.6	1.4' 60.6'
3	229°04.0	· · 37.3	200°45.7	2.9'	23°32.0	1.3' 60.6'
4	244°04.0	38.0	215°07.6	2.9'	23°33.3	1.1' 60.6'
5	259°03.9	38.8	229°29.5	2.9'	23°34.4	0.9' 60.5'
6	274°03.9	S16°39.5	243°51.4	3.0'	S23°35.3	0.7' 60.5'
7	289°03.8	40.2	258°13.4	3.0'	23°36.0	0.6' 60.5'
8	304°03.8	40.9	272°35.4	3.0'	23°36.6	0.4' 60.4'
9	319°03.7	· · 41.7	286°57.5	3.1'	23°37.0	0.2' 60.4'
10	334°03.7	42.4	301°19.5	3.1'	23°37.2	0.0' 60.4'
11	349°03.6	43.1	315°41.7	3.2'	23°37.2	-0.1' 60.4'
12	4°03.6	S16°43.8	330°03.9	3.2'	S23°37.0	-0.3' 60.3'
13	19°03.6	44.5	344°26.1	3.3'	23°36.7	-0.5' 60.3'
14	34°03.5	45.2	358°48.4	3.3'	23°36.2	-0.7' 60.3'
15	49°03.5	· · 46.0	13°10.7	3.4'	23°35.6	-0.8' 60.2'
16	64°03.4	46.7	27°33.1	3.4'	23°34.7	-1.0' 60.2'
17	79°03.4	47.4	41°55.5	3.5'	23°33.7	-1.2' 60.2'
18	94°03.3	S16°48.1	56°18.0	3.6'	S23°32.6	-1.3' 60.1'
19	109°03.3	48.8	70°40.5	3.6'	23°31.2	-1.5' 60.1'
20	124°03.2	49.6	85°03.1	3.7'	23°29.7	-1.7' 60.1'
21	139°03.2	· · 50.3	99°25.8	3.7'	23°28.0	-1.8' 60.0'
22	154°03.1	51.0	113°48.6	3.8'	23°26.2	-2.0' 60.0'
23	169°03.1	51.7	128°11.4	3.9'	23°24.2	-2.2' 60.0'
SD = 16.1' d = 0.7' SD = 16.5'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:31	07:58	09:40	13:46	15:28	16:55
N 70°	06:24	07:41	09:03	14:23	15:45	17:02
68°	06:18	07:27	08:38	14:49	15:59	17:08
66°	06:13	07:16	08:18	15:08	16:10	17:13
64°	06:08	07:07	08:02	15:24	16:20	17:18
62°	06:05	06:59	07:49	15:37	16:28	17:22
60°	06:01	06:51	07:38	15:48	16:35	17:25
N 58°	05:58	06:45	07:29	15:58	16:41	17:29
56°	05:55	06:39	07:20	16:06	16:47	17:32
54°	05:52	06:34	07:13	16:14	16:52	17:35
52°	05:49	06:30	07:06	16:21	16:57	17:37
50°	05:47	06:25	07:00	16:27	17:01	17:40
45°	05:41	06:16	06:47	16:40	17:11	17:46
N 40°	05:35	06:08	06:36	16:51	17:19	17:52
35°	05:30	06:00	06:27	17:00	17:27	17:57
30°	05:25	05:54	06:19	17:08	17:33	18:02
20°	05:15	05:42	06:05	17:23	17:46	18:12
N 10°	05:05	05:30	05:52	17:35	17:57	18:22
0°	04:54	05:19	05:40	17:47	18:09	18:34
S 10°	04:41	05:06	05:28	17:59	18:21	18:47
20°	04:25	04:52	05:15	18:12	18:35	19:03
30°	04:04	04:35	05:01	18:27	18:53	19:23
35°	03:52	04:25	04:52	18:36	19:03	19:37
40°	03:36	04:12	04:42	18:46	19:16	19:52
45°	03:16	03:57	04:30	18:58	19:31	20:12
S 50°	02:50	03:39	04:16	19:12	19:50	20:39
52°	02:37	03:30	04:09	19:19	19:59	20:52
54°	02:22	03:19	04:02	19:27	20:10	21:08
56°	02:03	03:08	03:54	19:35	20:21	21:27
58°	01:38	02:54	03:44	19:45	20:35	21:53
S 60°	01:04	02:38	03:33	19:56	20:52	22:30

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	10:22	■	■	13:21	■	■
66°	09:37	11:49	13:30	14:07	14:08	14:41
64°	09:07	10:57	12:19	14:37	15:00	15:52
62°	08:45	10:26	11:43	15:00	15:32	16:28
60°	08:27	10:02	11:17	15:19	15:56	16:54
N 58°	08:12	09:43	10:57	15:34	16:15	17:14
56°	07:59	09:27	10:40	15:48	16:31	17:31
54°	07:48	09:14	10:25	15:59	16:45	17:45
52°	07:39	09:02	10:13	16:10	16:57	17:58
50°	07:30	08:51	10:02	16:19	17:08	18:09
45°	07:12	08:29	09:39	16:38	17:30	18:32
N 40°	06:57	08:12	09:20	16:54	17:48	18:50
35°	06:44	07:57	09:05	17:07	18:03	19:06
30°	06:33	07:44	08:51	17:19	18:16	19:19
20°	06:14	07:22	08:28	17:39	18:39	19:42
N 10°	05:58	07:04	08:08	17:56	18:58	20:02
0°	05:43	06:46	07:50	18:13	19:17	20:20
S 10°	05:28	06:29	07:32	18:29	19:35	20:38
20°	05:12	06:10	07:12	18:47	19:54	20:58
30°	04:54	05:49	06:49	19:07	20:17	21:20
35°	04:44	05:36	06:36	19:19	20:30	21:34
40°	04:32	05:22	06:20	19:33	20:45	21:49
45°	04:18	05:05	06:02	19:49	21:03	22:07
S 50°	04:01	04:45	05:40	20:09	21:25	22:29
52°	03:53	04:35	05:29	20:18	21:36	22:40
54°	03:44	04:24	05:17	20:29	21:48	22:52
56°	03:34	04:11	05:03	20:41	22:02	23:06
58°	03:23	03:57	04:46	20:55	22:18	23:22
S 60°	03:10	03:40	04:27	21:12	22:38	23:41

Day	Sun		Mer. Pass hh:mm	Moon		Age 0-2 0-5%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
06	16:23	16:22	11:44	11:58	—:—	
07	16:20	16:19	11:44	13:01	00:29	
08	16:17	16:14	11:44	14:05	01:33	

November 09, 10, 11 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	48°27.4	134°56.5	S27°17.7	134°20.8	S24°42.9	190°43.5	S13°49.7	358°18.6	N15°50.8	Alpheratz	357°30.4	29°15.6											
1	63°29.8	149°56.5	17.7	149°21.2	42.9	205°45.5	49.8	13°21.3	50.7	Ankaa	353°03.1	-42°08.6											
2	78°32.3	164°56.4	17.6	164°21.6	42.8	220°47.4	50.0	28°23.9	50.7	Schedar	349°25.9	56°42.4											
3	93°34.8	179°56.4	•• 17.5	179°22.0	•• 42.8	235°49.3	•• 50.2	43°26.6	•• 50.6	Diphda	348°43.2	-17°49.3											
4	108°37.2	194°56.3	17.5	194°22.4	42.7	250°51.3	50.4	58°29.3	50.6	Achernar	335°17.0	-57°05.1											
5	123°39.7	209°56.3	17.4	209°22.8	42.7	265°53.2	50.5	73°31.9	50.5	Hamal	327°46.4	23°36.3											
6	138°42.2	224°56.2	S27°17.4	224°23.2	S24°42.6	280°55.1	S13°50.7	88°34.6	N15°50.5	Polaris	311°21.0	89°23.2											
7	153°44.6	239°56.2	17.3	239°23.6	42.6	295°57.1	50.9	103°37.3	50.4	Acamar	315°08.5	-40°11.1											
8	168°47.1	254°56.2	17.2	254°24.0	42.5	310°59.0	51.0	118°39.9	50.4	Menkar	314°01.8	4°12.5											
9	183°49.6	269°56.1	•• 17.2	269°24.4	•• 42.4	326°00.9	•• 51.2	133°42.6	•• 50.3	Mirfak	308°22.1	49°58.0											
10	198°52.0	284°56.1	17.1	284°24.7	42.4	341°02.9	51.4	148°45.3	50.3	Aldebaran	290°34.8	16°34.1											
11	213°54.5	299°56.1	17.0	299°25.1	42.3	356°04.8	51.6	163°47.9	50.2	Rigel	280°59.9	-8°10.0											
12	228°57.0	314°56.0	S27°17.0	314°25.5	S24°42.3	11°06.7	S13°51.7	178°50.6	N15°50.2	Capella	280°15.7	46°01.5											
13	243°59.4	329°56.0	16.9	329°25.9	42.2	26°08.6	51.9	193°53.3	50.1	Bellatrix	278°18.4	6°22.5											
14	259°01.9	344°56.0	16.8	344°26.3	42.2	41°10.6	52.1	208°55.9	50.1	Elnath	277°56.6	28°37.8											
15	274°04.3	359°55.9	•• 16.8	359°26.7	•• 42.1	56°12.5	•• 52.3	223°58.6	•• 50.0	Alnilam	275°33.5	-1°11.0											
16	289°06.8	14°55.9	16.7	14°27.1	42.0	71°14.4	52.4	239°01.3	50.0	Betelgeuse	270°47.6	7°24.7											
17	304°09.3	29°55.9	16.6	29°27.5	42.0	86°16.4	52.6	254°04.0	49.9	Canopus	263°50.3	-52°42.6											
18	319°11.7	44°55.8	S27°16.6	44°27.9	S24°41.9	101°18.3	S13°52.8	269°06.6	N15°49.9	Sirius	258°22.6	-16°45.4											
19	334°14.2	59°55.8	16.5	59°28.3	41.8	116°20.2	52.9	284°09.3	49.8	Adhara	255°02.6	-29°00.7											
20	349°16.7	74°55.8	16.4	74°28.7	41.8	131°22.2	53.1	299°12.0	49.8	Procyon	244°46.7	5°08.8											
21	4°19.1	89°55.8	•• 16.3	89°29.1	•• 41.7	146°24.1	•• 53.3	314°14.6	•• 49.7	Pollux	243°12.5	27°57.0											
22	19°21.6	104°55.8	16.3	104°29.5	41.7	161°26.0	53.5	329°17.3	49.7	Avior	234°12.9	-59°36.1											
23	34°24.1	119°55.7	16.2	119°29.8	41.6	176°28.0	53.6	344°20.0	49.6	Suhail	222°43.3	-43°33.0											
Mer.pass. 20:43				ν 0.0' d -0.1' m -4.66				ν 0.4' d -0.1' m 1.07				ν 1.9' d 0.2' m -1.69				ν 2.7' d -0.1' m -0.26							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	49°26.5	134°55.7	S27°16.1	134°30.2	S24°41.5	191°29.9	S13°53.8	359°22.6	N15°49.6										
1	64°29.0	149°55.7	16.0	149°30.6	41.5	206°31.8	54.0	14°25.3	49.5										
2	79°31.5	164°55.7	15.9	164°31.0	41.4	221°33.7	54.1	29°28.0	49.5										
3	94°33.9	179°55.7	•• 15.9	179°31.4	•• 41.3	236°35.7	•• 54.3	44°30.6	•• 49.4										
4	109°36.4	194°55.7	15.8	194°31.8	41.3	251°37.6	54.5	59°33.3	49.4										
5	124°38.8	209°55.6	15.7	209°32.2	41.2	266°39.5	54.7	74°36.0	49.3										
6	139°41.3	224°55.6	S27°15.6	224°32.6	S24°41.1	281°41.5	S13°54.8	89°38.6	N15°49.3										
7	154°43.8	239°55.6	15.5	239°33.0	41.1	296°43.4	55.0	104°41.3	49.2										
8	169°46.2	254°55.6	15.4	254°33.4	41.0	311°45.3	55.2	119°44.0	49.2										
9	184°48.7	269°55.6	•• 15.4	269°33.8	•• 40.9	326°47.3	•• 55.3	134°46.6	•• 49.1										
10	199°51.2	284°55.6	15.3	284°34.2	40.9	341°49.2	55.5	149°49.3	49.1										
11	214°53.6	299°55.6	15.2	299°34.5	40.8	356°51.1	55.7	164°52.0	49.0										
12	229°56.1	314°55.6	S27°15.1	314°34.9	S24°40.7	11°53.1	S13°55.9	179°54.7	N15°49.0										
13	244°58.6	329°55.6	15.0	329°35.3	40.7	26°55.0	56.0	194°57.3	48.9										
14	260°01.0	344°55.6	14.9	344°35.7	40.6	41°56.9	56.2	210°00.0	48.9										
15	275°03.5	359°55.6	•• 14.8	359°36.1	•• 40.5	56°58.8	•• 56.4	225°02.7	•• 48.8										
16	290°06.0	14°55.6	14.7	14°36.5	40.4	72°00.8	56.5	240°05.3	48.8										
17	305°08.4	29°55.6	14.6	29°36.9	40.4	87°02.7	56.7	255°08.0	48.7										
18	320°10.9	44°55.6	S27°14.5	44°37.3	S24°40.3	102°04.6	S13°56.9	270°10.7	N15°48.7										
19	335°13.3	59°55.6	14.4	59°37.7	40.2	117°06.6	57.1	285°13.3	48.6										
20	350°15.8	74°55.6	14.3	74°38.1	40.2	132°08.5	57.2	300°16.0	48.6										
21	5°18.3	89°55.6	•• 14.3	89°38.5	•• 40.1	147°10.4	•• 57.4	315°18.7	•• 48.5										
22	20°20.7	104°55.6	14.2	104°38.8	40.0	162°12.4	57.6	330°21.3	48.5										
23	35°23.2	119°55.6	14.1	119°39.2	39.9	177°14.3	57.7	345°24.0	48.4										
Mer.pass. 20:39				ν 0.0' d -0.1' m -4.67				ν 0.4' d -0.1' m 1.08				ν 1.9' d 0.2' m -1.69				ν 2.7' d -0.1' m -0.28			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	50°25.7	134°55.7	S27°14.0	134°39.6	S24°39.9	192°16.2	S13°57.9	0°26.7	N15°48.4										
1	65°28.1	149°55.7	13.9	149°40.0	39.8	207°18.2	58.1	15°29.3	48.3										
2	80°30.6	164°55.7	13.8	164°40.4	39.7	222°20.1	58.3	30°32.0	48.2										
3	95°33.1	179°55.7	•• 13.6	179°40.8	•• 39.6	237°22.0	•• 58.4	45°34.7	•• 48.2										
4	110°35.5	194°55.7	13.5	194°41.2	39.5	252°23.9	58.6	60°37.4	48.1										
5	125°38.0	209°55.8	13.4	209°41.6	39.5	267°25.9	58.8	75°40.0	48.1										
6	140°40.4	224°55.8	S27°13.3	224°42.0	S24°39.4	282°27.8	S13°58.9	90°42.7	N15°48.0										
7	155°42.9	239°55.8	13.2	239°42.3	39.3	297°29.7	59.1	105°45.4	48.0										
8	170°45.4	254°55.8	13.1	254°42.7	39.2	312°31.7	59.3	120°48.0	47.9										
9	185°47.8	269°55.8	•• 13.0	269°43.1	•• 39.2	327°33.6	•• 59.5	135°50.7	•• 47.9										
10	200°50.3	284°55.9	12.9	284°43.5	39.1	342°35.5	59.6	150°53.4	47.8										
11	215°52.8	299°55.9	12.8	299°43.9	39.0	357°37.5	13°59.8	165°56.0	47.8										
12	230°55.2	314°55.9	S27°12.7	314°44.3	S24°38.9	12°39.4	S14°00.0	180°58.7	N15°47.7										
13	245°57.7	329°56.0	12.6	329°44.7	38.8	27°41.3	00.1	196°01.4	47.7										
14	261°00.2	344°56.0	12.5	344°45.1	38.8	42°43.3	00.3	211°04.0	47.6										
15	276°02.6	359°56.0	•• 12.3	359°45.5	•• 38.7	57°45.2	•• 00.5	226°06.7	•• 47.6										
16	291°05.1	14°56.1	12.2	14°45.9	38.6	72°47.1	00.7	241°09.4	47.5										
17	306°07.6	29°56.1	12.1	29°46.2	38.5	87°49.1	00.8	256°12.0	47.5										
18	321°10.0	44°56.1	S27°12.0	44°46.6	S24°38.4	102°51.0	S14°01.0	271°14.7	N15°47.4										
19	336°12.5	59°56.2	11.9	59°47.0	38.3	117°52.9	01.2	286°17.4	47.4										
20	351°14.9	74°56.2	11.8	74°47.4	38.3	132°54.8	01.3	301°20.1	47.3										
21	6°17.4	89°56.3	•• 11.6	89°47.8	•• 38.2	147°56.8	•• 01.5	316°22.7	•• 47.3										
22	21°19.9	104°56.3	11.5	104°48.2	38.1	162°58.7	01.7	331°25.4	47.2										
23	36°22.3	119°56.4	11.4	119°48.6	38.0	178°00.6	01.8	346°28.1	47.2										
Mer.pass. 20:35				ν 0.0' d -0.1' m -4.68				ν 0.4' d -0.1' m 1.08				ν 1.9' d 0.2' m -1.69				ν 2.7' d -0.1' m -0.29			

Nov 09 Fri	SHA	Mer.pass
Venus	86°29.1	15:00
Mars	85°53.4	15:02
Jupiter	142°16.2	11:16
Saturn	309°51.2	00:07

Nov 10 Sat	SHA	Mer.pass
Venus	85°29.2	15:00
Mars	85°03.7	15:02
Jupiter	142°03.4	11:13
Saturn	309°56.1	00:02

Nov 11 Sun	SHA	Mer.pass
Venus	84°30.0	15:00
Mars	84°14.0	15:01
Jupiter	141°50.6	11:09
Saturn	310°01.0	23:54

Horizontal parallax		
Venus:	0.3	
Mars:	0.1	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	184°03.0	S16°52.4	142°34.3	4.0'	S23°22.0	-2.3'	59.9'
1	199°02.9	53.1	156°57.2	4.0'	23°19.7	-2.5'	59.9'
2	214°02.9	53.8	171°20.2	4.1'	23°17.2	-2.7'	59.9'
3	229°02.8	•• 54.5	185°43.3	4.2'	23°14.5	-2.8'	59.8'
4	244°02.8	55.3	200°06.5	4.3'	23°11.7	-3.0'	59.8'
5	259°02.7	56.0	214°29.8	4.3'	23°08.7	-3.1'	59.7'
6	274°02.7	S16°56.7	228°53.1	4.4'	S23°05.6	-3.3'	59.7'
7	289°02.6	57.4	243°16.5	4.5'	23°02.3	-3.4'	59.7'
8	304°02.6	58.1	257°40.0	4.6'	22°58.9	-3.6'	59.6'
9	319°02.5	•• 58.8	272°03.6	4.7'	22°55.3	-3.8'	59.6'
10	334°02.5	16°59.5	286°27.3	4.8'	22°51.5	-3.9'	59.6'
11	349°02.4	17°00.2	300°51.1	4.9'	22°47.6	-4.1'	59.5'
12	4°02.3	S17°00.9	315°14.9	4.9'	S22°43.6	-4.2'	59.5'
13	19°02.3	01.6	329°38.9	5.0'	22°39.4	-4.3'	59.4'
14	34°02.2	02.4	344°02.9	5.1'	22°35.0	-4.5'	59.4'
15	49°02.2	•• 03.1	358°27.0	5.2'	22°30.5	-4.6'	59.4'
16	64°02.1	03.8	12°51.3	5.3'	22°25.9	-4.8'	59.3'
17	79°02.1	04.5	27°15.6	5.4'	22°21.1	-4.9'	59.3'
18	94°02.0	S17°05.2	41°40.0	5.5'	S22°16.2	-5.1'	59.3'
19	109°01.9	05.9	56°04.5	5.6'	22°11.1	-5.2'	59.2'
20	124°01.9	06.6	70°29.1	5.7'	22°05.9	-5.3'	59.2'
21	139°01.8	•• 07.3	84°53.8	5.8'	22°00.6	-5.5'	59.1'
22	154°01.8	08.0	99°18.7	5.9'	21°55.1	-5.6'	59.1'
23	169°01.7	08.7	113°43.6	6.0'	21°49.5	-5.7'	59.1'
SD = 16.1'			SD = 16.3'				
d = 0.7'							
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	184°01.6	S17°09.4	128°08.6	6.1'	S21°43.7	-5.9'	59.0'
1	199°01.6	10.1	142°33.7	6.2'	21°37.8	-6.0'	59.0'
2	214°01.5	10.8	156°59.0	6.3'	21°31.8	-6.1'	58.9'
3	229°01.5	•• 11.5	171°24.3	6.4'	21°25.7	-6.3'	58.9'
4	244°01.4	12.2	185°49.7	6.5'	21°19.4	-6.4'	58.9'
5	259°01.3	12.9	200°15.3	6.7'	21°13.0	-6.5'	58.8'
6	274°01.3	S17°13.6	214°40.9	6.8'	S21°06.5	-6.6'	58.8'
7	289°01.2	14.3	229°06.7	6.9'	20°59.9	-6.8'	58.7'
8	304°01.1	15.0	243°32.5	7.0'	20°53.1	-6.9'	58.7'
9	319°01.1	•• 15.7	257°58.5	7.1'	20°46.2	-7.0'	58.7'
10	334°01.0	16.4	272°24.6	7.2'	20°39.3	-7.1'	58.6'
11	349°00.9	17.1	286°50.8	7.3'	20°32.1	-7.2'	58.6'
12	4°00.9	S17°17.8	301°17.1	7.4'	S20°24.9	-7.3'	58.5'
13	19°00.8	18.5	315°43.5	7.5'	20°17.6	-7.5'	58.5'
14	34°00.7	19.2	330°10.0	7.6'	20°10.1	-7.6'	58.5'
15	49°00.7	•• 19.9	344°36.7	7.7'	20°02.5	-7.7'	58.4'
16	64°00.6	20.5	359°03.4	7.8'	19°54.9	-7.8'	58.4'
17	79°00.5	21.2	13°30.2	8.0'	19°47.1	-7.9'	58.3'
18	94°00.5	S17°21.9	27°57.2	8.1'	S19°39.2	-8.0'	58.3'
19	109°00.4	22.6	42°24.3	8.2'	19°31.2	-8.1'	58.3'
20	124°00.3	23.3	56°51.5	8.3'	19°23.1	-8.2'	58.2'
21	139°00.3	•• 24.0	71°18.7	8.4'	19°14.9	-8.3'	58.2'
22	154°00.2	24.7	85°46.1	8.5'	19°06.6	-8.4'	58.1'
23	169°00.1	25.4	100°13.7	8.6'	18°58.2	-8.5'	58.1'
SD = 16.1'			SD = 16.1'				
d = 0.7'							
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	184°00.1	S17°26.1	114°41.3	8.7'	S18°49.7	-8.6'	58.0'
1	199°00.0	26.8	129°09.0	8.8'	18°41.2	-8.7'	58.0'
2	213°59.9	27.5	143°36.8	8.9'	18°32.5	-8.8'	58.0'
3	228°59.9	•• 28.1	158°04.8	9.1'	18°23.7	-8.9'	57.9'
4	243°59.8	28.8	172°32.8	9.2'	18°14.8	-9.0'	57.9'
5	258°59.7	29.5	187°01.0	9.3'	18°05.9	-9.0'	57.8'
6	273°59.6	S17°30.2	201°29.3	9.4'	S17°56.9	-9.1'	57.8'
7	288°59.6	30.9	215°57.7	9.5'	17°47.7	-9.2'	57.8'
8	303°59.5	31.6	230°26.1	9.6'	17°38.5	-9.3'	57.7'
9	318°59.4	•• 32.3	244°54.7	9.7'	17°29.2	-9.4'	57.7'
10	333°59.4	32.9	259°23.4	9.8'	17°19.8	-9.5'	57.6'
11	348°59.3	33.6	273°52.2	9.9'	17°10.4	-9.5'	57.6'
12	3°59.2	S17°34.3	288°21.1	10.0'	S17°00.8	-9.6'	57.6'
13	18°59.1	35.0	302°50.2	10.1'	16°51.2	-9.7'	57.5'
14	33°59.1	35.7	317°19.3	10.2'	16°41.5	-9.8'	57.5'
15	48°59.0	•• 36.4	331°48.5	10.3'	16°31.8	-9.8'	57.5'
16	63°58.9	37.0	346°17.8	10.4'	16°21.9	-9.9'	57.4'
17	78°58.8	37.7	0°47.3	10.5'	16°12.0	-10.0'	57.4'
18	93°58.8	S17°38.4	15°16.8	10.6'	S16°02.0	-10.1'	57.3'
19	108°58.7	39.1	29°46.4	10.7'	15°52.0	-10.1'	57.3'
20	123°58.6	39.8	44°16.1	10.8'	15°41.9	-10.2'	57.3'
21	138°58.5	•• 40.4	58°46.0	10.9'	15°31.7	-10.3'	57.2'
22	153°58.4	41.1	73°15.9	11.0'	15°21.4	-10.3'	57.2'
23	168°58.4	41.8	87°45.9	11.1'	15°11.1	-10.4'	57.1'
SD = 16.1'			SD = 15.8'				
d = 0.7'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:42	08:12	10:05	13:22	15:15	16:44
N 70°	06:34	07:53	09:20	14:07	15:34	16:53
68°	06:27	07:38	08:51	14:36	15:49	17:00
66°	06:21	07:25	08:29	14:58	16:02	17:06
64°	06:16	07:15	08:12	15:15	16:12	17:11
62°	06:11	07:06	07:58	15:29	16:21	17:15
60°	06:07	06:58	07:46	15:41	16:29	17:20
N 58°	06:04	06:51	07:36	15:52	16:36	17:23
56°	06:00	06:45	07:27	16:01	16:42	17:27
54°	05:57	06:40	07:19	16:09	16:48	17:30
52°	05:54	06:35	07:11	16:16	16:53	17:33
50°	05:51	06:30	07:05	16:22	16:57	17:36
45°	05:44	06:20	06:51	16:36	17:08	17:43
N 40°	05:38	06:11	06:40	16:48	17:17	17:49
35°	05:33	06:03	06:30	16:58	17:24	17:55
30°	05:27	05:56	06:21	17:06	17:32	18:00
20°	05:17	05:43	06:06	17:22	17:45	18:11
N 10°	05:06	05:31	05:53	17:35	17:57	18:22
0°	04:54	05:19	05:40	17:47	18:09	18:34
S 10°	04:40	05:06	05:28	18:00	18:22	18:48
20°	04:23	04:51	05:14	18:14	18:37	19:05
30°	04:02	04:33	04:59	18:30	18:56	19:26
35°	03:49	04:22	04:49	18:39	19:07	19:40
40°	03:32	04:09	04:39	18:50	19:20	19:57
45°	03:11	03:53	04:27	19:02	19:36	20:18
S 50°	02:44	03:33	04:11	19:17	19:56	20:46
52°	02:30	03:24	04:04	19:25	20:05	21:00
54°	02:13	03:13	03:56	19:33	20:16	21:17
56°	01:52	03:00	03:48	19:42	20:29	21:39
58°	01:25	02:46	03:37	19:52	20:44	22:08
S 60°	00:40	02:28	03:26	20:04	21:02	22:57

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	15:20	■	■	18:52
N 70°	■	■	14:35	■	■	19:36
68°	■	14:24	14:04	■	17:56	20:05
66°	13:45	13:44	13:42	16:35	18:35	20:27
64°	13:00	13:17	13:24	17:20	19:02	20:44
62°	12:30	12:55	13:09	17:49	19:23	20:57
60°	12:07	12:38	12:56	18:11	19:40	21:09
N 58°	11:49	12:23	12:46	18:30	19:54	21:19
56°	11:33	12:10	12:36	18:45	20:06	21:28
54°	11:20	11:59	12:28	18:58	20:16	21:36
52°	11:08	11:50	12:20	19:09	20:26	21:43
50°	10:58	11:41	12:13	19:19	20:34	21:49
45°	10:36	11:22	11:59	19:40	20:52	22:03
N 40°	10:19	11:07	11:47	19:57	21:06	22:14
35°	10:04	10:54	11:36	20:12	21:18	22:23
30°	09:51	10:43	11:27	20:24	21:29	22:31
20°	09:29	10:23	11:11	20:45	21:47	22:45
N 10°	09:10	10:06	10:57	21:04	22:02	22:57
0°	08:52	09:51	10:45	21:21	22:17	23:09
S 10°	08:34	09:35	10:32	21:38	22:31	23:20
20°	08:15	09:18	10:18	21:56	22:47	23:32
30°	07:53	08:58	10:02	22:16	23:04	23:46
35°	07:40	08:47	09:52	22:28	23:14	23:53
40°	07:25	08:34	09:42	22:42	23:26	•• ••
45°	07:08	08:18	09:29	22:58	23:40	•• ••
S 50°	06:46	07:59	09:14	23:19	23:56	•• ••
52°	06:35	07:50	09:07	23:28	•• ••	00:04
54°	06:23	07:39	08:59	23:39	•• ••	00:12
56°	06:10	07:28	08:50	23:51	•• ••	00:22
58°	05:54	07:14	08:39	•• ••	00:04	00:33
S 60°	05:35	06:58	08:27	•• ••	00:21	00:45

Day	Sun			Moon		Age 3-5 11-29%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	Mer. Pass hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	
09	16:12	16:09	11:44	15:06	02:36	
10	16:07	16:04	11:44	16:04	03:36	
11	16:00	15:57	11:44	16:57	04:31	

November 12, 13, 14 UT (Mon., Tue., Wed.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	51°24.8	134°56.4	S27°11.3	134°49.0	S24°37.9	193°02.6	S14°02.0	1°30.7	N15°47.1	Alpheratz	357°30.4	29°15.6											
1	66°27.3	149°56.5	11.2	149°49.3	37.8	208°04.5	02.2	16°33.4	47.1	Ankaa	353°03.2	-42°08.6											
2	81°29.7	164°56.5	11.0	164°49.7	37.7	223°06.4	02.4	31°36.1	47.0	Schedar	349°25.9	56°42.4											
3	96°32.2	179°56.6	•• 10.9	179°50.1	•• 37.6	238°08.4	•• 02.5	46°38.7	•• 47.0	Diphda	348°43.2	-17°49.3											
4	111°34.7	194°56.6	10.8	194°50.5	37.6	253°10.3	02.7	61°41.4	46.9	Achernar	335°17.0	-57°05.2											
5	126°37.1	209°56.7	10.7	209°50.9	37.5	268°12.2	02.9	76°44.1	46.9	Hamal	327°46.4	23°36.3											
6	141°39.6	224°56.7	S27°10.5	224°51.3	S24°37.4	283°14.2	S14°03.0	91°46.7	N15°46.8	Polaris	311°20.8	89°23.3											
7	156°42.1	239°56.8	10.4	239°51.7	37.3	298°16.1	03.2	106°49.4	46.8	Acamar	315°08.5	-40°11.1											
8	171°44.5	254°56.8	10.3	254°52.1	37.2	313°18.0	03.4	121°52.1	46.7	Menkar	314°01.8	4°12.5											
9	186°47.0	269°56.9	•• 10.1	269°52.5	•• 37.1	328°20.0	•• 03.5	136°54.7	•• 46.7	Mirfak	308°22.1	49°58.0											
10	201°49.4	284°57.0	10.0	284°52.8	37.0	343°21.9	03.7	151°57.4	46.6	Aldebaran	290°34.8	16°34.1											
11	216°51.9	299°57.0	09.9	299°53.2	36.9	358°23.8	03.9	167°00.1	46.6	Rigel	280°59.8	-8°10.0											
12	231°54.4	314°57.1	S27°09.7	314°53.6	S24°36.8	13°25.7	S14°04.1	182°02.8	N15°46.5	Capella	280°15.7	46°01.5											
13	246°56.8	329°57.2	09.6	329°54.0	36.7	28°27.7	04.2	197°05.4	46.5	Bellatrix	278°18.4	6°22.5											
14	261°59.3	344°57.2	09.5	344°54.4	36.6	43°29.6	04.4	212°08.1	46.4	Elnath	277°56.6	28°37.8											
15	277°01.8	359°57.3	•• 09.3	359°54.8	•• 36.6	58°31.5	•• 04.6	227°10.8	•• 46.4	Alnilam	275°33.5	-1°11.0											
16	292°04.2	14°57.4	09.2	14°55.2	36.5	73°33.5	04.7	242°13.4	46.3	Betelgeuse	270°47.6	7°24.7											
17	307°06.7	29°57.4	09.1	29°55.6	36.4	88°35.4	04.9	257°16.1	46.3	Canopus	263°50.3	-52°42.6											
18	322°09.2	44°57.5	S27°08.9	44°55.9	S24°36.3	103°37.3	S14°05.1	272°18.8	N15°46.2	Sirius	258°22.6	-16°45.4											
19	337°11.6	59°57.6	08.8	59°56.3	36.2	118°39.3	05.2	287°21.4	46.2	Adhara	255°02.6	-29°00.7											
20	352°14.1	74°57.7	08.7	74°56.7	36.1	133°41.2	05.4	302°24.1	46.1	Procyon	244°46.6	5°08.8											
21	7°16.5	89°57.7	•• 08.5	89°57.1	•• 36.0	148°43.1	•• 05.6	317°26.8	•• 46.1	Pollux	243°12.4	27°57.0											
22	22°19.0	104°57.8	08.4	104°57.5	35.9	163°45.1	05.8	332°29.4	46.0	Avior	234°12.8	-59°36.1											
23	37°21.5	119°57.9	08.2	119°57.9	35.8	178°47.0	05.9	347°32.1	46.0	Suhail	222°43.3	-43°33.1											
Mer.pass. 20:31				ν 0.0' d -0.1' m -4.69				ν 0.4' d -0.1' m 1.08				ν 1.9' d 0.2' m -1.69				ν 2.7' d -0.1' m -0.30							

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	52°23.9	134°58.0	S27°08.1	134°58.3	S24°35.7	193°48.9	S14°06.1	2°34.8	N15°45.9						
1	67°26.4	149°58.1	07.9	149°58.6	35.6	208°50.9	06.3	17°37.5	45.9						
2	82°28.9	164°58.2	07.8	164°59.0	35.5	223°52.8	06.4	32°40.1	45.8						
3	97°31.3	179°58.2	•• 07.7	179°59.4	•• 35.4	238°54.7	•• 06.6	47°42.8	•• 45.8						
4	112°33.8	194°58.3	07.5	194°59.8	35.3	253°56.7	06.8	62°45.5	45.7						
5	127°36.3	209°58.4	07.4	210°00.2	35.2	268°58.6	06.9	77°48.1	45.7						
6	142°38.7	224°58.5	S27°07.2	225°00.6	S24°35.1	284°00.5	S14°07.1	92°50.8	N15°45.6						
7	157°41.2	239°58.6	07.1	240°01.0	35.0	299°02.4	07.3	107°53.5	45.6						
8	172°43.7	254°58.7	06.9	255°01.4	34.9	314°04.4	07.5	122°56.1	45.5						
9	187°46.1	269°58.8	•• 06.8	270°01.7	•• 34.8	329°06.3	•• 07.6	137°58.8	•• 45.5						
10	202°48.6	284°58.9	06.6	285°02.1	34.7	344°08.2	07.8	153°01.5	45.4						
11	217°51.0	299°59.0	06.5	300°02.5	34.6	359°10.2	08.0	168°04.1	45.4						
12	232°53.5	314°59.1	S27°06.3	315°02.9	S24°34.5	14°12.1	S14°08.1	183°06.8	N15°45.3						
13	247°56.0	329°59.2	06.1	330°03.3	34.4	29°14.0	08.3	198°09.5	45.3						
14	262°58.4	344°59.3	06.0	345°03.7	34.3	44°16.0	08.5	213°12.2	45.2						
15	278°00.9	359°59.4	•• 05.8	0°04.1	•• 34.2	59°17.9	•• 08.6	228°14.8	•• 45.2						
16	293°03.4	14°59.5	05.7	15°04.4	34.1	74°19.8	08.8	243°17.5	45.1						
17	308°05.8	29°59.6	05.5	30°04.8	34.0	89°21.8	09.0	258°20.2	45.1						
18	323°08.3	44°59.7	S27°05.4	45°05.2	S24°33.9	104°23.7	S14°09.1	273°22.8	N15°45.0						
19	338°10.8	59°59.9	05.2	60°05.6	33.7	119°25.6	09.3	288°25.5	45.0						
20	353°13.2	75°00.0	05.0	75°06.0	33.6	134°27.6	09.5	303°28.2	44.9						
21	8°15.7	90°00.1	•• 04.9	90°06.4	•• 33.5	149°29.5	•• 09.6	318°30.8	•• 44.9						
22	23°18.2	105°00.2	04.7	105°06.8	33.4	164°31.4	09.8	333°33.5	44.8						
23	38°20.6	120°00.3	04.6	120°07.1	33.3	179°33.4	10.0	348°36.2	44.8						
Mer.pass. 20:27				ν 0.1' d -0.1' m -4.71				ν 1.9' d 0.2' m -1.69				ν 2.7' d -0.1' m -0.31			

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	53°23.1	135°00.4	S27°04.4	135°07.5	S24°33.2	194°35.3	S14°10.2	3°38.8	N15°44.7						
1	68°25.5	150°00.6	04.2	150°07.9	33.1	209°37.2	10.3	18°41.5	44.7						
2	83°28.0	165°00.7	04.1	165°08.3	33.0	224°39.2	10.5	33°44.2	44.6						
3	98°30.5	180°00.8	•• 03.9	180°08.7	•• 32.9	239°41.1	•• 10.7	48°46.9	•• 44.6						
4	113°32.9	195°00.9	03.7	195°09.1	32.8	254°43.0	10.8	63°49.5	44.5						
5	128°35.4	210°01.1	03.6	210°09.5	32.7	269°44.9	11.0	78°52.2	44.5						
6	143°37.9	225°01.2	S27°03.4	225°09.8	S24°32.5	284°46.9	S14°11.2	93°54.9	N15°44.4						
7	158°40.3	240°01.3	03.2	240°10.2	32.4	299°48.8	11.3	108°57.5	44.4						
8	173°42.8	255°01.5	03.0	255°10.6	32.3	314°50.7	11.5	124°00.2	44.3						
9	188°45.3	270°01.6	•• 02.9	270°11.0	•• 32.2	329°52.7	•• 11.7	139°02.9	•• 44.3						
10	203°47.7	285°01.7	02.7	285°11.4	32.1	344°54.6	11.8	154°05.5	44.2						
11	218°50.2	300°01.9	02.5	300°11.8	32.0	359°56.5	12.0	169°08.2	44.2						
12	233°52.6	315°02.0	S27°02.3	315°12.2	S24°31.9	14°58.5	S14°12.2	184°10.9	N15°44.1						
13	248°55.1	330°02.2	02.2	330°12.5	31.7	30°00.4	12.3	199°13.5	44.1						
14	263°57.6	345°02.3	02.0	345°12.9	31.6	45°02.3	12.5	214°16.2	44.0						
15	279°00.0	0°02.4	•• 01.8	0°13.3	•• 31.5	60°04.3	•• 12.7	229°18.9	•• 44.0						
16	294°02.5	15°02.6	01.6	15°13.7	31.4	75°06.2	12.9	244°21.5	43.9						
17	309°05.0	30°02.7	01.5	30°14.1	31.3	90°08.1	13.0	259°24.2	43.9						
18	324°07.4	45°02.9	S27°01.3	45°14.5	S24°31.2	105°10.1	S14°13.2	274°26.9	N15°43.8						
19	339°09.9	60°03.0	01.1	60°14.9	31.0	120°12.0	13.4	289°29.6	43.8						
20	354°12.4	75°03.2	00.9	75°15.2	30.9	135°13.9	13.5	304°32.2	43.7						
21	9°14.8	90°03.3	•• 00.7	90°15.6	•• 30.8	150°15.9	•• 13.7	319°34.9	•• 43.7						
22	24°17.3	105°03.5	00.6	105°16.0	30.7	165°17.8	13.9	334°37.6	43.6						
23	39°19.8	120°03.7	00.4	120°16.4	30.6	180°19.7	14.0	349°40.2	43.6						
Mer.pass. 20:23				ν 0.1' d -0.2' m -4.72				ν 1.9' d 0.2' m -1.69				ν 2.7' d -0.1' m -0.32			

Nov 12 Mon	SHA	Mer.pass
Venus	83°31.6	15:00
Mars	83°24.2	15:00
Jupiter	141°37.8	11:06
Saturn	310°05.9	23:50

Nov 13 Tue	SHA	Mer.pass
Venus	82°34.0	15:00
Mars	82°34.3	15:00
Jupiter	141°25.0	11:03
Saturn	310°10.8	23:45

Nov 14 Wed	SHA	Mer.pass
Venus	81°37.4	15:00
Mars	81°44.5	14:59
Jupiter	141°12.2	11:00
Saturn	310°15.8	23:41

Horizontal parallax		
Venus:	0.3	
Mars:	0.1	

h Sun							Moon						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°58.3	S17°42.5	102°16.1	11.2'	S15°00.7	-10.4'	57.1'						
1	198°58.2	43.1	116°46.3	11.3'	14°50.3	-10.5'	57.1'						
2	213°58.1	43.8	131°16.6	11.4'	14°39.8	-10.6'	57.0'						
3	228°58.1	44.5	145°47.0	11.5'	14°29.2	-10.6'	57.0'						
4	243°58.0	45.2	160°17.5	11.6'	14°18.6	-10.7'	57.0'						
5	258°57.9	45.8	174°48.1	11.7'	14°07.9	-10.7'	56.9'						
6	273°57.8	S17°46.5	189°18.8	11.8'	S13°57.2	-10.8'	56.9'						
7	288°57.7	47.2	203°49.6	11.9'	13°46.4	-10.8'	56.8'						
8	303°57.6	47.9	218°20.5	12.0'	13°35.6	-10.9'	56.8'						
9	318°57.6	48.5	232°51.4	12.1'	13°24.7	-10.9'	56.8'						
10	333°57.5	49.2	247°22.5	12.1'	13°13.7	-11.0'	56.7'						
11	348°57.4	49.9	261°53.6	12.2'	13°02.7	-11.0'	56.7'						
12	3°57.3	S17°50.5	276°24.8	12.3'	S12°51.7	-11.1'	56.7'						
13	18°57.2	51.2	290°56.1	12.4'	12°40.6	-11.1'	56.6'						
14	33°57.2	51.9	305°27.5	12.5'	12°29.5	-11.2'	56.6'						
15	48°57.1	52.5	319°59.0	12.6'	12°18.3	-11.2'	56.6'						
16	63°57.0	53.2	334°30.6	12.6'	12°07.1	-11.3'	56.5'						
17	78°56.9	53.9	349°02.2	12.7'	11°55.8	-11.3'	56.5'						
18	93°56.8	S17°54.5	3°34.0	12.8'	S11°44.5	-11.3'	56.4'						
19	108°56.7	55.2	18°05.8	12.9'	11°33.1	-11.4'	56.4'						
20	123°56.6	55.9	32°37.7	13.0'	11°21.8	-11.4'	56.4'						
21	138°56.6	56.5	47°09.6	13.0'	11°10.3	-11.5'	56.3'						
22	153°56.5	57.2	61°41.7	13.1'	10°58.9	-11.5'	56.3'						
23	168°56.4	57.9	76°13.8	13.2'	10°47.4	-11.5'	56.3'						
SD = 16.2' $d = 0.7'$							SD = 15.6'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°56.3	S17°58.5	90°46.0	13.3'	S10°35.8	-11.6'	56.2'						
1	198°56.2	59.2	105°18.2	13.3'	10°24.3	-11.6'	56.2'						
2	213°56.1	17°59.9	119°50.6	13.4'	10°12.7	-11.6'	56.2'						
3	228°56.0	18°00.5	134°23.0	13.5'	10°01.0	-11.7'	56.1'						
4	243°56.0	01.2	148°55.5	13.6'	09°49.4	-11.7'	56.1'						
5	258°55.9	01.9	163°28.0	13.6'	09°37.7	-11.7'	56.1'						
6	273°55.8	S18°02.5	178°00.7	13.7'	S09°25.9	-11.8'	56.0'						
7	288°55.7	03.2	192°33.3	13.8'	09°14.2	-11.8'	56.0'						
8	303°55.6	03.8	207°06.1	13.8'	09°02.4	-11.8'	56.0'						
9	318°55.5	04.5	221°38.9	13.9'	08°50.6	-11.8'	55.9'						
10	333°55.4	05.1	236°11.8	14.0'	08°38.8	-11.9'	55.9'						
11	348°55.3	05.8	250°44.8	14.0'	08°26.9	-11.9'	55.9'						
12	3°55.2	S18°06.5	265°17.8	14.1'	S08°15.0	-11.9'	55.9'						
13	18°55.1	07.1	279°50.9	14.1'	08°03.1	-11.9'	55.8'						
14	33°55.0	07.8	294°24.0	14.2'	07°51.2	-11.9'	55.8'						
15	48°55.0	08.4	308°57.2	14.3'	07°39.2	-12.0'	55.8'						
16	63°54.9	09.1	323°30.4	14.3'	07°27.3	-12.0'	55.7'						
17	78°54.8	09.7	338°03.7	14.4'	07°15.3	-12.0'	55.7'						
18	93°54.7	S18°10.4	352°37.1	14.4'	S07°03.3	-12.0'	55.7'						
19	108°54.6	11.0	7°10.5	14.5'	06°51.3	-12.0'	55.6'						
20	123°54.5	11.7	21°44.0	14.5'	06°39.2	-12.1'	55.6'						
21	138°54.4	12.4	36°17.5	14.6'	06°27.2	-12.1'	55.6'						
22	153°54.3	13.0	50°51.1	14.6'	06°15.1	-12.1'	55.6'						
23	168°54.2	13.7	65°24.8	14.7'	06°03.0	-12.1'	55.5'						
SD = 16.2' $d = 0.7'$							SD = 15.3'						
Wed	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°54.1	S18°14.3	79°58.4	14.7'	S05°50.9	-12.1'	55.5'						
1	198°54.0	15.0	94°32.2	14.8'	05°38.8	-12.1'	55.5'						
2	213°53.9	15.6	109°05.9	14.8'	05°26.7	-12.1'	55.4'						
3	228°53.8	16.3	123°39.8	14.9'	05°14.6	-12.1'	55.4'						
4	243°53.7	16.9	138°13.6	14.9'	05°02.4	-12.2'	55.4'						
5	258°53.6	17.5	152°47.5	15.0'	04°50.3	-12.2'	55.4'						
6	273°53.5	S18°18.2	167°21.5	15.0'	S04°38.1	-12.2'	55.3'						
7	288°53.4	18.8	181°55.5	15.0'	04°25.9	-12.2'	55.3'						
8	303°53.3	19.5	196°29.5	15.1'	04°13.8	-12.2'	55.3'						
9	318°53.2	20.1	211°03.6	15.1'	04°01.6	-12.2'	55.3'						
10	333°53.1	20.8	225°37.7	15.2'	03°49.4	-12.2'	55.2'						
11	348°53.0	21.4	240°11.9	15.2'	03°37.2	-12.2'	55.2'						
12	3°52.9	S18°22.1	254°46.1	15.2'	S03°25.0	-12.2'	55.2'						
13	18°52.8	22.7	269°20.3	15.3'	03°12.8	-12.2'	55.2'						
14	33°52.7	23.4	283°54.6	15.3'	03°00.6	-12.2'	55.1'						
15	48°52.6	24.0	298°28.9	15.3'	02°48.4	-12.2'	55.1'						
16	63°52.5	24.6	313°03.2	15.4'	02°36.2	-12.2'	55.1'						
17	78°52.4	25.3	327°37.6	15.4'	02°24.0	-12.2'	55.1'						
18	93°52.3	S18°25.9	342°12.0	15.4'	S02°11.8	-12.2'	55.0'						
19	108°52.2	26.6	356°46.4	15.5'	01°59.6	-12.2'	55.0'						
20	123°52.1	27.2	11°20.9	15.5'	01°47.4	-12.2'	55.0'						
21	138°52.0	27.8	25°55.4	15.5'	01°35.2	-12.2'	55.0'						
22	153°51.9	28.5	40°29.9	15.5'	01°23.0	-12.2'	55.0'						
23	168°51.8	29.1	55°04.4	15.6'	01°10.8	-12.2'	54.9'						
SD = 16.2' $d = 0.6'$							SD = 15.1'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:53	08:26	10:36	12:51	15:01	16:34
N 70°	06:44	08:05	09:38	13:49	15:23	16:44
68°	06:36	07:48	09:05	14:23	15:40	16:52
66°	06:29	07:34	08:41	14:47	15:53	16:58
64°	06:23	07:23	08:22	15:06	16:05	17:04
62°	06:18	07:13	08:06	15:21	16:14	17:10
60°	06:13	07:05	07:53	15:34	16:23	17:14
N 58°	06:09	06:58	07:42	15:45	16:30	17:19
56°	06:05	06:51	07:33	15:55	16:37	17:22
54°	06:02	06:45	07:24	16:04	16:43	17:26
52°	05:58	06:39	07:17	16:11	16:49	17:30
50°	05:55	06:34	07:10	16:18	16:54	17:33
45°	05:48	06:24	06:55	16:33	17:05	17:40
N 40°	05:41	06:14	06:43	16:45	17:14	17:47
35°	05:35	06:06	06:33	16:56	17:22	17:53
30°	05:29	05:58	06:24	17:05	17:30	17:59
20°	05:18	05:45	06:08	17:21	17:44	18:10
N 10°	05:06	05:32	05:54	17:35	17:57	18:22
0°	04:54	05:19	05:41	17:48	18:10	18:35
S 10°	04:39	05:05	05:28	18:01	18:23	18:49
20°	04:22	04:50	05:13	18:16	18:39	19:07
30°	04:00	04:31	04:57	18:32	18:58	19:29
35°	03:46	04:19	04:47	18:42	19:10	19:44
40°	03:29	04:06	04:36	18:53	19:24	20:01
45°	03:07	03:49	04:23	19:06	19:40	20:23
S 50°	02:38	03:28	04:07	19:23	20:01	20:53
52°	02:23	03:18	04:00	19:30	20:12	21:08
54°	02:04	03:07	03:51	19:39	20:23	21:27
56°	01:41	02:53	03:42	19:48	20:37	21:51
58°	01:09	02:38	03:31	19:59	20:53	22:24
S 60°	—:—	02:18	03:19	20:12	21:13	—:—

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	14:31	14:04	13:43	21:22	23:22	
N 70°	14:10	13:53	13:39	21:41	23:32	
68°	13:53	13:44	13:35	21:57	23:39	
66°	13:39	13:36	13:33	22:09	23:45	
64°	13:27	13:29	13:30	22:19	23:50	
62°	13:18	13:23	13:28	22:28	23:54	
60°	13:09	13:18	13:26	22:36	23:58	
N 58°	13:02	13:14	13:24	22:42		00:02
56°	12:55	13:10	13:23	22:48		00:05
54°	12:49	13:06	13:21	22:53		00:07
52°	12:44	13:03	13:20	22:58		00:10
50°	12:39	13:00	13:19	23:02		00:12
45°	12:28	12:54	13:16	23:11		00:17
N 40°	12:20	12:48	13:14	23:18		00:21
35°	12:12	12:44	13:12	23:25		00:24
30°	12:05	12:39	13:11	23:30		00:27
25°	11:54	12:32	13:08	23:40		00:32
N 10°	11:44	12:26	13:05	23:49		00:37
0°	11:34	12:20	13:03	23:56		00:41
S 10°	11:24	12:14	13:01		00:04	00:45
20°	11:14	12:07	12:58		00:12	00:49
30°	11:02	12:00	12:56		00:22	00:54
35°	10:56	11:56	12:54		00:27	00:57
40°	10:48	11:51	12:52	00:02	00:33	01:00
45°	10:39	11:45	12:50	00:13	00:40	01:04
S 50°	10:28	11:39	12:47	00:25	00:48	01:08
52°	10:22	11:36	12:46	00:31	00:52	01:10
54°	10:17	11:32	12:45	00:37	00:56	01:13
56°	10:10	11:28	12:43	00:44	01:01	01:15
58°	10:03	11:24	12:42	00:52	01:06	01:18
S 60°	09:55	11:19	12:40	01:01	01:12	01:21

November 15, 16, 17 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	54°22.2	135°03.8	S27°00.2	135°16.8	S24°30.4	195°21.7	S14°14.2	4°42.9	N15°43.5	Alpheratz	357°30.4	29°15.6					
1	69°24.7	150°04.0	27°00.0	150°17.2	30.3	210°23.6	14.4	19°45.6	43.5	Ankaa	353°03.2	-42°08.6					
2	84°27.1	165°04.1	26°59.8	165°17.5	30.2	225°25.5	14.5	34°48.2	43.4	Schedar	349°26.0	56°42.4					
3	99°29.6	180°04.3	· · 59.6	180°17.9	· · 30.1	240°27.5	· · 14.7	49°50.9	· · 43.4	Diphda	348°43.2	-17°49.3					
4	114°32.1	195°04.5	59.4	195°18.3	30.0	255°29.4	14.9	64°53.6	43.3	Achernar	335°17.0	-57°05.2					
5	129°34.5	210°04.6	59.3	210°18.7	29.8	270°31.3	15.0	79°56.2	43.3	Hamal	327°46.4	23°36.3					
6	144°37.0	225°04.8	S26°59.1	225°19.1	S24°29.7	285°33.3	S14°15.2	94°58.9	N15°43.2	Polaris	311°20.7	89°23.3					
7	159°39.5	240°05.0	58.9	240°19.5	29.6	300°35.2	15.4	110°01.6	43.2	Acamar	315°08.5	-40°11.1					
8	174°41.9	255°05.2	58.7	255°19.9	29.5	315°37.1	15.5	125°04.3	43.1	Menkar	314°01.7	4°12.5					
9	189°44.4	270°05.3	· · 58.5	270°20.2	· · 29.3	330°39.0	· · 15.7	140°06.9	· · 43.1	Mirfak	308°22.0	49°58.0					
10	204°46.9	285°05.5	58.3	285°20.6	29.2	345°41.0	15.9	155°09.6	43.0	Aldebaran	290°34.8	16°34.1					
11	219°49.3	300°05.7	58.1	300°21.0	29.1	0°42.9	16.0	170°12.3	43.0	Rigel	280°59.8	-8°10.1					
12	234°51.8	315°05.9	S26°57.9	315°21.4	S24°29.0	15°44.8	S14°16.2	185°14.9	N15°42.9	Capella	280°15.7	46°01.5					
13	249°54.3	330°06.0	57.7	330°21.8	28.8	30°46.8	16.4	200°17.6	42.9	Bellatrix	278°18.4	6°22.5					
14	264°56.7	345°06.2	57.5	345°22.2	28.7	45°48.7	16.5	215°20.3	42.8	Elnath	277°56.6	28°37.8					
15	279°59.2	0°06.4	· · 57.3	0°22.5	· · 28.6	60°50.6	· · 16.7	230°22.9	· · 42.8	Alnilam	275°33.5	-1°11.0					
16	295°01.6	15°06.6	57.1	15°22.9	28.5	75°52.6	16.9	245°25.6	42.7	Betelgeuse	270°47.6	7°24.7					
17	310°04.1	30°06.8	56.9	30°23.3	28.3	90°54.5	17.0	260°28.3	42.7	Canopus	263°50.2	-52°42.6					
18	325°06.6	45°07.0	S26°56.7	45°23.7	S24°28.2	105°56.4	S14°17.2	275°30.9	N15°42.6	Sirius	258°22.6	-16°45.4					
19	340°09.0	60°07.2	56.5	60°24.1	28.1	120°58.4	17.4	290°33.6	42.6	Adhara	255°02.5	-29°00.7					
20	355°11.5	75°07.4	56.3	75°24.5	27.9	136°00.3	17.5	305°36.3	42.5	Procyon	244°46.6	5°08.8					
21	10°14.0	90°07.6	· · 56.1	90°24.8	· · 27.8	151°02.2	· · 17.7	320°38.9	· · 42.5	Pollux	243°12.4	27°57.0					
22	25°16.4	105°07.8	55.9	105°25.2	27.7	166°04.2	17.9	335°41.6	42.4	Avior	234°12.8	-59°36.1					
23	40°18.9	120°08.0	55.7	120°25.6	27.5	181°06.1	18.0	350°44.3	42.4	Suhail	222°43.3	-43°33.1					
Mer.pass. 20:19			ν 0.2' d -0.2' m -4.73			ν 0.4' d -0.1' m 1.08			ν 1.9' d 0.2' m -1.69			ν 2.7' d -0.0' m -0.31					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	55°21.4	135°08.2	S26°55.5	135°26.0	S24°27.4	196°08.0	S14°18.2	5°47.0	N15°42.3					
1	70°23.8	150°08.4	55.3	150°26.4	27.3	211°10.0	18.4	20°49.6	42.3					
2	85°26.3	165°08.6	55.1	165°26.8	27.2	226°11.9	18.6	35°52.3	42.2					
3	100°28.7	180°08.8	· · 54.9	180°27.1	· · 27.0	241°13.8	· · 18.7	50°55.0	· · 42.2					
4	115°31.2	195°09.0	54.7	195°27.5	26.9	256°15.8	18.9	65°57.6	42.1					
5	130°33.7	210°09.2	54.5	210°27.9	26.8	271°17.7	19.1	81°00.3	42.1					
6	145°36.1	225°09.4	S26°54.3	225°28.3	S24°26.6	286°19.6	S14°19.2	96°03.0	N15°42.0					
7	160°38.6	240°09.6	54.0	240°28.7	26.5	301°21.6	19.4	111°05.6	42.0					
8	175°41.1	255°09.8	53.8	255°29.1	26.3	316°23.5	19.6	126°08.3	41.9					
9	190°43.5	270°10.0	· · 53.6	270°29.4	· · 26.2	331°25.4	· · 19.7	141°11.0	· · 41.9					
10	205°46.0	285°10.3	53.4	285°29.8	26.1	346°27.4	19.9	156°13.6	41.8					
11	220°48.5	300°10.5	53.2	300°30.2	25.9	1°29.3	20.1	171°16.3	41.8					
12	235°50.9	315°10.7	S26°53.0	315°30.6	S24°25.8	16°31.2	S14°20.2	186°19.0	N15°41.7					
13	250°53.4	330°10.9	52.8	330°31.0	25.7	31°33.2	20.4	201°21.6	41.7					
14	265°55.9	345°11.1	52.6	345°31.4	25.5	46°35.1	20.6	216°24.3	41.6					
15	280°58.3	0°11.4	· · 52.3	0°31.7	· · 25.4	61°37.0	· · 20.7	231°27.0	· · 41.6					
16	296°00.8	15°11.6	52.1	15°32.1	25.2	76°39.0	20.9	246°29.7	41.5					
17	311°03.2	30°11.8	51.9	30°32.5	25.1	91°40.9	21.1	261°32.3	41.5					
18	326°05.7	45°12.1	S26°51.7	45°32.9	S24°25.0	106°42.8	S14°21.2	276°35.0	N15°41.4					
19	341°08.2	60°12.3	51.5	60°33.3	24.8	121°44.8	21.4	291°37.7	41.4					
20	356°10.6	75°12.5	51.2	75°33.7	24.7	136°46.7	21.6	306°40.3	41.3					
21	11°13.1	90°12.8	· · 51.0	90°34.0	· · 24.5	151°48.6	· · 21.7	321°43.0	· · 41.3					
22	26°15.6	105°13.0	50.8	105°34.4	24.4	166°50.6	21.9	336°45.7	41.2					
23	41°18.0	120°13.3	50.6	120°34.8	24.3	181°52.5	22.1	351°48.3	41.2					
Mer.pass. 20:15			ν 0.2' d -0.2' m -4.74			ν 0.4' d -0.1' m 1.09			ν 1.9' d 0.2' m -1.69			ν 2.7' d -0.0' m -0.29		

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	56°20.5	135°13.5	S26°50.3	135°35.2	S24°24.1	196°54.4	S14°22.2	6°51.0	N15°41.1					
1	71°23.0	150°13.7	50.1	150°35.6	24.0	211°56.4	22.4	21°53.7	41.1					
2	86°25.4	165°14.0	49.9	165°36.0	23.8	226°58.3	22.6	36°56.3	41.0					
3	101°27.9	180°14.2	· · 49.7	180°36.3	· · 23.7	242°00.2	· · 22.7	51°59.0	· · 41.0					
4	116°30.3	195°14.5	49.4	195°36.7	23.5	257°02.2	22.9	67°01.7	40.9					
5	131°32.8	210°14.7	49.2	210°37.1	23.4	272°04.1	23.0	82°04.3	40.9					
6	146°35.3	225°15.0	S26°49.0	225°37.5	S24°23.2	287°06.0	S14°23.2	97°07.0	N15°40.8					
7	161°37.7	240°15.3	48.8	240°37.9	23.1	302°08.0	23.4	112°09.7	40.8					
8	176°40.2	255°15.5	48.5	255°38.3	23.0	317°09.9	23.5	127°12.3	40.7					
9	191°42.7	270°15.8	· · 48.3	270°38.6	· · 22.8	332°11.8	· · 23.7	142°15.0	· · 40.7					
10	206°45.1	285°16.0	48.1	285°39.0	22.7	347°13.8	23.9	157°17.7	40.6					
11	221°47.6	300°16.3	47.8	300°39.4	22.5	2°15.7	24.0	172°20.4	40.6					
12	236°50.1	315°16.6	S26°47.6	315°39.8	S24°22.4	17°17.6	S14°24.2	187°23.0	N15°40.5					
13	251°52.5	330°16.8	47.4	330°40.2	22.2	32°19.6	24.4	202°25.7	40.5					
14	266°55.0	345°17.1	47.1	345°40.6	22.1	47°21.5	24.5	217°28.4	40.4					
15	281°57.5	0°17.4	· · 46.9	0°40.9	· · 21.9	62°23.4	· · 24.7	232°31.0	· · 40.4					
16	296°59.9	15°17.6	46.7	15°41.3	21.8	77°25.4	24.9	247°33.7	40.3					
17	312°02.4	30°17.9	46.4	30°41.7	21.6	92°27.3	25.0	262°36.4	40.3					
18	327°04.8	45°18.2	S26°46.2	45°42.1	S24°21.5	107°29.2	S14°25.2	277°39.0	N15°40.2					
19	342°07.3	60°18.5	45.9	60°42.5	21.3	122°31.2	25.4	292°41.7	40.2					
20	357°09.8	75°18.7	45.7	75°42.9	21.2	137°33.1	25.5	307°44.4	40.1					
21	12°12.2	90°19.0	· · 45.5	90°43.2	· · 21.0	152°35.0	· · 25.7	322°47.0	· · 40.1					
22	27°14.7	105°19.3	45.2	105°43.6	20.9	167°37.0	25.9	337°49.7	40.0					
23	42°17.2	120°19.6	45.0	120°44.0	20.7	182°38.9	26.0	352°52.4	40.0					
Mer.pass. 20:11			ν 0.2' d -0.2' m -4.75			ν 0.4' d -0.1' m 1.09			ν 1.9' d 0.2' m -1.69			ν 2.7' d -0.0' m -0.28		

Nov 15 Thu			SHA	Mer.pass	
			Venus	80°41.6	15:00
			Mars	80°54.6	14:58
			Jupiter	140°59.4	10:57
			Saturn	310°20.7	23:37

Nov 16 Fri			SHA	Mer.pass	
			Venus	79°46.8	14:59
			Mars	80°04.6	14:58
			Jupiter	140°46.7	10:54
			Saturn	310°25.6	23:33

Nov 17 Sat			SHA	Mer.pass	
			Venus	78°53.0	14:59
			Mars	79°14.7	14:57
			Jupiter	140°33.9	10:51
			Saturn	310°30.5	23:28

Horizontal parallax			
Venus:			0.3
Mars:			0.1

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	183°51.7	S18°29.7	69°39.0	15.6'	S00°58.6	-12.2'	54.9'
1	198°51.6	30.4	84°13.5	15.6'	00°46.4	-12.2'	54.9'
2	213°51.5	31.0	98°48.1	15.6'	00°34.2	-12.2'	54.9'
3	228°51.4	· · 31.7	113°22.8	15.6'	00°22.0	-12.2'	54.8'
4	243°51.3	32.3	127°57.4	15.7'	S00°09.9	-12.2'	54.8'
5	258°51.2	32.9	142°32.1	15.7'	N00°02.3	12.1'	54.8'
6	273°51.1	S18°33.6	157°06.8	15.7'	N00°14.4	12.1'	54.8'
7	288°51.0	34.2	171°41.5	15.7'	00°26.6	12.1'	54.8'
8	303°50.9	34.8	186°16.2	15.7'	00°38.7	12.1'	54.7'
9	318°50.8	· · 35.5	200°50.9	15.8'	00°50.8	12.1'	54.7'
10	333°50.7	36.1	215°25.7	15.8'	01°02.9	12.1'	54.7'
11	348°50.5	36.7	230°00.4	15.8'	01°15.0	12.1'	54.7'
12	3°50.4	S18°37.3	244°35.2	15.8'	N01°27.1	12.1'	54.7'
13	18°50.3	38.0	259°10.0	15.8'	01°39.2	12.1'	54.6'
14	33°50.2	38.6	273°44.8	15.8'	01°51.2	12.0'	54.6'
15	48°50.1	· · 39.2	288°19.6	15.8'	02°03.3	12.0'	54.6'
16	63°50.0	39.9	302°54.4	15.8'	02°15.3	12.0'	54.6'
17	78°49.9	40.5	317°29.3	15.8'	02°27.3	12.0'	54.6'
18	93°49.8	S18°41.1	332°04.1	15.8'	N02°39.3	12.0'	54.6'
19	108°49.7	41.7	346°39.0	15.8'	02°51.3	12.0'	54.5'
20	123°49.6	42.4	1°13.8	15.9'	03°03.3	11.9'	54.5'
21	138°49.4	· · 43.0	15°48.7	15.9'	03°15.2	11.9'	54.5'
22	153°49.3	43.6	30°23.5	15.9'	03°27.1	11.9'	54.5'
23	168°49.2	44.2	44°58.4	15.9'	03°39.0	11.9'	54.5'
SD = 16.2' $d = 0.6'$			SD = 15.0'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°49.1	S18°44.9	59°33.2	15.9'	N03°50.9	11.9'	54.5'
1	198°49.0	45.5	74°08.1	15.9'	04°02.8	11.8'	54.4'
2	213°48.9	46.1	88°42.9	15.9'	04°14.6	11.8'	54.4'
3	228°48.8	· · 46.7	103°17.8	15.9'	04°26.4	11.8'	54.4'
4	243°48.7	47.4	117°52.7	15.9'	04°38.2	11.8'	54.4'
5	258°48.5	48.0	132°27.5	15.9'	04°50.0	11.7'	54.4'
6	273°48.4	S18°48.6	147°02.4	15.8'	N05°01.8	11.7'	54.4'
7	288°48.3	49.2	161°37.2	15.8'	05°13.5	11.7'	54.4'
8	303°48.2	49.8	176°12.0	15.8'	05°25.2	11.7'	54.3'
9	318°48.1	· · 50.4	190°46.9	15.8'	05°36.9	11.6'	54.3'
10	333°48.0	51.1	205°21.7	15.8'	05°48.5	11.6'	54.3'
11	348°47.8	51.7	219°56.5	15.8'	06°00.1	11.6'	54.3'
12	3°47.7	S18°52.3	234°31.3	15.8'	N06°11.7	11.6'	54.3'
13	18°47.6	52.9	249°06.1	15.8'	06°23.3	11.5'	54.3'
14	33°47.5	53.5	263°40.9	15.8'	06°34.8	11.5'	54.3'
15	48°47.4	· · 54.1	278°15.7	15.8'	06°46.3	11.5'	54.2'
16	63°47.3	54.8	292°50.5	15.8'	06°57.8	11.4'	54.2'
17	78°47.1	55.4	307°25.2	15.7'	07°09.2	11.4'	54.2'
18	93°47.0	S18°56.0	322°00.0	15.7'	N07°20.6	11.4'	54.2'
19	108°46.9	56.6	336°34.7	15.7'	07°32.0	11.3'	54.2'
20	123°46.8	57.2	351°09.4	15.7'	07°43.4	11.3'	54.2'
21	138°46.7	· · 57.8	5°44.1	15.7'	07°54.7	11.3'	54.2'
22	153°46.5	58.4	20°18.8	15.7'	08°06.0	11.2'	54.2'
23	168°46.4	59.0	34°53.4	15.6'	08°17.2	11.2'	54.2'
SD = 16.2' $d = 0.6'$			SD = 14.9'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°46.3	S18°59.6	49°28.1	15.6'	N08°28.4	11.2'	54.1'
1	198°46.2	19°00.3	64°02.7	15.6'	08°39.6	11.1'	54.1'
2	213°46.1	00.9	78°37.3	15.6'	08°50.7	11.1'	54.1'
3	228°45.9	· · 01.5	93°11.9	15.6'	09°01.8	11.1'	54.1'
4	243°45.8	02.1	107°46.4	15.5'	09°12.9	11.0'	54.1'
5	258°45.7	02.7	122°21.0	15.5'	09°23.9	11.0'	54.1'
6	273°45.6	S19°03.3	136°55.5	15.5'	N09°34.9	10.9'	54.1'
7	288°45.4	03.9	151°30.0	15.5'	09°45.8	10.9'	54.1'
8	303°45.3	04.5	166°04.4	15.4'	09°56.7	10.9'	54.1'
9	318°45.2	· · 05.1	180°38.9	15.4'	10°07.6	10.8'	54.1'
10	333°45.1	05.7	195°13.3	15.4'	10°18.4	10.8'	54.1'
11	348°44.9	06.3	209°47.7	15.4'	10°29.2	10.7'	54.1'
12	3°44.8	S19°06.9	224°22.0	15.3'	N10°39.9	10.7'	54.0'
13	18°44.7	07.5	238°56.4	15.3'	10°50.6	10.6'	54.0'
14	33°44.6	08.1	253°30.7	15.3'	11°01.2	10.6'	54.0'
15	48°44.4	· · 08.7	268°05.0	15.2'	11°11.8	10.6'	54.0'
16	63°44.3	09.3	282°39.2	15.2'	11°22.4	10.5'	54.0'
17	78°44.2	09.9	297°13.4	15.2'	11°32.9	10.5'	54.0'
18	93°44.1	S19°10.5	311°47.6	15.2'	N11°43.3	10.4'	54.0'
19	108°43.9	11.1	326°21.8	15.1'	11°53.7	10.4'	54.0'
20	123°43.8	11.7	340°55.9	15.1'	12°04.1	10.3'	54.0'
21	138°43.7	· · 12.3	355°30.0	15.1'	12°14.4	10.3'	54.0'
22	153°43.5	12.9	10°04.0	15.0'	12°24.7	10.2'	54.0'
23	168°43.4	13.5	24°38.0	15.0'	12°34.9	10.2'	54.0'
SD = 16.2' $d = 0.6'$			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:04	08:40	■	■	14:48	16:24
N 70°	06:53	08:16	09:58	13:31	15:12	16:35
68°	06:44	07:58	09:19	14:10	15:31	16:44
66°	06:37	07:43	08:52	14:37	15:45	16:52
64°	06:30	07:31	08:31	14:57	15:58	16:58
62°	06:25	07:21	08:15	15:14	16:08	17:04
60°	06:19	07:12	08:01	15:28	16:17	17:09
N 58°	06:15	07:04	07:49	15:40	16:25	17:14
56°	06:11	06:56	07:39	15:50	16:32	17:18
54°	06:07	06:50	07:30	15:59	16:39	17:22
52°	06:03	06:44	07:22	16:07	16:45	17:26
50°	06:00	06:39	07:15	16:14	16:50	17:30
45°	05:52	06:27	06:59	16:30	17:02	17:37
N 40°	05:45	06:17	06:46	16:43	17:12	17:45
35°	05:38	06:09	06:36	16:54	17:21	17:51
30°	05:32	06:01	06:26	17:03	17:29	17:58
20°	05:20	05:46	06:09	17:20	17:43	18:10
N 10°	05:07	05:33	05:55	17:35	17:57	18:22
0°	04:54	05:19	05:41	17:48	18:10	18:36
S 10°	04:39	05:05	05:27	18:02	18:25	18:51
20°	04:21	04:49	05:13	18:17	18:41	19:09
30°	03:58	04:29	04:55	18:35	19:01	19:32
35°	03:43	04:17	04:45	18:45	19:13	19:47
40°	03:25	04:03	04:34	18:57	19:27	20:05
45°	03:02	03:46	04:20	19:10	19:45	20:28
S 50°	02:32	03:24	04:03	19:28	20:07	21:00
52°	02:15	03:13	03:55	19:36	20:18	21:16
54°	01:56	03:01	03:46	19:45	20:30	21:36
56°	01:30	02:47	03:36	19:55	20:45	22:03
58°	00:52	02:30	03:25	20:06	21:02	22:44
S 60°	—:—	02:09	03:12	20:20	21:24	—:—

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	13:24	13:04	12:41	01:13	03:01	04:51
N 70°	13:26	13:13	12:58	01:14	02:55	04:36
68°	13:28	13:20	13:12	01:15	02:49	04:24
66°	13:29	13:26	13:23	01:16	02:45	04:14
64°	13:31	13:31	13:33	01:17	02:42	04:06
62°	13:32	13:36	13:41	01:17	02:38	03:59
60°	13:33	13:40	13:48	01:18	02:36	03:53
N 58°	13:34	13:43	13:54	01:18	02:33	03:48
56°	13:34	13:46	13:59	01:19	02:31	03:43
54°	13:35	13:49	14:04	01:19	02:29	03:39
52°	13:36	13:52	14:09	01:19	02:28	03:35
50°	13:36	13:54	14:13	01:20	02:26	03:32
45°	13:38	13:59	14:22	01:20	02:23	03:24
N 40°	13:39	14:04	14:29	01:21	02:20	03:18
35°	13:40	14:07	14:36	01:21	02:17	03:13
30°	13:41	14:11	14:41	01:22	02:15	03:08
20°	13:42	14:16	14:51	01:22	02:11	03:00
N 10°	13:44	14:22	15:00	01:23	02:08	02:53
0°	13:45	14:26	15:08	01:23	02:05	02:46
S 10°	13:46	14:31	15:17	01:24	02:02	02:40
20°	13:48	14:36	15:25	01:24	01:59	02:33
30°	13:49	14:42	15:36	01:25	01:55	02:25
35°	13:50	14:46	15:42	01:25	01:53	02:20
40°	13:51	14:50	15:48	01:26	01:50	02:15
45°	13:53	14:55	15:56	01:26	01:47	02:09
S 50°	13:54	15:00	16:06	01:26	01:44	02:02
52°	13:55	15:03	16:10	01:27	01:43	01:59
54°	13:56	15:05	16:15	01:27	01:41	01:55
56°	13:57	15:09	16:20	01:27	01:39	01:51
58°	13:57	15:12	16:27	01:27	01:37	01:47
S 60°	13:59	15:16	16:33	01:28	01:35	01:42

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	9-11
	mm:ss	mm:ss	mm:ss	hh:mm	hh:mm	69-85%
15	15:27	15:22	11:45	19:55	07:34	
16	15:16	15:11	11:45	20:36	08:16	
17	15:05	14:59	11:45	21:19	08:57	

November 18, 19, 20 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	57°19.6	135°19.9	S26°44.7	135°44.4	S24°20.5	197°40.8	S14°26.2	7°55.0	N15°39.9	Alpheratz	357°30.5	29°15.6	
1	72°22.1	150°20.2	44.5	150°44.8	20.4	212°42.8	26.4	22°57.7	39.9	Ankaa	353°03.2	-42°08.6	
2	87°24.6	165°20.5	44.2	165°45.2	20.2	227°44.7	26.5	38°00.4	39.8	Schedar	349°26.0	56°42.4	
3	102°27.0	180°20.8	· · 44.0	180°45.5	· · 20.1	242°46.6	· · 26.7	53°03.0	· · 39.8	Diphda	348°43.2	-17°49.3	
4	117°29.5	195°21.1	43.7	195°45.9	19.9	257°48.6	26.9	68°05.7	39.7	Achernar	335°17.0	-57°05.2	
5	132°32.0	210°21.4	43.5	210°46.3	19.8	272°50.5	27.0	83°08.4	39.7	Hamal	327°46.4	23°36.3	
6	147°34.4	225°21.7	S26°43.3	225°46.7	S24°19.6	287°52.4	S14°27.2	98°11.0	N15°39.7	Polaris	311°20.5	89°23.3	
7	162°36.9	240°22.0	43.0	240°47.1	19.5	302°54.4	27.4	113°13.7	39.6	Acamar	315°08.5	-40°11.1	
8	177°39.3	255°22.3	42.8	255°47.4	19.3	317°56.3	27.5	128°16.4	39.6	Menkar	314°01.7	4°12.5	
9	192°41.8	270°22.6	· · 42.5	270°47.8	· · 19.1	332°58.2	· · 27.7	143°19.0	· · 39.5	Mirfak	308°22.0	49°58.0	
10	207°44.3	285°22.9	42.3	285°48.2	19.0	348°00.2	27.9	158°21.7	39.5	Aldebaran	290°34.8	16°34.1	
11	222°46.7	300°23.2	42.0	300°48.6	18.8	3°02.1	28.0	173°24.4	39.4	Rigel	280°59.8	-8°10.1	
12	237°49.2	315°23.5	S26°41.7	315°49.0	S24°18.7	18°04.0	S14°28.2	188°27.1	N15°39.4	Capella	280°15.7	46°01.5	
13	252°51.7	330°23.8	41.5	330°49.4	18.5	33°06.0	28.3	203°29.7	39.3	Bellatrix	278°18.4	6°22.5	
14	267°54.1	345°24.1	41.2	345°49.7	18.3	48°07.9	28.5	218°32.4	39.3	Elnath	277°56.6	28°37.8	
15	282°56.6	0°24.4	· · 41.0	0°50.1	· · 18.2	63°09.8	· · 28.7	233°35.1	· · 39.2	Alnilam	275°33.5	-1°11.0	
16	297°59.1	15°24.8	40.7	15°50.5	18.0	78°11.8	28.8	248°37.7	39.2	Betelgeuse	270°47.6	7°24.6	
17	313°01.5	30°25.1	40.5	30°50.9	17.9	93°13.7	29.0	263°40.4	39.1	Canopus	263°50.2	-52°42.6	
18	328°04.0	45°25.4	S26°40.2	45°51.3	S24°17.7	108°15.6	S14°29.2	278°43.1	N15°39.1	Sirius	258°22.5	-16°45.5	
19	343°06.4	60°25.7	40.0	60°51.7	17.5	123°17.6	29.3	293°45.7	39.0	Adhara	255°02.5	-29°00.7	
20	358°08.9	75°26.1	39.7	75°52.0	17.4	138°19.5	29.5	308°48.4	39.0	Procyon	244°46.6	5°08.8	
21	13°11.4	90°26.4	· · 39.4	90°52.4	· · 17.2	153°21.4	· · 29.7	323°51.1	· · 38.9	Pollux	243°12.4	27°57.0	
22	28°13.8	105°26.7	39.2	105°52.8	17.0	168°23.4	29.8	338°53.7	38.9	Avior	234°12.8	-59°36.1	
23	43°16.3	120°27.0	38.9	120°53.2	16.9	183°25.3	30.0	353°56.4	38.8	Suhail	222°43.3	-43°33.1	
Mer.pass. 20:07		ν 0.3' d -0.2' m -4.77		ν 0.4' d -0.2' m 1.09		ν 1.9' d 0.2' m -1.69		ν 2.7' d -0.0' m -0.27		Miaplacidus	221°37.0	-69°50.2	
										Alphard	217°43.9	-8°47.3	
										Regulus	207°30.4	11°49.1	
										Dubhe	193°36.8	61°35.0	
										Denebola	182°21.2	14°24.2	
										Gienah	175°39.7	-17°42.4	
										Acrux	172°55.9	-63°15.7	
										Gacrux	171°47.5	-57°16.6	
										Alioth	166°10.2	55°47.6	
										Spica	158°18.4	-11°19.0	
										Alkaid	152°49.5	49°09.7	
										Hadar	148°30.8	-60°30.8	
										Menkent	147°53.3	-36°30.9	
										Arcturus	145°44.7	19°01.6	
										Rigil Kent.	139°35.4	-60°57.4	
										Kochab	137°21.4	74°01.9	
										Zuben'ubi	136°52.0	-16°09.8	
										Alphecca	126°00.9	26°36.9	
										Antares	112°11.4	-26°29.7	
										Atria	107°02.6	-69°04.7	
										Sabik	101°58.6	-15°45.5	
										Shaula	96°05.5	-37°07.4	
										Rasalhague	95°55.3	12°32.5	
										Eltanin	90°40.8	51°29.4	
										Kaus Aust.	83°27.7	-34°22.1	
										Vega	80°30.9	38°49.0	
										Nunki	75°43.2	-26°15.4	
										Altair	61°56.3	8°57.1	
										Peacock	53°00.0	-56°38.3	
										Deneb	49°23.2	45°23.7	
										Enif	33°35.0	10°01.0	
										Al Na'ir	27°28.2	-46°49.0	
										Fomalhaut	15°10.2	-29°27.8	
										Scheat	13°41.2	28°15.0	
										Markab	13°25.9	15°22.3	
										Nov 18 Sun		SHA	Mer.pass
										Venus	78°00.3	14:58	
										Mars	78°24.8	14:57	
										Jupiter	140°21.2	10:48	
										Saturn	310°35.4	23:24	
										Nov 19 Mon		SHA	Mer.pass
										Venus	77°08.6	14:58	
										Mars	77°34.8	14:56	
										Jupiter	140°08.5	10:45	
										Saturn	310°40.3	23:20	
										Nov 20 Tue		SHA	Mer.pass
										Venus	76°18.1	14:57	
										Mars	76°44.8	14:55	
										Jupiter	139°55.8	10:42	
										Saturn	310°45.2	23:16	
										Horizontal parallax			
										Venus:		0.3	
										Mars:		0.1	
Mer.pass. 20:00		ν 0.4' d -0.3' m -4.79		ν 0.4' d -0.2' m 1.09		ν 1.9' d 0.2' m -1.70		ν 2.7' d -0.0' m -0.25					

DUT1 = UT1-UTC = +0.2136 sec ΔT = TT-UT1 = +68.9704 sec

2029 November 18 to Nov. 20 UT

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	183°43.3	S19°14.1	39°12.0	14.9'	N12°45.0	10.1'	54.0'
1	198°43.1	14.7	53°46.0	14.9'	12°55.2	10.1'	54.0'
2	213°43.0	15.3	68°19.9	14.9'	13°05.2	10.0'	54.0'
3	228°42.9	· · 15.9	82°53.8	14.8'	13°15.2	9.9'	54.0'
4	243°42.8	16.5	97°27.6	14.8'	13°25.2	9.9'	54.0'
5	258°42.6	17.1	112°01.4	14.8'	13°35.0	9.8'	53.9'
6	273°42.5	S19°17.7	126°35.2	14.7'	N13°44.9	9.8'	53.9'
7	288°42.4	18.2	141°08.9	14.7'	13°54.7	9.7'	53.9'
8	303°42.2	18.8	155°42.6	14.6'	14°04.4	9.7'	53.9'
9	318°42.1	· · 19.4	170°16.2	14.6'	14°14.1	9.6'	53.9'
10	333°42.0	20.0	184°49.8	14.6'	14°23.7	9.5'	53.9'
11	348°41.8	20.6	199°23.4	14.5'	14°33.2	9.5'	53.9'
12	3°41.7	S19°21.2	213°56.9	14.5'	N14°42.7	9.4'	53.9'
13	18°41.6	21.8	228°30.4	14.4'	14°52.1	9.4'	53.9'
14	33°41.4	22.4	243°03.9	14.4'	15°01.5	9.3'	53.9'
15	48°41.3	· · 22.9	257°37.3	14.4'	15°10.8	9.2'	53.9'
16	63°41.2	23.5	272°10.6	14.3'	15°20.0	9.2'	53.9'
17	78°41.0	24.1	286°43.9	14.3'	15°29.2	9.1'	53.9'
18	93°40.9	S19°24.7	301°17.2	14.2'	N15°38.3	9.1'	53.9'
19	108°40.7	25.3	315°50.4	14.2'	15°47.4	9.0'	53.9'
20	123°40.6	25.9	330°23.6	14.1'	15°56.4	8.9'	53.9'
21	138°40.5	· · 26.5	344°56.8	14.1'	16°05.3	8.9'	53.9'
22	153°40.3	27.0	359°29.9	14.0'	16°14.2	8.8'	53.9'
23	168°40.2	27.6	14°02.9	14.0'	16°23.0	8.7'	53.9'
SD = 16.2'			SD = 14.7'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	183°40.1	S19°28.2	28°35.9	14.0'	N16°31.7	8.7'	53.9'
1	198°39.9	28.8	43°08.9	13.9'	16°40.3	8.6'	53.9'
2	213°39.8	29.4	57°41.8	13.9'	16°48.9	8.5'	53.9'
3	228°39.6	· · 29.9	72°14.6	13.8'	16°57.4	8.4'	53.9'
4	243°39.5	30.5	86°47.5	13.8'	17°05.9	8.4'	53.9'
5	258°39.4	31.1	101°20.2	13.7'	17°14.2	8.3'	53.9'
6	273°39.2	S19°31.7	115°53.0	13.7'	N17°22.5	8.2'	53.9'
7	288°39.1	32.2	130°25.6	13.6'	17°30.7	8.1'	53.9'
8	303°38.9	32.8	144°58.3	13.6'	17°38.9	8.1'	53.9'
9	318°38.8	· · 33.4	159°30.9	13.5'	17°47.0	8.0'	53.9'
10	333°38.7	34.0	174°03.4	13.5'	17°55.0	7.9'	53.9'
11	348°38.5	34.5	188°35.9	13.4'	18°02.9	7.8'	53.9'
12	3°38.4	S19°35.1	203°08.3	13.4'	N18°10.7	7.8'	53.9'
13	18°38.2	35.7	217°40.7	13.3'	18°18.5	7.7'	53.9'
14	33°38.1	36.3	232°13.1	13.3'	18°26.2	7.6'	53.9'
15	48°37.9	· · 36.8	246°45.4	13.2'	18°33.8	7.5'	53.9'
16	63°37.8	37.4	261°17.6	13.2'	18°41.4	7.5'	53.9'
17	78°37.7	38.0	275°49.8	13.1'	18°48.8	7.4'	53.9'
18	93°37.5	S19°38.5	290°21.9	13.1'	N18°56.2	7.3'	53.9'
19	108°37.4	39.1	304°54.0	13.1'	19°03.5	7.2'	53.9'
20	123°37.2	39.7	319°26.1	13.0'	19°10.7	7.1'	53.9'
21	138°37.1	· · 40.2	333°58.1	13.0'	19°17.8	7.0'	53.9'
22	153°36.9	40.8	348°30.1	12.9'	19°24.9	7.0'	53.9'
23	168°36.8	41.4	3°02.0	12.9'	19°31.8	6.9'	53.9'
SD = 16.2'			SD = 14.7'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	183°36.6	S19°41.9	17°33.8	12.8'	N19°38.7	6.8'	53.9'
1	198°36.5	42.5	32°05.6	12.8'	19°45.5	6.7'	53.9'
2	213°36.3	43.1	46°37.4	12.7'	19°52.2	6.6'	54.0'
3	228°36.2	· · 43.6	61°09.1	12.7'	19°58.8	6.5'	54.0'
4	243°36.0	44.2	75°40.8	12.6'	20°05.3	6.4'	54.0'
5	258°35.9	44.8	90°12.4	12.6'	20°11.8	6.4'	54.0'
6	273°35.7	S19°45.3	104°43.9	12.5'	N20°18.1	6.3'	54.0'
7	288°35.6	45.9	119°15.5	12.5'	20°24.4	6.2'	54.0'
8	303°35.4	46.5	133°46.9	12.4'	20°30.6	6.1'	54.0'
9	318°35.3	· · 47.0	148°18.4	12.4'	20°36.7	6.0'	54.0'
10	333°35.1	47.6	162°49.7	12.3'	20°42.6	5.9'	54.0'
11	348°35.0	48.1	177°21.1	12.3'	20°48.5	5.8'	54.0'
12	3°34.8	S19°48.7	191°52.3	12.2'	N20°54.3	5.7'	54.0'
13	18°34.7	49.2	206°23.6	12.2'	21°00.1	5.6'	54.0'
14	33°34.5	49.8	220°54.8	12.1'	21°05.7	5.5'	54.0'
15	48°34.4	· · 50.4	235°25.9	12.1'	21°11.2	5.4'	54.0'
16	63°34.2	50.9	249°57.0	12.1'	21°16.6	5.3'	54.0'
17	78°34.1	51.5	264°28.0	12.0'	21°22.0	5.2'	54.0'
18	93°33.9	S19°52.0	278°59.1	12.0'	N21°27.2	5.1'	54.0'
19	108°33.8	52.6	293°30.0	11.9'	21°32.3	5.0'	54.0'
20	123°33.6	53.1	308°00.9	11.9'	21°37.4	4.9'	54.0'
21	138°33.5	· · 53.7	322°31.8	11.8'	21°42.3	4.8'	54.0'
22	153°33.3	54.2	337°02.6	11.8'	21°47.2	4.7'	54.1'
23	168°33.2	54.8	351°33.4	11.7'	21°51.9	4.6'	54.1'
SD = 16.2'			SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:14	08:54	■	■	14:35	16:15
N 70°	07:02	08:28	10:20	13:10	15:02	16:27
68°	06:53	08:08	09:33	13:57	15:22	16:37
66°	06:44	07:52	09:03	14:27	15:38	16:46
64°	06:37	07:39	08:41	14:49	15:51	16:53
62°	06:31	07:28	08:23	15:07	16:03	16:59
60°	06:25	07:18	08:08	15:22	16:12	17:05
N 58°	06:20	07:09	07:56	15:34	16:21	17:10
56°	06:16	07:02	07:45	15:45	16:28	17:15
54°	06:11	06:55	07:36	15:55	16:35	17:19
52°	06:07	06:49	07:27	16:03	16:41	17:23
50°	06:04	06:43	07:19	16:11	16:47	17:27
45°	05:55	06:31	07:03	16:27	16:59	17:35
N 40°	05:48	06:21	06:50	16:41	17:10	17:43
35°	05:41	06:11	06:38	16:52	17:19	17:50
30°	05:34	06:03	06:29	17:02	17:28	17:57
20°	05:21	05:48	06:11	17:19	17:43	18:10
N 10°	05:08	05:34	05:56	17:35	17:57	18:23
0°	04:54	05:20	05:42	17:49	18:11	18:37
S 10°	04:39	05:05	05:27	18:04	18:26	18:52
20°	04:20	04:48	05:12	18:19	18:43	19:11
30°	03:56	04:28	04:54	18:37	19:04	19:35
35°	03:41	04:15	04:43	18:48	19:16	19:51
40°	03:22	04:00	04:31	19:00	19:31	20:10
45°	02:58	03:42	04:17	19:15	19:49	20:34
S 50°	02:26	03:19	03:59	19:32	20:13	21:07
52°	02:09	03:08	03:51	19:41	20:24	21:24
54°	01:47	02:55	03:42	19:50	20:37	21:46
56°	01:18	02:40	03:31	20:01	20:53	22:16
58°	00:29	02:22	03:19	20:13	21:11	23:12
S 60°	—:—	01:59	03:05	20:28	21:35	—:—

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	12:10	10:46	□	06:52	09:49	□
N 70°	12:40	12:12	□	06:23	08:24	□
68°	13:03	12:51	12:31	06:02	07:45	09:42
66°	13:21	13:19	13:18	05:45	07:19	08:55
64°	13:35	13:40	13:49	05:32	06:58	08:25
62°	13:47	13:57	14:12	05:20	06:42	08:03
60°	13:58	14:11	14:30	05:11	06:28	07:45
N 58°	14:07	14:23	14:46	05:02	06:17	07:30
56°	14:15	14:34	14:59	04:55	06:07	07:17
54°	14:22	14:43	15:10	04:48	05:58	07:06
52°	14:28	14:52	15:20	04:43	05:50	06:56
50°	14:34	14:59	15:30	04:37	05:43	06:47
45°	14:47	15:15	15:49	04:26	05:28	06:28
N 40°	14:57	15:29	16:05	04:17	05:15	06:13
35°	15:06	15:40	16:18	04:08	05:05	06:01
30°	15:14	15:50	16:30	04:01	04:55	05:50
20°	15:28	16:07	16:49	03:49	04:39	05:31
N 10°	15:40	16:22	17:07	03:39	04:26	05:14
0°	15:51	16:36	17:23	03:29	04:13	04:59
S 10°	16:03	16:51	17:40	03:19	04:00	04:43
20°	16:15	17:06	17:57	03:09	03:46	04:27
30°	16:29	17:23	18:18	02:57	03:31	04:08
35°	16:37	17:34	18:30	02:50	03:22	03:57
40°	16:47	17:45	18:43	02:42	03:11	03:45
45°	16:58	17:59	19:00	02:33	02:59	03:30
S 50°	17:11	18:16	19:20	02:22	02:45	03:12
52°	17:18	18:24	19:29	02:17	02:38	03:04
54°	17:25	18:33	19:40	02:12	02:31	02:54
56°	17:32	18:44	19:52	02:05	02:22	02:44
58°	17:41	18:55	20:07	01:59	02:13	02:32
S 60°	17:51	19:09	20:24	01:51	02:02	02:18

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12 ^h	Pass	Mer.Pass.	Lower	12-14
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	91-99%
18	14:53	14:47	11:45	22:02	09:40	
19	14:40	14:33	11:45	22:47	10:25	
20	14:27	14:19	11:46	23:35	11:11	

November 21, 22, 23 UT (Wed., Thu., Fri.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Wed	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	SHA Dec	SHA Dec			
0	60°17.0	135°45.8	S26°25.2	136°11.9	S24°08.2	200°00.1	S14°38.0	11°07.1	N15°36.4	Alpheratz	357°30.5	29°15.6					
1	75°19.5	150°46.3	24.9	151°12.3	08.0	215°02.1	38.2	26°09.7	36.4	Ankaa	353°03.2	-42°08.7					
2	90°22.0	165°46.7	24.6	166°12.7	07.8	230°04.0	38.3	41°12.4	36.3	Schedar	349°26.0	56°42.4					
3	105°24.4	180°47.1	•• 24.3	181°13.1	•• 07.6	245°05.9	•• 38.5	56°15.1	•• 36.3	Diphda	348°43.2	-17°49.3					
4	120°26.9	195°47.6	24.0	196°13.4	07.5	260°07.9	38.7	71°17.7	36.2	Achernar	335°17.0	-57°05.2					
5	135°29.4	210°48.0	23.7	211°13.8	07.3	275°09.8	38.8	86°20.4	36.2	Hamal	327°46.4	23°36.4					
6	150°31.8	225°48.5	S26°23.4	226°14.2	S24°07.1	290°11.7	S14°39.0	101°23.1	N15°36.1	Polaris	311°20.3	89°23.3					
7	165°34.3	240°48.9	23.1	241°14.6	06.9	305°13.7	39.2	116°25.7	36.1	Acamar	315°08.5	-40°11.1					
8	180°36.8	255°49.4	22.8	256°15.0	06.7	320°15.6	39.3	131°28.4	36.0	Menkar	314°01.7	4°12.4					
9	195°39.2	270°49.8	•• 22.5	271°15.4	•• 06.5	335°17.6	•• 39.5	146°31.1	•• 36.0	Mirfak	308°22.0	49°58.0					
10	210°41.7	285°50.3	22.2	286°15.7	06.3	350°19.5	39.6	161°33.7	35.9	Aldebaran	290°34.8	16°34.1					
11	225°44.2	300°50.7	21.9	301°16.1	06.1	5°21.4	39.8	176°36.4	35.9	Rigel	280°59.8	-8°10.1					
12	240°46.6	315°51.2	S26°21.6	316°16.5	S24°05.9	20°23.4	S14°40.0	191°39.1	N15°35.8	Capella	280°15.6	46°01.5					
13	255°49.1	330°51.7	21.3	331°16.9	05.7	35°25.3	40.1	206°41.7	35.8	Bellatrix	278°18.4	6°22.5					
14	270°51.5	345°52.1	21.0	346°17.3	05.5	50°27.2	40.3	221°44.4	35.8	Elnath	277°56.6	28°37.8					
15	285°54.0	0°52.6	•• 20.7	1°17.7	•• 05.3	65°29.2	•• 40.5	236°47.1	•• 35.7	Alnilam	275°33.5	-1°11.0					
16	300°56.5	15°53.1	20.4	16°18.0	05.1	80°31.1	40.6	251°49.7	35.7	Betelgeuse	270°47.6	7°24.6					
17	315°58.9	30°53.5	20.1	31°18.4	04.9	95°33.0	40.8	266°52.4	35.6	Canopus	263°50.2	-52°42.6					
18	331°01.4	45°54.0	S26°19.8	46°18.8	S24°04.7	110°35.0	S14°40.9	281°55.0	N15°35.6	Sirius	258°22.5	-16°45.5					
19	346°03.9	60°54.5	19.5	61°19.2	04.6	125°36.9	41.1	296°57.7	35.5	Adhara	255°02.5	-29°00.7					
20	1°06.3	75°54.9	19.1	76°19.6	04.4	140°38.8	41.3	312°00.4	35.5	Procyon	244°46.6	5°08.7					
21	16°08.8	90°55.4	•• 18.8	91°20.0	•• 04.2	155°40.8	•• 41.4	327°03.0	•• 35.4	Pollux	243°12.4	27°57.0					
22	31°11.3	105°55.9	18.5	106°20.3	04.0	170°42.7	41.6	342°05.7	35.4	Avior	234°12.7	-59°36.2					
23	46°13.7	120°56.4	18.2	121°20.7	03.8	185°44.7	41.7	357°08.4	35.3	Suhail	222°43.2	-43°33.1					
Mer.pass. 19:56			ν 0.4' d -0.3' m -4.80			ν 0.4' d -0.2' m 1.09			ν 1.9' d 0.2' m -1.70			ν 2.7' d -0.0' m -0.24					

Thu	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec					
0	61°16.2	135°56.9	S26°17.9	136°21.1	S24°03.6	200°46.6	S14°41.9	12°11.0	N15°35.3					
1	76°18.7	150°57.4	17.6	151°21.5	03.4	215°48.5	42.1	27°13.7	35.2					
2	91°21.1	165°57.9	17.3	166°21.9	03.2	230°50.5	42.2	42°16.4	35.2					
3	106°23.6	180°58.3	•• 17.0	181°22.2	•• 03.0	245°52.4	•• 42.4	57°19.0	•• 35.1					
4	121°26.0	195°58.8	16.6	196°22.6	02.8	260°54.3	42.6	72°21.7	35.1					
5	136°28.5	210°59.3	16.3	211°23.0	02.6	275°56.3	42.7	87°24.4	35.0					
6	151°31.0	225°59.8	S26°16.0	226°23.4	S24°02.4	290°58.2	S14°42.9	102°27.0	N15°35.0					
7	166°33.4	241°00.3	15.7	241°23.8	02.2	306°00.2	43.0	117°29.7	34.9					
8	181°35.9	256°00.8	15.4	256°24.2	01.9	321°02.1	43.2	132°32.4	34.9					
9	196°38.4	271°01.3	•• 15.0	271°24.5	•• 01.7	336°04.0	•• 43.4	147°35.0	•• 34.8					
10	211°40.8	286°01.9	14.7	286°24.9	01.5	351°06.0	43.5	162°37.7	34.8					
11	226°43.3	301°02.4	14.4	301°25.3	01.3	6°07.9	43.7	177°40.4	34.7					
12	241°45.8	316°02.9	S26°14.1	316°25.7	S24°01.1	21°09.8	S14°43.8	192°43.0	N15°34.7					
13	256°48.2	331°03.4	13.8	331°26.1	00.9	36°11.8	44.0	207°45.7	34.7					
14	271°50.7	346°03.9	13.4	346°26.5	00.7	51°13.7	44.2	222°48.3	34.6					
15	286°53.1	1°04.4	•• 13.1	1°26.8	•• 00.5	66°15.6	•• 44.3	237°51.0	•• 34.6					
16	301°55.6	16°05.0	12.8	16°27.2	00.3	81°17.6	44.5	252°53.7	34.5					
17	316°58.1	31°05.5	12.5	31°27.6	24°00.1	96°19.5	44.7	267°56.3	34.5					
18	332°00.5	46°06.0	S26°12.1	46°28.0	S23°59.9	111°21.5	S14°44.8	282°59.0	N15°34.4					
19	347°03.0	61°06.5	11.8	61°28.4	59.7	126°23.4	45.0	298°01.7	34.4					
20	2°05.5	76°07.1	11.5	76°28.8	59.5	141°25.3	45.1	313°04.3	34.3					
21	17°07.9	91°07.6	•• 11.2	91°29.1	•• 59.3	156°27.3	•• 45.3	328°07.0	•• 34.3					
22	32°10.4	106°08.1	10.8	106°29.5	59.1	171°29.2	45.5	343°09.7	34.2					
23	47°12.9	121°08.7	10.5	121°29.9	58.8	186°31.1	45.6	358°12.3	34.2					
Mer.pass. 19:52			ν 0.5' d -0.3' m -4.81			ν 0.4' d -0.2' m 1.10			ν 1.9' d 0.2' m -1.70			ν 2.7' d -0.0' m -0.23		

Fri	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec					
0	62°15.3	136°09.2	S26°10.2	136°30.3	S23°58.6	201°33.1	S14°45.8	13°15.0	N15°34.1					
1	77°17.8	151°09.8	09.8	151°30.7	58.4	216°35.0	45.9	28°17.7	34.1					
2	92°20.3	166°10.3	09.5	166°31.0	58.2	231°37.0	46.1	43°20.3	34.0					
3	107°22.7	181°10.8	•• 09.2	181°31.4	•• 58.0	246°38.9	•• 46.3	58°23.0	•• 34.0					
4	122°25.2	196°11.4	08.9	196°31.8	57.8	261°40.8	46.4	73°25.6	33.9					
5	137°27.6	211°11.9	08.5	211°32.2	57.6	276°42.8	46.6	88°28.3	33.9					
6	152°30.1	226°12.5	S26°08.2	226°32.6	S23°57.4	291°44.7	S14°46.7	103°31.0	N15°33.8					
7	167°32.6	241°13.1	07.9	241°33.0	57.2	306°46.6	46.9	118°33.6	33.8					
8	182°35.0	256°13.6	07.5	256°33.3	56.9	321°48.6	47.1	133°36.3	33.7					
9	197°37.5	271°14.2	•• 07.2	271°33.7	•• 56.7	336°50.5	•• 47.2	148°39.0	•• 33.7					
10	212°40.0	286°14.7	06.8	286°34.1	56.5	351°52.5	47.4	163°41.6	33.7					
11	227°42.4	301°15.3	06.5	301°34.5	56.3	6°54.4	47.6	178°44.3	33.6					
12	242°44.9	316°15.9	S26°06.2	316°34.9	S23°56.1	21°56.3	S14°47.7	193°47.0	N15°33.6					
13	257°47.4	331°16.4	05.8	331°35.3	55.9	36°58.3	47.9	208°49.6	33.5					
14	272°49.8	346°17.0	05.5	346°35.6	55.6	52°00.2	48.0	223°52.3	33.5					
15	287°52.3	1°17.6	•• 05.2	1°36.0	•• 55.4	67°02.1	•• 48.2	238°54.9	•• 33.4					
16	302°54.8	16°18.2	04.8	16°36.4	55.2	82°04.1	48.4	253°57.6	33.4					
17	317°57.2	31°18.8	04.5	31°36.8	55.0	97°06.0	48.5	269°00.3	33.3					
18	332°59.7	46°19.3	S26°04.1	46°37.2	S23°54.8	112°08.0	S14°48.7	284°02.9	N15°33.3					
19	348°02.1	61°19.9	03.8	61°37.6	54.6	127°09.9	48.8	299°05.6	33.2					
20	3°04.6	76°20.5	03.4	76°37.9	54.3	142°11.8	49.0	314°08.3	33.2					
21	18°07.1	91°21.1	•• 03.1	91°38.3	•• 54.1	157°13.8	•• 49.2	329°10.9	•• 33.1					
22	33°09.5	106°21.7	02.8	106°38.7	53.9	172°15.7	49.3	344°13.6	33.1					
23	48°12.0	121°22.3	02.4	121°39.1	53.7	187°17.6	49.5	359°16.3	33.0					
Mer.pass. 19:48			ν 0.5' d -0.3' m -4.82			ν 0.4' d -0.2' m 1.10			ν 1.9' d 0.2' m -1.70			ν 2.7' d -0.0' m -0.22		

Nov 21 Wed	SHA	Mer.pass
Venus	75°28.8	14:56
Mars	75°54.9	14:55
Jupiter	139°43.1	10:39
Saturn	310°50.0	23:11

Nov 22 Thu	SHA	Mer.pass
Venus	74°40.7	14:56
Mars	75°04.9	14:54
Jupiter	139°30.4	10:36
Saturn	310°54.9	23:07

Nov 23 Fri	SHA	Mer.pass
Venus	73°53.9	14:55
Mars	74°15.0	14:54
Jupiter	139°17.8	10:32
Saturn	310°59.7	23:03

Horizontal parallax		
Venus:	0.3	
Mars:	0.1	

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
Wed													
0	183°33.0	S19°55.3	6°04.2	11.7'	N21°56.6	4.5'	54.1'						
1	198°32.8	55.9	20°34.9	11.7'	22°01.1	4.4'	54.1'						
2	213°32.7	56.4	35°05.5	11.6'	22°05.5	4.3'	54.1'						
3	228°32.5	· · 57.0	49°36.1	11.6'	22°09.9	4.2'	54.1'						
4	243°32.4	57.5	64°06.7	11.5'	22°14.1	4.1'	54.1'						
5	258°32.2	58.1	78°37.2	11.5'	22°18.3	4.0'	54.1'						
6	273°32.1	S19°58.6	93°07.7	11.5'	N22°22.3	3.9'	54.1'						
7	288°31.9	59.2	107°38.2	11.4'	22°26.2	3.8'	54.1'						
8	303°31.8	19°59.7	122°08.6	11.4'	22°30.0	3.7'	54.1'						
9	318°31.6	20°00.3	136°39.0	11.3'	22°33.8	3.6'	54.1'						
10	333°31.4	00.8	151°09.3	11.3'	22°37.4	3.5'	54.1'						
11	348°31.3	01.4	165°39.6	11.3'	22°40.9	3.4'	54.2'						
12	3°31.1	S20°01.9	180°09.9	11.2'	N22°44.3	3.3'	54.2'						
13	18°31.0	02.4	194°40.1	11.2'	22°47.6	3.2'	54.2'						
14	33°30.8	03.0	209°10.3	11.2'	22°50.8	3.1'	54.2'						
15	48°30.6	· · 03.5	223°40.5	11.1'	22°53.8	3.0'	54.2'						
16	63°30.5	04.1	238°10.6	11.1'	22°56.8	2.9'	54.2'						
17	78°30.3	04.6	252°40.7	11.1'	22°59.7	2.8'	54.2'						
18	93°30.2	S20°05.1	267°10.8	11.0'	N23°02.4	2.6'	54.2'						
19	108°30.0	05.7	281°40.8	11.0'	23°05.1	2.5'	54.2'						
20	123°29.8	06.2	296°10.8	11.0'	23°07.6	2.4'	54.2'						
21	138°29.7	· · 06.8	310°40.7	10.9'	23°10.0	2.3'	54.2'						
22	153°29.5	07.3	325°10.7	10.9'	23°12.4	2.2'	54.3'						
23	168°29.3	07.8	339°40.6	10.9'	23°14.6	2.1'	54.3'						
SD = 16.2' d = 0.6'							SD = 14.7'						
Thu	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°29.2	S20°08.4	354°10.5	10.8'	N23°16.6	2.0'	54.3'						
1	198°29.0	08.9	8°40.3	10.8'	23°18.6	1.9'	54.3'						
2	213°28.8	09.4	23°10.1	10.8'	23°20.5	1.8'	54.3'						
3	228°28.7	· · 10.0	37°39.9	10.8'	23°22.2	1.6'	54.3'						
4	243°28.5	10.5	52°09.7	10.7'	23°23.9	1.5'	54.3'						
5	258°28.4	11.0	66°39.4	10.7'	23°25.4	1.4'	54.3'						
6	273°28.2	S20°11.6	81°09.2	10.7'	N23°26.8	1.3'	54.3'						
7	288°28.0	12.1	95°38.9	10.7'	23°28.1	1.2'	54.3'						
8	303°27.9	12.6	110°08.5	10.7'	23°29.3	1.1'	54.4'						
9	318°27.7	· · 13.2	124°38.2	10.6'	23°30.4	1.0'	54.4'						
10	333°27.5	13.7	139°07.8	10.6'	23°31.4	0.8'	54.4'						
11	348°27.4	14.2	153°37.4	10.6'	23°32.2	0.7'	54.4'						
12	3°27.2	S20°14.7	168°07.0	10.6'	N23°32.9	0.6'	54.4'						
13	18°27.0	15.3	182°36.6	10.6'	23°33.6	0.5'	54.4'						
14	33°26.9	15.8	197°06.2	10.5'	23°34.1	0.4'	54.4'						
15	48°26.7	· · 16.3	211°35.7	10.5'	23°34.4	0.3'	54.4'						
16	63°26.5	16.9	226°05.2	10.5'	23°34.7	0.2'	54.5'						
17	78°26.3	17.4	240°34.8	10.5'	23°34.9	0.0'	54.5'						
18	93°26.2	S20°17.9	255°04.3	10.5'	N23°34.9	-0.1'	54.5'						
19	108°26.0	18.4	269°33.7	10.5'	23°34.8	-0.2'	54.5'						
20	123°25.8	18.9	284°03.2	10.5'	23°34.6	-0.3'	54.5'						
21	138°25.7	· · 19.5	298°32.7	10.5'	23°34.3	-0.4'	54.5'						
22	153°25.5	20.0	313°02.1	10.4'	23°33.9	-0.5'	54.5'						
23	168°25.3	20.5	327°31.6	10.4'	23°33.3	-0.7'	54.5'						
SD = 16.2' d = 0.5'							SD = 14.8'						
Fri	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°25.2	S20°21.0	342°01.0	10.4'	N23°32.7	-0.8'	54.6'						
1	198°25.0	21.6	356°30.4	10.4'	23°31.9	-0.9'	54.6'						
2	213°24.8	22.1	10°59.8	10.4'	23°31.0	-1.0'	54.6'						
3	228°24.6	· · 22.6	25°29.3	10.4'	23°30.0	-1.1'	54.6'						
4	243°24.5	23.1	39°58.7	10.4'	23°28.9	-1.2'	54.6'						
5	258°24.3	23.6	54°28.1	10.4'	23°27.6	-1.4'	54.6'						
6	273°24.1	S20°24.1	68°57.5	10.4'	N23°26.2	-1.5'	54.6'						
7	288°23.9	24.7	83°26.9	10.4'	23°24.8	-1.6'	54.6'						
8	303°23.8	25.2	97°56.2	10.4'	23°23.2	-1.7'	54.7'						
9	318°23.6	· · 25.7	112°25.6	10.4'	23°21.5	-1.8'	54.7'						
10	333°23.4	26.2	126°55.0	10.4'	23°19.6	-1.9'	54.7'						
11	348°23.2	26.7	141°24.4	10.4'	23°17.7	-2.1'	54.7'						
12	3°23.1	S20°27.2	155°53.8	10.4'	N23°15.6	-2.2'	54.7'						
13	18°22.9	27.7	170°23.2	10.4'	23°13.4	-2.3'	54.7'						
14	33°22.7	28.2	184°52.6	10.4'	23°11.1	-2.4'	54.8'						
15	48°22.5	· · 28.8	199°22.0	10.4'	23°08.7	-2.5'	54.8'						
16	63°22.4	29.3	213°51.4	10.4'	23°06.2	-2.6'	54.8'						
17	78°22.2	29.8	228°20.8	10.4'	23°03.5	-2.8'	54.8'						
18	93°22.0	S20°30.3	242°50.2	10.4'	N23°00.8	-2.9'	54.8'						
19	108°21.8	30.8	257°19.7	10.4'	22°57.9	-3.0'	54.8'						
20	123°21.6	31.3	271°49.1	10.4'	22°54.9	-3.1'	54.8'						
21	138°21.5	· · 31.8	286°18.5	10.4'	22°51.8	-3.2'	54.9'						
22	153°21.3	32.3	300°48.0	10.5'	22°48.6	-3.3'	54.9'						
23	168°21.1	32.8	315°17.4	10.5'	22°45.2	-3.5'	54.9'						
SD = 16.2' d = 0.5'							SD = 14.9'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:24	09:09	■	■	14:23	16:07
N 70°	07:11	08:39	10:48	12:44	14:52	16:20
68°	07:01	08:18	09:48	13:43	15:14	16:31
66°	06:52	08:00	09:15	14:17	15:31	16:40
64°	06:44	07:46	08:50	14:41	15:45	16:48
62°	06:37	07:34	08:31	15:00	15:57	16:55
60°	06:31	07:24	08:16	15:16	16:08	17:01
N 58°	06:25	07:15	08:02	15:29	16:17	17:06
56°	06:20	07:07	07:51	15:41	16:25	17:11
54°	06:16	07:00	07:41	15:51	16:32	17:16
52°	06:12	06:54	07:32	16:00	16:38	17:20
50°	06:08	06:48	07:24	16:08	16:44	17:24
45°	05:59	06:35	07:07	16:25	16:57	17:33
N 40°	05:51	06:24	06:53	16:39	17:08	17:41
35°	05:43	06:14	06:41	16:51	17:18	17:49
30°	05:36	06:05	06:31	17:01	17:27	17:56
20°	05:23	05:50	06:13	17:19	17:43	18:09
N 10°	05:09	05:35	05:57	17:35	17:57	18:23
0°	04:55	05:21	05:43	17:50	18:12	18:37
S 10°	04:39	05:05	05:28	18:05	18:27	18:54
20°	04:19	04:48	05:12	18:21	18:45	19:13
30°	03:55	04:26	04:53	18:40	19:06	19:38
35°	03:39	04:13	04:42	18:51	19:19	19:54
40°	03:19	03:58	04:29	19:04	19:35	20:14
45°	02:55	03:39	04:14	19:19	19:54	20:39
S 50°	02:20	03:15	03:56	19:37	20:18	21:14
52°	02:02	03:04	03:47	19:46	20:30	21:32
54°	01:39	02:50	03:38	19:56	20:44	21:56
56°	01:06	02:34	03:26	20:07	21:00	22:31
58°	—:—	02:14	03:14	20:20	21:20	—:—
S 60°	—:—	01:50	02:59	20:35	21:46	—:—

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	13:21	13:42	14:50	10:32	11:56	12:33
64°	14:06	14:41	15:42	09:48	10:56	11:41
62°	14:36	15:15	16:13	09:19	10:23	11:09
60°	14:58	15:40	16:37	08:56	09:58	10:46
N 58°	15:17	15:59	16:56	08:38	09:38	10:27
56°	15:32	16:16	17:12	08:23	09:22	10:11
54°	15:45	16:30	17:25	08:10	09:08	09:57
52°	15:57	16:42	17:37	07:59	08:56	09:4

November 24, 25, 26 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	63°14.5	136°22.9	S26°02.1	136°39.5	S23°53.4	202°19.6	S14°49.6	14°18.9	N15°33.0	Alpheratz	357°30.5	29°15.6	
1	78°16.9	151°23.5	01.7	151°39.9	53.2	217°21.5	49.8	29°21.6	32.9	Ankaa	353°03.2	-42°08.7	
2	93°19.4	166°24.1	01.4	166°40.2	53.0	232°23.5	50.0	44°24.2	32.9	Schedar	349°26.0	56°42.4	
3	108°21.9	181°24.7	· · 01.0	181°40.6	· · 52.8	247°25.4	· · 50.1	59°26.9	· · 32.9	Diphda	348°43.2	-17°49.3	
4	123°24.3	196°25.3	00.7	196°41.0	52.6	262°27.3	50.3	74°29.6	32.8	Achernar	335°17.0	-57°05.2	
5	138°26.8	211°25.9	00.3	211°41.4	52.3	277°29.3	50.4	89°32.2	32.8	Hamal	327°46.4	23°36.4	
6	153°29.3	226°26.5	S26°00.0	226°41.8	S23°52.1	292°31.2	S14°50.6	104°34.9	N15°32.7	Polaris	311°20.2	89°23.3	
7	168°31.7	241°27.1	25°59.6	241°42.2	51.9	307°33.2	50.8	119°37.6	32.7	Acamar	315°08.5	-40°11.1	
8	183°34.2	256°27.7	59.3	256°42.5	51.7	322°35.1	50.9	134°40.2	32.6	Menkar	314°01.7	4°12.4	
9	198°36.6	271°28.4	· · 58.9	271°42.9	· · 51.4	337°37.0	· · 51.1	149°42.9	· · 32.6	Mirfak	308°22.0	49°58.1	
10	213°39.1	286°29.0	58.6	286°43.3	51.2	352°39.0	51.2	164°45.5	32.5	Aldebaran	290°34.8	16°34.1	
11	228°41.6	301°29.6	58.2	301°43.7	51.0	7°40.9	51.4	179°48.2	32.5	Rigel	280°59.8	-8°10.1	
12	243°44.0	316°30.2	S25°57.9	316°44.1	S23°50.8	22°42.8	S14°51.6	194°50.9	N15°32.4	Capella	280°15.6	46°01.5	
13	258°46.5	331°30.9	57.5	331°44.5	50.5	37°44.8	51.7	209°53.5	32.4	Bellatrix	278°18.4	6°22.5	
14	273°49.0	346°31.5	57.2	346°44.9	50.3	52°46.7	51.9	224°56.2	32.3	Elnath	277°56.6	28°37.8	
15	288°51.4	1°32.1	· · 56.8	1°45.2	· · 50.1	67°48.7	· · 52.0	239°58.9	· · 32.3	Alnilam	275°33.5	-1°11.1	
16	303°53.9	16°32.8	56.5	16°45.6	49.8	82°50.6	52.2	255°01.5	32.2	Betelgeuse	270°47.6	7°24.6	
17	318°56.4	31°33.4	56.1	31°46.0	49.6	97°52.5	52.3	270°04.2	32.2	Canopus	263°50.2	-52°42.6	
18	333°58.8	46°34.0	S25°55.8	46°46.4	S23°49.4	112°54.5	S14°52.5	285°06.8	N15°32.1	Sirius	258°22.5	-16°45.5	
19	349°01.3	61°34.7	55.4	61°46.8	49.2	127°56.4	52.7	300°09.5	32.1	Adhara	255°02.5	-29°00.8	
20	4°03.8	76°35.3	55.0	76°47.2	48.9	142°58.4	52.8	315°12.2	32.1	Procyon	244°46.5	5°08.7	
21	19°06.2	91°36.0	· · 54.7	91°47.5	· · 48.7	158°00.3	· · 53.0	330°14.8	· · 32.0	Pollux	243°12.3	27°57.0	
22	34°08.7	106°36.6	54.3	106°47.9	48.5	173°02.2	53.1	345°17.5	32.0	Avior	234°12.7	-59°36.2	
23	49°11.1	121°37.3	54.0	121°48.3	48.2	188°04.2	53.3	0°20.2	31.9	Suhail	222°43.2	-43°33.1	
Mer.pass. 19:44		ν 0.6' d -0.3' m -4.83		ν 0.4' d -0.2' m 1.10		ν 1.9' d 0.2' m -1.70		ν 2.7' d -0.0' m -0.22		Miaplacidus	221°36.9	-69°50.2	
										Alphard	217°43.8	-8°47.4	
										Regulus	207°30.3	11°49.1	
										Dubhe	193°36.7	61°35.0	
										Denebola	182°21.1	14°24.2	
										Gienah	175°39.7	-17°42.4	
										Acrux	172°55.8	-63°15.7	
										Gacrux	171°47.4	-57°16.6	
										Alioth	166°10.1	55°47.6	
										Spica	158°18.4	-11°19.0	
										Alkaid	152°49.5	-49°09.7	
										Hadar	148°30.8	-60°30.8	
										Menkent	147°53.3	-36°30.9	
										Arcturus	145°44.7	19°01.6	
										Rigil Kent.	139°35.3	-60°57.3	
										Kochab	137°21.4	74°01.8	
										Zuben'ubi	136°52.0	-16°09.8	
										Alphecca	126°00.9	26°36.9	
										Antares	112°11.4	-26°29.7	
										Atria	107°02.5	-69°04.7	
										Sabik	101°58.6	-15°45.5	
										Shaula	96°05.5	-37°07.4	
										Rasalhague	95°55.3	12°32.5	
										Eltanin	90°40.9	51°29.3	
										Kaus Aust.	83°27.7	-34°22.1	
										Vega	80°30.9	38°49.0	
										Nunki	75°43.2	-26°15.4	
										Altair	61°56.4	8°57.1	
										Peacock	53°00.0	-56°38.3	
										Deneb	49°23.2	45°23.6	
										Enif	33°35.0	10°01.0	
										Al Na'ir	27°28.2	-46°49.0	
										Fomalhaut	15°10.2	-29°27.8	
										Scheat	13°41.3	28°15.1	
										Markab	13°25.9	15°22.3	
										Nov 24 Sat		SHA	Mer.pass
										Venus	73°08.4	14:54	
										Mars	73°25.0	14:53	
										Jupiter	139°05.1	10:29	
										Saturn	311°04.4	22:59	
										Nov 25 Sun		SHA	Mer.pass
										Venus	72°24.3	14:53	
										Mars	72°35.1	14:52	
										Jupiter	138°52.5	10:26	
										Saturn	311°09.2	22:54	
										Nov 26 Mon		SHA	Mer.pass
										Venus	71°41.7	14:52	
										Mars	71°45.2	14:52	
										Jupiter	138°39.9	10:23	
										Saturn	311°13.9	22:50	
										Horizontal parallax			
										Venus:	0.3		
										Mars:	0.1		
Mer.pass. 19:36		ν 0.7' d -0.4' m -4.85		ν 0.4' d -0.2' m 1.10		ν 1.9' d 0.2' m -1.70		ν 2.7' d -0.0' m -0.20					

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°20.9	S20°33.3	329°46.9	10.5'	N22°41.8	-3.6'	54.9'
1	198°20.7	33.8	344°16.4	10.5'	22°38.2	-3.7'	54.9'
2	213°20.6	34.3	358°45.9	10.5'	22°34.5	-3.8'	54.9'
3	228°20.4	· · 34.8	13°15.4	10.5'	22°30.7	-3.9'	55.0'
4	243°20.2	35.3	27°44.9	10.5'	22°26.8	-4.0'	55.0'
5	258°20.0	35.8	42°14.4	10.5'	22°22.8	-4.1'	55.0'
6	273°19.8	S20°36.3	56°44.0	10.6'	N22°18.7	-4.3'	55.0'
7	288°19.7	36.8	71°13.5	10.6'	22°14.4	-4.4'	55.0'
8	303°19.5	37.3	85°43.1	10.6'	22°10.0	-4.5'	55.0'
9	318°19.3	· · 37.8	100°12.7	10.6'	22°05.6	-4.6'	55.1'
10	333°19.1	38.3	114°42.3	10.6'	22°01.0	-4.7'	55.1'
11	348°18.9	38.8	129°11.9	10.6'	21°56.3	-4.8'	55.1'
12	3°18.7	S20°39.3	143°41.6	10.7'	N21°51.5	-4.9'	55.1'
13	18°18.6	39.8	158°11.3	10.7'	21°46.5	-5.0'	55.1'
14	33°18.4	40.3	172°40.9	10.7'	21°41.5	-5.1'	55.2'
15	48°18.2	· · 40.8	187°10.6	10.7'	21°36.4	-5.3'	55.2'
16	63°18.0	41.3	201°40.4	10.7'	21°31.1	-5.4'	55.2'
17	78°17.8	41.8	216°10.1	10.8'	21°25.7	-5.5'	55.2'
18	93°17.6	S20°42.3	230°39.9	10.8'	N21°20.3	-5.6'	55.2'
19	108°17.4	42.8	245°09.7	10.8'	21°14.7	-5.7'	55.2'
20	123°17.3	43.3	259°39.5	10.8'	21°09.0	-5.8'	55.3'
21	138°17.1	· · 43.8	274°09.3	10.9'	21°03.2	-5.9'	55.3'
22	153°16.9	44.3	288°39.1	10.9'	20°57.3	-6.0'	55.3'
23	168°16.7	44.7	303°09.0	10.9'	20°51.3	-6.1'	55.3'
SD = 16.2'			SD = 15.0'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	183°16.5	S20°45.2	317°38.9	10.9'	N20°45.1	-6.2'	55.3'
1	198°16.3	45.7	332°08.8	10.9'	20°38.9	-6.3'	55.4'
2	213°16.1	46.2	346°38.8	11.0'	20°32.6	-6.4'	55.4'
3	228°15.9	· · 46.7	1°08.8	11.0'	20°26.2	-6.5'	55.4'
4	243°15.8	47.2	15°38.8	11.0'	20°19.6	-6.6'	55.4'
5	258°15.6	47.7	30°08.8	11.0'	20°13.0	-6.7'	55.4'
6	273°15.4	S20°48.1	44°38.8	11.1'	N20°06.2	-6.9'	55.5'
7	288°15.2	48.6	59°08.9	11.1'	19°59.4	-7.0'	55.5'
8	303°15.0	49.1	73°39.0	11.1'	19°52.4	-7.1'	55.5'
9	318°14.8	· · 49.6	88°09.1	11.2'	19°45.3	-7.2'	55.5'
10	333°14.6	50.1	102°39.3	11.2'	19°38.2	-7.3'	55.5'
11	348°14.4	50.6	117°09.4	11.2'	19°30.9	-7.4'	55.6'
12	3°14.2	S20°51.0	131°39.6	11.2'	N19°23.6	-7.5'	55.6'
13	18°14.0	51.5	146°09.9	11.3'	19°16.1	-7.6'	55.6'
14	33°13.8	52.0	160°40.1	11.3'	19°08.5	-7.7'	55.6'
15	48°13.7	· · 52.5	175°10.4	11.3'	19°00.9	-7.8'	55.7'
16	63°13.5	53.0	189°40.7	11.3'	18°53.1	-7.9'	55.7'
17	78°13.3	53.4	204°11.1	11.4'	18°45.3	-8.0'	55.7'
18	93°13.1	S20°53.9	218°41.4	11.4'	N18°37.3	-8.0'	55.7'
19	108°12.9	54.4	233°11.8	11.4'	18°29.3	-8.1'	55.7'
20	123°12.7	54.9	247°42.2	11.4'	18°21.1	-8.2'	55.8'
21	138°12.5	· · 55.3	262°12.7	11.5'	18°12.9	-8.3'	55.8'
22	153°12.3	55.8	276°43.2	11.5'	18°04.5	-8.4'	55.8'
23	168°12.1	56.3	291°13.7	11.5'	17°56.1	-8.5'	55.8'
SD = 16.2'			SD = 15.1'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	183°11.9	S20°56.8	305°44.2	11.6'	N17°47.6	-8.6'	55.9'
1	198°11.7	57.2	320°14.7	11.6'	17°39.0	-8.7'	55.9'
2	213°11.5	57.7	334°45.3	11.6'	17°30.3	-8.8'	55.9'
3	228°11.3	· · 58.2	349°15.9	11.6'	17°21.5	-8.9'	55.9'
4	243°11.1	58.6	3°46.6	11.7'	17°12.6	-9.0'	56.0'
5	258°10.9	59.1	18°17.2	11.7'	17°03.6	-9.1'	56.0'
6	273°10.7	S20°59.6	32°47.9	11.7'	N16°54.5	-9.2'	56.0'
7	288°10.5	21°00.0	47°18.6	11.7'	16°45.4	-9.2'	56.0'
8	303°10.3	00.5	61°49.4	11.8'	16°36.1	-9.3'	56.1'
9	318°10.1	· · 01.0	76°20.1	11.8'	16°26.8	-9.4'	56.1'
10	333°09.9	01.4	90°50.9	11.8'	16°17.4	-9.5'	56.1'
11	348°09.7	01.9	105°21.7	11.8'	16°07.9	-9.6'	56.1'
12	3°09.5	S21°02.4	119°52.6	11.9'	N15°58.3	-9.7'	56.2'
13	18°09.3	02.8	134°23.5	11.9'	15°48.6	-9.8'	56.2'
14	33°09.1	03.3	148°54.3	11.9'	15°38.8	-9.8'	56.2'
15	48°08.9	· · 03.8	163°25.3	11.9'	15°29.0	-9.9'	56.2'
16	63°08.7	04.2	177°56.2	12.0'	15°19.1	-10.0'	56.3'
17	78°08.5	04.7	192°27.2	12.0'	15°09.0	-10.1'	56.3'
18	93°08.3	S21°05.1	206°58.1	12.0'	N14°58.9	-10.2'	56.3'
19	108°08.1	05.6	221°29.2	12.0'	14°48.8	-10.3'	56.3'
20	123°07.9	06.1	236°00.2	12.1'	14°38.5	-10.3'	56.4'
21	138°07.7	· · 06.5	250°31.2	12.1'	14°28.2	-10.4'	56.4'
22	153°07.5	07.0	265°02.3	12.1'	14°17.8	-10.5'	56.4'
23	168°07.3	07.4	279°33.4	12.1'	14°07.3	-10.6'	56.4'
SD = 16.2'			SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:34	09:23	■	■	14:10	15:59
N 70°	07:20	08:51	■	■	14:43	16:13
68°	07:08	08:27	10:04	13:29	15:06	16:25
66°	06:58	08:09	09:26	14:08	15:25	16:35
64°	06:50	07:54	08:59	14:34	15:40	16:43
62°	06:43	07:41	08:39	14:54	15:53	16:51
60°	06:36	07:30	08:23	15:11	16:03	16:57
N 58°	06:30	07:21	08:09	15:25	16:13	17:03
56°	06:25	07:12	07:57	15:37	16:21	17:08
54°	06:20	07:05	07:46	15:47	16:29	17:13
52°	06:16	06:58	07:37	15:57	16:36	17:18
50°	06:12	06:52	07:29	16:05	16:42	17:22
45°	06:02	06:38	07:11	16:23	16:55	17:32
N 40°	05:54	06:27	06:56	16:37	17:07	17:40
35°	05:46	06:17	06:44	16:50	17:17	17:48
30°	05:38	06:08	06:33	17:00	17:26	17:55
20°	05:24	05:51	06:15	17:19	17:43	18:10
N 10°	05:10	05:36	05:59	17:35	17:58	18:24
0°	04:56	05:21	05:43	17:51	18:13	18:39
S 10°	04:39	05:05	05:28	18:06	18:29	18:55
20°	04:19	04:47	05:11	18:23	18:47	19:15
30°	03:53	04:25	04:52	18:42	19:09	19:41
35°	03:37	04:12	04:41	18:54	19:22	19:58
40°	03:17	03:56	04:28	19:07	19:38	20:18
45°	02:51	03:37	04:12	19:22	19:58	20:44
S 50°	02:15	03:12	03:53	19:42	20:23	21:20
52°	01:56	03:00	03:44	19:51	20:36	21:40
54°	01:30	02:45	03:34	20:01	20:50	22:07
56°	00:52	02:28	03:22	20:13	21:07	22:47
58°	—:—	02:07	03:09	20:27	21:29	—:—
S 60°	—:—	01:40	02:53	20:43	21:57	—:—

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	□	□	18:37	□	□	14:00
N 70°	□	16:50	19:11	□	14:03	13:24
68°	15:25	17:40	19:35	13:44	13:12	12:59
66°	16:28	18:11	19:53	12:40	12:41	12:39
64°	17:03	18:34	20:08	12:05	12:17	12:23
62°	17:28	18:52	20:21	11:39	11:58	12:10
60°	17:48	19:07	20:31	11:19	11:42	11:58
N 58°	18:04	19:20	20:40	11:03	11:29	11:48
56°	18:18	19:31	20:48	10:49	11:18	11:40
54°	18:29	19:40	20:55	10:36	11:07	11:32
52°	18:40	19:49	21:02	10:26	10:58	11:25
50°	18:49	19:57	21:07	10:16	10:50	11:19
45°	19:09	20:13	21:20	09:56	10:33	11:05
N 40°	19:25	20:26	21:30	09:39	10:19	10:54
35°	19:38	20:38	21:38	09:25	10:07	10:44
30°	19:50	20:47	21:46	09:13	09:56	10:36
20°	20:10	21:04	21:59	08:52	09:38	10:21
N 10°	20:27	21:19	22:10	08:34	09:22	10:08
0°	20:43	21:33	22:21	08:17	09:07	09:56
S 10°	20:59	21:46	22:32	08:00	08:52	09:44
20°	21:17	22:01	22:43	07:42	08:36	09:31
30°	21:36	22:18	22:56	07:21	08:17	09:15
35°	21:48	22:27	23:03	07:08	08:06	09:07
40°	22:01	22:38	23:11	06:54	07:54	08:57
45°	22:17	22:51	23:21	06:37	07:39	08:45
S 50°	22:36	23:07	23:33	06:16	07:21	08:30
52°	22:45	23:14	23:38	06:06	07:12	08:23
54°	22:55	23:22	23:44	05:54	07:02	08:16
56°	23:06	23:31	23:51	05:41	06:52	08:07
58°	23:19	23:41	23:58	05:26	06:39	07:58
S 60°	23:34	23:53	· · ·	05:08	06:24	07:47

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Upper hh:mm	Lower hh:mm	18-20 93-79%
24	13:24	13:15	11:47	02:05	14:30	
25	13:06	12:57	11:47	02:55	15:20	
26	12:48	12:38	11:47	03:44	16:09	

November 27, 28, 29 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	66°11.9	137°12.4	S25°35.7	137°07.2	S23°36.3	204°39.2	S15°01.1	17°30.5	N15°29.7	Alpheratz	357°30.5	29°15.6
1	81°14.4	152°13.2	35.3	152°07.6	36.0	219°41.2	01.2	32°33.2	29.6	Ankaa	353°03.2	-42°08.7
2	96°16.8	167°13.9	34.9	167°07.9	35.8	234°43.1	01.4	47°35.8	29.6	Schedar	349°26.0	56°42.4
3	111°19.3	182°14.7	· · 34.5	182°08.3	· · 35.5	249°45.1	· · 01.5	62°38.5	· · 29.5	Diphda	348°43.2	-17°49.3
4	126°21.7	197°15.5	34.1	197°08.7	35.3	264°47.0	01.7	77°41.2	29.5	Achernar	335°17.0	-57°05.2
5	141°24.2	212°16.3	33.7	212°09.1	35.0	279°48.9	01.8	92°43.8	29.4	Hamal	327°46.4	23°36.4
6	156°26.7	227°17.1	S25°33.3	227°09.5	S23°34.7	294°50.9	S15°02.0	107°46.5	N15°29.4	Polaris	311°20.4	89°23.4
7	171°29.1	242°17.9	32.9	242°09.9	34.5	309°52.8	02.1	122°49.1	29.3	Acamar	315°08.5	-40°11.1
8	186°31.6	257°18.7	32.5	257°10.3	34.2	324°54.8	02.3	137°51.8	29.3	Menkar	314°01.7	4°12.4
9	201°34.1	272°19.5	· · 32.1	272°10.6	· · 34.0	339°56.7	· · 02.5	152°54.4	· · 29.2	Mirfak	308°22.0	49°58.1
10	216°36.5	287°20.3	31.7	287°11.0	33.7	354°58.6	02.6	167°57.1	29.2	Aldebaran	290°34.8	16°34.1
11	231°39.0	302°21.1	31.4	302°11.4	33.4	10°00.6	02.8	182°59.8	29.2	Rigel	280°59.8	-8°10.1
12	246°41.5	317°21.9	S25°31.0	317°11.8	S23°33.2	25°02.5	S15°02.9	198°02.4	N15°29.1	Capella	280°15.6	46°01.5
13	261°43.9	332°22.7	30.6	332°12.2	32.9	40°04.5	03.1	213°05.1	29.1	Bellatrix	278°18.4	6°22.5
14	276°46.4	347°23.6	30.2	347°12.6	32.7	55°06.4	03.2	228°07.7	29.0	Elnath	277°56.5	28°37.8
15	291°48.8	2°24.4	· · 29.8	2°13.0	· · 32.4	70°08.4	· · 03.4	243°10.4	· · 29.0	Alnilam	275°33.4	-1°11.1
16	306°51.3	17°25.2	29.4	17°13.4	32.1	85°10.3	03.6	258°13.1	28.9	Betelgeuse	270°47.5	7°24.6
17	321°53.8	32°26.0	29.0	32°13.7	31.9	100°12.2	03.7	273°15.7	28.9	Canopus	263°50.1	-52°42.6
18	336°56.2	47°26.8	S25°28.6	47°14.1	S23°31.6	115°14.2	S15°03.9	288°18.4	N15°28.8	Sirius	258°22.5	-16°45.5
19	351°58.7	62°27.7	28.2	62°14.5	31.3	130°16.1	04.0	303°21.0	28.8	Adhara	255°02.5	-29°00.8
20	7°01.2	77°28.5	27.8	77°14.9	31.1	145°18.1	04.2	318°23.7	28.7	Procyon	244°46.5	5°08.7
21	22°03.6	92°29.3	· · 27.4	92°15.3	· · 30.8	160°20.0	· · 04.3	333°26.3	· · 28.7	Pollux	243°12.3	27°57.0
22	37°06.1	107°30.2	27.0	107°15.7	30.6	175°21.9	04.5	348°29.0	28.7	Avior	234°12.7	-59°36.2
23	52°08.6	122°31.0	26.6	122°16.1	30.3	190°23.9	04.7	3°31.7	28.6	Suhail	222°43.2	-43°33.1
Mer.pass. 19:32		ν 0.8' d -0.4' m -4.86		ν 0.4' d -0.3' m 1.10		ν 1.9' d 0.2' m -1.71		ν 2.7' d -0.0' m -0.20				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	67°11.0	137°31.9	S25°26.2	137°16.5	S23°30.0	205°25.8	S15°04.8	18°34.3	N15°28.6
1	82°13.5	152°32.7	25.8	152°16.8	29.8	220°27.8	05.0	33°37.0	28.5
2	97°16.0	167°33.6	25.4	167°17.2	29.5	235°29.7	05.1	48°39.6	28.5
3	112°18.4	182°34.4	· · 25.0	182°17.6	· · 29.2	250°31.7	· · 05.3	63°42.3	· · 28.4
4	127°20.9	197°35.3	24.6	197°18.0	29.0	265°33.6	05.4	78°44.9	28.4
5	142°23.3	212°36.1	24.2	212°18.4	28.7	280°35.5	05.6	93°47.6	28.3
6	157°25.8	227°37.0	S25°23.8	227°18.8	S23°28.4	295°37.5	S15°05.7	108°50.3	N15°28.3
7	172°28.3	242°37.9	23.4	242°19.2	28.1	310°39.4	05.9	123°52.9	28.2
8	187°30.7	257°38.7	23.0	257°19.6	27.9	325°41.4	06.1	138°55.6	28.2
9	202°33.2	272°39.6	· · 22.6	272°19.9	· · 27.6	340°43.3	· · 06.2	153°58.2	· · 28.2
10	217°35.7	287°40.5	22.2	287°20.3	27.3	355°45.3	06.4	169°00.9	28.1
11	232°38.1	302°41.3	21.8	302°20.7	27.1	10°47.2	06.5	184°03.5	28.1
12	247°40.6	317°42.2	S25°21.3	317°21.1	S23°26.8	25°49.1	S15°06.7	199°06.2	N15°28.0
13	262°43.1	332°43.1	20.9	332°21.5	26.5	40°51.1	06.8	214°08.9	28.0
14	277°45.5	347°44.0	20.5	347°21.9	26.3	55°53.0	07.0	229°11.5	27.9
15	292°48.0	2°44.9	· · 20.1	2°22.3	· · 26.0	70°55.0	· · 07.1	244°14.2	· · 27.9
16	307°50.5	17°45.8	19.7	17°22.7	25.7	85°56.9	07.3	259°16.8	27.8
17	322°52.9	32°46.6	19.3	32°23.0	25.4	100°58.8	07.5	274°19.5	27.8
18	337°55.4	47°47.5	S25°18.9	47°23.4	S23°25.2	116°00.8	S15°07.6	289°22.1	N15°27.8
19	352°57.8	62°48.4	18.5	62°23.8	24.9	131°02.7	07.8	304°24.8	27.7
20	8°00.3	77°49.3	18.1	77°24.2	24.6	146°04.7	07.9	319°27.5	27.7
21	23°02.8	92°50.2	· · 17.7	92°24.6	· · 24.3	161°06.6	· · 08.1	334°30.1	· · 27.6
22	38°05.2	107°51.1	17.3	107°25.0	24.1	176°08.6	08.2	349°32.8	27.6
23	53°07.7	122°52.1	16.8	122°25.4	23.8	191°10.5	08.4	4°35.4	27.5
Mer.pass. 19:28		ν 0.8' d -0.4' m -4.87		ν 0.4' d -0.3' m 1.10		ν 1.9' d 0.2' m -1.71		ν 2.7' d -0.0' m -0.19	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	68°10.2	137°53.0	S25°16.4	137°25.8	S23°23.5	206°12.5	S15°08.5	19°38.1	N15°27.5
1	83°12.6	152°53.9	16.0	152°26.1	23.2	221°14.4	08.7	34°40.7	27.4
2	98°15.1	167°54.8	15.6	167°26.5	22.9	236°16.3	08.9	49°43.4	27.4
3	113°17.6	182°55.7	· · 15.2	182°26.9	· · 22.7	251°18.3	· · 09.0	64°46.0	· · 27.4
4	128°20.0	197°56.6	14.8	197°27.3	22.4	266°20.2	09.2	79°48.7	27.3
5	143°22.5	212°57.6	14.4	212°27.7	22.1	281°22.2	09.3	94°51.4	27.3
6	158°24.9	227°58.5	S25°13.9	227°28.1	S23°21.8	296°24.1	S15°09.5	109°54.0	N15°27.2
7	173°27.4	242°59.4	13.5	242°28.5	21.6	311°26.1	09.6	124°56.7	27.2
8	188°29.9	258°00.4	13.1	257°28.9	21.3	326°28.0	09.8	139°59.3	27.1
9	203°32.3	273°01.3	· · 12.7	272°29.3	· · 21.0	341°29.9	· · 09.9	155°02.0	· · 27.1
10	218°34.8	288°02.2	12.3	287°29.6	20.7	356°31.9	10.1	170°04.6	27.0
11	233°37.3	303°03.2	11.9	302°30.0	20.4	11°33.8	10.2	185°07.3	27.0
12	248°39.7	318°04.1	S25°11.4	317°30.4	S23°20.1	26°35.8	S15°10.4	200°09.9	N15°27.0
13	263°42.2	333°05.1	11.0	332°30.8	19.9	41°37.7	10.6	215°12.6	26.9
14	278°44.7	348°06.0	10.6	347°31.2	19.6	56°39.7	10.7	230°15.3	26.9
15	293°47.1	3°07.0	· · 10.2	2°31.6	· · 19.3	71°41.6	· · 10.9	245°17.9	· · 26.8
16	308°49.6	18°07.9	09.8	17°32.0	19.0	86°43.6	11.0	260°20.6	26.8
17	323°52.1	33°08.9	09.3	32°32.4	18.7	101°45.5	11.2	275°23.2	26.7
18	338°54.5	48°09.9	S25°08.9	47°32.8	S23°18.4	116°47.4	S15°11.3	290°25.9	N15°26.7
19	353°57.0	63°10.8	08.5	62°33.2	18.2	131°49.4	11.5	305°28.5	26.6
20	8°59.4	78°11.8	08.1	77°33.5	17.9	146°51.3	11.6	320°31.2	26.6
21	24°01.9	93°12.8	· · 07.7	92°33.9	· · 17.6	161°53.3	· · 11.8	335°33.8	· · 26.6
22	39°04.4	108°13.7	07.2	107°34.3	17.3	176°55.2	11.9	350°36.5	26.5
23	54°06.8	123°14.7	06.8	122°34.7	17.0	191°57.2	12.1	5°39.1	26.5
Mer.pass. 19:24		ν 0.9' d -0.4' m -4.88		ν 0.4' d -0.3' m 1.11		ν 1.9' d 0.2' m -1.71		ν 2.7' d -0.0' m -0.19	

Nov 27 Tue	SHA	Mer.pass
Venus	71°00.5	14:50
Mars	70°55.3	14:51
Jupiter	138°27.3	10:20
Saturn	311°18.6	22:46

Nov 28 Wed	SHA	Mer.pass
Venus	70°20.8	14:49
Mars	70°05.4	14:51
Jupiter	138°14.8	10:17
Saturn	311°23.3	22:42

Nov 29 Thu	SHA	Mer.pass
Venus	69°42.8	14:48
Mars	69°15.6	14:50
Jupiter	138°02.3	10:14
Saturn	311°27.9	22:37

Horizontal parallax	
Venus:	0.3
Mars:	0.1

DUT1 = UT1-UTC = +0.2137 sec ΔT = TT-UT1 = +68.9703 sec

2029 November 27 to Nov. 29 UT

h		Sun			Moon		
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	183°07.1	S21°07.9	294°04.5	12.1'	N13°56.7	-10.6'	56.5'
1	198°06.9	08.3	308°35.6	12.2'	13°46.1	-10.7'	56.5'
2	213°06.7	08.8	323°06.8	12.2'	13°35.4	-10.8'	56.5'
3	228°06.5	· · 09.3	337°38.0	12.2'	13°24.6	-10.9'	56.6'
4	243°06.3	09.7	352°09.2	12.2'	13°13.7	-10.9'	56.6'
5	258°06.1	10.2	6°40.4	12.2'	13°02.8	-11.0'	56.6'
6	273°05.9	S21°10.6	21°11.6	12.2'	N12°51.7	-11.1'	56.6'
7	288°05.7	11.1	35°42.8	12.3'	12°40.7	-11.2'	56.7'
8	303°05.5	11.5	50°14.1	12.3'	12°29.5	-11.2'	56.7'
9	318°05.3	· · 12.0	64°45.4	12.3'	12°18.3	-11.3'	56.7'
10	333°05.1	12.4	79°16.7	12.3'	12°07.0	-11.4'	56.7'
11	348°04.8	12.9	93°48.0	12.3'	11°55.6	-11.4'	56.8'
12	3°04.6	S21°13.3	108°19.3	12.3'	N11°44.2	-11.5'	56.8'
13	18°04.4	13.8	122°50.6	12.3'	11°32.7	-11.6'	56.8'
14	33°04.2	14.2	137°22.0	12.4'	11°21.1	-11.6'	56.9'
15	48°04.0	· · 14.6	151°53.3	12.4'	11°09.5	-11.7'	56.9'
16	63°03.8	15.1	166°24.7	12.4'	10°57.8	-11.8'	56.9'
17	78°03.6	15.5	180°56.1	12.4'	10°46.0	-11.8'	56.9'
18	93°03.4	S21°16.0	195°27.5	12.4'	N10°34.2	-11.9'	57.0'
19	108°03.2	16.4	209°58.9	12.4'	10°22.3	-11.9'	57.0'
20	123°03.0	16.9	224°30.3	12.4'	10°10.4	-12.0'	57.0'
21	138°02.8	· · 17.3	239°01.7	12.4'	09°58.4	-12.1'	57.1'
22	153°02.5	17.7	253°33.1	12.4'	09°46.3	-12.1'	57.1'
23	168°02.3	18.2	268°04.6	12.4'	09°34.2	-12.2'	57.1'
SD = 16.2'		$d = 0.5'$	SD = 15.4'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	183°02.1	S21°18.6	282°36.0	12.4'	N09°22.0	-12.2'	57.2'
1	198°01.9	19.1	297°07.4	12.4'	09°09.7	-12.3'	57.2'
2	213°01.7	19.5	311°38.9	12.4'	08°57.4	-12.4'	57.2'
3	228°01.5	· · 19.9	326°10.3	12.4'	08°45.1	-12.4'	57.2'
4	243°01.3	20.4	340°41.8	12.4'	08°32.7	-12.5'	57.3'
5	258°01.1	20.8	355°13.2	12.5'	08°20.2	-12.5'	57.3'
6	273°00.9	S21°21.2	9°44.7	12.4'	N08°07.7	-12.6'	57.3'
7	288°00.6	21.7	24°16.1	12.4'	07°55.1	-12.6'	57.4'
8	303°00.4	22.1	38°47.6	12.4'	07°42.5	-12.7'	57.4'
9	318°00.2	· · 22.5	53°19.0	12.4'	07°29.8	-12.7'	57.4'
10	333°00.0	23.0	67°50.4	12.4'	07°17.1	-12.8'	57.5'
11	347°59.8	23.4	82°21.9	12.4'	07°04.3	-12.8'	57.5'
12	2°59.6	S21°23.8	96°53.3	12.4'	N06°51.5	-12.9'	57.5'
13	17°59.4	24.3	111°24.8	12.4'	06°38.6	-12.9'	57.6'
14	32°59.1	24.7	125°56.2	12.4'	06°25.7	-13.0'	57.6'
15	47°58.9	· · 25.1	140°27.6	12.4'	06°12.8	-13.0'	57.6'
16	62°58.7	25.6	154°59.0	12.4'	05°59.8	-13.0'	57.7'
17	77°58.5	26.0	169°30.4	12.4'	05°46.7	-13.1'	57.7'
18	92°58.3	S21°26.4	184°01.8	12.4'	N05°33.6	-13.1'	57.7'
19	107°58.1	26.8	198°33.2	12.4'	05°20.5	-13.2'	57.7'
20	122°57.8	27.3	213°04.5	12.4'	05°07.3	-13.2'	57.8'
21	137°57.6	· · 27.7	227°35.9	12.3'	04°54.1	-13.2'	57.8'
22	152°57.4	28.1	242°07.2	12.3'	04°40.9	-13.3'	57.8'
23	167°57.2	28.5	256°38.6	12.3'	04°27.6	-13.3'	57.9'
SD = 16.2'		$d = 0.4'$	SD = 15.6'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	182°57.0	S21°29.0	271°09.9	12.3'	N04°14.3	-13.4'	57.9'
1	197°56.7	29.4	285°41.2	12.3'	04°00.9	-13.4'	57.9'
2	212°56.5	29.8	300°12.4	12.3'	03°47.5	-13.4'	58.0'
3	227°56.3	· · 30.2	314°43.7	12.2'	03°34.1	-13.5'	58.0'
4	242°56.1	30.6	329°14.9	12.2'	03°20.6	-13.5'	58.0'
5	257°55.9	31.1	343°46.1	12.2'	03°07.1	-13.5'	58.1'
6	272°55.6	S21°31.5	358°17.3	12.2'	N02°53.6	-13.6'	58.1'
7	287°55.4	31.9	12°48.5	12.1'	02°40.0	-13.6'	58.1'
8	302°55.2	32.3	27°19.6	12.1'	02°26.5	-13.6'	58.2'
9	317°55.0	· · 32.7	41°50.8	12.1'	02°12.9	-13.6'	58.2'
10	332°54.8	33.1	56°21.8	12.1'	01°59.2	-13.7'	58.2'
11	347°54.5	33.6	70°52.9	12.0'	01°45.6	-13.7'	58.3'
12	2°54.3	S21°34.0	85°24.0	12.0'	N01°31.9	-13.7'	58.3'
13	17°54.1	34.4	99°55.0	12.0'	01°18.2	-13.7'	58.3'
14	32°53.9	34.8	114°25.9	11.9'	01°04.4	-13.8'	58.4'
15	47°53.6	· · 35.2	128°56.9	11.9'	00°50.7	-13.8'	58.4'
16	62°53.4	35.6	143°27.8	11.9'	00°36.9	-13.8'	58.4'
17	77°53.2	36.0	157°58.7	11.8'	00°23.1	-13.8'	58.5'
18	92°53.0	S21°36.4	172°29.5	11.8'	N00°09.3	-13.8'	58.5'
19	107°52.7	36.8	187°00.3	11.8'	S00°04.5	13.8'	58.5'
20	122°52.5	37.3	201°31.1	11.7'	00°18.3	13.9'	58.6'
21	137°52.3	· · 37.7	216°01.8	11.7'	00°32.2	13.9'	58.6'
22	152°52.1	38.1	230°32.5	11.7'	00°46.1	13.9'	58.6'
23	167°51.8	38.5	245°03.2	11.6'	00°59.9	13.9'	58.7'
SD = 16.2'		$d = 0.4'$	SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:43	09:37	■	■	13:58	15:52
N 70°	07:28	09:01	■	■	14:34	16:07
68°	07:15	08:36	10:20	13:15	14:59	16:20
66°	07:05	08:16	09:37	13:58	15:19	16:30
64°	06:56	08:00	09:08	14:27	15:35	16:39
62°	06:48	07:47	08:47	14:49	15:48	16:47
60°	06:41	07:36	08:29	15:06	16:00	16:54
N 58°	06:35	07:26	08:15	15:21	16:10	17:00
56°	06:30	07:17	08:02	15:34	16:18	17:06
54°	06:24	07:09	07:51	15:44	16:26	17:11
52°	06:20	07:02	07:42	15:54	16:33	17:16
50°	06:15	06:56	07:33	16:03	16:40	17:20
45°	06:05	06:42	07:15	16:21	16:54	17:30
N 40°	05:56	06:30	07:00	16:36	17:06	17:39
35°	05:48	06:19	06:47	16:49	17:16	17:48
30°	05:41	06:10	06:36	17:00	17:26	17:55
20°	05:26	05:53	06:17	17:19	17:43	18:10
N 10°	05:12	05:38	06:00	17:36	17:58	18:24
0°	04:56	05:22	05:44	17:52	18:14	18:40
S 10°	04:39	05:06	05:28	18:08	18:30	18:57
20°	04:19	04:47	05:11	18:25	18:49	19:18
30°	03:52	04:25	04:52	18:45	19:12	19:44
35°	03:36	04:11	04:40	18:57	19:25	20:01
40°	03:15	03:55	04:26	19:10	19:42	20:22
45°	02:48	03:35	04:10	19:26	20:02	20:49
S 50°	02:11	03:09	03:51	19:46	20:28	21:27
52°	01:50	02:56	03:41	19:56	20:41	21:48
54°	01:22	02:41	03:31	20:06	20:56	22:17
56°	00:37	02:23	03:18	20:19	21:15	23:06
58°	—:—	02:01	03:04	20:33	21:37	—:—
S 60°	—:—	01:31	02:48	20:50	22:08	—:—

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	20:54	23:00	· · ·	13:22	12:57	12:35
N 70°	21:11	23:06	· · ·	13:03	12:48	12:33
68°	21:24	23:11	· · ·	12:49	12:40	12:32
66°	21:34	23:15	· · ·	12:37	12:34	12:31
64°	21:43	23:19	· · ·	12:26	12:28	12:30
62°	21:51	23:22	· · ·	12:18	12:24	12:29
60°	21:57	23:25	· · ·	12:10	12:20	12:28
N 58°	22:03	23:27	· · ·	12:03	12:16	12:27
56°	22:08	23:29	· · ·	11:58	12:13	12:27
54°	22:12	23:31	· · ·	11:52	12:10	12:26
52°	22:17	23:33	· · ·	11:47	12:07	12:26
50°	22:20	23:35	· · ·	11:43	12:05	12:25
45°	22:28	23:38	· · ·	11:33	11:59	12:24
N 40°	22:35	23:41	· · ·	11:26	11:55	12:23
35°	22:40	23:43	· · ·	11:19	11:51	12:22
30°	22:45	23:46	· · ·	11:13	11:47	12:22
20°	22:54	23:49	· · ·	11:02	11:41	12:21
N 10°	23:01	23:53	· · ·	10:53	11:36	12:19
0°	23:08	23:56	· · ·	10:44	11:31	12:18
S 10°	23:15	23:59	· · ·	10:35	11:26	12:17
20°	23:23	· · ·	00:02	10:25	11:20	12:16
30°	23:31	· · ·	00:06	10:14	11:14	12:15
35°	23:36	· · ·	00:08	10:08	11:10	12:14
40°	23:42	· · ·	00:10	10:01	11:06	12:13
45°	23:48	· · ·	00:13	09:52	11:01	12:12
S 50°	23:56	· · ·	00:17	09:42	10:55	12:11
52°	23:59	· · ·	00:18	09:37	10:53	12:10
54°	· · ·	00:03	00:20	09:32	10:50	12:10
56°	· · ·	00:07	00:22	09:26	10:46	12:09
58°	· · ·	00:12	00:24	09:19	10:43	12:08
S 60°	00:07	00:17	00:26	09:12	10:39	12:07

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
27	12:28	12:19	11:48	04:32	16:56	
28	12:09	11:58	11:48	05:20	17:43	
29	11:48	11:37	11:48	06:07	18:31	

November 30, 01, 02 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	69°09.3	138°15.7	S25°06.4	137°35.1	S23°16.7	206°59.1	S15°12.2	20°41.8	N15°26.4	Alpheratz	357°30.5	29°15.6											
1	84°11.8	153°16.7	06.0	152°35.5	16.4	222°01.0	12.4	35°44.5	26.4	Ankaa	353°03.2	-42°08.7											
2	99°14.2	168°17.7	05.5	167°35.9	16.1	237°03.0	12.6	50°47.1	26.3	Schedar	349°26.0	56°42.4											
3	114°16.7	183°18.7	•• 05.1	182°36.3	•• 15.9	252°04.9	•• 12.7	65°49.8	•• 26.3	Diphda	348°43.2	-17°49.3											
4	129°19.2	198°19.7	04.7	197°36.7	15.6	267°06.9	12.9	80°52.4	26.3	Achernar	335°17.1	-57°05.2											
5	144°21.6	213°20.7	04.3	212°37.0	15.3	282°08.8	13.0	95°55.1	26.2	Hamal	327°46.4	23°36.4											
6	159°24.1	228°21.7	S25°03.8	227°37.4	S23°15.0	297°10.8	S15°13.2	110°57.7	N15°26.2	Polaris	311°20.7	89°23.4											
7	174°26.6	243°22.7	03.4	242°37.8	14.7	312°12.7	13.3	126°00.4	26.1	Acamar	315°08.5	-40°11.2											
8	189°29.0	258°23.7	03.0	257°38.2	14.4	327°14.7	13.5	141°03.0	26.1	Menkar	314°01.7	4°12.4											
9	204°31.5	273°24.7	•• 02.5	272°38.6	•• 14.1	342°16.6	•• 13.6	156°05.7	•• 26.0	Mirfak	308°22.0	49°58.1											
10	219°33.9	288°25.7	02.1	287°39.0	13.8	357°18.6	13.8	171°08.3	26.0	Aldebaran	290°34.8	16°34.1											
11	234°36.4	303°26.7	01.7	302°39.4	13.5	12°20.5	13.9	186°11.0	25.9	Rigel	280°59.8	-8°10.1											
12	249°38.9	318°27.7	S25°01.3	317°39.8	S23°13.2	27°22.4	S15°14.1	201°13.6	N15°25.9	Capella	280°15.6	46°01.5											
13	264°41.3	333°28.7	00.8	332°40.2	12.9	42°24.4	14.2	216°16.3	25.9	Bellatrix	278°18.3	6°22.5											
14	279°43.8	348°29.8	00.4	347°40.6	12.6	57°26.3	14.4	231°18.9	25.8	Elnath	277°56.5	28°37.8											
15	294°46.3	3°30.8	25°00.0	2°41.0	•• 12.3	72°28.3	•• 14.6	246°21.6	•• 25.8	Alnilam	275°33.4	-1°11.1											
16	309°48.7	18°31.8	24°59.5	17°41.3	12.1	87°30.2	14.7	261°24.3	25.7	Betelgeuse	270°47.5	7°24.6											
17	324°51.2	33°32.8	59.1	32°41.7	11.8	102°32.2	14.9	276°26.9	25.7	Canopus	263°50.1	-52°42.7											
18	339°53.7	48°33.9	S24°58.7	47°42.1	S23°11.5	117°34.1	S15°15.0	291°29.6	N15°25.6	Sirius	258°22.5	-16°45.5											
19	354°56.1	63°34.9	58.2	62°42.5	11.2	132°36.1	15.2	306°32.2	25.6	Adhara	255°02.4	-29°00.8											
20	9°58.6	78°36.0	57.8	77°42.9	10.9	147°38.0	15.3	321°34.9	25.6	Procyon	244°46.5	5°08.7											
21	25°01.0	93°37.0	•• 57.4	92°43.3	•• 10.6	162°39.9	•• 15.5	336°37.5	•• 25.5	Pollux	243°12.3	27°56.9											
22	40°03.5	108°38.0	56.9	107°43.7	10.3	177°41.9	15.6	351°40.2	25.5	Avior	234°12.6	-59°36.2											
23	55°06.0	123°39.1	56.5	122°44.1	10.0	192°43.8	15.8	6°42.8	25.4	Suhail	222°43.1	-43°33.1											
Mer.pass. 19:20				ν 1.0' d -0.4' m -4.88				ν 0.4' d -0.3' m 1.11				ν 1.9' d 0.2' m -1.71				ν 2.7' d -0.0' m -0.18							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	70°08.4	138°40.1	S24°56.1	137°44.5	S23°09.7	207°45.8	S15°15.9	21°45.5	N15°25.4	Denebola	182°21.1	14°24.1							
1	85°10.9	153°41.2	55.6	152°44.9	09.4	222°47.7	16.1	36°48.1	25.3	Gienah	175°39.7	-17°42.4							
2	100°13.4	168°42.3	55.2	167°45.3	09.1	237°49.7	16.2	51°50.8	25.3	Acrux	172°55.7	-63°15.7							
3	115°15.8	183°43.3	•• 54.8	182°45.6	•• 08.8	252°51.6	•• 16.4	66°53.4	•• 25.3	Gacrux	171°47.3	-57°16.6							
4	130°18.3	198°44.4	54.3	197°46.0	08.5	267°53.6	16.5	81°56.1	25.2	Alioth	166°10.1	55°47.6							
5	145°20.8	213°45.5	53.9	212°46.4	08.2	282°55.5	16.7	96°58.7	25.2	Spica	158°18.4	-11°19.0							
6	160°23.2	228°46.5	S24°53.5	227°46.8	S23°07.9	297°57.5	S15°16.8	112°01.4	N15°25.1	Alkaid	152°49.5	49°09.7							
7	175°25.7	243°47.6	53.0	242°47.2	07.6	312°59.4	17.0	127°04.0	25.1	Hadar	148°30.7	-60°30.8							
8	190°28.2	258°48.7	52.6	257°47.6	07.3	328°01.4	17.2	142°06.7	25.0	Menkent	147°53.2	-36°30.9							
9	205°30.6	273°49.8	•• 52.1	272°48.0	•• 07.0	343°03.3	•• 17.3	157°09.3	•• 25.0	Arcturus	145°44.7	19°01.6							
10	220°33.1	288°50.8	51.7	287°48.4	06.7	358°05.2	17.5	172°12.0	24.9	Rigil Kent.	139°35.3	-60°57.3							
11	235°35.5	303°51.9	51.3	302°48.8	06.4	13°07.2	17.6	187°14.6	24.9	Kochab	137°21.4	74°01.8							
12	250°38.0	318°53.0	S24°50.8	317°49.2	S23°06.1	28°09.1	S15°17.8	202°17.3	N15°24.9	Zuben'ubi	136°52.0	-16°09.8							
13	265°40.5	333°54.1	50.4	332°49.6	05.7	43°11.1	17.9	217°19.9	24.8	Alphecca	126°00.8	26°36.9							
14	280°42.9	348°55.2	49.9	347°50.0	05.4	58°13.0	18.1	232°22.6	24.8	Antares	112°11.4	-26°29.7							
15	295°45.4	3°56.3	•• 49.5	2°50.4	•• 05.1	73°15.0	•• 18.2	247°25.2	•• 24.7	Atria	107°02.5	-69°04.7							
16	310°47.9	18°57.4	49.1	17°50.7	04.8	88°16.9	18.4	262°27.9	24.7	Sabik	101°58.6	-15°45.5							
17	325°50.3	33°58.5	48.6	32°51.1	04.5	103°18.9	18.5	277°30.6	24.6	Shaula	96°05.5	-37°07.4							
18	340°52.8	48°59.6	S24°48.2	47°51.5	S23°04.2	118°20.8	S15°18.7	292°33.2	N15°24.6	Rasalhague	95°55.3	12°32.5							
19	355°55.3	64°00.7	47.7	62°51.9	03.9	133°22.8	18.8	307°35.9	24.6	Eltanin	90°40.9	51°29.3							
20	10°57.7	79°01.9	47.3	77°52.3	03.6	148°24.7	19.0	322°38.5	24.5	Kaus Aust.	83°27.7	-34°22.1							
21	26°00.2	94°03.0	•• 46.8	92°52.7	•• 03.3	163°26.7	•• 19.1	337°41.2	•• 24.5	Vega	80°30.9	38°49.0							
22	41°02.7	109°04.1	46.4	107°53.1	03.0	178°28.6	19.3	352°43.8	24.4	Nunki	75°43.3	-26°15.4							
23	56°05.1	124°05.2	46.0	122°53.5	02.7	193°30.6	19.4	7°46.5	24.4	Altair	61°56.4	8°57.1							
Mer.pass. 19:16				ν 1.1' d -0.4' m -4.89				ν 0.4' d -0.3' m 1.11				ν 1.9' d 0.2' m -1.71				ν 2.7' d -0.0' m -0.18			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	71°07.6	139°06.4	S24°45.5	137°53.9	S23°02.4	208°32.5	S15°19.6	22°49.1	N15°24.4	Fomalhaut	15°10.3	-29°27.8									
1	86°10.0	154°07.5	45.1	152°54.3	02.1	223°34.4	19.7	37°51.8	24.3	Scheat	13°41.3	28°15.0									
2	101°12.5	169°08.6	44.6	167°54.7	01.7	238°36.4	19.9	52°54.4	24.3	Markab	13°25.9	15°22.2									
3	116°15.0	184°09.8	•• 44.2	182°55.1	•• 01.4	253°38.3	•• 20.0	67°57.1	•• 24.2	Nov 30 Fri				SHA				Mer.pass			
4	131°17.4	199°10.9	43.7	197°55.5	01.1	268°40.3	20.2	82°59.7	24.2	Venus	69°06.4	14:46									
5	146°19.9	214°12.0	43.3	212°55.9	00.8	283°42.2	20.3	98°02.4	24.1	Mars	68°25.8	14:49									
6	161°22.4	229°13.2	S24°42.8	227°56.2	S23°00.5	298°44.2	S15°20.5	113°05.0	N15°24.1	Jupiter	137°49.8	10:11									
7	176°24.8	244°14.3	42.4	242°56.6	23°00.2	313°46.1	20.6	128°07.7	24.1	Saturn	311°32.5	22:33									
8	191°27.3	259°15.5	41.9	257°57.0	22°59.9	328°48.1	20.8	143°10.3	24.0	Dec 01 Sat				SHA				Mer.pass			
9	206°29.8	274°16.6	•• 41.5	272°57.4	•• 59.6	343°50.0	•• 20.9	158°13.0	•• 24.0	Venus	68°31.7	14:44									
10	221°32.2	289°17.8	41.0	287°57.8	59.2	358°52.0	21.1	173°15.6	23.9	Mars	67°36.0	14:49									
11	236°34.7	304°19.0	40.6	302°58.2	58.9	13°53.9	21.3	188°18.3	23.9	Jupiter	137°37.3	10:08									
12	251°37.1	319°20.1	S24°40.1	317°58.6	S22°58.6	28°55.9	S15°21.4	203°20.9	N15°23.8	Saturn	311°37.0	22:29									
13	266°39.6	334°21.3	39.7	332°59.0	58.3	43°57.8	21.6	218°23.6	23.8	Dec 02 Sun				SHA				Mer.pass			
14	281°42.1	349°22.5	39.2	347°59.4	58.0	58°59.8	21.7	233°26.2	23.8	Venus	67°58.8	14:42									
15	296°44.5	4°23.6	•• 38.8	2°59.8	•• 57.7	74°01.7	•• 21.9	248°28.8	•• 23.7	Mars	66°46.3	14:48									
16	311°47.0	19°24.8	38.3	18°00.2	57.3	89°03.7	22.0	263°31.5	23.7	Jupiter	137°24.9	10:05									
17	326°49.5	34°26.0	37.9	33°00.6	57.0	104°05.6	22.2	278°34.1	23.6	Saturn	311°41.5	22:25									
18	341°51.9	49°27.2	S24°37.4	48°01.0	S22°56.7	119°07.6	S15°22.3	293°36.8	N15°23.6	Horizontal parallax											
19	356°54.4	64°28.4	37.0	63°01.4	56.4	134°09.5	22.5	308°39.4	23.5	Venus:	0.4										
20	11°56.9	79°29.6	36.5	78°01.8	56.1	149°11.5	22.6	323°42.1	23.5	Mars:	0.1										
21	26°59.3	94°30.8	•• 36.1	93°02.2	•• 55.8	164°13.4	•• 22.8														

h	Sun				Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	182°51.6	S21°38.9	259°33.8	11.6'	S01°13.8	13.9'	58.7'	
1	197°51.4	39.3	274°04.4	11.5'	01°27.7	13.9'	58.7'	
2	212°51.2	39.7	288°34.9	11.5'	01°41.6	13.9'	58.8'	
3	227°50.9	· · 40.1	303°05.4	11.4'	01°55.6	13.9'	58.8'	
4	242°50.7	40.5	317°35.8	11.4'	02°09.5	13.9'	58.8'	
5	257°50.5	40.9	332°06.2	11.3'	02°23.4	13.9'	58.9'	
6	272°50.3	S21°41.3	346°36.6	11.3'	S02°37.3	13.9'	58.9'	
7	287°50.0	41.7	1°06.9	11.2'	02°51.3	13.9'	58.9'	
8	302°49.8	42.1	15°37.1	11.2'	03°05.2	13.9'	59.0'	
9	317°49.6	· · 42.5	30°07.3	11.1'	03°19.1	13.9'	59.0'	
10	332°49.3	42.9	44°37.5	11.1'	03°33.1	13.9'	59.0'	
11	347°49.1	43.3	59°07.6	11.0'	03°47.0	13.9'	59.1'	
12	2°48.9	S21°43.7	73°37.6	11.0'	S04°00.9	13.9'	59.1'	
13	17°48.7	44.1	88°07.6	10.9'	04°14.8	13.9'	59.1'	
14	32°48.4	44.5	102°37.5	10.9'	04°28.8	13.9'	59.1'	
15	47°48.2	· · 44.9	117°07.4	10.8'	04°42.7	13.9'	59.2'	
16	62°48.0	45.3	131°37.2	10.7'	04°56.6	13.9'	59.2'	
17	77°47.7	45.7	146°06.9	10.7'	05°10.4	13.9'	59.2'	
18	92°47.5	S21°46.1	160°36.6	10.6'	S05°24.3	13.9'	59.3'	
19	107°47.3	46.4	175°06.2	10.6'	05°38.2	13.8'	59.3'	
20	122°47.0	46.8	189°35.8	10.5'	05°52.0	13.8'	59.3'	
21	137°46.8	· · 47.2	204°05.3	10.4'	06°05.8	13.8'	59.4'	
22	152°46.6	47.6	218°34.7	10.4'	06°19.6	13.8'	59.4'	
23	167°46.3	48.0	233°04.1	10.3'	06°33.4	13.8'	59.4'	
SD = 16.2' $d = 0.4'$ SD = 16.0'								
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	182°46.1	S21°48.4	247°33.4	10.2'	S06°47.2	13.7'	59.5'	
1	197°45.9	48.8	262°02.6	10.2'	07°00.9	13.7'	59.5'	
2	212°45.6	49.2	276°31.8	10.1'	07°14.6	13.7'	59.5'	
3	227°45.4	· · 49.6	291°00.9	10.0'	07°28.3	13.7'	59.6'	
4	242°45.2	49.9	305°29.9	10.0'	07°42.0	13.6'	59.6'	
5	257°44.9	50.3	319°58.9	9.9'	07°55.6	13.6'	59.6'	
6	272°44.7	S21°50.7	334°27.8	9.8'	S08°09.2	13.6'	59.6'	
7	287°44.5	51.1	348°56.6	9.7'	08°22.8	13.5'	59.7'	
8	302°44.2	51.5	3°25.3	9.7'	08°36.4	13.5'	59.7'	
9	317°44.0	· · 51.9	17°54.0	9.6'	08°49.9	13.5'	59.7'	
10	332°43.8	52.2	32°22.5	9.5'	09°03.3	13.4'	59.8'	
11	347°43.5	52.6	46°51.0	9.4'	09°16.8	13.4'	59.8'	
12	2°43.3	S21°53.0	61°19.5	9.3'	S09°30.1	13.3'	59.8'	
13	17°43.1	53.4	75°47.8	9.3'	09°43.5	13.3'	59.8'	
14	32°42.8	53.8	90°16.1	9.2'	09°56.8	13.3'	59.9'	
15	47°42.6	· · 54.1	104°44.3	9.1'	10°10.1	13.2'	59.9'	
16	62°42.3	54.5	119°12.4	9.0'	10°23.3	13.2'	59.9'	
17	77°42.1	54.9	133°40.4	8.9'	10°36.5	13.1'	60.0'	
18	92°41.9	S21°55.3	148°08.3	8.9'	S10°49.6	13.1'	60.0'	
19	107°41.6	55.6	162°36.2	8.8'	11°02.6	13.0'	60.0'	
20	122°41.4	56.0	177°03.9	8.7'	11°15.6	13.0'	60.0'	
21	137°41.2	· · 56.4	191°31.6	8.6'	11°28.6	12.9'	60.1'	
22	152°40.9	56.7	205°59.2	8.5'	11°41.5	12.8'	60.1'	
23	167°40.7	57.1	220°26.7	8.4'	11°54.4	12.8'	60.1'	
SD = 16.2' $d = 0.4'$ SD = 16.2'								
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	182°40.4	S21°57.5	234°54.1	8.3'	S12°07.1	12.7'	60.1'	
1	197°40.2	57.9	249°21.5	8.2'	12°19.9	12.7'	60.2'	
2	212°40.0	58.2	263°48.7	8.2'	12°32.5	12.6'	60.2'	
3	227°39.7	· · 58.6	278°15.9	8.1'	12°45.1	12.5'	60.2'	
4	242°39.5	59.0	292°42.9	8.0'	12°57.6	12.5'	60.2'	
5	257°39.2	59.3	307°09.9	7.9'	13°10.1	12.4'	60.3'	
6	272°39.0	S21°59.7	321°36.8	7.8'	S13°22.5	12.3'	60.3'	
7	287°38.7	22°00.1	336°03.6	7.7'	13°34.8	12.2'	60.3'	
8	302°38.5	00.4	350°30.3	7.6'	13°47.0	12.2'	60.3'	
9	317°38.3	· · 00.8	4°56.9	7.5'	13°59.2	12.1'	60.4'	
10	332°38.0	01.2	19°23.4	7.4'	14°11.3	12.0'	60.4'	
11	347°37.8	01.5	33°49.8	7.3'	14°23.3	11.9'	60.4'	
12	2°37.5	S22°01.9	48°16.2	7.2'	S14°35.2	11.8'	60.4'	
13	17°37.3	02.2	62°42.4	7.1'	14°47.0	11.8'	60.4'	
14	32°37.0	02.6	77°08.5	7.0'	14°58.8	11.7'	60.5'	
15	47°36.8	· · 03.0	91°34.6	7.0'	15°10.5	11.6'	60.5'	
16	62°36.6	03.3	106°00.5	6.9'	15°22.0	11.5'	60.5'	
17	77°36.3	03.7	120°26.4	6.8'	15°33.5	11.4'	60.5'	
18	92°36.1	S22°04.0	134°52.2	6.7'	S15°44.9	11.3'	60.5'	
19	107°35.8	04.4	149°17.8	6.6'	15°56.2	11.2'	60.6'	
20	122°35.6	04.7	163°43.4	6.5'	16°07.4	11.1'	60.6'	
21	137°35.3	· · 05.1	178°08.9	6.4'	16°18.5	11.0'	60.6'	
22	152°35.1	05.5	192°34.3	6.3'	16°29.5	10.9'	60.6'	
23	167°34.8	05.8	206°59.6	6.2'	16°40.4	10.8'	60.6'	
SD = 16.2' $d = 0.4'$ SD = 16.4'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:52	09:51	■	■	13:46	15:46
N 70°	07:35	09:12	■	■	14:26	16:02
68°	07:22	08:44	10:38	13:00	14:53	16:16
66°	07:11	08:24	09:48	13:50	15:14	16:27
64°	07:02	08:07	09:17	14:21	15:31	16:36
62°	06:53	07:53	08:54	14:44	15:45	16:44
60°	06:46	07:41	08:35	15:02	15:57	16:52
N 58°	06:40	07:31	08:20	15:18	16:07	16:58
56°	06:34	07:22	08:07	15:31	16:16	17:04
54°	06:28	07:14	07:56	15:42	16:24	17:09
52°	06:23	07:06	07:46	15:52	16:32	17:14
50°	06:19	07:00	07:37	16:01	16:38	17:19
45°	06:08	06:45	07:18	16:20	16:53	17:30
N 40°	05:59	06:33	07:03	16:35	17:05	17:39
35°	05:51	06:22	06:50	16:48	17:16	17:47
30°	05:43	06:12	06:38	17:00	17:26	17:55
20°	05:28	05:55	06:19	17:19	17:43	18:10
N 10°	05:13	05:39	06:02	17:37	17:59	18:25
0°	04:57	05:23	05:45	17:53	18:15	18:41
S 10°	04:40	05:06	05:29	18:09	18:32	18:59
20°	04:19	04:47	05:12	18:27	18:51	19:20
30°	03:52	04:24	04:51	18:47	19:14	19:47
35°	03:34	04:10	04:39	18:59	19:28	20:04
40°	03:13	03:53	04:25	19:13	19:45	20:26
45°	02:46	03:33	04:09	19:30	20:06	20:53
S 50°	02:06	03:06	03:49	19:50	20:33	21:33
52°	01:44	02:53	03:39	20:00	20:46	21:56
54°	01:14	02:37	03:28	20:11	21:02	22:27
56°	00:15	02:18	03:15	20:24	21:21	23:37
58°	—:—	01:55	03:01	20:39	21:45	—:—
S 60°	—:—	01:22	02:43	20:56	22:19	—:—

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	01:04	03:15	05:43	12:13	11:48	11:12
N 70°	01:02	03:03	05:15	12:19	12:03	11:42
68°	01:00	02:53	04:55	12:24	12:15	12:04
66°	00:58	02:45	04:38	12:28	12:25	12:22
64°	00:57	02:38	04:25	12:31	12:33	12:36
62°	00:56	02:33	04:14	12:34	12:40	12:48
60°	00:55	02:28	04:05	12:37	12:46	12:59
N 58°	00:54	02:23	03:57	12:39	12:52	13:08
56°	00:53	02:20	03:50	12:41	12:57	13:16
54°	00:52	02:16	03:43	12:43	13:01	13:23
52°	00:52	02:13	03:38	12:44	13:05	13:30
50°	00:51	02:10	03:32	12:46	13:09	13:35
45°	00:50	02:04	03:21	12:49	13:17	13:48
N 40°	00:49	01:59	03:12	12:52	13:23	13:59
35°	00:48	01:55	03:04	12:55	13:29	14:08
30°	00:47	01:51	02:58	12:57	13:34	14:16
20°	00:46	01:45	02:46	13:01	13:43	14:29
N 10°	00:45	01:39	02:36	13:04	13:51	14:41
0°	00:44	01:34	02:26	13:07	13:58	14:53
S 10°	00:43	01:28	02:17	13:10	14:06	15:04
20°	00:42	01:23	02:07	13:14	14:13	15:16
30°	00:41	01:17	01:56	13:17	14:22	15:30
35°	00:40	01:13	01:49	13:20	14:28	15:38
40°	00:39	01:09	01:42	13:22	14:34	15:48
45°	00:38	01:05	01:34	13:25	14:40	15:59
S 50°	00:37	00:59	01:24	13:28	14:49	16:12
52°	00:37	00:57	01:19	13:30	14:53	16:18
54°	00:36	00:54	01:14	13:32	14:57	16:25
56°	00:36	00:51	01:08	13:34	15:02	16:33
58°	00:35	00:47	01:02	13:36	15:07	16:41
S 60°	00:35	00:44	00:55	13:38	15:13	16:51

Day	Sun			Moon		
	Eqn. of Time		Mer.	Mer. Pass.		Age
	00 ^h	12 ^h	Pass	Upper	Lower	24-26
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	39-18%
30	11:26	11:16	11:49	06:55	19:20	
01	11:04	10:53	11:49	07:46	20:12	
02	10:42	10:30	11:49	08:39	21:08	

December 03, 04, 05 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	SHA	Dec	
0	72°06.7	139°34.4	S24°34.7	138°03.3	S22°54.8	209°19.2	S15°23.2	23°52.7	N15°23.3	38°55.3	23.3	23°52.7	357°30.5	29°15.6	Alpheratz	357°30.5	29°15.6
1	87°09.2	154°35.6	34.3	153°03.7	54.5	224°21.2	23.4	38°55.3	23.3	53°58.0	23.3	38°55.3	353°03.2	-42°08.7	Ankaa	353°03.2	-42°08.7
2	102°11.6	169°36.8	33.8	168°04.1	54.2	239°23.1	23.5	53°58.0	23.3	69°00.6	23.2	53°58.0	349°26.0	56°42.4	Schedar	349°26.0	56°42.4
3	117°14.1	184°38.0	33.3	183°04.5	53.8	254°25.1	23.7	69°00.6	23.2	84°03.3	23.2	69°00.6	348°43.3	-17°49.3	Diphda	348°43.3	-17°49.3
4	132°16.6	199°39.2	32.9	198°04.9	53.5	269°27.0	23.8	84°03.3	23.2	99°05.9	23.1	84°03.3	335°17.1	-57°05.2	Achernar	335°17.1	-57°05.2
5	147°19.0	214°40.4	32.4	213°05.3	53.2	284°29.0	24.0	99°05.9	23.1	114°08.6	N15°23.1	99°05.9	327°46.4	23°36.4	Hamal	327°46.4	23°36.4
6	162°21.5	229°41.7	S24°32.0	228°05.7	S22°52.9	299°30.9	S15°24.1	114°08.6	N15°23.1	129°11.2	23.0	114°08.6	311°20.8	89°23.4	Polaris	311°20.8	89°23.4
7	177°24.0	244°42.9	31.5	243°06.1	52.5	314°32.9	24.3	129°11.2	23.0	144°13.9	23.0	129°11.2	315°08.5	-40°11.2	Acamar	315°08.5	-40°11.2
8	192°26.4	259°44.1	31.1	258°06.5	52.2	329°34.8	24.4	144°13.9	23.0	159°16.5	23.0	144°13.9	314°01.7	4°12.4	Menkar	314°01.7	4°12.4
9	207°28.9	274°45.4	30.6	273°06.9	51.9	344°36.8	24.6	159°16.5	23.0	174°19.2	22.9	159°16.5	308°22.0	49°58.1	Mirfak	308°22.0	49°58.1
10	222°31.4	289°46.6	30.1	288°07.3	51.6	359°38.7	24.7	174°19.2	22.9	189°21.8	22.9	174°19.2	290°34.7	16°34.1	Aldebaran	290°34.7	16°34.1
11	237°33.8	304°47.8	29.7	303°07.7	51.2	14°40.7	24.9	189°21.8	22.9	204°24.5	N15°22.8	189°21.8	280°59.8	-8°10.1	Rigel	280°59.8	-8°10.1
12	252°36.3	319°49.1	S24°29.2	318°08.1	S22°50.9	29°42.6	S15°25.0	204°24.5	N15°22.8	219°27.1	22.8	204°24.5	280°15.6	46°01.5	Capella	280°15.6	46°01.5
13	267°38.8	334°50.3	28.8	333°08.5	50.6	44°44.6	25.2	219°27.1	22.8	234°29.8	22.8	219°27.1	278°18.3	6°22.5	Bellatrix	278°18.3	6°22.5
14	282°41.2	349°51.6	28.3	348°08.9	50.3	59°46.5	25.3	234°29.8	22.8	249°32.4	22.7	234°29.8	277°56.5	28°37.8	Elnath	277°56.5	28°37.8
15	297°43.7	4°52.8	27.8	3°09.3	49.9	74°48.5	25.5	249°32.4	22.7	264°35.0	22.7	249°32.4	275°33.4	-1°11.1	Alnilam	275°33.4	-1°11.1
16	312°46.1	19°54.1	27.4	18°09.7	49.6	89°50.4	25.6	264°35.0	22.7	279°37.7	22.6	264°35.0	270°47.5	7°24.6	Betelgeuse	270°47.5	7°24.6
17	327°48.6	34°55.4	26.9	33°10.1	49.3	104°52.4	25.8	279°37.7	22.6	294°40.3	N15°22.6	279°37.7	263°50.1	-52°42.7	Canopus	263°50.1	-52°42.7
18	342°51.1	49°56.6	S24°26.5	48°10.5	S22°49.0	119°54.3	S15°25.9	294°40.3	N15°22.6	309°43.0	22.5	294°40.3	258°22.5	-16°45.5	Sirius	258°22.5	-16°45.5
19	357°53.5	64°57.9	26.0	63°10.9	48.6	134°56.3	26.1	309°43.0	22.5	324°45.6	22.5	309°43.0	255°02.4	-29°00.8	Adhara	255°02.4	-29°00.8
20	12°56.0	79°59.2	25.5	78°11.3	48.3	149°58.2	26.2	324°45.6	22.5	339°48.3	22.5	324°45.6	244°46.5	5°08.7	Procyon	244°46.5	5°08.7
21	27°58.5	95°00.4	25.1	93°11.7	48.0	165°00.2	26.4	339°48.3	22.5	354°50.9	22.4	339°48.3	243°12.3	27°56.9	Pollux	243°12.3	27°56.9
22	43°00.9	110°01.7	24.6	108°12.1	47.6	180°02.1	26.5	354°50.9	22.4	9°53.6	22.4	354°50.9	234°12.6	-59°36.2	Avior	234°12.6	-59°36.2
23	58°03.4	125°03.0	24.2	123°12.5	47.3	195°04.1	26.7	9°53.6	22.4			9°53.6	222°43.1	-43°33.1	Suhail	222°43.1	-43°33.1
Mer.pass. 19:08			ν 1.2' d -0.5' m -4.90			ν 0.4' d -0.3' m 1.11			ν 1.9' d 0.2' m -1.72			ν 2.6' d -0.0' m -0.17					

Tue	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	SHA	Dec	
0	73°05.9	140°04.3	S24°23.7	138°12.9	S22°47.0	210°06.0	S15°26.8	24°56.2	N15°22.3	39°58.9	22.3	24°56.2	175°39.6	-17°42.4	Gienah	175°39.6	-17°42.4
1	88°08.3	155°05.6	23.2	153°13.2	46.6	225°08.0	27.0	39°58.9	22.3	55°01.5	22.3	39°58.9	172°55.7	-63°15.7	Acrux	172°55.7	-63°15.7
2	103°10.8	170°06.9	22.8	168°13.6	46.3	240°09.9	27.1	55°01.5	22.3	70°04.2	22.2	55°01.5	171°47.3	-57°16.6	Gacrux	171°47.3	-57°16.6
3	118°13.3	185°08.2	22.3	183°14.0	46.0	255°11.9	27.3	70°04.2	22.2	85°06.8	22.2	70°04.2	166°10.1	55°47.6	Alioth	166°10.1	55°47.6
4	133°15.7	200°09.5	21.8	198°14.4	45.6	270°13.8	27.4	85°06.8	22.2	100°09.4	22.1	85°06.8	158°18.4	-11°19.0	Spica	158°18.4	-11°19.0
5	148°18.2	215°10.8	21.4	213°14.8	45.3	285°15.8	27.6	100°09.4	22.1	115°12.1	N15°22.1	100°09.4	152°49.4	49°09.6	Alkaid	152°49.4	49°09.6
6	163°20.6	230°12.1	S24°20.9	228°15.2	S22°45.0	300°17.7	S15°27.7	115°12.1	N15°22.1	130°14.7	22.1	115°12.1	148°30.7	-60°30.8	Hadar	148°30.7	-60°30.8
7	178°23.1	245°13.4	20.4	243°15.6	44.6	315°19.7	27.9	130°14.7	22.1	145°17.4	22.0	130°14.7	147°53.2	-36°30.9	Menkent	147°53.2	-36°30.9
8	193°25.6	260°14.7	20.0	258°16.0	44.3	330°21.6	28.0	145°17.4	22.0	160°20.0	22.0	145°17.4	145°44.7	19°01.6	Arcturus	145°44.7	19°01.6
9	208°28.0	275°16.0	19.5	273°16.4	44.0	345°23.6	28.2	160°20.0	22.0	175°22.7	21.9	160°20.0	139°35.3	-60°57.3	Rigel Kent.	139°35.3	-60°57.3
10	223°30.5	290°17.3	19.0	288°16.8	43.6	0°25.5	28.3	175°22.7	21.9	190°25.3	21.9	175°22.7	137°21.4	74°01.8	Kochab	137°21.4	74°01.8
11	238°33.0	305°18.6	18.6	303°17.2	43.3	15°27.5	28.5	190°25.3	21.9	205°28.0	N15°21.8	190°25.3	136°51.9	-16°09.8	Zuben'ubi	136°51.9	-16°09.8
12	253°35.4	320°20.0	S24°18.1	318°17.6	S22°43.0	30°29.4	S15°28.6	205°28.0	N15°21.8	220°30.6	21.8	205°28.0	126°00.8	26°36.9	Alphecca	126°00.8	26°36.9
13	268°37.9	335°21.3	17.6	333°18.0	42.6	45°31.4	28.8	220°30.6	21.8	235°33.3	21.8	220°30.6	112°11.4	-26°29.7	Antares	112°11.4	-26°29.7
14	283°40.4	350°22.6	17.2	348°18.4	42.3	60°33.3	28.9	235°33.3	21.8	250°35.9	21.7	235°33.3	107°02.5	-69°04.7	Atria	107°02.5	-69°04.7
15	298°42.8	5°24.0	16.7	3°18.8	41.9	75°35.3	29.1	250°35.9	21.7	265°38.5	21.7	250°35.9	101°58.6	-15°45.5	Sabik	101°58.6	-15°45.5
16	313°45.3	20°25.3	16.2	18°19.2	41.6	90°37.2	29.2	265°38.5	21.7	280°41.2	21.6	265°38.5	96°05.5	-37°07.3	Shaula	96°05.5	-37°07.3
17	328°47.7	35°26.6	15.8	33°19.6	41.3	105°39.2	29.4	280°41.2	21.6	295°43.8	N15°21.6	280°41.2	95°55.3	12°32.5	Rasalhague	95°55.3	12°32.5
18	343°50.2	50°28.0	S24°15.3	48°20.0	S22°40.9	120°41.1	S15°29.5	295°43.8	N15°21.6	310°46.5	21.6	295°43.8	90°40.9	51°29.3	Eltanin	90°40.9	51°29.3
19	358°52.7	65°29.3	14.8	63°20.4	40.6	135°43.1	29.7	310°46.5	21.6	325°49.1	21.5	310°46.5	83°27.7	-34°22.1	Kaus Aust.	83°27.7	-34°22.1
20	13°55.1	80°30.7	14.4	78°20.8	40.3	150°45.0	29.8	325°49.1	21.5	340°51.8	21.5	325°49.1	80°30.9	38°48.9	Vega	80°30.9	38°48.9
21	28°57.6	95°32.0	13.9	93°21.2	39.9	165°47.0	29.9	340°51.8	21.5	355°54.4	21.4	340°51.8	75°43.3	-26°15.4	Nunki	75°43.3	-26°15.4
22	44°00.1	110°33.4	13.4	108°21.6	39.6	180°48.9	30.1	355°54.4	21.4	10°57.0	21.4	355°54.4	61°56.4	8°57.1	Altair	61°56.4	8°57.1
2																	

Sun							Moon							
Mon	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
									Naut.	Civil			Civil	Naut.
0	182°34.6	S22°06.2	221°24.8	6.1'	S16°51.2	10.7'	60.7'	N 72°	07:59	10:05	■	■	13:35	15:41
1	197°34.3	06.5	235°49.9	6.0'	17°01.9	10.6'	60.7'	N 70°	07:42	09:21	■	■	14:19	15:58
2	212°34.1	06.9	250°14.9	5.9'	17°12.5	10.5'	60.7'	68°	07:28	08:52	10:57	12:43	14:48	16:12
3	227°33.9	· · 07.2	264°39.8	5.8'	17°23.0	10.4'	60.7'	66°	07:17	08:30	09:58	13:42	15:10	16:23
4	242°33.6	07.6	279°04.6	5.7'	17°33.3	10.3'	60.7'	64°	07:07	08:13	09:25	14:16	15:27	16:33
5	257°33.4	07.9	293°29.3	5.6'	17°43.6	10.1'	60.7'	62°	06:58	07:58	09:00	14:40	15:42	16:42
6	272°33.1	S22°08.3	307°54.0	5.5'	S17°53.7	10.0'	60.7'	60°	06:51	07:46	08:41	14:59	15:54	16:50
7	287°32.9	08.6	322°18.5	5.5'	18°03.8	9.9'	60.8'	N 58°	06:44	07:35	08:25	15:15	16:05	16:56
8	302°32.6	09.0	336°43.0	5.4'	18°13.7	9.8'	60.8'	56°	06:38	07:26	08:12	15:28	16:14	17:03
9	317°32.4	· · 09.3	351°07.3	5.3'	18°23.4	9.7'	60.8'	54°	06:32	07:18	08:00	15:40	16:23	17:08
10	332°32.1	09.7	5°31.6	5.2'	18°33.1	9.5'	60.8'	52°	06:27	07:10	07:50	15:50	16:30	17:13
11	347°31.9	10.0	19°55.8	5.1'	18°42.6	9.4'	60.8'	50°	06:22	07:03	07:41	16:00	16:37	17:18
12	2°31.6	S22°10.3	34°19.9	5.0'	S18°52.0	9.3'	60.8'	45°	06:11	06:48	07:21	16:19	16:52	17:29
13	17°31.4	10.7	48°43.9	4.9'	19°01.3	9.2'	60.8'	N 40°	06:02	06:35	07:06	16:35	17:05	17:39
14	32°31.1	11.0	63°07.8	4.8'	19°10.5	9.0'	60.8'	35°	05:53	06:24	06:52	16:48	17:16	17:47
15	47°30.9	· · 11.4	77°31.6	4.7'	19°19.5	8.9'	60.9'	30°	05:45	06:15	06:41	17:00	17:26	17:55
16	62°30.6	11.7	91°55.4	4.7'	19°28.4	8.8'	60.9'	20°	05:30	05:57	06:21	17:20	17:44	18:11
17	77°30.4	12.0	106°19.0	4.6'	19°37.1	8.6'	60.9'	N 10°	05:14	05:40	06:03	17:37	18:00	18:26
18	92°30.1	S22°12.4	120°42.6	4.5'	S19°45.8	8.5'	60.9'	0°	04:58	05:24	05:47	17:54	18:16	18:42
19	107°29.9	12.7	135°06.1	4.4'	19°54.2	8.3'	60.9'	S 10°	04:40	05:07	05:30	18:11	18:34	19:00
20	122°29.6	13.1	149°29.5	4.3'	20°02.6	8.2'	60.9'	20°	04:19	04:48	05:12	18:29	18:53	19:22
21	137°29.4	· · 13.4	163°52.8	4.2'	20°10.8	8.1'	60.9'	30°	03:51	04:24	04:51	18:50	19:17	19:50
22	152°29.1	13.7	178°16.1	4.2'	20°18.8	7.9'	60.9'	35°	03:34	04:10	04:39	19:02	19:31	20:07
23	167°28.9	14.1	192°39.2	4.1'	20°26.8	7.8'	60.9'	40°	03:12	03:53	04:25	19:16	19:48	20:29
SD = 16.2' d = 0.4'							SD = 16.5'							
Tue	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Moonrise		Moonset			
									Mon	Tue	Wed	Mon	Tue	Wed
0	182°28.6	S22°14.4	207°02.3	4.0'	S20°34.5	7.6'	60.9'	N 72°	■	■	■	■	■	■
1	197°28.3	14.7	221°25.4	3.9'	20°42.2	7.5'	60.9'	N 70°	07:57	■	■	11:00	■	■
2	212°28.1	15.1	235°48.3	3.9'	20°49.6	7.3'	60.9'	68°	07:10	10:19	■	11:48	10:48	■
3	227°27.8	· · 15.4	250°11.2	3.8'	20°57.0	7.2'	60.9'	66°	06:40	08:49	10:55	12:19	12:18	12:27
4	242°27.6	15.7	264°34.0	3.7'	21°04.1	7.0'	60.9'	64°	06:18	08:11	09:51	12:43	12:57	13:31
5	257°27.3	16.1	278°56.7	3.7'	21°11.2	6.9'	60.9'	62°	06:00	07:45	09:16	13:01	13:24	14:06
6	272°27.1	S22°16.4	293°19.4	3.6'	S21°18.0	6.7'	61.0'	60°	05:45	07:24	08:51	13:17	13:45	14:31
7	287°26.8	16.7	307°42.0	3.5'	21°24.7	6.6'	61.0'	N 58°	05:33	07:07	08:30	13:30	14:03	14:52
8	302°26.6	17.1	322°04.5	3.5'	21°31.3	6.4'	61.0'	56°	05:22	06:53	08:14	13:41	14:17	15:08
9	317°26.3	· · 17.4	336°27.0	3.4'	21°37.7	6.2'	61.0'	54°	05:13	06:40	08:00	13:52	14:30	15:23
10	332°26.1	17.7	350°49.4	3.3'	21°43.9	6.1'	61.0'	52°	05:04	06:29	07:47	14:00	14:41	15:35
11	347°25.8	18.0	5°11.7	3.3'	21°50.0	5.9'	61.0'	50°	04:57	06:20	07:36	14:09	14:51	15:46
12	2°25.5	S22°18.4	19°34.0	3.2'	S21°55.9	5.8'	61.0'	45°	04:41	06:00	07:14	14:26	15:12	16:09
13	17°25.3	18.7	33°56.2	3.2'	22°01.7	5.6'	61.0'	N 40°	04:28	05:43	06:55	14:40	15:29	16:27
14	32°25.0	19.0	48°18.4	3.1'	22°07.3	5.4'	61.0'	35°	04:17	05:30	06:40	14:52	15:44	16:43
15	47°24.8	· · 19.3	62°40.6	3.1'	22°12.7	5.3'	61.0'	30°	04:07	05:18	06:27	15:02	15:56	16:56
16	62°24.5	19.7	77°02.6	3.0'	22°17.9	5.1'	60.9'	20°	03:50	04:57	06:04	15:21	16:17	17:19
17	77°24.3	20.0	91°24.7	3.0'	22°23.0	4.9'	60.9'	N 10°	03:36	04:40	05:45	15:36	16:36	17:39
18	92°24.0	S22°20.3	105°46.7	2.9'	S22°27.9	4.8'	60.9'	0°	03:23	04:23	05:27	15:51	16:53	17:57
19	107°23.7	20.6	120°08.6	2.9'	22°32.7	4.6'	60.9'	S 10°	03:10	04:07	05:09	16:06	17:11	18:16
20	122°23.5	21.0	134°30.5	2.9'	22°37.3	4.4'	60.9'	20°	02:56	03:50	04:49	16:22	17:29	18:35
21	137°23.2	· · 21.3	148°52.4	2.8'	22°41.7	4.2'	60.9'	30°	02:40	03:30	04:27	16:40	17:51	18:58
22	152°23.0	21.6	163°14.2	2.8'	22°45.9	4.1'	60.9'	35°	02:31	03:19	04:14	16:51	18:03	19:11
23	167°22.7	21.9	177°36.0	2.8'	22°50.0	3.9'	60.9'	40°	02:20	03:05	03:59	17:04	18:18	19:27
SD = 16.2' d = 0.3'							SD = 16.6'							
Wed	GHA	Dec	GHA	ν	Dec	d	HP	Day	Sun		Mer.	Moon		Age
									Eqn.of Time	12 ^h	Pass	Mer.Pass.	Lower	27-0
									00 ^h	mm:ss	hh:mm	Upper	hh:mm	10-1%
0	182°22.5	S22°22.2	191°57.8	2.7'	S22°53.9	3.7'	60.9'	03	10:18	10:06	11:50	09:37	22:07	
1	197°22.2	22.5	206°19.5	2.7'	22°57.6	3.5'	60.9'	04	09:54	09:42	11:50	10:38	23:10	
2	212°21.9	22.9	220°41.2	2.7'	23°01.2	3.4'	60.9'	05	09:30	09:17	11:51	11:42	---	
3	227°21.7	· · 23.2	235°02.9	2.7'	23°04.5	3.2'	60.9'							
4	242°21.4	23.5	249°24.5	2.6'	23°07.7	3.0'	60.9'							
5	257°21.2	23.8	263°46.2	2.6'	23°10.8	2.8'	60.9'							
6	272°20.9	S22°24.1	278°07.8	2.6'	S23°13.6	2.7'	60.9'							
7	287°20.6	24.4	292°29.4	2.6'	23°16.3	2.5'	60.8'							
8	302°20.4	24.7	306°51.0	2.6'	23°18.8	2.3'	60.8'							
9	317°20.1	· · 25.0	321°12.6	2.6'	23°21.1	2.1'	60.8'							
10	332°19.9	25.4	335°34.2	2.6'	23°23.2	2.0'	60.8'							
11	347°19.6	25.7	349°55.7	2.6'	23°25.2	1.8'	60.8'							
12	2°19.3	S22°26.0	4°17.3	2.6'	S23°26.9	1.6'	60.8'							
13	17°19.1	26.3	18°38.9	2.6'	23°28.5	1.4'	60.8'							
14	32°18.8	26.6	33°00.5	2.6'	23°30.0	1.2'	60.8'							
15	47°18.5	· · 26.9	47°22.0	2.6'	23°31.2	1.1'	60.7'							
16	62°18.3	27.2	61°43.6	2.6'	23°32.3	0.9'	60.7'							
17	77°18.0	27.5	76°05.2	2.6'	23°33.1	0.7'	60.7'							
18	92°17.8	S22°27.8	90°26.8	2.6'	S23°33.9	0.5'	60.7'							
19	107°17.5	28.1	104°48.5	2.6'	23°34.4	0.4'	60.7'							
20	122°17.2	28.4	119°10.1	2.7'	23°34.7	0.2'	60.7'							
21	137°17.0	· · 28.7	133°31.8	2.7'	23°34.9	-0.0'	60.6'							
22	152°16.7	29.0	147°53.5	2.7'	23°34.9	-0.2'	60.6'							
23	167°16.4	29.3	162°15.2	2.7'	23°34.7	-0.4'	60.6'							
SD = 16.2' d = 0.3'							SD = 16.6'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:59	10:05	■	■	13:35	15:41
N 70°	07:42	09:21	■	■	14:19	15:58
68°	07:28	08:52	10:57	12:43	14:48	16:12
66°	07:17	08:30	09:58	13:42	15:10	16:23
64°	07:07	08:13	09:25	14:16	15:27	16:33
62°	06:58	07:58	09:00	14:40	15:42	16:42
60°	06:51	07:46	08:41	14:59	15:54	16:50
N 58°	06:44	07:35	08:2			

December 06, 07, 08 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	75°04.1	141°10.0	S24°01.1	138°32.0	S22°30.6	211°39.7	S15°33.9	27°03.1	N15°20.4	Alpheratz	357°30.5	29°15.6											
1	90°06.6	156°11.5	00.6	153°32.4	30.2	226°41.7	34.1	42°05.8	20.4	Ankaa	353°03.3	-42°08.7											
2	105°09.1	171°12.9	24°00.1	168°32.8	29.8	241°43.6	34.2	57°08.4	20.3	Schedar	349°26.0	56°42.4											
3	120°11.5	186°14.4	23°59.6	183°33.2	· · 29.5	256°45.6	· · 34.4	72°11.0	· · 20.3	Diphda	348°43.3	-17°49.3											
4	135°14.0	201°15.8	59.2	198°33.6	29.1	271°47.5	34.5	87°13.7	20.2	Achernar	335°17.1	-57°05.3											
5	150°16.5	216°17.3	58.7	213°34.0	28.8	286°49.5	34.7	102°16.3	20.2	Hamal	327°46.4	23°36.4											
6	165°18.9	231°18.8	S23°58.2	228°34.4	S22°28.4	301°51.4	S15°34.8	117°19.0	N15°20.2	Polaris	311°20.9	89°23.4											
7	180°21.4	246°20.3	57.7	243°34.8	28.1	316°53.4	35.0	132°21.6	20.1	Acamar	315°08.5	-40°11.2											
8	195°23.9	261°21.8	57.2	258°35.2	27.7	331°55.3	35.1	147°24.2	20.1	Menkar	314°01.7	4°12.4											
9	210°26.3	276°23.2	· · 56.8	273°35.6	· · 27.4	346°57.3	· · 35.3	162°26.9	· · 20.0	Mirfak	308°22.0	49°58.1											
10	225°28.8	291°24.7	56.3	288°36.0	27.0	1°59.2	35.4	177°29.5	20.0	Aldebaran	290°34.7	16°34.1											
11	240°31.2	306°26.2	55.8	303°36.4	26.6	17°01.2	35.6	192°32.2	20.0	Rigel	280°59.7	-8°10.1											
12	255°33.7	321°27.7	S23°55.3	318°36.8	S22°26.3	32°03.1	S15°35.7	207°34.8	N15°19.9	Capella	280°15.5	46°01.5											
13	270°36.2	336°29.2	54.8	333°37.2	25.9	47°05.1	35.9	222°37.4	19.9	Bellatrix	278°18.3	6°22.5											
14	285°38.6	351°30.7	54.4	348°37.6	25.6	62°07.1	36.0	237°40.1	19.8	Elnath	277°56.5	28°37.8											
15	300°41.1	6°32.2	· · 53.9	3°38.0	· · 25.2	77°09.0	· · 36.1	252°42.7	· · 19.8	Alnilam	275°33.4	-1°11.1											
16	315°43.6	21°33.7	53.4	18°38.4	24.9	92°11.0	36.3	267°45.4	19.8	Betelgeuse	270°47.5	7°24.6											
17	330°46.0	36°35.2	52.9	33°38.9	24.5	107°12.9	36.4	282°48.0	19.7	Canopus	263°50.1	-52°42.7											
18	345°48.5	51°36.8	S23°52.4	48°39.3	S22°24.1	122°14.9	S15°36.6	297°50.6	N15°19.7	Sirius	258°22.4	-16°45.5											
19	0°51.0	66°38.3	51.9	63°39.7	23.8	137°16.8	36.7	312°53.3	19.7	Adhara	255°02.4	-29°00.8											
20	15°53.4	81°39.8	51.5	78°40.1	23.4	152°18.8	36.9	327°55.9	19.6	Procyon	244°46.5	5°08.7											
21	30°55.9	96°41.3	· · 51.0	93°40.5	· · 23.0	167°20.7	· · 37.0	342°58.6	· · 19.6	Pollux	243°12.3	27°56.9											
22	45°58.4	111°42.9	50.5	108°40.9	22.7	182°22.7	37.2	358°01.2	19.5	Avior	234°12.6	-59°36.2											
23	61°00.8	126°44.4	50.0	123°41.3	22.3	197°24.6	37.3	13°03.8	19.5	Suhail	222°43.1	-43°33.1											
Mer.pass. 18:57				ν 1.5' d -0.5' m -4.91	ν 0.4' d -0.4' m 1.12	ν 2.0' d 0.1' m -1.72	ν 2.6' d -0.0' m -0.16																

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	76°03.3	141°45.9	S23°49.5	138°41.7	S22°22.0	212°26.6	S15°37.5	28°06.5	N15°19.5	Denebola	182°21.1	24°24.1	
1	91°05.7	156°47.5	49.0	153°42.1	21.6	227°28.6	37.6	43°09.1	19.4	Gienah	175°39.6	-17°42.5	
2	106°08.2	171°49.0	48.6	168°42.5	21.2	242°30.5	37.8	58°11.8	19.4	Acrux	172°55.6	-63°15.7	
3	121°10.7	186°50.6	· · 48.1	183°42.9	· · 20.9	257°32.5	· · 37.9	73°14.4	· · 19.3	Gacrux	171°47.2	-57°16.6	
4	136°13.1	201°52.1	47.6	198°43.3	20.5	272°34.4	38.1	88°17.0	19.3	Alioth	166°10.0	55°47.5	
5	151°15.6	216°53.7	47.1	213°43.7	20.1	287°36.4	38.2	103°19.7	19.3	Spica	158°18.3	-11°19.0	
6	166°18.1	231°55.3	S23°46.6	228°44.1	S22°19.8	302°38.3	S15°38.3	118°22.3	N15°19.2	Alkaid	152°49.4	49°09.6	
7	181°20.5	246°56.8	46.1	243°44.5	19.4	317°40.3	38.5	133°24.9	19.2	Hadar	148°30.6	-60°30.8	
8	196°23.0	261°58.4	45.6	258°44.9	19.0	332°42.2	38.6	148°27.6	19.1	Menkent	147°53.2	-36°30.9	
9	211°25.5	277°00.0	· · 45.2	273°45.3	· · 18.7	347°44.2	· · 38.8	163°30.2	· · 19.1	Arcturus	145°44.6	19°01.5	
10	226°27.9	292°01.6	44.7	288°45.7	18.3	2°46.2	38.9	178°32.9	19.1	Rigil Kent.	139°35.2	-60°57.3	
11	241°30.4	307°03.1	44.2	303°46.1	17.9	17°48.1	39.1	193°35.5	19.0	Kochab	137°21.3	74°01.7	
12	256°32.9	322°04.7	S23°43.7	318°46.5	S22°17.6	32°50.1	S15°39.2	208°38.1	N15°19.0	Zuben'ubi	136°51.9	-16°09.8	
13	271°35.3	337°06.3	43.2	333°46.9	17.2	47°52.0	39.4	223°40.8	19.0	Alphecca	126°00.8	26°36.9	
14	286°37.8	352°07.9	42.7	348°47.3	16.8	62°54.0	39.5	238°43.4	18.9	Antares	112°11.4	-26°29.7	
15	301°40.2	7°09.5	· · 42.2	3°47.7	· · 16.5	77°55.9	· · 39.7	253°46.0	· · 18.9	Atria	107°02.5	-69°04.7	
16	316°42.7	22°11.1	41.7	18°48.2	16.1	92°57.9	39.8	268°48.7	18.8	Sabik	101°58.6	-15°45.5	
17	331°45.2	37°12.7	41.2	33°48.6	15.7	107°59.8	39.9	283°51.3	18.8	Shaula	96°05.4	-37°07.3	
18	346°47.6	52°14.3	S23°40.8	48°49.0	S22°15.4	123°01.8	S15°40.1	298°54.0	N15°18.8	Rasalhague	95°55.3	12°32.5	
19	1°50.1	67°15.9	40.3	63°49.4	15.0	138°03.8	40.2	313°56.6	18.7	Eltanin	90°40.9	51°29.3	
20	16°52.6	82°17.5	39.8	78°49.8	14.6	153°05.7	40.4	328°59.2	18.7	Kaus Aust.	83°27.7	-34°22.0	
21	31°55.0	97°19.2	· · 39.3	93°50.2	· · 14.2	168°07.7	· · 40.5	344°01.9	· · 18.7	Vega	80°30.9	38°48.9	
22	46°57.5	112°20.8	38.8	108°50.6	13.9	183°09.6	40.7	359°04.5	18.6	Nunki	75°43.2	-26°15.4	
23	62°00.0	127°22.4	38.3	123°51.0	13.5	198°11.6	40.8	14°07.1	18.6	Altair	61°56.4	8°57.1	
Mer.pass. 18:53				ν 1.5' d -0.5' m -4.91	ν 0.4' d -0.4' m 1.12	ν 2.0' d 0.1' m -1.72	ν 2.6' d -0.0' m -0.15						

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	77°02.4	142°24.0	S23°37.8	138°51.4	S22°13.1	213°13.5	S15°41.0	29°09.8	N15°18.5	Dec 06 Thu	SHA		
1	92°04.9	157°25.7	37.3	153°51.8	12.7	228°15.5	41.1	44°12.4	18.5	Venus	66°05.9		
2	107°07.4	172°27.3	36.8	168°52.2	12.4	243°17.5	41.3	59°15.0	18.5	Mer.pass	14:34		
3	122°09.8	187°29.0	· · 36.3	183°52.6	· · 12.0	258°19.4	· · 41.4	74°17.7	· · 18.4	Mars	63°27.9		
4	137°12.3	202°30.6	35.9	198°53.0	11.6	273°21.4	41.5	89°20.3	18.4	Jupiter	136°35.6		
5	152°14.7	217°32.3	35.4	213°53.4	11.2	288°23.3	41.7	104°23.0	18.3	Saturn	311°59.0		
6	167°17.2	232°33.9	S23°34.9	228°53.8	S22°10.9	303°25.3	S15°41.8	119°25.6	N15°18.3		22:08		
7	182°19.7	247°35.6	34.4	243°54.3	10.5	318°27.2	42.0	134°28.2	18.3				
8	197°22.1	262°37.2	33.9	258°54.7	10.1	333°29.2	42.1	149°30.9	18.2	Dec 07 Fri	SHA		
9	212°24.6	277°38.9	· · 33.4	273°55.1	· · 09.7	348°31.2	· · 42.3	164°33.5	· · 18.2	Venus	65°42.7		
10	227°27.1	292°40.6	32.9	288°55.5	09.4	3°33.1	42.4	179°36.1	18.2	Mars	62°38.4		
11	242°29.5	307°42.2	32.4	303°55.9	09.0	18°35.1	42.6	194°38.8	18.1	Jupiter	136°23.3		
12	257°32.0	322°43.9	S23°31.9	318°56.3	S22°08.6	33°37.0	S15°42.7	209°41.4	N15°18.1	Saturn	312°03.2		
13	272°34.5	337°45.6	31.4	333°56.7	08.2	48°39.0	42.8	224°44.0	18.0		22:04		
14	287°36.9	352°47.3	30.9	348°57.1	07.9	63°40.9	43.0	239°46.7	18.0	Dec 08 Sat	SHA		
15	302°39.4	7°49.0	· · 30.4	3°57.5	· · 07.5	78°42.9	· · 43.1	254°49.3	· · 18.0	Venus	65°21.6		
16	317°41.9	22°50.7	29.9	18°57.9	07.1	93°44.9	43.3	269°51.9	17.9	Mars	61°49.0		
17	332°44.3	37°52.4	29.5	33°58.3	06.7	108°46.8	43.4	284°54.6	17.9	Jupiter	136°11.1		
18	347°46.8	52°54.1	S23°29.0	48°58.7	S22°06.3	123°48.8	S15°43.6	299°57.2	N15°17.9	Saturn	312°07.3		
19	2°49.2	67°55.8	28.5	63°59.1	05.9	138°50.7	43.7	314°59.8	17.8		21:59		
20	17°51.7	82°57.5	28.0	78°59.6	05.6	153°52.7	43.9	330°02.5	17.8				
21	32°54.2	97°59.2	· · 27.5	94°00.0	· · 05.2	168°54.6	· · 44.0	345°05.1	· · 17.7				
22	47°56.6	113°00.9	27.0	109°00.4	04.8	183°56.6	44.1	0°07.7	17.7				
23	62°59.1	128°02.6	26.5	124°00.8	04.4	198°58.6	44.3	15°10.4	17.7				
Mer.pass. 18:49				ν 1.6' d -0.5' m -4.91	ν 0.4' d -0.4' m 1.12	ν 2.0' d 0.1' m -1.73	ν 2.6' d -0.0' m -0.15						

Horizontal parallax		
Venus:	0.4	
Mars:	0.1	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	182°16.2	S22°29.6	176°37.0	2.8'	S23°34.4	-0.5'	60.6'
1	197°15.9	29.9	190°58.7	2.8'	23°33.8	-0.7'	60.6'
2	212°15.6	30.2	205°20.5	2.8'	23°33.1	-0.9'	60.5'
3	227°15.4	· · 30.5	219°42.4	2.9'	23°32.2	-1.1'	60.5'
4	242°15.1	30.8	234°04.3	2.9'	23°31.2	-1.2'	60.5'
5	257°14.9	31.1	248°26.2	3.0'	23°29.9	-1.4'	60.5'
6	272°14.6	S22°31.4	262°48.2	3.0'	S23°28.5	-1.6'	60.5'
7	287°14.3	31.7	277°10.2	3.1'	23°26.9	-1.8'	60.4'
8	302°14.1	32.0	291°32.3	3.1'	23°25.2	-1.9'	60.4'
9	317°13.8	· · 32.3	305°54.4	3.2'	23°23.3	-2.1'	60.4'
10	332°13.5	32.6	320°16.6	3.2'	23°21.2	-2.3'	60.4'
11	347°13.3	32.8	334°38.8	3.3'	23°18.9	-2.4'	60.3'
12	2°13.0	S22°33.1	349°01.1	3.3'	S23°16.5	-2.6'	60.3'
13	17°12.7	33.4	3°23.4	3.4'	23°13.8	-2.8'	60.3'
14	32°12.5	33.7	17°45.8	3.5'	23°11.1	-2.9'	60.3'
15	47°12.2	· · 34.0	32°08.3	3.5'	23°08.1	-3.1'	60.2'
16	62°11.9	34.3	46°30.8	3.6'	23°05.0	-3.3'	60.2'
17	77°11.7	34.6	60°53.4	3.7'	23°01.8	-3.4'	60.2'
18	92°11.4	S22°34.9	75°16.1	3.7'	S22°58.3	-3.6'	60.2'
19	107°11.1	35.1	89°38.8	3.8'	22°54.7	-3.8'	60.1'
20	122°10.8	35.4	104°01.7	3.9'	22°51.0	-3.9'	60.1'
21	137°10.6	· · 35.7	118°24.6	4.0'	22°47.1	-4.1'	60.1'
22	152°10.3	36.0	132°47.5	4.1'	22°43.0	-4.2'	60.0'
23	167°10.0	36.3	147°10.6	4.1'	22°38.8	-4.4'	60.0'
SD = 16.2'			SD = 16.5'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	182°09.8	S22°36.5	161°33.7	4.2'	S22°34.4	-4.5'	60.0'
1	197°09.5	36.8	175°56.9	4.3'	22°29.9	-4.7'	60.0'
2	212°09.2	37.1	190°20.2	4.4'	22°25.2	-4.8'	59.9'
3	227°09.0	· · 37.4	204°43.6	4.5'	22°20.3	-5.0'	59.9'
4	242°08.7	37.7	219°07.1	4.6'	22°15.3	-5.1'	59.9'
5	257°08.4	37.9	233°30.7	4.7'	22°10.2	-5.3'	59.8'
6	272°08.1	S22°38.2	247°54.3	4.7'	S22°04.9	-5.4'	59.8'
7	287°07.9	38.5	262°18.1	4.8'	21°59.4	-5.6'	59.8'
8	302°07.6	38.8	276°41.9	4.9'	21°53.9	-5.7'	59.7'
9	317°07.3	· · 39.0	291°05.9	5.0'	21°48.1	-5.9'	59.7'
10	332°07.1	39.3	305°29.9	5.1'	21°42.3	-6.0'	59.7'
11	347°06.8	39.6	319°54.0	5.2'	21°36.3	-6.1'	59.6'
12	2°06.5	S22°39.9	334°18.2	5.3'	S21°30.1	-6.3'	59.6'
13	17°06.2	40.1	348°42.6	5.4'	21°23.8	-6.4'	59.6'
14	32°06.0	40.4	3°07.0	5.5'	21°17.4	-6.6'	59.5'
15	47°05.7	· · 40.7	17°31.5	5.6'	21°10.9	-6.7'	59.5'
16	62°05.4	40.9	31°56.2	5.7'	21°04.2	-6.8'	59.5'
17	77°05.2	41.2	46°20.9	5.8'	20°57.3	-6.9'	59.4'
18	92°04.9	S22°41.5	60°45.8	5.9'	S20°50.4	-7.1'	59.4'
19	107°04.6	41.7	75°10.7	6.1'	20°43.3	-7.2'	59.4'
20	122°04.3	42.0	89°35.7	6.2'	20°36.1	-7.3'	59.3'
21	137°04.1	· · 42.3	104°00.9	6.3'	20°28.8	-7.5'	59.3'
22	152°03.8	42.5	118°26.2	6.4'	20°21.3	-7.6'	59.3'
23	167°03.5	42.8	132°51.5	6.5'	20°13.8	-7.7'	59.2'
SD = 16.2'			SD = 16.4'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	182°03.2	S22°43.1	147°17.0	6.6'	S20°06.1	-7.8'	59.2'
1	197°03.0	43.3	161°42.6	6.7'	19°58.3	-7.9'	59.2'
2	212°02.7	43.6	176°08.3	6.8'	19°50.3	-8.0'	59.1'
3	227°02.4	· · 43.8	190°34.1	6.9'	19°42.3	-8.2'	59.1'
4	242°02.1	44.1	205°00.1	7.0'	19°34.2	-8.3'	59.0'
5	257°01.9	44.3	219°26.1	7.1'	19°25.9	-8.4'	59.0'
6	272°01.6	S22°44.6	233°52.2	7.3'	S19°17.5	-8.5'	59.0'
7	287°01.3	44.9	248°18.5	7.4'	19°09.0	-8.6'	58.9'
8	302°01.0	45.1	262°44.8	7.5'	19°00.4	-8.7'	58.9'
9	317°00.8	· · 45.4	277°11.3	7.6'	18°51.7	-8.8'	58.9'
10	332°00.5	45.6	291°37.9	7.7'	18°43.0	-8.9'	58.8'
11	347°00.2	45.9	306°04.6	7.8'	18°34.1	-9.0'	58.8'
12	1°59.9	S22°46.1	320°31.4	7.9'	S18°25.1	-9.1'	58.7'
13	16°59.6	46.4	334°58.3	8.0'	18°16.0	-9.2'	58.7'
14	31°59.4	46.6	349°25.4	8.1'	18°06.8	-9.3'	58.7'
15	46°59.1	· · 46.9	3°52.5	8.3'	17°57.5	-9.4'	58.6'
16	61°58.8	47.1	18°19.8	8.4'	17°48.1	-9.5'	58.6'
17	76°58.5	47.4	32°47.2	8.5'	17°38.6	-9.6'	58.5'
18	91°58.3	S22°47.6	47°14.6	8.6'	S17°29.1	-9.6'	58.5'
19	106°58.0	47.9	61°42.2	8.7'	17°19.4	-9.7'	58.5'
20	121°57.7	48.1	76°09.9	8.8'	17°09.7	-9.8'	58.4'
21	136°57.4	· · 48.4	90°37.7	8.9'	16°59.9	-9.9'	58.4'
22	151°57.1	48.6	105°05.7	9.0'	16°50.0	-10.0'	58.4'
23	166°56.9	48.9	119°33.7	9.1'	16°40.0	-10.1'	58.3'
SD = 16.2'			SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:06	10:18	■	■	13:24	15:36
N 70°	07:48	09:30	■	■	14:13	15:54
68°	07:34	08:59	11:22	12:21	14:43	16:09
66°	07:22	08:36	10:08	13:35	15:06	16:21
64°	07:11	08:18	09:32	14:11	15:25	16:31
62°	07:02	08:03	09:06	14:36	15:40	16:40
60°	06:55	07:50	08:46	14:56	15:52	16:48
N 58°	06:48	07:39	08:30	15:13	16:03	16:55
56°	06:41	07:30	08:16	15:27	16:13	17:02
54°	06:35	07:21	08:04	15:39	16:22	17:07
52°	06:30	07:13	07:54	15:49	16:29	17:13
50°	06:25	07:06	07:44	15:59	16:37	17:18
45°	06:14	06:51	07:25	16:18	16:52	17:29
N 40°	06:04	06:38	07:08	16:35	17:05	17:39
35°	05:55	06:27	06:55	16:48	17:16	17:48
30°	05:47	06:17	06:43	17:00	17:26	17:56
20°	05:31	05:59	06:23	17:21	17:44	18:12
N 10°	05:16	05:42	06:05	17:38	18:01	18:27
0°	04:59	05:25	05:48	17:55	18:18	18:44
S 10°	04:41	05:08	05:31	18:12	18:35	19:02
20°	04:19	04:48	05:13	18:31	18:55	19:24
30°	03:51	04:24	04:51	18:52	19:19	19:52
35°	03:33	04:10	04:39	19:04	19:34	20:10
40°	03:11	03:52	04:24	19:19	19:51	20:32
45°	02:42	03:30	04:07	19:36	20:13	21:02
S 50°	02:00	03:02	03:46	19:58	20:41	21:44
52°	01:35	02:48	03:35	20:08	20:56	22:09
54°	00:59	02:31	03:24	20:20	21:13	22:46
56°	—:—	02:11	03:10	20:33	21:33	—:—
58°	—:—	01:45	02:55	20:49	22:00	—:—
S 60°	—:—	01:06	02:36	21:08	22:40	—:—

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	13:04	■	■	16:39
68°	■	12:57	12:23	■	14:48	17:20
66°	11:51	11:55	11:54	13:45	15:48	17:48
64°	10:53	11:20	11:32	14:42	16:23	18:09
62°	10:19	10:55	11:14	15:16	16:47	18:26
60°	09:54	10:35	10:59	15:41	17:07	18:40
N 58°	09:35	10:18	10:47	16:00	17:23	18:52
56°	09:18	10:04	10:36	16:16	17:37	19:02
54°	09:04	09:52	10:26	16:30	17:49	19:11
52°	08:52	09:41	10:18	16:42	17:59	19:19
50°	08:41	09:32	10:10	16:53	18:08	19:26
45°	08:18	09:11	09:53	17:15	18:28	19:42
N 40°	08:00	08:55	09:40	17:33	18:44	19:54
35°	07:45	08:41	09:28	17:48	18:57	20:05
30°	07:31	08:29	09:18	18:01	19:08	20:14
20°	07:09	08:08	09:01	18:24	19:28	20:30
N 10°	06:49	07:50	08:45	18:43	19:45	20:44
0°	06:31	07:33	08:31	19:01	20:01	20:57
S 10°	06:12	07:16	08:16	19:19	20:17	21:10
20°	05:53	06:58	08:01	19:38	20:34	21:23
30°	05:30	06:37	07:43	19:59	20:53	21:39
35°	05:17	06:24	07:33	20:12	21:04	21:47
40°	05:02	06:10	07:21	20:27	21:17	21:58
45°	04:44	05:53	07:07	20:44	21:32	22:09
S 50°	04:21	05:33	06:50	21:05	21:50	22:24
52°	04:10	05:23	06:41	21:16	21:58	22:30
54°	03:58	05:11	06:32	21:27	22:08	22:38
56°	03:44	04:59	06:22	21:40	22:19	22:46
58°	03:28	04:44	06:10	21:55	22:31	22:55
S 60°	03:08	04:26	05:57	22:13	22:45	23:05

Day	Sun		Mer. Pass hh:mm	Moon		Age 1-3 0-8%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
06	09:05	08:52	11:51	12:46	00:14	
07	08:39	08:26	11:52	13:47	01:17	
08	08:13	08:00	11:52	14:44	02:16	

Aries Venus Mars

<https://pypi.org/project/sfalmanac/>

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
0	181°56.6	S22°49.1	134°01.8	9.2'	S16°29.9	-10.1'	58.3'						
1	196°56.3	49.3	148°30.1	9.4'	16°19.8	-10.2'	58.2'						
2	211°56.0	49.6	162°58.4	9.5'	16°09.5	-10.3'	58.2'						
3	226°55.7	· · 49.8	177°26.9	9.6'	15°59.2	-10.4'	58.2'						
4	241°55.5	50.1	191°55.5	9.7'	15°48.9	-10.4'	58.1'						
5	256°55.2	50.3	206°24.1	9.8'	15°38.4	-10.5'	58.1'						
6	271°54.9	S22°50.5	220°52.9	9.9'	S15°27.9	-10.6'	58.0'						
7	286°54.6	50.8	235°21.8	10.0'	15°17.3	-10.7'	58.0'						
8	301°54.3	51.0	249°50.8	10.1'	15°06.7	-10.7'	58.0'						
9	316°54.1	· · 51.3	264°19.9	10.2'	14°56.0	-10.8'	57.9'						
10	331°53.8	51.5	278°49.1	10.3'	14°45.2	-10.8'	57.9'						
11	346°53.5	51.7	293°18.4	10.4'	14°34.3	-10.9'	57.8'						
12	1°53.2	S22°52.0	307°47.8	10.5'	S14°23.4	-11.0'	57.8'						
13	16°52.9	52.2	322°17.3	10.6'	14°12.5	-11.0'	57.8'						
14	31°52.7	52.4	336°47.0	10.7'	14°01.4	-11.1'	57.7'						
15	46°52.4	· · 52.7	351°16.7	10.8'	13°50.3	-11.1'	57.7'						
16	61°52.1	52.9	5°46.5	10.9'	13°39.2	-11.2'	57.6'						
17	76°51.8	53.1	20°16.4	11.0'	13°28.0	-11.2'	57.6'						
18	91°51.5	S22°53.4	34°46.4	11.1'	S13°16.8	-11.3'	57.6'						
19	106°51.2	53.6	49°16.5	11.2'	13°05.5	-11.4'	57.5'						
20	121°51.0	53.8	63°46.7	11.3'	12°54.1	-11.4'	57.5'						
21	136°50.7	· · 54.0	78°17.0	11.4'	12°42.7	-11.4'	57.4'						
22	151°50.4	54.3	92°47.4	11.5'	12°31.3	-11.5'	57.4'						
23	166°50.1	54.5	107°17.9	11.6'	12°19.8	-11.5'	57.4'						
SD = 16.2' $d = 0.2'$							SD = 15.9'						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	181°49.8	S22°54.7	121°48.4	11.7'	S12°08.2	-11.6'	57.3'						
1	196°49.5	54.9	136°19.1	11.8'	11°56.6	-11.6'	57.3'						
2	211°49.3	55.2	150°49.9	11.8'	11°45.0	-11.7'	57.2'						
3	226°49.0	· · 55.4	165°20.7	11.9'	11°33.3	-11.7'	57.2'						
4	241°48.7	55.6	179°51.7	12.0'	11°21.6	-11.8'	57.2'						
5	256°48.4	55.8	194°22.7	12.1'	11°09.9	-11.8'	57.1'						
6	271°48.1	S22°56.0	208°53.8	12.2'	S10°58.1	-11.8'	57.1'						
7	286°47.8	56.3	223°25.0	12.3'	10°46.3	-11.9'	57.1'						
8	301°47.6	56.5	237°56.3	12.4'	10°34.4	-11.9'	57.0'						
9	316°47.3	· · 56.7	252°27.6	12.4'	10°22.5	-11.9'	57.0'						
10	331°47.0	56.9	266°59.1	12.5'	10°10.6	-12.0'	56.9'						
11	346°46.7	57.1	281°30.6	12.6'	09°58.6	-12.0'	56.9'						
12	1°46.4	S22°57.3	296°02.2	12.7'	S09°46.6	-12.0'	56.9'						
13	16°46.1	57.6	310°33.9	12.8'	09°34.6	-12.0'	56.8'						
14	31°45.8	57.8	325°05.7	12.8'	09°22.6	-12.1'	56.8'						
15	46°45.6	· · 58.0	339°37.5	12.9'	09°10.5	-12.1'	56.8'						
16	61°45.3	58.2	354°09.5	13.0'	08°58.4	-12.1'	56.7'						
17	76°45.0	58.4	8°41.5	13.1'	08°46.3	-12.2'	56.7'						
18	91°44.7	S22°58.6	23°13.5	13.1'	S08°34.1	-12.2'	56.6'						
19	106°44.4	58.8	37°45.7	13.2'	08°21.9	-12.2'	56.6'						
20	121°44.1	59.0	52°17.9	13.3'	08°09.7	-12.2'	56.6'						
21	136°43.8	· · 59.2	66°50.2	13.4'	07°57.5	-12.2'	56.5'						
22	151°43.6	59.5	81°22.6	13.4'	07°45.3	-12.3'	56.5'						
23	166°43.3	59.7	95°55.0	13.5'	07°33.0	-12.3'	56.5'						
SD = 16.2' $d = 0.2'$							SD = 15.6'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	181°43.0	S22°59.9	110°27.5	13.6'	S07°20.7	-12.3'	56.4'						
1	196°42.7	23°00.1	125°00.1	13.6'	07°08.5	-12.3'	56.4'						
2	211°42.4	00.3	139°32.7	13.7'	06°56.2	-12.3'	56.4'						
3	226°42.1	· · 00.5	154°05.4	13.8'	06°43.8	-12.3'	56.3'						
4	241°41.8	00.7	168°38.1	13.8'	06°31.5	-12.3'	56.3'						
5	256°41.5	00.9	183°11.0	13.9'	06°19.1	-12.4'	56.3'						
6	271°41.3	S23°01.1	197°43.8	13.9'	S06°06.8	-12.4'	56.2'						
7	286°41.0	01.3	212°16.8	14.0'	05°54.4	-12.4'	56.2'						
8	301°40.7	01.5	226°49.8	14.1'	05°42.0	-12.4'	56.2'						
9	316°40.4	· · 01.7	241°22.9	14.1'	05°29.6	-12.4'	56.1'						
10	331°40.1	01.9	255°56.0	14.2'	05°17.2	-12.4'	56.1'						
11	346°39.8	02.1	270°29.1	14.2'	05°04.8	-12.4'	56.1'						
12	1°39.5	S23°02.3	285°02.4	14.3'	S04°52.4	-12.4'	56.0'						
13	16°39.2	02.5	299°35.7	14.3'	04°40.0	-12.4'	56.0'						
14	31°38.9	02.7	314°09.0	14.4'	04°27.6	-12.4'	56.0'						
15	46°38.6	· · 02.9	328°42.4	14.4'	04°15.1	-12.4'	55.9'						
16	61°38.4	03.0	343°15.8	14.5'	04°02.7	-12.4'	55.9'						
17	76°38.1	03.2	357°49.3	14.5'	03°50.3	-12.4'	55.9'						
18	91°37.8	S23°03.4	12°22.8	14.6'	S03°37.8	-12.4'	55.8'						
19	106°37.5	03.6	26°56.4	14.6'	03°25.4	-12.4'	55.8'						
20	121°37.2	03.8	41°30.0	14.7'	03°13.0	-12.4'	55.8'						
21	136°36.9	· · 04.0	56°03.7	14.7'	03°00.5	-12.4'	55.7'						
22	151°36.6	04.2	70°37.4	14.8'	02°48.1	-12.4'	55.7'						
23	166°36.3	04.4	85°11.2	14.8'	02°35.6	-12.4'	55.7'						
SD = 16.2' $d = 0.2'$							SD = 15.4'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:12	10:30	■	■	13:15	15:33
N 70°	07:54	09:38	■	■	14:08	15:52
68°	07:39	09:05	■	■	14:40	16:07
66°	07:26	08:42	10:16	13:29	15:04	16:19
64°	07:15	08:23	09:38	14:07	15:23	16:30
62°	07:06	08:08	09:12	14:34	15:38	16:39
60°	06:58	07:54	08:51	14:54	15:51	16:47
N 58°	06:51	07:43	08:34	15:11	16:02	16:55
56°	06:45	07:33	08:20	15:25	16:12	17:01
54°	06:39	07:25	08:08	15:38	16:21	17:07
52°	06:33	07:17	07:57	15:48	16:29	17:12
50°	06:28	07:09	07:48	15:58	16:36	17:18
45°	06:17	06:54	07:27	16:18	16:52	17:29
N 40°	06:07	06:41	07:11	16:35	17:05	17:39
35°	05:58	06:29	06:57	16:49	17:17	17:48
30°	05:49	06:19	06:45	17:01	17:27	17:57
20°	05:33	06:00	06:24	17:21	17:45	18:13
N 10°	05:17	05:43	06:06	17:40	18:02	18:29
0°	05:01	05:27	05:49	17:57	18:19	18:45
S 10°	04:42	05:09	05:32	18:14	18:37	19:04
20°	04:20	04:49	05:13	18:32	18:57	19:26
30°	03:51	04:25	04:52	18:54	19:21	19:55
35°	03:33	04:10	04:39	19:07	19:36	20:13
40°	03:11	03:52	04:25	19:21	19:54	20:35
45°	02:41	03:30	04:07	19:39	20:16	21:05
S 50°	01:58	03:01	03:45	20:01	20:45	21:49
52°	01:32	02:47	03:35	20:12	21:00	22:15
54°	00:53	02:29	03:23	20:24	21:17	22:55
56°	—:—	02:08	03:09	20:37	21:38	—:—
58°	—:—	01:41	02:53	20:54	22:06	—:—
S 60°	—:—	00:58	02:33	21:13	22:50	—:—

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	12:58	12:25	12:02	18:35	20:48	22:45
N 70°	12:30	12:10	11:55	19:02	21:01	22:49
68°	12:08	11:58	11:50	19:22	21:11	22:52
66°	11:51	11:48	11:45	19:38	21:19	22:55
64°	11:37	11:40	11:41	19:51	21:27	22:57
62°	11:25	11:32	11:38	20:02	21:33	22:59
60°	11:15	11:26	11:35	20:11	21:38	23:00
N 58°	11:06	11:21	11:32	20:19	21:42	23:02
56°	10:58	11:16	11:29	20:26	21:46	23:03
54°	10:52	11:11	11:27	20:32	21:50	23:04
52°	10:45	11:07	11:25	20:38	21:53	23:06
50°	10:40	11:03	11:23	20:43	21:56	23:07
45°	10:27	10:55	11:19	20:54	22:03	23:09
N 40°	10:17	10:48	11:16	21:03	22:08	23:10
35°	10:08	10:42	11:13	21:10	22:13	23:12
30°	10:00	10:37	11:11	21:17	22:17	23:13
20°	09:47	10:28	11:06	21:29	22:23	23:16
N 10°	09:35	10:20	11:02	21:39	22:29	23:18
0°	09:24	10:13	10:58	21:48	22:35	23:19
S 10°	09:13	10:06	10:55	21:57	22:41	23:21
20°	09:01	09:58	10:51	22:07	22:47	23:23
30°	08:47	09:48	10:46	22:18	22:53	23:25
35°	08:39	09:43	10:44	22:25	22:57	23:27
40°	08:30	09:37	10:41	22:32	23:01	23:28
45°	08:20	09:30	10:37	22:40	23:06	23:29
S 50°	08:07	09:22	10:33	22:50	23:12	23:31
52°	08:01	09:18	10:31	22:55	23:15	23:32
54°	07:54	09:13	10:29	23:00	23:18	23:33
56°	07:47	09:09	10:27	23:06	23:21	23:34
58°	07:38	09:03	10:24	23:12	23:25	23:35
S 60°	07:28	08:57	10:21	23:19	23:29	23:37

December 12, 13, 14 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars				
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	80°59.0	145°19.2	S22°49.9	139°31.0	S21°35.3	216°21.8	S15°54.6	33°22.3	N15°15.1	Alpheratz	357°30.5	29°15.6		
1	96°01.4	160°21.2	49.4	154°31.4	34.9	231°23.7	54.8	48°24.9	15.0	Ankaa	353°03.3	-42°08.7		
2	111°03.9	175°23.2	48.9	169°31.8	34.5	246°25.7	54.9	63°27.6	15.0	Schedar	349°26.1	56°42.5		
3	126°06.4	190°25.3	· · 48.4	184°32.3	· · 34.1	261°27.6	· · 55.1	78°30.2	· · 15.0	Diphda	348°43.3	-17°49.3		
4	141°08.8	205°27.3	47.9	199°32.7	33.6	276°29.6	55.2	93°32.8	14.9	Achernar	335°17.1	-57°05.3		
5	156°11.3	220°29.4	47.4	214°33.1	33.2	291°31.6	55.3	108°35.4	14.9	Hamal	327°46.4	23°36.4		
6	171°13.8	235°31.4	S22°46.9	229°33.5	S21°32.8	306°33.5	S15°55.5	123°38.1	N15°14.9	Polaris	311°22.0	89°23.4		
7	186°16.2	250°33.5	46.4	244°33.9	32.4	321°35.5	55.6	138°40.7	14.8	Acamar	315°08.5	-40°11.2		
8	201°18.7	265°35.5	45.9	259°34.4	32.0	336°37.5	55.8	153°43.3	14.8	Menkar	314°01.7	4°12.4		
9	216°21.2	280°37.6	· · 45.4	274°34.8	· · 31.6	351°39.4	· · 55.9	168°45.9	· · 14.8	Mirfak	308°22.0	49°58.1		
10	231°23.6	295°39.6	44.8	289°35.2	31.1	6°41.4	56.0	183°48.5	14.7	Aldebaran	290°34.7	16°34.1		
11	246°26.1	310°41.7	44.3	304°35.6	30.7	21°43.4	56.2	198°51.2	14.7	Rigel	280°59.7	-8°10.1		
12	261°28.5	325°43.8	S22°43.8	319°36.0	S21°30.3	36°45.3	S15°56.3	213°53.8	N15°14.7	Capella	280°15.5	46°01.5		
13	276°31.0	340°45.9	43.3	334°36.5	29.9	51°47.3	56.5	228°56.4	14.6	Bellatrix	278°18.3	6°22.5		
14	291°33.5	355°47.9	42.8	349°36.9	29.5	66°49.3	56.6	243°59.0	14.6	Elnath	277°56.5	28°37.8		
15	306°35.9	10°50.0	· · 42.3	4°37.3	· · 29.0	81°51.2	· · 56.7	259°01.7	· · 14.5	Alnilam	275°33.4	-1°11.1		
16	321°38.4	25°52.1	41.8	19°37.7	28.6	96°53.2	56.9	274°04.3	14.5	Betelgeuse	270°47.5	7°24.6		
17	336°40.9	40°54.2	41.3	34°38.1	28.2	111°55.2	57.0	289°06.9	14.5	Canopus	263°50.1	-52°42.7		
18	351°43.3	55°56.3	S22°40.8	49°38.6	S21°27.8	126°57.1	S15°57.2	304°09.5	N15°14.4	Sirius	258°22.4	-16°45.6		
19	6°45.8	70°58.4	40.3	64°39.0	27.4	141°59.1	57.3	319°12.2	14.4	Adhara	255°02.4	-29°00.8		
20	21°48.3	86°00.5	39.8	79°39.4	26.9	157°01.1	57.4	334°14.8	14.4	Procyon	244°46.4	5°08.7		
21	36°50.7	101°02.6	· · 39.3	94°39.8	· · 26.5	172°03.0	· · 57.6	349°17.4	· · 14.3	Pollux	243°12.2	27°56.9		
22	51°53.2	116°04.8	38.8	109°40.2	26.1	187°05.0	57.7	4°20.0	14.3	Avior	234°12.5	-59°36.3		
23	66°55.7	131°06.9	38.3	124°40.7	25.7	202°07.0	57.9	19°22.6	14.3	Suhail	222°43.0	-43°33.2		
Mer.pass. 18:33		ν 2.0' d -0.5' m -4.89		ν 0.4' d -0.4' m 1.13		ν 2.0' d 0.1' m -1.74		ν 2.6' d -0.0' m -0.13		Miaplacidus	221°36.6	-69°50.3		
Alphard		217°43.7		-8°47.4		Regulus		207°30.2		11°49.0		Dubhe	193°36.4	61°34.9
Denebola		182°21.0		14°24.1		Gienah		175°39.5		-17°42.5		Acrux	172°55.6	-63°15.7
Gacrux		171°47.2		-57°16.6		Alioth		166°10.0		55°47.5		Spica	158°18.3	-11°19.0
Alkaid		152°49.4		49°09.6		Hadar		148°30.6		-60°30.8		Menkent	147°53.1	-36°30.9
Arcturus		145°44.6		19°01.5		Rigel Kent.		139°35.1		-60°57.3		Kochab	137°21.3	74°01.7
Zuben'ubi		136°51.9		-16°09.9		Alphecca		126°00.8		26°36.8		Antares	112°11.3	-26°29.7
Atria		107°02.5		-69°04.6		Sabik		101°58.6		-15°45.5		Shaula	96°05.4	-37°07.3
Rasalhague		95°55.2		12°32.5		Eltanin		90°40.9		51°29.3		Kaus Aust.	83°27.7	-34°22.0
Vega		80°31.0		38°48.9		Nunki		75°43.2		-26°15.4		Altair	61°56.4	8°57.1
Peacock		53°00.1		-56°38.3		Deneb		49°23.3		45°23.6		Enif	33°35.1	10°01.0
Al Na'ir		27°28.3		-46°49.0		Fomalhaut		15°10.3		-29°27.8		Scheat	13°41.3	28°15.0
Markab		13°26.0		15°22.2										
Dec 12 Wed		SHA		Mer.pass										
Venus		64°20.2		14:17										
Mars		58°32.0		14:42										
Jupiter		135°22.8		09:33										
Saturn		312°23.3		21:43										
Dec 13 Thu		SHA		Mer.pass										
Venus		64°10.9		14:13										
Mars		57°43.0		14:41										
Jupiter		135°10.8		09:30										
Saturn		312°27.1		21:39										
Dec 14 Fri		SHA		Mer.pass										
Venus		64°04.1		14:10										
Mars		56°54.0		14:40										
Jupiter		134°58.9		09:27										
Saturn		312°30.9		21:34										
Horizontal parallax														
Venus:				0.4										
Mars:				0.1										

Sun							Moon							
Wed	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
									Naut.	Civil			Civil	Naut.
0	181°36.0	S23°04.6	99°45.0	14.8'	S02°23.2	-12.4'	55.6'	N 72°	08:18	10:41	■	■	13:07	15:31
1	196°35.7	04.7	114°18.8	14.9'	02°10.8	-12.4'	55.6'	N 70°	07:58	09:44	■	■	14:04	15:50
2	211°35.5	04.9	128°52.7	14.9'	01°58.4	-12.4'	55.6'	68°	07:43	09:11	■	■	14:38	16:06
3	226°35.2	· · 05.1	143°26.6	15.0'	01°46.0	-12.4'	55.5'	66°	07:30	08:46	10:24	13:25	15:02	16:19
4	241°34.9	05.3	158°00.6	15.0'	01°33.5	-12.4'	55.5'	64°	07:19	08:27	09:44	14:05	15:21	16:29
5	256°34.6	05.5	172°34.6	15.0'	01°21.1	-12.4'	55.5'	62°	07:10	08:11	09:16	14:32	15:37	16:39
6	271°34.3	S23°05.7	187°08.6	15.1'	S01°08.7	-12.4'	55.5'	60°	07:01	07:58	08:55	14:53	15:50	16:47
7	286°34.0	05.8	201°42.7	15.1'	00°56.4	-12.4'	55.4'	N 58°	06:54	07:46	08:38	15:10	16:02	16:54
8	301°33.7	06.0	216°16.7	15.1'	00°44.0	-12.4'	55.4'	56°	06:47	07:36	08:24	15:25	16:12	17:01
9	316°33.4	· · 06.2	230°50.9	15.2'	00°31.6	-12.4'	55.4'	54°	06:41	07:27	08:11	15:37	16:21	17:07
10	331°33.1	06.4	245°25.0	15.2'	00°19.3	-12.3'	55.3'	52°	06:36	07:19	08:00	15:48	16:29	17:13
11	346°32.8	06.6	259°59.2	15.2'	S00°06.9	-12.3'	55.3'	50°	06:31	07:12	07:50	15:58	16:36	17:18
12	1°32.5	S23°06.7	274°33.4	15.2'	N00°05.4	12.3'	55.3'	45°	06:19	06:56	07:30	16:19	16:52	17:29
13	16°32.2	06.9	289°07.7	15.3'	00°17.8	12.3'	55.3'	N 40°	06:09	06:43	07:13	16:35	17:06	17:40
14	31°31.9	07.1	303°41.9	15.3'	00°30.1	12.3'	55.2'	35°	06:00	06:31	06:59	16:49	17:17	17:49
15	46°31.7	· · 07.3	318°16.2	15.3'	00°42.4	12.3'	55.2'	30°	05:51	06:21	06:47	17:01	17:28	17:58
16	61°31.4	07.4	332°50.5	15.3'	00°54.6	12.3'	55.2'	20°	05:35	06:02	06:26	17:22	17:46	18:14
17	76°31.1	07.6	347°24.9	15.4'	01°06.9	12.2'	55.2'	N 10°	05:19	05:45	06:08	17:41	18:04	18:30
18	91°30.8	S23°07.8	1°59.2	15.4'	N01°19.2	12.2'	55.1'	0°	05:02	05:28	05:51	17:58	18:21	18:47
19	106°30.5	08.0	16°33.6	15.4'	01°31.4	12.2'	55.1'	S 10°	04:43	05:10	05:33	18:15	18:38	19:06
20	121°30.2	08.1	31°08.0	15.4'	01°43.6	12.2'	55.1'	20°	04:21	04:50	05:14	18:34	18:59	19:28
21	136°29.9	· · 08.3	45°42.5	15.4'	01°55.8	12.2'	55.1'	30°	03:52	04:25	04:53	18:56	19:23	19:57
22	151°29.6	08.5	60°16.9	15.5'	02°08.0	12.2'	55.0'	35°	03:34	04:10	04:40	19:09	19:39	20:15
23	166°29.3	08.6	74°51.4	15.5'	02°20.1	12.1'	55.0'	40°	03:11	03:52	04:25	19:24	19:57	20:38
	SD = 16.2'	d = 0.2'			SD = 15.2'			45°	02:41	03:30	04:07	19:42	20:19	21:08
0	181°29.0	S23°08.8	89°25.8	15.5'	N02°32.3	12.1'	55.0'	S 50°	01:56	03:01	03:45	20:04	20:48	21:53
1	196°28.7	09.0	104°00.3	15.5'	02°44.4	12.1'	55.0'	52°	01:30	02:46	03:34	20:15	21:03	22:20
2	211°28.4	09.1	118°34.8	15.5'	02°56.5	12.1'	54.9'	54°	00:47	02:28	03:22	20:27	21:21	23:03
3	226°28.1	· · 09.3	133°09.4	15.5'	03°08.6	12.1'	54.9'	56°	--	02:07	03:08	20:41	21:43	--
4	241°27.8	09.5	147°43.9	15.5'	03°20.6	12.0'	54.9'	58°	--	01:38	02:52	20:57	22:11	--
5	256°27.5	09.6	162°18.4	15.6'	03°32.7	12.0'	54.9'	S 60°	--	00:52	02:32	21:17	22:58	--
6	271°27.2	S23°09.8	176°53.0	15.6'	N03°44.7	12.0'	54.9'							
7	286°26.9	10.0	191°27.5	15.6'	03°56.7	12.0'	54.8'							
8	301°26.6	10.1	206°02.1	15.6'	04°08.6	11.9'	54.8'							
9	316°26.4	· · 10.3	220°36.7	15.6'	04°20.6	11.9'	54.8'							
10	331°26.1	10.4	235°11.3	15.6'	04°32.5	11.9'	54.8'							
11	346°25.8	10.6	249°45.9	15.6'	04°44.3	11.9'	54.7'							
12	1°25.5	S23°10.7	264°20.4	15.6'	N04°56.2	11.8'	54.7'							
13	16°25.2	10.9	278°55.0	15.6'	05°08.0	11.8'	54.7'							
14	31°24.9	11.1	293°29.6	15.6'	05°19.8	11.8'	54.7'							
15	46°24.6	· · 11.2	308°04.2	15.6'	05°31.6	11.7'	54.7'							
16	61°24.3	11.4	322°38.8	15.6'	05°43.3	11.7'	54.6'							
17	76°24.0	11.5	337°13.4	15.6'	05°55.0	11.7'	54.6'							
18	91°23.7	S23°11.7	351°48.0	15.6'	N06°06.7	11.6'	54.6'							
19	106°23.4	11.8	6°22.6	15.6'	06°18.4	11.6'	54.6'							
20	121°23.1	12.0	20°57.2	15.6'	06°30.0	11.6'	54.6'							
21	136°22.8	· · 12.1	35°31.8	15.6'	06°41.6	11.5'	54.6'							
22	151°22.5	12.3	50°06.4	15.6'	06°53.1	11.5'	54.5'							
23	166°22.2	12.4	64°41.0	15.6'	07°04.6	11.5'	54.5'							
	SD = 16.2'	d = 0.2'			SD = 15.0'									
Fri	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Moonrise		Moonset			
									Wed	Thu	Fri	Wed	Thu	Fri
0	181°21.9	S23°12.6	79°15.5	15.6'	N07°16.1	11.4'	54.5'	N 72°	11:42	11:23	11:01	· · ·	00:35	02:24
1	196°21.6	12.7	93°50.1	15.6'	07°27.6	11.4'	54.5'	N 70°	11:42	11:29	11:15	· · ·	00:31	02:13
2	211°21.3	12.9	108°24.7	15.5'	07°39.0	11.4'	54.5'	68°	11:42	11:34	11:27	· · ·	00:28	02:03
3	226°21.0	· · 13.0	122°59.2	15.5'	07°50.3	11.3'	54.5'	66°	11:42	11:39	11:36	· · ·	00:26	01:55
4	241°20.7	13.2	137°33.7	15.5'	08°01.7	11.3'	54.4'	64°	11:42	11:43	11:44	· · ·	00:24	01:49
5	256°20.4	13.3	152°08.3	15.5'	08°13.0	11.3'	54.4'	62°	11:42	11:46	11:51	· · ·	00:22	01:43
6	271°20.1	S23°13.5	166°42.8	15.5'	N08°24.2	11.2'	54.4'	60°	11:42	11:49	11:57	· · ·	00:20	01:38
7	286°19.8	13.6	181°17.3	15.5'	08°35.5	11.2'	54.4'	N 58°	11:42	11:52	12:02	· · ·	00:19	01:34
8	301°19.5	13.7	195°51.7	15.5'	08°46.7	11.1'	54.4'	56°	11:42	11:54	12:07	· · ·	00:17	01:30
9	316°19.2	· · 13.9	210°26.2	15.5'	08°57.8	11.1'	54.4'	54°	11:42	11:56	12:11	· · ·	00:16	01:27
10	331°18.9	14.0	225°00.7	15.4'	09°08.9	11.1'	54.3'	52°	11:42	11:58	12:15	· · ·	00:15	01:24
11	346°18.6	14.2	239°35.1	15.4'	09°20.0	11.0'	54.3'	50°	11:42	12:00	12:18	· · ·	00:14	01:21
12	1°18.3	S23°14.3	254°09.5	15.4'	N09°31.0	11.0'	54.3'	45°	11:42	12:03	12:26	· · ·	00:12	01:15
13	16°18.0	14.4	268°43.9	15.4'	09°42.0	10.9'	54.3'	N 40°	11:42	12:07	12:32	· · ·	00:11	01:10
14	31°17.7	14.6	283°18.3	15.4'	09°52.9	10.9'	54.3'	35°	11:42	12:09	12:38	· · ·	00:09	01:05
15	46°17.4	· · 14.7	297°52.7	15.3'	10°03.8	10.9'	54.3'	30°	11:42	12:12	12:42	· · ·	00:08	01:02
16	61°17.1	14.8	312°27.0	15.3'	10°14.6	10.8'	54.3'	20°	11:42	12:16	12:51	· · ·	00:06	00:55
17	76°16.8	15.0	327°01.3	15.3'	10°25.5	10.8'	54.3'	N 10°	11:42	12:20	12:59	· · ·	00:04	00:49
18	91°16.5	S23°15.1	341°35.6	15.3'	N10°36.2	10.7'	54.2'	0°	11:42	12:24	13:06	· · ·	00:02	00:44
19	106°16.2	15.2	356°09.9	15.3'	10°46.9	10.7								

December 15, 16, 17 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars				
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	83°56.4	147°56.2	S22°13.3	140°01.5	S21°04.4	218°43.4	S16°04.6	36°31.0	N15°12.7	Alpheratz	357°30.5	29°15.6		
1	98°58.9	162°58.6	12.8	155°01.9	04.0	233°45.4	04.7	51°33.6	12.7	Ankaa	353°03.3	-42°08.7		
2	114°01.3	178°00.9	12.3	170°02.4	03.5	248°47.4	04.9	66°36.2	12.6	Schedar	349°26.1	56°42.5		
3	129°03.8	193°03.3	· · 11.8	185°02.8	· · 03.1	263°49.3	· · 05.0	81°38.8	· · 12.6	Diphda	348°43.3	-17°49.3		
4	144°06.3	208°05.6	11.2	200°03.2	02.6	278°51.3	05.1	96°41.4	12.6	Achernar	335°17.2	-57°05.3		
5	159°08.7	223°08.0	10.7	215°03.7	02.2	293°53.3	05.3	111°44.0	12.5	Hamal	327°46.4	23°36.4		
6	174°11.2	238°10.3	S22°10.2	230°04.1	S21°01.8	308°55.3	S16°05.4	126°46.6	N15°12.5	Polaris	311°22.6	89°23.4		
7	189°13.6	253°12.7	09.7	245°04.5	01.3	323°57.2	05.6	141°49.3	12.5	Acamar	315°08.5	-40°11.2		
8	204°16.1	268°15.1	09.2	260°05.0	00.9	338°59.2	05.7	156°51.9	12.4	Menkar	314°01.7	4°12.4		
9	219°18.6	283°17.5	· · 08.7	275°05.4	· · 00.4	354°01.2	· · 05.8	171°54.5	· · 12.4	Mirfak	308°22.0	49°58.1		
10	234°21.0	298°19.9	08.2	290°05.8	21°00.0	9°03.2	06.0	186°57.1	12.4	Aldebaran	290°34.7	16°34.1		
11	249°23.5	313°22.3	07.7	305°06.2	20°59.5	24°05.1	06.1	201°59.7	12.3	Rigel	280°59.7	-8°10.1		
12	264°26.0	328°24.6	S22°07.2	320°06.7	S20°59.1	39°07.1	S16°06.2	217°02.3	N15°12.3	Capella	280°15.5	46°01.5		
13	279°28.4	343°27.0	06.7	335°07.1	58.6	54°09.1	06.4	232°04.9	12.3	Bellatrix	278°18.3	6°22.5		
14	294°30.9	358°29.4	06.1	350°07.5	58.2	69°11.0	06.5	247°07.6	12.2	Elnath	277°56.5	28°37.8		
15	309°33.4	13°31.9	· · 05.6	5°08.0	· · 57.7	84°13.0	· · 06.6	262°10.2	· · 12.2	Alnilam	275°33.4	-1°11.1		
16	324°35.8	28°34.3	05.1	20°08.4	57.3	99°15.0	06.8	277°12.8	12.2	Betelgeuse	270°47.5	7°24.6		
17	339°38.3	43°36.7	04.6	35°08.8	56.8	114°17.0	06.9	292°15.4	12.2	Canopus	263°50.1	-52°42.8		
18	354°40.7	58°39.1	S22°04.1	50°09.3	S20°56.4	129°18.9	S16°07.0	307°18.0	N15°12.1	Sirius	258°22.4	-16°45.6		
19	9°43.2	73°41.5	03.6	65°09.7	55.9	144°20.9	07.2	322°20.6	12.1	Adhara	255°02.4	-29°00.9		
20	24°45.7	88°44.0	03.1	80°10.1	55.5	159°22.9	07.3	337°23.2	12.1	Procyon	244°46.4	5°08.7		
21	39°48.1	103°46.4	· · 02.6	95°10.6	· · 55.0	174°24.9	· · 07.4	352°25.8	· · 12.0	Pollux	243°12.2	27°56.9		
22	54°50.6	118°48.8	02.1	110°11.0	54.6	189°26.8	07.6	7°28.5	12.0	Avior	234°12.5	-59°36.3		
23	69°53.1	133°51.3	01.5	125°11.4	54.1	204°28.8	07.7	22°31.1	12.0	Suhail	222°43.0	-43°33.2		
Mer.pass. 18:21		ν 2.3' d -0.5' m -4.86		ν 0.4' d -0.4' m 1.13		ν 2.0' d 0.1' m -1.74		ν 2.6' d -0.0' m -0.11		Miaplacidus	221°36.6	-69°50.3		
Alphard		217°43.7		-8°47.4		Regulus		207°30.1		11°49.0		Dubhe	193°36.4	61°34.9
Denebola		182°21.0		24°24.1		Gienah		175°39.5		-17°42.5		Acrux	172°55.5	-63°15.7
Gacrux		171°47.1		-57°16.6		Alioth		166°09.9		55°47.5		Spica	158°18.3	-11°19.0
Alkaid		152°49.3		49°09.6		Hadar		148°30.5		-60°30.8		Menkent	147°53.1	-36°30.9
Arcturus		145°44.6		19°01.5		Rigel Kent.		139°35.1		-60°57.3		Kochab	137°21.2	74°01.7
Zuben'ubi		136°51.9		-16°09.9		Alphecca		126°00.8		26°36.8		Antares	112°11.3	-26°29.7
Atria		107°02.4		-69°04.6		Sabik		101°58.6		-15°45.5		Shaula	96°05.4	-37°07.3
Rasalhague		95°55.2		12°32.4		Eltanin		90°40.9		51°29.2		Kaus Aust.	83°27.7	-34°22.0
Vega		80°31.0		38°48.9		Nunki		75°43.2		-26°15.4		Altair	61°56.4	8°57.1
Peacock		53°00.1		-56°38.3		Deneb		49°23.4		45°23.6		Enif	33°35.1	10°01.0
Al Na'ir		27°28.3		-46°49.0		Fomalhaut		15°10.3		-29°27.8		Scheat	13°41.3	28°15.0
Markab		13°26.0		15°22.2										
Dec 15 Sat		SHA		Mer.pass										
Venus		63°59.8		14:06										
Mars		56°05.1		14:39										
Jupiter		134°47.0		09:24										
Saturn		312°34.6		21:30										
Dec 16 Sun		SHA		Mer.pass										
Venus		63°58.2		14:02										
Mars		55°16.3		14:39										
Jupiter		134°35.2		09:21										
Saturn		312°38.2		21:26										
Dec 17 Mon		SHA		Mer.pass										
Venus		63°59.2		13:58										
Mars		54°27.6		14:38										
Jupiter		134°23.5		09:18										
Saturn		312°41.7		21:22										
Horizontal parallax		Venus:		0.5										
		Mars:		0.1										

Sun								Moon							
Sat	GHA	Dec	GHA	ν	Dec	d	HP								
0	181°14.7	S23°15.9	69°00.9	15.1'	N11°39.8	10.4'	54.2'								
1	196°14.4	16.0	83°35.1	15.1'	11°50.2	10.4'	54.2'								
2	211°14.1	16.2	98°09.2	15.1'	12°00.6	10.3'	54.2'								
3	226°13.8	· · 16.3	112°43.2	15.0'	12°10.9	10.3'	54.2'								
4	241°13.5	16.4	127°17.3	15.0'	12°21.2	10.2'	54.1'								
5	256°13.2	16.5	141°51.3	15.0'	12°31.4	10.2'	54.1'								
6	271°12.9	S23°16.7	156°25.3	14.9'	N12°41.6	10.1'	54.1'								
7	286°12.6	16.8	170°59.2	14.9'	12°51.7	10.1'	54.1'								
8	301°12.3	16.9	185°33.1	14.9'	13°01.8	10.0'	54.1'								
9	316°12.0	· · 17.0	200°07.0	14.8'	13°11.8	10.0'	54.1'								
10	331°11.7	17.1	214°40.8	14.8'	13°21.8	9.9'	54.1'								
11	346°11.4	17.3	229°14.7	14.8'	13°31.7	9.8'	54.1'								
12	1°11.1	S23°17.4	243°48.4	14.7'	N13°41.5	9.8'	54.1'								
13	16°10.8	17.5	258°22.2	14.7'	13°51.3	9.7'	54.1'								
14	31°10.5	17.6	272°55.9	14.7'	14°01.0	9.7'	54.1'								
15	46°10.2	· · 17.7	287°29.6	14.6'	14°10.7	9.6'	54.1'								
16	61°09.9	17.9	302°03.2	14.6'	14°20.3	9.6'	54.1'								
17	76°09.6	18.0	316°36.8	14.6'	14°29.9	9.5'	54.1'								
18	91°09.3	S23°18.1	331°10.4	14.5'	N14°39.4	9.4'	54.0'								
19	106°09.0	18.2	345°43.9	14.5'	14°48.8	9.4'	54.0'								
20	121°08.7	18.3	0°17.4	14.4'	14°58.2	9.3'	54.0'								
21	136°08.4	· · 18.4	14°50.8	14.4'	15°07.5	9.2'	54.0'								
22	151°08.1	18.5	29°24.2	14.4'	15°16.7	9.2'	54.0'								
23	166°07.8	18.6	43°57.6	14.3'	15°25.9	9.1'	54.0'								
		SD = 16.2'	$d = 0.1'$		SD = 14.8'										
Sun	GHA	Dec	GHA	ν	Dec	d	HP								
0	181°07.5	S23°18.8	58°30.9	14.3'	N15°35.0	9.1'	54.0'								
1	196°07.2	18.9	73°04.2	14.2'	15°44.1	9.0'	54.0'								
2	211°06.9	19.0	87°37.4	14.2'	15°53.1	8.9'	54.0'								
3	226°06.6	· · 19.1	102°10.6	14.1'	16°02.0	8.9'	54.0'								
4	241°06.3	19.2	116°43.7	14.1'	16°10.9	8.8'	54.0'								
5	256°06.0	19.3	131°16.8	14.1'	16°19.7	8.7'	54.0'								
6	271°05.7	S23°19.4	145°49.9	14.0'	N16°28.4	8.7'	54.0'								
7	286°05.4	19.5	160°22.9	14.0'	16°37.1	8.6'	54.0'								
8	301°05.1	19.6	174°55.9	13.9'	16°45.7	8.5'	54.0'								
9	316°04.8	· · 19.7	189°28.8	13.9'	16°54.2	8.4'	54.0'								
10	331°04.5	19.8	204°01.7	13.8'	17°02.6	8.4'	54.0'								
11	346°04.2	19.9	218°34.5	13.8'	17°11.0	8.3'	54.0'								
12	1°03.9	S23°20.0	233°07.3	13.7'	N17°19.3	8.2'	54.0'								
13	16°03.6	20.1	247°40.1	13.7'	17°27.5	8.2'	54.0'								
14	31°03.3	20.2	262°12.7	13.6'	17°35.7	8.1'	54.0'								
15	46°03.0	· · 20.3	276°45.4	13.6'	17°43.8	8.0'	54.0'								
16	61°02.7	20.4	291°18.0	13.6'	17°51.8	7.9'	54.0'								
17	76°02.3	20.5	305°50.5	13.5'	17°59.8	7.9'	54.0'								
18	91°02.0	S23°20.6	320°23.0	13.5'	N18°07.6	7.8'	54.0'								
19	106°01.7	20.7	334°55.5	13.4'	18°15.4	7.7'	54.0'								
20	121°01.4	20.8	349°27.9	13.4'	18°23.1	7.6'	54.0'								
21	136°01.1	· · 20.9	4°00.3	13.3'	18°30.7	7.6'	54.0'								
22	151°00.8	21.0	18°32.6	13.3'	18°38.3	7.5'	54.0'								
23	166°00.5	21.1	33°04.8	13.2'	18°45.8	7.4'	54.0'								
		SD = 16.2'	$d = 0.1'$		SD = 14.7'										
Mon	GHA	Dec	GHA	ν	Dec	d	HP								
0	181°00.2	S23°21.1	47°37.0	13.2'	N18°53.2	7.3'	54.0'								
1	195°59.9	21.2	62°09.2	13.1'	19°00.5	7.2'	54.0'								
2	210°59.6	21.3	76°41.3	13.1'	19°07.7	7.2'	54.0'								
3	225°59.3	· · 21.4	91°13.3	13.0'	19°14.9	7.1'	54.0'								
4	240°59.0	21.5	105°45.4	13.0'	19°21.9	7.0'	54.0'								
5	255°58.7	21.6	120°17.3	12.9'	19°28.9	6.9'	54.0'								
6	270°58.4	S23°21.7	134°49.2	12.9'	N19°35.8	6.8'	54.0'								
7	285°58.1	21.8	149°21.1	12.8'	19°42.6	6.7'	54.0'								
8	300°57.8	21.8	163°52.9	12.8'	19°49.4	6.6'	54.0'								
9	315°57.5	· · 21.9	178°24.6	12.7'	19°56.0	6.6'	54.0'								
10	330°57.2	22.0	192°56.3	12.7'	20°02.6	6.5'	54.0'								
11	345°56.9	22.1	207°28.0	12.6'	20°09.1	6.4'	54.1'								
12	0°56.6	S23°22.2	221°59.6	12.6'	N20°15.4	6.3'	54.1'								
13	15°56.3	22.2	236°31.2	12.5'	20°21.7	6.2'	54.1'								
14	30°56.0	22.3	251°02.7	12.5'	20°27.9	6.1'	54.1'								
15	45°55.6	· · 22.4	265°34.1	12.4'	20°34.1	6.0'	54.1'								
16	60°55.3	22.5	280°05.5	12.4'	20°40.1	5.9'	54.1'								
17	75°55.0	22.6	294°36.9	12.3'	20°46.0	5.8'	54.1'								
18	90°54.7	S23°22.6	309°08.2	12.3'	N20°51.9	5.8'	54.1'								
19	105°54.4	22.7	323°39.4	12.2'	20°57.6	5.7'	54.1'								
20	120°54.1	22.8	338°10.6	12.2'	21°03.3	5.6'	54.1'								
21	135°53.8	· · 22.8	352°41.8	12.1'	21°08.9	5.5'	54.1'								
22	150°53.5	22.9	7°12.9	12.1'	21°14.3	5.4'	54.1'								
23	165°53.2	23.0	21°43.9	12.0'	21°19.7	5.3'	54.1'								
		SD = 16.2'	$d = 0.1'$		SD = 14.7'										

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:22	10:50	■	■	13:01	15:30
N 70°	08:02	09:50	■	■	14:02	15:50
68°	07:46	09:15	■	■	14:36	16:05
66°	07:33	08:50	10:29	13:22	15:02	16:18
64°	07:22	08:30	09:48	14:04	15:21	16:29
62°	07:12	08:14	09:20	14:32	15:37	16:39
60°	07:04	08:01	08:58	14:53	15:51	16:47
N 58°	06:57	07:49	08:41	15:10	16:02	16:55
56°	06:50	07:39	08:26	15:25	16:12	17:01
54°	06:44	07:30	08:14	15:38	16:21	17:08
52°	06:38	07:22	08:03	15:49	16:30	17:13
50°	06:33	07:14	07:53	15:59	16:37	17:19
45°	06:21	06:58	07:32	16:19	16:53	17:30
N 40°	06:11	06:45	07:15	16:36	17:07	17:41
35°	06:01	06:33	07:01	16:50	17:18	17:50
30°	05:53	06:23	06:49	17:02	17:29	17:59
20°	05:36	06:04	06:28	17:24	17:48	18:15
N 10°	05:20	05:47	06:09	17:42	18:05	18:31
0°	05:03	05:29	05:52	17:59	18:22	18:48
S 10°	04:44	05:11	05:34	18:17	18:40	19:07
20°	04:22	04:51	05:16	18:36	19:00	19:30
30°	03:53	04:26	04:54	18:58	19:25	19:59
35°	03:34	04:11	04:41	19:11	19:41	20:17
40°	03:11	03:53	04:26	19:26	19:59	20:40
45°	02:41	03:30	04:08	19:44	20:21	21:11
S 50°						

December 18, 19, 20 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	86°53.8	150°56.7	S21°36.5	140°32.8	S20°31.4	221°05.6	S16°14.3	39°38.9	N15°10.5	Alpheratz	357°30.5	29°15.6
1	101°56.3	165°59.4	36.0	155°33.2	31.0	236°07.6	14.4	54°41.5	10.5	Ankaa	353°03.3	-42°08.7
2	116°58.7	181°02.1	35.5	170°33.7	30.5	251°09.6	14.5	69°44.1	10.5	Schedar	349°26.1	56°42.5
3	132°01.2	196°04.8	· · 35.0	185°34.1	· · 30.0	266°11.5	· · 14.7	84°46.7	· · 10.4	Diphda	348°43.3	-17°49.4
4	147°03.7	211°07.5	34.5	200°34.6	29.6	281°13.5	14.8	99°49.3	10.4	Achernar	335°17.2	-57°05.3
5	162°06.1	226°10.2	34.0	215°35.0	29.1	296°15.5	14.9	114°51.9	10.4	Hamal	327°46.4	23°36.4
6	177°08.6	241°12.9	S21°33.5	230°35.5	S20°28.6	311°17.5	S16°15.1	129°54.5	N15°10.4	Polaris	311°23.1	89°23.5
7	192°11.1	256°15.6	33.0	245°35.9	28.1	326°19.5	15.2	144°57.1	10.3	Acamar	315°08.5	-40°11.2
8	207°13.5	271°18.3	32.5	260°36.3	27.7	341°21.4	15.3	159°59.7	10.3	Menkar	314°01.7	4°12.4
9	222°16.0	286°21.0	· · 31.9	275°36.8	· · 27.2	356°23.4	· · 15.5	175°02.3	· · 10.3	Mirfak	308°22.0	49°58.1
10	237°18.5	301°23.7	31.4	290°37.2	26.7	11°25.4	15.6	190°04.9	10.2	Aldebaran	290°34.7	16°34.1
11	252°20.9	316°26.4	30.9	305°37.7	26.2	26°27.4	15.7	205°07.5	10.2	Rigel	280°59.7	-8°10.2
12	267°23.4	331°29.2	S21°30.4	320°38.1	S20°25.8	41°29.4	S16°15.9	220°10.1	N15°10.2	Capella	280°15.5	46°01.5
13	282°25.8	346°31.9	29.9	335°38.5	25.3	56°31.3	16.0	235°12.8	10.2	Bellatrix	278°18.3	6°22.5
14	297°28.3	1°34.6	29.4	350°39.0	24.8	71°33.3	16.1	250°15.4	10.1	Elnath	277°56.5	28°37.8
15	312°30.8	16°37.4	· · 28.9	5°39.4	· · 24.3	86°35.3	· · 16.3	265°18.0	· · 10.1	Alnilam	275°33.4	-1°11.1
16	327°33.2	31°40.1	28.4	20°39.9	23.8	101°37.3	16.4	280°20.6	10.1	Betelgeuse	270°47.5	7°24.6
17	342°35.7	46°42.9	27.9	35°40.3	23.4	116°39.3	16.5	295°23.2	10.0	Canopus	263°50.1	-52°42.8
18	357°38.2	61°45.6	S21°27.4	50°40.8	S20°22.9	131°41.2	S16°16.7	310°25.8	N15°10.0	Sirius	258°22.4	-16°45.6
19	12°40.6	76°48.4	26.9	65°41.2	22.4	146°43.2	16.8	325°28.4	10.0	Adhara	255°02.4	-29°00.9
20	27°43.1	91°51.2	26.3	80°41.6	21.9	161°45.2	16.9	340°31.0	10.0	Procyon	244°46.4	5°08.7
21	42°45.6	106°53.9	· · 25.8	95°42.1	· · 21.4	176°47.2	· · 17.0	355°33.6	· · 09.9	Pollux	243°12.2	27°56.9
22	57°48.0	121°56.7	25.3	110°42.5	21.0	191°49.2	17.2	10°36.2	09.9	Avior	234°12.4	-59°36.3
23	72°50.5	136°59.5	24.8	125°43.0	20.5	206°51.1	17.3	25°38.8	09.9	Suhail	222°43.0	-43°33.2
Mer.pass. 18:09		ν 2.7' d -0.5' m -4.79		ν 0.4' d -0.5' m 1.13		ν 2.0' d 0.1' m -1.75		ν 2.6' d -0.0' m -0.10		Miaplacidus	221°36.5	-69°50.3
										Alphard	217°43.7	-8°47.4
										Regulus	207°30.1	11°49.0
										Dubhe	193°36.3	61°34.9
										Denebola	182°21.0	14°24.1
										Gienah	175°39.5	-17°42.5
										Acrux	172°55.5	-63°15.7
										Gacrux	171°47.1	-57°16.7
										Alioth	166°09.9	55°47.5
										Spica	158°18.2	-11°19.0
										Alkaid	152°49.3	49°09.6
										Hadar	148°30.5	-60°30.8
										Menkent	147°53.1	-36°30.9
										Arcturus	145°44.6	19°01.5
										Rigel Kent.	139°35.1	-60°57.3
										Kochab	137°21.2	74°01.7
										Zuben'ubi	136°51.8	-16°09.9
										Alphecca	126°00.8	26°36.8
										Antares	112°11.3	-26°29.7
										Atria	107°02.4	-69°04.6
										Sabik	101°58.6	-15°45.5
										Shaula	96°05.4	-37°07.3
										Rasalhague	95°55.2	12°32.4
										Eltanin	90°40.9	51°29.2
										Kaus Aust.	83°27.7	-34°22.0
										Vega	80°31.0	38°48.9
										Nunki	75°43.2	-26°15.4
										Altair	61°56.4	8°57.1
										Peacock	53°00.1	-56°38.3
										Deneb	49°23.4	45°23.6
										Enif	33°35.1	10°01.0
										Al Na'ir	27°28.3	-46°49.0
										Fomalhaut	15°10.3	-29°27.8
										Scheat	13°41.4	28°15.0
										Markab	13°26.0	15°22.2
										Dec 18 TueSHA Mer.pass		
										Venus	64°02.9	13:54
										Mars	53°39.0	14:37
										Jupiter	134°11.8	09:14
										Saturn	312°45.1	21:18
										Dec 19 WedSHA Mer.pass		
										Venus	64°09.3	13:49
										Mars	52°50.5	14:37
										Jupiter	134°00.2	09:11
										Saturn	312°48.4	21:14
										Dec 20 ThuSHA Mer.pass		
										Venus	64°18.4	13:45
										Mars	52°02.0	14:36
										Jupiter	133°48.6	09:08
										Saturn	312°51.7	21:09
										Horizontal parallax		
										Venus:	0.5	
										Mars:	0.1	
										Mer.pass. 18:02		
										ν 2.9' d -0.5' m -4.74		
										ν 0.4' d -0.5' m 1.14		
										ν 2.0' d 0.1' m -1.76		
										ν 2.6' d -0.0' m -0.09		

2029 December 18 to Dec. 20 UT

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:25	10:56	■	■	12:58	15:30
N 70°	08:04	09:53	■	■	14:01	15:50
68°	07:49	09:18	■	■	14:37	16:06
66°	07:35	08:53	10:33	13:21	15:02	16:19
64°	07:24	08:33	09:51	14:04	15:22	16:30
62°	07:15	08:17	09:22	14:32	15:38	16:40
60°	07:06	08:03	09:01	14:54	15:51	16:48
N 58°	06:59	07:51	08:43	15:11	16:03	16:56
56°	06:52	07:41	08:29	15:26	16:13	17:02
54°	06:46	07:32	08:16	15:38	16:22	17:09
52°	06:40	07:24	08:05	15:50	16:31	17:14
50°	06:35	07:16	07:55	15:59	16:38	17:20
45°	06:23	07:00	07:34	16:20	16:54	17:31
N 40°	06:13	06:47	07:17	16:37	17:08	17:42
35°	06:03	06:35	07:03	16:51	17:19	17:51
30°	05:54	06:24	06:51	17:04	17:30	18:00
20°	05:38	06:05	06:29	17:25	17:49	18:16
N 10°	05:22	05:48	06:11	17:43	18:06	18:33
0°	05:05	05:31	05:53	18:01	18:23	18:50
S 10°	04:46	05:13	05:36	18:19	18:42	19:09
20°	04:23	04:52	05:17	18:38	19:02	19:31
30°	03:54	04:27	04:55	19:00	19:27	20:00
35°	03:35	04:12	04:42	19:13	19:42	20:19
40°	03:12	03:54	04:27	19:28	20:01	20:42
45°	02:42	03:31	04:09	19:46	20:23	21:13
S 50°	01:56	03:02	03:46	20:08	20:53	21:59
52°	01:28	02:46	03:35	20:19	21:08	22:27
54°	00:41	02:28	03:23	20:32	21:26	23:14
56°	—:—	02:06	03:09	20:46	21:48	—:—
58°	—:—	01:36	02:52	21:03	22:18	—:—
S 60°	—:—	00:46	02:31	21:23	23:09	—:—

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°						
N 70°						
68°						
66°	11:31	11:43	12:34	08:11	09:43	10:38
64°	12:10	12:38	13:31	07:32	08:47	09:41
62°	12:38	13:11	14:04	07:05	08:14	09:08
60°	12:59	13:36	14:28	06:44	07:50	08:43
N 58°	13:17	13:55	14:48	06:27	07:31	08:24
56°	13:31	14:11	15:04	06:13	07:15	08:08
54°	13:44	14:25	15:17	06:00	07:01	07:54
52°	13:55	14:37	15:30	05:49	06:49	07:42
50°	14:05	14:48	15:40	05:40	06:38	07:31
45°	14:26	15:10	16:02	05:19	06:16	07:08
N 40°	14:43	15:29	16:20	05:03	05:58	06:50
35°	14:57	15:44	16:35	04:49	05:43	06:35
30°	15:10	15:57	16:49	04:37	05:30	06:22
20°	15:31	16:19	17:11	04:16	05:08	05:59
N 10°	15:50	16:39	17:30	03:58	04:49	05:40
0°	16:07	16:57	17:48	03:42	04:31	05:22
S 10°	16:24	17:15	18:06	03:25	04:13	05:03
20°	16:43	17:35	18:25	03:07	03:54	04:44
30°	17:05	17:57	18:48	02:47	03:32	04:21
35°	17:17	18:11	19:01	02:35	03:19	04:08
40°	17:32	18:26	19:16	02:21	03:04	03:52
45°	17:50	18:44	19:34	02:05	02:46	03:34
S 50°	18:11	19:07	19:56	01:45	02:24	03:11
52°	18:22	19:18	20:06	01:36	02:13	03:01
54°	18:33	19:30	20:18	01:25	02:02	02:48
56°	18:47	19:44	20:32	01:13	01:48	02:34
58°	19:02	20:01	20:48	01:00	01:32	02:17
S 60°	19:21	20:21	21:07	00:44	01:13	01:57

Day	Sun			Moon		Age 13-15 92-99%
	Eqn.of Time	Mer.	Mer.Pass.	Upper	Lower	
	00 ^h mm:ss	12 ^h mm:ss	Pass hh:mm	hh:mm	hh:mm	
18	03:32	03:17	11:57	22:19	09:54	
19	03:02	02:47	11:57	23:09	10:44	
20	02:33	02:18	11:58	—:—	11:35	



December 21, 22, 23 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	89°51.2	154°21.5	S21°00.0	141°04.9	S19°56.5	223°28.3	S16°23.7	42°46.1	N15°08.6				Alpheratz	357°30.6	29°15.6		
1	104°53.7	169°24.5	20°59.5	156°05.4	56.0	238°30.3	23.8	57°48.7	08.6				Ankaa	353°03.3	-42°08.7		
2	119°56.2	184°27.5	59.0	171°05.8	55.5	253°32.3	23.9	72°51.3	08.5				Schedar	349°26.1	56°42.5		
3	134°58.6	199°30.5	•• 58.5	186°06.3	•• 55.0	268°34.3	•• 24.1	87°53.9	•• 08.5				Diphda	348°43.3	-17°49.4		
4	150°01.1	214°33.6	58.0	201°06.7	54.5	283°36.3	24.2	102°56.5	08.5				Achernar	335°17.2	-57°05.3		
5	165°03.6	229°36.6	57.5	216°07.2	54.0	298°38.3	24.3	117°59.1	08.5				Hamal	327°46.4	23°36.4		
6	180°06.0	244°39.6	S20°57.0	231°07.7	S19°53.5	313°40.2	S16°24.4	133°01.7	N15°08.4				Polaris	311°23.7	89°23.5		
7	195°08.5	259°42.7	56.5	246°08.1	52.9	328°42.2	24.6	148°04.2	08.4				Acamar	315°08.5	-40°11.2		
8	210°10.9	274°45.7	56.0	261°08.6	52.4	343°44.2	24.7	163°06.8	08.4				Menkar	314°01.7	4°12.4		
9	225°13.4	289°48.8	•• 55.5	276°09.0	•• 51.9	358°46.2	•• 24.8	178°09.4	•• 08.4				Mirfak	308°22.0	49°58.1		
10	240°15.9	304°51.8	55.0	291°09.5	51.4	13°48.2	25.0	193°12.0	08.3				Aldebaran	290°34.7	16°34.1		
11	255°18.3	319°54.9	54.5	306°09.9	50.9	28°50.2	25.1	208°14.6	08.3				Rigel	280°59.7	-8°10.2		
12	270°20.8	334°58.0	S20°54.0	321°10.4	S19°50.4	43°52.2	S16°25.2	223°17.2	N15°08.3				Capella	280°15.5	46°01.5		
13	285°23.3	350°01.0	53.5	336°10.8	49.9	58°54.1	25.3	238°19.8	08.3				Bellatrix	278°18.3	6°22.5		
14	300°25.7	5°04.1	53.0	351°11.3	49.4	73°56.1	25.5	253°22.4	08.2				Elnath	277°56.4	28°37.8		
15	315°28.2	20°07.2	•• 52.5	6°11.7	•• 48.9	88°58.1	•• 25.6	268°25.0	•• 08.2				Alnilam	275°33.4	-1°11.1		
16	330°30.7	35°10.3	51.9	21°12.2	48.4	104°00.1	25.7	283°27.6	08.2				Betelgeuse	270°47.4	7°24.6		
17	345°33.1	50°13.3	51.4	36°12.7	47.9	119°02.1	25.9	298°30.2	08.2				Canopus	263°50.0	-52°42.8		
18	0°35.6	65°16.4	S20°50.9	51°13.1	S19°47.4	134°04.1	S16°26.0	313°32.8	N15°08.2				Sirius	258°22.4	-16°45.6		
19	15°38.1	80°19.5	50.4	66°13.6	46.9	149°06.1	26.1	328°35.4	08.1				Adhara	255°02.3	-29°00.9		
20	30°40.5	95°22.6	49.9	81°14.0	46.4	164°08.1	26.2	343°37.9	08.1				Procyon	244°46.4	5°08.7		
21	45°43.0	110°25.7	•• 49.4	96°14.5	•• 45.9	179°10.0	•• 26.4	358°40.5	•• 08.1				Pollux	243°12.1	27°56.9		
22	60°45.4	125°28.9	48.9	111°14.9	45.4	194°12.0	26.5	13°43.1	08.1				Avior	234°12.4	-59°36.3		
23	75°47.9	140°32.0	48.4	126°15.4	44.9	209°14.0	26.6	28°45.7	08.0				Suhail	222°43.0	-43°33.2		
Mer.pass. 17:58			ν 3.0' d -0.5' m -4.70	ν 0.5' d -0.5' m 1.14	ν 2.0' d 0.1' m -1.76	ν 2.6' d -0.0' m -0.08											

Sat	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec									
0	90°50.4	155°35.1	S20°47.9	141°15.8	S19°44.4	224°16.0	S16°26.7	43°48.3	N15°08.0								
1	105°52.8	170°38.2	47.4	156°16.3	43.8	239°18.0	26.9	58°50.9	08.0								
2	120°55.3	185°41.4	46.9	171°16.8	43.3	254°20.0	27.0	73°53.5	08.0								
3	135°57.8	200°44.5	•• 46.4	186°17.2	•• 42.8	269°22.0	•• 27.1	88°56.1	•• 07.9								
4	151°00.2	215°47.6	45.9	201°17.7	42.3	284°24.0	27.3	103°58.7	07.9								
5	166°02.7	230°50.8	45.4	216°18.1	41.8	299°26.0	27.4	119°01.3	07.9								
6	181°05.2	245°53.9	S20°44.9	231°18.6	S19°41.3	314°27.9	S16°27.5	134°03.9	N15°07.9								
7	196°07.6	260°57.1	44.4	246°19.0	40.8	329°29.9	27.6	149°06.4	07.8								
8	211°10.1	276°00.2	43.9	261°19.5	40.3	344°31.9	27.8	164°09.0	07.8								
9	226°12.6	291°03.4	•• 43.4	276°20.0	•• 39.8	359°33.9	•• 27.9	179°11.6	•• 07.8								
10	241°15.0	306°06.6	42.9	291°20.4	39.3	14°35.9	28.0	194°14.2	07.8								
11	256°17.5	321°09.7	42.4	306°20.9	38.7	29°37.9	28.1	209°16.8	07.7								
12	271°19.9	336°12.9	S20°41.9	321°21.3	S19°38.2	44°39.9	S16°28.3	224°19.4	N15°07.7								
13	286°22.4	351°16.1	41.4	336°21.8	37.7	59°41.9	28.4	239°22.0	07.7								
14	301°24.9	6°19.3	40.9	351°22.2	37.2	74°43.9	28.5	254°24.6	07.7								
15	316°27.3	21°22.5	•• 40.4	6°22.7	•• 36.7	89°45.9	•• 28.6	269°27.2	•• 07.7								
16	331°29.8	36°25.6	39.9	21°23.2	36.2	104°47.8	28.8	284°29.7	07.6								
17	346°32.3	51°28.8	39.4	36°23.6	35.7	119°49.8	28.9	299°32.3	07.6								
18	1°34.7	66°32.0	S20°38.9	51°24.1	S19°35.1	134°51.8	S16°29.0	314°34.9	N15°07.6								
19	16°37.2	81°35.3	38.4	66°24.5	34.6	149°53.8	29.1	329°37.5	07.6								
20	31°39.7	96°38.5	37.9	81°25.0	34.1	164°55.8	29.3	344°40.1	07.5								
21	46°42.1	111°41.7	•• 37.4	96°25.5	•• 33.6	179°57.8	•• 29.4	359°42.7	•• 07.5								
22	61°44.6	126°44.9	36.9	111°25.9	33.1	194°59.8	29.5	14°45.3	07.5								
23	76°47.1	141°48.1	36.4	126°26.4	32.6	210°01.8	29.7	29°47.9	07.5								
Mer.pass. 17:54			ν 3.1' d -0.5' m -4.66	ν 0.5' d -0.5' m 1.14	ν 2.0' d 0.1' m -1.76	ν 2.6' d -0.0' m -0.08											

Sun	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec									
0	91°49.5	156°51.4	S20°35.9	141°26.8	S19°32.0	225°03.8	S16°29.8	44°50.4	N15°07.4								
1	106°52.0	171°54.6	35.4	156°27.3	31.5	240°05.8	29.9	59°53.0	07.4								
2	121°54.4	186°57.8	34.9	171°27.8	31.0	255°07.8	30.0	74°55.6	07.4								
3	136°56.9	202°01.1	•• 34.4	186°28.2	•• 30.5	270°09.8	•• 30.2	89°58.2	•• 07.4								
4	151°59.4	217°04.3	33.9	201°28.7	30.0	285°11.7	30.3	105°00.8	07.4								
5	167°01.8	232°07.6	33.4	216°29.1	29.5	300°13.7	30.4	120°03.4	07.3								
6	182°04.3	247°10.8	S20°32.9	231°29.6	S19°28.9	315°15.7	S16°30.5	135°06.0	N15°07.3								
7	197°06.8	262°14.1	32.5	246°30.1	28.4	330°17.7	30.7	150°08.5	07.3								
8	212°09.2	277°17.4	32.0	261°30.5	27.9	345°19.7	30.8	165°11.1	07.3								
9	227°11.7	292°20.6	•• 31.5	276°31.0	•• 27.4	0°21.7	•• 30.9	180°13.7	•• 07.2								
10	242°14.2	307°23.9	31.0	291°31.5	26.9	15°23.7	31.0	195°16.3	07.2								
11	257°16.6	322°27.2	30.5	306°31.9	26.3	30°25.7	31.2	210°18.9	07.2								
12	272°19.1	337°30.5	S20°30.0	321°32.4	S19°25.8	45°27.7	S16°31.3	225°21.5	N15°07.2								
13	287°21.5	352°33.7	29.5	336°32.8	25.3	60°29.7	31.4	240°24.1	07.2								
14	302°24.0	7°37.0	29.0	351°33.3	24.8	75°31.7	31.5	255°26.6	07.1								
15	317°26.5	22°40.3	•• 28.5	6°33.8	•• 24.2	90°33.7	•• 31.7	270°29.2	•• 07.1								
16	332°28.9	37°43.6	28.0	21°34.2	23.7	105°35.7	31.8	285°31.8	07.1								
17	347°31.4	52°46.9	27.5	36°34.7	23.2	120°37.6	31.9	300°34.4	07.1								
18	2°33.9	67°50.3	S20°27.0	51°35.2	S19°22.7	135°39.6	S16°32.0	315°37.0	N15°07.0								
19	17°36.3	82°53.6	26.5	66°35.6	22.1	150°41.6	32.2	330°39.6	07.0								
20	32°38.8	97°56.9	26.0	81°36.1	21.6	165°43.6	32.3	345°42.2	07.0								
21	47°41.3	113°00.2	•• 25.5	96°36.6	•• 21.1	180°45.6	•• 32.4	0°44.7	•• 07.0								
22	62°43.7	128°03.5	25.0	111°37.0	20.6	195°47.6	32.5	15°47.3	07.0								
23	77°46.2	143°06.9	24.5	126°37.5	20.0	210°49.6	32.7	30°49.9	06.9								
Mer.pass. 17:50			ν 3.2' d -0.5' m -4.62	ν 0.5' d -0.5' m 1.14	ν 2.0' d 0.1' m -1.77	ν 2.6' d -0.0' m -0.07											

Dec 21 Fri	SHA	Mer.pass
Venus	64°30.2	13:40
Mars	51°13.7	14:35
Jupiter	133°37.1	09:05
Saturn	312°54.9	21:05

Dec 22 Sat	SHA	Mer.pass
Venus	64°44.7	13:35
Mars	50°25.5	14:34
Jupiter	133°25.6	09:02
Saturn	312°57.9	21:01

Dec 23 Sun	SHA	Mer.pass
Venus	65°01.8	13:30
Mars	49°37.3	14:34
Jupiter	133°14.3	08:59
Saturn	313°00.9	20:57

Horizontal parallax	
Venus:	0.5
Mars:	0.1

h	Sun			Moon				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	180°30.7	S23°26.0		359°53.1	10.1'	N23°01.5	-2.9'	55.0'
1	195°30.4	26.0		14°22.2	10.1'	22°58.6	-3.0'	55.0'
2	210°30.1	26.0		28°51.2	10.1'	22°55.6	-3.1'	55.1'
3	225°29.8	•• 26.1		43°20.3	10.1'	22°52.5	-3.2'	55.1'
4	240°29.5	26.1		57°49.4	10.1'	22°49.3	-3.4'	55.1'
5	255°29.2	26.1		72°18.5	10.1'	22°45.9	-3.5'	55.1'
6	270°28.9	S23°26.1		86°47.6	10.1'	N22°42.4	-3.6'	55.1'
7	285°28.6	26.1		101°16.7	10.1'	22°38.8	-3.7'	55.1'
8	300°28.2	26.1		115°45.8	10.1'	22°35.1	-3.8'	55.2'
9	315°27.9	•• 26.1		130°14.9	10.1'	22°31.3	-3.9'	55.2'
10	330°27.6	26.1		144°44.1	10.1'	22°27.3	-4.1'	55.2'
11	345°27.3	26.1		159°13.2	10.2'	22°23.3	-4.2'	55.2'
12	0°27.0	S23°26.1		173°42.4	10.2'	N22°19.1	-4.3'	55.2'
13	15°26.7	26.1		188°11.5	10.2'	22°14.8	-4.4'	55.2'
14	30°26.4	26.1		202°40.7	10.2'	22°10.4	-4.5'	55.3'
15	45°26.1	•• 26.1		217°09.9	10.2'	22°05.8	-4.6'	55.3'
16	60°25.8	26.1		231°39.2	10.2'	22°01.2	-4.8'	55.3'
17	75°25.5	26.1		246°08.4	10.3'	21°56.4	-4.9'	55.3'
18	90°25.1	S23°26.1		260°37.6	10.3'	N21°51.6	-5.0'	55.3'
19	105°24.8	26.1		275°06.9	10.3'	21°46.6	-5.1'	55.4'
20	120°24.5	26.1		289°36.2	10.3'	21°41.5	-5.2'	55.4'
21	135°24.2	•• 26.1		304°05.5	10.3'	21°36.3	-5.3'	55.4'
22	150°23.9	26.1		318°34.8	10.3'	21°30.9	-5.4'	55.4'
23	165°23.6	26.1		333°04.2	10.4'	21°25.5	-5.5'	55.4'
	SD = 16.3'	$d = 0.0'$				SD = 15.0'		

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°23.3	S23°26.1	347°33.6	10.4'	N21°20.0	-5.7'	55.4'
1	195°23.0	26.1	2°03.0	10.4'	21°14.3	-5.8'	55.5'
2	210°22.7	26.0	16°32.4	10.4'	21°08.5	-5.9'	55.5'
3	225°22.4	• • 26.0	31°01.8	10.5'	21°02.7	-6.0'	55.5'
4	240°22.0	26.0	45°31.3	10.5'	20°56.7	-6.1'	55.5'
5	255°21.7	26.0	60°00.8	10.5'	20°50.6	-6.2'	55.5'
6	270°21.4	S23°26.0	74°30.3	10.5'	N20°44.4	-6.3'	55.6'
7	285°21.1	26.0	88°59.8	10.6'	20°38.0	-6.4'	55.6'
8	300°20.8	26.0	103°29.4	10.6'	20°31.6	-6.5'	55.6'
9	315°20.5	• • 26.0	117°58.9	10.6'	20°25.1	-6.6'	55.6'
10	330°20.2	25.9	132°28.6	10.6'	20°18.4	-6.7'	55.6'
11	345°19.9	25.9	146°58.2	10.7'	20°11.7	-6.9'	55.6'
12	0°19.6	S23°25.9	161°27.9	10.7'	N20°04.8	-7.0'	55.7'
13	15°19.3	25.9	175°57.6	10.7'	19°57.9	-7.1'	55.7'
14	30°19.0	25.9	190°27.3	10.8'	19°50.8	-7.2'	55.7'
15	45°18.6	• • 25.9	204°57.0	10.8'	19°43.7	-7.3'	55.7'
16	60°18.3	25.8	219°26.8	10.8'	19°36.4	-7.4'	55.7'
17	75°18.0	25.8	233°56.6	10.8'	19°29.0	-7.5'	55.8'
18	90°17.7	S23°25.8	248°26.5	10.9'	N19°21.6	-7.6'	55.8'
19	105°17.4	25.8	262°56.3	10.9'	19°14.0	-7.7'	55.8'
20	120°17.1	25.7	277°26.2	10.9'	19°06.3	-7.8'	55.8'
21	135°16.8	• • 25.7	291°56.2	11.0'	18°58.5	-7.9'	55.8'
22	150°16.5	25.7	306°26.1	11.0'	18°50.7	-8.0'	55.9'
23	165°16.2	25.7	320°56.1	11.0'	18°42.7	-8.1'	55.9'
SD = 16.3' $d = -0.0'$			SD = 15.1'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°15.9	S23°25.6	335°26.1	11.1'	N18°34.6	-8.2'	55.9'
1	195°15.5	25.6	349°56.2	11.1'	18°26.5	-8.3'	55.9'
2	210°15.2	25.6	4°26.3	11.1'	18°18.2	-8.4'	55.9'
3	225°14.9	•• 25.6	18°56.4	11.1'	18°09.8	-8.5'	56.0'
4	240°14.6	25.5	33°26.5	11.2'	18°01.4	-8.6'	56.0'
5	255°14.3	25.5	47°56.7	11.2'	17°52.8	-8.6'	56.0'
6	270°14.0	S23°25.5	62°26.9	11.2'	N17°44.2	-8.7'	56.0'
7	285°13.7	25.4	76°57.2	11.3'	17°35.4	-8.8'	56.0'
8	300°13.4	25.4	91°27.4	11.3'	17°26.6	-8.9'	56.0'
9	315°13.1	•• 25.4	105°57.8	11.3'	17°17.7	-9.0'	56.1'
10	330°12.8	25.3	120°28.1	11.4'	17°08.7	-9.1'	56.1'
11	345°12.4	25.3	134°58.5	11.4'	16°59.6	-9.2'	56.1'
12	0°12.1	S23°25.3	149°28.9	11.4'	N16°50.4	-9.3'	56.1'
13	15°11.8	25.2	163°59.3	11.5'	16°41.1	-9.4'	56.1'
14	30°11.5	25.2	178°29.8	11.5'	16°31.7	-9.5'	56.2'
15	45°11.2	•• 25.1	193°00.3	11.5'	16°22.3	-9.5'	56.2'
16	60°10.9	25.1	207°30.8	11.6'	16°12.7	-9.6'	56.2'
17	75°10.6	25.1	222°01.4	11.6'	16°03.1	-9.7'	56.2'
18	90°10.3	S23°25.0	236°31.9	11.6'	N15°53.4	-9.8'	56.2'
19	105°10.0	25.0	251°02.6	11.7'	15°43.6	-9.9'	56.3'
20	120°09.7	24.9	265°33.2	11.7'	15°33.7	-10.0'	56.3'
21	135°09.3	•• 24.9	280°03.9	11.7'	15°23.8	-10.0'	56.3'
22	150°09.0	24.8	294°34.6	11.7'	15°13.7	-10.1'	56.3'
23	165°08.7	24.8	309°05.4	11.8'	15°03.6	-10.2'	56.3'
SD = 16.3' $d = -0.0'$			SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:26	10:58	■	■	12:59	15:31
N 70°	08:06	09:55	■	■	14:02	15:51
68°	07:50	09:20	■	■	14:38	16:07
66°	07:37	08:54	10:35	13:22	15:03	16:20
64°	07:26	08:34	09:53	14:05	15:23	16:31
62°	07:16	08:18	09:24	14:33	15:39	16:41
60°	07:08	08:05	09:03	14:55	15:53	16:50
N 58°	07:00	07:53	08:45	15:12	16:04	16:57
56°	06:54	07:43	08:30	15:27	16:15	17:04
54°	06:47	07:34	08:18	15:40	16:24	17:10
52°	06:42	07:25	08:06	15:51	16:32	17:16
50°	06:36	07:18	07:56	16:01	16:39	17:21
45°	06:25	07:02	07:36	16:22	16:56	17:33
N 40°	06:14	06:48	07:19	16:39	17:09	17:43
35°	06:05	06:36	07:05	16:53	17:21	17:53
30°	05:56	06:26	06:52	17:05	17:32	18:01
20°	05:39	06:07	06:31	17:26	17:50	18:18
N 10°	05:23	05:50	06:12	17:45	18:08	18:34
0°	05:06	05:32	05:55	18:02	18:25	18:51
S 10°	04:47	05:14	05:37	18:20	18:43	19:10
20°	04:25	04:54	05:18	18:39	19:04	19:33
30°	03:55	04:29	04:56	19:01	19:29	20:02
35°	03:37	04:13	04:43	19:14	19:44	20:21
40°	03:13	03:55	04:28	19:29	20:02	20:44
45°	02:43	03:33	04:10	19:47	20:25	21:15
S 50°	01:57	03:03	03:47	20:10	20:55	22:00
52°	01:29	02:48	03:37	20:21	21:10	22:28
54°	00:42	02:30	03:24	20:33	21:28	23:15
56°	—:—	02:07	03:10	20:47	21:50	—:—
58°	—:—	01:37	02:53	21:04	22:20	—:—
S 60°	—:—	00:46	02:33	21:25	23:11	—:—

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	☐	☐	15:59	☐	☐	12:31
N 70°	☐	13:56	16:44	☐	12:49	11:45
68°	☐	15:14	17:13	☐	11:31	11:15
66°	14:07	15:51	17:35	10:51	10:53	10:52
64°	14:47	16:17	17:52	10:11	10:26	10:34
62°	15:14	16:37	18:06	09:43	10:05	10:19
60°	15:36	16:54	18:18	09:22	09:48	10:06
N 58°	15:53	17:08	18:28	09:04	09:34	09:55
56°	16:07	17:20	18:37	08:49	09:21	09:46
54°	16:20	17:30	18:45	08:37	09:11	09:37
52°	16:31	17:39	18:52	08:25	09:01	09:30
50°	16:41	17:47	18:58	08:15	08:52	09:23
45°	17:01	18:05	19:12	07:54	08:34	09:08
N 40°	17:18	18:19	19:23	07:37	08:19	08:56
35°	17:32	18:31	19:32	07:23	08:06	08:46
30°	17:44	18:42	19:41	07:10	07:55	08:36
20°	18:05	19:00	19:55	06:49	07:36	08:21
N 10°	18:23	19:15	20:07	06:30	07:19	08:07
0°	18:39	19:30	20:19	06:13	07:04	07:53
S 10°	18:56	19:44	20:30	05:55	06:48	07:40
20°	19:14	20:00	20:43	05:36	06:31	07:26
30°	19:34	20:17	20:57	05:15	06:11	07:10
35°	19:46	20:28	21:05	05:02	06:00	07:00
40°	20:00	20:39	21:14	04:47	05:47	06:49
45°	20:16	20:53	21:25	04:30	05:31	06:37
S 50°	20:36	21:10	21:38	04:08	05:12	06:21
52°	20:46	21:18	21:44	03:57	05:03	06:13
54°	20:57	21:27	21:50	03:46	04:52	06:05
56°	21:09	21:36	21:58	03:32	04:41	05:56
58°	21:22	21:48	22:06	03:16	04:27	05:45
S 60°	21:39	22:00	22:15	02:57	04:11	05:33

December 24, 25, 26 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	92°48.7	158°10.2	S20°24.0	141°37.9	S19°19.5	225°51.6	S16°32.8	45°52.5	N15°06.9	Alpheratz	357°30.6 29°15.6
1	107°51.1	173°13.5	23.5	156°38.4	19.0	240°53.6	32.9	60°55.1	06.9	Ankaa	353°03.3 -42°08.7
2	122°53.6	188°16.9	23.0	171°38.9	18.5	255°55.6	33.0	75°57.7	06.9	Schedar	349°26.2 56°42.5
3	137°56.0	203°20.2	· · 22.5	186°39.3	· · 17.9	270°57.6	· · 33.2	91°00.2	· · 06.8	Diphda	348°43.3 -17°49.4
4	152°58.5	218°23.6	22.0	201°39.8	17.4	285°59.6	33.3	106°02.8	06.8	Achernar	335°17.2 -57°05.3
5	168°01.0	233°27.0	21.6	216°40.3	16.9	301°01.6	33.4	121°05.4	06.8	Hamal	327°46.5 23°36.4
6	183°03.4	248°30.3	S20°21.1	231°40.7	S19°16.4	316°03.6	S16°33.5	136°08.0	N15°06.8	Polaris	311°24.6 89°23.5
7	198°05.9	263°33.7	20.6	246°41.2	15.8	331°05.6	33.7	151°10.6	06.8	Acamar	315°08.5 -40°11.3
8	213°08.4	278°37.0	20.1	261°41.7	15.3	346°07.6	33.8	166°13.1	06.7	Menkar	314°01.7 4°12.4
9	228°10.8	293°40.4	· · 19.6	276°42.1	· · 14.8	1°09.6	· · 33.9	181°15.7	· · 06.7	Mirfak	308°22.0 49°58.1
10	243°13.3	308°43.8	19.1	291°42.6	14.2	16°11.6	34.0	196°18.3	06.7	Aldebaran	290°34.7 16°34.1
11	258°15.8	323°47.2	18.6	306°43.1	13.7	31°13.6	34.1	211°20.9	06.7	Rigel	280°59.7 -8°10.2
12	273°18.2	338°50.6	S20°18.1	321°43.5	S19°13.2	46°15.5	S16°34.3	226°23.5	N15°06.7	Capella	280°15.5 46°01.6
13	288°20.7	353°54.0	17.6	336°44.0	12.6	61°17.5	34.4	241°26.1	06.6	Bellatrix	278°18.3 6°22.5
14	303°23.2	8°57.4	17.1	351°44.5	12.1	76°19.5	34.5	256°28.6	06.6	Elnath	277°56.4 28°37.8
15	318°25.6	24°00.8	· · 16.6	6°44.9	· · 11.6	91°21.5	· · 34.6	271°31.2	· · 06.6	Alnilam	275°33.3 -1°11.1
16	333°28.1	39°04.2	16.1	21°45.4	11.1	106°23.5	34.8	286°33.8	06.6	Betelgeuse	270°47.4 7°24.6
17	348°30.5	54°07.6	15.6	36°45.9	10.5	121°25.5	34.9	301°36.4	06.5	Canopus	263°50.0 -52°42.8
18	3°33.0	69°11.0	S20°15.1	51°46.3	S19°10.0	136°27.5	S16°35.0	316°39.0	N15°06.5	Sirius	258°22.4 -16°45.6
19	18°35.5	84°14.4	14.7	66°46.8	09.5	151°29.5	35.1	331°41.5	06.5	Adhara	255°02.3 -29°00.9
20	33°37.9	99°17.8	14.2	81°47.3	08.9	166°31.5	35.3	346°44.1	06.5	Procyon	244°46.4 5°08.7
21	48°40.4	114°21.3	· · 13.7	96°47.7	· · 08.4	181°33.5	· · 35.4	1°46.7	· · 06.5	Pollux	243°12.1 27°56.9
22	63°42.9	129°24.7	13.2	111°48.2	07.9	196°35.5	35.5	16°49.3	06.4	Avior	234°12.4 -59°36.3
23	78°45.3	144°28.1	12.7	126°48.7	07.3	211°37.5	35.6	31°51.9	06.4	Suhail	222°42.9 -43°33.2
Mer.pass. 17:46		ν 3.3' d -0.5' m -4.57		ν 0.5' d -0.5' m 1.14		ν 2.0' d 0.1' m -1.77		ν 2.6' d -0.0' m -0.07		Miaplacidus	221°36.4 -69°50.3
										Alphard	217°43.6 -8°47.5
										Regulus	207°30.1 11°49.0
										Dubhe	193°36.2 61°34.9
										Denebola	182°20.9 14°24.1
										Gienah	175°39.4 -17°42.5
										Acrux	172°55.4 -63°15.7
										Gacrux	171°47.0 -57°16.7
										Alioth	166°09.8 55°47.5
										Spica	158°18.2 -11°19.1
										Alkaid	152°49.2 49°09.5
										Hadar	148°30.4 -60°30.8
										Menkent	147°53.0 -36°30.9
										Arcturus	145°44.5 19°01.5
										Rigel Kent.	139°35.0 -60°57.3
										Kochab	137°21.1 74°01.6
										Zuben'ubi	136°51.8 -16°09.9
										Alphecca	126°00.7 26°36.8
										Antares	112°11.3 -26°29.7
										Atria	107°02.3 -69°04.6
										Sabik	101°58.5 -15°45.5
										Shaula	96°05.4 -37°07.3
										Rasalhague	95°55.2 12°32.4
										Eltanin	90°40.9 51°29.2
										Kaus Aust.	83°27.7 -34°22.0
										Vega	80°30.9 38°48.8
										Nunki	75°43.2 -26°15.4
										Altair	61°56.4 8°57.1
										Peacock	53°00.2 -56°38.2
										Deneb	49°23.4 45°23.6
										Enif	33°35.1 10°01.0
										Al Na'ir	27°28.3 -46°49.0
										Fomalhaut	15°10.4 -29°27.8
										Scheat	13°41.4 28°15.0
										Markab	13°26.0 15°22.2
										Dec 24 Mon	SHA Mer.pass
										Venus	65°21.5 13:24
										Mars	48°49.3 14:33
										Jupiter	133°03.0 08:55
										Saturn	313°03.8 20:53
										Dec 25 Tue	SHA Mer.pass
										Venus	65°43.8 13:19
										Mars	48°01.4 14:32
										Jupiter	132°51.7 08:52
										Saturn	313°06.6 20:49
										Dec 26 Wed	SHA Mer.pass
										Venus	66°08.4 13:13
										Mars	47°13.5 14:32
										Jupiter	132°40.5 08:49
										Saturn	313°09.4 20:45
										Horizontal parallax	
										Venus:	0.5
										Mars:	0.1
Mer.pass. 17:38		ν 3.5' d -0.5' m -4.46		ν 0.5' d -0.5' m 1.15		ν 2.0' d 0.1' m -1.78		ν 2.6' d -0.0' m -0.06			

DUT1 = UT1-UTC = +0.2140 sec ΔT = TT-UT1 = +68.9700 sec

2029 December 24 to Dec. 26 UT

h		Sun		Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°08.4	S23°24.7	323°36.1	11.8'	N14°53.4	-10.3'	56.4'
1	195°08.1	24.7	338°06.9	11.8'	14°43.1	-10.4'	56.4'
2	210°07.8	24.7	352°37.8	11.9'	14°32.7	-10.4'	56.4'
3	225°07.5	· · 24.6	7°08.7	11.9'	14°22.3	-10.5'	56.4'
4	240°07.2	24.6	21°39.5	11.9'	14°11.8	-10.6'	56.5'
5	255°06.9	24.5	36°10.5	12.0'	14°01.2	-10.7'	56.5'
6	270°06.6	S23°24.4	50°41.4	12.0'	N13°50.5	-10.7'	56.5'
7	285°06.2	24.4	65°12.4	12.0'	13°39.8	-10.8'	56.5'
8	300°05.9	24.3	79°43.4	12.0'	13°29.0	-10.9'	56.5'
9	315°05.6	· · 24.3	94°14.4	12.1'	13°18.1	-11.0'	56.6'
10	330°05.3	24.2	108°45.5	12.1'	13°07.2	-11.0'	56.6'
11	345°05.0	24.2	123°16.6	12.1'	12°56.1	-11.1'	56.6'
12	0°04.7	S23°24.1	137°47.7	12.1'	N12°45.0	-11.2'	56.6'
13	15°04.4	24.1	152°18.8	12.2'	12°33.9	-11.2'	56.6'
14	30°04.1	24.0	166°50.0	12.2'	12°22.6	-11.3'	56.7'
15	45°03.8	· · 23.9	181°21.2	12.2'	12°11.3	-11.4'	56.7'
16	60°03.5	23.9	195°52.4	12.2'	12°00.0	-11.4'	56.7'
17	75°03.2	23.8	210°23.6	12.3'	11°48.5	-11.5'	56.7'
18	90°02.8	S23°23.8	224°54.9	12.3'	N11°37.1	-11.6'	56.7'
19	105°02.5	23.7	239°26.2	12.3'	11°25.5	-11.6'	56.8'
20	120°02.2	23.6	253°57.5	12.3'	11°13.9	-11.7'	56.8'
21	135°01.9	· · 23.6	268°28.8	12.3'	11°02.2	-11.7'	56.8'
22	150°01.6	23.5	283°00.2	12.4'	10°50.5	-11.8'	56.8'
23	165°01.3	23.4	297°31.5	12.4'	10°38.6	-11.9'	56.8'
SD = 16.3'		$d = -0.0'$	SD = 15.4'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°01.0	S23°23.4	312°02.9	12.4'	N10°26.8	-11.9'	56.9'
1	195°00.7	23.3	326°34.3	12.4'	10°14.9	-12.0'	56.9'
2	210°00.4	23.2	341°05.7	12.4'	10°02.9	-12.0'	56.9'
3	225°00.1	· · 23.2	355°37.2	12.5'	09°50.9	-12.1'	56.9'
4	239°59.8	23.1	10°08.6	12.5'	09°38.8	-12.1'	57.0'
5	254°59.4	23.0	24°40.1	12.5'	09°26.6	-12.2'	57.0'
6	269°59.1	S23°23.0	39°11.6	12.5'	N09°14.4	-12.2'	57.0'
7	284°58.8	22.9	53°43.1	12.5'	09°02.2	-12.3'	57.0'
8	299°58.5	22.8	68°14.6	12.5'	08°49.9	-12.3'	57.0'
9	314°58.2	· · 22.7	82°46.1	12.5'	08°37.6	-12.4'	57.1'
10	329°57.9	22.7	97°17.7	12.6'	08°25.2	-12.4'	57.1'
11	344°57.6	22.6	111°49.2	12.6'	08°12.7	-12.5'	57.1'
12	359°57.3	S23°22.5	126°20.8	12.6'	N08°00.2	-12.5'	57.1'
13	14°57.0	22.4	140°52.4	12.6'	07°47.7	-12.6'	57.2'
14	29°56.7	22.4	155°24.0	12.6'	07°35.1	-12.6'	57.2'
15	44°56.4	· · 22.3	169°55.6	12.6'	07°22.5	-12.7'	57.2'
16	59°56.0	22.2	184°27.2	12.6'	07°09.8	-12.7'	57.2'
17	74°55.7	22.1	198°58.8	12.6'	06°57.1	-12.8'	57.2'
18	89°55.4	S23°22.0	213°30.4	12.6'	N06°44.3	-12.8'	57.3'
19	104°55.1	22.0	228°02.0	12.6'	06°31.5	-12.8'	57.3'
20	119°54.8	21.9	242°33.7	12.6'	06°18.7	-12.9'	57.3'
21	134°54.5	· · 21.8	257°05.3	12.6'	06°05.8	-12.9'	57.3'
22	149°54.2	21.7	271°36.9	12.6'	05°52.9	-13.0'	57.3'
23	164°53.9	21.6	286°08.6	12.6'	05°39.9	-13.0'	57.4'
SD = 16.3'		$d = -0.1'$	SD = 15.5'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179°53.6	S23°21.5	300°40.2	12.6'	N05°26.9	-13.0'	57.4'
1	194°53.3	21.5	315°11.9	12.6'	05°13.9	-13.1'	57.4'
2	209°53.0	21.4	329°43.5	12.6'	05°00.8	-13.1'	57.4'
3	224°52.7	· · 21.3	344°15.2	12.6'	04°47.7	-13.1'	57.5'
4	239°52.3	21.2	358°46.8	12.6'	04°34.6	-13.2'	57.5'
5	254°52.0	21.1	13°18.4	12.6'	04°21.4	-13.2'	57.5'
6	269°51.7	S23°21.0	27°50.1	12.6'	N04°08.2	-13.2'	57.5'
7	284°51.4	20.9	42°21.7	12.6'	03°55.0	-13.2'	57.5'
8	299°51.1	20.8	56°53.3	12.6'	03°41.8	-13.3'	57.6'
9	314°50.8	· · 20.7	71°24.9	12.6'	03°28.5	-13.3'	57.6'
10	329°50.5	20.6	85°56.5	12.6'	03°15.2	-13.3'	57.6'
11	344°50.2	20.5	100°28.1	12.6'	03°01.9	-13.3'	57.6'
12	359°49.9	S23°20.5	114°59.7	12.6'	N02°48.5	-13.4'	57.7'
13	14°49.6	20.4	129°31.3	12.6'	02°35.2	-13.4'	57.7'
14	29°49.3	20.3	144°02.9	12.6'	02°21.8	-13.4'	57.7'
15	44°49.0	· · 20.2	158°34.4	12.5'	02°08.4	-13.4'	57.7'
16	59°48.7	20.1	173°06.0	12.5'	01°54.9	-13.5'	57.8'
17	74°48.3	20.0	187°37.5	12.5'	01°41.5	-13.5'	57.8'
18	89°48.0	S23°19.9	202°09.0	12.5'	N01°28.0	-13.5'	57.8'
19	104°47.7	19.8	216°40.5	12.5'	01°14.5	-13.5'	57.8'
20	119°47.4	19.7	231°12.0	12.5'	01°01.0	-13.5'	57.8'
21	134°47.1	· · 19.6	245°43.5	12.4'	00°47.5	-13.5'	57.9'
22	149°46.8	19.5	260°14.9	12.4'	00°34.0	-13.5'	57.9'
23	164°46.5	19.3	274°46.3	12.4'	00°20.4	-13.6'	57.9'
SD = 16.3'		$d = -0.1'$	SD = 15.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:27	10:57	■	■	13:04	15:34
N 70°	08:07	09:55	■	■	14:05	15:54
68°	07:51	09:20	■	■	14:40	16:09
66°	07:38	08:55	■	■	15:05	16:22
64°	07:27	08:35	09:53	14:07	15:25	16:34
62°	07:17	08:19	09:25	14:36	15:41	16:43
60°	07:09	08:06	09:03	14:57	15:55	16:52
N 58°	07:01	07:54	08:46	15:15	16:06	16:59
56°	06:55	07:44	08:31	15:29	16:17	17:06
54°	06:49	07:35	08:19	15:42	16:26	17:12
52°	06:43	07:27	08:07	15:53	16:34	17:18
50°	06:38	07:19	07:58	16:03	16:41	17:23
45°	06:26	07:03	07:37	16:24	16:57	17:35
N 40°	06:15	06:50	07:20	16:40	17:11	17:45
35°	06:06	06:38	07:06	16:54	17:23	17:54
30°	05:57	06:27	06:54	17:07	17:33	18:03
20°	05:41	06:08	06:32	17:28	17:52	18:20
N 10°	05:25	05:51	06:14	17:47	18:09	18:36
0°	05:08	05:34	05:56	18:04	18:26	18:53
S 10°	04:49	05:16	05:39	18:21	18:45	19:12
20°	04:26	04:55	05:20	18:40	19:05	19:34
30°	03:57	04:30	04:58	19:02	19:30	20:03
35°	03:38	04:15	04:45	19:15	19:45	20:22
40°	03:15	03:57	04:30	19:30	20:03	20:45
45°	02:45	03:34	04:12	19:48	20:26	21:15
S 50°	01:59	03:05	03:49	20:11	20:55	22:01
52°	01:31	02:50	03:38	20:22	21:11	22:28
54°	00:45	02:32	03:26	20:34	21:28	23:14
56°	—:—	02:09	03:12	20:48	21:51	—:—
58°	—:—	01:40	02:55	21:05	22:20	—:—
S 60°	—:—	00:49	02:35	21:25	23:10	—:—

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	18:28	20:35	22:37	11:44	11:16	10:54
N 70°	18:48	20:44	22:37	11:21	11:05	10:50
68°	19:04	20:51	22:37	11:04	10:55	10:47
66°	19:17	20:57	22:38	10:50	10:47	10:44
64°	19:27	21:02	22:38	10:38	10:40	10:42
62°	19:36	21:07	22:38	10:28	10:34	10:40
60°	19:44	21:11	22:38	10:19	10:29	10:38
N 58°	19:51	21:14	22:39	10:12	10:25	10:36
56°	19:56	21:17	22:39	10:05	10:21	10:35
54°	20:02	21:20	22:39	09:59	10:17	10:34
52°	20:06	21:22	22:39	09:53	10:14	10:32
50°	20:11	21:24	22:39	09:48	10:11	10:31
45°	20:20	21:29	22:39	09:38	10:04	10:29
N 40°	20:28	21:33	22:40	09:29	09:59	10:27
35°	20:34	21:37	22:40	09:21	09:54	10:25
30°	20:40	21:40	22:40	09:14	09:50	10:24
20°	20:50	21:45	22:40	09:02	09:42	10:21
N 10°	20:59	21:50	22:41	08:52	09:36	10:19
0°	21:07	21:54	22:41	08:42	09:29	10:16
S 10°	21:15	21:58	22:41	08:32	09:23	10:14
20°	21:24	22:03	22:41	08:21	09:16	10:11
30°	21:33	22:08	22:42	08:09	09:09	10:08
35°	21:39	22:11	22:42	08:02	09:04	10:07
40°	21:45	22:14	22:42	07:54	08:59	10:05
45°	21:53	22:18	22:43	07:44	08:53	10:03
S 50°	22:02	22:23	22:43	07:33	08:46	10:00
52°	22:06	22:25	22:43	07:27	08:42	09:59
54°	22:10	22:27	22:44	07:21	08:39	09:57
56°	22:15	22:30	22:44	07:15	08:35	09:56
58°	22:21	22:33	22:44	07:07	08:30	09:54
S 60°	22:27	22:36	22:44	06:58	08:25	09:52

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
24	00:34	00:19	12:00	02:30	14:54	
25	00:04	00:11	12:00	03:18	15:42	
26	00:26	00:40	12:01	04:05	16:29	

December 27, 28, 29 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	95°46.1	162°21.4	S19°48.9	142°11.9	S18°40.7	228°15.5	S16°41.6	48°58.0	N15°05.5					Alpheratz	357°30.6	29°15.6	
1	110°48.5	177°25.1	48.4	157°12.3	40.2	243°17.5	41.7	64°00.6	05.5					Ankaa	353°03.4	-42°08.7	
2	125°51.0	192°28.7	47.9	172°12.8	39.6	258°19.5	41.8	79°03.2	05.4					Schedar	349°26.2	56°42.5	
3	140°53.5	207°32.4	· · 47.5	187°13.3	· · 39.1	273°21.5	· · 42.0	94°05.8	· · 05.4					Diphda	348°43.3	-17°49.4	
4	155°55.9	222°36.0	47.0	202°13.8	38.5	288°23.5	42.1	109°08.3	05.4					Achernar	335°17.2	-57°05.3	
5	170°58.4	237°39.6	46.5	217°14.3	37.9	303°25.5	42.2	124°10.9	05.4					Hamal	327°46.5	23°36.4	
6	186°00.9	252°43.3	S19°46.0	232°14.7	S18°37.4	318°27.5	S16°42.3	139°13.5	N15°05.4					Polaris	311°25.6	89°23.5	
7	201°03.3	267°47.0	45.5	247°15.2	36.8	333°29.5	42.4	154°16.0	05.3					Acamar	315°08.6	-40°11.3	
8	216°05.8	282°50.6	45.1	262°15.7	36.3	348°31.5	42.6	169°18.6	05.3					Menkar	314°01.7	4°12.4	
9	231°08.2	297°54.3	· · 44.6	277°16.2	· · 35.7	3°33.5	· · 42.7	184°21.2	· · 05.3					Mirfak	308°22.0	49°58.1	
10	246°10.7	312°58.0	44.1	292°16.6	35.2	18°35.6	42.8	199°23.8	05.3					Aldebaran	290°34.7	16°34.1	
11	261°13.2	328°01.6	43.6	307°17.1	34.6	33°37.6	42.9	214°26.3	05.3					Rigel	280°59.7	-8°10.2	
12	276°15.6	343°05.3	S19°43.1	322°17.6	S18°34.1	48°39.6	S16°43.0	229°28.9	N15°05.3					Capella	280°15.5	46°01.6	
13	291°18.1	358°09.0	42.7	337°18.1	33.5	63°41.6	43.2	244°31.5	05.2					Bellatrix	278°18.3	6°22.4	
14	306°20.6	13°12.7	42.2	352°18.6	33.0	78°43.6	43.3	259°34.0	05.2					Elnath	277°56.4	28°37.8	
15	321°23.0	28°16.4	· · 41.7	7°19.0	· · 32.4	93°45.6	· · 43.4	274°36.6	· · 05.2					Alnilam	275°33.3	-1°11.1	
16	336°25.5	43°20.0	41.2	22°19.5	31.8	108°47.6	43.5	289°39.2	05.2					Betelgeuse	270°47.4	7°24.6	
17	351°28.0	58°23.7	40.8	37°20.0	31.3	123°49.6	43.6	304°41.7	05.2					Canopus	263°50.0	-52°42.8	
18	6°30.4	73°27.4	S19°40.3	52°20.5	S18°30.7	138°51.6	S16°43.8	319°44.3	N15°05.2					Sirius	258°22.4	-16°45.6	
19	21°32.9	88°31.1	39.8	67°21.0	30.2	153°53.6	43.9	334°46.9	05.1					Adhara	255°02.3	-29°00.9	
20	36°35.4	103°34.8	39.3	82°21.4	29.6	168°55.6	44.0	349°49.4	05.1					Procyon	244°46.3	5°08.7	
21	51°37.8	118°38.6	· · 38.9	97°21.9	· · 29.0	183°57.6	· · 44.1	4°52.0	· · 05.1					Pollux	243°12.1	27°56.9	
22	66°40.3	133°42.3	38.4	112°22.4	28.5	198°59.6	44.2	19°54.6	05.1					Avior	234°12.4	-59°36.3	
23	81°42.7	148°46.0	37.9	127°22.9	27.9	214°01.6	44.3	34°57.1	05.1					Suhail	222°42.9	-43°33.2	
Mer.pass. 17:34			ν 3.6' d -0.5' m -4.40	ν 0.5' d -0.6' m 1.15	ν 2.0' d 0.1' m -1.78	ν 2.6' d -0.0' m -0.05											

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	96°45.2	163°49.7	S19°37.4	142°23.4	S18°27.4	229°03.6	S16°44.5	49°59.7	N15°05.0
1	111°47.7	178°53.4	37.0	157°23.9	26.8	244°05.6	44.6	65°02.3	05.0
2	126°50.1	193°57.1	36.5	172°24.3	26.2	259°07.6	44.7	80°04.8	05.0
3	141°52.6	209°00.9	· · 36.0	187°24.8	· · 25.7	274°09.6	· · 44.8	95°07.4	· · 05.0
4	156°55.1	224°04.6	35.5	202°25.3	25.1	289°11.7	44.9	110°10.0	05.0
5	171°57.5	239°08.3	35.1	217°25.8	24.6	304°13.7	45.1	125°12.5	05.0
6	187°00.0	254°12.1	S19°34.6	232°26.3	S18°24.0	319°15.7	S16°45.2	140°15.1	N15°04.9
7	202°02.5	269°15.8	34.1	247°26.7	23.4	334°17.7	45.3	155°17.7	04.9
8	217°04.9	284°19.6	33.6	262°27.2	22.9	349°19.7	45.4	170°20.2	04.9
9	232°07.4	299°23.3	· · 33.2	277°27.7	· · 22.3	4°21.7	· · 45.5	185°22.8	· · 04.9
10	247°09.8	314°27.1	32.7	292°28.2	21.8	19°23.7	45.7	200°25.4	04.9
11	262°12.3	329°30.8	32.2	307°28.7	21.2	34°25.7	45.8	215°27.9	04.9
12	277°14.8	344°34.6	S19°31.8	322°29.2	S18°20.6	49°27.7	S16°45.9	230°30.5	N15°04.8
13	292°17.2	359°38.4	31.3	337°29.7	20.1	64°29.7	46.0	245°33.1	04.8
14	307°19.7	14°42.1	30.8	352°30.1	19.5	79°31.7	46.1	260°35.6	04.8
15	322°22.2	29°45.9	· · 30.3	7°30.6	· · 18.9	94°33.7	· · 46.2	275°38.2	· · 04.8
16	337°24.6	44°49.7	29.9	22°31.1	18.4	109°35.7	46.4	290°40.8	04.8
17	352°27.1	59°53.5	29.4	37°31.6	17.8	124°37.8	46.5	305°43.3	04.8
18	7°29.6	74°57.2	S19°28.9	52°32.1	S18°17.2	139°39.8	S16°46.6	320°45.9	N15°04.7
19	22°32.0	90°01.0	28.5	67°32.6	16.7	154°41.8	46.7	335°48.5	04.7
20	37°34.5	105°04.8	28.0	82°33.0	16.1	169°43.8	46.8	350°51.0	04.7
21	52°37.0	120°08.6	· · 27.5	97°33.5	· · 15.5	184°45.8	· · 46.9	5°53.6	· · 04.7
22	67°39.4	135°12.4	27.1	112°34.0	15.0	199°47.8	47.1	20°56.2	04.7
23	82°41.9	150°16.2	26.6	127°34.5	14.4	214°49.8	47.2	35°58.7	04.7
Mer.pass. 17:30			ν 3.7' d -0.5' m -4.33	ν 0.5' d -0.6' m 1.15	ν 2.0' d 0.1' m -1.78	ν 2.6' d -0.0' m -0.05			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	97°44.3	165°20.0	S19°26.1	142°35.0	S18°13.8	229°51.8	S16°47.3	51°01.3	N15°04.7
1	112°46.8	180°23.8	25.6	157°35.5	13.3	244°53.8	47.4	66°03.8	04.6
2	127°49.3	195°27.6	25.2	172°36.0	12.7	259°55.8	47.5	81°06.4	04.6
3	142°51.7	210°31.4	· · 24.7	187°36.4	· · 12.1	274°57.8	· · 47.7	96°09.0	· · 04.6
4	157°54.2	225°35.2	24.2	202°36.9	11.6	289°59.9	47.8	111°11.5	04.6
5	172°56.7	240°39.0	23.8	217°37.4	11.0	305°01.9	47.9	126°14.1	04.6
6	187°59.1	255°42.9	S19°23.3	232°37.9	S18°10.4	320°03.9	S16°48.0	141°16.7	N15°04.6
7	203°01.6	270°46.7	22.8	247°38.4	09.8	335°05.9	48.1	156°19.2	04.5
8	218°04.1	285°50.5	22.4	262°38.9	09.3	350°07.9	48.2	171°21.8	04.5
9	233°06.5	300°54.3	· · 21.9	277°39.4	· · 08.7	5°09.9	· · 48.4	186°24.3	· · 04.5
10	248°09.0	315°58.2	21.4	292°39.9	08.1	20°11.9	48.5	201°26.9	04.5
11	263°11.5	331°02.0	21.0	307°40.3	07.6	35°13.9	48.6	216°29.5	04.5
12	278°13.9	346°05.8	S19°20.5	322°40.8	S18°07.0	50°15.9	S16°48.7	231°32.0	N15°04.5
13	293°16.4	1°09.7	20.1	337°41.3	06.4	65°18.0	48.8	246°34.6	04.5
14	308°18.8	16°13.5	19.6	352°41.8	05.8	80°20.0	48.9	261°37.1	04.4
15	323°21.3	31°17.4	· · 19.1	7°42.3	· · 05.3	95°22.0	· · 49.1	276°39.7	· · 04.4
16	338°23.8	46°21.2	18.7	22°42.8	04.7	110°24.0	49.2	291°42.3	04.4
17	353°26.2	61°25.1	18.2	37°43.3	04.1	125°26.0	49.3	306°44.8	04.4
18	8°28.7	76°28.9	S19°17.7	52°43.8	S18°03.6	140°28.0	S16°49.4	321°47.4	N15°04.4
19	23°31.2	91°32.8	17.3	67°44.3	03.0	155°30.0	49.5	336°49.9	04.4
20	38°33.6	106°36.7	16.8	82°44.7	02.4	170°32.0	49.6	351°52.5	04.3
21	53°36.1	121°40.5	· · 16.3	97°45.2	· · 01.8	185°34.1	· · 49.8	6°55.1	· · 04.3
22	68°38.6	136°44.4	15.9	112°45.7	01.3	200°36.1	49.9	21°57.6	04.3
23	83°41.0	151°48.3	15.4	127°46.2	00.7	215°38.1	50.0	37°00.2	04.3
Mer.pass. 17:26			ν 3.8' d -0.5' m -4.25	ν 0.5' d -0.6' m 1.15	ν 2.0' d 0.1' m -1.79	ν 2.6' d -0.0' m -0.04			

Dec 27 Thu	SHA	Mer.pass
Venus	66°35.4	13:07
Mars	46°25.8	14:31
Jupiter	132°29.4	08:46
Saturn	313°12.0	20:41

Dec 28 Fri	SHA	Mer.pass
Venus	67°04.5	13:01
Mars	45°38.2	14:30
Jupiter	132°18.4	08:43
Saturn	313°14.5	20:36

Dec 29 Sat	SHA	Mer.pass
Venus	67°35.6	12:55
Mars	44°50.6	14:29
Jupiter	132°07.5	08:39
Saturn	313°16.9	20:32

Horizontal parallax		
Venus:	0.5	
Mars:	0.1	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°46.2	S23°19.2	289°17.7	12.4'	N00°06.9	-13.6'	57.9'
1	194°45.9	19.1	303°49.1	12.4'	S00°06.7	13.6'	58.0'
2	209°45.6	19.0	318°20.5	12.3'	00°20.3	13.6'	58.0'
3	224°45.3	· · 18.9	332°51.8	12.3'	00°33.9	13.6'	58.0'
4	239°45.0	18.8	347°23.1	12.3'	00°47.5	13.6'	58.0'
5	254°44.7	18.7	1°54.4	12.3'	01°01.1	13.6'	58.0'
6	269°44.4	S23°18.6	16°25.6	12.2'	S01°14.7	13.6'	58.1'
7	284°44.0	18.5	30°56.8	12.2'	01°28.3	13.6'	58.1'
8	299°43.7	18.4	45°28.0	12.2'	01°41.9	13.6'	58.1'
9	314°43.4	· · 18.3	59°59.2	12.1'	01°55.5	13.6'	58.1'
10	329°43.1	18.1	74°30.3	12.1'	02°09.1	13.6'	58.2'
11	344°42.8	18.0	89°01.4	12.1'	02°22.7	13.6'	58.2'
12	359°42.5	S23°17.9	103°32.5	12.0'	S02°36.3	13.6'	58.2'
13	14°42.2	17.8	118°03.5	12.0'	02°49.9	13.6'	58.2'
14	29°41.9	17.7	132°34.5	12.0'	03°03.6	13.6'	58.3'
15	44°41.6	· · 17.6	147°05.5	11.9'	03°17.2	13.6'	58.3'
16	59°41.3	17.4	161°36.4	11.9'	03°30.7	13.6'	58.3'
17	74°41.0	17.3	176°07.3	11.8'	03°44.3	13.6'	58.3'
18	89°40.7	S23°17.2	190°38.1	11.8'	S03°57.9	13.6'	58.3'
19	104°40.4	17.1	205°08.9	11.8'	04°11.5	13.6'	58.4'
20	119°40.1	17.0	219°39.7	11.7'	04°25.0	13.5'	58.4'
21	134°39.8	· · 16.8	234°10.4	11.7'	04°38.6	13.5'	58.4'
22	149°39.5	16.7	248°41.0	11.6'	04°52.1	13.5'	58.4'
23	164°39.1	16.6	263°11.7	11.6'	05°05.6	13.5'	58.5'
SD = 16.3' $d = -0.1'$			SD = 15.8'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°38.8	S23°16.5	277°42.2	11.5'	S05°19.1	13.5'	58.5'
1	194°38.5	16.3	292°12.8	11.5'	05°32.6	13.5'	58.5'
2	209°38.2	16.2	306°43.3	11.4'	05°46.1	13.4'	58.5'
3	224°37.9	· · 16.1	321°13.7	11.4'	05°59.5	13.4'	58.5'
4	239°37.6	16.0	335°44.1	11.3'	06°12.9	13.4'	58.6'
5	254°37.3	15.8	350°14.4	11.3'	06°26.3	13.4'	58.6'
6	269°37.0	S23°15.7	4°44.7	11.2'	S06°39.7	13.4'	58.6'
7	284°36.7	15.6	19°14.9	11.2'	06°53.1	13.3'	58.6'
8	299°36.4	15.4	33°45.1	11.1'	07°06.4	13.3'	58.7'
9	314°36.1	· · 15.3	48°15.2	11.1'	07°19.7	13.3'	58.7'
10	329°35.8	15.2	62°45.2	11.0'	07°32.9	13.2'	58.7'
11	344°35.5	15.0	77°15.2	10.9'	07°46.2	13.2'	58.7'
12	359°35.2	S23°14.9	91°45.2	10.9'	S07°59.4	13.2'	58.7'
13	14°34.9	14.8	106°15.0	10.8'	08°12.6	13.1'	58.8'
14	29°34.6	14.6	120°44.9	10.8'	08°25.7	13.1'	58.8'
15	44°34.3	· · 14.5	135°14.6	10.7'	08°38.8	13.1'	58.8'
16	59°34.0	14.4	149°44.3	10.6'	08°51.9	13.0'	58.8'
17	74°33.7	14.2	164°13.9	10.6'	09°04.9	13.0'	58.9'
18	89°33.3	S23°14.1	178°43.5	10.5'	S09°17.9	13.0'	58.9'
19	104°33.0	13.9	193°13.0	10.4'	09°30.9	12.9'	58.9'
20	119°32.7	13.8	207°42.4	10.4'	09°43.8	12.9'	58.9'
21	134°32.4	· · 13.7	222°11.8	10.3'	09°56.7	12.8'	58.9'
22	149°32.1	13.5	236°41.1	10.2'	10°09.5	12.8'	59.0'
23	164°31.8	13.4	251°10.3	10.2'	10°22.3	12.7'	59.0'
SD = 16.3' $d = -0.1'$			SD = 15.9'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°31.5	S23°13.2	265°39.5	10.1'	S10°35.0	12.7'	59.0'
1	194°31.2	13.1	280°08.6	10.0'	10°47.7	12.6'	59.0'
2	209°30.9	12.9	294°37.6	9.9'	11°00.3	12.6'	59.0'
3	224°30.6	· · 12.8	309°06.6	9.9'	11°12.9	12.5'	59.1'
4	239°30.3	12.6	323°35.4	9.8'	11°25.4	12.5'	59.1'
5	254°30.0	12.5	338°04.2	9.7'	11°37.9	12.4'	59.1'
6	269°29.7	S23°12.3	352°33.0	9.6'	S11°50.3	12.4'	59.1'
7	284°29.4	12.2	7°01.6	9.6'	12°02.7	12.3'	59.2'
8	299°29.1	12.0	21°30.2	9.5'	12°15.0	12.2'	59.2'
9	314°28.8	· · 11.9	35°58.7	9.4'	12°27.2	12.2'	59.2'
10	329°28.5	11.7	50°27.1	9.3'	12°39.4	12.1'	59.2'
11	344°28.2	11.6	64°55.4	9.3'	12°51.5	12.0'	59.2'
12	359°27.9	S23°11.4	79°23.7	9.2'	S13°03.5	12.0'	59.3'
13	14°27.6	11.3	93°51.9	9.1'	13°15.5	11.9'	59.3'
14	29°27.3	11.1	108°20.0	9.0'	13°27.4	11.8'	59.3'
15	44°27.0	· · 11.0	122°48.0	8.9'	13°39.3	11.8'	59.3'
16	59°26.7	10.8	137°15.9	8.9'	13°51.0	11.7'	59.3'
17	74°26.4	10.6	151°43.8	8.8'	14°02.7	11.6'	59.3'
18	89°26.1	S23°10.5	166°11.6	8.7'	S14°14.4	11.5'	59.4'
19	104°25.8	10.3	180°39.3	8.6'	14°25.9	11.5'	59.4'
20	119°25.5	10.2	195°06.9	8.5'	14°37.4	11.4'	59.4'
21	134°25.2	· · 10.0	209°34.4	8.4'	14°48.8	11.3'	59.4'
22	149°24.9	09.8	224°01.8	8.4'	15°00.1	11.2'	59.4'
23	164°24.6	09.7	238°29.2	8.3'	15°11.3	11.1'	59.5'
SD = 16.3' $d = -0.1'$			SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:26	10:52	■	■	13:11	15:37
N 70°	08:07	09:54	■	■	14:10	15:57
68°	07:51	09:19	■	■	14:44	16:12
66°	07:38	08:55	10:33	13:30	15:09	16:25
64°	07:27	08:35	09:52	14:11	15:28	16:36
62°	07:18	08:19	09:25	14:39	15:44	16:46
60°	07:09	08:06	09:03	15:00	15:57	16:54
N 58°	07:02	07:55	08:46	15:17	16:09	17:01
56°	06:55	07:44	08:32	15:32	16:19	17:08
54°	06:49	07:35	08:19	15:44	16:28	17:14
52°	06:44	07:27	08:08	15:55	16:36	17:20
50°	06:38	07:20	07:58	16:05	16:44	17:25
45°	06:27	07:04	07:38	16:26	16:59	17:37
N 40°	06:17	06:51	07:21	16:42	17:13	17:47
35°	06:07	06:39	07:07	16:56	17:25	17:56
30°	05:59	06:28	06:55	17:09	17:35	18:05
20°	05:42	06:10	06:34	17:30	17:54	18:21
N 10°	05:26	05:52	06:15	17:48	18:11	18:37
0°	05:09	05:35	05:58	18:05	18:28	18:54
S 10°	04:50	05:17	05:40	18:23	18:46	19:13
20°	04:28	04:57	05:22	18:42	19:06	19:35
30°	03:59	04:32	05:00	19:04	19:31	20:04
35°	03:40	04:17	04:47	19:16	19:46	20:

December 30, 31, 01 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	98°43.5	166°52.1	S19°15.0	142°46.7	S18°00.1	230°40.1	S16°50.1	52°02.7	N15°04.3	Alpheratz	357°30.6	29°15.6
1	113°45.9	181°56.0	14.5	157°47.2	17°59.5	245°42.1	50.2	67°05.3	04.3	Ankaa	353°03.4	-42°08.7
2	128°48.4	196°59.9	14.0	172°47.7	59.0	260°44.1	50.3	82°07.9	04.3	Schedar	349°26.2	56°42.5
3	143°50.9	212°03.8	•• 13.6	187°48.2	•• 58.4	275°46.1	•• 50.5	97°10.4	•• 04.2	Diphda	348°43.3	-17°49.4
4	158°53.3	227°07.7	13.1	202°48.7	57.8	290°48.1	50.6	112°13.0	04.2	Achernar	335°17.3	-57°05.3
5	173°55.8	242°11.5	12.7	217°49.2	57.2	305°50.2	50.7	127°15.5	04.2	Hamal	327°46.5	23°36.4
6	188°58.3	257°15.4	S19°12.2	232°49.7	S17°56.6	320°52.2	S16°50.8	142°18.1	N15°04.2	Polaris	311°26.5	89°23.5
7	204°00.7	272°19.3	11.7	247°50.1	56.1	335°54.2	50.9	157°20.7	04.2	Acamar	315°08.6	-40°11.3
8	219°03.2	287°23.2	11.3	262°50.6	55.5	350°56.2	51.0	172°23.2	04.2	Menkar	314°01.7	4°12.4
9	234°05.7	302°27.1	•• 10.8	277°51.1	•• 54.9	5°58.2	•• 51.1	187°25.8	•• 04.2	Mirfak	308°22.0	49°58.1
10	249°08.1	317°31.0	10.4	292°51.6	54.3	21°00.2	51.3	202°28.3	04.1	Aldebaran	290°34.7	16°34.1
11	264°10.6	332°34.9	09.9	307°52.1	53.7	36°02.2	51.4	217°30.9	04.1	Rigel	280°59.7	-8°10.2
12	279°13.1	347°38.8	S19°09.5	322°52.6	S17°53.2	51°04.3	S16°51.5	232°33.4	N15°04.1	Capella	280°15.5	46°01.6
13	294°15.5	2°42.7	09.0	337°53.1	52.6	66°06.3	51.6	247°36.0	04.1	Bellatrix	278°18.3	6°22.4
14	309°18.0	17°46.7	08.5	352°53.6	52.0	81°08.3	51.7	262°38.6	04.1	Elnath	277°56.4	28°37.8
15	324°20.4	32°50.6	•• 08.1	7°54.1	•• 51.4	96°10.3	•• 51.8	277°41.1	•• 04.1	Anlilam	275°33.3	-1°11.1
16	339°22.9	47°54.5	07.6	22°54.6	50.8	111°12.3	52.0	292°43.7	04.1	Betelgeuse	270°47.4	7°24.6
17	354°25.4	62°58.4	07.2	37°55.1	50.3	126°14.3	52.1	307°46.2	04.0	Canopus	263°50.0	-52°42.8
18	9°27.8	78°02.3	S19°06.7	52°55.6	S17°49.7	141°16.3	S16°52.2	322°48.8	N15°04.0	Sirius	258°22.4	-16°45.6
19	24°30.3	93°06.3	06.3	67°56.1	49.1	156°18.4	52.3	337°51.3	04.0	Adhara	255°02.3	-29°00.9
20	39°32.8	108°10.2	05.8	82°56.5	48.5	171°20.4	52.4	352°53.9	04.0	Procyon	244°46.3	5°08.6
21	54°35.2	123°14.1	•• 05.4	97°57.0	•• 47.9	186°22.4	•• 52.5	7°56.4	•• 04.0	Pollux	243°12.1	27°56.9
22	69°37.7	138°18.1	04.9	112°57.5	47.3	201°24.4	52.6	22°59.0	04.0	Avior	234°12.4	-59°36.4
23	84°40.2	153°22.0	04.4	127°58.0	46.8	216°26.4	52.8	38°01.6	04.0	Suhail	222°42.9	-43°33.3
Mer.pass. 17:22		ν 3.9' d -0.5' m -4.17		ν 0.5' d -0.6' m 1.15		ν 2.0' d 0.1' m -1.79		ν 2.6' d -0.0' m -0.04		Miaplacidus	221°36.4	-69°50.4
Alphard		217°43.6		-8°47.5		Regulus		207°30.0		11°49.0		
Dubhe		193°36.2		61°34.9		Denebola		182°20.9		14°24.0		
Gienah		175°39.4		-17°42.5		Acrux		172°55.3		-63°15.7		
Gacrux		171°46.9		-57°16.7		Alioth		166°09.8		55°47.4		
Spica		158°18.1		-11°19.1		Alkaid		152°49.2		49°09.5		
Hadar		148°30.3		-60°30.8		Menkent		147°53.0		-36°30.9		
Arcturus		145°44.5		19°01.4		Rigel Kent.		139°34.9		-60°57.3		
Kochab		137°21.0		74°01.6		Zuben'ubi		136°51.8		-16°09.9		
Alphecca		126°00.7		26°36.7		Antares		112°11.2		-26°29.7		
Atria		107°02.3		-69°04.6		Sabik		101°58.5		-15°45.6		
Shaula		96°05.3		-37°07.3		Rasalhague		95°55.2		12°32.4		
Eltanin		90°40.9		51°29.2		Kaus Aust.		83°27.6		-34°22.0		
Vega		80°30.9		38°48.8		Nunki		75°43.2		-26°15.4		
Altair		61°56.4		8°57.0		Peacock		53°00.2		-56°38.2		
Deneb		49°23.4		45°23.5		Enif		33°35.1		10°01.0		
Al Na'ir		27°28.4		-46°49.0		Fomalhaut		15°10.4		-29°27.8		
Scheat		13°41.4		28°15.0		Markab		13°26.0		15°22.2		
Dec 30 Sun		SHA		Mer.pass		Venus		68°08.6		12:49		
Mars		44°03.2		14:28		Jupiter		131°56.6		08:36		
Saturn		313°19.3		20:28		Dec 31 Mon		SHA		Mer.pass		
Venus		68°43.3		12:43		Mars		43°15.9		14:28		
Jupiter		131°45.8		08:33		Saturn		313°21.5		20:24		
Jan 01 Tue		SHA		Mer.pass		Venus		69°19.4		12:37		
Mars		42°28.7		14:27		Jupiter		131°35.1		08:30		
Saturn		313°23.6		20:20		Horizontal parallax		Venus:		0.5		
								Mars:		0.1		

DUT1 = UT1-UTC = +0.2141 sec ΔT = TT-UT1 = +68.9699 sec

2029 December 30 to Jan. 01 UT

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	HP
0	179°24.3	S23°09.5		252°56.4	8.2'	S15°22.5	11.1'
1	194°24.0	09.3		267°23.6	8.1'	15°33.5	11.0'
2	209°23.7	09.2		281°50.7	8.0'	15°44.5	10.9'
3	224°23.4	· · 09.0		296°17.7	7.9'	15°55.4	10.8'
4	239°23.1	08.9		310°44.7	7.8'	16°06.2	10.7'
5	254°22.8	08.7		325°11.5	7.7'	16°16.9	10.6'
6	269°22.5	S23°08.5		339°38.2	7.7'	S16°27.5	10.5'
7	284°22.2	08.3		354°04.9	7.6'	16°38.0	10.4'
8	299°21.8	08.2		8°31.5	7.5'	16°48.5	10.3'
9	314°21.5	· · 08.0		22°58.0	7.4'	16°58.8	10.2'
10	329°21.2	07.8		37°24.3	7.3'	17°09.0	10.1'
11	344°20.9	07.7		51°50.7	7.2'	17°19.1	10.0'
12	359°20.6	S23°07.5		66°16.9	7.1'	S17°29.2	9.9'
13	14°20.3	07.3		80°43.0	7.0'	17°39.1	9.8'
14	29°20.0	07.1		95°09.1	7.0'	17°48.9	9.7'
15	44°19.7	· · 07.0		109°35.0	6.9'	17°58.6	9.6'
16	59°19.4	06.8		124°00.9	6.8'	18°08.2	9.5'
17	74°19.1	06.6		138°26.7	6.7'	18°17.7	9.4'
18	89°18.8	S23°06.4		152°52.4	6.6'	S18°27.1	9.3'
19	104°18.5	06.2		167°18.0	6.5'	18°36.3	9.2'
20	119°18.2	06.1		181°43.5	6.4'	18°45.5	9.0'
21	134°17.9	· · 05.9		196°08.9	6.4'	18°54.5	8.9'
22	149°17.6	05.7		210°34.3	6.3'	19°03.4	8.8'
23	164°17.3	05.5		224°59.5	6.2'	19°12.2	8.7'
SD = 16.3' d = -0.2'				SD = 16.2'			
Mon	GHA	Dec		GHA	ν	Dec	HP
0	179°17.0	S23°05.3		239°24.7	6.1'	S19°20.9	8.6'
1	194°16.8	05.2		253°49.8	6.0'	19°29.5	8.4'
2	209°16.5	05.0		268°14.8	5.9'	19°37.9	8.3'
3	224°16.2	· · 04.8		282°39.8	5.8'	19°46.2	8.2'
4	239°15.9	04.6		297°04.6	5.8'	19°54.4	8.1'
5	254°15.6	04.4		311°29.4	5.7'	20°02.4	7.9'
6	269°15.3	S23°04.2		325°54.0	5.6'	S20°10.4	7.8'
7	284°15.0	04.0		340°18.7	5.5'	20°18.1	7.7'
8	299°14.7	03.8		354°43.2	5.4'	20°25.8	7.5'
9	314°14.4	· · 03.7		9°07.6	5.4'	20°33.3	7.4'
10	329°14.1	03.5		23°32.0	5.3'	20°40.7	7.3'
11	344°13.8	03.3		37°56.3	5.2'	20°48.0	7.1'
12	359°13.5	S23°03.1		52°20.5	5.1'	S20°55.1	7.0'
13	14°13.2	02.9		66°44.6	5.1'	21°02.1	6.8'
14	29°12.9	02.7		81°08.7	5.0'	21°08.9	6.7'
15	44°12.6	· · 02.5		95°32.7	4.9'	21°15.6	6.5'
16	59°12.3	02.3		109°56.6	4.8'	21°22.1	6.4'
17	74°12.0	02.1		124°20.4	4.8'	21°28.5	6.3'
18	89°11.7	S23°01.9		138°44.2	4.7'	S21°34.8	6.1'
19	104°11.4	01.7		153°07.9	4.6'	21°40.9	6.0'
20	119°11.1	01.5		167°31.6	4.6'	21°46.9	5.8'
21	134°10.8	· · 01.3		181°55.1	4.5'	21°52.7	5.7'
22	149°10.5	01.1		196°18.7	4.5'	21°58.3	5.5'
23	164°10.2	00.9		210°42.1	4.4'	22°03.8	5.4'
SD = 16.3' d = -0.2'				SD = 16.3'			
Tue	GHA	Dec		GHA	ν	Dec	HP
0	179°09.9	S23°00.7		225°05.5	4.3'	S22°09.2	5.2'
1	194°09.6	00.5		239°28.8	4.3'	22°14.4	5.0'
2	209°09.3	00.3		253°52.1	4.2'	22°19.5	4.9'
3	224°09.0	23°00.1		268°15.3	4.2'	22°24.3	4.7'
4	239°08.7	22°59.9		282°38.5	4.1'	22°29.1	4.6'
5	254°08.4	59.7		297°01.6	4.1'	22°33.7	4.4'
6	269°08.1	S22°59.5		311°24.7	4.0'	S22°38.1	4.3'
7	284°07.8	59.3		325°47.7	4.0'	22°42.3	4.1'
8	299°07.5	59.1		340°10.7	3.9'	22°46.4	3.9'
9	314°07.2	· · 58.8		354°33.6	3.9'	22°50.4	3.8'
10	329°06.9	58.6		8°56.5	3.8'	22°54.2	3.6'
11	344°06.7	58.4		23°19.3	3.8'	22°57.8	3.4'
12	359°06.4	S22°58.2		37°42.1	3.8'	S23°01.2	3.3'
13	14°06.1	58.0		52°04.8	3.7'	23°04.5	3.1'
14	29°05.8	57.8		66°27.6	3.7'	23°07.6	3.0'
15	44°05.5	· · 57.6		80°50.2	3.7'	23°10.6	2.8'
16	59°05.2	57.4		95°12.9	3.6'	23°13.4	2.6'
17	74°04.9	57.1		109°35.5	3.6'	23°16.0	2.5'
18	89°04.6	S22°56.9		123°58.1	3.6'	S23°18.4	2.3'
19	104°04.3	56.7		138°20.7	3.6'	23°20.7	2.1'
20	119°04.0	56.5		152°43.3	3.5'	23°22.8	1.9'
21	134°03.7	· · 56.3		167°05.8	3.5'	23°24.8	1.8'
22	149°03.4	56.1		181°28.3	3.5'	23°26.6	1.6'
23	164°03.1	55.8		195°50.8	3.5'	23°28.2	1.4'
SD = 16.3' d = -0.2'				SD = 16.4'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:25	10:45	■	■	13:22	15:42
N 70°	08:05	09:51	■	■	14:16	16:01
68°	07:50	09:18	■	■	14:49	16:16
66°	07:38	08:53	10:29	13:37	15:13	16:29
64°	07:27	08:35	09:50	14:16	15:32	16:40
62°	07:18	08:19	09:23	14:43	15:48	16:49
60°	07:09	08:06	09:03	15:04	16:01	16:57
N 58°	07:02	07:54	08:46	15:21	16:12	17:04
56°	06:56	07:44	08:31	15:35	16:22	17:11
54°	06:50	07:36	08:19	15:47	16:31	17:17
52°	06:44	07:28	08:08	15:58	16:39	17:22
50°	06:39	07:20	07:59	16:08	16:46	17:27
45°	06:27	07:05	07:38	16:28	17:02	17:39
N 40°	06:17	06:51	07:22	16:45	17:15	17:49
35°	06:08	06:40	07:08	16:59	17:27	17:58
30°	06:00	06:29	06:56	17:11	17:37	18:07
20°	05:43	06:11	06:35	17:31	17:55	18:23
N 10°	05:27	05:54	06:17	17:50	18:12	18:39
0°	05:11	05:37	05:59	18:07	18:29	18:55
S 10°	04:52	05:19	05:42	18:24	18:47	19:14
20°	04:30	04:59	05:23	18:43	19:07	19:36
30°	04:01	04:34	05:02	19:04	19:32	20:05
35°	03:43	04:19	04:49	19:17	19:47	20:23
40°	03:20	04:01	04:34	19:32	20:05	20:46
45°	02:50	03:39	04:16	19:50	20:27	21:16
S 50°	02:06	03:10	03:54	20:12	20:56	22:00
52°	01:39	02:55	03:44	20:22	21:10	22:26
54°	00:58	02:38	03:32	20:34	21:28	23:06
56°	—:—	02:16	03:18	20:48	21:49	—:—
58°	—:—	01:48	03:01	21:04	22:17	—:—
S 60°	—:—	01:03	02:42	21:24	23:01	—:—

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	05:46	■	■	08:43	■	■
N 70°	04:55	■	■	09:36	■	■
68°	04:24	06:47	■	10:08	09:45	■
66°	04:01	06:02	08:07	10:33	10:31	10:32
64°	03:43	05:32	07:17	10:52	11:01	11:23
62°	03:28	05:10	06:46	11:07	11:24	11:54
60°	03:16	04:52	06:23	11:21	11:43	12:18
N 58°	03:05	04:37	06:04	11:32	11:58	12:37
56°	02:56	04:24	05:48	11:42	12:11	12:53
54°	02:48	04:13	05:35	11:51	12:23	13:07
52°	02:41	04:04	05:23	11:59	12:33	13:19
50°	02:34	03:55	05:12	12:06	12:42	13:29
45°	02:20	03:36	04:51	12:21	13:02	13:52
N 40°	02:09	03:22	04:33	12:34	13:17	14:10
35°	01:59	03:09	04:19	12:44	13:31	14:25
30°	01:50	02:58	04:06	12:54	13:42	14:38
20°	01:36	02:39	03:44	13:10	14:02	15:00
N 10°	01:23	02:23	03:25	13:24	14:20	15:19
0°	01:11	02:08	03:08	13:38	14:36	15:37
S 10°	01:00	01:53	02:51	13:51	14:53	15:56
20°	00:47	01:37	02:32	14:06	15:10	16:15
30°	00:33	01:19	02:11	14:22	15:30	16:37
35°	00:25	01:08	01:59	14:32	15:42	16:50
40°	00:16	00:56	01:44	14:43	15:56	17:05
45°	00:05	00:42	01:28	14:56	16:12	17:23
S 50°	· · ·	00:25	01:07	15:12	16:32	17:46
52°	· · ·	00:17	00:57	15:20	16:41	17:56
54°	· · ·	00:08	00:46	15:28	16:52	18:09
56°	23:58	· · ·	00:34	15:38	17:04	18:22
58°	23:47	· · ·	00:19	15:48	17:18	18:39
S 60°	23:34	· · ·	00:02	16:01	17:35	18:58

Day	Sun		Mer. Pass	Moon		Age 25-27 32-13%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
30	02:23	02:37	12:03	07:25	19:53	
31	02:52	03:06	12:03	08:22	20:52	
01	03:20	03:35	12:04	09:23	21:54	